



# Comprehensive **Algebra** VOL-2

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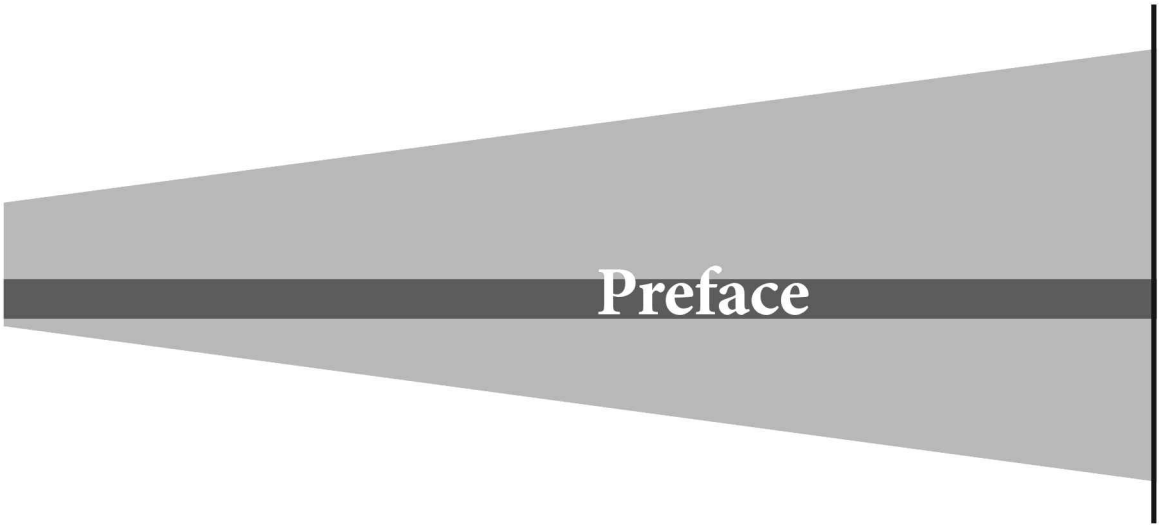
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## Preface

**A**lgebra is a vital topic of mathematics and comprises nearly 25% of questions asked in JEE. A thorough understanding and mastery of this topic is therefore essential for success in JEE and other engineering entrance examinations.

Due to the vastness of the topic, it has been divided into two volumes.

*Comprehensive Algebra Vol-2 for JEE (Main & Advanced)* has been conceived and developed as a textbook-cum-practice book. Starting off with an indepth treatment of the theory portion, it is supplemented with illustrative examples and practice problems (graded according to difficulty levels).

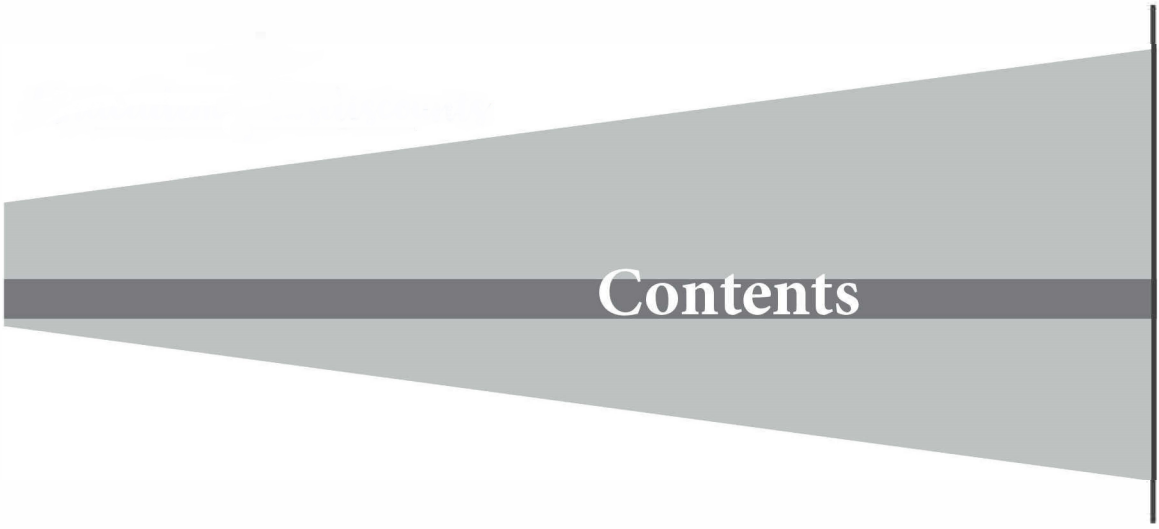
There are 4 chapters in this volume. Each chapter is dealt thoroughly, first with an extensive approach to theory followed by illustrative examples, practice problems, etc.

The collection of carefully selected Objective Questions will help in imparting practice in problem solving. The review exercises comprise problems of moderate level of difficulty while the Target Exercises are tougher problems. For tutors and trainers, the book would be of great help as it has numerous problems graded by difficulty levels which can be given as assignments.

My sincere thanks are due to all my teachers who have been my constant motivation and have given valuable suggestions and recommendations. I am especially thankful to Mr Devendra Kumar, Mr Shashyesh Bhardwaj and Mr Avinash Kumar Singh for their support and cooperation in bringing out this book.

Suggestions for improvement are always welcome and shall be gratefully acknowledged.

**VINAY KUMAR**



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# 1

# Complex Numbers

## 1.1 INTRODUCTION

The quadratic equation  $x^2 + 1 = 0$  has no solution in the real-number system because there is no real number whose square is  $-1$ . New types of numbers, called complex numbers, have been introduced to provide solutions to such equations. Complex numbers are important in solving algebraic equations and they also have an impact on differential and integral calculus.

As early as the 16th century, a symbol  $\sqrt{-1}$  was introduced to provide solutions of the quadratic equation  $x^2 + 1 = 0$ . This symbol, later denoted by the letter  $i$ , was regarded as a fictitious or imaginary number which could be manipulated algebraically like an ordinary real number, except that its square was  $-1$ . Thus, for example, the quadratic polynomial  $x^2 + 1$  was factored by writing  $x^2 + 1 = x^2 - i^2 = (x - i)(x + i)$ , and the solutions of the equation  $x^2 + 1 = 0$  were exhibited as  $x = \pm i$ , without any concern regarding the meaning or validity of such formula.

Complex numbers are formal expressions of the type  $a + bi$ , where  $a$  and  $b$  are real numbers and  $i = \sqrt{-1}$ . The set of all complex numbers is denoted by  $C$ . The term "formal" indicates that we do not attribute any

meaning to them, we do not ask what they might signify and we do not try to relate such expressions to any real thing. We regard them in a strictly formal fashion : to obtain such an expression we have to take two real numbers  $a$  and  $b$ , and, using the auxiliary symbols  $+$  and  $i$ , construct expressions of the kind indicated above. For instance,

$$2 + 3i, 2 + (-3i), 2 + 0i, 0 + 1i, (-\pi) + \sqrt{3}i$$

are called complex numbers. Likewise, we say nothing about the meaning of the auxiliary symbols  $+$  and  $i$ . The  $+$  sign here is not the sign of addition that we are used to. We regard this  $+$  sign as a formal symbol. Its sole purpose is to help in the construction of the formal expressions which we desire.

A complex number is usually denoted by  $z$

$$\text{i.e. } z = a + ib.$$

The number  $a$  is called the real part of the complex number  $z = a + bi$ .

This is denoted by  $R(z)$  i.e.  $R(z) = a$ .

The number  $b$  is called the imaginary part of the complex number  $a + bi$ .

This is denoted by  $I(z)$  i.e.  $I(z) = b$ .

For example, if  $z = 3 - 2i$  then  $R(z) = 3$  and  $I(z) = -2$ .

Note that the imaginary part of a complex number is real.

The complex number  $a + bi$  is called a real number if  $b = 0$ . Examples of real numbers are 1,  $-3$ , 0.

A complex number  $a + bi$  is called an imaginary number if  $b \neq 0$ . Examples of imaginary numbers are

$$2i, 1 - i, \sqrt{7} - i\sqrt{3}.$$

The complex number  $a + bi$  is called a purely imaginary number if  $a = 0$ . Examples of purely imaginary numbers are  $-2i$ ,  $\pi i$ , 0.

Note that the number 0 is both a real and a purely imaginary number, but is not an imaginary number.

We ordinarily make no distinction between the real number  $x$  and the complex number  $x + 0i$ . In particular, we write  $0 = 0 + 0i$ ,  $1 = 1 + 0i$ ,  $-1 = -1 + 0i$ , and so on. Thus, we can think of the complex number system as an extension of the real number system. Hence, the complete number system is

$$\mathbb{N} \subset \mathbb{W} \subset \mathbb{I} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}.$$

Gauss has proved that every polynomial equation of the form  $a_0 + a_1x + a_2x^2 + \dots + a_nx^n = 0$ , where  $a_0, a_1, \dots, a_n$  are arbitrary real numbers, with  $a_n \neq 0$ , has a solution among the complex numbers if  $n \geq 1$ . Moreover, even if the coefficients  $a_0, a_1, \dots, a_n$  are complex, a solution exists in the complex number system.

This fact is known as the fundamental theorem of algebra. It shows that there is no need to construct numbers more general than complex numbers to solve polynomial equations with complex coefficients.

## 1.2 ALGEBRAIC OPERATIONS

The algebraic operations on complex numbers are similar to those on real numbers treating  $i$  as a polynomial, replacing the product  $ii = i^2$  by  $-1$  whenever it occurs.

### 1. Equality in Complex Numbers

The expressions  $a + bi$  and  $c + di$  are considered equal if and only if their real and imaginary parts coincide i.e.  $a = c$  and  $b = d$ . The equality of the expressions  $a + bi$  and  $c + di$  is written as  $a + bi = c + di$ .

$a + bi$  and  $c + di$  are distinct or unequal if at least one of the inequalities  $a \neq c$ ,  $b \neq d$  is true.

Inequalities in complex numbers are not defined. Also there is no validity in saying that a complex number is positive or negative.

### 2. Conjugate Complex Numbers

The numbers  $x + yi$ ,  $x - yi$  are said to be conjugate complex numbers.

If  $z = x + iy$  then its conjugate complex is obtained by changing the sign of its imaginary part and is denoted by  $\bar{z}$ . i.e.  $\bar{z} = x - iy$ .

### 3. Addition of Complex Numbers

The rules for addition and multiplication of complex numbers is that we operate with complex numbers in exactly the same way as with real numbers, treating the symbol  $i$  as itself a number.

Let  $z_1 = x_1 + iy_1$ ,  $z_2 = x_2 + iy_2$ , then  
 $z_1 + z_2 = (x_1 + iy_1) + (x_2 + iy_2) = (x_1 + x_2) + i(y_1 + y_2)$

Thus,  $(2 + 3i) + (1 - \sqrt{3}i) = 3 + (3 - \sqrt{3})i$ .

### Properties of Addition of Complex Numbers

(i) **Closure law** : Closure law for addition holds in the set of complex numbers i.e. if  $z_1$  and  $z_2$  are any two complex numbers then  $z_1 + z_2$  is also a complex number.

Let  $z_1 = x_1 + iy_1$ ,  $z_2 = x_2 + iy_2$ , then  
 $z_1 + z_2 = (x_1 + iy_1) + (x_2 + iy_2)$   
 $= (x_1 + x_2) + i(y_1 + y_2)$   
 $= \text{a complex number}$

(ii) **Commutative law** : Commutative law for addition holds in the set of complex numbers i.e.

$$z_1 + z_2 = z_2 + z_1$$

Let  $z_1 = x_1 + iy_1$  and  $z_2 = x_2 + iy_2$   
 Now,  $z_1 + z_2 = (x_1 + iy_1) + (x_2 + iy_2)$   
 $= (x_1 + x_2) + i(y_1 + y_2)$   
 $= (x_2 + x_1) + i(y_2 + y_1)$   
 $= z_2 + z_1$

(iii) **Associative law** : Associative law holds in the set of complex numbers.

$$\text{i.e. } z_1 + (z_2 + z_3) = (z_1 + z_2) + z_3$$

Let  $z_1 = x_1 + iy_1$ ,  $z_2 = x_2 + iy_2$ ,  $z_3 = x_3 + iy_3$



$$\begin{aligned}
\text{Now, } z_1 + (z_2 + z_3) &= (x_1 + iy_1) + \{(x_2 + iy_2) + (x_3 + iy_3)\} \\
&= (x_1 + iy_1) + \{(x_2 + x_3) + i(y_2 + y_3)\} \\
&= \{(x_1 + (x_2 + x_3)) + i(y_1 + (y_2 + y_3))\} \\
&= \{(x_1 + x_2) + x_3 + i((y_1 + y_2) + y_3)\} \\
&= \{(x_1 + x_2) + i(y_1 + y_2)\} + (x_3 + iy_3) \\
&= \{(x_1 + iy_1) + (x_2 + iy_2)\} + (x_3 + iy_3) \\
&= (z_1 + z_2) + z_3
\end{aligned}$$

- (iv) **Additive Identity** : There exists a complex number  $0 + i0$  such that for every complex number  $x + iy$ ,  $(x + iy) + (0 + i0) = (x + 0) + i(y + 0) = x + iy$ . Here  $0 + i0$  is called the additive identity.
- (v) **Additive Inverse** : For every complex number  $x + iy$  there exists a complex number  $-x - iy$  such that  $(x + iy) + (-x - iy) = (x - x) + i(y - y) = 0 + i0$ .  $-x - iy$  is called the additive inverse of  $x + iy$ .

#### 4. Subtraction of Complex Numbers

$$\begin{aligned}
\text{Let } z_1 &= x_1 + iy_1, z_2 = x_2 + iy_2, \text{ then} \\
z_1 - z_2 &= (x_1 + iy_1) - (x_2 + iy_2) \\
&= (x_1 - x_2) + i(y_1 - y_2)
\end{aligned}$$

$$\text{Thus, } (2 + 3i) - (1 - \sqrt{3}i) = 1 + (3 + \sqrt{3})i.$$

#### 5. Multiplication of Complex Numbers

$$\text{Let } z_1 = x_1 + iy_1, z_2 = x_2 + iy_2$$

$$\begin{aligned}
\text{Now, } z_1 z_2 &= (x_1 + iy_1)(x_2 + iy_2) \\
&= x_1 x_2 + x_1 y_2 i + y_1 x_2 i + y_1 y_2 i^2, \text{ where } i^2 = -1 \\
&= (x_1 x_2 - y_1 y_2) + i(x_1 y_2 + x_2 y_1)
\end{aligned}$$

##### Properties of multiplication of complex numbers

- (i) **Closure property** : Closure law for multiplication holds in the set of complex numbers i.e. if  $z_1$  and  $z_2$  are any two complex numbers then  $z_1 z_2$  is also a complex number.

$$\begin{aligned}
\text{Let } z_1 &= x_1 + iy_1, z_2 = x_2 + iy_2 \\
\text{Now, } z_1 z_2 &= (x_1 + iy_1)(x_2 + iy_2) \\
&= (x_1 x_2 - y_1 y_2) + i(x_1 y_2 + x_2 y_1), \\
&\text{which is complex number.}
\end{aligned}$$

- (ii) **Commutative law** : Commutative law holds in the set of complex numbers i.e.  $z_1 z_2 = z_2 z_1$  for any two complex numbers  $z_1$  and  $z_2$ .

$$\begin{aligned}
\text{Let } z_1 &= x_1 + iy_1, z_2 = x_2 + iy_2 \\
\text{Now } z_1 z_2 &= (x_1 + iy_1)(x_2 + iy_2) \\
&= (x_1 x_2 - y_1 y_2) + i(x_1 y_2 + x_2 y_1)
\end{aligned}$$

$$\begin{aligned}
&= (x_2 x_1 - y_2 y_1) + i(x_2 y_1 + x_1 y_2) \\
&= (x_2 + iy_2)(x_1 + iy_1) = z_2 z_1
\end{aligned}$$

- (iii) **Associative law** : Associative law holds in the set of complex numbers i.e.  $z_1(z_2 z_3) = (z_1 z_2)z_3$  for any three complex numbers.

$$\begin{aligned}
\text{Let } z_1 &= x_1 + iy_1, z_2 = x_2 + iy_2, z_3 = x_3 + iy_3 \\
\text{Now, } z_1(z_2 z_3) &= (x_1 + iy_1)\{(x_2 + iy_2)(x_3 + iy_3)\} \\
&= (x_1 + iy_1)\{(x_2 x_3 - y_2 y_3) + i(x_2 y_3 + x_3 y_2)\} \\
&= x_1(x_2 x_3 - y_2 y_3) - y_1(x_2 y_3 + x_3 y_2) \\
&\quad + i\{x_1(x_2 y_3 + x_3 y_2) + y_1(x_2 x_3 - y_2 y_3)\} \\
&= (x_1 x_2 x_3 - x_1 y_2 y_3 - x_2 y_1 y_3 - x_3 y_1 y_2) \\
&\quad + i(x_1 x_2 y_3 + x_1 x_3 y_2 + x_2 x_3 y_1 - y_1 y_2 y_3)
\end{aligned}$$

Similarly we can show that

$$\begin{aligned}
(z_1 z_2)z_3 &= (x_1 x_2 x_3 - x_1 y_2 y_3 - x_2 y_1 y_3 - x_3 y_1 y_2) \\
&\quad + i(x_1 x_2 y_3 + x_1 x_3 y_2 + x_2 x_3 y_1 - y_1 y_2 y_3) \\
\text{Hence, } z_1(z_2 z_3) &= (z_1 z_2)z_3
\end{aligned}$$

- (iv) **Identity** : There exists complex number  $1 + i.0$  such that for every complex number  $x + iy$ ,  $(x + iy)(1 + i.0) = (x.1 - y.0) + i(x.0 + y.1) = x + iy$ ,  $1 + i.0$  is called the multiplicative identity.
- (v) **Inverse** : For every complex number  $z = x + iy \neq 0 + i.0$ ; there exists a complex number

$$\begin{aligned}
&\frac{x - iy}{x^2 + y^2} \text{ such that } (x + iy) \left( \frac{x - iy}{x^2 + y^2} \right) \\
&= (x + iy) \left( \frac{x}{x^2 + y^2} - i \frac{y}{x^2 + y^2} \right) \\
&= \left( \frac{x^2}{x^2 + y^2} + i \frac{y^2}{x^2 + y^2} \right) \\
&\quad + i \left( \frac{-xy}{x^2 + y^2} + \frac{xy}{x^2 + y^2} \right) \\
&= 1 + i.0 = \text{multiplicative identity}
\end{aligned}$$

$$\begin{aligned}
&\frac{x - iy}{x^2 + y^2} \text{ i.e. } \frac{x}{x^2 + y^2} - i \frac{y}{x^2 + y^2} \\
&\text{is called the multiplicative inverse of } x + iy.
\end{aligned}$$

- (vi) **Distributive law** : Distributive law holds in the set of complex numbers i.e. for any three complex number  $z_1, z_2, z_3$

$$\begin{aligned}
 z_1(z_2 + z_3) &= z_1 z_2 + z_1 z_3 \\
 \text{Let } z_1 &= x_1 + iy_1, z_2 = x_2 + iy_2, z_3 = x_3 + iy_3 \\
 \text{Now, } z_1(z_2 + z_3) &= (x_1 + iy_1) \{(x_2 + iy_2) + (x_3 + iy_3)\} \\
 &= (x_1 + iy_1) \{(x_2 + x_3) + i(y_2 + y_3)\} \\
 &= \{x_1(x_2 + x_3) - y_1(y_2 + y_3) + i\{x_1(y_2 + y_3) + y_1(x_2 + x_3)\}\} \\
 &= \{x_1 x_2 - y_1 y_2 + i(x_1 y_2 + x_2 y_1) + (x_1 x_3 - y_1 y_3) + i(x_1 y_3 + x_3 y_1)\} \\
 &= (x_1 + iy_1)(x_2 + iy_2) + (x_1 + iy_1)(x_3 + iy_3) \\
 &= z_1 z_2 + z_1 z_3.
 \end{aligned}$$



In real numbers if  $a^2 + b^2 = 0$  then we have  $a = 0 = b$ , but in complex numbers,

$$z_1^2 + z_2^2 = 0 \text{ does not imply } z_1 = z_2 = 0.$$

## 6. Division of Complex Numbers

$$\begin{aligned}
 \text{Let } z_1 &= x_1 + iy_1, z_2 = x_2 + iy_2 \\
 \text{Now, } \frac{x_1 + iy_1}{x_2 + iy_2} &= \frac{(x_1 + iy_1)(x_2 - iy_2)}{(x_2 + iy_2)(x_2 - iy_2)} \\
 &= \frac{x_1 x_2 + y_1 y_2 + (x_2 y_1 - x_1 y_2)i}{x_2^2 + y_2^2} \\
 &= \frac{x_1 x_2 + y_1 y_2}{x_2^2 + y_2^2} + \frac{(x_2 y_1 - x_1 y_2)i}{x_2^2 + y_2^2}
 \end{aligned}$$

## 7. Powers of i

Since  $i = \sqrt{-1}$ , we have  $i^2 = -1$ ;  $i^3 = -i$ ;  $i^4 = 1$  etc.

If  $n \in \mathbb{I}$ ,  $i^{4n} = 1$ ,  $i^{4n+1} = i$ ,  $i^{4n+2} = -1$ ,  $i^{4n+3} = -i$ .



The sum of four consecutive integral powers of  $i$  is zero.

For example

$$\sum_{r=1}^{4n+7} i^r = i + i^2 + i^3 + \sum_{r=4}^{4n+7} i^r = i - 1 - i + 0 = -1$$

Thus we see that the complex numbers obey the fundamental laws of algebra of real numbers and their algebra is identical in form, but not in meaning with algebra of real numbers.

► **Example 1.** Express  $\frac{(1+2i)^2}{(2+i)^2}$  in the form  $x + iy$ .

$$\begin{aligned}
 \text{► Solution } \frac{(1+2i)^2}{(2+i)^2} &= \frac{1+4i-4}{4+4i-1} \\
 &= \frac{-3+4i}{3+4i} = \frac{(-3+4i)(3-4i)}{(3+4i)(3-4i)} \\
 &= \frac{-9+16+24i}{9+16} = \frac{7}{25} + i \frac{24}{25}
 \end{aligned}$$

► **Example 2.** Show that a real value of  $x$  will satisfy

$$\frac{1-ix}{1+ix} = a-ib, \text{ if } a^2 + b^2 = 1.$$

► **Solution** We have  $\frac{1-ix}{1+ix} = a-ib$

$$\text{or } ix = \frac{1-(a-ib)}{1+(a-ib)} \quad [\text{by componendo and dividendo}],$$

$$\begin{aligned}
 \text{or } x &= \frac{1-a+ib}{b+i(1+a)} \\
 &= \frac{\{(1-a)+ib\}\{b-i(1+a)\}}{b^2+(1+a)^2} \\
 &= \frac{2b+i(a^2+b^2-1)}{b^2+(1+a)^2}
 \end{aligned}$$

Therefore,  $x$  will be real, if  $a^2 + b^2 = 1$ .

► **Example 3.** If  $f(x) = x^4 - 8x^3 + 4x^2 + 4x + 39$  and  $f(3+2i) = a+ib$ , find  $a : b$

► **Solution** Given,  $x = 3+2i \Rightarrow (x-3)^2 = 4i^2 = -4 \Rightarrow x^2 - 6x + 13 = 0 \quad \dots(1)$

We divide  $f(x)$  by  $x^2 - 6x + 13$ :

$$\begin{array}{r}
 x^2 - 6x + 13 \overline{) x^4 - 8x^3 + 4x^2 + 4x + 39} \\
 \underline{x^4 - 6x^3 + 13x^2} \phantom{+ 4x + 39} \\
 -2x^3 - 9x^2 + 4x + 39 \\
 \underline{-2x^3 + 12x^2 - 26x} \phantom{+ 39} \\
 + \phantom{-} - \phantom{+} +
 \end{array}$$

$$\begin{array}{r}
 -21x^2 + 30x + 39 \\
 -21x^2 + 126x - 273 \\
 + \quad - \quad + \\
 \hline
 -96x + 312
 \end{array}$$

Thus

$$\begin{aligned}
 x^4 - 8x^3 + 4x^2 + 4x + 39 &= (x^2 - 6x + 13)(x^2 - 2x - 21 - 96x + 312) \\
 \therefore f(3+2i) &= 0(x^2 - 2x - 21) - 96(3+2i) + 312 \\
 &= 24 - 192i = a + ib \\
 \therefore a &= 24 \text{ and } b = -192 \\
 \therefore \frac{a}{b} &= \frac{24}{-192} = -\frac{1}{8}
 \end{aligned}$$

### Square root of a complex number

To find the square root of  $a + ib$

assume  $\sqrt{a + ib} = x + iy$ , where  $x$  and  $y$  are real.

Squaring,  $a + ib = x^2 - y^2 + i2xy$ .

Equating real and imaginary parts,

$$a = x^2 - y^2 \quad \dots(1)$$

$$b = 2xy \quad \dots(2)$$

$$\text{Now } (x^2 + y^2)^2 = (x^2 - y^2)^2 + 4x^2y^2 = a^2 + b^2,$$

$$\text{or } x^2 + y^2 = \sqrt{a^2 + b^2} \quad \dots(3)$$

[ $\because$   $x$  and  $y$  are real, the sum of their squares must be positive]

From (1) and (3),

$$x^2 = \frac{\sqrt{a^2 + b^2} + a}{2} \quad \text{or}$$

$$x = \pm \sqrt{\frac{\sqrt{a^2 + b^2} + a}{2}}$$

$$\text{and } y^2 = \frac{\sqrt{a^2 + b^2} - a}{2} \quad \text{or}$$

$$y = \pm \sqrt{\frac{\sqrt{a^2 + b^2} - a}{2}}$$

From (2) we see that

if  $b$  is positive, both  $x$  and  $y$  have the same signs,

and if  $b$  is negative,  $x$  and  $y$  have opposite signs.



- (i)  $\sqrt{a + ib}$  denotes two complex numbers, except when  $b = 0$  and, in that case  $\sqrt{a}$  denotes the one complex number.

For example,  $\sqrt{8 + 6i} = \pm(3 + i)$

while  $\sqrt{4} = 2$  and  $\sqrt{-4} = 2i$

- (ii) The product  $\sqrt{a} \cdot \sqrt{-1}$  is equivalent to the quantity  $\sqrt{-a}$  except when  $a$  is a negative real number.

We calculate

$$\sqrt{-4} \sqrt{-1} = 2i \cdot i$$

$$= -2 \text{ while } \sqrt{(-4)(-1)} = \sqrt{4} = 2.$$

Hence  $\sqrt{a}\sqrt{b} = \sqrt{ab}$  only if at least one of  $a$  or  $b$  is non-negative.

► **Example 4.** Find the square root of  $8 - 15i$ .

► **Solution** Let  $\sqrt{8 - 15i} = \sqrt{a + ib} = x + iy$ , where  $x$  and  $y$  are real.  
Here  $a = 8$  and  $b = -15$ .

$$x = \pm \sqrt{\frac{\sqrt{a^2 + b^2} + a}{2}} = \pm \frac{5}{\sqrt{2}}$$

$$y = \pm \sqrt{\frac{\sqrt{a^2 + b^2} - a}{2}} = \pm \frac{3}{\sqrt{2}}$$

Since  $b = -15$  is negative,  $x$  and  $y$  have opposite signs.

$$\text{Thus, } \sqrt{8 - 15i} = \pm \frac{1}{\sqrt{2}}(5 - 3i)$$

**Alternative**

$$\begin{aligned}
 8 - 15i &= \frac{1}{2} [16 - 30i], \\
 &= \frac{1}{2} [5^2 + (3i)^2 - 2 \cdot 5 \cdot 3i] = \frac{1}{2} (5 - 3i)^2
 \end{aligned}$$

$$\therefore \sqrt{8-15i} = \pm \frac{1}{\sqrt{2}} (5-3i)$$

► **Example 5.** Find the square root of  $7-30\sqrt{-2}$ .

► **Solution**  $7-30\sqrt{-2} = 7-30\sqrt{2}i$   
 $= 5^2 + (3\sqrt{2}i)^2 - 2 \cdot 5 \cdot 3\sqrt{2}i$   
 $= (5-3\sqrt{2}i)^2$   
 $\therefore \sqrt{7-30\sqrt{-2}} = \pm (5-3\sqrt{2}i)$

► **Example 6.** Find the square root of  $x + \sqrt{x^4 + x^2 + 1}i$ .

► **Solution** Let  $E =$   
 $x + \sqrt{(x^2+x+1)(x^2-x+1)}i$   
 $= \frac{(x^2+x+1)}{2} + \frac{(x^2-x+1)}{2}i^2$   
 $+ \sqrt{(x^2+x+1)(x^2-x+1)}i$   
 $= \left( \sqrt{\frac{x^2+x+1}{2}} + \sqrt{\frac{x^2-x+1}{2}}i \right)^2$   
 $\Rightarrow \sqrt{E} = \pm \frac{1}{\sqrt{2}} (\sqrt{x^2+x+1} + \sqrt{x^2-x+1}i)$

### Practice Problems [A]

- If  $a + b\sqrt{2} + (c + d\sqrt{2})i = A + B\sqrt{2} + (C + D\sqrt{2})i$ , where  $a, b, c, d, A, B, C, D$  are real rational numbers, then prove that  $a=A, b=B, c=C, d=D$ .
- Given that  $x, y \in \mathbb{R}$ , solve  $(2+3i)x^2 - (3-2i)y = 2x - 3y + 5i$
- For what real values of  $x$  and  $y$  are the complex numbers  $z_1 = 9y^2 - 4 - 10xi$  and  $z_2 = 8y^2 + 20i^7$  conjugate to each other?
- Express the following complex numbers in the form  $a + bi$ .

- (i)  $1/i$  (ii)  $(2+3i)(3-4i)$   
 (iii)  $i^5 + i^{16}$  (iv)  $\frac{1}{2}(1+i)(1+i^8)$

5. Find the sum and the product of the complex numbers  $z_1$  and  $z_2$ , if :

- (i)  $z_1 = 5 + 4i, z_2 = -2 + 3i$ ;  
 (ii)  $z_1 = -8 - 7i, z_2 = -3i$ ;

6. Find the difference  $z_2 - z_1$  and the quotient  $\frac{z_2}{z_1}$ , if:

- (i)  $z_1 = -1 + \sqrt{3}i, z_2 = -\sqrt{2} + \sqrt{6}i$ ;  
 (ii)  $z_1 = a - \sqrt{b}i, z_2 = a + \sqrt{b}i$

7. Find the real part of the indicated complex number:

- (i)  $z = \frac{(1+2i)^3}{i} + i^{19}$   
 (ii)  $z = \frac{5+2i}{2-5i} - \frac{3-4i}{4+3i} + \frac{1}{i}$

8. Find the imaginary part of the given complex number :  $z = \frac{(1+2i)^2 - (1-i)^3}{(3+2i)^3 - (2+i)^2}$

9. In each case, determine all real numbers  $x$  and  $y$  which satisfy the given relation.

(i)  $x + iy = |x + iy|$

(ii)  $\frac{x+iy}{x-iy} = x-iy$  (iii)  $\sum_{k=0}^{100} i^k = x + iy$

10. Solve the following equation :

$$(i-z)(1+2i) + (1-iz)(3-4i) = 1 + 7i.$$

11. Solve the system of equations :

$$\begin{cases} z_1 + 2z_2 = 1 + i, \\ 3z_1 + iz_2 = 2 - 3i \end{cases}$$

12. Find the square root

(i)  $\sqrt{7-24i}$  (ii)  $\sqrt{1-i\sqrt{3}}$

13. Write the number  $z = \frac{\sqrt{5+12i} + \sqrt{5-12i}}{\sqrt{5+12i} - \sqrt{5-12i}}$  in the algebraic form, provided the real parts of  $\sqrt{5+12i}$  and  $\sqrt{5-12i}$  are negative.

### 1.3 GEOMETRIC REPRESENTATION OF A COMPLEX NUMBER

The notation  $z = x + iy$  is called the algebraic form of representation of a complex number  $z$ . Any complex number  $z = x + iy$  can also be represented by a point on the cartesian plane known as complex plane (Argand plane) by the ordered pair  $(x, y)$ .

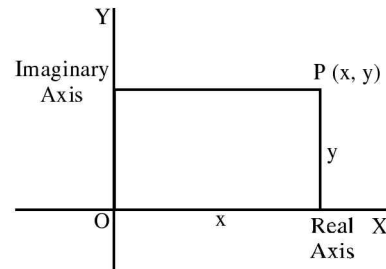
It is well known that it is possible to establish one-to-one correspondence between the set of real numbers and the set of points of a straight line.

When studying complex numbers, we may successfully use geometrical terminology and geometrical concepts, provided one-to-one correspondence between the set of complex numbers and the set of points in the coordinate plane is established. This correspondence can be established in the following way.

Each complex number  $z = x + iy$  is associated with a point  $P(x, y)$  of the coordinate plane, i.e. with a point whose abscissa is equal to the real part of the complex number, and the ordinate to its imaginary part. On the other hand, each point  $P(x, y)$  of the coordinate plane is associated with the complex number  $z = x + iy$ . Obviously, the correspondence thus obtained is one-to-one. It is customary to use the words complex number and point interchangeably. Thus, we refer to the point  $z$  rather than the point corresponding to the complex number  $z$ .

It enables us to interpret complex numbers as points of a certain plane on which the coordinate system is chosen. In this case, the coordinate plane is called the complex plane. The axis of abscissae is called the real axis, since it incorporates the points corresponding to complex numbers  $x + 0i$ , i.e. corresponding to real numbers. The axis of ordinates is called the imaginary axis, situated on it are the

points corresponding to purely imaginary complex numbers  $0 + yi$ .



It is well to note that for complex numbers the concepts of “more than” and “less than” cannot be reasonably defined because these numbers, in contrast to the real numbers, are not located on a straight line, whose points are naturally ordered, but in a plane. For this reason complex numbers as such (not their moduli) can never be connected by an inequality sign.

► **Example 1.** Find the set of points, belonging to the coordinate plane OXY, for which the real part of the complex number  $(1 + i)z^2$  is positive.

► **Solution** Let us represent the number  $z$  in the algebraic form :

$$z = x + iy. \text{ Then}$$

$$z^2 = x^2 - y^2 + i(2xy),$$

$$(1 + i)z^2 = (1 + i)[x^2 - y^2 + i(2xy)]$$

$$= (x^2 - 2xy - y^2) + i(x^2 + 2xy - y^2).$$

By the hypothesis, the real part of the complex number  $(1 + i)z^2$  is positive :

$$x^2 - 2xy - y^2 > 0 \quad \dots(1)$$

Assuming that  $y \neq 0$ , and dividing both sides of the inequality by  $y^2$ , we get an inequality

$$\left(\frac{x}{y}\right)^2 - 2\left(\frac{x}{y}\right) - 1 > 0.$$

Solving this quadratic inequality, we obtain

$$\frac{x}{y} > 1 + \sqrt{2}, \frac{x}{y} < 1 - \sqrt{2} \quad \dots(2)$$

For  $y > 0$ , we can write inequalities (2) in the form

$$y < \frac{1}{1 + \sqrt{2}}x, y < \frac{1}{1 - \sqrt{2}}x.$$

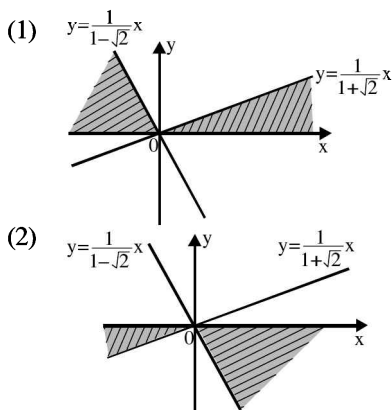
The set of points of the plane OXY which satisfy these conditions is hatched in figure (1).

## 1.8 Comprehensive Algebra Vol-2

For  $y < 0$ , inequalities (2) can be written in the form

$$y > \frac{1}{1+\sqrt{2}}x, y > \frac{1}{1-\sqrt{2}}x$$

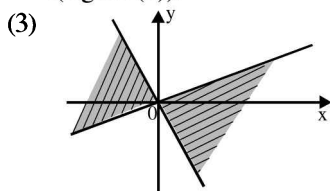
and the set of points of the coordinate plane Oxy which satisfy these inequalities is hatched in figure (2).



For  $y = 0$ , we cannot divide both sides of inequality (1) by  $y^2$ , but for  $y = 0$  inequality (1) itself turns into an inequality  $x^2 > 0$  whose solution is any real number  $x$  different from zero, i.e. any point on the  $x$ -axis except for zero is a solution of inequality (1).

Uniting these three cases, we definitely establish that the required set is constituted by the angles, without their boundaries, whose sides are straight lines

$y = \frac{1}{1+\sqrt{2}}x$  and  $y = \frac{1}{1-\sqrt{2}}x$  containing the  $x$ -axis (figure (3)).



### Practice Problems

[B]

- On the coordinate plane Oxy represent the set of points  $(p, q)$  such that the roots which may be complex of the equation  $x^2 + px + q = 0$  do not exceed unity in absolute value.
- Indicate all points of a complex plane such that
  - $z\alpha$
  - $z + \alpha$

are real numbers ( $\alpha = a + bi$  is a given complex number).

- On the plane OXY find the set of all points whose coordinates satisfy the following conditions: the number  $z^2 + z + 1$  is real and positive.
- Prove that the equation  $z^3 + iz - 1 = 0$  does not have real solutions.
- Let  $z = x + iy$  where  $x, y$  are variables. Prove that
  - If  $2|z - 1| = |z - 2|$ , then  $3(x^2 + y^2) = 4x$ ,
  - If  $|2z - 1| = |z - 2|$ , then  $x^2 + y^2 = 1$ .
- If  $z$  be a complex number and  $t$  be a real parameter. Find the curve represented by
 
$$z = 2 + it + (1 + i)t^2.$$

## 1.4 OTHER REPRESENTATIONS OF A COMPLEX NUMBER

### Modulus of a Complex Number

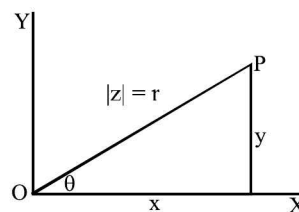
The modulus (or the absolute value) of a complex number  $z = x + iy$  is the distance of the point  $(x, y)$  from origin on the Argand plane.

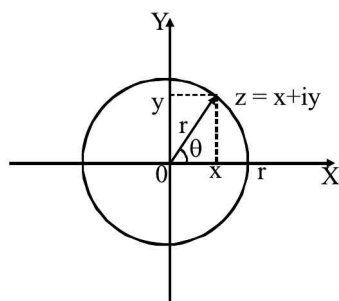
The modulus of the number  $z$  is denoted as  $|z|$  or  $r$ . By Pythagoras theorem, the modulus of the complex number  $z = x + iy$  is obtained as :

$$|z| = r = \sqrt{x^2 + y^2}.$$

For the real number  $z = a + 0i$ , the modulus coincides with the absolute value of the number

$$|z| = |a + 0i| = \sqrt{a^2} = |a|.$$





Complex numbers  $z$ , having one and the same modulus  $|z| = r$ , correspond to the points of the complex plane lying on the circle of radius  $r$  with centre at the point  $z = 0$ . If  $|z| \neq 0$ , then there exists infinite numbers with the given modulus. Only one complex number, namely  $z = 0$ , has a modulus equal to zero.

### Argument of a complex number

Geometrically it is obvious that, in order to single a concrete number out of the set of complex numbers with the given modulus  $|z| = r \neq 0$ , it is sufficient to specify the inclination of  $OP$  w.r.t. the real axis.

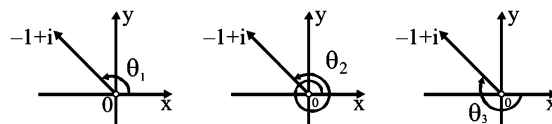
If  $P$  is the point corresponding to a complex number  $z$  ( $z \neq 0$ ) on the argand plane then the angle that  $OP$  makes with the positive direction of the real axis is called the argument (or amplitude) of the complex number. The angle is regarded as positive if it is measured anticlockwise, and negative if it is measured clockwise.

The argument of a complex number  $z = x + iy$  is denoted as  $\arg z$  or  $\text{amp } z$ . By specifying the modulus and argument, a complex number is defined uniquely. For the number  $z = 0$  the argument is not defined; in this case the number is specified only by its modulus.

The argument of a complex number, unlike its modulus, is not defined uniquely.  $\arg z$  is a many-valued function of  $x$  and  $y$ , having an infinity of values, which are angles differing by multiples of  $2\pi$ . For instance, the number  $-1 + i$  has the following arguments :

$$\theta_1 = \frac{3\pi}{4}, \theta_2 = \frac{3\pi}{4} + 2\pi, \theta_3 = \frac{3\pi}{4} - 2\pi \text{ and,}$$

in general, each of the angles  $\theta_k = \frac{3\pi}{4} + 2\pi k, k \in \mathbb{Z}$ .



Any two arguments of a complex number differ from each other by an integral multiple of  $2\pi$ . If  $\theta$  is the argument of a complex number then  $2n\pi + \theta, n \in \mathbb{I}$  will also be the argument of that complex number. The unique value of  $\theta$  such that  $-\pi < \theta \leq \pi$  is called the **principal value** of the argument. Unless otherwise stated,  $\arg z$  implies principal argument.

### Method to find principal argument

Let  $z = x + iy$ .

We first calculate  $\alpha = \tan^{-1} \left| \frac{y}{x} \right|$  and then find in

which quadrant the point  $z$  lies (by observing the signs of  $x$  and  $y$ ). Then the principal argument  $\theta$  of  $z$  is given as :

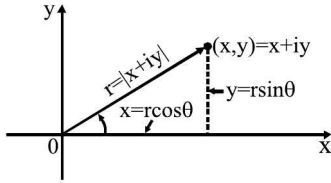
| Quadrant | $\arg(z)$                                                              |
|----------|------------------------------------------------------------------------|
| I        | $\theta = \alpha = \tan^{-1} \left  \frac{y}{x} \right $               |
| II       | $\theta = \pi - \alpha = \pi - \tan^{-1} \left  \frac{y}{x} \right $   |
| III      | $\theta = -\pi + \alpha = -\pi + \tan^{-1} \left  \frac{y}{x} \right $ |
| IV       | $\theta = -\alpha = -\tan^{-1} \left  \frac{y}{x} \right $             |

### Trigonometric or Polar form of representation of a complex number

Let us consider here the trigonometric form of representation of complex numbers.

The real and imaginary parts of the complex number  $z = x + iy$  are expressed by its modulus  $|z| = r$  and argument  $\theta$  in the following manner :

$$\begin{cases} x = r \cos \theta, \\ y = r \sin \theta. \end{cases}$$



This relation is obvious from the figure. Consequently, the complex number  $z$  can be written in the form

$$z = r(\cos \theta + i \sin \theta).$$

Here  $r$  is the modulus of the complex number  $z$ , and  $\theta$  is one of its arguments. This form is called the trigonometric form of representation of a complex number.

In order to pass from the algebraic notation of the complex number  $z = x + yi$  to its trigonometric form, it is sufficient to find its modulus and one of the arguments.

The modulus is determined by the formula

$$r = \sqrt{x^2 + y^2}.$$

Knowing the modulus  $r$ , we find the argument from the system

$$\begin{cases} \cos \theta = \frac{x}{r}, \\ \sin \theta = \frac{y}{r} \end{cases} \quad \dots(1)$$

► **Example 1.** Write the number  $z = 1 - i$  in the trigonometric form.

► **Solution** We first find the modulus

$$|z| = r = \sqrt{1^2 + (-1)^2} = \sqrt{2}.$$

For the argument  $\theta$  we get the system

$$\begin{cases} \cos \theta = \frac{1}{\sqrt{2}} \\ \sin \theta = -\frac{1}{\sqrt{2}} \end{cases}.$$

One of the solutions of this system will be, for instance,  $\theta = \frac{7\pi}{4}$  and, consequently, the given complex number will be written in the trigonometric form as follows:

$$1 - i = \sqrt{2} \left( \cos \frac{7\pi}{4} + i \sin \frac{7\pi}{4} \right).$$

The following

$$1 - i = \sqrt{2} \left( \cos \left( -\frac{\pi}{4} \right) + i \sin \left( -\frac{\pi}{4} \right) \right)$$

is also the trigonometric form of representation of a complex number in which the principal argument has been used.



Here we stress once again that for representation of the number  $x + iy$  in the trigonometric form, all the solutions of system (1) are not needed. It suffices to know one solution.

To solve system (1), in many cases it is convenient to pass over to the equation

$$\tan \theta = \frac{b}{a} \quad \dots(2)$$

This equation follows from system (1) and is obtained as the result of termwise division of the second equation of the system by the first one. Each solution of system (1) is a solution of equation (2). The converse is false and, consequently, equation (2) is not equivalent to system (1). But if the solutions of equation (2) are found, then it is very simple to pick those which satisfy system (1). To this effect it is necessary to determine (i.e. to look) in what quadrant of the complex plane the point  $z = x + iy$  is located. It is always easily found out from the algebraic form of representation of a complex number. Let us illustrate this procedure by the following example.

► **Example 2.** Write the number  $z = -1 - \sqrt{3}i$  in the trigonometric form.

► **Solution** We find the modulus of the given number

$$r = |-1 - \sqrt{3}i| = \sqrt{(-1)^2 + (-\sqrt{3})^2} = 2.$$

The arguments of the number  $z$  must satisfy the equation

$$\tan \theta = \sqrt{3}.$$

The solutions of this equation are

$$\theta = \frac{\pi}{3} + n\pi, \quad n \in \mathbb{Z}.$$



Since the number  $z = -1 - \sqrt{3}i$  is located in the third quadrant of the complex plane, the values

$$\theta = \frac{\pi}{3} + 2k\pi, \quad k \in \mathbb{Z},$$

should be rejected as they do not satisfy system (1).

We may take, for instance,  $\theta = \frac{\pi}{3} - \pi$  as the

argument (principal) of the number  $z = -1 - \sqrt{3}i$ . Thus,

$$-1 - \sqrt{3}i = 2 \left( \cos\left(-\frac{2\pi}{3}\right) + i \sin\left(-\frac{2\pi}{3}\right) \right).$$



Of course, one should not always pass from system (1) to equation (2).

For  $a=0$  either  $\theta = \frac{\pi}{2}$  (if  $b > 0$ ), or  $\theta = -\frac{\pi}{2}$  (if  $b < 0$ ) will be the argument of the given number.

► **Example 3.** Represent the complex number  $z = -3 + i$  in trigonometric form.

► **Solution** By the definition of the modulus of a complex number, we have

$$|z| = \sqrt{(-3)^2 + 1} = \sqrt{10}.$$

Designating the argument of the complex number as  $\theta$ , we get

$$\sin \theta = \frac{1}{\sqrt{10}}, \quad \cos \theta = -\frac{3}{\sqrt{10}},$$

whence it follows that the angle  $\theta$  belongs to the second quarter and is equal to  $\arccos(-3/\sqrt{10})$ . Consequently, the complex number  $z = -3 + i$  has a trigonometric form:

$$z = \sqrt{10} \left[ \cos\left(\arccos\left(-\frac{3}{\sqrt{10}}\right)\right) + i \sin\left(\arccos\left(-\frac{3}{\sqrt{10}}\right)\right) \right]$$

In many cases, the trigonometric form turns out to be more convenient for representation of a complex number than the algebraic form.

## Multiplication and Division of complex numbers represented in the trigonometric form

The trigonometric form of representation of complex numbers turns out to be very convenient for performing the operations of multiplication and division. Let

$$z_1 = r_1 (\cos \theta_1 + i \sin \theta_1) \quad \text{and}$$

$$z_2 = r_2 (\cos \theta_2 + i \sin \theta_2)$$

be two numbers written in the trigonometric form. Then

$$z_1 z_2 = r_1 r_2 \{ (\cos \theta_1 \cos \theta_2 - \sin \theta_1 \sin \theta_2) + i (\sin \theta_1 \cos \theta_2 + \cos \theta_1 \sin \theta_2) \}$$

be two numbers written in the trigonometric form.

Then  $z_1 z_2 = r_1 r_2 (\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2)) \dots (1)$

Thus, the following assertion holds true :

the modulus of a product of two complex numbers is equal to the product of the moduli of these numbers; the sum of the arguments of factors is the argument of a product, or differs from it by  $2\pi$ .

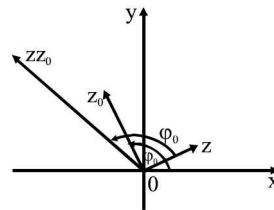
For example, if  $z = z' = -1 + i$ , the modulus of  $zz'$  is

$\sqrt{2} \cdot \sqrt{2} = 2$ . The principal values of the amplitudes of  $z$  and  $z'$  are each  $\frac{3}{4}\pi$ . One of the arguments of

$zz'$  is  $\frac{3}{4}\pi + \frac{3}{4}\pi = \frac{3}{2}\pi$ .

But the principal value of  $\arg(zz')$  is  $-\frac{1}{2}\pi$  and not  $\frac{3}{2}\pi$ .

Formula (1) enables us to give a geometrical interpretation of the operation of multiplication of the complex number  $z$  by the complex number  $z_0$ . The product of the numbers  $z$  and  $z_0$  is a vector which can be obtained by rotating the vector  $z$  through an angle equal to the argument of the number  $z_0$ , and stretching (contracting) it  $|z_0|$  times ( see fig.)



The particular case of multiplying the number  $z$  by the imaginary unit  $i$  means rotation of the vector  $z$  anticlockwise through an angle of  $\frac{\pi}{2}$ .

Now consider division of  $z_1$  by  $z_2$ .  
Multiplying the numerator and denominator by a number  $\cos \theta_2 - i \sin \theta_2$ , we get for the quotient

$$\begin{aligned} \frac{z_1}{z_2} &= \frac{r_1(\cos \theta_1 + i \sin \theta_1)(\cos \theta_2 - i \sin \theta_2)}{r_2(\cos \theta_2 - i \sin \theta_2)(\cos \theta_2 - i \sin \theta_2)} \\ &= \frac{r_1(\cos \theta_1 \cos \theta_2 + \sin \theta_1 \sin \theta_2 + i \sin \theta_1 \cos \theta_2 - i \cos \theta_1 \sin \theta_2)}{r_2(\cos^2 \theta_2 + \sin^2 \theta_2)}, \end{aligned}$$

$$\text{or } \frac{z_1}{z_2} = \frac{r_1}{r_2} (\cos(\theta_1 - \theta_2) + i \sin(\theta_1 - \theta_2)). \dots(2)$$

Hence, the modulus of a quotient of two complex numbers is equal to the quotient of the moduli of these numbers; the difference between the arguments of the dividend and divisor is the argument of the quotient, or differs from it by  $2\pi$ .

More generally

$$\begin{aligned} &r_1(\cos \theta_1 + i \sin \theta_1) \times r_2(\cos \theta_2 + i \sin \theta_2) \\ &\quad \times \dots \times r_n(\cos \theta_n + i \sin \theta_n) \\ &= r_1 r_2 \dots r_n \{ \cos(\theta_1 + \theta_2 + \dots + \theta_n) \\ &\quad + i \sin(\theta_1 + \theta_2 + \dots + \theta_n) \} \dots(3) \end{aligned}$$

A particularly interesting case is that in which

$$r_1 = r_2 = \dots r_n = 1, \theta_1 = \theta_2 = \dots = \theta_n = \theta.$$

We then obtain the equation

$$(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta, \dots(4)$$

where  $n$  is any positive integer; a result known as De Moivre's theorem.

It will sometimes be convenient, for the sake of brevity, to denote  $\cos \theta + i \sin \theta$  by  $\text{Cis } \theta$ : in this notation, De Moivre's theorem is expressed by the equation

$$(\text{Cis } \theta)^n = \text{Cis } n\theta.$$

Again  $(\cos \theta + i \sin \theta)^{-n} = (\cos \theta - i \sin \theta)^n$

$$= \{\cos(-\theta) + i \sin(-\theta)\}^n$$

$$= \cos(-n\theta) + i \sin(-n\theta). \dots(5)$$

Hence, De Moivre's theorem holds for all integral values of  $n$ , positive or negative.

► **Example 4.** Find the product of the numbers

$$z_1 = \sqrt{2} \left( \cos \frac{11\pi}{4} + i \sin \frac{11\pi}{4} \right)$$

$$\text{and } z_2 = \sqrt{8} \left( \cos \frac{3\pi}{8} + i \sin \frac{3\pi}{8} \right).$$

► **Solution** Since  $|z| = \sqrt{2} \cdot |z_2| \sqrt{8}$ , we have

$$|z_1 z_2| = \sqrt{2} \sqrt{8} = 4.$$

The argument of the product  $z_1 z_2$  of the given numbers is the sum  $\theta_1 + \theta_2 = \frac{11\pi}{4} + \frac{3\pi}{8} = \frac{25\pi}{8}$ .  
Consequently,

$$z_1 z_2 = 4(\cos(25\pi/8) + i \sin(25\pi/8)),$$

$$\text{or } z_1 z_2 = 4(\cos(9\pi/8) + i \sin(9\pi/8))$$

► **Example 5.** Express  $z = \frac{-1 + i\sqrt{3}}{1 + i}$  in polar

form and then find the modulus and argument of  $z$ .

Hence deduce the value of  $\cos \frac{5\pi}{12}$ .

► **Solution** Let  $-1 + i\sqrt{3} = r(\cos \theta + i \sin \theta)$   
Equating real and imaginary parts,  $r \cos \theta = -1$ ,  $r \sin \theta = \sqrt{3}$ .

$$\text{Now } r^2 = 1 + 3 = 4, r = 2, \cos \theta = -\frac{1}{2}, \sin \theta =$$

$$\frac{\sqrt{3}}{2}, \text{ or, } \theta = \frac{2\pi}{3} \text{ between } -\pi \text{ and } \pi.$$

$$\text{Consequently, } -1 + i\sqrt{3} = 2 \left( \cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right).$$

$$\text{Similarly, } 1 + i = \sqrt{2} \left( \cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right).$$

$$\therefore z = \frac{2 \left( \cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right)}{\sqrt{2} \left( \cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right)}$$

$$= \sqrt{2} \left( \cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right) \left( \cos \frac{\pi}{4} - i \sin \frac{\pi}{4} \right)$$

$$= \sqrt{2} \left\{ \left( \cos \frac{2\pi}{3} \cos \frac{\pi}{4} + \sin \frac{2\pi}{3} \sin \frac{\pi}{4} \right) \right.$$

$$+ i \left\{ \sin \frac{2\pi}{3} \cos \frac{\pi}{4} - \cos \frac{2\pi}{3} \sin \frac{\pi}{4} \right\}$$

$$= \sqrt{2} \left( \cos \frac{5\pi}{12} + i \sin \frac{5\pi}{12} \right)$$

It is the polar form of  $z$ .

Obviously,  $|z| = \sqrt{2}$  and  $\arg z = \frac{5\pi}{12}$  (principal value).

$$\text{Again, } z = \frac{-1 + i\sqrt{3}}{1 + i} = \frac{(-1 + i\sqrt{3})(1 - i)}{1^2 - i^2}$$

$$= \frac{1}{2} \{ (\sqrt{3} - 1) + i(\sqrt{3} + 1) \}.$$

$$\therefore \sqrt{2} \cos \frac{5\pi}{12} = \frac{\sqrt{3} - 1}{2} \text{ or } \cos \frac{5\pi}{12} = \frac{\sqrt{3} - 1}{2\sqrt{2}}.$$

► **Example 6.** Write the number

$$z = \frac{i - 1}{\cos(\pi/3) + i \sin(\pi/3)} \text{ in trigonometric form.}$$

► **Solution** We introduce the notation  $z_1 = i - 1$  and  $z_2 = \cos(\pi/3) + i \sin(\pi/3)$ . The number  $z_2$  is written in trigonometric form. It is evident that  $|z_2| = 1$  and  $\theta = \pi/3$ . Let us find the modulus and the argument

of the number  $z_1$  :  $|z_1| = \sqrt{2}$  and  $\theta_1 = 3\pi/4$ .

$$z = \frac{z_1}{z_2} = \sqrt{2} \left( \cos \left( \frac{3\pi}{4} - \frac{\pi}{3} \right) + i \sin \left( \frac{3\pi}{4} - \frac{\pi}{3} \right) \right),$$

that is,  $z = \sqrt{2} (\cos(5\pi/12) + i \sin(5\pi/12))$ .

► **Example 7.** Write the complex number

$$z = \frac{\left( \cos \frac{\pi}{3} - i \sin \frac{\pi}{3} \right) (\sqrt{3} + i)}{i - 1}$$

in the trigonometric form.

► **Solution** The number  $z_1 = \cos \frac{\pi}{3} - i \sin \frac{\pi}{3}$  has the modulus equal to 1, the argument  $\theta_1 = -\frac{\pi}{3}$ ; the number  $z_2 = \sqrt{3} + i$  has the modulus 2, the argument  $\theta_2 = \frac{\pi}{6}$ ; the number  $z_3 = i - 1$  has the modulus  $\sqrt{2}$ , the argument  $\theta_3 = \frac{3\pi}{4}$ . Therefore

$$|z| = \frac{|z_1| \cdot |z_2|}{|z_3|} = \frac{1 \cdot 2}{\sqrt{2}} = \sqrt{2}, \text{ the argument}$$

$$\theta = \theta_1 + \theta_2 - \theta_3 = -\frac{\pi}{3} + \frac{\pi}{6} - \frac{3\pi}{4} = -\frac{11}{12}\pi.$$

$$\text{Hence } z = \sqrt{2} \left( \cos \left( -\frac{11}{12}\pi \right) + i \sin \left( -\frac{11}{12}\pi \right) \right).$$

► **Example 8.** Simplify  $(\sqrt{3} - i)^9$ .

► **Solution** The modulus of the number  $z$  is equal to 2, and  $\theta = -\frac{\pi}{6}$  is one of its arguments, therefore the modulus of the number  $z^9$  is equal to 29, and the argument of the number  $z^9$  is equal to

$$9\theta = -\frac{3}{2}\pi. \text{ Consequently,}$$

$$\left( (\sqrt{3} - i)^9 = 2^9 \left( \cos -\frac{3\pi}{2} + i \sin \left( -\frac{3\pi}{2} \right) \right) \right) = 512i.$$

► **Example 9.** Find all the values of  $z$  satisfying the equality  $z^2 + |z| = 0$ .

► **Solution** Let  $r$  be the modulus and  $\theta$  the argument of the complex number  $z$  ( $r \geq 0$ ,  $0 \leq \theta < 2\pi$ ). Then  $z = r(\cos \theta + i \sin \theta)$  and the given equation takes the form  $r^2(\cos 2\theta + i \sin 2\theta) + r = 0$ .

It follows that either  $r = 0$  and  $z = z_1 = 0$  or  $r \cos 2\theta + 1 + i r \sin 2\theta = 0$ , and consequently,

$$\left. \begin{aligned} \sin 2\theta &= 0, \\ r \cos 2\theta + 1 &= 0 \end{aligned} \right\}.$$

The first equation is satisfied by the values  $\theta = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}$ , and since by virtue of the second equation we have  $\cos 2\theta < 0$ , only the values  $\theta = \frac{\pi}{2}$  and  $\theta = \frac{3\pi}{2}$  must be taken. In both cases we find from the second equation the value  $r = 1$ , which yields two more solutions :

$$z_2 = 1 \left( \cos \frac{\pi}{2} + i \sin \frac{\pi}{2} \right) = i,$$

$$z_3 = 1 \left( \cos \frac{3\pi}{2} + i \sin \frac{3\pi}{2} \right) = -i$$

### Exponential Representation

This representation was first given by Euler.

He proved that  $\cos \theta + i \sin \theta$  can be written as  $e^{i\theta}$ .

Thus,  $z = x + iy = r(\cos \theta + i \sin \theta) = re^{i\theta}$ .

Hence, every complex number  $z \neq 0$  can be expressed in the form  $z = re^{i\theta}$  where  $r = |z|$  and  $\theta = \arg(z)$ .

Also  $\bar{z} = x - iy = r(\cos \theta - i \sin \theta) = r(\cos(-\theta) + i \sin(-\theta)) = re^{-i\theta}$ .

Since  $\cos \theta + i \sin \theta = e^{i\theta}$

$\cos \theta - i \sin \theta = e^{-i\theta}$  we have

$$\cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2} \quad \text{and} \quad \sin \theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}$$

which are known as Euler's identities.

► **Example 10.** Represent the complex number

$$z = \frac{\sqrt{3}}{8} - \frac{1}{8}i \quad \text{in the exponential form.}$$

► **Solution** We find the modulus of the number

$$|z| = \sqrt{\frac{3}{64} + \frac{1}{64}} = \frac{1}{4}$$

and one of its arguments

$$\tan \phi = -\frac{1}{\sqrt{3}}, \quad \phi = -\frac{\pi}{6}$$

(since  $z$  is found in the fourth quadrant).

Consequently

$$z = \frac{1}{4}e^{-i\frac{\pi}{6}}.$$

The exponential form of representation of a complex number is more concise as compared with the trigonometric form.

### Complex power of the number $e$

Let us introduce the concept of the complex power of the number  $e$ . The operation of raising the number  $e$  to the complex power  $z = x + yi$  is defined by the formula

$$e^z = e^x(\cos y + i \sin y) \quad \dots(1)$$

Note that  $e^z = e^x$  when  $y = 0$ ; hence this exponential agrees with the usual exponential when  $z$  is real. We find that the law of exponents hold good.

If  $a$  and  $b$  are complex numbers, we have

$$e^a e^b = e^{a+b} \quad \dots(2)$$

**Proof:**

Writing  $a = x + iy$  and  $b = u + iv$ , we have

$$e^a = e^x(\cos y + i \sin y), \quad e^b = e^u(\cos v + i \sin v),$$

$$\text{so } e^a e^b = e^x e^u [\cos y \cos v - \sin y \sin v + i(\cos y \sin v + \sin y \cos v)].$$

Now we use the addition formulae for  $\cos(y + v)$  and  $\sin(y + v)$  and the law of exponents for real exponentials, and we see that the foregoing equation becomes

$$e^a e^b = e^{x+u} [\cos(y + v) + i \sin(y + v)] \quad \dots(3)$$

Since  $a + b = (x + u) + i(y + v)$ , the right member of (3) is  $e^{a+b}$ . This proves (2).

The representation of complex numbers in the exponential form is also useful in connection with multiplication and division of complex numbers. For example, if  $z_1 = r_1 e^{i\theta}$  and  $z_2 = r_2 e^{i\phi}$ , we have

$$z_1 z_2 = r_1 e^{i\theta} r_2 e^{i\phi} = r_1 r_2 e^{i(\theta+\phi)}. \quad \dots(5)$$

When  $z = re^{i\theta}$ , repeated application of (5) gives us the formula

$$z^n = r^n e^{in\theta} = r^n (\cos n\theta + i \sin n\theta), \quad \dots(6)$$

valid for any nonnegative integer  $n$ . This formula is also valid for negative integers  $n$  if we define  $z^{-m}$  to be  $(z^{-1})^m$  when  $m$  is positive integer.

Similarly, we have

$$\frac{z_1}{z_2} = \frac{r_1 e^{i\theta}}{r_2 e^{i\phi}} = \frac{r_1}{r_2} e^{i(\theta-\phi)}. \quad \dots(7)$$

► **Example 11.** Write the complex number

$$z = \frac{(-\sqrt{3} + i) \left( \cos \frac{\pi}{12} - i \sin \frac{\pi}{12} \right)}{1 - i}$$

in the exponential form.

► **Solution** We represent each of the numbers

$$-\sqrt{3} + i, \quad \cos \frac{\pi}{12} - i \sin \frac{\pi}{12}, \quad 1 - i$$

in the exponential form :

$$-\sqrt{3} + i = 2e^{i\frac{5\pi}{6}}; \quad \cos \frac{\pi}{12} - i \sin \frac{\pi}{12} = e^{-i\frac{\pi}{12}};$$

$$1 - i = \sqrt{2}e^{-i\frac{\pi}{4}}$$

Using formulae (1) and (2), we get

$$z = \frac{2e^{i\frac{5\pi}{6} - i\frac{\pi}{12}}}{\sqrt{2}e^{-i\frac{\pi}{4}}} = \sqrt{2}e^{i\left(\frac{5\pi}{6} - \frac{\pi}{12} + \frac{\pi}{4}\right)} = \sqrt{2}e^{i\pi}.$$

► **Example 12.** Raise the number  $e$  to the power  $z$  if

- (a)  $z = 1 + i$  (b)  $z = \frac{\pi}{2}i$ ,  
(c)  $z = \pi i$

► **Solution** By formula (1), we find

- (a)  $e^{1+i} = e(\cos 1 + i \sin 1)$ ,  
(b)  $e^{\frac{\pi}{2}i} = \cos \frac{\pi}{2} + i \sin \frac{\pi}{2} = i$   
(c)  $e^{\pi i} = \cos \pi + i \sin \pi = -1$

► **Example 13.** Find the modulus and arguments of the number  $e^z = e^{x+yi}$ .

► **Solution** The real and imaginary parts of the complex number  $e^z$  are equal to  $e^x \cos y$  and  $e^x \sin y$ , respectively. Consequently,

$$|e^z| = \sqrt{(e^x \cos y)^2 + (e^x \sin y)^2} = \sqrt{e^{2x}} = e^x.$$

The arguments are found from the system :

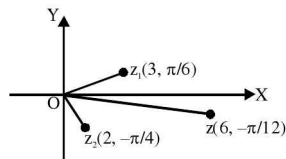
$$\begin{cases} \cos \theta = \frac{e^x \cos y}{e^x} = \cos y, \\ \sin \theta = \frac{e^x \sin y}{e^x} = \sin y. \end{cases}$$

Its solutions are  $\theta = y + 2n\pi$ ,  $n \in \mathbb{Z}$ .

Thus,  $|e^z| = e^x$ ,  $\arg e^z = y + 2n\pi$ ,  $n \in \mathbb{Z}$ .

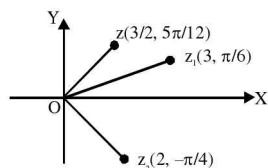
► **Example 14.** If  $z_1 = 3e^{i(\pi/6)}$  and  $z_2 = 2e^{i(-\pi/4)}$ , then plot the complex number  $z_1 z_2$  in the Argand plane.

► **Solution** We have  $z_1 z_2 = 6e^{i(\pi/6 - \pi/4)} = 6e^{i(-\pi/12)}$



► **Example 15.** If  $z_1 = 3e^{i(\pi/6)}$  and  $z_2 = 2e^{i(-\pi/4)}$ , then plot the complex number  $z_1/z_2$  in the Argand plane.

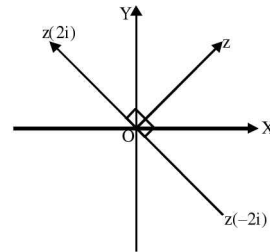
► **Solution** We have  $\frac{z_1}{z_2} = \frac{3}{2}e^{i(\pi/6 + \pi/4)} = \frac{3}{2}e^{i(5\pi/12)}$



It is simpler to add and subtract complex numbers in their  $x + iy$  form, whereas it is simpler to multiply and divide complex numbers in their  $re^{i\theta}$  form.

► **Example 16.** If a complex number  $z$  is multiplied by  $\pm 2i$ , find the modulus and the argument of the complex number thus obtained.

► **Solution** We have  $z(2i) = z \times 2e^{i(\pi/2)}$



which represents a complex number whose modulus

is  $2|z|$  and whose argument is  $\left\{ \arg(z) + \frac{\pi}{2} \right\}$ .

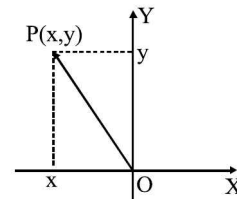
Similarly, we have  $z(-2i) = z \times 2e^{i(-\pi/2)}$

which represents a complex number whose modulus

is  $2|z|$  and whose argument is  $\left\{ \arg(z) - \frac{\pi}{2} \right\}$ .

## Vectorial Representation

No less important and convenient is the interpretation of the complex number  $z = x + iy$  as the vector  $\overrightarrow{OP}$ , i.e. a vector emanating from the origin  $O(0, 0)$  and directed towards the point  $P(x, y)$ . Instead of the vector  $\overrightarrow{OP}$ , we may take any vector equal to it. Obviously, to each vector of the plane with the origin at point  $O(0, 0)$  and terminus at the point  $P(x, y)$  there corresponds a complex number  $x + iy$ , and vice versa. The null vector corresponds to the complex number  $0 + 0i$ .



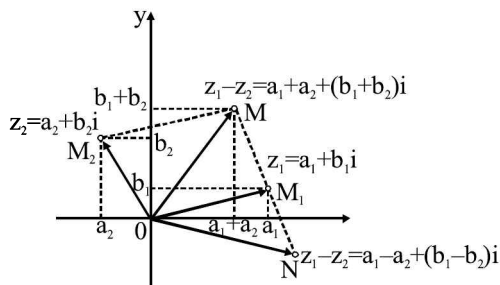
The correspondence established between the set of complex numbers, on one hand, and the sets of points

or vectors in the plane, on the other, enables us to refer to complex numbers as to points or vectors and to speak, for instance, of a vector  $a + bi$  or of a point  $a + bi$ .

Every complex number can be considered as if it is the position vector of that point. If the point P represents the complex number  $z$  then,  $\overrightarrow{OP} = z$  and  $|\overrightarrow{OP}| = |z|$ .

Representation of complex numbers by means of vectors makes it possible to render a simple geometrical interpretation to the operations of addition and subtraction of complex numbers.

As the numbers  $z_1 = a_1 + b_1i$  and  $z_2 = a_2 + b_2i$  are added, their real and imaginary parts are added (Fig.). As the vectors  $\overrightarrow{OM_1}$  and  $\overrightarrow{OM_2}$  corresponding to these numbers are added their coordinates are added. Therefore, with the correspondence established between complex numbers and vectors, to the sum  $z_1 + z_2$  of the numbers  $z_1$  and  $z_2$  there will correspond the vector  $\overrightarrow{OM}$  equal to the sum of the vectors  $\overrightarrow{OM_1}$  and  $\overrightarrow{OM_2}$ . Thus, the sum of complex numbers can be interpreted geometrically as a vector equal to the sum of the vectors corresponding to the complex numbers added. The sum  $z_1 + z_2$  is determined by the parallelogram law. The arrow from the origin to  $z_1 + z_2$  is a diagonal of the parallelogram determined by 0,  $z_1$ , and  $z_2$ ,



We now discuss the geometrical meaning of the difference between two complex numbers. It is established that the difference of the numbers  $z_1 - z_2$  corresponds to the vector  $\overrightarrow{M_2M_1}$  (or  $\overrightarrow{ON}$ ) which

is equal to the difference between the vectors  $\overrightarrow{OM_1}$  and  $\overrightarrow{OM_2}$  (Fig.). The arrow from  $z_1$  to  $z_2$  is parallel to and equal in length to the arrow from 0 to  $z_2 - z_1$ ; the arrow in the opposite direction, from  $z_2$  to  $z_1$ , is related in the same way to  $z_1 - z_2$ .

By definition of the modulus,  $|z_1 - z_2|$  is the length of the vector  $z_1 - z_2$ . This is the vector  $\overrightarrow{ON}$ , and its length gives the distance between the points  $M_1$  and  $M_2$ . Thus, the modulus of the difference between two complex numbers is the distance between the points of the complex plane which correspond to these numbers. This useful geometrical interpretation of the modulus of the difference between two complex numbers enables us to use simple geometric facts in solving certain problems.

### Angle bisector

Given in the complex plane are two points  $z_1$  and  $z_2$ . Let us find the complex numbers corresponding to the points lying on the bisector of the angle formed by the vector  $z_1$  and  $z_2$ .

We know that the diagonal of a rhombus bisects the angle at the vertex. If we add two unit vectors then the resultant is along the angle bisector.

Thus,  $\frac{z_1}{|z_1|}$  is a unit vector along  $z_1$  and similarly

$\frac{z_2}{|z_2|}$  is a unit vector along  $z_2$ .

Their sum  $\frac{z_1}{|z_1|} + \frac{z_2}{|z_2|}$  is a vector along the angle bisector.

A general vector along the angle bisector is

$t \left( \frac{z_1}{|z_1|} + \frac{z_2}{|z_2|} \right)$  where  $t$  is a positive real number.

► **Example 17.** A function  $f$  is defined on the complex number by  $f(z) = (a + bi)z$ , where 'a' and 'b' are positive numbers. This function has the property that the image of each point in the complex plane is equidistant from that point and the origin. Given

that  $|a + bi| = 8$  and that  $b^2 = \frac{u}{v}$  where  $u$  and  $v$  are coprimes. Find the value of  $(u + v)$ .

► **Solution** Given

$$\begin{aligned} |(a+bi)z - z| &= |(a+bi)z| \\ |z(a-1) + bzi| &= |az + bzi| \\ |z||a-1| + |b| &= |z||a+bi| \end{aligned}$$

$$\therefore (a-1)^2 + b^2 = a^2 + b^2$$

$$\therefore a = 1/2$$

$$\text{since } |a+bi| = 8$$

$$a^2 + b^2 = 64$$

$$b^2 = 64 - \frac{1}{4} = \frac{255}{4} = \frac{u}{v}$$

$$\therefore u = 255 \text{ and } v = 4$$

$$\therefore u + v = 259.$$

► **Example 18.** Show that  $\tan^{-1} \frac{1}{5}$  is nearly equal to  $\frac{\pi}{16}$ .

► **Solution** We have

$$(5+i) = \sqrt{26} (\cos\theta + i\sin\theta), \text{ where } \tan\theta = \frac{1}{5}$$

and therefore

$$(5+i)^4 = 676(\cos 4\theta + i\sin 4\theta).$$

$$\text{But } (5+i)^4 = (24+10i) = 476 + 480i;$$

hence we have

$$\cos 4\theta + 476/676, \sin 4\theta = 480/676, \text{ and } \tan 4\theta = 1, \text{ nearly.}$$

$$\therefore 4\theta = \pi/4 \text{ approximately.}$$

► **Example 19.** If  $z_1 = a+ib, z_2 = c+id$  are complex numbers such that  $|z_1| = |z_2| = 1$  and  $R(z_1 \bar{z}_2) = 0$ . Show that  $|w_1| = |w_2| = 1$  and  $R(w_1 \bar{w}_2) = 0$  where  $w_1 = a+ic, w_2 = b+id$  ( $a, b, c, d \in \mathbb{R}$ )

► **Solution**  $|a+ib| = 1 \Rightarrow a+ib$

$$= \cos\theta + i\sin\theta \quad \dots(1)$$

$$|c+id| = 1 \Rightarrow c+id = \cos\phi + i\sin\phi \quad \dots(2)$$

$$\Rightarrow a = \cos\theta, b = \sin\theta, c = \cos\phi, d = \sin\phi$$

$$\text{Now } z_1 \bar{z}_2 = (\cos\theta + i\sin\theta)(\cos\phi - i\sin\phi)$$

$$\Rightarrow R(z_1 \bar{z}_2) = \cos\theta \cos\phi + \sin\theta \sin\phi$$

$$\text{Thus } R(z_1 \bar{z}_2) = 0$$

$$\Rightarrow \cos(\theta - \phi) = 0 \quad \dots(3)$$

$$\Rightarrow \theta - \phi = \frac{\pi}{2} \text{ or } -\frac{\pi}{2} \quad \dots(4)$$

$$\text{Now, } |w_1|^2 = a^2 + c^2 = \cos^2\theta + \cos^2\phi$$

$$= \cos^2\left(\phi \pm \frac{\pi}{2}\right) + \cos^2\phi = \sin^2\phi + \cos^2\phi = 1$$

Thus  $|w_1| = 1$ , similarly  $|w_2| = 1$ .

$$\text{Now } w_1 \bar{w}_2 = (\cos\theta + i\cos\phi)(\sin\theta - i\sin\phi)$$

$$\begin{aligned} \Rightarrow R(w_1 \bar{w}_2) &= (\cos\theta \sin\theta + \cos\phi \sin\phi) \\ &= \frac{1}{2}(\sin 2\theta + \sin 2\phi) \\ &= \sin(\theta + \phi) \cos(\theta - \phi) = 0 \text{ from (3).} \end{aligned}$$

► **Example 20.** Find all complex numbers  $z$  which satisfy the equation,

$$\exp\left(\frac{|z|^2 - |z| + 4}{|z|^2 + 1} \cdot \ln 2\right) = \log_{\sqrt{2}} \left( \left| 3\sqrt{15} + 11i \right| \right).$$

► **Solution** Let  $|z| = r$

$$e^{\frac{r^2 - r + 4}{r^2 + 1} \ln 2} = \log_{\sqrt{2}} 16 = 8$$

$$\Rightarrow 2^{\frac{r^2 - r + 4}{r^2 + 1}} = 2^3$$

$$\Rightarrow r^2 - r + 4 = 3(r^2 + 1)$$

$$\Rightarrow r = \frac{1}{2} \text{ or } r = -1 \text{ (rejected)}$$

$$\text{Hence } z = \frac{1}{2} (\cos \theta + i \sin \theta)$$

► **Example 21.** Solve the equation  $2z = |z| + 2i$ .

► **Solution** We write the complex number  $z$  in the algebraic form  $z = x + iy$ , where  $x$  and  $y$  are real numbers.

$$\Rightarrow 2x + 2iy = \sqrt{x^2 + y^2} + 2i.$$

From the definition of the equality of two complex numbers we get a system of equations for determining  $x$  and  $y$ :

$$2x - \sqrt{x^2 + y^2} = 0.$$

$$2y - 2 = 0.$$

From the second equation we find  $y = 1$ . Substituting  $y = 1$  into the first equation of the system, we get an equation  $2x - \sqrt{x^2 + 1} = 0$  whose solution is  $x = \frac{1}{\sqrt{3}}$ . Thus the complex number  $z = \frac{1}{\sqrt{3}} + i$  is a solution of the given equation.

► **Example 22.** For every real number  $a > 0$  find all complex numbers  $z$  satisfying the equation  $z|z| + az + i = 0$ .

► **Solution** We write the complex number  $z$  in the algebraic form  $z = x + iy$ . Then the equation assumes the form  $(x + iy)\sqrt{x^2 + y^2} + a(x + iy) + i = 0$

$$\Leftrightarrow (x\sqrt{x^2 + y^2} + ax) + i(y\sqrt{x^2 + y^2} + ay + 1) = 0 + 0i.$$

It follows from the definition of the equality of two complex numbers that the last equation is equivalent to a system of two equations :

$$\begin{aligned} x\sqrt{x^2 + y^2} + ax &= 0, \\ y\sqrt{x^2 + y^2} + ay + 1 &= 0, \end{aligned} \quad \dots(1)$$

whose solutions are sought in the set of real numbers. It can be easily seen that the set of solutions of the first equation of system (1) can be found as a union of the sets of solutions of two equations :

$$x = 0, \quad \sqrt{x^2 + y^2} + a = 0.$$

The second equation has no solutions by virtue of the condition  $a > 0$ . Substituting  $x = 0$  into the second equation of system (1), we get an equation for the real number  $y$  :  $y|y| + ay + 1 = 0$ , whose set of solutions is obtained as a union of the sets of solutions of two systems:

$$\begin{aligned} y \geq 0, & \quad y < 0, \\ y^2 + ay + 1 = 0, & \quad -y^2 + ay + 1 = 0. \end{aligned}$$

Bearing in mind the condition  $a > 0$ , it is easy to verify that the first system has no solutions and the second system has a unique solution

$$y = \frac{a - \sqrt{a^2 + 4}}{2}.$$

Thus, the purely imaginary number

$$z = i \frac{a - \sqrt{a^2 + 4}}{2}$$

is a solution of the given equation.

### Practice Problems

[C]

1. Compute the absolute values of the following complex numbers.

(i)  $(1 + i)/(1 - i)$       (ii)  $i^7 + i^{10}$

2. Show that  $(x^2 + y^2)^5 = (x^5 - 10x^3y^2 + 5xy^4)^2 + (5x^4y - 10x^2y^3 + y^5)^2$

3. Represent the complex number  $z = -3 - 4i$  in trigonometric form.

4. Represent the following complex numbers in the algebraic and trigonometric forms :

(i)  $z = \frac{i\left(\cos \frac{5\pi}{3} + i \sin \frac{5\pi}{3}\right)}{\cos \frac{\pi}{6} + i \sin \frac{\pi}{6}}$

(ii)  $z = \frac{\left(\cos \frac{\pi}{3} - i \sin \frac{\pi}{3}\right)\left(\frac{1}{2} + i \frac{\sqrt{3}}{2}\right)}{i}$

5. Represent the following complex numbers in the trigonometric form :

(i)  $z = \frac{5(\cos 100^\circ + i \sin 100^\circ)i}{3(\cos 40^\circ - i \sin 40^\circ)}$

(ii)  $z = \frac{\sin \frac{2\pi}{5} + i\left(1 - \cos \frac{2\pi}{5}\right)}{i - 1}$

6. Simplify:

(i)  $z = (\cos 31^\circ + i \sin 31^\circ)^{-10}$

(ii)  $z = \left(\frac{i^8 + \sqrt{3}i^5}{4}\right)^5$

(iii)  $z = \frac{(2i)^7}{(-\sqrt{2} + i\sqrt{2})^6}$

7. Write the indicated complex numbers in the trigonometric form:

(i)  $z = (\tan 2 - i)^4$

(ii)  $z = \left(\sin \frac{6\pi}{5} + i\left(1 + \cos \frac{6\pi}{5}\right)\right)^5$

8. Represent the following complex numbers in the algebraic form:



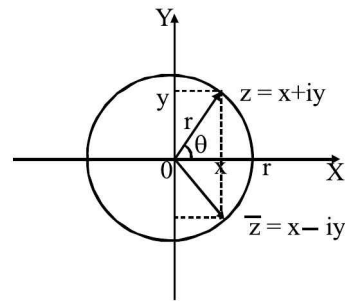
- (i)  $z = e^{2-i}$  (ii)  $z = e^{-\frac{3}{2}\pi i + 12\pi i}$
- (iii)  $z = e^{\frac{3i}{2} + 7 + 3\pi i - \frac{\pi}{2}i}$
9. Express each of the following complex numbers in the form  $a + bi$ .
- (i)  $e^{\pi i/4} - e^{-\pi i/4}$  (ii)  $\frac{1 - e^{\pi i/2}}{1 + e^{\pi i/2}}$
10. In each case, find all real  $x$  and  $y$  that satisfy the given relation
- (i)  $e^{x+iy} = -1$  (ii)  $e^{x+iy} = -1$
- (iii)  $\frac{1+i}{1-i} = xe^{iy}$
11. Represent the following complex numbers in the exponential form :
- (i)  $z = -\sqrt{12} - 2i$
- (ii)  $z = -\cos \frac{\pi}{7} + i \sin \frac{\pi}{7}$
12. Write the following complex numbers in the exponential and algebraic forms :
- (i)  $z = \left( \frac{1}{2} e^{i\frac{\pi}{12}} \right)^{-3}$
- (ii)  $z = \frac{e^{-i\frac{\pi}{3}} (1 + \sqrt{3}i)^7}{i}$
13. The point  $z_1 = 6 + 8i$ ,  $z_2 = 4 - 3i$  are given on a complex plane. Find the complex number corresponding to the bisector of the angle formed by  $z_1$  and  $z_2$  with  $O$ .
14. With the aid of the equality  $(e^{xi})^2 = e^{2xi}$  derive the formulae  
 $\sin 2x = 2 \sin x \cos x$  and  $\cos 2x = \cos^2 x - \sin^2 x$ .
15. Given in the complex plane are two points  $z_1 = 6 + 8i$  and  $z_2 = 4 - 3i$ . Find the complex numbers corresponding to the points lying on the bisector of the angle formed by the vector  $z_1$  and  $z_2$ .
16. Solve the following equations :
- (i)  $|z| - iz = 1 - 2i$  (ii)  $z^2 + 3|z| = 0$
- (iii)  $z^2 + |z|^2 = 0$ .

17. Find the range of real number  $\alpha$  for which the equation  $z + \alpha |z - 1| + 2i = 0$ ,  $z = x + iy$  has a solution. Find the solution.

## 1.5 PROPERTIES OF CONJUGATE

We already know that if  $z = x + iy$  then its conjugate complex is  $\bar{z}$  . i.e.  $\bar{z} = x - iy$ .

Further,  $|\bar{z}| = |z|$  and  $\arg(\bar{z}) = -\arg(z)$ .



The following are some properties of conjugate :

- (i)  $z + \bar{z} = 2 \operatorname{Re}(z)$  ;  
 $z - \bar{z} = 2i \operatorname{Im}(z)$
- If  $z = x + iy$  then  $x = \frac{z + \bar{z}}{2}$ ,  $y = \frac{z - \bar{z}}{2i}$ .
- (ii) If  $z$  is real then  $z = \bar{z}$  and viceversa.
- (iii) If  $z$  is purely imaginary then  $z = -\bar{z}$  and viceversa.
- (iv) Conjugate of the conjugate of a complex number  $z$  is the complex number itself, i.e.,  $\overline{(\bar{z})} = z$ .

*Proof:*

Let  $z = a + ib$ . then  $\bar{z} = a - ib$  and

$$\overline{(\bar{z})} = a + ib = z.$$

- (v) The conjugate of the sum of two complex numbers  $z_1, z_2$  is the sum of their conjugates, i.e.,  $\overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2$ .

*Proof:* Let  $z_1 = a + ib$  and  $z_2 = c + id$ . Then  
 $z_1 + z_2 = (a + ib) + (c + id)$   
 $= (a + c) + i(b + d),$

$$\bar{z}_1 = a - ib, \bar{z}_2 = c - id$$

and  $\bar{z}_1 + \bar{z}_2 = (a - ib) + (c - id) = (a + c) - i(b + d).$

Thus  $\overline{z_1 + z_2} = (a + c) - i(b + d) = \bar{z}_1 + \bar{z}_2$ .

This property extends to more than two complex numbers.

(vi) The conjugate of the difference of two complex numbers  $z_1, z_2$  is the difference of their conjugates, i.e.,  $\overline{z_1 - z_2} = \bar{z}_1 - \bar{z}_2$ .

(vii) The conjugate of the product of two complex numbers  $z_1, z_2$  is the product of their conjugates, i.e.,  $\overline{z_1 z_2} = \bar{z}_1 \bar{z}_2$ .

*Proof:*

Let  $z_1 = a + i b$  and  $z_2 = c + i d$ . Then  
 $z_1 z_2 = (a + i b)(c + i d) = (ac - bd) + i(ad + bc)$   
 $\bar{z}_1 = a - i b, \bar{z}_2 = c - i d$ .

and  $\bar{z}_1 \bar{z}_2 = (a - i b)(c - i d) = (ac - bd) - i(ad + bc)$

Thus  $\overline{z_1 z_2} = (ac - bd) - i(ad + bc) = \bar{z}_1 \bar{z}_2$ .

This property extends to more than two complex numbers.

Specially,  $\overline{(z^n)} = (\bar{z})^n$ .

(viii) The conjugate of the quotient of two complex numbers  $z_1, z_2 (z_2 \neq 0)$  is the quotient of their

conjugates, i.e.,  $\overline{\left(\frac{z_1}{z_2}\right)} = \frac{\bar{z}_1}{\bar{z}_2}$ .

*Proof:* Let  $z_1 = a + i b$  and  $z_2 = c + i d$ . Then

$$\begin{aligned} \frac{z_1}{z_2} &= \frac{a + i b}{c + i d} = \frac{(a + i b)(c - i d)}{(c + i d)(c - i d)} \\ &= \frac{ac + bd}{c^2 + d^2} - i \frac{ad - bc}{c^2 + d^2}. \end{aligned}$$

$$\text{so } \overline{\left(\frac{z_1}{z_2}\right)} = \frac{ac + bd}{c^2 + d^2} + i \frac{ad - bc}{c^2 + d^2}$$

$$\begin{aligned} \text{Hence } \frac{\bar{z}_1}{\bar{z}_2} &= \frac{a - i b}{c - i d} = \frac{(a - i b)(c + i d)}{(c - i d)(c + i d)} \\ &= \frac{ac + bd}{c^2 + d^2} + i \frac{ad - bc}{c^2 + d^2} = \overline{\left(\frac{z_1}{z_2}\right)}. \end{aligned}$$

► **Example 1.** If  $\left|\frac{z}{\bar{z}}\right| = 1 + |z|$ , then prove that  $z$  is a purely imaginary number.

► **Solution** Given that  $\left|\frac{z}{\bar{z}} - \bar{z}\right| = 1 + |z|$

$$\text{Put } z = re^{i\theta} \Rightarrow \bar{z} = re^{-i\theta}$$

$$\Rightarrow \left|\frac{z}{\bar{z}} - \bar{z}\right| = |e^{i\theta} - re^{-i\theta}| = 1 + r$$

$$\Rightarrow (1 - r)^2 \cos^2 \theta + (1 + r)^2 \sin^2 \theta = (1 + r)^2$$

$$\Rightarrow (1 - r)^2 \cos^2 \theta - (1 + r)^2 \cos^2 \theta = 0$$

$$\Rightarrow \cos^2 \theta = 0 \Rightarrow \operatorname{Re}(z) = 0$$

$\Rightarrow z$  is a purely imaginary number.

► **Example 2.** The complex numbers whose real and imaginary parts are integers and satisfy the relation  $z \bar{z}^3 + z^3 \bar{z} = 350$  forms a rectangle on the Argand plane. Find the length of diagonals.

► **Solution** Let  $z = x + iy$

$$z \bar{z}^3 + z^3 \bar{z} = 350$$

$$\Rightarrow z \bar{z} (\bar{z}^2 + z^2) = 2(x^2 + y^2)(x^2 - y^2) = 350$$

$$\Rightarrow (x^2 - y^2)(x^2 + y^2) = 175 = 35.5 = 25.7$$

$$\Rightarrow x^2 + y^2 = 25 \text{ and } x^2 - y^2 = 7$$

$$\Rightarrow x = \pm 4 \text{ \& } y = \pm 3 \text{ since } x \text{ and } y \text{ are integers.}$$

The vertices are  $(\pm 4, \pm 3)$

Hence the length of diagonal is 10.

► **Example 3.** If  $z_1, z_2, z_3$  are the points A, B, C in the Argand Plane then prove that ABC is an equilateral triangle if and only if

$$\frac{1}{z_2 - z_3} + \frac{1}{z_3 - z_1} + \frac{1}{z_1 - z_2} = 0.$$

► **Solution** Let  $z_2 - z_3 = p; z_3 - z_1 = q; z_1 - z_2 = r$

$$\Rightarrow p + q + r = 0 \quad \dots(1)$$

From the given condition,  $pq + qr + rp = 0$

$$\Rightarrow p(q + r) + qr = 0 \Rightarrow p(-p) + qr = 0$$

$$\Rightarrow p^2 = qr \quad (\bar{p})^2 = \bar{q} \bar{r}$$

$$(p\bar{p})^2 = (q\bar{q})(\bar{q}\bar{r}) \quad (p\bar{p})^3 = (p\bar{p})(q\bar{q})(\bar{q}\bar{r})$$

$$(p\bar{p})^3 = (p\bar{p})(q\bar{q})(\bar{p}\bar{q}\bar{r}) \quad \text{Similarly others.}$$

$$\text{Hence } p\bar{p} = q\bar{q} = r\bar{r} \Rightarrow |\bar{p}| = |\bar{q}| = |\bar{r}|.$$

Conversely, if the triangle is equilateral,

$$|z_1 - z_2| = |z_2 - z_3| = |z_3 - z_1|.$$

$$\begin{aligned} \text{Now } \frac{1}{z_1 - z_2} + \frac{1}{z_2 - z_3} + \frac{1}{z_3 - z_1} \\ = \frac{\bar{z}_1 - \bar{z}_2}{|z_1 - z_2|^2} + \frac{\bar{z}_2 - \bar{z}_3}{|z_2 - z_3|^2} + \frac{\bar{z}_3 - \bar{z}_1}{|z_3 - z_1|^2} = 0. \end{aligned}$$

Dividing the second of these equations by the first, we have  $z^2 = -1$  and, since  $z^3 = i$ , it follows that  $z = -i$ . Similarly, we find that if  $w = i$ , then  $z = i$ . Since in this approach we considered consequences from the original system instead of the original system of equations, we have to check to see that the values found do indeed satisfy the original system. This is done by direct substitution, which convinces us that the given system has two solutions :

$$z_1 = -i, w_1 = -i \text{ and } z_2 = i, w_2 = i$$

### The general quadratic equation with complex coefficients

Consider the equation

$$(a + Ai)z^2 + 2(b + Bi)z + (c + Ci) = 0.$$

The formulae for determination of roots, sum of roots and product of roots hold true as before.

Now, we have another way to find the roots. If  $a$  and  $A$  are not both zero we can divide through by  $a + iA$ . Hence we may consider

$$z^2 + (2(b + Bi)z + (c + Ci)) = 0 \quad \dots (1)$$

as the standard form of our equation. Putting  $z = x + yi$ , and equating real and imaginary parts, we obtain a pair of simultaneous equations for  $x$  and  $y$ , viz.

$$x^2 - y^2 + 2(bx - By) + c = 0, \quad 2xy + 2(by + Bx) + C = 0.$$

If we put

$$x + b = \alpha, \quad y + B = \beta, \quad b^2 - B^2 - c = h, \quad 2bB - C = k,$$

these equations becomes  $\alpha^2 - \beta^2 = h, \quad 2\alpha\beta = k$ .

Squaring and adding we obtain

$$\alpha^2 + \beta^2 = \sqrt{(h^2 + k^2)},$$

$$\alpha = \pm \sqrt{\frac{1}{2} \{ \sqrt{(h^2 + k^2)} + h \}},$$

$$\beta = \pm \sqrt{\frac{1}{2} \{ \sqrt{(h^2 + k^2)} - h \}}.$$

We must choose the signs so that  $\alpha\beta$  has the sign of  $k$  : i.e. if  $k$  is positive we must take like signs, if  $k$

is negative unlike signs. From here we can obtain the roots  $z = x + iy$ .

### Condition for equal roots

The two roots can only be equal if both the square roots above vanish, i.e. if  $h = 0, k = 0$ , or if  $c = b^2 - B^2, C = 2bB$ . These conditions are equivalent to the single condition  $c + Ci = (b + Bi)^2$ , which expresses the fact that the left-hand side of (1) is a perfect square.

### Condition for a real root

If  $x^2 + 2(b + Bi)x + (c + Ci) = 0$ , where  $x$  is real, then  $x^2 + 2bx + c = 0, 2Bx + C = 0$ .

Eliminating  $x$  we find that the required condition is  $C^2 - 4bBC + 4cB^2 = 0$ .

### Condition for a purely imaginary root

This is easily found to be  $C^2 - 4bBC - 4b^2c = 0$ .

### Condition for a pair of conjugate complex roots

Since the sum and the product of two conjugate complex numbers are both real,  $b + Bi$  and  $c + Ci$  must both be real, i.e.  $B = 0, C = 0$ . Thus the equation (1) can have a pair of conjugate complex roots only if its coefficients are real. The reader should verify this conclusion by means of the explicit expressions of the roots. Moreover, if  $b^2 \geq c$ , the roots will be real even in this case. Hence for a pair of conjugate imaginary roots we must have

$$B = 0, C = 0, b^2 < c.$$

► **Example 4.** Solve the equation

$$z^2 - (2 + i)z - 1 + 7i = 0.$$

► **Solution** By the formula for determining the roots of a quadratic equation, we have

$$\begin{aligned} z &= \frac{2 + i + \sqrt{(2 + i)^2 + 4(1 - 7i)}}{2} \\ &= \frac{2 + i + \sqrt{7 - 24i}}{2}. \end{aligned}$$

To determine all the values of  $\sqrt{7-24i}$  we set

$$\sqrt{7-24i} = x + yi$$

Then  $7-24i = x^2 + 2xyi - y^2$

and, consequently,  $x$  and  $y$  satisfy the system

$$\begin{cases} x^2 - y^2 = 7, \\ xy = -12, \end{cases}$$

$$z = \frac{2+i + \sqrt{(2+i)^2 + 4(1-7i)}}{2}$$

$$= \frac{2+i + \sqrt{7-24i}}{2}$$

where  $x$  and  $y$  are real numbers. The system has two real solutions :

$x = 4, y = -3$  and  $x = -4, y = 3$ . Therefore,

$$z_1 = \frac{2+i + 4-3i}{2}$$

$$= 3-i, z_2 = \frac{2+i - 4+3i}{2} = -1+2i.$$

► **Example 5.** Let 'A' denote the real part of the

complex number  $z = \frac{19+7i}{9-i} + \frac{20+5i}{7+6i}$

$$\begin{aligned} z &= \frac{2+i + \sqrt{(2+i)^2 + 4(1-7i)}}{2} \\ &= \frac{2+i + \sqrt{7-24i}}{2} \end{aligned}$$

'B' denote the sum of the imaginary parts of the roots of the equation  $z^2 - 8(1-i)z + 63 - 16i = 0$  and 'C' denote the sum of the series,

$1+i+i^2+i^3+\dots+i^{2008}$ , find the value of  $\frac{A-B}{2C}$ .

► **Solution**  $A = R(z)$

$$\begin{aligned} \text{Now } z &= \frac{(19+7i)(9+i)}{82} + \frac{(20+5i)(7-6i)}{85} \\ &= \frac{171+82i-7}{82} + \frac{140-120i+35i+30}{85} \end{aligned}$$

$$= \frac{164+82i}{82} + \frac{170-85i}{85} = 2+i+2-i$$

$$\Rightarrow z = 4+0i \Rightarrow A = 4$$

Let  $\alpha = x + iy, \beta = a + ib$

$$\alpha + \beta = (x+a) + i(y+b) = 8-8i$$

$$\therefore y+b = -8$$

$\therefore$  sum of the imaginary parts of the roots of the equation  $= -8$

$$\therefore B = -8$$

$$S = 1+i+i^2+i^3+\dots+i^{2008}$$

$$= \frac{(1-i^{2009})}{1-i} = \frac{1-i}{1-i} = 1 \Rightarrow C = 1$$

$$\text{Hence } \frac{A-B}{2C} = \frac{4+8}{2} = 6.$$

► **Example 6.** If  $(1-i)$  is a root of the equation,  $z^3 - 2(2-i)z^2 + (4-5i)z - 1 + 3i = 0$ , then find the other two roots.

► **Solution**  $z_1 + z_2 + z_3 = 2(2-i)$

$$\Rightarrow z_2 + z_3 = 3-i \quad (\text{since } z_1 = 1-i) \quad \dots(1)$$

$$\text{again } z_1 z_2 z_3 = 1-3i$$

$$\Rightarrow z_2 z_3 = \frac{1-3i}{1-i} = 2-i \quad \dots(2)$$

From (1) & (2)  $z_2 = 1$  &  $z_3 = 2-i$ .

► **Example 7.** If  $a, b$  are complex numbers and one of the roots of the equation  $x^2 + ax + b = 0$  is real where as the other is purely imaginary, prove that

$$a^2 - \bar{a}^2 = 4b.$$

► **Solution** Let  $\alpha$  be the real and  $i\beta$  be the imaginary roots of the given equation. Then

$$\alpha + i\beta = -a \Rightarrow \alpha - i\beta = -\bar{a}$$

$$\Rightarrow 2\alpha = -(a + \bar{a}) \text{ and } 2i\beta = -(a - \bar{a})$$

$$\text{so that } 4i\alpha\beta = a^2 - \bar{a}^2 \Rightarrow 4b = a^2 - \bar{a}^2.$$

**Alternative**

If one root is real and the other is purely imaginary, their product will be purely imaginary  $\Rightarrow b$  is purely imaginary.

Let  $b = ik$ , so that the equation  $x^2 + ax + ik = 0$  has one real root.

Let it be  $\alpha \Rightarrow \alpha^2 + a\alpha + ik = 0$   
and  $\alpha^2 + \bar{a}\alpha - ik = 0$ .

$$\text{Hence } \frac{\alpha^2}{ika - i\bar{a}k} = \frac{\alpha}{ik + ik} = \frac{1}{\bar{a} - a}$$

$$\Rightarrow \alpha^2 = \frac{ik(a + \bar{a})}{a - \bar{a}} \text{ and } \alpha = \frac{-2ik}{a - \bar{a}}, \text{ so that}$$

$$\frac{ik(a + \bar{a})}{a - \bar{a}} = \frac{-4k^2}{(a - \bar{a})}$$

$$\Rightarrow a^2 - \bar{a}^2 = 4ik = 4b.$$

► **Example 8.** If  $a, b, c$  are complex numbers and one root of the equation  $az^2 + bz + c = 0$  is purely imaginary, then prove that

$$(c\bar{a} - a\bar{c}) + (b\bar{c} + \bar{b}c)(a\bar{b} + \bar{a}b) = 0.$$

► **Solution**  $az^2 + bz + c = 0$  ... (1)

Taking conjugate of both sides of the equation, we get

$$\bar{a}z^2 + \bar{b}\bar{z} + \bar{c} = 0 \quad \dots (2)$$

However, the roots of (1) and (2) are same.

Now we replace  $\bar{z}$  by  $-z$  in (2). We get

$$\bar{a}z^2 - \bar{b}z + \bar{c} = 0 \quad \dots (3)$$

The roots of (3) are negative conjugate of the roots of (2) and hence (1). The negative conjugate of a purely imaginary root is same as itself. Hence, if (1) has a purely imaginary root then it is also a root of (3). The two equations have a common root.

Applying the condition of common root on (1) and (3) we get

$$(c\bar{a} - a\bar{c}) + (b\bar{c} + \bar{b}c)(a\bar{b} + \bar{a}b) = 0.$$

### Practice Problems

[D]

1. Prove that if the sum and product of two complex numbers are both real, then the two numbers must either be real or conjugate.

2. If the complex numbers  $z_1$  and  $z_2$  are such that the product  $z_1 \cdot z_2$  is a real number, are they conjugate complexes?

3. If the complex numbers  $z_1$  and  $z_2$  are such that the sum  $z_1 + z_2$  is a real number, are they conjugate complex?

4. Prove that if the complex numbers  $z_1$  and  $z_2$  with nonzero imaginary parts are such that the product  $z_1 z_2$  and the sum  $z_1 + z_2$  are real numbers, then  $z_1$  and  $z_2$  are conjugate complex numbers.

5. Solve  $z^2 + \bar{z} = 0$ .

6. Solve the following equations :

$$(i) z^2 - (5 + 2i)z + 21 + i = 0$$

$$(ii) z^2 - 8z - 3iz + 13 + 13i = 0$$

7. In the quadratic equation  $x^2 + (p + iq)x + 3i = 0$ ,  $p$  &  $q$  are real. If the sum of the squares of the roots is 8, then find all ordered pair  $(p, q)$ .

8. Given that  $1 + 2i$  is one root of the equation,  $x^4 - 3x^3 + 8x^2 - 7x + 5 = 0$ , find the other three roots.

9. For what real values of  $a$  will the roots of the quadratic equation  $(a + 5)x^2 + 2ax + (a - 1) = 0$  be imaginary?

10. For what  $k$  will the quadratic equation  $x^2 + (3 + i)x + k = 0$  have equal roots?

11. Show that the equation  $z^2 + \alpha z + \beta = 0$ , where  $\alpha$  and  $\beta$  are complex constants will have a real root if

$$(\bar{\alpha} - \alpha)(\alpha\bar{\beta} - \bar{\alpha}\beta) = (\beta - \bar{\beta})^2$$

## 1.6 PROPERTIES OF MODULUS

In the solution of many problems, one requires a geometric interpretation of complex numbers as points in a plane. Here, an essential role is played by the concept of the absolute value (or modulus) of a complex number  $z = x + iy$ , which is defined by

$$|z| = \sqrt{x^2 + y^2} \quad \dots (1)$$

The modulus is clearly a non-negative real number. The modulus is endowed with a simple geometric meaning :  $|z|$  is the distance from the origin to the point associated with the number  $z$ .

The properties of modulus are as follows :

$$(i) |z| = |\bar{z}| = |-z|$$

$$(ii) -|z| \leq R(z) \leq |z|$$

$$-|z| \leq I(z) \leq |z|$$

The real part  $x$  of a complex number  $x + iy$  can never exceed its absolute value since  $x \leq \sqrt{x^2 + y^2}$

$$(iii) z\bar{z} = |z|^2$$

(iv) The absolute value of a product of two complex numbers  $z_1$  and  $z_2$  is equal to the product of the absolute values of the numbers, i.e.,  $|z_1 z_2| = |z_1| |z_2|$

*Proof:* Since  $z\bar{z} = |z|^2$ , we have

$$\begin{aligned} |z_1 z_2|^2 &= (z_1 z_2) \overline{(z_1 z_2)} = (z_1 z_2) (\bar{z}_1 \bar{z}_2) \\ &= (z_1 \bar{z}_1) (z_2 \bar{z}_2) = |z_1|^2 |z_2|^2. \end{aligned}$$

Taking square root with positive sign on both sides, we have

$$|z_1 z_2| = |z_1| |z_2|.$$

Further if  $Z = z_1 z_2 \dots z_n$ , then

$$|Z| = |z_1| \cdot |z_2| \dots |z_n|.$$

Specially,  $|z^n| = |z|^n$ .

(v) The absolute value of a quotient of two complex numbers  $z_1$  and  $z_2$  ( $\neq 0$ ), is equal to the quotient of the absolute values of the dividend and the divisor, i.e.,

$$\left| \frac{z_1}{z_2} \right| = \frac{|z_1|}{|z_2|}$$

*Proof:* Since  $z_1 = \left( \frac{z_1}{z_2} \right) z_2$ , we have

$$|z_1| = \left| \frac{z_1}{z_2} \right| |z_2|.$$

$$\text{Hence } \left| \frac{z_1}{z_2} \right| = \frac{|z_1|}{|z_2|}.$$

(vi) The modulus of the sum of any number of complex numbers is less than or at most equal to the sum of their moduli, i.e.,  $|z_1 + z_2| \leq |z_1| + |z_2|$ . This inequality is called triangle inequality.

*Proof:* We have

$$\begin{aligned} |z_1 + z_2|^2 &= (z_1 + z_2) \overline{(z_1 + z_2)} \\ &= (z_1 + z_2) (\bar{z}_1 + \bar{z}_2) \\ &= z_1 \bar{z}_1 + z_1 \bar{z}_2 + z_2 \bar{z}_1 + z_2 \bar{z}_2 \\ &= |z_1|^2 + |z_2|^2 + 2R(z_1 \bar{z}_2) \\ &= |z_1|^2 + |z_2|^2 + 2R(z_1 \bar{z}_2) \quad \dots(1) \end{aligned}$$

The real part  $x$  of a complex number  $x + iy$  can never exceed its absolute value. Therefore

$$\begin{aligned} R(z_1 \bar{z}_2) &\leq |z_1 \bar{z}_2| = |z_1| |z_2| \\ &= |z_1| |z_2| \quad \dots(2) \end{aligned}$$

Using (2) in (1), we get

$$\begin{aligned} |z_1 + z_2|^2 &\leq |z_1|^2 + |z_2|^2 + 2|z_1| |z_2| \\ &= (|z_1| + |z_2|)^2. \end{aligned}$$

Hence, taking positive square roots, we get

$$|z_1 + z_2| \leq |z_1| + |z_2|.$$

#### Alternative

Let  $z_1$  and  $z_2$  be the two complex numbers and

$$z_1 = r_1(\cos \theta_1 + i \sin \theta_1),$$

$$z_2 = r_2(\cos \theta_2 + i \sin \theta_2).$$

Now  $|z_1 + z_2| = |(r_1 \cos \theta_1 + r_2 \cos \theta_2)$

$$+ i(r_1 \sin \theta_1 + r_2 \sin \theta_2)|$$

$$= \{(r_1 \cos \theta_1 + r_2 \cos \theta_2)^2 + (r_1 \sin \theta_1 + r_2 \sin \theta_2)^2\}^{1/2}$$

$$= \{r_1^2 + r_2^2 + 2r_1 r_2 \cos(\theta_1 - \theta_2)\}^{1/2}$$

$$\therefore |z_1 + z_2| \leq \{r_1^2 + r_2^2 + 2r_1 r_2\}^{1/2}$$

$$[\because \cos(\theta_1 - \theta_2) \leq 1],$$

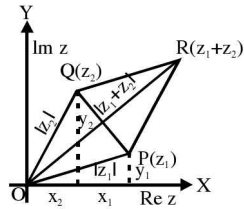
$$\leq r_1 + r_2 \leq |z_1| + |z_2|.$$

Equality holds when  $\theta_1 = \theta_2$ .

#### Geometrical Interpretation

Let points P and Q in the complex plane represent two complex numbers  $z_1 = x_1 + iy_1$  and  $z_2 = x_2 + iy_2$ , respectively, as shown in the figure. Join the origin O with the point P and Q and complete the

parallelogram OPRQ. It is clear from the figure that the coordinates of R are  $(x_1 + x_2, y_1 + y_2)$  and it represents the complex number  $(x_1 + x_2) + i(y_1 + y_2)$ , i.e.,  $z_1 + z_2$ .



The absolute values of  $z_1$ ,  $z_2$  and  $z_1 + z_2$  are geometrically given by

$$|z_1| = OP, |z_2| = OQ = PR \text{ and } |z_1 + z_2| = OR.$$

We know that the sum of any two sides of triangle is greater than the third. Hence, in  $\triangle ORP$ , we have

$OR \leq OP + PR$  which implies  $|z_1 + z_2| \leq |z_1| + |z_2|$ , with the equality holding only when O, P, Q lie on a ray. That is why this inequality for the absolute values of complex numbers is called the triangle inequality.

$$\text{Again, } |z_1 + z_2 + z_3| = |(z_1 + z_2) + z_3| \leq |z_1 + z_2| + |z_3| \leq |z_1| + |z_2| + |z_3|.$$

Proceeding in this way it can be found that

$$|z_1 + z_2 + z_3 + \dots + z_n| \leq |z_1| + |z_2| + |z_3| + \dots + |z_n|.$$

(vii) The modulus of the difference of the two complex numbers is greater than or equal to the difference of the moduli,

$$\text{i.e., } |z_1 - z_2| \geq ||z_1| - |z_2||$$

**Proof:** Let  $|z_1| \geq |z_2|$ . Now

$$z_1 = z_1 - z_2 + z_2, \text{ i.e., } |z_1| \leq |z_1 - z_2| + |z_2|$$

$$\text{Thus } |z_1| - |z_2| \leq |z_1 - z_2|. \quad \dots(1)$$

By interchanging  $z_1$  and  $z_2$ ; we get

$$|z_2| - |z_1| \leq |z_2 - z_1| = |z_1 - z_2| \quad \dots(2)$$

Combining the two inequalities (1) and (2), we get

$$||z_1| - |z_2|| \leq |z_1 - z_2|$$

$$\text{or } |z_1 - z_2| \geq ||z_1| - |z_2||.$$

**Alternative**

Let  $z_1$  and  $z_2$  be the two complex numbers and

$$z_1 = r_1(\cos \theta_1 + i \sin \theta_1),$$

$$z_2 = r_2(\cos \theta_2 + i \sin \theta_2).$$

We are to prove that  $|z_1 - z_2| \geq ||z_1| - |z_2||$ .

$$\begin{aligned} \text{Now } |z_1 - z_2| &= |(r_1 \cos \theta_1 - r_2 \cos \theta_2) \\ &\quad + i(r_1 \sin \theta_1 - r_2 \sin \theta_2)| \end{aligned}$$

$$\begin{aligned} &= \{(r_1 \cos \theta_1 - r_2 \cos \theta_2)^2 \\ &\quad + (r_1 \sin \theta_1 - r_2 \sin \theta_2)^2\}^{1/2} \\ &= \{r_1^2 + r_2^2 - 2r_1 r_2 \cos(\theta_1 - \theta_2)\}^{1/2} \\ &\geq \{r_1^2 + r_2^2 - 2r_1 r_2\}^{1/2} \quad [\because \cos(\theta_1 - \theta_2) \leq 1] \\ &\geq |r_1 - r_2| \end{aligned}$$

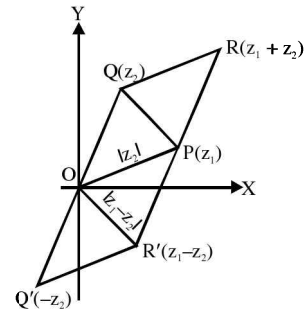
$$\therefore |z_1 - z_2| \geq ||z_1| - |z_2||.$$

### Geometrical Interpretation

In the figure,  $Q'$  represents the complex number  $-z_2$ . Completing the parallelogram  $OQ'R'P$ , we find that

$$OR' = |z_1 - z_2|, OQ' = |-z_2| = |z_2|$$

$$\text{and } Q'R' = OP = |z_1|.$$



The absolute difference of the two sides of a triangle is less than the third.

Hence, in  $\triangle OR'Q'$ , we find that

$$OR' \geq |Q'R' - OQ'|, \text{ which implies}$$

$$|z_1 - z_2| \geq ||z_1| - |z_2||$$

This inequality is also called triangle inequality.



$$(i) \quad ||z_1| - |z_2|| \leq |z_1 \pm z_2| \leq |z_1| + |z_2|$$

$$(ii) \quad |z_1 + z_2|^2 + |z_1 - z_2|^2 = 2 [|z_1|^2 + |z_2|^2]$$

**Proof:**  $|z_1 + z_2|^2 + |z_1 - z_2|^2$

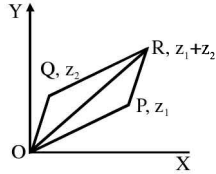
$$= (z_1 + z_2) \overline{(z_1 + z_2)} + (z_1 - z_2) \overline{(z_1 - z_2)}$$

$$= (z_1 + z_2) (\bar{z}_1 + \bar{z}_2) + (z_1 - z_2) (\bar{z}_1 - \bar{z}_2)$$

$$= 2z_1 \bar{z}_1 + 2z_2 \bar{z}_2 = 2(|z_1|^2 + |z_2|^2) \quad \dots(1)$$

### Geometrical Interpretation

Let the points P and Q in the Argand plane XOY represent the complex numbers  $z_1$  and  $z_2$  respectively. The parallelogram OPRQ is completed.



Here  $OP = |z_1|$ ,  $OQ = |z_2|$ ,  $OR = |z_1 + z_2|$ ,  $PQ = |z_1 - z_2|$ . Again  $OP = QR$  and  $OQ = PR$ .

Thus by result (1) the sum of squares of four sides of a parallelogram is equal to the sum of squares of its diagonals.

#### STUDY TIP

$$\begin{aligned} |z_1 \pm z_2|^2 &= (z_1 \pm z_2)(\bar{z}_1 \pm \bar{z}_2) \\ &= |z_1|^2 + |z_2|^2 \pm 2\operatorname{Re}(z_1 \bar{z}_2) \end{aligned}$$

► **Example 1.** If  $|z_1| = 1$ ,  $|z_2| = 2$ ,  $|z_3| = 3$  and  $|9z_1z_2 + 4z_1z_3 + 4z_2z_3 + z_1z_3z_2| = 12$  then find the value of  $|z_1 + z_2 + z_3|$ .

► **Solution**  $z_1 \bar{z}_1 = 1$ ,  $z_2 \bar{z}_2 = 4$ ,  $z_3 \bar{z}_3 = 9$   
 $\therefore |9z_1z_2 + 4z_1z_3 + z_2z_3|$   
 $= |z_1z_2z_3\bar{z}_3 + z_1z_3z_2\bar{z}_2 + z_2z_3z_1\bar{z}_1| = 12$   
 $\therefore |z_1z_2z_3| |\bar{z}_1 + \bar{z}_2 + \bar{z}_3| = 12$   
 $\Rightarrow 6|z_1 + z_2 + z_3| = 12$   
 $\Rightarrow |z_1 + z_2 + z_3| = 2.$

► **Example 2.** If  $\left| \frac{z_1 - z_2}{1 - \bar{z}_1 z_2} \right| < 1$  then prove that either  $|z_1| < 1$  and  $|z_2| < 1$ , or  $|z_1| > 1$  and  $|z_2| > 1$ .

► **Solution**  $\left| \frac{z_1 - z_2}{1 - \bar{z}_1 z_2} \right| < 1 \Leftrightarrow \left| \frac{z_1 - z_2}{1 - \bar{z}_1 z_2} \right|^2 < 1$   
 $\Leftrightarrow \frac{|z_1 - z_2|^2}{|1 - \bar{z}_1 z_2|^2} < 1 \Leftrightarrow |z_1 - z_2|^2 < |1 - \bar{z}_1 z_2|^2$

$$\Leftrightarrow (z_1 - z_2)(\bar{z}_1 - \bar{z}_2) < (1 - \bar{z}_1 z_2)(1 - \bar{z}_1 z_2)$$

$$\Leftrightarrow (z_1 - z_2)(\bar{z}_1 - \bar{z}_2) < (1 - \bar{z}_1 z_2)(1 - \bar{z}_1 z_2)$$

$$\Leftrightarrow z_1 \bar{z}_1 - z_2 \bar{z}_2 - z_1 \bar{z}_2 + z_2 \bar{z}_1 < 1 - \bar{z}_1 z_2 - z_1 \bar{z}_2 + z_1 \bar{z}_1 z_2 \bar{z}_2$$

$$\Leftrightarrow |z_1|^2 + |z_2|^2 < 1 + |z_1|^2 |z_2|^2$$

$$\Leftrightarrow 1 - |z_1|^2 - |z_2|^2 + |z_1|^2 |z_2|^2 > 0$$

$$\Leftrightarrow (1 - |z_1|^2) - |z_2|^2 (1 - |z_1|^2) > 0$$

$$\Leftrightarrow (1 - |z_1|^2)(1 - |z_2|^2) > 0$$

$$\Leftrightarrow (1 + |z_1|)(1 - |z_1|)(1 + |z_1|)(1 - |z_2|) > 0$$

$$\Leftrightarrow (1 - |z_1|)(1 - |z_2|) > 0 \quad [\because 1 + |z_1| > 0, 1 + |z_2| > 0]$$

which implies  $|z_1| < 1$  and  $|z_2| < 1$ , or  $|z_1| > 1$  and  $|z_2| > 1$ .

► **Example 3.** Find the maximum & minimum values of  $|z + 1|$  when  $|z + 3|$ .

► **Solution**  $|z + 3| \leq 3$  denotes set of points on or inside a circle with centre  $(-3, 0)$  and radius 3.  
 $|z + 1|$  denotes the distance of P(z) from A(-1).

The minimum and maximum distances are found along the diameter passing through A.

$|z + 1|_{\min} = 0$  when  $z = -1$  and  $|z + 1|_{\max} = 5$  when  $z = -6$ .

► **Example 4.** Let  $z_1, z_2$  be two complex numbers represented by points on the circle  $|z_1| = 1$  and  $|z_2| = 2$  respectively, then prove that

$$(i) \max |2z_1 + z_2| = 4, \quad (ii) \min |z_1 - z_2| = 1,$$

$$(iii) \left| z_2 + \frac{1}{z_1} \right| \leq 3.$$

► **Solution**  $|2z_1 + z_2| \leq 2|z_1| + |z_2|$   
 $= 2 \times 1 + 2 = 4$

$\therefore$  Maximum value of  $|2z_1 + z_2| = 4$

Clearly  $|z_1 - z_2|$  is least when 0,  $z_1, z_2$  are collinear. Then  $|z_1 - z_2| = 1$



$$\begin{aligned}
 \text{Again } \left| z_2 + \frac{1}{z_1} \right| &\leq |z_2| + \left| \frac{1}{z_1} \right| \\
 &= 2 + \frac{1}{|z_1|} = 2 + \frac{1}{1} = 3 \\
 \Rightarrow \left| z_2 + \frac{1}{z_1} \right| &\leq 3.
 \end{aligned}$$

► **Example 5.** If  $|z - 3i| < \sqrt{5}$  then prove that the complex number  $z$  also satisfies the inequality  $|\operatorname{Im}(z + 1) + 1| < 2\sqrt{5}$ .

► **Solution** Here  $i(z + 1) + 1$  has to be written as the sum of two complex number containing  $z - 3i$  because we have to use  $|z - 3i| < \sqrt{5}$ .

$$\begin{aligned}
 |\operatorname{Im}(z + 1) + 1| &= |\operatorname{Im}(z - 3i) + i - 2i| \\
 &\leq |\operatorname{Im}(z - 3i)| + |-2 + i| \text{ (by triangle inequality)} \\
 &= |\operatorname{Im}(z - 3i)| + \sqrt{5} \\
 &= 1 \cdot |z - 3i| + \sqrt{5} < \sqrt{5} + \sqrt{5} = 2\sqrt{5}.
 \end{aligned}$$

► **Example 6.** If  $n$  is a positive integer, prove that  $|\operatorname{Im}(z^n)| \leq n |\operatorname{Im}(z)| |z|^{n-1}$ .

► **Solution** We have

$$\begin{aligned}
 \left| \frac{\operatorname{Im}(z^n)}{\operatorname{Im} z} \right| &= \left| \frac{z^n - (\bar{z})^n}{z - \bar{z}} \right| \\
 &= |z^{n-1} + z^{n-2} \bar{z} + z^{n-3} (\bar{z})^2 + \dots + (\bar{z})^{n-1}| \\
 &\leq |z|^{n-1} + |z|^{n-2} |\bar{z}| + |z|^{n-3} |\bar{z}|^2 + \dots + |\bar{z}|^{n-1} \\
 &= |z|^{n-1} + |z|^{n-1} + |z|^{n-1} + \dots + |z|^{n-1} = n|z|^{n-1} \\
 \therefore \left| \frac{\operatorname{Im}(z^n)}{\operatorname{Im}(z)} \right| &\leq n |z|^{n-1} \\
 \Rightarrow |\operatorname{Im}(z^n)| &\leq n |\operatorname{Im} z| |z|^{n-1}.
 \end{aligned}$$

► **Example 7.** If  $\left| z + \frac{1}{z} \right| = a$  where  $z$  is a complex number and  $a$  is a real number, find the least and the greatest values of  $|z|$ . Also find  $a$  for which the greatest and the least values of  $|z|$  are equal.

► **Solution** Here  $a \geq 0$  because  $\left| z + \frac{1}{z} \right| \geq 0$ .

Let  $z = r(\cos \theta + i \sin \theta)$  where  $|z| = r$  and  $\arg z = \theta$ .

$$\begin{aligned}
 \therefore \frac{1}{z} &= \frac{1}{r(\cos \theta + i \sin \theta)} = \frac{1}{r} (\cos \theta + i \sin \theta)^{-1} \\
 &= \frac{1}{r} (\cos \theta - i \sin \theta).
 \end{aligned}$$

$$\therefore z + \frac{1}{z} = \left( r + \frac{1}{r} \right) \cos \theta + i \left( r - \frac{1}{r} \right) \sin \theta$$

$$\begin{aligned}
 \therefore \left| z + \frac{1}{z} \right|^2 &= \left( r + \frac{1}{r} \right)^2 \cos^2 \theta + \left( r - \frac{1}{r} \right)^2 \sin^2 \theta \\
 &= r^2 + \frac{1}{r^2} + 2(\cos^2 \theta - \sin^2 \theta) \\
 &= r^2 + \frac{1}{r^2} + 2\cos 2\theta.
 \end{aligned}$$

$$\text{or } a^2 = r^2 + \frac{1}{r^2} + 2 - 4 \sin^2 \theta$$

$$\text{or } \left( r + \frac{1}{r} \right)^2 = a^2 + 4 \sin^2 \theta \leq a^2 + 4$$

$$\text{or } r + \frac{1}{r} \leq \sqrt{a^2 + 4}$$

$$\text{or } r^2 - \sqrt{a^2 + 4} \cdot r + 1 \leq 0$$

$$\text{or } r^2 - 2 \cdot \frac{\sqrt{a^2 + 4}}{2} \cdot r + \frac{a^2 + 4}{4} + 1 - \frac{a^2 + 4}{4} \leq 0$$

$$\text{or } \left( r - \frac{\sqrt{a^2 + 4}}{2} \right)^2 - \left( \frac{a}{2} \right)^2 \leq 0$$

$$\text{or } \left( r - \frac{\sqrt{a^2 + 4} - a}{2} \right) \left( r - \frac{\sqrt{a^2 + 4} + a}{2} \right) \leq 0$$

$$\therefore r \text{ lies between } \frac{\sqrt{a^2 + 4} - a}{2} \text{ and } \frac{\sqrt{a^2 + 4} + a}{2},$$

$$\text{i.e., } \frac{\sqrt{a^2 + 4} - a}{2} \leq r \leq \frac{\sqrt{a^2 + 4} + a}{2}$$

for all real positive  $a$

$$\therefore \text{maximum value of } r = |z|_{\max} = \frac{\sqrt{a^2 + 4} + a}{2}$$

$$\text{minimum value of } r = |z|_{\min} = \frac{\sqrt{a^2 + 4} - a}{2}$$

This is true for all real  $a \geq 0$ .

$$|z|_{\max} = |z|_{\min} \text{ if}$$

$$\frac{\sqrt{a^2 + 4} + a}{2} = \frac{\sqrt{a^2 + 4} - a}{2}, \text{ i.e., } a = 0$$

$$\text{Then } |z|_{\max} = |z|_{\min} = \frac{\sqrt{4}}{2} = 1.$$

► **Example 8.**

$$\text{If } \left| \frac{2iz_1 - z_1 - z_2}{2iz_1 + z_1 + z_2} \right| = \left| \frac{\cos \theta + i \sin \theta}{\cos \theta - i \sin \theta} \right|, \text{ then}$$

prove that  $\frac{z_1}{z_2}$  is real.

► **Solution** The given relation can be written as

$$\left| \frac{z_1 + i \frac{z_1 + z_2}{2}}{z_1 - i \frac{z_1 + z_2}{2}} \right| = 1 \Rightarrow \left| \frac{\frac{2z_1}{z_1 + z_2} + i}{\frac{2z_1}{z_1 + z_2} - i} \right| = 1$$

We have  $|z + i| = |z - i|$  only when  $z$  is real.

$$\Rightarrow \frac{2z_1}{z_1 + z_2} \text{ is real}$$

$$\Rightarrow \frac{2z_1}{z_1 + z_2} = \frac{2\bar{z}_1}{\bar{z}_1 + \bar{z}_2}$$

$$\Rightarrow z_1(\bar{z}_1 + \bar{z}_2) = \bar{z}_1(z_1 + z_2)$$

$$\Rightarrow \frac{z_1}{z_2} = \frac{\bar{z}_1}{\bar{z}_2} \Rightarrow \frac{z_1}{z_2} \text{ is real.}$$

► **Example 9.** If  $|a_r| < 2$  for  $r = 1, 2, 3, \dots, n$  and  $1 + a_1 z + a_2 z^2 + \dots + a_n z^n = 0$ , show that  $z$  does not lie in the interior of the circle  $|z| = 1/3$ .

► **Solution** Given,  $|a_r| < 2$  for  $r = 1, 2, 3, \dots, n$   
Again given,  $1 + a_1 z + a_2 z^2 + \dots + a_n z^n = 0 \dots (1)$

$$\Rightarrow -1 = a_1 z + a_2 z^2 + \dots + a_n z^n \dots (2)$$

$$\Rightarrow |-1| = |a_1 z + a_2 z^2 + \dots + a_n z^n|$$

$$\begin{aligned} \Rightarrow 1 &\leq |a_1 z| + |a_2 z^2| + \dots + |a_n z^n| \\ &= |a_1| |z| + |a_2| |z|^2 + \dots + |a_n| |z|^n \\ &= |a_1| |z| + |a_2| |z|^2 + \dots + |a_n| |z|^n \end{aligned}$$

$$< 2 |z| + 2 |z|^2 + \dots + 2 |z|^n \quad [\text{From (1)}]$$

$$1 < 2 |z| + 2 |z|^2 + \dots + \text{to } \infty \quad [\text{From (2) } |z| \neq 0] \dots (3)$$

**Case I :** When  $|z| < 1$ .

$$\text{From (3), } \frac{2 |z|}{1 - |z|} \Rightarrow 1 - |z| < 2 |z|$$

$$\Rightarrow 3 |z| > 1 \Rightarrow |z| > 1/3$$

**Case II :** When  $|z| \geq 1$ . In this case (3) is obviously satisfied. Hence  $z$  does not lie in the interior of the circle  $|z| = 1/3$ .

**Practice Problems**

**[E]**

1. Prove that for any complex number  $z$ ,

$$|\operatorname{Re}(z)| + |\operatorname{Im}(z)| \leq |z| \sqrt{2}$$

2. Prove that  $|1 - \bar{z}_1 z_2|^2 - |z_1 - z_2|^2 = (1 - |z_1|^2)(1 - |z_2|^2)$

3. If  $\frac{5z_2}{7z_1}$  is a purely imaginary number, then

$$\text{prove that } \left| \frac{2z_1 + 3z_2}{2z_1 - 3z_2} \right| \text{ is equal to } 1.$$

4. If  $z$  is a complex number then show that the minimum value of

(a)  $|z - 1| + |z + 1|$  and

(b)  $|z| + |z - 1| + |2z - 3|$  is each 2.

5. If  $\left| z + \frac{2}{z} \right| = 2$ , then prove that the maximum value

$$\text{of } |z| \text{ is } \sqrt{3} + 1.$$

6. Find the greatest & the least values of,  $|z_1 + z_2|$  if  $z_1 = 24 + 7i$  and  $|z_2| = 6$ .

7. Show that there exist complex numbers  $z$  satisfying  $|z - a| + |z + a| = 2 |c|$  if and only if  $|a| \leq |c|$ . If this condition is fulfilled, what are the smallest and largest values of  $|z|$ ?

8. The only case in which  $|z| + |z'| + \dots = |z + z' + \dots|$  is that in which the numbers  $z, z', \dots$  have all the same amplitude. Prove this.

9. Prove that

$$|z_1 + \sqrt{z_1^2 - z_2^2}| + |z_1 - \sqrt{z_1^2 - z_2^2}| \\ = |z_1 + z_2| + |z_1 - z_2|, \quad z_1, z_2 \in \mathbb{C}.$$

## 1.7 PROPERTIES OF ARGUMENT

- (i)  $\arg(\bar{z}) = -\arg(z)$
- (ii)  $\arg(-z) = \pi + \arg(z) + 2k\pi$ , where  $k$  is a suitable integer to adjust the argument to principal value.
- (iii)  $\arg(z_1 \cdot z_2) = \arg z_1 + \arg z_2 + 2k\pi$ , where  $k$  is a suitable integer.
- (iv)  $\arg(z_1 z_2 \dots z_n) = \theta_1 + \theta_2 + \dots + \theta_n = \arg z_1 + \arg z_2 + \dots + \arg z_n + 2k\pi$ , where  $k$  is a suitable integer.
- (v)  $\arg(z^n) = n \arg(z) + 2k\pi$ , where  $k$  is a suitable integer.
- (vi)  $\arg\left(\frac{z_1}{z_2}\right) = \arg z_1 - \arg z_2 + 2k\pi$ , where  $k$  is a suitable integer.
- (vii) If  $z$  is real then  $\arg(z) = 0$  or  $\pi$ .  
If  $z$  is a positive real number then  $\arg(z) = 0$ .  
If  $z$  is a negative real number then  $\arg(z) = \pi$ .
- (viii) If  $z$  is a purely imaginary number then

$$\arg(z) = \pm \frac{\pi}{2}.$$

► **Example 1.** Find the complex number where the curves  $\arg(z-3i) = 3\pi/4$  &  $\arg(2z+1-2i) = \pi/4$  intersect.

► **Solution**  $\arg(z-3i) = \frac{3\pi}{4}$

$$\Rightarrow \frac{y-3}{x} = \tan \frac{3\pi}{4} \Rightarrow x+y=3 \quad \dots(1)$$

$$\arg(2z+1-2i) = \frac{\pi}{4}$$

$$\text{gives } \frac{2y-2}{2x+1} = \tan \frac{\pi}{4} \Rightarrow 2y-2x=3 \quad \dots(2)$$

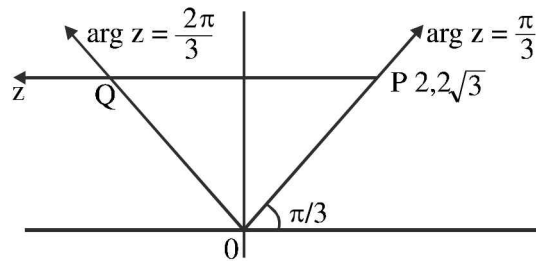
On solving (1) and (2), the point of intersection

$$\text{is } \frac{3}{4} + \frac{9}{4}i.$$

► **Example 2.** Find the area bounded by the curve,  $\arg z = \frac{\pi}{3}$ ,  $\arg z = \frac{2\pi}{3}$  and  $\arg(z-2-2\sqrt{3}i) = \pi$  in the complex plane.

► **Solution** The rays represented by these equations are drawn in the figure. The figure obtained is an equilateral triangle OPQ with side 4.

Hence, the required area =  $\frac{\sqrt{3}}{4} \cdot 16 = 4\sqrt{3}$ .



► **Example 3.** For  $|z-1|=1$ , show that

$$\tan\left(\frac{\arg(z-1)}{2}\right) - \frac{2i}{z} = -i.$$

► **Solution** Here  $z-1 = e^{i\theta}$  so that  $z = 1 + \cos \theta + i \sin \theta$

$$\Rightarrow z = 2 \cos^2 \frac{\theta}{2} + i 2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}$$

$$\Rightarrow z = 2 \cos \frac{\theta}{2} e^{i\theta/2}.$$

Hence

$$\tan\left(\frac{\arg(z-1)}{2}\right) - \frac{2i}{z} = \tan \frac{\theta}{2} - \frac{i}{\cos \theta/2} e^{-i\theta/2}$$

$$\Rightarrow \tan \frac{\theta}{2} - i \frac{\left(\cos \frac{\theta}{2} - i \sin \frac{\theta}{2}\right)}{\cos \frac{\theta}{2}} = -i.$$

► **Example 4.** If  $\arg(z^{3/8}) = \frac{1}{2} \arg(z^2 + \bar{z}z^{1/2})$ , then

show that  $|z| = 1$  or  $z = \bar{z}$ .

► **Solution** We have

$$\arg(z^{3/8}) = \frac{1}{2} \arg(z^2 + \bar{z}z^{1/2})$$

$$\Rightarrow 2 \arg(z^{3/8}) = \arg(z^2 + \bar{z}z^{1/2})$$

$$\Rightarrow \arg(z^{3/4}) - \arg(z^2 + \bar{z}z^{1/2}) = 0$$

$$[\because 2\arg(z) = \arg(z^2)]$$

$$\Rightarrow \arg\left(\frac{z^2 + \bar{z}z^{1/2}}{z^{3/4}}\right) = 0$$

$$\Rightarrow z^{5/4} + \frac{\bar{z}}{z^{1/4}} = \overline{\left(z^{5/4} + \frac{\bar{z}}{z^{1/4}}\right)}$$

$$\Rightarrow z^{5/4} + \frac{\bar{z}}{z^{1/4}} = (\bar{z})^{5/4} + \frac{z}{(\bar{z})^{1/4}}$$

$$\Rightarrow z^{5/4} + \frac{\bar{z}(\bar{z})^{1/4}}{|z|^{1/2}} = (\bar{z})^{5/4} + \frac{z z^{1/4}}{|z|^{1/2}}$$

$$\Rightarrow z^{5/4} - (\bar{z})^{5/4} = \frac{z^{5/4} - (\bar{z})^{5/4}}{|z|^{1/2}}$$

$$\Rightarrow \{z^{5/4} - (\bar{z})^{5/4}\} \left(1 - \frac{1}{|z|^{1/2}}\right) = 0$$

gives  $z = \bar{z}$

or  $|z|^{1/2} = 1$ , i.e.  $|z| = 1$ .

► **Example 5.** If  $|z| \leq 1$ ,  $|w| \leq 1$ , show that

$$|z - w|^2 \leq (|z| - |w|)^2 + (\text{Arg}z - \text{Arg}w)^2$$

► **Solution** Let  $|z| = r_1$ ,  $|w| = r_2$ ,  $\text{Arg}z = \theta_1$  and  $\text{Arg}w = \theta_2$

Then  $z = r_1 (\cos\theta_1 + i \sin\theta_1)$  and

$$w = r_2 (\cos\theta_2 + i \sin\theta_2)$$

$$\because |z| \leq 1 \quad \therefore r_1 \leq 1 \text{ and } |w| \leq 1 \quad \therefore r_2 \leq 1$$

Now  $|z - w|^2$

$$= (r_1 \cos\theta_1 - r_2 \cos\theta_2)^2 + (r_1 \sin\theta_1 - r_2 \sin\theta_2)^2$$

$$= r_1^2 + r_2^2 - 2r_1 r_2 \cos(\theta_1 - \theta_2)$$

$$= (r_1 - r_2)^2 + 2r_1 r_2 - 2r_1 r_2 \cos(\theta_1 - \theta_2)$$

$$= (r_1 - r_2)^2 + 2r_1 r_2 [1 - \cos(\theta_1 - \theta_2)]$$

$$= (r_1 - r_2)^2 + 2r_1 r_2 \cdot 2\sin^2 \frac{\theta_1 - \theta_2}{2}$$

$$\leq (r_1 - r_2)^2 + 2.1.1.2. \left(\frac{\theta_1 - \theta_2}{2}\right)^2$$

$$[\because \sin^2\theta \leq \theta^2 \text{ and } r_1 \leq 1, r_2 \leq 1]$$

$$= (|z| - |w|)^2 + (\theta_1 - \theta_2)^2$$

$$\text{Thus } |z - w|^2 \leq (|z| - |w|)^2 + (\text{Arg}z - \text{Arg}w)^2$$

### Practice Problems

[F]

1. What set of points of the complex plane is specified by the following condition :

(a) One of the arguments of the number  $z$  is equal to zero.

(b) One of the arguments is equal to  $\frac{\pi}{2}$ .

(c) One of the arguments  $\phi$  satisfies the inequalities  $2\pi < \phi < 3\pi$ .

(e) One of the arguments  $\phi$  satisfies the inequalities  $0 \leq \phi < 2\pi$  ?

2. If  $\frac{\tan\theta - i\left(\sin\frac{\theta}{2} + \cos\frac{\theta}{2}\right)}{1 + 2i\sin\frac{\theta}{2}}$  is purely imaginary,

then determine the general value of  $\theta$ .

3. Find the principal value of the  $\arg(z)$  of the complex number,

$$z = 1 + \cos\left(\frac{11\pi}{9}\right) + i \sin\left(\frac{11\pi}{9}\right).$$

4. If  $z = 2 + 3i$ , then find the principal argument of  $z^4 - z^3 + 10z^2 + 3z - 5$ .

5. Find the argument of the complex number

$$z_1 = z^2 + \bar{z} \text{ if } z = \cos\phi + i \sin\phi, 0 \leq \phi < 2\pi.$$

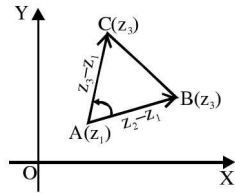
6. Prove that the least value of  $r$  for which the two curves  $\arg(z) = \pi/6$  and  $|z - 2\sqrt{3}i| = r$  intersect is 3.

7. Let  $z$  be a complex number having the argument  $\theta$ ,  $0 < \theta < \pi/2$  and satisfying the equality  $|z - 3i| = 3$ .  
Then prove that  $\cot \theta - 6/z$  is equal to  $i$ .

### 1.8 ROTATION THEOREM

If the vertices  $A, B, C$  of a  $\triangle ABC$  represent the complex number  $z_1, z_2, z_3$  respectively in the anti-clockwise sense and  $\angle BAC = \alpha$  then

$$\frac{z_3 - z_1}{|z_3 - z_1|} = \frac{z_2 - z_1}{|z_2 - z_1|} e^{i\alpha}$$



*Proof:* We have  $\frac{z_2 - z_1}{|z_2 - z_1|}$  as a unit vector along  $AB$ .

Also  $\frac{z_2 - z_1}{|z_2 - z_1|}$  is a unit vector along  $AB$ . If this vector is rotated by angle  $\alpha$  in the anti-clockwise sense then we obtain a unit vector along  $AC$ . If we multiply a complex number by  $e^{i\alpha}$  then it increases the argument by  $\alpha$  but does not change the modulus.

Hence  $\frac{z_3 - z_1}{|z_3 - z_1|} = \frac{z_2 - z_1}{|z_2 - z_1|} e^{i\alpha}$ . This can also

be written as

$$\frac{z_3 - z_1}{z_2 - z_1} = \frac{AC}{AB} e^{i\alpha}$$

or 
$$\frac{z_3 - z_1}{z_2 - z_1} = \frac{|z_3 - z_1|}{|z_2 - z_1|} \cdot e^{i\alpha}.$$

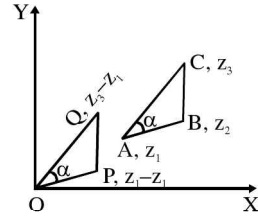
#### Alternative

Let, in Argand diagram, the points  $P, Q$  represent the complex numbers

$$z_2 - z_1, z_3 - z_1.$$

Now  $\triangle ABC$  and  $\triangle OPQ$  are congruent.

$$\therefore \angle BAC = \angle POQ = \alpha \text{ and } \frac{AC}{AB} = \frac{OQ}{OP}.$$

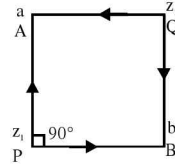


$$\left| \frac{z_3 - z_1}{z_2 - z_1} \right| = \frac{OQ}{OP} = \frac{AC}{AB} \text{ and } \arg \frac{z_3 - z_1}{z_2 - z_1} = \alpha.$$

$$\therefore \frac{z_3 - z_1}{z_2 - z_1} = \frac{AC}{AB} (\cos \alpha + i \sin \alpha) = \frac{AC}{AB} e^{i\alpha}.$$

► **Example 1.** If  $a$  &  $b$  are complex numbers then find the complex numbers  $z_1$  &  $z_2$  so that the points  $z_1, z_2$  and  $a, b$  be the corners of the diagonals of a square.

► **Solution**



$$a - z_1 = (b - z_1) e^{i\pi/2} = i(b - z_1)$$

$$a - i b = z_1 (1 - i)$$

$$z_1 = \frac{a - i b}{1 - i} = \frac{(a - i b)(1 + i)}{2}$$

$$z_1 = \frac{(a + b) + i(a - b)}{2}$$

$$\text{Similarly } b - z_2 = (a - z_2) e^{i\pi/2}$$

$$\Rightarrow z_2 = \frac{a + b}{2} - i \left( \frac{a - b}{2} \right)$$

#### Alternative

$$|z_1 - a|^2 = |z_2 - b|^2$$

$$\Rightarrow (z_1 - a)(\bar{z}_1 - \bar{a}) = (z_2 - b)(\bar{z}_2 - \bar{b})$$

$$\frac{\bar{z}_1 - \bar{b}}{\bar{z}_1 - \bar{a}} = \frac{z_1 - a}{z_1 - b} \quad \dots(1)$$

$$\text{Again } \arg \frac{z_1 - b}{z_1 - a} = \frac{\pi}{2}$$

$$\Rightarrow \frac{z_1 - b}{z_1 - a} \text{ is purely imaginary}$$

$$\text{Hence } \frac{z_1 - b}{z_1 - a} = -\frac{\bar{z}_1 - \bar{b}}{\bar{z}_1 - \bar{a}} \quad \dots(2)$$

From (1) & (2),

$$\frac{z_1 - a}{z_1 - b} = -\frac{z_1 - b}{z_1 - a} \text{ or } (z_1 - a)^2 + (z_1 - b)^2 = 0$$

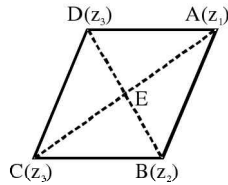
$$\text{or } 2z_1^2 - 2z_1(a+b) + a^2 + b^2 = 0$$

$$\text{or } z_1 = \frac{(a+b) + i(a-b)}{2}$$

$$z_2 = \frac{a+b}{2} - i\left(\frac{a-b}{2}\right)$$

► **Example 2.** ABCD is a rhombus described in the clockwise direction in the Argand diagram. Suppose that the vertices A, B, C, D are given by  $z_1, z_2, z_3, z_4$  respectively and the  $\angle CBA = 2\pi/3$ . Show that

$$2\sqrt{3}z_2 = (\sqrt{3} - i)z_1 + (\sqrt{3} + i)z_3$$



$$\text{and } 2\sqrt{3}z_4 = (\sqrt{3} + i)z_1 + (\sqrt{3} - i)z_3$$

► **Solution** We have  $\angle CBA = 2\pi/3$ . Therefore,

$$\frac{z_3 - z_2}{z_1 - z_2} = \left| \frac{CB}{AB} \right| \left[ \cos\left(\frac{2\pi}{3}\right) + i\sin\left(\frac{2\pi}{3}\right) \right]$$

$$\Rightarrow \frac{z_3 - z_2}{z_1 - z_2} = -\frac{1}{2} + \frac{i\sqrt{3}}{2} \quad [\because CB = AB]$$

$$\Rightarrow z_3 - z_2 = \left(-\frac{1}{2} + \frac{i\sqrt{3}}{2}\right)(z_1 - z_2)$$

$$\Rightarrow z_3 + \left(\frac{1}{2} - \frac{i\sqrt{3}}{2}\right)z_1 = \left(\frac{3}{2} - \frac{i\sqrt{3}}{2}\right)z_2$$

$$\Rightarrow \sqrt{3}z_2 = \frac{2}{\sqrt{3} - i}z_3 + \frac{1 - i\sqrt{3}}{\sqrt{3} - i}z_1$$

$$\Rightarrow z_3 + \left(\frac{1}{2} - \frac{i\sqrt{3}}{2}\right)z_1 = \left(\frac{3}{2} - \frac{i\sqrt{3}}{2}\right)z_2$$

$$\Rightarrow \sqrt{3}z_2 = \frac{2}{\sqrt{3} - i}z_3 + \frac{1 - i\sqrt{3}}{\sqrt{3} - i}z_1$$

$$\Rightarrow 2\sqrt{3}z_2 = (\sqrt{3} + i)z_3 + \frac{1}{2}(\sqrt{3} + i)(1 - i\sqrt{3})z_1$$

$$= (\sqrt{3} + i)z_3 + \frac{1}{2}(\sqrt{3} + \sqrt{3} + i - 3i)z_1$$

$$= (\sqrt{3} + i)z_3 + (\sqrt{3} - i)z_1$$

Also, since the diagonals of a rhombus bisect each other

$$\frac{z_1 + z_3}{2} = \frac{z_2 + z_4}{2} \Rightarrow z_4 = z_1 + z_3 - z_2$$

$$\Rightarrow 2\sqrt{3}z_4 = 2\sqrt{3}(z_1 + z_3) - 2\sqrt{3}z_2$$

$$= 2\sqrt{3}(z_1 + z_3) - [(\sqrt{3} - i)z_1 + (\sqrt{3} + i)z_3]$$

$$= (\sqrt{3} + i)z_1 + (\sqrt{3} - i)z_3$$

► **Example 3.** The points P, Q, R represent the numbers  $z_1, z_2, z_3$  respectively and the angles of the

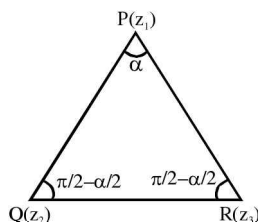
triangle PQR at Q and R are both  $\frac{1}{2}(\pi - \alpha)$ . Prove that

$$(z_3 - z_2)^2 = 4(z_3 - z_1)(z_1 - z_2) \sin^2\left(\frac{\alpha}{2}\right)$$

► **Solution** Since the angles at Q and R are both

$\frac{1}{2}(\pi - \alpha)$ , the triangle PQR is an isosceles triangle and  $\angle QPR = \alpha$ . We have

$$\arg\left(\frac{z_3 - z_1}{z_2 - z_1}\right) = \alpha$$



$$\Rightarrow \frac{z_3 - z_1}{z_2 - z_1} = \frac{PR}{PQ} (\cos \alpha + i \sin \alpha)$$

But  $PR = PQ$ .

$$\therefore \frac{z_3 - z_1}{z_2 - z_1} = \cos \alpha + i \sin \alpha$$

$$\Rightarrow \frac{z_3 - z_1}{z_2 - z_1} - 1 = (\cos \alpha - 1) + i \sin \alpha \quad \dots(1)$$

$$\Rightarrow \frac{z_3 - z_2}{z_2 - z_1} = -2 \sin^2 \left( \frac{\alpha}{2} \right) + 2i \sin \left( \frac{\alpha}{2} \right) \cos \left( \frac{\alpha}{2} \right)$$

$$= 2i \sin \left( \frac{\alpha}{2} \right) \left[ \cos \left( \frac{\alpha}{2} \right) + i \sin \left( \frac{\alpha}{2} \right) \right]$$

$$\Rightarrow \left( \frac{z_3 - z_2}{z_2 - z_1} \right)^2 = -4 \sin^2 \left( \frac{\alpha}{2} \right) \left[ \cos \left( \frac{\alpha}{2} \right) + i \sin \left( \frac{\alpha}{2} \right) \right]^2$$

$$= -4 \sin^2 \left( \frac{\alpha}{2} \right) [\cos \alpha + i \sin \alpha]$$

$$\Rightarrow \left( \frac{z_3 - z_1}{z_2 - z_1} \right)^2 = -4 \sin^2 \left( \frac{\alpha}{2} \right) \left( \frac{z_3 - z_1}{z_2 - z_1} \right)$$

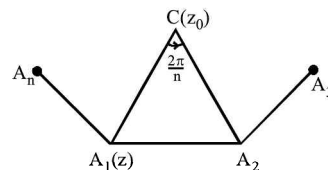
[Using (1)]

$$\Rightarrow (z_3 - z_2)^2 = 4(z_1 - z_2)(z_3 - z_1) \sin^2 \left( \frac{\alpha}{2} \right)$$



If  $z_0$  be the centre of a regular polygon of  $n$  sides and  $z$  be its one vertex  $A_1$ , then the vertices  $A_2, A_3, \dots, A_n$  (proceeding in anticlockwise direction. (taking actual position of points) will be

$$z_0 + (z - z_0) \alpha, \quad z_0 + (z - z_0) \alpha^2, \quad \dots, \quad z_0 + (z - z_0) \alpha^{n-1} \text{ respectively where } \alpha = e^{\frac{i2\pi}{n}}$$



But if the points are taken in clockwise direction then  $\alpha$  must be taken consecutive vertices taken in anticlockwise direction will be  $z \alpha, z \alpha^2, \dots, z \alpha^{n-1}$ ,

where  $\alpha = e^{\frac{i2\pi}{n}}$ .

If the vertices are taken in clockwise direction then

$\alpha$  must be taken as  $e^{-\frac{i2\pi}{n}}$ .

### Practice Problems

[G]

- When rotated clockwise through an angle of  $\frac{\pi}{2}$  and stretched two-fold, the vector  $z_1 = 2 + 5i$  goes into the vector  $z_2$ . Find the complex number corresponding to the vector  $z_2$ .
- The vector  $z = -2 + 3i$  is turned through an angle of  $180^\circ$  and stretched 1.5 times. Find the complex number corresponding to the newly obtained vector.
- Let  $z_1, z_2, z_3, z_4$  be the vertices A, B, C, D respectively of a square on the Argand diagram taken in anticlockwise direction then prove that :  
 (i)  $2z_2 = (1 + i)z_1 + (1 - i)z_3$   
 (ii)  $2z_4 = (1 - i)z_1 + (1 + i)z_3$
- P is a point on the Argand diagram. On the circle with OP as diameter two points Q & R are taken such that  $\angle POQ = \angle QOR = \theta$ . If 'O' is the

origin & P, Q & R are represented by the complex numbers  $Z_1$ ,  $Z_2$  &  $Z_3$  respectively, show that :

$$Z_2^2 \cdot \cos 2\theta = Z_1 \cdot Z_3 \cos^2 \theta.$$

5. If  $z_1, z_2, z_3$  are the vertices of an isosceles triangle, right angled at the vertex  $z_2$ , prove that  $z_1^2 + 2z_2^2 + z_3^2 = 2z_2(z_1 + z_3)$ .

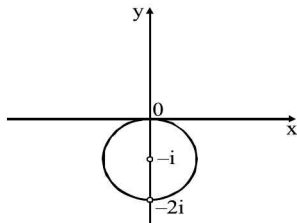
## 1.9 INTERPRETATION OF LOCII

Here we present different equations and inequations with their geometrical interpretations. This will develop the geometrical approach of handling problems in complex numbers.

1.  $|z - c| = a$  where  $a$  is a positive real number. This represents a circle with centre  $c$  and radius  $a$ . The values of  $z$  satisfying this equation lie on this circle and all points lying on this circle satisfies this equation.

For example, consider  $|z + i| = 1$ .

The relation is satisfied by those and only those points of the complex plane which are situated at a distance equal to unity from the point  $z_1 = -i$ . Such points lie on a unit circle with the point  $z = -i$  as centre.



2.  $|z - c| < a$  where  $a$  is a positive real number. This represents all points lying in the interior of the circle with centre  $c$  and radius  $a$ .
3.  $|z - c| > a$  where  $a$  is a positive real number. This represents all points lying on the exterior of the circle with centre  $c$  and radius  $a$ .

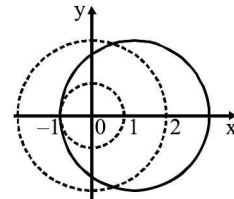
► **Example 1.** Let  $|z| = 2$ ; where are the points  $3z$  located ?

► **Solution** The points  $z$  which satisfy the condition  $|z| = 2$  are located on a circle of radius 2 with centre at the origin. The point  $3z$  is located on

the same ray as the point  $z$ , but is distant from the origin three times that of point  $z$ . For this reason, the points  $3z$ , where  $|z| = 2$ , are located on a circle of radius 6 with centre at the origin.

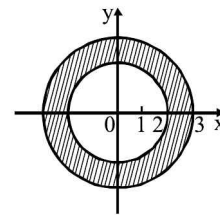
► **Example 2.** Let  $|z| = 1$ ; where are the points  $1 + 2z$  located?

► **Solution** The points  $z$  which satisfy the condition  $|z| = 1$  lie on a circle of radius 1 with centre at the origin. All points  $2z$ , where  $|z| = 1$ , are located on a circle of radius 2 with centre at the origin. The point  $2z + 1$  is obtained from point  $2z$  by a rightward shift of 1 unit. And so the points  $1 + 2z$ , where  $|z| = 1$ , are located on a circle of radius 2 with centre at the point  $(1, 0)$  (fig.).



► **Example 3.** Locate the points for which  $2 < |z| < 3$ .

► **Solution** We know that points satisfying the condition  $|z| = 2$  are located



on a circle of radius 2 with centre at the origin. Now points for which  $|z| > 2$  are located farther from the origin than the points of this circle, that is, outside the circle. Similarly, points satisfying the condition  $|z| < 3$  are located inside a circle of radius 3 centred at the origin.

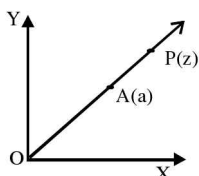
Hence, points which satisfy the condition  $2 < |z| < 3$  lie inside an annulus bounded by concentric circles centred at the origin having radii  $r_1 = 2$  and  $r_2 = 3$  (fig.)

4.  $|z + a| = |z| + |a|$

This represents a ray starting from origin and passing through point  $a$ .

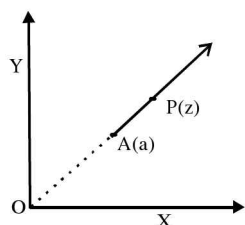


By the triangle inequality we know that equality holds when the arguments of  $z$  and  $a$  are equal.



5.  $|z - a| = |z| - |a|$

This represents a ray through origin and passing through point  $a$ , but starting from  $a$ .



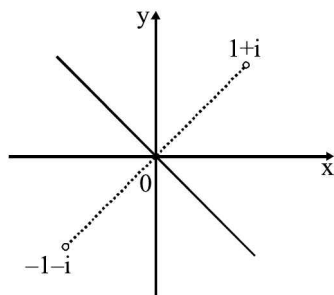
6.  $|z - z_1| = |z - z_2|$

This represents the perpendicular bisector of the line segment joining  $z_1$  and  $z_2$ .

For example consider

$$|z - 1 - i| = |z + 1 + i|$$

This condition is satisfied by those and only those points of the complex plane which are equidistant from the points  $z_1 = 1 + i$  and  $z_2 = -1 - i$ . The set of points in the plane equidistant from two given points is a straight line perpendicular to the segment joining these points and passing through its midpoint.



► **Example 4.** Locate the complex numbers  $z = a + bi$  for which  $\log_{1/2}|z - 2| > \log_{1/2}|z|$

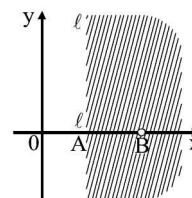
► **Solution** First of all, note that the left member of this inequality is meaningful for all complex numbers  $z$ , except  $z = 2$ , and the right member, for

all  $z \neq 0$ . Therefore, the expressions of this inequality have meaning simultaneously for all complex numbers  $z$  except  $z = 0, 2$ .

The inequality is equivalent to  $|z - 2| < |z|$ .

We know that the equation  $|z - 2| = |z|$  is satisfied by all complex numbers lying on a straight line  $\ell$  parallel to the  $y$ -axis and passing through the point  $A(1, 0)$  because all the points of this line are equidistant from two points:  $O(0, 0)$  and  $B(2, 0)$  (fig.). But our job is to find all those points in the plane which are closer to the point  $B(2, 0)$  than to the point  $O(0, 0)$ .

Quite obviously, these are the points of the plane to the right

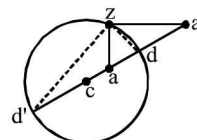


of  $\ell$  in the domain that includes point  $B$ . Thus all the points of the half plane to the right of  $\ell$  satisfy the condition  $|z - 2| < |z|$  (fig.) The points of the straight line  $\ell$  are excluded. Recall also that the point  $B(2, 0)$  located in the half-plane to the right of  $\ell$  must also be excluded.

To summarize, then, all points of the plane located to the right of the straight line parallel to the  $y$ -axis and passing through the point  $(1, 0)$ , with the exception of the point  $(2, 0)$ , satisfy the conditions of the problem.

7. If  $k$  is a positive real constant, and  $z$  satisfies

$$\left| \frac{z - a}{z - a'} \right| = k,$$



then the point  $z$  describes a circle of which  $A, A'$  are inverse points ;

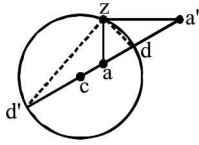
unless  $k = 1$ , in which case  $z$  describes the perpendicular bisector of  $AA'$ .

For let the bisectors of  $\angle AZA'$  meet  $AA'$  at  $D, D'$ . Then,  $AZ : A'Z = k : 1$ .

Therefore  $D, D'$  divide  $AA'$  internally and externally in the ratio  $k : 1$ , and are fixed points.

Therefore  $z$  describe the circle on  $DD'$  as diameter. Also if  $C$  is the mid-point of  $DD'$ , we have  $CA \cdot CA' = CD^2$ ; hence  $A, A'$  are inverse points with regard to the circle.

If  $k = 1$ , then  $AZ = A'Z$ ; and  $Z$  lies on the perpendicular of  $AA'$ .



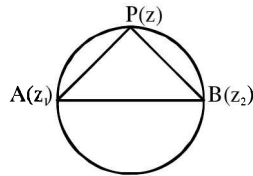
Conversely, if the point  $Z$  describes a circle of which

$A, A'$  are inverse points, then  $\left| \frac{z - a}{z - a'} \right| = k$ .

For, since  $CA \cdot CA' = CD^2$ , then  $DD'$  is divided internally and externally in the same ratio,  $k$  to 1, say. Hence,  $AZ : A'Z = k : 1$ .

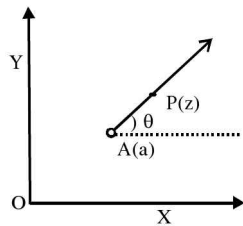
8.  $|z - z_1|^2 + |z - z_2|^2 = |z_1 - z_2|^2$

This is an equation of a circle with  $z_1$  &  $z_2$  as endpoints of a diameter. The condition is based on Pythagoras theorem applied on the right angled triangle  $PAB$ .



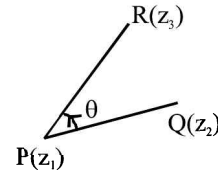
9.  $\arg(z - a) = \theta$ , where  $\theta$  is constant.

This represents a ray starting at  $A(a)$  and inclined at an angle of  $\theta$  with the positive real axis



10.  $\arg \left( \frac{z_3 - z_1}{z_2 - z_1} \right)$  represents the angle  $\theta$  formed between  $\vec{PQ}$  and  $\vec{PR}$ .

Here  $z_2 - z_1 = \vec{PQ}$  is the vector from which arrow starts and  $z_3 - z_1 = \vec{PR}$  is the vector from which arrow ends.



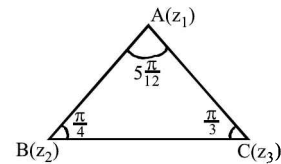
We mark the arrow in anticlockwise direction after plotting the points.

$$\arg \left( \frac{z_3 - z_1}{z_2 - z_1} \right) = \theta.$$



- (i) Here actual plot of points is required, to know whether  $PR$  is situated in the clockwise or anticlockwise direction of  $PQ$ .
- (ii)  $\theta$  will be taken as positive or negative according as it is measured in anticlockwise or clockwise direction.

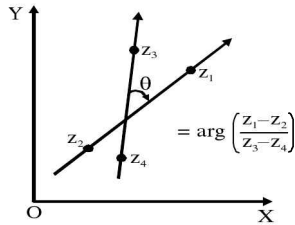
For instance in the figure given below



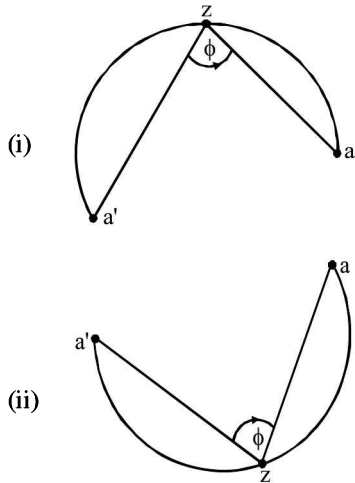
$$\arg \left( \frac{z_3 - z_2}{z_1 - z_2} \right) = -\frac{\pi}{4};$$

$$\arg \left( \frac{z_1 - z_2}{z_3 - z_2} \right) = \frac{\pi}{4} \text{ and } \arg \left( \frac{z_1 - z_3}{z_2 - z_3} \right) = -\frac{\pi}{3}$$

- (iii)  $\arg \left( \frac{z_1 - z_2}{z_3 - z_4} \right)$  represents the angle from vector  $z_3 - z_4$  towards vector  $z_1 - z_2$  as shown in the figure.



- (iv) If  $z$  varies so that  $\arg \frac{z - a}{z - a'} = \phi$ , where  $\phi$  is a constant angle, then the point  $Z$  describe an arc of a segment of a circle on  $AA'$ , containing an angle  $\phi$ .



$\angle A'ZA = \phi$ . The sign of  $\phi$  determines the side of  $AA'$  on which the segment lies. Thus  $\phi$  is positive in fig. (i) and negative in fig. (ii).

► **Example 5.** Find the locus of points  $z$  such that

$$\arg \left( \frac{z - i}{z + 1} \right) = \theta, \text{ when}$$

- (i)  $\theta = 0$  (ii)  $\theta = \pi$   
 (iii)  $\theta = \alpha$  where  $0 < \alpha < \pi/2$

► **Solution** Let  $z = x + iy$ . Then

$$\begin{aligned} \frac{z - i}{z + 1} &= \frac{x + i(y - 1)}{(x + 1) + iy} = \frac{[x + i(y - 1)][(x + 1) - iy]}{(x + 1)^2 + y^2} \\ &= \frac{(x^2 + y^2 + x - y) + i(y - x - 1)}{(x + 1)^2 + y^2} \end{aligned}$$

$$\text{So, } \theta = \arg \left( \frac{z - i}{z + 1} \right) = \tan^{-1} \left( \frac{y - x - 1}{x^2 + y^2 + x - y} \right)$$

$$\Rightarrow \frac{y - x - 1}{x^2 + y^2 + x - y} = \tan \theta \quad \dots(1)$$

(i) When  $\theta = 0$ , (1) gives us  $y - x - 1 = 0$  or  $x - y + 1 = 0$ . This represents a straight line passing through  $A(-1)$  and  $B(i)$ .

$$\text{But } \arg \left( \frac{z - i}{z + 1} \right) = 0$$

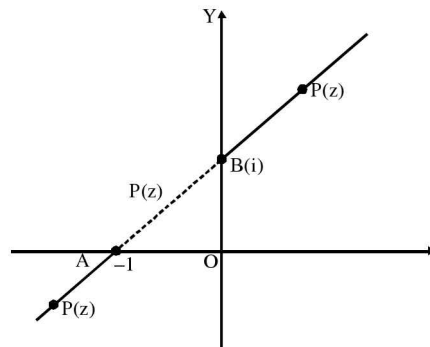
does not represent the complete line  $x - y + 1 = 0$ . In fact it represents two rays, one beginning at  $A$  (excluding  $A$ ) going downwards and other beginning at  $B$  (excluding  $B$ ) and going upwards. If we take any point  $P$  on the segment  $AB$ , then

$$\arg \left( \frac{z - i}{z + 1} \right) = \pi. \text{ Also, note that for any point } P(z)$$

on two rays beginning at  $A$  or  $B$

$$\arg \left( \frac{z - i}{z + 1} \right) = 0.$$

(ii) In case  $\theta = \pi$ ,  $\arg \left( \frac{z - i}{z + 1} \right) = \pi$ , represent the segment  $AB$  (excluding points  $A$  and  $B$ ). (see fig.)

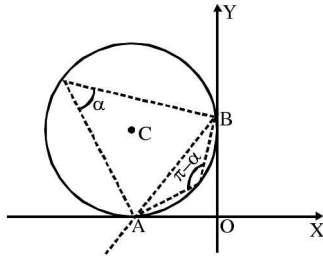


- (iii) When  $\theta = \alpha$  where  $0 < \alpha < \pi/2$ , (i) can be written as  
 $x^2 + y^2 + x - y = (\cot \alpha)(y - x - 1)$   
 or  $x^2 + y^2 + (1 + \cot \alpha)(x - y) + \cot \alpha = 0$ . ... (2)

This represents a circle with centre at

$$\left( -\frac{1 + \cot \alpha}{2}, \frac{1 + \cot \alpha}{2} \right).$$

But  $\arg \left( \frac{z - i}{z + 1} \right) = \alpha$  ( $0 < \alpha < \pi/2$ ) does not represent the whole circle given by (2). In fact it represents the major arc AB of the circle.



Equation of line through A(-1) and B(-i) is  $x - y + 1 = 0$ .

Since  $-\frac{1 + \cot \alpha}{2} - \frac{1 + \cot \alpha}{2} + 1 = -\cot \alpha < 0$   
 ( $\because 0 < \alpha < \pi/2$ )

and  $0 - 0 + 1 = 1 > 0$ ,  
 the origin and the centre C lie on the opposite sides of the line AB. Therefore, the arc of the circle which lies in the half plane  $x - y + 1 < 0$  is the major arc of the circle (2) and this is the arc of the circle that will contain angle  $\alpha$  with  $0 < \alpha < \pi/2$ .

Hence, locus of  $z$  such that  $\arg \left( \frac{z - i}{z + 1} \right) = \alpha$  represents the arc of the circle (2) that lies in the half plane  $x - y + 1 < 0$ .

► **Example 6.** Find the locus of  $z$  satisfying the equation  $\arg \left( \frac{z - 2}{z + 2} \right) = \frac{\pi}{3}$ .

► **Solution** Putting  $z = x + iy$ , the given complex number can be written as

$$\frac{z - 2}{z + 2} = \frac{(x + 2) + iy}{(x + 2) + iy} = \frac{(x^2 + y^2 - 4) + i(4y)}{(x^2 + 2)^2 + y^2}$$

Thus, we have

$$\arg \left( \frac{z - 2}{z + 2} \right) = \frac{4y}{x^2 + y^2 - 4}$$

Hence, according to the given condition, we have

$$\frac{4y}{x^2 + y^2 - 4} = \tan \left( \frac{\pi}{3} \right) = \sqrt{3}$$

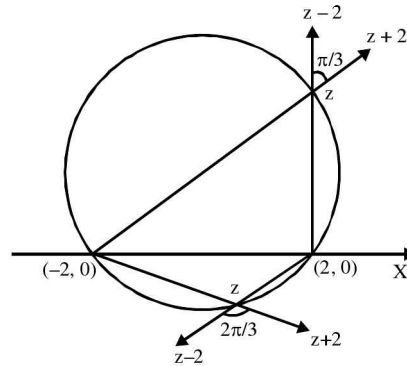
i.e.  $x^2 + y^2 - \frac{4}{\sqrt{3}}y - 4 = 0$

which represents a circle centred at  $\left( 0, \frac{2}{\sqrt{3}} \right)$  and

radius equal to  $\frac{4}{\sqrt{3}}$ . Drawing the above circle, we

can see that points lying on the portion of the circle above the X-axis only, satisfy the condition given in the problem

i.e.  $\arg \left( \frac{z - 2}{z + 2} \right) = \frac{\pi}{3}$



whereas for points lying on the portion of the circle below the X-axis, we have

$$\arg \left( \frac{z - 2}{z + 2} \right) = -\frac{2\pi}{3}$$

Hence the required locus is only the portion of the circle lying above the X-axis.

► **Example 7.** Locate the complex numbers  $z = x + iy$  for which

$$|z - 1| + |z + 1| \leq 4.$$

► **Solution**  $|z - 1| + |z + 1| = 4$

$$\text{or, } |x - 1 + iy| + |x + 1 + iy| = 4$$

$$\text{or, } \sqrt{(x-1)^2 + y^2} + \sqrt{(x+1)^2 + y^2} = 4$$

$$\text{or, } (x-1)^2 + y^2 = (x+1)^2 + y^2 - 8\sqrt{(x+1)^2 + y^2} + 16$$

$$\text{or, } x + 4 = 2\sqrt{(x+1)^2 + y^2}$$

$$\text{or, } x^2 + 8x + 16 = 4(x^2 + y^2 + 2x + 1)$$

$$\text{or, } 3x^2 + 4y^2 = 12$$

$$\text{or, } \frac{x^2}{4} + \frac{y^2}{3} = 1. \text{ It is an ellipse.}$$

Hence the points  $z$  satisfying  $|z - 1| + |z + 1| \leq 4$  lie in Argand plane on the boundary and in the interior

$$\text{of the ellipse } \frac{x^2}{4} + \frac{y^2}{3} = 1.$$

► **Example 8.** Among the complex numbers satisfying  $|z - 3 + 3i| = 3$ , find the numbers having

(i) least magnitude

(ii) greatest magnitude

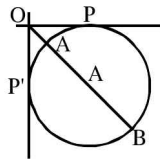
► **Solution** We have  $|z - (3 - 3i)| = 3$  which defines a circle with centre at  $(3, -3)$  and with radius 3. The required points are A and B where A and B lie on the line joining origin and centre of circle. The equation of line OC is  $y = -x$ . Putting  $z = x - ix$  in the given equation,

$$\text{We get } |x - ix - 3 + 3i| = 3$$

$$\Rightarrow (x-3)^2 + (-x+3)^2 = 9$$

$$\Rightarrow x = 3 \pm \frac{3}{\sqrt{2}}$$

Now as  $x$  coordinate of A must be smaller than that of B, we get A as



$$\left(3 - \frac{3}{\sqrt{2}}, -3 + \frac{3}{\sqrt{2}}\right) \text{ and}$$

$$B \text{ as } \left(3 + \frac{3}{\sqrt{2}}, -3 - \frac{3}{\sqrt{2}}\right) (\because y = -x)$$

Thus answers corresponding to (i) and (ii) are

$$3 - \frac{3}{\sqrt{2}} + i\left(-3 + \frac{3}{\sqrt{2}}\right)$$

$$\text{and } 3 + \frac{3}{\sqrt{2}} - i\left(3 + \frac{3}{\sqrt{2}}\right)$$

► **Example 9.** If  $z$  is any non zero complex number, show that

$$(i) \quad \left| \frac{z}{|z|} - 1 \right| \leq \arg z$$

$$(ii) \quad |z - 1| |z| - 1 \leq |z| \arg z$$

► **Solution** Let  $|z| = r$  and  $\arg z = \theta$ . Since  $z \neq 0$   
 $\therefore r \neq 0$

$$\text{Now } z = r(\cos \theta + i \sin \theta)$$

$$\therefore \frac{z}{|z|} = \cos \theta + i \sin \theta$$

$$\therefore \left| \frac{z}{|z|} - 1 \right| = |(\cos \theta - 1) + i \sin \theta|$$

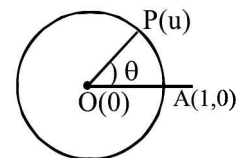
$$= \sqrt{(\cos \theta - 1)^2 + \sin^2 \theta}$$

$$= \sqrt{2 - 2\cos \theta} = \sqrt{4\sin^2 \frac{\theta}{2}}$$

$$= 2 \left| \sin \frac{\theta}{2} \right| \leq 2 \left| \frac{\theta}{2} \right| = |\theta| \quad [\because \sin \theta \leq \theta]$$

$$\text{Thus } \left| \frac{z}{|z|} - 1 \right| \leq \arg z$$

**Alternative**



$$\text{Let } \arg z = \theta, \text{ then } \arg \frac{z}{|z|} = \theta$$

$$\text{Let } u = \frac{z}{|z|}, \text{ then } |u| = 1 \text{ and } \arg u = \theta$$

Now  $|u| = 1 \Rightarrow u$  lies on the circle

whose centre is (0, 0) and radius is unity.

$$\text{Now } \left| \frac{z}{|z|} - 1 \right| = |u - 1| = AP$$

$$\text{Also arc } AP = 1 \cdot \theta = |\theta| \quad \left[ \because \theta = \frac{\ell}{r} \right]$$

Clearly chord  $AP \leq$  arc  $AP$  equality holds when  $\theta = 0$

$$\Rightarrow |u - 1| \leq |\theta| \Rightarrow \left| \frac{z}{|z|} - 1 \right| \leq \arg z.$$

(ii) Let  $|z| = r$  and  $\arg z = \theta$ , then  $z = r(\cos \theta + i \sin \theta)$

$$\text{Now, } |z - 1| = |(z - 1)| - (|z| - 1) \geq |z| - 1 - ||z| - 1|$$

$$\therefore |z - 1| - |z| - 1 \leq |z| - |z| \quad \dots(1)$$

$$\text{Now } |z - 1| = |r(\cos \theta + i \sin \theta) - 1|$$

$$= \sqrt{r^2(\cos \theta - 1)^2 + r^2 \sin^2 \theta}$$

$$= \sqrt{2r^2 - 2r^2 \cos \theta}$$

$$= 2r \left| \sin \frac{\theta}{2} \right| \leq 2r \left| \frac{\theta}{2} \right| \quad [\because |\sin \theta| \leq |\theta|]$$

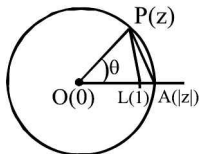
$$= r|\theta| = |z| \arg z \quad \dots(2)$$

From (1) and (2),

$$|z - 1| - ||z| - 1| \leq |z| \arg z$$

$$\Rightarrow |z - 1| \leq |z| - 1 + |z| \arg z$$

**Alternative**



$$|z - 1| = LP.$$

$$||z| - 1| = |r - 1| = LA \text{ and arc } AP = r|\theta|$$

From  $\triangle PLA$ ,  $PL \leq LA + AP \leq LA + \text{arc } AP$

$$\therefore |z - 1| \leq |r - 1| + r|\theta| \quad \dots(3)$$

$$\text{Hence } |z - 1| \leq ||z| - 1| + |z| \arg z$$

If L coincide with A or it lies outside the circle then also inequality (3) will hold.

### Practice Problems

[H]

- Let  $|z| = 5$ . Locate the points representing the complex numbers (a)  $-4z$ , (b)  $2 - z$ , (c)  $-1 + 3z$ .

- Locate the points representing the complex numbers, for which

- $|z + i| \leq 1$
- $|z + 2 - 2i| \leq 4$
- $1 < |z + 3 - 2i| < 2$
- $|z + i| = |z - 2|$
- $|z - 1| = |z - 3| = |z - i|$
- $\pi/3 < \arg z \leq 5\pi/6$ .

- Locate the points representing the complex numbers, for which

- $|z - 1| = |z + 1| = |z - i\sqrt{3}|$
- $|z - i\sqrt{2}| - |z + 4| = |z| - 1 = 0$ .
- $|z - 1|^2 + |z + 1|^2 = 4$ .

- Prove that the system of equations

$$\begin{cases} |z + 1 - i| = \sqrt{2}, \\ |z| = 3 \end{cases} \text{ has no solution.}$$

- If  $z$  and  $z'$  are two complex numbers with a non-real ratio, show that  $z$  and  $z'$  cannot lie on a straight line passing through the origin.

- If  $z = x + iy$ , prove that  $|z - 8| + |z + 8| = 20$  represents an ellipse.

- Solve the system of equations :

$$|z + 1 - i| = |3 + 2i - z| = |z + i|.$$

- Among complex numbers  $z$ , satisfying the condition  $|z + 1 - i| \leq 1$ , find the number having the greatest modulus.

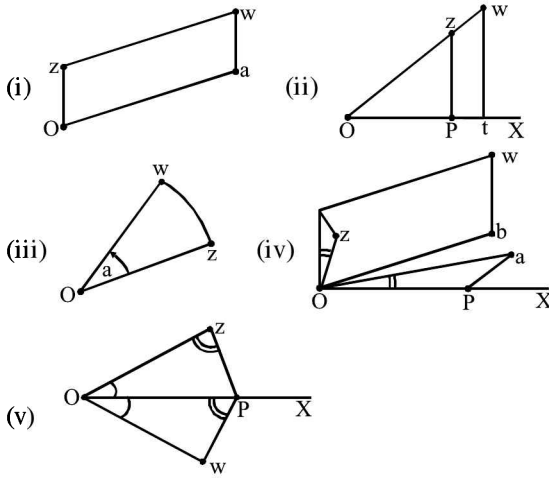
- Among complex numbers  $z$ , satisfying the condition  $|z - 5i| \leq 3$ , find the number having the least positive argument.

- What sets of points in the complex plane are specified by the indicated conditions :

$$(i) \log|z - 10i| < 1,$$

$$(ii) |z|^2 + 3z + 3\bar{z} = 0?$$

- Let  $z$ ,  $a$ ,  $b$  be complex numbers of which  $a$ ,  $b$  are constant and  $z$  varies. If  $w$  given in terms of  $z$  by one of the following equations, it is required to find the point  $w$  corresponding to a given point  $z$ .  $OP$  is of unit length.



12. Find the real values of the parameter  $a$  for which atleast one complex number

$$z = x + iy$$

satisfies both the equality

$$|z - a| = a + 4,$$

and the inequality  $|z - 2| < 1$ .

### 1.10 DE MOIVRE'S THEOREM

The Theorem of De Moivre tells us that the powers of a complex number  $\cos \theta + i \sin \theta$  is again a complex number of the same form namely,  $\cos \phi + i \sin \phi$  where  $\phi$  is only a suitable multiple of  $\theta$ . This leads to the following interesting geometrical interpretation. When a complex number on the unit circle is raised to a power say, the  $n$ th power, it is again a number on the same unit circle.

**Theorem 1 :** If  $n$  is an integer, positive or negative,  $(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta$  and if  $n$  is a rational number,  $\cos n\theta + i \sin n\theta$  is one of the values of  $(\cos \theta + i \sin \theta)^n$ .

*Proof:* In order to prove the Theorem we first note an elementary Lemma.

#### Lemma

$$\begin{aligned} & (\cos \alpha + i \sin \alpha)(\cos \beta + i \sin \beta) \\ &= (\cos \alpha \cos \beta - \sin \alpha \sin \beta) \\ & \quad + i(\sin \alpha \cos \beta + \cos \alpha \sin \beta) \end{aligned}$$

$$= \cos(\alpha + \beta) + i \sin(\alpha + \beta).$$

Consequently,

$$\begin{aligned} & (\cos \alpha + i \sin \alpha)(\cos \beta + i \sin \beta)(\cos \gamma + i \sin \gamma) \\ &= [\cos(\alpha + \beta) + i \sin(\alpha + \beta)][\cos \gamma + i \sin \gamma] \\ &= \cos(\alpha + \beta + \gamma) + i \sin(\alpha + \beta + \gamma) \end{aligned}$$

and so on. This can be extended to any number of quantities by induction. Now let us take the main theorem.

**Case I :** Let  $n$  be a positive integer. Consider  $n$  quantities  $\alpha_1, \alpha_2, \dots, \alpha_n$ . By Lemma, we have,

$$\begin{aligned} & (\cos \alpha_1 + i \sin \alpha_1)(\cos \alpha_2 + i \sin \alpha_2) \dots (\cos \alpha_n + i \sin \alpha_n) \\ &= \cos(\alpha_1 + \alpha_2 + \dots + \alpha_n) + i \sin(\alpha_1 + \alpha_2 + \dots + \alpha_n). \end{aligned}$$

Putting  $\alpha_1 = \alpha_2 = \dots = \alpha_n = \theta$ ,

we get

$$\begin{aligned} & (\cos \theta + i \sin \theta)(\cos \theta + i \sin \theta) \dots n \text{ times} \\ &= \cos(\theta + \theta + \dots n \text{ times}) + i \sin(\theta + \theta + \dots n \text{ times}). \end{aligned}$$

Hence  $(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta$ .

**Case II :** Let  $n$  be a negative integer and equal to  $-m$ , say, where  $m$  is a positive integer.

$$(\cos \theta + i \sin \theta)^n = (\cos \theta + i \sin \theta)^{-m}$$

$$= \frac{1}{(\cos \theta + i \sin \theta)^m} = \frac{1}{(\cos m\theta + i \sin m\theta)} \text{ by Case I.}$$

$$= \frac{\cos m\theta - i \sin m\theta}{(\cos m\theta + i \sin m\theta)(\cos m\theta - i \sin m\theta)}$$

$$\begin{aligned} &= \frac{\cos m\theta - i \sin m\theta}{\cos^2 m\theta + \sin^2 m\theta} \\ &= \cos m\theta - i \sin m\theta = \cos(-m\theta) + i \sin(-m\theta) \\ &= \cos n\theta + i \sin n\theta. \end{aligned}$$

**Case III :** Let  $n$  be a rational number and equal to  $p/q$  where  $p$  and  $q$  are integers.

By Cases 1 and 2, we have,

$$\begin{aligned} & [\cos(p/q)\theta + i \sin(p/q)\theta]^q \\ &= [\cos(q \cdot (p/q)\theta) + i \sin(q \cdot (p/q)\theta)] \\ &= \cos p\theta + i \sin p\theta \\ &= (\cos \theta + i \sin \theta)^p \end{aligned}$$

again, since  $p$  is an integer.

Thus,  $(\cos(p/q)\theta + i \sin(p/q)\theta)$

is one of the  $q$ th roots of

$$(\cos \theta + i \sin \theta)^p$$

that is one of the values of

$$((\cos \theta + i \sin \theta)^p)^{1/q}$$

i.e. of  $(\cos \theta + i \sin \theta)^{p/q}$ , which means that

$$\cos n\theta + i \sin n\theta \text{ is a value of } (\cos \theta + i \sin \theta)^n.$$



- (i) Using the notation of  $\text{cis } \theta$  for  $\cos \theta + i \sin \theta$ , the result of the theorem can be written as  $\text{cis } n\theta = \text{one of the values of } (\text{cis } \theta)^n$   
The result of the lemma will be  $\text{cis } \alpha_1 \cdot \text{cis } \alpha_2 \dots \text{cis } \alpha_n = \text{cis } (\alpha_1 + \alpha_2 + \dots + \alpha_n)$

(ii)  $\cos \theta - i \sin \theta = \cos(-\theta) + i \sin(-\theta)$   
 $= \text{cis}(-\theta) = [\text{cis } \theta]^{-1}$   
 $= \frac{1}{\cos \theta + i \sin \theta}$

Thus the conjugate numbers  $\cos \theta + i \sin \theta$  and  $\cos \theta - i \sin \theta$  are reciprocals of each other. This is also shown by the working :

$$\begin{aligned} &(\cos \theta + i \sin \theta)(\cos \theta - i \sin \theta) \\ &= \cos^2 \theta - (i \sin \theta)^2 \\ &= \cos^2 \theta + \sin^2 \theta = 1. \end{aligned}$$

- (iii) If  $z = \cos \theta + i \sin \theta$ ,  
 then  $(1/z) = \cos \theta - i \sin \theta$ .  
 so that  $z + (1/z) = 2 \cos \theta$  and  $z - (1/z) = 2i \sin \theta$ .  
 De Moivre's theorem will now give  
 $z^n = \cos n\theta + i \sin n\theta$   
 and  $(1/z)^n = \cos n\theta - i \sin n\theta$ .  
 so that  $z^n + (1/z)^n = 2 \cos n\theta$   
 and  $z^n - (1/z)^n = 2i \sin n\theta$ .  
 These results will be of immense use.

- (iv) Using Euler's representation, we can rewrite the formula for raising a complex number to a natural power in the following way :  $(re^{i\theta})^n = r^n e^{in\theta}$   
 The formula yielding all the solutions of the equation  $z^n = re^{i\theta}$  can also be rewritten in a more concise form:

$$z_k = \sqrt[n]{re^{i\theta}} = \sqrt[n]{r} e^{i\left(\frac{\theta}{n} + \frac{2\pi k}{n}\right)},$$

$$k = 0, 1, \dots, n-1.$$

- **Example 1.** Simplify :

$$\frac{(\cos 3\theta + i \sin 3\theta)^7 (\cos 5\theta - i \sin 5\theta)^4}{(\cos 4\theta + i \sin 4\theta)^{10} (\cos 13\theta - i \sin 13\theta)^3}$$

- **Solution** The given expression =

$$\frac{(\cos \theta + i \sin \theta)^{21} (\cos \theta + i \sin \theta)^{-20}}{(\cos \theta + i \sin \theta)^{40} (\cos \theta + i \sin \theta)^{-39}}$$

$$= \frac{(\cos \theta + i \sin \theta)}{(\cos \theta + i \sin \theta)} = 1$$

- **Example 2.** Find the general values of the real angle  $\theta$  which satisfies the equation

$$(\cos \theta + i \sin \theta) (\cos 2\theta + i \sin 2\theta) \dots (\cos n\theta + i \sin n\theta) = 1.$$

- **Solution** We have

$$(\cos \theta + i \sin \theta) (\cos 2\theta + i \sin 2\theta) \dots (\cos n\theta + i \sin n\theta) = 1,$$

$$\text{or } \cos(\theta + 2\theta + \dots + n\theta) + i \sin(\theta + 2\theta + \dots + n\theta) = 1,$$

$$\text{or } \cos\left\{\frac{1}{2}n(n+1)\theta\right\} + i \sin\left\{\frac{1}{2}n(n+1)\theta\right\} = 1.$$

$$\therefore \cos\left\{\frac{1}{2}n(n+1)\theta\right\} = 1 \text{ and}$$

$$\sin\left\{\frac{1}{2}n(n+1)\theta\right\} = 0$$

To satisfy the above two equations

$$\frac{1}{2}n(n+1)\theta = 2k\pi, \text{ or } \theta = \frac{4k\pi}{n(n+1)},$$

where  $k = \text{any integer}$ .

- **Example 3.**

If  $\cos \alpha + \cos \beta + \cos \gamma = 0 = \sin \alpha + \sin \beta + \sin \gamma$ ,  
 prove that

- (i)  $\Sigma \cos 3\alpha = 3 \cos(\alpha + \beta + \gamma)$ ,  
 (ii)  $\Sigma \sin 3\alpha = 3 \sin(\alpha + \beta + \gamma)$ ,  
 (iii)  $\Sigma \cos^2 \alpha = \Sigma \sin^2 \alpha = 3/2$ .

- **Solution** To prove (i) and (ii) we put

$$x = \cos \alpha + i \sin \alpha, y = \cos \beta + i \sin \beta$$

$$\text{and } z = \cos \gamma + i \sin \gamma.$$

$$x + y + z = (\cos \alpha + \cos \beta + \cos \gamma) + i(\sin \alpha + \sin \beta + \sin \gamma) = 0$$

$$\text{Now } x^3 + y^3 + z^3 = 3xyz \quad (\because x + y + z = 0)$$

$$\therefore (\cos \alpha + i \sin \alpha)^3 + (\cos \beta + i \sin \beta)^3 + (\cos \gamma + i \sin \gamma)^3 = 3(\cos \alpha + i \sin \alpha)(\cos \beta + i \sin \beta)(\cos \gamma + i \sin \gamma),$$

$$\text{or } (\cos 3\alpha + i \sin 3\alpha) + (\cos 3\beta + i \sin 3\beta) + (\cos 3\gamma + i \sin 3\gamma) = 3\{\cos(\alpha + \beta + \gamma) + i \sin(\alpha + \beta + \gamma)\}.$$

Equating the real and the imaginary parts, we get

$$\Sigma \cos 3\alpha = 3 \cos(\alpha + \beta + \gamma)$$



and  $\Sigma \sin 3\alpha = 3 \sin(\alpha + \beta + \gamma)$ .

(iii) We have  $\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = x^{-1} + y^{-1} + z^{-1}$

$$\text{or } \frac{xy + yz + zx}{xyz}$$

$$= (\cos\alpha - i \sin\alpha) + (\cos\beta - i \sin\beta) + (\cos\gamma - i \sin\gamma) \\ = (\cos\alpha + \cos\beta + \cos\gamma) - i(\sin\alpha + \sin\beta + \sin\gamma) = 0,$$

$$\text{i.e., } xy + yz + zx = 0.$$

$$\text{Now } (x + y + z)^2 = x^2 + y^2 + z^2 + 2(xy + yz + zx)$$

$$\therefore x^2 + y^2 + z^2 = 0 \quad (\because x + y + z = 0, xy + yz + zx = 0),$$

$$\text{or } \Sigma \cos 2\alpha + i \Sigma \sin 2\alpha = 0.$$

$$\text{It leads to } \Sigma \cos 2\alpha = 0,$$

$$\text{i.e., } \Sigma \cos^2 \alpha = \Sigma \sin^2 \alpha = k \text{ (say).}$$

$$\therefore 2k = \Sigma \cos^2 \alpha + \Sigma \sin^2 \alpha = 3, \text{ or } k = 3/2$$

► **Example 4.** If  $\alpha, \beta$  be the roots of the equation  $u^2 - 2u + 2 = 0$  and if  $\cot \theta = x + 1$ , then

$$\text{prove that } \frac{(x+\alpha)^n - (x+\beta)^n}{\alpha - \beta} = \frac{\sin n\theta}{\sin^n \theta}.$$

► **Solution**  $u^2 - 2u + 2 = 0$

$$\Rightarrow u = 1 \pm i$$

$$\text{LHS} = \frac{[(\cot \theta - 1) + (1 + i)]^n - [(\cot \theta - 1) + (1 - i)]^n}{2i}$$

$$= \frac{(\cos \theta + i \sin \theta)^n - (\cos \theta - i \sin \theta)^n}{\sin^n \theta \cdot 2i}$$

$$= \frac{2i \sin n\theta}{\sin^n \theta \cdot 2i} = \frac{\sin n\theta}{\sin^n \theta}$$

► **Example 5.** Show that the real and imaginary part of  $(1 + xi)^n$ , where  $x$  is real and  $n$  is a positive integer, will be equal if  $x = \tan (4r + 1) \pi/4n$ ,  $r$  being zero or any integer.

► **Solution**  $(1 + ix)^n = [\sqrt{1 + x^2} \text{cis} \theta]^n$

$$\text{where } \cos \theta = \frac{1}{\sqrt{1 + x^2}} \text{ and } \sin \theta = \frac{x}{\sqrt{1 + x^2}}$$

$$\text{So } (1 + xi)^n = (1 + x^2)^{n/2} \text{cis} n\theta.$$

$$\text{Hence real part} = (1 + x^2)^{n/2} \cos n\theta.$$

and imaginary part

$$= (1 + x^2)^{n/2} \sin n\theta = (1 + x^2)^{n/2} \cos((1/2)\pi - n\theta).$$

These are equal if

$$\cos n\theta = \cos((1/2)\pi - n\theta)$$

$$\therefore n\theta = 2r\pi \pm ((1/2)\pi - n\theta)$$

The lower sign gives no value for  $\theta$ .

$$\therefore n\theta = 2r\pi + (1/2)\pi - n\theta$$

$$\text{i.e., } 2n\theta = \frac{(4r + 1)\pi}{2}$$

$$\text{i.e., } \theta = \frac{(4r + 1)\pi}{4n}$$

$$\therefore x = \tan \theta = \tan \frac{(4r + 1)\pi}{4n}$$

### Solution of the equation $z^n = a$

$\sum_{k=0}^n$  Let it be required to extract the  $n$ th root of a

number  $a = \rho(\cos \phi + i \sin \phi)$ . Let us assume that this is possible and that we get the number  $z = r(\cos \theta + i \sin \theta)$ , that is  $z^n = a$

$$\text{or, } [r(\cos \theta + i \sin \theta)]^n = \rho(\cos \phi + i \sin \phi) \dots (1)$$

Then, by De Moivre's theorem,  $r^n = \rho$ , that is

$r = \sqrt[n]{\rho}$ , where the right member contains a uniquely determined positive value of the  $n$ th root of the positive real number  $\rho$ . On the other hand, the argument of the left member of (1) is  $n\theta$ . We cannot assert, however, that  $n\theta$  is equal to  $\phi$ , since these angles may actually differ by some integral multiple of  $2\pi$ . Therefore,  $n\theta = \phi + 2k\pi$ , where  $k$  is an integer, whence

$$\theta = \frac{\phi + 2k\pi}{n}$$

Conversely, if we take the number

$$\sqrt[n]{\rho} \left( \cos \frac{\phi + 2k\pi}{n} + i \sin \frac{\phi + 2k\pi}{n} \right), \text{ then for any}$$

integral  $k$ , positive or negative, the  $n$ th power of this

number is equal to  $a$ . Thus  $\sqrt[n]{\rho}(\cos \phi + i \sin \phi)$

$$= \sqrt[n]{\rho} \left( \cos \frac{\phi + 2k\pi}{n} + i \sin \frac{\phi + 2k\pi}{n} \right) \dots (2)$$

Assigning different values to  $k$ , we will not always get distinct values of the required root, Indeed, for  
 $k = 0, 1, 2, \dots, n-1$  ... (3)

we get  $n$  values of the root, all distinct, since increasing  $k$  by unity implies increasing the argument by  $\frac{2\pi}{n}$ .

Now let  $k$  be arbitrary. If  $k = nq + r$ ,  $0 \leq r \leq n-1$ , then

$$\frac{\varphi + 2k\pi}{n} = \frac{\varphi + 2(nq + r)\pi}{n} = \frac{\varphi + 2r\pi}{n} + 2q\pi$$

In other words, the value of the argument for our  $k$  differs from the value of the argument for  $k = r$  by a multiple of  $2\pi$ . We thus obtain the same value of the root as for the value of  $k$  equal to  $r$ , that is, such as lies in the set (3).

Thus, extracting the  $n$ th root of a complex number  $a$  is always possible and yields  $n$  distinct values. All values of the  $n$ th root lie on a circle of radius  $\sqrt[n]{|a|}$  with centre at zero and divide the circle into  $n$  equal parts. In particular, the  $n$ th root of a real number  $a$  also has  $n$  distinct values, of which two, one, or none will be real, depending on the sign of  $a$  and the parity of  $n$ .

Now consider the following examples :

$$\begin{aligned} \text{(i)} \quad \alpha &= \sqrt[3]{2 \left( \cos \frac{3}{4}\pi + i \sin \frac{3}{4}\pi \right)} \\ &= \sqrt[3]{2} \left( \cos \frac{\frac{3}{4}\pi + 2k\pi}{3} + i \sin \frac{\frac{3}{4}\pi + 2k\pi}{3} \right) \end{aligned}$$

$$k = 0 : \alpha_0 = \sqrt[3]{2} \left( \cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right);$$

$$k = 1 : \alpha_1 = \sqrt[3]{2} \left( \cos \frac{11}{12}\pi + i \sin \frac{11}{12}\pi \right);$$

$$k = 2 : \alpha_2 = \sqrt[3]{2} \left( \cos \frac{19}{12}\pi + i \sin \frac{19}{12}\pi \right).$$

$$\text{(ii)} \quad \alpha = \sqrt[3]{1} = \sqrt{\cos \frac{\pi}{2} + i \sin \frac{\pi}{2}}$$

$$= \cos \frac{\frac{\pi}{2} + 2k\pi}{2} + i \sin \frac{\frac{\pi}{2} + 2k\pi}{2};$$

$$\alpha_0 = \cos \frac{\pi}{4} + i \sin \frac{\pi}{4} = \frac{\sqrt{2}}{2} + i \frac{\sqrt{2}}{2};$$

$$\alpha_1 = \cos \frac{5}{4}\pi + i \sin \frac{5}{4}\pi = -\alpha_0.$$

$$\text{(iii)} \quad \alpha = \sqrt[3]{-8} = \sqrt[3]{8(\cos \pi + i \sin \pi)}$$

$$= 2 \left( \cos \frac{\pi + 2k\pi}{3} + i \sin \frac{\pi + 2k\pi}{3} \right);$$

$$\alpha_0 = 2 \left( \cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right) = 1 + i\sqrt{3};$$

$$\alpha_1 = 2 (\cos \pi + i \sin \pi) = -2;$$

$$\alpha_2 = 2 \left( \cos \frac{5\pi}{3} + i \sin \frac{5\pi}{3} \right) = 1 - i\sqrt{3}.$$

► **Example 6.** Calculate  $\sqrt[3]{i}$ .

► **Solution** We write the complex number  $i$  in trigonometric form

$$i = \cos \frac{\pi}{2} + i \sin \frac{\pi}{2}.$$

According to the definition of a root of degree  $n$  of the complex number,  $\sqrt[3]{i}$  are complex numbers  $z$  satisfying the equation

$$z^3 = i. \quad \dots(1)$$

Writing the required number  $z$  in the trigonometric form  $z = r (\cos \varphi + i \sin \varphi)$  and using de Moivre's formula, we write equation (1) in the form

$$r^3 (\cos 3\varphi + i \sin 3\varphi) = \cos \frac{\pi}{2} + i \sin \frac{\pi}{2}.$$

By the definition of the equality of two complex numbers written in trigonometric form, we obtain

$$r^3 = 1, \quad 3\varphi = \frac{\pi}{2} + 2\pi k, \quad k \in \mathbb{Z},$$

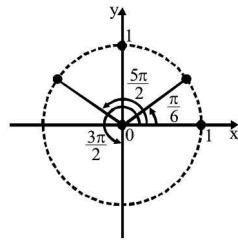
whence it follows that

$$r = \sqrt[3]{1} = 1, \quad \varphi = \frac{\pi}{6} + \frac{2\pi k}{3}.$$

Among the infinite set of numbers  $z$  with modulus 1 and argument  $\varphi = \frac{\pi}{6} + \frac{2\pi k}{3}$ , those numbers obtained for  $k = 0, k = 1, k = 2$  are distinct :

$$\cos \frac{\pi}{6} + i \sin \frac{\pi}{6} = \frac{\sqrt{3}}{2} + i \cdot \frac{1}{2},$$

$$\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6} = -\frac{\sqrt{3}}{2} + i \cdot \frac{1}{2},$$



$$\cos \frac{3\pi}{2} + i \sin \frac{3\pi}{2} = -i.$$

They are precisely the values  $\sqrt[3]{i}$ . In geometrical interpretation the complex numbers  $\sqrt[3]{i}$  can be represented as points lying on a unit circle (since  $r = 1$ ), whose radius vectors form the angles  $\pi/6, 5\pi/6$ , and  $3\pi/2$  with the positive direction of the  $x$ -axis (figure).

► **Example 7.** Find all the values of the number  $\sqrt[4]{-16}$ .

► **Solution** We write the number  $w = -16$  in the trigonometric form:

$$z^4 = 2 = -16 = 16 (\cos \pi + i \sin \pi).$$

This yields

$$z_k = 2 \left( \cos \left( \frac{\pi}{4} + \frac{2\pi k}{4} \right) + i \sin \left( \frac{\pi}{4} + \frac{2\pi k}{4} \right) \right)$$

$$(k = 0, 1, 2, 3).$$

Consequently,

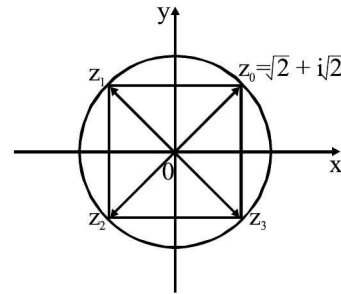
$$z_0 = 2 \left( \cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right) = \sqrt{2} + i\sqrt{2},$$

$$z_1 = 2 \left( \cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4} \right) = -\sqrt{2} + i\sqrt{2},$$

$$z_2 = 2 \left( \cos \frac{5\pi}{4} + i \sin \frac{5\pi}{4} \right) = -\sqrt{2} - i\sqrt{2}$$

$$z_3 = 2 \left( \cos \frac{7\pi}{4} + i \sin \frac{7\pi}{4} \right) = \sqrt{2} - i\sqrt{2}$$

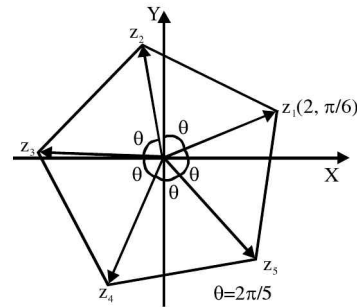
The figure represents all the four values of  $\sqrt[4]{-16}$ . The points corresponding to the numbers  $z_0, z_1, z_2, z_3$  are found at the vertices of the square inscribed into a circle of radius 2 with centre at the point  $z = 0$ .



► **Example 8.** Find the fifth roots of  $16(-\sqrt{3} + i)$  and plot them on the Argand plane.

► **Solution** Expressing it in its polar form,

$$\text{we have } z_0 = 16(-\sqrt{3} + i) = 32e^{i(5\pi/6 + 2k\pi)/5}, k \in \mathbb{I}$$



Therefore, we have

$$z_0^{(1/5)} = 2e^{i(\pi/6)} e^{i(2k\pi/5)}, k \in \mathbb{I} \quad \dots(1)$$

Putting  $k = 0, 1, 2, 3, 4$  in equation (1) gives five unique roots of  $z_0^{(1/5)}$

$$= \lambda, \lambda e^{i(2\pi/5)}, \lambda e^{i(4\pi/5)}, \lambda e^{i(6\pi/5)}, \lambda e^{i(8\pi/5)}$$

where  $\lambda = 2e^{i(\pi/5)}$

Note that the roots form the vertices of a 5-sided regular polygon.

► **Example 9.** For example let us write all the values of the root  $\sqrt[4]{\sqrt{3} + i}$  in the exponential form.

► **Solution** We represent the number  $\sqrt{3} + i$  in the exponential form and make use of the formula

$$\sqrt[4]{\sqrt{3} + i} = \sqrt[4]{2i^{\frac{\pi}{6}}} = \sqrt[4]{2}e^{i\left(\frac{\pi}{24} + \frac{\pi k}{2}\right)}, k = 0, 1, 2, 3.$$

### Practice Problems [1]

1. Simplify the following:

(a)  $\left(\frac{1+i}{1-i}\right)^{4n+1}$

(b)  $\frac{(1+i)^n}{(1-i)^{n-2}}$  ( $n$  is a positive integer)

2. If  $a = \cos \alpha - i \sin \alpha$  and  $b = \cos \beta - i \sin \beta$

prove that  $\frac{(a+b)(1-ab)}{(a-b)(1+ab)} = \frac{(\sin \alpha + \sin \beta)}{(\sin \alpha - \sin \beta)}$

3. If  $x = \cos \theta_1 + i \sin \theta_1$ ,  $y = \cos \theta_2 + i \sin \theta_2$ ,  $z = \cos \theta_3 + i \sin \theta_3$  and  $x + y + z = xyz$ , Show that  $\sum \cos(\theta_2 - \theta_3) = -1$ .

4. If  $\cos x + \cos y + \cos z = \sin x + \sin y + \sin z = 0$ , then prove that  $\cos 2x + \cos 2y + \cos 2z = 0$ .

5. If  $(1+x)^n = C_0 + C_1x + C_2x^2 + \dots + C_nx^n$  ( $n \in \mathbb{N}$ ), prove that

(i)  $C_2 + C_6 + C_{10} + \dots = \frac{1}{2} \left[ 2^{n-1} - 2^{n/2} \cos \frac{n\pi}{4} \right]$ ,

(ii)  $C_3 + C_7 + C_{11} + \dots = \frac{1}{2} \left[ 2^{n-1} - 2^{n/2} \sin \frac{n\pi}{4} \right]$

6. Calculate  $z^{14} + 1/z^{14}$  if  $z$  is a root of the equation  $z + 1/z = 1$ .

7. Find all the values of  $\sqrt[n]{w}$ , if

(i)  $w = -i$ ,  $n = 2$ ; (ii)  $w = 8i$ ,  $n = 3$

8. Solve the following equations:

(i)  $z^5 - 1 - i\sqrt{3} = 0$  (ii)  $z^6 + 64 = 0$

9. Find all the values of  $\sqrt[3]{w}$ , if  $w = -4 + \sqrt{48}i$

10. Compute  $\sqrt[4]{-81i}$

11. Solve the following system in complex numbers:

(a)  $z^5 w^7 = 1$ ,  $z^2 - \bar{w}^3 = 0$

(b)  $z^{13} w^{19} = 1$ ,  $z^2 + w^2 = -2$ .

12. Solve the equation  $z^3 = i(z-1)^3$  and show that the points on the Argand diagram corresponding to these roots are collinear.

13. Solve  $x^6 - 9x^3 + 8 = 0$

### 1.11 CUBE ROOTS OF UNITY

Let us now find the cube roots of unity.

For this we need to find the roots of the equation  $z^3 = 1$ .

$$z^3 = \cos 0 + i \sin 0$$

$$z^3 = \cos 2k\pi + i \sin 2k\pi$$

$$z = \cos \frac{2k\pi}{3} + i \sin \frac{2k\pi}{3}, \text{ where } k = 0, 1, 2.$$

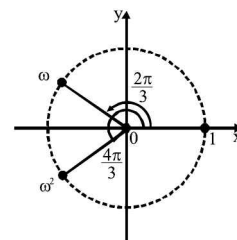
The cube roots of unity are:

$$\cos 0 + i \sin 0, \cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3},$$

$$\cos \frac{4\pi}{3} + i \sin \frac{4\pi}{3}$$

or,  $1, e^{i(2\pi/3)}, e^{i(4\pi/3)}$

or,  $1, \frac{-1+i\sqrt{3}}{2}, \frac{-1-i\sqrt{3}}{2}$ .



There is one real root and two imaginary roots. If we call one of the imaginary roots as  $\omega$ , then the cube roots of unity are  $1, \omega, \omega^2$ .

**Properties of Cube Root of Unity**

- (i) The three cube roots of unity when plotted on the Argand plane constitute the vertices of an equilateral triangle.
- (ii) If  $w$  is one of the imaginary cube roots of unity then  $1 + \omega + \omega^2 = 0$  and  $\omega^3 = 1$ .
- (iii) In general  $1 + \omega^r + \omega^{2r} = 0$  if  $r \in \mathbb{I}$  but is not the multiple of 3.  
 $= 3$  if  $r$  is a multiple of 3.
- (iv)  $z^3 - 1 = (z - 1)(z^2 + z + 1)$   
 $= (z - 1)(z - \omega)(z - \omega^2)$   
 $\Rightarrow z^2 + z + 1 = (z - \omega)(z - \omega^2)$
- (v)  $z^3 + 1 = (z + 1)(z^2 - z + 1)$   
 $= (z + 1)(z + \omega)(z + \omega^2)$   
 Thus the cube roots of  $-1$  are  $-1, -\omega, -\omega^2$ .
- (vi)  $a^3 - b^3 = (a - b)(a - \omega b)(a - \omega^2 b)$   
 $a^3 + b^3 = (a + b)(a + \omega b)(a + \omega^2 b)$   
 $a^2 + ab + b^2 = (a - \omega b)(a - \omega^2 b)$   
 $a^3 + b^3 + c^3 - 3abc = (a + b + c)(a + \omega b + \omega^2 c)(a + \omega^2 b + \omega c)$ .

► **Example 1.** Prove that  $a^3 + b^3 + c^3 - 3abc = (a + b + c)(a + b\omega + c\omega^2)(a + b\omega^2 + c\omega)$ , where  $\omega$  is an imaginary cube root of unity.

► **Solution** We have  $a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$   
 Now  $a^2 + b^2 + c^2 - ab - bc - ca$   
 $= a^2 + b^2 + c^2 + (\omega + \omega^2)ab + (\omega + \omega^2)bc + (\omega + \omega^2)ca$  [ $\because \omega + \omega^2 = -1$ ]  
 $= (a^2 + ab\omega + ac\omega^2) + (ba\omega^2 + b^2\omega^3 + bc\omega^4) + (ca\omega + cb\omega^2 + c^2\omega^3)$   
 $= a(a + b\omega + c\omega^2) + b\omega^2(a + b\omega + c\omega^2) + c\omega(a + b\omega + c\omega^2)$   
 $= (a + b\omega + c\omega^2)(a + b\omega^2 + c\omega)$   
 $\therefore a^3 + b^3 + c^3 - 3abc = (a + b + c)(a + b\omega + c\omega^2)(a + b\omega^2 + c\omega)$

► **Example 2.** If  $(1 + x + x^2)^{3n} = \sum_{r=0}^{6n} a_r x^r$ , then compute the value of  $a_0 + a_6 + a_{12} + \dots + a_{6n}$ .

► **Solution**  $(1 + x + x^2)^{3n} = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4 + a_5 x^5 + a_6 x^6 + \dots + a_{6n} x^{6n}$

Putting  $x = 1$

$$\begin{aligned} \Rightarrow 3^{3n} &= a_0 + a_1 + a_2 + a_3 + a_4 + a_5 + a_6 + \dots + a_{6n} \\ x &= w, w \text{ is an imaginary cube root of unity} \\ \Rightarrow 0 &= a_0 + a_1 w + a_2 w^2 + a_3 + a_4 w + a_5 w^2 + a_6 + \dots + a_{6n} \\ x &= w^2 \\ \Rightarrow 0 &= a_0 + a_1 w^2 + a_2 w + a_3 + a_4 w^2 + a_5 w + a_6 + \dots + a_{6n} \\ x &= -1 \Rightarrow 1 = a_0 - a_1 + a_2 - a_3 + a_4 - a_5 + a_6 - \dots + a_{6n} \\ x &= -w \\ \Rightarrow (-2)^{3n} &= a_0 - a_1 w + a_2 w^2 - a_3 + a_4 w - a_5 w^2 + a_6 - \dots + a_{6n} \\ x &= -w^2 \Rightarrow (-2)^{3n} = a_0 - a_1 w^2 + a_2 w - a_3 + a_4 w^2 - a_5 w + a_6 - \dots + a_{6n} \end{aligned}$$

$$\text{Adding } \frac{3^{3n} + 1 + (-1)^{3n} \cdot 2^{3n+1}}{6} = a_0 + a_6 + \dots + a_{6n}.$$

► **Example 3.** If ' $n$ ' is odd and not a multiple of 3, prove that  $x(x+1)(x^2+x+1)$  is a factor of  $(x+1)^n - x^n - 1, n \in \mathbb{N}$ .

► **Solution** Let  $f(x) = (x+1)^n - x^n - 1$   
 put  $x = 0$ ;  $f(0) = 0 \Rightarrow x$  is factor of  $f(x)$   
 put  $x = -1$ ;  $f(-1) = -(-1)^n - 1 = 1 - 1 = 0$   
 $\Rightarrow x+1$  is a factor of  $f(x)$ .

Put  $x = \omega$ ;  $f(\omega) = (1 + \omega)^n - \omega^n - 1$   
 $= (-\omega^2)^n - \omega^n - 1$   
 $= -\omega^{2n} - (\omega^n + 1)$   
 $= 0$  (since  $1 + \omega^n + \omega^{2n} = 0$ )

Similarly put  $x = \omega^2$  and proved  $f(\omega^2) = 0$ . Hence the expression

$x(x+1)(x^2+x+1) = x(x+1)(x-\omega)(x-\omega^2)$  divides  $f(x)$ .

► **Example 4.** If  $(1+x)^n = C_0 + C_1 x + C_2 x^2 + \dots + C_n x^n$  ( $n \in \mathbb{N}$ ;  $C_r = {}^nC_r$ )

and  $S_1 = C_0 + C_3 + C_6 + \dots$

$S_2 = C_1 + C_4 + C_7 + \dots$

$S_3 = C_2 + C_5 + C_8 + \dots$

Show that the values of  $S_1, S_2$  &  $S_3$  are

respectively given by,  $\frac{1}{3} \left( 2^n + 2 \cos \frac{r\pi}{3} \right)$

with,  $r = n$  for  $S_1$ ;  $r = n - 2$  for  $S_2$

and  $r = (n + 2)$  for  $S_3$ .

► **Solution** Putting  $x = 1, \omega$  &  $\omega^2$  in the expansion of  $(1+x)^n$  where  $\omega$  is the cube root of unity.

$$2^n = C_0 + C_1 + C_2 + \dots + C_n \quad \dots(1)$$

$$(1 + \omega)^n = C_0 + C_1 \omega + C_2 \omega^2 + \dots + C_n \omega^n \dots(2)$$

$$(1 + \omega^2)^n = C_0 + C_1 \omega^2 + C_2 \omega^4 + \dots + C_n \omega^{2n} \dots(3)$$

Adding

$$3S_1 = 2^n + (1 + \omega)^n + (1 + \omega^2)^n \\ = 2^n + 2 \text{ Real part of } (1 + \omega)^n$$

$$= 2^n + 2 \cos \frac{n\pi}{3} \quad (\text{verify})$$

$$\therefore S_1 = \frac{1}{3} \left( 2^n + 2 \cos \frac{r\pi}{3} \right) \quad \text{where } r = n$$

Again (1) + (2)  $\times \omega^2$  + (3)  $\times \omega$  gives

$$3S_2 = 2^n + \omega^2 (1 + \omega)^n + \omega (1 + \omega^2)^n \\ = 2^n + 2 \text{ Real part of } \omega^2 (1 + \omega)^n$$

$$= 2^n + 2 \cos \frac{n-2}{3} \pi$$

$$\therefore S_2 = \frac{1}{3} \left[ 2^n + 2 \cos \frac{r\pi}{3} \right] \quad \text{where } r = n-2.$$

Similarly  $S_3$  can be found out.

► **Example 5.** Prove that  $(x+y)^n - x^n - y^n$  is divisible by  $xy(x+y)(x^2+xy+y^2)$  if  $n$  is an odd number not divisible by 3.

► **Solution** Considering the given expression as a polynomial in  $y$ , let us put  $y=0$ . We see that at  $y=0$  the polynomial vanishes (for any  $x$ ). Therefore our polynomial is divisible by  $y$ . Since it is symmetrical both with respect to  $x$  and  $y$ , it is divisible by  $x$  as well. Thus, the polynomial is divisible by  $xy$ . To prove that it is divisible by  $x+y$ , let us put in it  $y=-x$ . It is evident that for odd  $n$  we have

$$(x-x)^n - x^n - (-x)^n = 0.$$

Consequently, our polynomial is divisible by  $x+y$ . It only remains to prove that divisibility of the polynomial by

$$x^2 + xy + y^2 = (y - x\omega)(y - x\omega^2).$$

where  $\omega^2 + \omega + 1 = 0$ .

For this purpose it only remains to replace  $y$  first by  $x\omega$  and then by  $x\omega^2$  and to make sure that with these substitutions the polynomial vanishes. Since, by

hypothesis,  $n$  is not divisible by three, it follows that  $n = 3l+1$  or  $3l+2$ . At  $y = x\omega$  the polynomial attains the following value  $(x+x\omega)^n - x^n(x\omega)^n = x^n \{ \omega^{2n} + 1 + \omega^n \} = x^n (1 + \omega + \omega^2) = 0$ . Likewise we prove that at  $y = x\omega^2$  the polynomial vanishes as well, and consequently, its divisibility by  $xy(x+y)(x^2+xy+y^2)$  is proved.

► **Example 6.** Let  $Z = 18 + i26$

where  $Z_0 = x_0 + i y_0$  ( $x_0, y_0 \in \mathbb{R}$ ) is the cube root of  $Z$  having least positive argument. Find the value of  $x_0 y_0 (x_0 + y_0)$ .

► **Solution** Let  $Z = 18 + 26i$

let  $r \cos \theta = 18$  and  $r \sin \theta = 26$

$$\therefore r^2 = 324 + 676 = 1000 \Rightarrow r = 10\sqrt{10}$$

$$\tan \theta = \frac{26}{18} = \frac{13}{9};$$

$$\text{hence } \theta \in \left(0, \frac{\pi}{2}\right); \frac{\theta}{3} \in \left(0, \frac{\pi}{6}\right)$$

$$\therefore Z^{1/3} = [10\sqrt{10} (\cos \theta + i \sin \theta)]^{1/3}$$

$$= 10 \left[ \cos \frac{\theta}{3} + i \sin \frac{\theta}{3} \right]$$

$$\text{now } \tan \theta = \frac{3 \tan(\theta/3) - \tan^3(\theta/3)}{1 - 3 \tan^2(\theta/3)} = \frac{3t - t^3}{1 - 3t^2}$$

$$\text{where } t = \tan \frac{\theta}{3}$$

$$13(1 - 3t^2) = 9(3t - t^3) \Rightarrow 13 - 39t^2 = 27t - 9t^3$$

$$\Rightarrow 9t^3 - 39t^2 - 27t + 13 = 0$$

$$3t^2(3t - 1) - 12t(3t - 1) - 13(3t - 1) = 0$$

$$\Rightarrow (3t - 1)(3t^2 - 12t - 13) = 0$$

$$\therefore \tan \frac{\theta}{3} = \frac{1}{3} \Rightarrow \frac{\theta}{3} \in \left(0, \frac{\pi}{6}\right)$$

$$\sin \frac{\theta}{3} = \frac{1}{\sqrt{10}} \quad \text{and} \quad \cos \frac{\theta}{3} = \frac{3}{\sqrt{10}}$$

$$\therefore Z^{1/3} = 1 + 3i \Rightarrow x_0 = 1 \text{ and } y_0 = 3$$

$$\Rightarrow x_0 y_0 (x_0 + y_0) = 12.$$

## Practice Problems

[J]

1. If  $\alpha$  and  $\beta$  are the complex cube roots of unity, show that  $\alpha^4 + \beta^4 + \alpha^{-1}\beta^{-1} = 0$ .
2. Prove that  $\frac{1}{1+2\omega} + \frac{1}{2+\omega} - \frac{1}{1+\omega} = 0$ , where  $\omega$  is imaginary cube root of unity.
3. Prove that the value of  $\sum_{n=1}^5 \left( x^n + \frac{1}{x^n} \right)^2$  when  $x^2 - x + 1 = 0$  is 8.
4. If  $x = a + b$ ;  $y = a\omega + b\omega^2$ ;  $z = a\omega^2 + b\omega$ ;  $\omega$  is cube root of unity show that
  - (i)  $xyz = a^3 + b^3$
  - (ii)  $x^2 + y^2 + z^2 = 6ab$
  - (iii)  $x^3 + y^3 + z^3 = 3(a^3 + b^3)$
5. If  $a, b$  are the roots of  $x^2 + px + q = 0$  then prove that  $(\omega a + \omega^2 b)(\omega^2 a + \omega b) = p^2 - 3q$  where  $\omega$  and  $\omega^2$  are imaginary cube roots of unity.
6. Prove that  $x^3 + y^3 - 3xy = -1$  if  $x = \frac{-1+i\sqrt{3}}{2}$ ,  $y = \frac{-1-\sqrt{3}i}{2}$ .
7. If  $x = 1 + i\sqrt{3}$ ;  $y = 1 - i\sqrt{3}$  &  $z = 2$ , then prove that  $x^p + y^p = z^p$  for every prime  $p > 3$ .
8. Prove that  $(2 - \omega)(2 - \omega^2)(2 - \omega^{10})(2 - \omega^{11}) = 49$ .
9. If  $1, \omega, \omega^2$  are the cube roots of unity, then prove that  $(3 + 3\omega + 5\omega^2)^6 - (2 + 6\omega + 2\omega^2)^3 = 0$ .
10. Prove that  $(1 - \omega + \omega^2)(1 - \omega^2 + \omega^4)(1 - \omega^4 + \omega^8) \dots$  to  $2n$  factors  $= 2^{2n}$ .
11. If  $1, \omega, \omega^2$  are the cube roots of unity, prove that
 
$$\frac{3}{x^3 - 1} = \frac{1}{x - 1} + \frac{1}{x\omega - 1} + \frac{1}{x\omega^2 - 1}$$
12. Find all non-zero complex numbers  $Z$  satisfying  $\bar{Z} = iZ^2$ .
13. Prove that the polynomial  $x^{3l} + x^{3m+1} + x^{3n+2}$  is exactly divisible by  $x^2 + x + 1$  if  $l, m$  &  $n$  are any non-negative integers.

1.12 N<sup>TH</sup> ROOTS OF UNITY

We start by solving  $z^n = a$  and later use  $a = 1$  to get  $n$ th roots of unity

Let  $a = \rho(\cos\phi + i\sin\phi)$ ,

where  $\rho$  is positive and  $\phi$  is an angle such that  $-\pi < \phi \leq \pi$ . If we put  $z = r(\cos\theta + i\sin\theta)$ , the equation takes the form

$$r^n(\cos n\theta + i\sin n\theta) = \rho(\cos\phi + i\sin\phi);$$

so that

$$r^n = \rho, \cos n\theta = \cos\phi, \sin n\theta = \sin\phi \quad \dots(1)$$

The only possible value of  $r$  is  $\sqrt[n]{\rho}$ , the ordinary arithmetical  $n$ th root of  $\rho$ ; and in order that the last two equations should be satisfied it is necessary and sufficient that  $n\theta = \phi + 2k\pi$ . Hence the equation

$$z^n = a = \rho(\cos\phi + i\sin\phi) \quad \dots(2)$$

has just  $n$  roots, given by  $z = r(\cos\theta + i\sin\theta)$ , where

$$r = \sqrt[n]{\rho}, \theta = (\phi + 2k\pi)/n, (k=0, 1, \dots, n-1) \quad \dots(3)$$

That the  $n$  roots are distinct is easily seen by plotting them on Argand's diagram.

The particular root  $\sqrt[n]{\rho} \{ \cos(\phi/n) + i\sin(\phi/n) \}$  is called the principal value of  $\sqrt[n]{a}$ .

The case in which  $a = 1, \rho = 1, \phi = 0$  is of particular interest.

The  $n$  roots of the equation  $z^n = 1$  are

$$\cos(2k\pi/n) + i\sin(2k\pi/n), (k = 0, 1, \dots, n-1). \quad \dots(4)$$

These numbers are called the  $n$ th roots of unity; the principal value is unity itself. If we write  $\alpha$  for  $\cos(2\pi/n) + i\sin(2\pi/n)$ , we see that the  $n$ th roots of unity are

$$1, \alpha, \alpha^2, \dots, \alpha^{n-1}.$$

For example, the two square roots of 1 are 1, -1; the

three cube roots are  $1, \frac{1}{2}(-1 + i\sqrt{3}), \frac{1}{2}(-1 - i\sqrt{3})$ ; the four fourth roots are 1,  $i, -1, -i$ ; and the five fifth roots are 1,

$$\frac{1}{4}[\sqrt{5} - 1 + i(\sqrt{10 + 2\sqrt{5}})],$$

$$\begin{aligned} & \frac{1}{4}[\sqrt{5} - 1 + i\sqrt{10 - 2\sqrt{5}}] \\ & \frac{1}{4}[-\sqrt{5} - 1 - i\sqrt{10 - 2\sqrt{5}}] \\ & \frac{1}{4}[\sqrt{5} - 1 - i\sqrt{10 + 2\sqrt{5}}] \\ & \frac{1}{4}[\sqrt{5} - 1 + i(\sqrt{10 + 2\sqrt{5}})] \end{aligned}$$

### Properties of nth roots of unity

- (i) The real values of the nth root of unity are obtained from formula (4) for the values  $k = 0$  and  $\frac{n}{2}$ , if  $n$  is even, and for  $k = 0$  if  $n$  is odd.
- (ii) In the complex plane, the nth roots of unity are located on the circumference of the unity circle and divide it into  $n$  equal arcs : one of the division points is the number 1.
- (iii) From this it follows that those of the nth roots of unity which are not real are situated symmetrically about the real axis (that is, are pairwise conjugate).
- (iv) All values of the nth root of a complex number  $a$  may be obtained by multiplying one of these values by all the nth roots of unity. Indeed, let  $\beta$  be one of the values of the nth root of the number  $a$ , i.e.,  $\beta^n = a$  and let  $\alpha$  be an arbitrary value of the nth root of unity, that is,  $\alpha^n = 1$ . Then  $(\beta\alpha)^n = \beta^n\alpha^n = a$ . Thus  $\beta\alpha$  is also one of the values for  $\sqrt[n]{a}$ . Multiplying  $\beta$  by each of the nth roots of unity, we get  $n$  distinct values of the nth root of the number  $a$ , that is, all the values of this root.

For instance, one of the values of the cube root of  $-8$  is 2.

The two others are the numbers  $-2\omega = 1 - i\sqrt{3}$  and  $-2\omega^2 = 1 + i\sqrt{3}$ .

$\sqrt[4]{81}$  has four values : 3,  $-3$ ,  $3i$ ,  $-3i$ .

- (v) The product of two nth roots of unity is itself an nth root of unity.

Indeed, if  $\alpha^n = 1$  and  $\beta^n = 1$ , then  $(\alpha\beta)^n = \alpha^n\beta^n = 1$ .

- (vi) Also, the reciprocal of an nth root of unity is itself that root.

Let  $\alpha^n = 1$ . Then from  $\alpha \cdot \alpha^{-1} = 1$  it follows that  $\alpha^n \cdot (\alpha^{-1})^n = 1$ , that is,  $(\alpha^{-1})^n = 1$ .

- (vii) Generally, any power of the nth root of unity is also an nth root of unity.
- (viii) Any kth root of unity will also be an lth root of unity for any  $l$  that is a multiple of  $k$ .

Whence it follows that if we regard the entire collection of nth roots of unity, then some of these roots will already be  $n'$ th roots of unity for some  $n'$  which are divisors of the number  $n$ .

- (ix) However, for any  $n$ , there exist nth roots of unity such that they are not any lesser roots of unity. These roots are termed **primitive** nth roots of unity. Their existence follows from formula (4): if the value of root corresponding to a given value of  $k$  is denoted by  $\alpha_k$  (so that  $\alpha_0 = 1$ ), then on the basis of De Moivre's theorem,  $\alpha_1^k = \alpha_k$ . Thus, no power of  $\alpha_1$  less than the nth will be

equal to 1, that is  $\alpha_1 = \cos \frac{2\pi}{n} + i \sin \frac{2\pi}{n}$  is a primitive root.

- (x) An nth root  $\alpha$  of unity is a primitive nth root if and only if its powers  $\alpha^k$ ,  $k = 0, 1, \dots, n-1$ , are distinct, that is, if they exhaust all the nth roots of unity.

Indeed, if all the indicated powers of the number  $\alpha$  are distinct, then  $\alpha$  is obviously an nth primitive root. But if, for example,  $\alpha^k = \alpha^l$  for  $0 \leq k < l \leq n-1$ , then  $\alpha^{l-k} = 1$ ; that is, because of the inequalities  $1 \leq l-k \leq n-1$ , the root  $\alpha$  will not be primitive.

The number  $\alpha_1$  found above is not, in the general case, the only primitive nth root. The following theorem is used to find all of these roots.

- (xi) If  $\alpha$  is a primitive nth root of unity, then the number  $\alpha^k$  is a primitive nth root if and only if  $k$  is relatively prime to  $n$ .

Let  $d$  be the largest common divisor of the numbers  $k$  and  $n$ . If  $d > 1$  and  $k = dk'$ ,  $n = dn'$ , then  $(\alpha^k)^{n'} = \alpha^{kn'} = \alpha^{k'n} = (\alpha^n)^{k'} = 1$

that is, the root  $\alpha^k$  is an  $n'$ th root of unity.

On the other hand, let  $d = 1$  and at the same time let the number  $\alpha^k$  be an  $m$ th root of unity,  $1 \leq m < n$ . Thus,  $(\alpha^k)^m = \alpha^{km} = 1$



Since the number  $\alpha$  is a primitive  $n$ th root of unity, that is, only its powers with exponents that are multiples of  $n$  can be equal to unity, it follows that the number  $k\alpha$  is a multiple of  $n$ . But since  $1 \leq k < n$ , the numbers  $k$  and  $n$  cannot be relatively prime; this contradicts the assumption.

(xii) Thus, the number of primitive  $n$ th roots of unity is equal to the number of positive integers  $k$  less than  $n$  and relatively prime to  $n$ . The expression for this number, then all these roots except unity itself will be primitive  $p$ th roots of unity. On the other hand,  $i$  and  $-i$  (not 1 and  $-1$ ) will be among the primitive fourth roots of unity.

(xiii) The  $n$ th roots of unity can also be written as :

$$1, e^{i\frac{2\pi}{n}}, e^{i\frac{4\pi}{n}}, \dots, e^{i\frac{2(n-1)\pi}{n}};$$

$$\text{or, } 1, \alpha, \alpha^2, \dots, \alpha^{n-1} \text{ where } \alpha = e^{i\frac{2\pi}{n}} \text{ since}$$

they are in G.P. with common ratio  $e^{i\frac{2\pi}{n}}$ .

or,  $1, \alpha, \alpha^2, \dots, \alpha^{n-1}, \bar{\alpha}^{n-1}, \bar{\alpha}$  since the imaginary roots appear in conjugate pair.

Sometimes, they are also denoted as  $1, \alpha_1, \alpha_2, \alpha_3, \dots, \alpha_{n-1}$ .

(xiv)  $z^n - 1 = (z - 1)(z^{n-1} + z^{n-2} + \dots + 1)$

$$\text{or, } z^n - 1 = (z - 1)(z - \alpha)(z - \alpha^2) \dots (z - \alpha^{n-1}).$$

Note that the roots of  $z^{n-1} + z^{n-2} + \dots + 1 = 0$  are the  $n$ th roots of unity except 1.

(xv) Sum of roots (using theory of equations)

$$1 + \alpha + \dots + \alpha^{n-2} + \alpha^{n-1} =$$

$$-\frac{\text{coefficient of } z^{n-1}}{\text{coefficient of } z^n} = 0.$$

$$\Rightarrow \sum_{k=0}^{n-1} \left( \cos \frac{2k\pi}{n} + i \sin \frac{2k\pi}{n} \right) = 0$$

$$\Rightarrow \sum_{k=0}^{n-1} \cos \frac{2k\pi}{n} + i \sum_{k=0}^{n-1} \sin \frac{2k\pi}{n} = 0$$

$$\Rightarrow \sum_{k=0}^{n-1} \cos \frac{2k\pi}{n} = 0, \quad \sum_{k=0}^{n-1} \sin \frac{2k\pi}{n} = 0$$

$$\text{(xvi)} \quad 1 + \alpha^p + \alpha^{2p} + \dots + \alpha^{p(n-1)}$$

$$= 1^p + \alpha_1^p + \alpha_2^p + \dots + \alpha_{n-1}^p$$

= 0 if  $p$  is not an integral multiple of  $n$   
or,  $n$  if  $p$  is an integral multiple of  $n$ .

(xvii) The product of the roots

$$= 1 \cdot \alpha_1 \cdot \alpha_2 \cdot \alpha_3 \cdot \dots \cdot \alpha_{n-1}$$

$$= 1 \cdot \alpha \cdot \alpha^2 \cdot \dots \cdot \alpha^{n-1}$$

$$= (-1)^n \frac{\text{coefficient of } x^0}{\text{coefficient of } x^n}$$

$$= \frac{(-1)^n (-1)}{1} = \frac{(-1)^{n+1}}{1}$$

= 1 or  $-1$  according as  $n$  is odd or even.

(xviii) By putting  $z = 1$  and  $-1$  in (xiv) we get :

$$(1 - \alpha_1)(1 - \alpha_2) \dots (1 - \alpha_{n-1}) = n$$

$$(1 + \alpha_1)(1 + \alpha_2) \dots (1 + \alpha_{n-1}) = 0$$

if  $n$  is even and 1 if  $n$  is odd.

### **$n$ th Roots of $(-1)$**

Since  $\cos \pi + i \sin \pi = -1$ ,

the  $n$ th roots are the values

$$\cos \frac{(2k+1)\pi}{n} + i \sin \frac{(2k+1)\pi}{n} \text{ where}$$

$$k = 0, 1, 2, \dots, n-1.$$

The principal  $n$ -th root of  $(-1)$  is  $\cos \frac{\pi}{n} + i \sin \frac{\pi}{n}$ ,

corresponding to  $k = 0$ .

If  $n$  is odd,  $k = \frac{1}{2}(n-1)$  gives  $(-1)$  as one root.

### **Factors of $x^n - 1$ and $x^n + 1$**

1. The  $n$  factors of  $x^n - 1$  are

$$x - \{\cos 2r\pi/n + i \sin 2r\pi/n\} \text{ where}$$

$$r = 0, 1, 2, \dots, n-1.$$

- (i) If  $n$  is even, factors  $x - 1$  and  $x + 1$  are given by  $r = 0, r = n/2$ . The remaining  $(n - 2)$  factors can be grouped in pairs as follows :

Since  $2r\pi/n + 2(n - r)\pi/n = 2\pi$ , the factors corresponding to  $r$  and  $n - r$  are

$$x - \left\{ \cos \frac{2r\pi}{n} + i \sin \frac{2r\pi}{n} \right\}$$

$$\text{and } x - \left\{ \cos \frac{2r\pi}{n} - i \sin \frac{2r\pi}{n} \right\}$$

And the product of these is

$$\begin{aligned} \left( x - \cos \frac{2r\pi}{n} \right)^2 + \sin^2 \frac{2r\pi}{n}; \\ = x^2 - 2x \cos \frac{2r\pi}{n} + 1; \end{aligned}$$

therefore,  $x^n - 1$

$$= (x-1)(x+1) \prod_{r=1}^{r=\frac{1}{2}(n-2)}$$

$$\left( x^2 - 2x \cos \frac{2r\pi}{n} + 1 \right) \quad \dots(1)$$

where  $\Pi$  denotes the product of the factors as indicated.

- (ii) If  $n$  is odd, the factor  $x - 1$  is given by  $r=0$ . The remaining  $n - 1$  factors can be grouped in pairs, and

$$\begin{aligned} x^n - 1 &= (x - 1) \prod_{r=1}^{r=\frac{1}{2}(n-1)} \\ &\left( x^2 - 2x \cos \frac{2r\pi}{n} + 1 \right) \quad \dots(2) \end{aligned}$$

2. Similarly, the  $n$  factors of  $x^n + 1$  are  $x - \{ \cos(2r + 1)\pi/n + i \sin(2r + 1)\pi/n \}$ , where  $r = 0, 1, 2, \dots (n - 1)$ .

- (i) If  $n$  is even

$$x^n + 1 = \prod_{r=0}^{r=\frac{1}{2}(n-2)}$$

$$\left( x^2 - 2x \cos \frac{(2r+1)\pi}{n} + 1 \right) \dots(3)$$

- (ii) If  $n$  is odd,

$$x^n + 1 = (x + 1) \prod_{r=0}^{r=\frac{1}{2}(n-3)}$$

$$\left( x^2 - 2x \cos \frac{(2r+1)\pi}{n} + 1 \right) \quad \dots(4)$$

- **Example 1.** Determine the value of  $z$  when

$$z^6 = \sqrt{3} + i$$

- **Solution**  $\sqrt{3} + i = 2 \left( \cos \frac{\pi}{6} + i \sin \frac{\pi}{6} \right)$

$$= 2 \left\{ \cos \left( 2k\pi + \frac{\pi}{6} \right) + i \sin \left( 2k\pi + \frac{\pi}{6} \right) \right\},$$

$k = \text{any integer.}$

$$\text{Now } z = (\sqrt{3} + i)^{1/6}$$

$$= 2^{1/6} \left\{ \cos \frac{2k\pi + \frac{\pi}{6}}{6} + i \sin \frac{2k\pi + \frac{\pi}{6}}{6} \right\}$$

where  $k = 0, 1, 2, 3, 4, 5$ .

- **Example 2.** If  $1, \alpha_1, \alpha_2, \dots, \alpha_{n-1}$  be the  $n$ th roots of unity, show that  $(1 - \alpha_1)(1 - \alpha_2) \dots (1 - \alpha_{n-1}) = n$

- **Solution** Let  $x$  be an  $n$ th root of unity, then

$$x^n = 1 \text{ or } x^n - 1 = 0 \quad \dots(1)$$

According to question 1,  $\alpha_1, \alpha_2, \dots, \alpha_{n-1}$  are all the  $n$ th roots of unity

$$\therefore x^n - 1 = (x - 1)(x - \alpha_1)(x - \alpha_2) \dots (x - \alpha_{n-1})$$

$$\Rightarrow (x - \alpha_1)(x - \alpha_2) \dots (x - \alpha_{n-1}) = \frac{x^n - 1}{x - 1}$$

$$\Rightarrow (x - \alpha_1)(x - \alpha_2) \dots (x - \alpha_{n-1}) = 1 + x + x^2 + \dots + x^{n-1} \quad \dots(2)$$

putting  $x = 1$  in (2), we get

$$(1 - \alpha_1)(1 - \alpha_2) \dots (1 - \alpha_{n-1}) = n.$$

- **Example 3.** Find the tenth roots of unity and show that the product of any two of them is again one of the tenth roots.

- **Solution** For  $r = 0, 1, 2, \dots, 9$ , the tenth roots of unity are given by

$$z^{10} = 1 = \cos(2r\pi) + i \sin(2r\pi)$$

So by De Moivre's theorem,

$$z = [\cos(2r\pi) + i \sin(2r\pi)]^{1/10}$$

$$= \cos \frac{2r\pi}{10} + i \sin \frac{2r\pi}{10} = \omega^r,$$

where  $\omega = \cos(\pi/5) + i \sin(\pi/5)$ . Let  $z_1 = \omega^r$  and  $z_2 = \omega^s$  ( $0 \leq r, s \leq 9$ ) be two of these tenth roots. Then

$$\begin{aligned} z_1 z_2 &= \omega^{r+s} = \left( \cos \frac{\pi}{5} + i \sin \frac{\pi}{5} \right)^{r+s} \\ &= \cos \left[ (r+s) \frac{\pi}{5} \right] + i \sin \left[ (r+s) \frac{\pi}{5} \right] \end{aligned}$$

If  $0 \leq r+s \leq 9$ , then  $z_1 z_2$  is also a tenth root of unity. On the other hand, if  $r+s \geq 10$ , let  $r+s = 10+k$ , where  $0 \leq k \leq 8$ , so that

$$\begin{aligned} z_1 z_2 &= \cos \left( 2\pi + \frac{k\pi}{5} \right) + i \sin \left( 2\pi + \frac{k\pi}{5} \right) \\ &= \cos \frac{k\pi}{5} + i \sin \frac{k\pi}{5}, \end{aligned}$$

showing that  $z_1 z_2$  is a tenth root of unit in general.

► **Example 4.** Find the real factors of  $x^6 + 1$ .

► **Solution** To factorise  $x^6 + 1$ , we first find the roots of  $x^6 + 1 = 0$ .

$$x^6 + 1 = 0$$

$$\text{or } x^6 = -1 = \cos(2k+1)\pi + i \sin(2k+1)\pi,$$

$$\text{or } x = \cos \frac{2k+1}{6} \pi + i \sin \frac{2k+1}{6} \pi, \text{ where}$$

$$k = 0, 1, 2, 3, 4, 5.$$

$$\therefore x = \begin{cases} \cos \frac{\pi}{6} + i \sin \frac{\pi}{6}, & \cos \frac{\pi}{2} + i \sin \frac{\pi}{2}, \\ & \cos \frac{5\pi}{6} - i \sin \frac{5\pi}{6} \\ \cos \frac{5\pi}{6} - i \sin \frac{5\pi}{6}, & \cos \frac{\pi}{2} - i \sin \frac{\pi}{2}, \\ & \cos \frac{\pi}{6} - i \sin \frac{\pi}{6} \end{cases}$$

$$\text{or } x = \frac{\sqrt{3}}{2} + i \frac{1}{2}, \frac{\sqrt{3}}{2} - i \frac{1}{2}, i,$$

$$-i, -\frac{\sqrt{3}}{2} + i \frac{1}{2}, -\frac{\sqrt{3}}{2} - i \frac{1}{2}$$

$$\text{Hence } x^6 - 1 = (x-i)(x+i)$$

$$\begin{aligned} &\left( x - \frac{\sqrt{3}}{2} - \frac{i}{2} \right) \left( x - \frac{\sqrt{3}}{2} + \frac{i}{2} \right) \\ &\quad \times \left( x + \frac{\sqrt{3}}{2} - \frac{i}{2} \right) \left( x + \frac{\sqrt{3}}{2} + \frac{i}{2} \right) \\ &= (x^2 + 1) \left\{ \left( x - \frac{\sqrt{3}}{2} \right)^2 + \frac{1}{4} \right\} \left\{ \left( x + \frac{\sqrt{3}}{2} \right)^2 + \frac{1}{4} \right\} \\ &= (x^2 + 1) (x^2 - \sqrt{3}x + 1) (x^2 + \sqrt{3}x + 1). \end{aligned}$$

► **Example 5.** Solve the equation

$$x^{12} - x^{11} + x^{10} - \dots - x + 1 = 0.$$

► **Solution** Multiplying by  $x+1$ , it becomes  $x^{13} + 1 = 0$ . All the roots of this equation, except  $-1$ , are the roots of the given equation.

$$\text{Now } x^{13} = -1 = \cos(2k+1)\pi + i \sin(2k+1)\pi, \\ k = 0, \pm 1, \pm 2, \pm 3, \pm 4, \pm 5, \pm 6.$$

$$\begin{aligned} \text{or } x &= \cos \frac{2k+1}{13} \pi \pm i \sin \frac{2k+1}{13} \pi, k \\ &= 0, 1, 2, 3, 4, 5, 6. \end{aligned}$$

$$\text{For } k = 6, x = -1.$$

Thus the required roots are

$$\begin{aligned} &\cos \frac{2k+1}{13} \pi \pm i \sin \frac{2k+1}{13} \pi, \text{ where} \\ &k = 0, 1, 2, 3, 4, 5. \end{aligned}$$

► **Example 6.** Find the value of:

$$\sum_{k=1}^{10} \left( \sin \frac{2\pi k}{11} - i \cos \frac{2\pi k}{11} \right)$$

► **Solution** Let

$$\begin{aligned} S &= \sum_{k=1}^{10} \left( \sin \frac{2\pi k}{11} - i \cos \frac{2\pi k}{11} \right) \\ &= \sum_{k=1}^{10} \left( -i^2 \sin \frac{2\pi k}{11} - i \cos \frac{2\pi k}{11} \right) \\ &= -i \sum_{k=1}^{10} \left( \cos \frac{2\pi k}{11} + i \sin \frac{2\pi k}{11} \right) = -i \sum_{k=1}^{10} e^{i \frac{2\pi k}{11}} \end{aligned}$$

$$= -i \left[ \sum_{k=0}^{10} e^{j\frac{2\pi k}{11}} - 1 \right] = -i$$

(sum of 11th roots of unit -1)  
 $= -i(0 - 1) = i$

► **Example 7.** If  $\omega^j, j = 0, 1, 2, \dots, n-1$  are the  $n$ th roots of unity, show that

$$\sum_{j=0}^{n-1} |z_1 + \omega^j z_2|^2 = n(|z_1|^2 + |z_2|^2)$$

for any two complex numbers  $z_1$  and  $z_2$ .

► **Solution** Since  $\omega^j, j = 0, 1, 2, \dots, n-1$  are  $n$ th roots of unity.

$$|\omega^j| = 1 \text{ for } j = 0, 1, 2, \dots, n-1 \text{ and } \sum_{j=0}^{n-1} \omega^j = 0$$

We know that if  $a, b$  are complex numbers,

$$|a + b|^2 = |a|^2 + |b|^2 + \bar{a}b + a\bar{b}$$

Thus,  $|z_1 + \omega^j z_2|^2$

$$= |z_1|^2 + |\omega^j z_2|^2 + \bar{z}_1 \omega^j z_2 + z_1 \overline{(\omega^j z_2)}$$

$$= |z_1|^2 + |z_2|^2 + \omega^j \bar{z}_1 z_2 + z_1 \bar{z}_2 \bar{\omega}^j$$

Therefore,

$$\sum_{j=0}^{n-1} |z_1 + \omega^j z_2|^2 = \sum_{j=0}^{n-1} \{ |z_1|^2 + |z_2|^2 + \omega^j \bar{z}_1 z_2 + \bar{\omega}^j z_1 \bar{z}_2 \}$$

$$= n|z_1|^2 + n|z_2|^2 + \bar{z}_1 z_2 \sum_{j=0}^{n-1} \omega^j + z_1 \bar{z}_2 \sum_{j=0}^{n-1} \bar{\omega}^j.$$

$$\text{But } \sum_{j=0}^{n-1} \omega^j = 0 \Rightarrow \left( \sum_{j=0}^{n-1} \omega^j \right) = 0 \Rightarrow \sum_{j=0}^{n-1} \bar{\omega}^j = 0$$

$$\text{Thus } \sum_{j=0}^{n-1} |z_1 + \omega^j z_2|^2 = n|z_1|^2 + n|z_2|^2.$$

► **Example 8.** Solve the equation  $z^8 + 1 = 0$  and deduce that

$$\cos 4\theta = 8 \left( \cos \theta - \cos \frac{\pi}{8} \right) \left( \cos \theta - \cos \frac{3\pi}{8} \right)$$

$$\left( \cos \theta - \cos \frac{5\pi}{8} \right) \left( \cos \theta - \cos \frac{7\pi}{8} \right)$$

► **Solution**  $z^8 + 1 = 0 \Rightarrow z^8 = -1 = \cos \pi + i \sin \pi$   
 $\therefore z = (\cos \pi + i \sin \pi)^{1/8}$

$$= \cos \frac{2r\pi + \pi}{8} + i \sin \frac{2r\pi + \pi}{8}, r = 0, 1, 2, \dots, 7$$

$$\therefore z = \cos \frac{\pi}{8} \pm i \sin \frac{\pi}{8}, \cos \frac{3\pi}{8} \pm i \sin \frac{3\pi}{8}$$

$$\cos \frac{5\pi}{8} \pm i \sin \frac{5\pi}{8}, \cos \frac{7\pi}{8} \pm i \sin \frac{7\pi}{8}$$

These are the roots of eqn.  $z^8 + 1 = 0$ .

Now quadratic eqn. whose roots are  $\cos \frac{\pi}{8} \pm i \sin \frac{\pi}{8}$  is

$$z^2 - 2 \cos \frac{\pi}{8} z + 1 = 0$$

Similarly we can find the quadratic equations whose roots are

$$\cos \frac{3\pi}{8} \pm i \sin \frac{3\pi}{8}, \cos \frac{5\pi}{8} \pm i \sin \frac{5\pi}{8}$$

$$\text{and } \cos \frac{7\pi}{8} \pm i \sin \frac{7\pi}{8}.$$

Thus

$$z^8 + 1 = \left( z^2 - 2 \cos \frac{\pi}{8} z + 1 \right) \left( z^2 - 2 \cos \frac{3\pi}{8} z + 1 \right)$$

$$\left( z^2 - 2 \cos \frac{5\pi}{8} z + 1 \right) \left( z^2 - 2 \cos \frac{7\pi}{8} z + 1 \right)$$

Dividing both sides by  $z^4$ , we get

$$z^4 + \frac{1}{z^4} = \left( z + \frac{1}{z} - 2 \cos \frac{\pi}{8} \right) \left( z + \frac{1}{z} - 2 \cos \frac{3\pi}{8} \right)$$

$$\left( z + \frac{1}{z} - 2 \cos \frac{5\pi}{8} \right) \left( z + \frac{1}{z} - 2 \cos \frac{7\pi}{8} \right)$$

Putting  $z = \cos \theta + i \sin \theta$ , so that  $z^n + \frac{1}{z^n} = 2 \cos n\theta$ , we get

$$2 \cos 4\theta = 2 \left( \cos \theta - \cos \frac{\pi}{8} \right) 2 \left( \cos \theta - \cos \frac{3\pi}{8} \right)$$

$$2 \left( \cos \theta - \cos \frac{5\pi}{8} \right) \left( 2 \cos \theta - \cos \frac{7\pi}{8} \right)$$

$$\therefore \cos 4\theta = 8 \left( \cos \theta - \cos \frac{\pi}{8} \right) \left( \cos \theta - \cos \frac{3\pi}{8} \right)$$

$$\left( \cos \theta - \cos \frac{5\pi}{8} \right) \left( \cos \theta - \cos \frac{7\pi}{8} \right)$$

► **Example 9.** Resolve  $z^7 - 1$  into linear and quadratic factors and hence deduce that,

$$\cos \frac{\pi}{7} \cdot \cos \frac{2\pi}{7} \cdot \cos \frac{4\pi}{7} = \frac{1}{8}.$$

► **Solution** 7<sup>th</sup> roots of unity are

$$1, \left( \cos \frac{2\pi}{7} + i \sin \frac{2\pi}{7} \right), \left( \cos \frac{4\pi}{7} + i \sin \frac{4\pi}{7} \right), \dots$$

$$\text{Hence } (z^7 - 1) = (z - 1) \left( z^2 - 2 \cos \frac{2\pi}{7} z + 1 \right)$$

$$\left( z^2 - 2 \cos \frac{4\pi}{7} z + 1 \right) \left( z^2 - 2 \cos \frac{6\pi}{7} z + 1 \right)$$

Put  $z = i, -(i + 1)$

$$= (i - 1) \left( -2 \cos \frac{2\pi}{7} i \right) \left( -2 \cos \frac{4\pi}{7} i \right) \left( -2 \cos \frac{6\pi}{7} i \right)$$

$$-(i + 1) = (-1 - i) 8 \cos \frac{2\pi}{7} \cos \frac{4\pi}{7} \cos \frac{6\pi}{7}$$

$$1 = 8 \cos \frac{\pi}{7} \cdot \cos \frac{2\pi}{7} \cdot \cos \frac{4\pi}{7}$$

► **Example 10.** Prove that the roots of the equation

$$8x^3 - 4x^2 - 4x + 1 = 0 \text{ are } \cos \frac{\pi}{7}, \cos \frac{3\pi}{7}, \cos \frac{5\pi}{7}.$$

► **Solution** Let  $z = \cos \theta + i \sin \theta$ , then  $z^7 = \cos 7\theta + i \sin 7\theta$

$$\text{if } \theta = \frac{\pi}{7}, \frac{3\pi}{7}, \frac{5\pi}{7}, \pi, \frac{9\pi}{7}, \frac{11\pi}{7}, \frac{13\pi}{7}$$

then  $z^7 = \cos 7\theta + i \sin 7\theta = -1$  or  $z^7 + 1 = 0 \dots (1)$

Thus  $z = \cos \theta + i \sin \theta$ , where

$$\theta = \frac{\pi}{7}, \frac{3\pi}{7}, \frac{5\pi}{7}, \pi, \frac{9\pi}{7}, \frac{11\pi}{7}, \frac{13\pi}{7}$$

are the roots of equation (1).

Also when  $\theta = \pi, z = \cos \pi + i \sin \pi = -1$

Now  $z^7 + 1 = 0 \Rightarrow (z + 1)(z^6 - z^5 + z^4 - z^3 + z^2 - z + 1) = 0$

Root of equation  $z + 1 = 0$  is  $\cos \theta + i \sin \theta$ , where  $\theta = \pi$

$\therefore$  Roots of equation

$$z^6 - z^5 + z^4 - z^3 + z^2 - z + 1 = 0 \dots (2)$$

are  $\cos \theta + i \sin \theta$ , where  $\theta = \frac{\pi}{7}, \frac{3\pi}{7}, \frac{5\pi}{7}, \frac{9\pi}{7}, \frac{11\pi}{7}, \frac{13\pi}{7}$

Let  $x = \cos \theta$ , then

$$z + \frac{1}{z} = \cos \theta + i \sin \theta + \frac{1}{\cos \theta + i \sin \theta} = 2 \cos \theta = 2x$$

Thus  $z + \frac{1}{z} = 2x$ ,

where  $x = \cos \theta$  and  $\theta = \frac{\pi}{7}, \frac{3\pi}{7}, \frac{5\pi}{7}, \frac{9\pi}{7}, \frac{11\pi}{7}, \frac{13\pi}{7}$

$$\text{But } \cos \frac{13\pi}{7} = \cos \left( 2\pi - \frac{\pi}{7} \right) = \cos \frac{\pi}{7},$$

$$\cos \frac{11\pi}{7} = \cos \frac{3\pi}{7}, \cos \frac{9\pi}{7} = \cos \frac{5\pi}{7}$$

$$\text{From (2), } z^3 - z^2 + z - 1 + \frac{1}{z} - \frac{1}{z^2} + \frac{1}{z^3} = 0$$

[Dividing by  $z^6$ ]

$$\Rightarrow \left( z^3 + \frac{1}{z^3} \right) - \left( z^2 + \frac{1}{z^2} \right) + \left( z + \frac{1}{z} \right) - 1 = 0$$

$$\Rightarrow \left( z + \frac{1}{z} \right)^3 - 3z \cdot \frac{1}{z} \left( z + \frac{1}{z} \right) - \left[ \left( z + \frac{1}{z} \right)^2 - 2z \cdot \frac{1}{z} \right] + z + \frac{1}{z} - 1 = 0$$

$$\Rightarrow 8x^3 - 3 \cdot 2x - (4x^2 - 2) + 2x - 1 = 0$$

$$\Rightarrow 8x^3 - 4x^2 - 4x + 1 = 0 \dots (3)$$

Thus the roots of equation (3)

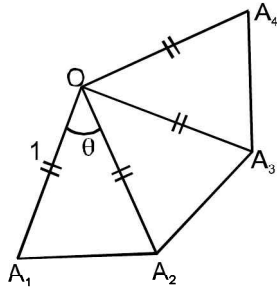
$$\text{are } \cos \frac{\pi}{7}, \cos \frac{3\pi}{7} \text{ and } \cos \frac{5\pi}{7}.$$

► **Example 11.** Let  $A_k$  ( $k = 1, 2, \dots, n$ ) be the vertices of a regular  $n$ -gon inscribed in a unit circle then prove that,  $\prod_{r=2}^{r=n} |A_1 A_r| = n$ .

► **Solution** If  $\alpha_1, \alpha_2, \dots, \alpha_{n-1}$  are  $n^{\text{th}}$  roots of unity then,

$$(1 - \alpha_1)(1 - \alpha_2) \dots (1 - \alpha_{n-1}) = n$$

$$|1 - \alpha_1|^2 |1 - \alpha_2|^2 \dots |1 - \alpha_{n-1}|^2 = n^2$$



when  $\alpha_1 = \cos \theta + i \sin \theta$  and  $\theta = 2\pi/n$

$$(1 - \alpha_1)(1 - \bar{\alpha}_1) \dots (1 - \alpha_{n-1})(1 - \bar{\alpha}_{n-1}) = n^2$$

$$\text{Now } (1 - \alpha_1)(1 - \bar{\alpha}_1) = 1 - (\alpha_1 + \bar{\alpha}_1) + \alpha_1 \bar{\alpha}_1$$

$$= 2(1 - \cos \theta) = 4 \sin^2 \frac{\theta}{2}$$

$$\Rightarrow 2 \sin \frac{\theta}{2} \cdot 2 \sin \frac{2\theta}{2} \cdot 2 \sin \frac{3\theta}{2} \dots 2 \sin \frac{(n-1)\theta}{2} = n$$

$$|A_1 A_2| |A_1 A_3| |A_1 A_4| \dots |A_1 A_n| = n$$

### Practice Problems

[K]

1. Solve the equation,  $x^{10} + x^5 + 1 = 0$ .
2. Find all possible roots of the equation  $z^{12} - 56z^6 - 512 = 0$  whose imaginary part is positive.
3. Show that

$$(i) \quad x^4 + 1 = (x^2 - \sqrt{2}x + 1)(x^2 + \sqrt{2}x + 1),$$

$$(ii) \quad x^5 - 1 = (x - 1)$$

$$? \left( x^2 + 2x \cos \frac{2\pi}{5} + 1 \right) \left( x^2 - 2x \cos \frac{2\pi}{5} + 1 \right).$$

4. Find the integral solution of the equation  $(1-i)^x = 2^x$ .

5. If  $Z_r, r = 1, 2, 3, \dots, 2m, m \in \mathbb{N}$  are the roots of the equation  $Z^{2m} + Z^{2m-1} + Z^{2m-2} + \dots$

$$+ Z + 1 = 0 \text{ then prove that } \sum_{r=1}^{2m} \frac{1}{Z_r - 1} = -m.$$

6. Given  $z = \cos \frac{2\pi}{2n+1} + i \sin \frac{2\pi}{2n+1}$ ,  $n$  is positive integer, find the equation whose roots are

$$\alpha = z + z^3 + \dots + z^{2n-1} \text{ and } \beta = z^2 + z^4 + \dots + z^{2n}.$$

7. Solve the equation,  $z^{n-1} = \bar{z}$ ,  $n \in \mathbb{N}$ .

8. Resolve  $z^5 + 1$  into linear and quadratic factors with real coefficients and deduce that :

$$4. \sin \frac{\pi}{10} \cdot \cos \frac{\pi}{5} = 1.$$

## 1.13 APPLICATIONS OF DE MOIVRE'S THEOREM

### Binomial Theorem

Let us now examine the raising of complex numbers to a power and the taking of roots. To raise a number  $\alpha = a + bi$  to a positive integral power  $n$ , it suffices to apply Newton's binomial theorem to the expression  $(a + bi)^n$  (this formula holds true for complex numbers as well, since its proof is based solely on the distributive law) and then take advantage of the equation

$$i^2 = -1, i^3 = -i, i^4 = 1.$$

In general, we have

$$i^{4k} = 1, i^{4k+1} = i, i^{4k+2} = -1, i^{4k+3} = -i.$$

Expansion of  $\cos n\theta$  and  $\sin n\theta$  when  $n$  is a positive integer and  $\theta$  is real.

By De Moivre's theorem, we have

$$\cos n\theta + i \sin n\theta = (\cos \theta + i \sin \theta)^n.$$

Expanding the right-hand member by the binomial theorem, we get

$$\begin{aligned} \cos n\theta + i \sin n\theta &= \cos^n \theta + {}^nC_1 \cos^{n-1} \theta (i \sin \theta) + {}^nC_2 \cos^{n-2} \theta (i \sin \theta)^2 \\ &\quad + {}^nC_3 \cos^{n-3} \theta (i \sin \theta)^3 + \dots \\ &\quad + {}^nC_r \cos^{n-r} \theta (i \sin \theta)^r + \dots + (i \sin \theta)^n \\ &= (\cos^n \theta - {}^nC_2 \cos^{n-2} \theta \sin^2 \theta + {}^nC_4 \cos^{n-4} \theta \sin^4 \theta - \dots) \\ &\quad + i({}^nC_1 \cos^{n-1} \theta \sin \theta - {}^nC_3 \cos^{n-3} \theta \sin^3 \theta \\ &\quad + {}^nC_5 \cos^{n-5} \theta \sin^5 \theta - \dots) \end{aligned}$$

Equating the real and imaginary parts, we have

$$\begin{aligned} \cos n\theta &= \cos^n \theta - {}^nC_2 \cos^{n-2} \theta \sin^2 \theta \\ &\quad + {}^nC_4 \cos^{n-4} \theta \sin^4 \theta - \dots \end{aligned}$$

$$\begin{aligned} \text{and } \sin n\theta &= {}^nC_1 \cos^{n-1} \theta \sin \theta - {}^nC_3 \cos^{n-3} \theta \sin^3 \theta \\ &\quad + {}^nC_5 \cos^{n-5} \theta \sin^5 \theta - \dots \end{aligned}$$

► **Example 1.** Expand  $\cos 6\theta$  in terms of powers of  $\cos \theta$ .

► **Solution**  $\cos 6\theta = \text{Real part of } \cos 6\theta + i \sin 6\theta$   
 $= \text{Real part of } (\cos \theta + i \sin \theta)^6$

Expanding the expression  $(\cos \theta + i \sin \theta)^6$  by Binomial theorem we get

$$\begin{aligned} \cos^6 \theta + 6 \cos^5 \theta \cdot i \sin \theta + {}^6C_2 \cos^4 \theta \cdot i^2 \sin^2 \theta \\ + {}^6C_3 \cos^3 \theta \cdot i^3 \sin^3 \theta \\ + {}^6C_4 \cos^2 \theta \cdot i^4 \sin^4 \theta + {}^6C_5 \cos \theta \cdot i^5 \sin^5 \theta + i^6 \sin^6 \theta. \end{aligned}$$

Since  $i^2 = -1, i^3 = -i, i^4 = 1$ , etc.,

the required real part

$$\begin{aligned} &= \cos^6 \theta - {}^6C_2 \cos^4 \theta \sin^2 \theta + {}^6C_4 \cos^2 \theta \sin^4 \theta - \sin^6 \theta \\ &= \cos^6 \theta - 15 \cos^4 \theta (1 - \cos^2 \theta) \\ &\quad + 15 \cos^2 \theta (1 - \cos^2 \theta)^2 - (1 - \cos^2 \theta)^3 \\ &= 32 \cos^6 \theta - 48 \cos^4 \theta + 16 \cos^2 \theta - 1. \end{aligned}$$

### Expansion of $\cos^n \theta$ in a series of cosines of multiples of $\theta$ when $n$ is a positive integer

Let  $z = \cos \theta + i \sin \theta$ .

$\therefore z^n = \cos n\theta + i \sin n\theta$  and  $z^{-n} = \cos n\theta - i \sin n\theta$ .

$$z^n + \frac{1}{z^n} = 2 \cos n\theta.$$

For  $n = 1, z + \frac{1}{z} = 2 \cos \theta$ . Now

$$2^n \cos^n \theta = \left( z + \frac{1}{z} \right)^n$$

$$= z^n + {}^nC_1 z^{n-2} + {}^nC_2 z^{n-4} + \dots$$

$$+ \dots + {}^nC_{n-1} \frac{1}{z^{n-2}} + \frac{1}{z^n}$$

$$= \left( z^n + \frac{1}{z^n} \right) + {}^nC_1 \left( z^{n-2} + \frac{1}{z^{n-2}} \right)$$

$$+ {}^nC_2 \left( z^{n-4} + \frac{1}{z^{n-4}} \right) + \dots$$

$$= 2 \cos n\theta + {}^nC_1 2 \cos(n-2)\theta$$

$$+ {}^nC_2 2 \cos(n-4)\theta + \dots$$

$$\text{or, } 2^{n-1} \cos^n \theta = \cos n\theta + {}^nC_1 \cos(n-2)\theta$$

$$+ {}^nC_2 \cos(n-4)\theta + \dots$$

### Expansion of $\sin^n \theta$ in a series of sines or cosines of multiples of $\theta$ when $n$ is a positive integer

Let  $z = \cos \theta + i \sin \theta$ , then  $z - \frac{1}{z} = 2i \sin \theta$ .

$$\text{We have } (2i \sin \theta)^n = \left( z - \frac{1}{z} \right)^n,$$

$$\begin{aligned} \text{or } 2^n i^n \sin^n \theta &= z^n - {}^nC_1 z^{n-2} + {}^nC_2 z^{n-4} - \dots \\ &\quad + (-1)^{n-1} {}^nC_{n-1} \frac{1}{z^{n-2}} + (-1)^n \frac{1}{z^n} \dots (1) \end{aligned}$$

**Case-I :  $n$  is even.**

From (1),  $(-1)^{n/2} 2^n \sin^n \theta$

$$= \left( z^n + \frac{1}{z^n} \right) - {}^nC_1 \left( z^{n-2} + \frac{1}{z^{n-2}} \right)$$

$$+ {}^nC_2 \left( z^{n-4} + \frac{1}{z^{n-4}} \right) - \dots + (-1)^{n/2} {}^nC_{n/2}$$

$$= 2 \cos n\theta - {}^nC_1 2 \cos(n-2)\theta$$

$$+ {}^nC_2 2 \cos(n-4)\theta - \dots + (-1)^{n/2} {}^nC_{n/2}$$

$$\text{or } (-1)^{n/2} 2^{n-1} \sin^n \theta = \cos n\theta - {}^nC_1 \cos(n-2)\theta$$

$$+ {}^nC_2 \cos(n-4)\theta - \dots + (-1)^{n/2} \cdot \frac{1}{2} \cdot {}^nC_{n/2} \dots (2)$$

**Case-II :  $n$  is odd**

From (1),  $2^n i(-1)^{(n-1)/2} \sin^n \theta$

$$= \left( z^n - \frac{1}{z^n} \right) - {}^nC_1 \left( z^{n-2} - \frac{1}{z^{n-2}} \right)$$

$$\begin{aligned}
& + {}^nC_2 \left( z^{n-4} - \frac{1}{z^{n-4}} \right) - \dots \\
& + (-1)^{(n-1)/2} {}^nC_{(n-1)/2} \left( z - \frac{1}{z} \right) \\
& = 2i \sin n\theta - {}^nC_1 2i \sin(n-2)\theta \\
& \quad + {}^nC_2 2i \sin(n-4)\theta - \dots \\
& \quad + (-1)^{(n-1)/2} {}^nC_{(n-1)/2} 2i \sin \theta, \\
\text{or, } & (-1)^{(n-1)/2} 2^{n-1} \sin^n \theta = \sin n\theta - {}^nC_1 \sin(n-2)\theta \\
& \quad + {}^nC_2 \sin(n-4)\theta - \dots (-1)^{(n-1)/2} {}^nC_{(n-1)/2} \sin \theta \dots (3)
\end{aligned}$$

Let us expand  $\sin^9 \theta$  in a series of sines of multiples of  $\theta$ .

$$\begin{aligned}
\text{By the formula (3), } & (-1)^{(9-1)/2} 2^{9-1} \sin^9 \theta \\
& = \sin 9\theta - {}^9C_1 \sin 7\theta \\
& \quad + {}^9C_2 \sin 5\theta - {}^9C_3 \sin 3\theta + {}^9C_4 \sin \theta \\
\text{or } 2^8 \sin^9 \theta & = \sin 9\theta - 9 \sin 7\theta \\
& \quad + 36 \sin 5\theta - 84 \sin 3\theta + 126 \sin \theta.
\end{aligned}$$

► **Example 2.** Show that  $\sin^4 \theta \cos^5 \theta = \frac{1}{256} (\cos 9\theta + \cos 7\theta - 4 \cos 5\theta - 4 \cos 3\theta + 6 \cos \theta)$ .

► **Solution** Let  $z = \cos \theta + i \sin \theta$ . Then  $z^n = \cos n\theta + i \sin n\theta$  and  $z^{-n} = \cos n\theta - i \sin n\theta$ .

$$\therefore z^n + \frac{1}{z^n} = 2 \cos n\theta.$$

$$\text{For } n = 1, z + \frac{1}{z} = 2 \cos \theta$$

$$\text{Again } z - \frac{1}{z} = \cos \theta + i \sin \theta - \cos \theta + i \sin \theta = 2i \sin \theta.$$

Thus

$$\begin{aligned}
& (2i \sin \theta)^4 \cdot (2 \cos \theta)^5 \\
& = \left( z - \frac{1}{z} \right)^4 \left( z + \frac{1}{z} \right)^5 = \left( z^2 - \frac{1}{z^2} \right)^4 \left( z + \frac{1}{z} \right) \\
& = \left( z^8 - 4z^4 + 6 - \frac{4}{z^4} + \frac{1}{z^8} \right) \left( z + \frac{1}{z} \right) \\
& = \left( z^9 + \frac{1}{z^9} \right) + \left( z^7 + \frac{1}{z^7} \right) - 4 \left( z^5 + \frac{1}{z^5} \right)
\end{aligned}$$

$$\begin{aligned}
& - 4 \left( z^3 + \frac{1}{z^3} \right) + 6 \left( z + \frac{1}{z} \right) \\
& = 2 \cos 9\theta + 2 \cos 7\theta - 8 \cos 5\theta - 8 \cos 3\theta \\
& \quad + 12 \cos \theta.
\end{aligned}$$

$$\begin{aligned}
\therefore \sin^4 \theta \cos^5 \theta & = \frac{1}{256} \\
& (\cos 9\theta + \cos 7\theta - 4 \cos 5\theta - 4 \cos 3\theta + 6 \cos \theta).
\end{aligned}$$

### Sum of Series

Using De Moivre's Theorem we can find the sum of several series. To understand the method consider the following example :

► **Example 3.** Find the sum of the series  $\cos \theta + \cos 2\theta + \cos 3\theta + \dots + \cos n\theta$  for  $0 < \theta < 2\pi$ .

► **Solution** Consider the complex number  $\cos \theta + i \sin \theta$ .

We have  $(\cos \theta + i \sin \theta)^k = \cos k\theta + i \sin k\theta$ , for any integer  $k$ . If we put  $z = \cos \theta + i \sin \theta$ , then  $\cos k\theta = \text{Real part of } z^k$ .

Hence we can write the given series in the form

$$\begin{aligned}
S_n & = \cos \theta + \cos 2\theta + \dots + \cos n\theta \\
& = \text{Real part of } (z + z^2 + \dots + z^n) \\
& = \text{Re}(1 + z + \dots + z^n - 1).
\end{aligned}$$

If  $0 < \theta < 2\pi$ , then  $z \neq 1$ . Hence, we get

$$S_n = \text{Re} \left( \frac{z^{n+1} - 1}{z - 1} - 1 \right)$$

Thus

$$\begin{aligned}
S_n & = \text{Re} \left[ \frac{\cos(n+1)\theta + i \sin(n+1)\theta - 1}{\cos \theta + i \sin \theta - 1} \right] - 1 \\
& = \text{Re} \left[ \frac{[(\cos \theta - 1) - i \sin \theta][\cos(n+1)\theta - 1 + i \sin(n+1)\theta]}{[(\cos \theta - 1) + i \sin \theta][(\cos \theta - 1) - i \sin \theta]} \right] - 1 \\
& = \frac{(\cos \theta - 1)(\cos(n+1)\theta - 1) + \sin \theta \sin(n+1)\theta}{(\cos \theta - 1)^2 + \sin^2 \theta} - 1 \\
& \quad \cos \theta \cos(n+1)\theta + \sin \theta \sin(n+1)\theta \\
& = - \frac{-\cos(n+1)\theta - \cos \theta + 1}{(\cos \theta - 1)^2 + \sin^2 \theta} - 1
\end{aligned}$$



$$= \frac{\cos n\theta - \cos(n+1)\theta - \cos\theta + 1}{2(1 - \cos\theta)} - 1$$

$$= \frac{\cos n\theta - \cos(n+1)\theta}{2(1 - \cos\theta)} - \frac{1}{2}$$

This can be further simplified to give

$$S_n = \frac{2 \sin\left(n + \frac{1}{2}\right) \theta \sin(\theta/2)}{4 \sin^2(\theta/2)} - \frac{1}{2}$$

$$= \frac{\sin\left(n + \frac{1}{2}\right) \theta}{2 \sin(\theta/2)} - \frac{1}{2}$$

$$= \frac{\sin\left(n + \frac{1}{2}\right) \theta - \sin \frac{\theta}{2}}{2 \sin(\theta/2)} = \frac{\sin \frac{n\theta}{2} \cos\left(\frac{n+1}{2}\right) \theta}{\sin(\theta/2)}$$



(i)  $\cos \theta + \cos 2\theta + \cos 3\theta + \dots + \cos n\theta$

$$= \frac{\sin \frac{n\theta}{2} \cos\left(\frac{n+1}{2}\right) \theta}{\sin(\theta/2)}.$$

(ii)  $\sin \theta + \sin 2\theta + \sin 3\theta + \dots + \sin n\theta$

$$= \frac{\sin \frac{n\theta}{2} \sin\left(\frac{n+1}{2}\right) \theta}{\sin(\theta/2)}.$$

If  $\theta = (2\pi/n)$  then the sum of the above series vanish.

► **Example 4.** If  $\theta \neq k\pi$ , show that  
 $\cos\theta \sin\theta + \cos^2\theta \sin 2\theta + \dots + \cos^n \theta \sin n\theta$   
 $= \cot\theta(1 - \cos^n \theta \cos n\theta)$

► **Solution**  $S = \cos\theta \sin\theta + \cos^2\theta \sin 2\theta + \dots$   
 $+ \cos^n \theta \sin n\theta$   
 and  $C = \cos\theta \cos\theta + \cos^2\theta \cos 2\theta + \dots$   
 $+ \cos^2\theta \cos n\theta$

so that

$$C + iS = \cos\theta z + \cos^2\theta z^2 + \dots + \cos^n \theta z^n,$$

where

$z = \cos\theta + i \sin\theta$ . In other words,

$$C + iS = \frac{\cos\theta z [1 - (\cos\theta z)^n]}{1 - \cos\theta z}$$

$$= \frac{\cos\theta(\cos\theta + i \sin\theta) [1 - \cos^n \theta (\cos n\theta + i \sin n\theta)]}{1 - \cos\theta(\cos\theta + i \sin\theta)} \quad [\theta \neq k\pi]$$

$$= \frac{\cos\theta(\cos\theta + i \sin\theta) [1 - \cos^n \theta (\cos n\theta + i \sin n\theta)]}{\sin^2 \theta - i \cos\theta \sin \theta}$$

$$= \frac{\cos\theta(\cos\theta + i \sin\theta) [1 - \cos^n \theta (\cos n\theta + i \sin n\theta)]}{-i \sin \theta (\cos\theta + i \sin \theta)}$$

$$= i \cot\theta (1 - \cos^n \theta \cos n\theta - i \cos^n \theta \sin n\theta)$$

Equating the imaginary parts we therefore get

$$S = \cot\theta (1 - \cos^n \theta \cos n\theta),$$

and hence the required result.

► **Example 5.** Find the sum of the infinite series,  
 $\sin\alpha + \frac{1}{2} \sin 2\alpha + \frac{1}{2^2} \sin 3\alpha + \frac{1}{2^3} \sin 4\alpha + \dots \infty$ .

► **Solution** Let  $S = \sin\alpha + \frac{1}{2} \sin 2\alpha + \dots$

$$\text{and } C = \cos\alpha + \frac{1}{2} \cos 2\alpha + \dots$$

$$r = \left| \frac{1}{2} e^{i\alpha} \right| = \frac{1}{2} < 1$$

Using sum of an infinite G.P., we get

$$C + iS = \frac{e^{i\alpha}}{1 - \frac{1}{2} e^{i\alpha}} = \frac{2e^{i\alpha}}{2 - e^{i\alpha}}$$

On simplifying we get

$$C + iS = \frac{4\cos\alpha - 2}{5 - 4\cos\alpha} + i \frac{4\sin\alpha}{5 - 4\cos\alpha}$$

On separating the real and imaginary parts we have

$$C = \frac{4\cos\alpha - 2}{5 - 4\cos\alpha} \text{ and } S = \frac{4\sin\alpha}{5 - 4\cos\alpha}$$

► **Example 6.** Use complex numbers to prove that the sum,

$$\sum_{r=0}^{n-1} \cos^2 \left( \alpha + \frac{\alpha\pi}{n} \right) = \frac{n}{2}$$

where  $n \in \mathbb{N}$ ,  $n \geq 2$ .

► **Solution** L.H.S. =  $\cos^2 \alpha + \cos^2(\alpha + \theta)$   
 $+ \cos^2(\alpha + 2\theta) + \dots + \cos^2(\alpha + (n-1)\theta)$

where  $\theta = \frac{\pi}{n} = \frac{\pi}{2} + \frac{\pi}{2}$

$$\left[ \cos 2\alpha + \cos 2(\alpha + \theta) + \cos 2(\alpha + 2\theta) \right. \\ \left. + \dots + \cos 2(\alpha + (n-1)\theta) \right] \dots (1)$$

Consider

$$C = \cos 2\alpha + \cos 2(\alpha + \theta) + \dots + \cos 2(\alpha + (n-1)\theta)$$

and

$$S = \sin 2\alpha + \sin 2(\alpha + \theta) + \dots + \sin 2(\alpha + (n-1)\theta)$$

$$\Rightarrow C + iS = e^{i2\alpha} \left[ 1 + e^{i2\theta} + e^{i4\theta} + \dots + e^{i2(n-1)\theta} \right]$$

$$= e^{i2\alpha} \frac{[e^{i2n\theta} - 1]}{e^{i2\theta} - 1}$$

$$= \frac{e^{i2\alpha} \cdot e^{in\theta} [e^{in\theta} - e^{-in\theta}]}{e^{i\theta} [e^{i\theta} - e^{-i\theta}]}$$

$$= e^{i(2\alpha + (n-1)\theta)} \frac{[2i \sin n\theta]}{2i \sin \theta}$$

Equating real part,

$$C = \frac{\sin n\theta}{\sin \theta} \left[ \cos(2\alpha + (n-1)\theta) \right] = 0 \text{ if } \theta = \frac{\pi}{n}.$$

Now using this in (1) we get the desired result.

► **Example 7.** If A and B are supplementary angles and n is odd integer, then prove that

$$\sum_{r=0}^n {}^nC_r \cos(nA + r(B-A)) = 0.$$

► **Solution**  $A + B = \pi$

$$e^{iA} + e^{iB} = 2i \sin A$$

$$\Rightarrow (e^{iA} + e^{iB}) = (2i \sin A)^n$$

As n is odd integer.

$$\Rightarrow \text{Real part of } (e^{iA} + e^{iB})^n = 0$$

$$\Rightarrow \text{Real part of } \sum_{r=0}^n {}^nC_r e^{i(n-r)A} e^{iBr} = 0$$

$$\Rightarrow \sum_{r=0}^n {}^nC_r \cos(nA + r(B-A)) = 0.$$

### Practice Problems

[L]

- From the formula  $\cos n\theta + i \sin n\theta = (\cos \theta + i \sin \theta)^n$ , where n is a positive integer, deduce that

$$\cos n\theta = \cos^n \theta - {}^nC_2 \cos^{n-2} \theta \sin^2 \theta \\ + {}^nC_4 \cos^{n-4} \theta \sin^4 \theta - \dots, \\ \sin n\theta = {}^nC_1 \cos^{n-1} \theta \sin \theta - {}^nC_3 \cos^{n-3} \theta \sin^3 \theta + \dots,$$

$$\tan n\theta = \frac{{}^nC_1 \tan \theta - {}^nC_3 \tan^3 \theta + \dots}{1 - {}^nC_2 \tan^2 \theta + {}^nC_4 \tan^4 \theta - \dots}$$

- Expand the following in powers of  $\cos \theta$  and  $\sin \theta$ .  
 (i)  $\cos 9\theta$ , (ii)  $\sin 10\theta$ , (iii)  $\operatorname{cosec} \theta \sin 7\theta$ .
- Prove that  $16 \cos^3 \theta \sin^2 \theta = -\cos 5\theta - \cos 3\theta + 2 \cos \theta$ , using complex methods.
- (i) Show that  $\cos 8\theta$

$$= 128 \cos^8 \theta - 256 \cos^6 \theta \\ + 160 \cos^4 \theta - 32 \cos^2 \theta + 1.$$

- (ii) Show that

$$\frac{1 + \cos 7\theta}{1 + \cos \theta} = (t^3 - t^2 - 2t + 1)^2$$

$$\text{where } 2 \cos \theta = t.$$

5. Sum  $\cos^n \alpha - n \cos^{n-1} \alpha \cos \alpha + \frac{n(n-1)}{2!} \cos^{n-2} \alpha \cos 2\alpha + \dots (n+1) \text{ terms}$
6. Sum :  $\cos y \sin y + \cos^2 y \sin 2y + \cos^3 y \sin 3y + \dots n \text{ terms.}$

## 1.14 STRAIGHT LINES IN COMPLEX NUMBERS

### Section Formula

If  $P(z)$  divides the line segment joining  $A(z_1)$  and  $B(z_2)$  internally in the ratio  $m : n$ ,

$$\text{then } z = \frac{mz_2 + nz_1}{m+n}.$$

$$\begin{array}{c} \text{m : n} \\ \text{A}(z_1) \quad \text{P}(z) \quad \text{B}(z_2) \end{array} \quad \frac{AP}{PB} = \frac{m}{n}$$

Let  $z_1 = x_1 + iy_1, z_2 = x_2 + iy_2$

then  $A \equiv (x_1, y_1)$  and  $B \equiv (x_2, y_2)$

Let  $z = x + iy$ , then  $P \equiv (x, y)$

$\therefore (m+n)x = mx_2 + nx_1$  and  $(m+n)y = my_2 + ny_1$

As in co-ordinate geometry,

$$x = \frac{mx_2 + nx_1}{m+n} \text{ and } y = \frac{my_2 + ny_1}{m+n}$$

Now  $(m+n)z = (m+n)(x+iy) = mx_2 + nx_1 + i(my_2 + ny_1)$

$$= m(x_2 + iy_2) + n(x_1 + iy_1) = mz_2 + nz_1$$

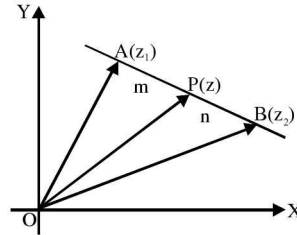
$$\text{Hence } z = \frac{mz_2 + nz_1}{m+n}$$

$$\text{If, then } z = \frac{mz_2 - nz_1}{m-n}$$

### Alternative

This can easily be proved using vectors by

observing that  $\frac{z - z_1}{z_2 - z} = \frac{m}{n}$  (see fig.)



which on simplification gives the desired result. Similarly, it can be shown that if  $P$  lies outside  $A$  and  $B$  i.e. the division is external, then

$$z = \frac{mz_2 - nz_1}{m-n}.$$

### Mid-point formula

In the special case, when  $P(z)$  is the mid-point of the line segment joining  $A(z_1)$  and  $B(z_2)$  so that the ratio is  $1 : 1$ ,

$$\text{then } z = \frac{z_1 + z_2}{2}.$$

► **Example 1.** Let  $z_1, z_2, z_3$  be three complex numbers and  $a, b, c$  be real numbers not all zero, such that  $a + b + c = 0$  and  $az_1 + bz_2 + cz_3 = 0$ . Show that  $z_1, z_2, z_3$  are collinear.

► **Solution** Given,  $a + b + c = 0$ . ... (1)

and  $az_1 + bz_2 + cz_3 = 0$  ... (2)

Since  $a, b, c$  are not all zero  $\therefore$  Let  $c \neq 0$

From (2)

$$az_1 + bz_2 - (a+b)z_3 = 0 \quad [\text{From (1), } c = -(a+b)]$$

$$\Rightarrow az_1 + bz_2 = (a+b)z_3 \Rightarrow z_3 = \frac{az_1 + bz_2}{a+b} \quad \dots (3)$$

From (3), it follows that  $z_3$  divides the line segment joining  $z_1$  and  $z_2$  internally in the ratio  $b : a$

If  $a$  and  $b$  are of the same sign, then division is in fact internal and if  $a$  and  $b$  are of opposite sign, then division is external in the ratio  $|b| : |a|$ .



If  $a, b, c$  are three real numbers such that

$$az_1 + bz_2 + cz_3 = 0;$$

where  $a + b + c = 0$  and  $a, b, c$  are not all simultaneously zero, then the complex numbers  $z_1$ ,  $z_2$  and  $z_3$  are collinear.

### Complex equations of straight line

#### 1. Equation of the line passing through the points $z_1$ and $z_2$

$$\text{is } \begin{vmatrix} z & \bar{z} & 1 \\ z_1 & \bar{z}_1 & 1 \\ z_2 & \bar{z}_2 & 1 \end{vmatrix} = 0 \quad \text{or, } \frac{z - z_1}{\bar{z} - \bar{z}_1} = \frac{z_1 - z_2}{\bar{z}_1 - \bar{z}_2}$$

*Proof:*

Let  $z_1 = x_1 + iy_1$  and  $z_2 = x_2 + iy_2$ .

Let A and B be the points representing  $z_1$  and  $z_2$  respectively.

Let P(z) be any point on the line joining A and B.

Let  $z = x + iy$ , then  $P \equiv (x, y)$ ,  $A \equiv (x_1, y_1)$  and  $B \equiv (x_2, y_2)$ . Points P, A, B are collinear.

$$\therefore \begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} = 0$$

$$\text{or, } \begin{vmatrix} z & y & 1 \\ z_1 & y_1 & 1 \\ z_2 & y_2 & 1 \end{vmatrix} = 0 \quad [C_1 \rightarrow C_1 + iC_2]$$

$$\text{or, } \begin{vmatrix} z & \frac{z - \bar{z}}{2i} & 1 \\ z_1 & \frac{z_1 - \bar{z}_1}{2i} & 1 \\ z_2 & \frac{z_2 - \bar{z}_2}{2i} & 1 \end{vmatrix} = 0$$

$$\text{or } \frac{1}{2i} \begin{vmatrix} z & z - \bar{z} & 1 \\ z_1 & z_1 - \bar{z}_1 & 1 \\ z_2 & z_2 - \bar{z}_2 & 1 \end{vmatrix} = 0 \quad [C_2 \rightarrow C_2 - C_1]$$

$$\text{or, } \begin{vmatrix} z & -\bar{z} & 1 \\ z_1 & -\bar{z}_1 & 1 \\ z_2 & -\bar{z}_2 & 1 \end{vmatrix} = 0 \quad [\text{Dividing by } -1]$$

$$\text{or, } \begin{vmatrix} z & \bar{z} & 1 \\ z_1 & \bar{z}_1 & 1 \\ z_2 & \bar{z}_2 & 1 \end{vmatrix} = 0 \quad \dots(1)$$

#### 2. General equation of straight line

is  $\bar{a}z + a\bar{z} + b = 0$

*Proof:*

$$\text{Again from (1), } \begin{vmatrix} z - z_1 & \bar{z} - \bar{z}_1 & 0 \\ z_1 - z_2 & \bar{z}_1 - \bar{z}_2 & 0 \\ z_2 & \bar{z}_2 & 1 \end{vmatrix} = 0$$

$$\text{or } (z - z_1)(\bar{z}_1 - \bar{z}_2) - (\bar{z} - \bar{z}_1)(z_1 - z_2) = 0$$

$$\text{or, } \frac{z - z_1}{\bar{z} - \bar{z}_1} = \frac{z_1 - z_2}{\bar{z}_1 - \bar{z}_2} \quad \dots(2)$$

(2) can also be written as

$$(\bar{z}_1 - \bar{z}_2)z - (z_1 - z_2)\bar{z} - z_1\bar{z}_1 + z_1\bar{z}_2 + z_2\bar{z}_1 - z_2\bar{z}_2 = 0$$

$$\text{or } (\bar{z}_1 - \bar{z}_2)z - (z_1 - z_2)\bar{z} + (z_1\bar{z}_2 - z_2\bar{z}_1) = 0$$

$$\text{or } i(\bar{z}_1 - \bar{z}_2)z - i(z_1 - z_2)\bar{z} + i(z_1\bar{z}_2 - z_2\bar{z}_1) = 0$$

$$\text{or } \bar{a}z + a\bar{z} + b = 0 \quad \dots(3)$$

$$\text{where } b = i(z_1\bar{z}_2 - z_2\bar{z}_1) = i(z_1\bar{z}_2 - z_1\bar{z}_2)$$

$$= i \cdot 2i I(z_1\bar{z}_2) = -2I(z_1\bar{z}_2) = \text{a real number}$$

and  $a \neq 0$  (as  $z_1 \neq z_2$ )

#### Complex Slope

If  $z_1$  and  $z_2$  are two unequal complex numbers represented by points P and Q, then  $\frac{z_1 - z_2}{\bar{z}_1 - \bar{z}_2}$  is called the complex slope of the line joining  $z_1$  and  $z_2$  (i.e. line PQ) and is denoted by w

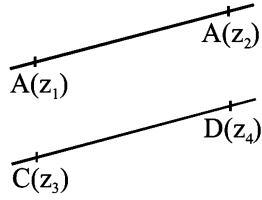
$$\text{Thus } w = \frac{z_1 - z_2}{\bar{z}_1 - \bar{z}_2}$$



The equation of the line PQ is  $z - z_1 = w(\bar{z} - \bar{z}_1)$

$$\text{Clearly } |w| = \frac{|z_1 - z_2|}{|\bar{z}_1 - \bar{z}_2|} = \frac{|z_1 - z_2|}{|z_1 - z_2|} = 1$$

(a) The two lines having complex slopes  $w_1$  and  $w_2$  are parallel if and only if  $w_1 = w_2$



$$w_1 = \frac{z_2 - z_1}{\bar{z}_1 - \bar{z}_2} \text{ (complex slope of AB)}$$

$$w_2 = \frac{z_4 - z_3}{\bar{z}_3 - \bar{z}_4} \text{ (complex slope of CD)}$$

$AB \parallel CD \Leftrightarrow$  angle between AB and CD = 0 or  $\pi$

$$\Leftrightarrow \arg \left( \frac{z_2 - z_1}{z_4 - z_3} \right) = 0 \text{ or } \pi$$

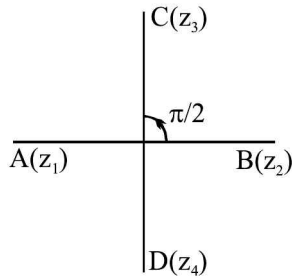
$$\Leftrightarrow \frac{z_2 - z_1}{z_4 - z_3} \text{ is real}$$

$$\Leftrightarrow \left( \frac{z_2 - z_1}{z_4 - z_3} \right) = \left( \frac{z_2 - z_1}{z_4 - z_3} \right)$$

$$\Leftrightarrow \left( \frac{z_1 - z_2}{z_3 - z_4} \right) = \frac{z_1 - z_2}{z_3 - z_4}$$

$$\Leftrightarrow \frac{z_1 - z_2}{\bar{z}_1 - \bar{z}_2} = \frac{z_3 - z_4}{\bar{z}_3 - \bar{z}_4} \Leftrightarrow w_1 = w_2$$

(b) The two lines having complex slope  $w_1$  and  $w_2$  are perpendicular if and only if  $w_1 = -w_2$



$AB \perp CD$

$$\Leftrightarrow \text{angle between AB and CD} = \frac{\pi}{2}$$

$$\Leftrightarrow \arg \left( \frac{z_3 - z_4}{z_2 - z_1} \right) = \pm \frac{\pi}{2}$$

$$\Leftrightarrow \frac{z_3 - z_4}{z_2 - z_1} \text{ is purely imaginary}$$

$$\Leftrightarrow \overline{\left( \frac{z_3 - z_4}{z_1 - z_2} \right)} = - \left( \frac{z_3 - z_4}{z_1 - z_2} \right)$$

$$\Leftrightarrow \frac{z_1 - z_2}{\bar{z}_1 - \bar{z}_2} = - \frac{z_3 - z_4}{\bar{z}_3 - \bar{z}_4} \Leftrightarrow w_1 = -w_2$$

### Two sides of a line

Two point  $P(z_1)$  and  $Q(z_2)$  lie on the same side or opposite side of the line  $\bar{a}z + a\bar{z} + b = 0$  according as  $\bar{a}z_1 + a\bar{z}_1 + b$  and  $\bar{a}z_2 + a\bar{z}_2 + b$  have same sign or opposite sign.

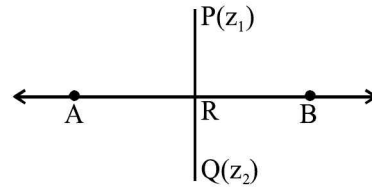
Let the equation of AB be  $\bar{a}z + a\bar{z} + b = 0$  ... (1)

Let AB divide PQ internally in the ratio  $m : n$  at R, then

$$R \equiv \frac{mz_2 + nz_1}{m + n}$$

Since R lies on line (1)

$$\therefore \bar{a} \left( \frac{mz_2 + nz_1}{m + n} \right) + a \left( \frac{n\bar{z}_2 + m\bar{z}_1}{m + n} \right) + b = 0$$



$$\Rightarrow m(\bar{a}z_2 + a\bar{z}_2 + b) + n(\bar{a}z_1 + a\bar{z}_1 + b) = 0$$

$$\Rightarrow \frac{m}{n} = \frac{\bar{a}z_1 + a\bar{z}_1 + b}{\bar{a}z_2 + a\bar{z}_2 + b}$$

P and Q will lie on the same side or opposite side of line AB according as division is external or internal

i.e. according as  $\frac{m}{n} < 0$  or  $> 0$

or according as  $-\frac{\bar{a}z_1 + a\bar{z}_1 + b}{\bar{a}z_2 + a\bar{z}_2 + b} < 0$  or  $> 0$

or according as  $\bar{a}z_1 + a\bar{z}_1 + b$  and  $\bar{a}z_2 + a\bar{z}_2 + b$  have same sign or opposite sign.

► **Example 2.** If points  $z_1, z_2, \dots, z_n$  all lie on the same side of a line passing through the origin, then show that the points  $\frac{1}{z_1}, \frac{1}{z_2}, \dots, \frac{1}{z_n}$  also lie on the same side of another line. Also show that

$$\sum z_i \neq 0 \text{ and } \sum \frac{1}{z_i} \neq 0, i = 1, 2, \dots, n.$$

► **Solution** Equation of any line through the origin can be written as

$$\bar{a}z + a\bar{z} = 0 \quad \dots(1)$$

Let us suppose that the points  $z_1, z_2, \dots, z_n$  lie in the half-plane  $\bar{a}z + a\bar{z} = 0$   $\bar{a}z + a\bar{z} > 0$ , then we have

$$\bar{a}z_i + a\bar{z}_i > 0, i = 1, 2, \dots, n \quad \dots(2)$$

$$\Rightarrow \frac{\bar{a}|z_i|^2}{\bar{z}_i} + \frac{a|z_i|^2}{z_i} > 0$$

$$\Rightarrow \bar{a}\left(\frac{1}{\bar{z}_i}\right) + a\left(\frac{1}{z_i}\right) > 0 \quad \dots(3)$$

which shows that the points  $\frac{1}{z_1}, \frac{1}{z_2}, \dots, \frac{1}{z_n}$

lie on the same side of the line  $az + \bar{a}\bar{z} = 0$ .

Also, from equation (2), we have

$$\bar{a}\sum_{i=1}^n z_i + a\sum_{i=1}^n \bar{z}_i > 0$$

$$\text{i.e. } \bar{a}X + a\bar{X} > 0 \text{ where } X = \sum_{i=1}^n z_i$$

which shows that  $X \neq 0$  (because if  $X = 0$ , then  $\bar{X} = 0$ )

$$\Rightarrow \bar{a}X + a\bar{X} = 0.$$

Similarly, using result (3), it can be shown that

$$\sum_{i=1}^n \frac{1}{z_i} \neq 0.$$

### Complex functions of a real variable

If  $f(t), \phi(t)$  are two real functions of a real variable  $t$  defined for a certain range of values of  $t$ , we call

$$z = f(t) + i\phi(t) \quad \dots(1)$$

a complex function of  $t$ . We can represent it graphically by drawing the curve

$$x = f(t), y = \phi(t). \quad \dots(2)$$

If  $z$  is a polynomial in  $t$ , or rational function of  $t$ , with complex coefficient, we can express it in the form (1) and so determine the curve represented by the function.

We can also eliminate  $t$  from the relations (2) to obtain a relation in  $x$  and  $y$ , which may be simpler to analyse.

For example, consider the curve

$$z = a + (b - a)t,$$

where  $a$  and  $b$  are complex numbers.

If  $a = \alpha + \alpha'i, b = \beta + \beta'i$ , then

$$x = \alpha + (\beta - \alpha)t, \quad y = \alpha' + (\beta' - \alpha')t.$$

The curve is the straight line joining the points  $z = a$  and  $z = b$ . The segment between the points corresponds to the range of values of  $t$  from 0 to 1.



- (i) The equation of a line joining  $z_1$  &  $z_2$  is given by  $z = z_1 + t(z_2 - z_1)$  where  $t$  is a parameter.
- (ii)  $z = z_1(1 + it)$  where  $t$  is a real parameter is a line through the point  $z_1$  & perpendicular to  $Oz_1$ .

### Length of perpendicular

Length of perpendicular from  $\omega$  to the line

$$\bar{a}z + a\bar{z} + C = 0 \quad (\text{where } C \text{ is real})$$

$$\text{is } |\bar{a}\omega + a\bar{\omega} + C| / 2|a|$$

*Proof :*

Taking  $a = \alpha + i\beta, z = x + iy$ , the given line becomes  $2(\alpha x + \beta y) + c = 0$ .

Length of perpendicular from  $\omega = x_1 + iy_1$  to the given line is

$$\frac{|2(\alpha x_1 + \beta y_1) + c|}{2\sqrt{\alpha^2 + \beta^2}} = \frac{|a\bar{\omega} + \bar{a}\omega + C|}{2|a|}$$

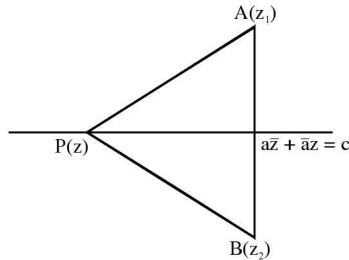
### Reflection points for a straight line

Two given points  $P$  &  $Q$  are the reflection points for a given straight line if the given line is the right

bisector of the segment PQ. Note that the two points denoted by the complex numbers  $z_1$  &  $z_2$  will be the reflection points for the straight line  $\bar{\alpha}z + \alpha\bar{z} + r = 0$  if and only if;  $\bar{\alpha}z_1 + \alpha\bar{z}_2 + r = 0$ , where  $r$  is real and  $\alpha$  is non zero complex constant.

► **Example 3.** Let  $\bar{a}z + a\bar{z} = c$ ,  $a \neq 0$  be a line in the complex plane. If a point  $z_1$  is the reflection of a point  $z_2$  is the reflection of a point  $z_2$  in the given line, then show that  $c = \bar{z}_1a + z_2\bar{a}$

► **Solution** Let  $P(z)$  be any point on the line  $\bar{a}z + a\bar{z} = c$



Then  $AP = BP$

$$\Rightarrow |z - z_1|^2 = |z - z_2|^2$$

$$\Rightarrow (z - z_1)(\bar{z} - \bar{z}_1) = (z - z_2)(\bar{z} - \bar{z}_2)$$

$$\Rightarrow (\bar{z}_2 - \bar{z}_1)z + (z_2 - z_1)\bar{z} = z_2\bar{z}_2 - z_1\bar{z}_1$$

The line is identical to the line

$$\bar{a}z + a\bar{z} = c$$

$$\therefore \frac{\bar{z}_2 - \bar{z}_1}{\bar{a}} = \frac{z_2 - z_1}{a} = \frac{z_2\bar{z}_2 - z_1\bar{z}_1}{c} = k(\text{say})$$

$$\Rightarrow \bar{z}_2 - \bar{z}_1 = k\bar{a}, \quad z_2 - z_1 = ka$$

$$\text{and } z_2\bar{z}_2 - z_1\bar{z}_1 = kc$$

$$\text{Now, } \bar{z}_1a + z_2\bar{a} = \frac{1}{k} [\bar{z}_1(z_2 - z_1) + z_2(\bar{z}_2 - \bar{z}_1)]$$

$$= \frac{1}{k} (\bar{z}_1z_2 - z_1\bar{z}_1 + z_2\bar{z}_2 - z_2\bar{z}_1)$$

$$= \frac{1}{k} (z_2\bar{z}_2 - z_1\bar{z}_1) = \frac{1}{k} (kc) = c.$$

### Practice Problems

[M]

1. If  $z_1 = 1 + 2i$ ,  $z_2 = 2 + 3i$ ,  $z_3 = 3 + 4i$ , then prove that  $z_1$ ,  $z_2$  and  $z_3$  are collinear.

2. If  $\alpha$  is a complex constant,  $z$  is a complex variable and  $r$  is a purely imaginary constant, does the equation

$$\alpha\bar{z} - \bar{\alpha}z + r = 0 \text{ represents a straight line?}$$

3. If  $c$  is real,  $b = p + iq$ ,  $z = x + iy$ , then show that the equation  $b(z + \bar{z}) + \bar{b}(\bar{z} - z) + c = 0$  represents two straight lines in the  $xy$  plane.

4. If the distinct points  $z_1, z_2, z_3$  are collinear, prove that  $z_1|z_2 - z_3| \pm z_2|z_3 - z_1| \pm z_3|z_1 - z_2| = 0$ .

5. If  $\text{amp} \frac{z-2}{2z+3i} = 0$  and  $z_0 = 3 + 4i$  then prove that  $z_0z + \bar{z}_0\bar{z} = 12$ .

6. If  $\text{Arg}(z_2 - z_1) + \text{Arg}(z_3 - z_1) = 2 \text{Arg}(z - z_1)$ , then prove that the locus of  $z$  is a straight line.

7.  $z_1$  and  $z_2$  are two distinct points in an argand plane. If  $a|z_1| = b|z_2|$ , then prove that the point

$$\frac{az_1}{bz_2} + \frac{bz_2}{az_1}, (a, b \in \mathbb{R}) \text{ lies on the line segment}$$

$$[-2, 2] \text{ of the real axis.}$$

8. If locus of  $z$  be a straight line the locus of  $2z + 1$  is a straight line which is parallel to the given line.

### 1.15 TRIANGLES IN COMPLEX NUMBERS

- (i) If the vertices  $A, B, C$  of a  $\Delta$  represent the complex nos.  $z_1, z_2, z_3$  respectively, then :

$$(a) \text{ Centroid of the } \Delta ABC \text{ is } \frac{z_1 + z_2 + z_3}{3}$$

$$(b) \text{ Orthocentre of the } \Delta ABC \text{ is}$$

$$\frac{(a \sec A)z_1 + (b \sec B)z_2 + (c \sec C)z_3}{a \sec A + b \sec B + c \sec C}$$

$$\text{or } \frac{z_1 \tan A + z_2 \tan B + z_3 \tan C}{\tan A + \tan B + \tan C}$$

(c) Incentre of the  $\Delta ABC$  is  
 $(az_1 + bz_2 + cz_3) \div (a + b + c).$

(d) Circumcentre of the  $\Delta ABC$  is  
 $(Z_1 \sin 2A + Z_2 \sin 2B + Z_3 \sin 2C) \div (\sin 2A + \sin 2B + \sin 2C).$

(ii) If  $z_1, z_2, z_3$  are the vertices of an equilateral triangle where  $z_0$  is its circumcentre then

$$(a) \quad z_1^2 + z_2^2 + z_3^2 - z_1 z_2 - z_2 z_3 - z_3 z_1 = 0$$

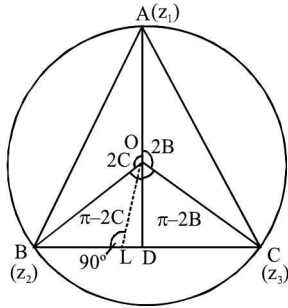
$$(b) \quad z_1^2 + z_2^2 + z_3^2 = 3 z_0^2$$

► **Example 1.** If the vertices of a triangle ABC are represented by  $z_1, z_2$  and  $z_3$  respectively, show that its circumcentre is

$$\frac{z_1 \sin 2A + z_2 \sin 2B + z_3 \sin 2C}{\sin 2A + \sin 2B + \sin 2C}.$$

► **Solution** Let O be the circumcentre of  $\Delta ABC$ .  
 where  $A \equiv z_1, B \equiv z_2$  and  $C \equiv z_3$ .

$$\text{Now } \frac{BD}{DC} = \frac{\frac{1}{2} BD \cdot OL}{\frac{1}{2} DC \cdot OL} = \frac{\text{area of } \Delta BOD}{\text{area of } \Delta COD}$$



$$= \frac{\frac{1}{2} OB \cdot OD \sin(\theta - 2C)}{\frac{1}{2} OC \cdot OD \sin(\pi - 2B)} = \frac{\sin 2C}{\sin 2B}$$

Thus D divides BC internally in the ratio  $\sin 2C : \sin 2B$

$$\therefore D \equiv \frac{z_3 \sin 2C + z_2 \sin 2B}{\sin 2C + \sin 2B}$$

$$\text{i.e. } D \equiv \frac{z_2 \sin 2B + z_3 \sin 2C}{\sin 2B + \sin 2C}$$

The complex number representing the point which divides AD internally in the ratio  $(\sin 2B + \sin 2C) : \sin 2A$  is

$$\frac{z_1 \sin 2A + z_2 \sin 2B + z_3 \sin 2C}{\sin 2A + \sin 2B + \sin 2C}$$

Let BO produced meet AC at E and CO produced meet AB at F. We can show that, the complex number representing the point dividing the line segment BE internally in the ratio  $(\sin 2C + \sin 2A) : \sin 2B$  and the complex number representing the point dividing the line segment CF internally in the ratio  $(\sin 2A + \sin 2B) : \sin 2C$  will be each

$$= \frac{z_1 \sin 2A + z_2 \sin 2B + z_3 \sin 2C}{\sin 2A + \sin 2B + \sin 2C}$$

Thus circumcentre of  $\Delta ABC$  is

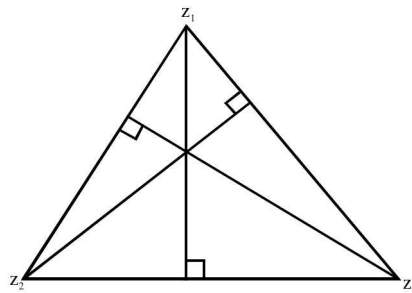
$$\frac{z_1 \sin 2A + z_2 \sin 2B + z_3 \sin 2C}{\sin 2A + \sin 2B + \sin 2C}$$

► **Example 2.** If  $z_1, z_2, z_3$  be the vertices of a triangle, then find its orthocentre and its circumcentre.

► **Solution Orthocentre:**

If  $z$  be the orthocentre, then we have

$$\arg \left( \frac{z_1 - z}{z_3 - z_2} \right) = \frac{\pi}{2} \quad (\text{see fig.})$$



$$\text{i.e. } \frac{z_1 - z}{z_3 - z_2} + \frac{\bar{z}_1 - \bar{z}}{\bar{z}_3 - \bar{z}_2} = 0$$

$$\text{i.e. } \bar{z}_1 - \bar{z} = \frac{(z - z_1)(\bar{z}_1 - \bar{z}_2)}{z_3 - z_2} \quad \dots(1)$$



Also, we have

$$\arg\left(\frac{z_2 - z}{z_1 - z_3}\right) = \frac{\pi}{2} \text{ (see fig.)}$$

$$\text{i.e. } \frac{z_2 - z}{z_1 - z_3} + \frac{\bar{z}_2 - \bar{z}}{\bar{z}_1 - \bar{z}_3} = 0$$

$$\text{i.e. } \bar{z}_2 - \bar{z} = \frac{(z - z_2)(\bar{z}_1 - \bar{z}_3)}{z_1 - z_3} \quad \dots(2)$$

Subtracting equation (1) from equation (2), we have

$$\text{i.e. } \bar{z}_1 - \bar{z}_2 = \frac{(z - z_1)(\bar{z}_3 - \bar{z}_2)}{z_3 - z_2} - \frac{(z - z_2)(\bar{z}_1 - \bar{z}_3)}{z_1 - z_3}$$

$$\begin{aligned} \text{i.e. } z[(z_1 - z_3)(\bar{z}_3 - \bar{z}_2) - (z_3 - z_2)(\bar{z}_1 - \bar{z}_3)] \\ = (z_1 - z_3)(z_3 - z_2)(\bar{z}_1 - \bar{z}_2) \\ + z_1(z_1 - z_3)(\bar{z}_3 - \bar{z}_2) - z_2(z_3 - z_2)(\bar{z}_1 - \bar{z}_3) \end{aligned}$$

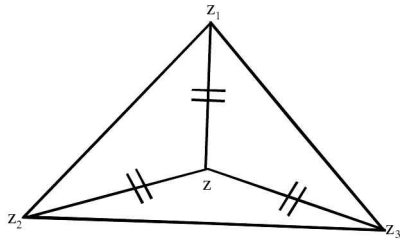
We solve for  $z$  in the above equation,

$$z = \frac{\sum z_1(\bar{z}_3 - \bar{z}_2) + \sum |z_1|^2(z_3 - z_2)}{\sum (z_1\bar{z}_2 - \bar{z}_1z_2)}$$

### Circumcentre

If  $z$  be the circumcentre, then we have

$$|z - z_1| = |z - z_2| \text{ (see fig.)}$$



$$\begin{aligned} \Rightarrow (z - z_1)(\bar{z} - \bar{z}_1) &= (z - z_2)(\bar{z} - \bar{z}_2) \\ \Rightarrow |z|^2 + |z_1|^2 - \bar{z}z_1 - z\bar{z}_1 &= |z|^2 + |z_2|^2 - \bar{z}z_2 - z\bar{z}_2 \\ \Rightarrow z(\bar{z}_1 - \bar{z}_2) + \bar{z}(z_1 - z_2) &= z_1\bar{z}_1 - z_2\bar{z}_2 \quad \dots(3) \end{aligned}$$

Also, we have  $|z - z_1| = |z - z_3|$

$$\begin{aligned} \Rightarrow z(\bar{z}_1 - \bar{z}_3) + \bar{z}(z_1 - z_3) &= z_1\bar{z}_1 - z_3\bar{z}_3 \quad \dots(4) \\ &\text{(proceeding as above)} \end{aligned}$$

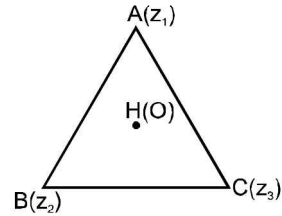
Equation (3)  $\times (z_1 - z_3)$  - equation (4)  $\times (z_1 - z_2)$  gives

$$\begin{aligned} z[(\bar{z}_1 - \bar{z}_2)(z_1 - z_3) - (\bar{z}_1 - \bar{z}_3)(z_1 - z_2)] \\ = z_1\bar{z}_1(z_2 - z_3) + z_2\bar{z}_2(z_3 - z_1) + z_3\bar{z}_3(z_1 - z_2) \\ \text{gives } z = \frac{\sum |z_1|^2(z_2 - z_3)}{\sum z_1(z_2 - z_3)} \end{aligned}$$

► **Example 3.** Let  $A(z_1)$ ,  $B(z_2)$  and  $C(z_3)$  forms an acute angled triangle in argand plane having origin as its orthocentre. Prove that

$$z_1\bar{z}_2 + \bar{z}_1z_2 = z_2\bar{z}_3 + \bar{z}_2z_3 = z_3\bar{z}_1 + \bar{z}_3z_1.$$

► **Solution** Clearly the angle between BC and AH is  $\frac{\pi}{2}$



$$\Rightarrow \frac{z_3 - z_2}{z_1} \text{ is purely imaginary,}$$

$$\Rightarrow \frac{z_3 - z_2}{z_1} + \frac{\bar{z}_3 - \bar{z}_2}{\bar{z}_1} = 0$$

$$\Rightarrow z_3\bar{z}_1 - z_2\bar{z}_1 + z_1\bar{z}_3 - z_2\bar{z}_1 = 0$$

$$\Rightarrow z_3\bar{z}_1 + z_1\bar{z}_3 = z_1\bar{z}_2 - z_2\bar{z}_1$$

$$\text{We also have, } \frac{z_3 - z_1}{z_2} + \frac{\bar{z}_3 - \bar{z}_1}{\bar{z}_2} = 0$$

$$\Rightarrow z_3 + \bar{z}_2 + z_2\bar{z}_3 = \bar{z}_1z_2 + z_1\bar{z}_2$$

$$\Rightarrow z_1\bar{z}_2 + z_1\bar{z}_1 = \bar{z}_2z_3 + z_3\bar{z}_2 = z_3\bar{z}_1 + z_1\bar{z}_3$$

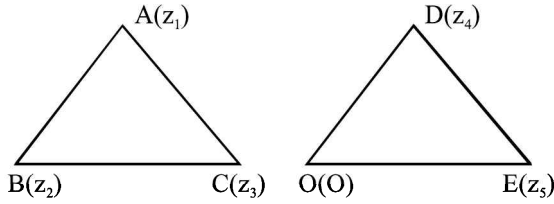
► **Example 4.** Let A, B, C, D, E be points on the complex plane representing the complex numbers  $z_1, z_2, z_3, z_4, z_5$  respectively. If  $(z_3 - z_2)z_4 = (z_1 - z_2)z_5$ , prove that  $\triangle ABC$  and  $\triangle DOE$  are similar.

► **Solution** Given,  $(z_3 - z_2)z_4 = (z_1 - z_2)z_5$

$\triangle ABC$  and  $\triangle DOE$  will be similar if

$$\frac{AC}{AB} = \frac{DE}{DO} \text{ and } \angle BAC = \angle ODE$$

$$\text{i.e. if } \left| \frac{z_3 - z_1}{z_2 - z_1} \right| = \left| \frac{z_5 - z_4}{0 - z_4} \right| \quad \dots(1)$$



$$\text{and } \arg \left( \frac{z_3 - z_1}{z_2 - z_1} \right) = \arg \left( \frac{z_5 - z_4}{0 - z_4} \right) \quad \dots(2)$$

From (1) and (2), we have

$$\frac{z_3 - z_1}{z_2 - z_1} = \frac{z_5 - z_4}{0 - z_4}$$

$$\text{or } -(z_3 - z_1)z_4 = (z_2 - z_1)z_5 - (z_2 - z_1)z_4$$

$$\text{or } (-z_3 + z_1 + z_2 - z_1)z_4 = (z_2 - z_1)z_5$$

$$\text{or } (z_2 - z_3)z_4 = (z_2 - z_1)z_5 \text{ or } (z_3 - z_2)z_4 = (z_1 - z_2)z_5$$

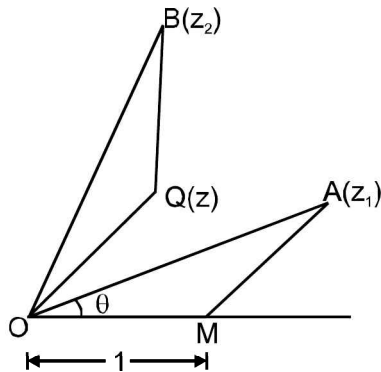
Thus  $\triangle ABC$  and  $\triangle DOE$  are similar if

$$(z_3 - z_2)z_4 = (z_1 - z_2)z_5$$

► **Example 5.** On the Argand plane point 'A' denotes a complex number  $z_1$ . A triangle OBQ is made directly similar to the triangle OAM, where  $OM = 1$  as shown in the figure. If the point B denotes the complex number  $z_2$ , then find the complex number corresponding to the point 'Q' in terms of  $z_1$  &  $z_2$ .

► **Solution**  $\frac{OB}{OQ} = \frac{OA}{OM} = OA \quad (\because OM = 1)$

$$OQ = \frac{OB}{OA} \text{ or } |z| = \frac{|z_2|}{|z_1|}$$



$$\begin{aligned} \text{Also } \arg \frac{\vec{OB}}{\vec{OA}} &= \arg \vec{OB} - \arg \vec{OA} \\ &= \angle BOM - \angle AOM = \angle BOM - \angle BOQ \\ &\quad (\angle AOM = \angle BOM = \theta) \\ &= \angle QOM = \arg \text{ of } z \end{aligned}$$

Hence complex number corresponding to the point Q =  $\frac{z_2}{z_1}$

► **Example 6.** The altitudes from the vertices A, B and C of the triangle ABC meet its circumcircle at D, E and F respectively. The complex numbers representing the points D, E and F are  $z_1, z_2$  and  $z_3$  respectively. If  $\frac{z_3 - z_1}{z_2 - z_1}$  is purely real then show that

triangle ABC is right angled at A.

► **Solution** The angles of  $\triangle DEF$  are  $\pi - 2A, \pi - 2B$  and  $\pi - 2C$  respectively.

Also it is given that  $\frac{z_3 - z_1}{z_2 - z_1}$  is purely real

$$\Rightarrow \arg \left( \frac{z_3 - z_1}{z_2 - z_1} \right) = 0 \text{ or } \pi \Rightarrow \pi - 2A = 0 \text{ or } \pi$$

$$\Rightarrow A = \frac{\pi}{2} \text{ or } 0 \text{ (not permissible)}$$

Hence triangle ABC is right angled at A.

► **Example 7.**  $z_1, z_2$  and  $z_3$  are the vertices of a triangle ABC such that  $|z_1| = |z_2| = |z_3|$  and  $AB = AC$ .

Prove that  $\frac{(z_1 + z_2)(z_1 + z_2)}{(z_2 + z_3)^2}$  is purely real.

► **Solution** Since  $|z_1| = |z_2| = |z_3|$

⇒ Circumcentre of  $\triangle ABC$  is at the origin.

⇒ Orthocentre ( $z_p$ ) of  $\triangle ABC$  is  $z_1 + z_2 + z_3$ .

Also angle subtended by AB and AC at the orthocentre are  $A + B$  and  $A + C$  respectively.

⇒ angles subtended by the sides AB and AC at the orthocentre are equal.

$$\Rightarrow \arg \left( \frac{z_1 + z_2 + z_3 - z_1}{z_1 + z_2 + z_3 - z_3} \right) = \arg \left( \frac{z_1 + z_2 + z_3 - z_2}{z_1 + z_2 + z_3 - z_1} \right)$$

$$\Rightarrow \arg\left(\frac{z_2 + z_3}{z_1 + z_2}\right) = \arg\left(\frac{z_1 + z_3}{z_2 + z_3}\right)$$

$$\Rightarrow \arg\left(\frac{(z_1 + z_3)(z_1 + z_2)}{(z_2 + z_3)^2}\right) = 0$$

$$\Rightarrow \frac{(z_1 + z_3)(z_1 + z_2)}{(z_2 + z_3)^2} \text{ is purely real.}$$

**Practice Problems****[N]**

1. The points  $A(z_1)$ ,  $B(z_2)$ ,  $C(z_3)$  are given on the argand plane.  $G$  is the centroid of the triangle  $ABC$ . If  $4z_1 + z_2 + z_3 = 0$ , show that origin is the mid-point of  $AG$ .
2. If  $z_1 + z_2 + z_3 = 0$  and  $|z_1| = |z_2| = |z_3| = 1$ , then prove that the points  $z_1, z_2, z_3$  are the vertices of an equilateral triangle inscribed in the unit circle.
3. Let  $A$  and  $B$  be two complex numbers such that  $A/B + B/A = 1$  then prove that the origin and the two points represented by  $A$  and  $B$  from vertices of an equilateral triangle.
4. Let the complex numbers  $z_1, z_2, z_3$  be the vertices of an equilateral triangle. Let  $z_0$  be the circumcentre of the triangle. Prove that  $z_1^2 + z_2^2 + z_3^2 = 3z_0^2$ .
5. If  $z_1^2 + z_2^2 \pm 2z_1z_2 \cos \theta = 0$ , prove that the points represented by  $z_1, z_2$  and the origin form an isosceles triangle.
6. If the complex numbers  $Z_i$  ( $i = 1, 2, 3, 4, 5, 6$ ) represent the vertices of a regular hexagon in the complex plane and  $Z_0$  represents the centre of the hexagon then prove that  $\sum_{i=1}^6 (Z_i)^2 = 6(Z_0)^2$ .
7. The three angular points of a triangle are given by  $z = \alpha, z = \beta, z = \gamma$ , where  $\alpha, \beta, \gamma$

are complex numbers. Establish the following propositions:

- (i) the centroid is given by  $z = \frac{1}{3}(\alpha + \beta + \gamma)$ ;
- (ii) the circum-centre is given by  $|z - \alpha| = |z - \beta| = |z - \gamma|$ ;
- (iii) the three perpendiculars from the angular points on the opposite sides meet in a point given by

$$R\left(\frac{z - \alpha}{\beta - \gamma}\right) = R\left(\frac{z - \beta}{\gamma - \alpha}\right) = R\left(\frac{z - \gamma}{\alpha - \beta}\right) = 0$$

**1.16 CIRCLES IN COMPLEX NUMBERS****1. General equation of circle**

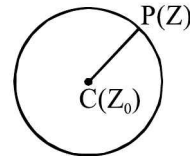
Let  $z_0$  be the centre of the circle and  $P(z)$  be any point on the circle.

$$\text{Then } |z - z_0| = r \text{ or } |z - z_0|^2 = r^2$$

$$\text{or } (z - z_0)(\bar{z} - \bar{z}_0) = r^2$$

$$\text{or } (z - z_0)(\bar{z} - \bar{z}_0) = r^2$$

$$\text{or } z\bar{z} - z_0\bar{z} - \bar{z}_0z + z_0\bar{z}_0 - r^2 = 0$$



$$\text{or } z\bar{z} - z_0\bar{z} - \bar{z}_0z + |z_0|^2 - r^2 = 0$$

$$\text{or } z\bar{z} + a\bar{z} + \bar{a}z + b = 0 \quad \dots(1)$$

where  $a = -z_0$ ,  $b = |z_0|^2 - r^2 = a \bar{a}$  is a real number

Clearly centre of the circle (1) is  $z_0$  i.e.  $-a$  and

$$\text{radius} = \sqrt{|z_0|^2 - r^2} = \sqrt{|a|^2 - r^2} = \sqrt{a\bar{a} - r^2}.$$

► **Example 1.** If  $r_1$  and  $r_2$  are the distances of points on the curve  $10(z\bar{z}) - 3i\{z^2 - (\bar{z})^2\} - 6 = 0$  which are at maximum and minimum distance from the origin then find the value of  $r_1 + r_2$ .

► **Solution** The equation is  $10(z\bar{z}) - 3i\{z^2 - (\bar{z})^2\} - 6 = 0$

or  $10(x^2 + y^2) - 3i(2x)(2iy) - 6 = 0$   
 $\Rightarrow 5(x^2 + y^2) + 6xy - 8 = 0 \quad \dots (1)$

Let  $(r \cos \theta, r \sin \theta)$  be a point on (1), then

$$5r^2 + 6r^2 \sin \theta \cos \theta - 8 = 0 \Rightarrow r^2 = \frac{8}{5 + 3 \sin 2\theta}$$

Clearly,  $1 \leq r^2 \leq 4 \Rightarrow 1 \leq |r| \leq 2$

$$\therefore r_1 = |r|_{\max} = 2 \text{ and } r_2 = |r|_{\min} = 1 \Rightarrow r_1 + r_2 = 3$$

► **Example 2.** Let  $\gamma$  be the imaginary part of  $(z-1)$   $(\cos \alpha + i \sin \alpha)^{-1} + (z-1)^{-1} (\cos \alpha + i \sin \alpha)$  where  $z (\neq 1)$  is complex and  $\alpha$  is real. If  $\gamma = 0$  then prove that  $z$  lies on a circle whose radius is 1.

► **Solution**  $1$ ; Let  $\alpha = \frac{z-1}{\cos \alpha + i \sin \alpha}$   

$$+ \frac{\cos \alpha + i \sin \alpha}{z-1}$$

Then  $\alpha = \bar{\alpha} = 0$

$$\Rightarrow (z-1)(\cos \alpha - i \sin \alpha) - \frac{\cos \alpha - i \sin \alpha}{\bar{z}-1} = 0$$

$$\Rightarrow \frac{\{(z-1)(\bar{z}-1)-1\}(\cos \alpha - i \sin \alpha)}{(\bar{z}-1)} + \frac{\{1-(z-1)(\bar{z}-1)\}(\cos \alpha + i \sin \alpha)}{z-1} = 0$$

$$\Rightarrow \frac{(z-1)(z-1)(\bar{z}-1)(\bar{z}-1)}{(z-1)(\bar{z}-1) - (\bar{z}-1)}$$

$$= \frac{\cos \alpha + i \sin \alpha}{\cos \alpha - i \sin \alpha}$$

$$\Rightarrow |z-1| |(z-1)(\bar{z}-1)-1| = |\bar{z}-1| |(z-1)(\bar{z}-1)-1|$$

$$\Rightarrow (z-1)(\bar{z}-1)-1=0 \Rightarrow z\bar{z}-z-\bar{z}=0$$

or  $x^2 + y^2 - 2x = 0$ ,

which represents equation of a circle whose radius is 1.

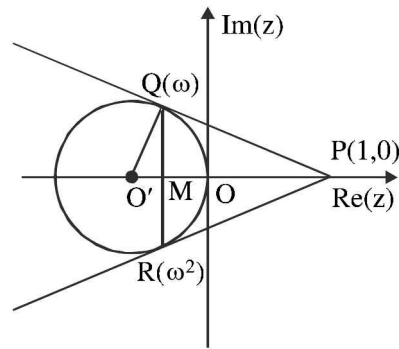
► **Example 3.** Find the equation of the circle in argand plane which passes through non real cube roots of unity and touches two sides of triangle with vertices as cube roots of unity.

► **Solution** Clearly triangle PQR is an equilateral triangle.

Now,  $QM = \frac{1}{2} |\omega - \omega^2| = \frac{1}{2} \left| 2 \frac{\sqrt{3}}{2} \right| = \frac{\sqrt{3}}{2}$  units.

In  $\Delta QO'M$

$$QO' = \frac{QM}{\sin 60^\circ} = \frac{\frac{\sqrt{3}}{2}}{\frac{\sqrt{3}}{2}} = 1 \text{ unit}$$



And  $Q'O = QO' = 1$  unit (radius of circle)

Point  $O'$  is given by  $(-1, 0)$

$$\Rightarrow \text{equation of circle } |z+1| = 1 \Rightarrow z\bar{z} + z + \bar{z} = 0$$

► **Example 4.** Show that the equation

$$|z - z_1|^2 + |z - z_2|^2 = k$$

(where  $k$  is a real number) will represent a circle

if  $k \geq \frac{1}{2} |z_1 - z_2|^2$

► **Solution** Given equation is  $|z - z_1|^2 + |z - z_2|^2 = k$

or  $|z|^2 + |z_1|^2 - 2\operatorname{Re}(z\bar{z}_1) + |z|^2 + |z_2|^2 - 2\operatorname{Re}(z\bar{z}_2) = k$

or  $2|z|^2 - 2\operatorname{Re}[z(\bar{z}_1 + \bar{z}_2)] = k - (|z_1|^2 + |z_2|^2)$

or  $|z|^2 - 2\operatorname{Re}\left[z\left(\frac{\bar{z}_1 + \bar{z}_2}{2}\right)\right] + \frac{1}{4}|z_1 + z_2|^2$

$$= \frac{k}{2} + \frac{1}{4}[|z_1 + z_2|^2 - 2|z_1|^2 - 2|z_2|^2]$$

or  $\left|z - \frac{z_1 + z_2}{2}\right|^2$

$$= \frac{1}{2}\left[k - \frac{1}{2}\{|z_1|^2 + |z_2|^2 - 2\operatorname{Re}(z_1\bar{z}_2)\}\right]$$

$$\text{or } \left| z - \frac{z_1 + z_2}{2} \right|^2 = \frac{1}{2} \left[ k - \frac{1}{2} |z_1 - z_2|^2 \right] \dots (1)$$

Equation (1) represents a circle with centre at

$$\frac{1}{2} (z_1 + z_2) \text{ and radius}$$

$$\frac{1}{2} \sqrt{2k - |z_1 - z_2|^2}, \text{ provided } k \geq \frac{1}{2} |z_1 - z_2|^2$$

2. The equation of the circle described on the line segment joining  $z_1$  and  $z_2$  as diameter is :

$$\arg \frac{z - z_2}{z - z_1} = \pm \frac{\pi}{2}$$

$$\text{or } (z - z_1)(\bar{z} - \bar{z}_2) + (z - z_2)(\bar{z} - \bar{z}_1) = 0.$$

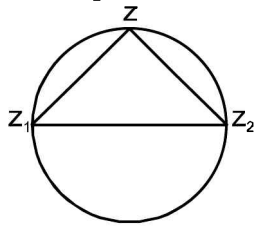
► **Example 5.** Show that the equation of the circle in the complex plane with  $z_1$  &  $z_2$  as its diameter can be expressed as,

$$2z\bar{z} - (\bar{z}_1 + \bar{z}_2)z - (z_1 + z_2)\bar{z} + z_1\bar{z}_2 + z_2\bar{z}_1 = 0.$$

► **Solution**  $\arg \frac{z - z_1}{z - z_2} = \pm \frac{\pi}{2}$

$$\Rightarrow \frac{z - z_1}{z - z_2} \text{ is purely imaginary.}$$

$$\Rightarrow \frac{z - z_1}{z - z_2} + \frac{\bar{z} - \bar{z}_1}{\bar{z} - \bar{z}_2} = 0 \Rightarrow \text{Result}$$

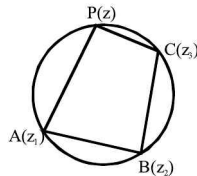


3. Equation of the circle through three non-collinear points  $z_1, z_2, z_3$  is

$$\left( \frac{z - z_1}{\bar{z} - \bar{z}_1} \right) \left( \frac{z_2 - z_3}{\bar{z}_2 - \bar{z}_3} \right) = \left( \frac{z - z_2}{\bar{z} - \bar{z}_2} \right) \left( \frac{z_1 - z_3}{\bar{z}_1 - \bar{z}_3} \right)$$

Let  $P(z)$  be any point on the circle through  $z_1, z_2$  and  $z_3$ , then

$$\arg \left( \frac{z_2 - z}{z_1 - z} \right) = \arg \left( \frac{z_2 - z_3}{z_1 - z_3} \right)$$



$$\text{or } \arg \left( \frac{z_2 - z}{z_1 - z} \right) \cdot \left( \frac{z_1 - z_3}{z_2 - z_3} \right) = 0$$

$$\left( \frac{z_2 - z}{z_1 - z} \right) \left( \frac{z_1 - z_3}{z_2 - z_3} \right) \text{ is real}$$

$$\text{or } \left( \frac{z_2 - z}{z_1 - z} \right) \left( \frac{z_1 - z_3}{z_2 - z_3} \right) = \left( \frac{z_2 - z}{z_1 - z} \right) \left( \frac{z_1 - z_3}{z_2 - z_3} \right)$$

$$\text{or } \frac{(\bar{z}_2 - \bar{z})(\bar{z}_1 - \bar{z}_3)}{(\bar{z}_1 - \bar{z})(\bar{z}_2 - \bar{z}_3)} = \frac{(z_2 - z)(z_1 - z_3)}{(z_1 - z)(z_2 - z_3)}$$

$$\text{or } \left( \frac{\bar{z} - \bar{z}_2}{\bar{z} - \bar{z}_1} \right) \left( \frac{\bar{z}_1 - \bar{z}_3}{\bar{z}_2 - \bar{z}_3} \right) = \left( \frac{z - z_2}{z - z_1} \right) \left( \frac{z_1 - z_3}{z_2 - z_3} \right)$$

$$\text{or } \left( \frac{z - z_1}{\bar{z} - \bar{z}_1} \right) \left( \frac{z_2 - z_3}{\bar{z}_2 - \bar{z}_3} \right) = \left( \frac{z - z_2}{\bar{z} - \bar{z}_2} \right) \left( \frac{z_1 - z_3}{\bar{z}_1 - \bar{z}_3} \right)$$

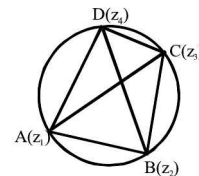
Thus the condition for four given points  $z_1, z_2, z_3$  and  $z_4$  to be concyclic is, the number  $\frac{z_1 - z_3}{z_2 - z_3} \cdot \frac{z_2 - z_4}{z_1 - z_4}$  is real.

► **Example 6.** Show that the points

$$3 + 4i, 3 - 4i, -4 + 3i, -4 - 3i \text{ are concyclic.}$$

► **Solution** Let  $z_1 = 3 + 4i, z_2 = 3 - 4i, z_3 = -4 + 3i$  and  $z_4 = -4 - 3i$

Let the point representing the complex numbers  $z_1, z_2, z_3$  and  $z_4$  be A, B, C and D respectively.



$$\text{Now } \angle ADB = \arg \left( \frac{z_2 - z_4}{z_1 - z_4} \right) = \arg \left( \frac{7 - i}{7 + 7i} \right)$$

$$\text{Now } \frac{z_2 - z_4}{z_1 - z_4} = \frac{7 - i}{7 + 7i} = \frac{(7 - i)(1 - i)}{7(1 + i)(1 - i)}$$

$$= \frac{6 - 8i}{7(1^2 + 1^2)} = \frac{3 - 4i}{7} = \frac{3}{7} - \frac{4}{7}i$$

$$\therefore \angle ADB = \left| \arg \left( \frac{z_2 - z_4}{z_1 - z_4} \right) \right|$$

$$= \left| -\tan^{-1} \frac{4}{3} \right| = \tan^{-1} \frac{4}{3} \quad \dots(1)$$

[Taking principal value since point  $\frac{3}{7} - \frac{4}{7}i$  lies in the 4th quadrant]

$$\begin{aligned} \text{Again } \frac{z_2 - z_3}{z_1 - z_3} &= \frac{7 - 7i}{7 + i} = \frac{(7 - 7i)(7 - i)}{(7 + i)(7 - i)} \\ &= \frac{42 - 56i}{50} = \frac{21}{25} - \frac{28}{25}i \end{aligned}$$

$$\begin{aligned} \therefore \angle ACB &= \left| \arg \left( \frac{z_2 - z_3}{z_1 - z_3} \right) \right| \\ &= \left| -\tan^{-1} \frac{28}{21} \right| = \tan^{-1} \left( \frac{4}{3} \right) \quad \dots(2) \end{aligned}$$

[Taking principal value since point  $\frac{21}{25} - \frac{28}{25}i$  lies in the 4th quadrant]

From (1) and (2),  $\angle ADB = \angle ACB$

$\therefore$  Points  $A(z_1)$ ,  $B(z_2)$ ,  $C(z_3)$  and  $D(z_4)$  are concyclic.

**Alternative**

$$\begin{aligned} \left( \frac{z_2 - z_4}{z_1 - z_4} \right) \left( \frac{z_1 - z_3}{z_2 - z_3} \right) &= \frac{(7 - i)(7 + i)}{(7 + 7i)(7 - 7i)} \\ &= \frac{50}{49 + 49} = \frac{25}{49} = \text{a real number.} \end{aligned}$$

Hence points  $A(z_1)$ ,  $B(z_2)$ ,  $C(z_3)$ ,  $D(z_4)$  are concyclic.

► **Example 7.** Prove that if the ratio  $\frac{z - i}{z - 1}$  is purely imaginary then the point  $z$  lies on the circle whose centre is at the point  $\frac{1}{2}(1 + i)$  and radius is  $\frac{1}{\sqrt{2}}$ .

► **Solution** Let  $z = x + iy$ .

$$\begin{aligned} \text{Then } \frac{z - i}{z - 1} &= \frac{x + i(y - 1)}{x - 1 + iy} \\ &= \frac{\{x + i(y - 1)\} \{(x - 1) - iy\}}{(x - 1)^2 + y^2} \\ &= \frac{x(x - 1) + y(y - 1)}{(x - 1)^2 + y^2} + i \frac{(x - 1)(y - 1) - xy}{(x - 1)^2 + y^2} \end{aligned}$$

Since  $\frac{z - i}{z - 1}$  is purely imaginary,

$$x(x - 1) + y(y - 1) = 0$$

$$\text{or } \left( x - \frac{1}{2} \right)^2 + \left( y - \frac{1}{2} \right)^2 = \frac{1}{2}$$

It is a circle with radius  $\frac{1}{\sqrt{2}}$  and centre  $\left( \frac{1}{2}, \frac{1}{2} \right)$ .

Therefore, the point  $z$  lies on a circle and the centre is  $\frac{1}{2}(1 + i)$ .

► **Example 8.** Prove that the equations,  $\left| \frac{z - 1}{z + 1} \right| = \text{constant}$  and  $\text{amp} \left( \frac{z - 1}{z + 1} \right) = \text{constant}$  on the argand plane represent the equations of the circles which are orthogonal.

► **Solution** Put  $\left| \frac{z - 1}{z + 1} \right| = \lambda$ .

$$\text{This simplifies to } x^2 + y^2 + 2 \frac{\lambda^2 + 1}{\lambda^2 - 1} x + 1 = 0$$

$$\text{or } x^2 + y^2 + 2gx + 1 = 0$$

Similarly the other equation,  $\text{amp} \left( \frac{z - 1}{z + 1} \right) = \mu$  reduces to,

$$x^2 + y^2 - \frac{2}{\mu} y + 1 = 0 \text{ or } x^2 + y^2 + 2fy - 1 = 0.$$

The two circles are orthogonal because

$$2 \cdot g \cdot 0 + 2 \cdot 0 \cdot f = 1 - 1 \text{ holds true.}$$

► **Example 9.** If complex number  $z$  lies on the curve  $|z - (-1 + i)| = 1$ , then find the locus of the complex

number  $w = \frac{z + i}{1 - i}$ ,  $i = \sqrt{-1}$ .

► **Solution**  $|z - (-1 + i)| = 1$

$$\Rightarrow |z + 1 - i| = 1 \quad \dots(1)$$

$$\text{Also } w = \frac{z + i}{1 - i}$$

$$\Rightarrow (1 - i)w = z + i \Rightarrow (1 - i)w - i = z$$

$$\Rightarrow |(1 - i)w - i + 1 - i| = |z + 1 - i|$$

$$\Rightarrow |1 - i| \left| w + \frac{1 - 2i}{1 - i} \right| = 1$$

$$\Rightarrow \left| w + \frac{(1-2i)(1+i)}{(1+i)(1-i)} \right| = \frac{1}{\sqrt{2}}$$

$$\Rightarrow \left| w + \frac{3+i}{2} \right| = \frac{1}{\sqrt{2}} \Rightarrow \left| w - \frac{-3+i}{2} \right| = \frac{1}{\sqrt{2}}$$

$$\Rightarrow \text{locus of } w \text{ is a circle centered at } \left( -\frac{3}{2}, \frac{1}{2} \right) \text{ and radius } \frac{1}{\sqrt{2}}.$$

### Inverse points w.r.t. a circle

Two points P & Q are said to be inverse w.r.t. a circle with centre 'O' and radius  $\rho$ , if :

- (i) the point O, P, Q are collinear and on the same side of O.
- (ii)  $OP \cdot OQ = \rho^2$ .



Two points  $z_1$  and  $z_2$  will be the inverse points w.r.t. the circle

$z\bar{z} + \bar{\alpha}z + \alpha\bar{z} + r = 0$  if and only if

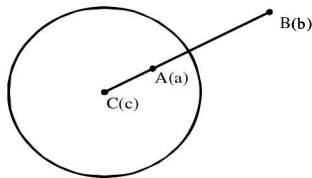
$$z_1\bar{z}_2 + \bar{\alpha}z_1 + \alpha\bar{z}_2 + r = 0.$$

► **Example 10.** C is the centre of a given circle  $|z - c| = r$ . A is a point represented by the complex number  $a$  inside the circle and B is a point represented by the complex number  $b$  outside the circle, such that C, A, B lie in a straight line and

$$CA \cdot CB = r^2. \text{ Show that } b = c + \frac{r^2}{(\bar{a} - \bar{c})}$$

► **Solution** We have  $CA \cdot CB = r^2$

$$\Rightarrow |(a - c)(b - c)| = r^2 \quad \dots(1)$$



Since, C, A, B are collinear points, therefore we have

$$\arg(b - c) = \arg(a - c) = \arg(\bar{a} - \bar{c})$$

$$[\because \arg(z) = -\arg(\bar{z})]$$

$$\arg(b - c) + \arg(\bar{a} - \bar{c}) = 0$$

$$\Rightarrow \arg\{(b - c)(\bar{a} - \bar{c})\} = 0$$

$$[\because \arg(z_1 z_2) = \arg(z_1) + \arg(z_2)]$$

$$\Rightarrow (b - c)(\bar{a} - \bar{c}) \text{ is purely real +ve number } \dots(2)$$

Therefore, equation (1) reduces to

$$|b - c| |\bar{a} - \bar{c}| = r^2 \quad [\because |z| = |\bar{z}|]$$

$$\Rightarrow (b - c)(\bar{a} - \bar{c}) = r^2$$

$[\because |z| = z, \text{ if } z \text{ is a real +ve number}] \text{ gives}$

$$b = c + \frac{r^2}{(\bar{a} - \bar{c})}.$$

► **Example 11.** Find all circles which are orthogonal to  $|z| = 1$  and  $|z - 1| = 4$

► **Solution** Let  $|z - \alpha| = k$  ... (1)  
(where  $\alpha = a + bi$  and  $a, b, k \in \mathbb{R}$ ) be a circle which cuts the circles

$$|z| = 1 \quad \dots(2)$$

$$\text{and } |z - 1| = 4 \text{ orthogonally. } \dots(3)$$

Then using the property that the sum of squares of their radii is equal to square of distance between centres. Thus the circle (1) will cut the circles (2) and (3) orthogonally if  $k^2 + 1 = |\alpha - 0|^2 = \alpha\bar{\alpha}$

$$\text{and } k^2 + 16 = |\alpha - 1|^2 = (\alpha - 1)(\bar{\alpha} - 1)$$

$$= \alpha\bar{\alpha} - (\alpha + \bar{\alpha}) + 1$$

$$\therefore 1 - (\alpha + \bar{\alpha}) - 15 = 0$$

$$\text{or } \alpha + \bar{\alpha} = -14$$

$$\therefore 2a = -14 \Rightarrow a = -7$$

$$\Rightarrow \alpha = a + ib = -7 + ib$$

$$\text{Also } k^2 = |\alpha|^2 - 1 = (-7)^2 + b^2 - 1 = b^2 + 48$$

$$\Rightarrow k = \sqrt{b^2 + 48}$$

$\therefore$  The required family of circles is given by

$$|z + 7 - bi| = \sqrt{48 + b^2}$$

### Ptolemy's Theorem

It states that the product of the lengths of the diagonals of a convex quadrilateral inscribed in a circle is equal to the sum of the products of the lengths of the two pairs of its opposite sides.

$$\text{i.e. } |z_1 - z_3| |z_2 - z_4| = |z_1 - z_2| |z_3 - z_4| + |z_1 - z_4| |z_2 - z_3|.$$

► **Example 12.** If A, B, C and D represent the complex numbers  $z_1, z_2, z_3$  and  $z_4$ , use the identity.

$$(z_1 - z_4)(z_2 - z_3) + (z_2 - z_4)(z_3 - z_1) + (z_3 - z_4)(z_1 - z_2) = 0$$

to show that  $AD \cdot BC \leq BD \cdot CA + CD \cdot AB$ .

► **Solution** The given identity can be rewritten

$$\begin{aligned} (z_1 - z_4)(z_2 - z_3) &= (z_4 - z_2)(z_3 - z_1) \\ &\quad + (z_4 - z_3)(z_1 - z_2) \\ \Rightarrow |(z_1 - z_4)(z_2 - z_3)| &= |(z_4 - z_2)(z_3 - z_1) \\ &\quad + (z_4 - z_3)(z_1 - z_2)| \\ \Rightarrow |z_1 - z_4| |z_2 - z_3| &\leq |z_4 - z_2| |z_3 - z_1| + |z_4 - z_3| |z_1 - z_2|, \end{aligned}$$

which proves the result, since

$$\begin{aligned} AD &= |z_1 - z_4|, \quad BD = |z_4 - z_2|, \quad CD = |z_3 - z_4|, \\ BD &= |z_2 - z_3|, \quad CA = |z_3 - z_1|, \quad AB = |z_1 - z_2|. \end{aligned}$$

### Practice Problems



1. If  $z(\overline{z + \alpha}) + \overline{z}(z + \alpha) = 0$ , where  $\alpha$  is a complex constant, then prove that  $z$  is represented by a point on a circle.
2. Prove that the points representing the complex numbers  $z$  for which  $|z + 4|^2 + |z - 4|^2 = 64$  lie on a circle with centre as origin.
3. If  $z_1, z_2, z_3$  &  $z_4$  are different complex numbers such that  $|z_1| = |z_2| = |z_3| = |z_4|$  then prove that the number  $\frac{(z_1 - z_2)(z_3 - z_4)}{(z_1 - z_4)(z_2 - z_3)}$  is a real positive number.
4. The point  $z$  in Argand's plane moves such that  $\operatorname{Re} \frac{iz + 1}{iz - 1} = 2$ . Find the locus of  $z$ .
5. In Argand plane prove that the locus of  $z \neq 1$  such that  $\arg \left[ \frac{2z^2 - 5z + 3}{3z^2 - z - 2} \right] = \frac{2\pi}{3}$  is a chord of a circle passing through  $z = -3/2, z = 2/3$ .
6. Find the number of solutions of the equation in  $z, z\bar{z} - (3 + i)z - (3 - i)\bar{z} - 6 = 0$ .

7. If  $z = 2 + k + i\sqrt{3 - k^2}$  where  $k$  is real such that

$k^2 < 3$ , prove that  $\left| \frac{z+1}{z-1} \right|$  is independent of  $k$ .

Also prove that the locus of the point  $z$  for different values of  $k$  is a part of the circle. What is its centre and radius?



### Target Problems for JEE Advanced

► **Problem 1.** If  $\frac{a}{b-c} + \frac{b}{c-a} + \frac{c}{a-b} = 0$ ,

where  $a, b, c$  are three distinct complex numbers, then find the value of

$$\frac{a^2}{(b-c)^2} + \frac{b^2}{(c-a)^2} + \frac{c^2}{(a-b)^2}.$$

► **Solution** We have

$$\begin{aligned} \frac{a}{b-c} + \frac{b}{c-a} + \frac{c}{a-b} &= 0 \\ \Rightarrow \left( \frac{a}{b-c} + \frac{b}{c-a} + \frac{c}{a-b} \right)^2 &= 0 \\ \Rightarrow \frac{a^2}{(b-c)^2} + \frac{b^2}{(c-a)^2} + \frac{c^2}{(a-b)^2} + 2 \\ &\Rightarrow \left[ \frac{ab}{(b-c)(c-a)} + \frac{bc}{(c-a)(a-b)} \right. \\ &\quad \left. + \frac{ca}{(a-b)(b-c)} \right] = 0 \\ \Rightarrow \frac{a^2}{(b-c)^2} + \frac{b^2}{(c-a)^2} + \frac{c^2}{(a-b)^2} + 2 \\ &\Rightarrow \left[ \frac{ab(a-b) + bc(b-c) + ca(c-a)}{(b-c)(c-a)(a-b)} \right] = 0 \dots (1) \end{aligned}$$

Now  $ab(a-b) + bc(b-c) + ca(c-a)$   
 $= a^2b - ab^2 + b^2c - bc^2 + ac^2 - a^2c$   
 $= a^2(b-c) - a(b^2 - c^2) + bc(b-c)$   
 $= (b-c)[a^2 - a(b+c) + bc]$   
 $= (b-c)(a-b)(a-c) = -(b-c)(c-a)(a-b)$



∴ From (1),

$$\frac{a^2}{(b-c)^2} + \frac{b^2}{(c-a)^2} + \frac{c^2}{(a-b)^2} 2(-1) = 0$$

$$\Rightarrow \frac{a^2}{(b-c)^2} + \frac{b^2}{(c-a)^2} + \frac{c^2}{(a-b)^2} = 2$$

► **Problem 2.** Compute the product

$$\left[1 + \left(\frac{1+i}{2}\right)\right] \left[1 + \left(\frac{1+i}{2}\right)^2\right] \left[1 + \left(\frac{1+i}{2}\right)^{2^2}\right] \dots \left[1 + \left(\frac{1+i}{2}\right)^{2^n}\right].$$

► **Solution** Putting  $\frac{1+i}{2} = z$ , the product

$$P = (1+z)(1+z^2) \dots (1+z^{2^n})$$

$$P = \frac{1-z^{2^{n+1}}}{1-z}.$$

Now we must substitute  $\frac{1+i}{2}$  for  $z$  into the above formula. We have

$$\frac{1}{1-z} = \frac{1}{1-\frac{1+i}{2}} = \frac{2}{1-i} = \frac{2(1+i)}{(1-i)(1+i)} = 1+i.$$

Furthermore, we find

$$1 - z^{2^{n+1}} = 1 - \left(\frac{1+i}{2}\right)^{2^{n+1}} = 1 - \left[\left(\frac{1+i}{2}\right)^2\right]^{2^n}$$

$$= 1 - \left(\frac{i}{2}\right)^{2^n} \dots (1)$$

Note that for  $n \geq 2$  we have  $i^{2^n} = (i^4)^{2^{n-2}} = 1$ .

Hence, by virtue of (1), for  $n \geq 2$  we have

$$1 - z^{2^{n+1}} = 1 - \frac{1}{2^{2^n}} \text{ and } P = (1+i) \left(1 - \frac{1}{2^{2^n}}\right).$$

For  $n = 1$  we obtain

$$1 - z^{2^{n+1}} = 1 - \left(\frac{i}{2}\right)^2 = \frac{5}{4} \text{ and } P = (1+i) \frac{5}{4}$$

► **Problem 3.** Let  $z$  and  $\omega$  be complex numbers such that  $|z| \leq 1$ ,  $|\omega| \leq 1$ ,  $|z + i\omega| = |z - i\bar{\omega}| = 2$  then show that  $z$  is either 1 or  $-1$ .

► **Solution** Let  $z = r(\cos \alpha + i \sin \alpha)$  ;

$$\omega = r'(\cos \beta + i \sin \beta)$$

$$z + i\omega = r(\cos \alpha + i \sin \alpha) + ir'(\cos \beta + i \sin \beta)$$

$$= (r \cos \alpha + r' \sin \beta) + i(r \sin \alpha - r' \cos \beta)$$

$$z - i\bar{\omega} = (r \cos \alpha - r' \sin \beta) + i(r \sin \alpha - r' \cos \beta)$$

On substituting in the given equality we get

$$r^2 + r'^2 + 2rr' \sin(\alpha - \beta) = 4$$

$$(1) \quad r^2 + r'^2 + 2rr' \sin(\alpha + \beta) = 4$$

$$(2) \quad \text{Equating LHS of (1) and (2) we get}$$

$$\sin(\alpha - \beta) + \sin(\alpha + \beta) = 0$$

$$\Rightarrow \sin(\alpha - \beta) + \sin(\alpha + \beta) = 0$$

Now (1) becomes

$$r^2 + r'^2 + 2rr' \sin \alpha \cos \beta = 4$$

Since  $|r| \leq 1$ ,  $|r'| \leq 1$ ,  $|2 \sin \alpha \cos \beta| \leq 2$

The last relation is possible only when

$$r^2 = r'^2 = 1, \sin \alpha = \pm 1, \cos \beta = \pm 1$$

$$\Rightarrow r \text{ or } r' = 1, \alpha = \pm \frac{\pi}{2}, \beta = 0 \text{ whence } z \text{ is } 1 \text{ or } -1.$$

► **Problem 4.** Let a sequence  $x_1, x_2, x_3, \dots$  of complex numbers be defined by  $x_1 = 0$ ,  $x_{n+1} = x_n^2 - i$  for  $n > 1$  where  $i^2 = -1$ . Find the distance of  $x_{2000}$  from  $x_{1997}$  in the complex plane.

► **Solution**  $x_1 = 0$ ,  $x_2 = 0^2 - i = -i$ ,

$$x_3 = (-i)^2 - i = -1 - i = -(1+i)$$

$$x_4 = [1(1+i)]^2 - i = 2i - i = i$$

$$x_5 = (i)^2 - i = -1 - i = x_3$$

$$x_6 = (-1-i)^2 - i = i = x_4$$

∴  $x_6 = x_4$  and hence  $x_7 = x_5$  and so on

$$x_{2n} = i \text{ for } n \geq 1, x_{2n+1} = -1 - i$$

So  $x_{2000} = i = (0, 1)$  in the complex plane.

$$x_{1997} = (1, -i) = (-1, -1) \text{ in the complex plane.}$$

So, distance between  $x_{2000}$  and  $x_{1997}$  is  $\sqrt{1^2 + 2^2} = \sqrt{5}$

► **Problem 5.** For every real number  $a \geq 0$ , find all the complex numbers  $z$  that satisfy the equation  $2|z| - 4az + 1 + i = 0$ .

► **Solution** We have  $2|z| - 4az + 1 + i = 0$

Put  $z = x + iy$

$$\text{We get, } 2\sqrt{x^2 + y^2} = 4ax - 1 \dots (1)$$

$$\text{or, } 4(x^2 + y^2) = (4ax - 1)^2$$

and  $a = 4ay$  (by separating imaginary and real parts)

$$\Rightarrow y = \frac{1}{2} \text{ and } 4x^2 + \frac{1}{4} - 16a^2x^2 - 1 + 8ax = 0$$

$$\Rightarrow x^2(16 - 64a^2) + 32ax - 3 = 0$$

$$\Rightarrow x = \frac{-4a \pm \sqrt{4a^2 + 3}}{4(1 - 4a^2)}$$

as  $x > \frac{1}{4a}$  (from equation (1))

$$\Rightarrow \text{either } \frac{4a + \sqrt{4a^2 + 3}}{16a^2 - 4} > \frac{1}{4a}$$

$$\Rightarrow \frac{3}{4(4a - \sqrt{4a^2 + 3})} > \frac{1}{4a}$$

$$\Rightarrow 3a > 4a - \sqrt{4a^2 + 3}$$

If  $a > \frac{1}{2}$ ,  $3(a^2 + 1) > 0$  is always true

If  $a < \frac{1}{2}$ ,  $4a^2 + 3 < a^2$  is never true

or,  $\frac{4a - \sqrt{4a^2 + 3}}{16a^2 - 4} > \frac{1}{4a}$

$$\Rightarrow \frac{3}{4(4a + \sqrt{4a^2 + 3})} > \frac{1}{4a}$$

$$\Rightarrow a + \sqrt{4a^2 + 3} < 0, \text{ which can never hold}$$

$$\Rightarrow x = \frac{4a + \sqrt{4a^2 + 3}}{16a^2 - 4} + \frac{i}{4} \text{ if } a > \frac{1}{2}.$$

$$\Rightarrow \text{no solution if } 0 \leq a \leq \frac{1}{2}$$

and  $z = \frac{4a + \sqrt{4a^2 + 3}}{16a^2 - 4} + \frac{1}{4}$  if  $a > \frac{1}{2}$ .

► **Problem 6.** Solve the following system of equations in terms of complex numbers :

$$z^3 + \bar{w}^7 = 0$$

$$z^5 \cdot w^{11} = 1$$

► **Solution** From the first equation we have  $z^3 = -\bar{w}^7$ ; from the second,  $z^5 = 1/w^{11}$ , whence we respectively obtain

$$z^{15} = -\bar{w}^{35}; \text{ and } z^{15} = 1/w^{32} \text{ and,}$$

$$\text{hence, } -\bar{w}^{35} = 1/w^{32}, \text{ or } w^{33}\bar{w}^{35} = -1.$$

From this equation it follows that  $|w^{33}\bar{w}^{35}| = 1$ .

Using the properties of moduli and conjugate numbers,

we obtain  $|w^{33}\bar{w}^{35}| = |w|^{33} \cdot |\bar{w}|^{35} = |w|^{68} = 1$  so that  $|w| = 1$ .

Reverting to the equation  $w^{33}\bar{w}^{35} = -1$ , we rewrite its left-hand member:

$$(w^{33}\bar{w}^{33})\bar{w}^2 = (w\bar{w})^{33}\bar{w}^2 = (|w|^2)^{33}\bar{w}^2 = \bar{w}^2$$

We have thus arrived at the equation  $\bar{w}^2 = -1$  or  $\bar{w}_1 = i$ ,  $\bar{w}_2 = -i$ , whence  $w_1 = -i$ ,  $w_2 = i$ .

Now compute the respective values of  $z$ .

If  $w = -i$ , then from the first equation of the origin system we have  $z^3 = -i^7 = i$

$$\text{and from the second, } z^5 = \frac{1}{(-i)^{11}} = \frac{1}{i} = -i$$

► **Problem 7.** Prove that if  $z_1$  and  $z_2$  are two complex numbers and  $c > 0$ , then

$$|z_1 + z_2|^2 \leq (1 + c)|z_1|^2 + (1 + c^{-1})|z_2|^2.$$

► **Solution**  $|z_1 + z_2|^2 = (z_1 + z_2)(\bar{z}_1 + \bar{z}_2)$

$$= |z_1|^2 + |z_2|^2 + 2\operatorname{Re}(z_1\bar{z}_2) \leq |z_1|^2 + |z_2|^2 + 2|z_1||z_2|$$

$$\text{i.e. } |z_1 + z_2|^2 \leq |z_1|^2 + |z_2|^2 + |z_1||z_2|. \quad [|z_1||z_2| = |z_2||z_1|]$$

Incorporating the number  $c > 0$ , the last term on the RHS can be written

$$2|z_1||z_2| = 2|\sqrt{c}z_1| \left| \frac{z_2}{\sqrt{c}} \right| \leq |\sqrt{c}z_1|^2 + \left| \frac{z_2}{\sqrt{c}} \right|^2$$

$$[2ab \leq a^2 + b^2]$$

$$\text{i.e., } 2|z_1||z_2| \leq c|z_1|^2 + \frac{|z_2|^2}{c}$$

$$\text{Hence } |z_1 + z_2|^2 \leq |z_1|^2 + |z_2|^2 + c|z_1|^2 + \frac{|z_2|^2}{c}$$

$$= (1 + c)|z_1|^2 + (1 + c^{-1})|z_2|^2$$

► **Problem 8.** Let  $z_1, z_2, z_3$  be the three non zero complex numbers such that  $z_2 \neq 1$ ,  $a = |z_1|$ ,  $b = |z_2|$  and  $c = |z_3|$ .

$$\text{If } \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = 0,$$

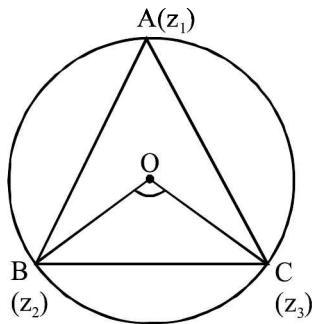
show that  $\arg\left(\frac{z_3}{z_2}\right) = \arg\left(\frac{z_3 - z_1}{z_2 - z_1}\right)^2$ .

► **Solution** Given,  $\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = 0$

$$\begin{aligned} \Rightarrow a^3 + b^3 + c^3 - 3abc &= 0 \\ \Rightarrow (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca) &= 0 \\ \Rightarrow \frac{1}{2}(a + b + c)[(a - b)^2 + (b - c)^2 + (c - a)^2] &= 0 \\ \Rightarrow (a - b)^2 + (b - c)^2 + (c - a)^2 &= 0 \\ \Rightarrow a = b = c \quad [\because a + b + c \neq 0, \therefore z_1 \neq 0, \\ \therefore |z_1| = a \neq 0 \text{ etc.}] \\ \Rightarrow OA = OB = OC, \text{ where } O \text{ is the origin and } A, \\ B, C \text{ are the points representing } z_1, z_2 \text{ and } z_3 \\ \text{respectively.} \\ \therefore O \text{ is the circumcentre of } \triangle ABC \end{aligned}$$

Now  $\arg\left(\frac{z_3}{z_2}\right) = \angle BOC \quad \dots(1)$

$$= 2\angle BAC$$



$$\begin{aligned} &= 2\arg\left(\frac{z_3 - z_1}{z_2 - z_1}\right) \quad \dots(2) \\ &= \arg\left(\frac{z_3 - z_1}{z_2 - z_1}\right)^2 \end{aligned}$$

Hence  $\arg\left(\frac{z_3}{z_2}\right) = \arg\left(\frac{z_3 - z_1}{z_2 - z_1}\right)^2$

► **Problem 9.** Find the real value of the parameter  $t$  for which there is at least one complex number  $z$

$= x + iy$  satisfying the condition  $|z + 3| = t^2 - 2t + 6$  and the inequality  $|z - 3\sqrt{3}i| < t^2$

► **Solution** Given,  $|z + 3| = t^2 - 2t + 6$   
 $\Rightarrow z$  lies on the circle whose centre is  $(-3, 0)$  and radius is  $t^2 - 2t + 6$ .

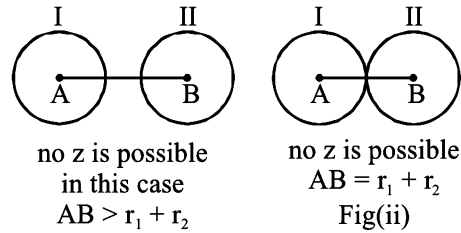
Let  $A \equiv (-3, 0)$  and  $r_1 = t^2 - 2t + 6$

Also  $|z - 3\sqrt{3}i| < t^2 \Rightarrow z$  lies in the interior of the circle whose centre is  $(0, 3\sqrt{3})$  and radius is  $t^2$ .

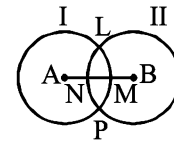
Let  $B \equiv (0, 3\sqrt{3})$ ,  $r_2 = t^2$

$$AB = \sqrt{3^2 + 27} = 6$$

Atleast one  $z$  will be there satisfying both the conditions in the following cases

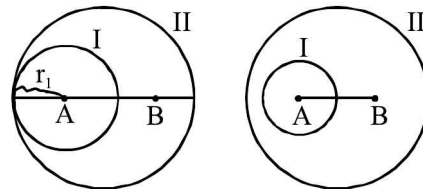


Fig(i)



Infinitely many  $z$  are possible lying on arc LMP.

Fig (iii)



Infinitely many  $z$  are possible lying on circle I  
 Fig (iv)

Infinitely many  $z$  are possible (all points on circle I)  
 Fig (v)

$$r_2 - r_1 = 2t - 6 = 2(t - 3)$$

**Case I :** When  $t > 3$

In this case atleast one  $z$  will be possible  $\dots(1)$   
 if  $AB < r_1 + r_2$  [fig. (iii), (iv) and (v)]

$$\text{or } 6 < 2(t^2 - t + 3)$$

$$\text{or } t^2 - t > 0 \quad \text{or} \quad t < 0 \text{ or } t > 1 \quad \dots(2)$$

From (1) and (2) common values of  $t$  are given by

$$3 < t < \infty \quad \dots(3)$$

**Case II :** When  $t \leq 3$  i.e.  $r_1 \geq r_2$   $\dots(4)$

In this case atleast one  $z$  will be possible if

$$|r_1 - r_2| \leq AB < r_1 + r_2$$

$$\text{or } 2(3 - t) \leq 6 < 2(t^2 - t + 3)$$

$$\text{or } 3 - t < 3 \leq t^2 - t + 3$$

$$\text{Now } 3 \geq 3 - 1 \Rightarrow t \leq 0 \quad \dots(5)$$

$$\text{and } 3 < t^2 - t + 3 \Rightarrow t^2 - t > 0 \Rightarrow t < 0$$

$$\text{or } t > 1 \quad \dots(6)$$

From (4), (5) and (6) common values of  $t$  are given

$$\text{by } 1 < t \leq 3 \quad \dots(7)$$

From (3) and (7) all possible values of  $t$  are given by  
 $1 < t < \infty$

► **Problem 10.** If  $\cos(\alpha - \beta) + \cos(\beta - \gamma) + \cos$

$$(\gamma - \alpha) = -\frac{3}{2} \text{ then prove that ,}$$

$$\Sigma \cos 4\alpha = 2 \Sigma \cos 2(\beta + \gamma) \text{ \& } \Sigma \sin 4\alpha = 2 \Sigma \sin 2(\beta + \gamma).$$

► **Solution** Let  $x = \cos \alpha + i \sin \alpha$ ,  $y = \cos \beta + i \sin \beta$ ,  $z = \cos \gamma + i \sin \gamma$   $\dots(1)$

$$\therefore x + y + z = (\cos \alpha + i \sin \alpha) + (\cos \beta + i \sin \beta) + (\cos \gamma + i \sin \gamma)$$

$$= (\cos \alpha + \cos \beta + \cos \gamma) + i(\sin \alpha + \sin \beta + \sin \gamma) = 0 + i \cdot 0 = 0 \text{ (as given)}$$

Now  $x + y + z = 0$  gives  $x + y = -z$

$$\text{or } (x + y)^2 = z^2, \quad (\text{squaring both sides})$$

$$\text{i.e. } x^2 + y^2 - z^2 = -2xy$$

$$\text{or } (x^2 + y^2 - z^2)^2 = 4x^2y^2,$$

(again squaring both sides)

$$\text{or } x^4 + y^4 + z^4 + 2x^2y^2 - 2x^2z^2 - 2y^2z^2 = 4x^2y^2$$

$$\text{or } x^4 + y^4 + z^4 = 2(x^2y^2 + y^2z^2 + z^2x^2)$$

$$\text{or } \Sigma x^4 = 2 \Sigma y^2z^2,$$

(expressing in the summation notation)

$$\begin{aligned} \text{or } \Sigma (\cos \alpha + i \sin \alpha)^4 &= 2 \Sigma (\cos \beta + i \sin \beta)^2 (\cos \gamma + i \sin \gamma)^2, \\ &\quad \text{putting for } x, y \text{ \& } z \text{ from (1)} \end{aligned}$$

$$\text{or } \Sigma (\cos 4\alpha + i \sin 4\alpha)$$

$$= 2 \Sigma (\cos 2\beta + i \sin 2\beta) (\cos 2\gamma + i \sin 2\gamma)$$

$$= 2 \Sigma [\cos 2(\beta + \gamma) + i \sin 2(\beta + \gamma)]$$

Equating real and imaginary parts on both sides, we get

$$\Sigma \cos 4\alpha = 2 \Sigma \cos 2(\beta + \gamma)$$

$$\text{and } \Sigma \sin 4\alpha = 2 \Sigma \sin 2(\beta + \gamma)$$

► **Problem 11.** Prove that if  $\cos \alpha + i \sin \alpha$  is the solution of the equation  $x^n + p_1x^{n-1} + \dots + p_n = 0$ , then  $p_1 \sin \alpha + p_2 \sin 2\alpha + \dots + p_n \sin n\alpha = 0$  ( $p_1, p_2, \dots, p_n$  are real).

► **Solution** Since  $\cos \alpha + i \sin \alpha$  is the root of the given equation, we have

$$\sum_{k=0}^n p_k (\cos \alpha + i \sin \alpha)^{n-k} = 0 \quad (p_0 = 1)$$

$$\text{or } (\cos \alpha + i \sin \alpha)^n \sum_{k=0}^n p_k (\cos \alpha + i \sin \alpha)^{-k} = 0.$$

But  $(\cos \alpha + i \sin \alpha)^{-1} = \cos \alpha - i \sin \alpha$ , therefore

$$\sum_{k=0}^n p_k (\cos \alpha - i \sin \alpha)^k = 0, \quad \sum_{k=0}^n p_k (\cos k\alpha - i \sin k\alpha) = 0.$$

Hence, indeed,

$$\sum_{k=0}^n p_k \sin k\alpha = p_1 \sin \alpha + p_2 \sin 2\alpha + \dots + p_n \sin n\alpha = 0.$$

► **Problem 12.** Prove that

(i) The polynomial  $(\cos \phi + x \sin \phi)^n - \cos n\phi - x \sin n\phi$  is divisible by  $x^2 + 1$ .

(ii) The polynomial  $x^n \sin \phi - \rho^{n-1} x \sin n\phi + \rho^n \sin(n-1)\phi$  is divisible by  $x^2 - 2\rho x \cos \phi + \rho^2$ .

► **Solution** (i) Let

$$f(x) = (\cos \phi + x \sin \phi)^n - \cos n\phi - x \sin n\phi.$$

But  $x^2 + 1 = (x + i)(x - i)$  and  $f(i)$

$$= (\cos \phi + i \sin \phi)^n - (\cos n\phi + i \sin n\phi) = 0$$

(by De Moivre's theorem). Likewise we make sure that  $f(-i) = 0$ , and our supposition is proved.

(ii) Resolve the polynomial  $x^2 - 2\rho x \cos \phi + \rho^2$  into factors linear in  $x$ . For this purpose find the roots of the quadratic equation  $x^2 - 2\rho x \cos \phi + \rho^2 = 0$ .

We get

$$x = \rho \cos \phi \pm \sqrt{\rho^2 \cos^2 \phi - \rho^2} = \rho(\cos \phi \pm i \sin \phi).$$

$$\text{Let } f(x) = x^n \sin \phi - \rho^{n-1} x \sin n\phi + \rho^n \sin(n-1)\phi.$$

We can hence prove that  $f[\rho(\cos \phi \pm i \sin \phi)] = 0$ .

► **Problem 13.** Show that

$$\sum_{p=1}^{32} (3p+2) \left( \sum_{q=1}^{10} \left( \sin \frac{2q\pi}{11} - i \cos \frac{2q\pi}{11} \right) \right)^{4p} = 1648,$$

where  $p$  and  $q$  are positive integers.

► **Solution**

$$\begin{aligned} \sin \frac{2q\pi}{11} - i \cos \frac{2q\pi}{11} &= -i \left( \cos \frac{2q\pi}{11} + i \sin \frac{2q\pi}{11} \right) \\ \Rightarrow \sum_{q=1}^{10} \left( \sin \frac{2q\pi}{11} - i \cos \frac{2q\pi}{11} \right) &= -i \left[ \left( \cos \frac{2\pi}{11} + i \sin \frac{2\pi}{11} \right) + \left( \cos \frac{4\pi}{11} + i \sin \frac{4\pi}{11} \right) \right. \\ &\quad \left. + \left( \cos \frac{6\pi}{11} + i \sin \frac{6\pi}{11} \right) + \dots \right. \\ &\quad \left. + \left( \cos \frac{20\pi}{11} + i \sin \frac{20\pi}{11} \right) \right] \end{aligned}$$

$= -i$  (Sum of eleventh roots of unity except unity itself)

$$= -i(-1) = i$$

$$\begin{aligned} \Rightarrow \text{The sum} &= \sum_{p=1}^{32} (3p+2)i^{4p} = \sum_{p=1}^{32} (3p+2) \\ &= 3 \sum_{p=1}^{32} p + 64 = 3 \times \frac{32 \times 33}{2} + 64 = 1648 \end{aligned}$$

► **Problem 14.** If  $1, a_1, a_2, a_3, \dots, a_{n-1}$  be the  $n$ th roots of unity and  $w$  be a non-real complex cube root of unity then prove that  $(w - a_1)(w - a_2) \dots (w - a_{n-1})$

$= 0$ , if  $n$  is of the form  $3m$

$1$ , if  $n$  is of the form  $3m+1$

$1+w$ , if  $n$  is of the form  $3m+2$ .

► **Solution** Here  $\sqrt[n]{1} = 1, a_1, a_2, \dots, a_{n-1}$

$$\therefore x^n - 1 = 0 \text{ has the roots } 1, a_1, a_2, \dots, a_{n-1}$$

$$\therefore x^n - 1 = (x-1)(x-a_1)(x-a_2) \dots (x-a_{n-1}).$$

Putting  $x = w$ , we get

$$w^n - 1 = (w-1)(w-a_1)(w-a_2) \dots (w-a_{n-1})$$

$$\therefore (w-a_1)(w-a_2)(w-a_3) \dots (w-a_{n-1}) = \frac{w^n - 1}{w - 1} \dots (1)$$

$$\text{If } n = 3m, w^n - 1 = w^{3m} - 1 = 1 - 1 = 0$$

$$\therefore \frac{w^n - 1}{w - 1} = 0.$$

$$\text{If } n = 3m+1, w^n - 1 = w^{3m+1} - 1 = w - 1$$

$$\therefore \frac{w^n - 1}{w - 1} = \frac{w - 1}{w - 1} = 1.$$

$$\text{If } n = 3m+2, w^n - 1 = w^{3m+2} - 1 = w^2 - 1$$

$$\therefore \frac{w^n - 1}{w - 1} = \frac{w^2 - 1}{w - 1} = 2 + 1.$$

Hence, the problem follows from (1).

► **Problem 15.** If the expression  $z^5 - 32$  can be factorised into linear and quadratic factors over real coefficients as

$$(z^5 - 32) = (z-2)(z^2 - pz + 4)(z^2 - qz + 4)$$

then find the value of  $(p^2 + 2p)$ .

► **Solution**  $z^5 - 32 = (z - z_0)(z - z_1)(z - z_2)(z - z_3)(z - z_4)$  where  $z_i$ 's,  $i = 0, 1, 2, 3, 4$  are given by

$$z_i = 2 \left( \cos \frac{2m\pi}{5} + i \sin \frac{2m\pi}{5} \right)$$

(using De Moivre's Theorem)

with  $m = 0, 1, 2, 3, 4$ , we get  $z_0 = 2$ ,

$$\text{and } z_1 = 2 \left( \cos \frac{2\pi}{5} + i \sin \frac{2\pi}{5} \right);$$

$$z_2 = 2 \left( \cos \frac{4\pi}{5} + i \sin \frac{4\pi}{5} \right);$$

$$z_3 = 2 \left( \cos \frac{6\pi}{5} + i \sin \frac{6\pi}{5} \right);$$

$$z_4 = 2 \left( \cos \frac{8\pi}{5} + i \sin \frac{8\pi}{5} \right)$$

hence,  $z^5 - 32$

$$\begin{aligned} &= (z-2) \left( z^2 - 4 \cos \frac{2\pi}{5} z + 4 \right) \\ &\quad \left( z^2 - 4 \cos \frac{4\pi}{5} z + 4 \right) \end{aligned}$$

$$\left[ \begin{array}{l} \text{using } (z-\alpha)(z-\bar{\alpha}) \\ = z^2 - (\alpha + \bar{\alpha})z + \alpha \bar{\alpha} \end{array} \right]$$

$$\therefore p = 4 \cos \frac{2\pi}{5} = 4 \cos 72^\circ = 4 \sin 18^\circ$$

$$\begin{aligned}\therefore p^2 + 2p &= [16\sin^2 18^\circ + 8\sin 18^\circ] \\ &= 8[1 - \cos 36^\circ + \sin 18^\circ] \\ &= 8\left[1 + \frac{\sqrt{5}-1}{4} - \frac{\sqrt{5}+1}{4}\right] = 4.\end{aligned}$$

► **Problem 16.** If  $z^2 - z + 1 = 0$ , then find the value of

$$\left(z + \frac{1}{z}\right)^2 + \left(z^2 + \frac{1}{z}\right)^2 + \left(z^3 + \frac{1}{z^3}\right)^2 + \dots + \left(z^{24} + \frac{1}{z^{24}}\right)^2.$$

► **Solution**  $z^2 - z + 1 = 0 \Rightarrow z = \frac{1 \pm i\sqrt{3}}{2}$

$$\text{Taking } z = \frac{1 + i\sqrt{3}}{2} = \cos \frac{\pi}{3} + i \sin \frac{\pi}{3}$$

$$\Rightarrow z^n = \cos \frac{n\pi}{3} + i \sin \frac{n\pi}{3}, n = 1, 2, \dots, 24$$

$$\therefore z^n + \frac{1}{z^n} = 2\cos \frac{n\pi}{3}$$

$$\therefore \left(z + \frac{1}{z}\right)^2 + \left(z^2 + \frac{1}{z^2}\right)^2 + \left(z^3 + \frac{1}{z^3}\right)^2 + \dots + \left(z^{24} + \frac{1}{z^{24}}\right)^2$$

$$= 2^2 \cos^2 \frac{\pi}{3} + 2^2 \cos^2 \frac{2\pi}{3} + 2^2 \cos^2 \frac{3\pi}{3} + \dots + 2^2 \cos^2 \frac{24\pi}{3}$$

$$= 2 \left[ \left(1 + \cos \frac{2\pi}{3}\right) + \left(1 + \cos \frac{4\pi}{3}\right) + \left(1 + \cos \frac{6\pi}{3}\right) + \dots + \left(1 + \cos \frac{48\pi}{3}\right) \right]$$

$$= 2 \left[ 24 + \frac{\cos \left\{ \frac{2\pi}{3} + \frac{23\pi}{3} \right\} \sin \frac{24\pi}{3}}{\sin \frac{\pi}{3}} \right]$$

$$= 2(24 + 0) = 48$$

Using the formula,

$$\cos \alpha + \cos(\alpha + \beta) + \cos(\alpha + 2\beta) + \dots + \cos\{\alpha + (n-1)\beta\}$$

$$= \frac{\cos \left\{ \alpha + \frac{(n-1)\beta}{2} \right\} \sin \frac{n\beta}{2}}{\sin \frac{\beta}{2}}$$

► **Problem 17.** Prove that the roots of the equation

$$\frac{1}{z-z_1} + \frac{1}{z-z_2} + \frac{1}{z-z_3} = 0$$

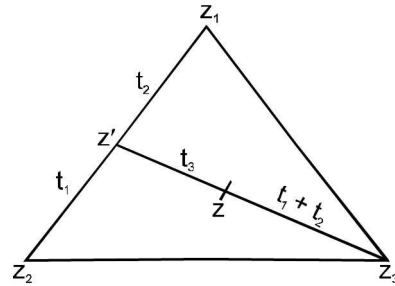
where  $z_1, z_2, z_3$  are pairwise distinct complex numbers, correspond to points on a complex plane which lie inside a triangle with vertices  $z_1, z_2, z_3$  or on its sides.

► **Solution**  $\frac{\bar{z}-\bar{z}_1}{|z-z_1|^2} + \frac{\bar{z}-\bar{z}_2}{|z-z_2|^2} + \frac{\bar{z}-\bar{z}_3}{|z-z_3|^2} = 0$

$$\frac{\bar{z}-\bar{z}_1}{a} + \frac{\bar{z}-\bar{z}_2}{b} + \frac{\bar{z}-\bar{z}_3}{c} = 0$$

where  $|z - z_1|^2 = a$  etc

$$\text{or } \frac{z - z_1}{a} + \frac{z - z_2}{b} + \frac{z - z_3}{c} = 0$$



$$bc(z - z_1) + zc(z - z_2) + ab(z - z_3) = 0$$

Let  $bc = t_1$ ;  $ca = t_2$ ;  $ab = t_3$

$$t_1(z - z_1) + t_2(z - z_2) + t_3(z - z_3) = 0$$

$$(t_1 + t_2 + t_3)z = t_1z_1 + t_2z_2 + t_3z_3$$

$$\Rightarrow z = \frac{t_1z_1 + t_2z_2 + t_3z_3}{t_1 + t_2 + t_3}$$

$$= \frac{t_1z_1 + t_2z_2}{t_1 + t_2} \cdot \frac{t_1 + t_2}{t_1 + t_2 + t_3} + \frac{t_3z_3}{t_1 + t_2 + t_3}$$

$$= \frac{t_1 + t_2}{t_1 + t_2 + t_3} z' + \frac{t_3z_3}{t_1 + t_2 + t_3}$$

$$\Rightarrow z = \frac{(t_1 + t_2)z' + t_3z_3}{t_1 + t_2 + t_3}$$

$$\Rightarrow z \text{ lies inside the } \Delta z_1 z_2 z_3$$

If  $t_1 = t_2 = t_3 \Rightarrow z$  is the centroid of the triangle.

Also if  $a = b = c$

$$\Rightarrow |z - z_1| = |z - z_2| = |z - z_3| \Rightarrow z$$

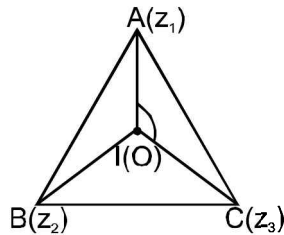
is the circumcentre

if  $t_3 = 0 \Rightarrow z$  lies on the line joining  $z_1$  and  $z_2$

► **Problem 18.**  $z_1, z_2$  and  $z_3$  are the vertices of an isosceles triangle in anticlockwise direction with

origin as incentre. If  $\arg \left( \frac{z_3 - z_1}{z_2 - z_1} \right) > \frac{\pi}{2}$ , then prove that  $z_2, z_1$  and  $kz_3$  are in G.P. where  $k \in \mathbb{R}^+$ .

► **Solution** Let  $z_1, z_2$  and  $z_3$  represent the vertices A, B and C respectively of triangle ABC.



Now  $\angle A > 90^\circ$  and triangle is isosceles

$$\Rightarrow \angle B = \angle C.$$

$$\angle AIC = \frac{\pi}{2} + \frac{B}{2} \text{ and } \angle BIA = \frac{\pi}{2} + \frac{C}{2}$$

$$\Rightarrow \frac{z_1}{z_3} = \left| \frac{z_1}{z_3} \right| e^{i\left(\frac{\pi+B}{2}\right)} \quad \dots(1)$$

$$\text{and } \frac{z_2}{z_1} = \left| \frac{z_2}{z_1} \right| e^{i\left(\frac{\pi+C}{2}\right)} \quad \dots(2)$$

on dividing equation (1) by equation (2) we get

$$\frac{z_1^2}{z_2 z_3} = \frac{|z_1|^2}{|z_2||z_3|} e^{i\left(\frac{B-C}{2}\right)}$$

$$\Rightarrow \frac{z_1^2}{z_2 z_3} = \frac{|z_1|^2}{|z_2||z_3|} = \text{positive real number}$$

( $\because \angle B = \angle C$ )

$$\Rightarrow z_1^2 = kz_2 z_3 \text{ where } k \in \mathbb{R}^+.$$

Hence  $z_2, z_1$  and  $kz_3$  are in G.P.

► **Problem 19.** In the equation,  $z^2 + 2\lambda z + 1 = 0$ ,  $\lambda$  is a parameter which can take any real value. Show that if  $-1 < \lambda < 1$ , the roots of this equation lie on a certain circle in the Argand diagram, but that if  $\lambda > 1$ , one root lies inside the unit circle and one outside. Prove that for very large values of  $\lambda$ , the roots are approximately  $-2\lambda, -1/2\lambda$ .

► **Solution**  $z^2 + 2\lambda z + 1 = 0$

$$\Rightarrow z = \frac{-2\lambda \pm \sqrt{4\lambda^2 - 4}}{2} \text{ or } z = -\lambda \pm \sqrt{\lambda^2 - 1}$$

$$\text{or } z = -\lambda \pm \sqrt{1 - \lambda^2} i \text{ (if } -1 < \lambda < 1)$$

$$\text{or } z = -\lambda \pm \mu i \text{ where } \sqrt{1 - \lambda^2} = \mu > 0$$

$$\therefore z + \lambda = \mu i \text{ or } z + \lambda = -\mu i \quad \overline{z + \lambda} = -\mu i$$

$$\text{or } \overline{z + \lambda} = \mu i. \text{ Hence } (z + \lambda)(\overline{z + \lambda}) = \mu^2$$

$$\text{or } (z + \lambda)(\overline{z + \lambda}) = \mu^2 \Rightarrow |z + \lambda| = \mu$$

Hence 'z' lies on a circle with centre  $-\lambda$  and radius  $\mu$ . Again let  $\lambda > 1$ , we have

$$z = -\lambda + \sqrt{\lambda^2 - 1} \text{ or } -\lambda - \sqrt{\lambda^2 - 1}$$

$$\text{Hence } z = -\lambda + \mu$$

$$\text{or } z = -\lambda - \mu \text{ where } \mu^2 = \lambda^2 - 1; z = -\lambda \pm \mu.$$

$$\text{If } z_1 \text{ \& } z_2 \text{ are the roots then, } |z_1| = |-\lambda + \mu|$$

$$\& \quad |z_2| = |\lambda + \mu|$$

$$|z_1||z_2| = |\mu^2 - \lambda^2| = 1$$

$$\Rightarrow \text{if } |z_1| < 1 \text{ then } |z_2| > 1$$

i.e. one root lies inside the unit circle and the other outside the unit circle. Further, if  $\lambda$  is large,

$$\lambda^2 = \mu^2 + 1 \Rightarrow \lambda \approx \mu$$

$$\text{Hence the roots are } z_1 = -\lambda - \mu \approx -2\lambda$$

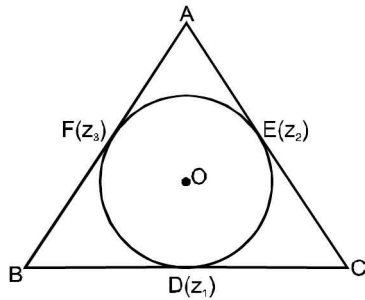
$$\text{and } z_2 = -\lambda + \mu = \frac{(-\lambda + \mu)(-\lambda - \mu)}{(-\lambda - \mu)}$$

$$= \frac{\lambda^2 - \mu^2}{-\lambda - \mu} = -\frac{1}{2\lambda}$$

► **Problem 20.** Complex numbers  $z_1, z_2, z_3$  are represented by the points of contact D, E, F of the incircle of triangle ABC, with the centre O of the incircle taken as the origin. If BO meets DE at G, find the complex number represented by G.

► **Solution** Let the incircle touch AB at F. Let O be the origin and let  $z_1, z_2, z_3$  be the complex numbers represented by D, E and F respectively. Since C is the point of intersection of the tangents to the circle at D and E, C represents the complex number  $\frac{2z_1z_2}{z_1+z_2}$ . Similarly A and B represent the complex numbers  $\frac{2z_2z_3}{z_2+z_3}$  and  $\frac{2z_1z_3}{z_1+z_3}$ . Let  $r$  be the radius of the incircle.

$$\Rightarrow |z_1| = |z_2| = |z_3| = r.$$



Equation of the line BO is

$$\frac{z}{\frac{2z_1z_3}{z_1+z_3}} = \frac{\bar{z}}{\frac{2\bar{z}_1\bar{z}_3}{\bar{z}_1+\bar{z}_3}} = \frac{\bar{z}}{r^2(\bar{z}_1+\bar{z}_3)}$$

$$\text{or } z = \sqrt{\frac{z_1\bar{z}_1z_3\bar{z}_3}{\bar{z}_1\bar{z}_3}}\bar{z}$$

$$\Rightarrow \sqrt{\bar{z}_1\bar{z}_3}z = \sqrt{z_1z_3}\bar{z} \quad \dots(1)$$

$$\text{Equation of line DE is } \frac{z-z_1}{z_2-z_1} = \frac{\bar{z}-\bar{z}_1}{\bar{z}_2-\bar{z}_1}.$$

Where it meets (1), we have

$$\frac{z-z_1}{z_2-z_1} = \frac{z \cdot \sqrt{\frac{\bar{z}_1\bar{z}_3}{z_1z_3}} - \bar{z}_1}{\bar{z}_2 - \bar{z}_1} = \frac{\frac{zr^2}{z_1z_3} - \frac{r^2}{z_1}}{\frac{r^2}{z_2} - \frac{r^2}{z_1}}$$

$$\Rightarrow z = \frac{(z_1-z_2)z_3}{z_2+z_3} \text{ which is represented by G.}$$

► **Problem 21.**  $A(z_a), B(z_b), C(z_c)$  are vertices of right angled triangle,  $z_c$  being the orthocentre. A circle is described on AC as diameter. Find the point of intersection of the circle with hypotenuse.

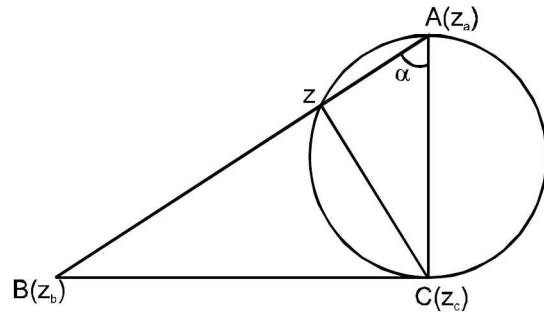
► **Solution** Let d be the point of intersection

$$\text{In } \triangle ADC \quad \frac{z_a-z}{|z_a-z|} = \frac{z_c-z}{|z_c-z|} e^{i\pi/2}$$

$$\text{In } \triangle ABC \quad \frac{z_b-z_c}{|z_b-z_c|} = \frac{z_a-z_c}{|z_a-z_c|} e^{i\pi/2}$$

Dividing both, we get

$$\frac{(z_a-z)|z_b-z_c|}{|z_a-z|(z_b-z_c)} = \frac{(z_c-z)|z_a-z_c|}{|z_c-z|(z_a-z_c)}$$



$$\tan \alpha = \frac{|z_c-z|}{|z-z_a|} = \frac{|z_b-z_c|}{|z_c-z_a|}$$

$$\Rightarrow \frac{(z_a-z)}{(z_b-z_c)} \times \frac{|z_b-z_c|^2}{|z_c-z_a|^2} = \frac{(z_c-z)}{(z_a-z_c)}$$

$$\Rightarrow (z_a-z) \left( \frac{\bar{z}_b-\bar{z}_c}{\bar{z}_a-\bar{z}_c} \right) = (z_c-z)$$

$$\Rightarrow z = \frac{z_a(\bar{z}_b-\bar{z}_c) - z_c(\bar{z}_a-\bar{z}_c)}{(z_b-z_a)}$$

► **Problem 22.** A, B, C are the vertices of a triangle inscribed in the circle  $|z| = 1$ . Altitude from A meets the circle again at D. If D, B, C represents complex



numbers  $z_1, z_2, z_3$  respectively, then prove that the complex number representing the reflection of D in the line BC, is  $\frac{z_1 z_2 + z_1 z_3 - z_2 z_3}{z_1}$ .

► **Solution** If the reflection D in BC is P( $z_5$ ), then

$$|z_2 - z_1| = |z_2 - z_5| \text{ and } |z_3 - z_1| = |z_3 - z_5|.$$

The first relation is

$$(z_2 - z_1)(\bar{z}_2 - \bar{z}_1) = (z_2 - z_5)(\bar{z}_2 - \bar{z}_5)$$

$$\text{or } 2 - z_1 \bar{z}_2 - \bar{z}_1 z_2 = 1 - \bar{z}_2 z_5 - \bar{z}_5 z_2 + z_5 \bar{z}_5$$

$$(\text{since } z_1 \bar{z}_1 = 1 = z_2 \bar{z}_2)$$

$$\text{or } 1 - \frac{z_1}{z_2} - \frac{z_2}{z_1} = z_5 \bar{z}_5 - \frac{z_5}{z_2} - \bar{z}_5 z_2$$

$$\text{or } z_1(z_2 + z_5) - (z_1^2 + z_2^2) = \bar{z}_5(z_5 - z_2)z_1 z_2 \quad \dots(1)$$

Similarly, form the second relation,

$$z_1(z_3 + z_5) - (z_1^2 + z_3^2) = \bar{z}_5(z_5 - z_3)z_1 z_3 \quad \dots(2)$$

Eliminating  $\bar{z}_5$  from (1) and (2), we get

$$z_3(z_5 - z_3)[z_1(z_2 + z_5) - (z_1^2 + z_2^2)] \\ = z_2(z_5 - z_2)[z_1(z_3 + z_5) - (z_1^2 + z_3^2)]$$

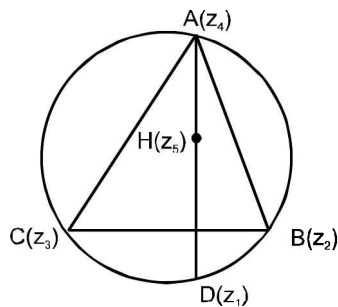
$$\text{or } z_5(z_2 - z_3)(z_1^2 - z_2 z_3 - z_1 z_5)$$

$$= (z_2 - z_3)(z_1^2(z_2 + z_3) - z_1 z_2 z_3 - z_1 z_5(z_2 + z_3))$$

$$\text{or } (z_5 - z_1)(-z_2 z_3 + z_1(z_2 + z_3) - z_1 z_5) = 0 \quad (z_2 \neq z_3).$$

$$\text{or } -z_2 z_3 + z_1(z_2 + z_3) - z_1 z_5 = 0 \quad (z_1 \neq z_3).$$

$$\text{Hence } z_5 = \frac{z_1 z_2 + z_1 z_3 - z_2 z_3}{z_1}$$



#### Alternative

Let H, G and O be respectively the orthocentre, the centroid and the circumcentre of  $\triangle ABC$ . Let A and H represent the complex numbers  $z_4$  and  $z_5$

$$\Rightarrow z_1 = -\frac{z_2 z_3}{z_4} \Rightarrow z_4 = -\frac{z_2 z_3}{z_1}$$

and the complex number associated with the G is

$$\frac{z_1 + z_3 + z_4}{3} = \frac{z_2 + z_3 - \frac{z_2 z_3}{z_1}}{3} \\ = \frac{z_1 z_2 + z_1 z_3 - z_2 z_3}{3z_1}.$$

Now the orthocentre is the reflection of D in the line BC, and G divides HO in the ratio 2 : 1. Since O represents the complex number zero,

$$\frac{z_1 z_2 + z_1 z_3 - z_2 z_3}{3z_1} = \frac{z_5}{3}$$

$$\Rightarrow z_5 = \frac{z_1 z_2 + z_1 z_3 - z_2 z_3}{z_1}$$

#### Things to Remember

1. If  $n \in \mathbb{I}$ ,  $i^{4n} = 1$ ,  $i^{4n+1} = i$ ,  $i^{4n+2} = -1$ ,  $i^{4n+3} = -i$ .

2.  $\sqrt{a + ib} = x + iy$ , where x and y are real numbers given by

$$x = \pm \sqrt{\frac{\sqrt{a^2 + b^2} + a}{2}},$$

$$y = \pm \sqrt{\frac{\sqrt{a^2 + b^2} - a}{2}} \text{ and if b is positive,}$$

both x and y have the same signs, and if b is negative, x and y have opposite signs.

3. To find the principal argument of  $z = x + iy$ , we first calculate  $\alpha = \tan^{-1} \left| \frac{y}{x} \right|$  and then find in which quadrant the point z lies (by observing the signs of x and y). Then the principal argument  $\theta$  of z in the various quadrant in order is given as :

$$\alpha, \pi - \alpha, -\pi + \alpha \text{ and } -\alpha.$$

4. If  $z_1 = r_1 (\cos \theta_1 + i \sin \theta_1)$

and  $z_2 = r_2 (\cos \theta_2 + i \sin \theta_2)$  then

$$z_1 z_2 = r_1 r_2 (\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2)),$$

$$\frac{z_1}{z_2} = \frac{r_1}{r_2} (\cos(\theta_1 - \theta_2) + i \sin(\theta_1 - \theta_2)).$$

5. Given in the complex plane are two points  $z_1$  and  $z_2$ . Then the complex numbers corresponding to the points lying on the bisector of the angle formed by the vector  $z_1$  and  $z_2$  are

$$t \left( \frac{z_1}{|z_1|} + \frac{z_2}{|z_2|} \right) \text{ where } t \text{ is a positive real number.}$$

6. The following are some properties of conjugate:

(i)  $z + \bar{z} = 2R(z)$  ;

$z - \bar{z} = 2iI(z)$

If  $z = x + iy$  then  $x = \frac{z + \bar{z}}{2}$ ,  $y = \frac{z - \bar{z}}{2i}$ .

(ii) If  $z$  is real then  $z = \bar{z}$  and viceversa.

(iii) If  $z$  is purely imaginary then  $z = -\bar{z}$  and vice versa.

(iv)  $\overline{\bar{z}} = z$ .

(v)  $\overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2$ .

(vi)  $\overline{z_1 - z_2} = \bar{z}_1 - \bar{z}_2$ .

(vii)  $\overline{z_1 z_2} = \bar{z}_1 \bar{z}_2$ , Specially,  $\overline{(z^n)} = (\bar{z})^n$ .

(viii)  $\overline{\left( \frac{z_1}{z_2} \right)} = \frac{\bar{z}_1}{\bar{z}_2}$ .

7. The properties of modulus are as follows :

(i)  $|z| = |\bar{z}| = |-z|$

(ii)  $-|z| \leq R(z) \leq |z|$   
 $-|z| \leq I(z) \leq |z|$

(iii)  $z \bar{z} = |z|^2$

(iv)  $|z_1 z_2| = |z_1| |z_2|$

Further if  $Z = z_1 z_2 \dots z_n$ , then

$$|Z| = |z_1| \cdot |z_2| \dots |z_n|.$$

Specially,  $|z^n| = |z|^n$ .

(v)  $\left| \frac{z_1}{z_2} \right| = \frac{|z_1|}{|z_2|}$

(vi)  $|z_1 + z_2| \leq |z_1| + |z_2|$ .

If  $z_1 = r_1 (\cos \theta_1 + i \sin \theta_1)$ ,

$z_2 = r_2 (\cos \theta_2 + i \sin \theta_2)$ .

$$|z_1 + z_2| = \{r_1^2 + r_2^2 + 2r_1 r_2 \cos(\theta_1 - \theta_2)\}^{1/2}$$

Equality holds when  $\theta_1 = \theta_2$ .

$$|z_1 + z_2 + z_3| \leq |z_1| + |z_2| + |z_3|.$$

$$|z_1 + z_2 + z_3 + \dots + z_n| \leq |z_1| + |z_2| + |z_3| + \dots + |z_n|.$$

(vii)  $|z_1 - z_2| \geq ||z_1| - |z_2||$

$$|z_1 - z_2| = \{r_1^2 + r_2^2 - 2r_1 r_2 \cos(\theta_1 - \theta_2)\}^{1/2}$$



(i)  $||z_1| - |z_2|| \leq |z_1 \pm z_2| \leq |z_1| + |z_2|$

(ii)  $|z_1 + z_2|^2 + |z_1 - z_2|^2 = 2[|z_1|^2 + |z_2|^2]$

$$|z_1 \pm z_2|^2 = (z_1 \pm z_2)(\bar{z}_1 \pm \bar{z}_2)$$

$$= |z_1|^2 + |z_2|^2 \pm 2R(z_1 \bar{z}_2)$$

8. The properties of argument are as follows :

(i)  $\arg(\bar{z}) = -\arg(z)$

(ii)  $\arg(-z) = \pi + \arg(z) + 2k\pi$ , where  $k$  is a suitable integer to adjust the argument to principal value.

(iii)  $\arg(z_1 \cdot z_2) = \arg z_1 + \arg z_2 + 2k\pi$ , where  $k$  is a suitable integer.

(iv)  $\arg(z_1 z_2 \dots z_n) = \theta_1 + \theta_2 + \dots + \theta_n$   
 $= \arg z_1 + \arg z_2 + \dots + \arg z_n + 2k\pi$ ,  
 where  $k$  is a suitable integer.

(v)  $\arg(z^n) = n \arg(z) + 2k\pi$ , where  $k$  is a suitable integer.

(vi)  $\arg\left(\frac{z_1}{z_2}\right) = \arg z_1 - \arg z_2 + 2k\pi$ ,

where  $k$  is a suitable integer.

(vii) If  $z$  is real then  $\arg(z) = 0$  or  $\pi$ .

If  $z$  is a positive real number then  $\arg(z) = 0$ .

If  $z$  is a negative real number then  $\arg(z) = \pi$ .

(viii) If  $z$  is a purely imaginary number then

$$\arg(z) = \pm \frac{\pi}{2}.$$

9. If the vertices A, B, C of a  $\Delta ABC$  represent the complex number  $z_1, z_2, z_3$  respectively in the anti-clockwise sense and  $\angle BAC = \alpha$  then

$$\frac{z_3 - z_1}{|z_3 - z_1|} = \frac{z_2 - z_1}{|z_2 - z_1|} e^{i\alpha}$$

10. (i) If  $n$  is an integer, positive or negative,  
 $(\cos\theta + i \sin\theta)^n = \cos n\theta + i \sin n\theta$  and if  
 $n$  is a rational number,  $\cos n\theta + i \sin n\theta$  is  
 one of the values of  $(\cos\theta + i \sin\theta)^n$ .

(ii)  $\cos\theta - i \sin\theta = \frac{1}{\cos\theta + i \sin\theta}$

- (iii) If  $z = \cos\theta + i \sin\theta$ .  
 then  $(1/z) = \cos\theta - i \sin\theta$ .  
 so that  $z + (1/z) = 2 \cos\theta$   
 and  $z - (1/z) = 2i \sin\theta$ .  
 so that  $z^n + (1/z)^n = 2 \cos n\theta$   
 and  $z^n - (1/z)^n = 2i \sin n\theta$ .

11. Properties of cube root of unity :

- (i) The three cube roots of unity when plotted on the Argand plane constitute the vertices of an equilateral triangle.  
 (ii) If  $\omega$  is one of the imaginary cube roots of unity then  $1 + \omega + \omega^2 = 0$  and  $\omega^3 = 1$ .  
 (iii) In general  $1 + \omega^r + \omega^{2r} = 0$  if  $r \in \mathbb{I}$  but is not the multiple of 3.  
 $= 3$  if  $r$  is a multiple of 3.  
 (iv)  $z^3 - 1 = (z - 1)(z^2 + z + 1)$   
 $= (z - 1)(z - \omega)(z - \omega^2)$   
 $\Rightarrow z^2 + z + 1 = (z - \omega)(z - \omega^2)$   
 (v)  $z^3 + 1 = (z + 1)(z^2 - z + 1)$   
 $= (z + 1)(z + \omega)(z + \omega^2)$   
 Thus the cube roots of  $-1$  are  $-1, -\omega, -\omega^2$ .

(vi)  $a^3 - b^3 = (a - b)(a - \omega b)(a - \omega^2 b)$   
 $a^3 + b^3 = (a + b)(a + \omega b)(a + \omega^2 b)$   
 $a^2 + ab + b^2 = (a - \omega b)(a - \omega^2 b)$   
 $a^3 + b^3 + c^3 - 3abc$   
 $= (a + b + c)(a + \omega b + \omega^2 c)(a + \omega^2 b + \omega c)$ .

12. The equation  $z^n = a = \rho(\cos\phi + i \sin\phi)$  has just  $n$  roots, given by  $z = r(\cos\theta + i \sin\theta)$ , where  $r = \sqrt[n]{\rho}$ ,  $\theta = (\phi + 2k\pi)/n$ , ( $k = 0, 1, \dots, n-1$ ).  
 The particular root  $\sqrt[n]{\rho} \{ \cos(\phi/n) + i \sin(\phi/n) \}$  is called the principal value of  $\sqrt[n]{a}$ .

13. The  $n$  roots of the equation  $z^n = 1$  are  
 $\cos(2k\pi/n) + i \sin(2k\pi/n)$ ,  
 $(k = 0, 1, \dots, n-1)$ . ... (4)

These numbers are called the  $n$ th roots of unity; the principal value is unity itself. If we write  $\alpha$  for  $\cos(2\pi/n) + i \sin(2\pi/n)$ , we see that the  $n$ th roots of unity are  $1, \alpha, \alpha^2, \dots, \alpha^{n-1}$ .

14. Properties of  $n$ th roots of unity

- (i) The real values of the  $n$ th root of unity are obtained for the values  $k = 0$  and  $\frac{n}{2}$ , if  $n$  is even, and for  $k = 0$  if  $n$  is odd.  
 (ii) In the complex plane, the  $n$ th roots of unity are located on the circumference of the unity circle and divide it into  $n$  equal arcs: one of the division points is the number 1.  
 (iii) From this it follows that those of the  $n$ th roots of unity which are not real are pairwise conjugate.  
 (iv) All values of the  $n$ th root of a complex number  $z$  may be obtained by multiplying one of these values by all the  $n$ th roots of unity.  
 (v) The product of two  $n$ th roots of unity is itself an  $n$ th root of unity.  
 (vi) Also, the reciprocal of an  $n$ th root of unity is itself that root.  
 (vii) Generally, any power of the  $n$ th root of unity is also an  $n$ th root of unity.  
 (viii) Any  $k$ th root of unity will also be an  $l$ th root of unity for any  $l$  that is a multiple of  $k$ .  
 (ix) However, for any  $n$ , there exist  $n$ th roots of unity such that they are not any lesser roots of unity. These roots are termed **primitive**  $n$ th roots of unity.  
 (x) An  $n$ th root  $\alpha$  of unity is a primitive  $n$ th root if and only if its powers  $\alpha^k$ ,  $k = 0, 1, \dots, n-1$ , are distinct, that is, if they exhaust all the  $n$ th roots of unity.  
 (xi) If  $\alpha$  is a primitive  $n$ th root of unity, then the number  $\alpha^k$  is a primitive  $n$ th root if and only if  $k$  is relatively prime to  $n$ .  
 (xii) The number of primitive  $n$ th roots of unity is equal to the number of positive integers  $k$  less than  $n$  and relatively prime to  $n$ .

- (xiii) The  $n$ th roots of unity can also be written as :

$$1, e^{i\frac{2\pi}{n}}, e^{i\frac{4\pi}{n}}, \dots, e^{i\frac{2(n-1)\pi}{n}};$$

$$\text{or, } 1, \alpha, \alpha^2, \dots, \alpha^{n-1}$$

where  $\alpha = e^{i\frac{2\pi}{n}}$  since they are in G.P.

with common ratio  $e^{i\frac{2\pi}{n}}$ .

or,  $1, \alpha, \alpha^2, \dots, \alpha^{n-1}$  since the imaginary roots appear in conjugate pair.

- (xiv)  $z^n - 1 = (z - 1)(z^{n-1} + z^{n-2} + \dots + 1)$

$$\text{or, } z^n - 1 = (z - 1)(z - \alpha)(z - \alpha^2)$$

$$\dots\dots\dots(z - \alpha^{n-1}).$$

Note that the roots of  $z^{n-1} + z^{n-2} + \dots + 1 = 0$  are the  $n$ th roots of unity except 1.

- (xv) Sum of roots:

$$1 + \alpha + \dots + \alpha^{n-2} + \alpha^{n-1} = 0.$$

$$\Rightarrow \sum_{k=0}^{n-1} \cos \frac{2k\pi}{n} = 0, \quad \sum_{k=0}^{n-1} \sin \frac{2k\pi}{n} = 0$$

- (xvi)  $1 + \alpha^p + \alpha^{2p} + \dots + \alpha^{p(n-1)}$   
 $= 0$  if  $p$  is not an integral multiple of  $n$  or,  $n$  if  $p$  is an integral multiple of  $n$ .

- (xvii) The product of the roots  
 $= 1$  or  $-1$  according as  $n$  is odd or even.

- (xviii)  $(1 - \alpha_1)(1 - \alpha_2) \dots (1 - \alpha_{n-1}) = n$   
 $(1 + \alpha_1)(1 + \alpha_2) \dots (1 + \alpha_{n-1}) = 0$   
 if  $n$  is even and 1 if  $n$  is odd.

15.  $x^n - 1 =$

$$(x - 1)(x + 1) \prod_{r=1}^{r=\frac{1}{2}(n-2)} \left( x^2 - 2x \cos \frac{2r\pi}{n} + 1 \right),$$

when  $n$  is even.

$$x^n - 1 =$$

$$(x - 1) \prod_{r=1}^{r=\frac{1}{2}(n-1)} \left( x^2 - 2x \cos \frac{2r\pi}{n} + 1 \right),$$

when  $n$  is odd,

$$x^n + 1 = \prod_{r=0}^{r=\frac{1}{2}(n-2)} \left( x^2 - 2x \cos \frac{(2r+1)\pi}{n} + 1 \right),$$

when  $n$  is even,

$$x^n + 1 =$$

$$(x + 1) \prod_{r=0}^{r=\frac{1}{2}(n-3)} \left( x^2 - 2x \cos \frac{(2r+1)\pi}{n} + 1 \right),$$

when  $n$  is odd.

$$16. \cos n\theta = \cos^n \theta - {}^nC_2 \cos^{n-2} \theta \sin^2 \theta + {}^nC_4 \cos^{n-4} \theta \sin^4 \theta - \dots$$

$$\text{and } \sin n\theta = {}^nC_1 \cos^{n-1} \theta \sin \theta -$$

$${}^nC_3 \cos^{n-3} \theta \sin^3 \theta + {}^nC_5 \cos^{n-5} \theta \sin^5 \theta - \dots$$

$$17. 2^{n-1} \cos^n \theta = \cos n\theta + {}^nC_1 \cos(n-2)\theta$$

$$+ {}^nC_2 \cos(n-4)\theta + \dots$$

$$(-1)^{n/2} 2^{n-1} \sin^n \theta = \cos n\theta - {}^nC_1 \cos(n-2)\theta$$

$$+ {}^nC_2 \cos(n-4)\theta - \dots + (-1)^{n/2} \cdot \frac{1}{2} \cdot {}^nC_{n/2}$$

$$\text{when } n \text{ is even, } (-1)^{(n-1)/2} 2^{n-1} \sin^n \theta$$

$$= \sin n\theta - {}^nC_1 \sin(n-2)\theta$$

$$+ {}^nC_2 \sin(n-4)\theta - \dots + (-1)^{(n-1)/2} {}^nC_{(n-1)/2} \sin \theta$$

when  $n$  is odd.

18. (i)  $\cos \theta + \cos 2\theta + \cos 3\theta + \dots + \cos n\theta$

$$= \frac{\sin \frac{n\theta}{2} \cos \left( \frac{n+1}{2} \theta \right)}{\sin(\theta/2)}.$$

$$(ii) \sin \theta + \sin 2\theta + \sin 3\theta + \dots + \sin n\theta$$

$$= \frac{\sin \frac{n\theta}{2} \sin \left( \frac{n+1}{2} \theta \right)}{\sin(\theta/2)}.$$

19. (i) If  $z_1$  &  $z_2$  are two complex numbers then the complex number  $z = \frac{n z_1 + m z_2}{m+n}$  divides the

joins of  $z_1$  &  $z_2$  in the ratio  $m : n$ .

- (ii) If  $a, b, c$  are three real numbers such that  $az_1 + bz_2 + cz_3 = 0$  ;

where  $a + b + c = 0$  and  $a, b, c$  are not all simultaneously zero, then the complex numbers  $z_1, z_2$  and  $z_3$  are collinear.

- (iii) If the vertices A, B, C of a  $\Delta$  represent the complex nos.  $z_1, z_2, z_3$  respectively, then :

$$(a) \text{ Centroid of the } \Delta ABC \text{ is } \frac{z_1 + z_2 + z_3}{3} :$$

(b) Orthocentre of the  $\Delta ABC$  is

$$\frac{(a \sec A)z_1 + (b \sec B)z_2 + (c \sec C)z_3}{a \sec A + b \sec B + c \sec C}$$

or 
$$\frac{z_1 \tan A + z_2 \tan B + z_3 \tan C}{\tan A + \tan B + \tan C}$$

(c) Incentre of the  $\Delta ABC$  is

$$(az_1 + bz_2 + cz_3) \div (a + b + c).$$

(d) Circumcentre of the  $\Delta ABC$  is

$$\frac{(Z_1 \sin 2A + Z_2 \sin 2B + Z_3 \sin 2C)}{\sin 2A + \sin 2B + \sin 2C}.$$

20. (i)  $\arg(z) = \theta$  is a ray emanating from the origin inclined at an angle  $\theta$  to the x-axis.

(ii)  $|z - a| = |z - b|$  is the perpendicular bisector of the line joining  $a$  to  $b$ .

(iii) The equation of a line joining  $z_1$  and  $z_2$  is given by  $z = z_1 + t(z_2 - z_1)$  where  $t$  is a parameter.

(iv)  $z = z_1(1 + it)$  where  $t$  is a real parameter is a line through the point  $z_1$  and perpendicular to  $Oz_1$ .

(v) The equation of a line passing through  $z_1$  &  $z_2$  can be expressed in the determinant form as

$$\begin{vmatrix} z & \bar{z} & 1 \\ z_1 & \bar{z}_1 & 1 \\ z_2 & \bar{z}_2 & 1 \end{vmatrix} = 0.$$

This is also the condition for three complex numbers to be collinear.

(vi) Complex equation of a straight line through two given points  $z_1$  &  $z_2$  can be written as

$$z(\bar{z}_1 - \bar{z}_2) - \bar{z}(z_1 - z_2) + (z_1 \bar{z}_2 - \bar{z}_1 z_2) = 0,$$

which on manipulating takes the form as

$$\bar{\alpha}z + \alpha\bar{z} + r = 0$$

where  $r$  is real and  $\alpha$  is a non zero complex constant.

21. (i) The equation of circle having centre  $z_0$  & radius  $\rho$  is:

$$|z - z_0| = \rho \text{ or } z\bar{z} - z_0\bar{z} - \bar{z}_0z + \bar{z}_0z_0 - \rho^2 = 0 \text{ which is of the form}$$

$$z\bar{z} + \bar{\alpha}z + \alpha\bar{z} + r = 0,$$

where  $r$  is real, with centre  $-\alpha$  and radius

$$\sqrt{\alpha\bar{\alpha} - r}.$$

Circle will be real if  $\alpha\bar{\alpha} - r \geq 0$ .

(ii) The equation of the circle described on the line segment joining  $z_1$  &  $z_2$  as diameter is:

$$(iii) \arg \frac{z - z_2}{z - z_1} = \pm \frac{\pi}{2}$$

$$\text{or } (z - z_1)(\bar{z} - \bar{z}_2) + (z - z_2)(\bar{z} - \bar{z}_1) = 0$$

(iv) Condition for four given points  $z_1, z_2, z_3$  &  $z_4$  to be concyclic is, the number

$$\frac{z_3 - z_1}{z_3 - z_2} \cdot \frac{z_4 - z_2}{z_4 - z_1} \text{ is real.}$$

Hence the equation of a circle through 3 non collinear points

$$z_1, z_2 \text{ \& } z_3 \text{ can be taken as } \frac{(z - z_2)(z_3 - z_1)}{(z - z_1)(z_3 - z_2)}$$

is real

$$\Rightarrow \frac{(z - z_2)(z_3 - z_1)}{(z - z_1)(z_3 - z_2)} = \frac{(\bar{z} - \bar{z}_2)(\bar{z}_3 - \bar{z}_1)}{(\bar{z} - \bar{z}_1)(\bar{z}_3 - \bar{z}_2)}$$

## 22. Reflection points for a straight line

Two given points  $P$  &  $Q$  are the reflection points for a given straight line if the given line is the right bisector of the segment  $PQ$ . Note that the two points denoted by the complex numbers  $z_1$  &  $z_2$  will be the reflection points for the straight line  $\bar{\alpha}z + \alpha\bar{z} + r = 0$  if and only if;  $\bar{\alpha}z_1 + \alpha\bar{z}_2 + r = 0$ , where  $r$  is real and  $\alpha$  is non zero complex constant.

## 23. Inverse points w.r.t. a circle

Two points  $P$  &  $Q$  are said to be inverse w.r.t. a circle with centre 'O' and radius  $\rho$ , if:

(i) the point  $O, P, Q$  are collinear and on the same side of  $O$ .

(ii)  $OP \cdot OQ = \rho^2$ .

Note that the two points  $z_1$  &  $z_2$  will be the inverse points w.r.t. the circle  $z\bar{z} + \bar{\alpha}z + \alpha\bar{z} + r = 0$  if and only if

$$z_1\bar{z}_2 + \bar{\alpha}z_1 + \alpha\bar{z}_2 + r = 0.$$

**24. Ptolemy's Theorem**

It states that the product of the lengths of the diagonals of a convex quadrilateral inscribed in a circle is equal to the sum of the lengths of the two pairs of its opposite sides. i.e.

$$|z_1 - z_3| |z_2 - z_4| = |z_1 - z_2| |z_3 - z_4| + |z_1 - z_4| |z_2 - z_3|.$$

**[Objective Exercises]****Single Correct Answer Type**

- If  $\prod_{r=1}^n (a_r + ib_r) = A + iB$  where  $\Pi$  indicates continued product from  $r = 1$  to  $r = n$ , then  $\sum \tan^{-1} \left( \frac{b_r}{a_r} \right)$  is equal to
  - $\tan^{-1} B/A$
  - $\tan^{-1} A/B$
  - zero
  - None
- If there is atleast one complex number  $z$  which satisfies both the equality  $|z - m| = m + 5$  & the inequality  $|z - 4| < 3$ , then the real values of the parameter  $m$  are given by
  - $-1 < m < 1$
  - $-2 < m < 2$
  - $-3 < m < 3$
  - None
- The number of 15th roots of unity which are also the 25th root of unity is
  - 3
  - 5
  - 10
  - None
- The complex number  $z$  satisfies the condition  $\left| z - \frac{25}{z} \right| = 24$ . The maximum distance from the origin of co-ordinates to the point  $z$  is
  - 25
  - 30
  - 32
  - None
- The points representing the complex number  $z$  for which  $|z + 5|^2 - |z - 5|^2 = 10$  lie on
  - a straight line
  - a circle
  - a parabola
  - the bisector of the line joining  $(5, 0)$  &  $(-5, 0)$
- If  $\cos \theta + i \sin \theta$  is a root of the equation  $x^n + a_1 x^{n-1} + a_2 x^{n-2} + \dots + a_{n-1} x + a_n = 0$  then the value of  $\sum_{r=1}^n a_r \cos r\theta =$ 
  - 0
  - 1
  - 1
  - None
- Let  $f(x)$  be a degree 2008 polynomial with real coefficients, having complex roots  $c_1, c_2, \dots, c_{2008}$ , such that the set  $\{|c_1|, |c_2|, \dots, |c_{2008}|\}$  consists of exactly 1008 distinct values. The minimum number of real roots of  $f(x)$  is
  - 1008
  - 1000
  - 4
  - 8
- If the area of the triangle on the complex plane formed by the points  $z, z + iz$  &  $iz$  is 50 then  $|z|$  is
  - 1
  - 5
  - 10
  - 15
- Let  $z$  be a root of  $x^5 - 1 = 0$  with  $z \neq 1$ . Then the value of  $z^{15} + z^{16} + z^{17} + \dots + z^{50} =$ 
  - 1
  - 1
  - 0
  - 5
- If  $x, y, z$  &  $\ell, m, n$  are real numbers satisfying  $x^3 + y^3 + z^3 = 3 (xyz - \ell^2 - m^2 - n^2)$  and  $x + y + z > 0$ , then the incorrect statement is
  - $\ell = m = n = 0$  &  $x = y = z$
  - $\ell = m = n = 0$  &  $x^2 + y^2 + z^2 - xy - yz - zx = 0$
  - $x^2 + y^2 + z^2 - xy - yz - zx = 0$
  - $x + y + z = 0$  &  $\ell = m = n = 0$
- If  $z = x + iy$ , then the equation  $\left| \frac{2z - i}{2z + i} \right| = k$  does not represent a circle when
  - $k = 1/2$
  - $k = 1$
  - $k = 2$
  - $k = 3$

12. The continued product of all values of  $(\cos \alpha + i \sin \alpha)^{3/5}$  is  
 (A) 1 (B)  $\cos \alpha + i \sin \alpha$   
 (C)  $\cos 3\alpha + i \sin 3\alpha$  (D)  $\cos 5\alpha + i \sin 5\alpha$
13. Set of all real  $x$  satisfying the inequality  $|4i - 1 - \log_2 x| \geq 5$  is  
 (A)  $\left(0, \frac{1}{16}\right] \cup [4, \infty)$  (B)  $\left(0, \frac{1}{16}\right]$   
 (C)  $[4, \infty)$  (D) None
14. If  $w$  is an imaginary cube root of unity, then the value of  $\sin \{(w^{10} + w^{23})\pi - \pi/4\}$  is  
 (A)  $-\sqrt{3}/2$  (B)  $-1/\sqrt{2}$   
 (C)  $1/\sqrt{2}$  (D)  $\sqrt{3}/2$
15. The centre of a square is at the point with complex number  $z_0 = 1 + i$  and one of its vertices is at the point  $z_1 = 1 - i$ . The complex number which does not correspond to the other vertices is  
 (A)  $-1 + i$  (B)  $1 + 3i$   
 (C)  $3 + i$  (D) None
16. Number of roots of the equation,  $\left(\frac{x+i}{x-i}\right)^n = 1$ , where  $n \in \mathbb{N}$  &  $x \in \mathbb{R}$  is  
 (A)  $n$  (B) 1  
 (C)  $n - 1$  (D) None
17. If  $\omega$  is an imaginary fifth root of 2 &  $x = \omega + \omega^2$ , then the value of  $x^5 - 10x^2 - 10x$   
 (A) 4 (B) 6  
 (C) 20 (D) None
18. Suppose  $A$  is a complex number &  $n \in \mathbb{N}$ , such that  $A^n = (A + 1)^n = 1$ , then the least value of  $n$  is  
 (A) 3 (B) 6  
 (C) 9 (D) 12
19. If one root of the quadratic equation  $(1 + i)x^2 - (7 + 3i)x + (6 + 8i) = 0$  is  $4 - 3i$ , then the other root must be -  
 (A)  $1 + i$  (B)  $4 + 3i$   
 (C)  $1 - i$  (D) None
20. If  $z = 1/2 (\sqrt{3}i - 1)$  then the value of,  $(z - z^2 + 2z^3)(2 - z + z^3)$  is  
 (A) 3 (B)  $-1$   
 (C) 5 (D) 7
21. If  $(\sqrt{3} + i)^{100} = 2^{99}(a + ib)$ , then  $a^2 + b^2$  is equal to  
 (A) 1 (B) 2  
 (C) 3 (D) 4
22. The locus represented by the equation,  $|z - 1| + |z + 1| = 2$  is  
 (A) an ellipse with foci  $(1, 0); (-1, 0)$   
 (B) one of the family of circles passing through the points of intersection of the circles  $|z - 1| = 1$  and  $|z + 1| = 1$   
 (C) the radical axis of the circles  $|z - 1| = 1$  and  $|z + 1| = 1$   
 (D) the portion of the real axis between the points  $(1, 0)$  &  $(-1, 0)$  including both points
23. The expression  $\left[\frac{1 + i \tan \alpha}{1 - i \tan \alpha}\right]^n - \frac{1 + i \tan n\alpha}{1 - i \tan n\alpha}$  when simplified reduces to -  
 (A) zero (B)  $2 \sin n\alpha$   
 (C)  $2 \cos n\alpha$  (D) None
24.  $\left[\frac{-1 + i\sqrt{3}}{2}\right]^6 + \left[\frac{-1 - i\sqrt{3}}{2}\right]^6 + \left[\frac{-1 + i\sqrt{3}}{2}\right]^5 + \left[\frac{-1 - i\sqrt{3}}{2}\right]^5$  is equal to -  
 (A) 1 (B)  $-1$   
 (C) 2 (D) None
25. Given that the equation  $z^2 + (p + iq)z + r + is = 0$  where  $p, q, r, s$  are non-zero has a real root. Then -  
 (A)  $pqr = r^2 + p^2s$  (B)  $prs = q^2 + r^2p$   
 (C)  $qrs = p^2 + s^2q$  (D)  $pqs = s^2 + q^2r$
26. The points of intersection of the two curves  $|z - 3| = 2$  and  $|z| = 2$  in an Argand plane are -  
 (A)  $\frac{1}{2}(7 \pm i\sqrt{3})$  (B)  $\frac{1}{2}(3 \pm i\sqrt{3})$   
 (C)  $\frac{3}{2} \pm i\sqrt{\frac{7}{2}}$  (D)  $\frac{7}{2} \pm i\sqrt{\frac{3}{2}}$

27. The complex number  $z = x + iy$  which satisfy the equation  $\left| \frac{z-5i}{z+5i} \right| = 1$  lie on -  
 (A) the x-axis  
 (B) the straight line  $y = 5$   
 (C) a circle passing through the origin  
 (D) the y-axis
28. Let  $S$  denote the set of complex number  $z$  such that  $\log_{1/3} (\log_{1/2} |z|^2 + 4|z| + 3) < 0$ , then  $s$  is contained in  
 (A)  $(0, 1)$  (B)  $\{z \mid \operatorname{Re}(z) > 0\}$   
 (C)  $|z| \operatorname{Re} > 3$  (D) None of these
29. The locus of the points representing the complex number  $z$  for which  $|z| - 2 = |z - i| - |z + 5i| = 0$  is  
 (A) a circle with centre at the origin  
 (B) a straight line passing thro' the origin  
 (C) the single point  $(0, -2)$   
 (D) none of these
30. Let  $\lambda \in \mathbb{R}$ . If the origin and the non real roots of  $2z^2 + 2z + \lambda = 0$  form the three vertices of an equilateral triangle in the Argand plane, then  $\lambda$  is  
 (A) 1 (B)  $2/3$   
 (C) 2 (D)  $-1$
31. If  $a, b, c, p, q, r$  are three complex numbers such that  $\frac{p}{a} + \frac{q}{b} + \frac{r}{c} = 1 + i$  and  $\frac{a}{p} + \frac{b}{q} + \frac{c}{r} = 0$ , then value of  $\frac{p^2}{a^2} + \frac{q^2}{b^2} + \frac{r^2}{c^2}$  is  
 (A) 0 (B)  $-1$   
 (C)  $2i$  (D)  $-2i$
32. If  $z_1 = -3 + 5i$ ;  $z_2 = -5 - 3i$  and  $z$  is a complex number lying on the line segment joining  $z_1$  &  $z_2$  then  $\arg$  can be  
 (A)  $-\frac{3\pi}{4}$  (B)  $-\frac{\pi}{4}$   
 (C)  $\frac{\pi}{6}$  (D)  $\frac{5\pi}{6}$
33. Let  $\alpha$  &  $\beta$  be the roots of the equation  $x^2 + x + 1 = 0$ . The equation whose roots are  $\alpha^{19}, \beta^7$  is  
 (A)  $x^2 - x - 1 = 0$  (B)  $x^2 - x + 1 = 0$   
 (C)  $x^2 + x - 1 = 0$  (D)  $x^2 + x + 1 = 0$
34. If  $x^2 - 2x \cos \theta + 1 = 0$  the value of  $x^{2n} - 2x^n \cos n\theta + 1$  is equal to  
 (A)  $\cos 2n\theta$   
 (B)  $\sin 2n\theta$   
 (C) 0  
 (D) some real number greater than 0
35. Points  $z_1$  &  $z_2$  are adjacent vertices of a regular octagon. The vertex  $z_3$  adjacent to  $z_3$  adjacent to  $z_2$  ( $z_3 \neq z_1$ ) is represented by  
 (A)  $z_2 + \frac{1}{\sqrt{2}} (1 \pm i) (z_1 + z_2)$   
 (B)  $z_2 + \frac{1}{\sqrt{2}} (1 \pm i) (z_1 + z_2)$   
 (C)  $z_2 + \frac{1}{\sqrt{2}} (1 \pm i) (z_2 + z_1)$   
 (D) None of these
36. The curve represented by  $\operatorname{Re} \left( \frac{1}{z} \right) = c$  (where  $z \neq 0$ ), is  
 (A) circle (B) st. line  
 (C) parabola (D) square
37. If  $1, \alpha_1, \alpha_2, \dots, \alpha_{n-1}$  are the  $n^{\text{th}}$  roots of unity, then  $(1 + \alpha_1)(1 + \alpha_2) \dots (1 + \alpha_{n-1})$  is equal to  
 (A)  $n - 1$  (B)  $n$   
 (C) 0 (D) None of these
38. Let  $|z_1| = |z_2| = \frac{c}{2}$  then  $|z_1 + z_2|^2 + |z_1 - z_2|^2 =$   
 (A)  $c^2$  (B)  $c^2/2$   
 (C)  $2c^2$  (D) None of these
39. If  $\omega$  is one of the imaginary cube root of unity then the value of the expression,  $(1 + 2\omega + 2\omega^2)^{10} + (2 + \omega + 2\omega^2)^{10} + (2 + 2\omega + 2\omega^2)^{10}$  is  
 (A) 0 (B) 1  
 (C)  $\omega$  (D)  $\omega^2$



40. Let  $\omega, \omega^2$  be complex cube roots of unity.

Then the determinant 
$$\begin{vmatrix} x+1 & \omega & \omega^2 \\ \omega & x+\omega^2 & 1 \\ \omega^2 & 1 & x+\omega \end{vmatrix}$$

reduces to

- (A)  $x^3$  (B)  $2x^3$   
(C)  $3x^3$  (D) None of these
41. If  $(1+x)^n = c_0 + c_1x + c_2x^2 + \dots + c_nx^n$ ,  $n \in \mathbb{N}$  then the value of  $c_0 + c_4 + c_8 + c_{12} + \dots =$

- (A)  $\frac{1}{2} \left( 2^{n-1} + 2^{n/2} \cos \frac{n\pi}{4} \right)$   
(B)  $\frac{1}{2} \left( 2^{n-1} + 2^{n/2} \sin \frac{n\pi}{4} \right)$   
(C)  $\frac{1}{2} \left( 2^{n-1} - 2^{n/2} \cos \frac{n\pi}{4} \right)$   
(D)  $\frac{1}{2} \left( 2^{n-1} - 2^{n/2} \sin \frac{n\pi}{4} \right)$

42. If  $x^2 + x + 1 = 0$  then the numerical value of

$$\left(x + \frac{1}{x}\right)^2 + \left(x^2 + \frac{1}{x^2}\right)^2 + \left(x^3 + \frac{1}{x^3}\right)^2 + \left(x^4 + \frac{1}{x^4}\right)^2 + \dots + \left(x^{27} + \frac{1}{x^{27}}\right)^2 =$$

(A) 54 (B) 36 (C) 27 (D) 18

43. If  $z_1$  &  $z_2$  are two complex numbers & if

$$\arg \frac{z_1 + z_2}{z_1 - z_2} = \frac{\pi}{2} \text{ but } |z_1 + z_2| \neq |z_1 - z_2| \text{ then the}$$

figure formed by the points represented by 0,  $z_1$ ,  $z_2$  &  $z_1 + z_2$  is

- (A) a parallelogram but not a rectangle or a rhombus  
(B) a rectangle but not a square  
(C) a rhombus but not a square  
(D) a square
44. The points  $z_1, z_2, z_3$  on the complex plane are the vertices of an equilateral triangle if and only if
- (A)  $\Sigma(z_1 - z_2)(z_2 - z_3) = 0$   
(B)  $z_1^2 + z_2^2 + z_3^2 = 2(z_1z_2 + z_2z_3 + z_3z_1)$

(C)  $z_1^2 + z_2^2 + z_3^2 = z_1z_2 + z_2z_3 + z_3z_1$

(D)  $2(z_1^2 + z_2^2 + z_3^2) = z_1z_2 + z_2z_3 + z_3z_1$

45. The centre of a regular polygon of  $n$  sides is located at the point  $z = 0$ , and one of its vertex  $z_1$  known. If  $z_2$  be the vertex adjacent to  $z_1$ , then  $z_2$  is equal to

- (A)  $z_1 \left( \cos \frac{2\pi}{n} \pm i \sin \frac{2\pi}{n} \right)$   
(B)  $z_1 \left( \cos \frac{\pi}{n} \pm i \sin \frac{\pi}{n} \right)$   
(C)  $z_1 \left( \cos \frac{\pi}{2n} \pm i \sin \frac{\pi}{2n} \right)$   
(D) None of these

### Multiple Correct Answer Type for JEE Advanced

46. Let  $P(x)$  and  $Q(x)$  be two polynomials. Suppose that  $f(x) = P(x^3) + xQ(x^3)$  is divisible by  $x^2 + x + 1$ , then

- (A)  $P(x)$  is divisible by  $(x-1)$  but  $Q(x)$  is not divisible by  $x-1$   
(B)  $Q(x)$  is divisible by  $(x-1)$  but  $P(x)$  is not divisible by  $x-1$   
(C) Both  $P(x)$  and  $Q(x)$  are divisible by  $x-1$   
(D)  $f(x)$  is divisible by  $x-1$ .

47. If  $z^2 / (z-1)$  is always real, then  $z$ , can lie on

- (A) real axis (B) a circle  
(C) imaginary axis (D) none of these

48. Let  $z_1, z_2$ , be two complex numbers represented by points on the circle  $|z| = 1$  and  $|z| = 2$  respectively then -

(A)  $\max |2z_1 + z_2| = 4$  (B)  $\min |z_1 - z_2| = 1$

(C)  $\left| z_2 + \frac{1}{z_1} \right| \leq 3$  (D) none of these

49. If  $z^3 - iz^2 - 2iz - 2 = 0$  then  $z$  can be equal to

- (A)  $1-i$  (B)  $i$   
(C)  $1+i$  (D)  $-1-i$

50. If  $\alpha$  is a complex constant such that  $\alpha z^2 + z + \bar{\alpha} = 0$  has a real root then -  
 (A)  $\alpha + \bar{\alpha} = 1$   
 (B)  $\alpha + \bar{\alpha} = 0$   
 (C)  $\alpha + \bar{\alpha} = -1$   
 (D) The absolute value of the real root is 1
51. A complex number  $z$  satisfies the equation  $|z|^2 - 2iz + 2c(1+i) = 0$ , where  $c$  is real. The value of  $c$  for which the above equation has no solution can be given by  
 (A)  $c \in (-\infty, -1 - \sqrt{2})$   
 (B)  $c \in [-1, \sqrt{2}, -1 + \sqrt{2}]$   
 (C)  $c \in (-1 + \sqrt{2}, \infty)$   
 (D)  $c \in \mathbb{R}$
52.  $z_1, z_2, z_3, z_4$  correspond to points A, B, C, D on a circle  $|z| = 1$ . If  $z_1 + z_2 + z_3 + z_4 = 0$ . Then ABCD is necessarily  
 (A) a rectangle (B) a square  
 (C) a rhombus (D) a parallelogram
53. If the lines  $a\bar{z} + \bar{a}z + b = 0$  and  $c\bar{z} + \bar{c}z + d = 0$  are mutually perpendicular, where  $a$  and  $c$  are non-zero complex numbers and  $b$  and  $d$  are real numbers, then  
 (A)  $a\bar{a} + c\bar{c} = 0$   
 (B)  $a\bar{c}$  is purely imaginary  
 (C)  $\arg\left(\frac{a}{c}\right) = \pm \frac{\pi}{2}$   
 (D)  $\frac{a}{\bar{a}} = \frac{c}{\bar{c}}$
54. Let  $z, iz$  and  $z + iz$  represent three vertices of a triangle then,  
 (A) area of  $\Delta$  is given by,  $\text{Area} = \frac{1}{2} |z|^2$   
 (B) area of  $\Delta$  is given by,  $\text{Area} = |z|^2$   
 (C) orthocentre of  $\Delta$  is given by  $z + iz$ .  
 (D) None of these
55. Let  $a\bar{z} + z\bar{a} + b = 0$  represent a straight line in argand plane, given that  $a \in \mathbb{Z}$  and  $b \in \mathbb{R}$ . Let  $z_1$  be a point on the argand plane and let  $z_2$  be the image of  $z_1$  in the line  $a\bar{z} + z\bar{a} + b = 0$ , then  
 (A)  $(a\bar{z}_1 + z_1\bar{a} + b) + (a\bar{z}_2 + z_2\bar{a} + b) = 0$   
 (B)  $\frac{a}{a} + \frac{(z_1 - z_2)}{(\bar{z}_1 - \bar{z}_2)} = 0$   
 (C)  $\frac{a}{a} + \frac{(z_1 - z_2)}{(\bar{z}_1 - \bar{z}_2)} = 1$   
 (D) None of these
56. Let  $z^{16} - 1 = 0$  have 16 roots, then  
 (A) if  $\alpha$  is one of the root of equation then  $\bar{\alpha}$  is also the root of same equation.  
 (B) let  $\alpha_1, \alpha_2, \alpha_3, \dots, \alpha_{16}$  be the roots of given equation, then  $\alpha_1\alpha_2 + \alpha_2\alpha_3 + \dots + \alpha_{15}\alpha_{16} = 0$   
 (C) area bounded by the polygon represented by the roots of given equation is  $8 \sin \frac{\pi}{8}$   
 (D)  $\frac{1}{\alpha_1} + \frac{1}{\alpha_2} + \dots + \frac{1}{\alpha_{16}} = 0$
57. If  $\alpha$  is the fifth root of unity then  
 (A)  $|1 + \alpha + \alpha^2 + \alpha^3 + \alpha^4| = 0$   
 (B)  $|1 + \alpha + \alpha^2 + \alpha^3| = 1$   
 (C)  $|1 + \alpha + \alpha^2| = 2 \cos \frac{\pi}{5}$   
 (D)  $|1 + \alpha| = 2 \cos \frac{\pi}{10}$
58. If the complex number  $z$  satisfies  $|iz + 2| = \text{Im}(z)$ , then  
 (A) minimum value of  $|z|$  is 1  
 (B) maximum value of  $|z|$  is  $\sqrt{5}$   
 (C) minimum value of  $\arg z$  is  $\frac{\pi}{4}$   
 (D) maximum value of  $\arg z$  is  $\frac{3\pi}{4}$

59. If  $z = x + iy$ , then the equation  $\left| \frac{2z-i}{z+1} \right| = p$  represents a circle when  $p =$   
 (A)  $1/2$  (B) 1  
 (C) 2 (D) 3
60. If  $z$  is the affix of a moving point in argand plane, then the equation  $z^2 + \bar{z}^2 - 2|z|^2 + z + \bar{z} = 0$  represents a  
 (A) straight line (B) conic  
 (C) hyperbola (D) parabola
61. Equation of the line in argand plane joining two points with affixes  $z_1$  and  $z_2$  must be  
 (A)  $z = tz_1 + (1-t)z_2, t \in \mathbb{R}$   
 (B)  $\arg \left( \frac{z - z_1}{z_2 - z_1} \right)$  is purely real  
 (C)  $\begin{vmatrix} z & \bar{z} & 1 \\ z_1 & \bar{z}_1 & 1 \\ z_2 & \bar{z}_2 & 1 \end{vmatrix} = 0$   
 (D)  $(z - z_1)(\bar{z} - \bar{z}_2) = (z - z_2)(\bar{z} - \bar{z}_1)$ .
62. If  $x, y, a, b$  are real numbers such that  $(x + iy)^{1/5} = a + ib$ , and  $p = x/a - y/b$ , then  
 (A)  $(a - b)$  is a factor of  $p$   
 (B)  $(a + b)$  is a factor of  $p$   
 (C)  $a + ib$  is a factor of  $p$   
 (D)  $a - ib$  is a factor of  $p$
63. If  $z_1, z_2, z_3$  are the affixes of vertices of an equilateral triangle and  $z_0$  is the affix of the circumcentre, then  
 (A)  $z_0 = z_1 + z_2 + z_3$   
 (B)  $|z_0 - z_1| = |z_0 - z_2| = |z_0 - z_3|$   
 (C)  $z_0^2 = z_1 z_2 + z_2 z_3 + z_3 z_1$   
 (D)  $z_0 = \frac{z_1 + z_2 + z_3}{3}$
64. Let  $S$  represents a circle on Argand Plane with  $z_1$  and  $z_2$  as the end points of diameter of  $S$ , then  
 (A) equation of  $S$  is  $(z - z_1)(\bar{z} - \bar{z}_2) + (z - z_2)(\bar{z} - \bar{z}_1) = 0$   
 (B) Let  $z$  be any complex number lying on circle  $S$  then  $\frac{(z_2 - z)}{|z_2 - z|} = \frac{(z - z_1)}{|z - z_1|} i$   
 (C) Center of  $S$  is represented by,  $(z_1 + z_2)$   
 (D) Equation of  $S$  is  $\left| z - \left( \frac{z_1 + z_2}{2} \right) \right| = \frac{1}{2} |z_1 - z_2|$
65. If points  $A$  and  $B$  are represented by the non-zero complex number  $z_1$  and  $z_2$  on the Argand plane such that  $|z_1 + z_2| = |z_1 - z_2|$  and  $O$  is the origin, then  
 (A) orthocentre of  $\triangle OAB$  lies at  $O$   
 (B) circumcentre of  $\triangle OAB$  is  $\frac{z_1 + z_2}{2}$   
 (C)  $\arg \left( \frac{z_1}{z_2} \right) = \pm \frac{\pi}{2}$   
 (D)  $\triangle OAB$  is isosceles
66. If  $z_1, z_2$ , and  $z_3$  are the vertices of a triangle  $ABC$  such that  $|z_1| = |z_2| = |z_3|$  and  $AB = AC$ , then  
 (A) orthocentre of  $\triangle ABC$  is  $z_1 + z_2 + z_3$   
 (B)  $\frac{(z_1 + z_3)(z_1 + z_2)}{(z_2 + z_3)^2}$  is purely real  
 (C)  $\frac{z_2 z_3}{z_1^2}$  is purely imaginary  
 (D) None of these
67.  $A, B, C$  are the points representing the complex numbers  $z_1, z_2, z_3$  respectively on the complex plane and the circumcentre of the triangle  $ABC$  lies at the origin. If the altitude  $AD$  of the triangle  $ABC$  meets the circumcircle again at  $P$ , then  $P$  represents the complex number  
 (A)  $-\bar{z}_1 z_2 z_3$  (B)  $-\frac{\bar{z}_1 z_2}{\bar{z}_3}$

$$(C) -\frac{\bar{z}_1 z_3}{z_2} \quad (D) -\frac{z_2 z_3}{z_1}$$

68. Let  $A(z_1)$ ,  $B(z_2)$  &  $C(z_3)$  represent three complex numbers on argand plane, then

$$(A) z = z_1 + t \left\{ \frac{(z_1 - z_2)}{|z_1 - z_2|} + \frac{z_1 - z_3}{|z_1 - z_3|} \right\}, t \in \mathbb{R}$$

is a real parameter, represents equation of one of the angle bisector of angle BAC

- (B) Centroid of  $\triangle ABC$  is given by,  $z_0$

$$= \frac{1}{3} (z_1 + z_2 + z_3)$$

- (C) Equation of perpendicular bisectors of

$$AB \text{ is } z(\bar{z}_1 - \bar{z}_2) + \bar{z}(z_1 - z_2) + |z_1|^2 - |z_2|^2 = 0$$

$$(D) \text{ If } \begin{vmatrix} z_1 & \bar{z}_1 & 1 \\ z_2 & \bar{z}_2 & 1 \\ z_3 & \bar{z}_3 & 1 \end{vmatrix} = 0 \text{ then A, B \& C will}$$

become collinear.

69. Let  $z$ ,  $\omega z$  and  $z + \omega z$  represent three vertices of  $\triangle ABC$ , where  $\omega$  is cube root unity, then

$$(A) \text{ centroid of } \triangle ABC \text{ is } \frac{2}{3} (z + \omega z)$$

$$(B) \text{ orthocentre of } \triangle ABC \text{ is } \frac{2}{3} (z + \omega z)$$

$$(C) \text{ circumcentre of } \triangle ABC \text{ is } \frac{2}{3} (z + \omega z)$$

$$(D) \text{ incentre of } \triangle ABC \text{ is } \frac{2}{3} (z + \omega z)$$

70. Let  $z_1$  &  $z_2$  be two complex numbers. Such that  $|z_1 + z_2|^2 = |z_1|^2 + |z_2|^2$ , then

$$(A) z_1 \cdot \bar{z}_2 \text{ is purely imaginary}$$

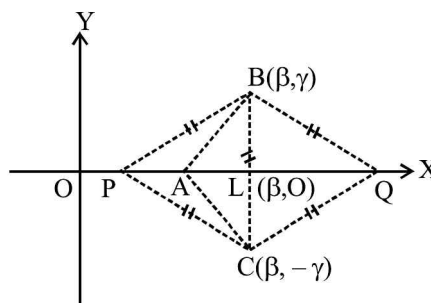
$$(B) z_1 \cdot \bar{z}_2 + \bar{z}_1 \cdot z_2 = 0$$

$$(C) O, z_1, z_2 \text{ are the vertices of a right triangle}$$

$$(D) z_1 \cdot \bar{z}_2 \text{ is purely imaginary}$$

### Comprehension - 1

A cubic equation  $f(x) = 0$  has one real and two complex roots  $\alpha$ ,  $\beta + i\gamma$  and  $\beta - i\gamma$  respectively on the Argand Plane. Points A, B, C represents roots  $\alpha$ ,  $\beta + i\gamma$  and  $\beta - i\gamma$  respectively. PBC and QBC are two equilateral triangles.



71. The distance PL is :

$$(A) \sqrt{3} |\beta|$$

$$(B) \sqrt{3} |\alpha|$$

$$(C) \sqrt{3} |\gamma|$$

$$(D) \text{ None of these}$$

72. The roots of the derived equation  $f'(x)$  are imaginary if:

$$(A) A \text{ falls inside one of the two equilateral triangles described on BC}$$

$$(B) A \text{ falls outside one of the two equilateral triangles described on BC}$$

$$(C) A \text{ forms an equilateral triangle with BC.}$$

$$(D) \text{ None of these}$$

73. The roots of the derived equation  $f'(x)$  are real and distinct if:

$$(A) A \text{ falls outside one of the two equilateral triangles described on BC.}$$

$$(B) A \text{ falls inside one of the two equilateral triangles described on BC.}$$

$$(C) A \text{ lies anywhere}$$

$$(D) \text{ None of these}$$

### Comprehension - 2

Let  $z_1, z_2, z_3, z_4, z_5, z_6$  be the vertices of a regular hexagon in the anticlockwise sense

74. The quadrilateral formed by the point  $z_1, z_3, z_4$  and  $z_6$  as vertices is

$$(A) \text{ A parallelogram} \quad (B) \text{ A square}$$

$$(C) \text{ A rhombus} \quad (D) \text{ A rectangle}$$

75.  $z_4$  is given by

$$(A) -i\sqrt{3} z_1 - (1 + \sqrt{3}) z_2$$

$$(B) i\sqrt{3} z_1 + (1 + i\sqrt{3}) z_2$$

$$(C) -i\sqrt{3}z_1 + (1+i\sqrt{3})z_2$$

$$(D) i\sqrt{3}z_1 - (1+i\sqrt{3})z_2$$

76.  $z_6$  is given by

$$(A) \frac{\sqrt{3}}{2}(\sqrt{3}+i)z_1 + \frac{1}{2}(1-i\sqrt{3})z_2$$

$$(B) \frac{\sqrt{3}}{2}(\sqrt{3}-i)z_1 - \frac{1}{2}(1-i\sqrt{3})z_2$$

$$(C) \frac{\sqrt{3}}{2}(\sqrt{3}-i)z_1 + \frac{1}{2}(1-i\sqrt{3})z_2$$

$$(D) \frac{\sqrt{3}}{2}(\sqrt{3}+i)z_1 - \frac{1}{2}(1-i\sqrt{3})z_2$$

### Comprehension - 3

In the argand plane  $z_1, z_2$  and  $z_3$  are respectively the vertices of an isosceles triangle ABC with  $AC = BC$  and equal angle  $\theta$ . If  $z_4$  is the incentre of triangle, then

77. The value of  $\left(\frac{AB}{IA}\right)^2 \cdot \left(\frac{AC}{AB}\right)$  is :

$$(A) \frac{(z_2 - z_1)(z_1 - z_3)}{(z_4 - z_1)^2}$$

$$(B) \frac{(z_2 - z_1)(z_3 - z_1)}{(z_4 - z_1)}$$

$$(C) \frac{(z_2 - z_1)(z_3 - z_1)}{(z_4 - z_1)^2}$$

(D) None of these

78. The value of  $(z_4 - z_1)^2 (1 + \cos \theta) \sec \theta$  is :

$$(A) (z_2 - z_1)(z_3 - z_1)$$

$$(B) \frac{(z_2 - z_1)(z_3 - z_1)}{(z_4 - z_1)}$$

$$(C) (z_2 - z_1)^2 (z_3 - z_1)$$

$$(D) (z_2 - z_1)(z_3 - z_1)^2$$

79. The value of  $\frac{(CD)(ID)}{(AD)^2}$  is

$$(A) \frac{(z_1 + z_2 + 2z_3)}{(z_2 - z_1)^2}$$

$$(B) \frac{(z_1 + z_2 - z_3)(z_1 + z_2 - 2z_4)}{(z_2 - z_1)^2}$$

$$(C) \frac{(z_1 + z_2 - z_3)(z_1 + z_2 - 2z_4)}{(z_2 - z_1)}$$

(D) None of these

### Comprehension - 4

Let  $b$  be a complex number lying on a circle

$|z| = \sqrt{2}a$ , then

80. The equation of tangent at point 'b' is :

$$(A) z\bar{b} + \bar{z}b = a^2 \quad (B) z\bar{b} + \bar{z}b = 2a^2$$

$$(C) z\bar{b} + \bar{z}b = 3a^2 \quad (D) z\bar{b} + \bar{z}b = 4a^2$$

81. The length of perpendicular from a given point  $z_0$  on the tangent at 'b' is

$$(A) \frac{|z_0\bar{b} + \bar{z}_0b - a^2|}{2\sqrt{2}a}$$

$$(B) \frac{|z_0\bar{b} + \bar{z}_0b - 2a^2|}{2\sqrt{2}a}$$

$$(C) \frac{|z_0\bar{b} + \bar{z}_0b - 3a^2|}{2\sqrt{2}a}$$

$$(D) \frac{|z_0\bar{b} + \bar{z}_0b - 4a^2|}{2\sqrt{2}a}$$

82. The equation of lines passing through the centre of the circle and making an angle  $\pi/4$  with the normal at 'b' are:

$$(A) z = \pm \frac{ib^2}{2a^2} \bar{z} \quad (B) z = \pm \frac{ib^2}{a^2} \bar{z}$$

$$(C) z = \pm \frac{ib^2}{3a^2} \bar{z} \quad (D) z = \pm \frac{ib^2}{4a^2} \bar{z}$$

**Comprehension - 5**

If  $A(\alpha)$ ,  $B(\beta)$ ,  $C(\gamma)$  are the vertices of a triangle where  $\alpha, \beta, \gamma$  are unimodular complex numbers.

83. The orthocentre of the triangle ABC must be represented by the complex number

$$(A) \sqrt{\alpha^2 + \beta^2 + \gamma^2 - \alpha\beta - \beta\gamma - \gamma\alpha}$$

$$(B) \sqrt{\alpha^2 + \beta^2 + \gamma^2}$$

$$(C) \frac{\alpha^2 + \beta^2 + \gamma^2}{3}$$

$$(D) \alpha + \beta + \gamma$$

84. The area of the triangle ABC must be

$$(A) i \begin{vmatrix} \alpha & \bar{\alpha} & 1 \\ \beta & \bar{\beta} & 1 \\ \gamma & \bar{\gamma} & 1 \end{vmatrix} \quad (B) \frac{i}{2} \begin{vmatrix} \alpha & \bar{\alpha} & 1 \\ \beta & \bar{\beta} & 1 \\ \gamma & \bar{\gamma} & 1 \end{vmatrix}$$

$$(C) \frac{i}{4} \begin{vmatrix} \alpha & \bar{\alpha} & 1 \\ \beta & \bar{\beta} & 1 \\ \gamma & \bar{\gamma} & 1 \end{vmatrix} \quad (D) \frac{i}{7} \begin{vmatrix} \alpha & \bar{\alpha} & 1 \\ \beta & \bar{\beta} & 1 \\ \gamma & \bar{\gamma} & 1 \end{vmatrix}$$

85. If a perpendicular AD from A is dropped on BC and is produced to a point E on the circle  $|z| = 1$ , then E must have affix

$$(A) \frac{-\beta\gamma}{\alpha} \quad (B) \frac{-\alpha\beta}{\gamma}$$

$$(C) \frac{-\alpha\gamma}{\beta} \quad (D) \frac{-\beta\gamma}{\alpha}$$

**Assertion(A) and Reason(R)**

- (A) Both A and R are true and R is the correct explanation of A.  
 (B) Both A and R are true but R is not the correct explanation of A.  
 (C) A is true, R is false.  
 (D) A is false, R is true.

86. **Assertion (A)** :  $\alpha_1, \alpha_2, \alpha_3, \alpha_4$  are roots of  $2x^4 - 2x^3 + 9x^2 - x + 12 = 0$  then value of

$$(3 + \alpha_1^2)(3 + \alpha_2^2)(3 + \alpha_3^2)(3 + \alpha_4^2) \text{ is } 21$$

**Reason (R)** :  $2(x - \alpha_1)(x - \alpha_2)(x - \alpha_3)(x - \alpha_4) = 2x^4 - 2x^3 + 9x^2 - x + 12$

Put  $x = \sqrt{3}i$  on both sides and take modulus of both sides

87. **Assertion (A)** : If A and B are real  $n \times n$  matrices such that  $A^2 + B^2 = AB$  and  $BA - AB$  is invertible matrix then n is divisible by n.

**Reason (R)** : Let  $w = -\frac{1}{2} + \frac{i\sqrt{3}}{2}$  and  $S = A + wB$ .

We have  $S\bar{S} = w(BA - AB)$ . Since  $\det(S\bar{S})$  is real and  $\det(BA - AB) \neq 0$  then  $\det w(BA - AB) = w^n \det(BA - AB)$  is real only when n is divisible by 3.

88. **Assertion (A)** : If two quadratic functions  $p(x) = a_1x^2 + b_1x + c_1$  and  $q(x) = a_2x^2 + b_2x + c_2$  satisfy  $p(w) = p(z)$  and  $q(w) = q(z)$  for same  $w, z \in \mathbb{R}$ ,  $w \neq z$  then  $b_1a_1 = b_2a_1$

**Reason (R)** :  $\frac{p(w) - p(z)}{w - z} = a_1(w + z) + b_1 = 0;$

$$\frac{q(w) - q(z)}{w - z} = a_2(w + z) + b_2 = 0;$$

$$w + z = \frac{-b_1}{a_1} = \frac{-b_2}{a_2} \Rightarrow b_1a_1 = b_2a_1.$$

89. **Assertion (A)** : The value of  $\sum_{n=0}^{\infty} \frac{\cos n\theta}{2^n}$ , where

$$\cos \theta = \frac{1}{5}, \text{ is } \frac{5}{6}.$$

**Reason (R) :** The above sum can be found as the real part of  $\sum_{n=0}^{\infty} \frac{e^{in\theta}}{2^n}$  whose sum is  $\frac{2}{2 - e^{i\theta}}$ .

90. **Assertion (A) :** If  $x$  is a fifth root of unity, then there are three possible values of

$$2x + \frac{1}{1+x} + \frac{x}{1+x^2} + \frac{x^2}{1+x^3} + \frac{x^3}{1+x^4}.$$

**Reason (R) :** Since  $x^5 = 1$ , the above expansion can be reduced to  $2\left(x + \frac{1}{x}\right)$ .

91. **Assertion (A) :** Let  $A_1, A_2, \dots, A_k$  be a regular  $k$ -gon inscribed in a circle of radius 1 and let  $P$  be the affix  $\theta$  a point lying on or inside the circumcircle. Then the maximum value of  $PA_1 \cdot PA_2 \cdot \dots \cdot PA_k$  is 2.

**Reason (R) :** Since  $x^k - 1 = (x-1)(x-w) \dots (x-w^{k-1})$  where  $1, w, \dots, w^{k-1}$  are the  $k^{\text{th}}$  roots of unity,  $(PA_1 \cdot PA_2 \cdot \dots \cdot PA_k)^2 = |p^k + 1|^2$  and therefore the maximum value of required product is 2.

92. **Assertion (A) :** The sum of the squares of the length of all the sides and diagonals of a regular  $k$ -gon inscribed in a circle of radius 1 is  $2k^2$ .

**Reason (R) :** The sum of squares of lengths of sides and diagonals that contain vertex 1 is

$$\sum_{i=0}^{k-1} |1 - w^i|^2 = \sum_{i=0}^{k-1} (2 - w^i - \bar{w}^i) = 2k;$$

where  $1, w, w^2, \dots, w^{k-1}$  are the  $k^{\text{th}}$  roots of unity placed at the vertices of the  $k$ -gon.

93. **Assertion (A) :** Let

$$P(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_0$$

be a polynomial with real coefficients,  $a_n \neq 0$ . If every root of  $P$  is a root of unity, but  $P(1) \neq 0$ , then  $a_n = a_0, a_{n-1} = a_1, \dots$  i.e. coefficients of  $P$  are symmetric.

**Reason (R) :** If  $x$  is root of unity then  $\bar{x} = \frac{1}{x}$ . Since coefficients are real, the imaginary roots appear in conjugate pair. Using transformation of roots and comparison of coefficients we find that the coefficients are symmetric.

94. **Assertion (A) :** The roots of

$$Q(z) = z^{10} + z^9 + z^6 + z^5 + z^4 + z + 1 = 0$$

are either 12<sup>th</sup> or 7<sup>th</sup> roots of unity.

**Reason (R) :**  $(z-1)Q(z) = (z^4 - z^2 + 1)(z^7 - 1)$ .

The first factor gives  $z^2 = e^{\pm i\frac{\pi}{3}} \Rightarrow z = e^{\pm i\frac{\pi}{6}}$  which are 12<sup>th</sup> roots of unity.

95. **Assertion (A) :** The product of real parts of roots

of the equation  $x^2 + 2x = i$  is  $1 - \sqrt{2}$ .

**Reason (R) :** The roots of the equation are  $-1 \pm \sqrt{2}e^{i\frac{\pi}{4}}$ . The required product is

$$\left(-1 + \sqrt{2}\cos\frac{\pi}{4}\right)\left(-1 - \sqrt{2}\cos\frac{\pi}{4}\right).$$

### Match the Columns for JEE Advanced

#### Column-I

96. (A) The roots of cubic equation  $(z + \alpha\beta)^3 = \alpha^3 (\alpha \neq 0)$  represent the vertices of a triangle of area equal to
- (B) If  $\alpha$  is a complex number then the radius of the circle  $\left|\frac{z-\alpha}{z-\bar{\alpha}}\right| = 2$  is equal to

#### Column-II

- (P)  $\tan \alpha$
- (Q)  $\frac{3\sqrt{3}}{4} |\alpha|^2$

- (C) If  $\arg z = \alpha$  and  $|z - 1| = 1$  then  $\left| \frac{z-2}{z} \right|$  is equal to (R)  $\frac{2}{3} |\alpha - \bar{\alpha}|$
- (D) Let A and B represent complex numbers  $z_1$  and  $z_2$ , which are roots of the equation  $z^2 + pz + q = 0$ . If  $\angle AOB = \alpha \neq 0$  and  $OA = OB$ , where O is the origin then  $\frac{p^2}{q}$  is equal to (S)  $4 \cos^2 \frac{\alpha}{2}$
97. If  $1, z_1, z_2, \dots, z_{10}$  are the 11th roots of unity then match the entries given in column A with one of the entries given in column B.
- | Column-I                                             | Column-II |
|------------------------------------------------------|-----------|
| (A) $(1 - z_1)(1 - z_2) \dots (1 - z_{10}) =$        | (P) 1     |
| (B) $z_1^{100} + z_2^{100} + \dots + z_{10}^{100} =$ | (Q) -1    |
| (C) $(1 + z_1)(1 + z_2) \dots (1 + z_{10}) =$        | (R) 0     |
| (D) $z_1 \cdot z_2 \cdot z_3 \dots z_{10} =$         | (S) 11    |
- 98.
- | Column-I                                                                                                           | Column-II |
|--------------------------------------------------------------------------------------------------------------------|-----------|
| (A) The small positive integer n for which $(1 + i)^{2n} = (1 - i)^{2n}$ is                                        | (P) 3     |
| (B) The value of x + y, if x and y satisfy the equation $\frac{(1+i)x - 2i}{3+i} + \frac{(2-3i)y + i}{3-i} = i$ is | (Q) 2     |
| (C) If $z = 3 - 4i$ , then $z^4 - 3z^3 + 3z^2 + 99z - 95 =$                                                        | (R) 4     |
| (D) If $z = x + iy$ , $z^{1/3} = a - ib$ and $\frac{x}{a} - \frac{y}{b} = k(a^2 - b^2)$ then the value of k is     | (S) 5     |
- 99.
- | Column-I                                                                                                                                                         | Column-II |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| (A) If $\left  \frac{z-1}{1-\bar{z}} \right  = \alpha$ , then $\alpha$ equals                                                                                    | (P) 1     |
| (B) Let the complex number $\frac{z-1}{z+1}$ be purely imaginary then $ z $ equals                                                                               | (Q) -1    |
| (C) If $ z + 6  =  2z + 3 $ , then $ z $ equals                                                                                                                  | (R) 3     |
| (D) Let $\omega \neq 1$ be a cube root of unity, then $\frac{a+b\omega+c\omega^2}{c+a\omega+b\omega^2} + \frac{a+b\omega+c\omega^2}{b+c\omega+a\omega^2}$ equals | (S) 2     |



100.  $z_1, z_2, z_3$  represents distinct points A, B, C in Argand Plane.

**Column-I**

- (A)  $2z_1z_3 = z_1z_2 + z_2z_3$   
 (B)  $|z_3 - z_1|^2 + |z_3 - z_2|^2 = |z_1 - z_2|^2$   
 (C)  $\lambda_1, \lambda_2, \lambda_3 \in \text{real}, \lambda_1z_1 + \lambda_2z_2 + \lambda_3z_3 = 0$   
 $\lambda_1 + \lambda_2 + \lambda_3 = 0$   
 (D)  $\frac{1}{z_1 - z_2} + \frac{1}{z_2 - z_3} + \frac{1}{z_3 - z_1}$

**Column-II**

- (P) A, B, C are collinear  
 (Q) A, B, C lie on circle whose diameter is AB  
 (R) A, B, C lie in circle which passes through O (0, 0)  
 (S) A, B, C make equilateral triangle

**Review Exercises for JEE Advanced**

- If  $z_1, z_2, z_3$  are three complex numbers, prove that  $z_1 \operatorname{Im}(\bar{z}_2z_3) + z_2 \operatorname{Im}(\bar{z}_3z_1) + z_3 \operatorname{Im}(\bar{z}_1z_2) = 0$
- Find all complex numbers satisfying the equation  $2|z|^2 + z^2 - 5 + i\sqrt{3} = 0$ .
- Show that the necessary and sufficient conditions that both the roots of the equation  $z^2 + az + b = 0$  should be of unit modulus are  $|a| \leq 2, |b| = 1, \arg b = 2\arg a$ .
- If  $t$  is a complex number such that  $|t| = 1$ , then prove that the point  $x = (at + b)/(t - c)$  describes a circle as  $t$  varies, unless  $|c| = 1$ , when it describes a straight line.
- Locate the complex numbers  $z = a + bi$  for which.
  - $\log_{\frac{1}{2}} \frac{|z - 1| + 4}{3|z - 1| - 2} > 1$
  - $\log_{\sqrt{3}} \frac{|z|^2 - |z| + 1}{2 + |z|} < 2$
- For every real number  $a \geq 1$  find all complex numbers  $z$  which satisfy the condition  $z + a|z| + 1 + i = 0$ .
- Solve the following system of equations in terms of complex numbers :  
 $z^3 + w^5 = 0$   
 $z^2 \cdot \bar{w}^4 = 1$
- If  $w = (1 + z)/(1 - z)$ , and show that if  $|z| = 1$ , the point  $w$  is on the  $y$ -axis.
- Solve the system of equations :  

$$\begin{cases} |z + 1| = |z + 2|, \\ |3z + 9| = |5z + 10i| \end{cases}$$
- Among the complex numbers  $z$  find all those for which  
 $\log_{14}(13 + |z|^2 - 4i)$   
 $+ \log_{196} \frac{1}{(13 + |z|^2 + 4i)^2} = 0$ .
- Find the real values of the parameter  $a$  for which atleast one complex number  $z = x + iy$  satisfies both the equality  $|z + \sqrt{2}| = a^2 - 3a + 2$  and the inequality  $|z + i\sqrt{2}| < a^2$ .
- If  $u = \sqrt{z_1z_2}$ , prove that  

$$|z_1| + |z_2| = \left| \frac{z_1 + z_2}{2} + u \right| + \left| \frac{z_1 + z_2}{2} - u \right|$$

13. Find all complex numbers  $z$  for which

$$\arg \left[ \frac{3z-6-3i}{2z-8-6i} \right] = \frac{\pi}{4} \text{ and } |z-3+i|=3.$$

14. Solve for  $x, y, z$  :

$$x+y+z=a$$

$$x+\omega y+\omega^2 z=b$$

$$x+\omega^2 y+\omega z=c, \text{ where } \omega \text{ is a cube root of unity.}$$

15. If  $z_n = \cos \frac{n\pi}{2^n} + i \sin \frac{n\pi}{2^n}$  then find

$$\lim_{n \rightarrow \infty} (z_1, z_2, z_3, \dots, z_n)$$

16. Find all real values of the parameter  $a$  for which the equation  $(a-1)z^4 - 4z^2 + a+2 = 0$  has only pure imaginary roots.

17.  $A(z_1), B(z_2), C(z_3)$  are vertices of a right angled

triangle where  $C = \frac{\pi}{2}$ . A circle is described

on  $AC$  as diameter. Prove that the point of intersection of the circle with  $AB$  is

$$\frac{z_1(\bar{z}_2 - \bar{z}_3) - z_3(z_1 - \bar{z}_3)}{(z_2 - z_1)}$$

18. A triangle with vertices represented by  $z_1, z_2$  and  $z_3$  has the opposite sides by lengths in the ratio  $2 : \sqrt{6} : \sqrt{3} - 1$  respectively prove that

$$(z_3 - z_2)^2 + (\sqrt{3} + 1)(z_1 - z_2)(z_3 - z_2) + (\sqrt{3} + 1)^2(z_1 - z_2)^2 = 0$$

19. If  $\cos \theta + i \sin \theta$  is a root of the equation  $a_0 x^n + a_1 x^{n-1} + \dots + a_n = 0$ , where  $a_1, a_2, \dots, a_n$  are real and  $a_0 \neq 0$ , show that  $a_0 + a_1 \cos \theta + a_2 \cos 2\theta + \dots + a_n \cos n\theta = 0$ .

20. Let  $f(z)$  be a polynomial in  $z$  having complex coefficients. When  $f(z)$  is divided by  $z^2 + z + 1$  and  $z - 1$ , remainders are respectively  $2\omega z + \omega$

and  $5 - \omega$ , where  $\omega = \frac{-1+i\sqrt{3}}{2}$ . Find the remainder when  $f(z)$  is divided by  $z^3 - 1$ .

21.  $z_1, z_2, z_3$  are complex numbers and  $p, q, r$  are real numbers such that

$$\frac{p}{|z_2 + z_3|} = \frac{q}{|z_3 - z_1|} = \frac{r}{|z_1 - z_2|}. \text{ Prove that}$$

$$\frac{p^2}{z_2 - z_3} + \frac{q^2}{z_3 - z_1} + \frac{r^2}{z_1 - z_2} = 0.$$

22. Find the complex number  $z$  which satisfies the condition  $|z - a + ai| = 1$  and has the greatest absolute value, where  $a$  is a real constant ( $a \neq 0$ ).

23. If  $A(z_1), B(z_2)$  and  $C(z_3)$  are the vertices of a triangle  $ABC$  and  $I(z_0)$  be its incentre, then show that

$$\frac{(z_0 - z_2)^2}{(z_3 - z_2)(z_1 - z_2)} \text{ is a purely real number.}$$

24. If  $\cos(\alpha - \beta) + \cos(\beta - \gamma) + \cos(\gamma - \alpha) = -3/2$  then prove that :

$$(i) \quad \Sigma \cos 2\alpha = 0 = \Sigma \sin 2\alpha$$

$$(ii) \quad \Sigma \sin 3\alpha = 3 \sin(\alpha + \beta + \gamma)$$

$$(iii) \quad \Sigma \cos 3\alpha = 3 \cos(\alpha + \beta + \gamma)$$

25. Equilateral triangles are drawn on the sides of a given triangle  $ABC$ , all outwards or all inwards. Prove that their centroids form an equilateral triangle.

26. If  $(1 + x + x^2)^{3n} = \sum_{r=0}^{6n} a_r x^r$ , then compute the value of,  $a_0 + a_6 + a_{12} + \dots + a_{6n}$ .

27. Find all the points in the complex plane which satisfy the equations

$$\log_5(|z| + 3) - \log_{\sqrt{5}} ||z| - 1| = 1 \text{ and}$$

$$\arg(z - 1 - i) = \frac{\pi}{4}.$$

28. Prove that:

$$(a) \quad \cos x + {}^nC_1 \cos 2x + {}^nC_2 \cos 3x + \dots + {}^nC_n \cos(n+1)x$$

$$= 2^n \cdot \cos^n \frac{x}{2} \cdot \cos \left( \frac{n+2}{2} x \right)$$

$$\begin{aligned}
 & \text{(b) } \cos\left(\frac{2\pi}{2n+1}\right) + \cos\left(\frac{4\pi}{2n+1}\right) \\
 & \quad + \cos\left(\frac{6\pi}{2n+1}\right) + \dots + \cos\left(\frac{2n\pi}{2n+1}\right) \\
 & = \frac{1}{2} \text{ When } n \in \mathbb{N}.
 \end{aligned}$$

29. Find all the roots of the equation  $(3z-1)^4 + (z-2)^4 = 0$  in the simplified form.

30. Prove that

$$\begin{aligned}
 \text{(i) } \frac{\sin 9\theta}{\sin \theta} &= 256 \sin^8 \theta - 576 \sin^6 \theta \\
 &\quad + 432 \sin^4 \theta - 120 \sin^2 \theta + 9 \\
 \text{(ii) } 128 \cos^3 \theta \sin^5 \theta &= 6 \sin 2\theta - 2 \sin 4\theta - 2 \sin 6\theta + \sin 3\theta
 \end{aligned}$$

31. Show that:

$$\begin{aligned}
 Z^9 - 1 &= (Z-1)(Z^2+Z+1)(Z^2-2\sin 10^\circ Z+1) \\
 &\quad (Z^2+2\cos 20^\circ Z+1)(Z^2-2\cos 40^\circ Z+1)
 \end{aligned}$$

32. Solve the equation  $z^{10} - 1 = 0$  and deduce that:

$$\sin 5\theta = 5\sin\theta \left[ 1 - \frac{\sin^2 \theta}{\sin^2 \pi/5} \right] \left[ 1 - \frac{\sin^2 \theta}{\sin^2 2\pi/5} \right]$$

33. Show that  $(x^{2m} - a^{2m})/(x^2 - a^2)$  is equal to

$$\begin{aligned}
 & \left( x^2 - 2ax \cos \frac{\pi}{m} + a^2 \right) \left( x^2 - 2ax \cos \frac{2\pi}{m} + a^2 \right) \dots \\
 & \quad \left( x^2 - 2ax \cos \frac{(m-1)\pi}{m} + a^2 \right)
 \end{aligned}$$

34. Find the area of the region in which the complex number  $z (= x + iy)$  lies, under the following

$$\begin{aligned}
 & \text{conditions: } z_1 = \frac{2z-i}{iz+2}, |x_1| < 1, y_1 > 0, |z_1| > 1, \\
 & \text{where } z_1 = x_1 + iy_1.
 \end{aligned}$$

35. Similar triangles BCX, CAY, ABZ are described on the sides of a triangle ABC. Show that the centroids of the triangles ABC and XYZ are coincident, using complex methods.



### Target Exercises for JEE Advanced

1. Let  $z_1$  &  $z_2$  be any two arbitrary complex numbers then prove that:

$$|z_1 + z_2| \geq \frac{1}{2} (|z_1| + |z_2|) \left| \frac{z_1}{|z_1|} + \frac{z_2}{|z_2|} \right|.$$

2. Prove that the equation  $z = a + 2bt + ct^2$  represents a parabola in general, and a straight line if  $b/c$  is real.

3. Prove that the equation  $Z = \frac{i+2t+t^2}{1+2t}$  represents a conic section.

4. If  $x + iy = \sqrt{a+ib}$  where  $a$  and  $b$  are real parameters, prove that  $a = \text{constant}$  and  $b = \text{constant}$  represent two systems of rectangular hyperbola cutting at right angles.

5. The roots of the equation  $t^3 + 3at^2 + 3bt + c = 0$  are  $z_1, z_2, z_3$  which represent the vertices of a triangle ABC on the Argand plane. Determine the centroid of the triangle. In case the triangle be equilateral then prove that  $a^2 = b$ .

6. If  $\cos(\alpha - \beta) + \cos(\beta - \gamma) + \cos(\gamma - \alpha) = -3/2$  then prove that:

$$\text{(i) } \Sigma \sin(\alpha + \beta) = 0 = \Sigma \cos(\alpha + \beta)$$

$$\text{(ii) } \Sigma \sin^2 \alpha = \Sigma \cos^2 \alpha = 3/2$$

$$\text{(iii) } \cos^3(\theta + \alpha) + \cos^3(\theta + \beta) + \cos^3(\theta + \gamma)$$

$$= 3 \cos(\theta + \alpha) \cdot \cos(\theta + \beta) \cdot \cos(\theta + \gamma)$$

where  $\theta \in \mathbb{R}$ .

7. Find the common roots of the equations

$$z^3 + 2z^2 + 2z + 1 = 0 \quad z^{1985} + z^{100} + 1 = 0$$

8. If  $1, \omega_1, \omega_2, \omega_3, \dots, \omega_{n-1}$  are the  $n^{\text{th}}$  root of unity then prove that

$$\frac{1}{2 - \omega_1} + \frac{1}{2 - \omega_2} + \frac{1}{2 - \omega_3} + \dots + \frac{1}{2 - \omega_{n-1}}$$

$$= \frac{(n-2)2^{n-1} + 1}{2^n - 1}$$

9. If  $r$  is less than  $n$  and prime to  $n$ , then

$$\cos \frac{2r\pi}{n} + i \sin \frac{2r\pi}{n} \text{ is a root of } x^n - 1 = 0, \text{ but}$$

it is not a root of any equation of lower degree, and of the same type. These type of roots are called primitive roots. What are the primitive roots of  $x^{15} - 1 = 0$ . Find the equation of which these are the roots. Show that this can be reduced to an equation of the fourth degree. What are its roots?

10.  $z$  is a root of  $az^2 + bz + c = 0$  where  $|z| = 1$  and  $a, b, c \in \mathbb{C}$ . Prove that  $(\bar{a}b - \bar{b}c)(a\bar{b} - b\bar{c}) = (a\bar{a} - c\bar{c})^2$ .

11. Prove that  $(x+y)^n - x^n - y^n$  is divisible by  $xy(x+y) \times (x^2 + xy + y^2)$  if  $n$  is an odd number not divisible by 3.

12. Show that the equation  $az^3 + bz^2 + \bar{b}z + \bar{a} = 0$  has a root  $\alpha$ , such that  $|\alpha| = 1$ .  $a, b, z$  and  $\alpha$  belong to the set of complex numbers.

13. If  $z_1, z_2, \dots, z_3$  are the roots of  $(z+1)^7 + z^7 = 0$

$$\text{find the value of } \sum_{k=1}^7 \operatorname{Re}(z_k)$$

14. If  $z = c + a + \frac{(1+it)}{(1-it)}$ , where  $t$  and  $a$  are real, show that as  $t$  varies from  $-\infty$  to  $+\infty$ , the point  $z$  describes once the circle with centre  $c$  and radius  $a$ .

15. If  $z$  is a complex number and  $\theta_0, \theta_1, \theta_2, \dots, \theta_{2n}$  are real then prove that all the roots of the equation  $\sin \theta_0 z^{2n} + \cos \theta_1 z^{2n-1} + \sin \theta_2 z^{2n-2} + \dots + \cos \theta_{2n-1} z + \sin \theta_{2n} = b$  lies in the region

$$|z| > 1 - \frac{1}{|b|}.$$

16. Given 3 non collinear distinct points  $A(z_1); B(z_2); C(z_3)$  on an argand plane, with equation of the line AB as  $\bar{b}z + b\bar{z} = c$  where  $b = i(z_2 - z_1)$  and  $c$

$= (z_1 \bar{z}_2 - \bar{z}_1 z_2)$  obtain the equation of the altitude CD of the triangle ABC. Hence show that the reflection of the centroid  $G(g)$  of

$\Delta ABC$  through CD is  $z_3 + \frac{b}{b}(\bar{g} - \bar{z}_3)$ .

17. Prove that  $\sin \frac{\pi}{2n} \sin \frac{2\pi}{2n} \dots$

$$\sin \frac{(n-1)\pi}{2n} = \frac{\sqrt{n}}{2^{n-1}};$$

18. Prove that

$$\cos \frac{2\pi}{2n+1} \cos \frac{4\pi}{2n+1} \dots \cos \frac{2n\pi}{2n+1} = \frac{(-1)^{\frac{n}{2}}}{2^n}$$

19. By considering  $\tan 7\theta = 0$ , show that  $x = \tan^2(3\pi/7)$  satisfies the cubic equation  $x^3 - 21x^2 + 35x - 7 = 0$ .

20. Sum  $\sin \theta \cos \theta + \frac{\sin 2\theta \cos^2 \theta}{2!} + \frac{\sin 3\theta \cos^3 \theta}{3!} + \dots$

21. Sum  $1 + \frac{1}{2} \cos \alpha + \frac{1 \cdot 3 \cdot 5}{2 \cdot 4 \cdot 6} \cos 3\alpha + \dots$

22. Show that  $x^{2n} - 2x^n a^n \cos \theta + a^{2n}$  is equal to

$$\left( x^2 - 2xa \cos \frac{\theta}{n} + a^2 \right) \left( x^2 - 2xa \cos \frac{\theta + 2\pi}{n} + a^2 \right) \dots$$

$$\left( x^2 - 2xa \cos \frac{\theta + 2(n-1)\pi}{n} + a^2 \right)$$

23. Find the roots of  $z^n = (z+1)^n$  and show that the points which represent them are collinear. Hence show that these roots are also the roots of the equation,

$$\left( 2 \sin \frac{m\pi}{n} \right)^2 \bar{z}^2 + \left( 2 \sin \frac{m\pi}{n} \right)^2 \bar{z} + 1 = 0$$

where  $m = 1, 2, 3, \dots, (n-1)$ .

24. Prove that  $\sin x + {}^nC_1 \sin 2x + {}^nC_2 \sin 3x + \dots$

$$+ {}^nC_n \sin (n+1)x = 2^n \cdot \cos^n \frac{x}{2} \cdot \sin \left( \frac{n+2}{2} \right) x$$

25. Let the complex numbers  $z_1, z_2, z_3$  form a triangle. Equilateral triangles are drawn on the sides of this triangle, all of them outwards. If  $G_1, G_2, G_3$  are the centroids of these triangles and  $G$  is the centroid of the triangle  $z_1 z_2 z_3$  such that  $G_1, G_2, G_3$  are occurring in the positive direction of rotation, then prove that :

(i)  $G$  is also the centroid of the  $\Delta G_1 G_2 G_3$ .

(ii)  $\vec{G_3 G_1} = w \vec{G_2 G_3}$  where  $w$  is the complex cube root of unity.

(iii)  $\frac{1}{\vec{G_1 G_2}} + \frac{1}{\vec{G_2 G_3}} + \frac{1}{\vec{G_3 G_1}} = 0$ .

26. If  $A X Y A' X' Y'$  is a regular hexagon and  $A, A'$  represent given complex numbers  $a, a'$ , then the numbers represented by  $X, X', Y, Y'$  are

given by  $\frac{1}{2}(a + a') + \frac{1}{2}(a - a')(\cos\theta + i\sin\theta)$ ,

where  $\theta$  has the values  $\pm \frac{\pi}{3}, \pm \frac{2\pi}{3}$ .

27. If  $\alpha_1, \alpha_2, \alpha_3, \dots, \alpha_n$  are the roots of the equation

$$z^n + a_1 z^{n-1} + a_2 z^{n-2} + \dots + a_{n-1} z + a_n = 0$$

where  $a_1, a_2, \dots, a_n$  are real, then prove that,

(i)  $\bar{\alpha}_1, \bar{\alpha}_2, \bar{\alpha}_3, \dots, \bar{\alpha}_n$

are also the roots of the same equation.

(ii)  $(1 + \alpha_1^2)(1 + \alpha_2^2)(1 + \alpha_3^2) \dots (1 + \alpha_n^2)$

$$= (1 - a_2 + a_4 \dots)^2 + (a_1 - a_3 + a_5 \dots)^2$$

28. By putting  $z = r(\cos\theta + i\sin\theta)$  in the identity  $1 + z + z^2 + \dots + z^{n-1} = (1 - z^n)/(1 - z)$ ,

Prove that

(i)  $1 + r \cos\theta + r^2 \cos 2\theta + \dots + r^{n-1} \cos(n-1)\theta$

$$= \frac{1 - r \cos\theta - r^n \cos n\theta + r^{n+1} \cos(n-1)\theta}{1 - 2r \cos\theta + r^2}$$

(ii)  $r \sin\theta + r^2 \sin 2\theta + \dots + r^{n-1} \sin(n-1)\theta$

$$= \frac{r \sin\theta - r^n \sin n\theta + r^{n+1} \sin(n-1)\theta}{1 - 2r \cos\theta + r^2}$$

29. If the points  $A_1, A_2$  are given by  $az^2 + 2bz + c = 0$  and the points  $A_3, A_4$  by  $a'z^2 + 2b'z + c' = 0$ ,  $O$  is the middle points of  $A_3 A_4$  and  $ac' + a'c - 2bb' = 0$  then prove that  $OA_1, OA_2$  are equally inclined to  $A_3 A_4$  and  $OA_1 \cdot OA_2 = OA_3^2 = OA_4^2$ .

30. Solve the equation  $z^7 + 1 = 0$  and deduce that

$$\tan \frac{\pi}{14} \cdot \tan \frac{3\pi}{14} \cdot \tan \frac{5\pi}{14} = \frac{1}{\sqrt{7}}$$

31. Complex numbers  $z_1, z_2, z_3$  (described ACW) form a right angled triangle with right angle at ' $z_1$ '. ' $z_4$ ' is the image of  $z_2$  about the line joining ' $z_1$ ' & the mid point of line joining  $z_2$  &  $z_3$ . Prove that

$$\tan \left[ \text{Arg} \left( \frac{z_2 - z_3}{z_1 - z_3} \right) \right] - \tan \left[ \text{Arg} \left( \frac{z_2 - z_3}{z_1 - z_3} \right) \right] \geq 2$$

32. Two points represented by complex number  $a, b$  lie on a circle with centre at the origin and radius  $r$ . The tangents at ' $a$ ' and ' $b$ ' intersect at

$z$ . Prove that  $z = \frac{2ab}{a+b}$ .

33. Let  $x_1, x_2, x_3, \dots, x_n$  be the roots of the equation  $x^n + x^{n-1} + \dots + x + 1 = 0$ .

Compute the expression

$$\frac{1}{x_1 - 1} + \frac{1}{x_2 - 1} + \dots + \frac{1}{x_n - 1}.$$

Hence or otherwise prove that  $\sum_{r=1}^n \cot \frac{r\pi}{n+1} = 0$

34. Form the equation whose roots are

$\cos \frac{\pi}{9}, \cos \frac{3\pi}{9}, \cos \frac{5\pi}{9}, \cos \frac{7\pi}{9}$  and hence prove

$$(i) \quad 8 \cos \frac{\pi}{9} \cdot \cos \frac{5\pi}{9} \cdot \cos \frac{7\pi}{9} = 1$$

$$= 8 \cos \frac{\pi}{9} \cos \frac{2\pi}{9} \cos \frac{4\pi}{9}.$$

$$(ii) \quad \sec^4 \frac{\pi}{9} + \sec^4 \frac{2\pi}{9} + \sec^4 \frac{4\pi}{9} = 1104.$$

### Previous Years' Questions (JEE Advanced)

#### A. Fill in the blanks

1. If the expression

$$\frac{\left[ \sin\left(\frac{x}{2}\right) + \cos\left(\frac{x}{2}\right) + i \tan(x) \right]}{\left[ 1 + 2i \sin\left(\frac{x}{2}\right) \right]}$$
 is real, then the

set of all possible values of  $x$  is...

[IIT - 1987]

2. For any two complex numbers  $z_1, z_2$  and any real number  $a$  and  $b$ .  $|az_1 - bz_2|^2 + |bz_1 + az_2|^2 = \dots\dots\dots$

[IIT - 1988]

3. If  $a, b, c$  are the numbers between 0 and 1 such that the points  $z_1 = a + i, z_2 = 1 + bi$  and  $z_3 = 0$  form an equilateral triangle, then  $a = \dots\dots\dots$  and  $b = \dots\dots\dots$

[IIT - 1989]

4. ABCD is a rhombus. Its diagonals AC and BD intersect at the point M and satisfy  $BD = 2AC$ . If the points D and M represent the complex numbers  $1 + i$  and  $2 - i$  respectively, then A represents the complex number.....or.....

[IIT - 1993]

5. Suppose  $Z_1, Z_2, Z_3$  are the vertices of an equilateral triangle inscribed in the circle;

$|Z| = 2$ . If  $Z_1 = 1 + i\sqrt{3}$  then  $Z_2 = \dots\dots\dots$ ,  $Z_3 = \dots\dots\dots$ . [IIT - 1994]

6. The value of the expression ;

$$1 \cdot (2 - w)(2 - w^2) + 2 \cdot (3 - w)(3 - w^2) + \dots\dots\dots + (n - 1) \cdot (n - w)(n - w^2),$$

where  $w$  is an imaginary cube root of unity is..... [IIT - 1996]

#### B. True/False

7. For complex number  $z_1 = x_1 + iy_1$  and  $z_2 = x_2 + iy_2$ , we write  $z_1 \cap z_2$ , if  $x_1 \leq x_2$  and  $y_1 \leq y_2$ . Then for all complex numbers  $z$  with  $1 \cap z$ , we have  $\frac{1 - z}{1 + z} \cap 0$ .

[IIT - 1981]

8. If the complex numbers,  $Z_1, Z_2$  and  $Z_3$  represent the vertices of an equilateral triangle such that  $|Z_1| = |Z_2| = |Z_3|$  then  $Z_1 + Z_2 + Z_3 = 0$ .

[IIT - 1984]

9. If three complex numbers are in A.P. then they lie on a circle in the complex plane.

[IIT - 1985]

10. The cube roots of unity when represented on Argand diagram form the vertices of an equilateral triangle.

[IIT - 1988]

#### C. Multiple Choice Questions with ONE correct answer

11. If the cube roots of unity are  $1, \omega, \omega^2$ , then the roots of the equation  $(x - 1)^3 + 8 = 0$  are

[IIT - 1979]

- (A)  $-1, 1 + 2\omega, 1 + 2\omega^2$   
(B)  $-1, 1 - 2\omega, 1 - 2\omega^2$   
(C)  $-1, -1, -1$   
(D) None of these

12. The smallest positive integer  $n$  for which

$$\left( \frac{1 + i}{1 - i} \right)^n = 1 \text{ is}$$

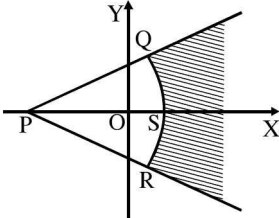
[IIT - 1980]

- (A)  $n = 8$  (B)  $n = 16$   
(C)  $n = 12$  (D) None of these

13. The complex numbers  $z = x + iy$  which satisfy the equation  $\left| \frac{z - 5i}{z + 5i} \right| = 1$  lie on [IIT - 1981]  
 (A) the x-axis  
 (B) the straight line  $y = 5$   
 (C) a circle passing through the origin  
 (D) None of these
14. If  $z = \left( \frac{\sqrt{3}}{2} + \frac{i}{2} \right)^5 + \left( \frac{\sqrt{3}}{2} - \frac{i}{2} \right)^5$ , then [IIT - 1982]  
 (A)  $\operatorname{Re}(z) = 0$   
 (B)  $\operatorname{Im}(z) = 0$   
 (C)  $\operatorname{Re}(z) > 0, \operatorname{Im}(z) > 0$   
 (D)  $\operatorname{Re}(z) > 0, \operatorname{Im}(z) < 0$
15. The inequality  $|z + 2| < |z - 2|$  represents the region given by [IIT - 1982]  
 (A)  $\operatorname{Re}(z) \geq 0$  (B)  $\operatorname{Re}(z) < 0$   
 (C)  $\operatorname{Re}(z) > 0$  (D) None of these
16. If  $z = x + iy$  and  $\omega = (1 - iz) / (z - i)$ , then  $|\omega| = 1$  implies that, in the complex plane [IIT - 1983]  
 (A)  $z$  lies on the imaginary axis  
 (B)  $z$  lies on the real axis  
 (C)  $z$  lies on the unit circle  
 (D) None of these
17. The points  $z_1, z_2, z_3, z_4$  in the complex plane are the vertices of a parallelogram taken in order if and only if [IIT - 1983]  
 (A)  $z_1 + z_4 = z_2 + z_3$  (B)  $z_1 + z_3 = z_2 + z_4$   
 (C)  $z_1 + z_2 = z_3 + z_4$  (D) None of these
18. If  $a, b, c$  and  $u, v, w$  are complex numbers representing the vertices of two triangles such that  $c = (1 - r)a + rb$  and  $w = (1 - r)u + rv$ , where  $r$  is a complex number, then the two triangles [IIT - 1985]  
 (A) have the same area (B) are similar  
 (C) are congruent (D) None of these
19. If  $z_1$  and  $z_2$  are two nonzero complex numbers such that  $|z_1 + z_2| = |z_1| + |z_2|$ , then  $\operatorname{Arg} z_1 - \operatorname{Arg} z_2$  is equal to [IIT - 1987]  
 (A)  $-\pi$  (B)  $-\frac{\pi}{2}$   
 (C)  $0$  (D)  $\frac{\pi}{2}$
20. The value of  $\sum_{k=1}^6 \left( \sin \frac{2\pi k}{7} - i \cos \frac{2\pi k}{7} \right)$  is :- [IIT - 1987]  
 (A)  $-1$  (B)  $0$   
 (C)  $-i$  (D)  $i$
21. The complex numbers  $\sin x + i \cos 2x$  and  $\cos x - i \sin 2x$  are conjugate to each other, for [IIT - 1988]  
 (A)  $x = n\pi$  (B)  $x = 0$   
 (C)  $x = (n + 1/2)\pi$  (D) no value of  $x$ .
22. The equation not representing a circle is given by - [IIT - 1991]  
 (A)  $\operatorname{Re} \left( \frac{1+z}{1-z} \right) = 0$   
 (B)  $z\bar{z} + iz - i\bar{z} + 1 = 0$   
 (C)  $\arg \left( \frac{z-1}{z+1} \right) = \frac{\pi}{2}$   
 (D)  $\left| \frac{z-1}{z+1} \right| = 1$
23. If  $z = -1$ , then the principal value of the  $\arg(z^{2/3})$  is equal to - [IIT - 1991]  
 (A)  $\frac{\pi}{3}$  (B)  $\frac{2\pi}{3}$  or  $0$   
 (C)  $\frac{10\pi}{3}$  (D)  $\pi$
24. If  $z$  is a complex number such that  $z \neq 0$  and  $\operatorname{Re}(z) = 0$ , then - [IIT - 1992]  
 (A)  $\operatorname{Re}(z^2) = 0$  (B)  $\operatorname{Im}(z^2) = 0$   
 (C)  $\operatorname{Re}(z^2) = \operatorname{Im}(z^2)$  (D) None of these
25. The complex number  $\frac{1+2i}{1-i}$  lies in the [IIT - 1992]  
 (A) Ist quadrant (B) IInd quadrant  
 (C) IIIrd quadrant (D) IVth quadrant

26. If  $\alpha$  and  $\beta$  are different complex numbers with  $|\beta| = 1$ , then  $\left| \frac{\beta - \alpha}{1 - \alpha\beta} \right|$  is equal to - [IIT-1992]  
 (A) 0 (B)  $1/2$  (C) 1 (D) 2
27.  $1, \omega, \omega^2$  are the cube roots of unity, then the value of  $(1 + \omega)^3 - (1 + \omega^2)^3$  is - [IIT-1993]  
 (A)  $2\omega$  (B) 2 (C)  $-2$  (D) 0
28. If  $\alpha$  and  $\beta$  are two fixed non-zero complex numbers and 'z' a variable complex number. If the lines  $\alpha \bar{z} + \bar{\alpha} z + 1 = 0$  and  $\beta \bar{z} + \bar{\beta} z - 1 = 0$  are mutually perpendicular, then- [IIT-1993]  
 (A)  $\alpha\beta + \bar{\alpha}\bar{\beta} = 0$  (B)  $\alpha\beta - \bar{\alpha}\bar{\beta} = 0$   
 (C)  $\bar{\alpha}\beta - \alpha\bar{\beta} = 0$  (D)  $\alpha\bar{\beta} + \bar{\alpha}\beta = 0$
29. If  $z_1, z_2, z_3$  are vertices of an equilateral triangle inscribed in the circle  $|z| = 2$  and if  $z_1 = 1 + i\sqrt{3}$ , then - [IIT-1994]  
 (A)  $z_2 = -2, z_3 = 1 - i\sqrt{3}$   
 (B)  $z_2 = 2, z_3 = 1 - i\sqrt{3}$   
 (C)  $z_2 = -2, z_3 = -1 - i\sqrt{3}$   
 (D)  $z_2 = 1 - i\sqrt{3}, z_3 = -1 - i\sqrt{3}$
30. If  $w (\neq 1)$  is a cube root of unity and  $(1 + w)^7 = A + Bw$ , then A & B are respectively the numbers [IIT-1995]  
 (A) 0, 1 (B) 1, 1  
 (C) 1, 0 (D)  $-1, 1$
31. Let  $z$  &  $\omega$  be two non zero complex numbers such that  $|z| = |\omega|$  and  $\text{Arg } z + \text{Arg } \omega = \pi$ . Then z equals: [IIT-1995]  
 (A)  $\omega$  (B)  $-\omega$   
 (C)  $\bar{\omega}$  (D)  $-\bar{\omega}$
32. Let  $z$  &  $\omega$  be two complex numbers such that  $|z| \leq 1, |\omega| \leq 1$  and  $|z + i\omega| = |z - i\bar{\omega}| = 2$ . Then z equals : [IIT-1995]  
 (A) 1 or i (B) i or  $-i$   
 (C) 1 or  $-1$  (D) i or  $-1$
33. For positive integers  $n_1, n_2$  the value of the expression;  
 $(1+i)^{n_1} + (1+i^3)^{n_1} + (1+i^5)^{n_2} + (1+i^7)^{n_2}$ ,  
 where  $i = \sqrt{-1}$  is a real number if & only if [IIT-1996]  
 (A)  $n_1 = n_2 + 1$  (B)  $n_1 = n_2 - 1$   
 (C)  $n_1 = n_2$  (D)  $n_1 > 0, n_2 > 0$
34. If  $\omega$  is an imaginary cube root of unity, then  $(1 + \omega - \omega^2)^7$  equals - [IIT-1998]  
 (A)  $128\omega$  (B)  $-128\omega$   
 (C)  $128\omega^2$  (D)  $-128\omega^2$
35. The value of the sum  $\sum_{n=1}^{13} (i^n + i^{n+1})$ , where  $i = \sqrt{-1}$  equals - [IIT-1998]  
 (A) i (B)  $i - 1$   
 (C)  $-i$  (D) 0
36.  $\begin{vmatrix} 6i & -3i & 1 \\ 4 & 3i & -1 \\ 20 & 3 & i \end{vmatrix} = x + iy$ , then - [IIT-1998]  
 (A)  $x = 3, y = 1$  (B)  $x = 1, y = 3$   
 (C)  $x = 0, y = 3$  (D)  $x = 0, y = 0$
37. If  $i = \sqrt{-1}$ , then  
 $4 + 5 \left( -\frac{1}{2} + \frac{i\sqrt{3}}{2} \right)^{334} + 3 \left( -\frac{1}{2} + \frac{i\sqrt{3}}{2} \right)^{365}$   
 is equal to: [IIT-1999]



- (A)  $1 - i\sqrt{3}$  (B)  $-1 + i\sqrt{3}$   
 (C)  $i\sqrt{3}$  (D)  $-i\sqrt{3}$
38. If  $\arg(z) < 0$ , then  $\arg(-z) - \arg(z) =$  [IIT - 2000]  
 (A)  $\pi$  (B)  $-\pi$   
 (C)  $-\frac{\pi}{2}$  (D)  $\frac{\pi}{2}$
39. If  $z_1, z_2, z_3$  are complex number such that  $|z_1| = |z_2| = |z_3| = \left| \frac{1}{z_1} + \frac{1}{z_2} + \frac{1}{z_3} \right| = 1$ , then  $|z_1 + z_2 + z_3|$  is [IIT - 2000]  
 (A) equal to 1 (B) less than 1  
 (C) greater than 3 (D) equal to 3
40. If  $z_1$  and  $z_2$  be the  $n^{\text{th}}$  roots of unity which subtend right angle at the origin. Then  $n$  must be of the form - [IIT - 2001]  
 (A)  $4k + 1$  (B)  $4k + 2$   
 (C)  $4k + 3$  (D)  $4k$
41. The complex numbers  $z_1, z_2$  and  $z_3$  satisfying  $\frac{z_1 - z_3}{z_2 - z_3} = \frac{1 - i\sqrt{3}}{2}$  are the vertices of a triangles which is - [IIT - 2001]  
 (A) of area zero  
 (B) right angled isosceles  
 (C) equilateral  
 (D) obtuse angled isosceles
42. For all complex numbers  $z_1, z_2$  satisfying  $|z_1| = 12$  and  $|z_2 - 3 - 4i| = 5$ , the minimum value of  $|z_1 - z_2|$  is - [IIT - 2002]  
 (A) 0 (B) 2 (C) 7 (D) 0.7
43. Let  $\omega = -\frac{1}{2} + i\frac{\sqrt{3}}{2}$ . Then the value of the determinant  $\begin{vmatrix} 1 & 1 & 1 \\ 1 & -1 - \omega^2 & \omega^2 \\ 1 & \omega^2 & \omega^4 \end{vmatrix}$  is - [IIT - 2002]  
 (A)  $3\omega$  (B)  $3\omega(\omega - 1)$   
 (C)  $2\omega^2$  (D)  $3\omega(1 - \omega)$
44. If  $|z| = 1$  and  $\omega = \frac{z-1}{z+1}$  (where  $z \neq -1$ ), then  $\text{Re}(\omega)$  is [IIT - 2003]  
 (A) 0 (B)  $-\frac{1}{|z+1|^2}$   
 (C)  $\left| \frac{1}{z+1} \right| \cdot \frac{1}{|z+1|^2}$  (D)  $\frac{\sqrt{2}}{|z+1|^2}$
45. If  $\omega (\neq 1)$  be a cube root of unity and  $(1 + \omega^2)^n = (1 + \omega^4)^n$ , then the least positive value of  $n$  is [IIT - 2004]  
 (A) 2 (B) 3  
 (C) 5 (D) 6
46. The shaded region, where  $P = (-1, 0)$ ,  $Q = (-1 + \sqrt{2}, \sqrt{2})$ ,  $R = (-1 + \sqrt{2}, -\sqrt{2})$ ,  $S = (1, 0)$  is represented by :- [IIT - 2005]
- 
- (A)  $z : |z + 1| > 2$  and  $\arg(z + 1) < \pi/4$   
 (B)  $z : |z + 1| < 2$  and  $\arg(z - 1) < \pi/2$   
 (C)  $z : |z + 1| > 2$  and  $\arg(z + 1) > \pi/4$   
 (D)  $z : |z - 1| < 2$  and  $\arg(z + 1) > \pi/2$
47.  $a, b, c$  are integers, not all simultaneously equal and  $\omega$  is cube root of unity ( $\omega \neq 1$ ), then minimum value of  $|a + b\omega + c\omega^2|$  is [IIT - 2005]  
 (A) 0 (B) 1 (C)  $\frac{\sqrt{3}}{2}$  (D)  $\frac{1}{2}$
48. If  $\frac{w - \bar{w}z}{1 - z}$  is purely real where  $w = \alpha + i\beta$ ,  $\beta \neq 0$  and  $z \neq 1$ , then the set of the values of  $z$  is [IIT - 2006]

- (A)  $\{z : |z| = 1\}$  (B)  $\{z : z = \bar{z}\}$   
 (C)  $\{z : z \neq 1\}$  (D)  $\{z : |z| = 1, z \neq 1\}$
49. A man walks a distance of 3 units from the origin towards the north-east ( $N 45^\circ E$ ) direction. From there, he walks a distance of 4 units towards the north-west ( $N 45^\circ W$ ) direction to reach a point P. Then the position of P in the Argand plane is [IIT - 2007]
- (A)  $3e^{i\pi/4} + 4i$  (B)  $(3 - 4i)e^{i\pi/4}$   
 (C)  $(4 + 3i)e^{i\pi/4}$  (D)  $(3 + 4i)e^{i\pi/4}$
50. If  $|z| = 1$  and  $z \neq \pm 1$ , then all the values of  $\frac{z}{1-z^2}$  lie on [IIT - 2007]
- (A) a line not passing through the origin  
 (B)  $|z| = \sqrt{2}$  (C) the x-axis  
 (D) the y-axis
51. A particle P starts from the point  $z_0 = 1 + 2i$ , where  $i = \sqrt{-1}$ . It moves first horizontally away from origin by 5 units and then vertically away from origin by 3 units to reach a point  $z_1$ . From  $z_1$  the particle moves  $\sqrt{2}$  units in the direction of the vector  $\hat{i} + \hat{j}$  and then it moves through an angle  $\frac{\pi}{2}$  in anticlockwise direction on a circle with centre at origin, to reach a point  $z_2$ . The point  $z_2$  is given by [IIT - 2008]
- (A)  $6 + 7i$  (B)  $-7 + 6i$   
 (C)  $7 + 6i$  (D)  $-6 + 7i$
52. Let  $z = x + iy$  be a complex number where  $x$  and  $y$  are integers. Then the area of the rectangle whose vertices are the roots of the equation  $\bar{z}z^3 + z\bar{z}^3 = 350$  is [IIT - 2009]
- (A) 48 (B) 32  
 (C) 40 (D) 80
53. Let  $z = \cos\theta + i \sin\theta$ . Then the value of  $\sum_{m=1}^{15} \text{Im}(z^{2m-1})$  at  $\theta = 2^\circ$  [IIT - 2009]
- (A)  $\frac{1}{\sin 2^\circ}$  (B)  $\frac{1}{3 \sin 2^\circ}$   
 (C)  $\frac{1}{2 \sin 2^\circ}$  (D)  $\frac{1}{4 \sin 2^\circ}$
54. Let  $z$  be a complex number such that the imaginary part of  $z$  is nonzero and  $a = z^2 + z + 1$  is real. Then  $a$  cannot take the value. [IIT - 2012]
- (A)  $-1$  (B)  $\frac{1}{3}$  (C)  $\frac{1}{2}$  (D)  $\frac{3}{4}$
- D. Multiple Choice Questions with ONE or MORE THAN ONE correct answer**
55. If  $z_1 = a + ib$  and  $z_2 = c + id$  are complex numbers such that  $|z_1| = |z_2| = 1$  and  $\text{Re}(z_1 \bar{z}_2) = 0$ , then the pair of complex numbers  $w_1 = a + ic$  and  $w_2 = b + id$  satisfies [IIT - 1985]
- (A)  $|w_1| = 1$  (B)  $|w_2| = 1$   
 (C)  $\text{Re}(w_1 \bar{w}_2) = 0$  (D) None of these
56. Let  $z_1$  and  $z_2$  be complex numbers such that  $z_1 \neq z_2$  and  $|z_1| = |z_2|$ . If  $z_1$  has positive real part and  $z_2$  has negative imaginary part, then  $\frac{z_1 + z_2}{z_1 - z_2}$  may be [IIT - 1986]
- (A) zero (B) real and positive  
 (C) real and negative (D) purely imaginary
57. Let  $z_1$  and  $z_2$  be two distinct complex numbers and let  $z = (1-t)z_1 + tz_2$  for some real number  $t$  with  $0 < t < 1$ . If  $\text{Arg}(w)$  denotes the principal argument of a nonzero complex number  $w$ , then [IIT - 2010]
- (A)  $|z - z_1| + |z - z_2| = |z_1 - z_2|$   
 (B)  $\text{Arg}(z - z_1) = \text{Arg}(z - z_2)$   
 (C)  $\left| \frac{z - z_1}{z_2 - z_1} \cdot \frac{\bar{z} - \bar{z}_1}{\bar{z}_2 - \bar{z}_1} \right| = 0$   
 (D)  $\text{Arg}(z - z_1) = \text{Arg}(z_2 - z_1)$

**E. Subjective Problems**

58. Express  $\frac{1}{1 - \cos \theta + 2i \sin \theta}$  in the form  $x + iy$ .

[IIT - 1978]

59. If  $x = a + b$ ,  $y = a\gamma + b\beta$  and  $z = a\beta + b\gamma$  where  $\gamma$  and  $\beta$  are the complex cube roots of unity, show that  $xyz = a^3 + b^3$ .

[IIT - 1978]

60. If  $x + iy = \sqrt{\frac{a + ib}{c + id}}$ , prove that  $(x^2 + y^2)^2 =$

$$\frac{a^2 + b^2}{c^2 + d^2}.$$

[IIT - 1979]

61. It is given that  $n$  is an odd integer greater than 3, but  $n$  is not a multiple of 3. Prove that  $x^3 + x^2 + x$  is a factor of  $(x + 1)^n - x^n - 1$ .

[IIT - 1980]

62. Find the real values of  $x$  and  $y$  for which the following equation is satisfied

$$\frac{(1+i)x - 2i}{3+i} + \frac{(2-3i)y + i}{3-i} = i$$

[IIT - 1980]

63. Let the complex number  $z_1$ ,  $z_2$  and  $z_3$  be the vertices of an equilateral triangle. Let  $z_0$  be the circumcentre of the triangle. Then prove that  $z_1^2 + z_2^2 + z_3^2 = 3z_0^2$ .

[IIT - 1981]

64. Prove that the complex numbers  $z_1$ ,  $z_2$  and the origin form an equilateral triangle only if  $z_1^2 + z_2^2 - z_1 z_2 = 0$ .

[IIT - 1983]

65. Show that the area of the triangle on the Argand diagram formed by the complex numbers  $z$ ,  $iz$

$$\text{and } z + iz \text{ is } \frac{1}{2} |z|^2.$$

[IIT - 1986]

66. Complex numbers  $z_1$ ,  $z_2$ ,  $z_3$  are the vertices A, B, C respectively of an isosceles right angled triangle with right angle at C. Show that  $(z_1 - z_2)^2 = 2(z_1 - z_3)(z_3 - z_2)$ .

[IIT - 1986]

67. Let  $z_1 = 10 + 6i$  and  $z_2 = 4 + 6i$ . If  $z$  is any complex number such that the argument of  $(z - z_1)/(z - z_2)$  is  $\pi/4$ , then prove that  $|z - 7 - 9i| = 3\sqrt{2}$

[IIT - 1990]

68. Write the correct letter from **column II** against the entry number in **column I** in your answer book,  $z \neq 0$  is a complex number

[IIT - 1992]

**Column I**(i)  $\operatorname{Re} z = 0$ (ii)  $\operatorname{Arg} z = \frac{\pi}{4}$ **Column II**(A)  $\operatorname{Re} z^2 = 0$ (B)  $\operatorname{Im} z^2 = 0$ (C)  $\operatorname{Re} z^2 = \operatorname{Im} z^2$ 

69. Prove that  $\sum_{r=1}^k (-3)^{r-1} {}^{3n}C_{2r-1} = 0$ , where  $k = \frac{3n}{2}$  &  $n$  is an even positive integer.

[IIT - 1993]

70. Find the equation in complex variable of all the circles which are orthogonal to  $|z| = 1$  and  $|z - 1| = 4$ .

[IIT - 1992]

71. If  $iz^3 + z^2 - z + i = 0$ , then show that  $|z| = 1$

[IIT - 1995]

72. If  $|z| \leq 1$ ,  $|w| \leq 1$ , show that

$$|z - w|^2 \leq (|z| - |w|)^2 + (\arg z - \arg w)^2$$

[IIT - 1995]

73. Find all non-zero complex numbers  $Z$  satisfying  $\bar{Z} = iZ^2$ .

[IIT - 1996]

74. Let  $\bar{b}z + b\bar{z} = c$ ;  $b \neq 0$ ; be a line in the complex plane where  $\bar{b}$  is the complex conjugate of  $b$ . If a point  $z$ , is the reflection of a point  $z_2$  through the line, then show that  $c = \bar{z}_1 b + z_2 \bar{b}$ .

[IIT - 1997]

75. Let  $z_1$  and  $z_2$  be roots of the equation  $z^2 + pz + q = 0$  where the coefficient  $p$  and  $q$  may be complex numbers. Let A and B represent  $z_1$  and  $z_2$  in the complex plane. If  $\angle AOB = \alpha \neq 0$  and  $OA = OB$ , where O is the origin prove that  $p^2 = 4q \cos^2(\alpha/2)$

[IIT - 1997]

76. For complex numbers  $z$  &  $\omega$ , prove that,

$$|z|^2 \omega - |\omega|^2 z = z - \omega \text{ if and only if, } z = \omega \text{ or } z\bar{\omega} = 1$$

[IIT - 1997]

77. Prove that  $\sum_{k=1}^{n-1} (n-k) \cos \frac{2k\pi}{n} = -\frac{n}{2}$  where  $n \geq 3$  is an integer. [IIT - 1997]

78. Find all the roots of the equation  $(3z-1)^4 + (z-2)^4 = 0$  in the simplified form of  $a+ib$ . [IIT - 1999]

79. Let a complex number  $\alpha$ ,  $\alpha \neq 1$ , be a root of the equation  $z^{p+q} - z^p - z^q + 1 = 0$ , where  $p, q$  are distinct primes. Show that either  $1 + \alpha + \alpha^2 + \dots + \alpha^{p-1} = 0$  or  $1 + \alpha + \alpha^2 + \dots + \alpha^{q-1} = 0$ , but not both together. [IIT - 2002]

80. Let  $z_1$  &  $z_2$  are two complex numbers such that  $|z_1| < 1 < |z_2|$ . Prove that  $\left| \frac{1 - z_1 \bar{z}_2}{z_1 - z_2} \right| < 1$ . [IIT - 2003]

81. Let  $a_r$ ,  $r = 1, 2, 3, \dots, n$  be complex numbers such that  $|a_r| < 2$ , for all  $r$ , then prove that there is no complex number such that  $|z| < 1/3$  &  $\sum_{r=1}^n a_r z^r = 1$ . [IIT - 2003]

82. Find the centre and radius of circle given by  $\left| \frac{z-\alpha}{z-\beta} \right| = k$ ,  $k \neq 1$  where,  $z = x + iy$ ,  $\alpha = \alpha_1 + i\alpha_2$ ,  $\beta = \beta_1 + i\beta_2$ . [IIT - 2004]

83. If one of the vertices of the square circumscribing the circle  $|z-1| = \sqrt{2}$  is  $2 + \sqrt{3}i$ . Find the other vertices of the square. [IIT - 2005]

84. Let  $\omega$  be the complex number  $\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}$ . Then the number of distinct complex

number  $z$  satisfying  $\begin{vmatrix} z+1 & \omega & \omega^2 \\ \omega & z+\omega^2 & 1 \\ \omega^2 & 1 & z+\omega \end{vmatrix} = 0$  is equal to [IIT - 2010]

## F. Integer Answer Type

85. If  $z$  is any complex number satisfying  $|z-3-2i| \leq 2$ , then the minimum value of  $|2z-6+5i|$  is [IIT - 2011]

## G. Comprehension

Let  $A, B, C$  be three sets of complex numbers as defined below [IIT - 2008]

$$A = \{z : \operatorname{Im} z \geq 1\}$$

$$B = \{z : |z-2-i| = 3\}$$

$$C = \{z : \operatorname{Re}((1-i)z) = \sqrt{2}\}.$$

86. The number of elements in the set  $A \cap B \cap C$  is (A) 0 (B) 1 (C) 2 (D)  $\infty$

87. Let  $z$  be any point in  $A \cap B \cap C$ . Then  $|z+1-i|^2 + |z-5-i|^2$  lies between (A) 25 and 29 (B) 30 and 34 (C) 35 and 39 (D) 40 and 44

88. Let  $z$  be any point in  $A \cap B \cap C$  and let  $w$  be any point satisfying  $|w-2-i| < 3$ . Then,  $|z| - |w| + 3$  lies between

- (A) -6 and 3 (B) -3 and 6  
(C) -6 and 6 (D) -3 and 9

## Paragraph for Q Nos 89 & 90

Let  $a, b$ , and  $c$  be three real numbers satisfying

$$[abc] \begin{bmatrix} 1 & 9 & 7 \\ 8 & 2 & 7 \\ 7 & 3 & 7 \end{bmatrix} = [000] \dots (E)$$

89. If the point  $P(a, b, c)$ , with reference to (E), lies on the plane  $2x + y + z = 1$ , then the value of  $7a + b + c$  is [IIT - 2011]

- (A) 0 (B) 12 (C) 7 (D) 6

90. Let  $\omega$  be a solution of  $x^3 - 1 = 0$  with  $\operatorname{Im}(\omega) > 0$ . If  $a = 2$  with  $b$  and  $c$  satisfying (E), then the value of [IIT - 2011]

$$\frac{3}{\omega^a} + \frac{1}{\omega^b} + \frac{3}{\omega^c}$$

is equal to

- (A) -2 (B) 2 (C) 3 (D) -3

**G. Match the Columns**

91. Match the statements in **Column-I** with those in **Column-II**.

[Note : Here  $z$  takes values in the complex plane and  $\text{Im } z$  and  $\text{Re } z$  denote, respectively, the imaginary part and the real part of  $z$ .] [IIT - 2010]

**Column-I**

- (A) The set of points  $z$  satisfying  $|z - i| = |z + i|$  is contained in or equal to
- (B) The set of points  $z$  satisfying  $|z + 4| + |z - 4| = 10$  is contained in or equal to
- (C) If  $|w| = 2$ , then the set of points  $z = w - \frac{1}{w}$  is contained in or equal to
- (D) If  $|w| = 1$ , then the set of points  $z = w + \frac{1}{w}$  is contained in or equal to

**Column-II**

- (p) an ellipse with eccentricity  $\frac{4}{5}$
- (q) the set of points  $z$  satisfying  $\text{Im } z = 0$
- (r) the set of point  $z$  satisfying  $|\text{Im } z| \leq 1$
- (s) the set of point  $z$  satisfying  $|\text{Re } z| \leq 2$
- (t) the set of point  $z$  satisfying  $|z| \leq 3$

**Answers****Practice Problems—A**

2.  $(1, 1) \left(0, \frac{5}{2}\right)$
3.  $(-2, -2), (-2, 2)$
4. (i)  $-i$  (ii)  $18 + i$  (iii)  $1 + i$  (iv)  $1 + i$
5. (i)  $3 + 7i, -22 + 7i$  (ii)  $-8 - 10i, 21 - 24i$
6. (i)  $1 - \sqrt{2} + (\sqrt{6} - \sqrt{3})i, \sqrt{2}$
- (ii)  $2\sqrt{b}i, \frac{a^2 - b}{a^2 + b} + \frac{2a\sqrt{b}}{a^2 + b}i$
7. (i)  $-2$  (ii)  $0$
8.  $-5/318$
9. (i)  $x > 0, y = 0$
- (ii)  $x = 1, y = 0; x = \frac{-1}{2}, y = \frac{\pm\sqrt{3}}{2}$
- (iii)  $x = 1, y = 0$
10.  $-1 - i$
11.  $z_1 = 1 - i, z_2 = i$

12. (i)  $\pm(4 - 3i)$  (ii)  $\pm \frac{\sqrt{2}}{2}(\sqrt{3} - i)$

13.  $-\frac{3}{2}i$ .

**Practice Problems—B**

1. The set of all points of the rectangle  $|q| \leq 1, |p| \leq 2$ .
2. (i) The straight lines defined by the equation  $ay + bx = 0$ .
- (ii) The straight lines defined by the equation  $y = -b$ .
3. The OX axis and points with coordinates which satisfy the conditions
- $$x = -\frac{1}{2}, -\frac{\sqrt{3}}{2} < y < \frac{\sqrt{3}}{2}.$$
6.  $y^2 - 2xy + 4y = 0$

**Practice Problems—C**

1. (i)  $1$  (ii)  $\sqrt{2}$
3.  $5 \left[ \cos \left( \pi + \arccos \frac{3}{5} \right) + i \sin \left( \pi + \arccos \frac{3}{5} \right) \right]$

4. (i)  $1 = \cos 0 + i \sin 0$   
 (ii)  $\frac{1}{2} - \frac{\sqrt{3}}{2}i = \cos(-\frac{2\pi}{3}) + i \sin(-\frac{2\pi}{3})$
5. (i)  $\frac{5}{3}[\cos(-130^\circ) + i \sin(-130^\circ)]$   
 (ii)  $\sqrt{2} \sin \frac{\pi}{5} [\cos(-\frac{11\pi}{20}) + i \sin(-\frac{11\pi}{20})]$
6. (i)  $\cos(50^\circ) + i \sin(50^\circ)$  (ii)  $\frac{1}{64} - \frac{\sqrt{3}}{64}i$   
 (iii)  $-2$
7. (i)  $\frac{1}{\cos^4 2} (\cos 8 + i \sin 8)$   
 (ii)  $-32 \cos^5 \frac{3\pi}{5} (\cos \frac{\pi}{2} + i \sin \frac{\pi}{2})$
8. (i)  $e^2 \cos 1 - i e^2 \sin 1$  (ii)  $i$   
 (iii)  $e^7 \cos 3 + i e^7 \sin 3$
9. (i)  $\sqrt{2}i$  (ii)  $-i$
10. (i)  $x = y = 0$   
 (ii)  $x = 0, y = (2n + 1)\pi$ , where  $n$  is any integer  
 (iii)  $x = 1, y = \frac{1}{2}\pi + 2n\pi$ , where  $n$  is any integer.
11. (i)  $4e^{\frac{7}{6}\pi i}$  (ii)  $e^{\frac{6}{7}\pi i}$
12. (i)  $8e^{-\frac{1}{4}\pi i} = 4\sqrt{2} - 4\sqrt{2}i$   
 (ii)  $2^7 e^{\frac{3}{2}\pi i} = -2^7 i$
13.  $t(7 + i), t > 0$   
 15.  $(7 + i)t, t > 0$ .
16. (i)  $2 - \frac{3}{2}i$  (ii)  $0, 3i, -3i$  (iii)  $ti$ ,  $t$  is real.
17.  $\alpha \in \left[-\frac{\sqrt{5}}{2}, \frac{\sqrt{5}}{2}\right]$  then  $x = \frac{\alpha^2 \pm \alpha\sqrt{5 - 4\alpha^2}}{\alpha^2 - 1}$

**Practice Problems—D**

5.  $0, -1, \frac{1}{2} - \frac{\sqrt{3}}{2}i, \frac{1}{2} + \frac{\sqrt{3}}{2}i$   
 6. (i)  $3 + 5i, 2 - 3i$  (ii)  $5 + i, 3 + 2i$   
 7.  $(3, 1)$  &  $(-3, -1)$   
 8.  $1 - 2i, \frac{1}{2} - \frac{\sqrt{3}}{2}i, \frac{1}{2} + \frac{\sqrt{3}}{2}i$   
 9.  $a > 1.25$   
 10.  $(4 + 3i)/2$

**Practice Problems—E**

6.  $31$  &  $19$   
 7.  $||c| - |a||, |c|$

**Practice Problems—F**

1. (a) positive real semi-axis.  
 (b) positive imaginary semi-axis.  
 (c) upper half plane leaving real axis.  
 (d) entire complex plane leaving the origin.
2.  $n\pi + (\pi/4)$
3.  $-\frac{7\pi}{18}$
4.  $\pi$
5.  $\begin{cases} \frac{\varphi}{2} & \text{if } 0 \leq \varphi < \frac{\pi}{3}, \pi < \varphi < \frac{5\pi}{3} \\ \frac{\varphi}{2} + \pi & \text{if } \frac{\pi}{3} < \varphi < \pi, \frac{5\pi}{3} < \varphi < 2\pi \end{cases}$ ; if  $\varphi = \pi/3$   
 or  $\pi$  or  $5\pi/3$  then  $z_1 = 0$ .

**Practice Problems—G**

1.  $-10 + 4i$   
 2.  $3 - \frac{9}{2}i$ .

**Practice Problems—H**

1. (a) A circle of radius 20 with centre at the origin;  
 (b) A circle of radius 5 with centre at the point  $(2, 0)$ ;

- (c) A circle of radius 15 with centre at the point  $(-1, 0)$
2. (i) On a circle and inside it : centre  $(0, -1)$  and  $r = 1$ ;  
 (ii) On a circle and inside it : centre  $(-2, 2)$  and  $r = 4$ ;  
 (iii) in a ring between the circles with centre  $(-3, 2)$  and  $r_1 = 1, r_2 = 2$ ;  
 (iv) a locus of points equidistant from the points  $(0, -1), (2, 0)$ ;  
 (v) a point equidistant from three points:  $(1, 0), (3, 0), (0, 1)$ .  
 (vi) The region between two rays from origin,  $\theta = \pi/3$  and  $\theta = 5\pi/6$ , excluding the ray  $\theta = \pi/3$ .
3. (i) point  $(0, 1/\sqrt{3})$ , (ii) no such point,  
 (iii) A circle of radius 1 with centre at the origin
5. If  $z$  and  $z'$  are two complex numbers with a non-real ratio, show that  $z$  and  $z'$  cannot lie on a straight line passing through the origin.
6. If  $z$  is a complex number and  $\frac{z+1}{z-i}$  is purely imaginary, show that  $z$  lies on the circle whose centre is  $\frac{1}{2}(-1+i)$  and radius is  $\frac{1}{\sqrt{2}}$ .
7.  $\frac{7}{6} + \frac{5}{6}i$
8.  $\left(-1 - \frac{1}{\sqrt{2}}\right) + i\left(1 + \frac{1}{\sqrt{2}}\right)$
9.  $\frac{12}{5} + \frac{16}{5}i$
10. (i) interior of a circle of radius 10 with centre at the point  $(0, 10)$ , except the centre;  
 (ii) a circle of radius 3 with centre at the point  $(-3, 0)$
11. (i)  $w = z + a$  (ii)  $w = tz$  where  $i$  is real  
 (iii)  $w = (\cos\alpha + i \sin\alpha)z$

$$(iv) w = az + b \quad (v) w = 1/z$$

$$12. -\frac{21}{10} < a < -\frac{5}{6}.$$

### Practice Problems—I

1. (a)  $i$  (b)  $2i^{n-1}$
6.  $-1$
7. (i)  $-\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}i, \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}i$   
 (ii)  $\pm\sqrt{3} + i, -2i$
8. (i)  $\sqrt[5]{2}e^{i\frac{k\pi}{15}}, k = 1, 7, 13, 19, 25$ .  
 (ii)  $\pm\sqrt{3} \pm i, \pm 2i$
9.  $2e^{\frac{2k\pi i}{9}}, k = 1, 4, 7$
10.  $\pm \frac{3}{\sqrt[4]{2}} \left( \sqrt{\frac{\sqrt{2}+1}{2}} - i\sqrt{\frac{\sqrt{2}-1}{2}} \right)$   
 and  $\pm \frac{3}{\sqrt[4]{2}} \left( \sqrt{\frac{\sqrt{2}-1}{2}} + i\sqrt{\frac{\sqrt{2}+1}{2}} \right)$
11. (a)  $(i, i), (-i, -i)$  (b)  $(i, i), (-i, -i)$
12.  $\frac{1}{2} - i\frac{1}{2} \cot \frac{4k+1}{12}\pi, k = 0, 1, 2$
13.  $1, \frac{1}{2}(-1 \pm i\sqrt{3}), 2 - 1 \pm i\sqrt{3}$ .

### Practice Problems—J

$$12. \frac{\sqrt{3}}{2} - \frac{i}{2}, -\frac{\sqrt{3}}{2} - \frac{i}{2}, i$$

### Practice Problems—K

$$1. \cos \frac{2r\pi}{15} \pm i \sin \frac{2r\pi}{15}, r = 1, 2, 4, 5, 7$$

2.  $(\sqrt{2}i), (-1+i\sqrt{3}), \left(\frac{1}{\sqrt{2}}+i\frac{\sqrt{3}}{\sqrt{2}}\right),$

$$\left(-\frac{1}{\sqrt{2}}+i\frac{\sqrt{3}}{\sqrt{2}}\right)$$

4. 0

6.  $z^2 + z + \frac{1}{4} \sec^2 \left( \frac{\pi}{2n+1} \right) = 0$

7.  $0, \cos \frac{2m\pi}{n} + i \sin \frac{2m\pi}{n}, m = 0, 1, \dots, n-1$

8.  $(Z+1)(Z^2-2Z \cos 36^\circ+1)(Z^2-2Z \cos 108^\circ+1)$

### Practice Problems—L

2. (i)  $\cos 9\theta = \cos^9 \theta - 36 \cos^7 \theta \sin^2 \theta + 126 \cos^5 \theta \sin^4 \theta - 84 \cos^3 \theta \sin^6 \theta + 9 \cos \theta \sin^8 \theta$

(ii)  $\sin 10\theta = 10 \cos^9 \theta \sin \theta - 120 \cos^7 \theta \sin^3 \theta + \dots + 10 \cos \theta \sin^9 \theta$

(iii)  $\operatorname{cosec} \theta \sin 7\theta = 7 \cos^6 \theta - 35 \cos^4 \theta \sin^2 \theta + 21 \cos^2 \theta \sin^4 \theta - \sin^6 \theta$

5.  $(-1)^{n/2} \sin^n \alpha$  when  $n$  is even ; 0 when  $n$  is odd.

6.  $\cot y (1 - \cos^n y \cos ny)$

### Practice Problems—M

2. Yes

### Practice Problems—N

4. a circle

6. infinite

7.  $(2, 0), \sqrt{3}$

### Objective Exercises

1. A

2. C

3. B

4. A

5. A

6. C

7. D

10. C

13. A

16. C

19. A

22. D

25. D

28. D

31. C

34. C

37. C

40. A

43. B

46. CD

49. BC

52. AD

55. AB

58. ACD

61. ABCD

64. ABCD

67. BCD

70. ABCD

73. A

76. B

79. B

82. A

85. A

88. A

91. C

94. A

8. C

11. C

14. C

17. B

20. D

23. A

26. B

29. C

32. D

35. C

38. A

41. A

44. A

47. AB

50. ACD

53. BC

56. AB

59. ABD

62. ABCD

65. ABC

68. ABCD

71. C

74. A

77. C

80. D

83. D

86. A

89. D

92. D

95. D

9. A

12. C

15. C

18. B

21. D

24. A

27. A

30. B

33. D

36. A

39. A

42. A

45. A

48. ABC

51. AC

54. AC

57. ABC

60. BD

63. BD

66. AB

69. ABCD

72. A

75. C

78. A

81. D

84. C

87. A

90. B

93. A

96. (A) – Q, (B) – R, (C) – P, (D) – S

97. (A) – S, (B) – Q, (C) – P, (D) – P

98. (A) – Q, (B) – Q, (C) – S, (D) – R

99. (A) – P, (B) – P, (C) – R, (D) – Q

100. (A) – R, (B) – Q, (C) – P, (D) – S

### Review Exercises for JEE Advanced

2.  $\pm \left( \frac{\sqrt{6}}{2} - \frac{1}{\sqrt{2}}i \right); \pm \left( \frac{1}{\sqrt{6}} - \frac{3}{2}i \right)$



5. (i) exterior to a circle of radius 10 with centre at the point (1, 0); (ii) inside a circle of radius 5 with centre at the origin.

6.  $z = -1 - i$  for  $a = 1$ ,  $z = \frac{-a^2 \pm \sqrt{2-a^2}}{a^2-1} - i$

for  $i < a \leq \sqrt{2}$ . For  $a > \sqrt{2}$  the equation has no solution.

7.  $z_1 = 1$ ,  $w_1 = -1$  and  $z_2 = -1$ ,  $w_2 = 1$

9.  $-\frac{3}{2} - \frac{17}{4}i, -\frac{3}{2} - 2i$

10. All real and all purely imaginary numbers.

11.  $a > 2$ .

13.  $\left(4 - \frac{4}{\sqrt{5}}\right) + i\left(1 + \frac{2}{\sqrt{5}}\right)$  and  $\left(4 + \frac{4}{\sqrt{5}}\right) + i\left(1 + \frac{2}{\sqrt{5}}\right)$

14.  $x = y = z = \frac{a+b+c}{3}$

15. 1

16.  $[-3, -2]$

20.  $\frac{1}{3} [(2\omega^2 - 2\omega + 7)z^2 + (2\omega + 5)z + \omega^2 + 6]$

22. For  $a > 0$ ,  $z = (\sqrt{2}a + 1)\left(\frac{1}{\sqrt{2}} - \frac{i}{\sqrt{2}}\right)$ ;

For  $a < 0$ ,  $z = |\sqrt{2}a - 1|\left(-\frac{1}{\sqrt{2}} + \frac{i}{\sqrt{2}}\right)$

25. Let P, Q, R be the centroids of the triangles drawn outwards on BC, CA, AB. Prove that

$$\begin{aligned}\overline{QA} &= \overline{CA} \cdot \frac{1}{\sqrt{3}}(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6}), \overline{AR} \\ &= \overline{AB} \cdot \frac{1}{\sqrt{3}}(\cos \frac{\pi}{6} - i \sin \frac{\pi}{6})\end{aligned}$$

Hence show that

$$\overline{QR} = \frac{1}{2}\overline{CB} + \frac{i}{2\sqrt{3}}(\overline{CA} - \overline{AB}), \text{ and}$$

that  $\overline{RP}$  has a similar value.

Hence show that  $\overline{RQ} = \overline{RP}(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3})$ .

26.  $4 + (-2)^{3n}$

27.  $z = 2e^{i\pi/4}$

29.  $z = \frac{5\sqrt{2}-7}{10\sqrt{2}-6} - \frac{5i}{10\sqrt{2}-6},$

$$\frac{5\sqrt{2}+7}{10\sqrt{2}+6} - \frac{5i}{10\sqrt{2}+6},$$

$$\frac{5\sqrt{2}+7}{10\sqrt{2}+6} + \frac{5i}{10\sqrt{2}+6},$$

$$\frac{5\sqrt{2}-7}{10\sqrt{2}-6} + \frac{5i}{10\sqrt{2}-6}$$

34.  $\left(3 - \frac{\pi}{2} - \frac{5}{4}\sin^{-1}\frac{4}{5}\right)$  units

### Target Exercises for JEE Advanced

5. a

7.  $\omega, \omega^2$

9.  $x^8 - x^7 + x^5 - x^4 + x^3 - x + 1 = 0,$

where  $x = e^{\frac{i2\pi}{15}}$ ,  $r = 1, 2, 4, 7, 8, 11, 13, 14$ .

$t^4 - t^3 - 4t^2 + 4t + 1 = 0$ , where  $t = 2 \cos \left(\frac{2r\pi}{15}\right)$ ,

$r = 1, 2, 4, 7$ .

13.  $-7/2$

16.  $\overline{bz} - b\overline{z} = \overline{bz_3} - b\overline{z_3}$

20.  $e^{\cos^2 \theta} \sin(\cos \theta \sin \theta)$

21.  $\left(2 \sin \frac{\alpha}{2}\right)^{-1/2} \cos \frac{\pi - \alpha}{4}$

33.  $-n/2$

34.  $16x^4 - 8x^3 - 12x^2 + 4x + 1 = 0$

### Previous Years' Questions (JEE Advanced)

1.  $2n\pi, n\pi + \frac{\pi}{4}$

2.  $(a^2 + b^2)(|z_1|^2 + |z_2|^2)$

3.  $2 - \sqrt{3}, 2 - \sqrt{3}$

4.  $\left(3 - \frac{1}{2}i\right)$  or  $\left(1 - \frac{3}{2}i\right)$

5.  $-2 + 0i$  &  $1 - i\sqrt{3}$

6.  $\left[\frac{n(n+1)}{2}\right]^2 - n$

7. T

8. T

9. F

10. T

11. B

12. D

13. A

14. B

15. B

16. B

17. B

18. B

19. C

20. D

21. D

22. D

23. B

24. B

25. B

26. C

27. D

28. D

29. A

30. B

31. D

32. C

33. D

34. D

35. B

36. D

37. C

38. A

39. A

40. D

41. C

42. B

43. B

44. A

45. B

46. A

47. B

48. D

49. D

50. D

51. D

52. A

53. D

54. D

55. ABC

56. AD

57. ACD

58.  $\left(\frac{1}{5+3\cos\theta}\right) + \left(\frac{-2\cot\theta/2}{5+3\cos\theta}\right)i$

62.  $x = 3 = -1$

68. (i) B (ii) A

70.  $|z - 7 - ib| = \sqrt{48 + b^2}, b \in \mathbb{R}$

73.  $\frac{\sqrt{3}}{2} - \frac{i}{2}, -\frac{\sqrt{3}}{2} - \frac{i}{2}, i$

78.  $z = \frac{(29 + 20\sqrt{2}) + i(\pm 15 + 25\sqrt{2})}{82};$

$$\frac{(290 - 20\sqrt{2}) + i(\pm 15 - 25\sqrt{2})}{82}$$

82. centre  $\frac{\alpha - k^2\beta}{1 - k^2}$ , radius =  $\frac{k}{|1 - k^2|} |\alpha - \beta|$

83.  $(1 - \sqrt{3}) + i, -i\sqrt{3}, (\sqrt{3} + 1) - i$

84. 1

85. 5

86. B

87. C

88. BD

89. D

90. A

91. (A) - QR, (B) - P, (C) - PST, (D) - QRST

# 2

## Determinant

### 2.1 INTRODUCTION

If the equations  $a_1x + b_1 = 0$  and  $a_2x + b_2 = 0$  have the same solution, then

$$-\frac{b_1}{a_1} = -\frac{b_2}{a_2}, \quad \text{or,} \quad \frac{a_1}{a_2} = \frac{b_1}{b_2}$$

$$\text{or,} \quad a_1 b_2 - a_2 b_1 = 0 \quad \dots(1)$$

Next we consider the system of equations

$$a_1x + b_1y + c_1 = 0,$$

$$a_2x + b_2y + c_2 = 0,$$

$$a_3x + b_3y + c_3 = 0.$$

If these three equations have the same solution, then the values of  $x$  and  $y$ , obtained from the last two, will satisfy the first.

From  $a_2x + b_2y + c_2 = 0$ ,  $a_3x + b_3y + c_3 = 0$ , we get by cross-multiplication,

$$\frac{x}{b_2c_3 - b_3c_2} = \frac{y}{c_2a_3 - c_3a_2} = \frac{1}{a_2b_3 - a_3b_2}$$

$$\text{or, } x = \frac{b_2c_3 - b_3c_2}{a_2b_3 - a_3b_2}, \quad y = \frac{c_2a_3 - c_3a_2}{a_2b_3 - a_3b_2}.$$

Putting the values of  $x$  and  $y$  in the first, we have

$$a_1(b_2c_3 - b_3c_2) + b_1(c_2a_3 - c_3a_2) + c_1(a_2b_3 - a_3b_2) = 0,$$

$$\text{or, } a_1b_2c_3 - a_1b_3c_2 + a_2b_3c_1 - a_2b_1c_3 + a_3b_1c_2 - c_3b_2c_1 = 0. \quad \dots(2)$$

The result (1) and (2) obtained by eliminating  $x$  or  $y$  from the given equations, are called eliminants. They may be written in a compact form which one can write from the given set of equations at a glance. A determinant of order two is written as

$$\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix} \text{ and is equal to } a_1 b_2 - a_2 b_1.$$

A determinant of order three is written as

$$\Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

where  $R_1 = [a_1, b_1, c_1]$ ,  $R_2 = [a_2, b_2, c_2]$ ,  $R_3 = [a_3, b_3, c_3]$  are its rows, and

$$C_1 = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix}, \quad C_2 = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}, \quad C_3 = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix}$$

are its columns. Its value is equal to

$$a_1b_2c_3 - a_1b_3c_2 + a_2b_3c_1 - a_2b_1c_3 + a_3b_1c_2 - c_3b_2c_1.$$

### Formation of a Determinant

Let us take nine quantities  $a_1, b_1, c_1, a_2, b_2, c_2, a_3, b_3, c_3$  and arrange them in a square with three horizontal rows and three vertical columns as follows :

$$\begin{array}{ccc} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{array}$$

This is a square array of the letters and since each row and each column contains three letters, we call this a square array of the third order. All the possible products of the nine quantities taken three at a time subject to the condition that, of the three quantities in each product one and only one is taken from each of the rows and one and only one is taken from each of the columns, and written in alphabetical order  $a, b, c$  are

$a_1 b_2 c_3, a_1 b_3 c_2, a_2 b_3 c_1, a_2 b_1 c_3, a_3 b_1 c_2$  and  $a_3 b_2 c_1$

Let us consider the sign of a product as positive or negative if the number of inversions of the natural order 1, 2 and 3 in the suffixes is even or odd.

$$a_1 b_2 c_3 - a_1 b_3 c_2 + a_2 b_3 c_1 - a_2 b_1 c_3 + a_3 b_1 c_2 - a_3 b_2 c_1 \dots (3)$$

for there is no inversion of the natural order of the suffixes in  $a_1 b_2 c_3$ , there is one inversion in  $a_1 b_3 c_2$  since 3 precedes 2, there are two inversions in  $a_2 b_3 c_1$  since 2 and 3 both precede 1 and so on.



The number of inversions in the suffixes is odd or even according as the number of interchanges of two suffixes necessary

to bring the suffixes in the natural order is odd or even.

The expression (3) is the same as the left-hand side of (2). The expression (3) is defined to be the determinant of the third order formed by the nine quantities  $a_1, b_1, c_1, a_2, b_2, c_2, a_3, b_3, c_3$  and is written as

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

The quantities  $a_1, b_1, c_1, \dots$  are known as elements or constituents and each of the six products, with the proper sign attached is called a term of the determinant.

Similarly, the expression  $a_1 b_2 - a_2 b_1$  which is the same as the left-hand side of (1) is defined to be a determinant of the second order and is written as

$$\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}$$

### General Term

The terms in a determinant consist of all possible products of the  $n$  elements of this matrix located in different rows and different columns, that is,

products of the form  $a_{1\alpha_1} a_{2\alpha_2} \dots a_{n\alpha_n}$  where the subscripts  $\alpha_1, \alpha_2, \dots, \alpha_n$  constitute an arrangement of the numbers 1, 2, ...,  $n$ . The number of such products is equal to the number of different arrangements of  $n$  symbols, or  $n!$ .

The  $n^{\text{th}}$ -order determinant associated with the matrix is the algebraic sum of  $n!$  terms which is constructed in the following fashion : the terms are all possible products of the  $n$  elements of the matrix taken one in each row and each column; in the term having a plus sign has an even number of inversion of natural order 1, 2, ...,  $n$ , in the suffixes The expanded form of the determinant has  $n!$  terms; half of these have the sign  $+$  and the rest half have the sign  $-$ .

### Determinant of A Square Matrix

To every square matrix  $A = [a_{ij}]$  of order  $n$ , we can associate a number (real or complex) called determinant of the square matrix  $A$ . A determinant is a polynomial of the elements of a square matrix. It is a scalar. It is the sum of certain products of the elements of the matrix from which it is derived, each product being multiplied by  $+1$  or  $-1$  according to certain rules.

Determinants are defined only for square matrices – the determinant of a non-square matrix is undefined. Determinant of a square matrix  $A$  is denoted by  $|A|$  or  $\det A$ .

To begin, we have the elementary result that the determinant of a  $1 \times 1$  matrix is the value of its sole element.

Let  $A = [a]_{1 \times 1}$  be an  $1 \times 1$  matrix. Determinant  $A$  is defined as  $|A| = a$

For example, if  $A = [-3]_{1 \times 1}$  then  $|A| = -3$

The value of a second-order determinant is calculated as follows.

For example, the determinant of

$$A = \begin{bmatrix} 7 & 3 \\ 4 & 6 \end{bmatrix} \text{ is written } |A| = \begin{vmatrix} 7 & 3 \\ 4 & 6 \end{vmatrix}$$

and is calculated as  $|A| = 7(6) - 3(4) = 30$ .

This illustrates the general result for expanding a second-order determinant: from the product of the diagonal terms subtract the product of the off-diagonal terms. In other words, the determinant of a  $2 \times 2$  matrix consists of the product (multiplied by +1) of the diagonal terms plus the product (multiplied by -1) of the off-diagonal terms. Hence, in general

$$\begin{aligned} |A| &= \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} + (-1) a_{12}a_{21} \\ &= a_{11}a_{22} - a_{12}a_{21} \end{aligned}$$

A third-order determinant can be expanded as a linear function of three second-order determinants derived from it. Their coefficients are elements of a row (or column) of the main determinant, each product being multiplied by +1 or -1. For example, the expansion of

$$|A| = \begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 10 \end{vmatrix}$$

based on the elements of the first row, 1, 2 and 3, is

$$\begin{aligned} |A| &= 1(+1) \begin{vmatrix} 5 & 6 \\ 8 & 10 \end{vmatrix} + 2(-1) \begin{vmatrix} 4 & 6 \\ 7 & 10 \end{vmatrix} + 3(+1) \begin{vmatrix} 4 & 5 \\ 7 & 8 \end{vmatrix} \\ &= 1(50 - 48) - 2(40 - 42) + 3(32 - 35) = -3. \end{aligned}$$

The determinant that multiplies each element of the chosen row (in this case the first row) is the determinant derived from  $|A|$  by crossing out the row

and column containing the element concerned. For example, the first element, 1, is multiplied by the

determinant  $\begin{vmatrix} 5 & 6 \\ 8 & 10 \end{vmatrix}$  which is obtained from  $|A|$

through crossing out the first row and column; and the element 2 is multiplied (apart from the factor of -1) by the determinant derived from  $|A|$  by deleting the row and column containing that element - namely, the first row and second column

$$\text{-- so leaving } \begin{vmatrix} 4 & 6 \\ 7 & 10 \end{vmatrix}.$$

The foregoing example illustrates the expansion of the general third-order determinant

$$|A| = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} \text{ as follows:}$$

$$\begin{aligned} |A| &= a_{11}(+1) \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} + a_{12}(-1) \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} \\ &\quad + a_{13}(+1) \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix} \end{aligned}$$

$$\begin{aligned} &= a_{11} a_{22} a_{33} - a_{11} a_{23} a_{32} + a_{12} a_{23} a_{31} - a_{12} a_{21} a_{33} \\ &\quad + a_{13} a_{21} a_{32} - a_{13} a_{22} a_{31}. \end{aligned}$$

### STUDY TIP

In a matrix the number of rows and the number of columns may be different. But in a determinant the number of rows is equal to the number of columns. Also a determinant is not simply a system of numbers. It has got numerical value. A matrix is just a system of numbers. It has got no numerical value.

For example, the value of the determinant  $\begin{vmatrix} 4 & 2 \\ 1 & 3 \end{vmatrix}$  is the number  $4 \times 3 - 1 \times 2$ , i.e., the number 10 while

the matrix  $\begin{pmatrix} 4 & 2 \\ 1 & 3 \end{pmatrix}$  has got no numerical value.

$$\text{Also } \begin{vmatrix} 4 & 2 \\ 1 & 3 \end{vmatrix} = \begin{vmatrix} 10 & 6 \\ 5 & 4 \end{vmatrix} = 10 \times 4 - 5 \times 6 = 10.$$

$$\therefore \begin{vmatrix} 4 & 2 \\ 1 & 3 \end{vmatrix} = \begin{vmatrix} 10 & 6 \\ 5 & 4 \end{vmatrix}.$$

But  $\begin{bmatrix} 4 & 2 \\ 1 & 3 \end{bmatrix} \neq \begin{bmatrix} 10 & 6 \\ 5 & 4 \end{bmatrix}.$

### Minor and Cofactor

There are simpler methods for evaluating determinants. They are based on the fact that a determinant of order  $n$  may be expressed in terms of a determinant of lower order. For this purpose we introduce the following notion.

#### Minor

Let there be a determinant  $\Delta$  of order  $n$ . Take an integer  $k$  which satisfies the condition  $1 \leq k \leq n-1$ , and in the determinant  $\Delta$  choose arbitrary  $k$  rows and  $k$  columns. The elements which lie at the intersection of these rows and columns, that is, which belong to one of the chosen rows and to one of the chosen columns will obviously form a matrix of order  $k$ . The determinant of this matrix is called a minor of order  $k$  of the determinant  $\Delta$ . We can also say that the  $k^{\text{th}}$  order minor is a determinant obtained by striking out  $n - k$  rows and  $n - k$  columns in  $\Delta$ . In particular, after striking out one row and one column in the determinant we obtain a minor of  $(n - 1)^{\text{th}}$  order; on the other hand, separate elements of determinant  $\Delta$  will be minors of the first order.

Consider the determinant

$$\Delta = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}.$$

If we leave the row and the column passing through the element  $a_{ij}$ , then the second order determinant thus obtained is called the minor of the element  $a_{ij}$  and we shall denote it by  $M_{ij}$ . In this way we can get 9 minors corresponding to the 9 elements of  $\Delta$ .

For example,

$$\text{the minor of the element } a_{21} = \begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix} = M_{21},$$

$$\text{the minor of the element } a_{32} = \begin{vmatrix} a_{11} & a_{13} \\ a_{21} & a_{23} \end{vmatrix} = M_{32},$$

$$\text{the minor of the element } a_{11} = \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} = M_{11},$$

and so on.

$$\text{For example if } \Delta = \begin{vmatrix} a & b & c \\ p & q & r \\ x & y & z \end{vmatrix}$$

$$\text{then } M_{11} = \begin{vmatrix} p & r \\ y & z \end{vmatrix} = qz - yr$$

$$M_{23} = \begin{vmatrix} a & b \\ x & y \end{vmatrix} = ay - bx.$$

Thus, if  $\Delta$  is a determinant, then the minor of element  $a_{ij}$ , denoted by  $M_{ij}$  is defined as the determinant of the submatrix obtained by deleting  $i^{\text{th}}$  row and  $j^{\text{th}}$  column of  $\Delta$ .

#### Cofactor

The minor  $M_{ij}$  multiplied by  $(-1)^{i+j}$  is called the cofactor of the element  $a_{ij}$ . We shall denote the cofactor of an element by the corresponding capital letter. With this notation, cofactor of  $a_{ij} = A_{ij} = (-1)^{i+j} M_{ij}$ . For example, the cofactor of the element

$$a_{21} = A_{21} = (-1)^{2+1} M_{21} = - \begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix},$$

$$\text{the cofactor of the element } a_{32} = A_{32} = - \begin{vmatrix} a_{11} & a_{13} \\ a_{21} & a_{23} \end{vmatrix},$$

$$\text{the cofactor of the element } a_{11} = A_{11} = - \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix},$$

and so on.

Thus the cofactor of any element  $a_{ij} = (-1)^{i+j} \times$  the determinant obtained by leaving the row and the column passing through that element.

$$\text{For example if } \Delta = \begin{vmatrix} a & b & c \\ p & q & r \\ x & y & z \end{vmatrix}$$

$$\text{then } A_{11} = (-1)^{1+1} \begin{vmatrix} q & r \\ y & z \end{vmatrix} = qz - yr = M_{11}$$

$$A_{21} = (-1)^{2+1} \begin{vmatrix} b & c \\ y & z \end{vmatrix} = -(bz - cy) = -M_{21}$$

### Evaluation of determinant using minors and cofactors

In terms of the notation of minors, the preceding expansion of

$$\begin{aligned} |A| &= \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} \\ &= a_{11} a_{22} a_{33} - a_{11} a_{23} a_{32} + a_{12} a_{23} a_{31} - a_{12} a_{21} a_{33} + a_{13} a_{21} a_{32} - a_{13} a_{22} a_{31} \end{aligned}$$

is same as

$$\begin{aligned} |A| &= a_{11}(+1) \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} + a_{12}(-1) \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} \\ &\quad + a_{13}(+1) \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix} \\ &= (-1)^{1+1} a_{11} M_{11} + (-1)^{1+2} a_{12} M_{12} + \\ &\quad (-1)^{1+3} a_{13} M_{13} \\ &= a_{11} M_{11} - a_{12} M_{12} + a_{13} M_{13}. \end{aligned}$$

This method of evaluating a determinant is known as expansion by the elements of a row (or column) or sometimes as expansion by minors. It has been illustrated using elements of the first row but can also be applied to the elements of any row (or column).

### Expansion along second row ( $R_2$ )

$$|A| = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

Expanding along  $R_2$ , we get

$$|A| = (-1)^{2+1} a_{21} \begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix}$$

$$\begin{aligned} &+ (-1)^{2+2} a_{22} \begin{vmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{vmatrix} + (-1)^{2+3} a_{23} \begin{vmatrix} a_{11} & a_{12} \\ a_{31} & a_{32} \end{vmatrix} \\ &= -a_{21} (a_{12} a_{33} - a_{32} a_{13}) + a_{22} (a_{11} a_{33} - a_{31} a_{13}) \\ &\quad - a_{23} (a_{11} a_{32} - a_{31} a_{12}) \\ |A| &= -a_{21} a_{12} a_{33} + a_{21} a_{32} a_{13} + a_{22} a_{11} a_{33} \\ &\quad - a_{22} a_{31} a_{13} - a_{23} a_{11} a_{32} + a_{23} a_{31} a_{12} \\ &= a_{11} a_{22} a_{33} - a_{11} a_{23} a_{32} - a_{12} a_{21} a_{33} + a_{12} a_{23} a_{31} \\ &\quad + a_{13} a_{21} a_{32} - a_{13} a_{22} a_{31} \end{aligned}$$

### Expansion along first column ( $C_1$ ):

$$|A| = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

By expanding along  $C_1$ , we get

$$\begin{aligned} |A| &= a_{11} (-1)^{1+1} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} \\ &\quad + a_{21} (-1)^{2+1} \begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix} \\ &\quad + a_{31} (-1)^{3+1} \begin{vmatrix} a_{12} & a_{13} \\ a_{22} & a_{23} \end{vmatrix} \\ &= a_{11} (a_{22} a_{33} - a_{23} a_{32}) - a_{21} (a_{12} a_{33} - a_{13} a_{32}) \\ &\quad + a_{31} (a_{12} a_{23} - a_{13} a_{22}) \\ |A| &= a_{11} a_{22} a_{33} - a_{11} a_{23} a_{32} - a_{21} a_{12} a_{33} + a_{21} a_{13} a_{32} \\ &\quad + a_{31} a_{12} a_{23} - a_{31} a_{13} a_{22} \\ &= a_{11} a_{22} a_{33} - a_{11} a_{23} a_{32} - a_{12} a_{21} a_{33} + a_{12} a_{23} a_{31} \\ &\quad + a_{13} a_{21} a_{32} - a_{13} a_{22} a_{31} \end{aligned}$$

Similarly, if we expand  $|A|$  along the second column, then

$$\begin{aligned} |A| &= -a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{22} \begin{vmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{vmatrix} \\ &\quad - a_{32} \begin{vmatrix} a_{11} & a_{13} \\ a_{21} & a_{23} \end{vmatrix} = -a_{12} M_{12} + a_{22} M_{22} - a_{32} M_{32}. \end{aligned}$$

The reader should verify that expansion by the elements of any other row or column leads to the same result. No matter by what row or column the expansion is made, the value of the determinant is the same. Note that once a row or column is decided on and the sign calculated for the product of the first element therein with its minor, the signs for the succeeding products alternate from plus to minus and minus to plus.

For example, the expansion of  $|A| = \begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 10 \end{vmatrix}$ ,

using elements of the second row gives

$$|A| = 4(-1) \begin{vmatrix} 2 & 3 \\ 8 & 10 \end{vmatrix} + 5(+1) \begin{vmatrix} 1 & 3 \\ 7 & 10 \end{vmatrix} + 6(-1) \begin{vmatrix} 1 & 2 \\ 7 & 8 \end{vmatrix}$$

$$= -4(-4) + 5(-11) - (-6) = -3$$

and using elements of the first column the expansion also gives the same result :

$$|A| = 1(+1) \begin{vmatrix} 5 & 6 \\ 8 & 10 \end{vmatrix} + 4(-1) \begin{vmatrix} 2 & 3 \\ 8 & 10 \end{vmatrix} + 7(+1) \begin{vmatrix} 2 & 3 \\ 5 & 6 \end{vmatrix}$$

$$= 1(2) - 4(-4) + 7(-3) = -3.$$

Thus we can express the determinant as a linear combination of the minors of the elements of any row or any column.

In terms of the notation of cofactors, we have

$$|A| = a_{11}A_{11} + a_{12}A_{12} + a_{13}A_{13}, \text{ (using first row)}$$

$$\text{or } |A| = a_{21}A_{21} + a_{22}A_{22} + a_{23}A_{23}, \text{ (using second column)}$$

$$\text{or } |A| = a_{13}A_{13} + a_{23}A_{23} + a_{33}A_{33}, \text{ (using third row)}$$

and so on.

Therefore, in a determinant the sum of the products of the elements of any row or column with the corresponding cofactors is equal to the value of the determinant.

### STUDY TIP

(i) For easier calculations, we shall expand the determinant along that row or column which contains maximum number of zeros.

(ii) While expanding, instead of multiplying by  $(-1)^{i+j}$ , we can multiply by +1 or -1 according to the value of  $(i+j)$  being even or odd.

► **Example 1.** Write the cofactors of the elements

of the determinant  $D = \begin{vmatrix} 2 & 3 & 2 \\ 1 & 4 & -1 \\ 5 & 6 & 8 \end{vmatrix}$  and hence

evaluate the determinant.

► **Solution**  $A_{11}$  = cofactor of the element 2 which is the first element of the first row is

$$= \begin{vmatrix} 4 & -1 \\ 6 & 8 \end{vmatrix} = 32 + 6 = 38.$$

$A_{12}$  = cofactor of the element 3 which is the second element of the first row

$$= - \begin{vmatrix} 1 & -1 \\ 5 & 8 \end{vmatrix} = -(8 + 5) = -13.$$

$A_{13}$  = cofactor of the element 2 which is the third

element of the first row =  $+ \begin{vmatrix} 1 & 4 \\ 5 & 6 \end{vmatrix} = 6 - 20 = -14.$

$A_{21}$  = A cofactor of the element 1 which is the first element of the second row

$$= - \begin{vmatrix} 3 & 2 \\ 6 & 8 \end{vmatrix} = -(24 - 12) = -12.$$

$A_{22}$  = cofactor of the element 4 which is the second

element of the second row =  $+ \begin{vmatrix} 2 & 2 \\ 5 & 8 \end{vmatrix} = 16 - 10 = 6.$

Similarly,

$$A_{23} = - \begin{vmatrix} 2 & 3 \\ 5 & 6 \end{vmatrix} = -(12 - 15) = 3,$$

$$A_{31} = + \begin{vmatrix} 3 & 2 \\ 4 & -1 \end{vmatrix} = -3 - 8 = -11$$

$$A_{32} = \begin{vmatrix} 2 & 2 \\ 1 & -1 \end{vmatrix} = -(-2 - 2) = 4,$$

$$A_{33} = + \begin{vmatrix} 2 & 3 \\ 1 & 4 \end{vmatrix} = 8 - 3 = 5.$$

We have

$$D = 2A_{11} + 3A_{12} + 2A_{13} = 2(38) + 3(-13) + 2(-14) = 9$$

$$D = 1A_{21} + 4A_{22} + (-1)A_{23} = 1(-12) + 4(6) + (-1)(3) = 9$$

$$D = 5A_{31} + 6A_{32} + 8A_{33} = 5(-11) + 6(4) + 8(5) = 9.$$

► **Example 2.** Find the value of the determinant

$$\Delta = \begin{vmatrix} 1 & 2 & 7 \\ 5 & 0 & 2 \\ 3 & -4 & 6 \end{vmatrix} \text{ by expanding along the second row.}$$



► **Solution** Expanding  $\Delta$  along the second row, we get

$$\Delta = -5 \begin{vmatrix} 2 & 7 \\ -4 & 6 \end{vmatrix} + 0 \begin{vmatrix} 1 & 7 \\ 3 & 6 \end{vmatrix} - 2 \begin{vmatrix} 1 & 2 \\ 3 & -4 \end{vmatrix}$$

$$= -5(12 + 28) + 0 - 2(-4 - 6) = -200 + 20 = -180.$$

The element 5 occurs in the second row and the first column. We have  $(-1)^{2+1} = (-1)^3 = -1$ . Therefore we have placed negative sign before 5. Then the sign before 0 will be positive and the sign before 2 will be negative.

► **Example 3.** If  $(x_1 - x_2)^2 + (y_1 - y_2)^2 = 9$ ,  
 $(x_2 - x_3)^2 + (y_2 - y_3)^2 = 16$  and  
 $(x_1 - x_3)^2 + (y_1 - y_3)^2 = 25$

then find the value of  $\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}^2$ .

► **Solution** Let  $A \equiv (x_1, y_1)$ ,  $B \equiv (x_2, y_2)$  and  $C \equiv (x_3, y_3)$

$$\therefore AB = 3, BC = 4 \text{ and } CA = 5$$

$$\therefore \text{area of } \Delta ABC = \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = \frac{1}{2} \times 3 \times 4$$

( $\Delta ABC$  is right angled)

$$\Rightarrow \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}^2 = (12)^2 = 144.$$

### Determinants of higher order

The determinant of any order is equal to the sum of the products of all the elements of an arbitrary row by their cofactors.

The expansion of the determinant of a square matrix  $A$  of order  $n$  in terms of the  $i^{\text{th}}$  row is

$$|A| = a_{i1}A_{i1} + a_{i2}A_{i2} + \dots + a_{in}A_{in}$$

A similar expansion of the determinant can also be obtained about any column.

$$D = \begin{vmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{vmatrix},$$

is called a determinant of order 4 and its value is the number

$$a_{11} \begin{vmatrix} a_{22} & a_{23} & a_{24} \\ a_{32} & a_{33} & a_{34} \\ a_{42} & a_{43} & a_{44} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} & a_{24} \\ a_{31} & a_{33} & a_{34} \\ a_{41} & a_{43} & a_{44} \end{vmatrix}$$

$$+ a_{13} \begin{vmatrix} a_{21} & a_{22} & a_{24} \\ a_{31} & a_{32} & a_{34} \\ a_{41} & a_{42} & a_{44} \end{vmatrix} - a_{14} \begin{vmatrix} a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \\ a_{41} & a_{42} & a_{43} \end{vmatrix}$$

This is the expansion of the determinant  $D$  along its first row. It can be expanded along any of its rows or columns as is the case of a determinant of order 3.

► **Example 4.** Show that  $\begin{vmatrix} 0 & a & b & c \\ -a & 0 & d & e \\ -b & -d & 0 & f \\ -c & -e & -f & 0 \end{vmatrix}$

$$= (af - be + cd)^2.$$

► **Solution** Expanding the determinant by the minor of the elements of the first column, we have

$$\Delta = a \begin{vmatrix} a & b & c \\ -d & 0 & f \\ -e & -f & 0 \end{vmatrix} - b \begin{vmatrix} a & b & c \\ 0 & d & e \\ -e & -f & 0 \end{vmatrix}$$

$$+ c \begin{vmatrix} a & b & c \\ 0 & d & e \\ -d & 0 & f \end{vmatrix}$$

$$= af(af - be + cd) - be(af - be + cd)$$

$$+ cd(af - be + cd) = (af - be + cd)^2.$$

### STUDY TIP

Instead of calculating  $(-1)^{i+j}$  for a given element, the following sign scheme can be used.

**For third-order determinant**

$+-+$

$-+-$

$+ - +$

**For fourth-order determinant**

$+-+-$

$-+-+$

$+ - + -$

$-+-+$

► **Example 5.** If all the elements of a determinant located on one side of the principal diagonal are equal to zero, then the determinant is equal to the product of the elements on the principal diagonal.

► **Solution** This assertion is obvious for a second-order determinant. We therefore prove it by induction, that is, we assume that for determinants of order  $(n - 1)$  it has been proved, and then we consider the  $n^{\text{th}}$ -order determinant

$$\Delta = \begin{vmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ 0 & a_{22} & a_{23} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & \dots & a_{nn} \end{vmatrix}$$

Expanding it in terms of the first column, we get

$$= a_{11} \begin{vmatrix} a_{22} & a_{23} & \dots & a_{2n} \\ 0 & a_{33} & \dots & a_{3n} \\ \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & a_{nn} \end{vmatrix}$$

But the induction hypothesis is applicable to the minor on the right-hand side :

it is equal to the product  $a_{22}a_{33} \dots a_{nn}$  and so

$$\Delta = a_{11}a_{22}a_{33} \dots a_{nn}$$



In a determinant the sum of the products of the elements of any row (column) with the cofactors of the corresponding elements of any other row (column) is zero.

For example,

$$\begin{aligned} \Delta &= a_{11}A_{21} + a_{12}A_{22} + a_{13}A_{23} \\ &= a_{11}(-1)^{1+1} \begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix} + a_{12}(-1)^{1+2} \begin{vmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{vmatrix} \\ &\quad + a_{13}(-1)^{1+3} \begin{vmatrix} a_{11} & a_{12} \\ a_{31} & a_{32} \end{vmatrix} \\ &= \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{11} & a_{12} & a_{13} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = 0 \text{ (since } R_1 \text{ and } R_2 \text{ are} \end{aligned}$$

identical)

Similarly, we can try for other rows and columns.

**Proof:**

Let  $\Delta = |a_{ij}|$  be a determinant of order  $n$ . Then

$$\Delta = a_{i1}A_{i1} + a_{i2}A_{i2} + \dots + a_{in}A_{in},$$

where  $A_{ij}$  is the cofactor of the element  $a_{ij}$  in the determinant  $\Delta$ .

Suppose we replaced the elements of the  $i^{\text{th}}$  row by the corresponding elements of the  $k^{\text{th}}$  row ( $i \neq k$ ). Then obviously the value of the new determinant is

$$= a_{k1}A_{i1} + a_{k2}A_{i2} + \dots + a_{kn}A_{in}.$$

Since in the new determinant two rows, i.e., the  $i^{\text{th}}$  and the  $k^{\text{th}}$ , are identical, therefore its value is zero.

Hence  $a_{k1}A_{i1} + a_{k2}A_{i2} + \dots + a_{kn}A_{in} = 0, i \neq k$ .

$$\text{Let } D = \begin{vmatrix} 2 & 3 & 2 \\ 1 & 4 & -1 \\ 5 & 6 & 8 \end{vmatrix}$$

For example, if we calculate the sum of the products of the elements of first row and the cofactors of second row we get

$$2A_{21} + 3A_{22} + 2A_{23} = 2(-12) + 3(6) + 2(3) = 0$$

Similarly

$$5A_{11} + 6A_{12} + 8A_{13} = 5(38) + 6(-13) + 8(-14) = 0$$

and so on.



- (i) Area of a triangle whose vertices are  $(x_r, y_r)$ ;  $r = 1, 2, 3$  is:

$$D = \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$$

If  $D = 0$  then the three points are collinear.

- (ii) Equation of a straight line passing through

$$(x_1, y_1) \text{ and } (x_2, y_2) \text{ is } \begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} = 0$$

- (iii) The lines:  $a_1x + b_1y + c_1 = 0 \dots \dots \dots (1)$   
 $a_2x + b_2y + c_2 = 0 \dots \dots \dots (2)$   
 $a_3x + b_3y + c_3 = 0 \dots \dots \dots (3)$

$$\text{are concurrent if, } \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0.$$

- (iv)  $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$  represents a pair of straight lines if

$$abc + 2fgh - af^2 - bg^2 - ch^2 = 0 = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix}$$

## Practice Problems

[A]

1. Evaluate  $\begin{vmatrix} 1 + \cos \theta & \sin \theta \\ \sin \theta & 1 - \cos \theta \end{vmatrix}$

2. Show that  $\begin{vmatrix} 4 & -7 \\ 5 & 2 \end{vmatrix} = \begin{vmatrix} 3 & 2 \\ -11 & 7 \end{vmatrix}$

3. Solve  $\begin{vmatrix} x^2 - 3 & 5 \\ -4 & 1 \end{vmatrix} + \begin{vmatrix} 5x & 3 \\ 4 & 2 \end{vmatrix} - 7x \begin{vmatrix} 8 & 3 \\ 5 & 2 \end{vmatrix} = 3$ .

4. Solve  $\begin{vmatrix} x & -3 \\ 8 & -2 \end{vmatrix} - \begin{vmatrix} x^2 & 3 & 2 \\ -2 & 1 & 2 \\ 3 & 1 & 1 \end{vmatrix} + \begin{vmatrix} 4 & 3 \\ 2 & 1 \end{vmatrix} - 8 = 0$

5. Show that no matter what may be the values of the x's and y's,

$$\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = \begin{vmatrix} a_{11} & a_{12} & 0 \\ a_{21} & a_{22} & 0 \\ x_1 & x_2 & 1 \end{vmatrix} = \begin{vmatrix} a_{11} & a_{12} & x_1 & y_1 \\ a_{21} & a_{22} & x_2 & y_2 \\ 0 & 0 & 1 & y_3 \\ 0 & 0 & 0 & 1 \end{vmatrix}$$

6. Compute the determinants:

(a)  $\begin{vmatrix} 1 & 3 & 4 & 1 \\ 1 & 1 & 2 & 5 \\ 0 & 1 & 1 & 6 \\ 0 & 0 & 0 & 2 \end{vmatrix}$

(b)  $\begin{vmatrix} 2 & 3 & 2 & 1 \\ 1 & 3 & 2 & 1 \\ 1 & 2 & 2 & 1 \\ 1 & 0 & 1 & 1 \end{vmatrix}$

 7. Compute the cofactors  $A_{12}$  and  $A_{43}$  in the

$$\text{determinant} \begin{vmatrix} 1 & 0 & 2 & 3 & 4 \\ 2 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 2 & 0 \\ 1 & 2 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 & 1 \end{vmatrix}$$

8. Determine the signs of the following terms of determinants:

(a)  $a_{12}a_{23}a_{31}a_{44}a_{55}$

(b)  $a_{16}a_{23}a_{32}a_{45}a_{54}a_{61}$

(c)  $a_{22}a_{33}a_{11}a_{44}$

(d)  $a_{21}a_{12}a_{34}a_{43}$

9. Evaluate (i)  $\begin{vmatrix} 6 & -3 & 2 \\ 2 & -1 & 2 \\ -10 & 5 & 2 \end{vmatrix}$

(ii)  $\begin{vmatrix} x+y & y+z & z+x \\ z & x & y \\ 1 & 1 & 1 \end{vmatrix}$

(iii)  $\begin{vmatrix} 0 & c & b \\ -c & 0 & a \\ -b & -a & 0 \end{vmatrix}$

(iv)  $\begin{vmatrix} 0 & p-q & p-r \\ q-p & 0 & q-r \\ r-p & r-q & 0 \end{vmatrix}$

10. Show that value of e is 0, if

$$ax^4 + bx^3 + cx^2 + dx + e = \begin{vmatrix} x^2 + 3x & x-1 & x+3 \\ x+1 & -2x & x-4 \\ x-3 & x+4 & 3x \end{vmatrix}$$

11. Prove that

$$\begin{vmatrix} \operatorname{cosec} x & 1 & 0 \\ 1 & 2 \operatorname{cosec} x & 1 \\ 0 & 1 & 2 \operatorname{cosec} x \end{vmatrix} \geq 1 \text{ for all } x \in (0, \pi).$$

12. If  $f(x) = \begin{vmatrix} 1+x^2 & x & 0 \\ x & 1+x^2 & x \\ 0 & x & 1+x^2 \end{vmatrix}$ , then find

 $f(x)$  and show that it is positive for all real x.

13. Show that,

$$\begin{vmatrix} \cos(x+x^2) & \sin(x+x^2) & -\cos(x+x^2) \\ \sin(x-x^2) & \cos(x-x^2) & \sin(x-x^2) \\ \sin 2x & 0 & \sin 2x^2 \end{vmatrix} = \sin(2x + 2x^2).$$

14. If  $(x_1 - x_2)^2 + (y_1 - y_2)^2 = a^2$ ,  $(x_2 - x_3)^2 + (y_2 - y_3)^2 = b^2$ ,  $(x_1 - x_3)^2 + (y_1 - y_3)^2 = c^2$  then prove that

$$4 \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}^2 = (a + b + c)(a + b - c)(a - b + c)(-a + b + c).$$

## 2.2 PROPERTIES OF DETERMINANTS

**Property 1.** The value of a determinant remains unaltered, if the rows and columns are interchanged, i.e., if  $A$  is an  $n$ -rowed square matrix, then  $|A| = |A'|$ .

For example  $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$

### Symmetric and Skew-Symmetric Determinants

The determinant of a symmetric matrix is called a symmetric determinant.

In a determinant  $\Delta$  if  $a_{ij} = a_{ji}$  for all  $i$  and  $j$ , then  $\Delta$  is said to be symmetric. Thus

$$\begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} \text{ and } \begin{vmatrix} a & h & g & \ell \\ h & b & f & m \\ g & f & c & n \\ \ell & m & n & d \end{vmatrix}$$

are symmetric determinants.

The determinant of a skew-symmetric matrix is called a skew-symmetric determinant.

That is, if for all  $i$  and  $j$  it is true that  $a_{ji} = -a_{ij}$ . For example

$$\begin{vmatrix} 0 & h & g & \ell \\ -h & 0 & f & m \\ -g & -f & 0 & n \\ -\ell & -m & -n & 0 \end{vmatrix} \text{ is a skew-symmetric determinant.}$$



- (i) The square of a symmetric or skew symmetric determinant is symmetric.
- (ii) A skew symmetric determinant of odd order is zero.  
Let  $\Delta$  be a skew symmetric determinant of order  $n$  and  $\Delta'$  obtained from  $\Delta$  by changing its rows to columns. Then  $\Delta = \Delta'$ . Again  $\Delta'$  is obtained from  $\Delta$  by multiplying each row of  $\Delta$  by  $-1$ . Since  $\Delta$  contains  $n$  rows,  $\Delta' = (-1)^n \Delta$ .  
Therefore,  $\Delta = (-1)^n \Delta$ .  
If  $n$  is odd,  $\Delta = -\Delta$ , or,  $\Delta = 0$ .
- (iii) A skew symmetric determinant of even order is a perfect square.

#### ► Example 1.

$$\text{Evaluate } \Delta = \begin{vmatrix} 0 & i-100 & i-500 \\ 100-i & 0 & 1000-i \\ 500-i & i-1000 & 0 \end{vmatrix}.$$

► **Solution** This is a skew-symmetric determinant of odd order and hence it is equal to 0.

**Property 2.** If any two rows (or columns) of a determinant be interchanged, the value of determinant is changed in sign only.

$$\text{Let } D_1 = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} \text{ and } D_2 = \begin{vmatrix} a_2 & b_2 & c_2 \\ a_1 & b_1 & c_1 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

Then  $D_2 = -D_1$ .

If  $R_i = i^{\text{th}}$  row and  $C_i = i^{\text{th}}$  column, then we can denote the interchange of rows by  $R_i \leftrightarrow R_j$  and interchange of columns by  $C_i \leftrightarrow C_j$ .

If a column (row) is moved past an even number of columns (row's), then the value of the determinant is unaltered; in particular

$$\begin{vmatrix} a_1 & b_1 & c_1 & d_1 \\ a_2 & b_2 & c_2 & d_2 \\ a_3 & b_3 & c_3 & d_3 \\ a_4 & b_4 & c_4 & d_4 \end{vmatrix} = \begin{vmatrix} c_1 & a_1 & b_1 & d_1 \\ c_2 & a_2 & b_2 & d_2 \\ c_3 & a_3 & b_3 & d_3 \\ c_4 & a_4 & b_4 & d_4 \end{vmatrix}.$$



If a column (row) is moved past an odd number of columns (rows), then the value of the determinant is thereby multiplied by  $-1$ .

For a column can be moved past an even (odd) number of columns by an even (odd) number of adjacent interchanges. In the particular example,  $a \ b \ c \ d$  can be changed into  $c \ a \ b \ d$  by first interchanging  $b$  and  $c$ , giving  $a \ c \ b \ d$ , and then interchanging  $a$  and  $c$ .

**Property 3.** If a determinant has any two rows (or columns) identical or proportional, then its value is zero.

Let  $D = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$  then it can be verified that  $D=0$ .

► **Example 2.** For positive numbers  $x, y$  and  $z$  find the value of the determinant

$$\begin{vmatrix} 1 & \log_x y & \log_x z \\ \log_y x & 1 & \log_y z \\ \log_z x & \log_z y & 1 \end{vmatrix}$$

► **Solution**  $D = \begin{vmatrix} 1 & \frac{\log y}{\log x} & \frac{\log z}{\log x} \\ \frac{\log x}{\log y} & 1 & \frac{\log z}{\log y} \\ \frac{\log x}{\log z} & \frac{\log y}{\log z} & 1 \end{vmatrix}$

On multiplying  $R_1$  by  $\log x$  and  $R_2$  by  $\log y$  we get two identical rows. Hence  $D=0$ .

**Property 4.** If all the elements of any row (or column) be multiplied by the same number, then the determinant is multiplied by that number.

If  $D = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$  and  $D' = \begin{vmatrix} Ka_1 & Kb_1 & Kc_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$

Then  $D' = KD$

Thus  $\begin{vmatrix} 3 & 0 & 27 \\ -9 & 3 & 0 \\ 15 & 6 & -3 \end{vmatrix} = 3^3 \begin{vmatrix} 1 & 0 & 9 \\ -3 & 1 & 0 \\ 5 & 2 & -1 \end{vmatrix}$

Let  $A = \begin{bmatrix} 2 & 2 \\ 4 & 0 \end{bmatrix}$  and  $B = \begin{bmatrix} 1 & 1 \\ 2 & 0 \end{bmatrix}$ . Here  $A = 2B$ .

Also  $|A| = 0 - 8 = -8$  and  $|B| = 0 - 2 = -2$ .

Observe that,  $|A| = 4(-2) = 2^2 \cdot |B|$

or  $|A| = 2^n |B|$ , where  $n = 2$  is the order of square matrices  $A$  and  $B$ .

In general, if  $A = kB$  where  $A$  and  $B$  are square matrices of order  $n$ , then  $|A| = k^n |B|$ .

► **Example 3.** If  $A = \begin{vmatrix} a & b & 2c \\ d & e & 2f \\ \ell & m & 2n \end{vmatrix}$  and

$B = \begin{vmatrix} f & 2d & e \\ 2n & 4\ell & 2m \\ c & 2a & b \end{vmatrix}$ , prove that  $2A - B = 0$ .

► **Solution**  $B = 2 \cdot 2 \begin{vmatrix} f & d & e \\ n & \ell & m \\ c & a & b \end{vmatrix}$

[Taking 2 common from  $R_2$  and  $C_2$ ]

$= 2 \begin{vmatrix} 2f & d & e \\ 2n & \ell & m \\ 2c & a & b \end{vmatrix} = 2 \begin{vmatrix} 2c & a & b \\ 2f & d & e \\ 2n & \ell & m \end{vmatrix}$

[ $R_3 \leftrightarrow R_2$ , then  $R_2 \leftrightarrow R_1$ ]

$= 2 \begin{vmatrix} a & b & 2c \\ d & e & 2f \\ \ell & m & 2n \end{vmatrix} = 2A$

[ $C_1 \leftrightarrow C_2$  and then  $C_2 \leftrightarrow C_3$ ]

## Practice Problems

[B]

1. If  $x = \begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{vmatrix}$  and  $y = \begin{vmatrix} 4 & 5 & 6 \\ 7 & 8 & 9 \\ 1 & 2 & 3 \end{vmatrix}$

2. Evaluate  $\begin{vmatrix} 4 & 8 & 16 \\ 2 & 3 & 4 \\ 5 & 10 & 20 \end{vmatrix}$

3. Show that  $\begin{vmatrix} a & b & c \\ x & y & z \\ p & q & r \end{vmatrix} = \begin{vmatrix} y & b & q \\ x & a & p \\ z & c & r \end{vmatrix}$

4. Find the value of the determinant

$$\begin{vmatrix} x_1 x_2 & y_1 y_2 & 1 \\ x_1 + x_2 & y_1 + y_2 & 2 \\ \sin(\pi x_1 x_2) & \cos\left(\frac{\pi}{2} y_1 y_2\right) & 1 \end{vmatrix}$$

where  $x_1, x_2$  and  $y_1, y_2$  are the roots of the equations

$$3x^2 - 18x + 9 = 0 \text{ and } y^2 - 4y + 2 = 0.$$

5. Without expanding the determinants, find the

value of  $\begin{vmatrix} 103 & 115 & 114 \\ 111 & 108 & 106 \\ 104 & 113 & 116 \end{vmatrix} + \begin{vmatrix} 113 & 116 & 104 \\ 108 & 106 & 111 \\ 115 & 114 & 103 \end{vmatrix}$

6. Without expanding, show that

$$\begin{vmatrix} 1 & a & a^2 - bc \\ 1 & b & b^2 - ca \\ 1 & c & c^2 - ab \end{vmatrix} = 0$$

7. Evaluate  $\begin{vmatrix} pa & qb & rc \\ qc & ra & pb \\ rb & pc & qa \end{vmatrix} = pqr \begin{vmatrix} a & b & c \\ c & c & b \\ b & a & a \end{vmatrix}$ .

8. Find the value of

$$\begin{vmatrix} (-1)^n a & (-1)^{n+1} b & (-1)^{n+1} c \\ a+1 & 1-b & 1+c \\ a-1 & b+1 & 1-c \end{vmatrix}$$

$$+ \begin{vmatrix} (-1)^{n+1} a & a+1 & a-1 \\ (-1)^n b & 1-b & b+1 \\ (-1)^{n+2} c & 1+c & 1-c \end{vmatrix}.$$

9. Without expanding the determinants, prove that

$$\begin{vmatrix} ax & by & cz \\ x^2 & y^2 & z^2 \\ 1 & 1 & 1 \end{vmatrix} = \begin{vmatrix} a & b & c \\ x & y & z \\ yz & zx & xy \end{vmatrix}$$

**Property 5.** If each element of any row (or column) can be expressed as a sum of two terms then the determinant can be expressed as the sum of two determinants.

For example

$$\begin{vmatrix} a_1 + x & b_1 + y & c_1 + z \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} + \begin{vmatrix} x & y & z \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

*Proof:*

$$\text{Let } \Delta = \begin{vmatrix} a_1 + \alpha_1 & b_1 & c_1 \\ a_2 + \alpha_2 & b_2 & c_2 \\ a_3 + \alpha_3 & b_3 & c_3 \end{vmatrix}.$$

Expanding  $\Delta$  along the first column, we get

$$\begin{aligned} D &= (a_1 + \alpha_1) \begin{vmatrix} b_2 & c_2 \\ b_3 & c_3 \end{vmatrix} - (a_2 + \alpha_2) \begin{vmatrix} b_1 & c_1 \\ b_3 & c_3 \end{vmatrix} \\ &\quad + (a_3 + \alpha_3) \begin{vmatrix} b_1 & c_1 \\ b_2 & c_2 \end{vmatrix} \\ &= \left\{ a_1 \begin{vmatrix} b_2 & c_2 \\ b_3 & c_3 \end{vmatrix} - a_2 \begin{vmatrix} b_1 & c_1 \\ b_3 & c_3 \end{vmatrix} + a_3 \begin{vmatrix} b_1 & c_1 \\ b_2 & c_2 \end{vmatrix} \right\} \\ &\quad + \left\{ \alpha_1 \begin{vmatrix} b_2 & c_2 \\ b_3 & c_3 \end{vmatrix} - \alpha_2 \begin{vmatrix} b_1 & c_1 \\ b_3 & c_3 \end{vmatrix} + \alpha_3 \begin{vmatrix} b_1 & c_1 \\ b_2 & c_2 \end{vmatrix} \right\} \\ &= \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} + \begin{vmatrix} \alpha_1 & b_1 & c_1 \\ \alpha_2 & b_2 & c_2 \\ \alpha_3 & b_3 & c_3 \end{vmatrix}. \end{aligned}$$

We also conclude that

$$\begin{vmatrix} a_1 + mb_1 & b_1 & c_1 \\ a_2 + mb_2 & b_2 & c_2 \\ a_3 + mb_3 & b_3 & c_3 \end{vmatrix} = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

*Proof:*

$$\text{We have } \begin{vmatrix} a_1 + mb_1 & b_1 & c_1 \\ a_2 + mb_2 & b_2 & c_2 \\ a_3 + mb_3 & b_3 & c_3 \end{vmatrix}$$

$$= \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} + \begin{vmatrix} mb_1 & b_1 & c_1 \\ mb_2 & b_2 & c_2 \\ mb_3 & b_3 & c_3 \end{vmatrix}$$

$$= \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} + m \begin{vmatrix} b_1 & b_1 & c_1 \\ b_2 & b_2 & c_2 \\ b_3 & b_3 & c_3 \end{vmatrix},$$

$$= \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}, \text{ since } \begin{vmatrix} b_1 & b_1 & c_1 \\ b_2 & b_2 & c_2 \\ b_3 & b_3 & c_3 \end{vmatrix} = 0$$

We can similarly prove that

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = \begin{vmatrix} a_1 + mb_1 + nc_1 & b_1 & c_1 \\ a_2 + mb_2 + nc_2 & b_2 & c_2 \\ a_3 + mb_3 + nc_3 & b_3 & c_3 \end{vmatrix}.$$

Also it should be marked that the numbers  $m$  and  $n$  may be positive or negative.

The sum of the determinants of each of two (or more) matrices generally does not equal the determinant of the sum.

$$|A| + |B| = \begin{vmatrix} a_{11} & b_{12} \\ a_{21} & b_{22} \end{vmatrix} + \begin{vmatrix} a_{11} & b_{12} \\ a_{21} & b_{22} \end{vmatrix}$$

$$= a_{11}a_{22} - a_{12}a_{21} + b_{11}b_{22} - b_{12}b_{21} \neq |A + B|$$

The same applies to the difference,

$$|A| - |B| \neq |A - B|$$

We may note in passing that both  $|A| + |B|$  and  $|A| - |B|$  have meaning even when  $A$  and  $B$  are square matrices of different orders, because the value of a determinant is a scalar. This contrasts with  $A + B$  and  $A - B$ , which have meaning only when the matrices are conformable for addition (have the same order).

Another point of interest is that although  $|A| + |B|$  does not generally equal  $|A + B|$  the latter can be

written as the sum of certain other determinants. For example,

$$|A + B| = \begin{vmatrix} a_{11} & b_{12} \\ a_{21} & b_{22} \end{vmatrix} + \begin{vmatrix} a_{11} & b_{12} \\ a_{21} & b_{22} \end{vmatrix}$$

$$= \begin{vmatrix} a_{11} + b_{11} & a_{12} + b_{12} \\ a_{21} + b_{21} & a_{22} + b_{22} \end{vmatrix}$$

$$= \begin{vmatrix} a_{11} & a_{12} \\ a_{21} + b_{21} & a_{22} + b_{22} \end{vmatrix} + \begin{vmatrix} b_{11} & b_{12} \\ a_{21} + b_{21} & a_{22} + b_{22} \end{vmatrix}$$

$$= \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} + \begin{vmatrix} a_{11} & a_{12} \\ b_{21} & b_{22} \end{vmatrix} + \begin{vmatrix} b_{11} & b_{12} \\ a_{21} & a_{22} \end{vmatrix} + \begin{vmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{vmatrix}$$

In general if  $A$  and  $B$  are  $n \times n$ ,  $|A + B|$  can be expanded as the sum of  $2^n$   $n$ -order determinants. The determinant of order  $n$

$$\begin{vmatrix} a_1 + \alpha_1 & b_1 + \beta_1 & \dots & k_1 + \kappa_1 \\ a_2 + \alpha_2 & b_2 + \beta_2 & \dots & k_2 + \kappa_2 \\ \dots & \dots & \dots & \dots \\ a_n + \alpha_n & b_n + \beta_n & \dots & k_n + \kappa_n \end{vmatrix}$$

is equal to the sum of the  $2^n$  determinants corresponding to the  $2^n$  different ways of choosing one letter from each column.

## Summation of Determinants

$$\text{Let } \Delta(r) = \begin{vmatrix} f(r) & g(r) & h(r) \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

where  $a_1, a_2, a_3, b_1, b_2, b_3$  are constants (independent

$$\text{of } r), \text{ then } \sum_{r=1}^n \Delta(r) = \begin{vmatrix} \sum_{r=1}^n f(r) & \sum_{r=1}^n g(r) & \sum_{r=1}^n h(r) \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

Here function of  $r$  can be the elements of only one row or column. None of the elements other than that row or column should be dependent on  $r$ . If more than one column or row have elements

dependent on  $r$  then first expand the determinant and then find the summation.

► **Example 4.** Evaluate  $\sum_{r=1}^n$

$$\begin{vmatrix} 2r-1 & {}^nC_r & 2^r \\ x & \cos^2 \theta & y \\ n^2 & 2^n-1 & 2^{n+1}-2 \end{vmatrix}$$

► **Solution**  $\sum_{r=1}^n D_r$

$$\begin{aligned} &= \begin{vmatrix} \sum_{r=1}^n (2r-1) & \sum_{r=1}^n {}^nC_r & \sum_{r=1}^n 2^r \\ x & \cos^2 \theta & y \\ n^2 & 2^n-1 & 2^{n+1}-2 \end{vmatrix} \\ &= \begin{vmatrix} n^2 & 2^n-1 & 2^{n+1}-2 \\ x & \cos^2 \theta & y \\ n^2 & 2^n-1 & 2^{n+1}-2 \end{vmatrix} = 0. \end{aligned}$$

► **Example 5.** If  $\Delta_r = \begin{vmatrix} r-1 & 1 & 0 \\ 2 & r & 3+r \\ r+1 & -1 & -2 \end{vmatrix}$ ,

find  $\sum_{r=1}^n \Delta_r$

► **Solution** On expanding the determinant, we get

$$\Delta_r = (r-1)(3-r) + 7 + r^2 + 4r = 8r + 4$$

$$\Rightarrow \sum_{r=1}^n \Delta_r = \sum_{r=1}^n (8r + 4) = 4n(n+2)$$

► **Example 6.** Let  $n$  be a positive integer and

$$\Delta_r = \begin{vmatrix} 2r-1 & {}^nC_r & 1 \\ n^2-1 & 2^n & n+1 \\ \cos^2(n^2) & \cos^2 n & \cos^2(n+1) \end{vmatrix}$$

then prove that  $\sum_{r=0}^n \Delta_r = 0$

► **Solution** Since

$$\Delta_r = \begin{vmatrix} 2r-1 & {}^nC_r & 1 \\ n^2-1 & 2^n & n+1 \\ \cos^2(n^2) & \cos^2 n & \cos^2(n+1) \end{vmatrix}$$

$$\sum_{r=0}^n \Delta_r = \begin{vmatrix} \sum_{r=0}^n (2r-1) & \sum_{r=0}^n {}^nC_r & \sum_{r=0}^n 1 \\ n^2-1 & 2^n & n+1 \\ \cos^2(n^2) & \cos^2 n & \cos^2(n+1) \end{vmatrix}$$

$$= \begin{vmatrix} -1+1+3+5+\dots+(2n-1) & 2^n & n+1 \\ n^2-1 & 2^n & n+1 \\ \cos^2(n^2) & \cos^2 n & \cos^2(n+1) \end{vmatrix}$$

$$= \begin{vmatrix} \frac{(n+1)}{2}(-1+2n-1) & 2^n & (n+1) \\ n^2-1 & 2^n & n+1 \\ \cos^2(n^2) & \cos^2 n & \cos^2(n+1) \end{vmatrix}$$

$$= \begin{vmatrix} n^2-1 & 2^n & (n+1) \\ n^2-1 & 2^n & (n+1) \\ \cos^2(n^2) & \cos^2 n & \cos^2(n+1) \end{vmatrix} = 0.$$

(since  $R_1$  and  $R_2$  are identical)

**Property 6.** The value of a determinant is not altered by adding to the elements of any row (or column) the same multiples of the corresponding elements of any other row (or column).

$$\text{Let } D_1 = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} \text{ and}$$

$$D_2 = \begin{vmatrix} a_1+ma_2 & b_1+mb_2 & c_1+mc_2 \\ a_2 & b_2 & c_2 \\ a_3+na_1 & b_3+nb_1 & c_3+nc_1 \end{vmatrix}.$$

Then  $D_2 = D_1$ .

► **Example 7.** Evaluate  $\Delta = \begin{vmatrix} 265 & 240 & 219 \\ 240 & 225 & 198 \\ 219 & 198 & 181 \end{vmatrix}$



► **Solution** Applying  $C_1 \rightarrow C_1 + C_3 - 2C_2$ , we get

$$\Delta = \begin{vmatrix} 4 & 240 & 219 \\ -12 & 225 & 198 \\ 4 & 198 & 181 \end{vmatrix} = 4 \begin{vmatrix} 1 & 240 & 219 \\ -3 & 225 & 198 \\ 1 & 198 & 181 \end{vmatrix},$$

taking 4 common from the first column

$$= 4 \times 3 \begin{vmatrix} 1 & 240 & 219 \\ -1 & 75 & 66 \\ 1 & 198 & 181 \end{vmatrix} \text{ applying}$$

$R_2 \rightarrow R_2 + R_1, R_3 \rightarrow R_3 - R_1$

$$= 12 \begin{vmatrix} 1 & 240 & 219 \\ 0 & 315 & 285 \\ 0 & -42 & -38 \end{vmatrix}$$

expanding the determinant along the first column

$$= 12 \begin{vmatrix} 315 & 285 \\ -42 & -38 \end{vmatrix},$$

taking 21 common from the first column and 19 from the second column

$$= 12 \times 21 \times 19 \begin{vmatrix} 15 & 15 \\ -2 & -2 \end{vmatrix} = 0$$



- (i) When a column (or row) is replaced by one which contains the elements of that column (or row) all multiplied by a number  $k$ , then the determinant is multiplied by  $k$ . For instance, if the second column is subtracted from the first, and the remainders are put in the second column, the sign of the determinant is changed.
- (ii) When more than one column (or row) is changed by a succession of changes of this kind, atleast one column (or row) must be left unaltered.

To understand this consider

$$\Delta = \begin{vmatrix} \alpha_1 + \lambda b_1 & b_1 + \mu c_1 & c_1 + \nu a_1 \\ \alpha_2 + \lambda b_2 & b_2 + \mu c_2 & c_2 + \nu a_2 \\ \alpha_3 + \lambda b_3 & b_3 + \mu c_3 & c_3 + \nu a_3 \end{vmatrix}$$

$$= \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} + \begin{vmatrix} \lambda b_1 & \mu c_1 & \nu a_1 \\ \lambda b_2 & \mu c_2 & \nu a_2 \\ \lambda b_3 & \mu c_3 & \nu a_3 \end{vmatrix};$$

all the other determinants such as

$$\begin{vmatrix} \lambda b_1 & b_1 & c_1 \\ \lambda b_2 & b_2 & c_2 \\ \lambda b_3 & b_3 & c_3 \end{vmatrix}, \text{ being zero.}$$

$$\text{Hence } \Delta = (1 + \lambda\mu\nu) \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}.$$

Now we see that atleast one of  $\lambda, \mu$  or  $\nu$  must be zero if  $\Delta$  is to have the same value as

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

- (iii) One of the more curious errors into which one is led by adding multiples of rows at random is a fallacious proof that  $\Delta = 0$ . In the example just given, 'subtract second column from first, add second and third, add first and third', corresponds to  $\lambda = -1, \mu = 1, \nu = 1$ , a choice of values that will (wrongly, of course) 'prove' that  $\Delta = 0$ : the manipulation of the columns has not left the value unaltered, but multiplied the determinant by a zero factor,  $1 + \lambda\mu\nu$ .
- (iv) The presence of 1 as the leading element of  $\Delta$  makes it easy to reduce the other elements of the first column to zero.

$$\begin{aligned} \Delta &= \begin{vmatrix} 1 & 4 & 9 \\ -4 & 7 & 25 \\ 7 & 5 & 2 \end{vmatrix} \\ &= \begin{vmatrix} 1 & 4 & 9 \\ -4 + 4(1) & 7 + 4(4) & 25 + 4(9) \\ 7 - 7(1) & 5 - 7(4) & 2 - 7(9) \end{vmatrix} \\ &= \begin{vmatrix} 1 & 4 & 9 \\ 0 & 23 & 61 \\ 0 & -23 & -61 \end{vmatrix} = 0 \end{aligned}$$

Let us now evaluate  $\Delta = \begin{vmatrix} x & y & y & y \\ y & x & y & y \\ y & y & x & y \\ y & y & y & x \end{vmatrix}$

Observe that every column of  $\Delta$  sums to  $x + 3y$ . Therefore add every row of  $\Delta$  to its first row and factor out  $x + 3y$ :

$$\Delta = (x + 3y) \begin{vmatrix} 1 & 1 & 1 & 1 \\ y & x & y & y \\ y & y & x & y \\ y & y & y & x \end{vmatrix}.$$

Now use the leading element of  $\Delta$  to reduce the remainder of the first row to zeros. This is achieved by subtracting the first column from each of the other columns :

$$D = (x + 3y) \begin{vmatrix} 1 & 0 & 0 & 0 \\ y & x - y & 0 & 0 \\ y & 0 & x - y & 0 \\ y & 0 & 0 & x - y \end{vmatrix}$$

$$= (x + 3y) (x - y)^3,$$

the last step being obtained by simple expansion by minors. It involves the more general result that the determinant of a diagonal matrix is the product of its diagonal elements.

► **Example 8.** Simplify  $\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$ .

► **Solution** Applying  $R_1 \rightarrow R_1 + R_2 + R_3$

$$\begin{vmatrix} a+b+c & a+b+c & a+b+c \\ b & c & a \\ c & a & b \end{vmatrix}$$

$$= (a + b + c) \begin{vmatrix} 1 & 1 & 1 \\ b & c & a \\ c & a & b \end{vmatrix}$$

Applying  $C_1 \rightarrow C_1 - C_2, C_2 \rightarrow C_2 - C_3$

$$= (a + b + c) \begin{vmatrix} 0 & 0 & 1 \\ b - c & c - a & a \\ c - a & a - b & b \end{vmatrix}$$

$$= (a + b + c) ((b - c)(a - b) - (c - a)^2)$$

$$= (a + b + c) (ab + bc - ca - b^2 - c^2 + 2ca - a^2)$$

$$= (a + b + c) (ab + bc + ca - a^2 - b^2 - c^2)$$

$$= 3abc - a^3 - b^3 - c^3$$

► **Example 9.** Prove that the value of the

determinant  $\begin{vmatrix} \cos \alpha & -\sin \alpha & 1 \\ \sin \alpha & \cos \alpha & 1 \\ \cos(\alpha + \beta) & -\sin(\alpha + \beta) & 1 \end{vmatrix}$

is independent of  $\alpha$ .

► **Solution**  $\begin{vmatrix} \cos \alpha & -\sin \alpha & 1 \\ \sin \alpha & \cos \alpha & 1 \\ \cos(\alpha + \beta) & -\sin(\alpha + \beta) & 1 \end{vmatrix}$

Applying  $R_3 \rightarrow R_3 - R_1 (\cos \beta) + R_2 (\sin \beta)$  we get

$$= \begin{vmatrix} \cos \alpha & -\sin \alpha & 1 \\ \sin \alpha & \cos \alpha & 1 \\ 0 & 0 & 1 + \sin \beta - \cos \beta \end{vmatrix}$$

$$= (1 + \sin \beta - \cos \beta) (\cos^2 \alpha + \sin^2 \alpha)$$

$$= 1 + \sin \beta - \cos \beta,$$

which is independent of  $\alpha$ .

► **Example 10.** If  $a, b$  and  $c$  are non-zero and

$$\begin{vmatrix} \frac{a^2 + b^2}{c} & c & c \\ a & \frac{b^2 + c^2}{a} & a \\ b & b & \frac{c^2 + a^2}{b} \end{vmatrix} = \alpha abc,$$

then find the values of  $\alpha$ .

► **Solution**  $\frac{1}{abc} \begin{vmatrix} a^2 + b^2 & c^2 & c^2 \\ a^2 & b^2 + c^2 & a^2 \\ b^2 & b^2 & c^2 + a^2 \end{vmatrix}$

$$\begin{aligned}
 R_1 &\rightarrow R_1 - (R_2 + R_3) \\
 &= \frac{1}{abc} \begin{vmatrix} 0 & -2b^2 & -2a^2 \\ a^2 & b^2 + c^2 & a^2 \\ b^2 & b^2 & c^2 + a^2 \end{vmatrix} \\
 R_2 &\rightarrow R_2 + 1/2 R_1 \quad \text{and} \quad R_3 \rightarrow R_3 + 1/2 R_1 \\
 &= \frac{1}{abc} \begin{vmatrix} 0 & -2b^2 & -2a^2 \\ a^2 & c^2 & 0 \\ b^2 & 0 & c^2 \end{vmatrix} \\
 &= \frac{1}{abc} [2b^2(a^2 c^2) - 2a^2(-b^2 c^2)] \\
 &= \frac{4a^2 b^2 c^2}{abc} = 4abc.
 \end{aligned}$$

Hence,  $\alpha$  is equal to 4.

► **Example 11.** Prove that the value of

$$\begin{vmatrix} 1 & bc & bc(b+c) \\ 1 & ca & ca(a+c) \\ 1 & ab & ab(a+b) \end{vmatrix}$$

is independent of  $a, b, c$ .

► **Solution** Multiplying  $R_1, R_2$  and  $R_3$  by  $a, b$ , and  $c$  respectively and hence divide it by  $a, b, c$

$$\begin{aligned}
 \therefore \Delta &= \frac{1}{abc} \begin{vmatrix} a & abc & abc(b+c) \\ b & abc & abc(c+a) \\ c & abc & abc(a+b) \end{vmatrix} \\
 &= \frac{(abc)(abc)}{abc} \begin{vmatrix} a & 1 & b+c \\ b & 1 & c+a \\ c & 1 & a+b \end{vmatrix} \\
 &= abc \begin{vmatrix} a & 1 & b+c \\ b & 1 & c+a \\ c & 1 & a+b \end{vmatrix} \\
 &= abc \begin{vmatrix} a+b+c & 1 & b+c \\ a+b+c & 1 & c+a \\ a+b+c & 1 & a+b \end{vmatrix} \\
 &\quad \text{(by using } C_1 \rightarrow C_1 + C_3) \\
 &= abc(a+b+c) \begin{vmatrix} 1 & 1 & b+c \\ 1 & 1 & c+a \\ 1 & 1 & a+b \end{vmatrix} = 0.
 \end{aligned}$$

Hence, the determinant is independent of  $a, b, c$ .

► **Example 12.** Show that

$$\begin{vmatrix} a+b+2c & a & b \\ c & b+c+2a & b \\ c & a & c+a+2b \end{vmatrix} = 2(a+b+c)^3$$

► **Solution**

$$\Delta = \begin{vmatrix} 2(a+b+c) & a & b \\ 2(a+b+c) & b+c+2a & b \\ 2(a+b+c) & a & c+a+2b \end{vmatrix}$$

using  $C_1 \rightarrow C_1 + C_2 + C_3$

$$= 2(a+b+c) \begin{vmatrix} 1 & a & b \\ 1 & b+c+2a & b \\ 1 & a & c+a+2b \end{vmatrix}$$

$$= 2(a+b+c) \begin{vmatrix} 1 & a & b \\ 0 & b+c+a & 0 \\ 0 & 0 & c+a+b \end{vmatrix}$$

$$\text{using } R_2 \rightarrow R_2 - R_1 \text{ and } R_3 \rightarrow R_3 - R_1$$

$$= 2(a+b+c)[1\{(b+c+a)^2 - 0\}]$$

$$= 2(a+b+c)(a+b+c)^2$$

$$= 2(a+b+c)^3.$$

► **Example 13.** Prove that

$$\begin{vmatrix} b+c & c & b \\ c & c+a & a \\ b & a & a+b \end{vmatrix} = 4abc.$$

$$\text{► **Solution** Let } \Delta = \begin{vmatrix} b+c & c & b \\ c & c+a & a \\ b & a & a+b \end{vmatrix}$$

$$\text{Applying } R_1 \rightarrow R_2 - (R_2 + R_3)$$

$$\therefore \Delta = \begin{vmatrix} 0 & -2a & -2a \\ c & c+a & a \\ b & a & a+b \end{vmatrix}$$

Taking  $-2a$  common from  $R_1$ ,

$$\Delta = (-2a) \begin{vmatrix} 0 & 1 & 1 \\ c & c+a & a \\ b & a & a+b \end{vmatrix}$$

Applying  $C_2 \rightarrow C_2 - C_3$

$$\therefore \Delta = -2a \begin{vmatrix} 0 & 0 & 1 \\ c & c & a \\ b & -b & a+b \end{vmatrix}$$

Expanding along  $R_1$ , we get

$$\Delta = -2a \cdot 1 \cdot \begin{vmatrix} c & c \\ b & -b \end{vmatrix} = (-2a)(-2bc)$$

Hence  $\Delta = 4abc$ .

► **Example 14.** Show that

$$\begin{vmatrix} -bc & ca+ab & ca+ab \\ ab+bc & -ca & ab+bc \\ bc+ca & bc+ca & -ab \end{vmatrix} = (\sum ab)^3.$$

► **Solution** Applying  $R_1 + R_2 + R_3$

$$D = \begin{vmatrix} \sum ab & \sum ab & \sum ab \\ ab+bc & -ca & ab+bc \\ bc+ca & bc+ca & -ab \end{vmatrix}$$

$$= \sum ab \begin{vmatrix} 1 & 1 & 1 \\ ab+bc & -ca & ab+bc \\ bc+ca & bc+ca & -ab \end{vmatrix}$$

(applying  $C_1 - C_2$  and  $C_2 - C_3$ )

$$= \sum ab \begin{vmatrix} 0 & 0 & 1 \\ \sum ab & -\sum ab & ab+bc \\ 0 & \sum ca & -ab \end{vmatrix} = (\sum ab)^3$$

► **Example 15.** Simplify  $\begin{vmatrix} a & b & c \\ a^2 & b^2 & c^2 \\ bc & ca & ab \end{vmatrix}$

► **Solution** Given determinant is equal to

$$= \frac{1}{abc} \begin{vmatrix} a^2 & b^2 & c^2 \\ a^3 & b^3 & c^3 \\ abc & abc & abc \end{vmatrix}$$

$$= \frac{abc}{abc} \begin{vmatrix} a^2 & b^2 & c^2 \\ a^3 & b^3 & c^3 \\ 1 & 1 & 1 \end{vmatrix}$$

Apply  $C_1 \rightarrow C_1 - C_2, C_2 \rightarrow C_2 - C_3$

$$= \begin{vmatrix} a^2 - b^2 & b^2 - c^2 & c^2 \\ a^3 - b^3 & b^3 - c^3 & c^3 \\ 0 & 0 & 1 \end{vmatrix}$$

$$\begin{aligned} &= (a-b)(b-c) \begin{vmatrix} a+b & b+c & c^2 \\ a^2+ab+b^2 & b^2+bc+c^2 & c^3 \\ 0 & 0 & 1 \end{vmatrix} \\ &= (a-b)(b-c) [ab^2 + abc + ac^2 + b^3 + b^2c \\ &\quad + bc^2 - a^2b - a^2c - ab^2 - abc - b^3 - b^2c] \\ &= (a-b)(b-c) [c(ab + bc + ca) - a(ab + bc + ca)] \\ &= (a-b)(b-c)(c-a)(ab + bc + ca) \end{aligned}$$

► **Example 16.** Prove that

$$\begin{vmatrix} a & a+b & a+b+c \\ 2a & 3a+2b & 4a+3b+2c \\ 3a & 6a+3b & 10a+6b+3a \end{vmatrix} = a^3.$$

► **Solution** Applying operations  $R_2 \rightarrow R_2 - 2R_1$  and  $R_3 \rightarrow R_3 - 3R_1$  to the given determinant  $\Delta$ , we have

$$\Delta = \begin{vmatrix} a & a+b & a+b+c \\ 0 & a & 2a+b \\ 0 & 3a & 7a+3b \end{vmatrix}$$

Now applying  $R_3 \rightarrow R_3 - 3R_2$ , we get

$$\Delta = \begin{vmatrix} a & a+b & a+b+c \\ 0 & a & 2a+b \\ 0 & 0 & a \end{vmatrix}$$

Expanding along  $C_1$ , we obtain

$$\Delta = a \begin{vmatrix} a & 2a+b \\ 0 & a \end{vmatrix} + 0 + 0$$

$$= a(a^2 - 0) = a(a^2) = a^3$$

► **Example 17.** Prove that

$$\begin{vmatrix} b^2+c^2 & a^2 & a^2 \\ b^2 & a^2+c^2 & b^2 \\ c^2 & c^2 & a^2+b^2 \end{vmatrix} = 4a^2b^2c^2$$

► **Solution** Applying  $R_1 \rightarrow R_1 - R_2 - R_3$ , we get

$$D = \begin{vmatrix} 0 & -2c^2 & -2b^2 \\ b^2 & a^2+c^2 & b^2 \\ c^2 & c^2 & a^2+b^2 \end{vmatrix}.$$

Taking 2 common out of  $R_1$  and then operating  $R_2 \rightarrow R_2 + R_1$  and  $R_3 \rightarrow R_3 + R_1$  we get

$$D = 2 \begin{vmatrix} 0 & -c^2 & -b^2 \\ b^2 & a^2 & 0 \\ c^2 & 0 & a^2 \end{vmatrix}$$

$$= 2 [c^2(a^2b^2 - 0) - b^2(0 - a^2c^2)] = 4a^2b^2c^2.$$

► **Example 18.** Suppose that digit numbers A28, 3B9 and 62C, where A, B and C are integers between 0 and 9, are divisible by a fixed integer k. Prove that the determinant

$$\begin{vmatrix} A & 3 & 6 \\ 8 & 9 & C \\ 2 & B & 2 \end{vmatrix} \text{ is also divisible by k.}$$

► **Solution** Given A28, 3B9 and 62C are divisible by K,

$$\text{then } A28 = n_1k = 100A + 20 + 8$$

$$3B9 = n_2k = 300 + 10B + 9$$

$$62C = n_3k = 600 + 20 + C$$

where  $n_1, n_2$  and  $n_3$  are integers.

$$\text{Let } \Delta = \begin{vmatrix} A & 3 & 6 \\ 8 & 9 & C \\ 2 & B & 2 \end{vmatrix}$$

Applying  $R_2 \rightarrow 100R_1 + 10R_3 + R_2$

$$\therefore \Delta = \begin{vmatrix} A & 3 & 6 \\ 100A+20+8 & 300+10B+9 & 600+20+C \\ 2 & B & 2 \end{vmatrix}$$

$$\therefore \Delta = \begin{vmatrix} A & 3 & 6 \\ n_1k & n_2k & n_3k \\ 2 & B & 2 \end{vmatrix} = k \begin{vmatrix} A & 3 & 6 \\ n_1 & n_2 & n_3 \\ 2 & B & 2 \end{vmatrix}$$

Hence  $\Delta$  is divisible by k.

► **Example 19.** Prove that

$$\Delta = \begin{vmatrix} a+bx & c+dx & p+qx \\ ax+b & cx+d & px+q \\ u & v & w \end{vmatrix}$$

$$= (1-x^2) \begin{vmatrix} a & c & p \\ b & d & q \\ u & v & w \end{vmatrix}$$

► **Solution** Applying  $R_1 \rightarrow R_1 - xR_2$  to  $\Delta$ , we get

$$\Delta = \begin{vmatrix} a(1-x^2) & c(1-x^2) & p(1-x^2) \\ ax+b & cx+d & px+q \\ u & v & w \end{vmatrix}$$

$$= (1-x^2) \begin{vmatrix} a & c & p \\ ax+b & cx+d & px+q \\ u & v & w \end{vmatrix}$$

Applying  $R_2 \rightarrow R_2 - xR_1$ , we get

$$\Delta = (1-x)^2 \begin{vmatrix} a & c & p \\ b & d & q \\ u & v & w \end{vmatrix}$$

► **Example 20.** Prove that

$$\Delta = \begin{vmatrix} a^2+1 & ab & ac \\ ab & b^2+1 & bc \\ ac & bc & c^2+1 \end{vmatrix}$$

$$= 1 + a^2 + b^2 + c^2.$$

► **Solution** Multiplying the 1st, 2nd and 3rd columns by a, b and c respectively, we get

$$\Delta = \frac{1}{abc} \begin{vmatrix} a(a^2+1) & ab^2 & ac^2 \\ a^2b & b(b^2+1) & bc^2 \\ a^2c & b^2c & c(c^2+1) \end{vmatrix}$$

taking  $a, b, c$  common from the first, second and third rows respectively.

$$= \frac{abc}{abc} \begin{vmatrix} a^2 + 1 & b^2 & c^2 \\ a^2 & b^2 + 1 & c^2 \\ a^2 & b^2 & c^2 + 1 \end{vmatrix},$$

applying  $C_1 \rightarrow C_1 + C_2 + C_3$

$$= \begin{vmatrix} 1 + a^2 + b^2 + c^2 & b^2 & c^2 \\ 1 + a^2 + b^2 + c^2 & b^2 + 1 & c^2 \\ 1 + a^2 + b^2 + c^2 & b^2 & c^2 + 1 \end{vmatrix},$$

$$= (1 + a^2 + b^2 + c^2) \begin{vmatrix} 1 & b^2 & c^2 \\ 1 & b^2 + 1 & c^2 \\ 1 & b^2 & c^2 + 1 \end{vmatrix},$$

applying  $R_2 \rightarrow R_2 - R_1 \rightarrow R_3 - R_1$

$$= (1 + a^2 + b^2 + c^2) \begin{vmatrix} 1 & b^2 & c^2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix},$$

$$= (1 + a^2 + b^2 + c^2)$$

► **Example 21.** Prove that

$$\Delta = \begin{vmatrix} (b+c)^2 & a^2 & a^2 \\ b^2 & (c+a)^2 & b^2 \\ c^2 & c^2 & (a+b)^2 \end{vmatrix}$$

$$= 2abc(a+b+c)^3.$$

► **Solution** Applying  $C_1 \rightarrow C_1 - C_3, C_2 \rightarrow C_2 - C_3$ , we get

$$\Delta = \begin{vmatrix} (b+c+a)(b+c-a) & 0 & a^2 \\ 0 & (c+a+b)(c+a-b) & b^2 \\ (c+a+b)(c-a-b) & (c+a+b)(c-a-b) & (a+b)^2 \end{vmatrix}$$

$$= (a+b+c)^2 \begin{vmatrix} b+c-a & 0 & a^2 \\ 0 & c+a-b & b^2 \\ c-a-b & c-a-b & (a+b)^2 \end{vmatrix}$$

taking  $a+b+c$  common from each of the columns  $C_1$  and  $C_2$

$$= (a+b+c)^2 \begin{vmatrix} b+c-a & 0 & a^2 \\ 0 & c+a-b & b^2 \\ -2b & -2a & 2ab \end{vmatrix}$$

by  $R_3 \rightarrow R_3 - R_2 - R_1$

$$= (a+b+c)^2 \begin{vmatrix} b+c & \frac{a^2}{b} & a^2 \\ \frac{b^2}{a} & c+a & b^2 \\ 0 & 0 & 2ab \end{vmatrix}$$

by  $C_1 \rightarrow C_1 + \frac{1}{a}C_3, C_2 \rightarrow C_2 + \frac{1}{b}C_3$

$$= (a+b+c)^2 2ab \begin{vmatrix} b+c & \frac{a^2}{b} \\ \frac{b^2}{a} & c+a \end{vmatrix}$$

$$= (a+b+c)^2 2ab \{ (b+c)(c+a) - ab \}$$

$$= 2abc(a+b+c)^3.$$

### Practice Problems

[C]

1. Prove the following :

$$(i) \begin{vmatrix} x+a & b & c \\ a & x+b & c \\ a & b & x+c \end{vmatrix} = x^2(x+a+b+c)$$

$$(ii) \begin{vmatrix} a+b+c & -c & -b \\ -c & a+b+c & -a \\ -b & -a & a+b+c \end{vmatrix}$$

$$= 2(a+b)(b+c)(c+a)$$

$$(iii) \begin{vmatrix} (x-2)^2 & (x-1)^2 & x^2 \\ (x-1)^2 & x^2 & (x+1)^2 \\ x^2 & (x+1)^2 & (x+2)^2 \end{vmatrix} = -8.$$

$$(iv) \begin{vmatrix} 1+a & 1 & 1 \\ 1 & 1+b & 1 \\ 1 & 1 & 1+c \end{vmatrix} = abc \left( 1 + \frac{1}{a} + \frac{1}{b} + \frac{1}{c} \right)$$

$$(iv) \begin{vmatrix} 15-2x & 11 & 10 \\ 11-3x & 17 & 16 \\ 7-x & 14 & 13 \end{vmatrix} = 0$$

2. Without computing the following determinants, prove that they are equal:

$$(i) \begin{vmatrix} 1 & 3 & 2 \\ 4 & 7 & 11 \\ 5 & 10 & 13 \end{vmatrix} = \begin{vmatrix} 3 & -1 & 2 \\ 2 & 4 & 6 \\ 5 & 6 & 11 \end{vmatrix}$$

$$(ii) \begin{vmatrix} 1 & -1 & 1 \\ 2 & 3 & -5 \\ 4 & 1 & -3 \end{vmatrix} = \begin{vmatrix} 2 & 1 & 5 \\ -1 & 3 & 1 \\ 1 & 2 & 4 \end{vmatrix}$$

3. Evaluate the following determinants :

$$(i) \begin{vmatrix} x-y-z & 2x & 2x \\ 2y & y-z-x & 2y \\ 2z & 2z & z-x-y \end{vmatrix}$$

$$(ii) \begin{vmatrix} x+4 & 2x & 2x \\ 2x & x+4 & 2x \\ 2x & 2x & x+4 \end{vmatrix}$$

$$(iii) \begin{vmatrix} a-b & b-c & c-a \\ b-c & c-a & a-b \\ c-a & a-b & b-c \end{vmatrix}$$

$$(iv) \begin{vmatrix} \sqrt{13}+\sqrt{3} & 2\sqrt{5} & \sqrt{5} \\ \sqrt{15}+\sqrt{26} & \frac{5}{\sqrt{15}} & \sqrt{10} \\ 3+\sqrt{65} & \sqrt{15} & 5 \end{vmatrix}.$$

4. Solve the equation :

$$(i) \begin{vmatrix} x & -6 & -1 \\ 2 & -3x & x-3 \\ -3 & 2x & x+2 \end{vmatrix} = 0$$

$$(ii) \begin{vmatrix} x-2 & 2x-3 & 3x-4 \\ x-4 & 2x-9 & 3x-16 \\ x-8 & 2x-27 & 2x-64 \end{vmatrix} = 0$$

$$(iii) \begin{vmatrix} x+3 & 5 & 2 \\ 1 & x+7 & 6 \\ 2 & 5 & x+3 \end{vmatrix} = 0$$

$$5. \text{ Find the value of } \begin{vmatrix} 67 & 19 & 21 \\ 39 & 13 & 14 \\ 81 & 24 & 26 \end{vmatrix}$$

$$6. \text{ Evaluate } \Delta = \begin{vmatrix} {}^xC_1 & {}^xC_2 & {}^xC_3 \\ {}^yC_1 & {}^yC_2 & {}^yC_3 \\ {}^zC_1 & {}^zC_2 & {}^zC_3 \end{vmatrix}$$

7. Prove the following:

$$(i) \begin{vmatrix} x+y+2z & x & y \\ z & y+z+2x & y \\ z & x & z+x+2y \end{vmatrix} = 2(x+y+z)^3$$

$$(ii) \begin{vmatrix} 1-x & a & a^2 \\ a & a^2-x & a^3 \\ a^2 & a^3 & a^4-x \end{vmatrix} = x^2(1+a^2+a^4)-x^3$$

$$(iii) \begin{vmatrix} a & b-c & c+b \\ a+c & b & c-a \\ a-b & b+a & c \end{vmatrix} = (a+b+c)(a^2+b^2+c^2)$$

$$8. \text{ Prove that } 2 \begin{vmatrix} 0 & e & d \\ e & 0 & f \\ d & f & 0 \end{vmatrix} = \begin{vmatrix} d+e & f & f \\ d & e+f & d \\ e & e & f+d \end{vmatrix}$$

9. Prove that

$$\begin{vmatrix} b+c & c+a & a+b \\ q+r & r+p & p+q \\ y+z & z+x & x+y \end{vmatrix} = 2 \begin{vmatrix} a & b & c \\ p & q & r \\ x & y & z \end{vmatrix}$$

10. Prove the following:

$$(i) \begin{vmatrix} a-b-c & 2a & 2a \\ 2b & b-c-a & 2b \\ 2c & 2c & c-a-b \end{vmatrix} = (a+b+c)^3$$

$$(ii) \begin{vmatrix} a^2 & bc & ac+c^2 \\ a^2+ab & b^2 & ac \\ ab & b^2+bc & c^2 \end{vmatrix} = 4a^2b^2c^2$$

$$(iii) \begin{vmatrix} a^2 + x & ab & ac \\ ab & b^2 + x & bc \\ ac & bc & c^2 + x \end{vmatrix} \\ = x^2 (a^2 + b^2 + c^2 + x)$$

$$(iv) \begin{vmatrix} 1+a^2-b^2 & 2ab & -2b \\ 2ab & 1-a^2+b^2 & 2a \\ 2b & -2a & 1-a^2-b^2 \end{vmatrix} \\ = (1+a^2+b^2)^3.$$

$$11. \text{ Prove that } \begin{vmatrix} 1+a_1 & a_2 & a_3 & a_4 \\ a_1 & 1+a_2 & a_3 & a_4 \\ a_1 & a_2 & 1+a_3 & a_4 \\ a_1 & a_2 & a_3 & 1+a_4 \end{vmatrix} \\ = 1 + a_1 + a_2 + a_3 + a_4.$$

12. If  $a, b, c$  be the sides of a triangle ABC, then prove that the determinant,

$$\begin{vmatrix} a^2 & b \sin A & c \sin A \\ b \sin A & 1 & \cos A \\ c \sin A & \cos A & 1 \end{vmatrix} \text{ vanishes.}$$

$$13. \text{ Prove that } \begin{vmatrix} a^2+2a & 2a+1 & 1 \\ 2a+1 & a+2 & 1 \\ 3 & 3 & 1 \end{vmatrix} > \text{ or } = \text{ or } < 0$$

according as  $a > 1$  or  $a = 1$  or  $a < 1$ .

$$14. \text{ Evaluate } \sum_{r=1}^n D_r \begin{vmatrix} r-1 & x & 6 \\ (r-1)^2 & y & 4n-2 \\ (r-1)^3 & z & 3n^2-3n \end{vmatrix}$$

$$15. \text{ If } \Delta_r = \begin{vmatrix} 2^{r-1} & 2(3^{r-1}) & 4(5^{r-1}) \\ x & y & z \\ 2^n-1 & 3^n-1 & 5^n-1 \end{vmatrix} \text{ show that}$$

$$\sum_{r=1}^n \Delta_r = 0.$$

### 2.3 FACTOR THEOREM

If the elements of a determinant  $\Delta$  are rational integral functions of  $x$  and two rows (or columns) become identical when  $x = a$ , then  $x - a$  is a factor of  $\Delta$ .

For  $\Delta$  can be expressed as a polynomial in  $x$ , and  $\Delta = 0$  when  $x = a$ ; so that the result follows by the Remainder theorem.

Again, if  $r$  rows become identical when  $a$  is substituted for  $x$ , then  $(x - a)^{r-1}$  is a factor of  $\Delta$ . For  $r - 1$  of these rows can be replaced by rows in which the elements are the differences of corresponding elements in the original rows. Hence  $x - a$  is a factor of all the elements in each of these  $(r - 1)$  substituted rows and the result follows.

► **Example 1.** Prove that  $\begin{vmatrix} a & b & c \\ a^2 & b^2 & c^2 \\ bc & ca & ab \end{vmatrix}$

$= (a - b)(b - c)(c - a)(ab + bc + ca)$  by using factor theorem.

► **Solution** Let  $a = b$

$$\Rightarrow D = \begin{vmatrix} a & b & c \\ a^2 & b^2 & c^2 \\ bc & ac & ab \end{vmatrix} = 0$$

Hence  $(a - b)$  is a factor of determinant.

Similarly, let  $b = c, D = 0$

$c = a, D = 0$

Hence,  $(a - b)(b - c)(c - a)$  is factor of determinant. But the given determinant is of fifth degree so

$$\begin{vmatrix} a & b & c \\ a^2 & b^2 & c^2 \\ bc & ca & ab \end{vmatrix} = (a - b)(b - c)(c - a).$$

$$(\lambda (a^2 + b^2 + c^2) + \mu (ab + bc + ca))$$

Since this is an identity so in order to find the values of  $\lambda$  and  $\mu$ , let  $a = 0, b = 1, c = -1$



$$\Rightarrow \begin{aligned} -2 &= (2) (2\lambda - \mu) \\ (2\lambda - \mu) &= -1. \end{aligned} \quad \dots(1)$$

$$\text{Let } a = 1, b = 2, c = 0$$

$$\begin{vmatrix} 1 & 2 & 0 \\ 1 & 4 & 0 \\ 0 & 0 & 2 \end{vmatrix} = (-1) 2 (-1) (5\lambda + 2\mu)$$

$$\Rightarrow 5\lambda + 2\mu = 2 \quad \dots(2)$$

From (1) and (2)  $\lambda = 0$  and  $\mu = 1$

$$\text{Hence } \begin{vmatrix} a & b & c \\ a^2 & b^2 & c^2 \\ bc & ca & ab \end{vmatrix}$$

$$= (a - b)(b - c)(c - a)(ab + bc + ca).$$

► **Example 2.** Prove that

$$\begin{vmatrix} -2a & a+b & a+c \\ b+a & -2b & b+c \\ c+a & c+b & -2c \end{vmatrix} = 4(b+c)(c+a)(a+b).$$

$$\text{► **Solution** Let } \Delta = \begin{vmatrix} -2a & a+b & a+c \\ b+a & -2b & b+c \\ c+a & c+b & -2c \end{vmatrix}.$$

Putting  $a + b = 0$  i.e.  $b = -a$ ,

$$\Delta = \begin{vmatrix} -2a & 0 & a+c \\ 0 & 2a & c-a \\ c+a & c-a & -2c \end{vmatrix}$$

Expanding along  $R_1$

$$\begin{aligned} &= -2a \{-4ac - (c-a)^2\} - 0 + (a+c) \{0 - 2a(c+a)\} \\ &= -2a(c+a)^2 - 2a(c+a)^2 = 0. \end{aligned}$$

Hence  $a + b$  is a factor of  $\Delta$ . Similarly  $b + c$  and  $c + a$  are factors of  $\Delta$ . On expansion of determinant we can see that each term of the determinant is a homogeneous expression in  $a, b$ , and  $c$  of degree 3 and also R.H.S. is a homogeneous expression of degree 3.

$$\text{Let } \Delta = k(a+b)(b+c)(c+a)$$

$$\text{or } \begin{vmatrix} -2a & a+b & a+c \\ b+a & -2b & b+c \\ c+a & c+b & -2c \end{vmatrix} = k(a+b)(b+c)(c+a)$$

Putting  $a = 0, b = 1, c = 2$ , we get

$$\begin{vmatrix} 0 & 1 & 2 \\ 0 & -2 & 3 \\ 2 & 3 & -4 \end{vmatrix} = k(0+1)(1+2)(2+0)$$

$$\Rightarrow 0 - 1(-4 - 6) + 2(3 + 4) = 6k \Rightarrow 24 = 6k$$

$$\therefore k = 4$$

$$\text{Hence } \begin{vmatrix} -2a & a+b & a+c \\ b+a & -2b & b+c \\ c+a & c+b & -2c \end{vmatrix} = 4(a+b)(b+c)(c+a).$$

### The product of differences : alternants

$$\text{► **Example 3.** Express } \begin{vmatrix} 1 & 1 & 1 \\ \alpha & \beta & \gamma \\ \alpha^2 & \beta^2 & \gamma^2 \end{vmatrix} \text{ in factors.}$$

► **Solution** We consider the determinant as a polynomial in  $\beta$ . For  $\beta = \gamma$ , the second and third columns become identical, so  $\beta - \gamma$  is a factor of the determinant. Similarly  $\gamma - \alpha$  and  $\alpha - \beta$  are two factors.

$$\therefore \Delta = k(\beta - \gamma)(\gamma - \alpha)(\alpha - \beta)$$

Since  $\Delta$  is a polynomial in  $\alpha, \beta, \gamma$  of degree 3,  $k$  is independent of  $\alpha, \beta, \gamma$ . Now by equating the coefficients of  $\beta\gamma^2$  from both sides we have  $k = 1$ .

$$\therefore \Delta = (\beta - \gamma)(\gamma - \alpha)(\alpha - \beta).$$

► **Example 4.** Prove that

$$\begin{vmatrix} 1 & 1 & 1 \\ \alpha & \beta & \gamma \\ \alpha^3 & \beta^3 & \gamma^3 \end{vmatrix} = (\beta - \gamma)(\gamma - \alpha)(\alpha - \beta)(\alpha + \beta + \gamma).$$

► **Solution** The first three factors are obtained using remainder theorem. The degree of the determinant in  $\alpha, \beta, \gamma$  is four; so the remaining factor must be linear in  $\alpha, \beta, \gamma$  and it must be unaltered by the interchange of any two letters (for both the determinant and the product of the first three factors are altered in sign by such an interchange).

Let it be  $k(\alpha + \beta + \gamma)$ , where  $k$  is a constant. We have then the identity

$$\begin{vmatrix} 1 & 1 & 1 \\ \alpha & \beta & \gamma \\ \alpha^3 & \beta^3 & \gamma^3 \end{vmatrix} = k(\alpha - \beta)(\beta - \gamma)(\gamma - \alpha)(\alpha + \beta + \gamma).$$

Putting  $\alpha = 0, \beta = 1, \gamma = 2$  in this identity, we have

$$\begin{vmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 1 & 8 \end{vmatrix} = k(-1)(-1)(2)(3)$$

or  $6k = 6$  or  $k = 1$ .

$$\therefore \Delta = (\alpha - \beta)(\beta - \gamma)(\gamma - \alpha)(\alpha + \beta + \gamma).$$



$$(a) \begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix} = (a - b)(b - c)(c - a)$$

$$(b) \begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^3 & b^3 & c^3 \end{vmatrix} = (a - b)(b - c)(c - a)(a + b + c)$$

$$(c) \begin{vmatrix} 1 & a & bc \\ 1 & b & ca \\ 1 & c & ab \end{vmatrix} = (a - b)(b - c)(c - a)$$

$$(d) \begin{vmatrix} a & b & c \\ a^2 & b^2 & c^2 \\ bc & ca & ab \end{vmatrix} = (a - b)(b - c)(c - a)(ab + bc + ca)$$

$$(e) \begin{vmatrix} a & b+c & a^2 \\ b & c+a & b^2 \\ c & a+b & c^2 \end{vmatrix} = -(a + b + c) - (a - b)(b - c)(c - a)$$

► **Example 5.** Prove that

$$\begin{vmatrix} (\beta + \gamma - \alpha - \delta)^4 & (\beta + \gamma - \alpha - \delta)^2 & 1 \\ (\gamma + \alpha - \beta - \delta)^4 & (\gamma + \alpha - \beta - \delta)^2 & 1 \\ (\alpha + \beta - \gamma - \delta)^4 & (\alpha + \beta - \gamma - \delta)^2 & 1 \end{vmatrix}$$

$$= 64(\alpha - \beta)(\alpha - \gamma)(\alpha - \delta)(\beta - \gamma)(\beta - \delta)(\gamma - \delta)$$

► **Solution** Put

$$(\beta + \gamma - \alpha - \delta)^2 = A, (\gamma + \alpha - \beta - \delta)^2 = B,$$

$$(\alpha + \beta - \gamma - \delta)^2 = C$$

so that

$$A - B = (\beta + \gamma - \alpha - \delta)^2 - (\gamma + \alpha - \beta - \delta)^2$$

$$= (2\gamma - 2\delta)(2\beta - 2\alpha) = -4(\alpha - \beta)(\gamma - \delta) \text{ etc.}$$

we get

$$D = \begin{vmatrix} A^2 & A & 1 \\ B^2 & B & 1 \\ C^2 & C & 1 \end{vmatrix} = -(A - B)(B - C)(C - A)$$

$$= -64(\alpha - \beta)(\alpha - \gamma)(\alpha - \delta)(\beta - \gamma)(\beta - \delta)(\gamma - \delta).$$

► **Example 6.** Express the determinant

$$\Delta_4 = \begin{vmatrix} \alpha^3 & \beta^3 & \gamma^3 & \delta^3 \\ \alpha^2 & \beta^2 & \gamma^2 & \delta^2 \\ \alpha & \beta & \gamma & \delta \\ 1 & 1 & 1 & 1 \end{vmatrix} \text{ as a product of}$$

differences.

► **Solution** On expanding  $\Delta_4$ , we see that it may be regarded

(a) as a homogeneous polynomial of degree 6 in the variables  $\alpha, \beta, \gamma, \delta$ ; the coefficients being  $\pm 1$ ;

(b) as a non-homogeneous polynomial of highest degree 3 in  $\alpha$ , the coefficients of the powers of  $\alpha$  being functions of  $\beta, \gamma, \delta$ .

The determinant vanishes when  $\alpha = \beta$ , since it then has two identical columns. Hence, by the remainder theorem applied to a polynomial in  $\alpha$ , the determinant has a factor  $\alpha - \beta$ . By a similar argument; the difference of any two of  $a, b, g, d$  is a factor of  $\Delta_4$ , so that

$$\Delta_4 = K(\alpha - \beta)(\alpha - \gamma)(\alpha - \delta)(\beta - \gamma)(\beta - \delta)(\gamma - \delta). \quad \dots(1)$$

Since  $\Delta_4$  is a homogeneous polynomial of degree 6 in  $\alpha, \beta, \gamma, \delta$ , the factor  $K$  must be independent of  $\alpha, \beta, \gamma, \delta$  and so is a numerical constant. The coefficient of  $\alpha^3\beta^2\gamma$  on the right-hand side of (1) is  $K$ , and in  $\Delta_4$  it is 1. Hence  $K = 1$  and

$$\Delta_4 = (\alpha - \beta)(\alpha - \gamma)(\alpha - \delta)(\beta - \gamma)(\beta - \delta)(\gamma - \delta).$$

The corresponding determinant of order  $n$ , is equal to the product of the differences that can be

formed from the letters  $a_1, a_2, \dots, a_n$  appearing in the determinant, due regard being paid to alphabetical order in the factors. It is called Vandermonde determinant.

$$\begin{vmatrix} 1 & 1 & 1 & \dots & 1 \\ a_1 & a_2 & a_3 & \dots & a_n \\ a_1^2 & a_2^2 & a_3^2 & \dots & a_n^2 \\ \dots & \dots & \dots & \dots & \dots \\ a_1^{n-1} & a_2^{n-1} & a_3^{n-1} & \dots & a_n^{n-1} \end{vmatrix}$$

For any  $n$ , the Vandermonde determinant is equal to the product of all possible differences

$$a_i - a_j, \text{ where } 1 \leq j < i \leq n.$$

► **Example 7.** Prove that

$$\Delta = \begin{vmatrix} 1 & bc+ad & b^2c^2+a^2d^2 \\ 1 & ca+bd & c^2a^2+b^2d^2 \\ 1 & ab+cd & a^2b^2+c^2d^2 \end{vmatrix}$$

$$= (a-b)(a-c)(a-d)(b-c)(b-d)(c-d).$$

► **Solution** Applying  $R_2 \rightarrow R_2 - R_1, R_3 \rightarrow R_3 - R_1$ , we get

$$\Delta = \begin{vmatrix} 1 & bc+ad & b^2c^2+a^2d^2 \\ 1 & (a-b)(c-d) & (a^2-b^2)(c^2-d^2) \\ 1 & (a-c)(b-d) & (a^2-c^2)(b^2-d^2) \end{vmatrix}$$

$$= \begin{vmatrix} (a-b)(c-d) & (a-b)(a+b)(c-d)(c+d) \\ (a-c)(b-d) & (a-c)(a+c)(b-d)(b+d) \end{vmatrix}$$

$$= (a-b)(c-d)(a-c)(b-d) \begin{vmatrix} 1 & (a+b)(c+d) \\ 1 & (a+c)(b+d) \end{vmatrix}$$

$$= (a-b)(c-d)(a-c)(b-d) [(a+c)(b+d) - (a+b)(c+d)]$$

$$= (a-b)(c-d)(a-c)(b-d)(ab+cd-ac-bd)$$

$$= (a-b)(a-c)(a-d)(b-c)(b-d)(c-d).$$

Alternatively

$$\begin{aligned} bc+ad &= x \\ \text{Let } ca+bd &= y \\ ab+cd &= z \end{aligned}$$

$$\Delta = \begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix} = (x-y)(y-z)(z-x).$$

$$\text{Hence } \Delta = (a-b)(b-c)(c-d)(a-d)(b-d)(a-c)$$

► **Example 8.** Without expanding show that

$$\begin{vmatrix} 1 & a^2 & a^3 \\ 1 & b^2 & b^3 \\ 1 & c^2 & c^3 \end{vmatrix} = (ab+bc+ca) \begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix}$$

► **Solution** Let  $D = \begin{vmatrix} 1 & a^2 & a^3 \\ 1 & b^2 & b^3 \\ 1 & c^2 & c^3 \end{vmatrix}$

$$\text{Then } D = abc \begin{vmatrix} 1/a & a & a^2 \\ 1/b & b & b^2 \\ 1/c & c & c^2 \end{vmatrix} = \begin{vmatrix} bc & a & a^2 \\ ac & b & b^2 \\ ab & c & c^2 \end{vmatrix}$$

(multiplying  $abc$  with  $C_1$ )

$$= \begin{vmatrix} bc-a^2 & a & a^2 \\ ac-b^2 & b & b^2 \\ ab-c^2 & c & c^2 \end{vmatrix} \quad (C_1 \rightarrow C_1 - C_3)$$

$$= \begin{vmatrix} bc+ac-a^2 & a & a^2 \\ ac+bc-b^2 & b & b^2 \\ ab & c & c^2 \end{vmatrix} \quad (C_1 \rightarrow C_1 + cC_2)$$

$$= \begin{vmatrix} bc+ac & a & a^2 \\ ac+bc+ab-b^2 & b & b^2 \\ ab & c & c^2 \end{vmatrix} \quad (C_1 \rightarrow C_1 + aC_2)$$

$$= \begin{vmatrix} ab+bc+ac & a & a^2 \\ ab+bc+ac & b & b^2 \\ ab+bc+ac & c & c^2 \end{vmatrix} \quad (C_1 \rightarrow C_1 + bC_2)$$

$$= (ab+bc+ac) \begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix} = \text{RHS}$$

► **Example 9.** Compute the determinant

$$D = \begin{vmatrix} 0 & x & y & z \\ x & 0 & z & y \\ y & z & 0 & x \\ z & y & x & 0 \end{vmatrix}.$$

► **Solution** Adding all columns to the first column, we find that the determinant is divisible by  $x + y + z$ ; adding to the first column the second column and subtracting the third and fourth yields the factor

$$y + z - x;$$

adding the third column to the first and subtracting the second and fourth columns yields the factor

$$x - y + z;$$

finally, adding the fourth column to the first and subtracting the second and third yields the factor

$$x + y - z.$$

Regarding  $x, y, z$  as independent unknowns, we conclude that all these four factors are relatively prime in pairs and, hence, the determinant is divisible by their product  $(x+y+z)(y-z-x)(x-y+z)(x+y-z)$ .

#### Circulant

Circulant is a determinant in which the elements of the rows (or columns) are in cyclic arrangement.

$$\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = -(a^3 + b^3 + c^3 - 3abc).$$

This is obtained by simply expanding along any row or column.

► **Example 10.** If  $\alpha, \beta, \gamma$  are the roots of  $x^3 + ax^2 + b = 0$ , then find the value of

$$\begin{vmatrix} \alpha & \beta & \gamma \\ \beta & \gamma & \alpha \\ \gamma & \alpha & \beta \end{vmatrix}$$

► **Solution** Since  $\alpha, \beta, \gamma$  are the roots of the given equation, therefore  $\alpha + \beta + \gamma = -a$ ,  $\alpha\beta + \beta\gamma + \gamma\alpha = 0$  and  $\alpha\beta\gamma = -b$ .

Now,

$$\begin{vmatrix} \alpha & \beta & \gamma \\ \beta & \gamma & \alpha \\ \gamma & \alpha & \beta \end{vmatrix}$$

$$\begin{aligned} &= -(\alpha + \beta + \gamma)(\alpha^2 + \beta^2 + \gamma^2 - \alpha\beta - \beta\gamma - \gamma\alpha) \\ &= -(\alpha + \beta + \gamma)((\alpha + \beta + \gamma)^2 - 3(\alpha\beta + \beta\gamma + \gamma\alpha)) \\ &= -(-a)\{a^2 - 0\} = a^3. \end{aligned}$$

► **Example 11.** If

$$\begin{vmatrix} x & 3 & 6 \\ 3 & 6 & x \\ 6 & x & 3 \end{vmatrix} = \begin{vmatrix} 2 & x & 7 \\ x & 7 & 2 \\ 7 & 2 & x \end{vmatrix} = \begin{vmatrix} 4 & 5 & x \\ 5 & x & 3 \\ x & 4 & 5 \end{vmatrix} = 0$$

then find the value of  $x$ .

► **Solution** Using the value of circulant, we must have  $a + b + c = 0$

$$x + 3 + 6 = x + 2 + 7 = x + 4 + 5 = 0 \Rightarrow x = -9.$$

► **Example 12.** If  $\omega$  is a cube root of unity, prove

$$\text{that } a + b\omega + c\omega^2 \text{ is a factor of } \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}.$$

Hence show that the determinant is equal to  $-(a^3 + b^3 + c^3 - 3abc)$ . Note that this determinant is called a circulant.

► **Solution** To the first column add  $\omega$  times the second and  $\omega^2$  times the third, therefore

$$\begin{aligned} \Delta &= \begin{vmatrix} a+b\omega+c\omega^2 & c & c \\ b+c\omega+a\omega^2 & b & a \\ c+a\omega+b\omega^2 & a & b \end{vmatrix} \\ &= (a + b\omega + c\omega^2) \begin{vmatrix} 1 & a & c \\ \omega^2 & b & a \\ \omega & a & b \end{vmatrix}; \end{aligned}$$

∴  $a + b\omega + c\omega^2$  is a factor of  $\Delta$ , similarly  $a + b\omega^2 + c\omega$  and  $a + b + c$  are factors.

#### Practice Problems

[D]

1. Without expanding, as far as possible, prove that

$$\begin{vmatrix} 1 & 1 & 1 \\ x & y & z \\ x^3 & y^3 & z^3 \end{vmatrix} = (x-y)(y-z)(z-x)(x+y+z).$$

2. Show that  $\begin{vmatrix} 1 & a & bc \\ 1 & b & ca \\ 1 & c & ab \end{vmatrix} = (a-b)(b-c)(c-a)$

3. Prove that 
$$\begin{vmatrix} x & x^2 & yz \\ y & y^2 & zx \\ z & z^2 & xy \end{vmatrix}$$

$$= (x-y)(y-z)(z-x)(xy+yz+zx)$$

4. Prove that 
$$\begin{vmatrix} x & x^2 & 1+px^3 \\ y & y^2 & 1+py^3 \\ z & z^2 & 1+pz^3 \end{vmatrix}$$

$$= (1+pxyz)(x-y)(y-z)(z-x)$$

5. Without expanding prove that

$$\begin{vmatrix} bc & a & a^2 \\ ca & b & b^2 \\ ab & c & c^2 \end{vmatrix} = \begin{vmatrix} 1 & a^2 & a^3 \\ 1 & b^2 & b^3 \\ 1 & c^2 & c^3 \end{vmatrix}.$$

6. Prove that

$$\begin{vmatrix} a^2 & b^2 & c^2 \\ (a+1)^2 & (b+1)^2 & (c+1)^2 \\ (a-1)^2 & (b-1)^2 & (c-1)^2 \end{vmatrix} = 4 \begin{vmatrix} a^2 & b^2 & c^2 \\ a & b & c \\ 1 & 1 & 1 \end{vmatrix}.$$

7. Prove that 
$$\begin{vmatrix} x^2 & x^2-(y-z)^2 & yz \\ y^2 & y^2-(z-x)^2 & zx \\ z^2 & z^2-(x-y)^2 & xy \end{vmatrix}$$

$$= (x-y)(y-z)(z-x)(x+y+z)(x^2+y^2+z^2).$$

8. Show that

$$\Delta = \begin{vmatrix} (b+c)^2 & a^2 & bc \\ (c+a)^2 & b^2 & ca \\ (a+b)^2 & c^2 & ab \end{vmatrix} = (a^2+b^2+c^2)(a+b+c)(a-b)(b-c)(c-a).$$

9. Prove that 
$$\begin{vmatrix} -2a & a+b & a+c \\ b+a & -2b & b+c \\ c+a & c+b & -2c \end{vmatrix} = 4(b+c)(c+a)(a+b)$$

10. Prove that 
$$\begin{vmatrix} b+c & a+b & a \\ c+a & b+c & b \\ a+b & c+a & c \end{vmatrix} = a^3+b^3+c^3-3abc.$$

11. Show that 
$$\begin{vmatrix} a-b+c & a+b-c & a-b-c \\ b-c+a & b+c-a & b-c-a \\ c-a+b & c+a-b & c-a-b \end{vmatrix} = 4(a^3+b^3+c^3-3abc).$$

## 2.4 MULTIPLICATION OF TWO DETERMINANTS

If A and B be two square matrices of the same order, then  $|AB| = |A| |B| = |BA|$ .

*Proof :*

If A and B are square matrices of order n, and if AB is the product of the two matrices, the determinant of the matrix AB is equal to the product of the determinants of the matrices A and B ; that is,

$$|AB| = |A| \times |B|.$$

$$|BA| = |B| \times |A|.$$

Since |A| and |B| are numbers, the commutative law holds for their product and  $|A| \times |B| = |B| \times |A|$ . Thus  $|AB| = |BA|$ , although the matrix AB is not the same as the matrix BA. This is due to fact that the value of a determinant is unaltered when rows and columns are interchanged, whereas such an interchange does not leave a general matrix unaltered.

Two determinants can be multiplied in a variety of ways. The natural way is to multiply them similar to matrices, i.e., row-by-column. The value of a determinant does not change by interchanging the rows and columns. Therefore while writing down the product of two determinants of the same order, we can also follow the row-by-row multiplication rule, or the column-by-row multiplication rule or column-by-column multiplication rule.

Thus 
$$\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix} \times \begin{vmatrix} l_1 & m_1 \\ l_2 & m_2 \end{vmatrix} = \begin{vmatrix} a_1 l_1 + b_1 l_2 & a_1 m_1 + b_1 m_2 \\ a_2 l_1 + b_2 l_2 & a_2 m_1 + b_2 m_2 \end{vmatrix} = \begin{vmatrix} a_1 l_1 + b_1 m_1 & a_1 l_2 + b_1 m_2 \\ a_2 l_1 + b_2 m_1 & a_2 l_2 + b_2 m_2 \end{vmatrix}$$

$$\begin{aligned}
 &= \begin{vmatrix} a_1 l_1 + a_2 m_1 & a_1 l_2 + a_2 m_2 \\ b_1 l_1 + b_2 m_1 & b_1 l_2 + b_2 m_2 \end{vmatrix} \\
 &= \begin{vmatrix} a_1 l_1 + a_2 l_2 & a_1 m_1 + a_2 m_2 \\ b_1 l_1 + b_2 l_2 & b_1 m_1 + b_2 m_2 \end{vmatrix}
 \end{aligned}$$

► **Example 1.** Show that

$$\begin{vmatrix} 1 & 2 \\ -1 & 3 \end{vmatrix} \times \begin{vmatrix} 3 & 0 \\ -1 & 4 \end{vmatrix} = \begin{vmatrix} 1 & 8 \\ -6 & 12 \end{vmatrix}.$$

► **Solution**  $\begin{vmatrix} 1 & 2 \\ -1 & 3 \end{vmatrix} \times \begin{vmatrix} 3 & 0 \\ -1 & 4 \end{vmatrix}$

$$\begin{aligned}
 &= \begin{vmatrix} 1 \times 3 - 2 \times (-1) & 1 \times 0 + 2 \times 4 \\ -1 \times 3 + 3 \times (-1) & -1 \times 0 + 3 \times 4 \end{vmatrix} \\
 &= \begin{vmatrix} 1 & 8 \\ -6 & 12 \end{vmatrix}
 \end{aligned}$$

Similarly two determinants of order three are multiplied.

Let  $A = \begin{pmatrix} 1 & -2 & 4 & 0 \\ 7 & 3 & 0 & 3 \\ -1 & 1 & -4 & 0 \\ 0 & 3 & 2 & 1 \end{pmatrix}$

and  $B = \begin{pmatrix} 2 & 16 & 17 & 4 \\ 0 & -3 & 22 & -3 \\ 0 & 0 & 6 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$

Verify that  $|AB| = |A| \cdot |B|$ .

We have  $AB = \begin{pmatrix} 2 & 22 & -3 & 10 \\ 14 & 103 & 185 & 22 \\ -2 & -19 & -19 & -7 \\ 0 & -9 & 78 & -8 \end{pmatrix}$

The operations  $7R_3 + R_2$  and  $R_3 + R_1$  show that

$$\det(AB) = \det \begin{pmatrix} 0 & 3 & -22 & 3 \\ 0 & -30 & 52 & -27 \\ -2 & -19 & -19 & -7 \\ 0 & -9 & 78 & -8 \end{pmatrix}$$

$$= -2 \det \begin{pmatrix} 3 & -22 & 3 \\ -30 & 52 & -27 \\ -9 & 78 & -8 \end{pmatrix}$$

Now using  $-C_1 + C_3$ ,  $10C_3 + C_1$ , and  $-3R_3 + R_1$ , in that order, we have

$$\det AB = -2 \det \begin{pmatrix} 0 & -256 & -3 \\ 0 & 52 & 3 \\ 1 & 78 & 1 \end{pmatrix}$$

$$= -2 \det \begin{pmatrix} -256 & -3 \\ 52 & 3 \end{pmatrix} = 1224$$

On the other hand,  $(\det A)(\det B) = (-34)(-36) = 1224$ .



If  $A$  is a matrix of order  $m \times n$  and  $B$  is a matrix of order  $n \times m$ , and if  $m > n$ , then  $\det AB = 0$ .

Let us consider

$$\begin{aligned}
 AB &= \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ a_{21} & \cdots & a_{2n} \\ \vdots & & \vdots \\ a_{m1} & \cdots & a_{mn} \end{bmatrix} \cdot \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1m} \\ \vdots & \vdots & \vdots & \vdots \\ b_{n1} & b_{n2} & \cdots & b_{nm} \end{bmatrix} \\
 &= \begin{bmatrix} a_{11} & \cdots & 0 & \cdots & 0 \\ a_{21} & \cdots & 0 & \cdots & 0 \\ \vdots & & \vdots & & \vdots \\ a_{m1} & \cdots & 0 & \cdots & 0 \end{bmatrix} \cdot \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1m} \\ \vdots & \vdots & \vdots & \vdots \\ b_{n1} & b_{n2} & \cdots & b_{nm} \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \cdots & 0 \end{bmatrix}
 \end{aligned}$$

$\underbrace{\hspace{10em}}_{m-n \text{ cols}}$

$\left. \begin{matrix} \vdots \\ b_{n1} & b_{n2} & \cdots & b_{nm} \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \cdots & 0 \end{matrix} \right\} m-n \text{ rows}$

Here the matrices of the last product are square and of order  $m$ . Since each of the latter matrices has determinant 0, we conclude that  $\det AB = 0$ .

► **Example 2.** Prove that

$$\begin{vmatrix} a_1 x_1 + b_1 y_1 & a_1 x_2 + b_1 y_2 & a_1 x_3 + b_1 y_3 \\ a_2 x_1 + b_2 y_1 & a_2 x_2 + b_2 y_2 & a_2 x_3 + b_2 y_3 \\ a_3 x_1 + b_3 y_1 & a_3 x_2 + b_3 y_2 & a_3 x_3 + b_3 y_3 \end{vmatrix} = 0$$

► **Solution** Given determinant can be split into product of two determinants

$$\begin{aligned} \text{i.e. } & \begin{vmatrix} a_1x_1 + b_1y_1 & a_1x_2 + b_1y_2 & a_1x_3 + b_1y_3 \\ a_2x_1 + b_2y_1 & a_2x_2 + b_2y_2 & a_2x_3 + b_2y_3 \\ a_3x_1 + b_3y_1 & a_3x_2 + b_3y_2 & a_3x_3 + b_3y_3 \end{vmatrix} \\ &= \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} \times \begin{vmatrix} x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \\ 0 & 0 & 0 \end{vmatrix} = 0 \end{aligned}$$

► **Example 3.** Let D

$$D = \begin{vmatrix} 2a_1b_1 & a_1b_2 + a_2b_1 & a_1b_3 + a_3b_1 \\ a_1b_2 + a_2b_1 & 2a_2b_2 & a_2b_3 + a_3b_2 \\ a_1b_3 + a_3b_1 & a_3b_2 + a_2b_3 & 2a_3b_3 \end{vmatrix}.$$

Express the determinant D as a product of two determinants. Hence or otherwise show that D = 0.

► **Solution** We have  $D = \begin{vmatrix} a_1 & b_1 & 0 \\ a_2 & b_2 & 0 \\ a_3 & b_3 & 0 \end{vmatrix}$

$$\times \begin{vmatrix} b_1 & a_1 & 0 \\ b_2 & a_2 & 0 \\ b_3 & a_3 & 0 \end{vmatrix}, \text{ as can be seen by applying}$$

row-by-row multiplication rule. Hence D = 0.

► **Example 4.** Prove that

$$\begin{vmatrix} x_1 & y_1 & 1 \\ ax_1 + bx_2 + cx_3 & ay_1 + by_2 + cy_3 & a + b + c \\ -ax_1 + bx_2 + cx_3 & -ay_1 + by_2 + cy_3 & -a + b + c \end{vmatrix} = 0.$$

► **Solution** Given that

$$\Delta = \begin{vmatrix} x_1 & y_1 & 1 \\ ax_1 + bx_2 + cx_3 & ay_1 + by_2 + cy_3 & a + b + c \\ -ax_1 + bx_2 + cx_3 & -ay_1 + by_2 + cy_3 & -a + b + c \end{vmatrix} = 0.$$

$$\begin{aligned} &= \begin{vmatrix} x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \\ 1 & 1 & 1 \end{vmatrix} \times \begin{vmatrix} 1 & 0 & 0 \\ a & b & c \\ -a & b & c \end{vmatrix} \\ &= \begin{vmatrix} x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \\ 1 & 1 & 1 \end{vmatrix} \times 0 = 0. \end{aligned}$$

► **Example 5.** If a, b, c and  $\alpha, \beta, \gamma$  are two triplets of distinct numbers, show that the value of the following determinant is not zero :

$$\begin{vmatrix} (a-\alpha)^2 & (\alpha-\beta)^2 & (\alpha-\gamma)^2 \\ (b-\alpha)^2 & (b-\beta)^2 & (b-\gamma)^2 \\ (c-\alpha)^2 & (c-\beta)^2 & (c-\gamma)^2 \end{vmatrix}$$

► **Solution** We have

$$\Delta = \begin{vmatrix} (a-\alpha)^2 & (\alpha-\beta)^2 & (\alpha-\gamma)^2 \\ (b-\alpha)^2 & (b-\beta)^2 & (b-\gamma)^2 \\ (c-\alpha)^2 & (c-\beta)^2 & (c-\gamma)^2 \end{vmatrix}$$

$$= \begin{vmatrix} a^2 \cdot 1 - 2a\alpha + 1\alpha^2 & a^2 \cdot 1 - 2a\beta + 1\beta^2 & a^2 \cdot 1 - 2a\gamma + 1\gamma^2 \\ b^2 \cdot 1 - 2b\alpha + 1\alpha^2 & b^2 \cdot 1 - 2b\beta + 1\beta^2 & b^2 \cdot 1 - 2b\gamma + 1\gamma^2 \\ c^2 \cdot 1 - 2c\alpha + 1\alpha^2 & c^2 \cdot 1 - 2c\beta + 1\beta^2 & c^2 \cdot 1 - 2c\gamma + 1\gamma^2 \end{vmatrix}$$

$$= \begin{vmatrix} a^2 & -2a & 1 \\ b^2 & -2b & 1 \\ c^2 & -2c & 1 \end{vmatrix} \times \begin{vmatrix} 1 & \alpha & \alpha^2 \\ 1 & \beta & \beta^2 \\ 1 & \gamma & \gamma^2 \end{vmatrix}$$

$$= 2 \begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix} \times \begin{vmatrix} 1 & \alpha & \alpha^2 \\ 1 & \beta & \beta^2 \\ 1 & \gamma & \gamma^2 \end{vmatrix}$$

(by interchanging the 1st and 3rd columns)

$$= 2(a-b)(b-c)(c-a)(\alpha-\beta)(\beta-\gamma)(\gamma-\alpha)$$

Since a, b, c and  $\alpha, \beta, \gamma$  are two distinct triplets none of the factors is zero. Therefore,  $\Delta \neq 0$ .

► **Example 6.** If  $\Sigma \cos^2 \alpha_1 = \Sigma \cos^2 \beta_1 = \Sigma \cos^2 \gamma_1 = 1$ ;  $\Sigma \cos \alpha_1 \cos \beta_1 = \Sigma \cos \beta_1 \cos \gamma_1 = \Sigma \cos \gamma_1 \cos \alpha_1 = 0$

then find the value of  $\begin{vmatrix} \cos \alpha_1 & \cos \alpha_2 & \cos \alpha_3 \\ \cos \beta_1 & \cos \beta_2 & \cos \beta_3 \\ \cos \gamma_1 & \cos \gamma_2 & \cos \gamma_3 \end{vmatrix}.$

► **Solution**

$$\begin{vmatrix} \cos \alpha_1 & \cos \alpha_2 & \cos \alpha_3 \\ \cos \beta_1 & \cos \beta_2 & \cos \beta_3 \\ \cos \gamma_1 & \cos \gamma_2 & \cos \gamma_3 \end{vmatrix} \times \begin{vmatrix} \cos \alpha_1 & \cos \alpha_2 & \cos \alpha_3 \\ \cos \beta_1 & \cos \beta_2 & \cos \beta_3 \\ \cos \gamma_1 & \cos \gamma_2 & \cos \gamma_3 \end{vmatrix}$$

$$= \begin{vmatrix} \Sigma \cos^2 \alpha_1 & \Sigma \cos \alpha_1 \cos \beta_1 & \Sigma \cos \alpha_1 \cos \gamma_1 \\ \Sigma \cos \beta_1 \cos \alpha_1 & \Sigma \cos^2 \beta_1 & \Sigma \cos \beta_1 \cos \gamma_1 \\ \Sigma \cos \alpha_1 \cos \gamma_1 & \Sigma \cos \gamma_1 \cos \beta_1 & \Sigma \cos^2 \gamma_1 \end{vmatrix}$$

(multiplying row by row)

$$= \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} = 1$$

**Practice Problems****[E]**

1. Express as a determinant  $\begin{vmatrix} 0 & c & b^2 \\ c & 0 & a \\ b & a & 0 \end{vmatrix}$ .

2. Express  $\Delta = \begin{vmatrix} (a-x)^2 & (a-y)^2 & (a-z)^2 \\ (b-x)^2 & (b-y)^2 & (b-z)^2 \\ (c-x)^2 & (c-y)^2 & (c-z)^2 \end{vmatrix}$  as the product of two determinants and evaluate it.

3. Prove that

$$\begin{vmatrix} 2bc - a^2 & c^2 & b^2 \\ c^2 & 2ca - b^2 & a^2 \\ b^2 & a^2 & 2ab - c^2 \end{vmatrix}$$

$$= (a^3 + b^3 + c^3 - 3abc)^2.$$

4. Show that

$$\begin{vmatrix} 1 & \cos(B-A) & \cos(C-A) \\ \cos(A-B) & 1 & \cos(C-B) \\ \cos(A-C) & \cos(B-C) & 1 \end{vmatrix} = 0$$

5. If  $\begin{vmatrix} a & a^3 & a^4 - 1 \\ b & b^3 & b^4 - 1 \\ c & c^3 & c^4 - 1 \end{vmatrix} = 0$

and  $a, b, c$  are all different, then prove that  $abc(ab + bc + ca) = a + b + c$ .

6. Express

$$\begin{vmatrix} 0 & (\alpha-\beta)^2 & (\alpha-\gamma)^2 \\ (\alpha-\beta)^2 & 0 & (\beta-\gamma)^2 \\ (\alpha-\gamma)^2 & (\beta-\gamma)^2 & 0 \end{vmatrix}$$

as a product of two determinants of the third order and hence find its value.

**2.5 ADJOINT OR ADJUGATE DETERMINANT**

The determinant  $\Delta' = |A_{ij}|$  where  $A_{ij}$  is the cofactor of  $a_{ij}$  in  $\Delta = |a_{ij}|$  is called the adjoint or adjugate of  $\Delta$ .

**Jacobi's Theorem**

If  $\Delta = |a_{ij}|$  is a determinant of  $n$ th order and  $\Delta' = |A_{ij}|$ , the adjoint of  $\Delta$ , then  $\Delta' = \Delta^{n-1}$  when  $\Delta \neq 0$ .

We have

$$\begin{aligned}
 \Delta \cdot \Delta' &= \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix} \\
 &\times \begin{vmatrix} A_{11} & A_{12} & \cdots & A_{1n} \\ A_{21} & A_{22} & \cdots & A_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ A_{n1} & A_{n2} & \cdots & A_{nn} \end{vmatrix} \\
 &= \begin{vmatrix} \sum_{r=1}^n a_{1r} A_{1r} & \sum_{r=1}^n a_{1r} A_{2r} & \cdots & \sum_{r=1}^n a_{1r} A_{nr} \\ \sum_{r=1}^n a_{2r} A_{1r} & \sum_{r=1}^n a_{2r} A_{2r} & \cdots & \sum_{r=1}^n a_{2r} A_{nr} \\ \cdots & \cdots & \cdots & \cdots \\ \sum_{r=1}^n a_{nr} A_{1r} & \sum_{r=1}^n a_{nr} A_{2r} & \cdots & \sum_{r=1}^n a_{nr} A_{nr} \end{vmatrix} \\
 &= \begin{vmatrix} \Delta & 0 & \cdots & 0 \\ 0 & \Delta & \cdots & 0 \\ \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & \cdots & \Delta \end{vmatrix}
 \end{aligned}$$



$$\text{since } \sum_{r=1}^n a_{ir} A_{jr} \begin{cases} = \Delta & \text{when } i = j \\ = 0 & \text{when } i \neq j \end{cases} = \Delta^n$$

$$\therefore \Delta' = \Delta^{n-1} \text{ when } \Delta \neq 0.$$



If  $\Delta = 0$ , then  $\Delta' = 0$ .

In this case  $\sum_{i=1}^n a_{ip} A_{pi} = 0$  for  $p = 1, 2, \dots, n$ .

Eliminating  $a_{11}, a_{12}, \dots, a_{1n}$  which are not all zero from the above  $n$  equations we have  $\Delta' = 0$ .

### Reciprocal determinant

If  $\Delta = |a_{ij}| \neq 0$ , then  $\Delta'' = \left| \frac{A_{ij}}{\Delta} \right|$  where  $A_{ij}$  is

the cofactor of  $a_{ij}$  is called the reciprocal determinant of  $\Delta$ .

$$\Delta'' = \left| \frac{A_{ij}}{\Delta} \right| = \frac{1}{\Delta^n} |A_{ij}| = \frac{\Delta'}{\Delta^n} = \frac{\Delta^{n-1}}{\Delta^n} = \frac{1}{\Delta}.$$

► **Example 1.** Show that

$$\begin{vmatrix} bc-a^2 & ca-b^2 & ab-c^2 \\ ca-b^2 & ab-c^2 & bc-a^2 \\ ab-c^2 & bc-a^2 & ca-b^2 \end{vmatrix}$$

$$= \begin{vmatrix} a^2 & c^2 & 2ac-b^2 \\ 2ab-c^2 & b^2 & a^2 \\ b^2 & 2bc-a^2 & c^2 \end{vmatrix}$$

► **Solution** Let  $\Delta = \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$

Replacing each element of  $\Delta$  by its cofactor, we get determinant of cofactors of  $\Delta$  as

$$\Delta_1 = \begin{vmatrix} bc-a^2 & ca-b^2 & ab-c^2 \\ ca-b^2 & ab-c^2 & bc-a^2 \\ ab-c^2 & bc-a^2 & ca-b^2 \end{vmatrix} \quad \dots(1)$$

$$\Delta^2 = \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}^2 = \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$$

$$= \begin{vmatrix} a & -c & b \\ b & -a & c \\ c & -b & a \end{vmatrix}$$

$$= \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} \begin{vmatrix} a & b & c \\ -c & -a & -b \\ b & c & a \end{vmatrix}$$

$$= \begin{vmatrix} a^2 & c^2 & 2ac-b^2 \\ 2ab-c^2 & b^2 & a^2 \\ b^2 & 2bc-a^2 & c^2 \end{vmatrix} \quad \dots(2)$$

From (1) and (2), we get the required result.

► **Example 2.** Show that

$$\begin{vmatrix} a & 0 & c \\ a & b & 0 \\ 0 & b & c \end{vmatrix}^2 = \begin{vmatrix} bc & -ca & ab \\ bc & ca & -ab \\ -bc & ca & ab \end{vmatrix}$$

$$= \begin{vmatrix} c^2-a^2 & a^2 & c^2 \\ a^2 & a^2+b^2 & b^2 \\ c^2 & b^2 & b^2+c^2 \end{vmatrix}.$$

► **Solution** The cofactors of the elements of

$$\Delta = \begin{vmatrix} a & 0 & c \\ a & b & 0 \\ 0 & b & c \end{vmatrix} \text{ are the corresponding elements}$$

of  $\begin{vmatrix} bc & -ca & ab \\ bc & ca & -ab \\ -bc & ca & ab \end{vmatrix}$

Therefore, adjoint of  $\Delta = \Delta^2$  ( $\because$  it is determinant of cofactors of third order determinant)

Again  $\begin{vmatrix} a & 0 & c \\ a & b & 0 \\ 0 & b & c \end{vmatrix} \times \begin{vmatrix} a & 0 & c \\ a & b & 0 \\ 0 & b & c \end{vmatrix}$

$$\begin{aligned}
&= \begin{vmatrix} a \cdot a + 0 \cdot 0 + c \cdot c & a \cdot a + 0 \cdot b + c \cdot 0 & a \cdot 0 + 0 \cdot b + c \cdot c \\ a \cdot a + b \cdot 0 + 0 \cdot c & a \cdot a + b \cdot b + 0 \cdot 0 & a \cdot 0 + b \cdot b + 0 \cdot c \\ 0 \cdot a + b \cdot 0 + c \cdot c & 0 \cdot a + b \cdot b + c \cdot 0 & 0 \cdot 0 + b \cdot b + c \cdot c \end{vmatrix} \\
&= \begin{vmatrix} c^2 + a^2 & a^2 & c^2 \\ a^2 & a^2 + b^2 & b^2 \\ c^2 & b^2 & b^2 + c^2 \end{vmatrix}
\end{aligned}$$

Hence the result follows.

► **Example 3.** Show that

$$\begin{vmatrix} yz - x^2 & zx - y^2 & xy - z^2 \\ zx - y^2 & xy - z^2 & yz - x^2 \\ xy - z^2 & yz - x^2 & zx - y^2 \end{vmatrix} = \begin{vmatrix} r^2 & u^2 & u^2 \\ u^2 & r^2 & u^2 \\ u^2 & u^2 & r^2 \end{vmatrix}$$

where  $r^2 = x^2 + y^2 + z^2$  and  $u^2 = xy + yz + zx$ .

► **Solution** Consider the determinant

$$\Delta = \begin{vmatrix} x & y & z \\ y & z & x \\ z & x & y \end{vmatrix}$$

We see that the L.H.S. determinant has its constituents which are the co-factor of  $\Delta$ . Hence L.H.S. determinant

$$\begin{aligned}
&= \begin{vmatrix} x & y & z \\ y & z & x \\ z & x & y \end{vmatrix} \begin{vmatrix} x & y & z \\ y & z & x \\ z & x & y \end{vmatrix} \\
&= \begin{vmatrix} x^2 + y^2 + z^2 & xy + yz + zx & xy + yz + zx \\ xy + yz + zx & y^2 + z^2 + x^2 & yz + zx + xy \\ zx + xy + yz & yz + xz + xy & z^2 + x^2 + y^2 \end{vmatrix} \\
&= \begin{vmatrix} r^2 & u^2 & u^2 \\ u^2 & r^2 & u^2 \\ u^2 & u^2 & r^2 \end{vmatrix}
\end{aligned}$$

► **Example 4.** Show that

$$\begin{vmatrix} a^2 + x^2 & ab - cx & ac + bx \\ ab + cx & b^2 + x^2 & bc - ax \\ ac - bx & bc + ax & c^2 + x^2 \end{vmatrix} = \begin{vmatrix} x & c & -b \\ -c & x & a \\ b & -a & x \end{vmatrix}^2$$

► **Solution** Let  $D = \begin{vmatrix} x & c & -b \\ -c & x & a \\ b & -a & x \end{vmatrix}$

Cofactors of 1st row of D are

$$x^2 + a^2, ab + cx, ac - bx,$$

cofactors of 2nd row of D are

$$ab - cx, x^2 + b^2, ax + bc$$

and cofactors of 3rd row of D are

$$ac + bx, bc - ax, x^2 + c^2$$

∴ Determinant of cofactors of D is

$$= \begin{vmatrix} x^2 + a^2 & ab + cx & ac - bx \\ ab - cx & x^2 + b^2 & ax + bc \\ ac + bx & bc - ax & x^2 + c^2 \end{vmatrix}$$

$$= \begin{vmatrix} a^2 + x^2 & ab - cx & ac + bx \\ ab + cx & b^2 + x^2 & bc - ax \\ ac - bx & bc + ax & c^2 + x^2 \end{vmatrix}$$

(interchanging rows into columns)

$$= D^2$$

$$= \begin{vmatrix} x & c & -b \\ -c & x & a \\ b & -a & x \end{vmatrix}^2$$

### The method of changing the elements of a determinant

This method is applied when changing all elements of a determinant by the same number yields a form such that the cofactors of all elements can readily be computed. The method is based on the following property :

If the same number  $x$  is added to all elements of the determinant  $D$ , the determinant is increased by the product of the number into the sum of the cofactors of all elements of  $D$ .

Let

$$D = \begin{vmatrix} a_{11} & \dots & a_{1n} \\ \dots & \dots & \dots \\ a_{n1} & \dots & a_{nn} \end{vmatrix}, \quad D' = \begin{vmatrix} a_{11} + x & \dots & a_{1n} + x \\ \dots & \dots & \dots \\ a_{n1} + x & \dots & a_{nn} + x \end{vmatrix}$$

Decompose  $D'$  into two determinants with respect to the first row, then each of them into two determinants with respect to the second row, and so on.

The terms containing more than one row of elements equal to  $x$  are equal to zero.

The terms containing one row of elements equal to  $x$  are then expanded in terms of that row to get

$$D' = D + x \sum_{i=1}^n A_{ij}, \text{ which is what we sought.}$$

Thus, computing the determinant  $D'$  reduces to computing  $D$  and the sum of its cofactors.

► **Example 5.** Compute the determinant

$$\Delta = \begin{vmatrix} a_1 & x & x \\ x & a_2 & x \\ x & x & a_3 \end{vmatrix}.$$

► **Solution** Subtracting  $x$  from all its elements, we obtain the determinant

$$D = \begin{vmatrix} a_1 - x & 0 & 0 \\ 0 & a_2 - x & 0 \\ 0 & 0 & a_3 - x \end{vmatrix}$$

The cofactors of the elements of  $D$  off the principle diagonal are equal to zero, and the cofactor of each element on the principal diagonal is equal to the product of the remaining elements of the principal diagonal. Therefore

$$\begin{aligned} \Delta &= (a_1 - x)(a_2 - x)(a_3 - x) + x\{(a_2 - x)(a_3 - x) \\ &\quad + (a_3 - x)(a_1 - x) + (a_1 - x)(a_2 - x)\} \\ &= x(a_1 - x)(a_2 - x)(a_3 - x) \\ &\quad \cdot \left( \frac{1}{x} + \frac{1}{a_1 - x} + \frac{1}{a_2 - x} + \frac{1}{a_3 - x} \right) \end{aligned}$$

## 2.6 DIFFERENTIATION AND INTEGRATION OF DETERMINANTS

### Differentiation of determinants

When the elements of a determinant are functions of a variable  $x$ , the rule for obtaining its differential coefficient is as follows. If  $\Delta$  denotes the determinant

$$\begin{vmatrix} a_1 & b_1 & \dots & k_1 \\ \dots & \dots & \dots & \dots \\ a_n & b_n & \dots & k_n \end{vmatrix}$$

and the elements are functions of  $x$ ,  $d\Delta/dx$  is the sum of the  $n$  determinants obtained by differentiating the elements of one row (or column) of  $\Delta$  and leaving the elements of the other  $n-1$  rows (or columns) unaltered. For example,

$$\begin{aligned} \frac{d}{dx} \begin{vmatrix} x^2 & x & 1 \\ x^3 & x^2 & x \\ x^4 & x^3 & x^2 \end{vmatrix} \\ = \begin{vmatrix} 2x & 1 & 0 \\ x^3 & x^2 & x \\ x^4 & x^3 & x^2 \end{vmatrix} + \begin{vmatrix} x^2 & x & 1 \\ 3x^2 & 2x & 1 \\ x^4 & x^3 & x^2 \end{vmatrix} + \begin{vmatrix} x^2 & x & 1 \\ x^3 & x^2 & x \\ 4x^3 & 3x^3 & 2x \end{vmatrix} \end{aligned}$$

If  $F(x) = \begin{vmatrix} f_1 & f_2 & f_3 \\ g_1 & g_2 & g_3 \\ h_1 & h_2 & h_3 \end{vmatrix}$  is a third order determinant

where  $f_1, f_2, f_3; g_1, g_2, g_3; h_1, h_2, h_3$  are the functions of  $x$ , the differentiation of determinant can be performed in two ways:

1. Row by row differentiation :

$$F'(x) = \begin{vmatrix} f'_1 & f'_2 & f'_3 \\ g_1 & g_2 & g_3 \\ h_1 & h_2 & h_3 \end{vmatrix} + \begin{vmatrix} f_1 & f_2 & f_3 \\ g'_1 & g'_2 & g'_3 \\ h_1 & h_2 & h_3 \end{vmatrix} + \begin{vmatrix} f_1 & f_2 & f_3 \\ g_1 & g_2 & g_3 \\ h'_1 & h'_2 & h'_3 \end{vmatrix}$$

2. Column by column differentiation :

$$F'(x) = \begin{vmatrix} f'_1 & f_2 & f_3 \\ g'_1 & g_2 & g_3 \\ h'_1 & h_2 & h_3 \end{vmatrix} + \begin{vmatrix} f_1 & f'_2 & f_3 \\ g_1 & g'_2 & g_3 \\ h_1 & h'_2 & h_3 \end{vmatrix} + \begin{vmatrix} f_1 & f_2 & f'_3 \\ g_1 & g_2 & g'_3 \\ h_1 & h_2 & h'_3 \end{vmatrix}$$

Thus to differentiate a determinant, we differentiate one row (or column) at a time, keeping others unchanged.

► **Example 1.** If  $F(x) = \begin{vmatrix} 1 & 2a & 3a^2 \\ x & x^2 & x^3 \\ e^{x-a} & e^{x^2-a^2} & e^{x^3-a^3} \end{vmatrix}$ ,

then find the value of  $F'(a)$ .

► **Solution**  $F'(x) = \begin{vmatrix} 0 & 0 & 0 \\ x & x^2 & x^3 \\ e^{x-a} & e^{x^2-a^2} & e^{x^3-a^3} \end{vmatrix}$

$$+ \begin{vmatrix} 1 & 2a & 3a^2 \\ 1 & 2x & 3x^2 \\ e^{x-a} & e^{x^2-a^2} & e^{x^3-a^3} \end{vmatrix}$$

$$+ \begin{vmatrix} 1 & 2a & 3a^2 \\ x & x^2 & x^3 \\ e^{x-a} & 2xe^{x^2-a^2} & 3x^2e^{x^3-a^3} \end{vmatrix}$$

(Differentiating row by row)

$$F'(a) = 0 + \begin{vmatrix} 1 & 2a & 3a^2 \\ 1 & 2a & 3a^2 \\ e^0 & e^0 & e^0 \end{vmatrix} + \begin{vmatrix} 1 & 2a & 3a^2 \\ a & a^2 & a^3 \\ 1 & 2a & 3a^2 \end{vmatrix}$$

$$= 0 + e^0 \begin{vmatrix} 1 & 2a & 3a^2 \\ 1 & 2a & 3a^2 \\ 1 & 1 & 1 \end{vmatrix} + \begin{vmatrix} 1 & 2a & 3a^2 \\ a & a^2 & a^3 \\ 1 & 2a & 3a^2 \end{vmatrix}$$

$$= 0 + 0 \text{ (} R_1 \text{ \& } R_2 \text{ are identical)} + 0 \text{ (} R_1 \text{ \& } R_3 \text{ are identical)} = 0.$$

► **Example 2.** If  $a, b, c$  are in A.P. and

$$f(x) = \begin{vmatrix} x+a & x^2+1 & 1 \\ x+b & 2x^2-1 & 1 \\ x+c & 3x^2-2 & 1 \end{vmatrix}, \text{ then find } f(x).$$

► **Solution**  $f(x) = \begin{vmatrix} x+a & x^2+1 & 1 \\ x+b & 2x^2-1 & 1 \\ x+c & 3x^2-2 & 1 \end{vmatrix}$

$$f(x) = \begin{vmatrix} 1 & x^2+1 & 1 \\ 1 & 2x^2-1 & 1 \\ 1 & 3x^2-1 & 1 \end{vmatrix} + \begin{vmatrix} x+a & 2x & 1 \\ x+b & 4x & 1 \\ x+c & 6x & 1 \end{vmatrix}$$

$$+ \begin{vmatrix} x+a & x^2+1 & 0 \\ x+b & 2x^2-1 & 0 \\ x+c & 3x^2-2 & 0 \end{vmatrix}$$

(Differentiating column by column)  
 $= 0$  ( $C_1$  &  $C_2$  are identical)

$$+ 2x \begin{vmatrix} x+a & 1 & 1 \\ b-a & 1 & 0 \\ c-b & 1 & 0 \end{vmatrix} \begin{pmatrix} R_2 \rightarrow R_2 - R_1 \\ R_3 \rightarrow R_3 - R_1 \end{pmatrix}$$

$$= 2x(2b - a - c)$$

$$= 2 \times 0 = 0 \quad (a, b, c \text{ are in A.P.})$$

► **Example 3.** If  $f(x) = \begin{vmatrix} 3 & 2 & 1 \\ 6x^2 & 2x^3 & x^4 \\ 1 & a & a^2 \end{vmatrix}$ , then find

the value of  $f''(a)$ .

► **Solution**  $f'(x) = \begin{vmatrix} 3 & 2 & 1 \\ 12x & 6x^2 & 4x^3 \\ 1 & a & a^2 \end{vmatrix}$

$$f''(x) = \begin{vmatrix} 3 & 2 & 1 \\ 12 & 12x & 12x^2 \\ 1 & a & a^2 \end{vmatrix}$$

$$\Rightarrow f''(a) = 12 \begin{vmatrix} 3 & 2 & 1 \\ 1 & a & a^2 \\ 1 & a & a^2 \end{vmatrix} = 0.$$

► **Example 4.** Prove that  $F$  is independent of  $a_1$ ,  $b_1$  and  $b_2$  where

$$F = \begin{vmatrix} 1 & 1 & 1 \\ x_1 + a_1 & x_2 + a_1 & x_3 + a_1 \\ x_1^2 + b_1x_1 + b_2 & x_2^2 + b_1x_2 + b_2 & x_3^2 + b_1x_3 + b_2 \end{vmatrix}$$

and hence simplify  $F$ .

► **Solution**  $\frac{dF}{da_1}$

$$= \begin{vmatrix} 0 & 0 & 0 \\ x_1 + a_1 & x_2 + a_1 & x_3 + a_1 \\ x_1^2 + b_1x_1 + b_2 & x_2^2 + b_1x_2 + b_2 & x_3^2 + b_1x_3 + b_2 \end{vmatrix}$$

$$+ \begin{vmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ x_1^2 + b_1 x_1 + b_2 & x_2^2 + b_1 x_2 + b_2 & x_3^2 + b_1 x_3 + b_2 \end{vmatrix} \\ + \begin{vmatrix} 1 & 1 & 1 \\ x_1 + a_1 & x_2 + a_1 & x_3 + a_1 \\ 0 & 0 & 0 \end{vmatrix} = 0$$

Hence F is independent of  $a_1$ .

$$\text{Similarly } \frac{dF}{db_1} = \frac{dF}{db_2} = 0.$$

Hence F is independent of  $b_1$  and  $b_2$  also.

So F is dependent only on  $x_1, x_2, x_3$

Put  $a_1 = 0, b_1 = 0, b_2 = 0$

$$\Rightarrow F = \begin{vmatrix} 1 & 1 & 1 \\ x_1 & x_2 & x_3 \\ x_1^2 & x_2^2 & x_3^2 \end{vmatrix} = (x_1 - x_2)(x_2 - x_3) \\ (x_3 - x_1).$$

► **Example 5.** If

$$\Delta = \begin{vmatrix} e^x & \sin x & 1 \\ \cos x & \log_e(1+x^2) & 1 \\ x & x^2 & 1 \end{vmatrix} = a + bx + cx^2$$

then find the value of b.

► **Solution** Now differentiating  $\Delta$  column by column

$$\Delta' = \begin{vmatrix} e^x & \sin x & 1 \\ -\sin x & \log_e(1+x^2) & 1 \\ 1 & x^2 & 1 \end{vmatrix} \\ + \begin{vmatrix} e^x & \cos x & 1 \\ \cos x & \frac{2x}{1+x^2} & 1 \\ x & 2x & 1 \end{vmatrix} + 0 = b + 2cx$$

Now putting  $x = 0$

$$b = \begin{vmatrix} 1 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 0 & 1 \end{vmatrix} + \begin{vmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 0 & 0 & 1 \end{vmatrix} = -1$$

## Integration of a Determinant

$$\text{If } F(x) = \begin{vmatrix} f & g & h \\ a & b & c \\ a_1 & b_1 & c_1 \end{vmatrix}$$

when f, g, h are the functions of x and a, b, c,  $a_1, b_1, c_1$  are constants, then

$$\int_{\lambda}^{\mu} F(x) dx = \begin{vmatrix} \int_{\lambda}^{\mu} f dx & \int_{\lambda}^{\mu} g dx & \int_{\lambda}^{\mu} h dx \\ a & b & c \\ a_1 & b_1 & c_1 \end{vmatrix}$$

$$\text{Also if } F(x) = \begin{vmatrix} f & a & a_1 \\ g & b & b_1 \\ h & c & c_1 \end{vmatrix}$$

$$\text{then } \int_{\lambda}^{\mu} F(x) dx = \begin{vmatrix} \int_{\lambda}^{\mu} f dx & a & a_1 \\ \int_{\lambda}^{\mu} g dx & b & b_1 \\ \int_{\lambda}^{\mu} h dx & c & c_1 \end{vmatrix}$$

► **Example 6.** If

$$f(x) = \begin{vmatrix} \sin^5 x & \ln \sin x & \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}} \\ n & \sum_{k=1}^n k & \prod_{k=1}^n k \\ \frac{8}{15} & \frac{\pi}{2} \ln\left(\frac{1}{2}\right) & \frac{\pi}{4} \end{vmatrix}$$

then find the value of  $\int_0^{\pi/2} f(x) dx$ .

► **Solution**  $\int_0^{\pi/2} f(x) dx =$

$$\begin{vmatrix} \int_0^{\pi/2} \sin^5 x dx & \int_0^{\pi/2} \ln \sin x dx & \int_0^{\pi/2} \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx \\ n & \sum_{k=1}^n k & \prod_{k=1}^n k \\ \frac{8}{15} & \frac{\pi}{2} \ln\left(\frac{1}{2}\right) & \frac{\pi}{4} \end{vmatrix}$$

$$= \begin{vmatrix} \frac{4}{5} \cdot \frac{2}{3} & -\frac{\pi}{2} \ln 2 & \frac{\pi}{4} \\ n & \sum_{k=1}^n k & \prod_{k=1}^n k \\ \frac{8}{15} & \frac{\pi}{2} \ln\left(\frac{1}{2}\right) & \frac{\pi}{4} \end{vmatrix}$$

$$= \begin{vmatrix} \frac{8}{15} & \frac{\pi}{2} \ln\left(\frac{1}{2}\right) & \frac{\pi}{4} \\ n & \sum_{k=1}^n k & \prod_{k=1}^n k \\ \frac{8}{15} & \frac{\pi}{2} \ln\left(\frac{1}{2}\right) & \frac{\pi}{4} \end{vmatrix} = 0$$

**Practice Problems****[F]**

1. If  $A_1, B_1, C_1, \dots$  are respectively the cofactors of the elements  $a_1, b_1, c_1, \dots$  of

the determinant  $\Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$  then prove

that

(i)  $\begin{vmatrix} B_2 & C_2 \\ B_3 & C_3 \end{vmatrix} = a_1 \Delta.$

(ii)  $\begin{vmatrix} A_1 & B_1 & C_1 \\ A_2 & B_2 & C_2 \\ A_3 & B_3 & C_3 \end{vmatrix} = \Delta^2$

2. If  $\Delta = \begin{vmatrix} x & 1 & x^2 \\ x+2 & 2x+3 & x \\ x^2 & x^3+1 & 2x^4+1 \end{vmatrix}$  find the

value of  $\frac{d\Delta}{dx}.$ 

3. If  $\Delta = \begin{vmatrix} x^4 & \sin 2x & e^x \\ 4x^3 & 2\cos 2x & e^x \\ 12x^2 & -4\sin 2x & e^x \end{vmatrix}$ , then show

that  $\frac{d\Delta}{dx} = \begin{vmatrix} x^4 & \sin 2x & e^x \\ 4x^3 & 2\cos 2x & e^x \\ 24x & -8\cos 2x & e^x \end{vmatrix}$

4. If  $\Delta(x) = \begin{vmatrix} x & x^2 & x^3 \\ 1 & 2x & 3x^2 \\ 0 & 2 & 6x \end{vmatrix}$ , evaluate  $\Delta'(x).$

5. If  $\begin{vmatrix} x & x-1 & x \\ -2x & x+1 & 1 \\ x+1 & 1 & x \end{vmatrix} = ax^3 + bx^2 + cx + d.$

Find (i) d (ii) a + b + c + d (iii) b

6. If  $\Delta = \begin{vmatrix} a & b & c \\ 6 & 4 & 3 \\ x & x^2 & x^3 \end{vmatrix}$ , evaluate  $\int_0^1 \Delta dx.$

7. If  $u, v, \omega$  be three functions of  $x$  and if suffixes denote differentiation w.r.t  $x$ , then prove that

$$\frac{d}{dx} \begin{vmatrix} u_1 & v_1 & \omega_1 \\ u_2 & v_2 & \omega_2 \\ u_3 & v_3 & \omega_3 \end{vmatrix} = \begin{vmatrix} u_1 & v_1 & \omega_1 \\ u_2 & v_2 & \omega_2 \\ u_4 & v_4 & \omega_4 \end{vmatrix}.$$

**Target Problems for JEE Advanced**

- **Problem 1.** If  $a^2 + b^2 + c^2 = -2$  and

$f(x) = \begin{vmatrix} 1+a^2x & (1+b^2)x & (1+c^2)x \\ (1+a^2)x & 1+b^2x & (1+c^2)x \\ (1+a^2)x & (1+b^2)x & 1+c^2x \end{vmatrix}$  then find the

degree of the polynomial  $f(x).$ 

- **Solution** Applying  $C_1 \rightarrow C_1 + C_2 + C_3$

$$\begin{vmatrix} 1+2x+x(a^2+b^2+c^2) & (1+b^2)x & (1+c^2)x \\ 1+2x+x(a^2+b^2+c^2) & 1+b^2x & (1+c^2)x \\ 1+2x+x(a^2+b^2+c^2) & (1+b^2)x & 1+c^2x \end{vmatrix}$$

$$= \begin{vmatrix} 1 & (1+b^2)x & (1+c^2)x \\ 1 & 1+b^2x & (1+c^2)x \\ 1 & (1+b^2)x & 1+c^2x \end{vmatrix}$$

$$R_2 \rightarrow R_2 - R_1 \text{ \& } R_3 \rightarrow R_3 - R_1$$

$$= \begin{vmatrix} 1 & (1+b^2)x & (1+c^2)x \\ 0 & 1-x & 0 \\ 0 & 0 & 1-x \end{vmatrix}$$

$$f(x) = (1-x)^2 = 1 - 2x + x^2$$

Hence the degree of the polynomial is 2.

► **Problem 2.** Find the positive integral solutions of the equation

$$\begin{vmatrix} x^3+1 & x^2y & x^2z \\ xy^2 & y^3+1 & y^2z \\ xz^2 & yz^2 & z^3+1 \end{vmatrix} = 11.$$

► **Solution**  $\begin{vmatrix} x^4+x & x^3y & x^3z \\ xy^3 & y^4+y & y^3z \\ xz^3 & yz^3 & z^4+z \end{vmatrix} = 11$

$$\frac{1}{xyz} \begin{vmatrix} x^3+1 & x^3 & x^3 \\ y^3 & y^3+1 & y^3 \\ z^3 & z^3 & z^3+1 \end{vmatrix} = 11$$

use  $R_1 \rightarrow R_1 + R_2 + R_3$

$$D = (x^3 + y^3 + z^3 + 1) \begin{vmatrix} 1 & 1 & 1 \\ y^3 & y^3+1 & y^3 \\ z^3 & z^3 & z^3+1 \end{vmatrix} = 11$$

$$\Rightarrow x^3 + y^3 + z^3 = 10$$

Hence, the positive integral solutions of the equation are  $(2, 1, 1)$ ,  $(1, 2, 1)$ ,  $(1, 1, 2)$ .

► **Problem 3.** Find the value of the determinant

$$\begin{vmatrix} \sqrt{13}+\sqrt{3} & 2\sqrt{5} & \sqrt{5} \\ \sqrt{15}+\sqrt{26} & 5 & \sqrt{10} \\ 3+\sqrt{65} & \sqrt{15} & 5 \end{vmatrix}$$

► **Solution** Let

$$\Delta = \begin{vmatrix} \sqrt{13}+\sqrt{3} & 2\sqrt{5} & \sqrt{5} \\ \sqrt{15}+\sqrt{26} & 5 & \sqrt{10} \\ 3+\sqrt{65} & \sqrt{15} & 5 \end{vmatrix}$$

$$= \begin{vmatrix} \sqrt{13} & 2\sqrt{5} & \sqrt{5} \\ \sqrt{15} & 5 & \sqrt{10} \\ 3 & \sqrt{15} & 5 \end{vmatrix} + \begin{vmatrix} \sqrt{13} & 2\sqrt{5} & \sqrt{5} \\ \sqrt{26} & 5 & \sqrt{10} \\ \sqrt{65} & \sqrt{15} & 5 \end{vmatrix}$$

Taking  $\sqrt{3}, \sqrt{5}, \sqrt{5}$  common from  $C_1, C_2, C_3$  respectively in the first determinant and taking  $\sqrt{13}, \sqrt{5}, \sqrt{5}$  common from  $C_1, C_2, C_3$  respectively in the second determinant, we get

$$= \sqrt{3} \times \sqrt{5} \times \sqrt{5} \begin{vmatrix} 1 & 2 & 1 \\ \sqrt{5} & \sqrt{5} & \sqrt{2} \\ \sqrt{3} & \sqrt{3} & \sqrt{5} \end{vmatrix}$$

$$+ \sqrt{13} \times \sqrt{5} \times \sqrt{5} \begin{vmatrix} 1 & 2 & 1 \\ \sqrt{2} & \sqrt{5} & \sqrt{2} \\ \sqrt{5} & \sqrt{3} & \sqrt{5} \end{vmatrix}$$

$$= \sqrt{3} \times 5 \begin{vmatrix} 1 & 2 & 1 \\ \sqrt{5} & \sqrt{5} & \sqrt{2} \\ \sqrt{3} & \sqrt{3} & \sqrt{5} \end{vmatrix} + 0$$

( $\because C_1$  and  $C_3$  are identical)

Applying  $C_2 \rightarrow C_2 - C_1$

$$\therefore \Delta = 5\sqrt{3} \begin{vmatrix} 1 & 1 & 1 \\ \sqrt{5} & 0 & \sqrt{2} \\ \sqrt{3} & 0 & \sqrt{5} \end{vmatrix}$$

Expanding along  $C_2$

$$\therefore \Delta = 5\sqrt{3} \cdot (-1) \begin{vmatrix} \sqrt{5} & \sqrt{2} \\ \sqrt{3} & \sqrt{5} \end{vmatrix} = -5\sqrt{3} (5 - \sqrt{6})$$

$$= -25\sqrt{3} + 15\sqrt{2} = 15\sqrt{2} - 25\sqrt{3}.$$

► **Problem 4.** Prove that

$$\begin{vmatrix} {}^x C_r & {}^x C_{r+1} & {}^x C_{r+2} \\ {}^y C_r & {}^y C_{r+1} & {}^y C_{r+2} \\ {}^z C_r & {}^z C_{r+1} & {}^z C_{r+2} \end{vmatrix} = \begin{vmatrix} {}^x C_r & {}^{x+1} C_{r+1} & {}^{x+2} C_{r+2} \\ {}^y C_r & {}^{y+1} C_{r+1} & {}^{y+2} C_{r+2} \\ {}^z C_r & {}^{z+1} C_{r+1} & {}^{z+2} C_{r+2} \end{vmatrix}$$

► **Solution** L.H.S. = 
$$\begin{vmatrix} {}^x C_r & {}^x C_{r+1} & {}^x C_{r+2} \\ {}^y C_r & {}^y C_{r+1} & {}^y C_{r+2} \\ {}^z C_r & {}^z C_{r+1} & {}^z C_{r+2} \end{vmatrix}$$

Applying  $C_2 \rightarrow C_2 + C_1$  and  $C_3 \rightarrow C_3 + C_1$

$$\begin{vmatrix} {}^x C_r & {}^x C_{r+1} + {}^x C_r & {}^x C_{r+2} + {}^x C_{r+1} \\ {}^y C_r & {}^y C_{r+1} + {}^y C_r & {}^y C_{r+2} + {}^y C_{r+1} \\ {}^z C_r & {}^z C_{r+1} + {}^z C_r & {}^z C_{r+2} + {}^z C_{r+1} \end{vmatrix}$$

$$= \begin{vmatrix} {}^x C_r & {}^{x+1} C_{r+1} & {}^{x+1} C_{r+2} \\ {}^y C_r & {}^{y+1} C_{r+1} & {}^{y+1} C_{r+2} \\ {}^z C_r & {}^{z+1} C_{r+1} & {}^{z+1} C_{r+2} \end{vmatrix}$$

$$(\because {}^n C_r + {}^n C_{r-1} = {}^{n+1} C_r)$$

Applying  $C_3 \rightarrow C_3 + C_2$

$$\begin{vmatrix} {}^x C_r & {}^{x+1} C_{r+1} & {}^{x+1} C_{r+1} + {}^{x+1} C_{r+2} \\ {}^y C_r & {}^{y+1} C_{r+1} & {}^{y+1} C_{r+1} + {}^{y+1} C_{r+2} \\ {}^z C_r & {}^{z+1} C_{r+1} & {}^{z+1} C_{r+1} + {}^{z+1} C_{r+2} \end{vmatrix}$$

$$= \begin{vmatrix} {}^x C_r & {}^{x+1} C_{r+1} & {}^{x+2} C_{r+2} \\ {}^y C_r & {}^{y+1} C_{r+1} & {}^{y+2} C_{r+2} \\ {}^z C_r & {}^{z+1} C_{r+1} & {}^{z+2} C_{r+2} \end{vmatrix} = \text{R.H.S.}$$

► **Problem 5.** Prove that

$$\begin{vmatrix} a-b-c & 2a & 2a \\ 2b & b-c-a & 2b \\ 2c & 2c & c-a-b \end{vmatrix} = (a+b+c)^3$$

► **Solution** Let

$$\Delta = \begin{vmatrix} a-b-c & 2a & 2a \\ 2b & b-c-a & 2b \\ 2c & 2c & c-a-b \end{vmatrix}$$

Since the determinant is  $(a+b+c)^3$ , we try to get  $(a+b+c)$ .

Applying  $R_1 \rightarrow R_1 + R_2 + R_3$ , then

$$\Delta = \begin{vmatrix} a+b+c & a+b+c & a+b+c \\ 2b & b-c-a & 2b \\ 2c & 2c & c-a-b \end{vmatrix}$$

Taking  $(a+b+c)$  common from  $R_1$ , we get

$$\Delta = (a+b+c) \begin{vmatrix} 1 & 1 & 1 \\ 2b & b-c-a & 2b \\ 2c & 2c & c-a-b \end{vmatrix}$$

Applying  $C_3 \rightarrow C_2 - C_1$  and  $C_3 \rightarrow C_3 - C_1$

$$\therefore \Delta = (a+b+c) \begin{vmatrix} 1 & 0 & 0 \\ 2b & -a-b-a & 0 \\ 2c & 2c & -c-a-b \end{vmatrix}$$

$$= (a+b+c) \cdot 1 \cdot (-a-b-c) \cdot (-c-a-b)$$

Hence  $\Delta = (a+b+c)^3$ .

► **Problem 6.** If  $a, b, c$  are all different and if

$$\begin{vmatrix} a & a^2 & 1+a^3 \\ b & b^2 & 1+b^3 \\ c & c^2 & 1+c^3 \end{vmatrix} = 0, \text{ prove that } abc = -1$$

► **Solution** Let  $D = \begin{vmatrix} a & a^2 & 1+a^3 \\ b & b^2 & 1+b^3 \\ c & c^2 & 1+c^3 \end{vmatrix}$

$$= \begin{vmatrix} a & a^2 & 1 \\ b & b^2 & 1 \\ c & c^2 & 1 \end{vmatrix} + \begin{vmatrix} a & a^2 & a^3 \\ b & b^2 & b^3 \\ c & c^2 & c^3 \end{vmatrix}$$

$$= \begin{vmatrix} a & a^2 & 1 \\ b & b^2 & 1 \\ c & c^2 & 1 \end{vmatrix} + abc \begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix}$$

$$= (-1) \begin{vmatrix} 1 & a^2 & a \\ 1 & b^2 & b \\ 1 & c^2 & c \end{vmatrix} + abc \begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix}$$

$[C_1 \leftrightarrow C_3 \text{ in the first det.}]$

$$= (-1)^2 \begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix} + abc \begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix}$$

$[C_2 \leftrightarrow C_3 \text{ in the first det.}]$



$$= \begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix} + abc \begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix}$$

$$= (1 + abc) \begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix}$$

$$= (1 + abc) \begin{vmatrix} 1 & a & a^2 \\ 0 & b-a & b^2-a^2 \\ 0 & c-a & c^2-a^2 \end{vmatrix}$$

$$[R_2 \rightarrow R_2 - R_1 \text{ and } R_3 \rightarrow R_3 - R_1]$$

$$= (1 + abc) \begin{vmatrix} 1 & a & a^2 \\ 0 & b-a & b^2-a^2 \\ 0 & c-a & c^2-a^2 \end{vmatrix}$$

(expanding along first column)

$$= (1 + abc) (b-a) (c-a) (b+c)$$

$$= (1 + abc) (b-a) (c-a)$$

$$\therefore D = (1 + abc) (a-b) (b-c) (c-a)$$

But given  $D = 0$

$$\therefore (1 + abc) (a-b) (b-c) (c-a) = 0$$

$$\therefore 1 + abc = 0$$

[since  $a, b, c$  are different  $a \neq b, b \neq c, c \neq a$ ]

Hence  $abc = -1$ .

► **Problem 7.** Without expanding a determinant at any stage show that

$$\begin{vmatrix} x^2+x & x+1 & x-2 \\ 2x^2+3x-1 & 3x & 3x-3 \\ x^2+2x+3 & 2x-1 & 2x-1 \end{vmatrix} = Ax + B,$$

where  $A$  and  $B$  are determinants of order 3 not involving  $x$ .

► **Solution** Let

$$\Delta = \begin{vmatrix} x^2+x & x+1 & x-2 \\ 2x^2+3x-1 & 3x & 3x-3 \\ x^2+2x+3 & 2x-1 & 2x-1 \end{vmatrix}$$

$$\text{Applying } R_1 \rightarrow R_1 + R_3 - R_2$$

$$\Delta = \begin{vmatrix} 4 & 0 & 0 \\ 2x^2+3x-1 & 3x & 3x-3 \\ x^2+2x+3 & 2x-1 & 2x-1 \end{vmatrix}$$

$$\text{Applying } C_1 \rightarrow C_1 - C_3 \text{ and } C_2 \rightarrow C_2 - C_3$$

$$\Delta = \begin{vmatrix} 4 & 0 & 0 \\ 2x^2+2 & 3 & 3x-3 \\ x^2+4 & 0 & 2x-1 \end{vmatrix}$$

$$\text{Applying } R_2 \rightarrow R_2 - \frac{x^2}{2} R_1 \text{ and}$$

$$R_3 \rightarrow R_3 - \frac{x^2}{4} R_1 \text{ then}$$

$$\Delta = \begin{vmatrix} 4 & 0 & 0 \\ 2 & 3 & 3x-3 \\ 4 & 0 & 2x-1 \end{vmatrix}$$

$$\Delta = \begin{vmatrix} 4 & 0 & 0 \\ 2 & 3 & 3x \\ 4 & 0 & 2x \end{vmatrix} + \begin{vmatrix} 4 & 0 & 0 \\ 2 & 3 & -3 \\ 4 & 0 & -1 \end{vmatrix}$$

$$\Delta = x \begin{vmatrix} 4 & 0 & 0 \\ 2 & 3 & 3 \\ 4 & 0 & 2 \end{vmatrix} + \begin{vmatrix} 4 & 0 & 0 \\ 2 & 3 & -3 \\ 4 & 0 & -1 \end{vmatrix} = xA + B.$$

Hence proved.

► **Problem 8.** If  $2s = a + b + c$ , show that

$$\begin{vmatrix} a^2 & (s-a)^2 & (s-a)^2 \\ (s-b)^2 & b^2 & (s-b)^2 \\ (s-c)^2 & (s-c)^2 & c^2 \end{vmatrix} = 2s^3 (s-a) (s-b) (s-c).$$

► **Solution** Let  $s-a = \alpha, s-b = \beta, s-c = \gamma$

Then  $\beta + \gamma = 2s - (b+c) = 2s - (2s-a) = a$

Similarly  $\gamma + \alpha = b$  and  $\alpha + \beta = c$  and

$$\alpha + \beta + \gamma = 3s - (a+b+c) = 3s - 2s = s$$

$$\begin{aligned} \text{Let } \Delta &= \begin{vmatrix} a^2 & (s-a)^2 & (s-a)^2 \\ (s-b)^2 & b^2 & (s-b)^2 \\ (s-c)^2 & (s-c)^2 & c^2 \end{vmatrix} \\ &= \begin{vmatrix} (\beta+\gamma)^2 & \alpha^2 & \alpha^2 \\ \beta^2 & (\gamma+\alpha)^2 & \beta^2 \\ \gamma^2 & \gamma^2 & (\alpha+\beta)^2 \end{vmatrix} \end{aligned}$$

Applying  $C_1 \rightarrow C_1 - C_2$  and  $C_2 \rightarrow C_2 - C_3$

$$\therefore \Delta = \begin{vmatrix} (\beta + \gamma)^2 - \alpha^2 & 0 & \alpha^2 \\ \beta^2 - (\gamma + \alpha)^2 & (\gamma + \alpha)^2 - \beta^2 & \beta^2 \\ 0 & \gamma^2 - (\alpha + \beta)^2 & (\alpha + \beta)^2 \end{vmatrix}$$

$$= \begin{vmatrix} (\beta + \gamma + \alpha)(\beta + \gamma - \alpha) & 0 & \alpha^2 \\ (\beta + \gamma + \alpha)(\beta - \gamma - \alpha) & (\alpha + \gamma + \beta)(\gamma + \alpha - \beta) & \beta^2 \\ 0 & (\gamma + \alpha + \beta)(\gamma - \alpha - \beta) & (\alpha + \beta)^2 \end{vmatrix}$$

Taking  $\alpha + \beta + \gamma$  common from  $C_1$  and  $C_2$ , we get

$$\Delta = (\alpha + \beta + \gamma)^2 \begin{vmatrix} \beta + \gamma - \alpha & 0 & \alpha^2 \\ \beta - \gamma - \alpha & \gamma + \alpha - \beta & \beta^2 \\ 0 & \gamma - \alpha - \beta & (\alpha + \beta)^2 \end{vmatrix}$$

Applying  $R_3 \rightarrow R_3 - (R_1 + R_2)$

$$\therefore \Delta = (\alpha + \beta + \gamma)^2 \begin{vmatrix} \beta + \gamma - \alpha & 0 & \alpha^2 \\ \beta - \gamma - \alpha & \gamma + \alpha - \beta & \beta^2 \\ 2\alpha - 2\beta & -2\alpha & 2\alpha\beta \end{vmatrix}$$

Taking 2 common from  $R_3$

$$\therefore \Delta = 2(\alpha + \beta + \gamma)^2 \begin{vmatrix} \beta + \gamma - \alpha & 0 & \alpha^2 \\ \beta - \gamma - \alpha & \gamma + \alpha - \beta & \beta^2 \\ \alpha - \beta & -\alpha & \alpha\beta \end{vmatrix}$$

Applying  $R_1 \rightarrow R_1 - \frac{\alpha}{\beta} R_3$  and  $R_2 \rightarrow R_2 - \frac{\beta}{\alpha} R_3$

$$\therefore \Delta = 2(\alpha + \beta + \gamma)^2 \begin{vmatrix} \beta + \gamma - \frac{\alpha^2}{\beta} & \frac{\alpha^2}{\beta} & 0 \\ -\gamma - \alpha + \frac{\beta^2}{\alpha} & \gamma + \alpha & 0 \\ \alpha - \beta & -\alpha & \alpha\beta \end{vmatrix}$$

Expanding along  $C_3$ , we get

$$\Delta = 2\alpha\beta (\alpha + \beta + \gamma)^2 \begin{vmatrix} \beta + \gamma - \frac{\alpha^2}{\beta} & \frac{\alpha^2}{\beta} \\ -\gamma - \alpha + \frac{\beta^2}{\alpha} & \gamma + \alpha \end{vmatrix}$$

Applying  $C_1 \rightarrow C_1 - C_2$ , then

$$\Delta = 2\alpha\beta (\alpha + \beta + \gamma)^2 \begin{vmatrix} \beta + \gamma & \frac{\alpha^2}{\beta} \\ \frac{\beta^2}{\alpha} & \gamma + \alpha \end{vmatrix}$$

$$= 2\alpha\beta (\alpha + \beta + \gamma)^2 \{(\beta + \gamma)(\gamma + \alpha) - \alpha\beta\}$$

$$= 2\alpha\beta (\alpha + \beta + \gamma)^2 \{\beta\gamma + \gamma^2 + \gamma\alpha\}$$

$$\Delta = 2\alpha\beta\gamma (\alpha + \beta + \gamma)^3$$

$$\text{Hence } \Delta = 2s^3 (s - a)(s - b)(s - c).$$

► **Problem 9.** Let  $\alpha, \beta$  be the roots of the equation  $ax^2 + bx + c = 0$ . Let  $S_n = \alpha^n + \beta^n$  for  $n \geq 1$ . Evaluate the determinant

$$\begin{vmatrix} 3 & 1 + S_1 & 1 + S_2 \\ 1 + S_1 & 1 + S_2 & 1 + S_3 \\ 1 + S_2 & 1 + S_3 & 1 + S_4 \end{vmatrix}$$

► **Solution** Since  $\alpha, \beta$  are the roots of the equation  $ax^2 + bx + c = 0$ .

$$\therefore \alpha + \beta = -\frac{b}{a} \text{ and } \alpha\beta = \frac{c}{a}.$$

$$\text{Let } \Delta = \begin{vmatrix} 3 & 1 + S_1 & 1 + S_2 \\ 1 + S_1 & 1 + S_2 & 1 + S_3 \\ 1 + S_2 & 1 + S_3 & 1 + S_4 \end{vmatrix}$$

$$= \begin{vmatrix} 3 & 1 + \alpha + \beta & 1 + \alpha^2 + \beta^2 \\ 1 + \alpha + \beta & 1 + \alpha^2 + \beta^2 & 1 + \alpha^3 + \beta^3 \\ 1 + \alpha^2 + \beta^2 & 1 + \alpha^3 + \beta^3 & 1 + \alpha^4 + \beta^4 \end{vmatrix}$$

$$= \begin{vmatrix} 1 & 1 & 1 \\ 1 & \alpha & \beta \\ 1 & \alpha^2 & \beta^2 \end{vmatrix} \times \begin{vmatrix} 1 & 1 & 1 \\ 1 & \alpha & \beta \\ 1 & \alpha^2 & \beta^2 \end{vmatrix}$$

$$= \Delta_1 \times \Delta_1 \text{ (assume)}$$

$$\therefore \Delta = \Delta_1^2 \quad \dots(1)$$

$$\text{Now } \Delta_1 = \begin{vmatrix} 1 & 1 & 1 \\ 1 & \alpha & \beta \\ 1 & \alpha^2 & \beta^2 \end{vmatrix}$$

Applying  $C_2 \rightarrow C_2 - C_1$  and  $C_3 \rightarrow C_3 - C_1$ ,

$$\Delta_1 = \begin{vmatrix} 1 & 0 & 0 \\ 1 & \alpha - 1 & \beta - 1 \\ 1 & \alpha^2 - 1 & \beta^2 - 1 \end{vmatrix}$$

Expanding along  $R_1$ ,

$$\begin{aligned}\therefore \Delta_1 &= \begin{vmatrix} \alpha-1 & \beta-1 \\ \alpha^2-1 & \beta^2-1 \end{vmatrix} \\ &= (\alpha-1)(\beta-1) \begin{vmatrix} 1 & 1 \\ \alpha+1 & \beta+1 \end{vmatrix} \\ &= \{\alpha\beta - (\alpha+\beta) + 1\} (\beta-\alpha) \\ &= \{\alpha\beta - (\alpha+\beta) + 1\} \sqrt{(\alpha+\beta)^2 - 4\alpha\beta} \\ &= \left(\frac{c}{a} + \frac{b}{a} + 1\right) \sqrt{\left(\frac{b^2}{a^2} - \frac{4c}{a}\right)} \\ \Delta_1 &= \frac{(a+b+c)\sqrt{(b^2-4ac)}}{a^2} \\ \therefore \Delta_1^2 &= \frac{(a+b+c)^2(b^2-4ac)}{a^4}\end{aligned}$$

$$\text{Hence } \Delta = \Delta_1^2 = \frac{(a+b+c)^2(b^2-4ac)}{a^4}.$$

► Problem 10.

$$\text{If } \Delta_r = \begin{vmatrix} 2r-1 & {}^m C_r & 1 \\ m^2-1 & 2^m & m+1 \\ \sin^2(m^2) & \sin^2(m) & \sin^2(m+1) \end{vmatrix}$$

then prove that  $\sum_{r=0}^m \Delta_r = 0$ .

► Solution

$$\begin{aligned}\Delta_r &= \begin{vmatrix} 2r-1 & {}^m C_r & 1 \\ m^2-1 & 2^m & m+1 \\ \sin^2(m^2) & \sin^2(m) & \sin^2(m+1) \end{vmatrix} \\ \Rightarrow \sum_{r=0}^m \Delta_r &= \begin{vmatrix} \sum_{r=0}^m (2r-1) & \sum_{r=0}^m {}^m C_r & \sum_{r=0}^m 1 \\ m^2-1 & 2^m & m+1 \\ \sin^2(m^2) & \sin^2(m) & \sin^2(m+1) \end{vmatrix} \\ &= \begin{vmatrix} m^2-1 & 2^m & m+1 \\ m^2-1 & 2^m & m+1 \\ \sin^2(m^2) & \sin^2(m) & \sin^2(m+1) \end{vmatrix} = 0\end{aligned}$$

► Problem 11. If  $\alpha, \beta$  are the roots of the equation  $ax^2 + bx + c = 0$  and  $s_n = \alpha^n + \beta^n$ , evaluate

$$\begin{vmatrix} 3 & 1+s_1 & 1+s_2 \\ 1+s_1 & 1+s_2 & 1+s_3 \\ 1-s_2 & 1+s_3 & 1+s_4 \end{vmatrix} \text{ in terms of } a, b, \text{ and } c \text{ only.}$$

► Solution

$$\begin{aligned}& \begin{vmatrix} 1+1+1 & 1+\alpha+\beta & 1+\alpha^2+\beta^2 \\ 1+\alpha+\beta & 1+\alpha^2+\beta^2 & 1+\alpha^3+\beta^3 \\ 1+\alpha^2+\beta^2 & 1+\alpha^3+\beta^3 & 1+\alpha^4+\beta^4 \end{vmatrix} \\ &= \begin{vmatrix} 1 & 1 & 1 \\ 1 & \alpha & \beta \\ 1 & \alpha^2 & \beta^2 \end{vmatrix} \times \begin{vmatrix} 1 & 1 & 1 \\ 1 & \alpha & \alpha^2 \\ 1 & \beta & \beta^2 \end{vmatrix} \\ &= ((1-\alpha)(\alpha-\beta)(\beta-1))^2 \\ &= (1-(\alpha+\beta)+\alpha\beta)^2 (\alpha-\beta)^2 \\ &= \left(1+\frac{b}{a}+\frac{c}{a}\right)^2 \left[\frac{b^2}{a^2}-\frac{4c}{a}\right] \\ &= \frac{(a+b+c)^2(b^2-4ac)}{a^4}.\end{aligned}$$

► Problem 12. If A, B and C are the angles of a triangle, show that

$$\begin{vmatrix} -1+\cos B & \cos C+\cos B & \cos B \\ \cos C+\cos A & -1+\cos A & \cos A \\ -1+\cos B & -1+\cos A & -1 \end{vmatrix} = 0$$

► Solution

$$\text{L.H.S.} = \begin{vmatrix} -1+\cos B & \cos C+\cos B & \cos B \\ \cos C+\cos A & -1+\cos A & \cos A \\ -1+\cos B & -1+\cos A & -1 \end{vmatrix}$$

Applying  $C_1 \rightarrow C_1 - C_3 : C_2 \rightarrow C_2 - C_3$

$$= \begin{vmatrix} -1 & \cos C & \cos B \\ \cos C & -1 & \cos A \\ \cos B & \cos A & -1 \end{vmatrix}$$

$$= \frac{1}{a} \begin{vmatrix} -a & \cos C & \cos B \\ a \cos C & -1 & \cos A \\ a \cos B & \cos A & -1 \end{vmatrix}$$

Applying  $C_1 \rightarrow C_1 + bC_2 + cC_3$

$$= \frac{1}{a} \begin{vmatrix} 0 & \cos C & \cos B \\ 0 & -1 & \cos A \\ 0 & \cos A & -1 \end{vmatrix} = 0 = \text{R.H.S.}$$

► **Problem 13.** Prove that 
$$\begin{vmatrix} -2a & a+b & a+c \\ b+a & -2b & b+c \\ c+a & c+b & -2c \end{vmatrix} = 4(b+c)(c+a)(a+b)$$

► **Solution**

$$\text{Let } \Delta = \begin{vmatrix} -2a & a+b & a+c \\ b+a & -2b & b+c \\ c+a & c+b & -2c \end{vmatrix}$$

Putting  $a+b=0 \Rightarrow b=-a$

$$\text{then } \Delta = \begin{vmatrix} -2a & 0 & a+c \\ 0 & 2a & c-a \\ c+a & c-a & -2c \end{vmatrix}$$

Expanding along  $R_1$

$$= -2a\{-4ac - (c-a)^2\} - 0 + (a+c)\{0 - 2a(c+a)\} \\ = 2a(c+a)^2 - 2a(c+a)^2 = 0$$

Hence  $a+b$  is a factor of  $\Delta$ . Similarly  $b+c$  and  $c+a$  are the factors of  $\Delta$ .

On expansion of determinant we can see that each term of the determinant is a homogeneous expression in  $a, b, c$  of degree 3 and also R.H.S is a homogeneous expression of degree 3.

Let  $\Delta = k(a+b)(b+c)(c+a)$

$$\text{or } \begin{vmatrix} -2a & a+b & a+c \\ b+a & -2b & b+c \\ c+a & c+b & -2c \end{vmatrix} = k(a+b)(b+c)(c+a)$$

If we choose  $a=0, b=1, c=2$ , we get

$$\Rightarrow 0 - 1(-4 - 6) + 2(3 + 4) = 6k$$

$$\Rightarrow k = 4$$

$$\text{Hence } \begin{vmatrix} -2a & a+b & a+c \\ b+a & -2b & b+c \\ c+a & c+b & -2c \end{vmatrix} = 4(a+b)(b+c)(c+a)$$

► **Problem 14.** Prove that

$$\begin{vmatrix} 2 & \alpha+\beta+\gamma+\delta \\ \alpha+\beta+\gamma+\delta & 2(\alpha+\beta)(\gamma+\delta) \\ \alpha\beta+\gamma\delta & \alpha\beta(\gamma+\delta)+\gamma\delta(\alpha+\beta) \\ \alpha\beta+\gamma\delta & \alpha\beta(\gamma+\delta)+\gamma\delta(\alpha+\beta) \\ 2\alpha\beta\gamma\delta \end{vmatrix} = 0$$

► **Solution** L.H.S.

$$= \begin{vmatrix} 2 & \alpha+\beta+\gamma+\delta \\ \alpha+\beta+\gamma+\delta & 2(\alpha+\beta)(\gamma+\delta) \\ \alpha\beta+\gamma\delta & \alpha\beta(\gamma+\delta)+\gamma\delta(\alpha+\beta) \\ \alpha\beta+\gamma\delta & \alpha\beta(\gamma+\delta)+\gamma\delta(\alpha+\beta) \\ 2\alpha\beta\gamma\delta \end{vmatrix}$$

$$= \begin{vmatrix} 1 & 1 & 0 \\ \alpha+\beta & \gamma+\delta & 0 \\ \alpha\beta & \gamma\delta & 0 \end{vmatrix} \times \begin{vmatrix} 1 & 1 & 0 \\ \gamma+\delta & \alpha+\beta & 0 \\ \gamma\delta & \alpha\beta & 0 \end{vmatrix}$$

(Row by Row)

$$= 0 \times 0 = 0 = \text{R.H.S.}$$

► **Problem 15.** Show that

$$\begin{vmatrix} 1 & \cos(\beta-\alpha) & \cos(\gamma-\alpha) \\ \cos(\alpha-\beta) & 1 & \cos(\gamma-\beta) \\ \cos(\alpha-\gamma) & \cos(\beta-\gamma) & 1 \end{vmatrix} = 0$$

► **Solution** L.H.S. =

$$\begin{vmatrix} 1 & \cos(\beta - \alpha) & \cos(\gamma - \alpha) \\ \cos(\alpha - \beta) & 1 & \cos(\gamma - \beta) \\ \cos(\alpha - \gamma) & \cos(\beta - \gamma) & 1 \end{vmatrix}$$

$$= \begin{vmatrix} \cos \alpha & \sin \alpha & 0 \\ \cos \beta & \sin \beta & 0 \\ \cos \gamma & \sin \gamma & 0 \end{vmatrix} \times \begin{vmatrix} \cos \alpha & \sin \alpha & 0 \\ \cos \beta & \sin \beta & 0 \\ \cos \gamma & \sin \gamma & 0 \end{vmatrix}$$

(Row by Row)

$$= 0 \times 0 = 0 = \text{R.H.S.}$$

► **Problem 16.** If  $a, b, c$  are non zero real numbers

such that  $\begin{vmatrix} bc & ca & ab \\ ca & ab & bc \\ ab & bc & ca \end{vmatrix} = 0$ , then show that

$$\frac{1}{a} + \frac{1}{b\omega} + \frac{1}{c\omega^2} = 0, \text{ or } \frac{1}{a} + \frac{1}{b\omega^2} + \frac{1}{c\omega} = 0,$$

$$\text{or } \frac{1}{a\omega} + \frac{1}{b\omega^2} + \frac{1}{c} = 0.$$

► **Solution** Given,  $\begin{vmatrix} bc & ca & ab \\ ca & ab & bc \\ ab & bc & ca \end{vmatrix} = 0$ .

$$\Rightarrow (ab)^3 + (bc)^3 + (ca)^3 - 3(ab)(bc)(ca) = 0$$

$$\Rightarrow (ab + bc\omega^2 + ca\omega)(ab\omega + bc\omega^2 + ca) \cdot (ab\omega^2 + bc\omega + ca) = 0$$

$$[\because (x + y\omega^2 + z\omega)(x\omega + y\omega^2 + z)(z\omega^2 + y\omega + z) = x^3 + y^3 + z^3 - 3xyz]$$

$$\Rightarrow ab + bc\omega^2 + ca\omega = 0, ab\omega + bc\omega^2 + ca\omega = 0, ab\omega^2 + bc\omega + ca = 0$$

$$\Rightarrow \frac{1}{c\omega^2} + \frac{1}{a} + \frac{1}{b\omega} = 0, \frac{1}{c\omega} + \frac{1}{a} + \frac{1}{b\omega^2} = 0,$$

$$\frac{1}{c} + \frac{1}{a\omega} + \frac{1}{b\omega^2} = 0$$

$$\Rightarrow \frac{1}{a} + \frac{1}{b\omega} + \frac{1}{c\omega^2} = 0, \frac{1}{a} + \frac{1}{b\omega^2} + \frac{1}{c\omega} = 0,$$

$$\frac{1}{a\omega} + \frac{1}{b\omega^2} + \frac{1}{c} = 0$$

► **Problem 17.** If all elements of a third order determinant are equal to 1 or  $-1$ , then prove that the determinant itself is an even number.

► **Solution** Let the third order determinant be

$$\Delta = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} \text{ where } a_{ij} = 1 \text{ or } -1,$$

then both  $\frac{a_{21}}{a_{11}}$  and  $\frac{a_{31}}{a_{11}}$  equal to  $\pm 1$

$$= \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ 0 & 2\alpha & 2\beta \\ 0 & 2\gamma & 2\delta \end{vmatrix}$$

$$R_2 \rightarrow R_2 - \left(\frac{a_{21}}{a_{11}}\right) R_1, R_3 \rightarrow R_3 - \left(\frac{a_{31}}{a_{11}}\right) R_1$$

where  $\alpha, \beta, \gamma$  and  $\delta$  all equal  $-1, 0$  or  $1$ .

Expand along  $C_1 = 4a_{11}(\alpha\delta - \beta\gamma)$  which is an even number.

► **Problem 18.** If  $a, b, c$  are sides of a triangle and

$$\begin{vmatrix} a^2 & b^2 & c^2 \\ (a+1)^2 & (b+1)^2 & (c+1)^2 \\ (a-1)^2 & (b-1)^2 & (c-1)^2 \end{vmatrix} = 0, \text{ then show that } \Delta$$

ABC is an isosceles triangle.

► **Solution**  $a \neq 0, b \neq 0, c \neq 0$

$$\therefore \begin{vmatrix} a^2 & b^2 & c^2 \\ (a+1)^2 & (b+1)^2 & (c+1)^2 \\ (a-1)^2 & (b-1)^2 & (c-1)^2 \end{vmatrix} = 0$$

$$= \begin{vmatrix} a^2 & b^2 - a^2 & c^2 - a^2 \\ (a+1)^2 & (b+1)^2 - (a+1)^2 & (c+1)^2 - (a+1)^2 \\ (a-1)^2 & (b-1)^2 - (a-1)^2 & (c-1)^2 - (a-1)^2 \end{vmatrix} = 0$$

$$\Rightarrow [C_2 \rightarrow C_2 - C_1; C_3 \rightarrow C_3 - C_1]$$

$$\begin{vmatrix} a^2 & (b+a)(b-a) & (c+a)(c-a) \\ (a+1)^2 & (b+a+2)(b-a) & (c+a+2)(c-a) \\ (a-1)^2 & (b+a-2)(b-a) & (c+a-2)(c-a) \end{vmatrix} = 0$$

$$\Rightarrow (b-a)(c-a) \begin{vmatrix} a^2 & b+a & c+a \\ (a+1)^2 & b+a+2 & c+a+2 \\ (a-1)^2 & b+a-2 & c+a-2 \end{vmatrix} = 0$$

$$[R_2 \rightarrow R_2 - R_1; R_3 \rightarrow R_3 - R_1]$$

$$\Rightarrow (b-a)(c-a) \begin{vmatrix} a^2 & b+a & c+a \\ 2a+1 & 2 & 2 \\ 2 & 0 & 0 \end{vmatrix} = 0$$

$$[R_3 \rightarrow R_3 + R_2]$$

Expand w.r.t. third row

$$\Rightarrow 2(b-a)(c-a) \begin{vmatrix} b+a & c+a \\ 2 & 2 \end{vmatrix} = 0$$

$$\Rightarrow 2(b-a)(c-a) \{2b+2a-2c-2a\} = 0$$

$$\Rightarrow 4(a-b)(b-c)(c-a) = 0$$

$$\therefore a-b=0 \text{ or } b-c=0 \text{ or } c-a=0$$

$$\therefore a=b \text{ or } b=c \text{ or } c=a$$

► **Problem 19.** If  $a, b, c, x, y, z \in \mathbb{R}$ , then prove that

$$\begin{vmatrix} (a-x)^2 & (b-x)^2 & (c-x)^2 \\ (a-y)^2 & (b-y)^2 & (c-y)^2 \\ (a-z)^2 & (b-z)^2 & (c-z)^2 \end{vmatrix}$$

$$= \begin{vmatrix} (1+ax)^2 & (1+bx)^2 & (1+cx)^2 \\ (1+ay)^2 & (1+by)^2 & (1+cy)^2 \\ (1+az)^2 & (1+bz)^2 & (1+cz)^2 \end{vmatrix}$$

► **Solution** L.H.S.

$$= \begin{vmatrix} (a-x)^2 & (b-x)^2 & (c-x)^2 \\ (a-y)^2 & (b-y)^2 & (c-y)^2 \\ (a-z)^2 & (b-z)^2 & (c-z)^2 \end{vmatrix}$$

$$= \begin{vmatrix} a^2 - 2ax + x^2 & b^2 - 2bx + x^2 & c^2 - 2cx + x^2 \\ a^2 - 2ay + y^2 & b^2 - 2by + y^2 & c^2 - 2cy + y^2 \\ a^2 - 2az + z^2 & b^2 - 2bz + z^2 & c^2 - 2cz + z^2 \end{vmatrix}$$

$$= \begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix} \times \begin{vmatrix} a^2 & -2a & 1 \\ b^2 & -2b & 1 \\ c^2 & -2c & 1 \end{vmatrix}$$

Multiplying row by row

$$= \begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix} \times (-1) \begin{vmatrix} a^2 & 2a & 1 \\ b^2 & 2b & 1 \\ c^2 & 2c & 1 \end{vmatrix}$$

$$= \begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix} \times (-1)(-1) \begin{vmatrix} 1 & 2a & a^2 \\ 1 & 2b & b^2 \\ 1 & 2c & c^2 \end{vmatrix}$$

$$(C_1 \leftrightarrow C_3)$$

$$= \begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix} \times \begin{vmatrix} 1 & 2a & a^2 \\ 1 & 2b & b^2 \\ 1 & 2c & c^2 \end{vmatrix}$$

Multiplying row by row

$$= \begin{vmatrix} 1+2ax+a^2x^2 & 1+2bx+b^2x^2 & 1+2cx+c^2x^2 \\ 1+2ay+a^2y^2 & 1+2by+b^2y^2 & 1+2cy+c^2y^2 \\ 1+2az+a^2z^2 & 1+2bz+b^2z^2 & 1+2cz+c^2z^2 \end{vmatrix}$$

$$= \begin{vmatrix} (1+ax)^2 & (1+bx)^2 & (1+cx)^2 \\ (1+ay)^2 & (1+by)^2 & (1+cy)^2 \\ (1+az)^2 & (1+bz)^2 & (1+cz)^2 \end{vmatrix} = \text{R.H.S.}$$

► **Problem 20.** If  $ax_1^2 + by_1^2 + cz_1^2 = ax_2^2 + by_2^2 + cz_2^2$

$$= ax_3^2 + by_3^2 + cz_3^2 = d, \quad ax_2x_3 + by_2y_3 + cz_2z_3$$

$$= ax_3x_1 + by_3y_1 + cz_3z_1 = ax_1x_2 + by_1y_2 + cz_1z_2 = f,$$

$$\text{then prove that } \begin{vmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix} = (d-f) \left\{ \frac{(d+2f)}{abc} \right\}^{1/2}$$

$$\text{► **Solution** Let } D = \begin{vmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix}$$

$$\therefore D^2 = \begin{vmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix} \times \begin{vmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix}$$

$$\begin{aligned}
 &= \frac{1}{abc} \begin{vmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix} \times \begin{vmatrix} ax_1 & by_1 & cz_1 \\ ax_2 & by_2 & cz_2 \\ ax_3 & by_3 & cz_3 \end{vmatrix} \\
 &= \frac{1}{abc} \begin{vmatrix} ax_1x_2 + by_1y_2 + cz_1z_2 & & \\ & ax_2^2 + by_2^2 + cz_2^2 & \\ & & ax_2x_3 + by_2y_3 + cz_2z_3 \end{vmatrix} \\
 &\quad \begin{vmatrix} ax_1x_2 + by_1y_2 + cz_1z_2 & ax_3x_1 + by_3y_1 + cz_3z_1 \\ ax_2^2 + by_2^2 + cz_2^2 & ax_2x_3 + by_2y_3 + cz_2z_3 \\ ax_2x_3 + by_2y_3 + cz_2z_3 & ax_3^2 + by_3^2 + cz_3^2 \end{vmatrix} \\
 &\text{(Multiplying Row by Row)} \\
 &= \frac{1}{abc} \begin{vmatrix} d & f & f \\ f & d & f \\ f & f & d \end{vmatrix} \\
 &= \frac{1}{abc} \begin{vmatrix} d+2f & f & f \\ d+2f & d & f \\ d+2f & f & d \end{vmatrix} \quad [C_1 \rightarrow C_1 + C_2 + C_3] \\
 &= \frac{(d+2f)}{abc} \begin{vmatrix} 1 & f & f \\ 1 & d & f \\ 1 & f & d \end{vmatrix} \\
 &= \frac{(d+2f)}{abc} \begin{vmatrix} 1 & f & f \\ 0 & d-f & 0 \\ 1 & f & d \end{vmatrix} \quad [R_2 \rightarrow R_2 - R_1] \\
 &= \frac{(d+2f)}{abc} \cdot (d-f) \begin{vmatrix} 1 & f \\ 1 & d \end{vmatrix} \\
 &= \frac{(d+2f)}{abc} \cdot (d-f) \cdot (d-f) \\
 &= \frac{(d+2f)(d-f)^2}{abc} \\
 \therefore D &= (d-f) \left\{ \frac{(d+2f)}{abc} \right\}^{1/2}. \text{ Hence proved.}
 \end{aligned}$$

► **Problem 21.** If

$$f(x) = \begin{vmatrix} (1+x)^{a_1b_1} & (1+x)^{a_1b_2} & (1+x)^{a_1b_3} \\ (1+x)^{a_2b_1} & (1+x)^{a_2b_2} & (1+x)^{a_2b_3} \\ (1+x)^{a_3b_1} & (1+x)^{a_3b_2} & (1+x)^{a_3b_3} \end{vmatrix}$$

(where  $a_i$ 's and  $b_j$ 's  $\in \mathbb{N}$ ), find the coefficient of  $x$  in the expansion of  $f(x)$ .

► **Solution** Let  $f(x) = c_0 + c_1x + c_2x^2 + \dots$

then,  $f'(x) = c_1 + 2c_2x + 3c_3x^2 + \dots$

Also  $f'(x) =$

$$\begin{aligned}
 &\begin{vmatrix} a_1b_1(1+x)^{a_1b_1-1} & a_1b_2(1+x)^{a_1b_2-1} & a_1b_3(1+x)^{a_1b_3-1} \\ (1+x)^{a_2b_1} & (1+x)^{a_2b_2} & (1+x)^{a_2b_3} \\ (1+x)^{a_3b_1} & (1+x)^{a_3b_2} & (1+x)^{a_3b_3} \end{vmatrix} \\
 &+ \begin{vmatrix} (1+x)^{a_1b_1} & (1+x)^{a_1b_2} & (1+x)^{a_1b_3} \\ a_2b_1(1+x)^{a_2b_1-1} & a_2b_2(1+x)^{a_2b_2-1} & a_2b_3(1+x)^{a_2b_3-1} \\ (1+x)^{a_3b_1} & (1+x)^{a_3b_2} & (1+x)^{a_3b_3} \end{vmatrix} \\
 &+ \begin{vmatrix} (1+x)^{a_1b_1} & (1+x)^{a_1b_2} & (1+x)^{a_1b_3} \\ (1+x)^{a_2b_1} & (1+x)^{a_2b_2} & (1+x)^{a_2b_3} \\ a_3b_1(1+x)^{a_3b_1-1} & a_3b_2(1+x)^{a_3b_2-1} & a_3b_3(1+x)^{a_3b_3-1} \end{vmatrix}
 \end{aligned}$$

$$\therefore c_1 = f'(0)$$

$$\begin{aligned}
 &= \begin{vmatrix} a_1b_1 & a_1b_2 & a_1b_3 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{vmatrix} + \begin{vmatrix} 1 & 1 & 1 \\ a_2b_1 & a_2b_2 & a_2b_3 \\ 1 & 1 & 1 \end{vmatrix} \\
 &+ \begin{vmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ a_3b_1 & a_3b_2 & a_3b_3 \end{vmatrix} \\
 &= 0 + 0 + 0 = 0
 \end{aligned}$$

► **Problem 22.** Prove that

$$\Delta = \begin{vmatrix} -bc & b^2 + bc & c^2 + bc \\ a^2 + ac & -ac & c^2 + ac \\ a^2 + ab & b^2 + ab & -ab \end{vmatrix} = (bc + ca + ab)^3.$$

► **Solution** Multiply  $R_1, R_2, R_3$  by  $a, b, c$  respectively and divide the determinant by  $abc$ ; then,

$$\Delta = \frac{1}{abc} \begin{vmatrix} -abc & ab^2 + abc & ac^2 + abc \\ a^2b + abc & -abc & bc^2 + abc \\ a^2c + abc & b^2c + abc & -abc \end{vmatrix}$$

Applying  $R_1 \rightarrow R_1 + R_2 + R_3$ , we get

$$\Delta = \begin{vmatrix} a[ab+ac+bc] & b[ab+bc+ac] & c[ac+bc+ab] \\ a^2b+abc & -abc & bc^2+abc \\ a^2c+abc & b^2c+abc & -abc \end{vmatrix}$$

$$\Delta = abc \frac{(ab+bc+ca)}{abc} \begin{vmatrix} 1 & 1 & 1 \\ ab+bc & -ac & bc+ab \\ ac+bc & bc+ac & -ab \end{vmatrix}$$

Applying  $C_2 \rightarrow C_2 - C_1$  and  $C_3 \rightarrow C_3 - C_1$   
 $\Delta = (ab+bc+ca) \cdot$

$$\begin{vmatrix} 1 & 0 & 0 \\ ab+bc & -(ac+bc+ab) & 0 \\ ac+bc & 0 & -(ab+ac+bc) \end{vmatrix}$$

$$= (ab+bc+ca)(ab+bc+ac)^2$$

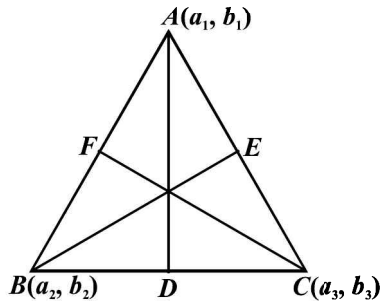
$$= (ab+bc+ca)^3.$$

► **Problem 23.** If  $(a_r, b_r)$ ,  $r = 1, 2, 3$ , be the vertices of a triangle, prove that  $\Delta =$

$$\begin{vmatrix} a_2 - a_3 & b_2 - b_3 & a_1(a_2 - a_3) + b_1(b_2 - b_3) \\ a_3 - a_1 & b_3 - b_1 & a_2(a_3 - a_1) + b_2(b_3 - b_1) \\ a_1 - a_2 & b_1 - b_2 & a_3(a_1 - a_2) + b_3(b_1 - b_2) \end{vmatrix} = 0 \quad \dots(1)$$

and hence show that the altitudes of a triangle are concurrent.

► **Solution**



Applying  $R_1 \rightarrow R_1 + R_2 + R_3$

$$\Delta = \begin{vmatrix} 0 & 0 & 0 \\ a_3 - a_1 & b_3 - b_1 & a_2(a_3 - a_1) + b_2(b_3 - b_1) \\ a_1 - a_2 & b_1 - b_2 & a_3(a_1 - a_2) + b_3(b_1 - b_2) \end{vmatrix} = 0$$

∴ Equation of altitude AD is

$$y - b_1 = -\frac{a_2 - a_3}{b_2 - b_3}(x - a_1)$$

$$\begin{aligned} \text{or } x(a_2 - a_3) + y(b_2 - b_3) \\ = a_1(a_2 - a_3) + b_1(b_2 - b_3) \quad \dots(2) \end{aligned}$$

Similarly equations of altitudes BE and CF are

$$\begin{aligned} x(a_3 - a_1) + y(b_3 - b_1) \\ = a_2(a_3 - a_1) + b_2(b_3 - b_1) \quad \dots(3) \end{aligned}$$

$$\begin{aligned} x(a_1 - a_2) + y(b_1 - b_2) \\ = a_3(a_1 - a_2) + b_3(b_1 - b_2) \quad \dots(4) \end{aligned}$$

Altitudes (2), (3), (4) are concurrent, since the determinant given by L.H.S of (1) is zero.

► **Problem 24.** If  $f(x)$  is a polynomial of degree  $< 3$ ,

$$\text{prove that } \begin{vmatrix} 1 & a & f(a)/(x-a) \\ 1 & b & f(b)/(x-b) \\ 1 & c & f(c)/(x-c) \end{vmatrix} \div \begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix}$$

$$= \frac{f(x)}{(x-a)(x-b)(x-c)}$$

► **Solution**

$$\frac{f(x)}{(x-a)(x-b)(x-c)} \equiv \frac{A}{(x-a)} + \frac{B}{(x-b)} + \frac{C}{(x-c)} \quad \dots(1)$$

On comparing the various powers of  $x$ , we get

$$A = -\frac{f(a)}{(a-b)(c-a)}$$

$$B = -\frac{f(b)}{(a-b)(b-c)}$$

$$C = -\frac{f(c)}{(b-c)(c-a)}$$



Now from (1) we have  $\frac{f(x)}{(x-a)(x-b)(x-c)}$

$$= \frac{(c-b)\frac{f(a)}{(x-a)} + (a-c)\frac{f(b)}{x-b} + (b-a)\frac{f(c)}{(x-c)}}{(a-b)(b-c)(c-a)}$$

$$= \begin{vmatrix} 1 & a & f(a)/(x-a) \\ 1 & b & f(b)/(x-b) \\ 1 & c & f(c)/(x-c) \end{vmatrix}$$

$$= \begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix}$$

### Things to Remember

1.  $|A| = a_{11}M_{11} - a_{12}M_{12} + a_{13}M_{13}$   
 $= a_{11}A_{11} + a_{12}A_{12} + a_{13}A_{13}$ .
2. The value of a determinant remains unaltered, if the rows and columns are inter changed, i.e., if  $A$  is an  $n$ -rowed square matrix, then  $|A| = |A'|$ .  
 A skew symmetric determinant of odd order is zero.
3. If any two rows (or columns) of a determinant are interchanged, the value of determinant is changed in sign only.
4. If a determinant has any two rows (or columns) identical or proportional, then its value is zero.
5. If all the elements of any row (or column) be multiplied by the same number, then the determinant is multiplied by that number.
6. If each element of any row (or column) can be expressed as a sum of two terms then the determinant can be expressed as the sum of two determinants.
7. The value of a determinant is not altered by adding to the elements of any row (or column) the same multiples of the corresponding elements of any other row (or column).

8. If the elements of a determinant  $\Delta$  are rational integral functions of  $x$  and two rows (or columns) become identical when  $x = a$ , then  $x - a$  is a factor of  $\Delta$ .

Again, if  $r$  rows become identical when  $a$  is substituted for  $x$ , then  $(x - a)^{r-1}$  is a factor of  $\Delta$ .

$$9. \begin{vmatrix} 1 & 1 & 1 \\ \alpha & \beta & \gamma \\ \alpha^2 & \beta^2 & \gamma^2 \end{vmatrix} = (\beta - \gamma)(\gamma - \alpha)(\alpha - \beta).$$

$$\begin{vmatrix} 1 & 1 & 1 \\ \alpha & \beta & \gamma \\ \alpha^3 & \beta^3 & \gamma^3 \end{vmatrix} = (\beta - \gamma)(\gamma - \alpha)(\alpha - \beta)$$

$$(\alpha + \beta + \gamma).$$

10. If  $A$  and  $B$  are two square matrices of the same order, then  $|AB| = |A| |B| = |BA|$ .
11. If  $\Delta = |a_{ij}|$  is a determinant of  $n^{\text{th}}$  order and  $\Delta' = |A_{ij}|$ , the adjoint of  $\Delta$ , then  $\Delta' = \Delta^{n-1}$  when  $\Delta \neq 0$ .
12. If the same number  $x$  is added to all elements of the determinant  $D$ , the determinant is increased by the product of the number into the sum of the cofactors of all elements of  $D$ .  

$$D' = D + x \sum_{i,j=1}^n A_{ij}$$
13.  $d\Delta/dx$  is the sum of the  $n$  determinants obtained by differentiating the elements of one row (or column) of  $\Delta$  and leaving the elements of the other  $n-1$  rows (or columns) unaltered.

### Objective Exercises

#### Single Correct Answer Type

1. The value of  $\begin{vmatrix} 1+x_1 & 1+x_1x & 1+x_1x^2 \\ 1+x_2 & 1+x_2x & 1+x_2x^2 \\ 1+x_3 & 1+x_3x & 1+x_3x^2 \end{vmatrix}$  depends upon  
 (A)  $x$  only (B)  $x_1$  only

- (C)  $x_2$  only (D) none of these
2. Let  $D_1 = \begin{vmatrix} a & b & a+b \\ c & d & c+d \\ a & b & a-b \end{vmatrix}$  and
- $D_2 = \begin{vmatrix} a & c & a+c \\ b & d & b+d \\ a & c & a+b+c \end{vmatrix}$  then the value of  $\frac{D_1}{D_2}$

where  $b \neq 0$  and  $ad \neq bc$ , is

- (A) -2 (B) 0  
(C) -2b (D) 2b
3. If A, B, C are the angles of a triangle and
- $$\begin{vmatrix} 1 & 1 & 1 \\ 1+\sin A & 1+\sin B & 1+\sin C \\ \sin A + \sin^2 A & \sin B + \sin^2 B & \sin C + \sin^2 C \end{vmatrix} = 0,$$
- then the triangle is
- (A) an equilateral (B) an isosceles  
(C) a right angled triangle (D) any triangle

4. If  $\begin{vmatrix} -a^2 & ab & ac \\ ab & -b^2 & bc \\ ac & bc & -c^2 \end{vmatrix} = ka^2 b^2 c^2$ , then  $k =$

- (A) 2 (B) 4  
(C) -4 (D) 8
5. The value of the determinant

$$\begin{vmatrix} a-b-c & 2a & 2a \\ 2b & b-c-a & 2b \\ 2c & 2c & c-a-b \end{vmatrix} \text{ is}$$

- (A)  $(a+b+c)^3$   
(B)  $(a-b-c)(a^2+b^2+c^2)$   
(C)  $(a+b+c)(ab+bc+ca)$   
(D) None of these

6. The determinant  $\begin{vmatrix} {}^x C_1 & {}^x C_2 & {}^x C_3 \\ {}^y C_1 & {}^y C_2 & {}^y C_3 \\ {}^z C_1 & {}^z C_2 & {}^z C_3 \end{vmatrix} =$

- (A)  $\frac{1}{3} xyz (x+y)(y+z)(z+x)$   
(B)  $\frac{1}{4} xyz (x+y-z)(y+z-x)$   
(C)  $\frac{1}{12} xyz (x-y)(y-z)(z-x)$   
(D) none

7. If a, b and c are the sides of a triangle ABC

$$\text{and } \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = 0 \text{ then } \sin^3 A + \sin^3 B +$$

$\sin^3 C$  is equal to

- (A)  $3 \sin A \cdot \sin B \cdot \sin C$   
(B)  $\sin 3A \cdot \sin 3B \cdot \sin 3C$   
(C)  $\sin A \cdot \sin B + \sin B \cdot \sin C + \sin C \cdot \sin A$   
(D) None of these
8. If the determinant

$$\begin{vmatrix} \cos 2x & \sin^2 x & \cos 4x \\ \sin^2 x & \cos 2x & \cos^2 x \\ \cos 4x & \cos^2 x & \cos 2x \end{vmatrix} \text{ is expanded in}$$

powers of  $\sin x$  then the constant term in the expansion is

- (A) 1 (B) 2  
(C) -1 (D) None of these
9. If  $f(x) = \begin{vmatrix} x+\lambda & x & x \\ x & x+\lambda & x \\ x & x & x+\lambda \end{vmatrix}$ , then  $f(3x) - f(x) =$
- (A)  $3x\lambda^2$  (B)  $6x\lambda^2$   
(C)  $x\lambda^2$  (D) None of these

10. If  $\alpha, \beta$  and  $\gamma$  are the roots of the equation  $x^3 + ax + b = 0$ , then the value of the determinant

$$\begin{vmatrix} \alpha - \beta - \gamma & 2\alpha & 2\alpha \\ 2\beta & \beta - \gamma - \alpha & 2\beta \\ 2\gamma & 2\gamma & \gamma - \alpha - \beta \end{vmatrix} \text{ is}$$

- (A) 0 (B) a  
(C) -b (D) None of these

11. If A, B and C are the angles of a triangle, then the value of the determinant

$$\begin{vmatrix} -1 + \cos B & \cos C + \cos B & \cos B \\ \cos C + \cos A & -1 + \cos A & \cos A \\ -1 + \cos B & -1 + \cos A & 1 \end{vmatrix}$$
 is

- (A) 0 (B) 1  
(C) -1 (D) 2

12. If  $p^{\text{th}}$ ,  $q^{\text{th}}$  and  $r^{\text{th}}$  terms of an H.P. be a, b, c respectively, then the determinant

$$\begin{vmatrix} bc & ca & ab \\ p & q & r \\ 1 & 1 & 1 \end{vmatrix}$$
 equals

- (A) -1  
(B)  $2abc / (a + b + c)$   
(C)  $pq + qb + rc$   
(D) none of these

13. If a, b, c are non-zero real numbers, then

$$\begin{vmatrix} 1 & ab & \frac{1}{a} + \frac{1}{b} \\ 1 & bc & \frac{1}{b} + \frac{1}{c} \\ 1 & ca & \frac{1}{c} + \frac{1}{a} \end{vmatrix}$$
 equals

- (A) 0 (B)  $bc + ca + ab$   
(C)  $a^{-1} + b^{-1} + c^{-1}$  (D) None of these

14. If  $\begin{vmatrix} 1 & x & x^2 \\ x & x^2 & 1 \\ x^2 & 1 & x \end{vmatrix} = 7$  and

$$\Delta = \begin{vmatrix} x^3 - 1 & 0 & x - x^4 \\ 0 & x - x^4 & x^3 - 1 \\ x - x^4 & x^3 - 1 & 0 \end{vmatrix}, \text{ then}$$

- (A)  $\Delta = 7$  (B)  $\Delta = 343$   
(C)  $\Delta = -49$  (D)  $\Delta = 49$

15. The value of the determinant

$$\begin{vmatrix} 1^2 & 2^2 & 3^2 & 4^2 \\ 2^2 & 3^2 & 4^2 & 5^2 \\ 3^2 & 4^2 & 5^2 & 6^2 \\ 4^2 & 5^2 & 6^2 & 7^2 \end{vmatrix}$$
 is

- (A) 1 (B) 0  
(C) 2 (D) None of these

16. In a third order determinant,  $a_{ij}$  denotes the element in the  $i$ th row and the  $j$ th column. If

$$a_{ij} = \begin{cases} 0 & , i = j \\ -1 & , i < j \\ 1 & , i > j \end{cases}$$
 then the value of the

determinant is

- (A) 0 (B) 1  
(C) -1 (D) none of these .

17. The determinant

$$\begin{vmatrix} \cos(x-y) & \cos(y-z) & \cos(z-x) \\ \cos(x+y) & \cos(y+z) & \cos(z+x) \\ \sin(x+y) & \sin(y+z) & \sin(z+x) \end{vmatrix} =$$

- (A)  $2 \sin(x-y) \sin(y-z) \sin(z-x)$   
(B)  $-2 \sin(x-y) \sin(y-z) \sin(z-x)$   
(C)  $2 \cos(x-y) \cos(y-z) \cos(z-x)$   
(D)  $-2 \cos(x-y) \cos(y-z) \cos(z-x)$

18. The value of the determinant

$$\begin{vmatrix} 1+x & 2 & 3 & 4 \\ 1 & 2+x & 3 & 4 \\ 1 & 2 & 3+x & 4 \\ 1 & 2 & 3 & 4+x \end{vmatrix}$$
 is

- (A)  $x^2(x+10)$  (B)  $x^3(x+10)$   
(C)  $x^4(x+10)$  (D) None of these

19. If A, B, C are angles of a triangle ABC, then

$$\begin{vmatrix} \sin \frac{A}{2} & \sin \frac{B}{2} & \sin \frac{C}{2} \\ \sin(A+B+C) & \sin \frac{B}{2} & \cos \frac{A}{2} \\ \cos \frac{(A+B+C)}{2} & \tan(A+B+C) & \sin \frac{C}{2} \end{vmatrix}$$
 is

less than or equal to

- (A)  $\frac{3\sqrt{3}}{8}$  (B)  $\frac{1}{8}$   
(C)  $2\sqrt{2}$  (D) 2

20. If a, b, c are the sides opposite to angles A, B, C respectively in a triangle

$$\Delta = \begin{vmatrix} 1 & \frac{4 \sin B}{b} & \cos A \\ 2a & 8 \sin A & 1 \\ 3a & 12 \sin A & \cos B \end{vmatrix}$$

then what is the value  $\Delta$  in the determinant?

- (A)  $\frac{1}{2} \cos 2A$  (B) 0  
(C)  $\frac{1}{2} \sin 2A$  (D)  $\frac{1}{2} (\cos^2 A + \cos^2 B)$

21. The determinant

$$\Delta = \begin{vmatrix} a^2(1+x) & ab & ac \\ ab & b^2(1+x) & bc \\ ac & bc & c^2(1+x) \end{vmatrix}$$

is divisible by

- (A)  $1+x$  (B)  $(1+x)^2$   
(C)  $x^2$  (D) none of these

22. If  $\alpha, \beta$  and  $\gamma$  are the roots of the equation  $x^3 + ax + b = 0$ , then the value of the determinant

$$\begin{vmatrix} \alpha - \beta - \gamma & 2\alpha & 2\alpha \\ 2\beta & \beta - \gamma - \alpha & 2\beta \\ 2\gamma & 2\gamma & \gamma - \alpha - \beta \end{vmatrix}$$
 is

- (A) 0 (B)  $\alpha$   
(C)  $-\beta$  (D) None of these

23. If every element of a third order determinant of value  $\Delta$  is multiplied by 5, then the value of the new determinant is

- (A)  $\Delta$  (B)  $5\Delta$   
(C)  $25\Delta$  (D)  $125\Delta$

$$24. \Delta = \begin{vmatrix} x & 3 & 2 \\ 3 & 2 & x \\ 2 & x & 3 \end{vmatrix} = \begin{vmatrix} 4 & x & 1 \\ x & 1 & 4 \\ 1 & 4 & x \end{vmatrix} = \begin{vmatrix} 100 & -95 & x \\ -95 & x & 100 \\ x & 100 & 95 \end{vmatrix},$$

$$= \begin{vmatrix} 1002 & -997 & x \\ x & 1002 & -997 \\ -997 & x & 1002 \end{vmatrix} = 0$$

then

- (A)  $x = 5$  (B)  $x = -5$   
(C)  $x = 0$  (D) None of these

25. If  $l_r^2 + m_r^2 + n_r^2 = 1$  where  $r = 1, 2, 3$  and  $l_1 l_2 + m_1 m_2 + n_1 n_2 = 0$  ...etc.

$$\text{then } \Delta = \begin{vmatrix} l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \\ l_3 & m_3 & n_3 \end{vmatrix} \text{ is equal to}$$

- (A) 0 (B) 2  
(C) 1 (D) 3

26. If  $f(x) = ax^2 + bx + c$ ,  $a, b, c \in \mathbb{R}$  and equation  $f(x) - x = 0$  has imaginary roots  $\alpha, \beta$  and  $\gamma, \delta$  be

$$\text{the roots of } f(f(x)) - x = 0 \text{ then } \begin{vmatrix} 2 & \alpha & \delta \\ \beta & 0 & \alpha \\ \gamma & \beta & 1 \end{vmatrix} \text{ is}$$

- (A) 0 (B) real  
(C) purely imaginary (D) None of these

$$27. \text{ If } \begin{vmatrix} \alpha & x & x & x \\ x & \beta & x & x \\ x & x & \gamma & x \\ x & x & x & \delta \end{vmatrix} = f(x) - xf'(x) \text{ then } f(x) \text{ is}$$

equal to

- (A)  $(x - \alpha)(x - \beta)(x - \gamma)(x - \delta)$   
(B)  $(x + \alpha)(x + \beta)(x + \gamma)(x + \delta)$

(C)  $2(x - \alpha)(x - \beta)(x - \gamma)(x - \delta)$

(D) None of these.

28. For positive numbers  $x, y, z$  the numerical value of the determinant

$$\begin{vmatrix} 1 & \log_x y & \log_x z \\ \log_y x & 3 & \log_y z \\ \log_z x & \log_z y & 5 \end{vmatrix}$$
 is

(A) 0

(B)  $\log x \log y \log z$

(C) 1

(D) 8

29. The equation  $\begin{vmatrix} (1+x)^2 & (1-x)^2 & -(2+x^2) \\ 2x+1 & 3x & 1-5x \\ x+1 & 2x & 2-3x \end{vmatrix} = 0$

$$\begin{vmatrix} (1+x)^2 & 2x+1 & x+1 \\ (1-x)^2 & 3x & 2x \\ 1-2x & 3x+2 & 2x-3 \end{vmatrix} = 0$$

(A) has no real solution

(B) has 4 real solutions

(C) has two real and two non-real solutions

(D) has infinite number of solutions, real or non-real

30. If  $f'(x)$

$$= \begin{vmatrix} mx & mx-p & mx+p \\ n & n+p & n-p \\ mx+2n & mx+2n+p & mx+2n-p \end{vmatrix}$$

then  $y = f(x)$  represents :

(A) a straight line parallel to x-axis

(B) a straight line parallel to y-axis

(C) parabola

(D) a straight line with negative slope

31. The value of the determinant

$$\begin{vmatrix} (a_1 - b_1)^2 & (a_1 - b_2)^2 & (a_1 - b_3)^2 & (a_1 - b_4)^2 \\ (a_2 - b_1)^2 & (a_2 - b_2)^2 & (a_2 - b_3)^2 & (a_2 - b_4)^2 \\ (a_3 - b_1)^2 & (a_3 - b_2)^2 & (a_3 - b_3)^2 & (a_3 - b_4)^2 \\ (a_4 - b_1)^3 & (a_4 - b_2)^2 & (a_4 - b_3)^2 & (a_4 - b_4)^2 \end{vmatrix}$$

is

(A) dependent on  $a_i, i = 1, 2, 3, 4$

(B) dependent on  $b_i, i = 1, 2, 3, 4$

(C) 1

(D) 0

32. If  $bc + qr = ca + rp = ab + pq = -1$ , then

$$\begin{vmatrix} ap & bq & cr \\ a & b & c \\ p & q & r \end{vmatrix}$$
 is equal to

(A)  $bc + ca + ab$

(B) 0

(C) 1

(D) 2

33. If  $a_1, a_2, a_3, \dots, a_{12}$  are in A.P. and

$$\Delta_1 = \begin{vmatrix} a_1 a_5 & a_1 & a_2 \\ a_2 a_6 & a_2 & a_3 \\ a_3 a_7 & a_3 & a_4 \end{vmatrix}, \Delta_2 = \begin{vmatrix} a_2 a_{10} & a_2 & a_3 \\ a_3 a_{11} & a_3 & a_4 \\ a_4 a_{12} & a_4 & a_5 \end{vmatrix}$$

then  $\Delta_1 : \Delta_2 =$

(A) 1 : 2

(B) 2 : 1

(C) 1 : 1

(D) None of these

34. If  $(1 + ax + bx^2)^4 = a_0 + a_1 x + a_2 x^2 + \dots + a_8 x^8$ , where  $a, b, a_0, a_1, \dots, a_8 \in \mathbb{R}$  such that

$$a_0 + a_1 + a_2 \neq 0 \text{ and } \begin{vmatrix} a_0 & a_1 & a_2 \\ a_1 & a_2 & a_0 \\ a_2 & a_0 & a_1 \end{vmatrix} = 0, \text{ then}$$

(A)  $a = 3/4, b = 5/8$

(B)  $a = 1/4, b = 5/2$

(C)  $a = 1, b = 2/3$

(D) None of these

35. The value of the determinant

$$\begin{vmatrix} 1+\alpha & 1+\alpha z & 1+\alpha z^2 \\ 1+\beta & 1+\beta z & 1+\beta z^2 \\ 1+\gamma & 1+\gamma z & 1+\gamma z^2 \end{vmatrix}$$
 is

(A)  $(\alpha - \beta)(\beta - \gamma)(\gamma - \alpha)$

(B) 0

(C)  $\alpha\beta\gamma$

(D) None of these

36. Let  $D_k = \begin{vmatrix} \alpha & \beta & \gamma \\ 2 \cdot 3^k & 16 \cdot 9^k & 26 \cdot 27^k \\ (3^{10} - 1) & 2(9^{10} - 1) & (27^{10} - 1) \end{vmatrix}$

then the value of  $\sum_{k=1}^{10} D_k$  is

- (A)  $2(\alpha + \beta + \gamma)$  (B)  $\alpha\beta + \alpha\gamma + \beta\gamma$   
 (C)  $\alpha\beta\gamma$  (D) 0

37. If  $\sum_{k=1}^n \alpha_k = pn^2 + qn + r$  and  $\alpha_1, \alpha_2, \alpha_3 \in \{1, 2, 3, \dots, 9\}$  and  $25\alpha_1, 37\alpha_2, 49\alpha_3$  be three-digit

numbers, then  $\begin{vmatrix} \alpha_1 & \alpha_2 & \alpha_3 \\ 5 & 7 & 9 \\ 25\alpha_1 & 37\alpha_2 & 49\alpha_3 \end{vmatrix}$  is equal to

- (A)  $\alpha_1 + \alpha_2 + \alpha_3$  (B)  $\alpha_1 - \alpha_2 + \alpha_3$   
 (C) 7 (D) 0

38. If  $\theta \in (0, \pi/2)$ , then value of

$\begin{vmatrix} (\sin \theta + \operatorname{cosec} \theta)^2 & (\sin \theta - \operatorname{cosec} \theta)^2 & 1 \\ (\cos \theta + \sec \theta)^2 & (\cos \theta - \sec \theta)^2 & 1 \\ (\tan \theta + \cot \theta)^2 & (\tan \theta - \cot \theta)^2 & 1 \end{vmatrix}$  is

- (A)  $\sin \theta + \cos \theta + \tan \theta$  (B) 1  
 (C) 0 (D) 4

39. If  $y = \sin \theta + \sqrt{3} \cos \theta$  and

$\begin{vmatrix} 1+y & 1-y & 1-y \\ 1-y & 1+y & 1-y \\ 1-y & 1-y & 1+y \end{vmatrix} = 0$ , then the number of

solutions in  $[0, 2\pi]$  is

- (A) one (B) two  
 (C) three (D) none

40. If  $[ ]$  denotes the greatest integer less than or equal to the real number under consideration, and  $-1 \leq x < 0$ ;  $0 \leq y < 1$ ;  $1 \leq z < 2$ , then the value of the determinant

$\begin{vmatrix} [x]+1 & [y] & [z] \\ [x] & [y]+1 & [z] \\ [x] & [y] & [z]+1 \end{vmatrix}$  is

- (A)  $[z]$  (B)  $[y]$   
 (C)  $[x]$  (D) None of these

41. If  $x, y, z$  are distinct and  $\begin{vmatrix} x^k & x^{k+2} & x^{k+3} \\ y^k & y^{k+2} & y^{k+3} \\ z^k & z^{k+2} & z^{k+3} \end{vmatrix}$

$= (x-y)(y-z)(z-x) \left( \frac{1}{x} + \frac{1}{y} + \frac{1}{z} \right)$ , then

- (A)  $k = -3$  (B)  $k = -1$   
 (C)  $k = 1$  (D)  $k = 3$

42. Let  $\Delta = \begin{vmatrix} \frac{1-a_1^3 b_1^3}{1-a_1 b_1} & \frac{1-a_1^3 b_2^3}{1-a_1 b_2} & \frac{1-a_1^3 b_3^3}{1-a_1 b_3} \\ \frac{1-a_2^3 b_1^3}{1-a_2 b_1} & \frac{1-a_2^3 b_2^3}{1-a_2 b_2} & \frac{1-a_2^3 b_3^3}{1-a_2 b_3} \\ \frac{1-a_3^3 b_1^3}{1-a_3 b_1} & \frac{1-a_3^3 b_2^3}{1-a_3 b_2} & \frac{1-a_3^3 b_3^3}{1-a_3 b_3} \end{vmatrix}$

then

- (A)  $\Delta = 0$   
 (B)  $\Delta$  depends only on  $a_1, a_2, a_3$   
 (C)  $\Delta$  depends only on  $b_1, b_2, b_3$   
 (D)  $\Delta$  depends on  $a_1, a_2, a_3, b_1, b_2, b_3$

43. The value of

$\Delta = \begin{vmatrix} \sin(2\alpha) & \sin(\alpha + \beta) & \sin(\alpha + \gamma) \\ \sin(\beta + \alpha) & \sin(2\beta) & \sin(\gamma + \beta) \\ \sin(\gamma + \alpha) & \sin(\gamma + \beta) & \sin(2\gamma) \end{vmatrix}$  is

- (A)  $\Delta = 0$   
 (B)  $\Delta = \sin^2 \alpha + \sin^2 \beta + \sin^2 \gamma$   
 (C)  $\Delta = 3/2$   
 (D) None of these

44. Let

$\Delta = \begin{vmatrix} 1+x_1 y_1 & 1+x_1 y_2 & 1+x_1 y_3 \\ 1+x_2 y_1 & 1+x_2 y_2 & 1+x_2 y_3 \\ 1+x_3 y_1 & 1+x_3 y_2 & 1+x_3 y_3 \end{vmatrix}$

then the value of  $\Delta$  is

- (A)  $x_1 x_2 x_3 + y_1 y_2 y_3$   
 (B)  $x_1 x_2 x_3 y_1 y_2 y_3$   
 (C)  $x_2 x_3 y_2 y_3 + x_3 x_1 y_3 y_1 + x_1 x_2 y_1 y_2$   
 (D) None of these

45. If  $D(x) = \begin{vmatrix} x-1 & (x-1)^2 & x^3 \\ x-1 & x^2 & (x+1)^3 \\ x & (x+1)^2 & (x+1)^3 \end{vmatrix}$  then the

coefficient of  $x$  in  $D(x)$  is –

- (A) 5 (B) –2  
 (C) 6 (D) 0

46.  $\begin{vmatrix} 1+a^2+a^4 & 1+ab+a^2b^2 & 1+ac+a^2c^2 \\ 1+ab+a^2b^2 & 1+b^2+b^4 & 1+bc+b^2c^2 \\ 1+ac+a^2c^2 & 1+bc+b^2c^2 & 1+c^2+c^4 \end{vmatrix} =$

- (A)  $(a+b+c)(ab+bc+ca)(a-b)(b-c)(c-a)$   
 (B)  $(a-b)^2(b-c)^2(c-a)^2$   
 (C)  $(a^2-b^2)(b^2-c^2)(c^2-a^2)$   
 (D)  $3abc(a^3+b^3+c^3)$

47. The determinant

$$\begin{vmatrix} 1+a+x & a+y & a+z \\ b+x & 1+b+y & b+z \\ c+x & c+y & 1+c+z \end{vmatrix} =$$

- (A)  $(1+a+b+c)(1+x+y+z) - 3(ax+by+cz)$   
 (B)  $a(x+y) + b(y+z) + c(z+x) - (xy+yz+zx)$   
 (C)  $x(a+b) + y(b+c) + z(c+a) - (ab+bc+ca)$   
 (D) None of these

48. Let  $N = \begin{vmatrix} 28 & 25 & 38 \\ 42 & 38 & 65 \\ 56 & 47 & 83 \end{vmatrix}$ , then the number of ways

is which  $N$  can be resolved as a product of two divisors which are relatively prime is

- (A) 4 (B) 8  
 (C) 9 (D) 16

49. In a third order determinant, each element of the first column consists of sum of two terms, each element of the second column consists of

sum of three terms and each element of the third column consists of sum of four terms. Then it can be decomposed into  $n$  determinants, where  $n$  has the value

- (A) 1 (B) 9  
 (C) 16 (D) 24

50. The value of the determinant of  $n^{\text{th}}$  order, being

given by  $\begin{vmatrix} x & 1 & 1 & \dots \\ 1 & x & 1 & \dots \\ 1 & 1 & x & \dots \\ \dots & \dots & \dots & \dots \end{vmatrix}$ , is

- (A)  $(x-1)^{n-1}(x+n-1)$   
 (B)  $(x-1)^n(x+n-1)$   
 (C)  $(1-x)^{n-1}(x+n-1)$   
 (D) None of these

### Multiple Correct Answer Type for JEE Advanced

51. The determinant

$$\begin{vmatrix} \cos(x-y) & \cos(y-z) & \cos(z-x) \\ \cos(x+y) & \cos(y+z) & \cos(z+x) \\ \sin(x+y) & \sin(y+z) & \sin(z+x) \end{vmatrix} =$$

- (A)  $2 \sin(x-y) \sin(y-z) \sin(z-x)$   
 (B)  $-2 \sin(x-y) \sin(y-z) \sin(z-x)$   
 (C)  $2 \cos(x-y) \cos(y-z) \cos(z-x)$   
 (D)  $-2 \cos(x-y) \cos(y-z) \cos(z-x)$

52. If  $p, q, r, s$  are in A.P. and

$$f(x) = \begin{vmatrix} p + \sin x & q + \sin x & p - r + \sin x \\ q + \sin x & r + \sin x & -1 + \sin x \\ r + \sin x & s + \sin x & s - q + \sin x \end{vmatrix}$$

such that

$$\int_0^2 f(x) dx = -4 \text{ then the common difference of the A.P. can be :}$$

- (A) –1 (B)  $\frac{1}{2}$   
 (C) 1 (D) None

53. The determinant  $\begin{vmatrix} a^2 & a^2 - (b-c)^2 & bc \\ b^2 & b^2 - (c-a)^2 & ca \\ c^2 & c^2 - (a-b)^2 & ab \end{vmatrix}$  is

divisible by:

- (A)  $a+b+c$   
 (B)  $(a+b)(b+c)(c+a)$   
 (C)  $a^2+b^2+c^2$   
 (D)  $(a-b)(b-c)(c-a)$

54. Let  $a, b > 0$  and  $\Delta = \begin{vmatrix} -x & a & b \\ b & -x & a \\ a & b & -x \end{vmatrix}$ , then

- (A)  $a+b-x$  is a factor of  $\Delta$   
 (B)  $x^2 + (a+b)x + a^2 + b^2 - ab$  is a factor of  $\Delta$   
 (C)  $\Delta = 0$  has two real roots if  $a = b$   
 (D) none of these

55. All the elements of a determinant  $D$  are integers, and determinant  $\Delta$  is formed by the co-factors of elements of corresponding determinant  $\Delta$  then

- (A)  $\Delta = D^2$   
 (B)  $D = 0$  implies  $\Delta = 0$   
 (C) if  $D = 27$ , then  $\Delta$  is a perfect cube  
 (D) None of these

56. Let  $f(x) = \begin{vmatrix} 2\sin x & \sin^2 x & 0 \\ 1 & 2\sin x & \sin^2 x \\ 0 & 1 & 2\sin x \end{vmatrix}$ , then

- (A)  $f(x)$  is independent of  $x$   
 (B)  $f'(\pi/2) = 0$   
 (C)  $\int_{-\pi/2}^{\pi/2} f(x) dx = 0$   
 (D) Tangent to the curve  $y = f(x)$  at  $x = 0$  is  $y = 0$

57. The determinant

$$\Delta = \begin{vmatrix} x & -\sin \theta & -\cos \theta \sin \phi \\ \sin \theta & x & -\cos \theta \cos \phi \\ \cos \theta \sin \phi & \cos \theta \cos \phi & x \end{vmatrix}$$

is independent of

- (A)  $x$  (B)  $\theta$   
 (C)  $\phi$  (D) None of these

58. Let  $\Delta = \begin{vmatrix} a & a^2 & 0 \\ 1 & 2a+b & (a+b)^2 \\ 0 & 1 & 2a+3b \end{vmatrix}$ , then

- (A)  $a+b$  is a factor of  $\Delta$   
 (B)  $a+2b$  is a factor of  $\Delta$   
 (C)  $2a+3b$  is a factor of  $\Delta$   
 (D)  $a^2$  is a factor of  $\Delta$

59. Let

$$\Delta = \begin{vmatrix} \sin(x+\alpha) & \cos(x+\alpha) & a+x \sin \alpha \\ \sin(x+\beta) & \cos(x+\beta) & b+x \sin \beta \\ \sin(x+\gamma) & \cos(x+\gamma) & c+x \sin \gamma \end{vmatrix}$$

Then

- (A)  $\Delta$  is independent of  $x$   
 (B)  $y = \Delta$  represents a straight line parallel to  $x$ -axis  
 (C)  $f(x) = \Delta$  has a local maximum at  $x = (\alpha + \beta + \gamma)/3$   
 (D)  $f(x) = \Delta$  has a local minimum at  $x = (\alpha + \beta + \gamma)/3$

60. Suppose  $x \in \mathbf{R}$ ,  $a_i, b_i, c_i \in \mathbf{R}$  for  $i = 1, 2, 3$ . The

determinant  $\begin{vmatrix} a_1 + b_1 x & a_1 x + b_1 & c_1 \\ a_2 + b_2 x & a_2 x + b_2 & c_2 \\ a_3 + b_3 x & a_3 x + b_3 & c_3 \end{vmatrix}$  is

equal to zero if

- (A)  $x = 1$  (B)  $x = -1$   
 (C)  $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$  (D) None of these



61. The value of the determinant

$$\Delta = \begin{vmatrix} \sin \alpha & \cos \alpha & 1 \\ \sin \beta & \cos \beta & 1 \\ \sin \gamma & \cos \gamma & 1 \end{vmatrix} \text{ equals}$$

- (A)  $-4 \sin \frac{\beta - \gamma}{2} \sin \frac{\gamma - \alpha}{2} \sin \frac{\alpha - \beta}{2}$   
 (B)  $\sin \alpha + \sin \beta + \sin \gamma$   
 (C)  $\sin (\beta - \gamma) + \sin (\gamma - \alpha) + \sin (\alpha - \beta)$   
 (D) none of these

62. If  $a^2 + b^2 + c^2 = 1$ , then

$$\begin{vmatrix} a^2 + (b^2 + c^2) \cos \phi & ab(1 - \cos \phi) & ba(1 - \cos \phi) & b^2 + (c^2 + a^2) \cos \phi & ca(1 - \cos \phi) & cb(1 - \cos \phi) \\ & ac(1 - \cos \phi) & bc(1 - \cos \phi) & & & \\ & & c^2 + (a^2 + b^2) \cos \phi & & & \end{vmatrix}$$

is independent of

- (A) a (B) b  
 (C) c (D) None of these

63. The determinant

$$\Delta = \begin{vmatrix} \cos \alpha & \sin \alpha \cos \beta & \sin \alpha \sin \beta \\ -\sin \alpha & \cos \alpha \sin \beta & \cos \alpha \cos \beta \\ 0 & -\sin \beta & \cos \beta \end{vmatrix} \text{ is}$$

- (A) independent of  $\alpha$   
 (B) independent of  $\beta$   
 (C)  $(\sin \alpha)x + (\cos \alpha)y = \Delta$  is a tangent line to the unit circle  $x^2 + y^2 = 1$   
 (D) none of these

64. If  $0 \leq x \leq 1$  and  $f(x) = \begin{vmatrix} x & 1 & 1 \\ -1 & x & 1 \\ -1 & -1 & x \end{vmatrix}$ , then

- (A) least value of  $f(x)$  is 0  
 (B) greatest value of  $f(x)$  is 4

(C)  $f(x)$  has a local maximum at  $x = 2/3$

(D)  $f(x)$  has a local minimum at  $x = 1/3$

65. Let  $f(x) = \begin{vmatrix} 2 \cos x & 1 & 0 \\ 1 & 2 \cos x & 1 \\ 0 & 1 & 2 \cos x \end{vmatrix}$ , then

(A)  $\int_0^\pi f(x) dx = 0$

(B) maximum value of  $f(x)$  is 4

(C)  $\lim_{x \rightarrow \pi/2} f(x) = 0$

(D)  $f'(0) = 0$

66. If A, B, and C are the angles of a triangle

then  $\Delta = \begin{vmatrix} -1 & \cos C & \cos B \\ \cos C & -1 & \cos A \\ \cos B & \cos A & -2 \end{vmatrix}$  is independent of

- (A) A (B) B  
 (C) C (D) none of these

67. Suppose x, y, z are positive and none of x, y, z is 1. If

$$\Delta = \begin{vmatrix} 1 & \log_x y & \log_x z \\ \log_y x & 1 & \log_y z \\ \sin(x+y) & -\cos(x+y) & \sin^2 z \end{vmatrix}$$

then  $\Delta$  is independent of

- (A) x (B) y  
 (C) z (D) None of these

68. Let  $\phi_1(x) = x + a_1$ ,  $\phi_2(x) = x^2 + b_1x + b_2$  and

$$\Delta = \begin{vmatrix} 1 & 1 & 1 \\ \phi_1(x_1) & \phi_1(x_2) & \phi_1(x_3) \\ \phi_2(x_1) & \phi_2(x_2) & \phi_2(x_3) \end{vmatrix}, \text{ then}$$

- (A)  $\Delta$  is independent of  $a_1$   
 (B)  $\Delta$  is independent of  $b_1$  and  $b_2$   
 (C)  $\Delta$  is independent of  $x_1, x_2$  and  $x_3$   
 (D) None of these

69. If  $\Delta = \begin{vmatrix} x & 2y-z & -z \\ y & 2x-z & -z \\ y & 2y-z & 2x-2y-z \end{vmatrix}$ , then

- (A)  $x - y$  is a factor of  $\Delta$   
 (B)  $(x - y)^2$  is a factor of  $\Delta$   
 (C)  $(x - y)^3$  is a factor of  $\Delta$   
 (D)  $\Delta$  is independent of  $z$

70. Which of the following determinant vanishes ?

(A)  $\begin{vmatrix} 1 & bc & bc(b+c) \\ 1 & ca & ca(c+a) \\ 1 & ab & ab(a+b) \end{vmatrix}$

(B)  $\begin{vmatrix} 1 & ab & \frac{1}{a} + \frac{1}{b} \\ 1 & bc & \frac{1}{b} + \frac{1}{c} \\ 1 & ca & \frac{1}{c} + \frac{1}{a} \end{vmatrix}$

(C)  $\begin{vmatrix} 0 & a-b & a-c \\ b-a & 0 & b-c \\ c-a & c-b & 0 \end{vmatrix}$

(D)  $\begin{vmatrix} \log_x xyz & \log_x y & \log_x z \\ \log_y xyz & 1 & \log_y z \\ \log_z xyz & \log_z y & 1 \end{vmatrix}$

71. Let  $\Delta = \begin{vmatrix} x_1 y_1 & x_1 y_2 & x_1 y_3 \\ x_1 y_2 & x_2 y_2 & x_2 y_3 \\ x_1 y_3 & x_2 y_3 & x_3 y_3 \end{vmatrix}$ , then

- (A)  $\Delta$  depends on  $x_1, x_2, x_3, y_1, y_2, y_3$   
 (B)  $x_1 y_3 - x_3 y_1$  is a factor of  $\Delta$   
 (C)  $\Delta$  can be written as product of four factors  
 (D) None of these

72. Let  $x_1, x_2, x_3 \neq 0$  and  $\Delta = \begin{vmatrix} a_1 + x_1 & a_2 & a_3 \\ -x_1 & x_2 & 0 \\ 0 & -x_2 & x_3 \end{vmatrix}$ ,

then

- (A)  $1 + a_1/x_1 + a_2/x_2 + a_3/x_3$  is a factor of  $\Delta$   
 (B)  $\Delta = 0 \Rightarrow (-x_1, -x_2, -x_3)$  lie on the surface  $a_1/x + a_2/y + a_3/z = 1$   
 (C)  $\Delta$  does not depend on  $x_1, x_2$  and  $x_3$   
 (D) None of these

73. Suppose  $a_1, a_2, a_3$  are in A.P. and  $b_1, b_2, b_3$  are in H.P. and let

$\Delta = \begin{vmatrix} a_1 - b_1 & a_1 - b_2 & a_1 - b_3 \\ a_2 - b_1 & a_2 - b_2 & a_2 - b_3 \\ a_3 - b_1 & a_3 - b_2 & a_3 - b_3 \end{vmatrix}$ , then

- (A)  $\Delta$  is independent of  $a_1, a_2, a_3, b_1, b_2, b_3$   
 (B)  $a_1 - \Delta, a_2 - 2\Delta, a_3 - 3\Delta$  are in A.P.  
 (C)  $b_1 + \Delta, b_2 + \Delta^2, b_3 + \Delta$  are in H.P.  
 (D) None of these

74. If  $\Delta_r = \begin{vmatrix} 2r & x & n(n+1) \\ 6r^2-1 & y & n^2(n+3) \\ 4r^3-2nr & z & n^3(n+1) \end{vmatrix}$ , then the value

of  $\sum_{r=1}^n \Delta_r$  is independent of -

- (A)  $x$  (B)  $y$   
 (C)  $z$  (D)  $n$

75. Let  $\Delta(x) = \begin{vmatrix} x+a & x+b & x+a-c \\ x+b & x+c & x-1 \\ x+c & x+d & x-b+d \end{vmatrix}$  and

$\int_0^2 \Delta(x) dx = -16$ , where  $a, b, c, d$  are in A.P.,

then the common difference of the AP is -

- (A) 1 (B) 2  
 (C) -2 (D) None of these

76. If both  $n$  and  $r$  be greater than 1 and if

$\Delta = \begin{vmatrix} {}^x C_r & {}^{n-1} C_r & {}^{n-1} C_{r-1} \\ {}^{x+1} C_r & {}^n C_r & {}^n C_{r-1} \\ {}^{x+2} C_r & {}^{n+1} C_r & {}^{n+1} C_{r-1} \end{vmatrix} = 0$ , the value of

$x$  is equal to

- (A)  $n$  (B)  $n+1$   
 (C)  $n-1$  (D) None of these

$$77. \text{ If } D = \begin{vmatrix} \frac{1}{z} & \frac{1}{z} & -\frac{(x+y)}{z^2} \\ -\frac{(y+z)}{x^2} & \frac{1}{x} & \frac{1}{x} \\ -\frac{y(y+z)}{x^2z} & \frac{x+2y+z}{xz} & -\frac{y(x+y)}{xz^2} \end{vmatrix}$$

then :

- (A) D is independent of x  
 (B) D is independent of y  
 (C) D is independent of z  
 (D) D is dependent on x, y, z

$$78. \text{ Let } \Delta = \begin{vmatrix} b+c & -b & 2c \\ -a & c+a & -c \\ 2a & 2b & a+b \end{vmatrix}, \text{ then}$$

- (A) c is a factor of  $\Delta$   
 (B)  $b-a$  is a factor of  $\Delta$   
 (C)  $a+b+c$  is a factor of  $\Delta$   
 (D)  $a^2+b^2$  is a factor of  $\Delta$

$$79. \text{ Let } f(x) = \begin{vmatrix} 1/x & \log x & x^n \\ 1 & -1/n & (-1)^n \\ 1 & a & a^2 \end{vmatrix}, \text{ then}$$

- (A)  $f^n(1)$  is independent of a  
 (B)  $f^n(1)$  is independent of n  
 (C)  $f^n(1)$  depends on a and n  
 (D)  $y = a(x - f^n(1))$  represents a straight line through the origin

80. Suppose  $f_1(x, y) = a_1x + a_2y$  and  $f_2(x, y) = b_1x^2 + b_2xy + b_3y^2$  and

$$\Delta = \begin{vmatrix} f_2(x_1, y_1) & y_1 f_1(x_1, y_1) & y_1^2 \\ f_2(x_2, y_2) & y_2 f_1(x_2, y_2) & y_2^2 \\ f_2(x_3, y_3) & y_3 f_1(x_3, y_3) & y_3^2 \end{vmatrix}$$

- (A)  $\Delta$  is independent of  $a_2, b_2$  and  $b_3$   
 (B)  $\Delta$  depends on  $x_1, x_2, x_3$  but not on  $y_1, y_2, y_3$   
 (C)  $\Delta$  depends on  $y_1, y_2, y_3$  but not on  $x_1, x_2, x_3$   
 (D) for  $1 \leq r, s \leq 3, r \neq s, x_r y_s - x_s y_r$  is a factor of  $\Delta$

### Comprehension-1

$$\text{If } f(x) = \begin{vmatrix} 5 + \sin^2 x & \cos^2 x & 4 \sin 2x \\ \sin^2 x & 5 + \cos^2 x & 4 \sin 2x \\ \sin^2 x & \cos^2 x & 5 + 4 \sin 2x \end{vmatrix}, \text{ then}$$

81. Number of values of x lying in the interval  $[-\pi, \pi]$  satisfying the equation  $f(x) = 250$  are  
 (A) 2 (B) 3  
 (C) 4 (D) 6
82. Maximum and minimum values of  $f(x)$  are respectively  
 (A) 1, -1 (B) 250, -50  
 (C) 0, -250 (D) 250, 50
83. Set of values of x satisfying  $f(x) < 25$  is  
 (A)  $(-\infty, \infty)$  (B)  $[1, 1]$   
 (C)  $(-\infty, -1] \cup [1, \infty)$  (D)  $\phi$

### Comprehension-2

At times it is more convenient to prove a determinant identity involving symmetric and homogeneous functions by factor theorem. A dimensional check and homogeneity becomes very helpful in solving problems. Recall that an identity is true for all values of the variables.

Let  $D_1, D_2, D_3$  be determinants given by

$$D_1 = \begin{vmatrix} 1 & a & a^3 \\ 1 & b & b^3 \\ 1 & c & c^3 \end{vmatrix}, D_2 = \begin{vmatrix} 1 & a & a^3 \\ 1 & b & b^3 \\ 1 & c & c^3 \end{vmatrix}$$

$$D_3 = \begin{vmatrix} 1 & a^2 & a^3 \\ 1 & b^2 & b^3 \\ 1 & c^2 & c^3 \end{vmatrix}$$

then it can be noted that  $D_1, D_2, D_3$  all will vanish for

$$a = b, b = c, c = a$$

$\Rightarrow (a-b)(b-c)(c-a)$  are factors of  $D_1, D_2, D_3$ .

84. The degree of  $D_1$  should be  
 (A) 3 (B) 4  
 (C) 5 (D) 6
85. The remaining factor of  $D_3$  should be  
 (A)  $(a^2 + b^2 + c^2)$  (B)  $(ab + bc + ac)$   
 (C)  $a^2 + b^2 - c^2$  (D)  $(a-c)(b-c)$
86. The remaining factor of  $D_2$  must be  
 (A) a (B) b  
 (C) c (D)  $a+b+c$

**Comprehension-3**

If we observe the expansion of a determinant  $D =$

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}, \text{ then we conclude that } D \text{ is a sum of 6}$$

terms which are of the type  $a_i a_j c_k$  ( $i \neq j \neq k$ ). We will observe that each permutation of 1, 2, 3, will occur exactly once. Half the terms will be positive, the other half will be negative.

$$D = a_1 b_2 c_3 + a_2 b_3 c_1 + a_3 b_1 c_2 - (a_1 b_3 c_2 + a_3 b_2 c_1 + a_2 b_1 c_3).$$

Note that each term is a product of three entries of the determinant. Each entry appears once in positive zone and once in negative zone. The above theory can be generalized to determinants of order  $n \times n$  ( $n$

$> 3$ ). More over, for any square matrix  $A$  of order  $n$ ,  $\det(-A) = (-1)^n \det(A)$

**87.** Let  $A$  be a  $n \times n$  matrix whose characteristic polynomial is  $f(\lambda)$ . If  $\text{tr } A$  denotes sum of diagonal elements of  $A$ , then coefficient of  $\lambda^{n-1}$  in  $f(\lambda)$  must be

- (A)  $\text{tr } A$  (B)  $-\text{tr } A$   
(C)  $(-1)^{n-1} \text{tr } A$  (D)  $2 \text{tr } A$

**88.** For  $n^{\text{th}}$  order determinant number of terms in the expansion

- (A)  $4n - 6$  (B)  $6(3n - 8)$   
(C)  ${}^n P_n$  (D) None of these

**89.** Given an  $n \times n$  matrix  $A$  with real entries such that  $A^2 = -I$ , then  $\det A$  must be equal to

- (A)  $-1$  or  $1$ ,  $n$  is even (B)  $1$ ,  $n$  is odd  
(C)  $0$  (D) None of these

**Match the Columns for JEE Advanced**

**90.** If  $a$  and  $b$  are two real numbers, then match the values of the determinants :

**Column - I**

(A)  $\begin{vmatrix} a & a+b & a+2b \\ -a & a & 0 \\ 0 & -a & a \end{vmatrix}$

(B)  $\begin{vmatrix} a & -(a+b) & -(a+2b) \\ a & -a & 0 \\ 0 & a & -a \end{vmatrix}$

(C)  $\begin{vmatrix} a & a^2 & 0 \\ 1 & 2a+b & (a+b)^2 \\ 0 & 1 & 2a+3b \end{vmatrix}$

(D)  $\begin{vmatrix} 1+a & 1 & 1 \\ 1 & 1+b & 1 \\ 1 & 1 & 3/2 \end{vmatrix}$

**Column - II**

(P)  $\frac{3}{2} ab + \frac{1}{2} a + \frac{1}{2} b$

(Q)  $a(a+b)(a+2b)$

(R)  $-a^2(a+3b)$

(S)  $3a^2(a+b)$

(T)  $(a^2 + b^2)b$

91. Let  $p$  be a positive integer,

**Column - I**

$$(A) \Delta_1 = \begin{vmatrix} a_0 & a_1 & a_2 \\ -1 & x & 0 \\ 0 & -1 & x \end{vmatrix}$$

$$(B) \Delta_2 = \begin{vmatrix} a_0 x^2 & a_1 x & a_2 \\ a_0 x & 1 & 0 \\ a_0 x^2 & a_1 x & 1 \end{vmatrix}$$

$$(C) \Delta_3 = \begin{vmatrix} a^p - x & a^{p+1} - x & a^{p+2} - x \\ a^{p+3} - x & a^{p+4} - x & a^{p+5} - x \\ a^{p+6} - x & a^{p+7} - x & a^{p+8} - x \end{vmatrix}$$

$$(D) \Delta_4 = \begin{vmatrix} 1-x & a & a^2 \\ a & a^2-x & a^3 \\ a^2 & a^3 & a^4-x \end{vmatrix}$$

**Column - II**

(P) is independent of  $a$  and  $x$

(Q)  $a_0 x^2 + a_1 x + a_2$

(R)  $a_0 (1 - a_1) (1 - a_2) x^2$

(S)  $a^{3p+12} - x^3 + 3a^{px^3}$

(T)  $x^2 (1 + a^2 + a^4) - x^3$

92. Let  $p$  be a positive integer,

**Column - I**

$$(A) \Delta_1 = \begin{vmatrix} {}^{p+2}C_2 & {}^{p+3}C_2 & {}^{p+4}C_2 \\ 1 & {}^{p+4}C_2 & {}^{p+5}C_2 \\ 1 & {}^{p+5}C_2 & {}^{p+6}C_2 \end{vmatrix}$$

$$(B) \Delta_2 = \begin{vmatrix} 1 & {}^pC_1 & {}^pC_2 \\ 1 & {}^{p+1}C_1 & {}^{p+1}C_2 \\ 1 & {}^{p+2}C_1 & {}^{p+2}C_2 \end{vmatrix}$$

$$(C) \Delta_3 = \begin{vmatrix} 2 & x_1 + x_2 & 1 \\ x_1 + x_2 & x_1^2 + x_2^2 & x \\ x_1^2 + x_2^2 & x_1^3 + x_2^3 & x^2 \end{vmatrix}$$

$$(D) \Delta_4 = \begin{vmatrix} 1+x_1 & 1+x_1x & 1+x_1x^2 \\ 1+x_2 & 1+x_2x & 1+x_2x^2 \\ 1+x_3 & 1+x_3x & 1+x_3x^2 \end{vmatrix}$$

**Column - II**

(P) 0

(Q)  $(x_2 - x_1)^2 (x - x_1) (x - x_2)$

(R) -1

(S) 1

(T)  $p - 1$

93. Let  $p$  be a positive integer,

**Column - I**

$$(A) \Delta_1 = \begin{vmatrix} x & y & x+y \\ y & x+y & x \\ x+y & x & y \end{vmatrix}$$

$$(B) \Delta_2 = \begin{vmatrix} 1 & x & y \\ 1 & x+y & y \\ 1 & x & x+y \end{vmatrix}$$

$$(C) \Delta_1 = \begin{vmatrix} \sin 2a & \sin(a+b) & \sin(a+c) \\ \sin(b+a) & \sin 2b & \sin(b+c) \\ \sin(c+a) & \sin(c+b) & \sin 2c \end{vmatrix}$$

$$(D) \Delta_2 = \begin{vmatrix} \cos(a-b) & \cos(b-c) & \cos(c-a) \\ \cos(a+b) & \cos(b+c) & \cos(c+a) \\ \sin(a+b) & \sin(b+c) & \sin(c+a) \end{vmatrix}$$

**Column - II**

(P)  $xy$

(Q)  $0$

(R)  $2 \sin(a-b) \sin(a-c) \sin(b-c)$

(S)  $-2(x^3 + y^3)$

(T)  $\cos(a+b+c)$

94.

**Column - I**

$$(A) [1 \times 1] \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 3 & 2 & 5 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = 0 \text{ then } x =$$

$$(B) \text{ If in the determinant } \begin{vmatrix} x & 3 & 3 \\ 3 & 3 & x \\ 2 & 3 & 3 \end{vmatrix}, C_{11} = C_{22}, \text{ then } x =$$

$$(C) \text{ If the determinant } \begin{vmatrix} \sin \theta & 1 & 0 \\ 1 & \cos \phi & -\cos \theta \\ \sin \phi & 0 & 1 \end{vmatrix} \text{ is}$$

a symmetric determinant then absolute value of minimum value of determinant is

$$(D) \text{ If } \begin{vmatrix} (b+c)^2 & a^2 & a^2 \\ b^2 & (c+a)^2 & b^2 \\ c^2 & c^2 & (a+b)^2 \end{vmatrix} = k abc (a+b+c)^3$$

then the value of  $k$  is

**Column - II**

(P)  $2$

(Q)  $-2$

(R)  $\frac{5}{2}$

(S)  $-\frac{9}{8}$

# Review Exercises for JEE Advanced

1. Show that

$$\Delta = \begin{vmatrix} (b+c)^2 & a^2 & bc \\ (c+a)^2 & b^2 & ca \\ (a+b)^2 & c^2 & ab \end{vmatrix}$$

$$= (a^2 + b^2 + c^2)(a+b+c)(a-b)(b-c)(c-a).$$

2. Show that

$$\Delta = \begin{vmatrix} a & b-c & c+b \\ a+c & b & c-a \\ a-b & b+a & c \end{vmatrix}$$

$$= (a+b+c)(a^2+b^2+c^2)$$

3. If  $s = a_1 + a_2 + a_3$ , then prove that

$$\begin{vmatrix} s-a_1 & s-a_2 & s-a_3 \\ s-a_3 & s-a_1 & s-a_2 \\ s-a_2 & s-a_3 & s-a_1 \end{vmatrix} = 2 \begin{vmatrix} a_1 & a_2 & a_3 \\ a_3 & a_1 & a_2 \\ a_2 & a_3 & a_1 \end{vmatrix}.$$

4. Prove, without expanding that

$$\begin{vmatrix} ah+bg & g & ab+ch \\ bf+ba & f & hb+bc \\ af+bc & c & bg+fc \end{vmatrix} = a \begin{vmatrix} ah+bg & a & h \\ bf+ba & h & b \\ af+bc & g & f \end{vmatrix}$$

5. If  $x, y$  and  $z$  are unequal and

$$\begin{vmatrix} x^3 & (x+a)^3 & (x-a)^3 \\ y^3 & (y+a)^3 & (y-a)^3 \\ z^3 & (z+a)^3 & (z-a)^3 \end{vmatrix} = 0$$

prove that  $a^2(x+y+z) = 3xyz$ .

6. Show that

$$\begin{vmatrix} a^2+x^2 & ab-cx & ac+bx \\ ab+cx & b^2+x^2 & bc+ax \\ ac-bx & bc-ax & c^2+x^2 \end{vmatrix} = \begin{vmatrix} x & c & -b \\ -c & x & a \\ b & -a & x \end{vmatrix}$$

7. Show that

$$\begin{vmatrix} yz-x^2 & zx-y^2 & xy-z^2 \\ zx-y^2 & xy-z^2 & yz-x^2 \\ xy-z^2 & yz-x^2 & zx-y^2 \end{vmatrix} = \begin{vmatrix} r^2 & u^2 & u^2 \\ u^2 & r^2 & u^2 \\ u^2 & u^2 & r^2 \end{vmatrix}$$

where  $r^2 = x^2 + y^2 + z^2$  and  $z^2 = yz + zx + xy$ .

8. If  $\alpha, \beta$  be the roots of the equation  $ax^2 + bx + c = 0$  and  $s_n = \alpha^n + \beta^n : n \geq 1$  evaluate

$$\begin{vmatrix} 3 & 1+s_1 & 1+s_2 \\ 1+s_1 & 1+s_2 & 1+s_3 \\ 1+s_2 & 1+s_3 & 1+s_4 \end{vmatrix}$$

9. If  $x_r \neq 0 ; r = 1, 2, 3$  then prove that

$$\begin{vmatrix} x_1 + a_1b_1 & a_1b_2 & a_1b_3 \\ a_2b_1 & x_2 + a_2b_2 & a_2b_3 \\ a_3b_1 & a_3b_2 & x_3 + a_3b_3 \end{vmatrix} = x_1x_2x_3 \left( 1 + \frac{a_1b_1}{x_1} + \frac{a_2b_2}{x_2} + \frac{a_3b_3}{x_3} \right)$$

10. Show that

$$\Delta = \begin{vmatrix} (b+c)^2 & a^2 & a^2 \\ b^2 & (c+a)^2 & b^2 \\ c^2 & c^2 & (a+b)^2 \end{vmatrix} = 2abc(a+b+c)^3$$

11. Find the value of  $f(\pi/6)$ , where

$$f(\theta) = \begin{vmatrix} \cos^2 \theta & \cos \theta \sin \theta & -\sin \theta \\ \cos \theta \sin \theta & \sin^2 \theta & \cos \theta \\ \sin \theta & -\cos \theta & 0 \end{vmatrix}$$

12. Show that

$$\Delta = \begin{vmatrix} a_1 & x & x \\ x & a_2 & x \\ x & x & a_3 \end{vmatrix} = xf'(x) - f(x)$$

where  $f(x) = (x-a_1)(x-a_2)(x-a_3)$

13. Show that

$$\begin{vmatrix} bc-a^2 & ca-b^2 & ab-c^2 \\ ca-b^2 & ab-c^2 & bc-a^2 \\ ab-c^2 & bc-a^2 & ca-b^2 \end{vmatrix}$$

$$= \begin{vmatrix} a^2 & c^2 & 2ac - b^2 \\ 2ab - c^2 & b^2 & a^2 \\ b^2 & 2bc - a^2 & c^2 \end{vmatrix}$$

14. Evaluate the determinant

$$\Delta = \begin{vmatrix} \sqrt{p} + \sqrt{q} & 2\sqrt{r} & \sqrt{r} \\ \sqrt{qr} + \sqrt{2p} & r & \sqrt{2r} \\ q + \sqrt{pr} & \sqrt{qr} & r \end{vmatrix}$$

where p, q and r are positive real numbers.

15. If  $a \neq p$ ,  $b \neq q$ ,  $c \neq r$  and

$$\Delta = \begin{vmatrix} p & b & c \\ a & q & c \\ a & b & r \end{vmatrix} = 0,$$

then find the value of  $\frac{p}{p-a} + \frac{q}{q-b} + \frac{r}{r-c}$

16. If a and x are real numbers and n is a positive integer prove that

$$\begin{vmatrix} a^n - x & a^{n+1} - x & a^{n+2} - x \\ a^{n+3} - x & a^{n+4} - x & a^{n+5} - x \\ a^{n+6} - x & a^{n+7} - x & a^{n+8} - x \end{vmatrix} = 0$$

17. Show that

$$\Delta = \begin{vmatrix} \cos(\alpha - \beta) & \cos(\beta - \gamma) & \cos(\gamma - \alpha) \\ \cos(\alpha + \beta) & \cos(\beta + \gamma) & \cos(\gamma + \alpha) \\ \sin(\alpha + \beta) & \sin(\beta + \gamma) & \sin(\gamma + \alpha) \end{vmatrix} \\ = -2 \sin(\beta - \gamma) \sin(\gamma - \alpha) \sin(\alpha - \beta)$$

18. Show that

$$\Delta = \begin{vmatrix} \sin 3\alpha & \sin 2\alpha & \sin \alpha \\ \sin 3\beta & \sin 2\beta & \sin \beta \\ \sin 3\gamma & \sin 2\gamma & \sin \gamma \end{vmatrix} \\ = 2^6 \sin \alpha \sin \beta \sin \gamma \sin \frac{\alpha + \beta}{2} \sin \frac{\beta + \gamma}{2} \\ \times \sin \frac{\gamma + \alpha}{2} \sin \frac{\alpha - \beta}{2} \sin \frac{\beta - \gamma}{2} \sin \frac{\gamma - \alpha}{2}$$

19. By differentiation or otherwise, show that the determinant

$$\begin{vmatrix} \sin(x + \alpha) & \cos(x + \alpha) & a + x \sin \alpha \\ \sin(x + \beta) & \cos(x + \beta) & b + x \sin \beta \\ \sin(x + \gamma) & \cos(x + \gamma) & c + x \sin \gamma \end{vmatrix}$$

is independent of x.

$$20. \text{ If } \Delta(x) = \begin{vmatrix} a_1 + x & b_1 + x & c_1 + x \\ a_2 + x & b_2 + x & c_2 + x \\ a_3 + x & b_3 + x & c_3 + x \end{vmatrix}$$

show that  $\Delta''(x) = 0$  and that  $\Delta(x) = \Delta(0) + Sx$ , where S denotes the sum of all the cofactors of the elements in  $\Delta(0)$ .

21. Show that the maximum value of a third order determinant whose elements are 1 or -1 is 4.

22. Without expanding, show that for all  $\alpha$ ,

$$\beta, \gamma, \delta; \begin{vmatrix} \sin^2 \alpha & \cos 2\alpha & \cos^2 \alpha \\ \sin^2 \beta & \cos 2\beta & \cos^2 \beta \\ \sin^2 \gamma & \cos 2\gamma & \cos^2 \gamma \end{vmatrix}$$

$$= \begin{vmatrix} \sin \alpha & \cos \alpha & \sin(\alpha + \delta) \\ \sin \beta & \cos \beta & \sin(\beta + \delta) \\ \sin \gamma & \cos \gamma & \sin(\gamma + \delta) \end{vmatrix}$$

23. Show that

$$\begin{vmatrix} a_1 \ell_1 + b_1 m_1 & a_1 \ell_2 + b_1 m_2 & a_1 \ell_3 + b_1 m_3 \\ a_2 \ell_1 + b_2 m_1 & a_2 \ell_2 + b_2 m_2 & a_2 \ell_3 + b_2 m_3 \\ a_3 \ell_1 + b_3 m_1 & a_3 \ell_3 + b_3 m_2 & a_3 \ell_3 + b_3 m_3 \end{vmatrix} \\ = 0.$$

24. If n is a positive integer, then find the value of

$$\begin{vmatrix} {}^{n+2}C_n & {}^{n+3}C_{n+1} & {}^{n+4}C_{n+2} \\ {}^{n+3}C_{n+1} & {}^{n+4}C_{n+2} & {}^{n+5}C_{n+3} \\ {}^{n+4}C_{n+2} & {}^{n+5}C_{n+3} & {}^{n+6}C_{n+4} \end{vmatrix}.$$

$$25. \text{ Prove that } \begin{vmatrix} \beta\gamma & \beta\gamma' + \beta'\gamma & \beta'\gamma' \\ \gamma\alpha & \gamma\alpha' + \gamma'\alpha & \gamma'\alpha' \\ \alpha\beta & \alpha\beta' + \alpha'\beta & \alpha'\beta' \end{vmatrix} \\ = (\alpha\beta' - \alpha'\beta)(\beta\gamma' - \beta'\gamma)(\gamma\alpha' - \gamma'\alpha)$$



26. Show that

$$\begin{vmatrix} 1 & \cos^2(\alpha - \beta) & \cos^2(\alpha - \gamma) \\ \cos^2(\beta - \alpha) & 1 & \cos^2(\beta - \gamma) \\ \cos^2(\gamma - \alpha) & \cos^2(\gamma - \beta) & 1 \end{vmatrix} = 2 \sin^2(\beta - \gamma) \sin^2(\gamma - \alpha) \sin^2(\alpha - \beta).$$

27. Show that

$$\begin{vmatrix} ax - by - cz & ay + bx & az + cx \\ bx + ay & by - cz - ax & bz + cy \\ cx + az & cy + bz & cz - ax - by \end{vmatrix} = (a^2 + b^2 + c^2)(ax + by + cz)(x^2 + y^2 + z^2)$$

28. Show that  $\begin{vmatrix} x & y & y \\ z & x & y \\ z & z & x \end{vmatrix} = \frac{z(x-y)^2 - y(x-z)^3}{z-y}$ .

29. If  $x$ ,  $y$  and  $z$  are unequal and

$$\begin{vmatrix} x^3 & (x+a)^3 & (x-a)^3 \\ y^3 & (y+a)^3 & (y-a)^3 \\ z^3 & (z+a)^3 & (z-a)^3 \end{vmatrix} = 0$$

prove that  $a^2(x + y + z) = 3xyz$ .

30. If  $a_1$ ,  $a_2$ , and  $a_3$  are distinct real roots of the equation  $px^3 + px^2 + qx + r = 0$  such that

$$\begin{vmatrix} 1+a_1 & 1 & 1 \\ 1 & 1+a_2 & 1 \\ 1 & 1 & 1+a_3 \end{vmatrix} = 0$$

then prove that  $\frac{r}{p} < 0$ .

## Target Exercises for JEE Advanced

1. Prove that if  $a$ ,  $b$ ,  $c$  are sides of  $\Delta ABC$  then  $\Delta^2 =$

$$\left| \frac{1}{16} \begin{vmatrix} -c^2 & c^2 & b^2 - a^2 \\ c^2 - b^2 & -a^2 & a^2 \\ 1 & 1 & 1 \end{vmatrix} \right|$$

where  $\Delta$  is area of the triangle.

2. Find the roots of the equation

$$\begin{vmatrix} 3x^2 & x^2 + x \cos \theta + \cos^2 \theta \\ x^2 + x \cos \theta + \cos^2 \theta & 3 \cos^2 \theta \\ x^2 + x \sin \theta + \sin^2 \theta & 1 + \frac{\sin 2\theta}{2} \\ x^2 + x \sin \theta + \sin^2 \theta & 1 + \frac{\sin 2\theta}{2} \\ & 3 \sin^2 \theta \end{vmatrix} = 0.$$

3. Show that the roots of the equation

$$\begin{vmatrix} x & m & n & 1 \\ a & x & n & 1 \\ a & b & x & 1 \\ a & b & c & 1 \end{vmatrix} = 0$$
 are independent of  $m$  and  $n$ .

4. Show that

$$\begin{vmatrix} -bc & b^2 + bc & c^2 + bc \\ a^2 + ac & -ac & c^2 + ac \\ a^2 + ab & b^2 + ab & -ab \end{vmatrix} = (bc + ca + ab)^3$$

5. If  $f(x) = \begin{vmatrix} x+c_1 & x+a & x+a \\ x+b & x+c_2 & x+a \\ x+b & x+b & x+c_3 \end{vmatrix}$  then show that  $f(x)$  is linear in  $x$ .

Hence deduce that  $f(0) = \frac{bg(a) - ag(b)}{(b-a)}$

where  $g(x) = (c_1 - x)(c_2 - x)(c_3 - x)$

6. Show that

$$\begin{vmatrix} -b^2c^2 & ab(c^2 + a^2) & ac(a^2 + b^2) \\ ba(b^2 + c^2) & -c^2a^2 & bc(a^2 + b^2) \\ ca(b^2 + c^2) & cb(c^2 + a^2) & -a^2b^2 \end{vmatrix} = (a^2b^2 + b^2c^2 + c^2a^2)^3.$$

7. If  $u = ax^2 + 2bxy + cy^2$ ,  $u' = a'x^2 + 2b'xy + c'y^2$  prove that

$$\begin{vmatrix} y^2 & -xy & x^2 \\ a & b & c \\ a' & b' & c' \end{vmatrix}$$

$$= \begin{vmatrix} ax+by & bx+cy \\ a'x+b'y & b'x+c'y \end{vmatrix}$$

$$= -\frac{1}{y} \begin{vmatrix} u & u' \\ ax+by & a'x+b'y \end{vmatrix}$$

8. Show that the value of the determinant

$$\begin{vmatrix} 3 & a+b+c & a^2+b^2+c^2 \\ a+b+c & a^2+b^2+c^2 & a^3+b^3+c^3 \\ a^2+b^2+c^2 & a^3+b^3+c^3 & a^4+b^4+c^4 \end{vmatrix}$$

is  $(a-b)^2(b-c)^2(c-a)^2$ .

9. Let  $a, b, c, x, y, z, \in \mathbf{R}$ . Prove that

$$\begin{vmatrix} (a-x)^2 & (b-x)^2 & (c-x)^2 \\ (a-y)^2 & (b-y)^2 & (c-y)^2 \\ (a-z)^2 & (b-z)^2 & (c-z)^2 \end{vmatrix}$$

$$= \begin{vmatrix} (1-ax)^2 & (1+bx)^2 & (1+cx)^2 \\ (1+ay)^2 & (1+by)^2 & (1+cy)^2 \\ (1+az)^2 & (1+bz)^2 & (1+cz)^2 \end{vmatrix}.$$

10. If  $f, g$  and  $h$  are differentiable functions of  $x$

$$\text{and } \Delta = \begin{vmatrix} f & g & h \\ (xf)' & (xg)' & (xh)' \\ (x^2f)'' & (x^2g)'' & (x^2h)'' \end{vmatrix}$$

$$\text{prove that } \Delta' = \begin{vmatrix} f & g & h \\ f' & g' & h' \\ (x^3f''')' & (x^3g''')' & (x^3h''')' \end{vmatrix}$$

11. Prove that

$$\begin{vmatrix} ax-by-cz & ay+bx & cx+az \\ ay+bx & by-cz-ax & bz+cy \\ cx+az & bz+cy & cz-ax-by \end{vmatrix}$$

$$= (x^2+y^2+z^2)(a^2+b^2+c^2)(ax+by+cz).$$

12. If  $x_r \neq 0$ ;  $r = 1, 2, 3$  then prove that

$$\begin{vmatrix} x_1+a_1b_1 & a_1b_2 & a_1b_3 \\ a_2b_1 & x_2+a_2b_2 & a_2b_3 \\ a_3b_1 & a_3b_2 & x_3+a_3b_3 \end{vmatrix}$$

$$= x_1x_2x_3 \left( 1 + \frac{a_1b_1}{x_1} + \frac{a_2b_2}{x_2} + \frac{a_3b_3}{x_3} \right).$$

$$13. \text{ Prove that } \begin{vmatrix} \sin \alpha & \sin \beta & \sin \gamma \\ \cos \alpha & \cos \beta & \cos \gamma \\ \cos 2\alpha & \sin 2\beta & \sin 2\gamma \end{vmatrix}$$

$$= 4 \sin \frac{1}{2}(\beta - \gamma) \cdot 4 \sin \frac{1}{2}(\gamma - \alpha) \cdot 4 \sin \frac{1}{2}(\alpha - \beta).$$

14. Prove that the value of

$$\begin{vmatrix} \cos(\theta + \phi) & -\sin(\theta + \phi) & \cos 2\phi \\ \sin \theta & \cos \theta & \sin \phi \\ -\cos \theta & \sin \theta & \lambda \cos \phi \end{vmatrix}$$

is independent of  $\theta$  for all real  $\lambda$  and independent of  $\theta$  and  $\phi$  when  $\lambda = -1$ .

15. If  $f(n)$  and  $h(n)$  are second degree polynomials in  $n$  having  $n+2$  as a common factor,

prove that  $\sum_{r=1}^n \Delta_r$  is independent of  $n$  where

$$\Delta_r = \begin{vmatrix} 2r+1 & 6n(n+2) & f(n) \\ 2^{r-1} & 3 \cdot 2^{n+1} - 6 & g(n) \\ r(n-r+1) & n(n+1)(n+2) & h(n) \end{vmatrix}.$$

16. Show that the value of the determinant

$$\begin{vmatrix} \tan(A+P) & \tan(B+P) & \tan(C+P) \\ \tan(A+Q) & \tan(B+Q) & \tan(C+Q) \\ \tan(A+R) & \tan(B+R) & \tan(C+R) \end{vmatrix}$$

vanishes for all values of  $A, B, C, P, Q$  &  $R$  where

$$A + B + C + P + Q + R = 0.$$

17. Show that,

$$\begin{vmatrix} a^2 + b^2 + c^2 & bc + c + ab & bc + ca + ab \\ bc + ca + ab & a^2 + b^2 + c^2 & bc + ca + ab \\ bc + ca + ab & bc + ca + ab & a^2 + b^2 + c^2 \end{vmatrix}$$

is always positive, where  $a, b, c$  are distinct real numbers.

18. If  $a, b, c$  are three non zero non equal numbers in  $A, P$  and  $\Delta =$

$$\begin{vmatrix} (b-c)(c-a) & (a-b)(c-a) & (a-b)(b-c) \\ (a-b)(c-a) & (a-b)(b-c) & (b-c)(c-a) \\ (a-b)(b-c) & (b-c)(c-a) & (a-b)(c-a) \end{vmatrix},$$

Prove that  $\Delta > 0$ .

19. If  $A, B$  and  $C$  are 3 distinct angles in the interval and  $\left(0, \frac{\pi}{3}\right)$  then prove that

$$\Delta = \begin{vmatrix} \sin^3 A & \sin^3 B & \sin^3 C \\ \sin A & \sin B & \sin C \\ \cos A & \cos B & \cos C \end{vmatrix} \neq 0.$$

20. Let

$$\Delta = \begin{vmatrix} 1+a^2-b^2 & 2ab & -2b \\ 2ab & 1-a^2+b^2 & 2a \\ 2b & -2a & 1-a^2-b^2 \end{vmatrix}.$$

Show that the minimum value of  $\Delta$  is  $27a^2b^2$ .

21. By differentiation or otherwise, show that the determinant

$$\begin{vmatrix} \sin(x+\alpha) & \cos(x+\alpha) & a+x \sin \alpha \\ \sin(x+\beta) & \cos(x+\beta) & b+x \sin \beta \\ \sin(x+\gamma) & \cos(x+\beta) & c+x \sin \gamma \end{vmatrix}$$

is independent of  $x$ .

22. If  $\Delta(x) = \begin{vmatrix} a_1+x & b_1+x & c_1+x \\ a_2+x & b_2+x & c_2+x \\ a_3+x & b_3+x & c_3+x \end{vmatrix}$

show that  $\Delta''(x) = 0$  and that  $\Delta(x) = \Delta(0) + Sx$ , where  $S$  denotes the sum of all the cofactors of the elements in  $\Delta(0)$ .

23. Prove that the product of two determinants

$$\begin{vmatrix} a+ib & c+id \\ -c+id & a-ib \end{vmatrix} \cdot \begin{vmatrix} \alpha+i\beta & \gamma-i\delta \\ -\gamma-i\delta & \alpha+i\beta \end{vmatrix}$$

can be written in the form  $\begin{vmatrix} A-iB & C-iD \\ -C-iD & A+iB \end{vmatrix},$

where  $i = \sqrt{-1}$ .

Hence show that the product of two sums each of four squares can be expressed as a sum of four squares.

Express  $(9^2 + 2^2 + 3^2 + 4^2) \cdot (5^2 + 6^2 + 7^2 + 8^2)$  as a sum of squares of four integers.

24. Let  $\alpha$  be a repeated root of the quadratic equation  $f(x) = 0$  and  $A(x), B(x)$  and  $C(x)$  polynomials of degrees 3, 4 and 5 respectively. Show that

$$\begin{vmatrix} A(x) & B(x) & C(x) \\ A(\alpha) & B(\alpha) & C(\alpha) \\ A'(\alpha) & B'(\alpha) & C'(\alpha) \end{vmatrix}$$

is divisible by  $f(x)$ , where prime denotes the derivative.

25. Prove that

$$\begin{vmatrix} (a_1-b_1)^2 & (a_1-b_2)^2 & (a_1-b_3)^2 & (a_1-b_4)^2 \\ (a_2-b_1)^2 & (a_2-b_2)^2 & (a_2-b_3)^2 & (a_2-b_4)^2 \\ (a_3-b_1)^2 & (a_3-b_2)^2 & (a_3-b_3)^2 & (a_3-b_4)^2 \\ (a_4-b_1)^2 & (a_4-b_2)^2 & (a_4-b_3)^2 & (a_4-b_4)^2 \end{vmatrix} = 0$$

26. Let  $m$  and  $p$  be two positive integers such that  $m > p + 2$ . Suppose.

$$\Delta(m, p) = \begin{vmatrix} {}^m C_p & {}^m C_{p+1} & {}^m C_{p+2} \\ {}^{m+1} C_p & {}^{m+1} C_{p+1} & {}^{m+1} C_{p+2} \\ {}^{m+2} C_p & {}^{m+2} C_{p+1} & {}^{m+2} C_{p+2} \end{vmatrix}$$

Show that  $\Delta(m, p) = \frac{{}^{m+2} C_3}{{}^{p+2} C_3} \Delta(m-1, p-1)$

Hence or otherwise, prove that

$$\Delta(m, p) = \frac{{}^{m+2}C_3 ({}^{m+1}C_3) \dots ({}^{m-p+2}C_3)}{{}^{p+2}C_3 ({}^{p+1}C_3) \dots ({}^3C_3)}$$

27. If the points  $\left(\frac{a^3}{a-1}, \frac{a^2-3}{a-1}\right),$

$$\left(\frac{b^3}{b-1}, \frac{b^2-3}{b-1}\right) \text{ and } \left(\frac{c^3}{c-1}, \frac{c^2-3}{c-1}\right)$$

are collinear for three distinct values of  $a, b, c$  then show that  $abc - (bc + ca + ab) + 3(a + b + c) = 0$ .

28. Show that

$$\begin{vmatrix} (a-a_1)^{-2} & (a-a_1)^{-1} & a_1^{-1} \\ (a-a_2)^{-2} & (a-a_2)^{-1} & a_2^{-1} \\ (a-a_3)^{-2} & (a-a_3)^{-1} & a_3^{-1} \end{vmatrix}$$

$$= \pm \frac{a^2 \prod (a_i - a_j)}{\prod a_i \prod (a - a_i)^2}$$

write out the terms of the product in the numerator and give the resulting expression its correct sign.

29. If  $f(x)$  is a polynomial of degree  $< 3$ , Prove that

$$\begin{vmatrix} 1 & a & f(a)/(x-a) \\ 1 & b & f(b)/(x-b) \\ 1 & c & f(c)/(x-c) \end{vmatrix} \div \begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix}$$

$$= \frac{f(x)}{(x-a)(x-b)(x-c)}$$

30. If  $\omega$  is a root of  $x^4 = 1$ , show that  $a + b\omega + c\omega^2 + d\omega^3$  is a factor of

$$\begin{vmatrix} a & b & c & d \\ b & c & d & a \\ c & d & a & b \\ d & a & b & c \end{vmatrix}.$$

Hence show that the determinant is equal to  $-(a+b+c+d)(a-b+c-d)\{(a-c)^2 + (b-d)^2\}$

### Previous Years' Questions (JEE Advanced)

#### A. Fill in the blanks

1. Let  $p\lambda^4 + q\lambda^3 + r\lambda^2 + s\lambda + t$

$$= \begin{vmatrix} \lambda^2 + 3\lambda & \lambda - 1 & \lambda + 3 \\ \lambda + 1 & -2\lambda & \lambda - 4 \\ \lambda - 3 & \lambda + 4 & 3\lambda \end{vmatrix} \text{ an identity in } \lambda,$$

where  $p, q, r, s$  and  $t$  are constant. Then, the value of  $t$  is..... [IIT - 1981]

2. The solution set of the equation

$$\begin{vmatrix} 1 & 4 & 20 \\ 1 & -2 & 5 \\ 1 & 2x & 5x^2 \end{vmatrix} = 0 \text{ is.....} \quad \text{[IIT - 1981]}$$

3. Given that  $x = -9$  is a root of  $\begin{vmatrix} x & 3 & 7 \\ 2 & x & 2 \\ 7 & 6 & x \end{vmatrix} = 0$  the

other two roots are .....and..... [IIT - 1983]

4. The system of equations

$$\lambda x + y + z = 0$$

$$-x + \lambda y + z = 0$$

$$-x - y + \lambda z = 0$$

will have a non-zero solution if real values of  $\lambda$  are given by ..... [IIT - 1984]

5. The value of the determinant  $\begin{vmatrix} 1 & a & a^2 - bc \\ 1 & b & b^2 - ca \\ 1 & c & c^2 - ab \end{vmatrix}$  is..... [IIT - 1988]

6. For positive numbers  $x, y$  &  $z$ , the numerical value of the determinant

$$\begin{vmatrix} 1 & \log_x y & \log_x z \\ \log_y x & 1 & \log_y z \\ \log_z x & \log_z y & 1 \end{vmatrix} \text{ is } \_\_\_\_\_\_ \quad \text{[IIT - 1993]}$$

**B. True / False**

7. The determinants  $\begin{vmatrix} 1 & a & bc \\ 1 & b & ca \\ 1 & c & ab \end{vmatrix}$  and  $\begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix}$  are not identically equal. [IIT - 1983]

8. If  $\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = \begin{vmatrix} a_1 & b_1 & 1 \\ a_2 & b_2 & 1 \\ a_3 & b_3 & 1 \end{vmatrix}$  then the two triangles with vertices  $(x_1, y_1), (x_2, y_2), (x_3, y_3)$  and  $(a_1, b_1), (a_2, b_2), (a_3, b_3)$  must be congruent. [IIT - 1985]

**C. Multiple Choice Questions with ONE correct answer**

9. Consider the set A of all determinants of order 3 with entries 0 or 1 only. Let B be the subset of A consisting of all determinants with value 1. Let C be the subset of A consisting of all determinants with value -1. Then [IIT - 1981]

- (A) C is empty  
(B) B has as many elements as C  
(C)  $A = B \cup C$   
(D) B has twice as many elements as elements as C

10. If  $\omega (\neq 1)$  is a cube root of unity, then

$$\begin{vmatrix} 1 & 1+i\omega^2 & \omega^2 \\ 1-i & -1 & \omega^2-1 \\ -i & -i+\omega-1 & -1 \end{vmatrix} =$$
 [IIT - 1995]

- (A) 0 (B) 1  
(C) i (D)  $\omega$

11. Let a, b, c be the real numbers. Then following system of equations in x, y & z.

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1; \quad \frac{x^2}{a^2} - \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1;$$

$$-\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1 \text{ has} \quad \text{[IIT - 1995]}$$

- (A) no solution  
(B) unique solution  
(C) infinitely many solutions  
(D) finitely many solutions

12. The determinant  $\begin{vmatrix} xp+y & x & y \\ py+z & y & z \\ 0 & xp+y & yp+z \end{vmatrix} = 0$  if - [IIT - 1997]

- (A) x, y, z are in A.P.  
(B) x, y, z are in G.P.  
(C) x, y, z are in H.P.  
(D) xy, yz, zx are in A.P.

13. The parameter on which the value of the

$$\text{determinant} \begin{vmatrix} 1 & a & a^2 \\ \cos(p-d)x & \cos px & \cos(p+d)x \\ \sin(p-d)x & \sin px & \sin(p+d)x \end{vmatrix}$$

does not depend upon is - [IIT - 1997]

- (A) a (B) p  
(C) d (D) x

14. If  $f(x)$

$$= \begin{vmatrix} 1 & x & x+1 \\ 2x & x(x-1) & (x+1)x \\ 3x(x-1) & x(x-1)(x-2) & (x+1)x(x-1) \end{vmatrix} \text{ then}$$

- $f(100)$  is equal to - [IIT - 1999]  
(A) 0 (B) 1  
(C) 100 (D) -100

15. If the system of equations

$x - ky - z = 0, kx - y - z = 0, x + y - z = 0$  has a non-zero solution then the possible values of k are [IIT - 2000]

- (A) -1, 2 (B) 1, 2  
(C) 0, 1 (D) -1, 1

16. The number of distinct real roots of

$$\begin{vmatrix} \sin x & \cos x & \cos x \\ \cos x & \sin x & \cos x \\ \cos x & \cos & \sin x \end{vmatrix} = 0 \text{ in the interval}$$

- $-\frac{\pi}{4} \leq x \leq \frac{\pi}{4}$  is [IIT - 2001]  
(A) 0 (B) 2  
(C) 1 (D) 3

17. Let  $\omega = -\frac{1}{2} + i\frac{\sqrt{3}}{2}$ . Then the value of the

determinant  $\begin{vmatrix} 1 & 1 & 1 \\ 1 & -1-\omega^2 & \omega^2 \\ 1 & \omega^2 & \omega^4 \end{vmatrix}$  is [IIT - 2002]

- (A)  $3\omega$  (B)  $3\omega(\omega-1)$   
(C)  $3\omega^2$  (D)  $3\omega(1-\omega)$

18. The number of values of  $k$  for which the system of equations  $(k+1)x + 8y = 4k$ ,  $kx + (k+3)y = 3k-1$  has infinitely many solutions is

[IIT - 2002]

- (A) 0 (B) 1  
(C) 2 (D) infinite

**D. Multiple Choice Questions with ONE or MORE THAN ONE correct answers**

19. The determinant  $\begin{vmatrix} a & b & a\alpha+b \\ b & c & b\alpha+c \\ a\alpha+b & b\alpha+c & 0 \end{vmatrix}$  is

equal to zero, if [IIT - 1986]

- (A)  $a, b, c$  are in A.P.  
(B)  $a, b, c$  are in G.P.  
(C)  $a, b, c$  are in H.P.  
(D)  $\alpha$  is a root of the equation  $ax^2 + bx + c = 0$   
(E)  $(x-\alpha)$  is a factor of  $ax^2 + 2bx + c$ .

20. If  $\begin{vmatrix} 6i & -3i & 1 \\ 4 & 3i & -1 \\ 20 & 3 & i \end{vmatrix} = x + iy$ , then [IIT - 1998]

- (A)  $x = 3, y = 1$  (B)  $x = 1, y = 3$   
(C)  $x = 0, y = 3$  (D)  $x = 0, y = 0$

**E. Subjective Problems**

21. For what value of  $k$  do the following system of equations possess a non trivial (i.e., not all zero) solution over the set of rationals  $Q$ ?

$$\begin{aligned} x + ky + 3z &= 0 \\ 3x + ky - 2z &= 0 \\ 2x + 3y - 4z &= 0 \end{aligned}$$

For what value of  $k$ , find all the solutions for the system. [IIT - 1979]

22. Let  $a, b, c$  be positive and not all equal. Show

that the value of the determinant  $\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$  is negative. [IIT - 1981]

23. Without expanding a determinant at any stage,

show that  $\begin{vmatrix} x^2+x & x+1 & x-2 \\ 2x^2+3x-1 & 3x & 3x-3 \\ x^2+2x+3 & 2x-1 & 2x-1 \end{vmatrix} = xA$

+ B, where A and B are determinants of order 3 not involving  $x$ . [IIT - 1982]

24. Show that the system of equation

$$\begin{aligned} 3x - y + 4z &= 3 \\ x + 2y - 3z &= -2 \\ 6x + 5y + \lambda z &= -3 \end{aligned}$$

has at least one solution for any real number  $\lambda \neq -5$ . Find the set of solutions if  $\lambda = -5$ .

[IIT - 1983]

25. Show that  $\begin{vmatrix} {}^x C_r & {}^x C_{r+1} & {}^x C_{r+2} \\ {}^y C_r & {}^y C_{r+1} & {}^y C_{r+2} \\ {}^z C_r & {}^z C_{r+1} & {}^z C_{r+2} \end{vmatrix}$

$$= \begin{vmatrix} {}^x C_r & {}^{x+1} C_{r+1} & {}^{x+2} C_{r+2} \\ {}^y C_r & {}^{y+1} C_{r+1} & {}^{y+2} C_{r+2} \\ {}^z C_r & {}^{z+1} C_{r+1} & {}^{z+2} C_{r+2} \end{vmatrix} \quad [\text{IIT - 1985}]$$

26. Consider the system of linear equations in  $x, y, z$ :

$$\begin{aligned} (\sin 3\theta) x - y + z &= 0 \\ (\cos 2\theta) x + 4y + 3z &= 0 \\ 2x + 7y + 7z &= 0 \end{aligned}$$

Find the values of  $\theta$  for which this system has nontrivial solutions. [IIT - 1986]

27. Let  $\Delta_a = \begin{vmatrix} a-1 & n & 6 \\ (a-1)^2 & 2n^2 & 4n-2 \\ (a-1)^3 & 3n^3 & 3n^2-3n \end{vmatrix}$ . Show that

$$\sum_{a=1}^n \Delta_a = c, \text{ a constant.} \quad [\text{IIT - 1989}]$$

28. Let the three digit numbers  $A28$ ,  $3B9$ , and  $62C$ , where  $A$ ,  $B$ , and  $C$  are integers between 0 and 9, be divisible by a fixed integer  $k$ . Show that

$$\text{the determinant } \begin{vmatrix} A & 3 & 6 \\ 8 & 9 & C \\ 2 & B & 2 \end{vmatrix} \text{ is divisible by } k.$$

[IIT - 1990]

29. If  $a \neq p$ ,  $b \neq q$ ,  $c \neq r$  and  $\begin{vmatrix} p & b & c \\ a & q & c \\ a & b & r \end{vmatrix} = 0$ , then find

$$\text{the value of } \frac{p}{p-a} + \frac{q}{q-b} + \frac{r}{r-c} \quad [\text{IIT - 1991}]$$

30. For a fixed positive integer  $n$ , if

$$D = \begin{vmatrix} n! & (n+1)! & (n+2)! \\ (n+1)! & (n+2)! & (n+3)! \\ (n+2)! & (n+3)! & (n+4)! \end{vmatrix} \text{ then show that}$$

$$\left[ \frac{D}{(n!)^3} - 4 \right] \text{ is divisible by } n \quad [\text{IIT - 1992}]$$

31. Let  $\lambda$  and  $\alpha$  be real. Find the set of all values of  $\lambda$  for which the system of linear equations  
 $\lambda x + (\sin \alpha) y + (\cos \alpha) z = 0$   
 $x + (\cos \alpha) y + (\sin \alpha) z = 0$   
 $-x + (\sin \alpha) y - (\cos \alpha) z = 0$   
 has a non-trivial solution. For  $\lambda = 1$ , find all values of  $\alpha$ . [IIT - 1993]

32. For all values of  $A, B, C$  &  $P, Q, R$  show that

$$\begin{vmatrix} \cos(A-P) & \cos(A-Q) & \cos(A-R) \\ \cos(B-P) & \cos(B-Q) & \cos(B-R) \\ \cos(C-P) & \cos(C-Q) & \cos(C-R) \end{vmatrix} = 0.$$

[IIT - 1994]

33. Let  $a > 0$ ,  $d > 0$ . Find the value of the determinant

$$\begin{vmatrix} \frac{1}{a} & \frac{1}{a(a+d)} & \frac{1}{(a+d)(a+2d)} \\ \frac{1}{a+d} & \frac{1}{(a+d)(a+2d)} & \frac{1}{(a+2d)(a+3d)} \\ \frac{1}{a+2d} & \frac{1}{(a+2d)(a+3d)} & \frac{1}{(a+3d)(a+4d)} \end{vmatrix}$$

[IIT - 1996]

$$\begin{vmatrix} bc & ca & ab \\ p & q & r \\ 1 & 1 & 1 \end{vmatrix}$$

34. Find the value of the determinant where  $a, b$  and  $c$  are respectively the  $p$ th,  $q$ th and  $r$ th terms of harmonic progression.

[IIT - 1997]

35. Let  $a, b, c, d$  be real numbers in G.P. If  $u, v, w$  satisfy the system of equations, [IIT - 1998]

$$u + 2v + 3w = 6$$

$$4u + 5v + 6w = 12$$

$$6u + 9v = 4$$

then show that the roots of the equation

$$\left( \frac{1}{u} + \frac{1}{v} + \frac{1}{w} \right) x^2 + [(b-c)^2 + (c-a)^2 +$$

$$(d-b)^2] x + u + v + w = 0 \text{ and } 20x^2 + 10(a-d)^2 x - 9 = 0 \text{ are reciprocals of each other.}$$

[IIT - 1999]

36. Prove that for all values of  $\theta$ ;

$$\begin{vmatrix} \sin \theta & \cos \theta & \sin 2\theta \\ \sin(\theta + 2\pi/3) & \cos(\theta + 2\pi/3) & \sin(2\theta + 4\pi/3) \\ \sin(\theta - 2\pi/3) & \cos(\theta - 2\pi/3) & \sin(2\theta - 4\pi/3) \end{vmatrix} = 0$$

[IIT - 2000]

37. Let  $a, b, c$  be real no with  $a^2 + b^2 + c^2 = 1$  then show that

$$\begin{vmatrix} ax - by - c & bx + ay & cx + a \\ bx + ay & -ax + by - c & cy + b \\ cx + a & cy + b & -ax - by + c \end{vmatrix}$$

represent straight line.

[IIT - 2001]

## ANSWERS

### Practice Problems—A

1. 0
2. -2, -1
3. 0, 2
4. (a) 0 (b) 1
5. 0, -2
6. (a) + (b) - (c) + (d) +
7. (i) 0 (ii) 0 (iii) 0 (iv) 0
8.  $1 + x^2 + x^4 + x^6$

**Practice Problems—B**

1. 0                      2. 0  
4. 0                      5. 0  
8. 0

**Practice Problems—C**

3. (i)  $(x + y + z)^3$                       (ii)  $(5x + 4)(4 - x)^2$   
      (iii) 0                      (iv)  $5(3\sqrt{2} - 5\sqrt{3})$   
4. (i) 1, 2, -3                      (ii) 4  
      (iii) 0, -1, -12                      (iv) 4  
5. -43

6.  $\frac{xyz}{12}(x - y)(y - z)(z - x)$

14. 0

**Practice Problems—E**

1. 
$$\begin{vmatrix} b^2 + c^2 & ab & ac \\ ba & c^2 + a^2 & bc \\ ca & cb & a^2 + b^2 \end{vmatrix}$$

2. 
$$\begin{vmatrix} x^2 & -2x & 1 \\ y^2 & -2y & 1 \\ z^2 & -2z & 1 \end{vmatrix} \begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix}$$

$= 2(a - b)(b - c)(c - a)(x - y)(y - z)(z - x)$

7.  $(\alpha - \beta)^2(\beta - \gamma)^2(\gamma - \alpha)^2$

**Practice Problems—F**

2. 
$$\begin{vmatrix} 1 & 0 & 2x \\ x+2 & 2x+3 & x \\ x^2 & x^3+1 & 2x^4+1 \end{vmatrix} + \begin{vmatrix} x & 1 & x^2 \\ 1 & 2 & 1 \\ x^2 & x^3+1 & 2x^4+1 \end{vmatrix}$$

$$+ \begin{vmatrix} x & 1 & x^2 \\ x+2 & 2x+3 & x \\ 2x & 3x^2 & 8x^3 \end{vmatrix}$$

4.  $6x^2$   
5. (i) -1    (ii) -5    (iii) -4  
6. 0

**Objective Exercises**

- |                |                |             |
|----------------|----------------|-------------|
| 1. D           | 2. A           | 3. B        |
| 4. B           | 5. A           | 6. C        |
| 7. A           | 8. C           | 9. B        |
| 10. A          | 11. A          | 12. D       |
| 13. A          | 14. D          | 15. B       |
| 16. A          | 17. B          | 18. B       |
| 19. B          | 20. B          | 21. C       |
| 22. A          | 23. D          | 24. B       |
| 25. C          | 26. B          | 27. A       |
| 28. D          | 29. D          | 30. A       |
| 31. D          | 32. B          | 33. C       |
| 34. B          | 35. B          | 36. D       |
| 37. D          | 38. C          | 39. B       |
| 40. A          | 41. B          | 42. D       |
| 43. A          | 44. D          | 45. A       |
| 46. B          | 47. A          | 48. B       |
| 49. D          | 50. A          | 51. A, B    |
| 52. A, B       | 53. A, C       | 54. A, B, C |
| 55. A, B, C    | 56. B, C, D    | 57. B, C    |
| 58. A, B       | 59. A, B       | 60. A, B, C |
| 61. A, C       | 62. A, B, C    | 63. A, B, C |
| 64. A, B       | 65. A, B, C    | 66. D       |
| 67. A, B, C    | 68. A, B       | 69. A, B    |
| 70. A, B, C, D | 71. A, C       | 72. A, B    |
| 73. A, B, C    | 74. A, B, C, D | 75. B, C    |
| 76. A, C       | 77. A, B, C    | 78. A, B, C |
| 79. A, B, D    | 80. A, D       | 81. A       |
| 82. D          | 83. A          | 84. A       |
| 85. B          | 86. D          | 87. C       |
| 88. C          | 89. A          |             |

90. (A) - (S), (B) - (R), (C) - (Q), (D) - (P)

91. (A) - (Q), (B) - (R), (C) - (P), (D) - (T)

92. (A) - (R), (B) - (S), (C) - (Q), (D) - (P)

93. (A) - (S), (B) - (P), (C) - (Q), (D) - (R)

94. (A) - (S), (B) - (R), (C) - (P), (D) - (P)

**Review Exercises for JEE Advanced**

8.  $\frac{(b^2 - 4ac)}{a^4}(a + b + c)^2$

24. -1



### Target Exercises for JEE Advanced

2.  $x = \cos\theta, \sin\theta$

### Previous Years' Questions (JEE Advanced)

- |                  |                |         |
|------------------|----------------|---------|
| 1. $t = 0$       | 2. $x = -1, 2$ | 3. 2, 7 |
| 4. $\lambda = 0$ | 5. 0           | 6. 0    |
| 7. F             | 8. F           | 9. B    |
| 10. A            | 11. B          | 12. B   |
| 13. B            | 14. A          | 15. D   |
| 16. C            | 17. B          | 18. B   |
| 19. BE           | 20. D          |         |

21.  $x = b, y = \frac{-2b}{15}, z = \frac{2b}{5}, b \in \mathbb{R}$

24.  $x = \frac{4-5k}{7}, \frac{13k-9}{7}, z = k$  (for  $\lambda = -5$ )

26.  $n\pi$  or  $n\pi + (-1)^n \pi/6, n \in \mathbb{Z}$

29. 2

31.  $\lambda = \sqrt{2} \sin(2\alpha + \pi/4); \alpha = \pi/8, 7\pi/8, 9\pi/8$

33. 
$$\frac{4d^4}{a(a+d)^2(a+2d)^3(a+3d)^2(a+4d)}$$

34. 0

# 3

## Matrices

### 3.1 INTRODUCTION

The development of concept of matrices is the result of an attempt to obtain compact and simple methods of solving system of linear equations. Matrices are not only used in solving system of linear equations, but they are used in electronic spreadsheet programs for personal computer, which in turn is used in different areas of business and science like budgeting, sales projection, cost estimation, analysing the results of an experiment etc. Also, many physical operations such as magnification, rotation and reflection through a plane can be represented mathematically by matrices.

Consider the system of equations

$$3x + 4y - 3z = 5$$

$$2x + 9y + 7z = 4$$

$$4x - 2y + z = 2$$

$$6x + 8y - 3z = 1.$$

Here  $x$ ,  $y$  and  $z$  are unknowns and their coefficients are all numbers. Arranging the coefficients in the order in which they occur in the equations and enclosing them in square brackets, we obtain a rectangular array of the form

$$\begin{bmatrix} 3 & 4 & -3 \\ 2 & 9 & 7 \\ 4 & -2 & 1 \\ 6 & 8 & -3 \end{bmatrix}.$$

This rectangular array is an example of a matrix. The horizontal lines ( $\rightarrow$ ) are called rows or row vectors and the vertical lines ( $\downarrow$ ) are called columns or column vectors of the matrix. There are 4 rows and 3 columns in this matrix. Therefore it is a matrix of the type of  $4 \times 3$ . The numbers 3, 4, -3, 2 etc., constituting this matrix are called its elements.

Let us have another example. Suppose we wish to express the information that Ram has 15 books. We may express it as [15] with the understanding that the number inside [ ] is the number of books that Ram has. Now, if we have to express that Ram has 15 books and 6 pens. We may express it as [15 6] with the understanding that first number inside [ ] is the number of books while the other one is the number of pens possessed by Ram. Let us now suppose that we wish to express the information of possession of books and pens by Ram and her two friends Prem and Ketan which is as follows :

### 3.2 Comprehensive Algebra Vol-2

Ram has 15 books and 6 pens,  
 Prem has 10 books and 2 pens,  
 Ketan has 13 books and 5 pens  
 Now this could be arranged in the tabular form as follows :

|       | Books | Pens |
|-------|-------|------|
| Ram   | 15    | 6    |
| Prem  | 10    | 2    |
| Ketan | 13    | 5    |

and this can be expressed as

$$\begin{array}{cc} \begin{bmatrix} 15 & 6 \\ 10 & 2 \\ 13 & 5 \end{bmatrix} & \begin{array}{l} \leftarrow \text{First row} \\ \leftarrow \text{Second row} \\ \leftarrow \text{Third row} \end{array} \\ \begin{array}{cc} \uparrow & \uparrow \\ \text{First} & \text{Second} \\ \text{Column} & \text{Column} \end{array} & \end{array}$$

or

|       | Ram | Prem | Ketan |
|-------|-----|------|-------|
| Books | 15  | 10   | 13    |
| Pens  | 6   | 2    | 5     |

which can be expressed as :

$$\begin{array}{ccc} \begin{bmatrix} 15 & 10 & 13 \\ 6 & 2 & 5 \end{bmatrix} & \begin{array}{l} \leftarrow \text{First row} \\ \leftarrow \text{Second row} \end{array} \\ \begin{array}{ccc} \uparrow & \uparrow & \uparrow \\ \text{First} & \text{Second} & \text{Third} \\ \text{Column} & \text{Column} & \text{Column} \end{array} & \end{array}$$

In the first arrangement the entries in the first column represent the number of books possessed by Ram, Prem and Ketan, respectively and the entries in the second column represent the number of pens possessed by Ram, Prem and Ketan, respectively. Similarly, in the second arrangement, the entries in the first row represent the number of books possessed by Ram, Prem and Ketan, respectively. The entries in the second row represent the number of pens possessed by Ram, Prem and Ketan, respectively. An arrangement of the above kind is called a matrix. We shall now define a matrix.

### 3.2 MATRIX

A set of  $mn$  numbers (real or complex) arranged in the form of a rectangular array having  $m$  rows and  $n$

columns is called an  $m \times n$  matrix [to be read as 'm by n' matrix].

An  $m \times n$  matrix is usually written as

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1j} & \dots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2j} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ a_{i1} & a_{i2} & a_{i3} & \dots & a_{ij} & \dots & a_{in} \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & a_{m3} & \dots & a_{mj} & \dots & a_{mn} \end{bmatrix}$$

In a compact form, the above matrix is represented by

$A = [a_{ij}]$ ,  $i = 1, 2, \dots, m$ ,  $j = 1, 2, \dots, n$  or by  $[a_{ij}]_{m \times n}$  or simply  $A_{m \times n}$ . We write the general element  $a_{ij}$  of the matrix and enclose it in brackets of the type  $[ ]$  or of the type  $( )$ .

Sometimes we also use the brackets  $( )$  or the double bars  $\| \|$ , in place of the square brackets  $[ ]$  to denote matrices. We shall use capital letters such as  $A$ ,  $B$ ,  $X$ ,  $Y$ , etc. to denote matrices. Thus

$$A = \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}, \quad B = \begin{pmatrix} 2+3i & 5 \\ -4 & 2-3i \end{pmatrix}, \quad C = \left\| \begin{array}{cc} 1 & 1 \\ 1 & 1 \end{array} \right\|$$

are all matrices.

The numbers  $a_{11}$ ,  $a_{12}$ , etc. of this rectangular array are called the elements of the matrix. The element  $a_{ij}$  belongs to the  $i^{\text{th}}$  row and the  $j^{\text{th}}$  column and is sometimes called the  $(i, j)^{\text{th}}$  element of the matrix. Thus  $a_{23}$  is the element in the second row and third column. The element in the first row and first column,  $a_{11}$ , the element in the upper left-hand corner of a matrix is called the **leading element** of the matrix.

The size of the matrix, i.e., the number of rows and columns, is referred to as its order, or as its dimensions. If a matrix has  $m$  rows and  $n$  columns then the **order** of the matrix is said to be  $m$  by  $n$  (denoted as  $m \times n$ ). The  $i^{\text{th}}$  row consists of the elements  $a_{i1}$ ,  $a_{i2}$ ,  $a_{i3}$ , ...,  $a_{in}$ , while the  $j^{\text{th}}$  column consists of the elements  $a_{1j}$ ,  $a_{2j}$ ,  $a_{3j}$ , ...,  $a_{mj}$ .

In a matrix, the number of rows and columns need not be equal. When a matrix has  $n$  rows and  $n$  columns then it is said to be a square matrix of order  $n$ .

A simple example of a matrix of order  $2 \times 3$  is

$$A_{2 \times 3} = \begin{bmatrix} 4 & 0 & -3 \\ -7 & 2.73 & 1 \end{bmatrix}$$

Further,  $B = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}_{3 \times 3},$

$$C = \begin{bmatrix} 1 & 0 & 2 & 5 \\ 0 & 1 & 3 & 7 \\ -1 & 2 & 4 & 2 \end{bmatrix}_{3 \times 4}$$

are matrices of the type  $3 \times 3$  and  $3 \times 4$  respectively.

With the help of matrices we can represent any point

$(x, y)$  in a plane by a matrix (column or row) as  $\begin{bmatrix} x \\ y \end{bmatrix}$  (or  $[x, y]$ ). For example point  $P(0, 1)$  as a matrix representation may be given as

$$P = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \text{ or } [0 \ 1].$$

Observe that in this way we can also express the vertices of a closed rectilinear figure in the form of a matrix. For example, consider a quadrilateral ABCD with vertices

$A(1, 0), B(3, 2), C(1, 3), D(-1, 2).$

Now, quadrilateral ABCD in the matrix form, can be represented as

$$X = \begin{bmatrix} A & B & C & D \\ 1 & 3 & 1 & -1 \\ 0 & 2 & 3 & 2 \end{bmatrix}_{2 \times 4} \text{ or } \begin{matrix} A \\ B \\ C \\ D \end{matrix} \begin{bmatrix} 1 & 0 \\ 3 & 2 \\ 1 & 3 \\ -1 & 2 \end{bmatrix}_{4 \times 2}$$

If all the elements of matrix belong to the field of real numbers, the matrix is said to be real. Throughout the present treatment, the elements of a matrix shall be assumed to be complex numbers unless stated otherwise.

The difference between a matrix and a number should be clearly understood. A matrix is not a number. It has got no numerical value. It is a new formation

with the help of numbers. It is just an ordered collection of numbers arranged in the form of a rectangular array. Simply 5 is a number. But in our notation of matrices  $[5]$  is a matrix of the type  $1 \times 1$  and we donot say  $5 = [5]$ . We cannot have a relation of equality between a matrix and a number.

► **Example 1.** If a matrix has 8 elements, what are the possible orders it can have ?

► **Solution** We know that if a matrix is of order  $m \times n$ , it has  $mn$  elements. Thus, to find all possible order of a matrix with 8 elements, we will find all ordered pairs of natural numbers, whose product is 8. Thus, all possible orders are  $1 \times 8, 8 \times 1, 4 \times 2, 2 \times 4$ .

► **Example 2.** Construct a  $3 \times 2$  matrix whose

elements are given by  $a_{ij} = \frac{1}{2} li - 3jl$ .

► **Solution** In general a  $3 \times 2$  matrix is given by

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix}$$

$$\text{Now } a_{ij} = \frac{1}{2} li - 3jl, i = 1, 2, 3 \text{ and } j = 1, 2$$

$$\text{Therefore } a_{11} = \frac{1}{2} |1 - 3 \times 1| = 1$$

$$a_{12} = \frac{1}{2} |1 - 3 \times 2| = \frac{5}{2}$$

$$a_{21} = \frac{1}{2} |2 - 3 \times 1| = \frac{1}{2}$$

$$a_{22} = \frac{1}{2} |2 - 3 \times 2| = 2$$

$$a_{31} = \frac{1}{2} |3 - 3 \times 1| = 0$$

$$a_{32} = \frac{1}{2} |3 - 3 \times 2| = \frac{3}{2}$$

Hence the required matrix is given by

$$A = \begin{bmatrix} 1 & \frac{5}{2} \\ \frac{1}{2} & 2 \\ 0 & \frac{3}{2} \end{bmatrix}.$$

### 3.3 TYPES OF MATRICES

(i) **Row Matrix** : A matrix having only one row is called as row matrix (or row vector).

A row matrix is in the form  $A = [a_{11}, a_{12}, a_{13}, \dots, a_{1n}]$

(ii) **Column Matrix** : A matrix having only one column is called as column matrix (or column vector)

A column matrix is in the form  $A = \begin{bmatrix} a_{11} \\ a_{21} \\ \dots \\ a_{m1} \end{bmatrix}$

For example,  $X = [2 \ 7 \ -8 \ 5 \ 11]_{1 \times 5}$  is a row matrix of the type  $1 \times 5$  while

$Y = \begin{bmatrix} 2 \\ -9 \\ 11 \end{bmatrix}_{3 \times 1}$  is a column matrix of the type  $3 \times 1$ .

(iii) **Square Matrix** : A matrix in which number of rows & columns are equal is called a square matrix. The general form of a square matrix is

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix}$$

which we denote as  $A = [a_{ij}]_n$ .

The elements  $a_{ij}$  of a square matrix  $A = [a_{ij}]_{n \times n}$ , for which  $i = j$  i.e., the elements  $a_{11}, a_{22}, a_{33}, \dots, a_{nn}$  are called the diagonal elements and the line along which they lie is called the principal or main diagonal of the matrix. Their sum is called as **trace of A** denoted as  $\text{tr}(A)$

$$\text{tr}(A) = a_{11} + a_{22} + a_{33} + \dots + a_{nn}$$

The elements  $a_{n1}, a_{n-1,2}, \dots, a_{1n}$  constitute its secondary diagonal.

The elements  $a_{ij}$  and  $a_{ji}$  are called conjugate elements.

For example, the matrix

$$A = \begin{bmatrix} 0 & 1 & 2 & 3 \\ 2 & 3 & 1 & 0 \\ 5 & 0 & 1 & 1 \\ 0 & 0 & 1 & 2 \end{bmatrix}_{4 \times 4}$$

is a square matrix of order 4. The elements 0, 3, 1, 2 constitute the principal diagonal of this matrix.

The trace of the matrix  $= 0 + 3 + 1 + 2 = 6$ .

(iv) **Zero Matrix** :  $A = [a_{ij}]_{m \times n}$  is called a zero matrix, if  $a_{ij} = 0 \ \forall \ i \text{ and } j$ .

A null matrix is simply denoted by the symbol  $O$  and read as 'zero'.

For example,

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}_{3 \times 5} \quad \text{and} \quad \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}_{3 \times 3}$$

are zero matrices of the types  $3 \times 5$  and  $3 \times 3$  respectively.

(v) **Upper Triangular Matrix** :  $A = [a_{ij}]_{m \times n}$  is said to be upper triangular, if  $a_{ij} = 0$  for  $i > j$  (i.e., all the elements below the diagonal element are zero).

The following matrices are upper triangular matrices :

$$\begin{pmatrix} 0 & 3 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 4 \\ 0 & 2 \end{pmatrix}, \begin{pmatrix} 1 & 7 & 6 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$$

Further,  $A = [a_{ij}]_{m \times n}$  is said to be strictly upper triangular, if  $a_{ij} = 0$  for  $i \geq j$ .

For example  $\begin{bmatrix} 0 & -9 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix}$  a strictly upper triangular

matrix.

(vi) **Lower Triangular Matrix** :  $A = [a_{ij}]_{m \times n}$  is said to be a lower triangular matrix, if  $a_{ij} = 0$  for  $i < j$  (i.e., all the elements above the diagonal elements are zero).

The following matrices are lower triangular :

$$\begin{pmatrix} -1 & 0 \\ -3 & 2 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 \\ 2-i & 1 & 0 \\ 0 & 2 & -8 \end{pmatrix}$$

Also,  $A = [a_{ij}]_{m \times n}$  is said to be strictly lower triangular, if  $a_{ij} = 0$  for  $i \leq j$ .



- (i) The minimum number of zeros in a triangular matrix of order

$$n = 1 + 2 + 3 + \dots + (n-1) = \frac{n(n-1)}{2}.$$

- (ii) The maximum number of distinct elements in a triangular matrix of order  $n$

$$= (1 + 2 + 3 + \dots + n) + 1 = \frac{n(n+1)}{2} + 1.$$

- (vii) **Diagonal Matrix** : A square matrix  $[a_{ij}]_{m \times n}$  is said to be a diagonal matrix if  $a_{ij} = 0$  for  $i \neq j$ . (i.e., all the elements of the square matrix other than diagonal elements are zero).

An  $n$ -rowed diagonal matrix whose diagonal elements in order are  $d_1, d_2, d_3, \dots, d_n$  will often be denoted by the symbol  $D = \text{diag}(d_1, d_2, \dots, d_n)$ .

The following matrices are diagonal matrices :

$$\begin{bmatrix} 2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 5 \end{bmatrix} \text{ and } \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

- (viii) **Scalar Matrix** : Scalar matrix is a diagonal matrix in which all the diagonal elements are same.  $A = [a_{ij}]_n$  is a scalar matrix, if (i)  $a_{ij} = 0$  for  $i \neq j$  and (ii)  $a_{ij} = k$  for  $i = j$ .

- (ix) **Unit Matrix (Identity Matrix)** : Unit matrix is a diagonal matrix in which all the diagonal elements are unity i.e.  $A = [a_{ij}]_n$  is a unit matrix when  $a_{ij} = 0$  for  $i \neq j$  &  $a_{ij} = 1$ . Unit matrix of order ' $n$ ' is denoted by  $I_n$  (or  $I$ ).

Identity matrices of order 2 and 3 are

$$I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Another way to represent the identity matrix is to use the Kronecker delta symbol  $\delta_{ij}$  which equals 1 if  $i = j$  and equals 0 if  $i \neq j$ . Thus  $I_n = (\delta_{ij})_{n \times n}$ .

- (x) **Tridiagonal Matrix** : A square matrix  $A_{n \times n}$ ,  $n \geq 3$ , is **tridiagonal** if  $a_{ij} = 0$  when  $i > j + 1$  or  $j > i + 1$ . That is, if the row number and column number of an entry differ by 2 or more, the entry must be zero. The other entries are unrestricted.

According to the definition, in a  $4 \times 4$  tridiagonal matrix

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{pmatrix}$$

the entries  $a_{13}, a_{14}, a_{24}, a_{31}, a_{41}$ , and  $a_{42}$  must all be zero.

Some  $4 \times 4$  tri-diagonal matrices are

$$\begin{pmatrix} 3 & 4 & 0 & 0 \\ 6 & -1 & 2 & 0 \\ 0 & -17 & 2 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 7-2i & 0 \\ 0 & i & 0 & 1 \\ 0 & 0 & 1+i & 1 \end{pmatrix}$$

All diagonal matrices are also tridiagonal.

- (xi) **Comparable Matrices** : Two matrices  $A$  and  $B$  are said to be comparable, if they have the same order (i.e., number of rows of  $A$  and  $B$  are same and also the number of columns).

- (xii) **Submatrix of a matrix** : Any matrix obtained by omitting some rows and columns from a given  $(m \times n)$  matrix  $A$  is called a submatrix of  $A$ . The matrix  $A$  itself is a sub-matrix of  $A$  as it can be obtained from  $A$  by omitting no row or columns.

The matrix  $\begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & 1 \end{bmatrix}$  is submatrix of the

matrix  $A = \begin{bmatrix} 1 & 2 & 3 & 9 \\ 7 & 11 & 6 & 5 \\ 0 & 2 & 1 & 8 \end{bmatrix}$  as it can be

obtained from A by omitting the second row and the fourth column.

► **Example 1.** Find the order of each matrix, and determine any special characteristics.

$$\begin{array}{ll} \text{(i)} \begin{bmatrix} 6 & 5 \\ 3 & 4 \\ 5 & -1 \end{bmatrix} & \text{(ii)} \begin{bmatrix} 5 & 8 & 9 \\ 0 & 5 & -3 \\ -4 & 0 & 5 \end{bmatrix} \\ \text{(iii)} [1 \ 6 \ 5 \ -2 \ 5] & \text{(iv)} \begin{bmatrix} 3 \\ -5 \\ 0 \\ 2 \end{bmatrix} \end{array}$$

► **Solution**

$$\begin{array}{ll} \text{(i)} \text{ The matrix } \begin{bmatrix} 6 & 5 \\ 3 & 4 \\ 5 & -1 \end{bmatrix} \text{ is a } 3 \times 2 \text{ matrix, because} \\ \text{it has 3 rows and 2 columns.} \\ \text{(ii)} \begin{bmatrix} 5 & 8 & 9 \\ 0 & 5 & -3 \\ -4 & 0 & 5 \end{bmatrix} \text{ is a } 3 \times 3 \text{ matrix. It is also a} \\ \text{square matrix.} \\ \text{(iii)} [1 \ 6 \ 5 \ -2 \ 5] \text{ is a } 1 \times 5 \text{ matrix. It is an example} \\ \text{of a row matrix.} \\ \text{(iv)} \begin{bmatrix} 3 \\ -5 \\ 0 \\ 2 \end{bmatrix} \text{ is a } 4 \times 1 \text{ column matrix.} \end{array}$$

### 3.4 OPERATIONS ON MATRICES

#### 1. Equality of Matrices

Two matrices A and B are said to be equal if they are of same order and all the corresponding elements are equal.

Let  $A = [a_{ij}]_{m \times n}$  and  $B = [b_{ij}]_{p \times q}$   
 $A = B$  if (i)  $m = p, n = q$

(ii)  $a_{ij} = b_{ij} \forall i \& j$ .

For example,  $\begin{pmatrix} 1 & 0 \\ 5 & 6 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 5 & 6 \end{pmatrix}$

but  $\begin{pmatrix} 1 & 0 \\ 5 & 6 \end{pmatrix} \neq \begin{pmatrix} 1 & 1 \\ 5 & 6 \end{pmatrix}$

and  $\begin{pmatrix} 1 & 0 \\ 5 & 6 \end{pmatrix} \neq \begin{pmatrix} 1 & 0 & 0 \\ 5 & 6 & 0 \end{pmatrix}$

In the last case the matrices are not equal because they are not the same order.

The properties of the equality of matrices are :

(a) If A and B are any two matrices, either  $A = B$  or  $A \neq B$  (the determinative property).

(b) If A is any matrix,  $A = A$  (the reflexive property).

(c) If  $A = B$ , then  $B = A$  (the symmetric property).

(d) If  $A = B$  and  $B = C$ , then  $A = C$  (the transitive property).

► **Example 1.** Find the values of a, b, c and d so that the matrices A and B may be equal, where

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, B = \begin{bmatrix} 1 & -1 \\ 0 & 3 \end{bmatrix}.$$

► **Solution** We see that the matrices A and B are of the same order  $2 \times 2$ . If  $A = B$ , then the corresponding elements of A and B must be equal.

∴ We will have  $A = B$  if  $a = 1, b = -1, c = 0, d = 3$ .

► **Example 2.** If

$$\begin{bmatrix} x+3 & z+4 & 2y-7 \\ -6 & a-1 & 0 \\ b-3 & -21 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 6 & 3y-2 \\ -6 & -3 & 2c+2 \\ 2b+4 & -21 & 0 \end{bmatrix}$$

Find the values of a, b, c, x, y and z.

► **Solution** As the given matrices are equal, therefore, their corresponding elements must be equal.

Comparing the corresponding elements, we get

$$\begin{array}{l} x+3=0, \quad z+4=6, \quad 2y-7=3y-2 \\ a-1=-3, \quad 0=2c+2 \quad b-3=2b+4, \end{array}$$

Simplifying, we get

$$\begin{array}{lll} a=-2, & b=-7, & c=-1, \\ x=-3, & y=-5, & z=2 \end{array}$$

#### 2. Addition of Matrices

Let A and B be two matrices of same order (i.e. they are comparable matrices).

Then  $A + B = [a_{ij}]_{m \times n} + [b_{ij}]_{m \times n}$ .

$$= [c_{ij}]_{m \times n} \text{ where } c_{ij} = a_{ij} + b_{ij} \forall i \& j.$$

$$\text{i.e. } A + B = \begin{pmatrix} a_{11} + b_{11} & \dots & a_{1n} + b_{1n} \\ \dots & a_{ij} + b_{ij} & \dots \\ a_{n1} + b_{n1} & \dots & a_{nn} + b_{nn} \end{pmatrix}$$

That is, the addition is performed entry by entry. For example, if

$$A = \begin{bmatrix} 3 & 2 & -1 \\ 4 & -3 & 1 \end{bmatrix}_{2 \times 3} \text{ and } B = \begin{bmatrix} 1 & -2 & 7 \\ 3 & 2 & -1 \end{bmatrix}_{2 \times 3}, \text{ then}$$

$$A + B = \begin{bmatrix} 3+1 & 2-2 & -1+7 \\ 4+3 & -3+2 & 1-1 \end{bmatrix}$$

$$= \begin{bmatrix} 4 & 0 & 6 \\ 7 & -1 & 0 \end{bmatrix}_{2 \times 3}.$$



- (i) If the matrices A and B are not of the same size, we cannot find their sum.
- (ii) Since the sum of any two  $m \times n$  matrices is again an  $m \times n$  matrix, we say that the set of all  $m \times n$  matrices is closed with respect to addition.

### Properties of Addition

#### (i) Matrix addition is commutative.

If A and B be two  $m \times n$  matrices, then  $A + B = B + A$ .

*Proof:*

Let  $A = [a_{ij}]_{m \times n}$  and  $B = [b_{ij}]_{m \times n}$ . Then

$$A + B = [a_{ij}]_{m \times n} + [b_{ij}]_{m \times n} = [a_{ij} + b_{ij}]_{m \times n}$$

[by definition of addition of two matrices]

$$= [b_{ij} + a_{ij}]_{m \times n}$$

[since  $a_{ij}$  and  $b_{ij}$  are numbers and addition of numbers is commutative]

$$= [b_{ij}]_{m \times n} + [a_{ij}]_{m \times n}$$

[by definition of addition of two matrices]

$$= B + A.$$

#### (ii) Matrix addition is associative.

If A, B, C be three matrices each of the type  $m$

$\times n$ , then  $(A + B) + C = A + (B + C)$ .

*Proof:*

Let  $A = [a_{ij}]_{m \times n}$ ,  $B = [b_{ij}]_{m \times n}$ ,  $C = [c_{ij}]_{m \times n}$ .

Then  $(A + B) + C = ([a_{ij}]_{m \times n} + [b_{ij}]_{m \times n}) + [c_{ij}]_{m \times n}$

$$= [a_{ij} + b_{ij}]_{m \times n} + [c_{ij}]_{m \times n}$$

[by definition of  $A + B$ ]

$$= [(a_{ij} + b_{ij}) + c_{ij}]_{m \times n}$$

[by definition of addition of matrices]

$$= [a_{ij} + (b_{ij} + c_{ij})]_{m \times n}$$

[since  $a_{ij}$ ,  $b_{ij}$ ,  $c_{ij}$  are numbers and addition of numbers is associative]

$$= [a_{ij}]_{m \times n} + [b_{ij} + c_{ij}]_{m \times n}$$

[by definition of addition of two matrices]

$$= [a_{ij}]_{m \times n} + ([b_{ij}]_{m \times n} + [c_{ij}]_{m \times n})$$

$$= A + (B + C).$$

#### (iii) Existence of additive identity.

If O be the  $m \times n$  matrix each of whose elements is zero, then  $A + O = A = O + A$  for every  $m \times n$  matrix A.

*Proof:*

Let  $A = [a_{ij}]_{m \times n}$ . Then

$$A + O = [a_{ij} + 0]_{m \times n} = [a_{ij}]_{m \times n} = A$$

$$O + A = [0 + a_{ij}]_{m \times n} = [a_{ij}]_{m \times n} = A.$$

Thus the null matrix O of the type  $m \times n$  acts as the identity element for addition in the set of all  $m \times n$  matrices.

#### (iv) Existence of the additive inverse.

##### Negative of a matrix.

Let  $A = [a_{ij}]_{m \times n}$ . Then the negative of the matrix A is defined as the matrix  $[-a_{ij}]_{m \times n}$  and is denoted by  $-A$ .

The matrix  $-A$  is the additive inverse of the matrix A. Obviously,  $-A + A = O = A + (-A)$ . Here O is the null matrix of the type  $m \times n$ . It is identity element for matrix addition.

#### (v) Cancellation laws hold good in the case of addition of matrices.

If A, B, C are three  $m \times n$  matrices, then

$$A + B = A + C \Rightarrow B = C$$

(left cancellation law)

$$\text{and } B + A = C + A \Rightarrow B = C$$

(right cancellation law)



*Proof:*

We have  $A + B = A + C$

$\Rightarrow -A + (A + B) = -A + (A + C)$ , adding  $-A$  to both sides

$\Rightarrow (-A + A) + B = (-A + A) + C$

[ $\because$  matrix addition is associative]

$\Rightarrow O + B = O + C$  [ $\because -A + A = O$ ]

$\Rightarrow B = C$ . [ $\because O + B = B$ ]

Similarly we can prove the right cancellation law.

► **Example 3.** If  $A = \begin{bmatrix} 1 & 0 \\ 2 & -1 \end{bmatrix}$ ,  $B = \begin{bmatrix} 3 & 7 \\ 4 & 8 \end{bmatrix}$ ,

$C = \begin{bmatrix} -1 & 1 \\ 0 & 0 \end{bmatrix}$  verify that  $A + (B + C) = (A + B) + C$ .

► **Solution** We have  $A + B = \begin{bmatrix} 1 & 0 \\ 2 & -1 \end{bmatrix} + \begin{bmatrix} 3 & 7 \\ 4 & 8 \end{bmatrix}$

$$= \begin{bmatrix} 1+3 & 0+7 \\ 2+4 & -1+8 \end{bmatrix} = \begin{bmatrix} 4 & 7 \\ 6 & 7 \end{bmatrix}$$

$$\therefore (A + B) + C = \begin{bmatrix} 4 & 7 \\ 6 & 7 \end{bmatrix} + \begin{bmatrix} -1 & 1 \\ 0 & 0 \end{bmatrix} \\ = \begin{bmatrix} 4-1 & 7+1 \\ 6+0 & 7+0 \end{bmatrix} = \begin{bmatrix} 3 & 8 \\ 6 & 7 \end{bmatrix}$$

$$\text{Also } B + C = \begin{bmatrix} 3 & 7 \\ 4 & 8 \end{bmatrix} + \begin{bmatrix} -1 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 3-1 & 7+1 \\ 4+0 & 8+0 \end{bmatrix} \\ = \begin{bmatrix} 2 & 8 \\ 4 & 8 \end{bmatrix}$$

$$\therefore A + (B + C) = \begin{bmatrix} 1 & 0 \\ 2 & -1 \end{bmatrix} + \begin{bmatrix} 2 & 8 \\ 4 & 8 \end{bmatrix} \\ = \begin{bmatrix} 1+2 & 0+8 \\ 2+4 & -1+8 \end{bmatrix} \\ = \begin{bmatrix} 3 & 8 \\ 6 & 7 \end{bmatrix} = (A + B) + C.$$

► **Example 4.** Find the additive inverse of the

matrix  $A = \begin{bmatrix} 2 & 3 & -1 & 1 \\ 3 & -1 & 2 & 2 \\ 1 & 2 & 8 & 7 \end{bmatrix}$ .

► **Solution** The additive inverse of the  $3 \times 4$  matrix  $A$  is the  $3 \times 4$  matrix each of whose elements is the

negative of the corresponding element of  $A$ . Therefore if we denote the additive inverse of  $A$  by  $-A$ , we have

$$-A = \begin{bmatrix} -2 & -3 & 1 & -1 \\ -3 & 1 & -2 & -2 \\ -1 & -2 & -8 & -7 \end{bmatrix}.$$

Obviously  $A + (-A) = (-A) + A = O$ , where  $O$  is the null matrix of the type  $3 \times 4$ .

### 3. Substraction of Matrices

Let  $A$  &  $B$  be two matrices of same order. Then  $A - B$  is defined as  $A + (-B)$ . This implies that the difference  $A - B$  may be found by subtracting corresponding elements.

For example,

$$\text{If } A = \begin{bmatrix} 2 & 7 \\ 9 & 8 \end{bmatrix}, B = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix},$$

$$A - B = A + (-B) = \begin{bmatrix} 2 & 7 \\ 9 & 8 \end{bmatrix} + \begin{bmatrix} -1 & -2 \\ 0 & -3 \end{bmatrix} \\ = \begin{bmatrix} 2-1 & 7-2 \\ 9+0 & 8-3 \end{bmatrix} = \begin{bmatrix} 1 & 5 \\ 9 & 5 \end{bmatrix}.$$

Or simply

$$\begin{bmatrix} 2 & 1 & -3 \\ -4 & 0 & 1 \end{bmatrix} - \begin{bmatrix} 1 & -1 & 2 \\ 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 2 & -5 \\ -4 & -1 & 1 \end{bmatrix}.$$

► **Example 5.** For the following pairs of matrices, determine the sum and difference, if they exist.

$$(i) \quad A = \begin{pmatrix} 1 & -1 & 2 \\ 0 & 1 & 3 \end{pmatrix} \quad B = \begin{pmatrix} 2 & 1.5 & 6 \\ -3 & 2+i & 0 \end{pmatrix}$$

$$(ii) \quad A = \begin{pmatrix} 1 & 0 \\ 3 & -4 \end{pmatrix} \quad B = \begin{pmatrix} 1 & 1 & 2 \\ 0 & -2 & 0 \end{pmatrix}$$

► **Solution** (i) Matrices  $A$  and  $B$  are  $2 \times 3$  and conformable for addition and subtraction.

$$A + B = \begin{pmatrix} 1+2 & -1+1.5 & 2+6 \\ 0+(-3) & 1+2+i & 3+0 \end{pmatrix} \\ = \begin{pmatrix} 3 & 0.5 & 8 \\ -3 & 3+i & 3 \end{pmatrix}$$

$$\begin{aligned} A - B &= \begin{pmatrix} 1-2 & -1-1.5 & 2-6 \\ 0-(-3) & 1-(2+i) & 3-0 \end{pmatrix} \\ &= \begin{pmatrix} -1 & -2.5 & -4 \\ 3 & -1-i & 3 \end{pmatrix} \end{aligned}$$

(ii) Matrix A is  $2 \times 2$ , and B is  $2 \times 3$ . Since A and B are not the same size, they are not conformable for addition or subtraction.

► **Example 6.** If  $A = \begin{bmatrix} 1 & 4 \\ 3 & 2 \\ 2 & 5 \end{bmatrix}$ ,  $B = \begin{bmatrix} -1 & -2 \\ 0 & 5 \\ 3 & 1 \end{bmatrix}$ , find

the matrix D such that  $A + B - D = 0$ .

► **Solution** We have  $A + B - D = 0$

$$\Rightarrow (A + B) + (-D) = 0 \Rightarrow A + B = (-D) = D$$

$$\begin{aligned} \text{Therefore } D = A + B &= \begin{bmatrix} 1 & 4 \\ 3 & 2 \\ 2 & 5 \end{bmatrix} + \begin{bmatrix} -1 & -2 \\ 0 & 5 \\ 3 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 2 \\ 3 & 7 \\ 5 & 6 \end{bmatrix}. \end{aligned}$$

#### 4. Scalar Multiplication of Matrices

Let  $\lambda$  be a scalar (real or complex number) and  $A = [a_{ij}]_{m \times n}$  be a matrix. Thus the product  $\lambda A$  is defined as

$$\lambda A = [b_{ij}]_{m \times n} \text{ where } b_{ij} = \lambda a_{ij} \quad \forall i \text{ \& } j.$$

**Note :** If S is a scalar matrix, then  $S = \lambda I$ , where  $\lambda$  is the diagonal element.

$$\text{For example, if } \lambda = 2 \text{ and } A = \begin{bmatrix} 3 & 2 & -1 \\ 4 & -3 & 1 \end{bmatrix}$$

then  $2A$

$$= \begin{bmatrix} 2 \times 3 & 2 \times 2 & 2 \times (-1) \\ 2 \times 4 & 2 \times (-3) & 2 \times 1 \end{bmatrix} = \begin{bmatrix} 6 & 4 & -2 \\ 8 & -6 & 2 \end{bmatrix}$$

#### Properties of Scalar Multiplication

(i)  $\lambda A = A\lambda$

(ii) If A be any  $m \times n$  matrix and  $\lambda$  be any scalar, then  $(-\lambda)A = -(\lambda A) = \lambda(-A)$

**Proof:**

Let  $A = [a_{ij}]_{m \times n}$ . Then

$$\begin{aligned} (-\lambda)A &= [(-\lambda)a_{ij}]_{m \times n} = [-(\lambda a_{ij})]_{m \times n} \\ &= -[\lambda a_{ij}]_{m \times n} = -(\lambda A). \end{aligned}$$

$$\begin{aligned} \text{Also } (-\lambda)A &= [(-\lambda)a_{ij}]_{m \times n} = [\lambda(-a_{ij})]_{m \times n} \\ &= \lambda[-a_{ij}]_{m \times n} = \lambda(-A). \end{aligned}$$

(iii) If  $\lambda_1$  and  $\lambda_2$  are two scalars and A is any  $m \times n$  matrix, then  $(\lambda_1 + \lambda_2)A = \lambda_1 A + \lambda_2 A$ .

**Proof:**

Let  $A = [a_{ij}]_{m \times n}$ . Then

$$\begin{aligned} (\lambda_1 + \lambda_2)A &= (\lambda_1 + \lambda_2)[a_{ij}]_{m \times n} = [(\lambda_1 + \lambda_2)a_{ij}]_{m \times n} \\ &= [\lambda_1 a_{ij} + \lambda_2 a_{ij}]_{m \times n} = \lambda_1 A + \lambda_2 A = [(\lambda_1 + \lambda_2)a_{ij}]_{m \times n} \\ [\because \text{multiplication of numbers is associative}] \\ &= (\lambda_1 + \lambda_2)[a_{ij}]_{m \times n} = (\lambda_1 + \lambda_2)A. \end{aligned}$$

(iv)  $\lambda(A + B) = \lambda A + \lambda B$

If A and B are two matrices each of the type  $m \times n$ , then  $\lambda(A + B) = \lambda A + \lambda B$  i.e., the scalar multiplication of matrices distributes over the addition of matrices.

**Proof:**

Let  $A = [a_{ij}]_{m \times n}$  and  $B = [b_{ij}]_{m \times n}$ . Then

$$\begin{aligned} \lambda(A + B) &= \lambda([a_{ij}]_{m \times n} + [b_{ij}]_{m \times n}) \\ &= \lambda(a_{ij} + b_{ij})_{m \times n} \\ &\quad [\text{by def. of addition of two matrices}] \\ &= [\lambda(a_{ij} + b_{ij})]_{m \times n} \\ &\quad [\text{by def. of scalar multiplication}] \\ &= [\lambda a_{ij} + \lambda b_{ij}]_{m \times n} \\ &\quad [\text{by the distributive law of numbers}] \\ &= [\lambda a_{ij}]_{m \times n} + [\lambda b_{ij}]_{m \times n} = \lambda[a_{ij}]_{m \times n} + \lambda[b_{ij}]_{m \times n} \\ &= \lambda A + \lambda B. \end{aligned}$$

(v)  $\lambda_1(\lambda_2 A) = (\lambda_1 \lambda_2)A$

(vi)  $A(\lambda B) = (\lambda A)B = \lambda(AB)$ .

► **Example 7.** If  $A = \begin{bmatrix} 3 & 9 & 0 \\ 1 & 8 & -2 \\ 7 & 5 & 4 \end{bmatrix}$ ,

$$B = \begin{bmatrix} 4 & 0 & 2 \\ 7 & 1 & 4 \\ 2 & 2 & 6 \end{bmatrix}, \text{ verify that } 3(A + B) = 3A + 3B.$$

► **Solution** We have  $A + B$

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$$\begin{aligned}
 &= \begin{bmatrix} 3+4 & 9+0 & 0+2 \\ 1+7 & 8+1 & -2+4 \\ 7+2 & 5+7 & 4+6 \end{bmatrix} \\
 &= \begin{bmatrix} 7 & 9 & 2 \\ 8 & 9 & 2 \\ 9 & 7 & 10 \end{bmatrix}
 \end{aligned}$$

$$\begin{aligned}
 \therefore 3(A+B) &= \begin{bmatrix} 3 \times 7 & 3 \times 9 & 3 \times 2 \\ 3 \times 8 & 3 \times 9 & 3 \times 2 \\ 3 \times 9 & 3 \times 7 & 3 \times 10 \end{bmatrix} \\
 &= \begin{bmatrix} 21 & 27 & 6 \\ 24 & 27 & 6 \\ 27 & 21 & 30 \end{bmatrix}.
 \end{aligned}$$

$$\begin{aligned}
 \text{Again } 3A &= 3 \begin{bmatrix} 3 & 9 & 0 \\ 1 & 8 & -2 \\ 7 & 5 & 4 \end{bmatrix} \\
 &= \begin{bmatrix} 3 \times 3 & 3 \times 9 & 3 \times 0 \\ 3 \times 1 & 3 \times 8 & 3 \times -2 \\ 3 \times 7 & 3 \times 5 & 3 \times 4 \end{bmatrix} \\
 &= \begin{bmatrix} 9 & 27 & 0 \\ 3 & 24 & -6 \\ 21 & 15 & 12 \end{bmatrix}
 \end{aligned}$$

$$\begin{aligned}
 \text{Also } 3B &= 3 \begin{bmatrix} 4 & 0 & 2 \\ 7 & 1 & 4 \\ 2 & 2 & 6 \end{bmatrix} = \begin{bmatrix} 3 \times 4 & 3 \times 0 & 3 \times 2 \\ 3 \times 7 & 3 \times 1 & 3 \times 4 \\ 3 \times 2 & 3 \times 2 & 3 \times 6 \end{bmatrix} \\
 &= \begin{bmatrix} 12 & 0 & 6 \\ 21 & 3 & 12 \\ 6 & 6 & 18 \end{bmatrix}
 \end{aligned}$$

$$\begin{aligned}
 \therefore 3A+3B &= \begin{bmatrix} 9 & 27 & 0 \\ 3 & 24 & -6 \\ 21 & 15 & 12 \end{bmatrix} + \begin{bmatrix} 12 & 0 & 6 \\ 21 & 3 & 12 \\ 6 & 6 & 18 \end{bmatrix} \\
 &= \begin{bmatrix} 9+12 & 27+0 & 0+6 \\ 3+21 & 24+3 & -6+12 \\ 21+6 & 15+6 & 12+18 \end{bmatrix} = \begin{bmatrix} 21 & 27 & 6 \\ 24 & 27 & 6 \\ 27 & 21 & 30 \end{bmatrix}
 \end{aligned}$$

$\therefore 3(A+B) = 3A + 3B$ , i.e. the scalar multiplication of matrices distributes over the addition of matrices.

► **Example 8.** Find the values of  $x$  and  $y$  from the

$$\text{equation : } 2 \begin{bmatrix} x & 5 \\ 7 & y-3 \end{bmatrix} + \begin{bmatrix} 3 & -4 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 7 & 6 \\ 15 & 14 \end{bmatrix}$$

► **Solution** We have

$$2 \begin{bmatrix} x & 5 \\ 7 & y-3 \end{bmatrix} + \begin{bmatrix} 3 & -4 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 7 & 6 \\ 15 & 14 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 2x & 10 \\ 14 & 2y-6 \end{bmatrix} + \begin{bmatrix} 3 & -4 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 7 & 6 \\ 15 & 14 \end{bmatrix}$$

$$\text{or } \begin{bmatrix} 2x+3 & 10-4 \\ 14+1 & 2y-6+2 \end{bmatrix} = \begin{bmatrix} 7 & 6 \\ 15 & 14 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 2x+3 & 6 \\ 15 & 2y-4 \end{bmatrix} = \begin{bmatrix} 7 & 6 \\ 15 & 14 \end{bmatrix}$$

$$\text{or } 2x+3=7 \text{ and } 2y-4=14 \text{ (Why ?)}$$

$$\text{or } 2x=7-3 \text{ and } 2y=18$$

$$\text{or } x=\frac{4}{2} \text{ and } y=\frac{18}{2}$$

$$\text{i.e. } x=2 \text{ and } y=9$$

### Practice Problems

[A]

1. A matrix  $A$  and an entry  $a_{ij}$  of the matrix are given. State the entry's row and column.

$$\text{(i) } \begin{pmatrix} 1 & -3 \\ 2 & 1 \end{pmatrix}; -3$$

$$\text{(ii) } \begin{pmatrix} 1-i & -1 & 2 \\ 0 & 1 & 1 \\ 3i & -6+i & 7 \end{pmatrix}; -6+i$$

$$\text{(iii) } \begin{pmatrix} -3 & 2 & 1 \\ 4 & 6 & 7 \end{pmatrix}; 4$$

$$(iv) \begin{pmatrix} 3 & 2 \\ 4 & 1 \\ 6 & 7 \\ 5 & 9 \end{pmatrix}; 5$$

2. If a matrix has 18 elements, what are the possible orders it can have? What, if it has 5 elements?
3. Find the value of a, b, c and d from the equation:

$$\begin{bmatrix} a-b & 2a+c \\ 2a-b & 3c+d \end{bmatrix} = \begin{bmatrix} -1 & 5 \\ 0 & 13 \end{bmatrix}$$

4. Simplify

$$\cos\theta \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix} + \sin\theta \begin{bmatrix} \sin\theta & -\cos\theta \\ \cos\theta & \sin\theta \end{bmatrix}$$

5. Compute  $A + 2B - 3C$  if

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 2 & 3 & 1 \end{pmatrix}, B = \begin{pmatrix} 1 & 2 & 0 \\ 3 & 1 & -1 \end{pmatrix},$$

$$C = \begin{pmatrix} -1 & -1 & 0 \\ 2 & 0 & -1 \end{pmatrix}$$

6. Find X and Y, if

$$(i) X+Y = \begin{bmatrix} 7 & 0 \\ 2 & 5 \end{bmatrix} \text{ and } X-Y = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}$$

$$(ii) 2X+3Y = \begin{bmatrix} 2 & 3 \\ 4 & 0 \end{bmatrix} \text{ and } 3X+2Y = \begin{bmatrix} 2 & -2 \\ -1 & 5 \end{bmatrix}$$

$$7. \text{ Given } 3 \begin{bmatrix} x & y \\ z & w \end{bmatrix} = \begin{bmatrix} x & 6 \\ -1 & 2w \end{bmatrix} +$$

$$\begin{bmatrix} 4 & x+y \\ z+w & 3 \end{bmatrix}, \text{ find the values of } x, y, z \text{ and } w.$$

### 3.5 MULTIPLICATION OF MATRICES

Let A and B be two matrices such that the number of columns of A is same as number of rows of B. i.e.,  $A = [a_{ij}]_{m \times p}$  and  $B = [b_{ij}]_{p \times n}$ .

Then  $AB = [c_{ij}]_{m \times n}$  where  $c_{ij} = \sum_{k=1}^p a_{ik} b_{kj}$

which is the dot product of  $i^{\text{th}}$  row vector of A and  $j^{\text{th}}$  column vector of B, i.e. the  $(i, j)^{\text{th}}$  element  $c_{ij}$  of the matrix AB is obtained by multiplying the corresponding elements of the  $i^{\text{th}}$  row of A and the  $j^{\text{th}}$  column of B and then adding the products.

The rule of multiplication is row by column multiplication i.e., in the process of multiplication we take the rows of A and the columns of B.

The element  $c_{11}$  of the matrix AB is obtained by adding the products of the corresponding elements of the first row of A and the first column of B. The element  $c_{12}$  of the matrix AB is obtained by adding the products of the corresponding elements of the first row of A and the second column of B. Similarly the element  $c_{21}$  of the matrix AB is obtained by adding the products of the corresponding elements of the second row of A and the first column of A. In this way we multiply two matrices A and B.

For example, if

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix}_{3 \times 2}, B = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}_{2 \times 2}$$

then AB

$$= \begin{bmatrix} a_{11}b_{11} + a_{12}b_{21} & a_{11}b_{12} + a_{12}b_{22} \\ a_{21}b_{11} + a_{22}b_{21} & a_{21}b_{12} + a_{22}b_{22} \\ a_{31}b_{11} + a_{32}b_{21} & a_{31}b_{12} + a_{32}b_{22} \end{bmatrix}_{3 \times 2}$$

In the product AB, the matrix A is called the pre-factor and the matrix B is called the post-factor. Also we say that the matrix A has been post-multiplied by the matrix B and the matrix B has been pre-multiplied by the matrix A.

Let us take another example. If  $A = \begin{bmatrix} 1 & -1 & 2 \\ 0 & 3 & 4 \end{bmatrix}$

and  $B = \begin{bmatrix} 2 & 7 \\ -1 & 1 \\ 5 & -4 \end{bmatrix}$ , then the product  $AB$  is

defined and is given by  $AB = \begin{bmatrix} 1 & -1 & 2 \\ 0 & 3 & 4 \end{bmatrix}$

$$\begin{bmatrix} 2 & 7 \\ -1 & 1 \\ 5 & -4 \end{bmatrix}.$$

This is a  $2 \times 2$  matrix in which each entry is the sum of the products across some row of  $A$  with the corresponding entries down some column of  $B$ . These four computations are

Entry in  
first row  
first column  $\begin{bmatrix} (1)(2) + (-1)(-1) + (2)(5) & ? \\ & ? & ? \end{bmatrix}$

Entry in  
first row  
second column  $\begin{bmatrix} 13 & (1)(7) + (-1)(1) + 2(-4) \\ ? & ? \end{bmatrix}$

Entry in  
second row  
first column  $\begin{bmatrix} 13 & -2 \\ 0(2) + 3(-1) + 4(5) & ? \end{bmatrix}$

Entry in  
second row  
second column  $\begin{bmatrix} 13 & -2 \\ 17 & 0(7) + 3(1) + 4(-4) \end{bmatrix}$

$$\text{Thus } AB = \begin{bmatrix} 13 & -2 \\ 17 & -13 \end{bmatrix}$$

► **Example 1.** Let

$$A = \begin{pmatrix} 1 & -1 & 2 \\ 0 & 1 & 3 \end{pmatrix} \text{ and } B = \begin{pmatrix} 1 & -1 & -3 \\ 1 & 2 & 4 \\ 1 & 3 & 6 \end{pmatrix}$$

Calculate  $(AB)_{12}$  and  $(AB)_{23}$ .

► **Solution** Now,  $(AB)_{12}$  denotes the row-column product of row 1 of  $A$  and column 2 of  $B$  :

$$(1 \ -1 \ 2) \begin{pmatrix} -1 \\ 2 \\ 3 \end{pmatrix} = 1(-1) + (-1)2 + 2(3) = 3$$

$$(AB)_{23} = (0 \ 1 \ 3) \begin{pmatrix} -3 \\ 4 \\ 6 \end{pmatrix} = 0(-3) + 1(4) + 3(6) = 22$$

### Existence of matrix products

A product  $AB$  can be written in the notation

$$A_{m \times p} B_{p \times n} = P_{m \times n},$$

a form which provides both for checking the conformability of  $A$  and  $B$  and for ascertaining the order of their product. Repeated use of this also simplifies determining the order of a matrix derived by multiplying several matrices together. Adjacent subscripts (which must be equal for conformability) "cancel out," leaving the first and last subscripts as the order of the product. For example, the product

$$A_{r \times c} C_{c \times s} D_{s \times u}$$

is a matrix of order  $r \times u$ .

This notation also demonstrates what is by now readily apparent from the definition of matrix multiplication, namely that the product  $BA$  does not necessarily exist, even if  $AB$  does. For  $BA$  can be written as  $B_{c \times s} A_{r \times c}$ , which we see at once is a legitimate product only if  $s = r$ . Otherwise  $BA$  is not defined.

For example if  $A$  is a  $4 \times 5$  matrix and  $B$  is a  $5 \times 3$  matrix, then the product  $AB$  exists while the product  $BA$  does not exist.

There are therefore three situations regarding the product of two matrices  $A$  and  $B$ . If  $A$  is of order  $r \times c$  :

- (i)  $AB$  exists only if  $B$  has  $c$  rows.
- (ii)  $BA$  exists only if  $B$  has  $r$  columns.
- (iii)  $AB$  and  $BA$  both exist only if  $B$  is  $c \times r$ .

A corollary to (iii) is that  $A^2$  exists only when  $A$  is square. Another corollary is that both  $AB$  and  $BA$  always exist and are of the same order when  $A$  and  $B$  are square and of the same order. But as shall be shown subsequently, the two products are not necessarily equal. Their inequality will be discussed

when considering the commutative law of multiplication, but meanwhile we simply state that they are not in general equal.

As a means of distinction,  $AB$  is described as  $A$  postmultiplied by  $B$ , or as  $A$  multiplied on the right by  $B$ ; and  $BA$  is either  $A$  premultiplied by  $B$ , or  $A$  multiplied on the left by  $B$ .

Even when  $AB$  and  $BA$  both exist and are of the same order, they are not in general equal; for example,

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 0 & -1 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 2 & -3 \\ 4 & -7 \end{bmatrix} \neq \begin{bmatrix} 0 & -1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \\ = \begin{bmatrix} -3 & -4 \\ -2 & -2 \end{bmatrix}$$

But in certain cases  $AB$  and  $BA$  are equal; for example,

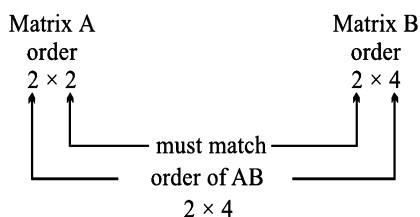
$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 0 & 2 \\ 3 & 3 \end{bmatrix} = \begin{bmatrix} 6 & 8 \\ 12 & 18 \end{bmatrix} = \begin{bmatrix} 0 & 2 \\ 3 & 3 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

### STUDY TIP

Since  $AB$  and  $BA$  are usually distinct, there can be no precision in the phrase 'multiply  $A$  by  $B$ ' until it is clear whether it shall mean  $AB$  or  $BA$ . Accordingly, we introduce the terms post-multiplication and pre-multiplication. The matrix  $A$  post-multiplied by  $B$  is the matrix  $AB$ ; the matrix  $A$  pre-multiplied by  $B$  is the matrix  $BA$ .

► **Example 2.** Suppose matrix  $A$  is  $2 \times 2$ , while matrix  $B$  is  $2 \times 4$ . Can the product  $AB$  be calculated? What is the order of the product?

► **Solution** The following diagram helps answer these questions.



The product  $AB$  can be calculated because  $A$  has two columns and  $B$  has two rows. The order of the product is  $2 \times 4$ . (However, the product  $BA$  could not be found.)

► **Example 3.** If

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 3 & 2 & 1 \\ 1 & 0 & 1 \end{bmatrix} \text{ and } B = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 0 & 1 & 2 \\ 3 & 1 & 0 & 5 \end{bmatrix}$$

then find  $AB$ . Does  $BA$  also exist?

► **Solution** The matrix  $A$  is of the type  $3 \times 3$  and the matrix  $B$  is of the type  $3 \times 4$ . Since the number of columns of  $A$  is equal to the number of rows of  $B$ , therefore  $AB$  is defined i.e., the product  $AB$  exists and it will be a matrix of the type  $3 \times 4$ .

$$\text{Let } AB = \begin{bmatrix} c_{11} & c_{12} & c_{13} & c_{14} \\ c_{21} & c_{22} & c_{23} & c_{24} \\ c_{31} & c_{32} & c_{33} & c_{34} \end{bmatrix}.$$

Then  $c_{11}$  = the sum of the products of the corresponding elements of the first row of  $A$  and the first column of  $B$ .

$c_{12}$  = the sum of the products of the corresponding elements of the first row of  $A$  and the second column of  $B$ .

$c_{13}$  = the sum of the products of the corresponding elements of the first row of  $A$  and the third column of  $B$ , and so on.

Therefore by the row by column rule of multiplication (rows of  $A$  multiplied by the columns of  $B$ ), we have

$$AB = \begin{bmatrix} 2 & 1 & 0 \\ 3 & 2 & 1 \\ 1 & 0 & 1 \end{bmatrix}_{3 \times 3} \times \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 0 & 1 & 2 \\ 3 & 1 & 0 & 5 \end{bmatrix}_{3 \times 4} \\ = \begin{bmatrix} 2.1 + 1.2 + 0.3 & 2.2 + 1.0 + 0.1 \\ 3.1 + 2.2 + 1.3 & 3.2 + 2.0 + 1.1 \\ 1.1 + 0.2 + 1.3 & 1.2 + 0.0 + 1.1 \\ 2.3 + 1.1 + 0.0 & 2.4 + 1.2 + 0.5 \\ 3.3 + 2.1 + 1.0 & 3.4 + 2.2 + 1.5 \\ 1.3 + 0.1 + 1.0 & 1.4 + 0.2 + 1.5 \end{bmatrix}$$

$$= \begin{bmatrix} 4 & 4 & 7 & 10 \\ 10 & 7 & 11 & 21 \\ 4 & 3 & 3 & 9 \end{bmatrix} 3 \times 4$$

Since the number of the columns of B is not equal to the number of rows of A, therefore the product BA does not exist.

► **Example 4.** A firm wants to promote its product in three ways : telephone, house calls, and letters. The cost per contact (in rupees) is given in matrix A as

$$A = \begin{bmatrix} 40 \\ 100 \\ 50 \end{bmatrix} \begin{array}{l} \text{Telephone} \\ \text{House call} \\ \text{Letter} \end{array}$$

The number of contacts of each type made in two cities X and Y is given by

$$B = \begin{bmatrix} 1000 & 500 & 5000 \\ 3000 & 1000 & 10,000 \end{bmatrix} \begin{array}{l} \text{Telephone} \quad \text{Housecall} \quad \text{Letter} \\ \rightarrow X \\ \rightarrow Y \end{array}$$

Find the total amount spent by the firm in the two cities X and Y.

► **Solution** We have

$$\begin{aligned} BA &= \begin{bmatrix} 40,000 + 50,000 + 250,000 \\ 120,000 + 100,000 + 500,000 \end{bmatrix} \begin{array}{l} \rightarrow X \\ \rightarrow Y \end{array} \\ &= \begin{bmatrix} 340,000 \\ 720,000 \end{bmatrix} \begin{array}{l} \rightarrow X \\ \rightarrow Y \end{array} \end{aligned}$$

So the total amount spent by the firm in the two cities is Rs. 340,000 and Rs. 720,000 respectively.

► **Example 5.** Express the following 3 equations in two unknowns in the form of a single matrix equation:

$$\begin{aligned} a_{11}x + a_{12}y &= b_1 \\ a_{21}x + a_{22}y &= b_2 \\ a_{31}x + a_{32}y &= b_3. \end{aligned}$$

► **Solution** Let  $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix} 3 \times 2$

and  $X = \begin{bmatrix} x \\ y \end{bmatrix} 2 \times 1$

$$\text{Then } AX = \begin{bmatrix} a_{11}x + a_{12}y \\ a_{21}x + a_{22}y \\ a_{31}x + a_{32}y \end{bmatrix} 3 \times 1 \quad \text{If } B = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} 3 \times 1,$$

then from our definition of equality of two matrices, the single matrix equation  $AX = B$  will be equivalent to the given system of 3 equations.

### Properties of Matrix Multiplication

- (i) If two matrices A and B square of order n then the product AB is also a square matrix of order n. This is called closure property of square matrices.
- (ii) Matrix multiplication is not commutative, in general i.e.  $AB \neq BA$ .
  - (a) Whenever AB and BA both exist, it is always not necessary that they should be matrices of the same order. For example if A be a  $5 \times 4$  matrix while B be a  $4 \times 5$  matrix, then AB exists and it is a  $5 \times 5$  matrix. In this case BA also exists and it is a  $4 \times 4$  matrix. Since the matrices AB and BA are not of the same order therefore we have  $AB \neq BA$ .
  - (b) If AB and BA are of different orders then naturally  $AB \neq BA$ . But one may think that perhaps AB and BA could be the same if they were of the same order. But it is not so, here we give an example to show that even if AB and BA are of same order they may not be same.

If  $A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$  and  $B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ , then

$$\begin{aligned} AB &= \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \\ &= \begin{bmatrix} 1.0 + 0.1 & 1.1 + 0.0 \\ 0.0 - 1.1 & 0.1 - 1.0 \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \end{aligned}$$

$$\text{and } BA = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} 0.1+1.0 & 0.0-1.1 \\ 1.1+0.0 & 1.0-0.1 \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

Hence  $AB \neq BA$ .

Thus, whenever  $AB$  and  $BA$  both exist and are matrices of the same order, it is not necessary that  $AB = BA$ .

- (c) It however does not imply that  $AB$  is never equal to  $BA$ . For example if

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 3 & 4 & 2 \\ 1 & 3 & 2 \end{bmatrix} \text{ and } B = \begin{bmatrix} 10 & -4 & -1 \\ -11 & 5 & 0 \\ 9 & -5 & 1 \end{bmatrix}$$

$$\text{then } AB = \begin{bmatrix} -3 & 1 & 0 \\ 4 & -2 & -1 \\ -5 & 1 & 1 \end{bmatrix} = BA.$$

**Whenever  $AB = BA$ , the matrices  $A$  and  $B$  are said to be commutative.**

**If  $AB = -BA$ , the matrices  $A$  and  $B$  are said to be anti-commutative.**

#### Caution

Since multiplication of square matrices is, in general, not commutative, be careful about the order when you multiply matrices.

When multiplying by matrices on both sides of a matrix equation, be careful to multiply in the same order on both sides of the equation, since multiplication of matrices is not commutative (unlike multiplication of real numbers).

- (iii) The multiplication of matrices is associative.

Let  $A = [a_{ij}]_{(m,n)}$ ,  $B = [b_{jk}]_{(n,p)}$ ,  $C = [c_{kr}]_{(p,q)}$ .

Then we have  $(AB)C = A(BC)$ .

- (iv) Matrix multiplication is distributive with respect to addition.

Let  $A = [a_{ik}]_{(m,n)}$ ,  $B = [b_{kj}]_{(n,p)}$ ,  $C = [c_{kj}]_{(n,p)}$ .

Here  $A$  is conformable to  $B$  and also to  $C$  for multiplication, and  $B$  is conformable to  $C$  for addition. Then we have  $A(B + C) = AB + AC$ . Also  $(A + B)(C + D) = AC + BC + AD + BD$ . To illustrate  $A(B + C) = AB + AC$ , let

$$A = \begin{pmatrix} 1 & -1 \\ 2 & 3 \end{pmatrix} \quad B = \begin{pmatrix} 0 & -2 \\ -4 & 1 \end{pmatrix} \quad C = \begin{pmatrix} -4 & 2 \\ 1 & 7 \end{pmatrix}$$

We have  $A(B + C)$

$$= \begin{pmatrix} 1 & -1 \\ 2 & 3 \end{pmatrix} \left( \begin{pmatrix} 0 & -2 \\ -4 & 1 \end{pmatrix} + \begin{pmatrix} -4 & 2 \\ 1 & 7 \end{pmatrix} \right)$$

$$= \begin{pmatrix} 1 & -1 \\ 2 & 3 \end{pmatrix} \begin{pmatrix} -4 & 0 \\ -3 & 8 \end{pmatrix} = \begin{pmatrix} -1 & -8 \\ -17 & 24 \end{pmatrix}$$

and  $AB + AC$

$$= \begin{pmatrix} 1 & -1 \\ 2 & 3 \end{pmatrix} \begin{pmatrix} 0 & -2 \\ -4 & 1 \end{pmatrix} +$$

$$\begin{pmatrix} 1 & -1 \\ 2 & 3 \end{pmatrix} \begin{pmatrix} -4 & 2 \\ 1 & 7 \end{pmatrix}$$

$$= \begin{pmatrix} 4 & -3 \\ -12 & 1 \end{pmatrix} + \begin{pmatrix} -5 & -5 \\ -5 & 25 \end{pmatrix} = \begin{pmatrix} -1 & -8 \\ -17 & 24 \end{pmatrix}$$

- (v) If  $A$  be an  $m \times n$  matrix,  $I_n$  denotes the  $n$ -rowed unit matrix, it can be easily seen that  $AI_n = A = I_m A$ .

Further, if  $S$  is a scalar matrix and  $A$  is any  $n$ -rowed square matrix, then  $AS = SA = kA$ , i.e., the pre-multiplication or the post-multiplication of  $A$  by  $S$  has the same effect as the multiplication of  $A$  by the scalar  $k$ .

$$\text{Let } A = \begin{pmatrix} 1 & -1 & 2 \\ 4 & -3 & 6 \end{pmatrix} \text{ and } I_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\text{Then } AI_3 = \begin{pmatrix} 1 & -1 & 2 \\ 4 & -3 & 6 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1+0+0 & 0-1+0 & 0+0+2 \\ 4+0+0 & 0-3+0 & 0+0+6 \end{pmatrix} = \begin{pmatrix} 1 & -1 & 2 \\ 4 & -3 & 6 \end{pmatrix} = A$$

Note that  $I_3 A$  is not defined in this example.

- (vi) If  $A$  be any  $m \times n$  matrix and  $O_{n \times p}$  be an  $n \times p$  null matrix, then  $AO_{n \times p} = O_{m \times p}$  is an  $m \times p$  null matrix. Similarly if  $O_{m \times n}$  be an  $m \times n$  null matrix and  $A$  be any  $n \times p$  matrix, then  $O_{m \times n} A = O_{m \times p}$ .



If  $A$  be any  $n$ -rowed square matrix, and  $O$  be an  $n$ -rowed null matrix then  $AO = OA = O$ .

- (vii) The equation  $AB = O$  does not necessarily imply that atleast one of the matrices  $A$  and  $B$  must be a zero matrix. In other words, the product of two matrices can be a zero matrix while neither of them is a zero matrix.

Let us find  $AB$ , if  $A = \begin{bmatrix} 0 & -1 \\ 0 & 2 \end{bmatrix}$  and

$$B = \begin{bmatrix} 3 & 5 \\ 0 & 0 \end{bmatrix}.$$

$$\text{We have } AB = \begin{bmatrix} 0 & -1 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 3 & 5 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

Thus, if the product of two matrices is a zero matrix, it is not necessary that one of the matrices is a zero matrix. We know that, for real numbers  $a, b$  if  $ab = 0$ , then either  $a = 0$  or  $b = 0$ . This need not be true for matrices, we will observe this through an example.

When a product  $AB = O$  but neither  $A$  nor  $B$  is  $O$ , then the factors  $A$  and  $B$  are called divisors of zero. Thus, in the algebra of matrices, there exist divisors of zero whereas in the algebra of complex numbers there do not.

- (viii) In the case of matrix multiplication if  $AB = O$ , then it does not necessarily imply that  $BA = O$ .

Let  $A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ , and  $B = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ . Then  $AB = O$ .

But  $BA$

$$\begin{aligned} &= \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1.0 + 0.0 & 1.1 + 0.0 \\ 0.0 + 0.0 & 0.1 + 0.0 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \neq \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}. \end{aligned}$$

Thus  $AB = O$  while  $BA \neq O$ .

- (ix) From  $AB = AC$  or  $BA = CA$  we cannot in general conclude that  $B = C$ , even if  $A \neq O$ ; that is, the cancellation law does not hold in general in multiplication.

We make another assault on the laws of scalar algebra by means of the following example.

$$\text{Let } A = \begin{bmatrix} 1 & 2 & 0 \\ 1 & 1 & 0 \\ -1 & 4 & 0 \end{bmatrix}, B = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 1 & -1 \\ 2 & 2 & 2 \end{bmatrix},$$

$$C = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 1 & -1 \\ 1 & 1 & 1 \end{bmatrix}.$$

$$\text{Then } AB = \begin{bmatrix} 3 & 4 & 1 \\ 2 & 3 & 2 \\ 3 & 2 & -7 \end{bmatrix} = AC.$$

Thus we can have  $AB = AC$  without having  $B = C$ . In other words, we cannot ordinarily cancel  $A$  from  $AB = AC$  even if  $A \neq O$ . However, there is an important special case when the cancellation is possible, as we shall see later.

A further illustration arises from observing that if  $A$  and  $B$  satisfy  $BA = 2B$ , then it can be rewritten as  $BA - 2B = 0$ , equivalent to  $B(A - 2I) = 0$ . But from this we cannot conclude either that  $A - 2I$  is  $O$  or that  $B = O$ .

Similarly, not even the equation  $X^2 = O$  means that  $X = O$ ; for example, with

$$X = \begin{bmatrix} 1 & 2 & 5 \\ 2 & 4 & 10 \\ -1 & -2 & -5 \end{bmatrix} \text{ we have } X^2 = O.$$

- (x)  $\text{tr}(AB) = \text{tr}(BA)$

We have  $AB = [c_{ij}]_{n \times n}$  where

$$c_{ij} = \sum_{k=1}^n a_{ik} b_{kj}.$$

Also  $BA = [d_{ij}]_{n \times n}$  where  $d_{ij} = \sum_{k=1}^n b_{ik} a_{kj}.$

$$\text{Now } \text{tr}(AB) = \sum_{i=1}^n c_{ii} = \sum_{i=1}^n \left( \sum_{k=1}^n a_{ik} b_{ki} \right)$$

$= \sum_{k=1}^n \sum_{i=1}^n a_{ik} b_{ki}$ , interchanging the order of summation in the last sum

$$= \sum_{k=1}^n \left( \sum_{i=1}^n b_{ki} a_{ik} \right) = \sum_{k=1}^n d_{kk} = \text{tr}(BA).$$

► **Example 6.** If  $A = \begin{bmatrix} 2 & 3 & 4 \\ 1 & 2 & 3 \\ -1 & 1 & 2 \end{bmatrix}$ ,

$$B = \begin{bmatrix} 1 & 3 & 0 \\ -1 & 2 & 1 \\ 0 & 0 & 2 \end{bmatrix}, \text{ the show that } AB \neq BA.$$

► **Solution** We have  $AB$

$$= \begin{bmatrix} 2 & 3 & 4 \\ 1 & 2 & 3 \\ -1 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 3 & 0 \\ -1 & 2 & 1 \\ 0 & 0 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 2.1 - 3.1 + 4.0 & 2.3 + 3.2 + 4.0 \\ 1.1 - 2.1 + 3.0 & 1.3 + 2.2 + 3.0 \\ -1.1 - 1.1 + 2.0 & -1.3 + 1.2 + 2.0 \end{bmatrix}$$

$$= \begin{bmatrix} 2.0 + 3.1 + 4.2 \\ 1.0 + 2.1 + 3.2 \\ -1.0 + 1.1 + 2.2 \end{bmatrix}$$

$$= \begin{bmatrix} -1 & 12 & 11 \\ -1 & 7 & 8 \\ -2 & -1 & 5 \end{bmatrix}$$

$$\text{Similarly, } BA = \begin{bmatrix} 1 & 3 & 0 \\ -1 & 2 & 1 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} 2 & 3 & 4 \\ 1 & 2 & 3 \\ -1 & 1 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 1.2 + 3.1 - 0.1 & 1.3 + 3.2 + 0.1 \\ -1.2 + 2.1 - 1.1 & -1.3 + 2.2 + 1.1 \\ 0.2 + 0.1 - 2.1 & 0.3 + 0.2 + 2.1 \end{bmatrix}$$

$$= \begin{bmatrix} 1.4 + 3.3 + 0.2 \\ -1.4 + 2.3 + 1.2 \\ 0.4 + 0.3 + 2.2 \end{bmatrix}$$

$$= \begin{bmatrix} 5 & 9 & 13 \\ -1 & 2 & 4 \\ -2 & 2 & 4 \end{bmatrix}$$

The matrix  $AB$  is of the type  $3 \times 3$  and the matrix  $BA$  is also of the type  $3 \times 3$ . But the corresponding elements of these matrices are not equal.

Hence  $AB \neq BA$ .

► **Example 7.** Under what conditions is the matrix equation  $A^2 - B^2 = (A - B)(A + B)$  is true ?

► **Solution** We have  $A^2 - B^2 = (A - B)(A + B)$

$$\Rightarrow A^2 - B^2 = A^2 + AB - BA - B^2$$

$$\Rightarrow A^2 - B^2 - (A^2 + AB - BA - B^2) = O$$

$$\Rightarrow BA - AB = O$$

$$\Rightarrow BA = AB.$$

Hence the given equation is true only if the matrices  $A$  and  $B$  commute.

► **Example 8.** If  $A$  and  $B$  are square matrices of order  $n$ , then prove that  $A$  and  $B$  will commute if and only if  $A - \lambda I$  and  $B - \lambda I$  commute for every scalar  $\lambda$ .

► **Solution** Suppose the matrices  $A$  and  $B$  commute i.e.,

$$AB = BA.$$

Then to prove that the matrices  $A - \lambda I$  and  $B - \lambda I$  commute for every scalar  $\lambda$ . We have.

$$(A - \lambda I)(B - \lambda I) = AB - \lambda AI - \lambda IB + \lambda^2 I^2$$

$$= AB - \lambda A - \lambda B + \lambda^2 I$$

$$= AB - \lambda(A + B) + \lambda^2 I$$

Similarly  $(B - \lambda I)(A - \lambda I) = BA - \lambda(A + B) + \lambda^2 I$ . But it is given that  $AB = BA$ . Therefore

$$(A - \lambda I)(B - \lambda I) = (B - \lambda I)(A - \lambda I) \text{ for every scalar } \lambda.$$

Conversely suppose that  $A - \lambda I$  and  $B - \lambda I$  commute for every scalar  $\lambda$ .

$$\text{Then } (A - \lambda I)(B - \lambda I) = (B - \lambda I)(A - \lambda I).$$

$$\therefore AB - \lambda(A + B) + \lambda^2 I = BA - \lambda(A + B) + \lambda^2 I$$

$$\Rightarrow AB = BA.$$

► **Example 9.** Show that for all values of  $p, q, r, s$

the matrices,  $P = \begin{bmatrix} p & q \\ -q & p \end{bmatrix}$ , and  $Q = \begin{bmatrix} r & s \\ -s & r \end{bmatrix}$

commute.

► **Solution** We have  $PQ$

$$= \begin{bmatrix} pr - qs & ps + qr \\ -qr - ps & -qs + pr \end{bmatrix}.$$

$$\text{Also } QP = \begin{bmatrix} r & s \\ -s & r \end{bmatrix} \begin{bmatrix} p & q \\ -q & p \end{bmatrix}$$

$$= \begin{bmatrix} rp - sq & rq + sp \\ -sp - rq & -sq + rp \end{bmatrix}$$

$$= \begin{bmatrix} pr - qs & ps + pq \\ -qr - ps & -qs + pr \end{bmatrix}$$

for all values of  $p, q, r, s$ .

Hence  $PQ = QP$ , for all values of  $p, q, r, s$ .

► **Example 10.** If  $A = \begin{bmatrix} 1 & -3 & 2 \\ 2 & 1 & -3 \\ 4 & -3 & -1 \end{bmatrix}$ ,

$$B = \begin{bmatrix} 1 & 4 & 1 & 0 \\ 2 & 1 & 1 & 1 \\ 1 & -2 & 1 & 2 \end{bmatrix} \text{ and } C$$

$$= \begin{bmatrix} 2 & 1 & -1 & -2 \\ 3 & -2 & -1 & -1 \\ 2 & -5 & -1 & 0 \end{bmatrix}$$

show that  $AB = AC$  though  $B \neq C$ .

► **Solution** We have

$$\begin{aligned} AB &= \begin{bmatrix} 1 & -3 & 2 \\ 2 & 1 & -3 \\ 4 & -3 & -1 \end{bmatrix} \times \begin{bmatrix} 1 & 4 & 1 & 0 \\ 2 & 1 & 1 & 1 \\ 1 & -2 & 1 & 2 \end{bmatrix} \\ &= \begin{bmatrix} 1.1 - 3.2 + 2.1 & 1.4 + 3.1 - 2.2 \\ 2.1 + 1.2 - 3.1 & 2.4 + 1.1 + 3.2 \\ 4.1 - 3.2 - 1.1 & 4.4 - 3.1 + 1.2 \end{bmatrix} \end{aligned}$$

$$\begin{bmatrix} 1.1 - 3.1 + 2.1 & 1.0 - 3.1 + 2.2 \\ 2.1 + 1.1 - 3.1 & 2.0 + 1.1 - 3.2 \\ 4.1 - 3.1 - 1.1 & 4.0 - 3.1 - 1.2 \end{bmatrix}$$

$$= \begin{bmatrix} -3 & -3 & 0 & 1 \\ 1 & 15 & 0 & -5 \\ -3 & 15 & 0 & -5 \end{bmatrix}.$$

$$\text{Also } AC = \begin{bmatrix} 1 & -3 & 2 \\ 2 & 1 & -3 \\ 4 & -3 & -1 \end{bmatrix} \times \begin{bmatrix} 2 & 1 & -1 & -2 \\ 3 & -2 & -1 & -1 \\ 2 & -5 & -1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} -3 & -3 & 0 & 1 \\ 1 & 15 & 0 & -5 \\ -3 & 15 & 0 & -5 \end{bmatrix}$$

∴  $AB = AC$ , though  $B \neq C$ .

► **Example 11.** Show that the pre-multiplication (or post-multiplication) of a square matrix  $A$  by a diagonal matrix  $D$  multiplies each row (or column) of  $A$  by the corresponding diagonal elements of  $D$ . Deduce that the only matrices commutative with a diagonal matrix with distinct diagonal elements are diagonal matrices.

► **Solution** Let  $A = [a_{ij}]_{n \times n}$  be a square matrix

$$\text{i.e. } A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix}.$$

$$\text{Let } D = \begin{bmatrix} b_{11} & 0 & 0 & 0 \\ 0 & b_{22} & 0 & 0 \\ \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & b_{nn} \end{bmatrix}$$

be a diagonal matrix of order  $n$ .

$$\text{Then } DA = \begin{bmatrix} b_{11}a_{11} & b_{11}a_{12} & \dots & b_{11}a_{1n} \\ b_{22}a_{21} & b_{22}a_{22} & \dots & b_{22}a_{2n} \\ \dots & \dots & \dots & \dots \\ b_{nn}a_{n1} & b_{nn}a_{n2} & \dots & b_{nn}a_{nn} \end{bmatrix}.$$

Thus in the product  $DA$  the first row of  $A$  has been multiplied by the corresponding diagonal element  $b_{11}$  of the first row of  $D$  and so on.

$$\text{Now } AD = \begin{bmatrix} a_{11}b_{11} & a_{12}b_{22} & \dots & a_{1n}b_{nn} \\ a_{21}b_{11} & a_{22}b_{22} & \dots & a_{2n}b_{nn} \\ \dots & \dots & \dots & \dots \\ b_{nn}a_{n1} & b_{nn}a_{n2} & \dots & b_{nn}a_{nn} \end{bmatrix}$$

From this it is obvious that in the product  $AD$  the first column of  $A$  has been multiplied by the corresponding diagonal element  $b_{11}$  of the first column of  $D$  and so on. Hence the given result is true for post-multiplication of  $A$  by  $D$ .

Second part : Let  $A = [a_{ij}]_{n \times n}$  commute with the diagonal matrix  $D$  of order  $n$  having its diagonal elements all distinct. Let  $D = \text{dia} [b_{11}, \dots, b_{nn}]$ .

The  $(i, j)^{\text{th}}$  element of  $DA = a_{ij}b_{ij}$ .

Also the  $(i, j)^{\text{th}}$  element of  $AD = a_{ij}b_{ij}$ .

But  $AD = DA$ .

$$\therefore a_{ij}b_{ii} = a_{ij}b_{jj} \quad \text{or} \quad a_{ij}(b_{ii} - b_{jj}) = 0.$$

But if  $i \neq j$ , then  $b_{ii} \neq b_{jj}$ . Therefore  $a_{ij} = 0$  if  $i \neq j$ .

Thus each non-diagonal element of  $A$  is zero.

Therefore  $A$  is a diagonal matrix.

► **Example 12.** Show that if a diagonal matrix is commutative with every matrix of the same order, then it is necessarily a scalar matrix.

► **Solution** Let  $D = \text{diag} [b_{11}, b_{22}, \dots, b_{nn}]$  be a diagonal matrix of order  $n$ . Let  $A = [a_{ij}]_{n \times n}$  be any square matrix of order  $n$ . Then it is given that  $AD = DA$ .

The  $(i, j)^{\text{th}}$  element of  $DA = a_{ij}b_{ii}$ .

Also the  $(i, j)^{\text{th}}$  element of  $AD = a_{ij}b_{jj}$ .

Since  $AD = DA$ , therefore

$a_{ij}b_{ii} = a_{ij}b_{jj}$  or  $a_{ij}(b_{ii} - b_{jj}) = 0$ . Since  $A$  is any square matrix of order  $n$ , therefore we can take  $a_{ij} \neq 0$ . Therefore we must have

$$b_{ii} - b_{jj} = 0 \quad \text{or} \quad b_{ii} = b_{jj} \quad \text{for each } i \text{ and } j.$$

Thus the diagonal elements of  $D$  are all equal. Therefore  $D$  is a scalar matrix.

## Practice Problems

[B]

1. Compute the indicated products.

$$(i) \begin{bmatrix} a & b \\ -b & a \end{bmatrix} \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$$

$$(ii) \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} [2 \ 3 \ 4]$$

$$(iii) \begin{bmatrix} 1 & -2 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{bmatrix}$$

$$(iv) \begin{bmatrix} 2 & 3 & 4 \\ 3 & 4 & 5 \\ 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} 1 & -3 & 4 \\ 0 & 2 & 4 \\ 3 & 0 & 5 \end{bmatrix}$$

$$2. \text{ If } A = \begin{bmatrix} 1 & -4 & 2 \\ -1 & 4 & -2 \end{bmatrix}, B = \begin{bmatrix} 1 & 2 \\ -1 & 3 \\ 5 & -2 \end{bmatrix},$$

$$C = \begin{bmatrix} 2 & 2 \\ 1 & -1 \\ 1 & -3 \end{bmatrix}, \text{ compute } B + C, AB,$$

$$BA, AC, CA, A(2B - 3C).$$

$$3. \text{ Let } A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, B = \begin{pmatrix} -2 & -2 \\ 4 & 4 \end{pmatrix},$$

$$C = \begin{pmatrix} -1 & -1 \\ 3 & 3 \end{pmatrix}$$

Show that  $AB = AC$  even though  $B \neq C$

4. Show that for all values of  $a, b, c, d$  the matrices

$$A = \begin{bmatrix} a & b \\ -b & a \end{bmatrix} \text{ and } B = \begin{bmatrix} c & d \\ -d & c \end{bmatrix} \text{ commute.}$$

5. Show that each of the matrices

$$\sigma_x = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix},$$

$$\sigma_y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}, \sigma_z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, (i^2 = -1),$$

anticommutes with the others.

6. Show that the product of two triangular matrices is itself triangular.  
7. Show that any two matrices which commute

with  $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ , commute with each other.

8. If  $A = \begin{bmatrix} \cos \theta_1 & -\sin \theta_1 \\ \sin \theta_1 & \cos \theta_1 \end{bmatrix}$ ,

$$B = \begin{bmatrix} \cos \theta_2 & -\sin \theta_2 \\ \sin \theta_2 & \cos \theta_2 \end{bmatrix}$$

show that  $AB = BA$ .

9. Using  $A = \begin{bmatrix} 1 & -1 & 1 \\ 2 & 0 & 1 \end{bmatrix}$ ,  $B = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 1 & 1 & 1 \end{bmatrix}$ ,

$$C = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}, \text{ test the rule } (AB)C = A(BC).$$

10. Let  $A, B$  be of order  $n$  and let  
 $C_1 = \alpha_1 A + \beta_1 B$ ,  
 $C_2 = \alpha_2 A + \beta_2 B$ ,  
where  $\alpha_1, \alpha_2, \beta_1, \beta_2$  are scalars such that  $\alpha_1 \beta_2 \neq \alpha_2 \beta_1$ . Show that  $C_1 C_2 = C_2 C_1$  if and only if  $AB = BA$ .

11. Find  $x$ , if  $[x \ -5 \ -1] \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix} \begin{bmatrix} x \\ 4 \\ 1 \end{bmatrix} = 0$

12. Find the matrix  $X$  so that  $X \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$

$$= \begin{bmatrix} -7 & -8 & -9 \\ 2 & 4 & 6 \end{bmatrix}$$

13. Given that  $A, B, C$  all have order  $n$ . Show that  
 $\text{tr } ABC = \text{tr } BCA = \text{tr } CAB = \text{tr } ACB$   
 $= \text{tr } BAC = \text{tr } CBA$ .

14. Let  $A = \begin{bmatrix} 0 & 1 \\ 0 & 2 \end{bmatrix}$ . Find all  $2 \times 2$  matrices  $B$

such that (a)  $AB = 0$ ; (b)  $BA = 0$ .

### 3.6 POSITIVE INTEGRAL POWERS OF MATRICES

The product  $AA$  is defined only when  $A$  is a square matrix. We shall denote this product by  $A^2$ . We shall write  $A^1 = A$ .

Now  $A^2 A = (AA)A = A(AA)$  [By associative law]  
 $= AA^2 = AAA$ .

We shall denote each of the above products by  $A^3$  so that  $A^3 = AAA = AA^2 = A^2 A$

The product of any number of matrices is associative i.e., it is independent of the manner in which the matrices may be bracketed for multiplication. Therefore it is quite legitimate to denote the product  $AAA \dots A$ ,  $m$  times by  $A^m$ , where  $m$  is any positive integer.

If  $m$  and  $n$  are any arbitrary positive integers, we have

$$\begin{aligned} A^m \cdot A^n &= (AA \dots A, m \text{ times}) \cdot (AAA \dots A, n \text{ times}) \\ &= AAA \dots A, m+n \text{ times [since the product of any number of matrices is associative]} \\ &= A^{m+n} \end{aligned}$$

Again  $(A^m)^n = A^m \cdot A^m \cdot A^m \dots A^m$ ,  $n$  times  
 $= (AAA \dots A, m \text{ times}) (AAA \dots A, m \text{ times}) \dots$   
upto  $n$  times  
 $= AAA \dots A, mn \text{ times} = (A)^{mn}$ .

The  $n$ th power of a square matrix  $A$  (where  $n$  is a positive integer) is written  $A^n$  and is defined as follows:

If  $n = 1$ ,  $A^1 = A$ ; for  $n \geq 2$ ,  $A^n = A(A^{n-1})$ .

► **Example 1.** Let  $A = \begin{pmatrix} 1 & 2 \\ 0 & 4 \end{pmatrix}$

► **Solution** Calculate  $A^2$ ,  $A^3$ , and  $A^6$ .

$$A^2 = \begin{pmatrix} 1 & 2 \\ 0 & 4 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 0 & 4 \end{pmatrix} = \begin{pmatrix} 1 & 10 \\ 0 & 16 \end{pmatrix}$$

$$A^3 = \begin{pmatrix} 1 & 2 \\ 0 & 4 \end{pmatrix} \begin{pmatrix} 1 & 10 \\ 0 & 16 \end{pmatrix} = \begin{pmatrix} 1 & 42 \\ 0 & 64 \end{pmatrix}$$

To calculate  $A^6$ , we try  $A^3 A^3$ , since laws of exponents hold for powers of matrices:

$$A^6 = A^3 A^3 = \begin{pmatrix} 1 & 42 \\ 0 & 64 \end{pmatrix} \begin{pmatrix} 1 & 42 \\ 0 & 64 \end{pmatrix} = \begin{pmatrix} 1 & 2730 \\ 0 & 4096 \end{pmatrix}$$

Notice that  $A$  is upper triangular and the powers we calculated are all upper triangular.

An important property of the power of a matrix is that  $A^m A^n = A^{m+n}$  for all positive integers  $m$  and  $n$ . This can be seen by using the associativity of multiplication and writing

$$A^m A^n = \underbrace{A \dots A}_{m \text{ times}} \underbrace{A \dots A}_{n \text{ times}} = \underbrace{A \dots A}_{m+n \text{ times}} = A^{m+n}$$

That  $(A^m)^n = A^{mn}$  for positive integers  $m$  and  $n$  is shown similarly.

► **Example 2.** If  $A_\alpha = \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix}$ , then show

that (i)  $(A_\alpha)^n = \begin{bmatrix} \cos n\alpha & \sin n\alpha \\ -\sin n\alpha & \cos n\alpha \end{bmatrix}$ , where  $n$  is a positive integer,

(ii)  $A_\alpha A_\beta = A_{\alpha+\beta}$ .

► **Solution**

(i) We shall prove the result by induction on  $n$ .

$$(A_\alpha)^1 = A_\alpha = \begin{bmatrix} \cos 1\alpha & \sin 1\alpha \\ -\sin 1\alpha & \cos 1\alpha \end{bmatrix}.$$

Thus the result is true when  $n = 1$ . Now suppose that the result is true for any positive integer  $n$ , i.e.,

$$(A_\alpha)^n = \begin{bmatrix} \cos n\alpha & \sin n\alpha \\ -\sin n\alpha & \cos n\alpha \end{bmatrix}.$$

We shall show that the result is true for  $n + 1$  if it is true for  $n$ .

We have,  $(A_\alpha)^{n+1} = (A_\alpha)^n A_\alpha$

$$\begin{aligned} &= \begin{bmatrix} \cos n\alpha & \sin n\alpha \\ -\sin n\alpha & \cos n\alpha \end{bmatrix} \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix} \\ &= \begin{bmatrix} \cos n\alpha \cos \alpha - \sin n\alpha \sin \alpha & \cos n\alpha \sin \alpha + \sin n\alpha \cos \alpha \\ -\sin n\alpha \cos \alpha - \cos n\alpha \sin \alpha & -\sin n\alpha \sin \alpha + \cos n\alpha \cos \alpha \end{bmatrix} \end{aligned}$$

$$= \begin{bmatrix} \cos(n+1)\alpha & \sin(n+1)\alpha \\ -\sin(n+1)\alpha & \cos(n+1)\alpha \end{bmatrix}.$$

Thus the result is true for  $n + 1$  if it is true for  $n$ . Now the proof is complete by induction.

$$\begin{aligned} \text{(ii) } A_\alpha A_\beta &= \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix} \begin{bmatrix} \cos \beta & \sin \beta \\ -\sin \beta & \cos \beta \end{bmatrix} \\ &= \begin{bmatrix} \cos \alpha \cos \beta - \sin \alpha \sin \beta & \cos \alpha \sin \beta + \sin \alpha \cos \beta \\ -\sin \alpha \cos \beta - \cos \alpha \sin \beta & -\sin \alpha \sin \beta + \cos \alpha \cos \beta \end{bmatrix} \\ &= \begin{bmatrix} \cos(\alpha + \beta) & \sin(\alpha + \beta) \\ \sin(\alpha + \beta) & \cos(\alpha + \beta) \end{bmatrix} = A_{\alpha+\beta}. \end{aligned}$$

► **Example 3.** If  $A = \begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix}$ , then prove that

$A^k = \begin{bmatrix} 1+2k & -4k \\ k & 1-2k \end{bmatrix}$ , where  $k$  is any positive integer.

► **Solution** We shall prove the result by induction on  $k$ .

$$\text{We have } A_1 = A = \begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 1+2.1 & -4.1 \\ 1 & 1-2.1 \end{bmatrix}.$$

Thus the result is true when  $k = 1$ .

Now suppose that the result is true for any positive

integer  $k$  i.e.,  $A^k = \begin{bmatrix} 1+2k & -4k \\ k & 1-2k \end{bmatrix}$  where  $k$  is any positive integer.

Now we shall show that the result is true for  $k + 1$  if it is true for  $k$ . We have

$$\begin{aligned} A^{k+1} &= A A^k = \begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1+2k & -4k \\ k & 1-2k \end{bmatrix} \\ &= \begin{bmatrix} 3+6k-4k & -12k-4+8k \\ 1+2k-k & -4k-1+2k \end{bmatrix} \\ &= \begin{bmatrix} 1+2+2k & -4-4k \\ 1+k & -2k-1 \end{bmatrix} \end{aligned}$$

$$= \begin{bmatrix} 1+2(k+1) & -4(1+k) \\ 1+k & 1-2(1+k) \end{bmatrix}.$$

Thus the result is true for  $k+1$  if it is true for  $k$ . But it is true for  $k=1$ . Hence by induction it is true for all positive integral value of  $k$ .

► **Example 4.** Find the possible square roots of the two rowed unit matrix  $I$ .

► **Solution** Let  $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$  be square root of the matrix  $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ . Then  $A^2 = I$ .

$$\text{i.e. } \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\text{i.e. } \begin{bmatrix} a^2 + bc & ab + bd \\ ac + cd & cb + d^2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Since the above matrices are equal, therefore

$$a^2 + bc = 1 \quad \dots(1)$$

$$ab + bd = 0 \quad \dots(2)$$

$$ac + cd = 0 \quad \dots(3)$$

$$cb + d^2 = 0 \quad \dots(4)$$

must hold simultaneously.

If  $a + d = 0$ , the above four equations hold simultaneously if  $d = -a$  and  $a^2 + bc = 1$ .

Hence one possible square root of  $I$  is

$$A = \begin{bmatrix} \alpha & \beta \\ \gamma & -\alpha \end{bmatrix} \text{ where } \alpha, \beta, \gamma \text{ are any three numbers related by the condition } \alpha^2 + \beta\gamma = 1.$$

If  $a + d \neq 0$ , the above four equations hold simultaneously if  $b = 0, c = 0, a = 1, d = 1$  or if  $b = 0, c = 0, a = -1, d = -1$ .

$$\text{Hence } \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$

i.e.  $\pm I$  are other possible square roots of  $I$ .

### Polynomial functions of matrices

Consider a polynomial

$f(x) = a_p x^p + a_{p-1} x^{p-1} + \dots + a_1 x + a_0$  of degree  $p$  in an indeterminate  $x$ , with either real or complex coefficients. With any such polynomial,

we can associate a polynomial function of an  $n \times n$  matrix  $A$ :

$f(A) = a_p A^p + a_{p-1} A^{p-1} + \dots + a_1 A + a_0 I_n$  simply by replacing the constant term  $a_0$  by the scalar matrix  $a_0 I_n$  and by replacing  $x$  by  $A$  throughout.

For example, if  $f(x) = x^2 - 3x - 2$ , then, for any matrix  $A$  of order  $n$ ,

$$f(A) = A^2 - 3A - 2I_n.$$

Now suppose  $f(x)$  and  $g(x)$  have only real coefficients. Then if  $f(x) + g(x) = h(x)$  and  $f(x) \cdot g(x) = q(x)$ ,  $h(x)$  and  $q(x)$  are again polynomials with real coefficients. Moreover, since scalars commute with matrices, since the distributive and associative laws hold, and since the exponent laws hold for positive integral powers of  $A$ , we may conclude also that

$$f(A) + g(A) = h(A) \text{ and } f(A) \cdot g(A) = q(A).$$

The matrix polynomial  $f(A)$  is factorable in the same manner as the scalar polynomial  $f(x)$ . For example, if

$$f(x) = x^2 - 3x + 2 = x(x-1)(x-2)$$

then  $f(A) = A^2 - 3A + 2I_n = A(A - I_n)(A - 2I_n)$ .

Now suppose that the solutions of

$$f(x) = a_p x^p + a_{p-1} x^{p-1} + \dots + a_1 x + a_0 = 0$$

are  $\alpha_1, \alpha_2, \dots, \alpha_p$  so that

$$f(x) \equiv a_p (x - \alpha_1)(x - \alpha_2) \dots (x - \alpha_p).$$

Hence, by  $f(A) \equiv a_p (A - \alpha_1 I)(A - \alpha_2 I) \dots (A - \alpha_p I)$ , the matrix equation  $f(A) = 0$  has for solutions the scalar matrices  $\alpha_1 I, \dots, \alpha_p I$ . Moreover, it has no other scalar matrices as solutions. However, since the product of matrices may be zero even though no factor is zero, the equation  $f(A) = 0$  may also have non-scalar matrices as solutions. For example, the equation

$$A^2 + I_2 = 0$$

has for solutions the scalar matrices

$$\begin{bmatrix} i & 0 \\ 0 & i \end{bmatrix}, \begin{bmatrix} -i & 0 \\ 0 & -i \end{bmatrix},$$

corresponding to the solutions  $i$  and  $-i$  of the equation  $x^2 + 1 = 0$ .

It also has (among others) the non-scalar solution

$$\begin{bmatrix} 1 & 2 \\ -1 & -1 \end{bmatrix}, \text{ as can be verified.}$$

The fact that in scalar algebra a polynomial

equation of degree  $p$  in a single unknown  $x$  has exactly  $p$  solutions therefore does not hold true in matrix algebra, where the number of solutions is always atleast  $p$ .

► **Example 5.** Find real numbers  $c_1$  and  $c_2$  so that

$I + c_1 M + c_2 M^2 = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$  where  $M = \begin{bmatrix} 1 & 3 \\ 0 & 2 \end{bmatrix}$  and  $I$  is the identity matrix.

► **Solution**  $M^2 = \begin{bmatrix} 1 & 3 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 9 \\ 0 & 4 \end{bmatrix}$

$$I + c_1 M + c_2 M^2 = \begin{bmatrix} 1+c_1+c_2 & 3c_1+9c_2 \\ 0 & 1+2c_1+4c_2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\Rightarrow c_1 + c_2 = -1 \text{ and } 3(c_1 + c_2) + 6c_2 = 0$$

$$\Rightarrow c_2 = 1/2, c_1 = -3/2.$$

► **Example 6.** If  $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$ , show that

$$A^2 - 4A - 5I = O.$$

► **Solution** We have  $A^2$

$$= \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 1+4+4 & 2+2+4 & 2+4+2 \\ 2+2+4 & 4+1+4 & 4+2+2 \\ 2+4+2 & 4+2+2 & 4+4+1 \end{bmatrix} = \begin{bmatrix} 9 & 8 & 8 \\ 8 & 9 & 8 \\ 8 & 8 & 9 \end{bmatrix}.$$

Now  $A^2 - 4A - 5I$

$$= \begin{bmatrix} 9 & 8 & 8 \\ 8 & 9 & 8 \\ 8 & 8 & 9 \end{bmatrix} - 4 \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix} - 5 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Now  $A^2 - 4A - 5I$

$$= \begin{bmatrix} 9 & 8 & 8 \\ 8 & 9 & 8 \\ 8 & 8 & 9 \end{bmatrix} - \begin{bmatrix} 4 & 8 & 8 \\ 8 & 4 & 8 \\ 8 & 8 & 4 \end{bmatrix} - \begin{bmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

### Binomial Theorem

If  $A$  and  $B$  be  $m$ -rowed square matrices which commute and  $n$  be a positive integer we have the binomial theorem :

$$(A+B)^n = {}^nC_0 A^n + {}^nC_1 A^{n-1} B + \dots + {}^nC_r A^{n-r} B^r + \dots + {}^nC_n B^n.$$

*Proof:*

We have  $A+B = A^1 + B^1$ .

Now  $(A+B)^2 = (A+B)(A+B)$

$$= A^2 + AB + BA + B^2, \quad \text{by distributive law}$$

$$= A^2 + 2AB + B^2, \text{ since } AB = BA$$

$$= {}^2C_0 A^2 + {}^2C_1 AB + {}^2C_2 B^2.$$

Thus the theorem is true for  $n=2$ .

Now assume that the theorem is true for  $n$  i.e.

$$(A+B)^n = {}^nC_0 A^n + {}^nC_1 A^{n-1} B + \dots + {}^nC_r A^{n-r} B^r + \dots + {}^nC_{r+1} A^{n-r-1} B^{r+1} + \dots + {}^nC_n B^n.$$

Then  $(A+B)^{n+1} = (A+B)(A+B)^n$

$$= (A+B)({}^nC_0 A^n + {}^nC_1 A^{n-1} B + \dots + {}^nC_r A^{n-r} B^r + \dots + {}^nC_{r+1} A^{n-r-1} B^{r+1} + \dots + {}^nC_n B^n)$$

$$= {}^nC_0 A^{n+1} + ({}^nC_0 BA^n + {}^nC_1 A^n B) + \dots$$

$$+ ({}^nC_r BA^{n-r} B^r + {}^nC_{r+1} A^{n-r} B^{r+1}) + \dots + {}^nC_n B^{n+1}.$$

Now  $AB = BA$ . We can prove by induction that for every positive integer  $n$ ,  $BA^n = A^n B$ .

$$\text{Again } AB^{n-r} B^r = (BA^{n-r}) B^r = (A^{n-r} B) B^r$$

$$= A^{n-r} BB^r = A^{n-r} B^{r+1}.$$

$$\text{Also } {}^nC_0 = {}^{n+1}C_0 = 1, {}^nC_n = {}^{n+1}C_{n+1} = 1,$$

$$\text{and } {}^nC_r + {}^nC_{r+1} = {}^{n+1}C_{r+1}.$$

$$\text{Hence } (A+B)^{n+1} = {}^{n+1}C_0 A^{n+1} + ({}^nC_0 + {}^nC_1)$$

$$A^n B + \dots$$

$$+ ({}^nC_r + {}^nC_{r+1}) A^{n-r} B^{r+1} + \dots + {}^{n+1}C_{n+1} B^{n+1}$$

$$= {}^{n+1}C_0 A^{n+1} + {}^{n+1}C_1 A^n B + \dots$$

$$+ {}^{n+1}C_{r+1} A^{n-r} B^{r+1} + \dots + {}^{n+1}C_{n+1} B^{n+1}.$$

Thus the theorem is true for  $n+1$ , if it is true for  $n$ .

But it is true for  $n=2$ . Hence it is true for all positive integral values of  $n$ .

► **Example 7.** If  $A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ , prove that

$(aI + bA)^n = a^n I + na^{n-1} bA$ . for " $a, b \in \mathbb{R}$  where  $I$  is the two rowed unit matrix  $n$  is a positive integer.

► **Solution**  $A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$



$$\Rightarrow A^2 = A \cdot A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} = 0$$

$$\Rightarrow A^3 = A^2 \cdot A = 0 \Rightarrow A^2 = A^3 = A^4 = \dots A^n = 0$$

Now by binomial theorem

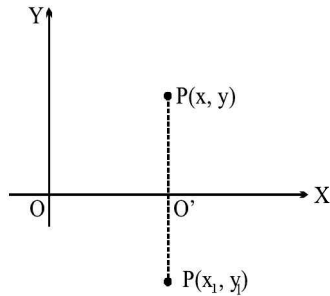
$$\begin{aligned} (aI + bA)^n &= (aI)^n + {}^nC_1(aI)^{n-1}bA \\ &+ {}^nC_2(aI)^{n-2}(bA)^2 + \dots + {}^nC_n(bA)^n \\ &= a^n I + {}^nC_1 a^{n-1} b IA + {}^nC_2 a^{n-2} b^2 I A^2 + \dots \\ &\quad \dots + {}^nC_n b^n A^n \\ &= a^n I + n a^{n-1} b A + 0 \dots \quad (\because A^n = 0) \\ \Rightarrow (aI + bA)^n &= a^n I + n a^{n-1} b A. \end{aligned}$$

### Transformation Matrices

(i) **Reflection in the x axis :** Let  $P(x, y)$  be any point and  $P'(x_1, y_1)$  be the image after reflection

in the x-axis, then  $\begin{cases} x_1 = x \\ y_1 = -y \end{cases}$  ( $O'$  is the midpoint of  $P$  &  $P'$ )

These may be rewritten as  $\begin{cases} x_1 = 1x + 0.y \\ y_1 = 0.x + (-1).y \end{cases}$



The system of equations in the matrix form are written as below :

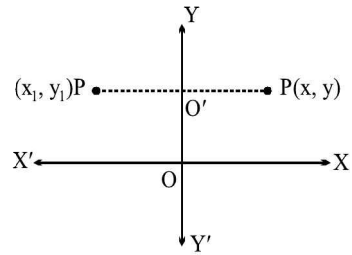
$$\begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}.$$

Thus, the matrix  $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$  describes the reflection of a point  $P(x, y)$  in the x-axis.

(ii) **Reflection in the y-axis :** Let  $P(x, y)$  be any point and  $P'(x_1, y_1)$  be its image after reflection

in the y-axis, then  $\begin{cases} x_1 = -x \\ y_1 = y \end{cases}$  ( $O'$  is the midpoint of  $P$  &  $P'$ )

These may be written as  $\begin{cases} x_1 = (-1).x + 0.y \\ y_1 = 0.x + 1.y \end{cases}$



These system of equations in the matrix form

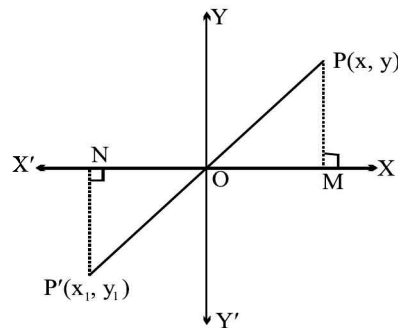
$$\text{are written as } \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

Thus the matrix  $\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$  describes the reflection of point  $P(x, y)$  in the y-axis.

(iii) **Reflection through the origin :** Let  $P(x, y)$  be any point and  $P'(x_1, y_1)$  be its image after reflection through the origin, then

$$\begin{cases} x_1 = -x \\ y_1 = -y \end{cases} \quad (O \text{ is the midpoint of } P \text{ & } P')$$

These may be written as  $\begin{cases} x_1 = (-1).x + 0.y \\ y_1 = 0.x + (-1).y \end{cases}$



These system of equations in the matrix form

are written as 
$$\begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

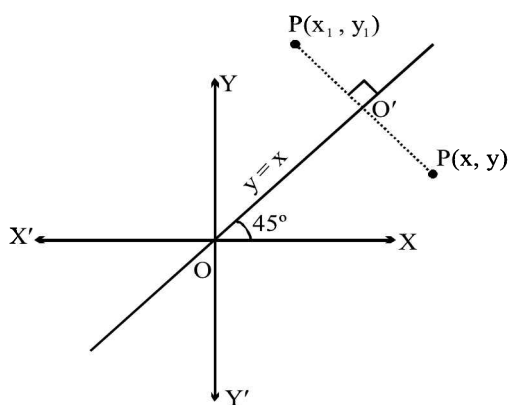
Thus the matrix  $\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$  describes the

reflection of a point of a point  $P(x, y)$  through the origin.

- (iv) **Reflection in the line  $y = x$  :** Let  $P(x, y)$  be any point and  $P'(x_1, y_1)$  be its image after reflection in the line  $y = x$ . then

$$\begin{cases} x_1 = y \\ y_1 = x \end{cases} \quad (O' \text{ is the midpoint of } P \text{ and } P')$$

These may be written as 
$$\begin{cases} x_1 = 0.x + 1.y \\ y_1 = 1.x + 0.y \end{cases}$$



These systems of equations in the matrix

form are written as 
$$\begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

Thus the matrix  $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$  describes the

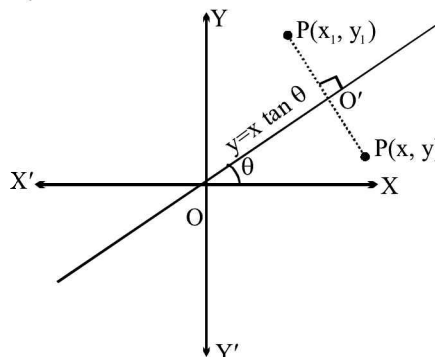
reflection of a point  $P(x, y)$  in the line  $y = x$ .

- (v) **Reflection in the line  $y = x \tan \theta$  :** Let  $P(x, y)$  be any point and  $P'(x_1, y_1)$  be its image after reflection in this line  $y = x \tan \theta$  .. then

$$\begin{cases} x_1 = x \cos 2\theta + y \sin 2\theta \\ y_1 = x \sin 2\theta - y \cos 2\theta \end{cases} \quad (O' \text{ is the midpoint of } P \text{ and } P').$$

These may be written as

$$\begin{cases} x_1 = x \cdot \cos 2\theta + y \cdot \sin 2\theta \\ y_1 = x \cdot \sin 2\theta + y \cdot (-\cos 2\theta) \end{cases}$$



These systems of equations in the matrix form

are written as 
$$\begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = \begin{bmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

Thus the matrix  $\begin{bmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{bmatrix}$  describes

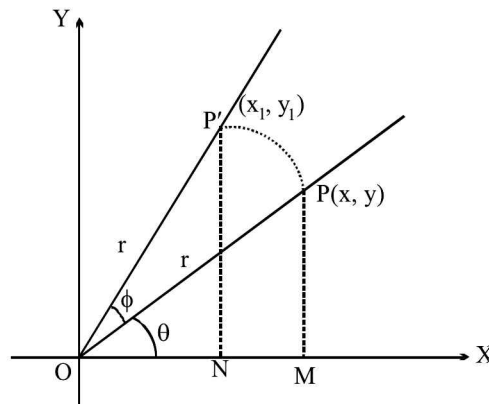
the reflection of a point  $P(x, y)$  in the line  $y = x \tan \theta$ .

- (vi) **Rotation Through an Angle  $\theta$**

Let  $P(x, y)$  be any point such that  $OP = r$  &  $\angle POX = \phi$ . Let  $OP$  rotate through an angle  $\theta$  in the anticlockwise direction such that  $P'(x_1, y_1)$  is the new position.

$$\therefore OP' = r \quad (\because OP = OP')$$

then 
$$\begin{cases} x_1 = x \cos \theta - y \sin \theta \\ y_1 = x \sin \theta + y \cos \theta \end{cases}$$



these system of equations in the matrix form

$$\text{are written as } \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

Thus the matrix  $\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$  describes a rotation of a line segment through an angle  $\theta$ .

► **Example 8.** Find the matrices of transformations  $T_1 T_2$  and  $T_2 T_1$ , when  $T_1$  is rotation through an angle  $60^\circ$  and  $T_2$  is the reflection in the  $y$ -axis. Also verify that  $T_1 T_2 \neq T_2 T_1$ .

► **Solution**  $T_1 = \begin{bmatrix} \cos 60^\circ & -\sin 60^\circ \\ \sin 60^\circ & \cos 60^\circ \end{bmatrix}$

$$= \begin{bmatrix} 1/2 & -\sqrt{3}/2 \\ \sqrt{3}/2 & 1/2 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 & -\sqrt{3} \\ \sqrt{3} & 1 \end{bmatrix}$$

and  $T_2 = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$

$$\therefore T_1 T_2 = \frac{1}{2} \begin{bmatrix} 1 & -\sqrt{3} \\ \sqrt{3} & 1 \end{bmatrix} \times \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} -1 & -\sqrt{3} \\ -\sqrt{3} & 1 \end{bmatrix} = \begin{bmatrix} -1/2 & -\sqrt{3}/2 \\ -\sqrt{3}/2 & 1/2 \end{bmatrix} \dots (1)$$

and  $T_2 T_1 = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \times \frac{1}{2} \begin{bmatrix} 1 & -\sqrt{3} \\ \sqrt{3} & 1 \end{bmatrix}$

$$= \frac{1}{2} \begin{bmatrix} -1 & \sqrt{3} \\ \sqrt{3} & 1 \end{bmatrix} = \begin{bmatrix} -1/2 & \sqrt{3}/2 \\ \sqrt{3}/2 & 1/2 \end{bmatrix} \dots (2)$$

It is clear from (1) and (2),  $T_1 T_2 \neq T_2 T_1$

► **Example 9.** Write down  $2 \times 2$  matrix  $A$  which corresponds to a counterclockwise rotation of  $60^\circ$  about the origin. A square OABC has vertices  $O(0, 0)$ ,  $A(2, 0)$ ,  $B(2, 2)$ ,  $O(0, 2)$ . The square is rotated counterclockwise about  $O$  through  $60^\circ$ . Find the co-ordinates of the vertices of the square after rotation.

► **Solution** The matrix describes a rotation through an angle  $60^\circ$  in counterclockwise direction is

$$\begin{bmatrix} \cos 60^\circ & -\sin 60^\circ \\ \sin 60^\circ & \cos 60^\circ \end{bmatrix} = \begin{bmatrix} 1/2 & -\sqrt{3}/2 \\ \sqrt{3}/2 & 1/2 \end{bmatrix}$$

$$= \frac{1}{2} \begin{bmatrix} 1 & -\sqrt{3} \\ \sqrt{3} & 1 \end{bmatrix}$$

Let after rotation  $A$  map into  $A'$ ,  $B$  map into  $B'$ ,  $C$  map into  $C'$  but the  $O$  map into itself.

If co-ordinates of  $A'$ ,  $B'$  &  $C'$  are  $(x', y')$ ,  $(x'', y'')$  and  $(x''', y''')$  respectively.

$$\therefore \begin{bmatrix} x' \\ y' \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 & -\sqrt{3} \\ \sqrt{3} & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 0 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 2 \\ 2\sqrt{3} \end{bmatrix} = \begin{bmatrix} 1 \\ \sqrt{3} \end{bmatrix}$$

$$\therefore x' = 1, y' = \sqrt{3}$$

$$\Rightarrow A(2, 0) \rightarrow A'(1, \sqrt{3})$$

$$\text{and } \begin{bmatrix} x'' \\ y'' \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 & -\sqrt{3} \\ \sqrt{3} & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 2 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 2 - 2\sqrt{3} \\ 2\sqrt{3} + 2 \end{bmatrix} = \begin{bmatrix} 1 - \sqrt{3} \\ \sqrt{3} + 1 \end{bmatrix}$$

$$\therefore x'' = 1 - \sqrt{3}, y'' = \sqrt{3} + 1$$

$$\Rightarrow B(2, 2) \rightarrow B'(1 - \sqrt{3}, \sqrt{3} + 1)$$

$$\begin{bmatrix} x''' \\ y''' \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 & -\sqrt{3} \\ \sqrt{3} & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 2 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} -2\sqrt{3} \\ 2 \end{bmatrix} = \begin{bmatrix} -\sqrt{3} \\ 1 \end{bmatrix}$$

$$\therefore x''' = -\sqrt{3}, y''' = 1$$

$$\Rightarrow C(0, 2) \rightarrow C'(-\sqrt{3}, 1)$$

### Practice Problems

[C]

1. (a) If  $A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ , show that  $A^2 = 0$ .

(b) If  $A = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$ , show that  $A^3 = 0$ .

- (c) Let  $A$  be  $n \times n$  with  $a_{ij} = 1$  if  $j = i + 1$  and  $a_{ij} = 0$  if  $j \neq i + 1$ . What can you say about  $A^n$ ?

2. Let  $A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$ . Verify that

$A^2 = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$ . Compute  $A^3$  and  $A^4$ . Guess a general formula for  $A^n$  and prove it by induction.

3. Evaluate  $\begin{pmatrix} \cos \theta + \sin \theta & \sqrt{2} \sin \theta \\ -\sqrt{2} \sin \theta & \cos \theta - \sin \theta \end{pmatrix}^n$
4. (a) Show that any two diagonal matrices of the same order commute.  
 (b) Show that if  $D$  is diagonal with non-negative elements, then  $AD^p = D^pA$  if and only if  $AD = DA$ .
5. If  $A$  and  $B$  are  $n$ -rowed square matrices. show that
- $(A + B)^2 = A^2 + AB + BA + B^2$ .
  - $(A + B)^3 = A^3 + BA^2 + ABA + B^2A + A^2B + BAB + AB^2 + B^3$
  - $(A + B)(A - B) = A^2 - AB + BA - B^2$
  - $(A - B)^2 = A^2 - AB - BA + B^2$
  - $(A - B)(A + B) = A^2 + AB - BA - B^2$ .
- What do these formulae become when  $A$  and  $B$  commute?

6. Explain why in matrix algebra  
 $(A + B)^2 \neq A^2 + 2AB + B^2$ ,  
 $(A + B)(A - B) \neq A^2 - B^2$ ,  
 except in special cases. Under what circumstances would equality hold?
7. Prove by induction that if  $B, C$  are of order  $n$  and if  $A = B + C$ ,  $C^2 = 0$ , and  $BC = CB$ , then for every positive integer  $k$ ,  
 $A^{k+1} = B^k(B + (k + 1)C)$ .
8. Show by induction that, if  $A$  is square and  $AB = \lambda B$ , where  $\lambda$  is a scalar, then  $A^p B = \lambda^p B$  for every positive integer  $p$ .

9. Given that  $AB = BA$ , show that, for all positive integers  $r$  and  $s$ ,  $A^r B^s = B^s A^r$ .

10. Find the matrix polynomial  $f(A)$  where  $f(x) = 3x^2 - 2x + 5$  of the

matrix  $A = \begin{pmatrix} 1 & -2 & 3 \\ 2 & -4 & 1 \\ 3 & -5 & 2 \end{pmatrix}$ .

11. If  $A = \begin{bmatrix} 3 & -2 \\ 4 & -2 \end{bmatrix}$  and  $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ , find  $k$  so that  $A^2 = kA - 2I$

12. Show that any two polynomial functions of a square matrix  $A$  commute.
13. If  $B = \lambda A + \mu I$ , where  $\lambda$  and  $\mu$  are numbers, then prove that  $BA = AB$ .

14. Let  $A = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}$ . Prove that  $A^2 = 2A - I$  and compute  $A^{100}$ .

15. Find all scalar matrices which satisfy the matrix equations :

- $A^2 - 5A + 4I_2 = 0$
- $2A^3 + 3A^2 - 4A - 6I_2 = 0$
- $A^3 - I_n = 0$

16. Show that the nonscalar matrix

$$A = \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}$$

is a solution of the equation  
 $A^2 - 5A + 7I_2 = 0$

### 3.7 TRANSPOSE OF A MATRIX

Let  $A = [a_{ij}]_{m \times n}$ . Then the transpose of  $A$  is denoted by  $A'$  and is defined as

$$A' = [b_{ij}]_{n \times m} \text{ where } b_{ij} = a_{ji} \forall i \text{ and } j.$$

i.e.  $A'$  is obtained by rewriting all the rows of  $A$  as columns (or by rewriting all the columns of  $A$  as rows).

For example, the transpose of the  $3 \times 4$  matrix

$$A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \\ 3 & 4 & 2 & 1 \end{bmatrix}_{3 \times 4}$$

is the  $4 \times 3$  matrix

$$A' = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 4 & 2 \\ 4 & 1 & 1 \end{bmatrix}_{4 \times 3}$$

The first row of A is the first column of A'. The second row of A is the second column of A'. The third row of A is the third column of A'.

Whenever matrices are related as  $A' = B$ , each is said to be the transpose of the other.

The transpose is written as A', although the notations  $A^t$  and  $A^T$  are also seen in the literature.

An obvious consequence of the transpose operation is that the rows of A' are the same as the columns of A, and if A is  $m \times n$ , the order of A' is  $n \times m$ .

► **Example 1.** Calculate  $A^T$  for the following matrices:

$$(i) A = \begin{pmatrix} 1 & -2 & 2 \\ 3 & 0 & 4 \\ -6 & \pi & 2 \end{pmatrix} \quad (ii) A = \begin{pmatrix} 3 & 1 & 1 \\ 0 & 2 & 5 \end{pmatrix}$$

► **Solution**

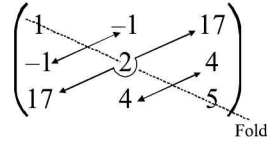
$$(i) A^T = \begin{pmatrix} 1 & 3 & -6 \\ -2 & 0 & \pi \\ 2 & 4 & 2 \end{pmatrix}$$

1st row of A      2nd row of A      3rd row of A

$$(ii) A^T = \begin{pmatrix} 3 & 0 \\ 1 & 2 \\ 1 & 5 \end{pmatrix}$$

Note that A was  $2 \times 3$  and  $A^T$  is  $3 \times 2$ .

If a matrix is symmetric we notice that it has a "mirror image" property i.e. if it is folded along the main diagonal, the numbers on top of each other are equal.



### Properties of Transpose

- (i) For any matrix  $A = [a_{ij}]_{m \times n}$ ,  $(A')' = A$
- (ii) Let  $\lambda$  be a scalar and A be a matrix. Then  $(\lambda A)' = \lambda A'$
- (iii)  $(A + B)' = A' + B'$  and  $(A - B)' = A' - B'$  for two comparable matrices A and B.

*Proof :*

Let A and B be  $m \times n$ . The  $ij$  entry of  $(A + B)^T = ji$  entry of  $A + B = ji$  entry of A +  $ji$  entry of B =  $ij$  entry of  $A^T + ij$  entry of  $B^T = ij$  entry of  $A^T + B^T$ .

- (iv)  $(A_1 \pm A_2 \pm \dots \pm A_n)' = A_1' \pm A_2' \pm \dots \pm A_n'$ , where  $A_j$  are comparable matrices.
- (v) Let  $A = [a_{ij}]_{m \times p}$  &  $B = [b_{ij}]_{p \times n}$ , then  $(AB)' = B'A'$

*Proof:*

Let A be  $m \times n$  and let B be  $n \times p$ .

$$\begin{aligned} ij \text{ entry of } (AB)^T &= ji \text{ entry of } AB = \sum_{k=1}^n a_{jk} b_{ki} \\ &= \sum_{k=1}^n b_{ki} a_{jk} = \sum_{k=1}^n [(ik \text{ entry of } B^T) (kj \text{ entry of } A^T)] \\ &= ij \text{ entry of } B^T A^T \end{aligned}$$

$$\text{Let } A = \begin{pmatrix} 1 & 0 \\ -1 & 2 \end{pmatrix} \quad B = \begin{pmatrix} 3 & 7 \\ 6 & 4 \end{pmatrix}$$

$$\text{Then } (AB)^T = \begin{pmatrix} 3 & 7 \\ 9 & 1 \end{pmatrix}^T = \begin{pmatrix} 3 & 9 \\ 7 & 1 \end{pmatrix}$$

$$B^T A^T = \begin{pmatrix} 3 & 6 \\ 7 & 4 \end{pmatrix}^T = \begin{pmatrix} 1 & -1 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} 3 & 9 \\ 7 & 1 \end{pmatrix}$$

The above law is called the reversal law for transpose i.e. the transpose of the product is the product of the transposes taken in the reverse order.

► **Example 2.** If  $A = \begin{bmatrix} 2 & 3 \\ 0 & 1 \end{bmatrix}$ ,  $B = \begin{bmatrix} 3 & 4 \\ 2 & 1 \end{bmatrix}$ , verify

that  $(AB)' = B'A'$ .

► **Solution** We have

$$\begin{aligned}
 AB &= \begin{bmatrix} 2 & 3 \\ 0 & 1 \end{bmatrix} \times \begin{bmatrix} 3 & 4 \\ 2 & 1 \end{bmatrix} \\
 &= \begin{bmatrix} 2.3 + 3.2 & 2.4 + 3.1 \\ 0.3 + 1.2 & 0.4 + 1.1 \end{bmatrix} \\
 &= \begin{bmatrix} 12 & 11 \\ 2 & 1 \end{bmatrix} \\
 \therefore (AB)' &= \begin{bmatrix} 12 & 2 \\ 11 & 1 \end{bmatrix} \\
 \text{Also } B'A' &= \begin{bmatrix} 3 & 2 \\ 4 & 1 \end{bmatrix} \times \begin{bmatrix} 2 & 0 \\ 3 & 1 \end{bmatrix} \\
 &= \begin{bmatrix} 3.2 + 2.3 & 3.0 + 2.1 \\ 4.2 + 1.3 & 4.0 + 1.1 \end{bmatrix} = \begin{bmatrix} 12 & 2 \\ 11 & 1 \end{bmatrix} \\
 \text{Hence } (AB)' &= B'A'.
 \end{aligned}$$

(vi)  $(A_1 A_2 \dots A_n)' = A_n' \cdot A_{n-1}' \dots A_2' \cdot A_1'$ ,  
provided the product is defined.

Recall that if a sum of squares of real numbers is zero, then each of the numbers is zero i.e. for real numbers  $x_1, x_2, \dots, x_n$ ,  $\sum_{i=1}^n x_i^2 = 0$  implies  $x_1 = 0 = x_2 = \dots = x_n$ . This is the basis for proving, for any real matrix  $A$ , that

$$A'A = O \text{ implies } A = O \quad \dots(1)$$

$$\text{and } \text{tr}(A'A) = 0 \text{ implies } A = O \quad \dots(2)$$

Result (1) is true because  $A'A = O$  implies that every diagonal element of  $A'A$  is zero; and for the  $j$ th such element this means, by, that the inner product of the  $j$ th column of  $A$  with itself is zero; i.e., for

$A$  of order  $m \times n$ , that  $\sum_{k=1}^m a_{kj}^2 = 0$  for  $j = 1, \dots, n$ . Hence  $a_{kj} = 0$  for  $k = 1, \dots, m$ , and this is true for  $j = 1, \dots, n$ . Therefore,  $A = O$ .

Proof of (2) is similar :

$$\begin{aligned}
 \text{tr}(A'A) &= \sum_{j=1}^n (\text{jth diagonal element of } A'A) = \\
 &= \sum_{j=1}^n \sum_{k=1}^m a_{kj}^2
 \end{aligned}$$

and so  $\text{tr}(A'A) = 0$  implies that every  $a_{kj}$  is zero; i.e., that every element of  $A$  is zero. Hence  $A = O$ . The above result is often helpful.

For example, by means of (1) we can prove, for real matrices  $P, Q$  and  $X$ , that

$$PXX' = QXX' \text{ implies } PX = QX \quad \dots(3)$$

The proof consists of observing that

$$(PXX' - QXX')(P' - Q') \equiv (PX - QX)(PX - QX)' \quad \dots(4)$$

Hence if  $PXX' = QXX'$ , the left-hand side of (4) is null and so, therefore, is the right-hand side; hence by (1),  $PX - QX = 0$ ; i.e.,  $PX = QX$ , and (3) is established.

### Symmetric and Skew-Symmetric Matrices

A square matrix  $A$  is said to be symmetric if  $A' = A$  i.e. if  $A = [a_{ij}]_n$  then  $A$  is symmetric iff  $a_{ij} = a_{ji} \forall i \& j$ . A square matrix  $A$  is said to be skew-symmetric if  $A' = -A$  i.e. if  $A = [a_{ij}]_n$  then  $A$  is skew-symmetric iff  $a_{ij} = -a_{ji} \forall i \& j$ .

e.g.  $A = \begin{bmatrix} a & h & g \\ h & b & f \\ g & f & c \end{bmatrix}$  is a symmetric matrix.

$$B = \begin{bmatrix} 0 & x & y \\ -x & 0 & z \\ -y & -z & 0 \end{bmatrix} \text{ is a skew-symmetric}$$

matrix.



(i) In skew-symmetric matrix all the diagonal elements are zero.

$$\because a_{ij} = -a_{ji} \Rightarrow a_{ii} = 0$$

(ii) For any square matrix  $A$ ,  $A + A'$  is symmetric &  $A - A'$  is skew-symmetric.

(iii) Every square matrix can be uniquely expressed as a sum of two square matrices of which one is symmetric and the other is skew-symmetric.

$$A = B + C, \text{ where } B = \frac{1}{2} (A + A') \&$$

$$C = \frac{1}{2} (A - A')$$

- (iv) The sum of skew-symmetric matrices is also skew-symmetric.  
 (v) The square of a skew-symmetric matrix is not skew-symmetric.  
 (vi) The cube of a skew-symmetric matrix is also skew-symmetric.  
 (vii) The product of a skew-symmetric and a symmetric matrix is skew-symmetric when the product is commutative.  
 (viii) If  $A$  is a symmetric (skew-symmetric) matrix, then  $kA$  is also symmetric (skew-symmetric).

► **Example 3.** Prove that, if  $A, B$  are symmetric (skew-symmetric), then so is also  $A + B$ .

► **Solution** (i) Let  $A$  and  $B$  be two symmetric matrices of the same order. Then  $A' = A$  and  $B' = B$ .

$$\text{Now } (A + B)' = A' + B' = A + B.$$

Since  $(A + B)' = A + B$ , therefore  $A + B$  is a symmetric matrix.

- (ii) Let  $A$  and  $B$  be two skew-symmetric matrices of the same order. Then  $A' = -A$  and  $B' = -B$ .

$$\text{Now } (A + B)' = A' + B' = (-A) + (-B) = -(A + B).$$

Since  $(A + B)' = -(A + B)$ , therefore  $A + B$  is a skew-symmetric matrix.

► **Example 4.** Show that every square matrix is uniquely expressible as the sum of a symmetric matrix and a skew-symmetric matrix.

► **Solution** Let  $A$  be any square matrix. We can write

$$A = \frac{1}{2} (A + A') + \frac{1}{2} (A - A') = P + Q, \text{ say,}$$

$$\text{where } P = \frac{1}{2} (A + A') \text{ and } Q = \frac{1}{2} (A - A').$$

$$\text{We have } P' = \left\{ \frac{1}{2} (A + A') \right\}'$$

$$= \frac{1}{2} (A + A')' \quad [\because (kA)' = kA']$$

$$= \frac{1}{2} \{A' + (A')'\} \quad [\because (A + B)' = A' + B']$$

$$= \frac{1}{2} (A' + A) \quad [\because (A')' = A]$$

$$= \frac{1}{2} (A + A') = P.$$

Therefore  $P$  is a symmetric matrix.

$$\text{Again } Q' = \left\{ \frac{1}{2} (A - A') \right\}' = \frac{1}{2} (A - A')' = \frac{1}{2} \{A' - (A')'\}$$

$$= \frac{1}{2} (A' - A) = -\frac{1}{2} (A - A') = -Q.$$

therefore  $Q$  is a skew-symmetric matrix.

Thus we have expressed the square matrix  $A$  as the sum of a symmetric and a skew-symmetric matrix.

To prove that the representation is unique, let  $A = R + S$  be another such representation of  $A$ , where  $R$  is symmetric and  $S$  skew-symmetric. Then to prove that  $R = P$  and  $S = Q$ .

$$\begin{aligned} \text{We have } A' &= (R + S)' = R' + S' \\ &= R - S \quad [\because R' = R \text{ and } S' = -S]. \end{aligned}$$

$$\therefore A + A' = 2R \text{ and } A - A' = 2S.$$

$$\text{This gives } R = \frac{1}{2} (A + A') \text{ and } S = \frac{1}{2} (A - A').$$

Thus  $R = P$  and  $S = Q$ . Therefore the representation is unique.

► **Example 5.** Express the matrix  $B$

$$B = \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & 0 \end{bmatrix} \text{ as the sum of a symmetric and a skew symmetric matrix.}$$

$$\text{► **Solution** Here } B' = \begin{bmatrix} 2 & -1 & 1 \\ -2 & 3 & -2 \\ -4 & -2 & 0 \end{bmatrix}$$

$$\text{Let } P = \frac{1}{2} (B + B') = \frac{1}{2} \begin{bmatrix} 4 & -3 & -3 \\ -3 & 6 & 2 \\ -3 & 2 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & \frac{-3}{2} & \frac{-3}{2} \\ \frac{-3}{2} & 3 & 1 \\ \frac{-3}{2} & 1 & 0 \end{bmatrix}$$

$$\text{Now } P' = \begin{bmatrix} 2 & \frac{-3}{2} & \frac{-3}{2} \\ \frac{-3}{2} & 3 & 1 \\ \frac{-3}{2} & 1 & 0 \end{bmatrix} = P$$

Thus  $P = \frac{1}{2}(B + B')$  is a symmetric matrix.

$$\text{Also, let } Q = \frac{1}{2}(B - B') = \frac{1}{2} \begin{bmatrix} 0 & -1 & -5 \\ 1 & 0 & 6 \\ 5 & -6 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & \frac{-1}{2} & \frac{-5}{2} \\ \frac{1}{2} & 0 & 3 \\ \frac{5}{2} & -3 & 0 \end{bmatrix}$$

$$\text{Then } Q' = \begin{bmatrix} 0 & \frac{1}{2} & \frac{5}{2} \\ \frac{-1}{2} & 0 & -3 \\ \frac{-5}{2} & -3 & 0 \end{bmatrix} = -Q$$

Thus  $Q = \frac{1}{2}(B - B')$  is a skew symmetric matrix.

$$\begin{aligned} \text{Now } P + Q &= \begin{bmatrix} 2 & \frac{-3}{2} & \frac{-3}{2} \\ \frac{-3}{2} & 3 & 1 \\ \frac{-3}{2} & 1 & 0 \end{bmatrix} + \begin{bmatrix} 0 & \frac{-1}{2} & \frac{-5}{2} \\ \frac{1}{2} & 0 & 3 \\ \frac{5}{2} & -3 & 0 \end{bmatrix} \\ &= \begin{bmatrix} 2 & -2 & 4 \\ -1 & 3 & 4 \\ 1 & -2 & 0 \end{bmatrix} = B \end{aligned}$$

Thus, B is represented as the sum of a symmetric and a skew symmetric matrix.

► **Example 6.** If A and B are symmetric matrices, then show that AB is symmetric if and only if A and B commute i.e.,  $AB = BA$ .

► **Solution** It is given that A and B are two symmetric matrices.

Therefore  $A' = A$  and  $B' = B$ .

Now suppose that  $AB = BA$ .

Then to prove that AB is symmetric.

We have  $(AB)' = B'A'$

$$= BA \quad [\because A' = A, B' = B]$$

$$= AB \quad [\because AB = BA]$$

Since  $(AB)' = AB$ , therefore AB is a symmetric matrix.

Conversely suppose that AB is a symmetric matrix. Then to prove that  $AB = BA$ .

We have  $AB = (AB)' \quad [\because AB \text{ is a symmetric matrix}]$

$$= B'A' = BA.$$

$$\text{With } A = \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix} \text{ and } B = \begin{bmatrix} 3 & 7 \\ 7 & 6 \end{bmatrix},$$

$$(AB)' = \begin{bmatrix} 17 & 19 \\ 27 & 32 \end{bmatrix} = \begin{bmatrix} 17 & 27 \\ 19 & 32 \end{bmatrix} = BA \neq AB$$

► **Example 7.** If A is any matrix, then prove that  $AA'$  and  $A'A$  are both symmetric matrices.

► **Solution** Let A be any matrix.

We have  $(AA')' = (A')' A'$  [By the reversal law for transpose]

$$= AA' \quad [\because (A')' = A]$$

Since  $(AA')' = AA'$ , therefore  $AA'$  is a symmetric matrix.

Again  $(A'A)' = A' (A')' = A'A$ .

Since  $(A'A)' = A'A$ , therefore  $A'A$  is a symmetric matrix.

► **Example 8.** Show that the matrix  $B'AB$  is symmetric or skew-symmetric according as A is symmetric or skew-symmetric.

► **Solution**

**Case I.** Let A be a symmetric matrix.

Then  $A' = A$ .

$$\text{Now } (B'AB)' = B'A'(B')' = B'A'B = B'AB$$

Hence  $B'AB$  is a symmetric matrix.

**Case II.** Let A be a skew-symmetric matrix.

Then  $A' = -A$ .

$$\text{Now } (B'AB)' = B'A'(B')' = B'A'B = B'(-A)B$$

$$= -(B'A)B = -B'AB.$$

Hence  $B'AB$  is a skew-symmetric matrix.

► **Example 9.** Show that all positive integral powers of a symmetric matrix are symmetric.

► **Solution** Let A be a symmetric matrix of order n. Then

$A^m = AAA \dots A$  upto m times, where m is a positive integer.

$$\text{Now } (A^m)' = (AAA \dots A \text{ upto m times})'$$

$$= (A'A'A' \dots A' \text{ upto m times})$$

$$= (AAA \dots A \text{ upto m times}) \quad [\because A' = A]$$

$$= A^m.$$

Hence  $A^m$  is also a symmetric matrix.



► **Example 10.** Show that positive odd integral powers of a skew-symmetric matrix are skew-symmetric while positive even integral powers are symmetric.

► **Solution** Let  $A$  be a skew-symmetric matrix. Then  $A' = -A$ .

Now let  $m$  be a positive integer. We have

$(A^m)' = (AAA \dots \text{upto } m \text{ times})' = A'A'A' \dots \text{upto } m \text{ times}$

$$= (-A)(-A) \dots \text{upto } m \text{ times} = (-1)^m A^m$$

$$= -A^m \text{ or } A^m \text{ according as } m \text{ is odd or even.}$$

Thus if  $m$  is an odd positive integer, then

$(A^m)' = -A^m$  and so  $A^m$  is skew-symmetric. If  $m$  is an even positive integer, then  $(A^m)' = A^m$  and so  $A^m$  is symmetric.

### Conjugate of a matrix

The matrix obtained from any given matrix  $A$  on replacing its elements by the corresponding conjugate complex numbers is called the conjugate of  $A$  and is denoted by  $\bar{A}$ .

Thus if  $A = [a_{ij}]_{m \times n}$ , then  $\bar{A} = [\bar{a}_{ij}]_{m \times n}$  where  $\bar{a}_{ij}$  denotes the conjugate complex of  $a_{ij}$ .

$$\text{If } A = \begin{bmatrix} 2+3i & 4-7i & 8 \\ -i & 6 & 9+i \end{bmatrix},$$

$$\text{then } \bar{A} = \begin{bmatrix} 2-3i & 4+7i & 8 \\ -i & 6 & 9-i \end{bmatrix}$$

If  $\bar{A}$  and  $\bar{B}$  be the conjugates of  $A$  and  $B$  respectively, then

- (i)  $\overline{(\bar{A})} = A$ ;
- (ii)  $\overline{(A+B)} = \bar{A} + \bar{B}$ ;
- (iii)  $\overline{(kA)} = k\bar{A}$ ,  $k$  being any complex number;
- (iv)  $\overline{(AB)} = \bar{A}\bar{B}$ ,  $A$  and  $B$  being conformable to multiplication.

### Transposed conjugate of matrix

The transpose of the conjugate of a matrix  $A$  is called transposed conjugate of  $A$  and is denoted by  $A^\theta$  or by  $A^*$ .

Obviously the conjugate of the transpose of  $A$  is the same as the transpose of the conjugate of  $A$  i.e.,

$$\overline{(A')} = (\bar{A})' = A^\theta.$$

If  $A = [a_{ij}]_{m \times n}$ , then  $A^\theta = [b_{ji}]_{n \times m}$  where  $b_{ji} = \bar{a}_{ij}$  i.e., the  $(j, i)^{\text{th}}$  element of  $A^\theta$  = the conjugate of the  $(i, j)^{\text{th}}$  element of  $A$ .

$$\text{If } A = \begin{bmatrix} 1+2i & 2-3i & 3+4i \\ 4-5i & 5+6i & 6-7i \\ 8 & 7+8i & 7 \end{bmatrix},$$

$$\text{then } A' = \begin{bmatrix} 1+2i & 4-5i & 8 \\ 2-3i & 5+6i & 7+8i \\ 3+4i & 6-7i & 7 \end{bmatrix}$$

$$\text{and } \overline{(A')} = A^\theta = \begin{bmatrix} 1-2i & 4+5i & 8 \\ 2+3i & 5-6i & 7-8i \\ 3-4i & 6+7i & 7 \end{bmatrix}.$$

If  $A^\theta$  and  $B^\theta$  be the transposed conjugates of  $A$  and  $B$  respectively, then

- (i)  $(A^\theta)^\theta = A$ ;
- (ii)  $(A+B)^\theta = A^\theta + B^\theta$ ,  $A$  and  $B$  being of the same order.
- (iii)  $(kA)^\theta = \bar{k}A^\theta$ ,  $k$  being any complex number;
- (iv)  $(AB)^\theta = B^\theta A^\theta$ ,  $A$  and  $B$  being conformable to multiplication.

► **Example 11.** For the following matrices, calculate the conjugate and conjugate transpose.

$$A = \begin{pmatrix} 2 & 1+i \\ 1-i & 3 \end{pmatrix} \quad B = \begin{pmatrix} 1 & -2 & 0 \\ 7 & -2 & 6 \\ 1 & 5 & 0 \end{pmatrix}$$

$$\text{► Solution } \bar{A} = \begin{pmatrix} \bar{2} & \overline{1+i} \\ \overline{1-i} & \bar{3} \end{pmatrix} = \begin{pmatrix} 2 & 1-i \\ 1+i & 3 \end{pmatrix}$$

$$A^* = \bar{A}^T = \begin{pmatrix} 2 & 1+i \\ 1-i & 3 \end{pmatrix}$$

$$\bar{B} = B, \quad B^* = \bar{B}^T = B^T = \begin{pmatrix} 1 & 7 & 1 \\ -2 & -2 & 5 \\ 0 & 6 & 0 \end{pmatrix}$$

### Hermitian matrix

A square matrix  $A = [a_{ij}]$  is said to be Hermitian if the  $(i, j)^{\text{th}}$  element of  $A$  is equal to the conjugate of the  $(j, i)^{\text{th}}$  element of  $A$ .

$i$ th element of  $A$  i.e., if  $a_{ij} = \bar{a}_{ij}$  for all  $i$  and  $j$ .

For example,

$$\begin{bmatrix} a & b+ic \\ b-ic & d \end{bmatrix}, \begin{bmatrix} 1 & 2-3i & 3+4i \\ 2+3i & 0 & 4-5i \\ 3-4i & 4+5i & 2 \end{bmatrix}$$

are Hermitian matrices.

If  $A$  is a Hermitian matrix, then  $a_{ij} = \bar{a}_{ij}$ , by definition.

$\therefore a_{ii}$  is real for all  $i$ . Thus every diagonal element of a Hermitian matrix must be real.

A Hermitian matrix over the field of real numbers is nothing but a real symmetric matrix.

Obviously a necessary and sufficient condition for a matrix  $A$  to be Hermitian is that  $A = A^H$ .

► **Example 12.** Prove that

$$A = \begin{bmatrix} \ell & \alpha + i\beta & p + iq \\ \alpha - i\beta & m & a + ib \\ p - iq & a - ib & n \end{bmatrix} \text{ is a hermitian matrix.}$$

► **Solution**  $(A^T) = \begin{bmatrix} \ell & \alpha - i\beta & p - iq \\ \alpha + i\beta & m & a - ib \\ p + iq & a + ib & n \end{bmatrix}$

$$\Rightarrow (\bar{A}^T) = \begin{bmatrix} \ell & \alpha - i\beta & p + iq \\ \alpha - i\beta & m & a + ib \\ p - iq & a - ib & n \end{bmatrix} = A$$

Hence  $A$  is hermitian.

### Skew-Hermitian matrix

A square matrix  $A = [a_{ij}]$  is said to be skew-Hermitian if the  $(i, j)$ th element of  $A$  is equal to the negative of the conjugate of the  $(j, i)$ th element of  $A$  i.e., if

$$a_{ij} = -\bar{a}_{ji} \text{ for all } i \text{ and } j.$$

If  $A$  is a skew-Hermitian matrix, then

$$a_{ii} = \bar{a}_{ii}, \text{ by definition.}$$

$$\therefore a_{ii} + \bar{a}_{ii} = 0$$

i.e.,  $a_{ii}$  must be either a purely imaginary number or must be zero. Thus the diagonal elements of a skew-Hermitian matrix must be purely imaginary numbers or zero.

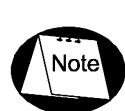
For example,

$$\begin{pmatrix} 0 & -2-i \\ 2-i & 0 \end{pmatrix}, \begin{pmatrix} -i & 3+4i \\ -3+4i & 0 \end{pmatrix} \text{ are}$$

skew-hermitian matrices.

A skew-hermitian matrix over the field of real numbers is nothing but a real skew-symmetric matrix.

Obviously a necessary and sufficient condition for a matrix  $A$  to be skew-Hermitian is that  $\bar{A}^T = -A$ .



Every square matrix (with complex elements) can be uniquely expressed as the sum of a hermitian and skew hermitian matrices

$$\text{i.e. } A = \frac{1}{2}(A + \bar{A}^T) + \frac{1}{2}(A - \bar{A}^T)$$

► **Example 13.** Prove that

$$A = \begin{bmatrix} 2i & -2-3i & -2+i \\ 2-3i & -i & 3i \\ 2+i & 3i & 0 \end{bmatrix}$$

is a skew-hermitian matrix.

► **Solution**  $A^T = \begin{bmatrix} 2i & -2-3i & 2+i \\ -2-3i & -i & 3i \\ -2+i & 3i & 0 \end{bmatrix}$

$$\therefore \bar{A}^T = \begin{bmatrix} -2i & 2+3i & 2-i \\ -2+3i & -i & -3i \\ -2-i & -3i & 0 \end{bmatrix}$$

$$= - \begin{bmatrix} 2i & -2-3i & -2+i \\ 2-3i & -i & 3i \\ 2+i & 3i & 0 \end{bmatrix} = -A$$

Hence  $A$  is a skew hermitian matrix.

### Practice Problems

[D]

1. If  $B = \begin{bmatrix} 1/\sqrt{6} & 1/\sqrt{3} & 1/\sqrt{2} \\ -2/\sqrt{6} & 1/\sqrt{3} & 0 \\ 1/\sqrt{6} & 1/\sqrt{3} & -1/\sqrt{2} \end{bmatrix}$ , then show that  $BB' = B'B = I_3$ ;

2. If  $A = \begin{bmatrix} 3 & -4 \\ 1 & 1 \\ 2 & 0 \end{bmatrix}$ ,  $B = \begin{bmatrix} 2 & 1 & 2 \\ 1 & 3 & 4 \end{bmatrix}$ ,

verify that  $(AB)' = B'A'$ .

3. What are the transposes of

$A = \begin{bmatrix} 1 & 2 & 3 \\ 6 & 8 & 4 \end{bmatrix}$  and  $B = \begin{bmatrix} 1 & 6 \\ 8 & 9 \\ 2 & -7 \end{bmatrix}$  ?

Calculate  $A + B'$  and  $A' + B$ , and explain the relationship between these two sums.

4. Prove that the matrix  $A^2$  is symmetric if either  $A$  is symmetric or  $A$  is skew-symmetric.
5. Explain why  $AVA' = BVB'$  implies that  $A$  and  $B$  have the same order and  $V$  is square.
6. Express the following matrices as the sum of a symmetric and a skew symmetric matrix :

(i)  $\begin{bmatrix} 3 & 5 \\ 1 & -1 \end{bmatrix}$  (ii)  $\begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$

7. Show that if  $X'X = X$ , then  $X = X' = X^2$ .
8. Explain why each of the following statements is true.
  - (a)  $(A + A')B = AB + A'B$ .
  - (b)  $\text{tr}(B'B) = \text{tr}(BB')$
  - (c)  $x'B'Bx = (Bx)'Bx$ .

Verify each statement for the following values of  $A$ ,  $B$  and  $x$  :

$A = \begin{bmatrix} -1 & 2 \\ 7 & 3 \end{bmatrix}$ ,  $B = \begin{bmatrix} 6 & 8 & -1 & 0 \\ 2 & 3 & 1 & 4 \end{bmatrix}$ ,

$x' = [1 \ 1 \ 2 \ 3]$ .

9. (a) When does  $(A + B)(A - B) = A^2 - B^2$  ?
- (b) When  $A = A'$ , prove that  $\text{tr}(AB) = \text{tr}(AB')$ .
- (c) When  $X'XGX'X = X'X$ , prove that  $X'XGX'X = X'X$ .
10. Show that if  $A$  and  $B$  are symmetric square matrices of the same order, then the matrix  $C = ABAB \dots ABA$  is symmetric.

11. If  $A$  and  $B$  are skew-symmetric matrices of order  $n$ , then show that  $AB$  is symmetric if and only if  $A$  and  $B$  commute.

12. If  $A$  and  $B$  are symmetric matrices of order  $n$ , then show that  $AB + BA$  is symmetric and  $AB - BA$  is skew-symmetric.

### 3.8 FURTHER TYPES OF MATRICES

#### 1. Involutory Matrix:

A square matrix  $A$  is called involutory provided it satisfies the relation  $A^2 = I$ , where  $I$  is identity matrix.

For instance,  $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$  is an

involutory matrix.

► **Example 1.** Show that the matrix

$A = \begin{bmatrix} -5 & -8 & 0 \\ 3 & 5 & 0 \\ 1 & 1 & -1 \end{bmatrix}$  is involutory.

► **Solution**  $A^2 = A \cdot A = \begin{bmatrix} -5 & -8 & 0 \\ 3 & 5 & 0 \\ 1 & 1 & -1 \end{bmatrix} \times \begin{bmatrix} -5 & -8 & 0 \\ 3 & 5 & 0 \\ 1 & 1 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I$

Hence the given matrix  $A$  is involutory.

► **Example 2.** Prove that a matrix  $A$  is involutory, if and only if  $(I + A)(I - A) = O$ .

► **Solution** Since  $A$  is an involutory matrix  $A^2 = I$ .

$\therefore I - A^2 = O$

$\therefore I^2 - A^2 = O \quad [\because I^2 = I]$

$\therefore I^2 - A^2 = O. [\because AI = IA]$

Conversely if  $(I - A)(I - A) = O$

then  $I^2 - IA + AI - A^2 = O$

or  $I - A^2 + AI - AI = O$

$[\because AI = IA]$

or  $I - A^2 + O = O$

$$\begin{aligned}\text{or } I - A^2 &= O \\ \text{or } A^2 &= I.\end{aligned}$$

## 2. Periodic Matrix:

A square matrix  $A$  is called periodic, if  $A^{k+1} = A$ , where  $k$  is a positive integer. If  $k$  is the least positive integer for which  $A^{k+1} = A$ , then  $k$  is said to be period of  $A$ . For  $k = 1$ , we get  $A^2 = A$  and we called it to be idempotent matrix.

► **Example 3.** Show that the matrix

$$A = \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix} \text{ is periodic.}$$

► **Solution** We have  
 $A^2 = AA$

$$\begin{aligned}&= \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix} \times \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix} \\&= \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix} = A.\end{aligned}$$

Since  $A^2 = A$ , therefore the matrix  $A$  is periodic with period 1.

## 3. Idempotent matrix:

If  $A^2 = A$ , then  $A$  is called idempotent matrix.

(a) If  $AB = A$  and  $BA = B$ , then both of  $A$  and  $B$  are idempotent.

By associative law  $ABA = (AB)A = AA = A^2$  and  $ABA = A(BA) = AB = A$ .

∴  $A^2 = A$  and  $A$  is idempotent.

Similarly  $BAB = (BA)B = BB = B^2$  and

$BAB = B(AB) = BA = B$

∴  $B^2 = B$  and  $B$  is idempotent.

(b) If  $A$  is idempotent, then  $B = I - A$  is idempotent and  $AB = BA = O$ . Here  $A^2 = A$ .

$$\begin{aligned}\text{Now } B^2 &= (I - A)^2 = I^2 - IA - AI + A^2 \\ &= I - 2A + A^2\end{aligned}$$

$$= I - 2A + A = I - A = B.$$

Therefore,  $B$  is idempotent.

Again  $AB = AI - A^2 = A - A = O$  and

$$BA = IA - A^2 = A - A = O.$$

When  $K$  is idempotent, all powers of  $K$  equal  $K$ ; i.e.  $K^r = K$  for  $r$  being a positive integer,

and  $(I - K)$  is idempotent. Thus

$$K^2 = K \text{ implies } (I - K)^2 = I - K,$$

but  $K - I$  is not idempotent.

A product of two idempotent matrices is idempotent if the matrices commute in multiplication.

► **Example 4.** Show that  $A = \begin{bmatrix} 2 & -3 & -5 \\ -1 & 4 & 5 \\ 1 & -3 & -4 \end{bmatrix}$

is idempotent.

► **Solution** Here

$$\begin{aligned}A^2 &= \begin{bmatrix} 2 & -3 & -5 \\ -1 & 4 & 5 \\ 1 & -3 & -4 \end{bmatrix} \begin{bmatrix} 2 & -3 & -5 \\ -1 & 4 & 5 \\ 1 & -3 & -4 \end{bmatrix} \\&= \begin{bmatrix} 2 \cdot 2 + 3 \cdot 1 - 5 \cdot 1 & -2 \cdot 3 - 3 \cdot 4 + 5 \cdot 3 & -2 \cdot 5 - 3 \cdot 5 + 5 \cdot 4 \\ -1 \cdot 2 - 4 \cdot 1 + 5 \cdot 1 & 1 \cdot 3 + 4 \cdot 4 - 5 \cdot 3 & 1 \cdot 5 + 4 \cdot 5 - 5 \cdot 4 \\ 1 \cdot 2 + 3 \cdot 1 - 4 \cdot 1 & -1 \cdot 3 - 3 \cdot 4 + 4 \cdot 3 & -1 \cdot 5 - 3 \cdot 5 + 4 \cdot 4 \end{bmatrix} \\&= \begin{bmatrix} 2 & -3 & -5 \\ -1 & 4 & 5 \\ 1 & -3 & -4 \end{bmatrix} = A.\end{aligned}$$

Therefore,  $A$  is idempotent.

► **Example 5.** Determine all the idempotent diagonal matrices of order  $n$ .

► **Solution** Let  $A = \text{diag. } [d_1, d_2, d_3, \dots, d_n]$  be an idempotent matrix of order  $n$ .

Then  $A^2 = A$ .

$$\begin{aligned}\text{Now } A^2 &= \begin{bmatrix} d_1 & 0 & 0 & \dots & 0 \\ 0 & d_2 & 0 & \dots & 0 \\ \dots & \dots & \dots & \dots & 0 \\ 0 & 0 & 0 & \dots & d_n \end{bmatrix} \\&= \begin{bmatrix} d_1 & 0 & 0 & \dots & 0 \\ 0 & d_2 & 0 & \dots & 0 \\ \dots & \dots & \dots & \dots & 0 \\ 0 & 0 & 0 & \dots & d_n \end{bmatrix}\end{aligned}$$

$$= \begin{bmatrix} d_1^2 & 0 & 0 & \dots & 0 \\ 0 & d_2^2 & 0 & \dots & 0 \\ \dots & \dots & \dots & \dots & 0 \\ 0 & 0 & 0 & \dots & d_n^2 \end{bmatrix}.$$

$\therefore A^2 = A$  gives

$$d_1^2 = d_1, d_2^2 = d_2, \dots, d_n^2 = d_n$$

i.e.  $d_1 = 0, 1; d_2 = 0, 1; \dots; d_n = 0, 1$ .

Hence diag.  $[d_1, d_2, d_3, \dots, d_n]$  where  $d_1, d_2, \dots, d_n = 0, 1$  are the required idempotent diagonal matrices of order  $n$ .

► **Example 6.** Prove that if  $A$  and  $B$  are two matrices such that  $AB = A$  and  $BA = B$ , then  $A^T, B^T$  are idempotent.

► **Solution** From  $AB = A, (AB)^T = A^T$ ,  
or  $B^T A^T = A^T$ . ... (1)

Since  $B = BA, (BA)^T A^T = A^T$ ,

$$\text{or, } A^T B^T A^T = A^T$$

$$\text{or, } A^T A^T = A^T \text{ by (1)}$$

$$\text{or, } (A^T)^2 = A^T.$$

From  $BA = B, (BA)^T = B^T$ ,

$$\text{or } A^T B^T = B^T \text{ ... (2)}$$

Since  $A = AB, (AB)^T B^T = B^T$ ,

$$\text{or } B^T A^T B^T = B^T = B^T$$

$$\text{or } B^T B^T = B^T \text{ by (2)}$$

$$\text{or } (B^T)^2 = B^T$$

Therefore,  $A^T$  and  $B^T$  are idempotent.

#### 4. Nilpotent Matrix:

A square matrix  $A$  is called nilpotent matrix of order  $m$  provided it satisfies the relation  $A^k = 0$  and  $A^{k-1} \neq 0$ , where  $k$  is positive integer and  $O$  is null matrix and  $k$  is the order or index of the nilpotent matrix  $A$ .

A square matrix  $A$  is said to be nilpotent of order 2 if,  $A^2 = O$ .

► **Example 7.** Show that  $\begin{bmatrix} 1 & 1 & 3 \\ 5 & 2 & 6 \\ -2 & -1 & -3 \end{bmatrix}$  is

nilpotent matrix of order 3.

► **Solution** Let  $A = \begin{bmatrix} 1 & 1 & 3 \\ 5 & 2 & 6 \\ -2 & -1 & -3 \end{bmatrix}$

$$A^2 = A \cdot A = \begin{bmatrix} 1 & 1 & 3 \\ 5 & 2 & 6 \\ -2 & -1 & -3 \end{bmatrix} \times \begin{bmatrix} 1 & 1 & 3 \\ 5 & 2 & 6 \\ -2 & -1 & -3 \end{bmatrix}$$

$$= \begin{bmatrix} 1+5-6 & 1+2-3 & 3+6-9 \\ 5+10-12 & 5+4-6 & 15+12-18 \\ -2-5+6 & -2-2+3 & -6-6+9 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 0 \\ 3 & 3 & 9 \\ -1 & -1 & -3 \end{bmatrix}$$

$$A^3 = A^2 \cdot A = \begin{bmatrix} 0 & 0 & 0 \\ 3 & 3 & 9 \\ -1 & -1 & -3 \end{bmatrix} \times \begin{bmatrix} 1 & 1 & 3 \\ 5 & 2 & 6 \\ -2 & -1 & -3 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = O$$

$\therefore A^3 = O$

Hence  $A$  is nilpotent of order 3.

► **Example 8.** Show that  $A(I \pm A)^n = A$  when  $n$  is a positive integer and  $A$  is a nilpotent matrix of index 2.

► **Solution** Here  $A^2 = O$

$\therefore A^3 = A^2 \cdot A = O \cdot A = O, A^4 = A^3 \cdot A = O \cdot A = O$  and so on.

Now  $A(I \pm A)^n$

$$= A(I^n \pm {}^nC_1 I^{n-1} A + {}^nC_2 I^{n-2} A^2 \pm {}^nC_3 I^{n-3} A^3 + \dots)$$

$$= A(I \pm nA + O + O + \dots) = A(I \pm nA) = A \pm nA^2$$

$$= A \pm n \cdot O = A.$$

#### Practice Problems

[E]

1. If  $C = \begin{bmatrix} 6 & -4 \\ 9 & -6 \end{bmatrix}$ , then prove that  $C^2$  is null.

2. Show that  $\begin{bmatrix} 1 & 1 & 3 \\ 5 & 2 & 6 \\ -2 & -1 & -3 \end{bmatrix}$  is nilpotent of index 3.

3. Show that  $\begin{bmatrix} 1 & -2 & -6 \\ -3 & 2 & 9 \\ 2 & 0 & -3 \end{bmatrix}$  is periodic.
4. If  $A = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$ , then prove that A is idempotent.
5. Show that  $\begin{bmatrix} 0 & 4 & 3 \\ 1 & -3 & -3 \\ -1 & 4 & 4 \end{bmatrix}$  is involuntary.
6. If B is an idempotent matrix, show that  $A = I - B$  is also idempotent and that  $AB = BA = O$ .
7. Find the number of idempotent diagonal matrices of order n.
8. Prove that the matrix  $\begin{bmatrix} \lambda_1^2 & \lambda_1\lambda_2 & \lambda_1\lambda_3 \\ \lambda_1\lambda_2 & \lambda_2^2 & \lambda_2\lambda_3 \\ \lambda_1\lambda_3 & \lambda_2\lambda_3 & \lambda_3^2 \end{bmatrix}$  is idempotent, where  $\lambda_1, \lambda_2, \lambda_3$  are the direction cosines i.e.,  $\lambda_1^2 + \lambda_2^2 + \lambda_3^2 = 1$ .
9. Find all  $2 \times 2$  matrices A such that  $A^2 = O$ .
10. If  $A^2 = A$ , prove that  $(A + I)^k = I + (2^k - 1)A$ .
11. A square matrix P is said to be idempotent if  $P^2 = P$ . Show that if P is idempotent, then  $E = 2P - I$  is involuntary, and conversely, if E is involuntary, then  $P = 1/2 (I + E)$  is idempotent.

### 3.9 INVERSE OF A MATRIX

#### Singular matrices

A square matrix A is said to be singular or non-singular according as  $|A|$  is zero or non-zero respectively.

If A and B are nonsingular matrices of the same order, then AB and BA are also nonsingular matrices of the same order.

#### Cofactor Matrix & Adjoint Matrix

Let  $A = [a_{ij}]_n$  be a square matrix. The matrix obtained by replacing each element of A by corresponding cofactor is called as cofactor matrix of A, called as cofactor A. The transpose of cofactor matrix of A is called as adjoint of A, denoted as  $\text{adj } A$ .

i.e. if  $A = [a_{ij}]_n$  then cofactor  $A = [A_{ij}]_n$  when  $A_{ij}$  is the cofactor of  $a_{ij} \forall i \& j$ .

$\text{Adj } A = [b_{ij}]_n$  where  $b_{ij} = A_{ji} \forall i \& j$ .

$$\text{If } A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix},$$

then  $\text{adj } A$  is the transpose of the matrix

$$\begin{bmatrix} A_{11} & A_{12} & \dots & A_{1n} \\ A_{21} & A_{22} & \dots & A_{2n} \\ \dots & \dots & \dots & \dots \\ A_{n1} & A_{n2} & \dots & A_{nn} \end{bmatrix}$$

$$\text{i.e. the matrix } \begin{bmatrix} A_{11} & A_{12} & \dots & A_{1n} \\ A_{12} & A_{22} & \dots & A_{2n} \\ \dots & \dots & \dots & \dots \\ A_{n1} & A_{n2} & \dots & A_{nn} \end{bmatrix}$$



Sometimes the adjoint of a matrix is also called the adjugate of the matrix.

**Theorem :** If A is a square matrix of order n, then  $A \cdot (\text{adj } A) = |A| I_n = (\text{adj } A) \cdot A$ .

*Proof:*

$$A \cdot (\text{adj } A) = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix}$$

$$\times \begin{bmatrix} A_{11} & A_{12} & \cdots & A_{1n} \\ A_{21} & A_{22} & \cdots & A_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ A_{n1} & A_{n2} & \cdots & A_{nn} \end{bmatrix}$$

$$= \begin{bmatrix} \sum_{r=1}^n a_{1r} A_{1r} & \sum_{r=1}^n a_{1r} A_{2r} & \cdots & \sum_{r=1}^n a_{1r} A_{nr} \\ \sum_{r=1}^n a_{2r} A_{1r} & \sum_{r=1}^n a_{2r} A_{2r} & \cdots & \sum_{r=1}^n a_{2r} A_{nr} \\ \cdots & \cdots & \cdots & \cdots \\ \sum_{r=1}^n a_{nr} A_{1r} & \sum_{r=1}^n a_{nr} A_{2r} & \cdots & \sum_{r=1}^n a_{nr} A_{nr} \end{bmatrix}$$

$$= \begin{bmatrix} |A| & 0 & \cdots & 0 \\ 0 & |A| & \cdots & 0 \\ \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & \cdots & |A| \end{bmatrix}$$

since  $\sum_{r=1}^n a_{ir} A_{jr} = |A|$  when  $i = j$   
 $= 0$  when  $i \neq j$

$$= |A| \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & \cdots & 1 \end{bmatrix} = |A| I_n$$

Similarly we can prove  $(\text{Adj} A) \cdot A = |A| I_n$

► **Example 1.** Find the adjoint of the matrix

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 5 & 0 \\ 2 & 4 & 3 \end{bmatrix} \text{ and verify the theorem}$$

$$A \cdot (\text{adj} A) = |A| I = (\text{adj} A) \cdot A.$$

► **Solution** We have  $|A| = \begin{vmatrix} 1 & 2 & 3 \\ 0 & 5 & 0 \\ 2 & 4 & 3 \end{vmatrix}$

The cofactors of the elements of the first row of  $|A|$

are  $\begin{vmatrix} 5 & 0 \\ 4 & 3 \end{vmatrix}, -\begin{vmatrix} 0 & 0 \\ 2 & 3 \end{vmatrix}, \begin{vmatrix} 0 & 5 \\ 2 & 4 \end{vmatrix}$  i.e., are 15, 0, -10 respectively.

The cofactors of the elements of the second row of

$|A|$  are  $-\begin{vmatrix} 2 & 3 \\ 4 & 3 \end{vmatrix}, \begin{vmatrix} 1 & 3 \\ 2 & 3 \end{vmatrix}, -\begin{vmatrix} 1 & 2 \\ 2 & 4 \end{vmatrix}$  i.e. are 6, -3, 0 respectively.

The cofactors of the elements of the third row of  $|A|$  are

$\begin{vmatrix} 2 & 3 \\ 5 & 0 \end{vmatrix}, -\begin{vmatrix} 1 & 3 \\ 0 & 0 \end{vmatrix}, \begin{vmatrix} 1 & 2 \\ 0 & 5 \end{vmatrix}$  i.e., are -15, 0, 5 respectively.

Therefore the  $\text{adj. } A$  = the transpose of the cofactor matrix  $B$  where

$$B = \begin{bmatrix} 15 & 0 & -10 \\ 6 & -3 & 0 \\ -15 & 0 & 5 \end{bmatrix}.$$

$$\therefore \text{adj} A = \begin{bmatrix} 15 & 6 & -15 \\ 0 & -3 & 0 \\ -10 & 0 & 5 \end{bmatrix}.$$

$$|A| = 1(15) - 2(0) + 3(-10) = -15$$

$$A \cdot (\text{adj} A) = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 5 & 0 \\ 2 & 4 & 3 \end{bmatrix} \begin{bmatrix} 15 & 6 & -15 \\ 0 & -3 & 0 \\ -10 & 0 & 5 \end{bmatrix}$$

$$= \begin{bmatrix} -15 & 0 & 0 \\ 0 & -15 & 0 \\ 0 & 0 & -15 \end{bmatrix} = -15 I = |A| I.$$

### Properties of adjoint matrix

- (i)  $|\text{adj} A| = |A|^{n-1}$ , where  $n$  is order of  $A$ .  
 In particular, for  $3 \times 3$  matrix,  $|\text{adj} A| = |A|^2$   
 We have  $A \cdot \text{adj} A = |A| I_n$ .  
 $\therefore |A \cdot \text{adj} A| = |A| |I_n|$   
 $\therefore |A| \cdot |\text{adj} A| = |A|^n |I_n|$   
 [Since  $|AB| = |A| |B|$  and  $|kA| = k^n |A|$ ]  
 $\therefore |A| |\text{adj} A| = |A|^n [\because |I_n| = 1]$   
 $\therefore |\text{adj} A| = |A|^{n-1}$ , if  $|A| \neq 0$ .

If  $|A| = 0$ , then  $|\text{adj } A| = 0$ . So in this case also  $|\text{adj } A| = |A|^{n-1}$



We conclude that if  $|A| \neq 0$  i.e.  $A$  is a non-singular matrix, then  $\text{adj } A$ , is also non-singular.

(ii) If  $|A| = 0$ , then  $|\text{adj } A| = 0$ .

If  $A$  is singular, then  $\text{adj } A$  is also singular.

► **Solution** Let  $A$  be a square matrix of the type  $n \times n$ . Then  $A(\text{adj } A) = \text{a null matrix}$

If  $A$  is a null matrix,  $\text{adj } A$  is also a null matrix and so  $|A| = |\text{adj } A| = 0$

If  $A$  is not a null matrix, then

$A \text{ adj } A = \text{a null matrix}$  holds either if  $\text{adj } A$  is a null matrix or if  $\text{adj } A$  is a singular matrix. In either case  $|\text{adj } A| = 0$ .

(iii)  $\text{adj } (kA) = k^{n-1}(\text{adj } A)$

(iv)  $\text{adj } A' = (\text{adj } A)'$

Thus if  $A$  is a symmetric matrix, then  $\text{adj } A$  is also symmetric.

Since  $(\text{adj } A)' = \text{adj } A' = \text{adj } A$  we say that  $\text{adj } A$  is symmetric.

If  $A$  is skew-symmetric, then  $\text{adj } A$  is symmetric or skew-symmetric according as the order of  $A$  is odd or even.

Let  $A = [a_{ij}]_{n \times n}$ .

$(\text{adj } A)' = \text{adj } A' = \text{adj } (-A)$

$$= \begin{cases} \text{adj } A & \text{when } n \text{ is odd} \\ -\text{adj } A & \text{when } n \text{ is even} \end{cases} \quad \text{using (iii)}$$

(v)  $\text{adj } I = I$

(vi)  $\text{adj}(AB) = (\text{adj } B)(\text{adj } A)$ .

**Case I.**  $A$  and  $B$  are nonsingular.

We have  $(AB) \text{ adj } (AB) = |AB| I$

$$\therefore (\text{adj } B)(\text{adj } A)(AB) \text{ adj } (AB) = (\text{adj } B)(\text{adj } A) |AB| \quad \dots(1)$$

L.H.S of (1)

$$\begin{aligned} &= (\text{adj } B) I |AB| B \text{ adj } (AB) \\ &= |AB| \{(\text{adj } B)B\} \text{ adj } (AB) \\ &= |AB| |B| I \text{ adj } (AB) \\ &= |AB| \text{ adj } (AB) \end{aligned}$$

R.H.S of (1)

$$= (\text{adj } B)(\text{adj } A) |AB|,$$

$$= |AB| (\text{adj } B)(\text{adj } A),$$

$$\Rightarrow \text{adj } (AB) = (\text{adj } B)(\text{adj } A) \quad (\because |AB| \neq 0)$$

**Case II.**  $AB$  is singular, i.e., atleast one of  $A$  and  $B$  is singular.

In this case we can find out an  $x$  such that  $(A - xI)$  and  $(B - xI)$  are nonsingular. By Case I we have.

$$\text{adj } \{(A - xI)(B - xI)\} = \text{adj } (B - xI) \text{ adj } (A - xI).$$

Since this result is true for an infinite number of values of  $x$ , the coefficients of the various powers of  $x$  on the two sides must be equal. Hence  $\text{adj } (AB) = (\text{adj } B)(\text{adj } A)$ .

(vii) If  $A$  is a non-singular matrix, then

$$\text{adj } (\text{adj } A) = |A|^{n-2} A.$$

$$\text{We have } A(\text{adj } A) = |A| I_n. \quad \dots(1)$$

If we take  $\text{adj } A$  in place of  $A$ , then (1) gives

$$(\text{adj } A)(\text{adj } \text{adj } A) = |\text{adj } A| I_n$$

$$\text{or } (\text{adj } A)(\text{adj } \text{adj } A) = |A|^{n-1} I_n$$

$$I_n [\because |\text{adj } A| = |A|^{n-1}]$$

Pre-multiplying both sides of this last relation by  $A$ , we get

$$A \{(\text{adj } A)\} (\text{adj } \text{adj } A) = A \{|A|^{n-1} I_n\}$$

$$\text{or } (A \text{ adj } A)(\text{adj } \text{adj } A) = |A|^{n-1} (A I_n)$$

$$[\because \text{matrix multiplication is associative}]$$

$$\text{or } (|A| I_n)(\text{adj } \text{adj } A) = |A|^{n-1} A \quad [\because A I_n = A]$$

$$\text{or } |A| (I_n \text{ adj } \text{adj } A) = |A|^{n-1} A$$

$$\text{or } |A| \text{ adj } \text{adj } A = |A|^{n-1} A. \quad \dots(2)$$

Since  $A$  is non-singular, therefore  $|A| \neq 0$ .

So cancelling  $|A|$  from both sides of (2), we get  $\text{adj } \text{adj } A = |A|^{n-2} A$ .

$$\text{(viii) } |\text{adj } (\text{adj } A)| = |A|^{(n-1)^2}$$

► **Example 2.** If  $A$  be an  $n$ -square matrix and  $B$  be its adjoint, then show that  $\text{Det } (AB + K I_n)$

$= [\text{Det } (A) + K]^n$ , where  $K$  is a scalar quantity.

► **Solution** We have,  $A B = A (\text{adj } A) = \text{Det } (A) \cdot I_n$

$$\Rightarrow A B + K I_n = \text{Det } (A) I_n + K I_n$$

$$\Rightarrow \text{Det } (A B + K I_n) = \text{Det } (\text{Det } (A) I_n + K I_n)$$

$$= (\text{Det } (A) + K)^n \quad (\because \text{Det } (\alpha I_n) = \alpha^n)$$

$$\Rightarrow \text{Det } (A B + K I_n) = [\text{Det } (A) + K]^n.$$

### Inverse of a matrix

Let  $A$  be a square matrix of order  $n$ . Then a matrix  $B$ , if it exists, is called the inverse of  $A$  if  $AB = BA = I_n$



B is also called as Reciprocal matrix and is denoted by  $A^{-1}$

$$\text{Thus } AA^{-1} = A^{-1}A = I_n$$

$$\text{Since } \begin{pmatrix} 4 & -1 \\ -3 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 3 & 4 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I_2$$

$$\text{and } \begin{pmatrix} 1 & 1 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 4 & -1 \\ -3 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I_2$$

$$\text{we say that } \begin{pmatrix} 4 & -1 \\ -3 & 1 \end{pmatrix}$$

$$\text{is the inverse of } \begin{pmatrix} 1 & 1 \\ 3 & 4 \end{pmatrix}.$$



Every invertible matrix possesses a unique inverse.

*Proof:*

Let A be an  $n \times n$  invertible matrix. Let B and C be two inverses of A.

$$\text{Then } AB = BA = I_n \quad \dots(1)$$

$$AC = CA = I_n \quad \dots(2)$$

From (1), we have  $AB = I_n$

Pre-multiplication with C gives  $C(AB) = CI_n = C$

From (2), we have  $CA = I_n$

Pre-multiplication with C gives

$$C(AB) = CI_n = C \quad \dots(3)$$

From (2), we have  $CA = I_n$

Post-multiplication with B gives

$$(CA)B = I_n B = B \quad \dots(4)$$

Since  $C(AB) = (CA)B$ , therefore from (3) and (4), we have  $B = C$ . Hence an invertible matrix possesses a unique inverse.

### Inverse using adjoint of a matrix

Let A be a non-singular matrix. Then the matrix

$$\frac{1}{|A|} \text{adj } A \text{ is the inverse of } A.$$

$$\text{We have } A(\text{adj } A) = |A| I_n = (\text{adj } A)A$$

$$\Rightarrow A \left( \frac{1}{|A|} \text{adj } A \right) = I_n = \left( \frac{1}{|A|} \text{adj } A \right) A,$$

when A is non-singular.

$$\Rightarrow A^{-1} = \frac{1}{|A|} \text{adj } A.$$

For example, if  $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$  and  $|A| \neq 0$  then

$$A^{-1} = \frac{1}{|A|} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \text{ or } A^{-1} = \begin{bmatrix} \frac{d}{ad-bc} & \frac{-b}{ad-bc} \\ \frac{-c}{ad-bc} & \frac{a}{ad-bc} \end{bmatrix}$$

For a second order matrix one simply interchanges the diagonal elements and changes the sign of the off-diagonal elements and then divide each element by  $|A|$ .

The necessary and sufficient condition for a square matrix A to possess the inverse is that  $|A| \neq 0$ .

*Proof:*

The condition is necessary.

Let A be an  $n \times n$  matrix and let B be the inverse of A.

$$\text{Then } AB = I_n.$$

$$\therefore |AB| = |I_n| = 1,$$

$$\therefore |A| \text{ must be different from } 0.$$

Conversely, the condition is also sufficient.

If  $|A| \neq 0$ , then let us define a matrix B by the relation

$$B = \frac{1}{|A|} (\text{adj } A).$$

$$\text{Then } AB = A \left( \frac{1}{|A|} \text{adj } A \right) = \frac{1}{|A|} (A \text{adj } A) = \frac{1}{|A|} |A| I_n$$

$$= I_n.$$

$$\text{Similarly } BA = \left( \frac{1}{|A|} \text{adj } A \right) A = \frac{1}{|A|} (\text{adj } A) A$$

$$= \frac{1}{|A|} \cdot |A| I_n = I_n.$$

$$\text{Thus } AB = BA = I_n.$$

Hence the matrix A invertible and B is the inverse of A.

A square matrix need not have a multiplicative inverse.

$$\text{For example, consider } A = \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix}$$

and try to find a multiplicative inverse

$$M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

Since  $AM$  has to be equal to  $I$ , we must have

$$\text{or } \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\text{or } \begin{pmatrix} 2a & 2b \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$2a = 1 \quad 2b = 0 \quad 0 = 0 \quad 0 = 1$$

Since this last equation can never hold, there is no choice for  $a$ ,  $b$ ,  $c$  and  $d$  for which  $AM = I$ .

That an inverse does not always exist is shown by the following example. The product

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \cdot \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & a \\ 0 & c \end{bmatrix}$$

shows that there is no way to choose  $a$ ,  $b$ ,  $c$ , and  $d$  so as to make the right member equal to  $I_2$ . Hence

the matrix  $\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$  has no inverse.

► **Example 3.** Find the inverse of the matrix

$$A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{bmatrix}.$$

► **Solution** We have  $|A| = \begin{vmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{vmatrix} = \begin{vmatrix} 0 & 1 & 0 \\ 1 & 2 & -1 \\ 3 & 1 & -1 \end{vmatrix},$

applying  $C_3 \rightarrow C_3 \rightarrow 2C_2$

$$= -1 \begin{vmatrix} 1 & -1 \\ 3 & -1 \end{vmatrix}, \text{ expanding the determinant along}$$

the first row  $= -2$ .

Since  $|A| \neq 0$ , therefore  $A^{-1}$  exists.

Now the cofactors of the elements of the first row of

$|A|$  are  $\begin{vmatrix} 2 & 3 \\ 1 & 1 \end{vmatrix}, -\begin{vmatrix} 1 & 3 \\ 3 & 1 \end{vmatrix}, \begin{vmatrix} 1 & 2 \\ 3 & 1 \end{vmatrix}$  i.e., are  $-1, 8, -5$  respectively.

The cofactors of the elements of the second row of  $|A|$  are  $-\begin{vmatrix} 1 & 2 \\ 1 & 1 \end{vmatrix}, -\begin{vmatrix} 0 & 2 \\ 3 & 1 \end{vmatrix}, -\begin{vmatrix} 0 & 1 \\ 3 & 1 \end{vmatrix}$  i.e. are  $1, -6, 3$  respectively.

The cofactors of the elements of the third row of  $|A|$  are  $\begin{vmatrix} 1 & 2 \\ 2 & 3 \end{vmatrix}, -\begin{vmatrix} 3 & 2 \\ 1 & 3 \end{vmatrix}, \begin{vmatrix} 0 & 1 \\ 1 & 2 \end{vmatrix}$  i.e. are  $-1, 2, -1$  respectively.

Therefore the  $\text{Adj. } A$  = the transpose of the matrix  $B$  where

$$B = \begin{bmatrix} -1 & 8 & -5 \\ 1 & -6 & 3 \\ -1 & 2 & -1 \end{bmatrix}.$$

$$\therefore \text{Adj. } A = \begin{bmatrix} -1 & 1 & -1 \\ 8 & -6 & 2 \\ 5 & 3 & -1 \end{bmatrix}.$$

Now  $A^{-1} = \frac{1}{|A|} \text{Adj. } A$  and here  $|A| = -2$ .

$$\therefore A^{-1} = -\frac{1}{2} \begin{bmatrix} -1 & 1 & -1 \\ 8 & -6 & 2 \\ -5 & 3 & -1 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ -4 & 3 & -1 \\ \frac{5}{2} & -\frac{3}{2} & \frac{1}{2} \end{bmatrix}.$$

However, from a wider viewpoint there is a more basic question : "Is there a matrix  $L$  whose products with the matrix  $A$  are  $I$ ?" There are three answers to this, depending on the characteristics of  $A$ : (i) in some cases  $L$  exists and is unique for a given  $A$ ; (ii) sometimes numerous matrices  $L$  exist for a particular  $A$ , i.e.,  $L$  exists but is not unique; and (iii) in some instances  $L$  does not exist at all. Examples of these three situations are now given.

$$(i) \text{ If } A_1 = \begin{bmatrix} 2 & 5 \\ 3 & 8 \end{bmatrix} \text{ and } L_1 = \begin{bmatrix} 8 & -5 \\ -3 & 2 \end{bmatrix}$$

$L_1 A_1 = I$ , and  $L_1$  is the unique matrix  $L$  for this particular  $A$  such that  $LA = I$ . Furthermore,  $A_1 L_1 = I$ .

$$(ii) \text{ For } A_2 = \begin{bmatrix} 1 & 1 \\ -1 & 0 \\ 3 & -1 \end{bmatrix}$$

there is an infinite number of matrices  $L$  such that  $LA_2 = I$ ; e.g.,

$$L_2 = \begin{bmatrix} 1 & 3 & 1 \\ 2 & 5 & 1 \end{bmatrix} \text{ and } L_2^* = \begin{bmatrix} 4 & 15 & 4 \\ 7 & 25 & 6 \end{bmatrix}.$$

(iii) When  $A_3 = \begin{bmatrix} 0 & 3 & 7 \\ 0 & 2 & 5 \end{bmatrix}$

there is no matrix  $L$  such that  $LA_3 = I$ , for the leading element of  $LA_3$  will always be zero and so the product matrix can never equal  $I$ .

One might notice a further characteristic of these three situations :

- (i)  $A_1L_1 = I$  as well as  $L_1A_1 = I$ .
- (ii)  $A_2L_2 \neq I$  even though  $L_2A_2 = I$ .
- (iii) There are matrices  $R$  such that  $A_3R = I$  even though there is no matrix  $L$  such that  $LA_3 = I$ ; e.g.,

$$R = \begin{bmatrix} 4 & 8 \\ 5 & -7 \\ -2 & 3 \end{bmatrix}.$$

The  $A^{-1}$  established in each example was a matrix such that  $A^{-1}A = I$ . But we have just illustrated how  $LA = I$  does not necessarily yield a matrix  $L$  that is unique for a given  $A$ , nor one for which  $AL$  necessarily equals  $I$  even if  $LA$  does. Only in the case of  $A_1$  did  $L_1$  have both these properties ; and they seem reasonable properties to ask of an inverse – that it be unique for a given  $A$ , and that both premultiplication and postmultiplication by  $A$  lead to the identity matrix. Analogous to the familiar reciprocal of a number in scalar algebra, these properties seem appropriate for the inverse (reciprocal) of a matrix. Accordingly, when possible, we will derive a matrix from  $A$  which has these properties. The derived matrix will be the inverse of  $A$ , and instead of satisfying only  $A^{-1}A = I$  it will have the two properties

$$A^{-1}A = AA^{-1} = I \quad (a)$$

$$\text{and } A^{-1} \text{ unique for given } A \quad (b)$$

The equality of  $A^{-1}A$  and  $AA^{-1}$  in (a) requires that both products exist and have the same order. This occurs only when  $A$  and  $A^{-1}$  are both square, of the same order. This means that an  $A^{-1}$  such that (a) is true exists only when  $A$  is square. What we are

about to describe as the inverse of a matrix can therefore exist only if that matrix is square (and its inverse will also be square, of the same order). In contrast, rectangular matrices do not have inverses, although some of them have what is called either a left or right inverse.

### Properties of inverse matrix

- (i)  $(A^{-1})^{-1} = A$ , if  $A$  is non-singular.  
 $A^{-1}$  is always non-singular.

(ii)  $|A^{-1}| = \frac{1}{|A|}$

Since  $AA^{-1} = I$ ,  $|AA^{-1}| = |I|$ . This means that  $|A||A^{-1}| = 1$ .

The invertibility of  $A$  implies that  $|A| \neq 0$ , so we can divide by  $|A|$  to find

$$|A^{-1}| = \frac{1}{|A|}$$

- (iii)  $(A')^{-1} = (A^{-1})'$

Here  $|A| \neq 0$ . Since  $|A'| = |A|$ ,  $|A'| \neq 0$ .

Therefore,  $(A')^{-1}$  exists.

Again  $A^{-1}A = AA^{-1} = I$ .

$$\therefore (A^{-1}A)' = (AA^{-1})' = I',$$

$$\text{or, } A'(A^{-1})' = (A^{-1})'A' = I$$

Thus  $(A^{-1})'$  is the inverse of  $A'$ , i.e.,  $(A')^{-1} = (A^{-1})'$ .

Note that  $A$  is non-singular and symmetric

$$\Rightarrow A^{-1} \text{ is symmetric.}$$

- (iv) If  $k \neq 0$  and  $A$  is a nonsingular matrix then

$$(kA)^{-1} = \frac{1}{k} A^{-1}.$$

- (v)  $I^{-1} = I$

- (vi) If  $D = \text{diag}(a_{11}, a_{22}, \dots, a_{nn})$  where  $a_{ii} \neq 0 \forall$

$$i \text{ then } D^{-1} = \text{diag}\left(\frac{1}{a_{11}}, \frac{1}{a_{22}}, \dots, \frac{1}{a_{nn}}\right).$$

- (vii) We define  $A^{-n} = (A^{-1})^n$  when  $A$  is non-singular and  $n$  is a positive integer. We define  $A^0 = I$  for an arbitrary square matrix  $A$ , singular or not. Also the laws of exponents  $A^mA^n = A^{m+n}$  and  $(A^m)^n = A^{mn}$  now apply for all integral values of  $m$  and  $n$  when  $A$  is non-singular.

- (viii)  $\text{adj } A^{-1} = (\text{adj } A)^{-1} = \frac{A}{|A|}.$

We have  $AA^{-1} = I = A^{-1}A$

$$\begin{aligned}\therefore \operatorname{adj} I &= \operatorname{adj} A \operatorname{adj} A^{-1}, \text{ or, } \operatorname{adj} A^{-1} \operatorname{adj} A = I \\ &= \operatorname{adj} A \operatorname{adj} A^{-1} \\ \therefore \operatorname{adj} A^{-1} &= (\operatorname{adj} A)^{-1}.\end{aligned}$$

Again  $A^{-1} = \frac{\operatorname{adj} A}{|A|}$ . Therefore,  $\operatorname{adj} A = |A|A^{-1}$ .

$$\therefore \operatorname{adj} A^{-1} = [|A|A^{-1}] = |A|^{-1} (A^{-1})^{-1} = \frac{A}{|A|}$$

- (ix) If  $A, B$  be two  $n$ -rowed non-singular matrices, then  $AB$  is also non-singular and  $(AB)^{-1} = B^{-1}A^{-1}$ .

This is shown by actual multiplication by the candidate  $B^{-1}A^{-1}$  for an inverse of  $AB$ .

$$(AB)(B^{-1}A^{-1}) = A(BB^{-1})$$

$$A^{-1} = AIA^{-1} = AA^{-1} = I$$

$$(B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)$$

$$B = B^{-1}IB = B^{-1}B = I$$

- (x) If  $A, B$  are square matrices and  $AB = O$ , then  $A = O$ , or  $B = O$ , or  $A$  and  $B$  are both singular. For suppose that  $A \neq O$ ,  $B \neq O$ . If atleast one of  $A, B$  is non-singular, let, for instance,  $|A| \neq 0$ . Then  $AB = O$  implies  $A^{-1}AB = O$ , i.e.  $B = O$ , and we have a contradiction.

The relations  $AX = AY$ ,  $A \neq O$ , where  $A, X, Y$  are square matrices, do not imply that  $X = Y$ . Similarly, the relations  $XB = YB$ ,  $B \neq O$ , where  $B$  is also square, do not imply that  $X = Y$ .

- (xi) Let  $A$  be a non-singular matrix. Then  $AB = AC \Rightarrow B = C$  and  $BA = CA \Rightarrow B = C$ . Division by a matrix never occurs. Multiplication by the corresponding inverse (when it exists) is used in its place. But, as usual, distinction must always be made between premultiplication and postmultiplication.

The nonexistence of division, and the existence of inverse matrices only in special cases, also means that in matrix algebra and equation such as  $PK = QK$  does not always imply that  $P$  equals  $Q$ . An example of this is

$$\begin{bmatrix} 7 & 5 & 4 \\ 3 & 3 & 10 \end{bmatrix} \begin{bmatrix} 1 & 2 & 2 \\ -1 & 0 & 1 \\ 0 & 2 & 3 \end{bmatrix} = \begin{bmatrix} 3 & 1 & 8 \\ 2 & 2 & 11 \end{bmatrix} \begin{bmatrix} 1 & 2 & 2 \\ -1 & 0 & 1 \\ 0 & 2 & 3 \end{bmatrix} \dots (1)$$

The reason that the equality (1) does not imply equality of the two  $2 \times 3$  matrices (which would clearly be false) is that the  $3 \times 3$  matrix has no inverse. It is only the existence of  $K^{-1}$  that permits the equation  $PK = QK$  to imply  $P = Q$ . This is so because when  $K^{-1}$  exists, the equation  $PK = QK$  can be postmultiplied on both sides by  $K^{-1}$  to yield  $PKK^{-1} = QKK^{-1}$ , i.e.  $PI = QI$ , or  $P = Q$ ... (2) But when  $K^{-1}$  does not exist as in (1), then (2) cannot be used.

► **Example 4.** If the matrices  $A$  and  $B$  commute, then  $A^{-1}$  and  $B^{-1}$  also commute.

► **Solution** Since  $A$  and  $B$  commute, therefore  $AB = BA$ .

$$\text{Now } (AB)^{-1} = B^{-1}A^{-1}.$$

$$\text{Also } (AB)^{-1} = (BA)^{-1} = A^{-1}B^{-1}.$$

$$\therefore B^{-1}A^{-1} = A^{-1}B^{-1}.$$

Thus  $A^{-1}$  and  $B^{-1}$  also commute.

► **Example 5.** Prove that if a matrix  $A$  is non-singular, then  $AB = AC + AD$  implies  $B = C + D$ , where  $B, C$  and  $D$  are square matrices of the same order as  $A$ .

► **Solution** Since the matrix  $A$  is non-singular, therefore  $A^{-1}$  exists.

$$\text{Hence } AB = AC + AD$$

$$\Rightarrow A^{-1}(AB) = A^{-1}(AC + AD)$$

$$\Rightarrow (A^{-1}A)B = (A^{-1}A)(C + D)$$

$$\Rightarrow I_n B = I_n (C + D) \Rightarrow B = C + D.$$

► **Example 6.** Determine the  $2 \times 2$  matrix  $X$  from the equation

$AX = O$ , where  $A$  is a given  $2 \times 2$  matrix, in each of the following cases:

- (i)  $|A| \neq 0$ ; (ii)  $|A| = 0$ , but  $A \neq O$  and the ratio of the elements of the first column of  $A$  to the corresponding elements of the second column is a non-zero scalar  $k$ .

► **Solution** (i) Here  $A^{-1}$  exists. Now  $A^{-1}AX = O$ , or  $IX = O$ , or  $X = O$ .  $X$  is a null matrix.

$$(ii) \text{ Let } A = \begin{bmatrix} ka & a \\ kb & b \end{bmatrix} \text{ and } X = \begin{bmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{bmatrix}$$

Now  $AX = O$ ,

$$\text{or, } \begin{bmatrix} a(kx_{11} + x_{21}) & a(kx_{12} + x_{22}) \\ b(kx_{11} + x_{21}) & b(kx_{12} + x_{22}) \end{bmatrix} = O$$

$$\begin{aligned}\therefore a(kx_{11} + x_{21}) &= 0, & a(kx_{12} + x_{22}) &= 0, \\ b(kx_{11} + x_{21}) &= 0, & b(kx_{12} + x_{22}) &= 0. \\ a \neq 0, b \neq 0. \text{ Therefore, } kx_{11} + x_{21} &= 0 \text{ and } kx_{12} \\ + x_{22} &= 0.\end{aligned}$$

$$\therefore X = \begin{bmatrix} x_{11} & x_{12} \\ -kx_{11} & -kx_{12} \end{bmatrix}$$



(i) Any polynomial identity between scalar polynomials remains valid for the corresponding matrix polynomials.

(ii) Any two polynomials in  $A$  commute with each other.

Again, let  $f(x), g(x)$  be two scalar polynomials and suppose that the square matrix  $A$  satisfies the relation  $|g(A)| \neq 0$ . We have just seen that  $f(A)g(A) = g(A)f(A)$ .

Hence, premultiplying and postmultiplying by  $\{g(A)\}^{-1}$ , we obtain  $\{g(A)\}^{-1} f(A) = f(A)\{g(A)\}^{-1}$ .

This identity enables us to define rational functions of  $A$ .

(iii) Let  $f(x), g(x)$  be scalar polynomials and let  $A$  be any square matrix such that  $|g(A)| \neq 0$ . Then the matrix appearing on either side of  $\{g(A)\}^{-1} f(A) = f(A)\{g(A)\}^{-1}$  is known as the quotient of  $f(A)$  by  $g(A)$ , and is denoted by or  $f(A)/g(A)$ .

(iv) A further example of the algebra of inverse is a matrix analogue of the scalar result that for scalar  $x \neq 1$

$$1 + x + x^2 + \dots + x^n = (x^n - 1)/(x - 1).$$

A matrix counterpart is that, provided

$(X - I)^{-1}$  exists

$$I + X + X^2 + \dots + X^{n-1} = (X^n - I)(X - I)^{-1}. \quad \dots(1)$$

This is established by noting that the product  $(I + X + X^2 + \dots + X^{n-1})(X - I)$

$$\begin{aligned}&= X + X^2 + X^3 + \dots + X^{n-1} + X^n \\ &\quad - (I + X + X^2 + \dots + X^{n-1}) = X^n - I\end{aligned} \quad \dots(2)$$

and therefore if  $(X - I)^{-1}$  exists,

postmultiplication of both sides of (1) yield (2).

$$\begin{aligned}\text{Also } (I + X + X^2 + \dots + X^{n-1}) \\ = (X - I)^{-1} (X^n - I)\end{aligned}$$

(v) The only idempotent matrices that are nonsingular are identity matrices. But idempotent matrices can involve nonsingular matrices; e.g.,  $P(QP)^{-1}Q$  is idempotent when  $(QP)^{-1}$  exists.

### Practice Problems

[F]

1. If  $P = \begin{bmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{bmatrix}$ , show that

$$\text{adj}P = \begin{bmatrix} -bc & -ca & -ab \\ b+c & c+a & a+b \\ -1 & -1 & -1 \end{bmatrix} \begin{bmatrix} b-c & 0 & 0 \\ 0 & c-a & 0 \\ 0 & 0 & a-b \end{bmatrix}$$

2. If  $\begin{bmatrix} 1 & 2 \\ -3 & 6 \end{bmatrix}$  is the inverse of  $A$ , what is the inverse of  $3A$ ?

3. Show that if

$$(i) A = \begin{bmatrix} -1 & 3 & 0 \\ 0 & 2 & 1 \\ 1 & 0 & 4 \end{bmatrix}, \text{ then } |A| = -5,$$

$$A^{-1} = \frac{1}{5} \begin{bmatrix} -8 & 12 & -3 \\ -1 & 4 & -1 \\ 2 & -3 & 2 \end{bmatrix} \text{ and}$$

$$A^{-1}A = AA^{-1} = I;$$

$$(ii) C = \frac{1}{10} \begin{bmatrix} 0 & -6 & 8 \\ -10 & 0 & 0 \\ 0 & -8 & -6 \end{bmatrix}, \text{ then}$$

$$|C| = 1, \quad C^{-1} = C' \text{ and } CC' = C'C = I$$

4. Find the inverse of

$$\begin{bmatrix} 0 & 0 & 0 & 3 \\ 0 & 0 & 7 & 0 \\ 0 & 4 & 0 & 0 \\ 5 & 0 & 0 & 0 \end{bmatrix}$$

5. Assuming that each of the stated inverses exist, prove the following statements.

(i)  $(I + A^{-1})^{-1} = A(A + I)^{-1}$ .

(ii)  $(A + BB')^{-1}B = A^{-1}B(I + B'A^{-1}B)^{-1}$ .

(iii)  $(A^{-1} + B^{-1})^{-1} = A(A + B)^{-1}$

$B = B(A + B)^{-1}A$ .

(iv)  $A - A(A + B)^{-1}A = B - B(A + B)^{-1}B$

(v)  $A^{-1} + B^{-1} = A^{-1}(A + B)B^{-1}$ .

(vi)  $(A + B)^{-1} = A^{-1} + B^{-1}$  implies  
 $AB^{-1}A = BA^{-1}B$ .

(vii)  $(I + AB)^{-1} = I - A(I + BA)^{-1}B$ .

(viii)  $(I + AB)^{-1}A = A(I + BA)^{-1}$ .

6. Prove that if A is idempotent and  $A \neq I$ , then A is singular.

7. Suppose A is invertible and B is not. Show that AB is not invertible.

8. Prove that, if  $AB = BA$  and  $S^2 = B$ , then also  $(A^{-1}SA)^2 = B$

9. Given that  $AB = BA$  and that  $B^{-1}$  exists, prove that  $AB^r = B^rA$  for all integer values of r. If  $A^{-1}$  also exists, prove that  $A^rB^s = B^sA^r$  for all integers r and s.

10. Prove that, if A is nonsingular, then from  $AB = AC$  we can conclude  $B = C$ . (B and C need not be square, of course.)

11. Prove that, if  $RAC = I$ , where all four matrices have order n, then  $A^{-1}$  exists and  $A^{-1} = CR$ .

12. (i) Show that if the nonsingular symmetric matrices A and B commute, then  $A^{-1}B$ ,  $AB^{-1}$  and  $A^{-1}B^{-1}$  are symmetric.

(ii) If A and B are both square matrices of order n and A has an inverse, show that  
 $(A + B)A^{-1}(A - B) = (A - B)A^{-1}(A + B)$ .

Also the determinant

$$|A - \lambda I| = \begin{vmatrix} a_{11} - \lambda & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} - \lambda & \cdots & a_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} - \lambda \end{vmatrix}$$

which is an ordinary polynomial in  $\lambda$  of degree n, is called the characteristic polynomial of A.

The equation  $|A - \lambda I| = 0$ , is called the characteristic equation of A.

The Cayley–Hamilton Theorem : Every square matrix satisfies its characteristic equation

i.e., if for a square matrix A of order n,

$$|A - \lambda I| = (-1)^n [\lambda^n + a_1\lambda^{n-1} + a_2\lambda^{n-2} + \cdots + a_n],$$

then the matrix equation

$$X^n + a_1X^{n-1} + a_2X^{n-2} + \cdots + a_nI = O$$

is satisfied by  $X = A$

$$\text{i.e., } A^n + a_1A^{n-1} + a_2A^{n-2} + \cdots + a_nI = O.$$

► **Example 1.** Obtain the characteristic equation of

the matrix  $A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix}$  and verify that it is

satisfied by A and hence find its inverse.

► **Solution** We have

$$\begin{aligned} |A - \lambda I| &= \begin{vmatrix} 1-\lambda & 0 & 2 \\ 0 & 2-\lambda & 1 \\ 2 & 0 & 3-\lambda \end{vmatrix} \\ &= (1-\lambda)(2-\lambda)(3-\lambda) + 2[0 - 2(2-\lambda)] \\ &= (2-\lambda)[(1-\lambda)(3-\lambda) - 4] \\ &= (2-\lambda)[\lambda^2 - 4\lambda - 1] \\ &= -(\lambda^3 - 6\lambda^2 + 7\lambda + 2). \end{aligned}$$

∴ the characteristic equation of A is

$$\lambda^3 - 6\lambda^2 + 7\lambda + 2 = 0 \quad \dots(1)$$

By the Cayley–Hamilton theorem

$$A^3 - 6A^2 + 7A + 2I = O. \quad \dots(2)$$

Verification: We have  $A^2$

$$= \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix} \times \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix} = \begin{bmatrix} 5 & 0 & 8 \\ 2 & 4 & 5 \\ 8 & 0 & 13 \end{bmatrix}.$$

### 3.10 OTHER METHODS OF FINDING INVERSE

#### Cayley–Hamilton Theorem

Let  $A = [a_{ij}]_{n \times n}$  be any n-rowed square matrix and  $\lambda$  be an unknown. The matrix  $A - \lambda I$  is called the characteristic matrix of A where I is the unit matrix of order n.

$$\text{Also } A^3 = A \cdot A^2 = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix} \times \begin{bmatrix} 5 & 0 & 8 \\ 2 & 4 & 5 \\ 8 & 0 & 13 \end{bmatrix}$$

$$= \begin{bmatrix} 21 & 0 & 34 \\ 12 & 8 & 23 \\ 34 & 0 & 55 \end{bmatrix}.$$

Now  $A^2 - 6A + 7I$

$$= \begin{bmatrix} 21 & 0 & 34 \\ 12 & 8 & 23 \\ 34 & 0 & 55 \end{bmatrix} - 6 \begin{bmatrix} 5 & 0 & 8 \\ 2 & 4 & 5 \\ 8 & 0 & 13 \end{bmatrix}$$

$$+ 7 \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix}$$

$$+ 2 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = O.$$

Hence Cayley-Hamilton theorem is verified.

Now we shall compute  $A^{-1}$ .

Multiplying (2) by  $A^{-1}$ , we get

$$A^2 - 6A + 7I + 2A^{-1} = O.$$

$$\therefore A^{-1} = -\frac{1}{2}(A^2 - 6A + 7I)$$

$$= -\frac{1}{2} \begin{bmatrix} 5 & 0 & 8 \\ 2 & 4 & 5 \\ 8 & 3 & 13 \end{bmatrix} + 3 \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix}$$

$$- \frac{7}{2} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} -3 & 0 & 2 \\ -1 & \frac{1}{2} & \frac{1}{2} \\ 2 & 0 & -1 \end{bmatrix}.$$

► **Example 2.** Use Cayley-Hamilton Theorem to express  $2A^5 - 3A^4 + A^2 - 4I$  as a linear polynomial in

$$A, \text{ when } A = \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}$$

► **Solution** Now let us find the characteristic equation of the matrix  $A$ . We have

$$|A - \lambda I| = \begin{vmatrix} 3 - \lambda & 1 \\ -1 & 2 - \lambda \end{vmatrix} = (3 - \lambda)(2 - \lambda) + 1$$

$$= \lambda^2 - 5\lambda + 7.$$

The characteristic equation of  $A$  is  $|A - \lambda I| = 0$  i.e., is

$$\lambda^2 - 5\lambda + 7 = 0 \quad \dots(1)$$

By Cayley-Hamilton theorem, the matrix  $A$  must satisfy (1).

Therefore we have

$$A^2 - 5A + 7I = O \quad \dots(2)$$

From (2), we get

$$A^2 = 5A - 7I \quad \dots(3)$$

Multiplying both sides of (3) by  $A$ , we get

$$A^3 = 5A^2 - 7A \quad \dots(4)$$

$$\therefore A^4 = 5A^3 - 7A^2 \quad \dots(5)$$

$$\text{and } A^5 = 5A^4 - 7A^3 \quad \dots(6)$$

$$\text{Now } 2A^5 - 3A^4 + A^2 - 4I = 2(5A^4 - 7A^3) - 3A^4 + A^2 - 4I$$

[Substituting for  $A^5$  from (6)]

$$= 7A^4 - 14A^3 + A^2 - 4I$$

$$= 7(5A^3 - 7A^2) - 14A^3 + A^2 - 4I \quad [\text{by (5)}]$$

$$= 21A^3 - 48A^2 - 4I$$

$$= 21(5A^2 - 7A) - 48A^2 - 4I \quad [\text{by (4)}]$$

$$= 57A^2 - 147A - 4I$$

$$= 57(5A - 7I) - 147A - 4I \quad [\text{by (3)}]$$

$$= 138A - 403I, \text{ which is a linear polynomial in } A.$$

### Elementary Transformations of a Matrix

There are six operations (transformations) on a matrix, three of which are due to rows and three due to columns, which are known as elementary operations or transformation.

- (i) The interchange of any two rows or two columns. Symbolically the interchange of  $i^{\text{th}}$  and  $j^{\text{th}}$  rows is denoted by  $R_i \leftrightarrow R_j$  and interchange of  $i^{\text{th}}$  and  $j^{\text{th}}$  column is denoted by  $C_i \leftrightarrow C_j$ .

For example, applying  $R_1 \leftrightarrow R_2$  to  $A$

$$= \begin{bmatrix} 1 & 2 & 1 \\ -1 & \sqrt{3} & 1 \\ 5 & 6 & 7 \end{bmatrix}, \text{ we get } \begin{bmatrix} -1 & \sqrt{3} & 1 \\ 1 & 2 & 1 \\ 5 & 6 & 7 \end{bmatrix}.$$

- (ii) The multiplication of the elements of any row or column by a non zero number. Symbolically, the multiplication of each element of the  $i^{\text{th}}$  row by  $k$ , where  $k \neq 0$  is denoted by  $R_i \rightarrow kR_i$ . The corresponding column operation is denoted by  $C_i \rightarrow kC_i$ .

For example, applying  $C_3 \rightarrow -\frac{1}{7}C_3$ ,

$$\text{to } B = \begin{bmatrix} 1 & 2 & 1 \\ -1 & \sqrt{3} & 1 \end{bmatrix}, \text{ we get } \begin{bmatrix} 1 & 2 & \frac{1}{7} \\ -1 & \sqrt{3} & \frac{1}{7} \end{bmatrix}$$

- (iii) The additions to the elements of any row or column, the corresponding elements of any other row or column multiplied by any non zero number. Symbolically, the addition to the elements of  $i^{\text{th}}$  row, the corresponding elements of  $j^{\text{th}}$  row multiplied by  $k$  is denoted by  $R_i \rightarrow R_i + kR_j$ . The corresponding column operation is denoted by  $C_i \rightarrow C_i + kC_j$ . For example, applying

$$R_2 \rightarrow R_1 - 2R_1, \text{ to } C = \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}, \text{ we get}$$

$$\begin{bmatrix} 1 & 2 \\ 0 & -5 \end{bmatrix}.$$

Two matrices  $A$  &  $B$  are said to be equivalent if one is obtained from other using elementary transformations. We write  $A \sim B$ . In short, the following operations on a matrix are called as elementary row transformations.

- Interchanging two rows.
- Multiplications of all the elements of row by a non-zero scalar.
- Addition of constant multiple of a row to another row.

### Inverse of a matrix by elementary transformations

Let  $X$ ,  $A$  and  $B$  be matrices of, the same order such that  $X = AB$ . In order to apply a sequence of

elementary row operations on the matrix equation  $X = AB$ , we will apply these row operations simultaneously on  $X$  and on the first matrix  $A$  of the product  $AB$  on RHS.

Similarly, in order to apply a sequence of elementary column operations on the matrix equation  $X = AB$ , we will apply, these operations simultaneously on  $X$  and on the second matrix  $B$  of the product  $AB$  on RHS.

In view of the above discussion, we conclude that if  $A$  is a matrix such that  $A$  exists, then to find  $A^{-1}$  using elementary row operations, write  $A = IA$  and apply a sequence of row operation on  $A = IA$  till we get,  $I = BA$ . The matrix  $B$  will be the inverse of  $A$ . Similarly, if we wish to find  $A^{-1}$  using column operations, then, write  $A = AI$  and apply a sequence of column operations on

$A = AI$  till we get,  $I = AB$ .

**Remark** In case, after applying one or more elementary row (column) operations on  $A = IA$  ( $A = AI$ ), If we obtain all zeros in one or more rows the matrix  $A$  on L.H.S., then  $A^{-1}$  does not exist.

► **Example 3.** By using elementary operations, find

$$\text{the inverse of the matrix } A = \begin{bmatrix} 1 & 2 \\ 0 & -1 \end{bmatrix}.$$

► **Solution** In order to use elementary row operations we may write  $A = IA$ .

$$\begin{aligned} \text{or } \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix} &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} A, \text{ then } \begin{bmatrix} 1 & 2 \\ 0 & -5 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix} A \text{ (applying } R_2 \rightarrow R_2 - 2R_1) \end{aligned}$$

$$\begin{aligned} \text{or } \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} &= \begin{bmatrix} 1 & 0 \\ \frac{2}{5} & \frac{-1}{5} \end{bmatrix} \\ &A \text{ (applying } R_2 \rightarrow -\frac{1}{5} R_2) \end{aligned}$$

$$\begin{aligned} \text{or } \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} &= \begin{bmatrix} \frac{1}{5} & \frac{2}{5} \\ \frac{2}{5} & \frac{-1}{5} \end{bmatrix} \\ &A \text{ (applying } R_1 \rightarrow R_1 - 2R_2) \end{aligned}$$



$$\text{Thus } A^{-1} = \begin{bmatrix} \frac{1}{5} & \frac{2}{5} \\ \frac{2}{5} & \frac{-1}{5} \end{bmatrix}$$

Alternatively, in order to use elementary column operations, we write  $A = AI$ , i.e.,

$$\begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix} = A \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Applying  $C_2 \rightarrow C_2 - 2C_1$ , we get

$$\begin{bmatrix} 1 & 0 \\ 2 & -5 \end{bmatrix} = A \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix}$$

Now applying  $C_2 \rightarrow -\frac{1}{5} C_2$ , we have

$$\begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} = A \begin{bmatrix} 1 & \frac{2}{5} \\ 0 & \frac{-1}{5} \end{bmatrix}$$

Finally, applying  $C_1 \rightarrow C_1 - 2C_2$ , we obtain

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = A \begin{bmatrix} \frac{1}{5} & \frac{2}{5} \\ \frac{2}{5} & \frac{-1}{5} \end{bmatrix}$$

$$\text{Hence } A^{-1} = \begin{bmatrix} \frac{1}{5} & \frac{2}{5} \\ \frac{2}{5} & \frac{-1}{5} \end{bmatrix}$$

► **Example 4.** Find  $P^{-1}$ , if it exists, given

$$P = \begin{bmatrix} 10 & -2 \\ -5 & 1 \end{bmatrix}$$

► **Solution** We have  $P = IP$ , i.e.,  $\begin{bmatrix} 10 & -2 \\ -5 & 1 \end{bmatrix}$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} P.$$

$$\text{or } \begin{bmatrix} 1 & \frac{-1}{5} \\ -5 & 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{10} & 0 \\ 0 & 1 \end{bmatrix} P$$

$$(\text{applying } R_1 \rightarrow \frac{1}{10} R_1)$$

$$\text{or } \begin{bmatrix} 1 & \frac{-1}{5} \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} \frac{1}{10} & 0 \\ \frac{1}{2} & 1 \end{bmatrix} P$$

(applying  $R_2 \rightarrow R_2 + 5R_1$ )

We have all zeros in the second row of the left hand side matrix of the above equation. Therefore,  $P^{-1}$  does not exist.

### Practice Problems

[G]

1. Find the characteristic equation of the matrix

$$A = \begin{bmatrix} 1 & 3 & 7 \\ 4 & 2 & 3 \\ 0 & 2 & 1 \end{bmatrix}$$

and show that it is satisfied by  $A$ . Hence obtain the inverse of the given matrix  $A$ .

2. If  $A = \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}$ , show that  $A^2 - 5A + 7I = O$ . Hence find  $A^{-1}$ .

$$3. \text{ For the matrix } A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & -3 \\ 2 & -1 & 3 \end{bmatrix}$$

Show that  $A^3 - 6A^2 + 5A + 11I = O$ . Hence, find  $A^{-1}$ .

$$4. \text{ If } A = \begin{bmatrix} 1 & 0 & -1 \\ 2 & 3 & 1 \\ -1 & 2 & 3 \end{bmatrix} \text{ and } B = \begin{bmatrix} 3 & 0 & 2 \\ -1 & 2 & 0 \\ 3 & -1 & 1 \end{bmatrix},$$

then solve for  $X$  in the following equations :

$$(i) A + X = B, \quad (ii) XB = A.$$

- 5 Using elementary transformations, find the inverse of the matrices, if it exists :

$$(i) \begin{bmatrix} 2 & -3 \\ -1 & 2 \end{bmatrix} \quad (ii) \begin{bmatrix} 2 & 1 \\ 4 & 2 \end{bmatrix}$$

$$(iii) \begin{bmatrix} 2 & -3 & 3 \\ 2 & 2 & 3 \\ 3 & -2 & 2 \end{bmatrix}$$

6. Let  $A = \begin{bmatrix} 2 & -1 \\ 3 & 4 \end{bmatrix}$ ,  $B = \begin{bmatrix} 5 & 2 \\ 7 & 4 \end{bmatrix}$ ,  $C = \begin{bmatrix} 2 & 5 \\ 3 & 8 \end{bmatrix}$ .

Find a matrix  $D$  such that  $CD - AB = O$ .

7. If  $A = \begin{bmatrix} 1 & 2 \\ 5 & 4 \end{bmatrix}$ , find a nonsingular matrix  $P$

such that  $P^{-1}AP = \begin{bmatrix} 6 & 0 \\ 0 & -1 \end{bmatrix}$ .

8. Suppose  $KA = O$  and  $K$  is idempotent. Define  $G = (A - K)^{-1}$ . Prove that
- (a)  $AG = I - K$       (b)  $AGA = A$   
 (c)  $AGK = O$ .

### 3.11 ORTHOGONAL MATRIX

A square matrix  $A$  is said to be orthogonal if  $A'A = I$ .

A square matrix  $A$  is said to be unitary if  $A(\bar{A})' = I$ ,

where  $\bar{A}$  is the complex conjugate of  $A$ .

#### Properties of orthogonal matrix

- (i) Every orthogonal matrix is nonsingular.  
 $A'A = I$ .  
 $\therefore |A'A| = |I|$ , or,  $|A'| |A| = 1$ .  
 $\therefore |A| \neq 0$ . Thus  $A$  is nonsingular.
- (ii)  $I$  is an orthogonal matrix.  
 $I'I = II' = I$ . Hence  $I$  is orthogonal.
- (iii)  $|A|$  is either 1 or  $-1$  when  $A$  is orthogonal.  
 We have  $A'A = I$ . Then  
 $\therefore |A'A| = |I|$ , or,  $|A'| |A| = 1$ , or,  $|A|^2 = 1$ .  
 $(\because |A'| = |A|)$   
 $\therefore |A| = \pm 1$ .
- (iv) The product of two orthogonal matrices is orthogonal.  
 Let  $A$  and  $B$  be two orthogonal matrices.  
 Therefore,  $A'A = I$ ,  $B'B = I$  and  
 $(AB)'(AB) = (B'A')(AB) = B'(A'A)B = B'IB = B'B = I$ .  
 Hence the result follows.

- (v) The inverse of an orthogonal matrix is orthogonal.

Let  $A$  be an orthogonal matrix and  $B$  be the inverse of it.

Therefore,  $A'A = I$ ,  $AB = BA = I$  and

$(AB)'(AB) = (B'A')(AB) = B'(A'A)B = B'IB = B'B$

But  $(AB)'(AB) = I'I = I$ .

$\therefore B'B = I$

Therefore,  $B$  is orthogonal.

- (vi) The transpose of an orthogonal matrix is orthogonal.

Let us consider a matrix  $B$  such that  $AB = I$  when  $A$  is orthogonal.

$\therefore A'A = I$ ,  $A' = A'I = A'(AB) = (A'A)B = IB = B$ .

$\therefore AB = AA' = I$ .

Now  $(AA') = I'$ , or  $(A')'A' = I$ .

Hence  $A'$  is orthogonal.

► **Example 1.** Verify that the matrix

$\frac{1}{3} \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ -2 & 2 & -1 \end{bmatrix}$  is orthogonal.

► **Solution** Let  $A = \frac{1}{3} \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ -2 & 2 & -1 \end{bmatrix}$ .

Then  $A' = \frac{1}{3} \begin{bmatrix} 1 & 2 & -2 \\ 2 & 1 & 2 \\ 2 & -2 & -1 \end{bmatrix}$ .

We have  $AA' = \frac{1}{9} \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ -2 & 2 & -1 \end{bmatrix} \begin{bmatrix} 1 & 2 & -2 \\ 2 & 1 & 2 \\ 2 & -2 & -1 \end{bmatrix}$

$= \frac{1}{9} \begin{bmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I_3$

Hence the matrix  $A$  is orthogonal.

► **Example 2.** Determine the values of  $\alpha, \beta, \gamma$  where

$\begin{bmatrix} 0 & 2\beta & \gamma \\ \alpha & \beta & -\gamma \\ \alpha & -\beta & \gamma \end{bmatrix}$  is orthogonal.

► **Solution** Let  $A = \begin{bmatrix} 0 & 2\beta & \gamma \\ \alpha & \beta & -\gamma \\ \alpha & -\beta & \gamma \end{bmatrix}$ .

$$\therefore A'A = \begin{bmatrix} 0 & \alpha & \alpha \\ 2\beta & \beta & -\beta \\ \gamma & -\gamma & \gamma \end{bmatrix} \times \begin{bmatrix} 0 & 2\beta & \gamma \\ \alpha & \beta & -\gamma \\ \alpha & -\beta & \gamma \end{bmatrix}$$

$$= \begin{bmatrix} 2\alpha^2 & 0 & 0 \\ 0 & 6\beta^2 & 0 \\ 0 & 0 & 3\gamma^2 \end{bmatrix}.$$

Since A is orthogonal,  $2\alpha^2 = 1$ ,  $6\beta^2 = 1$  and  $3\gamma^2 = 1$ .

$$\therefore \alpha = \pm \frac{1}{\sqrt{2}}, \beta = \pm \frac{1}{\sqrt{6}}, \gamma = \pm \frac{1}{\sqrt{3}}.$$

► **Example 3.** A and B are orthogonal and  $|A| + |B| = 0$ . Prove that A + B is singular.

► **Solution** Since A and B are orthogonal,  $|A| \neq 0$ ,  $|B| \neq 0$ .

$$\text{Let } A^T + B^T = C^T.$$

It implies that  $I + AB^T = AC^T$  and  $B + A = AC^T B$ .

$$\therefore |A + B| = |A| |C^T| |B| = -|A|^2 |C^T|$$

$$(\because |A| + |B| = 0)$$

$$= -|C^T| = -|A^T + B^T| = -|A + B|.$$

$$\therefore |A + B| = 0 \therefore A + B \text{ is singular.}$$

► **Example 4.** If  $(I - A)(I + A)^{-1}$  is orthogonal, prove that A is skew-symmetric.

► **Solution** Let  $B = (I - A)(I + A)^{-1}$ .

Since B is orthogonal,  $B'B = I$ ,

$$\text{or } \{(I + A)^{-1}\}'(I - A)'(I - A)(I + A)^{-1} = I,$$

$$\text{or } \{(I + A)'\}^{-1}(I - A)'(I - A)(I + A)^{-1} = I,$$

$$\text{or } (I - A')(I - A) = (I + A')(I + A),$$

$$\text{or } I - A - A' + A'A = I + A + A' + A'A,$$

$$\text{or } 2(A + A') = O, \text{ or, } A' = -A.$$

Therefore, A is skew-symmetric.

2. Compute

$$D = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \phi & -\sin \phi \\ 0 & \sin \phi & \cos \phi \end{pmatrix} \begin{pmatrix} \cos \psi & -\sin \psi & 0 \\ \sin \psi & \cos \psi & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

and show that D is an orthogonal matrix.

3. (i) If A is skew symmetric prove that  $(I - A)(I + A)^{-1}$  is orthogonal.

(ii) Show that  $AB - BA \neq I$  for  $2 \times 2$  matrices.

4. Prove that the minor of any order of an orthogonal matrix A is equal to its cofactor taken with its sign if  $|A| = 1$  and with sign reversed if  $|A| = -1$ .

5. Show that each of the following three properties of a square matrix follows from the other two : symmetry, orthogonality, involutoriness.

6. Determine the values of the elements of

$$\begin{bmatrix} \alpha_1 & \beta_1 \\ \alpha_2 & \beta_2 \end{bmatrix} \text{ when it is orthogonal.}$$

7. Prove the following statements.

(a) If A and B are symmetric, then  $[(AB)']^{-1} = A^{-1}B^{-1}$ .

(b)  $C = X(X'X)^{-1}X'$  is symmetric and idempotent.

(c) All powers of a symmetric orthogonal matrix are the matrix itself, or an identity matrix.

8. Prove that there is only one form of matrix that is orthogonal and idempotent, and find that form.

### 3.12 ECHELON FORM OF A MATRIX

#### Echelon form of a matrix

A matrix is said to be in Echelon form if it satisfies the following :

- (a) The first non-zero element in each row is 1 & all the other elements in the corresponding column (i.e. the column where 1 appears) are zeroes.

#### Practice Problems

[H]

1. Prove that

$$P = \frac{1}{15} \begin{bmatrix} 5 & -14 & 2 \\ -10 & -5 & -10 \\ 10 & 2 & -11 \end{bmatrix} \text{ is in orthogonal matrix.}$$

- (b) The number of zeros before the first non zero element in any non zero row is less than the number of such zeroes in succeeding non zero rows.

Examples of echelon form of matrices :

$$\begin{bmatrix} 2 & 1 & 5 & 6 \\ 0 & 3 & 2 & 0 \\ 0 & 0 & 1 & 6 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} -2 & 7 & 8 \\ 0 & 0 & 5 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix},$$

$$\begin{bmatrix} 0 & 3 & 8 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

► **Example 1.** Convert

$$A = \begin{bmatrix} 1 & 4 & 6 \\ 4 & -1 & 2 \\ -7 & 6 & 2 \\ 2 & 8 & 13 \end{bmatrix}$$

into echelon form using elementary row transformations.

► **Solution** The operations

$$R_2 - 4R_1$$

$$R_3 + 7R_1 \quad \text{give } A \sim \begin{bmatrix} 1 & 4 & 6 \\ 0 & -17 & -22 \\ 0 & 34 & 44 \\ 0 & 0 & 1 \end{bmatrix}$$

$$R_4 - 2R_1 \quad \text{and then}$$

$$R_3 + 2R_2 \quad \text{yields } A \sim \begin{bmatrix} 1 & 4 & 6 \\ 0 & -17 & -22 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Finally, in order to have all non-null rows preceding null rows, interchange rows 3 and 4 to get

$$A \sim \begin{bmatrix} 1 & 4 & 6 \\ 0 & -17 & -22 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

## Rank of a matrix

Let  $A = [a_{ij}]_{m \times n}$  be a given matrix. A natural number  $\rho$  is said to be the rank of  $A$  if  $A$  has a nonsingular submatrix of order  $\rho$  and it has no non-singular submatrix of order more than  $\rho$ .

Rank of zero matrix is regarded to be zero.

For example  $A = \begin{bmatrix} 3 & -1 & 2 & 5 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 5 & 0 \end{bmatrix}$

we have  $\begin{bmatrix} 3 & 2 \\ 0 & 2 \end{bmatrix}$  as a non-singular submatrix.

The square matrices of order 3 are

$$\begin{bmatrix} 3 & -1 & 2 \\ 0 & 0 & 2 \\ 0 & 0 & 5 \end{bmatrix}, \begin{bmatrix} 3 & -1 & 5 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix},$$

$$\begin{bmatrix} 3 & 2 & 5 \\ 0 & 2 & 0 \\ 0 & 5 & 0 \end{bmatrix}, \begin{bmatrix} -1 & 2 & 5 \\ 0 & 2 & 0 \\ 0 & 5 & 0 \end{bmatrix}$$

and all these are singular. Hence rank of  $A$  is 2.

Another example :

The matrix  $\begin{bmatrix} 4-x & 2\sqrt{5} & 0 \\ 2\sqrt{5} & 4-x & \sqrt{5} \\ 0 & \sqrt{5} & 4-x \end{bmatrix}$

has rank less than 3 if and only if its determinant, which is the polynomial  $(4-x)^3 - 25(4-x)$ , is zero. This polynomial vanishes when  $x = 9, 4$ , or  $-1$ . When  $x = 9$ , we have the singular matrix

$$\begin{bmatrix} -5 & 2\sqrt{5} & 0 \\ 2\sqrt{5} & -5 & \sqrt{5} \\ 0 & \sqrt{5} & -5 \end{bmatrix},$$

which has, among other, the nonsingular submatrix

$$\begin{bmatrix} -5 & \sqrt{5} \\ \sqrt{5} & -5 \end{bmatrix},$$

so that its rank is 2. It is easy to verify that the rank

is 2 also when  $x = 4$  or  $-1$ . For any other value of  $x$ , the rank is, of course, 3.



Rank of a matrix is the number of non-zero rows (i.e. number of rows with atleast one non-zero element) in the echelon form of the matrix.

Hence to find the rank of a given matrix we may reduce it to Echelon form using elementary row transformations and then count the number of non zero rows. Elementary transformation of matrix does not affect its rank.

$$\text{Let } A = \begin{bmatrix} 1 & 2 & 4 & 3 \\ 3 & -1 & 2 & -2 \\ 5 & -4 & 0 & -7 \end{bmatrix}$$

Carrying out the operations

$$R_2 - 3R_1 \text{ gives } A \sim \begin{bmatrix} 1 & 2 & 4 & 3 \\ 0 & -7 & -10 & -11 \\ 0 & -14 & -20 & -22 \end{bmatrix}$$

$$R_3 - 5R_1$$

and then on this matrix the operation

$$R_3 - 2R_2 \text{ gives } A \sim \begin{bmatrix} 1 & 2 & 4 & 3 \\ 0 & -7 & -10 & -11 \\ 0 & 0 & 0 & 0 \end{bmatrix} = B$$

B is in echelon form, its rank is 2 and that is the rank of A.

► **Example 2.** Determine the rank of the following matrix for different values of  $\lambda$ .

$$\begin{bmatrix} \lambda & 1 & 0 \\ 3 & \lambda - 2 & 1 \\ 3(\lambda + 1) & 0 & \lambda + 1 \end{bmatrix}.$$

► **Solution**

$$\begin{bmatrix} \lambda & 1 & 0 \\ 3 & \lambda - 2 & 1 \\ 3(\lambda + 1) & 0 & \lambda + 1 \end{bmatrix} \sim \begin{bmatrix} \lambda & 1 & 0 \\ 0 & \lambda - 2 & 1 \\ 0 & 0 & \lambda + 1 \end{bmatrix}$$

$$\text{Now } \begin{bmatrix} \lambda & 1 & 0 \\ 0 & \lambda - 2 & 1 \\ 0 & 0 & \lambda + 1 \end{bmatrix} = \lambda(\lambda - 2)(\lambda + 1).$$

If  $\lambda \neq 0, 2, -1$ , the determinant is not zero and the rank of the given matrix is 3.

If  $\lambda = 0$ , one minor of the equivalent matrix

$$\begin{vmatrix} 1 & 0 \\ -2 & 1 \end{vmatrix} \neq 0.$$

Therefore, the rank of the given matrix is 2. Similarly the rank is 2 for  $\lambda = 2$  or  $-1$ .

**Note :** In computing the **rank** of a matrix, move from minors of smaller order to minors of greater order. If a non-zero  $k$ th-order minor  $D$  has already been found, then only the  $(k + 1)$ th-order minors bordering minor  $D$  need be computed ; if they are all zero, the rank of the matrix is  $k$ .

► **Example 3.** Find the rank of the matrix

$$A = \begin{bmatrix} 2 & -4 & 3 & 1 & 0 \\ 1 & -2 & 1 & -4 & 2 \\ 0 & 1 & -1 & 3 & 1 \\ 4 & -7 & 4 & -4 & 5 \end{bmatrix}$$

► **Solution** The second-order minor in the upper left corner of this matrix is zero. However, the matrix also contains non-zero minors of order two, for instance,

$$d = \begin{vmatrix} -4 & 3 \\ -2 & 1 \end{vmatrix} \neq 0$$

$$\text{The third-order minor } d' = \begin{vmatrix} 2 & -4 & 3 \\ 1 & -2 & 1 \\ 0 & 1 & 1 \end{vmatrix}$$

bordering minor  $d$  is different from zero,  $d' = 1$ , but both fourth-order minors bordering minor  $d'$  are zero:

$$\begin{vmatrix} 2 & -4 & 3 & 1 \\ 1 & -2 & 1 & -4 \\ 0 & 1 & -1 & 3 \\ 4 & -7 & 4 & -4 \end{vmatrix} = 0, \quad \begin{vmatrix} 2 & -4 & 3 & 0 \\ 1 & -2 & 1 & 2 \\ 0 & 1 & -1 & 1 \\ 4 & -7 & 4 & 5 \end{vmatrix} = 0$$

Thus, the rank of matrix A is three.



- (i) For any non-zero matrix  $A$  of type  $m \times n$ , the rank cannot exceed  $m$  or  $n$ , i.e.  
 $0 < R(A) \leq \min(m, n)$ .
- (ii) It is equally obvious that, for a square matrix  $A$  of order  $n$ ,  $R(A) < n$  or  $R(A) = n$  according as  $A$  is singular or non-singular.
- (iii) If  $A$  and  $B$  are matrices (of the same type), then  $R(A + B) \leq R(A) + R(B)$ .
- (iv) The rank of a product of two matrices is not greater than the rank of either factor, i.e. if  $AB$  exists, then  $R(AB) \leq \min\{R(A), R(B)\}$ .  
 In the case of singular matrices, the multiplication theorem for determinants does not lead to any utterance beyond the fact that their product will also be singular, although singular square matrices can be distinguished according to rank as well. Note that there is no completely definite relationship between the ranks of the factors and the rank of the product, as is evident from a glance at the following examples:

$$\begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix} \cdot \begin{pmatrix} 3 & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 6 & 0 \\ 0 & 0 \end{pmatrix},$$

$$\begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix} \cdot \begin{pmatrix} 0 & 0 \\ 0 & 3 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

In both cases, matrices of rank 1 are multiplied, but in one case the product has rank 1, in the other, rank 0. It is only the following theorem which holds true (and not only for square but for rectangular matrices as well).

The rank of a product of matrices does not exceed the rank of each of the factors.

- (v) The rank of the product obtained by pre or post-multiplication of an arbitrary matrix  $A$  by a nonsingular square matrix  $Q$  is equal to the rank of matrix  $A$ .

For example, suppose

$$AQ = C \quad \dots(1)$$

From the preceding theorem it follows that the rank of matrix  $C$  is not greater than the rank of matrix  $A$ . However, multiplying (1) on the right by  $Q^{-1}$ , we arrive at the equation

$$A = CQ^{-1}$$

and for this reason, again on the basis of the preceding theorem, the rank of  $A$  does not exceed that of  $C$ . A comparison of these two results proves the coincidence of the ranks of matrices  $A$  and  $C$ .

### Practice Problems

[1]

1. Find the values of  $\lambda$  such that the matrix

$$\begin{pmatrix} 3 - \lambda & 2 \\ 1 & 1 - \lambda \end{pmatrix} \text{ is of rank 1.}$$

2. Find  $x$  such that

$$(i) \text{ rank } \begin{bmatrix} 2 & 1 & 4 \\ 1 & x & 2 \\ 4 & 0 & x + 2 \end{bmatrix} = 2,$$

$$(ii) \text{ rank } \begin{bmatrix} 1 & x & x & x \\ x & 1 & x & x \\ x & x & 1 & x \\ x & x & x & 1 \end{bmatrix} < 4.$$

$$3. \text{ For } A = \begin{bmatrix} 1 & 2 & 4 & 0 \\ -2 & -3 & -1 & 1 \\ 0 & 1 & 7 & 1 \\ -2 & -2 & 6 & 2 \\ -3 & -6 & -12 & 0 \end{bmatrix},$$

show the following :

- (a) The rank of  $A$  is 2.  
 (b) All minors of  $A$  of order 3 or more are zero.
4. Show that the rank of a skew-symmetric matrix cannot be odd.
5. Given two matrices find whether the matrices have the same rank:

$$(i) \begin{pmatrix} 1 & 5 & 4 \\ 0 & 2 & 7 \\ 0 & 0 & 3 \\ 0 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 5 & 4 & 2 \\ 0 & 2 & 7 & 1 \\ 0 & 0 & 3 & 8 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$(ii) \begin{pmatrix} 1 & 3 & 0 & 2 \\ 0 & 1 & 1 & 4 \\ 0 & 0 & 1 & 8 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 3 & 0 & 2 & 0 \\ 0 & 1 & 1 & 4 & 0 \\ 0 & 0 & 1 & 8 & 1 \\ 0 & 0 & 0 & 0 & 2 \end{pmatrix}$$

6. Determine the rank of the matrix

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 3 & -2 & x \\ 2 & 2x-2 & -x-2 & 3x-1 \\ 3 & x+2 & -3 & 2x+1 \end{bmatrix}$$

for different values of  $x$ .

### 3.13 SYSTEM OF LINEAR EQUATIONS

A simple example of a linear equation is

$$ax + by = c$$

where  $x$  and  $y$  are real variables and  $a$ ,  $b$ , and  $c$  are given real numbers. If  $a$  and  $b$  are not both zero, then the graph of this equation is a straight line. The linear equation with three real variables  $x$ ,  $y$  and  $z$

$$ax + by + cz = d$$

where  $a$ ,  $b$ ,  $c$  and  $d$  are real numbers, not all zero, represents a plane in the standard XYZ coordinate system.

A linear equation in the  $n$  variables  $x_1, x_2, \dots, x_n$  is any equation of the form

$$c_1x_1 + c_2x_2 + \dots + c_nx_n = r$$

where  $c_1, c_2, \dots, c_n$  and  $r$  are real or complex numbers.

An equation which is not in the form given in definition is called nonlinear. For example

$$x_1^2 + x_2^2 = 4$$

$$-\sqrt{x} + y = 2$$

and  $x_1 + |x_2| - 3x_3 = 7$  are nonlinear equations.

A solution of the single equation

$$c_1x_1 + c_2x_2 + \dots + c_nx_n = r$$

is an ordered  $n$ -tuple of complex numbers  $(s_1, s_2, \dots, s_n)$  which when substituted ( $s_1$  for  $x_1$ ,  $s_2$  for  $x_2$ , and so on) into the equation make it a true mathematical statement.

We consider the problem of solving the system of linear equations with  $m$  equations in  $n$  unknowns

$$x_1, x_2, \dots, x_n,$$

$$a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = b_2 \quad \dots(1)$$

$$\dots\dots\dots$$

$$a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n = b_m$$

When atleast one of the "right-hand side" values  $b_1, b_2, \dots, b_m$  is not equal to zero, the system of equation is called nonhomogeneous.

If  $b_1 = b_2 = \dots = b_m = 0$ , the system of equations is called homogeneous.

Now there are two issues :

1. To determine whether solutions exist, where a solution is an  $n$ -tuple of numbers which satisfies all the equations in (1) simultaneously.
2. To compute all of them, if solutions exist.

### Methods of solving a system of linear equations in two variables

#### The Substitution Method

In a system of two equations with two variables, the substitution method involves using one equation to find an expression for one variable in terms of the other, then substituting into the other equation of the system.

► **Example 1.** Solve the system

$$3x + 2y = 11 \quad \dots(1)$$

$$-x + y = 3 \quad \dots(2)$$

► **Solution** We solve equation (2) for  $y$  to get  $y = x + 3$ . We substitute  $x + 3$  for  $y$  in equation (1) to get

$$3x + 2(x + 3) = 11 \quad \dots(3)$$

Now we solve equation (3) for  $x$ .

$$3x + 2x + 6 = 11$$

$$x = 1$$

Replace  $x$  with 1 in  $y = x + 3$  to find that  $y = 1 + 3 = 4$ . The solution for this system is (1, 4).

## The Addition Method

Another method of solving systems of two equations is the addition method. With this method, we first multiply the equations on both sides by suitable numbers, so that when they are added, one variable is eliminated. The result is an equation in one variable that can be solved easily. The solution is then substituted into one of the original equations, making it possible to solve for the other variable.

► **Example 2.** Solve the system

$$3x - 4y = 1 \quad \dots(1)$$

$$2x + 3y = 12 \quad \dots(2)$$

► **Solution** To eliminate  $x$ , multiply both sides of equation (1) by  $-2$  and both sides of equation (2) by  $3$  to get equations (3) and (4).

$$-6x + 8y = -2 \quad \dots(3)$$

$$6x + 9y = 36 \quad \dots(4)$$

Although this new system is not the same as the given system, it will have the same solution set.

Now, add the two equations to eliminate  $x$ , and then solve the result for  $y$ .

$$17y = 34 \Rightarrow y = 2$$

Substitute  $2$  for  $y$  in equation (1) or (2). Choosing equation (1) gives.

$$3x - 4(2) = 1 \Rightarrow x = 3$$

The solution of the given system is  $(3, 2)$ .



Two systems of linear equations (in the same unknowns) are equivalent if they have the same solutions. In the process of solving, the given system should be replaced by new systems that have the same solution set as the original system. Systems that have the same solution set are called equivalent systems.

## Nature of Solution

If a system of linear equations has solution(s), it is termed consistent. If there is no solution then the system is said to be inconsistent. A consistent system is called determinate (independent) if it has a unique solution and indeterminate (dependent) if there are more than one solution. As we shall learn later on, there may even be an infinite number of solutions.

There are three possibilities which can occur in case of nonhomogeneous system of two linear equations in two unknowns.

The system

$$x - y = 0 \quad \dots(1)$$

$$x + y = 2$$

is consistent and determinate since it has only one solution  $x = 1, y = 1$ . This solution is unique.

And the system

$$x + y = 2 \quad \dots(2)$$

$$x + y = 3$$

is inconsistent since it has no solution. The system

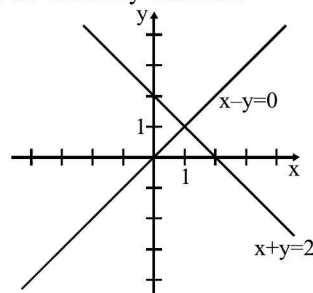
$$x + y = 2 \quad \dots(3)$$

$$2x + 2y = 4$$

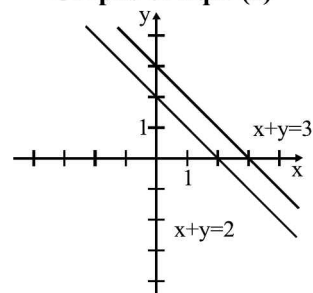
is consistent and indeterminate since it has infinitely many solutions of the form

$$x = k, \quad y = 2 - k$$

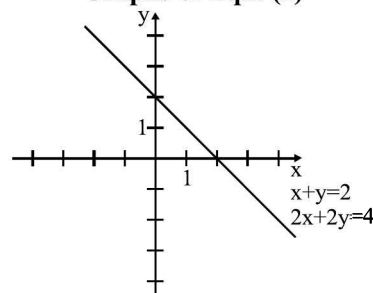
where  $k$  is an arbitrary number.



Graphs of Eqn. (1)



Graphs of Eqn. (2)



Graphs of Eqn. (3)



The possible graphs of two linear equations in  $x$  and  $y$  are as follows :

1. The graphs of the two equations intersect in a single point. The coordinates of this point gives the solution of the system. See figure (1).
2. The graphs are distinct parallel lines which do not intersect. In this case, the system is said to be inconsistent. That is, there is no solution common to both equations. See Figure (2).
3. The graphs are the same line. In this case, the equations are said to be dependent, and any solution of one equation is also a solution of the other. Thus, there are infinitely many solutions. See Figure (3).

Let  $a_1x + b_1y + c_1 = 0$   
 $a_2x + b_2y + c_2 = 0$  then

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$$

$\Rightarrow$  the given equations have unique solution

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$$

$\Rightarrow$  the given equations are inconsistent

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

$\Rightarrow$  the given equations are dependent.

A certain amount of complication is introduced into the study of linear equations by the possible presence of **redundant equations**. Consider, for instance, the system of equations

$$x + y = 5, \quad 2x + y = 3, \quad 3x + 2y = 8.$$

It is clear that the last equation adds nothing to the information provided by the first two, since it can be obtained by adding these equations. Such an equation of the system is called **redundant** (with respect to the system in question.) It is easy to see that redundant equations can be discarded.

Now let us consider a system of homogeneous linear equations.

It has either exactly one solution namely,  $x = y = 0$ , called the trivial solution or an infinite number of solutions (including the

trivial solution) called as nontrivial solution. Note that a solution is nontrivial if atleast one of the unknowns is not equal to zero.

For example, the system

$$x + y = 0$$

$$x - y = 0$$

has only the solution  $x = 0, y = 0$ , while the system

$$x + y = 0$$

$$2x + 2y = 0$$

has nontrivial solutions  $x = k, y = -k$ , for any number  $k$  (if  $k = 0$ , we get the trivial solution).

### A system of linear equations in three variables

Our work with systems of equations so far has dealt strictly with systems in two variables. We can extend the ideas of systems of equations to linear equations of the form  $ax + by + cz = d$ . The method for solving a system of linear equations in three variables is similar to that used for systems of linear equations in two variables. We use addition to eliminate any variable, reducing the system to two equations in two variables. Once we obtain a system of two equations in two variables, we use the addition method again to solve the system.

A solution of such an equation is called an ordered triple, and is denoted  $(x, y, z)$ . For example,  $(1, 2, -4)$  is a solution of  $2x + 5y - 3z = 24$ . The solution set of such an equations is an infinite set of ordered triples. In geometry, the graph of a linear equation in three variables is a plane representing three equations in three unknowns shows that the solution set of such a system may be either a single ordered triple  $(x, y, z)$ , an infinite set of ordered triples (dependent equations), or the empty set (an inconsistent system).

For an example involving three unknowns, consider

$$x - z = 0$$

$$y - z = 0$$

where we require the solution to be real. Each equation is that of a plane passing through the origin. The solution is geometrically the straight line of the intersection of the planes, which is analytically  $x = y = z = k$ , where  $k$  is any real number.

Consider the system:

$$x + y = 5$$

$$2x + 2y = 11.$$

Obviously if one is true the other cannot be, for the second is incompatible, or inconsistent, with the first. Considered as a set of linear equations they are said to be inconsistent.

Similarly, consider

$$2x_1 + 3x_2 + x_3 = 14$$

$$x_1 + x_2 + x_3 = 6$$

$$3x_1 + 5x_2 + x_3 = 19 ;$$

Subtracting twice the first from the sum of the second and third leads to the meaningless result  $0 = -3$ . This is because the left-hand side of the third equation is twice that of the first minus that of the second, but without the same relationship being true of the right-hand sides. This incompatibility leads to the equation being described as inconsistent.

► **Example 3.** Solve the system of linear equations.

$$x + y - 3z = -1 \quad \dots(1)$$

$$y - z = 0 \quad \dots(2)$$

$$-x + 2y = 1 \quad \dots(3)$$

► **Solution** We convert the system as follows.

$$x + y - 3z = -1 \quad \text{Adding the first equation to}$$

$$y - z = 0 \quad \text{the third equation produces}$$

$$3y - 3z = 0 \quad \text{a new third equation}$$

$$x + y - 3z = -1 \quad \text{Adding } -3 \text{ times the second}$$

$$y - z = 0 \quad \text{equation to the third equation}$$

$$0 = 0 \quad \text{produces a new third equation}$$

This means that equation 3 is dependent on equation 1 and 2 in the sense that it gives no additional information about the variables. Thus, the original system is equivalent to the system

$$x + y - 3z = -1$$

$$y - z = 0.$$

In this last equation, solve for  $y$  in terms of  $z$  to obtain  $y = z$ . Back-substituting for  $y$  into the previous equation, you can find  $x$  in terms of  $z$ , as follows.

$$x + z - 3z = -1$$

$$x - 2z = -1$$

$$x = 2z - 1$$

Finally, letting  $z = a$ , the solutions to the given system are all of the form  $x = 2a - 1$ ,  $y = a$ ,  $z = a$ , where  $a$  is a real number. Therefore, every ordered triple that is in the form  $(2a - 1, a, a)$ , where  $a$  is a real number, is a solution of the system.

### Practice Problems

[J]

1. Classify the following equations as linear or nonlinear.

$$(i) \quad x^2 + y - z = 7$$

$$(ii) \quad -|x| + y = 0$$

$$(iii) \quad 3x + 2y - z = 6$$

$$(iv) \quad y + \sqrt{x^2} = 4$$

2. A consulting firm bills clients for computer time  $C$ , automobile travel  $T$ , and hours of actual work  $H$ . The rates are, respectively, Rs. 400 per hour, Rs. 0.35 per mile, and Rs. 120 per hour. Write a linear equation for the billing amount  $B$  in terms of  $C$ ,  $T$ , and  $H$ .

$$3. \text{ Solve the system } \begin{cases} 2x + 3y = 13, \\ 4x + 6y = 20 \end{cases}$$

$$4. \text{ Solve the system } \begin{cases} 4x - 3y = 7, \\ 20x - 15y = 35 \end{cases}$$

5. Show that the systems  $x + y + z = 0$ ,  $2x + y - 2z = 1$ , and  $x + y + z = 0$ ,  $3x + 2y - z = 1$  are equivalent.

### 3.14 CRAMER'S RULE

Let the  $n$  linear simultaneous equations in  $n$  unknowns  $x_1, x_2, x_3, \dots, x_n$  be

$$a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1,$$

$$a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = b_2,$$

$$\dots \dots \dots \dots \dots$$

$$\dots \dots \dots \dots \dots$$

$$a_{n1}x_1 + a_{n2}x_2 + \dots + a_{nn}x_n = b_n$$

$$\text{Let } \Delta = \begin{vmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{vmatrix} \neq 0.$$

Suppose  $A_{11}, A_{12}, A_{13}, \dots$  etc. denote the cofactors of  $a_{11}, a_{12}, a_{13}, \dots$  etc. in  $\Delta$ . Then multiplying the given equations respectively by  $A_{11}, A_{21}, A_{31}, \dots, A_{n1}$  and adding, we get

$$\begin{aligned} & x_1(a_{11}A_{11} + a_{21}A_{21} + \dots + a_{n1}A_{n1}) \\ & + x_2(0) + x_3(0) \\ & + \dots + x_n(0) = b_1A_{11} + b_2A_{21} + \dots + b_nA_{n1} \end{aligned}$$

or  $x_1\Delta = \Delta_1$ ,

where  $\Delta_1$  is the determinant obtained by replacing the elements in the first column of  $\Delta$  by the elements  $b_1, b_2, \dots, b_n$ .

Again multiplying the given equations respectively by  $A_{12}, A_{22}, \dots, A_{n2}$  and adding, we get

$$x_2\Delta = \Delta_2,$$

where  $\Delta_2$  is the determinant obtained by replacing the elements in the second column of  $\Delta$  by the elements  $b_1, b_2, \dots, b_n$ .

Similarly, we get

$$x_3\Delta = \Delta_3$$

.....

$$x_n\Delta = \Delta_n$$

This method of solving  $n$  simultaneous equations in  $n$  unknowns is known as Cramer's rule.

Thus by Cramer's rule, if  $\Delta \neq 0$ , we have

$$x_i = \frac{\Delta_i}{\Delta}, i = 1, 2, \dots, n,$$

where  $\Delta_i$  is the determinant obtained by replacing the  $i$ th column in  $\Delta$  by the elements  $b_1, b_2, \dots, b_n$ .

### Simultaneous Equations Involving Three Unknowns

$$\text{Let } a_1x + b_1y + c_1z = d_1 \quad \dots(1)$$

$$a_2x + b_2y + c_2z = d_2 \quad \dots(2)$$

$$a_3x + b_3y + c_3z = d_3 \quad \dots(3)$$

$$\text{Then, } x = \frac{\Delta_1}{\Delta}, y = \frac{\Delta_2}{\Delta}, z = \frac{\Delta_3}{\Delta}.$$

$$\text{where } \Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}; \Delta_1 = \begin{vmatrix} d_1 & b_1 & c_1 \\ d_2 & b_2 & c_2 \\ d_3 & b_3 & c_3 \end{vmatrix};$$

$$\Delta_2 = \begin{vmatrix} a_1 & d_1 & c_1 \\ a_2 & d_2 & c_2 \\ a_3 & d_3 & c_3 \end{vmatrix} \& \Delta_3 = \begin{vmatrix} a_1 & b_1 & d_1 \\ a_2 & b_2 & d_2 \\ a_3 & b_3 & d_3 \end{vmatrix}$$

(a) If  $\Delta \neq 0$  and atleast one of  $\Delta_1, \Delta_2, \Delta_3 \neq 0$ , then the given system of equations are consistent and have unique non-trivial solution.

(b) If  $\Delta \neq 0$  &  $\Delta_1 = \Delta_2 = \Delta_3 = 0$ , then the given system of equations are consistent and have trivial solution only.

(c) If  $\Delta = \Delta_1 = \Delta_2 = \Delta_3 = 0$ , then the given system of equations either have infinitely many solutions or no solution.

To distinguish find the cofactors of  $\Delta$ .

(i) If atleast one of the cofactors is non-zero then the system has infinitely many solutions.

(ii) If all the cofactors of  $\Delta$  are zero then find the cofactors of  $\Delta_1, \Delta_2, \Delta_3$ .

If all the cofactors of  $\Delta_1, \Delta_2, \Delta_3$  are zero then the system has infinitely many solutions.

If atleast one of the cofactors of  $\Delta_1, \Delta_2, \Delta_3$  is non-zero then the system has no solution.

(d) If  $\Delta = 0$  but atleast one of  $\Delta_1, \Delta_2, \Delta_3$  is not zero then the equations are inconsistent and have no solution.

$$\text{Let } a_1x + b_1y + c_1 = 0 \quad \dots(1)$$

$$a_2x + b_2y + c_2 = 0 \quad \dots(2)$$

$$a_3x + b_3y + c_3 = 0 \quad \dots(3)$$

If  $\Delta = 0$  the system is consistent provided

$$\frac{a_1}{b_1} = \frac{a_2}{b_2} = \frac{a_3}{b_3} \text{ does not hold true.}$$

If  $\Delta \neq 0$  the system is inconsistent.

### Study Tip

Cramer's rule does not apply if  $\Delta = 0$ . When  $\Delta = 0$ , the system either is inconsistent or contains infinitely many solutions. To find the nature of solution and the solution itself when  $\Delta = 0$ , we should go to the method of elimination for quicker results. The significance of Cramer's rule lies mainly in the fact that for cases when it is applicable it offers an explicit expression of the solution of the system in terms of the coefficients of the system. However,

Cramer's rule involves very unwieldy computations; in the case of a system of  $n$  linear equations in  $n$  unknowns, one has to compute  $n + 1$  determinants of the  $n$ th order. The method of elimination of unknowns is much more convenient in this respect since the computations involved here are actually equivalent to those required in the evaluation of a single determinant of the  $n$ th order.

► **Example 1.** Solve the following system of two equations in two unknowns:

$$\begin{cases} 5x - 3y = 16, \\ x + 2y = 11 \end{cases}$$

► **Solution** Let us write down and compute  $\Delta$ ,  $\Delta_1$  and  $\Delta_2$ :

$$\begin{aligned} \Delta &= 5 \times 2 - 1 \times (-3) = 13, \\ \Delta_1 &= 16 \times 2 - 11 \times (-3) = 65, \\ \Delta_2 &= 5 \times 11 - 16 \times 1 = 39; \end{aligned}$$

$$\text{then } x = \frac{\Delta_1}{\Delta} = \frac{65}{13} = 5, \quad y = \frac{\Delta_2}{\Delta} = \frac{39}{13} = 3.$$

► **Example 2.** Solve the given system of two homogeneous equations in two unknowns

$$\begin{cases} 5x + 3y = 0, \\ 2x - 4y = 0. \end{cases}$$

► **Solution** Compute  $\Delta = 5 \times (-4) - 2 \times 3 = -26$ . Since  $\Delta \neq 0$ , the system has a trivial (unique) solution:  $x = 0$ ,  $y = 0$ .

► **Example 3.** Solve the system  $\begin{cases} 3x + 5y = 0, \\ 9x + 15y = 0. \end{cases}$

► **Solution** Since  $\Delta = 3 \times 15 - 9 \times 5 = 0$ , the given system has an infinite number of solutions:

$$\left(x, -\frac{3}{5}x\right), \text{ where } x \in \mathbb{R}.$$

► **Example 4.** Solve the system of equations

$$\begin{cases} 2x_1 - x_2 + x_3 = 0, \\ 3x_1 + 2x_2 - 5x_3 = 1, \\ x_1 + 3x_2 - 2x_3 = 4 \end{cases}$$

► **Solution** The determinant of the coefficients is non-zero:

$$\Delta = \begin{vmatrix} 2 & -1 & 1 \\ 3 & 2 & -5 \\ 1 & 3 & -2 \end{vmatrix} = 28$$

so the Cramer rule is applicable. The numerators for the unknowns are

$$\Delta_1 = \begin{vmatrix} 0 & -1 & 1 \\ 1 & 2 & -5 \\ 4 & 3 & -2 \end{vmatrix} = 13,$$

$$\Delta_2 = \begin{vmatrix} 2 & 0 & 1 \\ 3 & 1 & -5 \\ 1 & 4 & -2 \end{vmatrix} = 47.$$

$$\Delta_3 = \begin{vmatrix} 2 & -1 & 0 \\ 3 & 2 & 1 \\ 1 & 3 & 4 \end{vmatrix} = 21$$

Hence, the following numbers constitute the solution of the system :

$$x_1 = \frac{13}{28}, x_2 = \frac{47}{28}, x_3 = \frac{21}{28} = \frac{3}{4}$$

► **Example 5.** Using Cramer's rule, solve the system of equations

$$\begin{cases} 2x_1 - x_2 = 1, \\ -x_2 + 2x_2 - x_3 = 0, \\ -x_2 + 2x_3 - x_4 = 0, \\ -x_3 + 2x_4 = 0. \end{cases}$$

► **Solution** We compute the determinant  $\Delta$  using the expansion by the first row:

$$\begin{aligned} \Delta &= \begin{vmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & 1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{vmatrix} \\ &= 2 \begin{vmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & 1 & 2 \end{vmatrix} - (-1) \begin{vmatrix} 1 & -1 & 0 \\ 0 & 2 & -1 \\ 0 & -1 & 2 \end{vmatrix} \end{aligned}$$

$$= 2(8 - 2 - 2) - 1(4 - 1) = 8 - 3 = 5.$$

We then form the determinants  $\Delta_i$  ( $i = 1, 2, 3, 4$ ) and compute them in succession by expanding by the  $i$ th column :

$$\Delta_1 = \begin{vmatrix} 1 & -1 & 0 & 0 \\ 0 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{vmatrix} = 1 \begin{vmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{vmatrix} \\ = 1(8 - 2 - 2) = 4$$

$$\Delta_2 = \begin{vmatrix} 2 & 1 & 0 & 0 \\ -1 & 0 & -1 & 0 \\ 0 & 0 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{vmatrix} = -1 \begin{vmatrix} -1 & -1 & 0 \\ 0 & 2 & -1 \\ 0 & -1 & 2 \end{vmatrix} \\ = 1(4 - 1) = 3$$

$$\Delta_3 = \begin{vmatrix} 2 & -1 & 1 & 0 \\ -1 & 2 & 0 & 0 \\ 0 & -1 & 0 & -1 \\ 0 & 0 & 0 & 2 \end{vmatrix} = 1 \begin{vmatrix} -1 & 2 & 0 \\ 0 & -1 & -1 \\ 0 & 0 & 2 \end{vmatrix} = 2$$

$$\Delta_4 = \begin{vmatrix} 2 & -1 & 0 & 1 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & 0 \\ 0 & 0 & -1 & 0 \end{vmatrix} = -1 \begin{vmatrix} -1 & 2 & -1 \\ 0 & -1 & 2 \\ 0 & 0 & -1 \end{vmatrix} = 1$$

Consequently,  $x_1 = 4/5$ ,  $x_2 = 3/5$ ,  $x_3 = 2/5$ , and  $x_4 = 1/5$ .

► **Example 6.** Find the value of  $\lambda$ , if the following equations are consistent :

$$x + y - 3 = 0$$

$$(1 + \lambda)x + (2 + \lambda)y - 8 = 0$$

$$x - (1 + \lambda)y + (2 + \lambda) = 0$$

► **Solution** The given equations in two unknowns are consistent, when  $\Delta = 0$

$$\text{i.e., } \begin{vmatrix} 1 & 1 & -3 \\ 1 + \lambda & 2 + \lambda & -8 \\ 1 & -(1 + \lambda) & 2 + \lambda \end{vmatrix} = 0$$

Applying  $C_2 \rightarrow C_2 - C_1$  and  $C_3 \rightarrow C_3 + 3C_1$

$$\therefore \begin{vmatrix} 1 & 0 & 0 \\ 1 + \lambda & 1 & 3\lambda - 5 \\ 1 & -2 - \lambda & 5 + \lambda \end{vmatrix} = 0$$

$$\Rightarrow 1 \begin{vmatrix} 1 & 3\lambda - 5 \\ -2 - \lambda & 5 + \lambda \end{vmatrix} = 0$$

$$\Rightarrow (5 + \lambda) - (3\lambda - 5)(-2 - \lambda) = 0$$

$$\Rightarrow 5 + \lambda + 6\lambda + 3\lambda^2 - 10 - 5\lambda = 0$$

$$\Rightarrow 3\lambda^2 + 2\lambda - 5 = 0$$

$\therefore \lambda = 1, -5/3$  (with these values of

$$\lambda, \frac{a_1}{b_1} = \frac{a_2}{b_2} = \frac{a_3}{b_3} \text{ does not hold true})$$

Hence the system is consistent.

### Practice Problems

[K]

1. Use Cramer's rule to solve the following systems.

$$\begin{array}{ll} \text{(i) } 3x - 2y = 7 & \text{(ii) } x - 2y + z = 3 \\ 4x + y = 8 & -x - y - z = 5 \\ 3x + y - 7 & z = -3 \end{array}$$

2. Solve the system  $\begin{cases} x - (\alpha - 1)y = 1, \\ \alpha x - 2y = 4 - \alpha, \alpha \in \mathbb{R} \end{cases}$ .

3. Find the equation of the parabola  $y = ax^2 + bx + c$  that passes through  $(2, 4)$ ,  $(-1, 1)$ , and  $(-2, 5)$ .

4. Solve the system

$$\frac{1}{x} + \frac{1}{y} - \frac{1}{z} = \frac{1}{4}$$

$$\frac{2}{x} - \frac{1}{y} + \frac{3}{z} = \frac{9}{4}$$

$$-\frac{1}{x} - \frac{2}{y} + \frac{4}{z} = 1$$

5. For what value of  $k$  will the system

$$2x_1 + x_2 = 5, \quad x_1 - 3x_2 = -1, \quad 3x_1 + 4x_2 = k$$

be solvable? Solve the system for that value of  $k$ .

6. For what of  $k$  the following system has non-trivial solution and find the solution for  $k = -1$ .

$$x + 2y + 3z = kx, \quad 2x + y + 3z = ky, \\ 2x + 3y + z = kz.$$

### 3.15 SOLVING SYSTEM OF LINEAR EQUATIONS USING INVERSE

Consider the system

$$a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = b_2$$

$$\dots\dots\dots$$

$$a_{n1}x_1 + a_{n2}x_2 + \dots + a_{nn}x_n = b_n.$$

$$\text{Let } A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix}, X = \begin{bmatrix} x_1 \\ x_2 \\ \dots \\ x_n \end{bmatrix} \text{ and } B = \begin{bmatrix} b_1 \\ b_2 \\ \dots \\ b_n \end{bmatrix}$$

Then the above system can be expressed in the matrix form as  $AX = B$ .

Since the above system consist of  $n$  equations in  $n$  unknowns,  $A$  is a square matrix.

If  $A$  is non-singular, i.e.  $|A| \neq 0$ , then  $A^{-1}$  exists.

Hence, multiplying both sides of  $AX = B$  from the left by  $A^{-1}$ , we get  $X = A^{-1}B$ .

Thus the solution is given by  $X = A^{-1}B$ .

If  $A$  is a singular matrix, i.e.  $|A| = 0$ , we calculate  $(\text{adj } A)B$ .

If  $(\text{adj } A)B \neq O$ , ( $O$  being zero matrix), then solution does not exist and the system of equations is called inconsistent.

If  $(\text{adj } A)B = O$ , then system may have either infinitely many solutions or no solution.

To find the nature of solution and the solution itself when  $|A| = 0$ , we should go to the method of elimination for quicker results.

#### Solving Homogeneous System using Matrix Inverse

If the above system is homogeneous,  $n$  equations in  $n$  unknowns, then in the matrix form it is  $AX = O$ . ( $\because$  in this case  $b_1 = b_2 = \dots = b_n = 0$ ), where  $A$  is a square matrix.

If  $A$  is non-singular, the system has only the trivial solution (zero solution)  $X = 0$ .

If  $A$  is singular, then the system has infinitely many solutions (including the trivial solution) and hence it has non-trivial solutions.

The solutions to equations of the following forms are :

$$(i) \quad XA = B \Rightarrow X = BA^{-1}.$$

If we multiply both sides of the original equation by  $A^{-1}$  from the right, then we get  $X = BA^{-1}$ .

$$(ii) \quad AXB = C \Rightarrow X = A^{-1}CB^{-1}$$

It suffices to multiply both sides of this equation by  $A^{-1}$  from the left and by  $B^{-1}$  from the right.



#### CAUTION

Always evaluate the determinant of the coefficient matrix before using the inverse matrix method. If the determinant is 0, the system either is inconsistent or has dependent equations.

► **Example 1.** Solve  $x_1 + x_2 = 2$   
 $3x_1 + 4x_2 = 1$   
by the matrix inverse method.

► **Solution** Since  $A = \begin{pmatrix} 1 & 1 \\ 3 & 4 \end{pmatrix}$

we know that  $A^{-1} = \begin{pmatrix} 4 & -1 \\ -3 & 1 \end{pmatrix}$

Therefore

$$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = A^{-1} \begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 4 & -1 \\ -3 & 1 \end{pmatrix} \begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 7 \\ -5 \end{pmatrix}$$

and  $x_1 = 7$  and  $x_2 = -5$ .

► **Example 2.** Given the matrices

$$A = \begin{pmatrix} 3 & 2 \\ 4 & 3 \end{pmatrix}, \quad B = \begin{pmatrix} -1 & 7 \\ 3 & 5 \end{pmatrix} \quad \text{find the}$$

solutions of the equations  $AX = B$ ,  $YA = B$ .

► **Solution** The matrix  $A$  is nonsingular, and hence

$$A^{-1} = \begin{pmatrix} 3 & -2 \\ -4 & 3 \end{pmatrix}$$

$$X = A^{-1}B \text{ and } Y = BA^{-1}.$$

Therefore the following matrices are solutions of the equations  $AX = B$ ,  $YA = B$  :

$$X = \begin{pmatrix} 3 & -2 \\ -4 & 3 \end{pmatrix} \cdot \begin{pmatrix} -1 & 7 \\ 3 & 5 \end{pmatrix} = \begin{pmatrix} -9 & 11 \\ 13 & -13 \end{pmatrix},$$

$$Y = \begin{pmatrix} -1 & 7 \\ 3 & 5 \end{pmatrix} \cdot \begin{pmatrix} 3 & -2 \\ -4 & 3 \end{pmatrix} = \begin{pmatrix} -31 & 23 \\ -11 & 9 \end{pmatrix}$$

► **Example 3.** Solve the following matrix equations :

$$(a) \begin{pmatrix} 3 & 2 \\ 1 & 1 \end{pmatrix} X = \begin{pmatrix} 1 & 2 & 7 \\ 0 & 4 & 8 \end{pmatrix}$$

$$(b) X \begin{pmatrix} 3 & 2 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 4 & 7 \\ 0 & 1 \\ 2 & 3 \\ 3 & 4 \end{pmatrix},$$

$$(c) \begin{pmatrix} 4 & 3 \\ 1 & 1 \end{pmatrix} X \begin{pmatrix} 6 & 8 \\ 2 & 1 \end{pmatrix} = \begin{pmatrix} 5 & 4 \\ -2 & 0 \end{pmatrix}$$

► **Solution** (a) This equation is of the form  $AX = B$ , its solution is  $X = A^{-1}B$ . We find  $\det A = 1$ ,

$$A^{-1} = \frac{1}{\det A} \begin{pmatrix} 1 & -2 \\ -1 & 3 \end{pmatrix} = \begin{pmatrix} 1 & -2 \\ -1 & 3 \end{pmatrix}$$

Hence

$$X = \begin{pmatrix} 1 & -2 \\ -1 & 3 \end{pmatrix} \begin{pmatrix} 1 & 2 & 7 \\ 0 & 4 & 8 \end{pmatrix} = \begin{pmatrix} 1 & -6 & -9 \\ -1 & 10 & 17 \end{pmatrix}.$$

(b) This is an equation of the form  $XA = B$ , its solution is  $X = BA^{-1}$ . We have  $A^{-1}$

$$= \begin{pmatrix} 1 & -2 \\ -1 & 3 \end{pmatrix}.$$

Consequently,

$$X = \begin{pmatrix} 4 & 7 \\ 0 & 1 \\ 2 & 3 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 1 & -2 \\ -1 & 3 \end{pmatrix} = \begin{pmatrix} -3 & 13 \\ -1 & 3 \\ -1 & 5 \\ -1 & 6 \end{pmatrix}.$$

(c) This is an equation of the form  $AXB = C$ , its solution is  $X = A^{-1}CB^{-1}$ . We find

$$A^{-1} = \begin{pmatrix} 1 & -3 \\ -1 & 4 \end{pmatrix}, \quad B^{-1} = -\frac{1}{10} \begin{pmatrix} 1 & -8 \\ -2 & 6 \end{pmatrix}$$

Therefore

$$X = \begin{pmatrix} 1 & -3 \\ -1 & 4 \end{pmatrix} \begin{pmatrix} 5 & 4 \\ -2 & 0 \end{pmatrix} \begin{pmatrix} 1 & -8 \\ -2 & 6 \end{pmatrix} \begin{pmatrix} -\frac{1}{10} \end{pmatrix}$$

$$X = \begin{pmatrix} 1 & -3 \\ -1 & 4 \end{pmatrix} \begin{pmatrix} 5 & 4 \\ -2 & 0 \end{pmatrix} \begin{pmatrix} 1 & -8 \\ -2 & 6 \end{pmatrix} \begin{pmatrix} -\frac{1}{10} \end{pmatrix}$$

$$= -\frac{1}{10} \begin{pmatrix} 1 & -3 \\ -1 & 4 \end{pmatrix} \begin{pmatrix} -3 & -16 \\ -2 & 16 \end{pmatrix}$$

$$= -\frac{1}{10} \begin{pmatrix} 3 & -64 \\ -5 & 80 \end{pmatrix} = \begin{pmatrix} -0.3 & 6.4 \\ -0.5 & -8 \end{pmatrix}$$



If  $AX = B$  and  $AY = 0$ , then  $A(X + Y) = B$ . That is, if  $X$  is a solution of the nonhomogeneous equation and  $Y$  is a solution of the homogeneous equation,

their sum is a solution of the nonhomogeneous equation. Moreover every solution of  $AX = B$  is of this form.

*Proof :*

To show that  $X + Y$  is a solution, calculate

$$A(X + Y) = AX + AY = B + 0 = B$$

Now let  $Z$  be any solution of the nonhomogeneous equation, that is,  $AZ = B$ . We can write  $Z = (Z - Y) + Y$ , and  $Z - Y$  is a solution of the nonhomogeneous problem because  $A(Z - Y) = AZ - AY = B - 0 = B$ .

Consider the system

$$AX = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 4 \\ 7 \\ 10 \end{pmatrix} = B$$

$$\text{which has the solution } \begin{pmatrix} -2 + k \\ 3 - 2k \\ k \end{pmatrix}$$

where  $k$  is arbitrary. This can be written as

$$\begin{pmatrix} -2 \\ 3 \\ 0 \end{pmatrix} + \begin{pmatrix} k \\ -2k \\ k \end{pmatrix}$$

where the second term in the sum solves  $AX = 0$  and the first term solves  $AX = B$ . This "split" of the solution is interesting.

## Practice Problems

[L]

1. If  $A = \begin{bmatrix} 2 & -1 \\ -2 & 3 \end{bmatrix}$  and  $B = \begin{bmatrix} 7 & 6 \\ 9 & 8 \end{bmatrix}$ , find  $2 \times 2$  matrices  $C$  and  $D$  such that  $AC = B$  and  $DA = B$ .

2. If  $|A| = 2$  and  $\text{adj } A = \begin{bmatrix} -2 & 3 & -1 \\ 6 & -8 & -2 \\ -4 & 7 & 1 \end{bmatrix}$ , solve the system of equations  $3x + 2y + z = 1$ ,  $x + y + z = 2$ ,  $5x + y - z = 3$ .

3. Solve the matrix equations :

(i)  $AX = B$  if  $A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$ ,  $B = \begin{pmatrix} 1 & 0 & 7 \\ 8 & 1 & 2 \end{pmatrix}$ .

(ii)  $XA = B$  if  $A = \begin{pmatrix} 4 & 3 \\ 1 & 1 \end{pmatrix}$ ,  $B = \begin{pmatrix} 2 & 8 \\ 1 & 1 \\ 0 & -1 \end{pmatrix}$ .

(iii)  $AXB = C$  if

$A = \begin{pmatrix} 2 & 3 \\ 5 & 8 \end{pmatrix}$ ,  $B = \begin{pmatrix} 5 & 4 & 1 \\ 1 & 1 & 7 \\ 6 & 5 & 9 \end{pmatrix}$ ,  $C = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}$

## 3.16 AUGMENTED MATRIX METHOD

A linear system of equations can be solved by repeated use of the addition method, using the following transformations.

1. Exchanging any two equations.
2. Multiplying both sides of an equation by any nonzero real number.
3. Replacing any equation by nonzero multiple of that equation plus a nonzero multiple of any other equation.

**Solving a system of three equations by the echelon method.**

A systematic approach for using the three transformations to solve a system is called the

**echelon method.** The goal of the echelon method is to use the transformations to rewrite the equations of the system until the system has a triangular form. For a system of three equations in three variables, for example, the system should have the form

$$x + ay + bz = 0$$

$$y + dz = e$$

$$z = f$$

where  $a, b, c, d, e$  and  $f$  are constants. Then the value of  $z$  from the third equation can be substituted back into the second equation to find  $y$ , and the values of  $y$  and  $z$  can be substituted into the first equation to find  $x$ . This is called back-substitution.

► **Example 1.** Solve the system

$$2x + y - z = 2 \quad \dots(1)$$

$$x + 3y + 2z = 1 \quad \dots(2)$$

$$x + y + z = 2 \quad \dots(3)$$

► **Solution** It is desired that in the first equation,  $x$  should have a coefficient of 1. We can use transformation 1 to exchange equations (1) and (2) to get the system

$$x + 3y + 2z = 1 \quad \dots(2)$$

$$2x + y - z = 2 \quad \dots(1)$$

$$x + y + z = 2 \quad \dots(3)$$

Next we use transformation 3 to eliminate the  $x$ -term in equations (1) and (3). Multiply equation (2) by  $-2$  and add the results to equation (1) to get an equivalent system. We will indicate this process by the notation  $-2R_1 + R_2 \rightarrow R_2$ . ( $R$  stands for row.) Similarly, multiplying equation (2) by  $-1$  and adding the results to equation (3) eliminates the  $x$ -term there.

$$x + 3y + 2z = 1 \quad \dots(2)$$

$$-2R_1 + R_2 \rightarrow R_2 \quad -5y - 5z = 0 \quad \dots(4)$$

$$-1R_1 + R_3 \rightarrow R_3 \quad -2y - z = 1 \quad \dots(5)$$

Use transformation 2 to get a coefficient of 1 for  $y$  in equation (4); multiply by  $-1/5$ .

$$x + 3y + 2z = 1 \quad \dots(2)$$

$$(-1/5)R_2 \rightarrow R_2 \quad y + z = 0 \quad \dots(6)$$

$$-2y - z = 1 \quad \dots(5)$$

To eliminate the  $y$ -term in equation (5), multiply equation (6) by 2 and add the results to equation (5).

$$x + 3y + 2z = 1 \quad \dots(2)$$

$$y + z = 0 \quad \dots(6)$$

$$2R_2 + R_3 \rightarrow R_3 \quad z = 1$$



The system is now in triangular form. Substitute 1 for  $z$  in equation (6) to get  $y = -1$ . Finally, substitute 1 for  $z$  and  $-1$  for  $y$  in equation (2) to get  $x = 2$ . The solution of the system is  $(2, -1, 1)$ .

► **Example 2.** Solve the system.

$$x + 2y + z = 4 \quad \dots(1)$$

$$3x - y - 4z = -9 \quad \dots(2)$$

► **Solution** Geometrically, the solution is the intersection of the two planes given by equations (1) and (2). The intersection of two different nonparallel planes is a line. Thus, there will be an infinite number of ordered triples in the solution set, representing the points on the line of intersection. To describe these ordered triples, use the echelon method to get the system in triangular form as much as possible.

$$x + 2y + z = 4 \quad \dots(1)$$

$$-3R_1 + R_2 \rightarrow R_2 \quad -7y - 7z = -21 \quad \dots(2)$$

$$x + 2y + z = 4 \quad \dots(3)$$

$$(-1/7) R_2 \rightarrow R_2 \quad y + z = 3 \quad \dots(4)$$

This is as far as we can go with the echelon method. Solve  $y + z = 3$  for  $y$  to get  $y = 3 - z$ . Now, express  $x$  also in terms of  $z$  by solving equation (1) for  $x$  and substituting  $3 - z$  for  $y$ .

$$x + 2y + z = 4$$

$$x = -2y - z + 4$$

$$x = -2(3 - z) - z + 4$$

$$x = z - 2$$

The solution set is written  $\{(z - 2, 3 - z, z)\}$ . The system has an infinite number of solutions. For any arbitrary value of  $z$ ,  $y = 3 - z$  and  $x = z - 2$ . For example, if  $z = 1$ ,  $y = 3 - 1 = 2$  and  $x = 1 - 2 = -1$ , giving the solution  $(-1, 2, 1)$ . Verify that another solution is  $(0, 1, 2)$ .

A system like the one in the previous example occurs when two of the equations in a system of three equations with three variables are dependent. In such a case, there are really only two equations in three variables. On the other hand, an inconsistent system is indicated by a false statement at some point in the solution.

### Gaussian Elimination

Let  $a_1x + b_1y + c_1z = d_1 \quad \dots(1)$

$$a_2x + b_2y + c_2z = d_2 \quad \dots(2)$$

$$a_3x + b_3y + c_3z = d_3 \quad \dots(3)$$

This system can be written in an abbreviated form as

$$\left[ \begin{array}{ccc|c} a_1 & b_1 & c_1 & d_1 \\ a_2 & b_2 & c_2 & d_2 \\ a_3 & b_3 & c_3 & d_3 \end{array} \right]$$

The constants in the last column of the matrix has been set apart from the coefficients of the variables with a vertical line. It is called as augmented matrix. (Because the matrix of coefficient has an extra column determined by the constants of the system, the coefficient matrix is augmented.)

As an example, the system

$$x + 3y + 2z = 1$$

$$2x + y - z = 2$$

$$x + y + z = 2$$

has the augmented matrix

$$\left[ \begin{array}{ccc|c} 1 & 3 & 2 & 1 \\ 2 & 1 & -1 & 2 \\ 1 & 1 & 1 & 2 \end{array} \right]$$

### STUDY TIP

When forming either the coefficient matrix or the augmented matrix of a system, you should begin by vertically aligning the variables in the equations and using 0's for the missing variables.

#### System

$$x + 3y = 9$$

$$-y + 4z = -2$$

$$x - 5z = 0$$

#### Lined up Variables

$$x + 3y = 9$$

$$-y + 4z = -2$$

$$x - 5z = 0$$

#### Augmented Matrix

$$\left[ \begin{array}{ccc|c} 1 & 3 & 0 & 9 \\ 0 & -1 & 4 & -2 \\ 1 & 0 & -5 & 0 \end{array} \right]$$

The rows of this augmented matrix can be treated just like the equations of the system of linear equations. Since the augmented matrix is nothing more than a short form of the system, any transformation of the matrix that results in an equivalent system of equations can be performed. Operations that produce such transformations are given below.

For any augmented matrix of a system of linear equations, the following row transformations will result in the matrix of an equivalent system.

1. Any two rows may be interchanged.
2. The elements of any row may be multiplied by a nonzero real number.
3. Any row may be changed by adding to its elements a multiple of the corresponding elements of another row.

The elementary row operations correspond to the algebraic operations which obtain equivalent equation from a given set of equations.

### Gaussian elimination

### Example

A system is given :

A system is given :

$$a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1$$

$$x_2 - x_3 + x_4 = 0$$

$$a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = b_2$$

$$2x_1 - x_2 + 3x_3 - x_4 = 6$$

$$4x_1 + 4x_2 + 4x_4 = 12$$

$$a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n = b_m$$

$$3x_1 - 3x_2 + 6x_3 - 3x_4 = 9$$

1. Represent the system in an augmented matrix

$$\left( \begin{array}{cccc|c} a_{11} & a_{12} & \dots & a_{1n} & b_1 \\ a_{21} & a_{22} & \dots & a_{2n} & b_2 \\ \dots & \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & \dots & a_{mn} & b_m \end{array} \right) \rightarrow \left( \begin{array}{cccc|c} 0 & 1 & -1 & 1 & 0 \\ 2 & -1 & 3 & -1 & 6 \\ 4 & 4 & 0 & 4 & 12 \\ 3 & -3 & 6 & -3 & 9 \end{array} \right)$$

2. Make sure that the number in the first column is nonzero, by interchanging two rows if necessary.

Pivot number

$$\left( \begin{array}{cccc|c} 2 & -1 & 3 & -1 & 6 \\ 0 & 1 & -1 & 1 & 0 \\ 4 & 4 & 0 & 4 & 12 \\ 3 & -3 & 6 & -3 & 9 \end{array} \right)$$

(This corresponds to interchanging two equations.) This number is called a pivot number. If the first column contains all zeroes, move to the second column.

3. Obtain a pivot number of 1 by multiply row 1 by  $\frac{1}{2}$ .

multiplication.

Actually this step is not necessary.

However, it makes the arithmetic simpler

Pivot

$$\left( \begin{array}{cccc|c} 1 & -1/2 & 3/2 & -1/2 & 3 \\ \blacksquare & 1 & -1 & 1 & 0 \\ 4 & 4 & 0 & 4 & 12 \\ 3 & -3 & 6 & -3 & 9 \end{array} \right)$$

in many problems.

4. Obtain a zero below the pivot in each row below the pivot's row. To do this Multiply row 1 by  $-4$ , and add to row 3. Multiply row 1 by  $-3$ , and add to row 4. in a row (called the object row), multiply

$$\left( \begin{array}{cccc|c} 1 & -1/2 & 3/2 & -1/2 & 3 \\ \blacksquare & 1 & -1 & 1 & 0 \\ \blacksquare & 6 & -6 & 6 & 0 \\ \blacksquare & -3/2 & 3/2 & -3/2 & 0 \end{array} \right)$$

the pivot row by a constant and add the result to the object row.

5. Restrict attention to the subarray of the augmented matrix obtained by ignoring

$$\left( \begin{array}{cccc|c} 1 & -1/2 & 3/2 & -1/2 & 3 \\ \blacksquare & 1 & -1 & 1 & 0 \\ \blacksquare & 6 & -6 & 6 & 0 \\ \blacksquare & -3/2 & 3/2 & -3/2 & 0 \end{array} \right)$$

Work with this

all previous pivot rows and columns.

6. Go back to step 2 and begin working on the subarray, unless the subarray has no nonzero pivots. Multiply row 2 by  $-6$ , and add to row 3. Multiply row 2 by  $3/2$ , and add to row 4

$$\left( \begin{array}{cccc|c} 1 & -1/2 & 3/2 & -1/2 & 3 \\ \blacksquare & 1 & -1 & 1 & 0 \\ \blacksquare & 0 & 0 & 0 & 0 \\ \blacksquare & 0 & 0 & 0 & 0 \end{array} \right)$$

no nonzero pivots

7. When finished, write the solution. We have two equations in four unknowns. Thus two of the unknowns can be arbitrarily assigned.

$$\begin{aligned}
 &\text{Let } x_4 = t \text{ and } x_3 = s. \text{ Then} \\
 &x_1 - 1/2x_2 + 3/2s - 1/2t = 3 \\
 &x_2 - s + t = 0 \\
 &x_2 = s - t \quad x_1 = 1/2(s - t) - 3/2s + 1/2t + 3 \\
 &= -s + 3
 \end{aligned}$$

► **Example 3.** Solve the following system.

$$\begin{aligned}
 y + z - 2w &= -3 \\
 x + 2y - z &= 2 \\
 2x + 4y + z - 3w &= -2 \\
 x - 4y - 7z - w &= -19
 \end{aligned}$$

► **Solution** The augmented matrix for this system is

$$\left[ \begin{array}{cccc|c} 0 & 1 & 1 & -2 & -3 \\ 1 & 2 & -1 & 0 & 2 \\ 2 & 4 & 1 & -3 & -2 \\ 1 & -4 & -7 & 1 & -19 \end{array} \right].$$

Begin by obtaining a leading 1 in the upper left corner by interchanging the first and second rows, and then proceed to obtain zeros elsewhere in the first column.

$$\begin{array}{l} \curvearrowright R_2 \\ \curvearrowleft R_1 \end{array} \left[ \begin{array}{cccc|c} 1 & 2 & -1 & 0 & 2 \\ 0 & 1 & 1 & -2 & -3 \\ 2 & 4 & 1 & -3 & -2 \\ 1 & -4 & -7 & 1 & -19 \end{array} \right]$$

**First column has leading 1 in upper left corner.**

$$\begin{array}{l} -2R_2 + R_2 \rightarrow \\ -R_1 + R_4 \rightarrow \end{array} \left[ \begin{array}{cccc|c} 1 & 2 & -1 & 0 & 2 \\ 0 & 1 & 1 & -2 & -3 \\ 0 & 0 & 3 & -3 & -6 \\ 0 & -6 & -6 & -1 & -21 \end{array} \right]$$

**First column has zeros below its leading 1**

Now that the first column is already in the desired form, you can change the second, third, and fourth columns as follows.

$$6R_2 + R_4 \rightarrow \left[ \begin{array}{cccc|c} 1 & 2 & -1 & 0 & 2 \\ 0 & 1 & 1 & -2 & -3 \\ 0 & 0 & 3 & -3 & -6 \\ 0 & 0 & 0 & -13 & -39 \end{array} \right]$$

**Second column has zeros below its leading 1**

$$\frac{1}{3}R_4 \rightarrow \left[ \begin{array}{cccc|c} 1 & 2 & -1 & 0 & 2 \\ 0 & 1 & 1 & -2 & -3 \\ 0 & 0 & 1 & -1 & -2 \\ 0 & 0 & 0 & -13 & -39 \end{array} \right]$$

**Third column has zeros below its leading 1**

$$\frac{1}{13}R_4 \rightarrow \left[ \begin{array}{cccc|c} 1 & 2 & -1 & 0 & 2 \\ 0 & 1 & 1 & -2 & -3 \\ 0 & 0 & 1 & -1 & -2 \\ 0 & 0 & 0 & -1 & 3 \end{array} \right]$$

**Fourth column has a leading 1**

The matrix is now in echelon form, and the corresponding system of linear equations is

$$\begin{aligned}
 x + 2y - z &= 2 \\
 y + z - 2w &= -3 \\
 z - w &= -2 \\
 w &= 3.
 \end{aligned}$$

Using back-substitution, you can determine that the solution is  $x = -1$ ,  $y = 2$ ,  $z = 1$ , and  $w = 3$ .

When solving a system of linear equations, remember that it is possible for the system to have no solution. If, in the elimination process, you obtain a row with zeros except for the last entry, it is unnecessary to continue the elimination process. You can simply

**STUDY TIP** conclude that the system is inconsistent. For instance, applying Gaussian elimination to the system

$$\begin{aligned}
 x - y + 2z &= 4 \\
 x + z &= 6 \\
 2x - 3y + 5z &= 4 \\
 3x + 2y - z &= 1
 \end{aligned}$$

produces

$$\left[ \begin{array}{ccc|c} 1 & -1 & 2 & 4 \\ 0 & 1 & -1 & 2 \\ 0 & 0 & 0 & -2 \\ 0 & 5 & -7 & -11 \end{array} \right]$$

Note the third row of this matrix consists of zeros except for the last entry.

This means that the original system of linear equations is inconsistent.

### Gauss-Jordan Elimination

Using Gaussian elimination, matrix row operations are applied to an augmented matrix to obtain a row-equivalent matrix corresponding to a linear system in echelon form. A second method, called Gauss-Jordan elimination, continues the process until a matrix with 1s down the main diagonal from left to right and 0s in every position above and below each 1 is found. For three linear equations in three variables,  $x$ ,  $y$  and  $z$ , we try to get the augmented matrix into form

$$\left[ \begin{array}{ccc|c} 1 & 0 & 0 & A \\ 0 & 1 & 0 & B \\ 0 & 0 & 1 & C \end{array} \right]$$

from which we conclude that  $x = A$ ,  $y = B$ , and  $z = C$ .

► **Example 4.** Use Gauss-Jordan elimination to solve the system :

$$3x + y + 2z = 31$$

$$x + y + 2z = 19$$

$$x + 3y + 2z = 25$$

► **Solution** Using Gaussian elimination we obtain the following matrix :

$$\left[ \begin{array}{ccc|c} 1 & 1 & 2 & 19 \\ 0 & 1 & 2 & 13 \\ 0 & 0 & 1 & 5 \end{array} \right]$$

Using Gauss-Jordan elimination, the boxed entries should be made 0.

Rather than using back-substitution, we now will apply matrix row operations to get 0s above the 1s in the main diagonal. To obtain 0 in row 1, column 2 (where there is now 1), we multiply each entry in row 2 by  $-1$  and add this result to the corresponding entry in row 1.

$$\left[ \begin{array}{ccc|c} 1 & 0 & 0 & 6 \\ 0 & 1 & 2 & 13 \\ 0 & 0 & 1 & 5 \end{array} \right] \longleftarrow -1R_2 + R_1$$

The boxed entries should be 0s.

Since we already have a 0 in row 1, column 3, we need only concentrate on obtaining a 0 in row 2, column 3 (where there is now a 2). To obtain 0, we multiply row 2 by  $-2$  and add the resulting products to row 2.

$$\left[ \begin{array}{ccc|c} 1 & 0 & 0 & 6 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 5 \end{array} \right] \longleftarrow -2R_3 + R_2$$

This last matrix corresponds to  $x = 6$ ,  $y = 3$ ,  $z = 5$ .

The solution set is  $(6, 3, 5)$ .

### STUDY TIP

The advantage to Gauss-Jordan elimination is that from the augmented matrix we can simply read the solution. The disadvantage is that we must continue row operations in the augmented matrix from the Gaussian elimination process, and it's fairly easy to make computational errors.

$$x_2 - x_3 = -9$$

► **Example 5.** Solve  $2x_1 - x_2 + 4x_3 = 29$

$$x_1 + x_2 - 3x_3 = -20$$

► **Solution** 
$$\left( \begin{array}{ccc|c} 0 & 1 & -1 & -9 \\ 2 & -1 & 4 & 29 \\ 1 & 1 & -3 & -20 \end{array} \right)$$

$$\xrightarrow{R_1 \leftrightarrow R_3} \left( \begin{array}{ccc|c} 1 & 1 & -3 & -20 \\ 2 & -1 & 4 & 29 \\ 0 & 1 & -1 & -9 \end{array} \right)$$

$$\xrightarrow{-2R_1 + R_2} \left( \begin{array}{ccc|c} 1 & 1 & -3 & -20 \\ 0 & -3 & 10 & 69 \\ 0 & 1 & -1 & -9 \end{array} \right)$$

$$\xrightarrow{-\frac{1}{3}R_2} \left( \begin{array}{ccc|c} 1 & 1 & -3 & -20 \\ 0 & 1 & -\frac{10}{3} & -23 \\ 0 & 1 & -1 & -9 \end{array} \right)$$

$$\xrightarrow{-1R_2 + R_3} \left( \begin{array}{ccc|c} 1 & 1 & -3 & -20 \\ 0 & 1 & -\frac{10}{3} & -23 \\ 0 & 0 & \frac{7}{3} & 14 \end{array} \right)$$

$$\xrightarrow{\frac{3}{7}R_3} \left( \begin{array}{ccc|c} 1 & 1 & -3 & -20 \\ 0 & 1 & -\frac{10}{3} & -23 \\ 0 & 0 & 1 & 6 \end{array} \right)$$

$$\xrightarrow{\begin{array}{l} \frac{10}{3}R_3 + R_2 \\ 3R_3 + R_1 \\ -1R_2 + R_1 \end{array}} \left( \begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -3 \\ 0 & 0 & 1 & 6 \end{array} \right)$$

It is easy to see that  $x_1 = 1$ ,  $x_2 = -3$ ,  $x_3 = 6$ .

#### Nature of solution based on rank of matrix

Let the system be  $AX = B$  where  $A$  is an  $m \times n$  matrix,  $X$  is the  $n$ -column vector &  $B$  is the  $m$  column vector. Let  $[AB]$  denote the **augmented matrix** (i.e. matrix obtained by accepting elements of  $B$  as  $n + 1^{\text{th}}$  column & first  $n$  columns are that of  $A$ ).

**Kronecker-Capelli theorem :** A system of linear equations is consistent if and only if the rank of the augmented matrix  $A$  is equal to the rank of the matrix  $A$ .

A consistent system has a unique solution if and only if the rank of matrix  $A$  is equal to the number of unknowns. It has infinitely many solutions if the rank of matrix  $A$  is less than the number of unknowns.

Let  $\rho(A)$  denote rank of  $A$  and  $\rho([AB])$  denote rank of the augmented matrix.

Clearly  $\rho(A) \leq \rho([AB])$ . In all there are three possibilities :

- If  $\rho(A) < \rho([AB])$  then the system has no solution (i.e. system is inconsistent).
- If  $\rho(A) = \rho([AB]) = \text{number of unknowns}$ , then the system has unique solution. (and hence is consistent)
- If  $\rho(A) = \rho([AB]) < \text{number of unknowns}$ , then the systems has infinitely many solutions (and so is consistent).

#### Homogeneous System

Let the homogenous system be  $AX = 0$ ,  $m$  equations in ' $n$ ' unknowns in this case  $B = 0$  and

so  $\rho(A) = \rho([AB])$ .

Hence if  $\rho(A) = n$ , then the system has only the trivial solution. If  $\rho(A) < n$ , then the system has infinitely many solutions.



If one has a homogeneous system of linear equations in which the number of unknowns exceeds the number of equations, then the system has nontrivial solution.

► **Example 6.** Show that the equations

$$x + 2y - z = 3,$$

$$3x - y + 2z = 1,$$

$$2x - 2y + 3z = 2,$$

$$x - y + z = 1 \text{ are consistent and solve them.}$$

► **Solution** The given system of equations is equivalent to the single matrix equation

$$AX = \begin{bmatrix} 1 & 2 & -1 \\ 3 & -1 & 2 \\ 2 & -2 & 3 \\ 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \\ 2 \\ -1 \end{bmatrix} = B.$$

The augmented matrix

$$[AB] = \begin{bmatrix} 1 & 2 & -1 & : & 3 \\ 3 & -1 & 2 & : & 1 \\ 2 & -2 & 3 & : & 2 \\ 1 & -1 & 1 & : & -1 \end{bmatrix}.$$

Performing  $R_2 \rightarrow R_2 - 3R_1$ ,  $R_3 \rightarrow R_3 - 2R_1$ ,  $R_4 \rightarrow R_4 - R_1$ , we get

$$[A \ B] \sim \begin{bmatrix} 1 & 2 & -1 & 3 \\ 0 & -7 & 5 & -8 \\ 0 & -6 & 5 & -4 \\ 0 & -3 & 2 & -4 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 2 & -1 & 3 \\ 0 & -1 & 0 & -4 \\ 0 & -6 & 5 & -4 \\ 0 & -3 & 2 & -4 \end{bmatrix} \text{ by } R_2 \rightarrow R_2 - R_3$$

$$\sim \begin{bmatrix} 1 & 2 & -1 & 3 \\ 0 & -1 & 0 & -4 \\ 0 & 0 & 5 & 20 \\ 0 & 0 & 2 & 8 \end{bmatrix}$$

by  $R_3 \rightarrow R_3 - 6R_2$ ,  $R_4 \rightarrow R_4 - 3R_2$ ,

$$\sim \begin{bmatrix} 1 & 2 & -1 & 3 \\ 0 & -1 & 0 & -4 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 1 & 4 \end{bmatrix},$$

by  $R_3 \rightarrow \frac{1}{5} R_3$ ,  $R_4 \rightarrow \frac{1}{2} R_4$

$$\sim \begin{bmatrix} 1 & 2 & -1 & 3 \\ 0 & -1 & 0 & -4 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix},$$

by  $R_4 \rightarrow R_4 - R_3$ .

Thus the matrix  $[A \ B]$  has been reduced to Echelon form. We have  $\text{rank } [A \ B] = \text{the number of non-zero rows in this Echelon form} = 3$ . Also

$$A \sim \begin{bmatrix} 1 & 2 & -1 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

We have  $\text{rank } A = 3$ . Since  $\text{rank } [A \ B] = \text{rank } A$ , therefore the given equations are consistent. Since  $\text{rank } A = 3 = \text{the number of unknowns}$ , therefore the given equations have unique solution. The given equations are equivalent to the equations

$$x + 2y - z = 3, -y = -4, z = 4.$$

These give  $z = 4$ ,  $y = 4$ ,  $x = -1$ .

► **Example 7.** Solve completely the system of equations

$$\begin{aligned} x + y + z &= 0, & 2x - y - 3z &= 0, \\ 3x - 5y + 4z &= 0, & x + 17y + 4z &= 0, \end{aligned}$$

► **Solution** The given system of equations is equivalent to the single matrix equation

$$AX = \begin{bmatrix} 1 & 1 & 1 \\ 2 & -1 & -3 \\ 3 & -5 & 4 \\ 1 & 17 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = O.$$

We shall first find the rank of the coefficient matrix  $A$  by reducing it to Echelon form by applying elementary row transformations only. Applying  $R_2 \rightarrow R_2 - 2R_1$ ,  $R_3 \rightarrow R_3 - 3R_1$ ,  $R_4 \rightarrow R_4 - R_1$ , we get

$$A \sim \begin{bmatrix} 1 & 1 & 1 \\ 0 & -3 & -5 \\ 0 & -8 & 1 \\ 0 & 16 & 3 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 1 & 1 \\ 0 & -3 & -5 \\ 0 & -24 & 3 \\ 0 & 48 & 9 \end{bmatrix} \text{ by } R_3 \rightarrow 3R_3, R_4 \rightarrow 3R_4$$

$$\sim \begin{bmatrix} 1 & 1 & 1 \\ 0 & -3 & -5 \\ 0 & 0 & 43 \\ 0 & 0 & -71 \end{bmatrix} \text{ by } R_3 \rightarrow R_3 - 8R_2, R_4 \rightarrow R_4 + 16R_2.$$

$$\sim \begin{bmatrix} 1 & 1 & 1 \\ 0 & -3 & -5 \\ 0 & 0 & 43 \\ 0 & 0 & 0 \end{bmatrix} \text{ by } R_4 \rightarrow R_4 + \frac{71}{43} R_3.$$

Above is the Echelon form of the coefficient matrix A. We have rank A = the number of non zero rows in this Echelon form = 3. The number of unknowns is also 3. Since rank A is equal to the number of unknowns, therefore the given system of equations possesses no non-zero solution. Hence the zero solution i.e.  $x = y = z = 0$  is the only solution of the given system of equations.

### Practice Problems

[M]

1. Use Gaussian elimination to determine whether the following systems have no solutions, exactly one solution, or an infinite number of solutions. State the solutions, if they exist.

(i)  $x_1 - x_2 + x_3 = 6$

$$x_1 + x_2 + 2x_3 = 8$$

$$2x_1 - 3x_2 - x_3 = 1$$

(ii)  $2x_1 - 3x_2 + x_3 = 5$

$$x_1 + x_2 - x_3 = 3$$

$$4x_1 - x_2 - x_3 = 11$$

(iii)  $x_1 - 3x_2 + x_3 = 6$

$$2x_1 + x_2 - 3x_3 = -2$$

$$x_1 + 4x_2 - 4x_3 = 0$$

(iv)  $x_1 - x_2 = 4$

$$2x_1 + x_2 = 7$$

$$5x_1 - 2x_2 = 19$$

(v)  $3x_1 - 2x_2 + x_3 = 12$

$$x_1 - 6x_2 + 4x_3 = 6$$

(vi)  $x_1 + 2x_2 = 6$

$$3x_1 + 6x_2 = 8$$

$$5x_1 + 10x_2 = 12$$

2.  $3x + 2y - w = 0$

$$2x + z + 2w = 5$$

$$x + 2y - z = -2$$

$$2x - y + z + w = 2$$

3. Solve the system of the following equations

$$\frac{2}{x} + \frac{3}{y} + \frac{10}{z} = 4$$

$$\frac{2}{x} + \frac{3}{y} + \frac{10}{z} = 4$$

$$\frac{6}{x} + \frac{9}{y} - \frac{20}{z} = 2$$

4. Discuss the nature of solutions of the system  $ax + y + z = 1$ ,  $x + ay + z = a$ ,  $x + y + az = a^2$  for different values of a.

5. Find the value of  $\lambda$  for which the equations.

$$x + \lambda y - z = 1$$

$$2x + y + 2z = 5\lambda + 1$$

$$x - y + 3z = 4\lambda + 2,$$

$$x - 2\lambda y + 7z = 10\lambda - 1 \quad \text{are consistent.}$$

6. Solve the systems of homogeneous equation

$$(i) \begin{cases} x_1 + x_2 - 3x_3 = 0, \\ x_1 - x_2 + x_3 + 2x_4 = 0, \\ 3x_1 + 3x_2 - 9x_3 = 0 \\ x_1 + 2x_2 - 5x_3 - x_4 = 0 \end{cases}$$

$$(ii) \begin{cases} x_1 + x_2 + x_3 = 0 \\ 2x_1 + 3x_2 + 3x_3 = 0 \\ 3x_1 - x_2 - x_3 = 0 \end{cases}$$

7. Find the general solution of the system

$$3x + 2y + z = 1$$

$$5x + 3y + 3z = 2$$

$$7x + 4y + 5z = 3$$

$$x + y - z = 0$$



### Target Problems for JEE Advanced

- **Problem 1.** The set of natural numbers N is partitioned into arrays of rows and columns in the

form of matrices as  $M_1 = (1)$ ,  $M_2 = \begin{pmatrix} 2 & 3 \\ 4 & 5 \end{pmatrix}$ ,

$$M_3 = \begin{pmatrix} 6 & 7 & 8 \\ 9 & 10 & 11 \\ 12 & 13 & 14 \end{pmatrix}, \dots, \dots,$$

$M_n = ( )$  and so on. Find the sum of the elements of the diagonal in  $M_n$ .

► **Solution** Let  $M_n = (a_{ij})$  where  $i, j = 1, 2, 3, \dots, n$ .

We first find out  $a_{ii}$  for the  $n^{\text{th}}$  matrix; which is the  $n^{\text{th}}$  term in the series ;  $1, 2, 6, \dots$

Let  $S = 1 + 2 + 6 + 15 + \dots + T_{n-1} + T_n$ .

Again writing

$$S = 1 + 2 + 6 + \dots + T_{n-1} + T_n$$

$$\begin{aligned} \Rightarrow 0 + 1 + 1 + 4 + 9 + \dots + (T_n - T_{n-1}) - T_n \\ \Rightarrow T_n = 1 + (1 + 4 + 9 + \dots \text{ upto } (n-1) \text{ terms}) \\ = 1 + (1^2 + 2^2 + 3^2 + 4^2 + \dots + (n-1)^2) \\ = 1 + \frac{n(n-1)(2n-1)}{6} \end{aligned}$$

Now, observing carefully, the consecutive distance between the elements of the diagonal of the  $n^{\text{th}}$  matrix is  $n+1$ .

Therefore first term is  $1 + \frac{n(n-1)(2n-1)}{6}$  and

common difference  $= n+1$ .

Hence the required sum

$$\begin{aligned} M_n &= \frac{n}{2} \left[ 2 \left( 1 + \frac{n(n-1)(2n-1)}{6} \right) + (n-1)(n+1) \right] \\ &= \frac{n}{6} [6 + (n-1)(2n^2 + 2n + 3)] \\ &= \frac{n}{6} [2n^3 + n + 3]. \end{aligned}$$

► **Problem 2.** If  $\begin{bmatrix} 1 & 2 & a \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix}^n = \begin{bmatrix} 1 & 18 & 2007 \\ 0 & 1 & 36 \\ 0 & 0 & 1 \end{bmatrix}$

then find the value of  $(n+a)$ .

► **Solution** Consider  $\begin{bmatrix} 1 & 2 & a \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & a \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix}$

$$= \begin{bmatrix} 1 & 4 & 2a+8 \\ 0 & 1 & 8 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & a \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 6 & 3a+24 \\ 0 & 1 & 12 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\therefore \begin{bmatrix} 1 & 2 & a \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix}^n = \begin{bmatrix} 1 & 2n & na + 8 \sum_{k=0}^{n-1} k \\ 0 & 1 & 4n \\ 0 & 0 & 1 \end{bmatrix}$$

hence  $n=9$  and  $2007 = 9a + 8 \sum_{k=0}^8 k = 9a + 8 \left( \frac{8 \cdot 9}{2} \right)$

$$2007 = 9a + 32 \cdot 9 = 9(a + 32)$$

$$a + 32 = 223 \Rightarrow a = 191$$

Hence  $a + n = 200$ .

► **Problem 3.** Let  $A = \begin{bmatrix} 2 & 1 \\ 1 & 0 \end{bmatrix}^n = (a_{ij}(n))$ .

If  $\lim_{n \rightarrow \infty} \frac{a_{12}(n)}{a_{22}(n)} = l$  where  $l^2 = \sqrt{a} + \sqrt{b}$

( $a, b \in \mathbb{N}$ ), find the value of  $(a+b)$ .

► **Solution** Suppose

$$A_1 = \begin{bmatrix} 2 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = I + B \text{ (say)}$$

hence  $A = \begin{bmatrix} 2 & 1 \\ 1 & 0 \end{bmatrix}^n = (I + B)^n$

$$\therefore A = \begin{bmatrix} 2 & 1 \\ 1 & 0 \end{bmatrix}^n = {}^nC_0 I + {}^nC_1 B + {}^nC_2 B^2 + {}^nC_3 B^3 + {}^nC_4 B^4 + \dots \dots \dots (1)$$

now  $B^2 = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = 2I$

Hence  $B^{2k} = 2^k I$  and  $B^{2k+1} = B^{2k} B = 2^k B$

$$\begin{aligned} \text{now } \begin{bmatrix} 2 & 1 \\ 1 & 0 \end{bmatrix}^n &= \underbrace{({}^nC_0 + {}^nC_2 \cdot 2 + {}^nC_4 \cdot 2^2 + \dots)}_{\text{'X' say}} I \\ &+ \underbrace{({}^nC_1 + {}^nC_3 \cdot 2 + {}^nC_5 \cdot 2^2 + \dots)}_{\text{'Y' say}} B \end{aligned}$$

$$\begin{aligned} \therefore \begin{bmatrix} 2 & 1 \\ 1 & 0 \end{bmatrix}^n &= \begin{bmatrix} X & 0 \\ 0 & X \end{bmatrix} + \begin{bmatrix} Y & Y \\ Y & -Y \end{bmatrix} \\ &= \begin{bmatrix} X+Y & Y \\ Y & X-Y \end{bmatrix} \end{aligned}$$

Hence  $a_{12} \text{ in } \begin{bmatrix} 2 & 1 \\ 1 & 0 \end{bmatrix}^n = Y$

$$\begin{aligned} \therefore a_{12} &= {}^nC_1 + {}^nC_3 \cdot 2 + {}^nC_5 \cdot 2^2 + {}^nC_7 \cdot 2^3 + \dots \\ &= \frac{1}{\sqrt{2}} [{}^nC_1 \cdot \sqrt{2} + {}^nC_3 (\sqrt{2})^3 + {}^nC_5 (\sqrt{2})^5 + \dots] \end{aligned}$$



$$\begin{aligned}
 &= \frac{1}{\sqrt{2}} \left[ \frac{(1+\sqrt{2})^n - (1-\sqrt{2})^n}{2} \right] \text{III ly } a_{22} = X - Y \\
 &= ({}^nC_0 + {}^nC_2 \cdot 2 + {}^nC_4 \cdot 2^2 + {}^nC_6 \cdot 2^3 + \dots) \\
 &\quad - ({}^nC_1 + {}^nC_3 \cdot 2 + {}^nC_5 \cdot 2^2 + {}^nC_7 \cdot 2^3 + \dots) \\
 &= \frac{(1+\sqrt{2})^n + (1-\sqrt{2})^n}{2} - \frac{(1+\sqrt{2})^n - (1-\sqrt{2})^n}{2\sqrt{2}} \\
 &= \frac{\sqrt{2}[(1+\sqrt{2})^n + (1-\sqrt{2})^n] - [(1+\sqrt{2})^n - (1-\sqrt{2})^n]}{2\sqrt{2}} \\
 a_{22} &= \frac{(\sqrt{2}-1)(1+\sqrt{2})^n - (\sqrt{2}+1)(1-\sqrt{2})^n}{2\sqrt{2}} \\
 \therefore \lim_{n \rightarrow \infty} \frac{a_{12}}{a_{22}} &= \lim_{n \rightarrow \infty} \frac{(1+\sqrt{2})^n - (1-\sqrt{2})^n}{(\sqrt{2}-1)(1+\sqrt{2})^n + (\sqrt{2}+1)(1-\sqrt{2})^n} \\
 &= \lim_{n \rightarrow \infty} \frac{1 - \left(\frac{1-\sqrt{2}}{1+\sqrt{2}}\right)^n}{(\sqrt{2}-1) + (\sqrt{2}+1)\left(\frac{1-\sqrt{2}}{1+\sqrt{2}}\right)^n} \\
 &= \frac{1-0}{\sqrt{2}-1} = 1 + \sqrt{2};
 \end{aligned}$$

hence  $l^2 = (1 + \sqrt{2})^2 = 3 + 2\sqrt{2} = \sqrt{9} + \sqrt{8}$ .  
Hence  $a + b = 9 + 8 = 17$ .

► **Problem 4.** Given that,  $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ ,  $abcd \neq 0$

Show that  $AA' - A'A = (b - c) \begin{bmatrix} b + c & d - a \\ d - a & -b - c \end{bmatrix}$

Further show that  $AA' = A'A \Rightarrow A$  is symmetric or  $A - aI$  is skew symmetric.

► **Solution** We have

$$\begin{aligned}
 AA' - A'A &= \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} a & c \\ b & d \end{bmatrix} - \begin{bmatrix} a & c \\ b & d \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} \\
 &= \begin{bmatrix} a^2 + b^2 & ac + bd \\ ac + bd & c^2 + d^2 \end{bmatrix} - \begin{bmatrix} a^2 + c^2 & ab + cd \\ ab + cd & b^2 + d^2 \end{bmatrix}
 \end{aligned}$$

$$\begin{aligned}
 &= \begin{bmatrix} b^2 - c^2 & ac + bd - ab - cd \\ ac + bd - ab - cd & c^2 - b^2 \end{bmatrix} \\
 &= (b - c) \begin{bmatrix} b + c & d - a \\ d - a & -b - c \end{bmatrix}
 \end{aligned}$$

Now  $AA' = A'A \Rightarrow$  either  $b = c$  or  $b + c, d$

$-a$ , both are zero. If  $b = c$  then  $A = \begin{bmatrix} a & b \\ b & d \end{bmatrix}$

which is symmetric if  $b + c, d - a$  both are zero then  $b = -c, d = a$

whence  $A = \begin{bmatrix} a & -c \\ c & a \end{bmatrix} \Rightarrow A - aI = \begin{bmatrix} 0 & -c \\ c & 0 \end{bmatrix}$

$\Rightarrow A - aI$  is skew symmetric.

► **Problem 5.** Show that the matrix

$$A = \begin{bmatrix} ab & b^2 \\ -a^2 & -ab \end{bmatrix}$$

is nilpotent such that  $A^2 = O$ . Conversely, show that all two rowed nilpotent matrices such that  $A^2 = O$  are of the above form.

► **Solution** If  $A = \begin{bmatrix} ab & b^2 \\ -a^2 & -ab \end{bmatrix}$

$$\begin{aligned}
 \text{then } A^2 &= AA = \begin{bmatrix} ab & b^2 \\ -a^2 & -ab \end{bmatrix} \begin{bmatrix} ab & b^2 \\ -a^2 & -ab \end{bmatrix} \\
 &= \begin{bmatrix} a^2b^2 - b^2a^2 & ab^3 - ab^3 \\ -a^3b + a^3b & -a^2b^2 + a^2b^2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.
 \end{aligned}$$

Hence the matrix  $A$  is nilpotent with index 2.

Conversely suppose  $A = \begin{bmatrix} \alpha & \beta \\ \gamma & \delta \end{bmatrix}$  is two rowed

nilpotent matrix such that  $A^2 = O$ ,

i.e.  $\begin{bmatrix} \alpha^2 + \beta\gamma & \alpha\beta + \beta\delta \\ \alpha\gamma + \gamma\delta & \gamma\beta + \delta^2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ .

If the above equation holds,

$$\alpha^2 + \beta\gamma = 0 \quad \dots(1) \quad \alpha\gamma + \gamma\delta = 0 \quad \dots(3)$$

$$\alpha\beta + \beta\delta = 0 \quad \dots(2) \quad \gamma\beta + \delta^2 = 0 \quad \dots(4)$$

must hold simultaneously.

From (1) and (4), it is obvious that the above four equations will hold simultaneously if  $\alpha^2 = \delta^2$  i.e.  $\alpha = \pm \delta$ . If  $\alpha = -\delta$ , the above four equations will hold simultaneously for all values of the numbers  $\alpha, \beta, \gamma$  provided they are related by the condition  $\alpha^2 + \beta\gamma = 0$ . If  $\alpha = \delta$  and either of them is not equal to 0, we have from (2) and (3)  $\beta = 0$  and  $\gamma = 0$ .

Then from (1) and (4), we shall have  $\alpha = \delta = 0$ . Hence if  $\alpha = \delta \neq 0$ , the above four equations cannot hold simultaneously.

If  $\alpha = \delta = 0$ , the equations can hold simultaneously, but this solution can be included in the solution  $\alpha = -\delta, \alpha^2 + \beta\gamma = 0$ .

Thus the matrix A is nilpotent if

$$A = \begin{bmatrix} \alpha & \beta \\ \gamma & -\alpha \end{bmatrix} \text{ and } \alpha, \beta, \gamma \text{ are any numbers related}$$

by the condition  $\alpha^2 + \beta\gamma = 0$ .

Obviously the matrix A is of the given form.

► **Problem. 6** Let a, b, c be real numbers with  $a^2 + b^2 + c^2 = 1$ . Show that if

$$\begin{vmatrix} ax - by - c & bx + ay & cx + a \\ bx + ay & -ax + by - c & cy + b \\ cx + a & cy + b & -ax - by + c \end{vmatrix} = 0$$

then  $ax + by + c = 0$

► **Solution** Let  $a \neq 0$ , applying the operation  $C_1 \rightarrow aC_1 + bC_2 + cC_3$  the given determinant

$$D = \frac{1}{a} \begin{vmatrix} (a^2 + b^2 + c^2)x & bx + ay & cx + a \\ (a^2 + b^2 + c^2)y & -ax + by - c & cy + b \\ a^2 + b^2 + c^2 & cy + b & -ax - by + c \end{vmatrix}$$

$$= \frac{1}{a} (a^2 + b^2 + c^2) \begin{vmatrix} x & bx + ay & cx + a \\ y & -ax + by - c & cy + b \\ 1 & cy + b & -ax - by + c \end{vmatrix}$$

$$= \frac{1}{a} \begin{vmatrix} x & bx + ay & cx + a \\ y & -ax + by - c & cy + b \\ 1 & cy + b & -ax - by + c \end{vmatrix} \quad (\because a^2 + b^2 + c^2 = 1)$$

On applying  $C_2 \rightarrow C_2 - bC_1, C_3 \rightarrow C_3 - cC_1$ ,

Finally on applying  $R_3 \rightarrow R_3 + xR_1 + yR_2$ ,

$$D = \begin{vmatrix} x & ay & a \\ y & -ax - c & b \\ x^2 + y^2 + 1 & 0 & 0 \end{vmatrix} = a(x^2 + y^2 + 1)(ax + by + c)$$

Now, as we are given  $D = 0$  the result follows since  $x^2 + y^2 + 1 \neq 0, a \neq 0$ . If  $a = 0$  then either b or c must be non-zero and then following the same set of steps the result follows.

► **Problem 7.** If A, B and C are the angles of a triangle and

$$\begin{vmatrix} 1 & 1 & 1 \\ 1 + \sin A & 1 + \sin B & 1 + \sin C \\ \sin A + \sin^2 A & \sin B + \sin^2 B & \sin C + \sin^2 C \end{vmatrix} = 0$$

Prove that  $\Delta ABC$  must be isosceles.

► **Solution** Let

$$\Delta = \begin{vmatrix} 1 & 1 & 1 \\ 1 + \sin A & 1 + \sin B & 1 + \sin C \\ \sin A + \sin^2 A & \sin B + \sin^2 B & \sin C + \sin^2 C \end{vmatrix}$$

Applying  $C_2 \rightarrow C_2 - C_1$  and  $C_3 \rightarrow C_3 - C_1$  then

$$\Delta = \begin{vmatrix} 1 & 0 & 0 \\ a + \sin A & \sin B - \sin A & \sin C - \sin A \\ \sin A + \sin^2 A & (\sin B - \sin A)(\sin B + \sin A + 1) & (\sin C - \sin A)(\sin C + \sin A + 1) \end{vmatrix}$$

Expanding along  $R_1$

$$\begin{aligned} \therefore \Delta &= \begin{vmatrix} \sin B - \sin A & \sin C - \sin A \\ (\sin B - \sin A)(\sin B + \sin A + 1) & (\sin C - \sin A)(\sin C + \sin A + 1) \end{vmatrix} \\ &= (\sin B - \sin A)(\sin C - \sin A) \\ &\quad \begin{vmatrix} 1 & 1 \\ \sin B + \sin A + 1 & \sin C + \sin A + 1 \end{vmatrix} \end{aligned}$$

$$= (\sin B - \sin A) (\sin C - \sin A) (\sin C - \sin B)$$

But given  $\Delta = 0$

$$\therefore (\sin B - \sin A) (\sin C - \sin A) (\sin C - \sin B) = 0$$

$$\therefore \sin B - \sin A = 0 \text{ or } \sin C - \sin A = 0 \\ \text{or } \sin C - \sin B = 0$$

$$\Rightarrow \sin B = \sin A \text{ or } \sin C = \sin A \text{ or } \sin C = \sin B$$

$$\Rightarrow B = A \text{ or } C = A \text{ or } C = B$$

In the three cases, we will have an isosceles triangle.

► **Problem 8.** Show that

$$\begin{vmatrix} 1 & \cos x - \sin x & \cos x + \sin x \\ 1 & \cos y - \sin y & \cos y + \sin y \\ 1 & \cos z - \sin z & \cos z + \sin z \end{vmatrix} \\ = 2 \begin{vmatrix} 1 & \cos x & \sin x \\ 1 & \cos y & \sin y \\ 1 & \cos z & \sin z \end{vmatrix}$$

► **Solution** We have

$$\Delta = \begin{vmatrix} 1 & \cos x & \cos x \\ 1 & \cos y & \cos y \\ 1 & \cos z & \cos z \end{vmatrix} + \begin{vmatrix} 1 & \cos x & \sin x \\ 1 & \cos y & \sin y \\ 1 & \cos z & \sin z \end{vmatrix} \\ + \begin{vmatrix} 1 & -\sin x & \cos x \\ 1 & -\sin y & \cos y \\ 1 & -\sin z & \cos z \end{vmatrix} + \begin{vmatrix} 1 & -\sin x & \sin x \\ 1 & -\sin y & \sin y \\ 1 & -\sin z & \sin z \end{vmatrix}$$

The first determinant is zero because the second and third columns are identical.

The fourth determinant is zero because the second column is a multiple of the third.

$$\therefore \Delta = \begin{vmatrix} 1 & \cos x & \sin x \\ 1 & \cos y & \sin y \\ 1 & \cos z & \sin z \end{vmatrix} + \begin{vmatrix} 1 & -\sin x & \cos x \\ 1 & -\sin y & \cos y \\ 1 & -\sin z & \cos z \end{vmatrix} \\ = \begin{vmatrix} 1 & \cos x & \sin x \\ 1 & \cos y & \sin y \\ 1 & \cos z & \sin z \end{vmatrix} - \begin{vmatrix} 1 & \sin x & \cos x \\ 1 & \sin y & \cos y \\ 1 & \sin z & \cos z \end{vmatrix} \\ = \begin{vmatrix} 1 & \cos x & \sin x \\ 1 & \cos y & \sin y \\ 1 & \cos z & \sin z \end{vmatrix} + \begin{vmatrix} 1 & \cos x & \sin x \\ 1 & \cos y & \sin y \\ 1 & \cos z & \sin z \end{vmatrix}$$

$$= 2 \begin{vmatrix} 1 & \cos x & \sin x \\ 1 & \cos y & \sin y \\ 1 & \cos z & \sin z \end{vmatrix}$$

► **Problem 9.** If  $p + q + r = 0$ , prove that

$$\begin{vmatrix} pa & qb & rc \\ qc & ra & pb \\ rb & pc & qa \end{vmatrix} = pqr \begin{vmatrix} a & b & c \\ c & a & b \\ b & c & a \end{vmatrix}$$

► **Solution**  $p + q + r = 0$

$$\therefore p^3 + q^3 + r^3 = 3pqr$$

$$\Rightarrow p^3 + q^3 + r^3 - 3pqr = 0 \quad \dots(1)$$

$$\text{R.H.S} = pqr \begin{vmatrix} a & b & c \\ c & a & b \\ b & c & a \end{vmatrix}$$

Applying  $C_1 \rightarrow C_1 + C_2 + C_3$

$$= pqr \begin{vmatrix} a+b+c & b & c \\ a+b+c & a & b \\ a+b+c & c & a \end{vmatrix}$$

$$= pqr(a+b+c) \begin{vmatrix} 1 & b & c \\ 1 & a & b \\ 1 & c & a \end{vmatrix}$$

Applying  $R_2 \rightarrow R_2 - R_1$  and  $R_3 \rightarrow R_3 - R_1$

$$= pqr(a+b+c) \begin{vmatrix} 1 & b & c \\ 0 & a-b & b-c \\ 0 & c-b & a-c \end{vmatrix}$$

$$= pqr(a+b+c)(a^2 + b^2 + c^2 - ab - bc - ca)$$

$$= pqr(a^3 + b^3 + c^3 - 3abc)$$

$$= pqr(a^3 + b^3 + c^3 - 3abc) - abc(p^3 + q^3 + r^3 - 3pqr) \quad \text{from (1)}$$

$$= pqr(a^3 + b^3 + c^3) - abc(p^3 + q^3 + r^3)$$

$$= (pqra^3 - abcp^3) - (abcq^3 - pqr b^3) + (pqrc^3 - abcr^3)$$

$$= (ap) \{(ra)(aq) - (pb)(pc)\} - (qb) \{(pc)(qa) - (pb)(rb)\} + (rc) \{(pc)(qc) - (ra)(rb)\}$$

$$= \begin{vmatrix} pa & qb & rc \\ qc & ra & pb \\ rb & pc & qa \end{vmatrix} = \text{L.H.S.}$$

► **Problem 10.** Find the value of the determinant

$$\begin{vmatrix} {}^nC_{r-1} & {}^nC_r & (r+1)^{n+2}C_{r+1} \\ {}^nC_r & {}^nC_{r+1} & (r+2)^{n+2}C_{r+2} \\ {}^nC_{r+1} & {}^nC_{r+2} & (r+3)^{n+2}C_{r+3} \end{vmatrix}$$

► **Solution** Operating  $C_1 \rightarrow C_1 + C_2$  and using

$${}^nC_r = \frac{n}{r} {}^{n-1}C_{r-1} \text{ in } C_3, \text{ we get}$$

$$\begin{vmatrix} {}^{n+1}C_r & {}^nC_r & (n+2)^{n+1}C_r \\ {}^{n+1}C_{r+1} & {}^nC_{r+1} & (n+2)^{n+1}C_{r+1} \\ {}^{n+1}C_{r+2} & {}^nC_{r+2} & (n+2)^{n+1}C_{r+2} \end{vmatrix} = 0.$$

(as  $C_1$  and  $C_3$  are identical)

► **Problem 11.** Prove that

$$\begin{vmatrix} 2 & \alpha + \beta + \gamma + \delta \\ \alpha + \beta + \gamma + \delta & 2(\alpha + \beta)(\gamma + \delta) \\ \alpha\beta + \gamma\delta & \alpha\beta(\gamma + \delta) + \gamma\delta(\alpha + \beta) \end{vmatrix} = 0$$

► **Solution** L.H.S

$$\begin{aligned} &= \begin{vmatrix} 2 & \alpha + \beta + \gamma + \delta \\ \alpha + \beta + \gamma + \delta & 2(\alpha + \beta)(\gamma + \delta) \\ \alpha\beta + \gamma\delta & \alpha\beta(\gamma + \delta) + \gamma\delta(\alpha + \beta) \end{vmatrix} \\ &= \begin{vmatrix} 1 & 1 & 0 \\ \alpha + \beta & \gamma + \delta & 0 \\ \alpha\beta & \gamma\delta & 0 \end{vmatrix} \times \begin{vmatrix} 1 & 1 & 0 \\ \gamma + \delta & \alpha + \beta & 0 \\ \gamma\delta & \alpha\beta & 0 \end{vmatrix} \quad (\text{Row by Row}) \\ &= 0 \times 0 = 0 = \text{R.H.S} \end{aligned}$$

► **Problem 12.** Prove that

$$\begin{vmatrix} \cos(A-P) & \cos(A-Q) & \cos(A-R) \\ \cos(B-P) & \cos(B-Q) & \cos(B-R) \\ \cos(C-P) & \cos(C-Q) & \cos(C-R) \end{vmatrix} = 0$$

► **Solution** L.H.S.

$$\begin{aligned} &= \begin{vmatrix} \cos(A-P) & \cos(A-Q) & \cos(A-R) \\ \cos(B-P) & \cos(B-Q) & \cos(B-R) \\ \cos(C-P) & \cos(C-Q) & \cos(C-R) \end{vmatrix} \\ &= \begin{vmatrix} \cos A & \sin A & 0 \\ \cos B & \sin B & 0 \\ \cos C & \sin C & 0 \end{vmatrix} \times \begin{vmatrix} \cos P & \sin P & 0 \\ \cos Q & \sin Q & 0 \\ \cos R & \sin R & 0 \end{vmatrix} \\ &(\text{Row by Row}) \\ &= 0 \times 0 = 0 = \text{R.H.S.} \end{aligned}$$

► **Problem 13.** If  $\Delta = \begin{vmatrix} h & a & 0 \\ \frac{1}{h} & \frac{1}{b} & \frac{1}{f} \\ 0 & c & f \end{vmatrix}$  and

$$\Delta' = \begin{vmatrix} \frac{1}{bc} - \frac{1}{f^2} & -\frac{1}{ch} & \frac{1}{fh} \\ af & -fh & ch \\ \frac{1}{fh} & -\frac{1}{af} & \frac{1}{ab} - \frac{1}{h^2} \end{vmatrix}$$

find the value of  $\frac{\Delta'}{\Delta^2}$  in its simplest form.

► **Solution** Here

$$\begin{aligned} \text{adj } \Delta &= \begin{vmatrix} \frac{f}{b} - \frac{c}{f} & -\frac{f}{h} & \frac{c}{h} \\ -af & hf & -ch \\ \frac{a}{f} & -\frac{h}{f} & \frac{h}{b} - \frac{a}{h} \end{vmatrix} \\ &= -caf h \begin{vmatrix} \frac{1}{bc} - \frac{1}{f^2} & -\frac{1}{ch} & \frac{1}{fh} \\ af & -hf & ch \\ \frac{1}{fh} & -\frac{1}{af} & \frac{1}{ab} - \frac{1}{h^2} \end{vmatrix} \end{aligned}$$

(taking out  $fc, -1, ah$  from the 1st, 2nd and 3rd row respectively)

$$\therefore \Delta^2 = -caf h \Delta' \text{ or } \frac{\Delta'}{\Delta^2} = -\frac{1}{caf h}.$$

► **Problem 14.** Let  $\alpha$  be a repeated root of the quadratic equation  $f(x) = 0$  and  $A(x)$ ,  $B(x)$  and  $C(x)$  are polynomials of degree 3, 4 and 5 respectively.

$$\text{Show that the determinant } \begin{vmatrix} A(x) & B(x) & C(x) \\ A(\alpha) & B(\alpha) & C(\alpha) \\ A'(\alpha) & B'(\alpha) & C'(\alpha) \end{vmatrix}$$

is divisible by  $f(x)$  where  $A'(\alpha) = \left( \frac{dA}{dx} \right)_{x=\alpha}$ .

► **Solution** We know that  $\alpha$  is a root of  $f(x) = 0$ . This means  $(x - \alpha)$  is a factor of  $f(x)$ . If  $\alpha$  is a repeated root of  $f(x) = 0$  then  $f(x)$  has a repeated factor  $(x - \alpha)$ , i.e.,  $(x - \alpha)^2$  is a factor of  $f(x)$ . But here  $f(x)$  is quadratic.

$$\therefore f(x) = \lambda(x - \alpha)^2 \quad \dots(1)$$

where  $\lambda$  is a constant.

$$\text{Let } \Delta(x) = \begin{vmatrix} A(x) & B(x) & C(x) \\ A(\alpha) & B(\alpha) & C(\alpha) \\ A'(\alpha) & B'(\alpha) & C'(\alpha) \end{vmatrix}$$

which is of the degree 5 atmost and 3 atleast.  
Clearly,  $\Delta(\alpha) = 0$  ...(2)  
Differentiating  $\Delta(x)$  w.r.t.  $x$ ,

$$\Delta'(x) = \begin{vmatrix} A'(x) & B'(x) & C'(x) \\ A(\alpha) & B(\alpha) & C(\alpha) \\ A'(\alpha) & B'(\alpha) & C'(\alpha) \end{vmatrix}$$

because derivatives of constants = 0

$$\therefore \Delta'(\alpha) = \begin{vmatrix} A'(\alpha) & B'(\alpha) & C'(\alpha) \\ A(\alpha) & B(\alpha) & C(\alpha) \\ A'(\alpha) & B'(\alpha) & C'(\alpha) \end{vmatrix} = 0 \quad \dots(3)$$

because  $R_1$  is identical with  $R_3$ .

We know that

$$\begin{aligned} \phi(\alpha) = 0 &\Rightarrow (x - \alpha) \text{ is a factor of } \phi(x) \\ \text{and } \phi'(\alpha) = 0 &\Rightarrow (x - \alpha)^2 \text{ is a factor of } \phi(x) \\ \therefore \text{ from (2) and (3), } \Delta(x) &\text{ has a factor } (x - \alpha)^2. \\ \therefore \Delta(x) &= (x - \alpha)^2 \cdot F(x) \end{aligned}$$

$$= \frac{1}{\lambda} \lambda(x - \alpha)^2 \cdot F(x) = \frac{1}{\lambda} f(x) \cdot F(x), \text{ using (1)}$$

$$\therefore \Delta(x) \text{ is divisible by } f(x).$$

► **Problem 15.** If  $l_r, m_r, n_r, r = 1, 2, 3$  be the direction cosines of three mutually perpendicular lines referred to an orthogonal cartesian co-ordinate system, then prove that

$$\begin{bmatrix} l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \\ l_3 & m_3 & n_3 \end{bmatrix} \text{ is an orthogonal matrix.}$$

► **Solution** Let  $A = \begin{bmatrix} l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \\ l_3 & m_3 & n_3 \end{bmatrix}$ .

$$\text{Then } A' = \begin{bmatrix} l_1 & l_2 & l_3 \\ m_1 & m_2 & m_3 \\ n_1 & n_2 & n_3 \end{bmatrix}.$$

$$\text{We have } AA' = \begin{bmatrix} l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \\ l_3 & m_3 & n_3 \end{bmatrix} \begin{bmatrix} l_1 & l_2 & l_3 \\ m_1 & m_2 & m_3 \\ n_1 & n_2 & n_3 \end{bmatrix}.$$

$$= \begin{bmatrix} l_1^2 + m_1^2 + n_1^2 & l_1 l_2 + m_1 m_2 + n_1 n_2 & l_1 l_3 + m_1 m_3 + n_1 n_3 \\ l_2 l_1 + m_2 m_1 + n_2 n_1 & l_2^2 + m_2^2 + n_2^2 & l_2 l_3 + m_2 m_3 + n_2 n_3 \\ l_3 l_1 + m_3 m_1 + n_3 n_1 & l_3 l_2 + m_3 m_2 + n_3 n_2 & l_3^2 + m_3^2 + n_3^2 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I_3.$$

Hence the matrix  $A$  is orthogonal.

► **Problem 16.** If  $ax + by + cz = 1, cx + ay + bz = 0, bx + cy + az = 0$ , find  $x, y, z$  and hence prove that

$$\begin{vmatrix} x & y & z \\ z & x & y \\ y & z & x \end{vmatrix} \text{ and } \begin{vmatrix} a & c & b \\ b & a & c \\ c & b & a \end{vmatrix}$$

have reciprocal values.

► **Solution** We have the determinant of the system

$$\Delta = \begin{vmatrix} a & b & c \\ c & a & b \\ b & c & a \end{vmatrix}$$

and we suppose that  $\Delta \neq 0$ .

By Cramer's rule

$$\frac{x}{\begin{vmatrix} 1 & b & c \\ 0 & a & b \\ 0 & c & a \end{vmatrix}} = \frac{y}{\begin{vmatrix} a & 1 & c \\ c & 0 & b \\ b & 0 & a \end{vmatrix}} = \frac{z}{\begin{vmatrix} a & b & 1 \\ c & a & 0 \\ b & c & 0 \end{vmatrix}} = \frac{1}{\Delta}$$

$$\therefore x = \frac{A}{\Delta}, \quad y = \frac{B}{\Delta}, \quad z = \frac{C}{\Delta}$$

where A, B, C are cofactors of a, b, c in  $\Delta$ .

Now

$$\begin{vmatrix} x & y & z \\ z & x & y \\ y & z & x \end{vmatrix} = \begin{vmatrix} \frac{A}{\Delta} & \frac{B}{\Delta} & \frac{C}{\Delta} \\ \frac{C}{\Delta} & \frac{A}{\Delta} & \frac{B}{\Delta} \\ \frac{B}{\Delta} & \frac{C}{\Delta} & \frac{A}{\Delta} \end{vmatrix} = \frac{1}{\Delta^3} \begin{vmatrix} A & B & C \\ C & A & B \\ B & C & A \end{vmatrix} \\ = \frac{\Delta^2}{\Delta^3} = \frac{1}{\Delta}.$$

► **Problem 17.** If the system of following equations in x and y

$$2x + 3y + 4 = 0 ; \quad 3x + 5y + 6 = 0 ;$$

$$2x^2 + 6xy + 5y^2 + 8x + 12y + 1 = t$$

consists of a solution, then find the value of t.

► **Solution** Clearly  $2x^2 + 6xy + 5y^2 + 8x + 12y + 1 = t$

$$\Rightarrow x(2x + 3y + 4) + y(3x + 5y + 6) + 4x + 6y + 1 - t = 0 \quad \dots(1)$$

The values of x, y that satisfy

$$2x + 3y + 4 = 0 \quad \dots(2)$$

$$3x + 5y + 6 = 0 \quad \dots(3)$$

will satisfy the equation (1) if

$$4x + 6y + 1 - t = 0 \quad \dots(4)$$

The linear equations (2), (3), (4) will have a solution if they are consistent

$$\text{i.e. } \begin{vmatrix} 2 & 3 & 4 \\ 3 & 5 & 6 \\ 4 & 6 & 1-t \end{vmatrix} = 0 \quad \Rightarrow t = -4$$

► **Problem 18.** By the method of matrix inversion, solve the system.

$$\begin{bmatrix} 1 & 1 & 1 \\ 2 & 5 & 7 \\ 2 & 1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 9 \\ 52 \\ 0 \end{bmatrix}$$

► **Solution** We have

$$\begin{bmatrix} 1 & 1 & 1 \\ 2 & 5 & 7 \\ 2 & 1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 9 \\ 52 \\ 0 \end{bmatrix}$$

$$\text{or } AX = B$$

$$\text{or } X = A^{-1}B \quad \dots(1)$$

Where

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 5 & 7 \\ 2 & 1 & -1 \end{bmatrix} \quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$\text{and } B = \begin{bmatrix} 9 \\ 52 \\ 0 \end{bmatrix}$$

$$\therefore |A| = 1(-5-7) - 1(-2-14) + 1(2-10) \\ = -12 + 16 - 8 \\ = -4 \neq 0$$

Let C be the matrix of cofactors of elements of |A|.

$$C = \begin{bmatrix} C_{11} & C_{12} & C_{13} \\ C_{21} & C_{22} & C_{23} \\ C_{31} & C_{32} & C_{33} \end{bmatrix}$$

$$= \begin{bmatrix} \begin{vmatrix} 5 & 7 \\ 1 & -1 \end{vmatrix} & -\begin{vmatrix} 2 & 7 \\ 2 & -1 \end{vmatrix} & \begin{vmatrix} 2 & 5 \\ 2 & 1 \end{vmatrix} \\ -\begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix} & \begin{vmatrix} 1 & 1 \\ 2 & -1 \end{vmatrix} & -\begin{vmatrix} 1 & 1 \\ 2 & 1 \end{vmatrix} \\ \begin{vmatrix} 1 & 1 \\ 5 & 7 \end{vmatrix} & -\begin{vmatrix} 1 & 7 \\ 2 & 7 \end{vmatrix} & \begin{vmatrix} 1 & 1 \\ 2 & 5 \end{vmatrix} \end{bmatrix}$$

$$= \begin{bmatrix} -12 & 16 & -8 \\ 2 & -3 & 1 \\ 2 & -5 & 3 \end{bmatrix}$$

$$\therefore \text{Adj } A = C' = \begin{bmatrix} -12 & 2 & 2 \\ 16 & -3 & -5 \\ -8 & 1 & 3 \end{bmatrix}$$

$$\therefore A^{-1} = \frac{\text{Adj } A}{|A|} = -\frac{1}{4} \begin{bmatrix} -12 & 2 & 2 \\ 16 & -3 & -5 \\ -8 & 1 & 3 \end{bmatrix}$$

$$\text{Now, } A^{-1}B = -\frac{1}{4} \begin{bmatrix} -12 & 2 & 2 \\ 16 & -3 & -5 \\ -8 & 1 & 3 \end{bmatrix} \times \begin{bmatrix} 9 \\ 52 \\ 0 \end{bmatrix} \\ = -\frac{1}{4} \begin{bmatrix} -4 & 4 \\ -12 & -8 \\ -20 & -4 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 3 & 2 \\ 5 & 1 \end{bmatrix}$$

From (1)  $X = A^{-1}B$

$$\Rightarrow \begin{bmatrix} x & u \\ y & v \\ z & w \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 3 & 2 \\ 5 & 1 \end{bmatrix}$$

On equating the corresponding elements, we have

$$x = 1, u = -1$$

$$y = 3, v = 2$$

$$z = 5, w = 1$$

► **Problem 19.** Solve completely the system of equations

$$4x + 2y + z + 3u = 0,$$

$$6x + 3y + 4z + 7u = 0,$$

$$2x + y + u = 0.$$

► **Solution** The matrix form of the given system is

$$\begin{bmatrix} 4 & 2 & 1 & 3 \\ 6 & 3 & 4 & 7 \\ 2 & 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ u \end{bmatrix} = O$$

or  $\begin{bmatrix} 1 & 2 & 4 & 3 \\ 4 & 3 & 6 & 7 \\ 0 & 1 & 2 & 1 \end{bmatrix} \begin{bmatrix} z \\ y \\ x \\ u \end{bmatrix} = O$ , interchanging the

variables  $x$  and  $z$ .

Performing  $R_2 \rightarrow R_2 - 4R_1$ , we get

$$\begin{bmatrix} 1 & 2 & 4 & 3 \\ 0 & -5 & -10 & -5 \\ 0 & 1 & 2 & 1 \end{bmatrix} \begin{bmatrix} z \\ y \\ x \\ u \end{bmatrix} = O.$$

Performing  $R_2 \rightarrow -\frac{1}{5}R_2$ , we get

$$\begin{bmatrix} 1 & 2 & 4 & 3 \\ 0 & 1 & 2 & 1 \\ 0 & 1 & 2 & 1 \end{bmatrix} \begin{bmatrix} z \\ y \\ x \\ u \end{bmatrix} = O.$$

Performing  $R_3 \rightarrow R_3 - R_2$ , we get

$$\begin{bmatrix} 1 & 2 & 4 & 3 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} z \\ y \\ x \\ u \end{bmatrix} = O.$$

= O.

The coefficient matrix is of rank 2 and therefore the given system will have  $4 - 2$  i.e. 2 linearly

independent solutions. The given system of equations is equivalent to

$$z + 2y + 4x + 3u = 0,$$

$$y + 2x + u = 0.$$

$$\therefore y = -2x - u, z = -4x - 3u + 4x + 2u = -u.$$

$$\therefore x = c_1, u = c_2, y = -2c_1 - c_2, z = c_2$$

constitute the general solution where  $c_1$  and  $c_2$  are arbitrary constants.

► **Problem 20.** Solve the equations

$$\lambda x + 2y - 2z - 1 = 0,$$

$$4x + 2\lambda y - z - 2 = 0,$$

$$6x + 6y + \lambda z - 3 = 0.$$

► **Solution** The matrix form of the given system is

$$\begin{bmatrix} \lambda & 2 & -2 \\ 4 & 2\lambda & -1 \\ 6 & 6 & \lambda \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \quad \dots(1)$$

The given system of equations will have a unique solution if and only if the coefficient matrix is nonsingular, i.e., iff

$$\begin{vmatrix} \lambda & 2 & -2 \\ 4 & 2\lambda & -1 \\ 6 & 6 & \lambda \end{vmatrix} \neq 0$$

$$\text{i.e., iff } \lambda^3 + 11\lambda - 30 \neq 0$$

$$\text{i.e., iff } (\lambda - 2)(\lambda^2 + 2\lambda + 15) \neq 0.$$

Now the only real root of the equation

$$(\lambda - 2)(\lambda^2 + 2\lambda + 15) \neq 2 = 0 \text{ is } \lambda = 2.$$

Therefore if  $\lambda \neq 2$ , the given system of equations will have a unique solution given by

$$\begin{array}{c} x \\ \begin{vmatrix} 1 & 2 & -2 \\ 2 & 2\lambda & -1 \\ 3 & 6 & \lambda \end{vmatrix} \end{array} = \begin{array}{c} y \\ \begin{vmatrix} \lambda & 1 & -2 \\ 4 & 2 & -1 \\ 6 & 3 & \lambda \end{vmatrix} \end{array} = \begin{array}{c} z \\ \begin{vmatrix} \lambda & 2 & 1 \\ 4 & 2\lambda & 2 \\ 6 & 6 & 3 \end{vmatrix} \end{array}$$

$$= \begin{array}{c} z \\ \begin{vmatrix} \lambda & 2 & 1 \\ 4 & 2\lambda & 2 \\ 6 & 6 & 3 \end{vmatrix} \end{array}$$

In case  $\lambda = 2$ , the equation (1) becomes

$$\begin{bmatrix} 2 & 2 & -2 \\ 4 & 4 & -1 \\ 6 & 6 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}.$$

Performing  $R_2 \rightarrow R_2 - 2R_1$ ,  $R_3 \rightarrow R_3 - 3R_1$ , we get

$$\begin{bmatrix} 2 & 2 & -2 \\ 0 & 0 & 3 \\ 0 & 0 & 8 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

The above system of equations is equivalent to  
 $8z = 0, 3z = 0, 2x + 2y - 2z = 1.$

$\therefore x = \frac{1}{2} - c, y = c, z = 0$  constitute the general solution of the given system of equations in case  $\lambda = 2$ .

► **Problem 21.** Consider the following system of 3 equations in 5 unknowns:

$$2x - 5y + 4z + u - v = -3$$

$$x - 2y + z - u + v = 5$$

$$x - 4y + 6z + 2u - v = 10$$

► **Solution** The corresponding augmented matrix is

$$\left[ \begin{array}{ccccc|c} 2 & -5 & 4 & 1 & -1 & -3 \\ 1 & -2 & 1 & -1 & 1 & 5 \\ 1 & -4 & 6 & 2 & -1 & 10 \end{array} \right].$$

We perform row operations to finally arrive at the augmented matrix

$$\left[ \begin{array}{ccccc|c} 1 & 0 & 0 & -16 & 19 & 124 \\ 0 & 1 & 0 & -9 & 11 & 75 \\ 0 & 0 & 1 & -3 & 4 & 31 \end{array} \right].$$

The corresponding system of equations can be solved for  $x, y,$  and  $z$  in terms of  $u$  and  $v$ , giving us

$$x = 124 + 16u - 19v$$

$$y = 75 + 9u - 11v$$

$$z = 31 + 3u - 4v.$$

If we let  $u = t_1$  and  $v = t_2$ , where  $t_1$  and  $t_2$  are arbitrary real numbers, and determine  $x, y, z$  by these equations, the vector  $(x, y, z, u, v)$  given by  $(x, y, z, u, v) = (124 + 16t_1 - 19t_2, 75 + 9t_1 - 11t_2, 31 + 3t_1 - 4t_2, t_1, t_2)$

is a solution. By separating the parts involving  $t_1$  and  $t_2$ , we can rewrite this as follows :

$$(x, y, z, u, v) = (124, 75, 31, 0, 0)$$

$$+ t_1(16, 9, 3, 1, 0) + t_2(-19, -11, -4, 0, 1).$$

This equation gives the general solution of the system. The vector  $(124, 75, 31, 0, 0)$  is a particular solution of the non-homogeneous system. The two

vectors  $(16, 9, 3, 1, 0)$  and  $(-19, -11, -4, 0, 1)$  are solutions of the corresponding homogeneous system.

### ● Things to Remember

1. The square array of scalars is the matrix, whereas the determinant is just one scalar, associated with the matrix according to a definite rule.
2. The trace of a square matrix is given by  $\text{tr}(A) = a_{11} + a_{22} + a_{33} + \dots + a_{nn}$
3. In the process of multiplication  $AB$  we take the rows of  $A$  and the columns of  $B$ , which is possible when the number of columns of  $A$  is same as number of rows of  $B$ .
  - (i) Matrix multiplication is not commutative, in general.
  - (ii) From  $AB = AC$  or  $BA = CA$  we cannot in general conclude that  $B = C$ , even if  $A \neq O$ .
4. Let  $A = [a_{ij}]_{m \times p}$  and  $B = [b_{ij}]_{p \times n}$ , then  $(AB)' = B'A'$
5. (i) A square matrix  $A$  is said to be involutory if  $A^2 = I$ 
  - (ii) A square matrix  $A$  is called periodic, if  $A^{k+1} = A$ , where  $k$  is a positive integer. If  $k$  is the least positive integer for which  $A^{k+1} = A$ , then  $k$  is said to be period of  $A$ .
  - (iii) If  $A^2 = A$ , then  $A$  is called idempotent matrix.
  - (iv) A square matrix  $A$  is called nilpotent matrix of order  $m$  provided it satisfies the relation  $A^k = 0$  and  $A^{k-1} \neq 0$ .
6. If  $A = [a_{ij}]_n$  then cofactor  $A = [A_{ij}]_n$  when  $A_{ij}$  is the cofactor of  $a_{ij} \forall i \& j$ .  
 $\text{Adj } A = [b_{ij}]_n$  where  $b_{ij} = A_{ji} \forall i \& j$ .
7. If  $A$  is a square matrix of order  $n$ , then  $A \cdot (\text{adj } A) = |A| I_n = (\text{adj } A) \cdot A$ .
8. (i)  $|\text{adj } A| = |A|^{n-1}$ , where  $n$  is order of  $A$ .  
 (ii) If  $|A| = 0$ , then  $|\text{adj } A| = 0$ .  
 (iii)  $\text{adj}(kA) = k^{n-1}(\text{adj } A)$



- (iv)  $\text{adj } A' = (\text{adj } A)'$   
 (v)  $\text{adj } I = I$   
 (vi)  $\text{adj}(AB) = (\text{adj } B)(\text{adj } A)$ .  
 (vii) If  $A$  is a non-singular matrix, then  
 $\text{adj}(\text{adj } A) = |A|^{n-2} A$ .  
 (viii)  $|\text{adj}(\text{adj } A)| = |A|^{(n-1)^2}$
9.  $A^{-1} = \frac{1}{|A|} \text{adj } A$ .
10. (i)  $(A^{-1})^{-1} = A$ , if  $A$  is non-singular.  
 (ii)  $|A^{-1}| = \frac{1}{|A|}$   
 (iii)  $(A')^{-1} = (A^{-1})'$   
 (iv) If  $k \neq 0$  and  $A$  is a nonsingular matrix then  
 $(kA)^{-1} = \frac{1}{k} A^{-1}$ .  
 (v)  $I^{-1} = I$   
 (vi) If  $D = \text{diag}(a_{11}, a_{22}, \dots, a_{nn})$  where  $a_{ii} \neq 0 \forall i$  then  
 $D^{-1} = \text{diag}\left(\frac{1}{a_{11}}, \frac{1}{a_{22}}, \dots, \frac{1}{a_{nn}}\right)$ .  
 (vii) We define  $A^{-n} = (A^{-1})^n$  when  $A$  is non-singular and  $n$  is a positive integer. We define  $A^0 = I$  for an arbitrary square matrix  $A$ , singular or not. Also the laws of exponents  $A^m A^n = A^{m+n}$  and  $(A^m)^n = A^{mn}$  apply for all integral values of  $m$  and  $n$  when  $A$  is non-singular.  
 (viii)  $\text{adj } A^{-1} = (\text{adj } A)^{-1} = \frac{A}{|A|}$ .  
 (ix) If  $A, B$  be two  $n$ -rowed non-singular matrices, then  $AB$  is also non-singular and  $(AB)^{-1} = B^{-1} A^{-1}$ .  
 (x) If  $A, B$  are square matrices and  $AB = O$ , then  $A = O$ , or  $B = O$ , or  $A$  and  $B$  are both singular.  
 (xi) Let  $A$  be a non-singular matrix. Then  $AB = AC \Rightarrow B = C$  and  $BA = CA \Rightarrow B = C$ .
11. The Cayley–Hamilton Theorem : Every square matrix satisfies its characteristic equation i.e., if for a square matrix  $A$  of order  $n$ ,  $|A - \lambda I| = (-1)^n [\lambda^n + a_1 \lambda^{n-1} + a_2 \lambda^{n-2} + \dots + a_n]$ , then the matrix equation  
 $X^n + a_1 X^{n-1} + a_2 X^{n-2} + \dots + a_n I = O$   
 is satisfied by  $X = A$   
 i.e.,  $A^n + a_1 A^{n-1} + a_2 A^{n-2} + \dots + a_n I = O$ .
12. A square matrix  $A$  is said to be orthogonal if  $A'A = I$ .  
 (i) Every orthogonal matrix is nonsingular.  
 (ii)  $I$  is an orthogonal matrix.  
 (iii)  $|A|$  is either 1 or  $-1$  when  $A$  is orthogonal.  
 (iv) The product of two orthogonal matrices is orthogonal.  
 (v) The inverse of an orthogonal matrix is orthogonal.  
 (vi) The transpose of an orthogonal matrix is orthogonal.
13. Let  $A = [a_{ij}]_{m \times n}$  be a given matrix. A natural number  $\rho$  is said to be the rank of  $A$  if  $A$  has a non singular submatrix of order  $\rho$  and it has no non-singular submatrix of order more than  $\rho$ . Rank of zero matrix is regarded to be zero.
14. Cramer's rule: If  $\Delta \neq 0$ , we have  
 $x_i = \frac{\Delta_i}{\Delta}, i = 1, 2, \dots, n$ ,  
 where  $\Delta_i$  is the determinant obtained by replacing the  $i$ th column in  $\Delta$  by the elements  $b_1, b_2, \dots, b_n$ .
15. **Non-Homogeneous System**  
 Let  $\rho(A)$  denote rank of  $A$  and  $\rho([AB])$  denote rank of the augmented matrix. Clearly  $\rho(A) \leq \rho([AB])$ . In all there are three possibilities :  
 (a) If  $\rho(A) < \rho([AB])$  then the system has no solution (i.e. system is inconsistent).  
 (b) If  $\rho(A) = \rho([AB]) = \text{number of unknowns}$ , then the system has uniwue solution. (and hence is consistent)  
 (c) If  $\rho(A) = \rho([AB]) < \text{number of unknowns}$ , then the systems has infinitely many solutions (and so is consistent).
16. **Homogeneous System**  
 Let the homogenous system be  $AX = 0$ ,  $m$  equations in ' $n$ ' unknowns. in this case  $B = 0$  and so  $\rho(A) = \rho([AB])$ . Hence if  $\rho(A) = n$ , then the system has only the trivial solution. If  $\rho(A) < n$ , then the system has infinitely many solutions.

## [Objective Exercises]

## Single Correct Answer Type

1. Which of the given values of  $x$  and  $y$  make the following pair of matrices equal

$$\begin{bmatrix} 3x+7 & 5 \\ y+1 & 2-3x \end{bmatrix}, \begin{bmatrix} 0 & y-2 \\ 8 & 4 \end{bmatrix}$$

(A)  $x = \frac{-1}{3}, y = 7$  (B) Not possible to find

(C)  $y = 7, x = \frac{-2}{3}$  (D)  $x = \frac{-1}{3}, y = \frac{-2}{3}$

2. The number of all possible matrices of order  $3 \times 3$  with each entry 0 or 1 is :

(A) 27 (B) 18  
(C) 81 (D) 512

3. If  $A, B$  are symmetric matrices of same order, then  $AB - BA$  is a

- (A) Skew symmetric matrix  
(B) Symmetric matrix  
(C) Zero matrix  
(D) Identity matrix

4. If  $A = \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix}$ , then  $A + A' = I$ , if the value of  $\alpha$  is

(A)  $\frac{\pi}{6}$  (B)  $\frac{\pi}{3}$   
(C)  $\pi$  (D)  $\frac{3\pi}{2}$

5. If  $A$  is square matrix such that  $A^2 = A$ , then  $(I + A)^3 - 7A$  is equal to

- (A)  $A$  (B)  $I - A$   
(C)  $I$  (D)  $3A$

6. If  $\begin{vmatrix} x & 2 \\ 18 & x \end{vmatrix} = \begin{vmatrix} 6 & 2 \\ 18 & 6 \end{vmatrix}$ , then  $x$  is equal to

(A) 6 (B)  $\pm 6$   
(C)  $-6$  (D) 0

7. The value of determinant

$$\Delta = \begin{vmatrix} 2a_1b_1 & a_1b_2 + a_2b_1 & a_1b_3 + a_3b_1 \\ a_1b_2 + a_2b_1 & 2a_2b_2 & a_2b_3 + a_3b_2 \\ a_1b_3 + a_3b_1 & a_3b_2 + a_2b_3 & 2a_3b_3 \end{vmatrix}$$
 is

- (A) 1 (B)  $-1$   
(C) 0 (D)  $a_1a_2a_3b_1b_2b_3$

8. If  $\alpha + \beta + \gamma = \pi$ , then the value of

$$\begin{vmatrix} \sin(\alpha + \beta + \gamma) & \sin \beta & \cos \gamma \\ -\sin \beta & 0 & \tan \alpha \\ \cos(\alpha + \beta) & -\tan \alpha & 0 \end{vmatrix}$$
 is :

- (A) 0 (B) 1  
(C) 2  
(D)  $2 \cdot \sin \beta \cdot \cos \gamma \cdot \tan \alpha$

9. Let  $D = \begin{vmatrix} x^2 + 4x + 5 & x + 2 & 5 \\ 2x^2 + 6x + 10 & 2x + 3 & 10 \\ 4x^2 - 2x + 20 & 4x - 1 & 20 \end{vmatrix}$ , then

- (A)  $y = \Delta$  represents a parabola a passing through the origin  
(B)  $y = \Delta$  represents a straight line through the origin  
(C)  $\Delta = 0$  has two distinct real roots  
(D)  $\Delta = 0$  has exactly one real root.

10. For three squares matrix  $A, B$  &  $C$  if  $ABC = 0$  &  $|A| \neq 0, B$  is non-zero singular matrix then

- (A)  $C$  must be zero matrix  
(B)  $C$  must be non-singular matrix  
(C)  $C$  must be singular matrix  
(D) None of these

11.  $A = \begin{bmatrix} x & x+2 \\ 2x-1 & 3x \end{bmatrix}$  is a skew symmetric matrix,

then  $x =$

- (A) 1 (B) 0  
(C) 3 (D) does not exist

12. If a square matrix of order 3 such that  $|A| = 2$  then  $\text{adj } A^{-1}|$  is

(A) 2 (B) 4  
(C)  $\frac{1}{2}$  (D)  $\frac{1}{4}$

13. If A is  $4 \times 4$  matrix and if  $\|A\| \operatorname{adj}(|A| A) = |A|^n$ , then n is

(A) 11 (B) 13  
(C) 17 (D) 19

14. Let A be a idempotent square matrix then  $(I + A)^4$  is

(A)  $I - A$  (B)  $I + A$   
(C)  $I + 15A$  (D) None

15. The rank of the matrix  $\begin{bmatrix} 1 & 2 & 3 \\ \lambda & 2 & 4 \\ 2 & -3 & 1 \end{bmatrix}$  is 3 if

(A)  $\lambda \neq \frac{18}{11}$  (B)  $\lambda = 2$   
(C)  $\lambda = -\frac{18}{11}$  (D) None of these

16. If  $\Delta_r = \begin{vmatrix} 2^r & \frac{1}{(r+1)(r+2)} & r \\ a & b & c \\ 2^{n+1} - 1 & \frac{n+1}{n+2} & \frac{n(n+1)}{2} \end{vmatrix}$ ,

then  $\sum_{r=0}^n \Delta_r$ , is equal to

(A) 0 (B) 1  
(C) -1 (D) None

17. Let X be the solution set of the equation  $A^x = I$ ,

where  $A = \begin{bmatrix} 2 & -1 \\ 3 & -2 \end{bmatrix}$  and I is the corresponding

unit matrix and  $x \in W$  then the minimum value

of  $\Sigma(\cos^x \theta + \sin^x \theta)$ ,  $\theta \in R - \frac{n\pi}{2}$ , is

(A) 4 (B) 2  
(C) 1 (D)  $1/2$

18. Let the matrix A and B be defined as A

$= \begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix}$  and  $B = \begin{bmatrix} 3 & 1 \\ 7 & 3 \end{bmatrix}$  then the value of  $\operatorname{Det.}(2A^9 B^{-1})$ , is

(A) 2 (B) 1  
(C) -1 (D) -2

19. The matrix A has x rows and  $(x + 5)$  columns. The matrix B has y rows and  $(11 - y)$  columns. Both AB and BA exist. The value of x and y is

(A) 8, 3 (B) 3, 4  
(C) 3, 8 (D) 8, 8

20. There are two possible values of A in the solution of the matrix equation

$$\begin{bmatrix} 2A+1 & -5 \\ -4 & A \end{bmatrix}^{-1} \begin{bmatrix} A-5 & B \\ 2A-2 & C \end{bmatrix} = \begin{bmatrix} 14 & D \\ E & F \end{bmatrix}$$

where A, B, C, D, E, F are real numbers. The absolute value of the difference of these two solutions, is

(A)  $\frac{8}{3}$  (B)  $\frac{11}{3}$   
(C)  $\frac{1}{3}$  (D)  $\frac{19}{3}$

21. If the system of equation  $x + 2y - 3z = 1$ ,  $(p + 2)z = 3$ ,  $(2p + 1)y + z = 2$  is inconsistent, then the value of p is

(A) -2 (B)  $-1/2$   
(C) 0 (D) 2

22. If  $A = \begin{bmatrix} 1 & 2 & x \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$  and  $B = \begin{bmatrix} 1 & -2 & y \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$  and

$AB = I_3$ , then x + y equals

(A) 0 (B) -1  
(C) 2 (D) None

23. If A is square matrix of order  $2 \times 2$  such that  $A^2 = 0$ , then

(A)  $A = \begin{bmatrix} \alpha & \beta \\ \gamma & -\alpha \end{bmatrix}$  where  $\alpha, \beta, \gamma$  are numbers such that  $\alpha^2 + \beta\gamma = 0$

(B)  $A = \begin{bmatrix} \alpha & \beta \\ \beta & -\alpha \end{bmatrix}$  with  $\alpha = \pm\beta$

(C)  $A = \begin{bmatrix} \alpha & -\alpha \\ -\beta & \beta \end{bmatrix}$  with  $\alpha^2 + \beta^2 = 1$

(D) None of these

24. Let  $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ ,  $B = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ ,  $C = \begin{bmatrix} 0 & i \\ i & 0 \end{bmatrix}$ ,

then which of the following is not true ?

- (A)  $A^2 = B^2 = C^2 = I$   
 (B)  $AB + BA = BC + CB = AC + CA = 0$   
 (C)  $AB$  is skew symmetric  
 (D) None of these

25. If  $x_1, x_2$  &  $y_1, y_2$  are the roots of the equations  $3x^2 - 18x + 9 = 0$  &  $y^2 - 4y + 2 = 0$ , then the value of determinant

$$\begin{vmatrix} x_1 x_2 & y_1 y_2 & 1 \\ x_1 + x_2 & y_1 + y_2 & 2 \\ \sin(\pi x_1 x_2) & \cos\left(\frac{\pi}{2} x_1 x_2\right) & 1 \end{vmatrix}$$
 is

- (A) 0 (B) 1  
 (C) C (D) None

26. Let  $x > 0, y > 0, z > 0$  are respectively the 2nd, 3rd, 4th terms of a G.P. and

$$\Delta = \begin{vmatrix} x^k & x^{k+1} & x^{k+2} \\ y^k & y^{k+1} & y^{k+2} \\ z^k & z^{k+1} & z^{k+2} \end{vmatrix} = (r-1)^2 \left(1 - \frac{1}{r^2}\right)$$

(where  $r$  is the common ratio). Then

- (A)  $k = -1$  (B)  $k = 1$   
 (C)  $k = 0$  (D) None of these

27. If the value of  $\Delta = \begin{vmatrix} {}^{10}C_4 & {}^{10}C_5 & {}^{11}C_m \\ {}^{11}C_6 & {}^{11}C_7 & {}^{12}C_{m+2} \\ {}^{12}C_8 & {}^{12}C_9 & {}^{13}C_{m+4} \end{vmatrix}$  is

equal to zero, then  $m$  is

- (A) 6 (B) 4  
 (C) 5 (D) None

28. If  $\sin \beta \neq \cos \beta$  and  $x, y, z$  satisfy the equations  
 $x \cos \alpha - y \sin \alpha + z = \cos \beta + 1$   
 $x \sin \alpha + y \cos \alpha + z = -\sin \beta + 1$   
 $x \cos(\alpha + \beta) - y \sin(\alpha + \beta) + z = 2$   
 then  $(x^2 + y^2 + z^2)$  is

- (A) 1 (B) 0  
 (C) 2 (D)  $\Sigma \frac{\sin \alpha \cos \beta}{\sin(\alpha + \beta)}$

29. The number of values of 't' for which the system of equations  $(a-t)x + by + cz = 0$  ;  $bx + (c-t)y + az = 0$  ;  $cx + ay + (b-t)z = 0$  has non trivial solution is

- (A) 3 (B) 4  
 (C) 5 (D) 6

30. If  $\alpha$  and  $\beta$  are the roots of the equation  $ax^2 + bx + c = 0$  &  $S_n = \alpha^n + \beta^n$  for  $n \geq 1$  and

$$\Delta = \begin{vmatrix} 3 & 1 + S_1 & 1 + S_2 \\ 1 + S_1 & 1 + S_2 & 1 + S_3 \\ 1 + S_2 & 1 + S_3 & 1 + S_4 \end{vmatrix}, \text{ then } \Delta \text{ is}$$

(A)  $\frac{(a+b+c)^2 (b^2 - 4ac)}{a^4}$

(B)  $\frac{(a+b+c)^4 (b^2 - 4ac)}{a^3}$

(C)  $\frac{(a+b+c)^4 (b^2 - 4ac)}{a^2}$

(D) None of these

31. The system of equations  $6x + 5y + \lambda z = 0$  ;  $3x - y + 4z = 0$  ;  $x + 2y - 3z = 0$ , has

- (A) Only one trivial solution for  $\lambda \in \mathbb{R}$   
 (B) Exactly one non trivial solution for some real  $\lambda$   
 (C) Infinite number of non trivial solutions for one value of  $\lambda$   
 (D) Only one solution for  $\lambda \neq -5$

32. If  $\begin{vmatrix} x & 2 & x \\ x^2 & x & 6 \\ x & x & 6 \end{vmatrix} = Ax^4 + Bx^3 + Cx^2 + Dx + E$

then the value of  $5A + 4B + 3C + 2D + E$  is equal to

- (A) 0 (B) -16  
 (C) 16 (D) None

33. If  $f(x) = \begin{vmatrix} 3 & 2 & 1 \\ 6x^2 & 2x^3 & x^4 \\ 1 & a & a^2 \end{vmatrix}$ , then  $f(a)$  is equal to
- (A) 1 (B) 2  
(C) 3 (D\*) 0

34. If  $\Delta^2 = \begin{vmatrix} \sin \theta & \cos \theta & 0 \\ -\cos \theta & \sin \theta & 1 \\ \sin \theta & \cos \theta & 2 \end{vmatrix}$ , then which of the following equation will be satisfied by  $\Delta$
- (A)  $\Delta^4 - 8\Delta^2 + 16 = 0$   
(B)  $\Delta^2 - 4\Delta + 1 = 0$   
(C)  $\Delta^4 + 8\Delta^2 + 16 = 0$   
(D) None

35. The number of triplets of  $a, b$  &  $c$  for which the system of equations,  $ax - by = 2a - b$  and  $(c+1)x + cy = 10 - a + 3b$  has infinitely many solutions and  $x = 1, y = 3$  is one of the solutions, is:
- (A) Exactly one (B) Exactly two  
(C) Exactly three (D) Infinitely many

### Multiple Correct Answer Type for JEE Advanced

36. Which one of the following is/are true?
- (A) The augmented matrix for a system of three linear equations in three variables is always a matrix with three rows and four columns, described as a  $3 \times 4$  matrix.  
(B) The notation  $-2R_1 + R_3$  means to replace all entries in the third row of a matrix by  $-2$  times the entries in the first row.  
(C) The solution set to the system  
 $y = 4x - 3$   
 $y = 4x + 5$  is the empty set.  
(D) One solution to the system in three variables whose augmented matrix is given by

$$\left[ \begin{array}{ccc|c} 1 & 1 & -10 & -4 \\ 0 & 1 & -3 & 1 \\ 0 & 0 & 0 & 0 \end{array} \right] \text{ is } (9, 7, 2).$$

37. If  $A$  and  $B$  be  $3 \times 3$  real matrices such that  $AB + A + B = 0$ , then
- (A)  $AB = BA$   
(B)  $(A + I)(B + I) = I$   
(C)  $(A + I)(B + I) = (B + I)(A + I)$   
(D) None of these

38. Let  $A, B$  and  $C$  be  $2 \times 2$  matrices with entries from the set of real numbers. Define  $*$  as follows

$$(A) B = \frac{1}{2}(AB + BA), \text{ then}$$

- (A)  $A * B = B * A$   
(B)  $A * A = A^2$   
(C)  $A * (B + C) = A * B + A * C$   
(D)  $A * I = A$

39. If  $A(\alpha) = \begin{pmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{pmatrix}$

$$B(\beta) = \begin{pmatrix} \cos \beta & 0 & \sin \beta \\ 0 & 1 & 0 \\ -\sin \beta & 0 & \cos \beta \end{pmatrix}. \text{ Then}$$

- (A)  $(A(\alpha))^{-1} = A(-\alpha)$   
(B)  $A(\alpha)A(\beta) = A(\alpha + \beta)$   
(C)  $B(\alpha)B(\beta) = B(\alpha + \beta)$   
(D)  $(B(\beta))^{-1} = B(-\beta)$

40. Let  $A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$  and  $A^{-n}$  is defined  $(A^{-1})^n$ , then

$$(A) A^{-n} = \begin{bmatrix} 1 & 0 \\ -n & 1 \end{bmatrix} \quad "n \in \mathbb{N}$$

$$(B) \lim_{n \rightarrow \infty} \frac{1}{n} A^{-n} = \begin{bmatrix} 0 & 0 \\ -1 & 0 \end{bmatrix} \quad n \in \mathbb{N}$$

$$(C) \lim_{n \rightarrow \infty} \frac{1}{n^2} A^{-n} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \quad "n \in \mathbb{N}$$

- (D) None of these

41. The determinant

$$\begin{vmatrix} \cos(\theta + \phi) & -\sin^2(\theta + \phi) & \cos 2\phi \\ \sin \theta & \cos \theta & \sin \phi \\ -\cos \theta & \sin \theta & \cos \phi \end{vmatrix} \text{ is}$$

- (A) Positive  
 (B) Independent of  $\theta$   
 (C) Independent of  $\phi$   
 (D) Independent of  $\theta$  and  $\phi$  both

42. The value of  $x$  for which

$$\begin{vmatrix} x & 2 & 2 \\ 3 & x & 2 \\ 3 & 3 & x \end{vmatrix} + \begin{vmatrix} 1-x & 2 & 4 \\ 2 & 4-x & 8 \\ 4 & 8 & 16-x \end{vmatrix} > 33 \text{ is}$$

- (A)  $0 < x < 1$  (B)  $-\frac{1}{2} < x < \frac{1}{2}$   
 (C)  $x < -\frac{1}{7}$  (D)  $x > 1$

43. The value of  $\left( \begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^4 & b^4 & c^4 \end{vmatrix} - \begin{vmatrix} 2 & 2 & 2 \\ a^2 & b^2 & c^2 \\ a^3 & b^3 & c^3 \end{vmatrix} \right)$

is

- (A)  $(a-b)(b-c)(c-a)(a^2+b^2+c^2-ab-bc-ca)$   
 (B)  $(a-b)(b-c)(c-a)(a+b\omega+c\omega^2)(a+b\omega^2+c\omega)$   
 (C)  $(a-b)(b-c)(c-a)(a^2+b^2+c^2)^2$   
 (D) None of these

44. Let  $f(n) = \begin{vmatrix} n & n+1 & n+2 \\ {}^nP_n & {}^{n+1}P_{n+1} & {}^{n+2}P_{n+2} \\ {}^nC_n & {}^{n+1}C_{n+1} & {}^{n+2}C_{n+2} \end{vmatrix}$  where

the symbols have their usual meanings. The  $f(n)$  is divisible by

- (A)  $n^2 + n + 1$  (B)  $(n+1)!$   
 (C)  $n!$  (D) None of these

45. If  $f(x) = \begin{vmatrix} (1+x)^a & (1+2x)^b & 1 \\ 1 & (1+x)^a & (1+2x)^b \\ (1+2x)^b & 1 & (1+x)^b \end{vmatrix}$  a, b

being positive integers, then

- (A) constant term of  $f(x)$  is 0  
 (B) coefficient of  $x$  in  $f(x)$  is 0  
 (C) constant term in  $f(x)$  is  $a-b$   
 (D) constant term in  $f(x)$  is  $a+b$

46. If a square matrix  $A = [a_{ij}]$ ,  $a_{ij} = i^2 - j^2$  is of even order then

- (A)  $A$  is a skew-symmetric  
 (B)  $|A|$  is a perfect square  
 (C)  $A$  is symmetric and  $|A| = 0$   
 (D)  $A$  is neither symmetric nor skew-symmetric

47. If maximum and minimum values of the

$$\text{determinant } \begin{vmatrix} 1 + \sin^2 x & \cos^2 x & \sin 2x \\ \sin^2 x & 1 + \cos^2 x & \sin 2x \\ \sin^2 x & \cos^2 x & 1 + \sin 2x \end{vmatrix}$$

are  $\alpha$  and  $\beta$ , then

- (A)  $\alpha + \beta^{99} = 4$   
 (B)  $\alpha^3 - \beta^{17} = 26$   
 (C)  $(\alpha^{2n} - \beta^{2n})$  is always an even integer for  $n \in \mathbb{N}$   
 (D) a triangle can be constructed having its sides as  $\alpha - \beta$ ,  $\alpha + \beta$  and  $\alpha + 3\beta$

48. If  $\begin{vmatrix} a & a+d & a+2d \\ a^2 & (a+d)^2 & (a+2d)^2 \\ 2a+3d & 2(a+d) & 2a+d \end{vmatrix} = 0$ , then

- (A)  $d = 0$  (B)  $a + d = 0$   
 (C)  $a + d = 1$  (D) None

49. If  $A = \begin{vmatrix} 1 & -1 & 1 \\ 2 & -1 & 0 \\ 1 & 0 & 0 \end{vmatrix}$ , then

- (A)  $A^3 = I$  (B)  $A^{-1} = A^2$   
 (C)  $A^n = A \forall n \neq 4$  (D) All above

50. Let  $A$

$$= \begin{vmatrix} 0 & \sin \alpha & \sin \alpha \sin \beta \\ -\sin \alpha & 0 & \cos \alpha \cos \beta \\ -\sin \alpha \sin \beta & -\cos \alpha \cos \beta & 0 \end{vmatrix} \text{ then}$$

- (A)  $|A|$  is independent of  $\alpha$  and  $\beta$   
 (B)  $A^{-1}$  depends only on  $\alpha$   
 (C)  $A^{-1}$  depends only on  $\beta$   
 (D)  $|A|$  is independent of  $\alpha$ .

51. If  $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$ , then

- (A)  $A^2 - 4A - 5I_3 = 0$   
 (B)  $A^{-1} = \frac{1}{5}(A - 4I_3)$   
 (C)  $A^3$  is not invertible  
 (D)  $A^2$  is invertible
52. Let  $\begin{bmatrix} 8x + 3y & 6z & 3z \\ 4 & 12 & 26x - 5y \end{bmatrix}$   
 $= \begin{bmatrix} 4 & 1 & 2 \\ 0 & 5 & 3 \end{bmatrix} \begin{bmatrix} 3 & 4 & 5 \\ -1 & 0 & -2 \\ 3 & 4 & 7 \end{bmatrix}$   
 Then which of the following is/are correct?  
 (A)  $x = 1$  (B)  $y = 3$   
 (C)  $z = 5$  (D)  $x + y + z = 8$
53. If  $A$  and  $B$  are square matrices of the same order such that  $A^2 = A$ ,  $B^2 = B$ ,  $AB = BA = 0$ , then  
 (A)  $AB^2 = 0$   
 (B)  $(A + B)^2 = A + B$   
 (C)  $(A - B)^2 = A - B$   
 (D) All above
54. If  $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ , (where  $b \neq c$ ) satisfies the equation  $x^2 + k = 0$  then  
 (A)  $a + d = 0$  (B)  $k = -|A|$   
 (C)  $k = |A|$  (D) All above
55. If  $A = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$ , then  
 (A)  $\text{Adj } A$  is zero matrix  
 (B)  $\text{adj } A = \begin{bmatrix} 0 & 0 & -1 \\ 0 & -1 & 0 \\ -1 & 0 & 0 \end{bmatrix}$   
 (C)  $A^{-1} = A$   
 (D)  $A^2 = I$
56. Let  $X$  be an unknown matrix of order 2 and  $I$  be identity matrix of order 2, then the equation  $X^2 = I$   
 (A) is satisfied if  $X = I$   
 (B) is satisfied if  $X = \begin{pmatrix} 7i & 10 \\ 5 & -7i \end{pmatrix}$   
 (C) is satisfied if  $X = \begin{pmatrix} \pm\sqrt{1-\lambda\mu} & \lambda \\ \mu & \mp\sqrt{1-\lambda\mu} \end{pmatrix}$   
 (D) has no solution except identity matrix.
57. For all  $x$  in  $(-1, 1)$ , let  $A(x)$  be a matrix defined by  $A(x) = \sqrt{1-x^2} \begin{pmatrix} 1 & -x \\ -x & 1 \end{pmatrix}$  then  
 (A)  $A(x)A(y) = A(xy)$   
 (B)  $A(x)A(y) = A\left(\frac{x+y}{1+xy}\right)$   
 (C)  $A(x)A(y) = A\left(\frac{x-y}{1+xy}\right)$   
 (D)  $(A(x))^{-1} = A(-x)$
58. If  $A$  and  $B$  are symmetric non-singular matrices,  $AB = BA$  and then  
 (A)  $A^{-1}B$  is symmetric  
 (B)  $AB^{-1}$  is symmetric  
 (C)  $A^{-1}B^{-1}$  is symmetric  
 (D) All above
59. Let  $f(x) = \begin{vmatrix} 1/x & \log x & x^n \\ 1 & -1/n & (-1)^n \\ 1 & a & a^2 \end{vmatrix}$ , then  
 (A)  $n$ th derivative of  $f(x)$  at 1 is independent of  $a$   
 (B)  $n$ th derivative of  $f(x)$  at 1 is independent of  $n$   
 (C)  $n$ th derivative of  $f(x)$  at 1 depends upon  $a$  and  $n$   
 (D) All above are correct
60. The value of the determinant  $D = \begin{vmatrix} \sin \alpha & \cos \alpha & 1 \\ \sin \beta & \cos \beta & 1 \\ \sin \gamma & \cos \gamma & 1 \end{vmatrix}$  equals  
 (A)  $-4 \sin \frac{\beta-\gamma}{2} \sin \frac{\gamma-\alpha}{2} \sin \frac{\alpha-\beta}{2}$   
 (B)  $\sin \alpha + \sin \beta + \sin \gamma$

- (C)  $\sin(\beta - \gamma) + \sin(\gamma - \alpha) + \sin(\alpha - \beta)$   
 (D) All above are correct.

61. If  $f_r(x)$ ,  $g_r(x)$ ,  $h_r(x)$ ,  $r = 1, 2, 3$ , are polynomials in  $x$  such that  $f_r(a) = g_r(a) = h_r(a)$ ,  $r = 1, 2, 3$  and

$$F(x) = \begin{vmatrix} f_1(x) & f_2(x) & f_3(x) \\ g_1(x) & g_2(x) & g_3(x) \\ h_1(x) & h_2(x) & h_3(x) \end{vmatrix} \text{ then } F'(x) \text{ at } x = a$$

is

- (A) finite (B) 0  
 (C) 2 (D) -1

62. Let  $A = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}$ , then

- (A)  $A^2 = 2A - I$  (B)  $A^2 = 3A - I$   
 (C)  $A^2 = 3A - I$  (D)  $A^{100} = \begin{bmatrix} 1 & 0 \\ -100 & 1 \end{bmatrix}$

63. Let  $D = \begin{vmatrix} a & a^2 & 0 \\ 1 & 2a+b & (a+b)^2 \\ 0 & 1 & 2a+3b \end{vmatrix}$  then

- (A)  $a+b$  is a factor of  $D$   
 (B)  $a+2b$  is a factor of  $D$   
 (C)  $2a+3b$  is a factor of  $D$   
 (D)  $a^2$  is a factor of  $D$

64. Let  $D(x) = \begin{vmatrix} 1 & x & x^2 \\ x^2 & 1 & x \\ x & x^2 & 1 \end{vmatrix}$ , then

- (A)  $1 - x^3$  is a factor of  $D(x)$   
 (B)  $(1 - x^3)^2$  is a factor of  $D(x)$   
 (C)  $D(x) = 0$  has 4 real roots  
 (D)  $D'(1) = 0$

65. The value of the

determinant  $\begin{vmatrix} 1+a & 1 & 1 \\ 1 & 1+a & 1 \\ 1 & 1 & 1+a \end{vmatrix}$  is zero, if

- (A)  $a = -3$  (B)  $a = 0$   
 (C)  $2a = 1$  (D)  $a = 3$

66. The value of  $\theta$  lying between  $\theta = 0$  and  $\theta = \frac{\pi}{2}$  and satisfying the equation

$$\begin{vmatrix} 1 + \sin^2 \theta & \cos^2 \theta & 4 \sin 4\theta \\ \sin^2 \theta & 1 + \cos^2 \theta & 4 \sin 4\theta \\ \sin^2 \theta & \cos^2 \theta & 1 + 4 \sin 4\theta \end{vmatrix} = 0, \text{ are}$$

- (A)  $\frac{7\pi}{24}$  (B)  $\frac{5\pi}{24}$   
 (C)  $\frac{11\pi}{24}$  (D)  $\frac{\pi}{24}$

67.  $\begin{vmatrix} bc & bc' + b'c & b'c' \\ ca & ca' + c'a & c'a' \\ ab & ab' + a'b & a'b' \end{vmatrix}$  must be equal to

- (A)  $(abc + a'b'c')^2$   
 (B)  $(abc)(a'b'c')$   
 (C)  $(bc' - b'c)(ca' - ac')(ab' - ba')$   
 (D) (A) and (B) are false.

68. Let  $A_n$  be an  $n \times n$  matrix in which diagonal elements are  $1, 2, 3, \dots, n$  (i.e.,  $a_{11} = 1, a_{22} = 2, \dots, a_{ii} = i, \dots, a_{nn} = n$ ) and all other elements are equal to  $n$

- (A)  $A_n$  is singular for all  $n$   
 (B)  $A_n$  is non singular for all  $n$   
 (C)  $\det A_n = (-1)^n n!$   
 (D)  $\det A_n = (-1)^{n+1} n!$

69. If  $A$  is a  $3 \times 3$  hermitian matrix with complex element while  $P$  and  $Q$  be two  $3 \times 3$  matrices with real elements if  $A = P + iQ$  then

- (A)  $P$  must be symmetric  
 (B)  $Q$  must be symmetric  
 (C)  $P$  must be skew symmetric  
 (D)  $Q$  must be skew symmetric

70. The values of  $\alpha$  for which the system of equation  $x + y + z = 1$ ,  $x + 2y + 4z = \alpha$ ,  $x + 4y + 10z = \alpha^2$  is consistent, are given by

- (A) 1, 2 (B) 1, -2  
 (C) -1, 2 (D) None



**Comprehension - 1**

$$\text{If } D = \begin{vmatrix} x^2 + x + 2 & 2x^2 + 3x + 1 \\ 20x^2 + 20x + 59 & 40x^2 + 60x + 20 \\ 2x^2 + 2x + 6 & 4x^2 + 6x + 2 \end{vmatrix}$$

$$= ax^2 + bx + c, \text{ then}$$

71. The value of a is

$$\begin{array}{ll} \text{(A)} \begin{vmatrix} 1 & 2 & 3 \\ 1 & 0 & 1 \\ 2 & 1 & 1 \end{vmatrix} & \text{(B)} \begin{vmatrix} 1 & 2 & 3 \\ 1 & 1 & 1 \\ 2 & 0 & 1 \end{vmatrix} \\ \text{(C)} \begin{vmatrix} 1 & 2 & 3 \\ 1 & 0 & 1 \\ 2 & 0 & 1 \end{vmatrix} & \text{(D)} \begin{vmatrix} 1 & 2 & 3 \\ 1 & 0 & 1 \\ 0 & 2 & 1 \end{vmatrix} \end{array}$$

72. The value b is

$$\begin{array}{ll} \text{(A)} \begin{vmatrix} 1 & 3 & 5 \\ 1 & 0 & 1 \\ 2 & 0 & 1 \end{vmatrix} & \text{(B)} \begin{vmatrix} 1 & 3 & 5 \\ 1 & 0 & 1 \\ 2 & 1 & 1 \end{vmatrix} \\ \text{(C)} \begin{vmatrix} 1 & 3 & 5 \\ 2 & 0 & 1 \\ 1 & 1 & 1 \end{vmatrix} & \text{(D)} \begin{vmatrix} 1 & 3 & 5 \\ 1 & 1 & 1 \\ 2 & 0 & 1 \end{vmatrix} \end{array}$$

73. The value of c is

$$\begin{array}{ll} \text{(A)} \begin{vmatrix} 2 & 1 & 3 \\ 1 & 0 & 1 \\ 2 & 1 & 1 \end{vmatrix} & \text{(B)} \begin{vmatrix} 2 & 1 & 3 \\ 1 & 0 & 1 \\ 2 & 0 & 1 \end{vmatrix} \\ \text{(C)} \begin{vmatrix} 2 & 1 & 1 \\ 3 & 2 & 0 \\ 1 & 1 & 1 \end{vmatrix} & \text{(D)} \begin{vmatrix} 2 & 1 & 3 \\ 1 & 0 & 1 \\ 2 & 1 & 0 \end{vmatrix} \end{array}$$

**Comprehension - 2**

Let  $A * B = A^T B^{-1}$  where  $A^T$  represents transpose of matrix A and  $B^{-1}$  represents inverse of square matrix B. This operation is defined when the number of rows of A is equal to the number of rows of B.

74. Which of the following operations are always conformable

- (A)  $(A * B) * A$       (B)  $(A * B) + A$   
 (C)  $(A * B) + B$       (D)  $(A * B) + A^T$

75. Choose the correct statement :

- (A)  $(A * B)^{-1} = B * A$  if A is symmetric matrix  
 (B)  $(A * B)^{-1} = B * A^{-1}$  if B is symmetric matrix  
 (C)  $(A * B)^T = (B^{-1}) * A$  if A is orthogonal matrix  
 (D) None of these

76. If X is a variable matrix in the equation  $X * A = B$  where A and B are invertible matrices then which of the following matrices is not necessarily a solution of the equation

- (A)  $A^T B^T$       (B)  $A * (B^{-1})^T$   
 (C)  $B^T A^T$       (D)  $A * (B^T)^{-1}$

**Comprehension - 3**

Consider an  $n \times n$  matrix with entries 1, 2, ...,  $n^2$  written in order starting top left and moving along each row in turn left to right. We choose n entries of the matrix such that exactly one entry is chosen in each row and each column. Let S be the sum of the selected entries.

77. The number of ways of selecting the n entries is

- (A)  $\frac{n}{2}$       (B)  $n^n$   
 (C)  $n^2$       (D) None of these

78. The correct statement is

- (A)  $S = \frac{n(n^2 + 1)}{2}$   
 (B)  $S = \frac{n(2n + 1)}{2}$   
 (C)  $S = \frac{n(n + 1)(2n + 1)}{4}$   
 (D) S is not unique

79. The maximum value of n such that  $S \leq 5n$  is

- (A) 3      (B) 4  
 (C) 5      (D) None

**Comprehension - 4**

Let  $A = [a_{ij}]_3$  be a square matrix of order 3 whose elements are distinct integers from 1, 2, ..., 9. The matrix is formed so that the sum of numbers in every row, column and diagonal is a multiple of 9.

80. The number of possible combinations of three distinct numbers from 1 to 9 that have a sum of 9 or 18 is  
 (A) 6 (B) 7  
 (C) 8 (D) 9
81. The element  $a_{22}$  must be a multiple of  
 (A) 2 (B) 3  
 (C) 4 (D) 9
82. The maximum value of trace of the matrix A is  
 (A) 18 (B) 19  
 (C) 12 (D) None

**Comprehension - 5**

Let  $A = [a_{ij}]$  and  $B = [b_{ij}]$  be two  $3 \times 3$  matrices with

$$a_{ij} = \begin{cases} 2 & \text{if } i = j \\ 1 & \text{if } i - j = 3k + 2, k \in I \\ 0 & \text{otherwise} \end{cases}$$

$$b_{ij} = \begin{cases} 1 & \text{if } i - j = 3m \pm 1, m \in I \\ 0 & \text{otherwise} \end{cases}$$

83. Which of the following does not hold true ?  
 (A)  $\det A = \det B + 3$   
 (B)  $\det A = 2 \det B$   
 (C)  $\det A = (\det B)^2$   
 (D) None of these
84. The incorrect statement is  
 (A)  $A = B^2$   
 (B)  $A B$  is commutative  
 (C)  $(AB)^{-1} = A^{-1} B^{-1}$   
 (D)  $(B + I)^3 = AB + 6A + I$
85. The matrix B satisfies the polynomial equation  
 (A)  $x^3 - 3x + 2 = 0$   
 (B)  $x^3 - 3x - 2 = 0$   
 (C)  $-x^3 - 3x + 2 = 0$   
 (D) None of these

**Assertion (A) and Reason (R)**

- (A) Both A and R are true and R is the correct explanation of A.  
 (B) Both A and R are true but R is not the correct explanation of A.  
 (C) A is true, R is false.  
 (D) A is false, R is true.

86. **Assertion (A)** : Let A and B be two square matrices of order 2 and assume that there exist three different real numbers  $t_1, t_2, t_3$  such that the matrices  $C_i = A + t_i B, i = 1, 2, 3$  are nilpotent i.e.  $(C_i)^2 = 0$ , both A and B are nilpotent.

**Reason (R)** :  $(A + tB)^2 = A^2 + t(AB + BA) + t^2 B^2$ : Comparing the  $(i - j)^{\text{th}}$  element, we get a polynomial equation of degree 2 which has 3 distinct roots. Hence all its coefficients vanish which implies that  $A^2 = 0, B^2 = 0$ .

87. **Assertion (A)** : For any  $m \in \mathbb{N}$  there exists a real  $m \times m$  matrix A such that  $A^3 = A + I$ , where I is the identity matrix.

**Reason (R)** : Since  $\lambda^3 - \lambda - 1 = 0$  has real solution, the matrix  $\lambda I$  satisfies the equation  $A^3 = A + I$ .

88. **Assertion (A)** : The area of a triangle with integral coordinates where sum of coordinates of each vertex is even, is always an integer.

**Reason (R)** :  $2\Delta = \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$  is even is such

a situation since the determinant is unaltered by  $C_1 \rightarrow C_1 + C_2$ .

89. **Assertion (A)** : If the three digit numbers abc,

pqr,  $\alpha\beta\gamma$  are all divisible by k, then  $\begin{vmatrix} a & c & b \\ p & r & q \\ \alpha & \gamma & \beta \end{vmatrix}$

is divisible by k.

**Reasons (R)** :  $C_i \rightarrow C_i + \lambda C_j$  does not change value of determinant.

**90. Assertion (A) :** The matrices  $A = [a_{ij}]$  where  $a_{ij}$

$$= \begin{cases} (-1)^{i+j} & \text{if } i \neq j \\ 2, & \text{if } i = j \end{cases} \text{ and } B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 4 \end{bmatrix} \text{ have}$$

the same rank.

**Reason (R):** Matrix A can be transformed into matrix B using elementary transformation and such matrices have the same rank.

**91. Assertion (A):** Let  $AX = 0$  be a matrix equation in four knowns where  $A = [a_{ij}]$ ,  $a_{ij} = i + j \forall i, j = 1, 2, 3, 4$ . Then X has non-trivial solution with one disposable unknown.

**Reason (R):** Rank  $A = 2$ , which is less than 4. Hence the equation has non-trivial solution.

**92. Assertion (A):** In a third order determinant  $|A|$  each term multiplied by 3, then the value of determinant so formed is  $27|A|$

**Reasons (R):** In an  $n^{\text{th}}$  order determinant  $|A|$  if each term multiplied by  $k$ , then the value of determinant so formed is equal to  $kn^2|A|$

**93. Assertion (A):** If  $r \in \mathbb{Q}$  and  $ad - bc = 1$ , then

there exists a matrix  $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \neq \pm I_2$  with integer

entries such that  $\frac{ar+b}{cr+d} = r$

**Reason (R):** If  $r \in \mathbb{Q}$ , we can choose an integer

$t$  such that  $r^2 t \in \mathbb{I}$  and set  $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$

$$= \begin{pmatrix} 1+rt & -r^2t \\ t & 1-rt \end{pmatrix}$$

**94. Assertion (A):** Let  $n \geq 2$  be an integer. The maximum possible rank of an  $n \times n$  matrix whose  $n^2$  entries are precisely the numbers  $1, 2, \dots, n^2$  is  $n$ .

**Reason (R):** We can form a non-singular matrix of order  $n$ . Put odd numbers into the diagonal, only even number above the diagonal and arrange the entries under the diagonal arbitrarily. Then the determinant of the matrix is odd.

**95. Assertion (A):** Let  $n \geq 2$  be an integer. Then the minimum rank of an  $n \times n$  matrix A where  $n^2$  entries are precisely the numbers  $1, 2, \dots, n^2$ , is 2.

**Reason (R):** The rank of the matrix

$$A = \begin{bmatrix} 1 & 2 & \dots & n \\ n+1 & n+2 & \dots & 2n \\ \dots & \dots & \dots & \dots \\ n^2 - n + 1 & n^2 - n + 2 & \dots & n^2 \end{bmatrix} \text{ is } 3.$$

### Match the Columns for JEE Advanced

| 96. | Column-I                                                                                                                                                                                                 | Column-II |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| (A) | The value of $n$ for which length of subnormal at any point of the curve $xy^n = 2$ is constant, is                                                                                                      | (P) 2     |
| (B) | Number of points of maxima for a continuous function, defined on an open interval, increasing in three different intervals and decreasing in two intervals (function is not constant in any interval) is | (Q) 3     |
| (C) | Maximum rank of a matrix having product of number of rows and number of columns 150                                                                                                                      | (R) -2    |
| (D) | Number of times the value of determinant gets multiplied on applying following transformations                                                                                                           | (S) 10    |

$$(i) R_1 \rightarrow 2R_1 - R_2 \quad (ii) R_2 \rightarrow -2R_3 + 3R_2 \quad (iii) R_3 \rightarrow \frac{R_3}{2}$$

## 97. Column-I

## Column-II

(A)  $M_r$  is defined as  $M_r = \begin{bmatrix} (r-1) & \frac{1}{r} \\ 1 & \frac{1}{(r-1)^2} \end{bmatrix}$

(P) 16

then  $\lim_{n \rightarrow \infty} (\det M_2 + \det M_3 + \dots + \det M_n)^{\log n} =$

(B) If  $\begin{vmatrix} 1 & \cos \alpha & \cos \beta \\ \cos \alpha & 1 & \cos \gamma \\ \cos \beta & \cos \gamma & 1 \end{vmatrix} = \begin{vmatrix} 0 & \cos \alpha & \cos \beta \\ \cos \alpha & 0 & \cos \gamma \\ \cos \beta & \cos \gamma & 0 \end{vmatrix}$

(Q) 4

then  $\sin^2 \alpha + \sin^2 \beta + \sin^2 \gamma =$

(C) If  $A = \begin{bmatrix} 3 & 1 \\ -1 & 1 \end{bmatrix}$  and a matrix C is defined as

(R) 2

$C = (BAB^{-1})(B^{-1}A^TB)$ , where  $|B| \neq 0$ , then  $\det C$  is a square of natural number equal to

(D) If  $A = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$  and  $A^4 = -\lambda I$  then  $\lambda$  equals

(S) 1

## 98. Column-I

## Column-II

(A) If  $f(x) = \begin{vmatrix} a & -1 & 0 \\ ax & a & -1 \\ ax^2 & ax & a \end{vmatrix}$ , then  $f(x) - f(-x)$  is a polynomial

(P) 4

of degree.

(B) The A.M. of two numbers exceeds the G.M. by  $3/2$  and the G.M. exceeds the H.M. by  $6/5$ , then the difference of numbers is

(Q) -2

(C) Area of quadrilateral formed by lines  $|x| + |y| = \sqrt{2}$  is

(R) 1

(D) Sum of the abscissa of points on the line  $x + y = 4$  which lie at a unit distance from the line  $4x + 3y = 11$

(S) 9

## 99. Column-I

## Column-II

(A) If  $ax + by - 5 = 0$  is the equation of the shortest chord of the circle  $(x-3)^2 + (y-4)^2 = 4$  [passing through the point (2, 3)], then  $|a + b|$  is

(P) 9

(B) Exponent of 12 in  $(50)!$  is

(Q) 1

(C) Let  $A = \begin{bmatrix} 3 & 2 & 1 \\ 1 & -1 & 0 \\ 1 & 2 & 0 \end{bmatrix}$ . If  $a_{ij}$  are the elements of  $A$  and  $C_{ij}$  is

(R) 2

cofactor of  $a_{ij}$ , for all  $i$  and  $j$ , then  $\sum_{1 \leq i, j \leq 3} a_{ij} C_{ij} =$

(D) If  $\alpha = 15^\circ$ , then value of  $\log_{\cot \alpha} (\tan \alpha + 2 \tan 2\alpha + 4 \tan 4\alpha + 8 \cot 8\alpha)$  is

(S) 22

## 100. Column-I

## Column-II

(A) If  $[1 \times 1] \begin{bmatrix} 1 & 3 & 2 \\ 2 & 5 & 1 \\ 15 & 3 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ x \end{bmatrix} = 0$ , then a value of  $1 - x$  is

(P) 3

(B) Let  $A = \begin{bmatrix} 3-x & 2 & 2 \\ 2 & 4-x & 1 \\ -2 & -4 & -1-x \end{bmatrix}$ , then a value of  $x$  for

(Q) 0

which  $A^{-1}$  does not exist

(C) Let  $A = \begin{bmatrix} 0 & 2y & z \\ x & y & -z \\ x & -y & z \end{bmatrix}$  and  $A^* A = I$ , then the value of

(R) 1

$x^2 + y^2 + z^2$  is

(D) If  $y = f(x)$  is continuous and differentiable function and three consecutive roots of  $f(x) = 0$  are  $\alpha, \beta, \gamma$ , ( $\alpha < \beta < \gamma$ ), then minimum number of roots of  $f'(x) = 0$  between  $\alpha$  and  $\gamma$  is

(S) 2

## Review Exercises for JEE Advanced

1. A manufacturer produces three products  $x, y, z$  which he sells in two markets. Annual sales are indicated below:

| Market | Products |              |
|--------|----------|--------------|
| I      | 10,000   | 2,000 18,000 |
| II     | 6,000    | 20,000 8,000 |

- (i) If unit sale prices of  $x, y$  and  $z$  are Rs 2.50, Rs 1.50 and Rs 1.00, respectively, find the total revenue in each market with the help of matrix algebra.

- (ii) If the unit costs of the above three commodities are Rs 2.00, Rs 1.00 and 50 paise respectively. Find the gross profit.

2. In each case find  $a, b, c, d$  to satisfy the given equation.

(a)  $\begin{bmatrix} 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} = \begin{bmatrix} 1 \\ 9 \\ 6 \\ 5 \end{bmatrix}$

$$(b) \begin{bmatrix} a & b & c & d \\ 1 & 4 & 9 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0 & 2 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 6 & 6 \\ 1 & 9 & 8 & 4 \end{bmatrix}$$

3. Prove that the equality  $AB - BA = I$  does not hold no matter what the matrices  $A$  and  $B$ .
4. A store has in stock 30 dozen shirts, 16 dozen trousers and 25 dozen pairs of socks. If the selling prices are Rs. 50 per shirt, Rs. 90 per trouser and Rs. 12 per pair of socks, then find the total amount the store owner get after selling all the items in the stock.
5. (a) Prove that a  $2 \times 2$  matrix  $A$  commutes with every  $2 \times 2$  matrix if and only if  $A$  commutes with each of the four matrices.

$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}.$$

- (b) Find all such matrices  $A$ .

$$6. \text{ Let } P = \begin{bmatrix} p^2 & 2pq & q^2 \\ p^2 & 2pq & q^2 \\ p^2 & 2pq & q^2 \end{bmatrix}, T = \begin{bmatrix} p & q & 0 \\ \frac{1}{2}p & \frac{1}{2} & \frac{1}{2}q \\ 0 & p & q \end{bmatrix}$$

$$\text{and } S = \frac{1}{4}I + \frac{1}{2}T + \frac{1}{4}P.$$

With  $p + q = 1$ , and with the vector  $x' = [1 \ 1 \ 1]$ , show that

$$(i) P = x[p^2 \ 2pq \ q^2] \quad (ii) Px = x$$

$$(iii) Tx = x \quad (iv) PT = P = TP$$

$$(v) T^2 = \frac{1}{2}(P + T) \quad (vi) P^2 = P$$

$$(vii) T^n = P + \left(\frac{1}{2}\right)^{n-1} (T - P)$$

$$(viii) ST = TS = T^2$$

$$(ix) S^2 = \frac{1}{16} (I + 6T + 9P)$$

7. (a) Show that the only real symmetric matrix whose square is null is the null matrix itself.

- (b) Explain why  $X'XGX'X = X'X$  implies  $XGX'X = X$ .

8. If  $B, C$  are  $n$ -rowed square matrices and if  $A = B + C, BC = CB, C^2 = O$ , then show that for every positive integer  $p$ ,  $A^{p+1} = B^p[B + (p+1)C]$ .

9. Find all matrices that commute with the matrix  $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$

10. Find all the numbers  $c$  which, when multiplied by the nonsingular matrix  $A$  of order 3, do not change the determinant of the matrix.

11. Let  $A$  be a second-order matrix and  $k$  an integer exceeding two. Prove that  $A^k = 0$  if and only if  $A^2 = 0$ .

12. Prove that

$$\begin{vmatrix} \alpha & \alpha^2 & \beta + \gamma \\ \beta & \beta^2 & \gamma + \alpha \\ \gamma & \gamma^2 & \alpha + \beta \end{vmatrix} = (\beta - \gamma)(\gamma - \alpha)(\alpha - \beta)(\alpha + \beta + \gamma)$$

13. Prove that  $M \begin{vmatrix} \sin^3 \alpha & \sin^3 \beta & \sin^3 \gamma \\ \sin \alpha & \sin \beta & \sin \gamma \\ \cos \alpha & \cos \beta & \cos \gamma \end{vmatrix}$   
 $= -\sin(\beta - \gamma) \sin(\gamma - \alpha) \sin(\alpha - \beta) \sin(\alpha + \beta + \gamma)$

14. Prove that if all elements of some row (or column) of a determinant are equal to unity, then the sum of the cofactors of all elements of the determinant is equal to the determinant itself.

15. Prove that a skew-symmetric determinant of even order remains unchanged if the same number is added to all elements.

16. Let  $S$  be symmetric and nonsingular. Prove that there exists a non-singular matrix  $C$  such that  $S^{-1} = C^T C$ .

17. Find the largest value of a third-order determinant provided that the elements are equal to  $\pm 1$ .

18. Find the value of  $\text{adj}(P^{-1})$  in terms of  $P$  where  $P$  is a non-singular matrix and hence show that  $\text{adj}(Q^{-1}BP^{-1}) = PAQ$ , given that  $\text{adj} B = A$  and  $|P| = |Q| = 1$ .

19. Determine the value of

$$\alpha, \beta, \gamma \in (0, \frac{\pi}{2}) \text{ when } \begin{bmatrix} 0 & 2\sin\beta & \tan\gamma \\ \cos\alpha & \sin\beta & -\tan\gamma \\ \cos\alpha & -\sin\beta & \tan\gamma \end{bmatrix} \text{ is orthogonal.}$$

20. Prove that, if  $U$  is nilpotent and if  $U^n = 0$  but  $U^{n-1} \neq 0$ , then  $I + U$  has an inverse, namely,  $(I + U)^{-1} = I - U + U^2 - U^3 + \dots + (-1)^{n-1} U^{n-1}$ .

21. Prove that, if  $U$  is nilpotent and the polynomial  $f(x)$  has no constant term, then  $f(U)$  is nilpotent.

22. An integral square matrix is said to be unimodular if its determinant is equal to  $\pm 1$ . Prove that an integral matrix has an integral inverse if and only if the given matrix is unimodular.

23. Prove that the sum of the squares of all second-order minors lying in two rows (or columns) of an orthogonal matrix is equal to unity.

24. Show that the sum and the product of two matrices of the type  $\begin{pmatrix} a & b \\ -b & a \end{pmatrix}$  are again matrices of the same type.

25. Show that, if  $P^{-1}AP = Q^{-1}BQ$ , then there exist matrices  $R, S$  such that  $A = RS, B = SR$ .

26. Show that the sum of two idempotent matrices  $A$  and  $B$  is idempotent if and only if  $AB = BA = O$ .

27. By reducing a matrix to echelon form, compute the ranks of the following matrices.

$$(a) \begin{pmatrix} -2 & 3 & 0 & -1 \\ -3 & -1 & 3 & 2 \\ -8 & 1 & 6 & 3 \end{pmatrix}$$

$$(b) \begin{pmatrix} 5 & 13 & 3 & 7 \\ 16 & 40 & 10 & 22 \\ 22 & 54 & 14 & 30 \end{pmatrix}$$

$$(c) \begin{pmatrix} -1 & -2 & 3 \\ -1 & 1 & -3 \\ 1 & 4 & 5 \\ 0 & 0 & 1 \end{pmatrix}$$

$$(d) \begin{pmatrix} 1 & 2 & 3 \\ 3 & 5 & 8 \\ 5 & 8 & 13 \\ 7 & 11 & 20 \end{pmatrix}$$

28. If in a determinant  $\Delta$ , when  $x = a$ :

(i)  $p$  rows become the same, then prove that  $(x - a)^{p-1}$  is a factor of  $\Delta$ ;

(ii)  $p$  rows become the same and  $q$  columns become the same, then prove that  $(x - a)^{p+q-2}$  is a factor, provided that the common elements are zero or are divisible by  $(x - a)^2$ .

29. With regard to the equations

$$4x + 7y - 14z = -24, 2x + 3y - 4z = -4, x + y + z = 6, \text{ show that the first can be put in the form}$$

$$\ell(2x + 3y - 4z + 4) + m(x + y + z - 6) = 0, \text{ and find } \ell : m. \text{ Hence show that the equations are not independent.}$$

30. Having given the equations

$$x = cy + bz, y = az + cx, z = bx + ay$$

where  $x, y, z$  are not all zero, prove that

$$a^2 + b^2 + c^2 + 2abc = 1$$

$$\text{and } x^2/(1 - a^2) = y^2/(1 - b^2) = z^2/(1 - c^2).$$

31. If  $a, b, c$  be the  $p$ th,  $q$ th and  $r$ th terms of an H.P. respectively, prove that the system of equations  
 $x + y + z = 0$   
 $px + qy + rz = 0$   
 $bcx + cay + abz = 0$   
 has infinite number of solutions.

32. A manufacturer produces chairs in 3 styles : 104, 110, and 118. Each chair requires time on 3 machines, as shown in the following chart. The total available time per week for each machine is also shown in the chart. All available time must be used each week. How many of each style chair can be made under these conditions each week ?

| Style                | Number Made Each Week | Hours on Machine A,B,C |
|----------------------|-----------------------|------------------------|
| 104                  | $x$                   | 1, 3, 2                |
| 110                  | $y$                   | 2, 2, 3                |
| 118                  | $z$                   | 1, 2, 1                |
| Total hours per week |                       | 30, 56, 44             |

33. Prove that if the rank of a homogeneous system of linear equations is less by unity than the number of unknowns, then any two solutions of the system are proportional, that is, differ by only numerical factor (which may be zero).
34. Solve the following system of equations over the field of rational numbers and find all integer solutions.  
 $x_1 + 7x_2 + 4x_3 = 0,$   
 $3x_1 + x_2 + 12x_3 = 0,$   
 $3x_1 + 7x_2 + 12x_3 = 0,$   
 $3x_1 + 12x_3 = 0.$
35. For what values of  $\lambda$  and  $\mu$  the following system of equations  
 $-4x + 2y - 9z = 2, \quad 3x + 4y + z = 5,$   
 $3x + 4y + \lambda z = \mu$   
 has (a) no solution, (b) a unique solution and (c) an infinite number of solutions.



### Target Exercises for JEE Advanced

1. Prove that, when  $s_r = \alpha^r + \beta^r + \gamma^r$ ,

$$\begin{vmatrix} 1 & 1 & 1 \\ \alpha & \beta & \gamma \\ \alpha^3 & \beta^3 & \gamma^3 \end{vmatrix} \times \begin{vmatrix} 1 & 1 & 1 \\ \alpha & \beta & \gamma \\ \alpha^2 & \beta^2 & \gamma^2 \end{vmatrix} = \begin{vmatrix} s_0 & s_1 & s_2 \\ s_1 & s_2 & s_3 \\ s_3 & s_4 & s_5 \end{vmatrix}$$

2. The matrix  $A = \begin{bmatrix} a & i \\ i & b \end{bmatrix}$ , where  $i^2 = -1$ ,  $a = \frac{1}{2}(1$

$$+ \sqrt{5}), \text{ and } b = \frac{1}{2}(1 - \sqrt{5}), \text{ has the property}$$

that  $A^2 = A$ . Describe completely all  $2 \times 2$  matrices  $A$  with complex entries such that  $A^2 = A$ .

3. The equation  $A^2 = I$  is satisfied by each of the  $2 \times 2$  matrices

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ c & -1 \end{bmatrix}, \begin{bmatrix} 1 & b \\ 0 & -1 \end{bmatrix},$$

where  $b$  and  $c$  are arbitrary real numbers. Find all  $2 \times 2$  matrices  $A$  such that  $A^2 = I$ .

4. A Seifert matrix is a square matrix  $V$  which satisfies  $V - V^T = F$  where  $F$  is a given matrix. For the following  $F$ 's find  $V$  (if it exists).

$$(i) F = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad (ii) F = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

$$(iii) F = O_{n \times n}$$

5. How will the product  $AB$  of matrices  $A$  and  $B$  change if we

- Interchange the  $i$ th and  $j$ th rows of  $A$ ?
- Add the  $j$ th row multiplied by the scalar  $c$  to the  $i$ th row of  $A$ ?
- Interchange the  $i$ th and  $j$ th columns of  $B$ ?
- Add to the  $i$ th column of  $B$  the  $j$ th column multiplied by the scalar  $c$ ?

6. (i) The  $i$ th and  $j$ th rows of the product are interchanged;  
 (ii) to the  $i$ th row of the product is added the  $j$ th row multiplied by  $c$ ;



- (iii) the  $i$ th and  $j$ th columns of the product are interchanged;  
 (iv) to the  $i$ th column of the product is added the  $j$ th column multiplied by  $c$ .

7. If  $\Delta$  denotes the determinant

$$\begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix}$$

and  $G, F, C$  are the co-factors of  $g, f, c$  in  $\Delta$ , prove that  
 $aG^2 + 2hFG + bF^2 + 2gGC + 2fFC + cC^2 = C\Delta$ .

8. Find the rank of the matrix

$$\begin{bmatrix} 0 & c & -b & \alpha \\ -c & 0 & \alpha & \beta \\ b & -a & 0 & \gamma \\ -\alpha & -\beta & -\gamma & 0 \end{bmatrix}$$

where  $a, b, c$  are all positive numbers and  $a\alpha + b\beta + c\gamma = 0$ .

9. Prove that  $\begin{vmatrix} \sin \alpha & \cos \alpha & \cos(\alpha + \delta) \\ \sin \beta & \cos \beta & \cos(\beta + \delta) \\ \sin \gamma & \cos \gamma & \cos(\gamma + \delta) \end{vmatrix} = 0$

10. Let  $\Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$  and  $A_i, B_i, C_i$

be the co-factors of  $a_i, b_i, c_i$  respectively and  $\alpha_i, \beta_i, \gamma_i$  be the co-factors of  $A_i, B_i, C_i$  respectively, where  $i = 1, 2, 3$ , show that

$$\begin{vmatrix} A_1 & B_1 & C_1 \\ A_2 & B_2 & C_2 \\ A_3 & B_3 & C_3 \end{vmatrix} \times \begin{vmatrix} \alpha_1 & \beta_1 & \gamma_1 \\ \alpha_2 & \beta_2 & \gamma_2 \\ \alpha_3 & \beta_3 & \gamma_3 \end{vmatrix} = \Delta^6$$

11. If  $\Delta(x) = \begin{vmatrix} \alpha + x & \theta + x & \lambda + x \\ \beta + x & \phi + x & \mu + x \\ \gamma + x & \psi + x & v + x \end{vmatrix}$

Show that  $\Delta''(x) = 0$  and  $\Delta(x) = \Delta(0) + Sx$ , where  $S$  denotes the sum of cofactors of all elements in  $\Delta(0)$  and dash denotes the derivative.

12. Prove that if  $B = P^{-1}AP$ , then  $B^k = P^{-1}A^kP$ . Hence, if  $f(A) = a_p A^p + a_{p-1} A^{p-1} + \dots + a_1 A + a_0 I = O$ , and  $B = P^{-1}AP$ , then prove that  $f(B) = O$ .

13. Prove that, if  $U$  is nilpotent and  $(I + U)^p = I$ , then  
 $U(I + f(U)) = O$ , where  $f(U)$  is a polynomial with constant term zero.

14.  $A$  is a given square matrix. Writing  $A_t = (tI - A)^{-1}$  whenever  $tI - A$  is non-singular, prove that  
 $(t - u)A_t A_u = A_u - A_t$ .

15. Let  $f(t), g(t)$  be polynomials whose highest common factor is 1; let  $M$  be an  $n \times n$  matrix; and write  $A = f(M), B = g(M)$ . Show that every solution of the equation  $ABx = 0$  can be written in the form  $x = y + z$ , where  $By = Az = O$ .

16. Prove that the rectangular cartesian coordinates  $(X_1, X_2, X_3)$  of the orthogonal projection of  $(\bar{X}_1, \bar{X}_2, \bar{X}_3)$  of the orthogonal projection of  $(x_1, x_2, x_3)$  on a line through the origin having directions cosines  $(\ell_1, \ell_2, \ell_3)$  are given by

$$X = \begin{bmatrix} \ell_1^2 & \ell_1 \ell_2 & \ell_1 \ell_3 \\ \ell_1 \ell_2 & \ell_2^2 & \ell_2 \ell_3 \\ \ell_1 \ell_3 & \ell_2 \ell_3 & \ell_3^2 \end{bmatrix} x,$$

where  $X, x$  are single column matrices.

17. Show that, if  $P_1$  is a solution of  $AX = B_1$  and  $P_2$  is a solution of  $AX = B_2$  then  $\lambda_1 P_1 + \lambda_2 P_2$  is a solution of  $AX = \lambda_1 B_1 + \lambda_2 B_2$ . Then show how one may use this fact to write the complete solution of this last equation.

18. Prove that, if

$$A_j = \begin{bmatrix} a_j & d & g \\ b_j & e & h \\ c_j & f & k \end{bmatrix}, j = 1, 2, \dots, p,$$

$$\text{then } \det \left( \sum_{j=1}^p A_j \right) = p^2 \sum_{j=1}^p (\det A_j).$$

19. Show that if the determinant  $D$  is equal to zero then all the rows (and also the columns) of the adjoint determinant are proportional.

20. Find the values of  $\lambda$  for which the matrix

$$\begin{pmatrix} 3 & 1 & 1 & 4 \\ \lambda & 4 & 10 & 1 \\ 1 & 7 & 17 & 3 \\ 2 & 2 & 4 & 3 \end{pmatrix} \text{ has the lowest rank.}$$

What is the rank for the values of  $\lambda$  thus found, and what is it for other values of  $\lambda$ ?

21. If  $A(v) = \left(1 - \frac{v^2}{c^2}\right)^{-1/2} \begin{bmatrix} 1 & -v \\ -v/c^2 & 1 \end{bmatrix}$

where  $-c < v < c$ ,  $c$  being a positive constant, show that  $A(v_1) \cdot A(v_2) = A(v_3)$  where

$$v_3 = \frac{v_1 + v_2}{1 + \frac{v_1 v_2}{c^2}}. \text{ Deduce that } [A(v)]^{-1} = A(-v).$$

22. (i) If a square matrix  $L$  is such that  $LL'$  is nonsingular, prove that  $L$  is nonsingular.  
 (ii) For symmetric  $V$  prove that  $K = X'(XV^{-1}X')^{-1}X$  is symmetric and that the transpose of  $D = KV^{-1}$  is  $V^{-1}DV$ .  
 (iii) Prove  $(A + UB)V^{-1} = A^{-1} - A^{-1}UBV(I + A^{-1}UBV)^{-1}A^{-1}$ .

23. Prove that two quadratic equations  $a_0x^2 + a_1x + a_2 = 0$ ,  $a_0 \neq 0$ ,  $b_0x^2 + b_1x + b_2 = 0$ ,  $b_0 \neq 0$ , have a common root if and only if the determinant

$$\begin{vmatrix} a_0 & a_1 & a_2 & 0 \\ 0 & a_0 & a_1 & a_2 \\ b_0 & b_1 & b_2 & 0 \\ 0 & b_0 & b_1 & b_2 \end{vmatrix}$$

vanishes. Use this fact to show that, for any values of  $\alpha$  and  $\beta$ , the equations  $\alpha x^2 + x + (1 - \alpha) = 0$ ,  $(1 - \beta)x^2 + x + \beta = 0$ , will have a common root.

24. Show that, if  $A = [a_{ij}]_n$  and if  $A_{ij}$  is the cofactor of  $a_{ij}$  in  $\det A$ , then

$$\det[a_{ij} + x]_n = \det A + x \sum_{i,j=1}^n A_{ij}.$$

25. Prove that, if  $C_{ji}$  denotes a cofactor of the adjoint matrix  $C$  of  $A$ , then  $C_{ji} = a_{ij} \cdot (\det A)^{n-2}$ .

26. Given that  $D$  is diagonal and nonsingular and that  $D = (I + A)^{-1} \cdot A$ , Prove that  $A$  is diagonal also.

27. Prove that, if  $S_m = I + A + A^2 + \dots + A^m$ , and if  $I - A$  is non-singular, then

$$S_m = (I - A^{m+1})(I - A)^{-1}. \text{ See if you can}$$

define  $\lim_{m \rightarrow \infty} A^{m+1}$  and then show that, when

this limit is the zero matrix,

$$\lim_{m \rightarrow \infty} S_m = (I - A)^{-1}.$$

28. How will the inverse matrix  $A^{-1}$  change if in the given matrix  $A$  we

- (i) Interchange the  $i$ th and  $j$ th rows ?  
 (ii) Multiply the  $i$ th row by a nonzero scalar  $c$ ?  
 (iii) Add to the  $i$ th row the  $j$ th row multiplied by a scalar  $c$ , or do the same with respect to the columns ?

29. Find the value of  $a, b, c, d$  in the matrix

$$H = \begin{bmatrix} a & b & c \\ d & e & -4 \\ 5 & -6 & 7 \end{bmatrix} \text{ so that } GH = HG \text{ where}$$

$$G = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

and then find the value of  $e$  so that the rank of the product  $GH$  is 2.

30. Prove that

$$\begin{vmatrix} \ell_2 - t & m_1 & n_1 \\ \ell_2 & m_2 - t & n_2 \\ \ell_3 & m_3 & n_3 - t \end{vmatrix} \text{ is equal to}$$

$$-t^3 + t^2 (\ell_1 + m_2 + n_3) - t(L_1 + M_2 + N_3) + (\ell_1 m_2 n_3),$$

where  $L_1, M_2, N_3$  are the cofactors of  $\ell_1, m_2, n_2$  in  $(\ell_1 m_2 n_3)$ .

31. For each system state conditions (if any) on a, b, and c which force either no solution or an infinite number of solutions for the system.

(i)  $2x - 3y = 0$

$x + 4y = b$

(ii)  $x + y = a$

$2x + 2y = b$

(iii)  $x - y + z = 3$

$x + y - z = a$

$2x - y + 2z = b$

(iv)  $x - 2y + z = a$

$-x + y + 3z = b$

$-2x + y + 10z = c$

32. Solve

$$x_1 + 2x_2 + 3x_3 = 0$$

$$4x_1 + 5x_2 + 6x_3 = 0$$

$$7x_1 + 8x_2 + 9x_3 = 0$$

$$10x_1 + 11x_2 + 12x_3 = 0$$

33. State conditions on a, b, c, d, e, f, g, and h which will guarantee a unique solution for

$$ax_1 + bx_2 = r \quad a \neq 0$$

$$cx_1 + dx_2 = s \quad e \neq 0$$

$$ex_3 + fx_4 = t$$

$$gx_3 + hx_4 = u$$

34. Let  $x = r, y = s$  be a solution of

$$ax + by = 0$$

$$cx + dy = 0 \text{ and}$$

$$x = u, y = v \text{ be a solution of}$$

$$ax + by = m$$

$$cx + dy = n$$

Show that  $x = u + r, y = v + s$  is also a solution of the second set of equations.

35. Find the general solution of the system

$$x + y - 3z + u = 5$$

$$2x - y + z - 2u = 2$$

$$7x + y - 7z + 3u = 3$$

36. Prove that the system  $x + y + 2z = 2, 2x - y + 3z = 2, 5x - y + az = 6$ , has a unique solution if  $a \neq 8$ . Find all solutions when  $a = 8$ .

### Previous Years' Questions (JEE Advanced)

#### A. Multiple Choice Questions with ONE correct answer

1. If A and B are square matrices of equal degree, then which one is correct among the following?  
[IIT - 1995]

(A)  $A + B = B + A$  (B)  $A + B = A - B$

(C)  $A - B = B - A$  (D)  $AB = BA$

2. If  $A = \begin{bmatrix} \alpha & 0 \\ 1 & 1 \end{bmatrix}$  and  $B = \begin{bmatrix} 1 & 0 \\ 5 & 1 \end{bmatrix}$ , then value of  $\alpha$  for which  $A^2 = B$ , is -  
[IIT - 2003]

(A) 1 (B) -1

(C) 4 (D) no real values

3. If the system of equations  $x + ay = 0, az + y = 0$ , and  $ax + z = 0$  has infinite solutions, then the value of a is -  
[IIT - 2003]

(A) -1 (B) 1

(C) 0 (D) no real values

4. Given

$$2x - y + 2z = 2, x - 2y + z = -4, x + y + \lambda z = 4$$

then the value of  $\lambda$  such that the given system of equation has no solution, is- [IIT - 2004]

(A) 3 (B) 1

(C) 0 (D) -3

5. If  $A = \begin{bmatrix} \alpha & 2 \\ 2 & \alpha \end{bmatrix}$  and  $|A^3| = 125$  then the value of  $\alpha$  is  
[IIT - 2004]

(A)  $\pm 1$  (B)  $\pm 2$

(C)  $\pm 3$  (D)  $\pm 5$

6.  $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & -2 & 4 \end{bmatrix}$  and  $I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$  and

$$A^{-1} = \left[ \frac{1}{6}(A^2 + cA + dI) \right], \text{ then the value of } c$$

and d are

(A)  $(-6, -11)$  (B)  $(6, 11)$

(C)  $(-6, 11)$  (D)  $(6, -11)$

7. If  $P = \begin{bmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}$  and  $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$  and  $Q = PAP^T$

and  $x = P^T Q^{2005} P$  then  $x$  is equal to

[IIT - 2005]

(A)  $\begin{bmatrix} 1 & 2005 \\ 0 & 1 \end{bmatrix}$

(B)  $\begin{bmatrix} 4 + 2005\sqrt{3} & 6015 \\ 2005 & 4 - 2005\sqrt{3} \end{bmatrix}$

(C)  $\frac{1}{4} \begin{bmatrix} 2 + \sqrt{3} & 1 \\ -1 & 2 - \sqrt{3} \end{bmatrix}$

(D)  $\frac{1}{4} \begin{bmatrix} 2005 & 2 - \sqrt{3} \\ 2 + \sqrt{3} & 2005 \end{bmatrix}$

8. The number of  $3 \times 3$  matrices  $A$  whose entries are either 0 or 1 and for which the system  $A$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \text{ has exactly two distinct solutions, is}$$

[IIT - 2010]

- (A) 0 (B)  $2^9 - 1$   
(C) 168 (D) 2

9. If matrix  $A = \begin{bmatrix} a & b & c \\ b & c & a \\ c & a & b \end{bmatrix}$  where  $a, b, c$  are real

positive numbers,  $abc = 1$  and  $A^T A = I$ , then find the value of  $a^3 + b^3 + c^3$ . [IIT - 2003]

10. Let  $\omega \neq 1$  be a cube root of unity and  $S$  be the set of all non-singular matrices of the form

$$\begin{bmatrix} 1 & a & b \\ \omega & 1 & c \\ \omega & \omega & 1 \end{bmatrix},$$

where each of  $a, b$  and  $c$  is either  $\omega$  or  $\omega^2$ . Then the number of distinct matrices in the set  $S$  is

- (A) 2 (B) 6  
(C) 4 (D) 8

[IIT - 2011]

11. Let  $P = [a_{ij}]$  be a  $3 \times 3$  matrix and let  $Q = [b_{ij}]$ , where  $b_{ij} = 2^{i+j} a_{ij}$  for  $1 \leq i, j \leq 3$ . If the determinant of  $P$  is 2, then the determinant of the matrix  $Q$  is

- (A)  $2^{10}$  (B)  $2^{11}$   
(C)  $2^{12}$  (D)  $2^{13}$

[IIT - 2012]

### B. Subjective Questions

12. If  $P$  is a  $3 \times 3$  matrix such that  $P^T = 2P + I$ , where  $P^T$  is the transpose of  $P$  and  $I$  is the  $3 \times 3$  identity matrix, then there exists a column matrix

$$X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \neq \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \text{ such that}$$

(A)  $PX = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$  (B)  $PX = X$

(C)  $PX = 2X$  (D)  $PX = -X$

[IIT - 2012]

13. If the adjoint of a  $3 \times 3$  matrix  $P$  is  $\begin{bmatrix} 1 & 4 & 4 \\ 2 & 1 & 7 \\ 1 & 1 & 3 \end{bmatrix}$ ,

then the possible value(s) of the determinant of  $P$  is(are)

- (A) -2 (B) -1  
(C) 1 (D) 2

[IIT - 2012]

14. If  $M$  is  $3 \times 3$  matrix, where  $\det M = 1$  and  $MM^T = I$ , where ' $I$ ' is an identity matrix, prove that  $\det(M - I) = 0$ . [IIT - 2004]

15. If  $A = \begin{bmatrix} a & 0 & 1 \\ 1 & c & b \\ 1 & d & b \end{bmatrix}$ ,  $B = \begin{bmatrix} a & 1 & 1 \\ 0 & d & c \\ f & g & h \end{bmatrix}$ ,  $U = \begin{bmatrix} f \\ g \\ h \end{bmatrix}$

$$V = \begin{bmatrix} a^2 \\ 0 \\ 0 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

and  $AX = U$  has infinite solutions, prove that  $BX = V$ , can not have a unique solution. If  $a \neq 0$  then prove that  $Bx = V$  has no solution.

[IIT - 2004]

### C. Comprehension - 1

Let  $A = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{bmatrix}$ , and  $U_1, U_2$  and  $U_3$  are columns

of a  $3 \times 3$  matrix  $U$ . If column matrices  $U_1, U_2$  and  $U_3$

satisfying  $AU_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ ,  $AU_2 = \begin{bmatrix} 2 \\ 3 \\ 0 \end{bmatrix}$ ,  $AU_3 = \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix}$  evaluate

as directed in the following questions.

16. The value  $|U|$  is [IIT - 2006]

(A) 3 (B) -3

(C)  $\frac{3}{2}$  (D) 2

17. The sum of the elements of the matrix  $U^{-1}$  is [IIT - 2006]

(A) -1 (B) 0

(C) 1 (D) 3

18. The value of  $[3 \ 2 \ 0] U \begin{bmatrix} 3 \\ 2 \\ 0 \end{bmatrix}$  is [IIT - 2006]

(A) 5 (B)  $5/2$

(C) 4 (D)  $3/2$

### Comprehension - 2

Let  $A$  be the set of all  $3 \times 3$  symmetric matrices all of whose entries are either 0 to 1. Five of these entries are 1 and four of them are 0. [IIT - 2009]

19. The number of matrices in  $A$  is:

(A) 12 (B) 6

(C) 9 (D) 3

20. The number of matrices  $A$  in  $A$  for which the

system of linear  $A \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$  equations  $A$  has

a unique solution, is:

(A) Less than 4

(B) At least 4 but less than 7

(C) At least 7 but less than 10

(D) At least 10

21. The number of matrices  $A$  in  $A$  for which the

system of linear equation  $A \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$  is

inconsistent is:

(A) 0

(B) More than 2

(C) 2

(D) 1

### Comprehension - 3

Let  $p$  be an odd prime number and  $T_p$  be the following set of  $2 \times 2$  matrices: [IIT - 2010]

$$T_p = \left\{ A = \begin{bmatrix} a & b \\ c & a \end{bmatrix} : a, b, c \in \{0, 1, 2, \dots, p-1\} \right\}$$

22. The number of  $A$  in  $T_p$  such that  $A$  is either symmetric or skew-symmetric or both, and  $\det(A)$

(A) divisible by  $p$  is

(A)  $(p-1)^2$  (B)  $2(p-1)$

(C)  $(p-1)^2 + 1$  (D)  $2p-1$

23. The number of  $A$  in  $T_p$  such that the trace of  $A$  is not divisible by  $p$  but  $\det(A)$  is divisible by  $p$  is [Note : The trace of matrix is the sum of its diagonal entries.]

(A)  $(p-1)(p^2-p+1)$

(B)  $p^3 - (p-1)^2$

(C)  $(p-1)^2$  (D)  $(p-1)(p-1)^2$

24. The number of  $A$  in  $T_p$  such that  $\det(A)$  is not divisible by  $p$  is

(A)  $2p^2$

(B)  $p^3 - 5p$

(C)  $p^3 - 3p$

(D)  $p^3 - p^2$

**D. Assertion (A) and Reason (R)**

25. Consider the system of equations [IIT - 2008]

$$x - 2y + 3z = -1$$

$$-x + y - 2z = k$$

$$x - 3y + 4z = 1.$$

**Assertion (A):** The system of equation has no solution for  $k \neq 3$ .

**Reason (R):** The determinant  $\begin{vmatrix} 1 & 3 & -1 \\ -1 & -2 & k \\ 1 & 4 & 1 \end{vmatrix} \neq 0$ ,

for  $k \neq 3$

(A) Both A and R are true and R is the correct explanation of A.

(B) Both A and R are true but R is not the correct explanation of A.

(C) A is true, R is false.

(D) A is false, R is true.

**E. Integer Answer Type**

26. Let  $k$  be a positive real number and let

$$A = \begin{bmatrix} 2k-1 & 2\sqrt{k} & 2\sqrt{k} \\ 2\sqrt{k} & 1 & -2k \\ -2\sqrt{k} & 2k & -1 \end{bmatrix} \text{ and}$$

$$B = \begin{bmatrix} 0 & 2k-1 & \sqrt{k} \\ 1-2k & 0 & 2\sqrt{k} \\ -\sqrt{k} & -2\sqrt{k} & 0 \end{bmatrix}. \text{ If } \det(\text{adj}A) + \det$$

$(\text{adj} B) = 10^6$ , then  $[k]$  is equal to

(Note :  $\text{adj} M$  denotes the adjoint of a square matrix  $M$  and  $[k]$  denotes the largest integer less than or equal to  $k$ ). [IIT - 2009]

27. Let  $M$  be a  $3 \times 3$  matrix satisfying

$$M \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \\ 3 \end{bmatrix}, M \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}, \text{ and}$$

$$M \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 12 \end{bmatrix}.$$

Then the sum of the diagonal entries of  $M$  is [IIT - 2011]

**ANSWERS****Practice Problems—A**

- (i)  $a_{12}$  (ii)  $a_{32}$   
(iii)  $a_{21}$  (iv)  $a_{41}$
- $1 \times 18, 2 \times 9, 3 \times 6, 6 \times 3, 9 \times 2, 18 \times 1$ ;  
 $1 \times 5, 5 \times 1$
- $a = 1, b = 2, c = 3, d = 4$
- $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$  5.  $\begin{pmatrix} 5 & 8 & 0 \\ 2 & 5 & 2 \end{pmatrix}$
- (i)  $X = \begin{bmatrix} 5 & 0 \\ 1 & 4 \end{bmatrix}, Y = \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix}$   
(ii)  $X = \begin{bmatrix} 2/5 & -12/5 \\ -11/5 & 3 \end{bmatrix}, Y = \begin{bmatrix} 2/5 & 13/5 \\ 14/5 & -2 \end{bmatrix}$
- $x = 2, y = 4, z = 1, w = 3$

**Practice Problems—B**

- (i)  $\begin{bmatrix} a^2 + b^2 & 0 \\ 0 & a^2 + b^2 \end{bmatrix}$   
(ii)  $\begin{bmatrix} 2 & 3 & 4 \\ 4 & 6 & 8 \\ 6 & 9 & 12 \end{bmatrix}$   
(iii)  $\begin{bmatrix} -3 & -4 & 1 \\ 8 & 13 & 9 \end{bmatrix}$  (iv)  $\begin{bmatrix} 14 & 0 & 42 \\ 18 & -1 & 53 \\ 22 & -2 & 66 \end{bmatrix}$

$$2. B + C = \begin{bmatrix} 3 & 4 \\ 0 & 2 \\ 6 & -5 \end{bmatrix}, AB = \begin{bmatrix} 15 & -14 \\ -15 & 14 \end{bmatrix},$$

$$BA = \begin{bmatrix} -1 & 4 & -2 \\ -4 & 16 & -8 \\ 7 & -28 & 14 \end{bmatrix},$$

$$AC = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}, CA = \begin{bmatrix} 0 & 0 & 0 \\ 2 & -8 & 4 \\ 4 & -16 & 8 \end{bmatrix},$$

$$A(2B - 3C) = \begin{bmatrix} 30 & -28 \\ -30 & 28 \end{bmatrix}$$

$$11. x = \pm 4\sqrt{3} \quad 12. X = \begin{bmatrix} 1 & -2 \\ 2 & 0 \end{bmatrix}$$

$$14. (a) \begin{bmatrix} a & b \\ 0 & 0 \end{bmatrix}, a \text{ and } b \text{ arbitrary}$$

$$(b) \begin{bmatrix} -2a & a \\ -2b & b \end{bmatrix}, a \text{ and } b \text{ arbitrary}$$

### Practice Problems—C

$$1. A^n = O. \quad 2. A^n = \begin{bmatrix} 1 & n & \frac{n(n+1)}{2} \\ 0 & 1 & n \\ 0 & 0 & 1 \end{bmatrix}$$

$$3. \begin{pmatrix} \cos n\theta + \sin n\theta & \sqrt{2} \sin n\theta \\ -\sqrt{2} \sin n\theta & \cos n\theta - \sin n\theta \end{pmatrix}$$

6. Equality holds when  $AB = BA$ .

$$10. f(A) = \begin{pmatrix} 21 & -23 & 15 \\ -13 & 4 & 10 \\ -9 & 22 & 25 \end{pmatrix} \quad 11. k = 1$$

$$14. \begin{bmatrix} 1 & 0 \\ -100 & 1 \end{bmatrix}$$

$$15. (i) I_2, 4I_2$$

$$(ii) -3/2 I_2, -\sqrt{2} I_2, \sqrt{2} I_2$$

$$(iii) I_n, \omega I_n, \omega^2 I_n$$

### Practice Problems—D

$$3. (A + B)' = A' + B'$$

$$6. (i) \begin{bmatrix} 3 & 3 \\ 3 & -1 \end{bmatrix} + \begin{bmatrix} 0 & 2 \\ -2 & 0 \end{bmatrix}$$

$$(ii) \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$9. (a) AB = BA$$

### Practice Problems—E

$$7. 2^n$$

$$9. \begin{bmatrix} a & b \\ c & -a \end{bmatrix}, \text{ where } b \text{ and } c \text{ are arbitrary, and } a$$

is any solution of the equation  $a^2 = -bc$ .

### Practice Problems—F

$$2. \frac{1}{3} \begin{pmatrix} 1 & 2 \\ -3 & 6 \end{pmatrix} \quad 4. \begin{bmatrix} 0 & 0 & 0 & -5 \\ 0 & 0 & -4 & 0 \\ 0 & -7 & 0 & 0 \\ -3 & 0 & 0 & 0 \end{bmatrix}$$

### Practice Problems—G

$$1. \lambda^3 - 4\lambda^2 - 13\lambda - 40 = 0; A^{-1}$$

$$= \frac{1}{40} \begin{bmatrix} -4 & 11 & -5 \\ -4 & 1 & 25 \\ 8 & -2 & -10 \end{bmatrix}$$

$$2. A^{-1} = \frac{1}{7} \begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix}$$

$$3. A^{-1} = \frac{1}{11} \begin{bmatrix} -3 & 4 & 5 \\ 9 & -1 & -4 \\ 5 & -3 & -1 \end{bmatrix}$$

4. (i)  $\begin{bmatrix} 2 & 0 & 3 \\ -3 & -1 & -1 \\ 4 & -3 & -2 \end{bmatrix};$

(ii)  $\begin{bmatrix} -7/4 & 5/4 & 5/2 \\ -1/2 & 5/2 & 2 \\ 15/4 & -5/4 & -9/2 \end{bmatrix}$

5. (i)  $\begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix}$

(ii) Inverse does not exist

(iii)  $\begin{bmatrix} -2/5 & 0 & 3/5 \\ -1/5 & 1/5 & 0 \\ 2/5 & 1/5 & -2/5 \end{bmatrix}$

6.  $\begin{bmatrix} -191 & -110 \\ 77 & 44 \end{bmatrix}$  7.  $P = \begin{bmatrix} 2 & 1 \\ 5 & -1 \end{bmatrix}$

### Practice Problems—H

2.  $\begin{pmatrix} \cos \phi & -\sin \phi & 0 \\ \cos \phi \sin \psi & \cos \phi \cos \psi & -\sin \psi \\ \sin \phi \sin \psi & \sin \phi \cos \psi & \cos \phi \end{pmatrix}$

6.  $\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}, \begin{bmatrix} \cos \theta & -\sin \theta \\ -\sin \theta & -\cos \theta \end{bmatrix}$

8. Identity matrix

### Practice Problems—I

1.  $2 \pm \sqrt{3}$

2. (i) 6, 1/2 (ii) 1, -1/3

5. (i) yes (ii) no

6. rank = 2 for  $x = 5$ ; rank = 3 for  $x = 1$ ;  
otherwise rank = 4.

### Practice Problems—J

1. (i) Nonlinear (ii) Nonlinear  
(iii) Linear (iv) Nonlinear

2.  $B = 400C + 0.35T + 120H$

3. no solution.

4. Infinitely many solutions  $\left(x, \frac{4x-7}{3}\right),$

where  $x \in \mathbb{R}.$

### Practice Problems—K

1. (i)  $x = \frac{16}{11} + \frac{7i}{11}, y = \frac{24}{11} - \frac{28i}{11}$

(ii)  $(x, y, z) = \left(-\frac{5}{3}, -\frac{8}{3}, -\frac{2}{3}\right)$

2. If  $\alpha \neq -1, \alpha \neq 2$ , then  $x = \frac{3-\alpha}{\alpha+1}, y = \frac{2}{\alpha+1}.$

3.  $y = \frac{5}{4}x^2 - \frac{1}{4}x - \frac{1}{2}$

4. (2, 4, 2)

5.  $k = 10, x_1 = 2, x_2 = 1$

6. Nontrivial solution for  $k = -1, -2, 6$ ;  
for  $k = -1$ , the solution is  $(-5\lambda, 2\lambda, 2\lambda)$

### Practice Problems—L

1.  $C = \begin{bmatrix} \frac{15}{2} & \frac{13}{2} \\ 8 & 7 \end{bmatrix}, D = \begin{bmatrix} \frac{33}{4} & \frac{19}{4} \\ \frac{43}{4} & \frac{25}{4} \end{bmatrix}$

2.  $(7/2, -8, 13/2)$

3. (i)  $X = \begin{pmatrix} -7 & -1 & 5 \\ 15 & 2 & -3 \end{pmatrix}$

(ii)  $X = \begin{pmatrix} -6 & 26 \\ 0 & 1 \\ 1 & -4 \end{pmatrix}$

(iii)  $X = \begin{pmatrix} -40 & -50 & 43 \\ 29 & 36 & -31 \end{pmatrix}$

### Practice Problems—M

1. (i)  $x_1 = 1, x_2 = -1, x_3 = 4$ ;

(ii)  $x_1 = (14 + 2k)/5, x_2 = (1 + 3k)/5, x_3 = k,$   
 $k$  arbitrary;

(iii) No solution; (iv)  $x_1 = 11/3, x_2 = -1/3;$

(v)  $x_1 = (30 + 25k)/8, x_2 = (11k - 6)/16,$   
 $x_3 = k, k$  arbitrary; (vi) No solution

2.  $(-1, 2, 5, 1)$

3.  $x = 2, y = 3, z = 5.$



4. Unique solution when  $a \neq 1$  and  $a \neq -2$ ;  
infinitely many solutions for  $a = 1$ ; no solution  
for  $a = -2$
5.  $\lambda = 2$
6. (i)  $X = k(1, 2, 1, 0) + m(-1, 1, 0, 1)$   
(ii)  $X = k(0, -1, 1)$
7.  $(x, y, z) = (1, -1, 0) + t(-3, 4, 1)$

**Objective Exercises**

- |          |          |          |
|----------|----------|----------|
| 1. B     | 2. D     | 3. A     |
| 4. B     | 5. C     | 6. B     |
| 7. C     | 8. B     | 9. B     |
| 10. C    | 11. D    | 12. D    |
| 13. D    | 14. C    | 15. A    |
| 16. A    | 17. B    | 18. D    |
| 19. C    | 20. D    | 21. A    |
| 22. A    | 23. A    | 24. D    |
| 25. A    | 26. A    | 27. C    |
| 28. C    | 29. A    | 30. A    |
| 31. D    | 32. D    | 33. D    |
| 34. A    | 35. B    | 36. ABCD |
| 37. ABC  | 38. ABCD | 39. ABCD |
| 40. ABC  | 41. AB   | 42. CD   |
| 43. AB   | 44. AC   | 45. AB   |
| 46. AB   | 47. ABC  | 48. AB   |
| 49. AB   | 50. AD   | 51. ABCD |
| 52. ABD  | 53. AB   | 54. ABCD |
| 55. BC   | 56. ABC  | 57. BD   |
| 58. ABCD | 59. AB   | 60. AC   |
| 61. AB   | 62. ABCD | 63. AB   |
| 64. ABD  | 65. AB   | 66. AC   |
| 67. CD   | 68. BD   | 69. AC   |
| 70. AB   | 71. C    | 72. A    |
| 73. B    | 74. D    | 75. D    |
| 76. C    | 77. A    | 78. A    |
| 79. C    | 80. D    | 81. B    |
| 82. A    | 83. A    | 84. D    |
| 85. B    | 86. A    | 87. A    |
| 88. A    | 89. A    | 90. A    |

91. D                      92. C                      93. A
94. A                      95. C
96. (A) – (R), (B) – (P), (C) – (S), (D) – (Q)
97. (A) – (S), (B) – (R), (C) – (Q), (D) – (Q)
98. (A) – (R), (B) – (S), (C) – (P), (D) – (Q)
99. (A) – (R), (B) – (S), (C) – (P), (D) – (Q)
100. (A) – (P), (B) – (PQ), (C) – (R), (D) – (S)

**Review Exercises for JEE Advanced**

1. (i) Rs. 46000 and Rs. 53000.  
(ii) Rs 15000, Rs. 17000
2. (a)  $a = 9, b = 6, c = 1, d = 5$   
(b)  $a = 1, b = 6, c = 0, d = -2$
4. Rs. 31,800
5. (b)  $\begin{bmatrix} a & 0 \\ 0 & a \end{bmatrix}$ , where  $a$  is arbitrary.
9.  $\begin{pmatrix} a & 2b \\ 3b & a+3b \end{pmatrix}$  where  $a$  and  $b$  are arbitrary numbers.
10.  $c = 1, \omega, \omega^2$ , where  $\omega$  is an imaginary cube root of unity.
17. 2                      19.  $\frac{\pi}{4}, \sin^{-1} \frac{1}{\sqrt{6}}, \frac{\pi}{6}$
27. (a) 2, (b) 2, (c) 3, (d) 3
29.  $-3/2$
32. The manufacturer can produce 8 style 104 chairs, 6 style 110 chairs, and 10 style 118 chairs each week.
34.  $(-4\lambda, 0, \lambda)$  for  $\lambda \in \mathbb{Q}$ , for integral solution  $\lambda \in \mathbb{I}$ .
35. (a)  $\lambda = 1, \mu \neq 5$  (b)  $\lambda \neq 1$  (c)  $\lambda = 1, \mu = 5$

**Target Exercises for JEE Advanced**

2.  $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ , and  $\begin{bmatrix} a & b \\ c & 1-a \end{bmatrix}$ , where  $b$  and  $c$  are arbitrary and  $a$  is any solution of the quadratic equation  $a^2 - a + bc = 0$
3.  $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$ , and  $\begin{bmatrix} a & b \\ c & -a \end{bmatrix}$ , where  $b$  and

$c$  are arbitrary and  $a$  is any solution of the equation  $a^2 = 1 - bc$ .

4. (i) No solution (ii)  $\begin{pmatrix} a & c-1 \\ c & d \end{pmatrix}$

(iii) any symmetric matrix.

8. 2

20. For  $\lambda = 0$ , the rank is 2; for  $\lambda \neq 0$ , it is 3.

28. In the matrix  $A^{-1}$ : (i) the  $i$ th and  $j$ th columns will be interchanged; (ii) the  $i$ th column will be multiplied by  $1/c$ ; (iii) the  $i$ th column multiplied by  $c$  will be subtracted from the  $j$ th column multiplied by  $c$ .

29.  $a = 7$ ,  $b = -6$ ,  $c = 5$ ,  $d = -4$ ,  $e = 4$

31. (i) Unique solution for any  $a$  and  $b$ ;  
 (ii) If  $-2a + b \neq 0$  then no solution ;  
       if  $-2a + b = 0$  then infinitely many solutions;  
 (iii) Unique solution for any  $a$  and  $b$ ;  
 (iv) If  $-a - 3b + c \neq 0$  then no solution;  
       if  $-a - 3b + c = 0$  then infinitely many solutions.

32.  $(k, -2k, k)$  where  $k$  is arbitrary.

33.  $ad - bc \neq 0$  and  $eh - fg \neq 0$

35.  $(x, y, z, u) = (1, 8, 0, -4) + t(2, 7, 3, 0)$

36.  $(x, y, z) = \left(\frac{4}{3}, \frac{2}{3}, 0\right) + t(5, 1, -3)$

### Previous Years' Questions

#### (JEE Advanced)

- |          |       |       |
|----------|-------|-------|
| 1. A     | 2. D  | 3. A  |
| 4. D     | 5. C  | 6. C  |
| 7. A     | 8. A  | 9. D  |
| 10. A    | 11. D | 12. D |
| 13. A, D | 16. A | 17. B |
| 18. A    | 19. A | 20. B |
| 21. B    | 22. D | 13. C |
| 24. D    | 25. A | 26. 4 |
| 27. 9    |       |       |

# 4

# Probability

## 4.1 INTRODUCTION

We often hear statements of the following kind : “It is likely to rain today,” “I have a fair chance of passing this course,” “There is an even chance that a coin will come up heads,” etc. In each case our statement refers to a situation in which we are not certain of the outcome, but we express some degree of confidence that our prediction will be verified. The theory of probability provides a mathematical framework for such assertions.

### The Concept of a Random Experiment

There are various phenomena in nature which lead to outcomes which cannot be predicted in advance. For example in tossing a coin, it cannot be predicted whether a head or tail will appear in a given toss. The theory of probability aims at measuring the uncertainties of such outcomes.

**Probability** is the study of mathematical models of random phenomena. It studies models of experiments with random outcomes. Only experiments that can be repeated (reproduced) an arbitrary number of times (atleast theoretically) under an invariable set of conditions are considered.

A fair coin is tossed. We assume that the coin does not stand on its edge. On any toss, it is known that either head or tail will appear but it is not known exactly what the outcome will be. Obviously, the coin may be tossed any number of times.

A fair die is rolled. It is known that any of the six possible outcomes will occur but it is not known exactly what the outcome will be. The die may be rolled any number of times.

A **random experiment** is one in which all the outcomes of the experiment are known in advance but the exact outcome of any specific performance is not known in advance. The object of observation in a random experiment may be a certain process, physical phenomenon or operational system. Any observed result is interpreted as a random outcome of the experiment. The event may or may not occur as a result of the experiment. In the mathematical formalization of the model of a random experiment the concept to start with is that of the set of elementary outcomes (events) designated as  $S$  connected with the experiment. By this is meant a set of mutually exclusive outcomes, such that the experiment always results in one and only one outcome.

## Sample Space

Consider an experiment whose outcome is not predictable with certainty. However, although the outcome of the experiment will not be known in advance, let us suppose that the set of all possible outcomes is known. This set of all possible outcomes of an experiment is known as the sample space of the experiment and is denoted by  $S$ . Some examples are as follows:

- (i) If the experiment consists of flipping two coins, then the sample space consists of the following four points:

$$S = \{(H, H), (H, T), (T, H), (T, T)\}$$

The outcome will be (H, H) if both coins are heads, (H, T) if the first coin is heads and the second tails,

(T, H) if the first is tails and the second heads and (T, T) if both coins are tails.

- (ii) If the experiment consists of tossing two dice, then the sample space consists of the 36 points  $S = \{(i, j) : i, j = 1, 2, 3, 4, 5, 6\}$  where the outcome (i, j) is said to occur if i appears on the leftmost dice and j on the other dice.

- (iii) If the experiment consists of measuring (in hours) the lifetime of a transistor, then the sample space consists of all nonnegative real numbers. That is,

$$S = \{x : 0 \leq x < \infty\}$$

Given any set  $S$  of objects of any kind, the size of  $S$  is defined as the number of members of  $S$ . Sets are said to be of finite size if their size is one of the finite numbers  $\{1, 2, 3, \dots\}$ . Examples of sets of finite size are the following : the set of all the continents in the world, which has size 7 ; the set of all the planets in the universe, which has size 9 ; the set  $\{1, 2, 3, 5, 7, 11, 13\}$  of all prime numbers from 1 to 15, which has size 7 ; the set  $\{(1, 4), (2, 3), (3, 2), (4, 1)\}$  of 2-tuples of whole numbers between 1 and 6 whose sum is 5, which has size 4.

However, there are also sets of infinite size. Examples are the set of all prime numbers  $\{1, 2, 3, 5, 7, 11, 13, 17, \dots\}$  and the set of all points on the real line between the numbers 0 and 1, called the interval between 0 and 1.

If a set  $S$  has as many members as there are integers 1, 2, 3, 4, ..... (by which is meant that a one-to-one correspondence may be set up between the members of  $S$  and the members of the set  $\{1, 2, 3, \dots\}$  of all integers) then  $S$  is said to be **countably infinite**. The set of even integers  $\{2, 4, 6, 8, \dots\}$  contains a countable infinity of members, as does the set of odd integers  $\{1, 3, 5, \dots\}$  and the set of primes. A set that is neither finite nor countably infinite is said to be **noncountably infinite**. An interval on the real line, say the interval between 0 and 1, contains a noncountable infinity of members.

The set  $S$  for a given experiment may be either **discrete or continuous**. Among the former are finite or countable sets of elementary outcomes, among the latter are sets of the continuum type (any finite or infinite interval on the number line is an example of a continuum type set).

The sample space  $S_1 = \{1, 2, \dots, 100\}$  is finite and contains 100 elements. This set is discrete. The sample space  $S_2$  of points within or on a circle of radius

$$R > 0 : S_2 = \{x^2 + y^2 \leq R^2\}$$

is infinite and uncountable (its elements cannot be enumerated). This set is continuous.

## Sample Point

Each element of the sample space is called a sample point.

When a die is thrown the sample space is  $S = \{1, 2, 3, 4, 5, 6\}$ . Here 1, 2, 3, 4, 5, and 6 are the sample points. A **sample** is something that has been observed. A sample description is the name of something that is observable.

## Trial

When an experiment is repeated under similar conditions and it does not give the same result each time but may result in any one of the several possible outcomes, the experiment is called a trial and the outcomes are called cases. The number of times the experiment is repeated is called the number of trials.

## For example :

- (i) One toss of a coin is a trial when the coin is tossed 3 times.
- (ii) One throw of a die is a trial when the die is thrown 4 times.

## Event

Any subset  $E$  of the sample space is known as an event. That is an event is a set consisting of possible outcomes of the experiment. If the outcome of the experiment is contained in  $E$ , then we say that  $E$  has occurred.

Some examples of events are as follows :

In example (i), if  $E = \{H, H\}, (H, T)\}$ , then  $E$  is the event that a head appears on the first coin.

In example (ii), if  $E = \{(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)\}$ , then  $E$  is the event that the sum of the dice equals 7.

In example (iii), if  $E = \{x : 0 \leq x \leq 4\}$ , then  $E$  is the event that the transistor does not last longer than 4 hours.

## Occurrence of an Event

For a random experiment, let  $E$  be an event.

Let  $E = \{a, b, c\}$ . If the outcome of the experiment is  $a$  or  $b$  or  $c$  then we say that event  $E$  has occurred.

When a die is thrown, let  $E = \{1, 3, 5\}$

If the face 1 or 3 or 5 turns up, we say that event  $E$  has occurred.

An event  $E$  is called **elementary event** if it consists of only one element, i.e. when  $E$  is singleton. An event which is not elementary is called **compound event**. The event coinciding with the empty set  $\phi$  is called an **impossible event**, and the event coinciding with the entire set  $S$  is called a certain or **sure event**. The collection of all observed events makes up the field of events for the experiment. An example of a certain event : {a score not exceeding 6 when a die is tossed} : an example of an impossible event : {a score of 7 on a face of a die}.

Two events  $A$  and  $B$  are said to be **joint or compatible** (disjoint or incompatible) if the experiment can (cannot) result in their joint occurrence. In other words, the events  $A$  and  $B$  are compatible if the corresponding sets  $A$  and  $B$  have elements in common and incompatible otherwise.

A mathematical formalization of a random experiment model comprises : (a) constructing the set of elementary outcomes  $S$ , (b) describing the field

of events of the experiment, and (c) assigning the probability distribution on the field of events.

The construction of  $S$  (if it is not given in the description of the experiment) is carried out in practice, proceeding from the requirement that it should be possible for all the results of interest in the experiment to be uniquely described on the basis of the constructed set  $S$ . In other words. If we are interested in events  $A, B, C$ , etc., which are the observed events in the experiment, then the set  $S$  must consist of such outcomes that there should be subsets of  $S$  equivalent to the events  $A, B, C$ , etc.

Since the concept of an "elementary outcome" is not strictly defined, the above problem allows a non-unique solution and depends on the set of events we are concerned with. If we require that the unique description of an elementary outcome becomes more definite. That is, in the collection of all subsets of  $Q$  making up the field of events, elementary outcomes are one-element subsets.

For instance, consider  $E$  : Toss a coin twice:

$$S = \{(T, T), (T, H), (H, T), (H, H)\}$$

The sample space of a random experiment need not be unique. The particular sample space considered is a matter of convenience and expediency. For example, the sample space for the experiment  $E$  may well be represented by  $S' = \{0, 1, 2\}$ , where 0 means no head, 1 means one head and 2 means two heads.

► **Example 1.** An experiment consists in throwing a balanced hexahedral die once. Let  $X$  be the number of points turning up on the upper face of the die. Describe the set of elementary outcomes  $S$  and show the composition of the subsets corresponding to the following events:

$$A = \{X \text{ is a multiple of three}\},$$

$$B = \{X \text{ is odd}\}, C = \{X > 3\}.$$

$$D = \{X < 7\}, E = \{X \text{ is fractional}\},$$

$$F = \{0.5 < X < 1.5\}. \text{ Find pairs of joint events.}$$

► **Solution** For the events observed in the experiment we let :  $\omega_k = \{X = k\}$ ,  $k = 1, 2, 3, \dots, 6$  :

$$\omega^{(1)} = \{X \text{ is an odd number}\},$$

$$\text{and } \omega^{(2)} = \{X \text{ is an event.... number}\}.$$

On the basis of these outcomes we can construct two certain events  $S_1 = \{\omega_1, \omega_2, \dots, \omega_6\}$  and  $S_2 = \{\omega^{(1)}, \omega^{(2)}\}$ . Which of them is more suitable for the role of the set of elementary outcomes?

Clearly  $S_2$  should be rejected since the observed events  $\omega_1, \omega_2, \dots, \omega_6, A, B, D, E$ , for example, are not subsets of the set  $S_2$ . On the other hand, all the enumerated events can be described as subsets of  $S_1$ . Indeed,  $A = \{\omega_2, \omega_6\}$ ,  $B = \omega^{(1)} = \{\omega_1, \omega_2, \omega_5\}$ ,

$$C = \{\omega_4, \omega_5, \omega_6\}, D = \{\omega_1, \omega_2, \dots, \omega_6\} = S_1, \\ E = \Phi, B = \omega_1.$$

From these equations we see, in particular, that the outcomes  $\omega^{(1)}$  and  $\omega^{(2)}$  are decomposable into elements that are themselves outcomes of the experiment.

Thus the outcomes  $\omega_1, \omega_2, \dots, \omega_6$  are "more elementary" than the outcomes  $\omega^{(1)}$  and  $\omega^{(2)}$ .

Comparing the events pairwise and checking for the presence of common elements we find pairs of joint events: A and B, A and C, A and D, B and C, B and D, B and F, C and D, D and F.

► **Example 2.** A flying target is detected by a radar. The observed result is the position of the light spot (of the pulse reflected from the target) on the screen of the target indicator, shaped as a circle of 10 cm radius, in rectangular coordinates with origin at the screen centre.

Describe the set of elementary outcomes and the composition of the subsets corresponding to the following events:  $A = \{\text{the target is in the first quadrant}\}$ ,  $B = \{\text{the target is in the circle of radius 5 cm whose centre is at the screen centre}\}$ ,  $C = \{\text{the target is in the circle of radius 2.5 cm whose centre is shifted 5 cm along the x-axis in the negative direction}\}$ . Are the pairs of events A and B, A and C, B and C joint?

► **Solution** All the observed events of interest in the experiment involve recording the position of the light spot on the indicator screen. A convenient form of mathematical description of an elementary outcome is, in this corresponds, say, to the centre of the spot (assumed to be a circle of small enough radius). Although it is obviously impossible for a point (as an abstract mathematical notion) to be physically observed on the indicator screen, this idealized way

of describing an elementary outcome simplifies the mathematical formalization of this experiment.

Thus the set S is continuous and may be written as  $S = \{(x, y) \mid x^2 + y^2 \leq 100\}$ .

The subsets equivalent to the indicated events are of the form

$$A = \{(x, y) \mid x^2 + y^2 \leq 100, x > 0, y > 0\}, \\ B = \{(x, y) \mid x^2 + y^2 \leq 25\}, \\ C = \{(x, y) \mid (x + 5)^2 + y^2 \leq 6.25\}.$$

By definition, events are joint if the corresponding subsets have elements in common (intersect), otherwise they are disjoint. Therefore events A and B, B and C are joint, and events A and C are disjoint.

► **Example 3.** A football match is played between Tigers and Royals. The events of interest are:

$$A = \{\text{Tigers wins}\}, \\ B = \{\text{one of the team wins}\}, \\ C = \{\text{Royals win 3 to 1}\}, \\ D = \{\text{atleast 3 goals are scored in the game}\}.$$

Describe the set of elementary outcomes and show the composition of the subsets corresponding to the indicated events.

► **Solution** We write the set S as follows:

$$S = \{(x, y) \mid x \in Z_0, y \in Z_0\},$$

where x is the number of goals scored by Tigers, y is the number of goals scored by Royals, and  $Z_0$  is the set of non-negative integers.

We are forced to consider the ranges of x and y to be unlimited, atleast theoretically, since there are no grounds for setting any explicit precise limit on the possible score before the game. Practically, no indefinitely large score is feasible in any game.

The subsets corresponding to the events we are concerned with have the form:

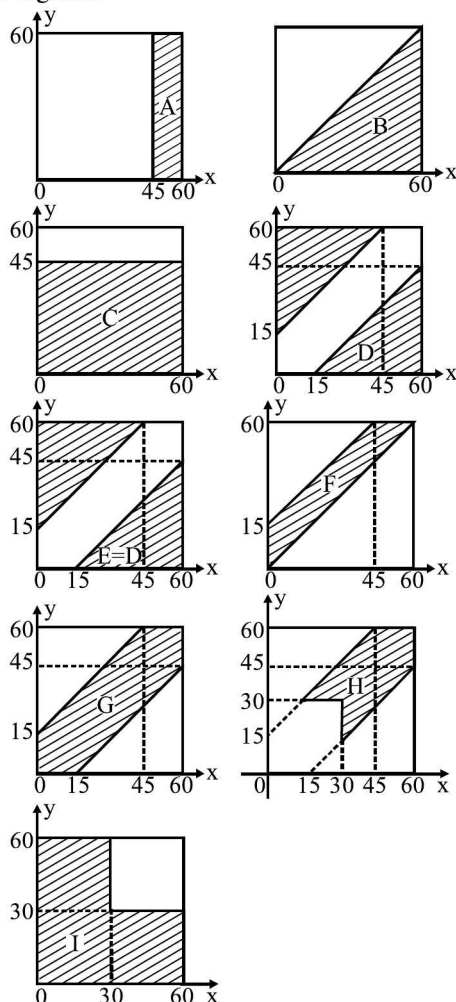
$$A = \{(x, y) \mid x \in Z_0, y \in Z_0, x > y\}, \\ B = \{(x, y) \mid x \in Z_0, y \in Z_0, x \neq y\}, \\ C = \{(x, y) \mid x = 1, y = 3\} = \{(1, 3)\}, \\ D = \{(x, y) \mid x \in Z_0, x + y \geq 3\}.$$

► **Example 4.** Mona and Piya agree to meet at a certain place between eleven and twelve. Each comes at a random moment of this time interval and waits for the other to appear until the hour elapses but no longer than 15 minutes, after which he leaves. The result to be observed is a pair of numbers (x, y), where x is the time of Piya's arrival and y is the time of

Mona's arrival (the time is calculated in minutes starting from eleven o'clock). The events are :

- A = {Piya comes after 11.45 a.m.},
- B = {Piya comes after Mona},
- C = {Mona comes before 11.45 a.m.},
- D = {the meeting does not take place},
- E = {Piya waits for Mona all the agreed time but fails to meet him},
- F = {Mona meets Piya without having to wait},
- G = {the meeting takes place},
- H = {the meeting takes place after 11.30 a.m.},
- I = {the first to comes before 11.30 a.m.}.

► **Solution**  $S = \{(x, y) \mid 0 \leq x, y \leq 60\}$ . The sets corresponding to the indicated events are represented in the figures.



## Statistical Definition of Probability

The question of what probability value should be assigned to events in real experiments is dealt with by methods of mathematical statistics. Let experiment S be repeated  $n$  times, with event E occurring  $m$  times. The ratio  $m/n$  is called the relative frequency of event E and denoted by  $P^*(E)$ . Before a series of experiments is carried out the frequency  $P^*(E)$  is a random number which cannot be predicted precisely for any finite number  $n$  of trial. However, the regularities inherent in random phenomena are such that in practice as the number of duplicate tests increases the relative frequency tends to become increasingly less random and get stabilized about some value of  $P(E)$  which is constant (and non-random) for a given event E in a given experiment S. Let a random experiment be repeated  $n$  times under similar conditions i.e. let  $n$  trials of a random experiment be made, where  $n$  is very large and an event E occurs  $r$  times in  $n$  trials, Then

$$P(E) = \lim_{n \rightarrow \infty} \frac{r}{n}. \text{ Sometimes, } P(E) \text{ is also denoted by } p.$$

If  $p$  is the probability that an event happens in a single trial, then if the number of trials is indefinitely increased, it becomes a certainty that the limit of the ratio of the number of successes to the number of trials is equal to  $p$ ; in other words, if the number of trials is  $N$ , the number of successes may be taken to be  $pN$ . On the other hand, when the probability of occurrence of event E in which we are interested is known, it is possible to predict with a certain accuracy the relative frequency of the given event if a large enough number of real tests is carried out. Such a problem of scientific forecasting becomes especially urgent when estimating the probability from frequency involves costly and time-consuming experiments. Problems solved in probability theory are just those of determining from given probabilities of some simple events the probabilities of compound events of interest.

**Practice Problems****[A]**

1. A die is thrown twice. The result observed is a pair of numbers corresponding to the numbers of points turning up on the first and the second throw. The events are :  $A = \{\text{both times the number of points is a multiple of three}\}$ ,  $B = \{\text{no six has turned up either time}\}$ ,  $C = \{\text{both times the number of points is greater than three}\}$ ,  $D = \{\text{the same number of points has turned up both times}\}$ .
2. A coin is tossed three times. The result observed is the occurrence of heads (h) or tails (t) on the upward side of the coin. The events are :  $A = \{\text{exactly one heads turned up}\}$ ,  $B = \{\text{no tails turned up}\}$ ,  $C = \{\text{more heads than tails turned up}\}$ ,  $D = \{\text{heads turned up atleast twice in succession}\}$ .
3. A coin is tossed until the first occurrence of a head. The result to be observed is the total number of tosses. The events are :  $A = \{\text{heads turns up on the third throw}\}$ ,  $B = \{\text{heads turns up not earlier than on the third throw}\}$ .
4. A flat rectangular target  $(-2 \leq x \leq 2, -1 \leq y \leq 1)$  is shot at. The result to be observed is the co-ordinates of the point of strike in the Cartesian plane. By the rules of the shooting there are no misses. The events are :  $A = \{\text{the abscissa of the point of strike is no less than its ordinate}\}$ ,  $B = \{\text{the product of the coordinates of the point is non-negative}\}$ ,  $C = \{\text{the sum of the absolute values of the coordinates of the point is greater than unity}\}$ .

**4.2 ALGEBRA OF EVENTS**

The theory of probability is based on indirect rather than direct methods of calculating probabilities when the probability of an event is expressed in terms of the probabilities of other related events. It is necessary, first of all, to know how to express an event in terms of other events using the so-called algebra of events. We now introduce the concepts of "a sum

of events" and "a product of events" and operations on them. Note that these concepts are only introduced for events which are subsets of the same space of elementary events  $S$ .

Since in our set-theoretical representation events are sets, the actions performed on them (addition and multiplication) are defined as those corresponding to set operations.

Nevertheless, we shall repeat certain definitions and give them a geometric interpretation. We shall depict the space of elementary events as a rectangle and the events as parts of the rectangle.

As soon as we define 'events' as subsets of the sample space we can transfer the language of set theory to the world of 'events'. In proving some statement in the algebra of events it is very useful to illustrate events geometrically, treating every event as a point falling or placed in a region corresponding to that event. Agreeing to represent events as different regions in the plane we obtain the so called Venn diagrams.

Venn diagram is a graphical representation that is very useful for illustrating logical relations among events. The sample space  $S$  is represented as consisting of all the outcomes in a large rectangle, and the events  $E, F, G, \dots$  are represented as consisting of all the outcomes in given circles within the rectangle. Events of interest can then be indicated by shading appropriate regions of the diagram.

Since events are identified with sets, we can perform all operations on events that can be performed on sets. In particular, the following operations and relations between events are defined :

**1. Subevent**

An event  $A$  is said to be a subevent of an event  $B$ , if the occurrence of  $A$  necessarily implies the occurrence of  $B$ . In other words, events  $B$  occurs whenever event  $A$  does.

For any two events  $A$  and  $B$ , if all of the outcomes in  $A$  are also in  $B$ , then we say that  $A$  is contained in  $B$  and write  $A \subset B$  (or equivalently,  $B \supset A$ ).

**2. Equal Events**

If  $A \subset B$  and  $B \subset A$ , we say that  $A$  and  $B$  are equal and write  $A = B$ .



$A = B$  (equivalence of sets) means event  $A$  is identical with event  $B$ .

### 3. Intersection of Events

For any two events  $A$  and  $B$  we may also define the new event  $AB$ , called the intersection of  $A$  and  $B$ , to consist of all outcomes that are both in  $A$  and in  $B$ . That is, the event  $AB$  (sometimes written  $A \cap B$ ) will occur only if both  $A$  and  $B$  occur.  $AB$  is a product of events. See figure 1.

For instance, in the toss of two coins if

$$A = \{(H, H), (H, T), (T, H)\}$$

is the event that atleast 1 head occurs, and

$$B = \{(H, T), (T, H), (T, T)\}$$

is the event that atleast 1 tail occurs, then

$$AB = \{(H, T), (T, H)\}$$

is the event that exactly 1 head and 1 tail appear.

In the rolling of two dice, if  $A = \{(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)\}$  is the event that the sum of the dice is 7 and  $B = \{(1, 5), (2, 4), (3, 3), (4, 2), (5, 1)\}$  is the event that the sum is 6, then event  $AB$  does not contain any outcomes and hence could not occur to give such an event a name, we shall refer to it as the null event and denote it by  $\emptyset$  (that is,  $\emptyset$  refers to the event consisting of no outcomes). If  $AB = \emptyset$ , then  $A$  and  $B$  are said to be mutually exclusive.

**Compound Event :** If  $A$  and  $B$  are two given events then  $A \cap B$  is called a compound event and is denoted by  $A \cap B$  or  $AB$ .

**Mutually Exclusive Events :** Two events are said to be mutually exclusive (or disjoint or incompatible) if the occurrence of one precludes (rules out) the simultaneous occurrence of the other. Thus events  $A$  and  $B$  are disjoint if  $AB = \phi$ .

### 4. Union of Events

For any two events  $A$  and  $B$  of a sample space  $S$ , we define the new event  $A \cup B$  to consist of all outcomes that are either in  $A$  or in  $B$  or in both  $A$  and  $B$ . The event  $A \cup B$  is called the union of the event  $A$  and the event  $B$ . That is, the event  $A \cup B$  (sometimes written  $A + B$ ) will occur if atleast one of the two events  $A$  or  $B$  occurs.  $A \cup B$  is a sum of events. See figure 2.

In the toss of two coins, if  $A = \{(H, H), (H, T)\}$  and  $B = \{(T, H)\}$ , then

$$A \cup B = \{(H, H), (H, T), (T, H)\}$$

Thus  $A \cup B$  would occur if a head appeared on either coin.

### 5. Complement of an event

$\bar{A} = S - A$  (complement of the set  $A$  to  $S$ ) is an opposite or contrary event. This is an event implying that  $A$  does not occur. The set of all out comes which are in  $S$  but not in  $A$  is called the complement of the event  $A$ .

If  $A$  is the event,  $\{2, 3, 4\}$  in the rolling of a die then not- $A$  means the nonoccurrence of 2, 3, 4 that is, the occurrence of  $\{1, 5, 6\}$ . So  $A$  and not- $A$  (which is written as  $A^c$  or  $\bar{A}$  or  $A'$ ) are complements of each other.  $A^c$  occurs if  $A$  does not occur.

### 6. Difference of Events

$A - B$  (a set of elements that are in  $A$  but not in  $B$ ) is a difference between events. This is an event implying that  $A$  occurs and  $B$  does not. Clearly

$$A - B = A \cap B' \quad \text{and} \quad B - A = B \cap A'$$

$$P(A - B) = P(A) - P(A \cap B)$$

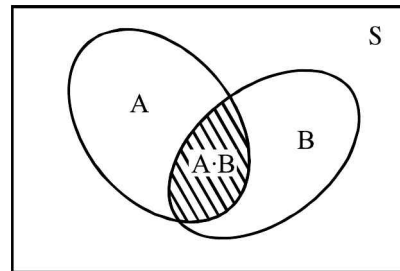


Fig. 1

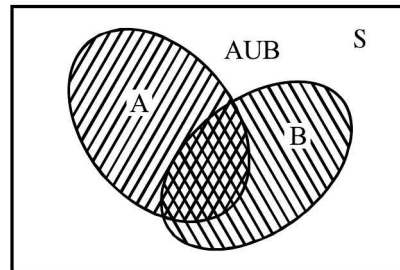


Fig. 2

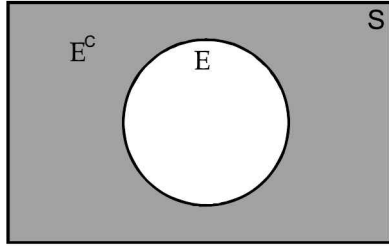


Fig. 3

**Equally Likely Events :** Events are said to be equally likely when each event is as likely to occur as any other event.

**Exhaustive Events :** Events A, B, C, ..., G are said to be exhaustive events if no event outside this set can result as an outcome of an experiment. For example, if A and B are two events defined on a sample space S, then A and B are exhaustive when  $A \cup B = S$ .

For example, in the experiment of tossing a die, the events  $A = \{1, 2, \dots\}$ ,  $B = \{2, 3, 4\}$  and  $C = \{4, 5, 6\}$  form an exhaustive group of events.

$$A \cup B \cup C = \{1, 2, 3, 4, 5, 6\} = S.$$

The operations of forming unions, intersections, and complements of events obey certain rules similar to the rules of algebra. The rules of the event algebra make it possible to combine various simple events and form, in that way, other more complex, events.

We list a few of these rules which follow from the definitions of sum and product of events:

$$A \cup S = S, \quad A S = A,$$

$$A \cup \Phi = A, \quad A \Phi = \Phi.$$

$$\text{If } A \subseteq B, \text{ then } A \cup B = B, \quad A B = A.$$

$$(\overline{\overline{A}}) = A, \quad \overline{S} = \Phi, \quad \overline{\Phi} = S.$$

### Commutative laws

$$A \cup B = B \cup A \quad AB = BA$$

### Associative laws

$$(A \cup B) \cup C = A \cup (B \cup C) \quad (AB)C = A(BC)$$

### Distributive laws

$$(A \cup B)C = AC \cup BC \quad AB \cup C = (A \cup C)(B \cup C)$$

### Idempotency law

$$A \cup A = A \quad AA = A$$

### De Morgan's Law

If A & B are two subsets of the sample space S, then

$$(a) (A \cup B)^c = A^c \cap B^c \quad \text{and}$$

$$(b) (A \cap B)^c = A^c \cup B^c \quad \text{and for } n \text{ events,}$$

$$A_1, A_2, \dots, A_n$$

$$(c) (A_1 \cup A_2 \cup \dots \cup A_n)^c = A_1^c A_2^c \dots A_n^c$$

$$(d) (A_1 A_2 \dots A_n)^c = A_1^c \cup A_2^c \cup \dots \cup A_n^c.$$

An intuitive justification for these results may be obtained by considering Venn diagrams.

Event may be defined verbally, and it is important to be able to express them in terms of the event operations. For example, let us consider two events, A and B.

The event that exactly one of the events, A and B, will occur is equal to  $AB^c \cup A^c B$ ; the event that exactly none of the events, A and B, will occur is equal to  $A^c B^c$ .

The event that atleast one (one or more) of the events, A or B, will occur is equal to  $A \cup B$ .

The event that atmost one (one or less) of the events will occur is equal to  $(AB)^c = A^c \cup B^c$ .

► **Example 1.** Prove that the operations of addition and multiplication of events have the following properties :

$$(a) A \cup B = B \cup A, AB = BA, AB = BA \quad (\text{commutative property});$$

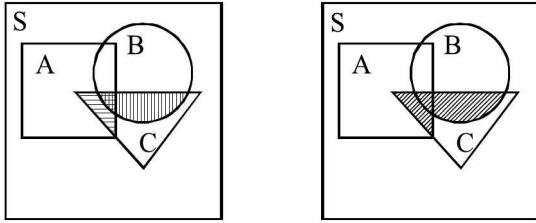
$$(b) (A \cup B) \cup C = A \cup (B \cup C), A(BC), A(BC) = (AB)C \quad (\text{associative property});$$

$$(c) (A \cup B)C = AC \cup BC \quad (\text{distributive property of multiplication with respect to addition}).$$

► **Solution** Properties (a) and (b) follow immediately from the definition of the operations of addition and multiplication, since the events of "atleast one of them occurring" and "all of them occurring together" do not depend on the order of the participating events.

We prove that event  $(A \cup B)C$  is identical with event  $AC + BC$ . Let  $\omega \in (A \cup B)C$ . This means that  $\omega \in C$  is at least in one of the events  $A$  or  $B$ . But then  $\omega$  is in at least one of the events  $AC$  or  $BC$ , i.e.,  $\omega \in AC + BC$ .

Conversely, let  $\omega \in AC \cup BC$ . Then  $\omega \in AC$  or  $\omega \in BC$ . Consequently,  $\omega \in C$  and is, in addition, in at least one of the events  $A$  or  $B$ , i.e.,  $\omega \in (A \cup B)C$ . The identity of the left-hand and the right hand side is thus proved.



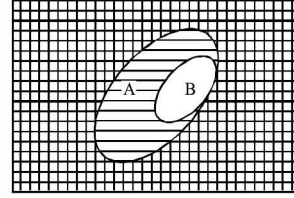
We illustrate property (c) with Venn diagrams. Let the events be :  $A = \{\text{falling in a square}\}$ ,  $B = \{\text{falling in a circle}\}$ ,  $C = \{\text{falling in a triangle}\}$ . The above figures represent the corresponding regions. The horizontal shading marks the region corresponding to event  $AC$ , the vertical one the region corresponding to event  $BC$ , and the oblique one corresponding to event  $AC \cup BC$ .

► **Example 2.** An event  $B$  is a special case of an event  $A : B \subseteq A$ , i.e., it follows from the occurrence of the event  $B$  that the event  $A$  has occurred. Does it follow from  $\bar{B}$  that  $\bar{A}$  has occurred?

► **Solution** No, it does not. For instance, an experiment may consist in sending two messages; and event

$A = \{\text{at least one message is distorted}\}$  and an event  $B = \{\text{both messages are distorted}\}$ . If an event  $\bar{B}$  has occurred,  $\bar{B} = \{\text{less than two messages are distorted}\}$ , it does not follow that neither of the messages is distorted (event  $\bar{A}$ ). On the contrary,  $\bar{B}$  follows from  $\bar{A} (\bar{A} \subseteq \bar{B})$ .

Figure shows events  $A$  and  $B$ ,  $B \subseteq A$ , and the complementary events  $\bar{A}$  (vertical hatching) and  $\bar{B}$  (horizontal hatching). It can be immediately seen that  $\bar{A} \subseteq \bar{B}$ .



► **Example 3.** The destruction of warplane may occur either as a result of its both engines being damaged (events  $D_1$  and  $D_2$ ) or as a result of the pilot's cabin being hit (event  $K$ ). The plane is fired at many times by an anti-aircraft gun. Any hit in the corresponding unit causes it to be damaged. Let event  $A = \{\text{the destruction of the plane}\}$ .

- Describe the set of elementary outcomes.
- Write  $A$  in the algebra of events both directly using events  $D_1$ ,  $D_2$  and  $K$  and in terms of elementary outcomes.
- Obtain the first representation from the second by allowable algebraic transformations.

► **Solution**

- $S = \{\omega_k, k = 1, 2, \dots, 8\}$  where

$$\omega_1 = \bar{D}_1 \bar{D}_2 \bar{K}, \omega_2 = D_1 \bar{D}_2 \bar{K}, \omega_3 = \bar{D}_1 D_2 \bar{K},$$

$$\omega_4 = \bar{D}_1 \bar{D}_2 K, \omega_5 = \bar{D}_1 D_2 K, \omega_6 = D_1 \bar{D}_2 K,$$

$$\omega_7 = D_1 D_2 \bar{K}, \omega_8 = D_1 D_2 K.$$

- $A = D_1 D_2 \cup K = \omega_4 + \omega_5 + \dots + \omega_8$

- $\omega_4 + \omega_5 + \dots + \omega_8$

$$= D_1 D_2 (K \cup \bar{K}) \cup$$

$$K (\bar{D}_1 \bar{D}_2 \cup \bar{D}_1 D_2 \cup D_1 \bar{D}_2 \cup D_1 D_2)$$

$$= D_1 D_2 \cup K (\bar{D}_1 \cup D_2 \cup D_1 \bar{D}_2)$$

$$= D_1 D_2 \cup K (\bar{D}_1 \bar{D}_2 \cup D_1 D_2) = D_1 D_2 \cup K.$$

**Practice Problems****[B]**

- Find out whether the events indicated in each example form an exhaustive group of events for the given experiment (answer yes or no).
  - An experiment involves throwing two dice; the events are  
 $C_1 = \{\text{each die comes up 6}\}$  ;  
 $C_2 = \{\text{none of the dice comes up 6}\}$  ;  
 $C_3 = \{\text{one die comes up 6 and the other does not}\}$ .
  - Three messages are sent over a communication channel; the events are  
 $E_1 = \{\text{the three messages are transmitted without an error}\}$  ;  
 $E_2 = \{\text{the three messages are transmitted with errors}\}$  ;  
 $E_3 = \{\text{two messages are transmitted with errors and one without an error}\}$ .
- Regarding each group of events say whether they are mutually exclusive in the given experiment (yes, no).
  - An experiment involves tossing two coins; the events are  
 $B_1 = \{\text{the first coin comes up heads}\}$  ;  
 $B_2 = \{\text{the second coin comes up heads}\}$ .
  - Two shots are fired at a target; the events are  
 $C_0 = \{\text{no hits}\}$  ;  $C_1 = \{\text{one hit}\}$  ;  
 $C_2 = \{\text{two hits}\}$ .
  - Two cards are selected from a pack; the events are  
 $E_1 = \{\text{both cards are from black suits}\}$  ;  
 $E_2 = \{\text{there is a queen of clubs among the selected cards}\}$  ;  
 $E_3 = \{\text{there is an ace of spades among the selected cards}\}$ .
- Regarding each group of events say whether they are equally likely in the given experiment (answer yes or no).
  - An experiment involves tossing two coins; the events are  
 $C_1 = \{\text{two heads}\}$  ;  $C_2 = \{\text{two tails}\}$  ;  
 $C_3 = \{\text{one head and one tail}\}$ .
  - A card is selected from a pack at random; the events are  
 $D_1 = \{\text{a heart}\}$  ;  $D_2 = \{\text{a diamond}\}$  ;  
 $D_3 = \{\text{a club}\}$  ;  $D_4 = \{\text{a spade}\}$ .
  - An experiment involves throwing a die; the events are  
 $E_1 = \{\text{the score is no less than 3}\}$  ;  
 $E_2 = \{\text{the score is no more than 3}\}$ .
- A die is tossed once. The result to be observed is the number of points on the upward face of the die. The events A, B, C, D, E, F are described as :  
 $A = \{X \text{ is a multiple of three}\}$   
 $B = \{X \text{ is odd}\}$ ,  
 $C = \{X > 3\}$ ,  $D = \{X < 7\}$ ,  
 $E = \{X \text{ is fractional}\}$ ,  
 $F = \{0.5 < X < 1.5\}$ . Describe the composition and find out the meaning of the following events :  
 $E_1 = \overline{A}$ ,  $E_2 = \overline{C}$ ,  $E_3 = AB$ ,  $E_4 = A \cup B$ ,  $E_5 = A - \overline{B}$ ,  $E_6 = E + D$ ,  $E_7 = EF$ .
- Let A, B, C be three events observable in a given experiment. Express the following events in terms of the algebra of events:  
 $E_1 = \{\text{exactly one event of the three events A, B, C will occur}\}$ ,  
 $E_2 = \{\text{atleast one event of the three events A, B, C will occur}\}$ ,  
 $E_3 = \{\text{exactly two events of the three will occur}\}$ ,  
 $E_4 = \{\text{atleast two of the three events will occur}\}$ ,  
 $E_5 = \{\text{none of the three events will occur}\}$ ,  
 $E_6 = \{\text{atleast two events of the three will take place}\}$ ,  
 $E_7 = \{\text{atleast one event of the three will not occur}\}$ .
- An experiment consists of tossing two coins. The following events are considered :  
 $A = \{\text{the first coin comes up heads}\}$  ;  
 $B = \{\text{the first coin comes up tails}\}$  ;  
 $C = \{\text{the second coin comes up heads}\}$  ;  
 $D = \{\text{the second coin comes up tails}\}$  ;  
 $E = \{\text{atleast one head}\}$  ;

$F = \{\text{atleast one tail}\};$

$G = \{\text{no heads}\};$

$K = \{\text{two heads}\}.$

Determine the events to which the following events are equivalent :

- (i)  $A \cup C$ , (ii)  $AC$ ,  
 (iii)  $\bar{E}$ , (iv)  $G \cup E$ ,  
 (v)  $GE$ , (vi)  $BD$ ,  
 (vii)  $E \cup K$ .

7. Prove  $(A \cup B) - B = A - AB = A\bar{B} = A - B$ .

8. Let  $A$ ,  $B$  and  $C$  be events observable in an experiment, with  $A$  and  $B$  disjoint. Show that events  $AC$  and  $BC$  are also disjoint.

9. Show that (i) if  $A \subset B$ , then  $AB = A$ ,  $A + B = B$ ,  $\bar{B} \subset \bar{A}$ , (ii) by any of the relations in (1),  $A \subset B$ .

10. Prove that

(i)  $(\bar{A} \cup BC)(\bar{B} \cup AC)(\bar{C} \cup AB)$   
 $= ABC \cup \bar{A}\bar{B}\bar{C},$   
 (ii)  $(A - B) \cup (A - C) = A - BC.$

11. A symmetrical difference between two events,  $A \Delta B$ , is given as follows :

$A \Delta B = (A - B) \cup (B - A).$

Prove the following identities :

- (i)  $A \Delta B = (A \cup B) - AB$   
 (ii)  $\overline{A \Delta B} = AB \cup \bar{A}\bar{B}$   
 (iii)  $A \Delta B = \overline{(AB)} \Delta \overline{(AB)}.$

12. Find the random event  $X$  from the equation  $\overline{X \cup A} \cup \overline{X \cup \bar{A}} = B.$

13. From a box containing 10 parts of which 3 are defective, parts are drawn one at a time without replacement until the first defective part appears, after which the experiment is stopped. Let the outcome of the  $i$ th trial be  $\omega_i = \{\text{a defective part appears on the } i\text{th trail}\}$ . Consider event  $A = \{\text{a third extraction of part has to be done}\}.$

- (i) Construct elementary outcomes of the experiment using algebraic operations on the outcomes  $w_i$ ,  $i = 1, 2, \dots$

- (ii) Write event  $A$  in terms of elementary outcomes and simplify the representation by means of algebraic transformations.

14. Two basket-ball players in turn throw a ball until the first hit. The first player to throw the ball into the basket wins. The events are :  $A_k = \{\text{the first player throws the ball in on his } k\text{th throw}\}$ ,  $B_k = \{\text{the second player throws the ball in on his } k\text{th throw}\}$ ;  $A = \{\text{the first player wins}\}$ ,  $B = \{\text{the second player wins}\}$ . The first basket-ball player is the first to throw. Find the composition of the set of elementary outcomes and write events  $A$  and  $B$  in terms of the algebra of events.

### 4.3 CLASSICAL DEFINITION OF PROBABILITY

#### Sample Space Having Equally Likely Outcomes

For many experiments it is natural to assume that all outcomes in the sample space are equally likely to occur. That is, consider an experiment whose sample space  $S$  is a finite set, say

$$S = \{1, 2, \dots, N\}.$$

Then it is often natural to assume that

$$P(\{1\}) = P(\{2\}) = \dots = P(\{N\})$$

$$P(\{i\}) = 1/N, \quad i = 1, 2, \dots, N$$

From this, it follows that for any event  $A$ ,

$$P(A) = \frac{\text{number of points in } A}{\text{number of points in } S} = \frac{n(A)}{n(S)}.$$

In words, if we assume that all outcomes of an experiment are equally likely to occur, then the probability of any event  $A$  equals the proportion of outcomes in the sample space that are contained in  $A$ .

#### Classical Definition of Probability

If  $n$  represents the total number of equally likely, mutually exclusive and exhaustive outcomes of an experiment and  $m$  of them are favourable to the

happening of the event A, then the probability of happening of the event A is given by

$$P(A) = \frac{n(A)}{n(S)} = \frac{m}{n}.$$

### Laplace's First Principle

If an event can happen in a ways and fail in b ways, and there is nothing to lead us to believe that one of these ways should occur rather than another, that is to say, all the ways are equally likely, then the chance that the event happens is

$$a/(a + b),$$

the chance that the event fails is

$$b/(a + b),$$

the odds in favour of the event are as  $a : b$ ,

the odds against the event are as  $b : a$ .

If a dice is rolled and we suppose that all six sides are equally likely to appear, then we would have

$$P(\{1\}) = P(\{2\}) = P(\{3\}) = P(\{4\})$$

$$= P(\{5\}) = P(\{6\}) = \frac{1}{6}.$$

The probability of rolling an even number would equal

$$P(\{2, 4, 6\}) = P(\{2\}) + P(\{4\}) + P(\{6\}) = \frac{3}{6} = \frac{1}{2}$$

► **Example 1.** Fifteen of a class of 20 students are members of a mathematics society. What is the probability that a randomly chosen student is a member of the society?

► **Solution** Assume that event A is the randomly chosen student being a member of the society. Then the number of elementary events favourable to the event A is 15, while the number of all the elementary events in this case is 20. Consequently, the probability is

$$P(A) = 15/20 = 3/4.$$

► **Example 2.** A set E contains the first ten letters of the English alphabet. How many different groups of three letters can be made from the given set of letters? What is the probability that a randomly chosen group will contain the letter a?

► **Solution** The number of different alphabets is equal to the number of 3-element subsets of the set E (the number of combinations of 10 elements 3 at a time);

$$n(S) = {}^{10}C_3 = \frac{10 \times 9 \times 8}{1 \times 2 \times 3} = 120.$$

Event A = {the randomly chosen alphabet of three letters contains a}. The number of elements of the set A equals the number of all possible ways of selecting two letters from nine (the letter a, which is a member of A, is excluded from the ten letters), i.e., the number of combinations of 9 elements 2 at a time ;

$$n(A) = {}^9C_2 = \frac{9 \times 8}{2} = 36.$$

$$\text{Thus } P(A) = \frac{n(A)}{n(S)} = \frac{36}{120} = 0.3.$$

► **Example 3.** Find the chance of throwing more than 15 in one throw with 3 dices.

► **Solution** A throw of 18 can be made up of 6, 6, 6 which can occur in 1 way; 17 can be made up of 6, 6, 5 which can occur in 3 ways; 16 may be made up of 6, 6, 4 and 6, 5, 5 each of which arrangements can occur in 3 ways.

Therefore the number of favourable cases  
 $= 1 + 3 + 3 + 3 = 10.$

And the total number of cases is  $6^3$ , or 216. Therefore

$$\text{the required chance} = \frac{10}{216} = \frac{5}{108}.$$

► **Example 4.** Two dice are tossed. Which event is more probable : a score of 11 or a score of 4?

► **Solution** We associate the outcome of the experiment with an ordered pair of numbers (x, y), where x is the number of points on the second die. Thus the sample space consists of a set of pairs (x, y), where x and y assume the value from 1 to 6. The number of such pairs is 36. Two elementary events to which the pairs (6, 5) and (5, 6) correspond are favourable to event A, which is that the points on the two dice sum to 11. Three elementary events to which the pairs (1, 3), (3, 1), (2, 2) correspond are favourable to the event B (that the points on the two dice sum to 4).

The probabilities of events A and B are thus

$P(A) = 2/36 = 1/18$  and  $P(B) = 3/36 = 1/12$   
and consequently event B is more probable.

### Odds in Favour and Odds Against an Event

Let S be the sample space and A be an event. Let A' denote the complement of event A, then

(i) Odds in favour of event A

$$= \frac{\text{number of cases favourable to event A}}{\text{number of cases against A}}$$

$$= \frac{n(A)}{n(A')} = \frac{n(A)/n(S)}{n(A')/n(S)} = \frac{P(A)}{P(A')}$$

(ii) Odds against an event A

$$= \frac{\text{number of cases against the event A}}{\text{number of cases favourable to event A}}$$

$$= \frac{n(A')}{n(A)} = \frac{P(A')}{P(A)}$$

**For example :**

(i) If  $P(A) = \frac{2}{7}$ , then odds in favour of A =  $\frac{2}{5}$

$$\text{and odds against A} = \frac{5}{2}$$

(ii) If odds against A =  $\frac{3}{11}$ , then odds in favour of

$$A = \frac{11}{3} \text{ and } P(A) = \frac{11}{14}$$

(iii) If odds in favour of A =  $\frac{3}{8}$ , then odds against

$$A = \frac{8}{3} \text{ and } P(A) = \frac{3}{11}$$

► **Example 5.** In a single cast with two dice, what are the odds against throwing 7, i.e. against two numbers of which the sum is 7?

► **Solution** Out of 36 possible cases, those in favour of the event are

(6, 1), (1, 6), (5, 2), (2, 5), (4, 3), (3, 4),

and the number of these is 6, therefore

the chance of throwing 7 is  $\frac{6}{36} = \frac{1}{6}$ ,

and the odds against throwing 7 are as 5 : 1.

### Axioms of Probability

The association of probability P(A) to every event A has to satisfy the following three Axioms, in order to be both meaningful and useful.

**Axiom 1**  $0 \leq P(A) \leq 1$

In other words, the probability of every event, that is, an outcome or a set of outcomes, is a number between 0 and 1, both inclusive.

**Axiom 2**  $P(S) = 1$

In other words, the probability of the whole sample space considered as an event (= subset of itself), has to be 1. It is therefore called the sure event since one of the outcomes listed under S is bound to happen.

**Axiom 3**

For any sequence (finite or infinite) of mutually exclusive events  $E_1, E_2, \dots$ , that is, events for which the intersection of any two of them is empty,

$$P\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} P(E_i)$$

Let us establish these axioms.

(i) To prove that the probability of occurrence of an event is a number lying between 0 and 1. Let S be the sample space and E be an event.

Then  $\phi \subseteq A \subseteq S$

$$\therefore n(\phi) \leq n(A) \leq n(S)$$

$$\text{or } 0 \leq n(A) \leq n(S)$$

$$\text{or } \frac{0}{n(S)} \leq \frac{n(A)}{n(S)} \leq \frac{n(S)}{n(S)} \quad [\text{Dividing by } n(S)]$$

$$\text{or } 0 \leq P(A) \leq 1 \quad [\because P(A) = \frac{n(A)}{n(S)}]$$



If  $\phi$  is the impossible event,

$$\text{then } P(\phi) = \frac{n(\phi)}{n(S)} = \frac{0}{n(S)} = 0$$

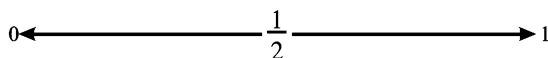
(ii) If  $S$  be the sure event, then  $P(S) = \frac{n(S)}{n(S)} = 1$ .



$$\frac{n(A)}{n(S)} \quad P(A) = 0 \Leftrightarrow A = \phi \text{ and } P(A) = 1 \Leftrightarrow A = S$$

### 'Scale' of Probability

We have seen that probability can never be less than 0 (absolute impossibility). Nor can it be greater than 1 (absolute certainty). This is because the number of outcomes given the result we are looking for can never exceed the total number of possible outcomes. Thus, when we measure probability we are using a 'scale' that runs from zero to unity :



In the experiment of tossing of a single coin,  $S = \{H, T\}$ . So the events are  $\phi$  (= the empty set);  $\{H\}$ ;  $\{T\}$  and the whole space  $S = \{H, T\}$ .

We define  $P(H) = 1/2 = P(T)$ . Already  $P(S) = 1$ .

Why did we define  $P(H)$  and  $P(T)$  as  $1/2$ ? Here lies the intuitive assumption that 'heads' and 'tails' are equally likely whenever an unbiased coin is tossed. In other words, the two events constituting the sample space are assumed to have the same probability. Incidentally, the events that constitute the sample space are called 'elementary events'; — 'elementary', in the sense that they are not made up of other events. Note that the two probabilities  $P(H)$  and  $P(T)$  add up to 1.

$$P(H \cup T) = P(H) + P(T) = \frac{1}{2} + \frac{1}{2} = 1$$

and this confirms with  $P(S) = 1$ .

This connection is what is postulated in the Third Axiom of probability. Note that  $\{H\}$  and  $\{T\}$  are 'mutually exclusive' events.

Now let us make an important assumption that the elementary events of these sample spaces are equally likely. This enables us to assign an equal probability to the elementary events of each of these sample spaces. Thus we assign the probability

$$\frac{1}{\text{Total number of elementary events}} \\ \text{(in the sample space)}$$

to each elementary event.

This gives.

In toss of a coin :  $P(H) = 1/2 = P(T)$

In rolling of a die :  $P(1) = 1/6 = P(2) = P(3) \\ = P(4) = P(5) = P(6)$

In toss of two coins :  $P(H, H) = 1/4 = P(H, T) \\ = P(T, H) = P(T, T)$  and

In rolling of two dice :  $P(i, j) = 1/36$  where  
 $i = 1, 2, \dots, 6$  and  
 $j = 1, 2, \dots, 6$ .

The event  $\{1, 2, 3\}$  in the throw of a single dice is actually the union of three events  $\{1\}$ ,  $\{2\}$  and  $\{3\}$ . Whereas, the events  $\{1\}$ ,  $\{2\}$ ,  $\{3\}$ ,  $\{4\}$ ,  $\{5\}$  and  $\{6\}$  stand by themselves and make up the sample space. They are the 'elementary events' of the experiment.

Let  $A_i \in F$ ,  $i = 1, \dots, n$ , and the collection  
 $\{A_1, A_2, \dots, A_n\}$

be a partition of the set  $S$ .

Then by axioms (ii) and (iii)

$$P(A_1) + P(A_2) + \dots + P(A_n) = 1.$$

This means that the unity probability of certain event is distributed over a set of disjoint events making up an exhaustive group. In general the correspondence between the events of some field of events and their probabilities is called probability distribution. Thus the probability  $P(A)$  as a function of a set  $A \in F$  gives the probability distribution on  $F$ .

The triple  $(S, F, P)$ , where  $F$  is the algebra of subsets of the set  $S$  and  $P$  is a numerical function satisfying axioms (i)-(iii), is called the probability space of a random experiment.

Let us find the probability of throwing an even number using a die?

$p(2 \text{ or } 4 \text{ or } 6)$

$$= \frac{\text{No. of outcomes giving 2 or 4 or 6}}{\text{Total no. of equally likely outcomes}} = \frac{3}{6} = \frac{1}{2}$$



Again notice that the total probability of getting one or other of these results is equal to the SUM of their separate probabilities :

$$\left. \begin{array}{l} p(2) = \frac{1}{6} \\ p(4) = \frac{1}{6} \\ p(6) = \frac{1}{6} \end{array} \right\} p(2 \text{ or } 4 \text{ or } 6) = \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = \frac{3}{6} = \frac{1}{2}$$

This is because the results are mutually exclusive. That is, if any one of these results happens it automatically prevents the other from happening at the same time. If the number thrown is a 4, it can't at the same time be a 2 or a 6. With these three results, each excludes the others.

#### Laplace's second principle

If an event can happen in several different ways, one only of which can occur on the same occasion, the chance of its happening is the sum of the chances of its happening in the several different ways.

We conduct an experiment which consists of throwing two dice and calculating the probability of an event C being that the score does not exceed three. The space of elementary events occurring as a result of the experiment can be represented as ordered pairs of integers varying from 1 to 6. There are 36 pairs of this kind. Among these events, the events (1, 1), (1, 2), (2, 1) are favourable to the event C. Thus, we infer that the probability of the event C is

$$P(C) = 3/36 = 1/12.$$

Let us consider now the event C as a combination of more simple events. For that purpose we note that the event C occurs if an event A – the score is equal to two, or an event B – the score is equal to three, occurs. Thus the event C is the sum of the events A and B :  $C = A + B$ . From the original space of elementary events only the pair (1, 1) is favourable to the event A and the pairs (1, 2) and (2, 1) are favourable to the event B. Consequently, the probabilities of the events A and B are equal to

$$P(A) = 1/36, \quad P(B) = 1/18$$

respectively. Thus, in the given case, there holds an equality  $P(C) = P(A) + P(B)$ .

Note that in this example the events A and B are incompatible (the total cannot equal two and three at the same time).

#### Addition Rule for Mutually Exclusive events

If two or more results are mutually exclusive, the probability that one or the other will happen is the sum of their separate probabilities.  $P(A \cup B) = P(A) + P(B)$  if  $A \cap B = \phi$ .

Further,  $P(A) + P(\bar{A}) = 1$ , where  $\bar{A}$  is the complement of A.

► **Example 6.** Use the addition rule to find the probability of throwing either a 5 or an even number.

► **Solution** Since the results are mutually exclusive, the total probability of one or other happening is the sum of their separate probabilities :

$$\begin{aligned} \left. \begin{array}{l} p(5) = \frac{1}{6} \\ p(\text{even}) = \frac{1}{2} \end{array} \right\} p(5 \text{ or even}) \\ = \frac{1}{6} + \frac{1}{2} = \frac{1+3}{6} = \frac{4}{6} = \frac{2}{3} \\ p(5 \text{ or even}) = \frac{2}{3}. \end{aligned}$$

You can check the result we have got from the addition rule by using the formula for probability

$$= \frac{\text{No. of outcomes giving result (5 or even)}}{\text{Total no. of equally likely outcomes}} = \frac{4}{6}$$

► **Example 7.** One ticket is drawn at random out of 30 tickets numbered from 1 to 30. Find the probability that the number on the ticket is a multiple of 5 or 7.

► **Solution** Here random experiment is : Drawing one ticket out of 30 tickets numbered from 1 to 30.

$$S = \{1, 2, 3, \dots, 30\} \quad \therefore n(S) = 30$$

Let A = event of drawing a ticket having a number which is a multiple of 5

B = event of drawing a ticket having a number of which is a multiple of 7.

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Then  $A = \{5, 10, 15, 20, 25, 30\} \therefore n(A) = 6$

$B = \{7, 14, 21, 28\} \therefore n(B) = 4$

Clearly  $A \cap B = \emptyset$ , hence events A and B are mutually exclusive.

$$\text{Now } P(A \cup B) = P(A) + P(B) = \frac{6}{30} + \frac{4}{30} = \frac{1}{3}$$

► **Example 8.** Suppose that we toss two coins and suppose that each of the four points in the sample space  $S = \{(H, H), (H, T), (T, H), (T, T)\}$  is equally likely and hence has probability  $1/4$ .

► **Solution** Let  $A = \{(H, H), (H, T)\}$   
and  $B = \{(H, H), (T, H)\}$

That is, A is the event that the first coin falls head, and B is the event that the second coin falls head.

$P(A \cup B)$ , the probability that either the first or second coin falls head, is given by

$$P(A \cup B) = P(A) + P(B) - P(AB)$$

$$= \frac{1}{2} + \frac{1}{2} - P(\{(H, H)\}) = 1 - \frac{1}{4} = \frac{3}{4}$$

This probability could, of course, have been computed directly because

$$P(A \cup B) = P(\{(H, H), (H, T), (T, H)\}) = \frac{3}{4}$$

► **Example 9.** In a certain country, the numerals in car registration marks range from 1 to 999. Find the probability that the first local car which you see while visiting that country have atleast two digits the same in its registration mark.

► **Solution** (i) For two digit number = 9  
(11, 22, .....)

(ii) For three digit number all same = 9 (111, 222, ....)

(iii) For 3 digit when 2 digit same and 1 different.

$$(a) \begin{array}{|c|c|c|} \hline \times & \times & 0 \\ \hline \end{array} \text{ or } \begin{array}{|c|c|c|} \hline \times & 0 & \times \\ \hline \end{array} = 9 + 9 = 18$$

$$(b) \text{ without zero } {}^3C_1 \cdot 9 \cdot 8 = 216$$

$$\text{Total} = 216 + 36 + 9 = 261$$

$$P(A) = \frac{261}{999} = \frac{87}{333} = \frac{29}{111}$$

► **Example 10.** The digits of a number are 1, 2, 3, 4, 5, 6, 7, 8 & 9 written at random in any order. Find the probability that the order is divisible by 11. Find the odds in favour of the event.

► **Solution** 1, 2, 3, 4, 5, 6, 7, 8, 9

$$x + y = 45; x - y = 11$$

$$\Rightarrow x = 28; y = 17$$

Now to realize a sum 17 using 4 digits we can have different cases,

$$\begin{array}{l} \left[ \begin{array}{cccc} 9 & 4 & 3 & 1 \\ 9 & 5 & 2 & 1 \end{array} \right]; \left[ \begin{array}{cccc} 8 & 6 & 2 & 1 \\ 8 & 5 & 3 & 1 \\ 8 & 4 & 3 & 2 \end{array} \right]; \end{array}$$

$$\left[ \begin{array}{cccc} 7 & 6 & 3 & 1 \\ 7 & 5 & 4 & 1 \\ 7 & 5 & 3 & 2 \end{array} \right];$$

$$\left[ \begin{array}{cccc} 6 & 5 & 4 & 2 \end{array} \right] \text{ (9 cases)}$$

If we use five digits then

$$7, 1, 2, 3, 4 \text{ (2 cases)}$$

$$6, 5, 3, 2, 1$$

$$\text{Hence } p = \frac{4! \times 5! \times 9 + 5! \times 4! \times 2}{9!}$$

$$= \frac{11 \times 5! \times 4!}{9!} = \frac{11}{126}$$

The odds in favour of the event are

$$= 11 : (126 - 11) = 11 : 115.$$

► **Example 11.** In a single cast with three dice, what is the chance of throwing (i) four-five-six, (ii) eleven, (iii) less than eleven, (iv) more than ten?

► **Solution** The number of possible cases is  $6^3$ .

(i) Among these four-five-six occurs  $3!$  times;

$$\therefore \text{ chance of throwing four-five-six} = \frac{3!}{6^3} = \frac{1}{36}.$$

(ii) Let a be the number of cases in which 11 is thrown, then

$$a = \text{coeff. of } x^{11} \text{ in } (x + x^2 + x^3 + \dots + x^6)^3 = 27;$$

$$\therefore \text{ chance of throwing 11 is } \frac{27}{6^3} = \frac{1}{8}.$$

- (iii) Let  $b$  be the number of cases in which less than 11 is thrown. These cases are those in which exactly 3, 4, 5, ... 10 are thrown. Therefore  $b$  is the sum of the coefficients of  $x^3, x^4, x^5, \dots, x^{10}$  in the expansion of  $(x + x^2 + x^3 + \dots + x^6)^3$ ; hence,

$$\begin{aligned} b &= \text{coefficient of } x^{10} \text{ in } x^3(1-x^6)^3(1-x)^{-4} \\ &= \text{coefficient of } x^7 \text{ in } (1-x)^{-4}(1-3x^6) \\ &= \text{coefficient of } x^7 \text{ in} \end{aligned}$$

$$\begin{aligned} &\left(4x + \frac{8 \cdot 9 \cdot 10}{1 \cdot 2 \cdot 3}x^7\right)(1-3x^6) \\ &= 120 - 12 = 108; \end{aligned}$$

$$\therefore \text{ chance of throwing less than 11 is } \frac{108}{6^2} = \frac{1}{2}.$$

- (iv) The total number of possible cases is made up of those in which more than 10 are thrown and those in which less than 11 are thrown;  
 $\therefore$  chance of throwing more than

$$10 = 1 - \frac{1}{2} = \frac{1}{2}$$

► **Example 12.** Words are formed using all the letters of the word “CANARDS”. A word is selected from these words randomly. What is the probability that this word contains neither the pattern “CAN” nor “RAD”.

► **Solution** There are 7 letters in the word CANARDS of which there are 2 A's, 1 C, 1 N, 1 R, 1 D and 1 S.

$$\text{Total number of words without any restriction} = \frac{7!}{2!}.$$

$$\text{Number of words containing “CAN”} = 5!$$

$$\text{Number of words containing “RAD”} = 5!$$

$$\text{Number of words containing both “CAN” and “RAD”} = 3!$$

$$\Rightarrow \text{Total number of words neither containing}$$

$$\text{“CAN” nor “RAD”} = \frac{7!}{2!} - (5! + 5! - 3!)$$

$$\Rightarrow \text{The probability that the word contains neither the pattern “CAN” nor “RAD.”}$$

$$\begin{aligned} &= \frac{\frac{7!}{2!} - (2(5!) - 3!)}{\frac{7!}{2!}} = \frac{127}{140}. \end{aligned}$$

► **Example 13.** Out of  $3n$  consecutive numbers, 3 are selected at random. Find the chance that their sum is divisible by 3.

► **Solution** Let the consecutive numbers be  $x, x+1, x+2, \dots, x+3n-1$ .

We arrange them in three rows, each consisting of  $n$  numbers as follows :

$$R_1 : x, x+3, x+6, \dots, x+3n-3.$$

$$R_2 : +1, x+4, x+7, \dots, x+3n-2.$$

$$R_3 : x+2, x+5, x+8, \dots, x+3n-1.$$

Sum of three numbers shall be divisible by 3 if all three numbers are from the same row or one each from  $R_1, R_2, R_3$ .

- (i) The number of selections of 3 numbers from  $R_1$ , or  $R_2$ , or  $R_3$  is

$${}^nC_3 + {}^nC_3 + {}^nC_3 = 3{}^nC_3 = \frac{1}{2}n(n-1)(n-2)$$

- (ii) The number of selections of 1 number from each of  $R_1, R_2, R_3$  is  ${}^nC_1 {}^nC_1 {}^nC_1 = n^3$

Thus, the total number of favourable cases

$$= n^3 + \frac{1}{2}n(n-1)(n-2) = \frac{1}{2}n(2n^2 - 3n + 2).$$

The exhaustive number of cases

$$= {}^{3n}C_3 = \frac{1}{2}n(3n-1)(3n-2)$$

$$\therefore p = (3n^2 - 3n + 2)/(9n^2 - 9n + 2)$$

► **Example 14.** A coin is tossed until two heads or two tails are obtained in succession. Construct the sample space for the experiment and isolate the subset corresponding to the event  $A = \{\text{no more than three tossings will be necessary}\}$ . Can the probability of the event be found as the ratio of the number of elementary events favourable to  $A$  to the total number of elementary events, and if not why?

► **Solution** The elementary events which form the set  $S$  (the heads are denoted by the letter “H” and the tails by “T”) are :

$$\begin{aligned}\omega_1 &= \{H, H\}, \omega_2 = \{T, T\}, \omega_3 = \{H, T, T\}, \\ \omega_4 &= \{T, H, H\}; \omega_5 = \{H, T, H, H\}, \\ \omega_6 &= \{T, H, T, T\} \dots\end{aligned}$$

The number of events is infinite but countable :

$$S = \{\omega_1, \omega_2, \omega_3, \dots\}.$$

The subset of elementary events favourable to A is

$$A = \{\omega_1, \omega_2, \omega_3, \omega_4\}.$$

It is impossible to find here the probability of the event A as the ratio of the number of elementary events favourable to A to the total number of elementary events since the elementary events  $\omega_1, \omega_2, \omega_3, \dots$  are not equally likely : each pair is less probable than its predecessor.

► **Example 15.** For a post three persons A, B and C appear in the interview. The probability of A being selected is twice that of B and the probability of B being selected is thrice that of C, what are the individual probabilities of A, B, C being selected ?

► **Solution** Let  $E_1, E_2, E_3$  be the events of selection of A, B and C respectively.

$$\text{Let } P(E_3) = x$$

$$\text{then } P(E_2) = 3P(E_3) = 3x$$

$$\text{and } P(E_1) = 2P(E_2) = 6x$$

since  $E_1, E_2, E_3$  are mutually exclusive and exhaustive events

$$\therefore P(E_1 \cup E_2 \cup E_3) = P(E_1) + P(E_2) + P(E_3) = 1$$

$$\therefore P(E_1) + P(E_2) + P(E_3) = 1$$

$$\Rightarrow 6x + 3x + x = 1$$

$$\therefore x = \frac{1}{10}$$

$$\text{Hence } P(E_1) = 6x = \frac{6}{10} = \frac{3}{5}; \quad P(E_2) = 3x = \frac{3}{10}$$

$$\text{and } P(E_3) = x = \frac{1}{10}$$

► **Example 16.** Suppose one is drawing a sample of size 2 from an urn containing white and red balls. Suppose that as the sample description space of the experiment one takes  $S = \{(W, W), (W, R), (R, W), (R,$

$R)\}$ . To specify a probability function P on S, one may specify the values of P on the single-member events by a table :

| x      | (W, W)         | (W, R)         | (R, W)         | (R, R)         |
|--------|----------------|----------------|----------------|----------------|
| P[(x)] | $\frac{6}{15}$ | $\frac{4}{15}$ | $\frac{4}{15}$ | $\frac{1}{15}$ |

Find the probability that the ball drawn on the first draw is white.

► **Solution** Let E be the event that the ball drawn on the first draw is white. The event E may be represented as a set of descriptions by  $E = \{(W, W), (W, R)\}$ . Then,

$$P[E] = P[\{(W, W)\}] + P[\{(W, R)\}] = \frac{2}{3}$$

► **Example 17.** A man attends a race involving three horses A, B, and C. He feels that A and B have the same chance of wining but that A (and hence also B) is twice as likely to win as C is. What is the probability that A or C wins?

► **Solution** We take S as the set {A, B, C}. If we were to assign weight a to the outcome C, then we would assign weight 2a to each of the outcomes A and B. Since the sum of the weights must be 1, we have

$$2a + 2a + a = 1, \text{ or } a = \frac{1}{5}.$$

Hence we assign weights  $\frac{2}{5}, \frac{2}{5}, \frac{1}{5}$  to the outcomes

A, B, and C, respectively. The truth set of the statement "Horse A or C wins" is {A, C}. The sum of

the weights of the elements of this set is  $\frac{2}{5} + \frac{1}{5} = \frac{3}{5}$ .

Hence the probability that A or C wins is  $\frac{3}{5}$ .

► **Example 18.** A closet contains 12 pairs of shoes. If 8 shoes are randomly selected what is the probability that there will be (a) no complete pair, and (b) exactly one complete pair?

► **Solution** (a) The number of elements in the sample space is  ${}^{24}C_8$ . The event that no complete pair is in the choice happens as follows. Choose 8 pairs out of the 12 pairs. This can be done in  ${}^{12}C_8$  ways. For each such pair choose only one of the pair, avoiding the other. This can be done in  $2^8$  ways.

$$\text{Thus } P(\text{no complete pair}) = \frac{{}^{12}C_8 \times 2^8}{{}^{24}C_8}.$$

(b) The sample space again has  ${}^{24}C_8$  points. To choose 8 shoes with only one complete pair, first choose the complete pair, in  ${}^{12}C_1$  ways. Keep it aside. From the remaining 11 pair choose 6 pairs from which you then choose 6 single (unmatching) shoes from each pair. This can be done in  ${}^{11}C_6 \times 2^6$  ways.

Thus we get the required probability as

$$\frac{12 \times {}^{11}C_6 \times 2^6}{{}^{24}C_8}.$$

► **Example 19.** A determinant of the second order is formed with elements 0 and 1. What is the probability that the determinant formed is non-negative ?

► **Solution** The number of determinants of the second order that can be formed with

$$0, 1 = 2 \times 2 \times 2 \times 2 = 16$$

( $\because$  each of the four places of elements can be filled in 2 ways : by 0 or 1).  $\therefore n(S) = 16$ .

Let  $E$  = the event of getting non-negative determinants. Then  $E'$  = the event of getting negative determinants. Clearly, negative determinants of the second order that can be made with 0, 1 are

$$\begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix}, \begin{vmatrix} 0 & 1 \\ 1 & 1 \end{vmatrix}, \begin{vmatrix} 1 & 1 \\ 1 & 0 \end{vmatrix}.$$

$$\therefore P(E') = \frac{n(E')}{n(S)} = \frac{3}{16}$$

$$\therefore P(E) = 1 - P(E') = 1 - \frac{3}{16} = \frac{13}{16}.$$

$$\text{Hence, the required probability} = \frac{13}{16}.$$

► **Example 20.** Ten men come to a hotel, Mr. Kumar and Mr. Sharma among them. There are two rooms for three persons each and one room for four persons to accommodate them. How many ways are there of accommodating the men ? What is the probability of Kumar and Sharma finding themselves in the room for four ?

► **Solution** In this experiment  $n(S) = \frac{40!}{3!3!4!} = 4200$ .

Event  $A = \{\text{Kumar and Sharma find themselves in the room for four}\}$ .

Outcomes favourable to the event  $A$  correspond to

$$n(A) = \frac{8!}{3!3!2!} = 560.$$

$$\text{The required probability } P(A) = \frac{560}{4200} = \frac{2}{45}.$$

► **Example 21.** A technical library has books on mathematics, physics, chemistry, etc., on 16 branches of science in all. Another four orders for books have arrived. Assuming that any composition of the ordered literature is equally possible, find the probabilities of the following events:  $A = \{\text{books on different branches of science are ordered}\}$ ,  $B = \{\text{books on the same branch of science are ordered}\}$ .

► **Solution** The number of equiprobable outcomes of this experiment is obviously equal to the number of combinations with repetitions of 16 elements 4 at a time, i.e.

$$n(S) = {}^{16+4-1}C_4 = {}^{19}C_4$$

The number of outcomes favourable to event  $A$  equals the number of ways of choosing without replacement four elements out of 16, so

$$\begin{aligned} P(A) &= \frac{n(A)}{n(S)} = \frac{{}^{16}C_4}{{}^{19}C_4} = \frac{15 \times 14 \times 13}{19 \times 18 \times 17} \\ &= \frac{455}{969} \approx 0.47 \end{aligned}$$

The number of outcomes favourable to event B equals the number of ways of choosing one element out of sixteen, so

$$P(B) = \frac{n(B)}{n(S)} = \frac{{}^{16}C_1}{{}^{19}C_4} = \frac{4}{969} \approx 0.004.$$

► **Example 22.** If 3 balls are randomly drawn from a bowl containing 6 white and 5 black balls, what is the probability that one of the drawn balls is white and the other two black?

► **Solution** If we regard the order in which the balls are selected as being relevant, then the sample space consists of  $11 \cdot 10 \cdot 9 = 990$  outcomes.

Furthermore, there are  $6 \cdot 5 \cdot 4 = 120$  outcomes in which the first ball selected is white and the other two black;

$5 \cdot 6 \cdot 4 = 120$  outcomes in which the first is black, the second white and the third black;  
and  $5 \cdot 4 \cdot 6 = 120$  in which the first two are black and the third white.

Hence, assuming that randomly drawn means that each outcome in the sample space is equally likely to occur, we see that the desired probability is

$$\frac{120 + 120 + 120}{990} = \frac{4}{11}$$

This problem could also have been solved by regarding the outcome of experiment as the unordered set of drawn balls. From this point of view, there are  ${}^{11}C_3 = 165$  outcomes in the sample space. Now, each set of 3 balls corresponds to  $3!$  outcomes when the order of selection is noted. As a result, if all outcomes are assumed equally likely when the order of selection is noted, then it follows that they remain equally likely when the outcome is taken to be the unordered set of selected balls. Hence, using the latter representation of the experiment, we see that the desired probability is

$$\frac{{}^6C_1 {}^5C_2}{{}^{11}C_3} = \frac{4}{11}$$

which, of course, agrees with the answer obtained previously.

► **Example 23.** A basket ball team consist of 6 frontcourt and 4 backcourt players. If players are divided into room-mates at random, what is the probability that there will be exactly two room-mate pairs made up of a backcourt and a frontcourt player ?

► **Solution** There are  $(10)!/2^5$  different divisions of the 10 players into a first room-mate pair, a second roommate pair, and so on. Hence, there are  $(10)!/(5!2^5)$  divisions into 5 room-mate pairs. There are  ${}^6C_2 {}^4C_2$  ways of choosing the frontcourt and backcourt players to be in the mixed room-mate pairs, and then 2 ways of pairing them up. As there is then 1 way to pair up the remaining two backcourt players, and  $4!(2!2^2) = 3$  ways of making two room-mate pairs from the remaining four frontcourt players, we see that the desired probability is

$$P\{2 \text{ mixed pairs}\} = \frac{{}^6C_2 {}^4C_2 (2)(3)}{(10)!/(5!2^5)} = .5714$$

► **Example 24.** A car C is parked among N cars in a row, not at either end. On his return, the owner finds that exactly r of the N places are still occupied. What is the probability that both neighbouring places are empty?

► **Solution** On his return, the owner finds that exactly r of N places are still occupied. This includes the space occupied by car C as well. This means, the remaining  $(r - 1)$  cars are parked in the  $(N - 1)$  slots. This provides the total choices  $n(S) = {}^{N-1}C_{r-1}$ . Since the neighbouring slots to car C are empty, it follows that the remaining  $(r - 1)$  cars are parked among the  $(N - 3)$  slots. This provides the favourable choices as  $n(A) = {}^{N-3}C_{r-1}$ . Consequently,

$$\begin{aligned} p &= \frac{n(A)}{n(S)} = \frac{{}^{N-3}C_{r-1}}{{}^{N-1}C_{r-1}} \\ &= \frac{(N-3)!(N-r)!}{(N-1)!(N-r-2)!} \\ &= \frac{(N-r)(N-r-1)}{(N-1)(N-2)} \end{aligned}$$

**STUDY TIP**

The following table gives a summary of the number of ways of drawing a sample in various situations.

| The Number of ways in which $n$ balls may be distributed into $M$ distinguishable urns                        |                                       |                                             |                     |
|---------------------------------------------------------------------------------------------------------------|---------------------------------------|---------------------------------------------|---------------------|
| Balls exclusion                                                                                               | Distinguishable balls                 | Indistinguishable balls                     |                     |
| Without exclusion                                                                                             | $M^n$<br>Maxwell-Boltzmann statistics | $M + n - 1 C_n$<br>Bose-Einstein statistics | With replacement    |
| With exclusion                                                                                                | $M!$                                  | $M C_n$<br>Fermi-Dirac statistics           | Without replacement |
|                                                                                                               | Ordered samples                       | Unordered samples                           | Samples drawn       |
| The Number of ways in which samples of size $n$ may be drawn from an urn containing $M$ distinguishable balls |                                       |                                             |                     |

**Practice Problems****[C]**

- What is the probability that a leaf from a new calendar torn out at random corresponds to the first day of month? (The year is assumed not to be a leap-year.)
- What is the probability of a randomly chosen two-digit number being divisible by 3?
- Two boys and two girls are placed at random in a row for a picture. What is the probability that the boys and girls alternate in the picture?
- A coin is thrown thrice. Which of the events is more probable : event A consisting in the tails occurring all three times or event B consisting in the tails occurring twice and the head's occurring once ? Calculate the probability of these events.
- If a number  $x$  between 1 and 200 (both inclusive) is picked at random what is the probability that  $\gcd(x, 6) > 1$ ?
- Out of 20 consecutive numbers, two are chosen at random : prove that the chance that their sum is odd is  $\frac{10}{19}$ .
- Nine cards are labelled 0, 1, 2, 3, 4, 5, 6, 7, 8. Two cards are drawn at random and put on a table in a successive order, and then the resulting number is read, say, 07(seven), 14(fourteen) and so on. Find the probability that the number is even.
- A five-digit number is chosen at random. What is the probability of the following events :  $A = \{\text{the number reads alike from left to right and from right to left (as number 13531)}\}$ ,  $B = \{\text{the number is multiple of five}\}$ ,  $C = \{\text{the number consists of odd digits}\}$ .

9. A box contains 10 balls numbered 1 through 10. Five balls are drawn successively at random and without replacement. Let A be the event that exactly two odd-numbered balls are drawn and that they occur on odd numbered draws from the box. Evaluate  $P(A)$ .
10. A letter is taken at random out of "ASSISTANT" and a letter is taken out of "STATISTICS". Show that the chance that they are the same letter is  $19/90$ .
11. Four different objects 1, 2, 3, 4 are distributed at random on four places marked 1, 2, 3, 4. Prove that the probability that none of the objects occupies the place corresponding to its number is  $3/8$ .
12. Out of 10 tickets, 5 are blank and the rest are marked 1, 2, 3, 4, 5. Show that the chances of drawing a total of 10 in three trials when (a) tickets are replaced after every trial (b) tickets are not replaced are (a)  $33/1000$ , (b)  $1/60$ .
13. Nine cards are drawn from a set of cards. Each is marked with one of the numbers 1, 0, -1 and it is equally likely that any of the three numbers will be drawn. Prove that the chance of obtaining a total zero is  $3139/(3)^9$ .
14. A number consists of 7 digits whose sum is 59. Prove that the chance of its being divisible by 11 is  $4/21$ .
15. The chance of first event happening is the square of the chance of a second event, but the odds against the first are the cube of the odds against the second. Find the chance of each.
16. A die is loaded in such a way that the probability of each face is proportional to the number of dots on that face. (For instance, a 6 is three times as probable as a 2.) What is the probability of getting an even number in one throw?
17. Out of the set of numbers  $E = \{1, 2, \dots, n\}$  two numbers are chosen. What is the probability of the second number being greater than the first if sampling is performed : (a) without replacement, (b) with replacement?

18. Seven identical balls are thrown at random into four holes (any one hole may hold any number of balls). How many ways are there of distributing seven balls among four holes? What is the probability of the first (and possibly some other) hole being found empty?

#### 4.4 COMPLEMENT AND ADDITION RULES

##### Theorem 1

If  $A^c$  is the event complementary to A, then

$$P(A^c) = 1 - P(A).$$

*Proof :*

A and  $A^c$  (not-A) are mutually exclusive.

Therefore  $P(A \cup A^c) = P(A) + P(A^c)$

i.e.,  $1 = P(S) = P(A) + P(A^c)$  which gives

$$P(A^c) = 1 - P(A).$$

As an illustration, the probability of throwing a sum not equal to 5 with two dices

$$= 1 - P(\text{throw of a sum} = 5).$$

$$= P(\{1, 4\} \cup \{2, 3\} \cup \{3, 2\} \cup \{4, 1\})$$

$$= 1/36 + 1/36 + 1/36 + 1/36 = 4/36 = 1/9.$$

So  $P(\text{Sum} \neq 5) = 1 - 1/9 = 8/9$ .

Again,  $P(\text{sum} \geq 6)$

$$= 1 - P(\text{sum} = 2, 3, 4 \text{ or } 5)$$

$$= 1 - (1/36 + 2/36 + 3/36 + 4/36)$$

$$= 26/36 = 13/18.$$

##### Corollary

$P(\phi)$  is always zero. From axiom  $3P(S) = 1$ .

But  $\phi = S^c$ ; so  $P(\phi) = P(S^c) = 1 - P(S) = 1 - 1 = 0$ .

If 'p' is the probability of a result happening, the probability of the alternative – failure to happen – is often written as 'q',

p = probability of 'success'

q = probability of 'failure'.

Since the result we are looking for in any probability situation must either happen or else fail to happen, the two fractions must add up to 1. That is

$$p + q = 1.$$

Suppose the chances of an astronaut returning safely

from a mission to the moon are  $p = \frac{9}{11}$ . What would

q mean? And what would be its value ?



$q$  would be the probability of the astronaut failing to return.

Since, in this case,  $p = \frac{9}{11}$ ,  $q = 1 - \frac{9}{11} = \frac{2}{11}$ .

► **Example 1.**  $n$  pairs of shoes are in a closet but the  $2n$  individual shoes are thoroughly mixed.  $r$  shoes are selected at random. Find the probability that there is atleast one pair among the  $r$  selected.

► **Solution** Let  $A$  denote the event that there is atleast one pair among  $r$  shoes selected. Then  $A'$  denotes the event that there are no pairs among the  $r$  selected shoes. We investigate  $P(A')$ .

Observe that  $r$  shoes can be chosen out of  $2n$  shoes in  ${}^{2n}C_r$  ways. This is precisely the number of subsets of size  $r$  obtainable from a set of size  $2n$ .

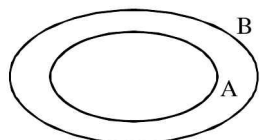
Let us choose  $r$  pairs out of  $n$  pairs, this can be accomplished in  ${}^nC_r$  ways. From each of these  $r$  pairs, we select only one of these two shoes. Obviously, there are  $2^r$  ways of doing it.

Hence  $n(A') = {}^nC_r 2^r$

Thus  $P(A') = {}^nC_r 2^r / {}^{2n}C_r$ ,  $P(A) = 1 - {}^nC_r 2^r / {}^{2n}C_r$ .

### Theorem 2

If  $A \subset B$ , then  $P(A) \leq P(B)$ .



*Proof:*

Either with the help of the figure or otherwise one can prove that

$$B = A \cup (A^c \cap B) = A \cup A^c B$$

Since  $A$  and  $A^c B$  are mutually exclusive (non-overlapping) we have

$$P(B) = P(A \cup A^c B) = P(A) + P(A^c B)$$

This shows that  $P(B) \geq P(A)$  since  $P(A^c B) \geq 0$ .

For example,

- (i) Probability of drawing two kings from a pack of 52 cards is  $\geq$  probability of drawing two aces, two kings, two queens and two jacks out of the pack.

The former is,  $\frac{{}^4C_2}{{}^{52}C_2}$  while the latter is  $\frac{({}^4C_2)^4}{{}^{52}C_8}$ ,

- (ii) In five tosses of an unbiased coin,

$$P(\text{head appearing at least twice}) \leq P(\text{head appearing at least once}) \quad \dots(1)$$

This is verified because,

$$P(\text{head never appearing})$$

$$= P(T, T, T, T, T) = 1/32.$$

(Consider all elementary events of the sample space).

$$P(\text{head appearing precisely once})$$

$$= 5 \times 1/32 = 5/32$$

So  $P(2H's \text{ or } 3H's \text{ or } 4H's \text{ or } 5H's)$

$$= 1 - (1/32 + 5/32) = 26/32$$

and  $P(1H \text{ or } 2H's \text{ or } 3H's \text{ or } 4H's \text{ or } 5H's)$

$$= 1 - 1/32 = 31/32.$$

Hence the inequality (1)

### Theorem 3

**Addition law for two events :** If  $A$  and  $B$  are any two events in a sample space  $S$ , then

$$P(A \cup B) = P(A) + P(B) - P(AB) \quad \dots(1)$$

*Proof:*

Denote the number of cases favourable to the events  $A$ ,  $B$ ,  $A \cup B$  and  $A \cap B$  by  $n(A) = n(B)$ ,  $n(A \cup B)$  and  $n(AB)$  respectively.

By counting the points in various sets of  $S$ , we see that

$$n(A) = n(AB) + n(A \bar{B}), \quad n(B) = n(AB) + n(\bar{A} B),$$

$$n(A \cup B) = n(AB) + n(A \bar{B}) + n(\bar{A} B).$$

Eliminating  $n(A \bar{B})$ ,  $n(\bar{A} B)$  provides

$$n(A \cup B) = n(A) + n(B) - n(AB).$$

Divide this equation throughout by  $N = n(S)$ , to get

$$\frac{n(A \cup B)}{N} = \frac{n(A)}{N} + \frac{n(B)}{N} - \frac{n(AB)}{N}.$$

By classical definition of probability, this equation is equivalent to

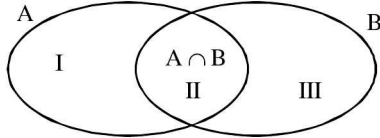
$$P(A \cup B) = P(A) + P(B) - P(AB).$$

For any two events  $A$  and  $B$ ,

$$P(A \cup B) = P(A) + P(B) - P(AB).$$

**Alternative Proof :**

Look at the events as subsets of the sample space.



The portions  $A - (A \cap B)$ ,  $A \cap B$  and  $B - (A \cap B)$  are marked as sets I, II and III. These three are mutually exclusive. So

$$P(A \cup B) = P(I) + P(II) + P(III)$$

But  $P(A) = P(I) + P(II)$

and  $P(B) = P(III) + P(II)$

$$\begin{aligned} \text{So } P(A) + P(B) &= P(I) + P(III) + 2P(II) \\ &= P(I) + P(III) + P(II) + P(II) \\ &= P(A \cup B) + P(II) \\ &= P(A \cup B) + P(AB). \end{aligned}$$

$$\text{So } P(A \cup B) = P(A) + P(B) - P(AB).$$

For example,  $P(\text{throwing either an even number or a sum of 2, 3, 4, with two dices})$  may be calculated as follows.

Let  $A = \text{event } \{2, 4, 6\}$ ;  $B = \text{event } \{2, 3, 4\}$ .

Then  $A \cup B = \{2, 3, 4, 6\}$

$$P(A) = 9/36; P(B) = 6/36;$$

$$P(AB) = P(2, 4) = 4/36.$$

$$\begin{aligned} P(A) + P(B) - P(AB) &= (9 + 6 - 4)/36 \\ &= 11/36 = P(A \cup B). \end{aligned}$$

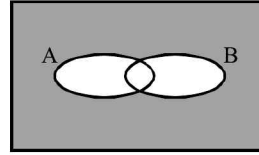
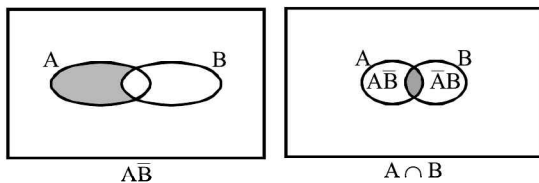
**Corollary**

If  $A \cap B = \Phi$  i.e. if the events are mutually exclusive, then there is no point favourable to the simultaneous occurrence of A and B so that  $P(AB) = 0$ . Equation (1) then reduces to

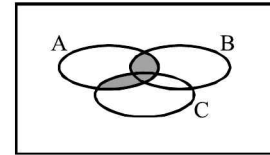
$$P(A \cup B) = P(A) + P(B).$$

**Venn Diagrams**

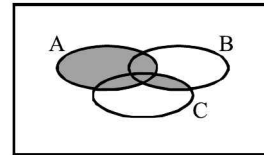
Some simple combinations of the events (shaded), and their expressions are demonstrated below; the rectangle denoting the sample space S.



$$\bar{A} \cap \bar{B} = \overline{(A \cup B)}$$



$$AB \cup AC = (A \cup B) \cap (A \cap C)$$



$$A \cap (B \cap C) = (A \cap B) \cap (A \cap C)$$

**Extension of Addition Rule to three events :** For three events A, B, C

$$\begin{aligned} P(A \cup B \cup C) &= P(A) + P(B) + P(C) - [P(AB) \\ &\quad + P(BC) + P(AC)] + P(ABC). \end{aligned}$$

*Proof :* Replace B by  $B \cup C$  in (1) to obtain

$$\begin{aligned} P(A \cup B \cup C) &= P(A) + P(B \cup C) - P(A \cap (B \cup C)) \\ &= P(A) + P(B \cup C) - P(AB \cup AC), \text{ [distributive law]} \end{aligned}$$

Apply  $AB \cap AC = ABC$ , to obtain

$$\begin{aligned} P(A \cup B \cup C) &= P(A) + [P(B) + P(C) - P(BC)] \\ &\quad - [P(AB) + P(AC) - P(ABC)] \end{aligned}$$

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - [P(AB) + P(BC) + P(AC)] + P(ABC). \quad \dots(2)$$

If  $A_i, i = 1, \dots, n$ , are events then  $\bigcup_{i=1}^n A_i$ , called the union of these events, consists of all outcomes that are in atleast one of the events  $A_i, i = 1, \dots, n$ .

Similarly,  $\bigcap_{i=1}^n A_i$ , sometimes written as  $A_1, \dots, A_n$ , is called the intersection of the events  $A_i$ , and consists of all outcomes that are in all of the events  $A_i, i = 1, \dots, n$ . For mutually exclusive events  $A_i, i \geq 1$ ,

$$P\left(\bigcup_{i=1}^n A_i\right) = \sum_{i=1}^n P(A_i)$$

If the events are not necessarily mutually then the generalized result for union of events is

$$P\left(\bigcup_{i=1}^n A_i\right) = \sum_{i=1}^n P(A_i) - \sum_{i < j} P(A_i A_j) + \sum_{i < j < k} P(A_i A_j A_k) + \dots + (-1)^{n+1} P(A_1 \dots A_n).$$

**STUDY TIP** If three events A, B and C are pair-wise mutually exclusive then they must be mutually exclusive . i.e

$$P(A \cap B) = P(B \cap C) = P(C \cap A) = 0$$

$$\Rightarrow P(A \cap B \cap C) = 0.$$

However the converse of this is not true.



(i) The opposite of “atleast A or B” is neither A nor B i.e.  $\overline{A \cup B} = \bar{A} \cap \bar{B}$

$$P(A \cup B) + P(\overline{A \cup B}) = 1.$$

$$P(A \cup B) + P(\bar{A} \cap \bar{B}) = 1.$$

(ii) For any two events A and B, P(exactly one of A, B occurs)

$$= P(A \cap \bar{B}) + P(B \cap \bar{A})$$

$$= P(A) + P(B) - 2P(A \cap B)$$

$$= P(A \cup B) - P(A \cap B)$$

$$= P(A^c \cup B^c) - P(A^c \cap B^c)$$

(iii) P (atleast two of A, B, C occur)

$$= P(B \cap C) + P(C \cap A) + P(A \cap B) - 2P(A \cap B \cap C)$$

(iv) P(exactly two of A, B, C occur)

$$= P(B \cap C) + P(C \cap A) + P(A \cap B) - 3P(A \cap B \cap C)$$

(v) P(exactly one of A,B,C occurs)

$$= P(A) + P(B) + P(C) - 2P(B \cap C) - 2P(C \cap A) - 2P(A \cap B) + 3P(A \cap B \cap C)$$

► **Example 2.** The probability that a contractor will get a plumbing contract is  $2/3$  and the probability

that he will get an electric contract is  $5/9$ . If the probability of getting atleast one contract is  $4/5$ , what is the probability that he will get both the contracts?

► **Solution** Let A = the event that the contractor will get plumbing contract

B = the event that the contractor will get an electric contract.

Then  $A \cup B$  = the event that the contractor will get atleast one contract.

and  $A \cap B$  = the event that the contractor will get both the contracts.

$$\text{Here } P(A) = \frac{2}{3}, P(B) = \frac{5}{9}, P(A \cup B) = \frac{4}{5}$$

$$\therefore P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$\therefore P(A \cap B) = P(A) + P(B) - P(A \cup B)$$

$$= \frac{2}{3} + \frac{5}{9} - \frac{4}{5} = \frac{30 + 25 - 36}{45} = \frac{19}{45}$$

► **Example 3.** Five balls are randomly chosen, without replacement, from an urn that contains 5 red, 6 white, and 7 blue balls. Find the probability that atleast one ball of each colour is chosen.

► **Solution** Let R, W, B denote, respectively, the events that there are no red, no white, and no blue balls chosen. Then

$$P(R \cup W \cup B) = P(R) + P(W)$$

$$+ P(B) - P(RW) - P(RB) - P(WB) + P(RWB)$$

$$= \frac{{}^{13}C_5}{{}^{18}C_5} + \frac{{}^{12}C_5}{{}^{18}C_5} + \frac{{}^{11}C_5}{{}^{18}C_5} - \frac{{}^7C_5}{{}^{18}C_5} - \frac{{}^6C_5}{{}^{18}C_5} - \frac{{}^5C_5}{{}^{18}C_5}$$

Thus, the probability that all colours appear in the chosen subset is approximately  $1 - 0.2933 = 0.7067$ .

► **Example 4.** Given the probabilities :

$p_1 = P(A), p_2 = P(B), p_3 = P(AB), [p_1, p_2, p_3 > 0]$  express the following in terms of  $p_1, p_2, p_3$ .

$$(a) P(A \cup B) \quad (b) P(\bar{A} \cup \bar{B})$$

$$(c) P(\bar{A}B) \quad (d) P(\bar{A} \cup B)$$

$$(e) P(\bar{A} \cap \bar{B}) \quad (f) P(A \cap \bar{B})$$

$$(g) P[\bar{A} \cap (A \cup B)] \quad (h) P[A \cup (\bar{A} \cap B)]$$

► **Solution** (a)  $P(\overline{A \cup B}) = 1 - P(A \cup B) = 1 - [P(A) + P(B) - P(AB)] = 1 - p_1 - p_2 + p_3$ .

(b)  $P(\overline{A} \cap \overline{B}) = P(\overline{A \cap B}) = 1 - P(A \cap B) = 1 - p_3$ .

(c)  $P(\overline{AB}) = P(B - AB) = P(B) - P(AB) = p_2 - p_3$ .

(d)  $P(\overline{A} \cup B) = P(\overline{A}) + P(B) - P(\overline{A} \cap B)$   
 $= 1 - p_1 + p_2 - (p_2 - p_3) = 1 - p_1 + p_3$ . [by (c)]

(e)  $P(\overline{A} \cap \overline{B}) = P(\overline{A \cup B}) = 1 - p_1 - p_2 + p_3$ . [by (a)]

(f)  $P(A \cap \overline{B}) = P(A) - P(AB) = p_1 - p_3$ .

(g)  $P[\overline{A} \cap (A \cup B)] = P[\overline{A} \cap A] \cup (\overline{A} \cap B)$   
 $= P(\emptyset \cup \overline{A} \cap B) = P(\overline{A} \cap B) = p_2 - p_3$  [by (c)]

(h)  $P[A \cup (\overline{A} \cap B)] = P(A \cup B) = P(A) + P(B) - P(AB)$   
 $= p_1 + p_2 - p_3$ .

► **Example 5.** Two dices are rolled simultaneously. We define the events

A : Sum of the numbers appearing on the upper faces is 9.

B : Sum is divisible by 4.

C : 5 has appeared on one of the dice.

Find  $P(A \cup (B \cap C))$ .

► **Solution**  $(B \cap C) \Rightarrow$  sum is either 4, 8 or 12 and 5 has appeared on one of the dice i.e. (5, 3), (3, 5).

$A \cup (B \cap C)$   
 $\Rightarrow$  either (5, 3) or (3, 5) or sum is 9.  
 i.e. (6, 3), (3, 6), (5, 4), (4, 5), (5, 3), (3, 5).

$$P(A \cup (B \cap C)) = \frac{6}{36} = \frac{1}{6}$$

► **Example 6.** Three die are rolled simultaneously and events A, B and C are defined as sum of the numbers on the upper faces of die is 9, sum is divisible by 3 and 5 has appeared on one of the die respectively. Find  $P(A \cap B \cap C)$ .

► **Solution** Since event C is the case when 5 has appeared on one of the die, the favourable cases for  $(A \cap B \cap C)$  are

(5, 1, 3), (5, 3, 1), (1, 3, 5), (1, 5, 3), (3, 1, 5), (3, 5, 1) and (5, 2, 2), (2, 5, 2), (2, 2, 5).

$$\Rightarrow P(A \cap B \cap C) = \frac{9}{6^3} = \frac{1}{24}.$$

► **Example 7.** Two balls are drawn with replacement (without replacement) from an urn containing six balls, of which four are white and two are red. Find the probability that (i) both balls will be white, (ii) both balls will be the same color, (iii) atleast one of the balls will be white.

► **Solution** To set up a mathematical model for the experiment described, assume that the balls in the urn are distinguishable; in particular, assume that they are numbered 1 to 6. Let the white balls bear numbers 1 to 4, and let the red balls be numbered 5 and 6. Let us first consider that the balls are drawn without replacement.

$S = \{(z_1, z_2); \text{ for } i = 1, 2, z_i = 1, \dots, 6, \text{ but } z_1 \neq z_2\}$ . S is the set of all 2-tuples  $(z_1, z_2)$  whose components are any numbers, 1 to 6, subject to the restriction that no two components of a 2-tuple are equal. The jth component  $z_j$  of a description represents the number of the ball drawn on the jth draw. Now let A be the event that both balls drawn are white, let B be the event that both balls drawn are red, and let C be the event that atleast one of the balls drawn is white. The problem is to find

(i)  $P[A]$ , (ii)  $P[A \cup B]$ , (iii)  $P[C]$ .

It should be noted that  $C = B^c$ , so that  $P[C] = 1 - P[B]$ .

Further, A and B are mutually exclusive, so that

$$P[A \cup B] = P[A] + P[B].$$

Now  $A = \{(1, 2), (1, 3), (1, 4), (2, 1), (2, 3), (2, 4), (3, 1), (3, 2), (3, 4), (4, 1), (4, 2), (4, 3)\}$  where as  $B = \{(5, 6), (6, 5)\}$ .

Let us assume that all descriptions in S are equally likely. Then

$$P[A] = \frac{n[A]}{n[S]} = \frac{4.3}{6.5} = 0.4,$$

$$P[B] = \frac{2.1}{6.5} = 0.066$$

In the case of sampling without replacement, we have (i)  $P[A] = 0.4$ , (ii)  $P[A \cup B] = 0.466$ , (iii)  $P[C] = 0.933$ . These probabilities have been obtained under the assumption that the balls in the urn may be regarded as numbered (distinguishable) and that all descriptions in the sample description space S are equally

likely. In the case of sampling with replacement, a similar analysis may be carried out ; one obtains the answers

$$P[A] = \frac{4.4}{6.6} = 0.444, P[B] = \frac{2.2}{6.6} = 0.11,$$

$$P[A \cup B] = 0.555, P[C] = 0.888$$

### Practice Problems

[D]

- Let A and B be two events in the field of events for a given experiment. Show that  
 $P(\overline{A} \cup \overline{B}) = P(A) + P(B) - 2P(AB).$
- Prove that if A and B are observable events in an experiment, then  
 $P(\overline{AB}) - P(\overline{A} \overline{B}) = P(A) - P(B).$
- Events A and B are such that  
 $P(A \cup B) = 3/4, P(AB) = 1/4$  and  $P(\overline{A}) = 2/3$ .  
 Show that  $P(B) = 2/3$  and  $P(A \overline{B}) = 1/12$ .
- B is the set of all functions from an n-set A to a 3-set  $B = \{a, b, c\}$ . In a random experiment of selection of functions from f assume that every  $f \in B$  is equally likely. What is the probability that such a function has a in its range?
- A man offers 1 : 3 odds that A will occur, 1 : 2 odds that B will occur. He knows that A and B cannot both occur. What odds should he give that A or B will occur?
- A student is worried about his grades in English and Art. He estimates that the probability of passing English is 0.4, that he will pass atleast one course with probability 0.6, but that he has only probability 0.1 of passing both courses. What is the probability that he will pass Art?
- There are 15 text-books on a book-shelf, 5 of which are bound. Three text-books are selected at random. What is the probability of atleast one of them being bound?
- If two balls are drawn from a bag containing 2 white, 4 red and 5 black balls, what is the chance that
  - they are both red?
  - one is red and the other black?
- An urn A contains four tickets numbered 1, 2, 3, 4 and the urn B contains six tickets numbered 2, 4, 6, 7, 8, 9. One of the two urns is chosen at random and a ticket is drawn at random from it. Show that the probability that the ticket drawn bears the number
  - 2 or 4 is  $5/12$
  - 3 is  $1/8$
  - 1 or 9 is  $5/24$ .
- A card is drawn at random from a pack of playing cards.
  - What is the probability that it is either a heart or the king of clubs?
  - What is the probability that it is either the queen of hearts or an honor card (i.e., ten, jack, queen, king, or ace)?
- Suppose that the following probabilities have been assigned to the possible results of putting a penny in a certain defective peanut-vending machine :  
 The probability that nothing comes out is  $1/2$ .  
 The probability that either you get your money back or you get peanuts (but not both) is  $1/3$ .
  - What is the probability that you get your money back and also get peanuts?
  - From the information given, is it possible to find the probability that you get peanuts?
- An experiment has three possible outcomes, a b and c. Let p be the statement “the outcome is a or b,” and q be the statement “the outcome is b or c.” Assume that weights have been assigned to the three outcomes so that  $P[p] = 2/3$  and  $P[q] = 5/6$ . Find the weights.
- Consider an experiment that consists of six horses, numbered 1 through 6, running a race and suppose that the sample space consists of the  $6!$  possible orders in which the horses finish. Let A be the event that the number 1 horse is among the top three finishers, and let B be the event that the number 2 horse comes in second. How many outcomes are in the event  $A \cup B$ ?

14. Twenty cars were driven to a service station. Five of them had a fault in the running gear, eight had a fault in the motor and ten had no faults. What is the probability that a car has faults both in the running gear and in the motor?
15. Three items are chosen at random from a lot of ten, with three defective, and inspected. Find the probabilities of the following events :  $A = \{\text{the sample obtained contains atleast one defective item}\}$ ,  $B = \{\text{all items of the sample are defective}\}$ ,  $C = \{\text{there are exactly two defective in the sample}\}$ .
16. In a residential university a survey revealed that 60% of the students read magazines in the regional language, 50% read English Magazines. 30% read magazines in a foreign language. Also 30% read magazines in English as well as in the regional language, 20% read English magazines as well as foreign language magazines and 16% read regional language magazines as well as foreign language magazines, while a bare 5% read all three kinds of magazines. If a student is randomly selected from the rolls of the University what is the probability that he does not read any magazine?
17. Two dice are rolled. If  $A$  denotes the event of getting odd sum of the points on the upturned faces and  $B$  denotes the event that there is atleast one 3 shown, show that  $P(A \cup B)$ ,  $P(AB)$ ,  $P(A - B)$  and  $P(\bar{B} \cup \bar{A})$  are respectively equal to  $23/36$ ,  $1/6$ ,  $1/3$ ,  $5/6$ .
18. From a group of 8 children, 5 boys and 3 girls, three children are chosen at random. Show that the probabilities that the elected group contains :  
(a) no girl, (b) only one girl, (c) one particular girl, (d) atleast one girl, (e) more girls than boys, are respectively  $10p$ ,  $30p$ ,  $21p$ ,  $46p$ ,  $16p$  where  $p = 1/56$ .

## 4.5 INEQUALITIES IN PROBABILITY

We have the following inequalities which can be verified using venn diagram.

Let  $A$  and  $B$  be two events on a probability space.

1. If  $A \subset B$ , then  $P(A) \leq P(B)$ .
2.  $P[AB] \leq P[A] \leq P[A \cup B] \leq P[A] + P[B]$ .
3.  $\max\{P(A), P(B)\} \leq P(A \cup B) \leq \min\{1, P(A) + P(B)\}$ .
4. **Bon-ferroni's inequality :**  
 $P(A) + P(B) - 1 \leq P(AB) \leq \min\{P(A), P(B)\}$
5.  $\max\{0, 1 - P(\bar{A}) - P(\bar{B})\} \leq P(AB)$
6. **Boole's inequality :** For a finite set of events,  $A_1, A_2, \dots, A_n$ ,  
 $P[A_1 \cup A_2 \cup \dots \cup A_n] \leq P[A_1] + P[A_2] + \dots + P[A_n]$ .

► **Example 1.** If  $P(A) = 0.9$ ,  $P(B) = 0.8$ , show that  $P(AB) \geq 0.7$ .

► **Solution**  $P(AB) \geq P(A) + P(B) - 1 = 0.9 + 0.8 - 1 = 0.7$

► **Example 2.** If  $P(A) = 3/5$  and  $P(B) = 2/3$ , then prove that

- (i)  $P(A \cup B) \geq 2/3$
- (ii)  $4/15 \leq P(A \cap B) \leq 3/5$
- (iii)  $2/5 \leq P(A / B) \leq 9/10$
- (iv)  $P(A \cap B') \leq 1/3$

► **Solution** Since  $P(A \cup B) \geq P(B)$

$$\therefore P(A \cup B) \geq 2/3 \quad \dots(1)$$

Since  $P(A \cap B) \leq P(A)$

$$\therefore P(A \cap B) \leq 3/5 \quad \dots(2)$$

$$P(A \cap B) = P(A) + P(B) - P(A \cup B) \geq P(A) + P(B) - 1$$

$$= \frac{3}{5} + \frac{2}{3} - 1 = \frac{4}{15} \quad \dots(3)$$

$$\text{From (2) and (3), } 4/15 \leq P(A \cap B) \leq 3/5 \quad \dots(4)$$

From (4),

$$\frac{4}{15P(B)} \leq \frac{P(A \cap B)}{P(B)} \leq \frac{3}{5P(B)}$$

$$\Rightarrow \frac{4}{15} \cdot \frac{3}{2} \leq P(A / B) \leq \frac{3}{5} \cdot \frac{3}{2}$$

$$\Rightarrow \frac{2}{5} \leq P(A / B) \leq \frac{9}{10} \quad \dots(5)$$

$$P(A \cap B') = P(A) - P(A \cap B) \leq \frac{3}{5} - \frac{4}{15} = \frac{1}{3} \quad \text{[From (3)]}$$

► **Example 3.** If A, B, C are three events such that  $P(A) = 0.3$ ,  $P(B) = 0.4$ ,  $P(C) = 0.8$ ,  $P(AB) = 0.08$ ,  $P(AC) = 0.28$ ,  $P(ABC) = 0.09$ .

If  $P(A \cup B \cup C) \geq 0.75$   
then show that  $P(BC)$  lies in the interval  $0.23 \leq x \leq 0.48$

► **Solution** Let  $P(BC) = x$   
Since  $P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(AB) - P(BC) - P(CA) + P(ABC)$   
 $\therefore = 0.3 + 0.4 + 0.8 - 0.08 - x - 0.28 + 0.09$   
 $= 1.23 - x$

But given

$$\begin{aligned} P(A \cup B \cup C) &\geq 0.75 \text{ and } P(A \cup B \cup C) \leq 1 \\ \therefore 0.75 &\leq 1.23 - x \leq 1 \\ \Rightarrow -0.75 &\geq 1.23 - x \geq -1 \\ \text{or } 1.23 - 0.75 &\geq x \geq 1.23 - 1 \\ \text{or } 0.23 &\leq x \leq 0.48 \end{aligned}$$

► **Example 4.** Let A, B, C be three events and assume that the simultaneous occurrence of A and B implies occurrence of C then prove that

$$P(C) \geq P(A) + P(B) - 1, \text{ or } P(\bar{C}) \leq P(\bar{A}) + P(\bar{B}).$$

► **Solution** From stated condition :  $A \cap B \subset C$ , we get  $P(AB) \leq P(C)$

$$\begin{aligned} \therefore P(A \cup B) &= P(A) + P(B) - P(AB) \leq 1 \\ \Rightarrow P(A) + P(B) &\leq 1 + P(AB) \leq 1 + P(C). \\ \text{i.e. } P(C) &\geq P(A) + P(B) - 1. \end{aligned}$$

By negation :

$$\begin{aligned} 1 - P(\bar{C}) &\geq 1 - P(\bar{A}) + 1 - P(\bar{B}) - 1 \\ \Rightarrow P(\bar{C}) &\leq P(\bar{A}) + P(\bar{B}). \end{aligned}$$

► **Example 5.** A social worker reports that his survey of a community reveals that the chance of getting a church going person is 90%, a person who has mathophobia (extreme fear of numbers) is 80% and a person who is church-going, having mathophobia is only 20%. Should you question the result of his survey ? Give reasons in your answer.

► **Solution** Let C and M denote the events : church-going person and having mathophobia.  
Then  $P\{C\} = 0.90$ ,  $P\{M\} = 0.80$ ,  $P\{C \cap M\} = 0.20$ .  
 $P\{C \cup M\} = P\{C\} + P\{M\} - P\{C \cap M\}$   
 $= 0.90 + 0.80 - 0.20 = 1.50 > 1$

The given data is certainly questionable.

## Practice Problems

[E]

- Events A and B are such that  $P(A) = 3/4$  and  $P(B) = 5/8$ . Show that
  - $P(A \cup B) \geq 3/4$
  - $3/8 \leq P(AB) \leq 5/8$
  - $1/8 \leq P(\bar{A} \bar{B}) \leq 3/8$
- Given  $P(A) = 3/4$ ,  $P(B) = 3/8$ . Show that  $P(A \cup B) \geq 3/4$  and  $0 \leq P(AB) \leq 1/4$
- Show that  $|P(A) - P(B)| \leq P(\bar{A} \bar{B} \cup \bar{A} B)$ .
- If  $P(A) = 0.4$ ,  $P(B) = 0.7$ , determine the maximum and minimum possible values of  $P(AB)$  and the conditions under which each of these values is attained.
- Examine the consistency of the following data:  $P(A) = 0.4$ ,  $P(B) = 0.3$ ,  $P(C) = 0.7$ ,  $P(AB) = 0.2$ ,  $P(BC) = 0.2$ ,  $P(AC) = 0.4$ ,  $P(\bar{A} \bar{B} \bar{C}) = 0.5$ .
- Let observable events A, B, C be such that  $C \supset AB$ . Show that  $P(AB + AC + BC) \leq P(A + B)$ .
- Prove that  $P(ABC) \leq P(AB) \leq P(A \cup B) \leq P(A \cup B \cup C) \leq P(A) + P(B) + P(C)$ .

## 4.6 CONDITIONAL PROBABILITY

Given two events, A and B, by the conditional probability of the event B, given the event A, denoted by  $P[B|A]$ , we mean intuitively the probability that B will occur, under the assumption that A has occurred. In other words,  $P[B|A]$  represents our re-evaluation of the probability of B in the light of the information that A has occurred.

Suppose that we toss 2 dice and suppose that each of the 36 possible outcomes is equally likely to occur and hence has probability  $1/36$ . Suppose further that we observe that the first die is a 3. Then, given this information, what is the probability that the sum of the 2 dice equals 8 ? To calculate this probability, we reason as follows : given that the initial dice is a 3, it follows that there can be at most 6 possible outcomes of our experiment, namely, (3, 1), (3, 2),

(3, 3), (3, 4), (3, 5), and (3, 6). Since each of these outcomes originally had the same probability occurring, the outcomes should still have equal probabilities. That is given that the first dice is a 3, the (conditional) probability of each of the outcomes (3, 1), (3, 2), (3, 3), (3, 4), (3, 5), and (3, 6) is  $1/6$ , whereas the (conditional) probability of the other 30 point in the sample space is 0. Hence the desired probability will be  $1/6$ .

If we let A and B denote, respectively, the event that the sum of the dice is 8 and the event that the first die is a 3, then the probability just obtained is called the conditional probability that B occurs given that A has occurred and is denoted by  $P(B|A)$ .

A general formula for  $P(B|A)$  that is valid for all events A and B is derived as follows :

If the event B occurs, then in order for B to occur it is necessary that the actual occurrence be a point in both B and in A; that is, it must be in  $A \cap B$ . The event  $A \cap B$ , which corresponds to the intersection of A and B, is favourable to the event B on the hypothesis that the event A is realized.

Now, as we know that A has occurred, it follows that A becomes our new or reduced sample space; hence the probability that the event A|B occurs will equal the probability of  $A \cap B$  relative to the probability of A.

That is, we have the following definition.

$$\text{If } P(A) > 0, \text{ then } P(B|A) = \frac{P(A \cap B)}{P(A)} \quad \dots(1)$$

Note that (1) is valid only when  $P(A) \neq 0$ .

This formula essentially reduces calculating the conditional probability to calculating two unconditional probabilities given in the probability space.

Now consider the experiment of tossing three fair coins. The sample space of the experiment is  $S = \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\}$ . Since the coins are fair, we can assign the probability to  $1/8$  to each sample point. Let B be the event 'atleast two heads appear' and A be the event 'first coin shows tail'.

Then  $B = \{HHH, HHT, HTH, THH\}$

and  $A = \{THH, THT, TTH, TTT\}$

Therefore

$$P(B) = P(\{HHH\}) + P(\{HHT\}) + P(\{HTH\}) + P(\{THH\})$$

$$= \frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8} = \frac{1}{2} \text{ (Why ?)}$$

$$\text{and } P(A) = P(\{THH\}) + P(\{THT\}) + P(\{TTH\}) + P(\{TTT\})$$

$$= \frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8} = \frac{1}{2}$$

Also  $B \cap A = \{THH\}$

with  $P(B \cap A) = P(\{THH\}) = 1/8$ .

Now, suppose we are given that the first coin shows tail, i.e. A occurs, then what is the probability of occurrence of B ? With the information of occurrence of A, we are sure that the cases in which first coin does not result into a tail should not be considered while finding the probability of B. This information reduces our sample space from the set S to its subset A for the event B. In other words, the additional information really amounts to telling us that the situation may be considered as being that of a new random experiment for which the sample space consists of all those outcomes only which are favourable to the occurrence of the event A.

Now, the sample point of A which is favourable to event B is THH. Thus, Probability of B considering A as the sample space =  $1/4$ ,

or Probability of B given that the event A has occurred =  $1/4$ .

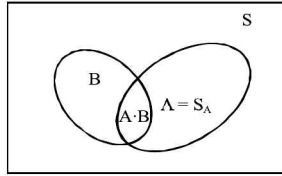
This probability of the event B is the conditional probability of B given that A has already occurred, and is denoted by  $P(B|A) = 1/4$ .

In some problems, another method of calculating the conditional probability, which is similar to the one used in considering the classical scheme, turns out to be useful.

Advancing a hypothesis that the event A has occurred is equivalent to changing the conditions of the experiment, i.e. retaining only those elementary events which are favourable to the event A and removing all other events. It follows that instead of



the space of elementary events  $S$  we have a new space  $S_A$  corresponding to the event  $A$ .



We can thus write the conditional probability of  $B$  given that  $A$  has occurred as

$$P(B|A) = \frac{n(A \cap B)}{n(A)} \quad \dots(2)$$

From here also, we can obtain the formula (1). Dividing the numerator and the denominator by total number of elementary events of the sample space, we see that  $P(B|A)$  can also be written as

$$P(B|A) = \frac{\frac{n(A \cap B)}{n(S)}}{\frac{n(A)}{n(S)}} = \frac{P(A \cap B)}{P(A)}$$

► **Example 1.** A bag contains 5 black and 3 red balls. A ball is taken out of the bag and is not returned to it. (i) What is the probability of drawing a red ball in the second draw if the ball in the first draw is red? (ii) What is the probability of drawing a black ball in the second draw if the ball in the first draw is red?

► **Solution** Let  $A$  = the event of drawing a red ball in the first draw

$B$  = the event of drawing a red ball in the second draw  
 $C$  = the event of drawing a black ball in the second draw.

- (i) After drawing a red ball in the first draw, number of red balls left is 2  
 and total number of balls left is 7.

$$\therefore P(B|A) = 2/7$$

- (ii)  $P(C|A) = 5/7$

**Alternative**

$A \cap B$  = the event of drawing a red ball in both draws

$$\therefore n(A \cap B) = {}^3C_1 \times {}^2C_1 = 6$$

$n(A)$  = number of ways of drawing red balls in both the draws + number of ways of drawing a red ball in the first draw and a black ball in the second draw  
 $= {}^3C_1 \cdot {}^2C_1 + {}^3C_1 \cdot {}^5C_1 = 21$

$$\therefore P(B|A) = \frac{n(A \cap B)}{n(A)} = \frac{6}{21} = \frac{2}{7}$$

$$\begin{aligned} \text{Similarly } P(C|A) &= \frac{n(A \cap C)}{n(A)} \\ &= \frac{{}^3C_1 \cdot {}^5C_1}{21} = \frac{15}{21} = \frac{5}{7} \end{aligned}$$

► **Example 2.** A coin is flipped twice. If we assume that all four points in the sample space  $S = \{(H, H), (H, T), (T, H), (T, T)\}$  are equally likely, what is the conditional probability that both flips result in heads, given that the first flip does?

► **Solution** If  $B = \{(H, H)\}$  denotes the event that both flips land heads, and  $A = \{(H, H), (H, T)\}$  the event that the first flip lands heads, then the desired probability is given by

$$P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{P(\{(H, H)\})}{P(\{(H, H), (H, T)\})} = \frac{\frac{1}{4}}{\frac{2}{4}} = \frac{1}{2}$$

► **Example 3.** A family has two children. What is the probability that both the children are boys given that atleast one of them is a boy?

► **Solution** Let  $b$  stand for boy and  $g$  for girl. The sample space of the experiment is

$$S = \{(b, b), (g, b), (b, g), (g, g)\}$$

Let  $E$  and  $F$  denote the following events :

$E$  : 'both the children are boys'

$F$  : 'atleast one of the child is a boy'

Then  $E = \{(b, b)\}$  and  $F = \{(b, b), (g, b), (b, g)\}$

Now  $E \cap F = \{(b, b)\}$

$$\text{Thus } P(F) = \frac{3}{4} \text{ and } P(E \cap F) = \frac{1}{4}$$

Therefore

$$P(E|F) = \frac{P(E \cap F)}{P(F)} = \frac{\frac{1}{4}}{\frac{3}{4}} = \frac{1}{3}$$

► **Example 4.** Ten cards numbered 1 to 10 are placed in a box, mixed up thoroughly and then one card is drawn randomly. If it is known that the number on the drawn card is more than 3, what is the probability that it is an even number?

► **Solution** Let A be the event 'the number on the card drawn is even' and B the event 'the number on the card drawn is greater than 3'. We have to find  $P(A|B)$ .

Now, the sample space of the experiment is

$$S = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$$

Then  $A = \{2, 4, 6, 8, 10\}$ ,  $B = \{4, 5, 6, 7, 8, 9, 10\}$  and  $A \cap B = \{4, 6, 8, 10\}$

$$\text{Also } P(A) = \frac{5}{10}, P(B) = \frac{7}{10} \text{ and } P(A \cap B) = \frac{4}{10}$$

$$\text{Then } P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{\frac{4}{10}}{\frac{7}{10}} = \frac{4}{7}$$

► **Example 5.** A die is thrown twice and the sum of the numbers appearing is observed to be 6. What is the conditional probability that the number 4 had appeared atleast once ?

► **Solution** Let E be the event that 'number 4 appears atleast once, and B be the event that 'the sum of the numbers appearing is 6'.

Then,  $E = \{(4, 1), (4, 2), (4, 3), (4, 4), (4, 5), (4, 6), (1, 4), (2, 4), (3, 4), (5, 4), (6, 4)\}$

and  $F = \{(1, 5), (2, 4), (3, 3), (4, 2), (5, 1)\}$

We have

$$P(E) = \frac{11}{36} \text{ and } P(F) = \frac{5}{36}$$

Also  $E \cap F = \{(2, 4), (4, 2)\}$

Therefore

$$P(E \cap F) = \frac{2}{36}$$

Hence, the required probability

$$P(E|F) = \frac{P(E \cap F)}{P(F)} = \frac{\frac{2}{36}}{\frac{5}{36}} = \frac{2}{5}$$

### Properties of conditional probability

Let A and B be events of a sample space S of an experiment, then we have

#### Property 1 :

$$P(S|A) = P(A|A) = 1$$

We know that

$$P(S|A) = \frac{P(S \cap A)}{P(A)} = \frac{P(A)}{P(A)} = 1$$

$$\text{Also } P(A|A) = \frac{P(A \cap A)}{P(A)} = \frac{P(A)}{P(A)} = 1$$

Thus  $P(S|A) = P(A|A) = 1$ .

#### Property 2 :

If A and B are any two events of a sample space S and B is an event of S such that  $P(A) \neq 0$ , then

$$P((A \cup B)|A) = P(A|A) + P(B|A) - P((A \cap B)|A)$$

In particular, if A and B are disjoint events, then

$$P((A \cup B)|A) = P(A|A) + P(B|A)$$

We have

$$P((A \cup B)|A) = \frac{P[(A \cup B) \cap A]}{P(A)}$$

$$= \frac{P[(A \cap A) \cup (B \cap A)]}{P(A)}$$

(by distributive law of union of sets over intersection)

$$= \frac{P(A \cap A) + P(B \cap A) - P(A \cap B \cap A)}{P(A)}$$

$$= \frac{P(A \cap A)}{P(A)} + \frac{P(B \cap A)}{P(A)} - \frac{P[(A \cap B) \cap A]}{P(A)}$$

$$= P(A|A) + P(B|A) - P((A \cap B)|A)$$

When A and B are disjoint events, then

$$P((A \cap B)|A) = 0$$

$$\Rightarrow P((A \cup B)|A) = P(A|A) + P(B|A)$$

#### Property 3 :

$$= \frac{P(A \cap A)}{P(A)} + \frac{P(B \cap A)}{P(A)} - \frac{P[(A \cap B) \cap A]}{P(A)}$$

$$P(B'|A) = 1 - P(B|A)$$

From property 1, we know that  $P(S|A) = 1$

$$\Rightarrow P(B \cup B'|A) = 1 \quad \text{since } S = B \cup B'$$

$$\Rightarrow P(B|A) + P(B'|A) = 1$$

since B and B' are disjoint events

$$\text{Thus, } P(B'|A) = 1 - P(B|A)$$



For the possible events A and B, the

following can be easily established:

- (a)  $P(A|B) = P(B|A) \Leftrightarrow P(A) = P(B)$
- (b)  $P(A|B) = P(A) \Leftrightarrow P(B|A) = P(B)$ .
- (c)  $P(A|B) > P(A) \Leftrightarrow P(B|A) > P(B)$
- (d)  $P(A|B) < P(A) \Leftrightarrow P(B|A) < P(B)$ .

We say : B is favourable to A if  $P(A|B) > P(A)$  and B is unfavourable to A if  $P(A|B) < P(A)$ . Thus, B carries positive information about A is  $P(A|B) > P(A)$  and B carries negative information about A if  $P(A|B) < P(A)$

Part (c) says : B is (un)favourable to A if A is (un)favourable to B.

► **Example 6.** In terms of probabilities :  $p_1 = P(A)$ ,  $p_2 = P(B)$ ,  $p_3 = P(AB)$ ,  $[p_1, p_2, p_3 > 0]$  express the following in terms of  $p_1, p_2, p_3$ .

- (i)  $P(A|B)$     (ii)  $P(B|\bar{A})$     (iii)  $P(\bar{A}|\bar{B})$

► **Solution** (i)  $P(A|B) = P(AB)/P(B) = p_3/p_2$ .

$$(ii) \quad P(\bar{A}B) = P(B - AB) = P(B) - P(AB) = p_2 - p_3.$$

$$P(B|\bar{A}) = P(\bar{A}B)/P(\bar{A}) = (p_2 - p_3)/(1 - p_1)$$

$$(iii) \quad P(\bar{A} \cup \bar{B}) = 1 - P(A \cup B) \\ = 1 - [P(A) + P(B) - P(AB)] = 1 - p_1 - p_2 + p_3. \\ P(\bar{A} \cap \bar{B}) = P(\overline{A \cup B}) = 1 - p_1 - p_2 + p_3. \\ P(\bar{A}|\bar{B}) = P(\bar{A} \cap \bar{B}) / P(\bar{B}) \\ = (1 - p_1 - p_2 + p_3) / (1 - p_2)$$

► **Example 7.** Let  $P(A) = p$ ,  $P(A|B) = q$ ,  $P(B|A) = r$ . Find the relations between the numbers p, q, r for the following cases :

- (a) Events A and B are mutually exclusive.
- (b) A and B are m.e. and collectively exhaustive.
- (c) A is a subevent of B ; B is a subevent of A.
- (d)  $\bar{A}$  and  $\bar{B}$  are mutually exclusive.

► **Solution** From given data :

$$P(AB) = rp, \quad P(B) = rp/q, \quad P(A) = p.$$

$$(a) \quad P(AB) = 0 \quad \Rightarrow \quad rp = 0.$$

$$(b) \quad P(A) + P(B) = 1 \\ \Rightarrow \quad p(q + r) = q ; rp = 0. \Rightarrow \quad pq = q.$$

$$(c) \quad A \subseteq B \quad \Rightarrow \quad A \cap B = A, \\ \text{so } P(AB) = P(A) \quad \Rightarrow \quad rp = q. \\ B \subseteq A \Rightarrow AB = B, \text{ so } P(AB) = P(B) \\ \Rightarrow \quad rp = (rp/q) \text{ or } rp(1 - q) = 0.$$

$$(d) \quad 0 = (\bar{A} \cap \bar{B}) = 1 - P(A \cup B) = 1 - [P(A) + P(B)] \\ = 1 - p - (rp/q) + rp. \\ \text{Thus } p(q + r) = q(1 + pr).$$

► **Example 8.** If  $P(A) = a$ ,  $P(B) = b$ , then  $P(A|B) \geq (a + b - 1)/b$ .

► **Solution** From  $P(A \cup B) \leq 1$  we obtain :

$$P(A) + P(B) - P(AB) \leq 1. \quad \dots(1)$$

Since  $P(AB)P(A|B) = bP(A|B)$ , relation (1) reduces to

$$a + b - bP(A|B) \leq 1 \Rightarrow P(A|B) \geq (a + b - 1)/b.$$

► **Example 9.** Show that if A, B, C are three events such that  $ABC \neq \emptyset$  and  $P(C|AB) = P(C|B)$ , then  $P(A|BC) = P(A|B)$ .

► **Solution**  $P(C|AB) = P(C|B)$

$$\Rightarrow \frac{P(ABC)}{P(AB)} = \frac{P(BC)}{P(B)}, \text{ i.e. } \frac{P(ABC)}{P(BC)} = \frac{P(AB)}{P(B)} \\ \Rightarrow P(A|BC) = P(A|B).$$

► **Example 10.** Suppose  $B_1 \cap B_2 \neq \emptyset$ , and  $A_1 \cap A_2 \subset B_1 B_2$ . Show that

$$P(A_1|B_1)P(A_2|B_2) = P(A_1|B_2)P(A_2|B_1).$$

► **Solution** From  $A_1 \subseteq A_1 \cup A_2 \subset B_1 B_2 \subset B_1$ , it follows that  $A_1 \subset B_1$ . Similarly,  $A_2 \subset B_2$ . Now  $P(A|B) = P(AB)/P(B) = P(A)/P(B)$

$$[(\because A \cap B = A, \text{ as } A \subset B)] \quad \dots(1)$$

Using (1) we obtain

$$P(A_1|B_1)P(A_2|B_2) \\ = \frac{P(A_1)}{P(B_1)} \cdot \frac{P(A_2)}{P(B_2)} = \frac{P(A_1)}{P(B_2)} \cdot \frac{P(A_2)}{P(B_1)} \\ = P(A_1|B_2)P(A_2|B_1). \text{ [by (1)]}$$

► **Example 11.** If A and B are two events, prove that

$$P(A|\bar{B}) = [P(A) - P(AB)]/[1 - P(B)]. \quad [P(B) \neq 1].$$

Deduce that  $P(AB) \geq P(A) + P(B) - 1$ . Also show that

$P(A) > P(A|B)$  iff  $P(A|\bar{B}) > P(A)$ .

► **Solution** (i) By definition,

$$P(A|\bar{B}) = P(A\bar{B})/P(\bar{B}) \\ = [P(A) - P(AB)]/[1 - P(B)]. \quad \dots(1)$$

Using  $P(A|\bar{B}) \leq 1$  provides

$$P(A) - P(AB) \leq 1 - P(B) \text{ or } P(AB) \geq P(A) + P(B) - 1.$$

$$(ii) \quad P(A|\bar{B}) > P(A)$$

$$\Leftrightarrow P(A) - P(AB) > P(A) \quad [1 - P(B)] \quad [\text{by (1)}].$$

$$\Leftrightarrow P(AB) < P(A)P(B) \Leftrightarrow P(A|B) < P(A).$$

Similarly,  $P(A|\bar{B}) < P(A) \Leftrightarrow P(A|B) > P(A).$

### Practice Problems

[F]

- If  $P(A) = 0.8$ ,  $P(B) = 0.5$  and  $P(B|A) = 0.4$ , find  
(i)  $P(A \cap B)$  (ii)  $P(A|B)$  (iii)  $P(A \cup B)$
- If  $P(A) = 6/11$ ,  $P(B) = 5/11$  and  $P(A \cup B) = 7/11$ , find  
(i)  $P(A \cap B)$  (ii)  $P(A|B)$  (iii)  $P(B|A)$
- A coin is tossed three times, where  
(i) E : head on third toss, F : heads on first two tosses  
(ii) E : atleast two heads, F : atmost two heads  
(iii) E : atmost two tails, F : atleast one tail  
Determine  $P(E|F)$  for each case.
- A black and a red dice are rolled.  
(i) Find the conditional probability of obtaining a sum greater than 9, given that the black die resulted in a 5.  
(ii) Find the conditional probability of obtaining the sum 8, given that the black die resulted in a number less than 4.
- A fair die is rolled. Consider events  $E = \{1, 3, 5\}$ ,  $F = \{2, 3\}$  and  $G = \{2, 3, 4, 5\}$ . Find  
(i)  $P(E|F)$  and  $P(F|E)$   
(ii)  $P(E|G)$  and  $P(G|E)$   
(iii)  $P((E \cup F)|G)$  and  $P((E \cap F)|G)$
- An instructor has a question bank consisting of 300 easy true/false questions, 200 difficult true/false questions, 500 easy multiple choice questions and 400 difficult multiple choice questions. If a question is selected at random from the question bank, what is the probability that it will be an easy question given that it is a multiple choice question?
- Given that the two numbers appearing on throwing two dice are different. Find the probability of the event 'the sum of numbers on the dice is 4'.
- In a class there are 13 girls and 14 boys. Two students are selected one by one without replacement for the recitation of a lesson. Show that the probability of selecting a girl on the second occasion given that  
(i) a boy was selected on the first occasion is  $1/2$ .  
(ii) a girl was selected on the first occasion is  $6/13$ .
- An experiment resulted in six outcomes  $e_1, e_2, \dots, e_6$  with respective probabilities 0.1, 0.2, 0.3, 0.2, 0.1, 0.1. If  $A = \{e_1, e_3, e_4\}$ ,  $B = \{e_2, e_4, e_6\}$  and  $C = \{e_2, e_4, e_5\}$  show that  
(i)  $P(A \cup B \cup C) = 1$   
(ii)  $P(A|B) = 2/5$   
(iii)  $P(\bar{B}|AC) = 0$ .
- Consider the experiment of throwing a die; if a multiple of 3 comes up, throw the die again and if any other number comes, toss a coin. Find the conditional probability of the event 'the coin shows a tail', given that 'atleast one die shows a 3'.
- A die is thrown twice. What is the probability that the sum of the faces which turn up is greater than 10, given that one of them is a 6? Given that the first throw is a 6?
- A coin is weighted so that heads has probability 0.7, tails has probability 0.2, and it stands on edge with probability 0.1. What is the probability that it does not come up heads, given that it does not come up tails?
- It is known that A and B are observable events in an experiment, with  $P(B) = 0.4$ ,  $P(A/\bar{B}) = 0.2$ . Find  $P(A)$ ,  $P(\bar{A}\bar{B})$  and  $P(\bar{A} + \bar{B})$ .
- If  $A \cap B = \emptyset$ , then  
(a)  $P(A|\bar{B}) = P(A)/[1 - P(B)]$ ,  $P(B) \neq 1$

- (b)  $P(A|A \cup B) = P(A)/[P(A) + P(B)]$ ,  $P(A \cup B) \neq 0$ .

Show that the estimate of  $P(A|B)$  from above and from below given that

$P(A) = p$ ,  $P(B) = 1 - \epsilon$ , where  $\epsilon$  is small is  
 $(p - \epsilon)/(1 - \epsilon) \leq P(A|B) \leq p/(1 - \epsilon)$ .

15. Let A, B, C be three events such that :  
 $P(A) = 0.3$ ,  $P(B) = 0.4$ ,  $P(C) = 0.5$ ,  $P(AB) = 0.2$ ,  
 $P(BC) = 0.3$ ,  $P(A'B'C') = 0.3$ ,  $P(AB|C') = 0.1$ .  
 Find  $P(B|C')$ ,  $P(A|B)$  and  $P(A|C')$ .

#### 4.7 MULTIPLICATION THEOREM

The multiplication theorem calculates the probability of joint occurrence of events A and B when the conditional probability is assumed to be known (from additional experiments).

Let A and B be two events associated with a sample space S. Clearly, the set  $A \cap B$  denotes the event that both A and B have occurred. In other words,  $A \cap B$  is also written as AB.

We know that the conditional probability of event B given that A has occurred is denoted by  $P(B|A)$  and is given by

$$P(B|A) = \frac{P(A \cap B)}{P(A)}, P(A) \neq 0$$

From this result, we can write

$$P(A \cap B) = P(A) \cdot P(B|A) \quad \dots(1)$$

Also, we know that

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, P(B) \neq 0$$

$$\text{Thus, } P(A \cap B) = P(B) \cdot P(A|B) \quad \dots(2)$$

Combining (1) and (2), we find that

$$P(A \cap B) = P(A) \cdot P(B|A) = P(B) \cdot P(A|B)$$

provided  $P(A) \neq 0$  and  $P(B) \neq 0$ .

The above result is known as the multiplication rule of probability.

In words, the above equation states that the probability that both A and B occur is equal to the probability that A occurs multiplied by the conditional probability of B given that A occurred. The above equation is often quite useful in

computing the probability of the intersection of events.

► **Example 1.** Suppose that an urn contains 8 red balls and 4 white balls. We draw 2 balls from the urn without replacement. If we assume that at each draw each ball in the urn is equally likely to be chosen, what is the probability that both balls drawn are red?

► **Solution** Let  $R_1$  and  $R_2$  denote, respectively, the events that the first and second balls drawn are red. Now, given that the first ball selected is red, there are 7 remaining red balls and 4 white balls, and so  $P(R_2 | R_1) = 7/11$ .

As  $P(R_1)$  is clearly  $8/12$ , the desired probability is

$$P(R_1 \cap R_2) = P(R_1)P(R_2|R_1) = \left(\frac{2}{3}\right)\left(\frac{7}{11}\right) = \frac{14}{33}$$

Of course, this probability could also have been

$$\text{computed by } P(R_1 \cap R_2) = \frac{{}^8C_2}{{}^{12}C_2}.$$

► **Example 2.** An urn contains 10 black and 5 white balls. Two balls are drawn from the urn one after the other without replacement. What is the probability that both drawn balls are black ?

► **Solution** Let E and F denote respectively the events that first and second ball drawn are black. We have to find  $P(E \cap F)$  or  $P(EF)$ .

Now  $P(E) = P(\text{black ball in first draw}) = 10/15$

Also given that the first ball drawn is black, i.e., event E has occurred, now there are 9 black balls and five white balls left in the urn. Therefore, the probability that the second ball drawn is black, given that the ball in the first draw is black, is nothing but the conditional probability of F given that E has occurred.

$$\text{i.e. } P(F|E) = 9/14$$

By multiplication rule of probability, we have

$$P(E \cap F) = P(E) P(F|E) = \frac{10}{15} \times \frac{9}{14} = \frac{3}{7}$$

► **Example 3.** A bag contains 15 items, of which 4 are defective. The items are selected at random, one by one and examined. The one examined are not put back. What is the chance that the 10<sup>th</sup> one examined is the last defective?

► **Solution** We define the events :

$A = \{\text{Exactly 3 defectives obtained in 9 examined}\}$

$B = \{10^{\text{th}} \text{ examined piece is defective}\}$

$$\therefore P(A) = \frac{{}^4C_3 \cdot {}^{11}C_6}{{}^{15}C_9} = \frac{24}{65},$$

$$P(B|A) = 1/6, \text{ so } P(AB) = P(A) P(B|A) = 4/65.$$

► **Example 4.** Consider a family with two children. Assume that each child is as likely to be a boy as it is to be a girl. What is the conditional probability that both children are boys, given that (i) the older child is a boy, (ii) at least one of the children is a boy?

► **Solution** Let  $A$  be the event that the older child is a boy, and let  $B$  be the event that the younger child is a boy. Then  $A \cup B$  is the event that at least one of the children is a boy, and  $AB$  is the event that both children are boys. The probability that both children are boys, given that the older is a boy, is equal to

$$P[A \cap B | A] = \frac{P[A \cap B]}{P[A]} = \frac{1/4}{1/2} = \frac{1}{2}.$$

The probability that both children are boys, given that at least one of them is a boy, is equal to since

$$(A \cap B) \cap (A \cup B) = A \cap B$$

$$P[A \cap B | A \cup B] = \frac{P[A \cap B]}{P[A \cup B]} = \frac{1/4}{3/4} = \frac{1}{3}.$$

► **Example 5.** Three cards are drawn successively, without replacement from a pack of 52 well shuffled cards. What is the probability that first two cards are kings and the third card drawn is an ace ?

► **Solution** Let  $K$  denote the event that the card drawn is king and  $A$  be the event that the card drawn is an ace. Clearly, we have to find  $P(KKA)$

Now  $P(K) = 4/52$

Also,  $P(K|E)$  is the probability of second king with the condition that one king has already been drawn. Now there are three kings in  $(52 - 1) = 51$  cards.

Therefore  $P(K|K) = 3/51$

Lastly,  $P(A|KK)$  is the probability of third drawn card to be an ace, with the condition that two kings have already been drawn. Now there are four aces in left 50 cards.

Therefore  $P(A|KK) = 4/50$

By multiplication law of probability, we have

$$P(KKA) = P(K) P(K|K) P(A|KK)$$

$$= \frac{4}{52} \times \frac{3}{51} \times \frac{4}{50} = \frac{2}{5525}$$

► **Example 6.** An urn contains ten balls of which three are black and 7 are white. The following game is played: at each trial a ball is selected at random, its colour noted, and it is replaced along with two additional balls of the same colour. What is the probability that a black ball is selected in each of the first three trials ?

► **Solution** Let  $B_i$  denote the event that a black ball is selected on the  $i^{\text{th}}$  trial. We need

$$P(B_1 B_2 B_3) = P(B_1) P(B_2 | B_1) P(B_3 | B_1 B_2)$$

$$= \frac{3}{10} \cdot \frac{5}{12} \cdot \frac{7}{14} = \frac{1}{16}.$$

### The generalized multiplication rule

The multiplication formula for a arbitrary number of events is written as follows :

$$P(E_1 E_2 E_3 \dots E_n) = P(E_1) P(E_2 | E_1) P(E_3 | E_1 E_2) \dots P(E_n | E_1 \dots E_{n-1})$$

This formula is true if all the conditional probabilities at the right are defined.

To prove the multiplication rule, we just apply the definition of conditional probability to its right-hand side. This gives

$$\begin{aligned} P(E_1) \frac{P(E_1 E_2)}{P(E_1)} \frac{P(E_1 E_2 E_3)}{P(E_1 E_2)} \dots \frac{P(E_1 E_2 \dots E_n)}{P(E_1 E_2 \dots E_{n-1})} \\ = P(E_1 E_2 \dots E_n) \end{aligned}$$

► **Example 7.** An ordinary deck of 52 playing cards is randomly divided into 4 piles of 13 cards each. Compute the probability that each pile has exactly 1 ace.

► **Solution** Define events  $E_i = i = 1, 2, 3, 4$  as follows :

$E_1 = \{\text{the ace of spades is in any one of the piles}\}$

$E_2 = \{\text{the ace of spades and the ace of hearts are in different piles}\}$

$E_3 = \{\text{the aces of spades, hearts, and diamonds are all in different piles}\}$

$E_4 = \{\text{all 4 aces are in different piles}\}$

The probability desired is  $P(E_1 E_2 E_3 E_4)$  and by the multiplication rule

$$P(E_1 E_2 E_3 E_4)$$

$$= P(E_1) P(E_2 | E_1) P(E_3 | E_1 E_2) P(E_4 | E_1 E_2 E_3)$$

Now,  $P(E_1) = 1$

since  $E_1$  is the sample space  $S$ .

$$P(E_2 | E_1) = 39/51$$

since the pile containing the ace of spades will receive 12 of the remaining 51 cards.

$$P(E_3 | E_1 E_2) = 26/50$$

since the piles containing the aces of spades and hearts will receive 24 of the remaining 50 cards ; and finally,

$$P(E_4 | E_1 E_2 E_3) = 13/49$$

Therefore, we obtain that the probability that each pile has exactly 1 ace is

$$P(E_1 E_2 E_3 E_4) = \frac{39 \cdot 26 \cdot 13}{51 \cdot 50 \cdot 49} \approx .105$$

## Joint Probability

In the section of addition rule we considered how to combine separate probabilities in order to arrive at total probability — the probability that one or other of several possible results would occur. We noticed that this kind of probability is sometimes called 'either .... or' probability.

In this section we'll look at another kind of combined probability — the probability that two or more results will happen together. This is sometimes called 'both .... and' probability or, more properly, joint probability.

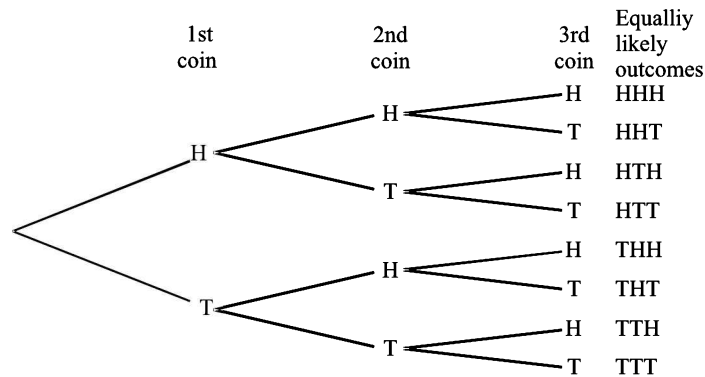
When two or more events occur in connection with each other, the joint occurrence is called a **compound event**.

For example, suppose we have a bag containing 5 white and 8 black balls, and two drawings, each of three balls, are made from it successively. If we wish to estimate the chance of drawing first 3 white and then 3 black balls, we should be dealing with compound event.

What is the probability of getting head and head and head in three tosses?

Let us write out all the equally likely outcomes.

There are 8 equally likely outcomes i.e. there are eight possible sequences in which the three coins can fall.



Only one of the eight equally likely outcomes gives HHH, so the probability of throwing three heads in a row is  $1/8$ .

Here, we notice that the combined probability of throwing three heads in a row is the product of the individual probabilities of throwing head in each toss

$$\text{i.e. } \frac{1}{8} = \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2}.$$

Given two or more separate results that are NOT mutually exclusive, the probability of them ALL happening is the PRODUCT of their separate probabilities.

**STUDY TIP** Notice that the rule will not apply to mutually exclusive results. As you'll remember, if one such result happens, it prevents the others from happening. So the probability of two or more mutually exclusive results happening is zero (e.g. a coin can't land with both heads and tails uppermost).

**CAUTION** Suppose you had to calculate the probability of each pair of result below. In which case, (a) or (b) would you use the multiplication rule?

(a) Drawing a King or a Queen from a pack of cards?  
OR

(b) Drawing a King from one pack of cards and a Queen from a second pack?

Only in case (b) would you use the multiplication rule. Here you are trying to calculate a joint probability and the two separate results do not exclude one another. (In case (a) however, it was a matter of choosing one card or the other – mutually exclusive possibilities, for which the addition rule would be used.)

Suppose you have those two packs of 52 cards each. Since there are 4 Kings and 4 Queens in each pack, you have a  $1/13$  chance of getting a King and a  $1/13$  chance of getting a Queen.

The probability of drawing a King from the first pack and a Queen from the second is  $1/169$ . (Since  $P(K) = 1/13$  and  $P(Q) = 1/13$ , and these are quite separate results, we find the probability that both will happen

by using the multiplication rule :  $\frac{1}{13} \times \frac{1}{13} = \frac{1}{169}$  .

► **Example 8.** A factory worker looks after three separate machines. Each has a different probability of breaking down during a shift

Machine A :  $2/15$       Machine B :  $1/10$

Machine C :  $3/11$

- (i) What is the probability that all three machines will break down during one shift?
- (ii) What is the probability that none of the machines will break down during a particular shift?

► **Solution** (i) To find the probability that all three machines will break down during a shift, we must combine their separate probabilities of breaking down. And, since it is the joint probability we are looking for, we'll use the multiplication rule :

$$\frac{2}{15} \times \frac{1}{10} \times \frac{3}{11} = \frac{1}{275}$$

The the probability that all three machines will break down during one shift is  $1/275$ .

- (ii) The probability that a particular machine will not break down is  $1 - P(\text{breakdown})$

The probability that none of the machines will break down during a particular shift

$$= \frac{13}{15} \times \frac{9}{10} \times \frac{8}{11} = \frac{156}{275}$$

The probability of any one of the results did not depend on whether one of the other results happened. But you'll often need to calculate the joint probability of separate results that are not independent. In such cases, the probability of any given result will differ according to which other results are taking place.

There are five people in a doctor's waiting-room : 2 men and 3 women. What is the probability that the first two people called in to see the doctor will be a man and a woman in that order?

There are 2 men and 3 women. So the probability of a man being called in first is  $2/5$ . But, once he has been called in, there are only 4 people left in the waiting-room, 3 of them women. So the probability of a woman being called in next is  $3/4$  (NOT  $3/5$ ). So

the combined (joint) probability is  $\frac{2}{5} \times \frac{3}{4} = \frac{3}{10}$  .

Hence,  $P(\text{man then woman}) = \frac{3}{10}$

### With or Without Replacement?

Clearly, this is a vital question in dealing with the probabilities of choosing. If the chosen items are not replaced, the probabilities change with each choice. Here is another example.



A box contains 6 red balls and 4 white ones. What is the probability of drawing out (at random) three white balls in succession, if :

- (i) each ball is replaced before the next draw?
- (ii) the balls are not replaced?

(i)  $P(\text{white}) = \frac{4}{10} = \frac{2}{5}$  each time.

Probability of all three balls chosen being white

$$= \frac{2}{5} \times \frac{2}{5} \times \frac{2}{5} = \frac{8}{125}$$

(ii)  $P(\text{white}) = \frac{4}{10}$  for first draw

$$= \frac{3}{9} \text{ for second draw}$$

$$= \frac{2}{8} \text{ for third draw}$$

$$P(\text{three whites}) = \frac{4}{10} \times \frac{3}{9} \times \frac{2}{8} = \frac{24}{720} = \frac{1}{30}$$

As you can see, then, in choosing two or more items from a small group, it makes quite a difference to the probabilities if the items are not replaced between choices. Always lookout for this when using the multiplication rule.

► **Example 9.** Balls are drawn one at a time without replacement from a box containing 6 red and 4 white balls. What is the probability of drawing out the balls in the order red-white-red-white etc., until only red balls (two of them) are left in the box?

► **Solution** The required probability is

R W R W R W R W

$$\frac{6}{10} \times \frac{4}{9} \times \frac{5}{8} \times \frac{3}{7} \times \frac{4}{6} \times \frac{2}{5} \times \frac{3}{4} \times \frac{1}{3} = \frac{1}{120}$$

The total number of balls, and the number of red or white balls, reduces by one with each choice, so the fraction changes each time.

► **Example 10.** What is the probability of selecting 1 bad egg, and 2 good ones out of a dozen eggs, three of which are bad.

► **Solution** In selecting 3 eggs from 12, 3 of which are bad :

$$P(\text{1st bad, other two good}) = \frac{3}{12} \times \frac{9}{11} \times \frac{8}{10} = \frac{216}{1320}$$

$P(\text{2nd bad, other two good})$

$$= \frac{9}{12} \times \frac{3}{11} \times \frac{8}{10} = \frac{216}{1320}$$

$$P(\text{3rd bad, other two good}) = \frac{9}{12} \times \frac{8}{11} \times \frac{3}{10} = \frac{216}{1320}$$

So there are three alternative ways to get one bad egg in a selection of three, and each possible way has a

probability of  $\frac{216}{1320}$ .

How do we combine these probabilities to find the total probability of getting one bad egg and two good ones? add them? or multiply them?

We add the probabilities. The bad egg would be either first, second, or third. If it was in one position, it couldn't be in any of the others. So the three possible outcomes (BGG, or GBG, or GGB) are mutually exclusive. And the probability of getting one bad egg and two good eggs is

$$\frac{216 + 216 + 216}{1320} = \frac{648}{1320}$$

► **Example 11.** In a single throw of two dice, what is the probability that the total will be an even number and that a 6 will appear on just one of the dice?

► **Solution** The multiplication rule will not give us the answer here. You'll remember that the rule requires that the two (or more) results be separate. These two results (an event total and a 6 on just one die) are NOT separate. Some of the outcomes that would give an even total would also have given a 6 on one die, e.g. 6 + 2, 4 + 6, etc. So one of the results is included in the other.

If we had tried to use the multiplication rule, what would have happened? You would have seen that eighteen of the thirty-six possible totals are even.

Thus

$$P(\text{even}) = \frac{18}{36} = \frac{1}{2}.$$

Similarly, you would have noticed that ten of the thirty-six outcomes showed a 6 on one die only. Thus  $P(\text{one } 6) = 10/36$ . But it would have been quite WRONG to multiply these probabilities and get

$$\frac{1}{2} \times \frac{10}{36} = \frac{5}{36}.$$

In fact, it is clear that only four outcomes would give the results we are looking for : 6 on the first die together with 2 or 4 on the second ; or 6 on the second with 2 or 4 on the first. So, only 4 out of 36 equally likely outcomes give the results we are looking for and even total and a 6 on just one die (i.e.  $6 + 2$ ,  $6 + 4$ ,  $2 + 6$  and  $4 + 6$ ). So the probability is  $\frac{4}{36} = \frac{1}{9}$ .

### Alternative

Let us try to solve it now using the multiplication and addition rules together.

The solitary 6 must appear on either the first die or the second. (It can't be on both.) And, for the total to be an even number, the other die must score 2 or 4. (But not 6, because 6 is to appear on one die only.) The probability of 6 on the first die is  $1/6$ . The probability of 2 or 4 (two of six equally likely outcomes) on the second die is  $2/6$ . So, by the multiplication rule, the joint probability of this

sequence is  $\frac{1}{6} \times \frac{2}{6} = \frac{2}{36}$ . Similarly, the probability of 2 or 4 on the first die, with 6 appearing on the

second die, is  $\frac{2}{6} \times \frac{1}{6} = \frac{2}{36}$ .

These are mutually exclusive results - either is possible, but not both. So the total probability of one sequence or the other happening is found by the addition rule:

$$\frac{2}{36} + \frac{2}{36} = \frac{4}{36} = \frac{1}{9}.$$

► **Example 12.** Jayesh is being interviewed for three jobs. At the first interview there are three candidates altogether, at the second there are five, and at the third

there are four candidates. If all candidates are equally likely to be selected, what is the probability that Jayesh would be offered :

- (i) one and only one of the jobs?
- (ii) atleast one of the jobs?

► **Solution** (i) If he is offered one job only, it must be either the first :

$$p(\text{offered 1st job only}) = \frac{1}{3} \times \frac{4}{5} \times \frac{3}{4} = \frac{12}{60}$$

or the second  $p(\text{offered 2nd job only})$

$$= \frac{2}{3} \times \frac{1}{5} \times \frac{3}{4} = \frac{6}{60}$$

or the third :

$$p(\text{offered 3rd job only}) = \frac{2}{3} \times \frac{4}{5} \times \frac{1}{4} = \frac{8}{60}$$

Since these are mutually exclusive events, the probability that one or the other will happen is the sum of the separate probabilities :

$$P(\text{offered one job only}) = \frac{12}{60} + \frac{6}{60} + \frac{8}{60} = \frac{26}{60}$$

- (ii) The probability that Jayesh will be offered

$$\text{none of the jobs is } \frac{2}{3} \times \frac{4}{5} \times \frac{3}{4} = \frac{24}{60}.$$

And since he must either be offered no job or atleast one of them, i.e.

$P(\text{none}) + P(\text{atleast one}) = 1$ , then the probability of him being offered atleast one job is :

$$1 - \frac{24}{60} = \frac{36}{60}$$

► **Example 13.** A bag contains 5 white and 7 black balls; if two balls are drawn what is the chance that one is white and the other black?

► **Solution** (i) Regarding the occurrence as a simple event, the chance

$$= (5 \times 7) / {}^{12}C_2 = \frac{35}{66}.$$

- (ii) The occurrence may be regarded as the happening of one or other of the two following compound events:

- (a) drawing a white and then a black ball, the chance of which is

$$\frac{5}{12} \times \frac{7}{11} \text{ or } \frac{35}{132};$$

- (b) drawing a black and then a white ball, the chance of which is

$$\frac{7}{12} \times \frac{5}{11} \text{ or } \frac{35}{132}$$

And since these events are mutually exclusive, the required chance

$$= \frac{35}{132} + \frac{35}{132} = \frac{35}{66}.$$

It will be noticed that we have here assumed that the chance of drawing two specified balls successively is the same as if they were drawn simultaneously. A little consideration will show that this must be the case.

### Practice Problems

[G]

- Two cards are drawn at random and without replacement from a pack of 52 playing cards. Find the probability that both the cards are black.
- A box of oranges is inspected by examining three randomly selected oranges drawn without replacement. If all the three oranges are good, the boxes approved for sale, otherwise, it is rejected. Find the probability that a box containing 15 oranges out of which 12 are good and 3 are bad ones will be approved for sale.
- In toss of three coins one after another, what is the probability of getting :
  - Three tails in a row?
  - Two tails and a head (in that order)?
  - Two tails and a head (in any order)?
- What is the probability of drawing three hearts in a row if the cards are not replaced in the pack once they've been chosen?

5. Four boys are taken at random. What is the probability that they were :

- all born on the same day of the week.
- all born on different days of the week?

6. A school bicycle shed contains 25 bicycles, seven of which have defective brakes. If three bicycles are selected at random for testing, what is the probability that exactly one will have defective brakes?

7. Suppose you toss a coin three times. What is the probability of throwing atleast one head?

8. A pair of dice is tossed. If the numbers appearing are different, find the probability that the sum is even.

9. If  $P(B) = P(A|B) = P(C|AB) = 1/2$ , show that  $P(ABC) = 1/8$ .

10. A tray of 20 components contains 3 defective. If 3 components are selected at random, what is the probability that they will include no defectives?

### 4.8 TOTAL PROBABILITY

Let A and B be two events. We can compute  $P(B)$  by “conditioning” on whether A occurs.

We may express B as

$$B = (B \cap A) \cup (B \cap A^c)$$

for in order for an outcome to be in B, it must either be in both B and A or be in B but not in A (See Figure).

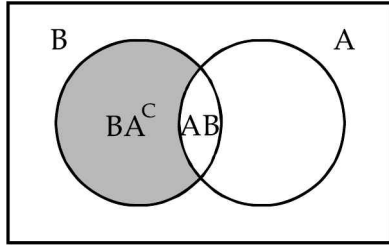
As  $B \cap A$  and  $B \cap A^c$  are clearly mutually exclusive, we have

$$P(B) = P(B \cap A) + P(B \cap A^c)$$

$$P(B) = P(B|A)P(A) + P(B|A^c)P(A^c)$$

where  $P(A^c) = 1 - P(A)$

This is called as theorem of total probability. The above equation states that the probability of the event B is a weighted average of the conditional probability of B given that A has occurred and the conditional probability of B given that A has not occurred - each conditional probability being given as



$$B = BA \cup BA^C$$

much weight as the event on which it is conditioned has of occurring. This is an extremely useful formula because it use often enables us to determine the probability of an event by first “conditioning” upon whether or not some second event has occurred. That is there are many instances where it is difficult to compute the probability of an event directly, but it is straightforward to compute it once we know whether or not some second event has occurred.

### Partition of a sample space

A set of events  $E_1, E_2, \dots, E_n$  is said to represent a partition of the sample space  $S$  if

- (a)  $E_i \cap E_j = \phi, i \neq j, i, j = 1, 2, 3, \dots, n$
- (b)  $E_1 \cup E_2 \cup \dots \cup E_n = S$  and
- (c)  $P(E_i) > 0$  for all  $i = 1, 2, \dots, n$ .

In other words, the events  $E_1, E_2, \dots, E_n$  represent a partition of the sample space  $S$  if they are pairwise disjoint, exhaustive and have non-zero probabilities.

As an example, we see that any non-empty event  $E$  and its complement  $E'$  form a partition of the sample space  $S$  since they satisfy  $E \cap E' = \phi$  and  $E \cup E' = S$ .

From the Venn diagram, one can easily observe that if  $E$  and  $B$  are any two events associated with a sample space  $S$ , then the set  $\{E \cap B, E \cap B', E' \cap B, E' \cap B'\}$  is a partition of the sample space  $S$ . It may be mentioned that the partition of a sample space is not unique. There can be several partitions of the same sample space.

### Theorem of total probability

Let  $\{A_1, A_2, \dots, A_n\}$  be a partition of the sample space  $S$ , and suppose that each of the events  $A_1, A_2, \dots, A_n$  has non-zero probability of occurrence. Let  $B$  be any event associated with  $S$ , then

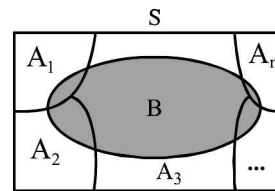
$$P(B) = P(A_1) P(B|A_1) + P(A_2) P(B|A_2) + \dots + P(A_n) P(B|A_n)$$

$$= \sum_{j=1}^n P(A_j) P(B|A_j)$$

*Proof:* Given that  $A_1, A_2, \dots, A_n$  is a partition of the sample space  $S$  (figure). Therefore,

$$S = A_1 \cup A_2 \cup \dots \cup A_n \quad \dots(1)$$

and  $A_i \cap A_j = \phi, i \neq j, i, j = 1, 2, \dots, n$



Now, we know that for any event  $B$ ,

$$\begin{aligned} B &= B \cap S \\ &= B \cap (A_1 \cup A_2 \cup \dots \cup A_n) \\ &= (B \cap A_1) \cup (B \cap A_2) \cup \dots \cup (B \cap A_n) \end{aligned}$$

Also  $B \cap A_i$  and  $B \cap A_j$  are respectively the subsets of  $A_i$  and  $A_j$ . We know that  $A_i$  and  $A_j$  are disjoint, for  $i \neq j$ , therefore,  $B \cap A_i$  and  $B \cap A_j$  are also disjoint for all  $i \neq j, i, j = 1, 2, \dots, n$ .

$$\begin{aligned} \text{Thus, } P(B) &= P[(B \cap A_1) \cup (B \cap A_2) \cup \dots \cup (B \cap A_n)] \\ &= P(B \cap A_1) + P(B \cap A_2) + \dots + P(B \cap A_n) \end{aligned}$$

Now, by multiplication rule of probability, we have

$$\begin{aligned} P(B \cap A_i) &= P(A_i) P(B|A_i) \text{ as } P(A_i) \neq 0 \quad \forall i \\ &= 1, 2, \dots, n \end{aligned}$$

Therefore,

$$P(B) = P(A_1) P(B|A_1) + P(A_2) P(B|A_2) + \dots + P(A_n) P(B|A_n)$$

$$\text{or } P(B) = \sum_{j=1}^n P(A_j) P(B|A_j)$$

► **Example 1.** A person has undertaken a construction job. The probabilities are 0.65 that there will be strike, 0.80 that the construction job will be completed on time if there is no strike, and 0.32 that the construction job will be completed on time if there is a strike. Determine the probability that the construction job will be completed on time.

► **Solution** Let A be the event that the construction job will be completed on time, and B be the event that there will be a strike. We have to find  $P(A)$ .

We have

$$P(B) = 0.65, P(\text{no strike}) = P(B') = 1 - P(B) \\ = 1 - 0.65 = 0.35$$

$$P(A|B) = 0.32, P(A|B') = 0.80$$

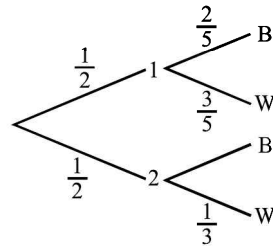
Since events B and B' form a partition of the sample space S, therefore, by theorem on total probability, we have

$$P(A) = P(B) P(A|B) + P(B') P(A|B') \\ = 0.65 \times 0.32 + 0.35 \times 0.8 \\ = 0.208 + 0.28 = 0.488$$

Thus, the probability that the construction job will be completed in time is 0.488.

► **Example 2.** Suppose that we have two urns. Urn 1 contains two black balls and three white balls. Urn 2 contains two black balls and one white ball. An urn is chosen at random and a ball chosen from this urn at random. What is the probability that a white ball is chosen?

► **Solution** A hasty answer might be  $1/2$ , since there are an equal number of black and white balls involved and everything is done at random. However, it is hasty answers like this (which is wrong) which show the need for a more careful analysis. We are considering two experiments.



The first consists in choosing the urn and the second in choosing the ball. There are two possibilities for the first experiment, and we assign  $p_1 = p_2 = 1/2$  for the probabilities of choosing the first and the second urn, respectively. We then assign  $p_{1,W} = 3/5$  for the probability that a white ball is chosen,

under the assumption that urn 1 is chosen. Similarly

we assign  $p_{1,B} = \frac{2}{5}$ ,  $p_{2,W} = \frac{1}{3}$ ,  $p_{2,B} = \frac{2}{3}$ . We indicate

these probabilities on the possibility tree in figure 4. The probability tree in figure . The probability that a white ball is drawn is then found from the tree as the sum of the weights assigned to paths which lead to a

choice of a white ball. This is  $\frac{1}{2} \cdot \frac{3}{5} + \frac{1}{2} \cdot \frac{1}{3} = \frac{7}{15}$ .

► **Example 3.** A batch of transistors among which one out of ten is defective are to be tested. The testing scheme is such that there is a 0.95 probability of detecting a defect (if any) and a probability of 0.03 that a good transistor will be recognized as defective. What is the probability that a randomly chosen transistor will be recognized as defective?

► **Solution** It is clear from the statement of the problem that closely connected with event  $A = \{\text{a randomly chosen transistor is recognized as defective}\}$  to be considered are two hypotheses :

$H_1 = \{\text{the transistor to be tested is defective}\},$

$H_2 = \bar{H}_1 = \{\text{the transistor to be tested is good}\}.$

The unconditional a priori probabilities of these hypotheses are easy to calculate:

$$P(H_1) = 0.1, P(H_2) = 0.9.$$

The conditional probabilities are given in the statement of the problem :

$$P(A|H_1) = 0.95, P(A|H_2) = 0.03.$$

Applying the total probability formula we get

$$P(A) = 0.1 \times 0.95 + 0.9 \times 0.03 = 0.122.$$

► **Example 4.** One purse contains 1 rupee and 3 nP's, a second purse contains 2 rupees and 4 nP's and a third contains 3 rupees and 1 nP. If a coin is taken out of one of the purses selected at random, find the chance that it is a rupee.

► **Solution** Since each purse is equally likely to be taken, the chance of selecting the first is  $1/3$ ; hence the chance of drawing a rupee is  $1/4$ ; hence the chance of drawing a rupee so far as it depends upon

the first purse is  $\frac{1}{3} \times \frac{1}{4}$ , or  $\frac{1}{12}$ . Similarly the chance

of drawing a rupee so far as it depends on the second purse is  $\frac{1}{3} \times \frac{2}{6}$ , or  $\frac{1}{9}$ ; and from the third purse the chance of drawing a rupee is  $\frac{1}{3} \times \frac{3}{4}$ , or  $\frac{1}{4}$ ;  
 $\therefore$  the required chance =  $\frac{1}{12} + \frac{1}{9} + \frac{1}{4} = \frac{4}{9}$

The probability of an event may sometimes be considered as the sum of the probabilities of two or more separate events; but it is very important to notice that the probability of one or other of a series of events is the sum of the probabilities of the separate events only when the events are mutually exclusive, that is, when the occurrence of one is incompatible with the occurrence of any of the others.

► **Example 5.** You ask your neighbour to water a sickly plant while you are on vacation. Without water it will die with probability .8; with water it will die with probability .15. You are 90 percent certain that your neighbour will remember to water the plant.

- What is the probability that the plant will be alive when you return?
- If it is dead, what is the probability your neighbour forgot to water it?

► **Solution** Let A denote the event that the plant is alive and let W be the event that it was watered.

$$(a) P(A) = P(A|W)P(W) + P(A|W^c)P(W^c) \\ = (.85)(.9) + (.2)(.1) = .785$$

$$(b) P(W^c|A^c) = \frac{P(A^c|A^c)P(W^c)}{P(A^c)} \\ = \frac{(.8)(.1)}{.215} = \frac{16}{43}$$

► **Example 6.** If the probability that a family has exactly n children is  $(1/2)^n$ ,  $n = 1, 2, 3, \dots$  and if all  $2^n$  permutations of the sexes of the n children are equally likely, find the probability that a family will have no boys.

► **Solution** Let  $C_n$  be the event that a family has exactly n children and  $B_0$  be the event that a family has no boy. Then by hypothesis,  $P[C_n] = (1/2)^n$  and given  $C_n$ ,  $B_0$  requires n girls, we must have

$$P\{B_0|C_n\} = (1/2)^n, \text{ for every } n = 1, 2, \dots$$

By Total Probability theorem,

$$P(B_0) = \sum_{n=1}^{\infty} P(C_n) P(B_0|C_n) \\ = \sum_{n=1}^{\infty} (1/2)^{2n} = \frac{1}{4} \sum_{n=0}^{\infty} (1/2)^n = \frac{1}{3}$$

► **Example 7.** An insurance company believes that people can be divided into two classes : those who are accident prone and those who are not. Their statistics show that an accident-prone person will have an accident at some time within a fixed 1-year period with probability .4, whereas this probability decreases to .2 for a non-accident-prone person. If we assume that 30 percent of the population is accident prone, what is the probability that a new policyholder will have an accident within a year of purchasing a policy?

► **Solution** We shall obtain the desired probability by first conditioning upon whether or not the policyholder is accident prone. Let  $A_1$  denote the event that the policyholder will have an accident within a year of purchase; and let A denote the event that the policyholder is accident prone. Hence the desired probability,  $P(A_1)$ , is given by

$$P(A_1) = P(A_1|A)P(A) + P(A_1|A^c)P(A^c) \\ = (.4)(.3) + (.2)(.7) = .26$$

### Practice Problems

[H]

- There are two balls in an urn whose colours are not known (each ball can be either white or black). We put a white ball in the urn. What is now the probability of drawing a white ball ?
- One gun is selected from five guns among which there are 3 sniper guns, and a shot is fired from that gun. Find the probability of hitting the target if the probability of hitting when firing the sniper gun is 0.95 and that using the ordinary gun is 0.7.

3. A device carried on board a plane can operate under two sets of conditions : under conditions of normal cruising flight and under load factor conditions at takeoff and landing. Cruising flight accounts for 80% of the entire flight time, load factor conditions accounting for 20%. The probability of the device failing under normal flight conditions is 0.1, with a 0.4 probability of failing under load factor conditions. Calculate the reliability of the device during the flight.
4. TV sets produced by three factories are on sale. The output of one of the factories contains 20% of TV sets with a hidden flaw, that of another 10% and that of a third 5%. What is the probability of buying a good TV set if the shop has received 30% of the sets from the first factory 20% from the second and 50% from the third?
5. The probability that a certain beginner at golf gets a good shot if he uses the correct club is  $1/3$ , and the probability of a good shot with an incorrect club is  $1/4$ . In his bag are five different clubs, only one of which is correct for the shot in question. If he chooses a club at random and takes a stroke, show that the probability that he gets a good shot is  $4/15$ .
6. In a purse are 10 coins, all five nP.'s except one which is a rupee; in another are ten coins all five nP.'s. Nine coins are taken from the former purse and put into the latter, and then nine coins are taken from the latter and put into the former. Find the chance that the rupee is still in the first purse.
7. A is one of 6 horses entered for a race, and is to be ridden by one of two jockeys B and C. It is 2 to 1 that B rides A, in which case all the horses are equally likely to win; if C rides A, his chance is trebled. What are the odds against his winning?
8. A person throws two dices, one the common cube, and the other a regular tetrahedron, the

number on the lowest face being taken in the case of the tetrahedron. What is the chance that the sum of the numbers thrown is not less than 5?

## 4.9 INDEPENDENT EVENTS

The previous examples of this chapter show that  $P(B|A)$ , the conditional probability of B given A, is not generally equal to  $P(B)$ , the unconditional probability of B. In other words, knowing that A has occurred generally changes the chances of B's occurrence. In the special cases where  $P(B|A)$  does in fact equal  $P(B)$ , we say that B is independent of A. That is, B is independent of A if knowledge that A has occurred does not change the probability that B occurs.

Since  $P(B|A) = P(A \cap B)/P(A)$ , we see that B is independent of A if

$$P(A \cap B) = P(A) P(B)$$

As the above equation is symmetric in A and B, it shows that whenever A is independent of B, B is also independent of A. We thus have the following definition.

Let A and B be two events, then A and B are said to be independent if

$$P(A \cap B) = P(A) \cdot P(B)$$

Two events A and B that are not independent are said to be dependent

i.e. if  $P(A \cap B) \neq P(A) \cdot P(B)$



Two events A and B are said to be independent, if

$$P(A|B) = P(A) \text{ provided } P(B) \neq 0$$

and  $P(A|B) = P(A)$  provided  $P(B) \neq 0$

Thus, in this definition we need to have  $P(A) \neq 0$  and  $P(B) \neq 0$ .

Consider the experiment of drawing a card from a deck of 52 plying cards, in which the elementary events are assumed to be equally likely. If A and B denote the events 'the card drawn is a spade' and 'the card drawn is an ace' respectively, then

$$P(A) = \frac{13}{52} = \frac{1}{4} \text{ and } P(B) = \frac{4}{52} = \frac{1}{13}$$

Also A and B is the event 'the card drawn is the ace of spades' so that

$$P(A \cap B) = \frac{1}{52}$$

$$\text{Hence } P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{\frac{1}{52}}{\frac{1}{13}} = \frac{1}{4}$$

Since  $P(A) = \frac{1}{4} = P(A|B)$ , we can say that the occurrence of event B has not affected the probability of occurrence of the event A.

We also have

$$P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{\frac{1}{52}}{\frac{1}{4}} = \frac{1}{13} = P(B)$$

Again,  $P(B) = \frac{1}{13} = P(B|A)$  shows that occurrence of event A has not affected the probability of occurrence of the event B.

Thus, A and B are two events such that the probability of occurrence of one of them is not affected by occurrence of the other. Such events are called independent event.

Examples of independent events:

- (i) A card is selected at random from an ordinary deck of 52 playing cards. If A is the event that the selected card is an ace and B is the event that it is a spade, then A and B are independent.

This follows because  $P(A \cap B) = 1/52$ , whereas  $P(A) = 4/52$  and  $P(B) = 13/52$ . We have  $P(A \cap B) = P(A) \cdot P(B)$

- (ii) Two coins are flipped, and all 4 outcomes are assumed to be equally likely. If A is the event that the first coin lands heads and B the event that the second lands tails, then A and B are independent, since

$$P(A \cap B) = P(\{(H, T)\}) = 1/4; \text{ whereas}$$

$$P(A) = P(\{(H, H), (H, T)\}) = 1/2$$

$$\text{and } P(B) = P(\{(H, T), (T, T)\}) = 1/2.$$

$$\text{We have } P(A \cap B) = P(A) \cdot P(B).$$

- (iii) Of the twenty students in a lecture-room eight smoke, 12 wear spectacles and six both smoke and wear spectacles. One of the students is called at random to the blackboard. The events are as follows : A = {the student is a smoker}, B = {the student wears spectacles}. Determine whether or not the events A and B are dependent.

Since  $P(AB) = 3/10 \neq P(A)P(B) = 8/20 \times 12/20$ , the independence condition is not satisfied and hence the events A and B are dependent.

- (iv) It is natural to consider as independent the results of firing from two guns simultaneously at a target or the events associated with the appearance of defective articles of a certain sort that come of two production lines at different factories, etc.

► **Example 1.** A die is thrown. If A is the event 'the number appearing is a multiple of 3' and B be the event 'the number appearing is even' then find whether A and B are independent?

► **Solution** We know that the sample space is

$$S = \{1, 2, 3, 4, 5, 6\}$$

Now  $A = \{3, 6\}$ ,  $B = \{2, 4, 6\}$  and  $A \cap B = \{6\}$

$$\text{Then } P(A) = \frac{2}{6} = \frac{1}{3}, P(B) = \frac{3}{6} = \frac{1}{2}$$

$$\text{and } P(A \cap B) = \frac{1}{6}$$

$$\text{Clearly } P(A \cap B) = P(A) \cdot P(B)$$

Hence A and B are independent events.

► **Example 2.** An unbiased die is thrown twice. Let the event A be 'odd number on the first throw' and B the event 'odd number on the second throw'. Check the independence of the events A and B.

► **Solution** If all the 36 elementary events of the experiment are considered to be equally likely, we

$$\text{have } P(A) = \frac{18}{36} = \frac{1}{2} \text{ and } P(B) = \frac{18}{36} = \frac{1}{2}$$

$$\text{Also } P(A \cap B) = P(\text{odd number on both throws})$$



$$= \frac{9}{36} = \frac{1}{4}$$

Now  $P(A)P(B) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$

Clearly  $P(A \cap B) = P(A) \times P(B)$

Thus, A and B are independent events

## Properties of independent events

### 1. The events A and $\phi$ are independent.

*Proof:*  $A \cap \phi = \phi$  ... (1)

$\therefore P(A \cap \phi) = P(\phi) = 0$  ... (2)

Also  $P(A) \cdot P(\phi) = P(A) \cdot 0 = 0$

From (1) and (2),  $P(A \cap \phi) = P(A) \cdot P(\phi)$

Hence A and S are independent events.

### 2. The events A and S are independent.

*Proof:*

$\therefore A \cap S = A$  ... (1)

$\therefore P(A \cap S) = P(A)$

$P(A \cap S) = P(A) \cdot 1 = P(A) \cdot P(S)$

Hence A and S are independent events.

### 3. If any one of the pairs {A, B}, {A, B<sup>c</sup>}, {A<sup>c</sup>, B}, or {A<sup>c</sup>, B<sup>c</sup>} is an independent pair, then all the pairs are independent pairs.

*Proof:*

To establish independence we must establish the appropriate product rules. First we note that

$$P(AB^c) = P(A) - P(AB) = P(A) - P(A)P(B)$$

$$= P(A)[1 - P(B)]$$

$$= P(A)P(B^c) \text{ iff the pair } \{A, B\} \text{ is independent}$$

By interchanging the role of A and B in the previous argument, we obtain  $P(A^cB) = P(A^c)P(B)$  iff {A, B} is an independent pair. On replacing B by B<sup>c</sup>, we may use the last result to assert

$$P(A^cB^c) = P(A^c)P(B^c) \text{ iff } \{A, B^c\}$$

is an independent pair. But this pair is independent iff {A, B} is independent.

### 4. If A and B are independent events with non-zero probabilities, the corresponding sets of the sample space must have at least one common point.

*Proof:*

Let A and B, be the sets of the sample space corresponding to the events A and B, which are such that

$$P(AB) = P(A) \cdot P(B), \text{ with } P(A) \neq 0, P(B) \neq 0.$$

If  $A \cap B = \phi$  then  $P(A \cap B) = P(\phi) = 0$ . So either  $P(A)$  or  $P(B)$  is zero. This contradicts the hypothesis that neither of  $P(A)$ ,  $P(B)$  is zero. This contradicts the hypothesis that neither of  $P(A)$ ,  $P(B)$  is zero. Hence  $A \cap B \neq \phi$ . Hence the theorem is proved.

In the dealing of a pack of 52 cards,  $P(\text{dealing an ace}) = 4/52 = 1/13$  and  $P(\text{dealing a spade or a club}) = 26/52 = 1/2$ . Now if we call these two events A and P,  $P(AB) = P(\text{ace of spades or ace of clubs}) = 2/52 = 1/26$ .

So  $P(AB) = P(A) \cdot P(B)$ . This means that A and B are independent. The sets corresponding to A and B are {4 cards of ace}, {all 13 spades and all 13 clubs}. Their intersection is {ace of spades, ace of clubs}.

### 5. Mutually exclusive events (none of which is an impossible event) are not independent and non-impossible independent events are not mutually exclusive.

*Proof:*

Let A and B be two non impossible mutually exclusive events.

$$\text{Then } P(A \cup B) = P(A) + P(B) \quad \dots(1)$$

$$\text{Now } P(A \cap B) = P(A) + P(B) - P(A \cup B) = 0$$

$$[\text{from (1)}] \quad \dots(2)$$

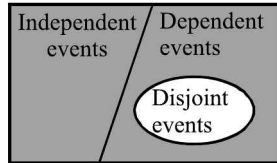
$$\text{But } P(A) > 0, P(B) > 0 [\therefore A \neq \phi, B \neq \phi]$$

$$\therefore P(A) \cdot P(B) > 0$$

From (2) and (3), it is clear that

$$P(A \cap B) \neq P(A) \cdot P(B)$$

Hence A and B are not independent events.



### Second Part :

Let A and B be non-impossible independent events, then

$$P(A \cap B) = P(A) \cdot P(B) \quad \dots(1)$$

$$\begin{aligned} \text{Now } P(A \cap B) &= P(A) + P(B) - P(A \cup B) \\ &= P(A) + P(B) - P(A) \cdot P(B) \end{aligned}$$

[From (1)]

$$\neq P(A) + P(B) \quad [\because P(A) > 0 \text{ and } P(B) > 0]$$

$$\therefore P(A) \cdot P(B) > 0]$$

Hence A and B are not mutually exclusive.

If A and B are mutually exclusive events and each has positive probability, they cannot be independent; if they have positive probabilities and are independent, they cannot be mutually exclusive.

Note also the distinction between mutually exclusive events and independent events. The former corresponds to non-overlapping sets. The latter has no corresponding concept in set theory.

In the experiment of throwing two dices, let, event A be the throw of 10. This consists of the following points of the sample space

$$\{(4, 6), (5, 5), (6, 4)\}$$

and so has the probability  $3/36 = 1/12$ . Let event B be the event that no dice shows 5. The points of the sample space corresponding to this are  $\{(x, y); x \neq 5, y \neq 5\}$ . The probability of this event is  $1 - 11/36 = 25/36$ . The simultaneous event  $AB = A \cap B$  is  $\{(4, 6), (6, 4)\}$  has probability  $2/36 = 1/18$ . Clearly  $P(AB) \neq P(A) \cdot P(B)$ .

So A and B are dependent events.

### STUDY TIP

Events can be dependent without being disjoint. In fact, the disjoint sets of events can be depicted as a subset of the dependent sets of events of the sample space. Independence and mutually exclusiveness may be present in the same problem :

Let us define the events for a single roll of a die :

$$A = \{1, 3, 5\}, B = \{2, 4, 6\},$$

$$C = \{5, 6\}. \text{ Then, } A \cap B = \emptyset,$$

$$B \cap C = \{6\} \neq \emptyset. \text{ Now,}$$

$$(i) P(AB) = 0 \neq P(A) P(B) = 1/4, \text{ Thus A and B are disjoint but not independent.}$$

$$(ii) P(B \cap C) = \frac{1}{6} = \frac{1}{2} \cdot \frac{1}{3} = P(B) \cdot P(C). \text{ Thus B and C are independent but not disjoint.}$$

► **Example 3.** A single card is drawn from an ordinary deck. Give examples of events A and B associated with this experiment that are

- (a) mutually exclusive but not independent
- (b) independent but not mutually exclusive
- (c) independent and mutually exclusive
- (d) neither independent nor mutually exclusive.

► **Solution** We define the following events :

$$S = \{\text{Spade is obtained}\}, H = \{\text{Heart is obtained}\},$$

$$A = \{\text{Ace is obtained}\}, D = \{S \cup H\}$$

$$(a) P(S \cap H) = 0 \neq P(S) P(H)$$

$$(b) P(S \cap A) = 1/52 = P(S) P(A) = [(1/3) (1/4)]$$

$$(c) \text{ The classes of mutually exclusive and independent events are disjoint.}$$

$$(d) P(S \cap D) = P(S) = 1/4, P(S) \cdot P(D) = (1/4) (1/2) = 1/8.$$

### Independence of three or more events

In order to generalize the formula  $P(A \cap B) = P(A) \cdot P(B)$  for the case of three and more events, it is necessary to introduce a new notion – the notion of mutual (collective) independence.

Events  $A_1, A_2, \dots, A_m$  ( $m > 2$ ) are called collectively independent if each of these events and the event which is the intersection of any number of the remaining events are independent.

Three events A, B and C are said to be mutually independent, if

$$P(A \cap B) = P(A)P(B) \quad \dots(1)$$

$$P(A \cap C) = P(A)P(C) \quad \dots(2)$$

$$P(B \cap C) = P(B)P(C) \quad \dots(3)$$

$$\text{and } P(A \cap B \cap C) = P(A)P(B)P(C) \quad \dots(4)$$

If atleast one of the above is not true for three given events, we say that the events are not independent. If only conditions (1), (2), (3) hold true for three given events, we say that the events are pairwise independent.

► **Example 4.** An urn contains four tickets having numbers 112, 121, 211, 222 written on them. If one ticket is drawn at random and  $A_i (i = 1, 2, 3)$  be the event that the  $i$ th digit from left of the number on ticket drawn is 1, test whether the events  $A_1, A_2, A_3$  are mutually independent or not.

► **Solution** Here random experiment is : Drawing one ticket from the urn containing 4 tickets having numbers 112, 121, 211, 222.

Let S be the sample space, then  $n(S) = 4$

given,  $A_i$  = the event that the  $i$ th digit from left of the number on ticket drawn is 1.

$$\therefore A_1 = \{11, 121\}, A_2 = \{112, 211\}, A_3 = \{121, 211\}$$

$$\therefore P(A_1) = \frac{2}{4} = \frac{1}{2}, P(A_2) = \frac{2}{4} = \frac{1}{2}, P(A_3) = \frac{2}{4} = \frac{1}{2}$$

$$\therefore P(A_1) \cdot P(A_2) \cdot P(A_3) = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8} \quad \dots(1)$$

$$\therefore A_1 \cap A_2 \cap A_3 = \emptyset \therefore n(A_1 \cap A_2 \cap A_3) = 0$$

$$\therefore P(A_1 \cap A_2 \cap A_3) = \frac{n(A_1 \cap A_2 \cap A_3)}{n(S)} = \frac{0}{4} = 0 \quad \dots(2)$$

From (1) and (2), it is clear that

$$P(A_1 \cap A_2 \cap A_3) \neq P(A_1) \cdot P(A_2) \cdot P(A_3)$$

Hence  $A_1, A_2, A_3$  are not mutually independent events.



$$(A_1 \cap A_2) = \{112\}, A_2 \cap A_3 = \{211\}$$

$$\text{and } A_1 \cap A_3 = \{121\}$$

$$\therefore P(A_1 \cap A_2) = \frac{1}{4} = P(A_1) \cdot P(A_2)$$

$$P(A_2 \cap A_3) = \frac{1}{4} = P(A_2) \cdot P(A_3)$$

$$\text{and } P(A_1 \cap A_3) = \frac{1}{4} = P(A_1) \cdot P(A_3)$$

Hence  $A_1, A_2, A_3$  are pairwise independent.

Let us explain the difference between pairwise independence and mutual independence by considering a particular example dealing with three events. For the events A, B, C to be pairwise independent, it is necessary that the pair of events A and B, B and C, C and A be independent. But this is not enough for mutual independence, the latter requires in addition that three more pairs of events:

A and  $B \cap C$ , B and  $C \cap A$ , C and  $A \cap B$  be independent. The requirement of mutual independence is stronger than that of pairwise independence.

It should be noted that if A, B, and C are mutually independent, then A will be independent of any event formed from B and C. For instance, A is independent of  $B \cup C$ , since

$$\begin{aligned} P[A \cap (B \cup C)] &= P[(A \cap B) \cup (A \cap C)] \\ &= P(A \cap B) + P(A \cap C) - P(A \cap B \cap C) \\ &= P(A)P(B) + P(A)P(C) - P(A)P(B \cap C) \\ &= P(A)[P(B) + P(C) - P(B \cap C)] \\ &= P(A)P(B \cup C) \end{aligned}$$

Let us consider another illustration.

A box contains four balls : a red, a yellow, a green, and a ball coloured in all the three colours. One ball is taken out of the box at a time. Consider the following events:

A (a red ball is taken out),

B (a yellow ball is taken out),

C (a green ball is taken out).

Will the events A, B, C be pairwise independent?

Obviously, there are four elementary events in this trial:

$U_1$  (a red ball is taken out),  
 $U_2$  (a yellow ball is taken out),  
 $U_3$  (a green ball is taken out),  
 $U_4$  (a ball coloured in all the three colours is taken out).

The event A is favoured by  $U_1$  and  $U_4$ , therefore,

$P(A) = \frac{2}{4} = \frac{1}{2}$ . Let us compute the conditional probability of the event A on the hypothesis that the event B occurs. The occurrence of the event B means that the event  $U_2$  or  $U_4$  is realized of which only  $U_4$  is favourable to the event A.

Consequently,  $P(A|B) = \frac{1}{2}$ . Thus, the events A and

B are independent. It is clear that, proceeding in the same way, we can obtain that

$$P(B) = P(B|C) = \frac{1}{2} \text{ and } P(C) = P(C|A) = \frac{1}{2}$$

and get convinced that two other pairs of events B and C, C and A are also independent. Hence, the events A, B, C are independent pairwise.

Will the formula  $P(A \cap B \cap C) = P(A) \cdot P(B) \cdot P(C)$  be true for these events? Obviously, not, since

$$P(A \cap B \cap C) = \frac{1}{4} \text{ and } P(A) \cdot P(B) \cdot P(C) = \frac{1}{8}.$$

But the event  $A \cap B \cap C$  is favoured only by the elementary event  $U_4$ .

The events dealt in this example are pairwise independent, but they are not collectively independent. Indeed, the events A and  $B \cap C$  are dependent on each other, since the occurrence of the event  $B \cap C$  means the realization of the elementary event  $U_4$  in which the event A takes place necessarily, i.e.

$$P(A | B \cap C) = 1$$

and, consequently,  $P(A) \neq P(A | B \cap C)$ .

Thus, the events  $A_1, A_2, \dots, A_n$  are said to be mutually independent iff

$$P(A_i A_j) = P(A_i) P(A_j), \forall i, j, i \neq j.$$

$$P(A_i A_j A_k) = P(A_i) P(A_j) P(A_k), \forall i, j, k, i \neq j \neq k. \dots(5)$$

$$\dots \dots \dots \dots \dots \dots \dots \dots$$

$$P(A_1 A_2 \dots A_n) = P(A_1) P(A_2) \dots P(A_n).$$

If n events are completely independent, then every combination of n events, taken any number of times, should be independent. It follows that, the number of relations in the (n – 1) equations listed in (5) have  ${}^nC_2 + {}^nC_3 + \dots + {}^nC_n$  terms respectively. Thus, the total number of conditions among the relations (5) are

$${}^nC_2 + {}^nC_3 + \dots + {}^nC_n = (1 + 1)^n - {}^nC_0 - {}^nC_1 = 2^n - 1 - n.$$

**► Example 5.** Three coins are tossed simultaneously. Consider the event A ‘three heads or three tails’, B ‘atleast two heads’ and C ‘atmost two heads’. Of the pairs (A, B), (A, C) and (B, C), which are independent? which are dependent?

**► Solution** The sample space of the experiment is given by

$$S = \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\}$$

Clearly

$$A = \{HHH, TTT\} \quad B = \{HHH, HHT, HTH, THH\}$$

$$\text{and } C = \{HHT, HTH, THH, HTT, THT, TTH, TTT\}$$

Also

$$A \cap B = \{HHH\}, A \cap C = \{TTT\}, B \cap C = \{HHT, HTH, THH\}$$

Therefore

$$P(A) = \frac{2}{8} = \frac{1}{4}, P(B) = \frac{4}{8} = \frac{1}{2}, P(C) = \frac{7}{8}$$

$$\text{and } P(A \cap B) = \frac{1}{8}, P(A \cap C) = \frac{1}{8}, P(B \cap C) = \frac{3}{8}$$

Also

$$P(A \cap B) = \frac{1}{4} \times \frac{1}{2} = \frac{1}{8}, P(A) \cdot P(B) = \frac{1}{4} \times \frac{1}{2} = \frac{1}{8}$$

$$\text{and } P(B) \cdot P(C) = \frac{1}{2} \times \frac{7}{8} = \frac{7}{16}$$

$$\text{Thus } P(A \cap B) = P(A) \cdot P(B)$$

$$P(A \cap C) \neq P(A) \cdot P(C)$$

$$\text{and } P(B \cap C) \neq P(B) \cdot P(C)$$

Hence, the events (A and B) are independent, and the events (A and C) are (B and C) are dependent.

► **Example 6.** The event  $A$  is known to be independent of the events  $B$ ,  $B \cup C$  and  $B \cap C$ . Then prove that  $A$  is also independent of  $C$ .

► **Solution** We are given that, by independence,  
 $P(AB) = P(A)P(B)$ ,  $P[A(B \cup C)] = P(A)P(B \cup C)$ ,  
 $P[A(BC)] = P(A)P(BC)$  ... (1)

$$\begin{aligned}\text{Now } P(AB \cup AC) &= P(AB) + P(AC) - P[A(BC)] \\ &= P(A)P(B) + P(AC) - P(A)P(BC) \text{ [by (1)]} \\ P[A(B \cup C)] &= P(A)P(B \cup C) \\ &= P(A)[P(B) + P(C) - P(BC)] \quad \dots (2)\end{aligned}$$

Equating (i) and (ii) [because  $A(B \cup C) = AB \cup AC$ ] we get

$$P(AC) = P(A)P(C) \Rightarrow A \text{ is independent of } C.$$

We throw two dice  $D_1$  and  $D_2$  and define the events :

$$A = \{D_1 \text{ shows a } 6\} = \{(6, 1), (6, 2), (6, 3), (6, 4), (6, 5), (6, 6)\}; P(A) = 1/6$$

$$B = \{D_2 \text{ shows a } 6\} = \{(6, 1), (2, 6), (3, 6), (4, 6), (5, 6), (6, 6)\}; P(B) = 1/6$$

$$A = \{D_1, D_2 \text{ show the same number}\} = \{(1, 1), (2, 2), \dots, (6, 6)\}; P(C) = 1/6$$

$$A \cap B = B \cap C = C \cap A = A \cap B \cap C = \{(6, 6)\}$$

$$\Rightarrow P(AB) = P(BC) = P(CA) = P(ABC)$$

Obviously,  $P(AB) = 1/36 = P(A)P(B)$ , etc.

$$P(ABC) = 1/36 \neq P(A)P(B)P(C) = 1/216.$$

Consequently, these events are not independent.

► **Example 7.** Of three events  $A$ ,  $B$ ,  $C$  suppose that  $P(AB) = P(A)P(B)$ ;  $P(ABC) = P(A)P(B)P(C)$ . Show that  $AB$  and  $C$  are independent but  $A$  and  $C$  need not be independent.

► **Solution**  $P[(AB) \cap C] = P(ABC) = P(A)P(B)P(C)$   
 $= P(AB)P(C)$  since,  $P(AB) = P(A)P(B)$ .

This shows that  $AB$  and  $C$  are independent.

Now let  $B = \emptyset$  and suppose  $A$  and  $C$  are dependent events. Then

$$\begin{aligned}P(AB) &= P(\emptyset) = 0 = P(A)P(B); P(ABC) \\ &= P(\emptyset) = 0 = P(A)P(B)P(C).\end{aligned}$$

It follows that, the events  $A$  and  $C$  are dependent though the above conditions hold.

► **Example 8.** (i) If  $A$  and  $B$  are independent events and  $A \subseteq B$ , prove that either  $P(A) = 0$  or  $P(B) = 1$ .

(ii) If an event  $A$  is independent of itself, (i.e.  $A$  and  $A$  are indep.) show that  $P(A) = 0$  or  $P(A) = 1$ .

(iii) If  $P(A) = 0$  or  $1$ , show that  $A$  is independent with any event  $B$ , in particular  $A$  and  $A$  are independent.

► **Solution** (i) Here  $P(A \cap B) = P(A)P(B)$ . ... (1)  
 Since  $A \subseteq B$ , we get  $A \cap B = A$ , and (1) provides  $P(A)[1 - P(B)] = 0 \Rightarrow P(A) = 0$  or  $P(B) = 1$ .

(ii) Since  $A \cap A = A$ , (1) gives  $P(A) = P(A)P(A)$  so that  $P(A)[1 - P(A)] = 0$   
 $\Rightarrow P(A) = 0$  or  $P(A) = 1$ .

(iii) Let  $P(A) = 1$ . Since  $P(A \cup B) \geq P(A)$  and the probability measure cannot exceed unity, this really reads

$$1 = P(A \cup B) = P(A) + P(B) - P(AB)$$

$$\Rightarrow P(B) = P(AB).$$

$$\text{So } P(AB) = 1 \cdot P(B) = P(A) \cdot P(B)$$

$$\Rightarrow A \text{ and } B \text{ are independent.}$$

Let  $P(A) = 0$ . Since  $AB \subseteq A$ , we do have  $P(AB) \leq P(A) = 0$ .

Since probability measure is non-negative, this really gives

$$P(AB) = 0 \Rightarrow P(A)P(B) = 0. P(A)P(B) = P(AB)$$

This shows that  $A$  and  $B$  are independent.

In particular, replacing  $B$  by  $A$ , we see that  $A$  and  $A$  are independent.

► **Example 9.** If events  $A$ ,  $B$ ,  $C$  are independent with  $P(A) = p$ ,  $P(B) = q$ ,  $P(C) = r$ , find the probability that atleast two will occur among the three events.

► **Solution**  $P(AB \cup BC \cup CA) = P(AB) + P(BC) + P(CA) - 3P(ABC) + P(ABC)$   
 $= P(A)P(B) + P(B)P(C) + P(C)P(A) - 2P(A)P(B)P(C)$   
 $= pq + qr + rp - 2pqr.$

► **Example 10.** Let  $A$  and  $B$  be independent events. Find whether  $A \cup B$  and  $A \cap B$  are independent events.

► **Solution** Let  $A \cup B = C$ ,  $A \cap B = D$ , then using  
 $P(AB) = P(A)P(B)$ ,  
 $P(C) = P(A) + P(B) - P(A)P(B)$ ;  
 $P(D) = P(A)P(B)$ ,  
 $P(C \cap D) = P(D) \neq P(C)P(D)$  [ $\because C \cap D = D$ ]  
 This establishes that  $C$  and  $D$  are not independent events.

► **Example 11.** If A is independent of B and B is independent of C and also A is independent of BC, prove that C is independent of AB.

► **Solution**  $P(C|AB) = \frac{P(ABC)}{P(AB)} = \frac{P(A) P(BC)}{P(A) P(B)}$

[since A is independent of B and A is independent of BC]

$$= \frac{P(A) \cdot P(B) \cdot P(C)}{P(A) \cdot P(B)} = P(C)$$

[since B is independent of C]

### Practice Problems

[1]

1. A fair coin and an unbiased die are tossed. Let A be the event 'head appears on the coin' and B be the event '3 on the die'. Check whether A and B are independent events or not.
2. A die marked 1, 2, 3 in red and 4, 5, 6 in green is tossed. Let A be the event, 'the number is even', and B be the event, 'the number is red'. Are A and B independent?
3. Given that the events A and B are such that

$$P(A) = \frac{1}{2}, P(A \cup B) = \frac{3}{5} \text{ and } P(B) = p.$$

Find p if they are (i) mutually exclusive (ii) independent.

4. Events A and B are such that  $P(A) = 1/2$ ,  $P(B) = 7/12$  and  $P(\text{not } A \text{ or not } B) = 1/4$ . State whether A and B are independent?
5. In a high school public examination 15% of the students failed in Mathematics and 12% of the students failed in English. Also 3% of the students failed in both Mathematics and English. In an experiment of random selection of students from the school consider the two events; "failure in Maths" and "failure in English". Are these two events independent?
6. Let A and B be possible outcomes of an experiment and suppose that  $P(A) = 0.4$ ,  $P(B) = p$ ,  $P(A \cup B) = 0.7$

Find p if

- (a) A and B are disjoint
- (b) A and B are independent
- (c) A is a subevent of B,
- (d)  $P(A|B) = 0.4$ .

7. Find  $P(A)$ ,  $P(B)$  and  $P(C)$ , where mutually independent events A, B, C satisfy the relations :

$$P(AB) = 1/6, P(B \bar{C}) = \frac{1}{2} P(A), 3P(B) = 1/P(C)$$

8. Let event A involve event B, with  $P(A) = \frac{1}{3}$   $P(B)$ . Calculate the conditional probabilities  $P(A/B)$  and  $P(B/A)$  and determine whether or not these events are independent.
9. Let events A and B be disjoint, with  $P(A) \neq 0$  and  $P(B) \neq 0$ . Prove that A and B are dependent. It follows, in particular, that elementary outcomes of any probability experiment are dependent.
10. Three dice are tossed at random. Determine whether the following events are dependent or independent :  $C = \{\text{atleast two units will turn up}\}$  and  $D = \{\text{atmost two six will turn up}\}$  and  $D = \{\text{atmost two sixes will turn up}\}$ . Calculate the conditional probability  $P(D/C)$ .

### 4.10 ALGEBRA OF INDEPENDENT EVENTS

1. If A and B are two independent events then
  - (i)  $P(A \cup B) = P(A) + P(B) - P(A \cap B)$   
 $= P(A) + P(B) - P(A) \cdot P(B)$
  - (ii)  $P(A \cup B) = 1 - P(\bar{A} \cap \bar{B})$   
 $= 1 - P(\bar{A})P(\bar{B})$   
 $= 1 - P(\bar{A})P(\bar{B})$
2. If A, B and C are three independent events then
  - (i) the probability of intersection of these is equal to the product of the probabilities of these events, i.e.  
 $P(A \cap B \cap C) = P(A) P(B) P(C)$ .

- (ii) the probability that atleast one of the three events A, B or C occur is

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A)P(B) - P(B)P(C) - P(C)P(A) + P(A)P(B)P(C).$$

3. If  $A_1, A_2, \dots, A_n$  are  $n$  independent events then

$$P[A_1 \cup A_2 \cup \dots \cup A_n] = 1 - P(\overline{A_1} \cap \overline{A_2} \cap \dots \cap \overline{A_n}) \\ = 1 - P(\overline{A_1})P(\overline{A_2})\dots P(\overline{A_n})$$

4. Let the events  $A_1, A_2, \dots, A_n$  be independent and  $P(A_i) = p_i$  for  $i = 1, \dots, n$ .

- the probability that  $A_1$  happens and  $A_2$  fails is  $p_1(1 - p_2)$ .
- the probability that  $A_2$  happens and  $A_1$  fails is  $p_2(1 - p_1)$ .
- the probability that both  $A_1$  and  $A_2$  fail is  $(1 - p_1)(1 - p_2)$ .
- the probability that one and only one of the events  $A_1$  and  $A_2$  happen is  $p_1(1 - p_2) + p_2(1 - p_1) = p_1 + p_2 - 2p_1p_2$ .
- the probability that all the events  $A_1, A_2, \dots, A_n$  occur is  $p_1 p_2 \dots p_n$ .
- the probability that none of the events  $A_1, A_2, \dots, A_n$  occur is  $(1 - p_1)(1 - p_2)\dots(1 - p_n)$ .
- the probability that the first two events happen and rest fail is  $p_1 p_2(1 - p_3)(1 - p_4)\dots(1 - p_n)$ .

### Laplace's third principle

The probability of the concurrence of several independent events is the product of their separate probabilities.

► **Example 1.** If A and B are two independent events, then the probability of occurrence of atleast one of A and B is given by  $1 - P(A')P(B')$

► **Solution** We have

$P(\text{atleast one of}$

$$\begin{aligned} A \text{ and } B) &= P(A \cup B) \\ &= P(A) + P(B) - P(A \cap B) \\ &= P(A) + P(B) - P(A)P(B) \end{aligned}$$

$$\begin{aligned} &= P(A) + P(B) [1 - P(A)] \\ &= P(A) + P(B) \cdot P(A') \\ &= 1 - P(A') + P(B)P(A') \\ &= 1 - P(A') [1 - P(B)] \\ &= 1 - P(A')P(B') \end{aligned}$$

► **Example 2.** Supposing that it is 9 to 7 against a person A who is now 25 year of age living till he is 55, and 3 to 2 against a person B now 35 living till he is 65; find the chance that one atleast of these persons will be alive 30 years hence.

► **Solution** The chance that A will die within 30 years is  $9/16$ ;

the chance that B will die within 30 years is  $3/5$ ; therefore the chance that both will die is

$$\frac{9}{16} \times \frac{3}{5} = \frac{27}{80};$$

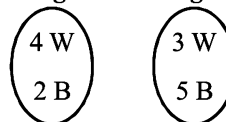
therefore the chance that both will not be dead,

$$\text{that is that atleast one will be alive, is } 1 - \frac{27}{80} = \frac{53}{80}.$$

► **Example 3.** A bag contains 4 white and 2 black balls. Another contains 3 white and 5 black balls. If one ball is drawn from each bag, find the probability that

- both are white;
- both are black;
- One is white and one is black

► **Solution** **Bag I** **Bag II**



Let  $A$  = the event of drawing a white ball from first bag

$B$  = the event of drawing a white ball from second bag

Then  $A'$  = the event of drawing a black ball from first bag and  $B'$  = the event of drawing a black ball from second bag.

Let  $E_1 = A \cap B$ ,  $E_2 = A' \cap B'$ ,  $E_3 = A \cap B'$ ,

$$E_4 = A' \cap B \text{ and } E = E_3 \cup E_4$$

Clearly,

$$P(A) = \frac{4}{6} = \frac{2}{3} \text{ and } P(B) = \frac{3}{8}, P(A') = \frac{1}{3}, P(B') = \frac{5}{8}$$

$$\begin{aligned} \text{(i)} \quad P(E_1) &= P(A \cap B) \\ &= P(A) \cdot P(B) \\ &[\because A \text{ and } B \text{ are independent events}] \end{aligned}$$

$$= \frac{2}{3} \cdot \frac{3}{8} = \frac{1}{4}$$

$$\begin{aligned} \text{(ii)} \quad P(E_2) &= P(A' \cap B') \\ &= P(A') \cdot P(B') \\ &[\because A' \text{ and } B' \text{ are independent events}] \end{aligned}$$

$$= \frac{1}{3} \cdot \frac{5}{8} = \frac{5}{24}$$

$$\begin{aligned} \text{(iii)} \quad P(E) &= P(E_3 \cup E_4) \\ &= P(E_3) + P(E_4) \\ &[\because E_3 \text{ and } E_4 \text{ are mutually exclusive}] \\ &= P(A \cap B') + P(A' \cap B) \\ &= P(A) \cdot P(B') + P(A') \cdot P(B) \\ &[\because A, B' \text{ are independent and } A', B \text{ are independent}] \end{aligned}$$

$$= \frac{2}{3} \cdot \frac{5}{8} + \frac{1}{3} \cdot \frac{3}{8} = \frac{13}{24}$$

► **Example 4.** A can hit a target 3 times in 5 shots, B : 2 times in 5 shots, C : 3 times in 4 shots. They fire a volley. Find the chance that:

- 2 shots hit
- Atleast 2 shots hit
- Exactly one of them hits.
- If only one of them hits the target, what is the chance that it was the first man?
- And if two hit, what is the chance that it is C who missed?

► **Solution** Here  $P(A) = p_1 = 3/5$ ,  $P(B) = p_2 = 2/5$ ,  $P(C) = p_3 = 3/4$ .

$$P(ABC) = p_1 p_2 p_3 = \frac{3}{5} \cdot \frac{2}{5} \cdot \frac{3}{4} = \frac{18}{100}$$

[all three hit the target independently]

$$P(\bar{A}BC) = q_1 p_2 p_3 = \frac{12}{100},$$

$$P(A\bar{B}C) = p_1 q_2 p_3 = \frac{27}{100}$$

$$P(AB\bar{C}) = p_1 p_2 q_3 = 6/100.$$

$$\begin{aligned} \text{(i)} \quad P(E_2) &= P\{\text{exactly one of the shots hit}\} \\ &= P(\bar{A}BC) + P(A\bar{B}C) + P(AB\bar{C}) \\ &= q_1 p_2 p_3 + p_1 q_2 p_3 + p_1 p_2 q_3 \\ &= (12 + 27 + 6)/100 = 45/100. \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad P(E_1) &= P\{\text{Exactly one of the shots hit}\} \\ &= P(\bar{A}\bar{B}C) + P(\bar{B}B\bar{C}) + P(A\bar{B}\bar{C}) \\ &= q_1 q_2 p_3 + q_1 p_2 p_3 + p_1 q_2 q_3 \\ &= (18 + 4 + 9)/100 = 31/100. \end{aligned}$$

$$\begin{aligned} \text{(iii)} \quad P\{L_2\} &= P\{\text{atleast 2 shots hit}\} \\ &= P(E_2) + P(ABC) \\ &= P(E_2) + p_1 p_2 p_3 = P(E_2) + P(ABC) \\ &= 0.45 + 0.18 = 0.63. \end{aligned}$$

$$\begin{aligned} \text{(iv)} \quad P\{A|E_1\} &= P(A \cap E_1)/P(E_1) \\ &= P(\bar{B}\bar{C}A)/P(E_1) \\ &= (9/100)/(31/100) = 9/31. \end{aligned}$$

$$\begin{aligned} \text{(v)} \quad P\{\bar{C}|E_2\} &= P\{AB\bar{C}\}/P(E_2) = 6/45 \\ &= 2/15. \end{aligned}$$

► **Example 5.** There are a white, b black and c red balls in an urn. Three of them are drawn at random. Find the probability that atleast two of them are of the same colour.

► **Solution** To find the probability of an event consisting in atleast two balls being the same colour, we pass to the complementary event  $\bar{A} = \{\text{all the balls are different in colour}\}$ .

$$\begin{aligned} P(\bar{A}) &= P\{\underbrace{wbr + wrb + rbw + \dots}_{6 \text{ combinations}}\} \\ &= 6 \frac{a}{a+b+c} \frac{b}{a+b+c-1} \frac{c}{a+b+c-2}. \end{aligned}$$

Hence

$$\begin{aligned} P(A) &= 1 - P(\bar{A}) \\ &= 1 - \frac{6abc}{(a+b+c)(a+b+c-1)(a+b+c-2)}. \end{aligned}$$

► **Example 6.** There is a target at which n shots are fired, every shot hitting the target independently with probability p. If the probability of hitting the target is to be not less than P, then find the value of n.



► **Solution** Probability of hitting the target  $\geq P$

$$1 - (1 - p)^n \geq P$$

$$1 - P \geq (1 - p)^n$$

$$(1 - p)^n \leq 1 - P$$

$$n \geq \frac{\ell n (1 - P)}{\ell n (1 - p)}$$

[Note that  $\ln(1 - p)$  &  $\ln(1 - P)$  are negative

► **Example 7.** A tennis match of best of 5 sets is played by two players 'A' and 'B'. The probability that first set is won by A is  $1/2$  and if he loses the first then probability of his winning of next set is  $1/4$  otherwise it remains same. Find the probability that A wins the match.

► **Solution** In the tennis match of best of 5 sets. A can win the match. If score of A against the score of B is (3, 0), (3, 1) or (3, 2).

The probability of A's doing the score of (3, 0) is  $\frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8}$ .

The probability of A's winning by the score of (3, 1)

$$= \frac{1}{4} \left( \frac{1}{2} \right)^3 + \frac{1}{2} \cdot \left( \frac{1}{2} \right) \cdot \left( \frac{1}{4} \right) \left( \frac{1}{2} \right) + \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{4} = \frac{3}{32}.$$

The probability of A's winning by the score of (3, 2) =  $P(\text{A loses I}^{\text{st}} \text{ and II}^{\text{nd}} \text{ sets}) + P(\text{A loses I}^{\text{st}} \text{ and III}^{\text{rd}} \text{ sets}) + P(\text{A loses I}^{\text{st}} \text{ and IV}^{\text{th}} \text{ sets}) + P(\text{A loses II}^{\text{nd}} \text{ and III}^{\text{rd}} \text{ set}) + P(\text{A loses II}^{\text{nd}} \text{ and IV}^{\text{th}} \text{ sets}) + P(\text{A loses III}^{\text{rd}} \text{ and IV}^{\text{th}} \text{ sets})$

$$\begin{aligned} & \frac{1}{2} \left( \frac{3}{4} \right) \left( \frac{1}{4} \right) \left( \frac{1}{2} \right)^2 + \frac{1}{2} \left( \frac{1}{4} \right) \left( \frac{1}{2} \right) \left( \frac{1}{4} \right) \left( \frac{1}{2} \right) \\ & + \frac{1}{2} \left( \frac{1}{4} \right) \left( \frac{1}{2} \right) \left( \frac{1}{2} \right) \left( \frac{1}{4} \right) + \frac{1}{2} \left( \frac{1}{2} \right) \left( \frac{3}{4} \right) \left( \frac{1}{4} \right) \frac{1}{2} \\ & + \frac{1}{2} \left( \frac{1}{2} \right) \left( \frac{1}{4} \right) \left( \frac{1}{2} \right) \left( \frac{1}{4} \right) + \frac{1}{2} \left( \frac{1}{2} \right) \left( \frac{1}{2} \right) \left( \frac{3}{4} \right) \frac{1}{4} \\ & = \frac{3}{128} + \frac{1}{128} + \frac{1}{128} + \frac{3}{128} + \frac{1}{128} + \frac{3}{128} = \frac{12}{128} \end{aligned}$$

The probability that A wins the match

$$= \frac{1}{8} + \frac{3}{32} + \frac{12}{128} = \frac{16 + 12 + 12}{128} = \frac{40}{128} = \frac{5}{16}.$$

► **Example 8.** Of the total amount of parts produced by a factory 5% are defective. Twenty parts are chosen and inspected. What is the probability of there being atleast one defective among them?

► **Solution** For any part produced at the factory, the probability of being defective is by hypothesis  $p = 0.05 = P(A_k)$ ,  $k = 1, 2, \dots, 20$ , where the event  $A_k = \{\text{the } k\text{th drawn out part is defective}\}$ . Obviously we are interested in the event  $A_1 \cup A_2 \cup \dots \cup A_{20}$ . It may be assumed under the conditions of an adjusted manufacturing method that events  $A_0, A_2, \dots, A_{20}$  are collectively independent. We have

$$\begin{aligned} P(A_1 \cup A_2 \cup \dots \cup A_{20}) &= 1 - \prod_{k=1}^{20} P(\bar{A}_k) \\ &= 1 - 0.95^{20} \approx 0.64. \end{aligned}$$

► **Example 9.** There are ten pairs of shoes in a bag. First A draws one shoe from the bag, then B draws one shoe, then A draws one and finally B draws one shoe. Drawn shoes are not replaced. Show that the chance of A drawing a pair is same as the chance of B drawing a pair. Also find probability that neither draws a pair.

► **Solution** If  $E_1$  be the event that A draws a pair and  $E_2$  be the event that B draws a pair, then

$$\begin{aligned} P(E_1) &= \frac{20}{20} \cdot \frac{18}{19} \cdot \frac{1}{18} = \frac{1}{19} & P(E_2) \\ &= \frac{20}{20} \cdot \frac{18}{19} \cdot \frac{17}{18} \cdot \frac{1}{17} = \frac{1}{19} \end{aligned}$$

Clearly,  $P(E_1) = P(E_2)$

$$P(E_1 \cap E_2) = \frac{20}{20} \cdot \frac{18}{19} \cdot \frac{1}{18} \cdot \frac{1}{17} = \frac{1}{323}$$

$$\begin{aligned} \therefore \text{The probability, that neither A nor B draws a pair} \\ &= P(E_1 \cup E_2)' = 1 - P(E_1 \cup E_2) \\ &= 1 - [P(E_1) + P(E_2) - P(E_1 \cap E_2)] \\ &= 1 - \frac{1}{19} - \frac{1}{19} + \frac{1}{323} = \frac{290}{323} \end{aligned}$$

► **Example 10.** A fair coin is tossed until one of the two sides occurs twice in a row. Prove that the probability that the number of tosses required is even is  $2/3$ .

► **Solution** Let  $P(HH)$  denote the probability of occurrence of head in first and second tosses.

Similarly, let  $P(HTHH)$  denote the probability of occurrence of head, tail, head and head in first, second, third and fourth tosses respectively.

Let  $A$  = the event of occurrence of two consecutive heads in the last two tosses and non-occurrence of two consecutive heads before the last two tosses.

$B$  = the event of occurrence of two consecutive tails in the last two tosses and non-occurrence of two consecutive tails before the last two tosses.

Then  $A = \{HH, HTHH, HTHTHH, \dots\}$

and  $B = \{TT, THTT, THTHTT, \dots\}$

Now  $P(A) = P(HH) + P(HTHH) + P(HTHTHH) + \dots$

$$= \frac{1}{2^2} + \frac{2}{2^4} + \frac{1}{2^6} + \dots = \frac{1/4}{1 - (1/4)} = \frac{1}{3}.$$

Similarly  $P(B) = 1/3$ . Since,  $A$  and  $B$  are mutually exclusive, therefore, the required probability,

$$P(A \cap B) = P(A) + P(B) = \frac{1}{3} + \frac{1}{3} = \frac{2}{3}$$

► **Example 11.** Compare the probability of at least one six in 4 tosses of a fair die with the probability of at least one double-six in 24 tosses of two fair dice.

► **Solution** Let  $A = \{\text{at least one 6 in 4 tosses of a single die}\}$ ,

$B = \{\text{at least one double 6 in 24 tosses of 2 dice}\}$ .

Then  $P(A) = 1 - P(\bar{A}) = 1 - P(\text{no six in 4 tosses of one die}) = 1 - (5/6)^4 = 0.518$ .

$P(B) = 1 - P(\bar{B}) = 1 - P(\text{no double 6 in 24 tosses of dice-pair}) = 1 - (35/36)^{24} = 0.491$ .

► **Example 12.** A significant message is sent simultaneously over  $n$  communication channels and is repeated  $k$  times over each channel for the sake of reliability. During one transmission a message (independently of the other messages) is distorted with probability  $p$ . Each communication channel (independently of the other channels) is "blocked up" with noise with probability  $Q$  : a "blocked-up"

channel cannot transmit any messages. Find the probability of the event  $A = \{\text{a message is transmitted in a correct form atleast once}\}$ .

► **Solution** We introduce the event

$B = \{\text{a message is transmitted over one communication channel without distortion atleast once}\}$ .

For the event  $B$  to occur, first the channel must not be "blocked up" with noise and, second, at least one message sent over it must not be distorted :

$$P(B) = (1 - Q)(1 - p^k).$$

The probability of the event  $A$ , which is that the event  $B$  occurs atleast over one channel, is

$$P(A) = 1 - [1 - P(B)]^n = 1 - [1 - (1 - Q)(1 - p^k)]^n.$$

► **Example 13.** An air battle between a bomber and two attacking fighters is going on. The bomber is the first to fire; it fires once at each fighter and brings it down with probability  $p_1$ . If a fighter is not brought down, it fires at the bomber irrespective of the destiny of the other fighter and brings it down with probability  $p_2$ . Find the probabilities of the following outcomes :

$A = \{\text{the bomber is brought down}\}$ ;

$B = \{\text{both fighters are brought down}\}$ ;

$C = \{\text{at least one fighter is brought down}\}$ ;

$D = \{\text{at least one aircraft is brought down}\}$ ;

$F = \{\text{exactly one fighter is brought down}\}$ ;

$B = \{\text{exactly one aircraft is brought down}\}$ ;

► **Solution** The probability of one of the fighters bringing down the bomber is  $(1 - p_1)p_2$  ; the probability that neither of them brings down the bomber is  $[1 - (1 - p_1)p_2]^2$ .

Hence  $P(A) = 1 - [1 - (1 - p_1)p_2]^2$ ,  $P(B) = p_1^2$ ,

$$P(C) = 1 - (1 - p_1)^2,$$

$$P(D) = 1 - (1 - p_1)^2(1 - p_2)^2,$$

$$P(E) = 2p_1(1 - p_1).$$

The event  $F$  can be represented in the form

$$F = F_1 \cup F_2 \cup F_3, \text{ where}$$

$F_1 = \{\text{the bomber is brought down and both fighters are safe}\}$ ;

$F_2 = \{\text{the first fighter is brought down and the second fighter and the bomber are safe}\}$ ;

$F_3 = \{\text{the second fighter is brought down and the first fighter and the bomber are safe}\}.$

$$\begin{aligned} P(F_1) &= (1 - p_1)^2 [1 - (1 - p_2)^2], \\ P(F_2) &= P(F_3) = p_1 (1 - p_1) (1 - p_2), \\ P(F) &= (1 - p_1)^2 [1 - (1 - p_2)^2] \\ &\quad + 2p_1(1 - p_1) (1 - p_2). \end{aligned}$$

► **Example 14.** An instrument is being tested. Upon each trial the instrument fails with probability  $p$ . After the first failure the instrument is repaired, after the second failure it is considered to be unfit for operation. Find the probability that the instrument is rejected exactly in the  $k$ th trial.

► **Solution** For the given event to occur, it is necessary, first, that the instrument should fail in the  $k$ th trial, the probability of that event being  $p$ . In addition, it is necessary that in the preceding  $k-1$  trials the instrument should fail exactly once; the probability of that event being  $(k-1)p(1-p)^{k-2}$ . The required probability is equal to  $(k-1)p^2(1-p)^{k-2}$ .

► **Example 15.** An instrument consisting of  $k$  units is in use for a period  $t$ . During that time, the first unit may fail with probability  $q_1$ , the second with probability  $q_2$  and so on. A repairman is summoned to check the instrument. He checks each unit and either locates and clears the fault, if one exists, with probability  $p$  or passes it as faultless with probability  $q = 1 - p$ . Find the probability of the event  $A = \{\text{at least one unit remains faulty after the inspections}\}$ .

► **Solution** The probability that the  $i$ th unit remains faulty is equal to the probability that it became faulty during the time  $t$  multiplied by the probability that the repairman failed to locate the fault:  $q_i q$ . The probability that this event will occur in at least one unit is

$$P(A) = 1 - \prod_{i=1}^k (1 - q_i q).$$

► **Example 16.** At a railway station a passenger leaves his luggage in a locker which is opened by dialling a three-digit code (say, 253, 009, 325, etc.). The passenger chooses the code, closes the locker and leaves for the town. A strange man, who does not know the code, tries to open the locker by dialling

three digits at random. Find the probabilities of the following events:

$A = \{\text{the locker opens on the first trial}\};$

$B = \{\text{the locker opens after } k \text{ trials}\}.$

► **Solution**  $P(A) = 0.1 \cdot 0.1 \cdot 0.1 = 0.001$ .

If  $k$  trials are made, then it is natural to assume that the unsuccessful combinations are not repeated. We pass from  $B$  to a complementary event  $\bar{B} = \{\text{all } k \text{ trials are unsuccessful}\}$ .

Then  $P(B) = 1 - P(\bar{B})$

$$\begin{aligned} &= 1 - \frac{999}{1000} \cdot \frac{998}{999} \cdots \frac{1000-k+1}{1000-k+2} \cdot \frac{1000-k}{1000-k+1} \\ &= 1 - \frac{1000-k}{1000} = \frac{k}{1000}. \end{aligned}$$

It stands to reason that formula is meaningful only for  $k < 1000$ ;

for  $k \geq 1000$  the probability of finding the correct combination  $P(B) = 1$ .

► **Example 17.** A pair of dice is thrown until either a 4 or 6 appears. Find the probability that a 6 occurs first.

► **Solution** Let  $E_n$  denote the event that a 6 occurs in the  $n$ th throw and no 4 or 6 occurs in the first  $(n-1)$  throws. The probability for this is calculated as follows.

Sample points corresponding to 4 are

$(1, 3); (2, 2); (3, 1)$

Sample points corresponding to 6 are

$(1, 5); (2, 4); (3, 3); (4, 2); (5, 1).$

$$\text{So } P(E_n) = (1 - 8/36)^{n-1} \cdot 5/36 = \left(\frac{7}{9}\right)^{n-1} \times \frac{5}{36}$$

The probability that 6 occurs first

$$\begin{aligned} &= P(E_1 \cup E_2 \cup E_3 \cup \dots) \\ &= P(E_1) + P(E_2) + P(E_3) + \dots \text{ since } E_i \text{'s are m.e.} \\ &= 5/36 + 7/9 \times 5/36 + (7/9)^2 \times 5/36 + \dots \end{aligned}$$

This is an infinite geometric series with first term  $= 5/36$  and common ratio  $= 7/9$ .

Therefore, its sum is

$$= \frac{5/36}{1 - 7/9} = \frac{5}{36} \times \frac{9}{2} = \frac{5}{8}.$$

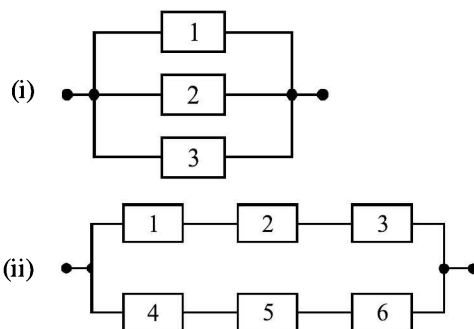
This is the required probability.

## Practice Problems

[J]

1. A town has two doctors A and B operating independently. If the probability that doctor A is available is 0.9 and that for B is 0.8, show that the probability that atleast one doctor is available when needed, is 0.98.
2. The probability that a husband will live 15 more years is  $\frac{1}{4}$ , and that his wife will live 15 more years is  $\frac{1}{3}$ . Find the probability that in 15 years hence, (i) both will be alive (ii) neither will be alive, (iii) atleast one will be alive, (iv) only the wife will be alive (v) only the husband will be alive.
3. A language class has only three students A, B, C and they independently attend the class. The chances of attendance of A, B, C on any given day are  $\frac{1}{2}$ ,  $\frac{2}{3}$  and  $\frac{3}{4}$  respectively. Find the chance that the total number of attendances in two consecutive days is exactly 3.
4. Two balls are drawn at random with replacement from a box containing 10 black and 8 red balls. Find the probability that
  - (i) Both balls are red,
  - (ii) First ball is black and second is red,
  - (iii) One of them is black and other is red.
5. Probability of solving a specific problem independently by A and B are  $\frac{1}{2}$  and  $\frac{1}{3}$  respectively. If both try to solve the problem independently, find the probability that
  - (i) the problem is solved
  - (ii) exactly one of them solves the problem.
6. In a hostel, 60% of the students read Hindi newspaper, 40% read English newspaper and 20% read both Hindi and English newspapers. A student is selected at random.
  - (i) Find the probability that he reads neither Hindi nor English newspapers.
  - (ii) If he reads Hindi newspaper, find the probability that he reads English newspaper.
  - (iii) If he reads English newspaper, find the probability that he reads Hindi news paper.
7. The outcome of an experiment is equally likely to be one of the four points in 3-dimensional space with coordinates (1, 0, 0), (0, 1, 0), (0, 0, 1) and (1, 1, 1). Let A, B, C be the events x-coordinate = 1, y-coordinate = 1, z-coordinate = 1 respectively. Show that the events A, B, C are pairwise independent.
8. Of three independent events A, B, C, the probability that A only should happen is  $\frac{1}{4}$ , the probability that B only should happen is  $\frac{1}{8}$  and the probability that C only should happen is  $\frac{1}{12}$ . Show that the (unconditional) probabilities of the three events are  $\frac{1}{2}$ ,  $\frac{1}{3}$ ,  $\frac{1}{4}$  respectively.
9. An experiment consists of tossing a coin and a die. If H is the event that head comes up in tossing the coin and D is the event that 3 or 6 comes up in tossing the die, find the probability for the following events:  
 $P(H)$ ,  $P(\bar{D})$ ,  $P(\overline{HD})$ ,  $P(H/D)$ ,  $P(\bar{H} \cup \bar{D})$
10. If A and B are independent events such that  $AB \subset C$ , and  $\bar{A} \bar{B} \subset \bar{C}$ , show that  $P(A)P(C) \leq P(AB)$ .
11. A and B alternate shots in a duel. The probability of A hitting B is  $p_1$  and probability of B hitting A is  $p_2$ . If A fires first, show that the chance that A wins the duel is  $p_1/(1 - q_1q_2)$ .
12. A man alternately tosses a coin and throws a die, beginning with a coin. Show that the chance that he will get a head before he gets a "5 or 6" on the die is  $\frac{3}{5}$ .
13. A and B play for a prize; A is to throw a dice first, and is to win if the throws 6. If he fails B is to throw, and to win if he throws 6 or 5. If he fails, A is to throw again and to win with 6 or 5 or 4, and so on. Find the chance of each player.

14. When the ignition is turned on, the engine picks up with probability  $p$ . Find (i) the probability that the engine picks up when the ignition is turned on for the second time (ii) the probability that it is necessary to switch on the ignition no more than two times for the engine to begin working.
15. From an urn containing six white and four black balls, balls are drawn one at a time at random until the first black ball appears. Find the probability that a fourth drawing will have to be done if the sampling is performed:
- with replacement;
  - without replacement.
16. A workshop produces picture tubes for TV sets, with 70% of all tubes intended for colour TV sets and 50% of all the output are exported, with 40% of all picture tubes intended for colour TV sets exported. Find the probability of a randomly chosen and inspected picture tube being intended for a black and white TV set and it is to be exported.
17. The following problems give connections of elements which form a circuit with one input and one output. It is assumed that failures of elements are collectively independent events. The reliability  $p_k$  of the  $k$ th elements is assumed to be known (correspondingly  $q_k = 1 - p_k$  is the probability of the failure of the  $k$ th element). A failure of any one of the elements results in a signal being interrupted in the branch of the circuit containing the element. Calculate the reliability  $p$  of each of the circuit.



18. A person takes 4 tests in succession. The probability of his passing first test is  $p$ , that of his passing each succeeding test is  $p$  or  $p/2$  according as he passes or fails the preceding one. He qualifies provided he passes atleast 3 tests. Show that his chance of qualifying is  $p^3$

$$+ \frac{3}{2} p^3 q.$$

## 4.11 RANDOM VARIABLES

### Random Variables

It is frequently the case when an experiment is performed that we are mainly interested in some function of the outcome as opposed to the actual outcome itself. For instance, in tossing dice we are often interested in the sum of the two dice and are not really concerned about the separate value of each dice. That is, we may be interested in knowing that the sum is 7 and not be concerned over whether the actual outcome was (1, 6) or (2, 5) or (3, 4) or (4, 3) or (5, 2) or (6, 1). Also, in coin flipping, we may be interested in the total number of heads that occur and not care at all about the actual head-tail sequence that results.

In all of the above experiments, we have a rule which assigns to each outcome of the experiment a single real number. This single real number may vary with different outcomes of the experiment. Hence, it is a variable. Also its value depends upon the outcome of a random experiment and, hence, is called random variable. A random variable is usually denoted by  $X$ . If you recall the definition of a function, you will realise that the random variable  $X$  is really speaking a function whose domain is the set of outcomes (or sample space) of a random experiment. A random variable can take any real value, therefore, its co-domain is the set of real numbers. Hence, a random variable can be defined as follows:

A random variable is a real valued function whose domain is the sample space of a random experiment. For example, let us consider the experiment of tossing a coin two times in succession. The sample space of the experiment is  $S = \{HH, HT, TH, TT\}$ . If  $X$  denotes the number of heads obtained, then  $X$  is a random variable and for each outcome, its value is as given below :

$X(HH) = 2, X(HT) = 1, X(TH) = 1, X(TT) = 0$   
 More than one random variables can be defined on the same sample space. For example, let  $Y$  denote the number of heads minus the number of tails for each outcome of the above sample space  $S$ .

Then  $Y(HH) = 2, Y(HT) = 0, Y(TH) = 0, Y(TT) = -2$ . Thus,  $X$  and  $Y$  are two different random variables defined on the same sample space  $S$ .

Discrete random variable is a random variable that can take on at most a countable number of possible values is said to be discrete.

► **Example 1.** A person plays a game of tossing a coin thrice. For each head, he is given Rs 2 by the organiser of the game and for each tail, he has to give Rs 1.50 to the organiser. Let  $X$  denote the amount gained or lost by the person. Show that  $X$  is a random variable and exhibit it as a function on the sample space of the experiment.

► **Solution**  $X$  is a number whose values are defined on the outcomes of a random experiment. Therefore,  $X$  is a random variable.

Now, sample space of the experiment is  
 $S = \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\}$   
 Then  $X(HHH) = \text{Rs}(2 \times 3) = \text{Rs } 6$

$$\begin{aligned} X(HHT) &= X(HTH) = X(THH) \\ &= \text{Rs}(2 \times 2 - 1 \times 1.50) = \text{Rs } 2.50 \end{aligned}$$

$$\begin{aligned} X(HTT) &= X(THT) = X(TTH) \\ &= \text{Rs}(1 \times 2) - (2 \times 1.50) = -\text{Rs } 1 \end{aligned}$$

$$\text{and } X(TTT) = -\text{Rs}(3 \times 1.50) = -\text{Rs } 4.50$$

where, minus sign shows the loss to the player. Thus, for each element of the sample space,  $X$  takes a unique value, hence,  $X$  is a function on the sample space whose range is

$$\{-1, 2.50, -4.50, -4.50, 6\}$$

To give a complete characteristic of a random variable, it is first of all necessary to know the values it can attain. But this is, of course, not sufficient. In addition, we must know the probability with which a random variable takes on this or that value.

We shall denote a random variable by  $X$ , its values by  $x_1, x_2, \dots, x_n$ ,

and the respective probabilities with which these values are attained by

$$p_1, p_2, \dots, p_n.$$

If for a certain random variable  $X$  all the values  $x_1, x_2, \dots, x_n$  which can be attained by it, and all the probabilities  $p_1, p_2, \dots, p_n$  with which these values

are attained are known, then we say that the law of distribution of the random variable  $X$  or, simply, the distribution of the variable  $X$  is specified.

It is convenient to write the distribution law in a tabular form:

$$\begin{array}{ccccccc} x_1 & x_2 & x_3 & \dots & x_k & \dots & x_n \\ p_1 & p_2 & p_3 & \dots & p_k & \dots & p_n \end{array} \quad \dots(1)$$

Consider  $n$  random events :

$A_1$  – the random variable  $X$  attained the value  $x_1$ ,

$A_2$  – the random variable  $X$  attained the value  $x_2$ ,

.....

$A_n$  – the random variable  $X$  attained the value  $x_n$ .

The events  $A_1, A_2, \dots, A_n$  are mutually exclusive in a single realization of the trial, since the random variable  $X$  can take on only one of the values  $x_1, x_2, \dots, x_n$ . It is also obvious that the sum of the events  $A_1, A_2, \dots, A_n$  is a certain event, i.e.

$$A_1 \cup A_2 \cup \dots \cup A_n = S,$$

since one of the values  $x_1, x_2, \dots, x_n$  is necessarily attained by the random variable.

Therefore, by the addition theorem for mutually exclusive events, we obtain

$$\begin{aligned} P(A_1) + P(A_2) + \dots + P(A_n) &= P(S) \\ &= 1, p_1 + p_2 + \dots + p_n = 1 \end{aligned}$$

or in a briefer notation

$$\sum_{k=1}^n p_k = 1 \quad \dots(2)$$

Consequently, the sum of all numbers standing in the second row of table (1) representing the distribution law for the random variable  $X$  must be equal to unity.

A **probability distribution** or probability mass function spells out how a total probability of 1 is distributed over several values of a random variable.

Suppose that our experiment consists of tossing 3 fair coins. If we let  $Y$  denote the number of heads appearing, then  $Y$  is a random variable taking on one of the values 0, 1, 2, 3 with respectively probabilities

$$P\{Y = 0\} = P\{(T, T, T)\} = \frac{1}{8}$$

$$P\{Y = 1\} = P\{(T, T, H), (T, H, T), (H, T, T)\} = \frac{3}{8}$$

$$P\{Y = 2\} = P\{(T, H, H), (H, T, H), (H, H, T)\} = \frac{3}{8}$$

$$P\{Y = 3\} = P\{(H, H, H)\} = \frac{1}{8}$$

Since  $Y$  must take on one of the values 0 through 3, we must have

$$1 = P\left(\bigcup_{i=0}^3 \{Y = i\}\right) = \sum_{i=0}^3 P\{Y = i\}.$$

It is often instructive to present the probability mass function in a graphical format by plotting  $p(x_i)$  on

the y-axis against  $x_i$  on the x-axis. For instance, if the probability mass function of  $X$  is

$$p(0) = \frac{1}{4} \quad p(1) = \frac{1}{2} \quad p(2) = \frac{1}{4}$$

We can represent this graphically as shown in Figure 1. Similarly, a graph of the probability mass function of the random variable representing the sum when two dice are rolled looks like the one shown in Figure 2.

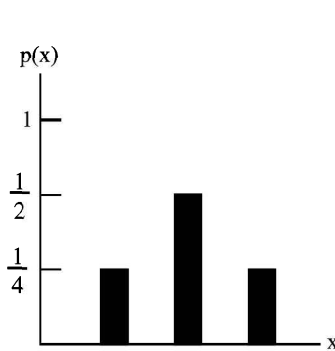


Figure 1

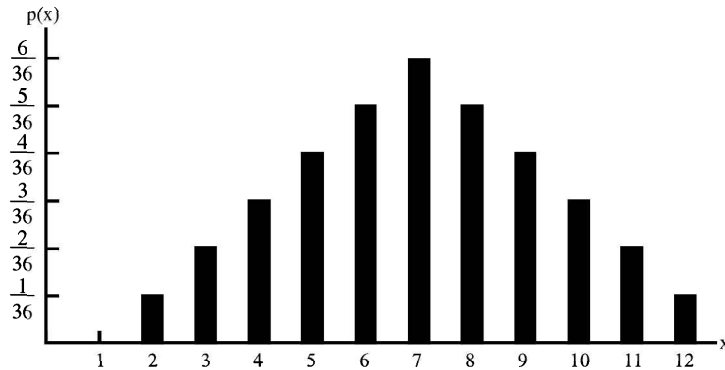


Figure 2

► **Example 2.** Two cards are drawn successively with replacement from a well-shuffled deck of 52 cards. Find the probability distribution of the number of aces.

► **Solution** The number of aces is a random variable. Let it be denoted by  $X$ . Clearly,  $X$  can take the values 0, 1, or 2.

Now, since the draws are done with replacement, therefore, the two draws form independent experiments. Therefore,

$$\begin{aligned} P(X = 0) &= P(\text{non-ace and non-ace}) \\ &= P(\text{non-ace}) \times P(\text{non-ace}) \\ &= \frac{48}{52} \times \frac{48}{52} = \frac{144}{169} \end{aligned}$$

$$\begin{aligned} P(X = 1) &= P(\text{ace and non-ace or non-ace and ace}) \\ &= P(\text{ace and non-ace}) + P(\text{non-ace and ace}) \\ &= P(\text{ace}) \cdot P(\text{non-ace}) + P(\text{non-ace}) \cdot P(\text{ace}) \\ &= \frac{4}{52} \times \frac{48}{52} = \frac{1}{169} \end{aligned}$$

Thus, the required probability distribution is

|        |                   |                  |                 |
|--------|-------------------|------------------|-----------------|
| $X$    | 0                 | 1                | 2               |
| $P(X)$ | $\frac{144}{169}$ | $\frac{24}{169}$ | $\frac{1}{169}$ |

► **Example 3.** Find the probability distribution of number of doublets in three throws of a pair of dice.

► **Solution** Let  $X$  denote the number of doublets. Possible doublets are

(1, 1), (2, 2), (3, 3), (4, 4), (5, 5), (6, 6)

Clearly,  $X$  can take the value 0, 1, 2, or 3.

$$\text{Probability of getting a doublet} = \frac{6}{36} = \frac{1}{6}$$

$$\text{Probability of not getting a doublet} = 1 - \frac{1}{6} = \frac{5}{6}$$

$$\text{Now } P(X = 0) = P(\text{no doublet}) = \frac{5}{6} \times \frac{5}{6} \times \frac{5}{6} = \frac{125}{216}$$

$$\begin{aligned} P(X = 1) &= P(\text{one doublet and two non-doublets}) \\ &= \frac{1}{6} \times \frac{5}{6} \times \frac{5}{6} + \frac{5}{6} \times \frac{1}{6} \times \frac{5}{6} + \frac{5}{6} \times \frac{5}{6} \times \frac{1}{6} \end{aligned}$$

$$= 3 \left( \frac{1}{6} \times \frac{5^2}{6^2} \right) = \frac{75}{216}$$

$P(X = 2) = P(\text{two doublets and one non-doublet})$

$$= \frac{1}{6} \times \frac{1}{6} \times \frac{5}{6} + \frac{1}{6} \times \frac{5}{6} \times \frac{1}{6} + \frac{5}{6} \times \frac{1}{6} \times \frac{1}{6}$$

$$= 3 \left( \frac{1}{6} \times \frac{5}{6} \right) = \frac{15}{216}$$

and  $P(X = 3) = P(\text{three doublets})$

$$= \frac{1}{6} \times \frac{1}{6} \times \frac{1}{6} = \frac{1}{216}$$

Thus, the required probability distribution is

|      |                   |                  |                  |                 |
|------|-------------------|------------------|------------------|-----------------|
| X    | 0                 | 1                | 2                | 3               |
| P(X) | $\frac{215}{216}$ | $\frac{75}{216}$ | $\frac{15}{216}$ | $\frac{1}{216}$ |

Verification : Sum of the probabilities.

$$\sum_{i=1}^n p_i = \frac{125}{216} + \frac{75}{216} + \frac{15}{216} + \frac{1}{216}$$

$$= \frac{125 + 75 + 15 + 1}{216} = \frac{216}{216} = 1$$

► **Example 4.** Let  $X$  denote the number of hours you study during a randomly selected school day. The probability that  $X$  can take the values  $x$ , has the following form, where  $k$  is some unknown constant.

$$P(X = x) = \begin{cases} 0.1, & \text{if } x = 0 \\ kx, & \text{if } x = 1 \text{ or } 2 \\ k(5 - x), & \text{if } x = 3 \text{ or } 4 \\ 0, & \text{otherwise} \end{cases}$$

- Find the value of  $k$
- What is the probability that you study atleast two hours ? Exactly two hours ? Atmost two hours ?

► **Solution** The probability distribution of  $X$  is

|      |     |     |      |      |     |
|------|-----|-----|------|------|-----|
| X    | 0   | 1   | 2    | 3    | 4   |
| P(X) | 0.1 | $k$ | $2k$ | $2k$ | $k$ |

- We know that  $\sum_{i=1}^n p_i = 1$

Therefore  $0.1 + k + 2k + 2k + k = 1$

i.e.  $k = 0.15$

- $P(\text{you study atleast two hours})$   
 $= P(X \geq 2) = P(X = 2) + P(X = 3) + P(X = 4)$   
 $= 2k + 2k + k = 5k = 5 \times 0.15 = 0.75$   
 $P(\text{you study exactly two hours})$   
 $= P(X = 2) = 2k = 2 \times 0.15 = 0.3$   
 $P(\text{you study atmost two hours})$   
 $= P(X \leq 2) = P(X = 0) + P(X = 1) + P(X = 2)$   
 $= 0.1 + k + 2k = 0.1 + 3k = 0.1 + 3 \times 0.15$   
 $= 0.55$

### Practice Problems

[K]

- State which of the following are not the probability distributions of a random variable. Give reasons for your answer.
  - |      |     |     |     |
|------|-----|-----|-----|
| X    | 0   | 1   | 2   |
| P(X) | 0.4 | 0.4 | 0.2 |
  - |      |     |     |     |      |     |
|------|-----|-----|-----|------|-----|
| X    | 0   | 1   | 2   | 3    | 4   |
| P(X) | 0.1 | 0.5 | 0.2 | -0.1 | 0.3 |
  - |      |     |     |     |
|------|-----|-----|-----|
| Y    | -1  | 0   | 1   |
| P(Y) | 0.6 | 0.1 | 0.2 |
  - |      |     |     |     |     |      |
|------|-----|-----|-----|-----|------|
| Z    | 3   | 2   | 1   | 0   | -1   |
| P(Z) | 0.3 | 0.2 | 0.4 | 0.1 | 0.05 |
- An urn contains 5 red and 2 black balls. Two balls are randomly drawn. Let  $X$  represent the number of black balls. What are the possible values of  $X$  ? Is  $X$  a random variable ?
- Let  $X$  represent the difference between the number of heads and the number of tails obtained when a coin is tossed 6 times. What are possible values of  $X$  ?
- Find the probability distribution of the number of successes in two tosses of a die, where a success is defined as
  - number greater than 4
  - six appears on atleast one die
- From a lot of 30 bulbs which include 6 defectives, a sample of 4 bulbs is drawn at random with replacement. Find the probability distribution of the number of defective bulbs.
- A coin is biased so that the head is 3 times as likely to occur as tail. If the coin is tossed twice,



find the probability distribution of number of tails.

7. A random variable  $X$  has the following probability distribution :

|        |   |     |      |      |      |       |        |          |
|--------|---|-----|------|------|------|-------|--------|----------|
| $X$    | 0 | 1   | 2    | 3    | 4    | 5     | 6      | 7        |
| $P(X)$ | 0 | $k$ | $2k$ | $2k$ | $3k$ | $k^2$ | $2k^2$ | $7k^2+k$ |

Determine

- (i)  $k$  (ii)  $P(X < 3)$   
 (iii)  $P(X > 6)$  (iv)  $P(0 < X < 3)$
8. The random variable  $X$  has a probability distribution  $P(X)$  of the following form, where  $k$  is some number :

$$P(X) = \begin{cases} k, & \text{if } x = 0 \\ 2k, & \text{if } x = 1 \\ 3k, & \text{if } x = 2 \\ 0, & \text{otherwise} \end{cases}$$

- (i) Determine the value of  $k$   
 (ii) Find  $P(X < 2)$ ,  $P(X \leq 2)$ ,  $P(X \geq 2)$ .

## 4.12 EXPECTATION AND VARIANCE

### Expectation

One of the most important concepts in probability theory is that of the expectation of a random variable. Expectation or mean is a measure of location or central tendency in the sense that it roughly locates a middle or average value of the random variable.

Let  $X$  be a random variable whose possible values  $x_1, x_2, x_3, \dots, x_n$  occur with probabilities  $p_1, p_2, p_3, \dots, p_n$ , respectively.

The mean of  $X$ , denoted by  $\mu$ , is the number  $\sum_{i=1}^n x_i p_i$

i.e. the mean of  $X$  is the weighted average of the possible values of  $X$ , each value being weighted by its probability with which it occurs.

The mean of a random variable  $X$  is also called the expectation of  $X$ , denoted by  $E(X)$ .

Thus,  $E(X) = \mu = \sum_{i=1}^n x_i p_i = x_1 p_1 + x_2 p_2 + \dots + x_n p_n$ .

In other words, the mean or expectation of a random variable  $X$  is the sum of the products of all possible values of  $X$  by their respective probabilities.

For instance, if the probability mass function of  $X$

is given by  $p(0) = \frac{1}{2} = p(1)$

$$\text{then } E[X] = 0 \left( \frac{1}{2} \right) + 1 \left( \frac{1}{2} \right) = \frac{1}{2}$$

is just the ordinary average of the two possible values 0 and 1 that  $X$  can assume.

On the other hand, if

$$p(0) = \frac{1}{3} \quad p(1) = \frac{2}{3} \quad \text{then}$$

$$E[X] = 0 \left( \frac{1}{3} \right) + 1 \left( \frac{2}{3} \right) = \frac{2}{3}$$

is a weighted average of the two possible values 0 and 1 where the value 1 is given twice as much weight as the value 0, since  $p(1) = 2p(0)$ .

Now, consider a random variable  $X$  that must take on one of the values  $x_1, x_2, \dots, x_n$  with respectively probabilities  $p(x_1), p(x_2), \dots, p(x_n)$ ; and think of  $X$  as representing our winnings in a single game of chance. That is, with probability  $p(x_i)$  we shall win  $x_i$  units  $i = 1, 2, \dots, n$ . Now by the frequency interpretation, it follows that if we continually play this game, then the proportion of time that we win  $x_i$  will be  $p(x_i)$ . As this is true for all  $i$ ,  $i = 1, 2, \dots, n$ , it follows that our average winnings per game will be

$$\sum_{i=1}^n x_i p(x_i) = E[X]$$

► **Example 1.** Find  $E[X]$  where  $X$  is the outcome when we roll a fair die.

► **Solution** Since

$$p(1) = p(2) = p(3) = p(4) = p(5) = p(6) = \frac{1}{6},$$

we obtain that

$$\begin{aligned} E[X] &= 1 \left( \frac{1}{6} \right) + 2 \left( \frac{1}{6} \right) + 3 \left( \frac{1}{6} \right) + 4 \left( \frac{1}{6} \right) \\ &\quad + 5 \left( \frac{1}{6} \right) + 6 \left( \frac{1}{6} \right) = \frac{7}{2} \end{aligned}$$

► **Example 2.** A coin that when flipped comes up heads with probability  $p$  is flipped until either heads or tails has occurred twice. Find the expected number of flips.

► **Solution** Letting  $X$  be the number of flips, then the probability mass function of  $X$  is

$$p_2 = p^2 + (1-p)^2, p_3 = 1 - p_2 = 2p(1-p)$$

Hence,  $E[X] = 2p_2 + 3(1 - p_2) = 3 - p^2 - (1 - p)^2$ .

► **Example 3.** A person draws 3 balls from a bag containing 3 white and 5 black balls. If he is to receive Rs.10 for every white balls which he draws and Re1 for every black ball, what is his expectation?

► **Solution** The number of ways in which 2 ball can be drawn is  ${}^8C_2 = 28$ .

Of these the number of ways in which  
2 white balls can be drawn is  ${}^3C_2 = 3$ ,  
1 white and 1 black ball can be drawn is  $3 \cdot 5 = 15$ ,  
2 black balls can be drawn is  ${}^5C_2 = 10$ .

Hence his chances of receiving Rs. 20, Rs. 11 and Rs. 2 are respectively  $3/28$ ,  $15/28$ ,  $10/28$ ;

$$\therefore \text{Expectation} = \left(\frac{3}{28} \cdot 20 + \frac{15}{28} \cdot 11 + \frac{10}{28} \cdot 2\right) = \text{Rs. } 8.75$$

► **Example 4.** A school class of 120 students are driven in 3 buses to a symphonic performance. There are 36 students in one of the buses, 40 in another, and 44 in the third bus. When the buses arrive, one of the 120 students is randomly chosen. Let  $X$  denote the number of students on the bus of that randomly chosen student, and find  $E[X]$ .

► **Solution** Since the randomly chosen student is equally likely to be any of the 120 students, it follows that

$$\begin{aligned} P\{X = 36\} &= \frac{36}{120} P\{X = 40\} \\ &= \frac{40}{120} P\{X = 44\} = \frac{44}{120} \end{aligned}$$

Hence

$$\begin{aligned} E[X] &= 36 \left(\frac{3}{10}\right) + 40 \left(\frac{1}{3}\right) + 44 \left(\frac{11}{30}\right) = \frac{1208}{30} \\ &= 40.2667 \end{aligned}$$

On the other hand, the average number of students on a bus is  $120/3 = 40$ , showing that the expected number of students on the bus of a randomly chosen student is larger than the average number of students on a bus. This is a general phenomenon and occurs because the more students there are on a bus, then the more likely a randomly chosen student would have been on that bus. As a result, buses with many students are given more weight than those with fewer students.

► **Example 5.** A and B throw with one dice for a stake of Rs. 11 which is to be won by the player who first throws 6. If A has the first throw, what are their respective expectations?

► **Solution** In his first throw A's chance is  $\frac{1}{6}$ ; in his second it is  $\frac{5}{6} \times \frac{5}{6} \times \frac{1}{6}$ , because each player must have failed once before A can have a second throw; in his third throw his chance is  $\left(\frac{5}{6}\right)^4 \times \frac{1}{6}$  because each player must have failed twice; and so on. Thus A's chance is the sum of the infinite series

$$\frac{1}{6} \left\{ 1 + \left(\frac{5}{6}\right)^2 + \left(\frac{5}{6}\right)^4 + \dots \right\}$$

Similarly B's chance is the sum of the infinite series

$$\frac{5}{6} \cdot \frac{1}{6} \left\{ 1 + \left(\frac{5}{6}\right)^2 + \left(\frac{5}{6}\right)^4 \right\};$$

$\therefore$  A's chance is to B's as 6 is to 5; their respective chance are therefore  $\frac{6}{11}$  and  $\frac{5}{11}$ , and their expectations are Rs. 6 and Rs. 5 respectively.

### Variance of a random variable

The mean of a random variable does not give us information about the variability in the values of the random variable. In fact, if the variance is small, then the values of the random variable are close to the mean. Also random variables with different

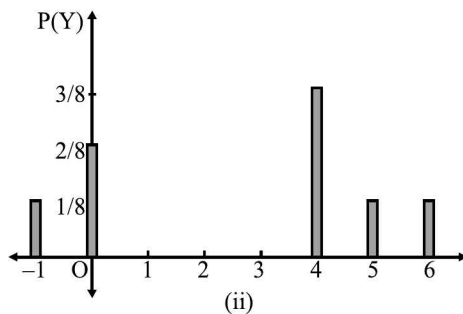
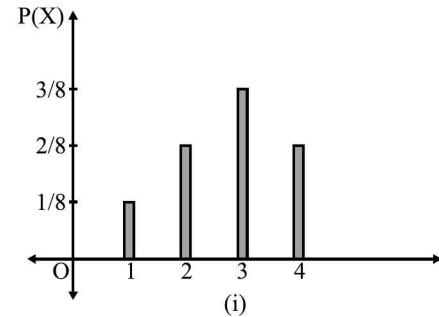
probability distributions can have equal means, as shown in the following distributions of X and Y.

|      |               |               |               |               |               |
|------|---------------|---------------|---------------|---------------|---------------|
| X    | 1             | 2             | 3             | 4             |               |
| P(X) | $\frac{1}{8}$ | $\frac{2}{8}$ | $\frac{3}{8}$ | $\frac{2}{8}$ |               |
| Y    | -1            | 0             | 4             | 5             | 6             |
| P(Y) | $\frac{1}{8}$ | $\frac{2}{8}$ | $\frac{3}{8}$ | $\frac{1}{8}$ | $\frac{1}{8}$ |

$$\begin{aligned}\text{Clearly } E(X) &= 1 \times \frac{1}{8} + 2 \times \frac{2}{8} + 3 \times \frac{3}{8} + 4 \times \frac{2}{8} \\ &= \frac{22}{8} = 2.75\end{aligned}$$

$$\begin{aligned}\text{and } E(Y) &= -1 \times \frac{1}{8} + 0 \times \frac{2}{8} + 4 \times \frac{3}{8} + 5 \times \frac{1}{8} + 6 \times \frac{1}{8} \\ &= 6 \times \frac{1}{8} = \frac{22}{8} = 2.75\end{aligned}$$

The variables X and Y are different, however their means are same. It is also easily observable from the diagrammatic representation of these distributions (Figure).



To distinguish X from Y, we require a measure of the extent to which the values of the random variables

spread out. In Statistics, we know that the variance is a measure of the spread or scatter in data. Likewise, the variability or spread in the values of a random variable may be measured by variance.

Let X be a random variable whose possible values  $x_1, x_2, \dots, x_n$  occur with probabilities  $p(x_1), p(x_2), \dots, p(x_n)$  respectively.

Let  $\mu = E(X)$  be the mean of X. The variance of X, denoted by  $\text{Var}(X)$  or  $\sigma_x^2$  is defined as

$$\sigma_x^2 = \text{Var}(X) = \sum_{i=1}^n (x_i - \mu)^2 p(x_i)$$

or equivalently  $\sigma_x^2 = E(X - \mu)^2$

The non-negative number

$$\sigma_x = \sqrt{\text{Var}(X)} = \sqrt{\sum_{i=1}^n (x_i - \mu)^2 p(x_i)}$$

is called the standard deviation of the random variable X.

Another formula to find the variance of a random variable. We know that,

$$\begin{aligned}\text{Var}(X) &= \sum_{i=1}^n (x_i - \mu)^2 p(x_i) \\ &= \sum_{i=1}^n (x_i^2 + \mu^2 - 2\mu x_i) p(x_i) \\ &= \sum_{i=1}^n x_i^2 p(x_i) + \sum_{i=1}^n \mu^2 p(x_i) - \sum_{i=1}^n 2\mu x_i p(x_i) \\ &= \sum_{i=1}^n x_i^2 p(x_i) + \mu^2 \sum_{i=1}^n p(x_i) - 2\mu \sum_{i=1}^n x_i p(x_i) \\ &= \sum_{i=1}^n x_i^2 p(x_i) + \mu^2 - 2\mu^2 \\ &\quad \left[ \text{since } \sum_{i=1}^n p(x_i) = 1 \text{ and } \mu = \sum_{i=1}^n x_i p(x_i) \right] \\ &= \sum_{i=1}^n x_i^2 p(x_i) - \mu^2\end{aligned}$$

$$\text{or } \text{Var}(X) = \sum_{i=1}^n x_i^2 p(x_i) - \left( \sum_{i=1}^n x_i p(x_i) \right)^2$$

$$\text{or } \text{Var}(X) = E(X^2) - [E(X)]^2, \text{ where } E(X^2)$$

$$= \sum_{i=1}^n x_i^2 p(x_i)$$

► **Example 6.** Find the variance of the number obtained on a throw of an unbiased die.

► **Solution** The sample space of the experiment is  $S = \{1, 2, 3, 4, 5, 6\}$ .

Let  $X$  denote the number obtained on the throw. Then  $X$  is a random variable which can take values 1, 2, 3, 4, 5, or 6.

Also  $P(1) = P(2) = P(3) = P(4) = P(5) = P(6) = 1/6$

Therefore, the probability distribution of  $X$  is

|        |               |               |               |               |               |               |
|--------|---------------|---------------|---------------|---------------|---------------|---------------|
| $X$    | 1             | 2             | 3             | 4             | 5             | 6             |
| $P(X)$ | $\frac{1}{6}$ | $\frac{1}{6}$ | $\frac{1}{6}$ | $\frac{1}{6}$ | $\frac{1}{6}$ | $\frac{1}{6}$ |

$$\text{Now } E(X) = \sum_{i=1}^n x_i p(x_i)$$

$$= 1 \times \frac{1}{6} + 2 \times \frac{1}{6} + 3 \times \frac{1}{6} + 4 \times \frac{1}{6} + 5 \times \frac{1}{6} + 6 \times \frac{1}{6} = \frac{21}{6}$$

$$\text{Also } E(X^2) = 1^2 \times \frac{1}{6} + 2^2 \times \frac{1}{6} + 3^2 \times \frac{1}{6} + 4^2 \times \frac{1}{6} + 5^2 \times \frac{1}{6} + 6^2 \times \frac{1}{6} = \frac{91}{6}$$

$$\text{Thus, } \text{Var}(X) = E(X^2) - (E(X))^2$$

$$= \frac{91}{6} - \left( \frac{21}{6} \right)^2 = \frac{91}{6} - \frac{441}{36} = \frac{35}{12}$$

### Practice Problems

[L]

- Two dice are thrown simultaneously. If  $X$  denotes the number of sixes, find the expectation of  $X$ .

- Two numbers are selected at random (without replacement) from the first six positive integers. Let  $X$  denote the larger of the two numbers obtained. Find  $E(X)$ .

- Let  $X$  denote the sum of the numbers obtained when two fair dice are rolled. Find the variance and standard deviation of  $X$ .

- There are 9 coins in a bag, 5 of which are rupees and the rest are unknown coins of equal value. Find what they must be if the probable value of a draw is  $60 \text{ nP}$ .

- Suppose that  $P\{X=0\} = 1 - P\{X=1\}$ . If  $E[X] = 3\text{Var}(X)$ , find  $P\{X=0\}$ .

- A random variable  $X$  is distributed according to the law

|               |               |               |               |               |
|---------------|---------------|---------------|---------------|---------------|
| 2             | 4             | 6             | 8             | 10            |
| $\frac{1}{4}$ | $\frac{1}{8}$ | $\frac{1}{4}$ | $\frac{1}{8}$ | $\frac{1}{4}$ |

Find mean and variance.

- Two balls are drawn at random from an urn containing two white and three black balls. Find mean and variance, if  $X$  is the number of the white balls taken out of the urn.

- There are four electric lamps each of which has a defect with the probability  $1/3$ . When screwing a defective lamp into a lamp-holder, it burns out immediately and then the next lamp is screwed up. Consider the random variable  $X$  representing the number of lamps screwed into the lamp-holder. Find the law of distribution, expectation, and variance of the random variable  $X$ .

- Suppose that A tosses two coins and receives Rs. 2 if two heads appear, Rs. 1 if one head appears, and nothing if no heads appear. What is the expected value of the game to him?

- In a game, a man wins a rupee for a six and loses a rupee for any other number when a fair die is thrown. The man decided to throw a die thrice

but to quit as and when he gets a six. Find the expected value of the amount he wins/loses.

11. Prove that if the expected value of a given experiment is  $E$ , and if a constant  $c$  is added to each of the outcomes, the expected value of the new experiment is  $E + c$ .
12. Prove that, if the expected value of a given experiment is  $E$ , and if each of the possible outcomes is multiplied by a constant  $k$ , the expected value of the new experiment is  $k \cdot E$ .

### 4.13 BINOMIAL RANDOM VARIABLE

#### Bernoulli trials

Many experiments are dichotomous in nature. For example, a tossed coin shows a 'head' or 'tail', a manufactured item can be 'defective' or 'non-defective', the response to a question might be 'yes' or 'no', an egg has 'hatched' or 'not hatched', the decision is 'yes' or 'no' etc. In such cases, it is customary to call one of the outcomes a 'success' and the other 'not success' or 'failure'. For example, in tossing a coin, if the occurrence of the head is considered a success, then occurrence of tail is a failure.

Each time we toss a coin or roll a die or perform any other experiment, we call it a trial. If a coin is tossed, say, 4 times, the number of trials is 4, each having exactly two outcomes, namely, success or failure. The outcome of any trial is independent of the outcome of any other trial. In each of such trials, the probability of success or failure remains constant. Such independent trials which have only two outcomes usually referred as 'success' or 'failure' are called Bernoulli trials.

Trials of a random experiment are called Bernoulli trials, if they satisfy the following conditions :

- (i) There should be a finite number of trials.
- (ii) The trials should be independent.
- (iii) Each trial has exactly two outcomes; success or failure.
- (iv) The probability of success remains the same in each trial.

For example, throwing a die 50 times is a case of 50 Bernoulli trials, in which each trial results in success (say an even number) or failure (an odd number) and the probability of success ( $p$ ) is same for all 50 throws. Obviously, the successive throws of the die are independent experiments. If the die is fair and have six numbers 1 to 6 written on six faces, then

$$p = 1/2 \text{ and } q = 1 - p = 1/2 \\ = \text{probability of failure.}$$

► **Example 1.** Six balls are drawn successively from an urn containing 7 red and 9 black balls. Tell whether or not the trials of drawing balls are Bernoulli trials when after each draw the ball drawn is

- (i) replaced
- (ii) not replaced in the urn.

► **Solution** (i) The number of trials is finite. When the drawing is done with replacement, the probability of success (say, red ball) is  $p = 7/16$  which is same for all six trials (draws). Hence, the drawing of balls with replacements are Bernoulli trials.

- (ii) When the drawing is done without replacement, the probability of success (i.e., red ball) in first trial is  $7/16$ , in 2nd trial is  $6/15$  if the first ball drawn is red or  $7/15$  if the first ball drawn is black and so on. Clearly, the probability of success is not same for all trials, hence the trials are not Bernoulli trials.

#### Binomial distribution

Consider the experiment of tossing a coin in which each trial results in success (say, heads) or failure (tails). Let  $S$  and  $F$  denote respectively success and failure in each trial. Suppose we are interested in finding the ways in which we have one success in six trials.

Clearly, six different cases are there as listed below: SFFFFF, FSFFFF, FFSFFF, FFFSFF, FFFFSF, FFFFFS.

Similarly, two successes and four failures can have

$\frac{6!}{4! \times 2!}$  combinations. It will be lengthy job to list

all of these ways. Therefore, calculation of probabilities of 0, 1, 2, ...,  $n$  number of successes may be lengthy and time consuming. To avoid the lengthy

calculations and listing of all the possible cases, for the probabilities of number of successes in  $n$ -Bernoulli trials, a formula is derived. For this purpose, let us take the experiment made up of three Bernoulli trials with probabilities  $p$  and  $q = 1 - p$  for success and failure respectively in each trial. The sample space of the experiment is the set

$$S = \{SSS, SSF, SFS, FSS, SFF, FSF, FFS, FFF\}$$

The number of successes is a random variable  $X$  and can take values 0, 1, 2, or 3. The probability distribution of the number of successes is as below:

$$\begin{aligned} P(X = 0) &= P(\text{no success}) \\ &= P(\{FFF\}) = P(F) P(F) P(F) \\ &= q \cdot q \cdot q = q^3 \end{aligned}$$

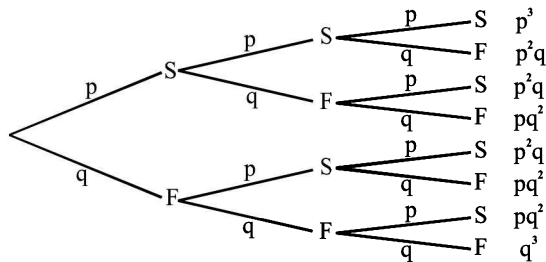
since the trials are independent

$$\begin{aligned} P(X = 1) &= P(\text{one success}) \\ &= P(\{SFF, FSF, FFS\}) \\ &= P(\{SFF\}) + P(\{FSF\}) + P(\{FFS\}) \\ &= P(S) P(F) P(F) + P(F) P(S) P(F) \\ &\quad + P(F) P(F) P(S) \\ &= p \cdot q \cdot q + q \cdot p \cdot q + q \cdot q \cdot p = 3pq^2 \end{aligned}$$

$$\begin{aligned} P(X = 2) &= P(\text{two successes}) \\ &= P(\{SSF, SFS, FSS\}) \\ &= P(\{SSF\}) + P(\{SFS\}) + P(\{FSS\}) \\ &= P(S) P(S) P(F) + P(S) P(F) P(S) \\ &\quad + P(F) P(S) P(S) \\ &= p \cdot p \cdot q + p \cdot q \cdot p + q \cdot p \cdot p = 3p^2q \end{aligned}$$

$$\text{and } P(X = 3) = P(\text{three successes}) = P(\{SSS\}) \\ = P(S) \cdot P(S) \cdot P(S) = p^3$$

The tree diagram for the experiments is shown in figure. The weights assigned to each path are indicated at the end of the path.



Thus, the probability distribution of  $X$  is

|        |       |         |         |       |
|--------|-------|---------|---------|-------|
| $X$    | 0     | 1       | 2       | 3     |
| $P(X)$ | $q^3$ | $3q^2p$ | $3qp^2$ | $p^3$ |

Also, the binomial expansion of

$$(q + p)^3 \text{ is } q^3 + 3q^2p + 3qp^2 + p^3$$

Note that the probabilities of 0, 1, 2 or 3 successes are respectively the 1st, 2nd, 3rd and 4th term in the expansion of  $(q + p)^3$ .

Also, since  $q + p = 1$ , it follows that the sum of these probabilities, as expected, is 1.

Thus, we may conclude that in an experiment of  $n$ -Bernoulli trials, the probabilities of 0, 1, 2, ...,  $n$  successes can be obtained as 1st, 2nd, ...,  $(n + 1)^{\text{th}}$  terms in the expansion of  $(q + p)^n$ . To prove this assertion (result), let us find the probability of  $x$ -successes in an experiment of  $n$ -Bernoulli trials.

Clearly, in case of  $x$  successes ( $S$ ), there will be  $(n - x)$  failures ( $F$ ).

Now,  $x$  successes ( $S$ ) and  $(n - x)$  failures ( $F$ ) can be

obtained in  $\frac{n!}{x!(n - x)!}$  ways.

In each of these ways, the probability of  $x$  successes and  $(n - x)$  failures is  $= P(x \text{ successes}) \cdot P(n - x \text{ failures})$  is

$$\underbrace{P(S) \cdot P(S) \dots P(S)}_{x \text{ times}} \cdot \underbrace{P(F) \cdot P(F) \dots P(F)}_{(n - x) \text{ times}} = p^x q^{n-x}$$

Thus, the probability of  $x$  successes in  $n$ -Bernoulli

trials is  $\frac{n!}{x!(n - x)!} p^x q^{n-x}$  or  ${}^nC_x p^x q^{n-x}$

Thus  $P(x \text{ successes})$

$$= {}^nC_x p^x q^{n-x}, x = 0, 1, 2, \dots, n. (q = 1 - p)$$

Clearly,  $P(x \text{ successes})$ , i.e.  ${}^nC_x p^x q^{n-x}$  is the  $(x + 1)^{\text{th}}$  term in the binomial expansion of  $(q + p)^n$ .

Thus, the probability distribution of number of successes in an experiment consisting of  $n$  Bernoulli trials may be obtained by the binomial expansion of  $(q + p)^n$ . Hence, this distribution of number of successes  $X$  can be written as

$$\begin{array}{ccccccc} X & 0 & 1 & 2 & \dots & x & \dots & n \\ P(X) & {}^nC_0 q^n & {}^nC_1 q^{n-1} p^1 & {}^nC_2 q^{n-2} p^2 & \dots & {}^nC_x q^{n-x} p^x & \dots & {}^nC_n p^n \end{array}$$

The above probability distribution is known as binomial distribution with parameters  $n$  and  $p$ ,

because for given values of  $n$  and  $p$ , we can find the complete probability distribution.

The probability of  $x$  successes  $P(X = x)$  is also denoted by  $P(x)$  and is given by

$$P(x) = {}^nC_x q^{n-x} p^x, x = 0, 1, \dots, n. (q = 1 - p)$$

This  $P(x)$  is called the probability function of the binomial distribution.

A binomial distribution with  $n$ -Bernoulli trials and probability of success in each trial as  $p$ , is denoted by  $B(n, p)$ .



- (i) The probability of atleast  $r$  successes in  $n$  trials  
 $= {}^nC_r p^r q^{n-r} + {}^nC_{r+1} p^{r+1} q^{n-(r+1)} + \dots + {}^nC_n p^n$ .
- (ii) The probability of atmost  $r$  successes in  $n$  trials  
 $= {}^nC_0 q^n + {}^nC_1 p q^{n-1} + \dots + {}^nC_r p^r q^{n-r}$ .

Let us now take up some examples.

From a huge box of marbles, 25% of them red and the rest white we draw a sample of 3 marbles and write down the binomial probability distribution.

The probability of getting a red marble is  $p = 1/4$

The probability of getting a white marble is  $q = 3/4$

Consider the binomial expansion

$$(q + p)^3 = q^3 + 3q^2p + 3qp^2 + p^3$$



1. Each term stands for the  
 $q^3 \quad 3q^2p \quad 3qp^2 \quad p^3$   
probability of one possible  
0 Red 1 Red 2 Red 3 Red  
combination  
3 White 2 White 1 White 0 White
2. The number of combinations is 1 more than the number of items in the sample.
3. The indices above the  $p$ 's and  $q$ 's tell us what the combination is e.g. in  $3q^2p$  we know that  $q^2p$  stands for 2 White and 1 Red.
4. The coefficient tells us how many ways that combination can arise, e.g. the 3 in  $3q^2p$  says that  $q^2p$  can arise in three ways.

Further,  $q^3 + 3q^2p + 3qp^2 + p^3$

$$= \left(\frac{3}{4}\right)^3 + 3\left(\frac{3}{4}\right)^2\left(\frac{1}{4}\right) + 3\left(\frac{3}{4}\right)\left(\frac{1}{4}\right)^2 + \left(\frac{1}{4}\right)^3$$

$$= \frac{27}{64} + \frac{27}{64} + \frac{9}{64} + \frac{1}{64} = 1$$

Now consider a factory having five telephone lines and the probability that any one of them is engaged at a given moment during the day is  $2/7$ . Calculate the probability that atleast three of the lines will be engaged if we contact the factory now.

From the Binomial Theorem we get the coefficients:

$$\begin{array}{cccc} q^5 & 5q^4p & 10q^3p^2 & 10q^3p^2 \\ 10q^2p^3 & 5qp^4 & p^5 & \end{array}$$

Since we are looking for the probability that atleast 3 (that is, 3, 4 or 5) lines are engaged, we need to evaluate:

$$\begin{aligned} & 10q^2p^3 + 5qp^4 + p^5 \\ &= 10 \left(\frac{5}{7}\right)^2 \left(\frac{2}{7}\right)^3 + 5 \left(\frac{5}{7}\right) \left(\frac{2}{7}\right)^4 + \left(\frac{2}{7}\right)^5 \\ &= \frac{2000}{16807} + \frac{400}{16807} + \frac{32}{16807} \\ &= \frac{2432}{16807} = 0.145 \end{aligned}$$

► **Example 1.** In four throws with a pair of dices, what is the chance of throwing doublets twice atleast?

► **Solution** In a single throw the chance of doublets is  $\frac{6}{36}$ , or  $\frac{1}{6}$ ; and the chance of failing to throw

doublets is  $\frac{5}{6}$ . Now the required event follows if

doublets are thrown four times, three times, or twice; therefore the required chance is the sum of the first

three terms of the expansion of  $\left(\frac{1}{6} + \frac{5}{6}\right)^4$ .

$$\text{Thus the chance} = \frac{1}{6^4} (1 + 4 \cdot 5 + 6 \cdot 5^2) = \frac{19}{144}.$$

► **Example 2.** Three shots are fired at a target, the probability of hit being equal to 0.8 for each shot.

The number of hits is considered to be a random variable  $X$ . Find the law of its distribution.

► **Solution** In this case the random variable  $X$  may attain the following values :

$$x_1 = 0, x_2 = 1, x_3 = 2, x_4 = 3.$$

The corresponding probabilities can be determined in a very simple way : form the generating polynomial  $(0.2 + 0.8x)^3$  and write its expansion

$$0.2^3 + 3 \cdot 0.2^2 \cdot 0.8x + 3 \cdot 0.2 \cdot 0.8^2 x^2 + 0.8^3 x^3.$$

As is known, the coefficient of  $x^k$  ( $k = 0, 1, 2, 3$ ) yields the probability of the event that  $k$  shots will hit the target, i.e. the probability that the random variable  $X$  will take on a value equal to  $k$ .

$$\begin{aligned} \text{Thus, } p_1 &= 0.2^3 = 0.008, \\ p_2 &= 3 \cdot 0.2^2 \cdot 0.8 = 0.096, \\ p_3 &= 3 \cdot 0.2 \cdot 0.8^2 = 0.384, \\ p_4 &= 0.8^3 = 0.512. \end{aligned}$$

Hence, the law of distribution of the random variable  $X$  has the form

|       |       |       |       |
|-------|-------|-------|-------|
| 0     | 1     | 2     | 3     |
| 0.008 | 0.096 | 0.384 | 0.512 |

► **Example 3.** If a fair coin is tossed 10 times, find the probability of

- (i) exactly six heads
- (ii) atleast six heads
- (iii) atmost six heads

► **Solution** The repeated tosses of a coin are Bernoulli trials. Let  $X$  denote the number of heads in an experiment of 10 trials.

Clearly,  $X$  has the binomial distribution with  
 $n = 10$  and  $p = 1/10$

Therefore  $P(X = x) = {}^nC_x q^{n-x} p^x$ ,  $x = 0, 1, 2, \dots, n$

$$\text{Here } n = 10, p = \frac{1}{2}, q = 1 - p = \frac{1}{2}$$

$$\text{Therefore, } P(X = x) = {}^{10}C_x \left(\frac{1}{2}\right)^{10-x} \left(\frac{1}{2}\right)^x = {}^{10}C_x \left(\frac{1}{2}\right)^{10}$$

Now

$$(i) P(X = 6) = {}^{10}C_6 \left(\frac{1}{2}\right)^{10} = \frac{10!}{6! \times 4!} \frac{1}{2^{10}} = \frac{105}{512}$$

$$(ii) P(\text{atleast six heads}) = P(X \geq 6) \\ = P(X = 6) + P(X = 7) + P(X = 8) + P(X = 9)$$

$$+ P(X = 10)$$

$$\begin{aligned} &= {}^{10}C_6 \left(\frac{1}{2}\right)^{10} + {}^{10}C_7 \left(\frac{1}{2}\right)^{10} + {}^{10}C_8 \left(\frac{1}{2}\right)^{10} \\ &\quad + {}^{10}C_9 \left(\frac{1}{2}\right)^{10} + {}^{10}C_{10} \left(\frac{1}{2}\right)^{10} \\ &= \left[ \left( \frac{10!}{6! \times 4!} \right) + \left( \frac{10!}{7! \times 3!} \right) + \left( \frac{10!}{8! \times 2!} \right) \right. \\ &\quad \left. + \left( \frac{10!}{9! \times 1!} \right) + \left( \frac{10!}{10!} \right) \right] \end{aligned}$$

$$\frac{1}{2^{10}} = \frac{193}{512}$$

$$(iii) P(\text{atmost six heads}) = P(X \leq 6) \\ = P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3) \\ \quad + P(X = 4) + P(X = 5) + P(X = 6)$$

$$\begin{aligned} &= \left(\frac{1}{2}\right)^{10} + {}^{10}C_1 \left(\frac{1}{2}\right)^{10} + {}^{10}C_2 \left(\frac{1}{2}\right)^{10} + {}^{10}C_3 \left(\frac{1}{2}\right)^{10} \\ &\quad + {}^{10}C_4 \left(\frac{1}{2}\right)^{10} + {}^{10}C_5 \left(\frac{1}{2}\right)^{10} + {}^{10}C_6 \left(\frac{1}{2}\right)^{10} \\ &= \frac{848}{1024} = \frac{53}{64} \end{aligned}$$

► **Example 4.** Ten eggs are drawn successively with replacement from a lot containing 10% defective eggs. Find the probability that there is atleast one defective egg.

► **Solution** Let  $X$  denote the number of defective eggs in the 10 eggs drawn. Since the drawing is done with replacement, the trials are Bernoulli trials. Clearly,  $X$  has the binomial distribution with

$$n = 10 \quad \text{and} \quad p = \frac{10}{100} = \frac{1}{10}.$$

$$\text{Therefore, } q = 1 - p = \frac{9}{10}$$

$$\text{Now } P(\text{atleast one defective egg}) = P(X \geq 1) \\ = 1 - P(X = 0)$$

$$= 1 - {}^{10}C_0 \left(\frac{9}{10}\right)^{10} = 1 - \frac{9^{10}}{10^{10}}$$



► **Example 5.** A communication system consists of  $n$  components, each of which will independently function with probability  $p$ . The total system will be able to operate effectively if atleast one-half of its components function. For what values of  $p$  is a 5-component system more likely to operate effectively than a 3-component system ?

► **Solution** As the number of functioning components is a binomial random variable with parameters  $(n, p)$ , it follows that the probability that a 5-component system will be effective is

$${}^5C_3 p^3 (1-p)^2 + {}^5C_4 p^4 (1-p) + p^5$$

whereas the corresponding probability for a 3-component system is  ${}^3C_2 p^2 (1-p) + p^3$

Hence, the 5-component system is better if

$$10p^3(1-p)^2 + 5p^4(1-p) + p^5 > 3p^2(1-p) + p^3$$

which reduces to

$$3(p-1)^2(2p-1) > 0 \quad \text{or} \quad p > \frac{1}{2}$$

► **Example 6.** An unbiased normal coin is tossed  $n$  times. Let

$E_1$  : event that both heads and tails are present in  $n$  tosses.

$E_2$  : event that the coin shows up heads atmost once.

Find the value of ' $n$ ' for which  $E_1$  and  $E_2$  are independent.

► **Solution**  $P(E_1) = 1 - [P(\text{all heads}) + P(\text{all tails})]$

$$= 1 - \left[ \frac{1}{2^n} + \frac{1}{2^n} \right] = 1 - \frac{1}{2^{n-1}}$$

$$P(E_2) = P(\text{no head}) + P(\text{exactly one head})$$

$$= \frac{1}{2^n} + {}^nC_1 \cdot \frac{1}{2^n} = \frac{n+1}{2^n}$$

$$P(E_1 \cap E_2) = (\text{exactly one head} \& (n-1) \text{ tail})$$

$$= {}^nC_1 \cdot \frac{1}{2} \cdot \frac{1}{2^{n-1}} = \frac{n}{2^n}$$

If  $E_1$  &  $E_2$  are independent, then

$$\frac{n}{2^n} = \left(1 - \frac{1}{2^{n-1}}\right) \left(\frac{n+1}{2^n}\right)$$

$$\frac{n}{2^n} = \frac{n}{2^n} + \frac{1}{2^n} - \frac{n}{2^{2n-1}} - \frac{1}{2^{2n-1}}$$

$$n+1 = \frac{2^{2n-1}}{2^n} = 2^{n-1} \quad \Rightarrow n=3$$

### Mean of Binomial Distribution

We will now examine the properties of a binomial random variable with parameters  $n$  and  $p$ . Let us now find the mean or expectation of the number of occurrences of event  $A$  in a sequence of  $n$  independent trials in each of which the event  $A$  occurs with probability  $p$ .

$$E[X^k] = \sum_{i=0}^n i^k {}^nC_i p^i (1-p)^{n-i}$$

$$= \sum_{i=1}^n i^k {}^nC_i p^i (1-p)^{n-i}$$

Using the identity

$$i {}^nC_i = n {}^{n-1}C_{i-1}$$

$$E[X^k] = np \sum_{i=1}^n i^{k-1} {}^{n-1}C_{i-1} p^{i-1} (1-p)^{n-i}$$

$$= np \sum_{j=0}^{n-1} (j+1)^{k-1} {}^{n-1}C_j p^j (1-p)^{n-1-j}$$

by letting

$$j = i-1$$

$$= np E[(Y+1)^{k-1}]$$

where  $Y$  is a binomial random variable with parameters  $n-1, p$ .

Setting  $k=1$  in the preceding equation yields

$$E[X] = np$$

That is, the expected number of successes that occur in  $n$  independent trials when each is a success with probability  $p$  is equal to  $np$ . Setting  $k=2$  in the preceding equation, and using the preceding formula for the expected value of a binomial random variable, gives that

$$E[X^2] = npE[Y+1]$$

$$= np[(n-1)p + 1]$$

Since  $E[X] = np$ , we obtain

$$\begin{aligned}\text{Var}(X) &= E[X^2] - (E[X])^2 \\ &= np[(n-1)p + 1] - (np)^2 \\ &= np(1-p)\end{aligned}$$

If  $X$  is a binomial random variable with parameters  $n$  and  $p$ , then

$$\begin{aligned}E[X] &= np \\ \text{Var}(X) &= np(1-p).\end{aligned}$$

### Alternative method to compute expectation and variance:

The probability that the random variable  $X$  will take on the value  $k$  is equal to  $C_n^k q^{n-k} p^k$ .

Consequently,

$$E(X) = \sum_{k=0}^n k C_n^k q^{n-k} p^k$$

In order to simplify the obtained expression, let us

take advantage of identity  $\sum_{k=0}^n p_k = 1$ .

$$(q + px)^n = \sum_{k=0}^n C_n^k q^{n-k} p^k x^k.$$

Differentiating both sides of this identity with respect to  $x$ , we get

$$n(q + px)^{n-1} p = \sum_{k=0}^n C_n^k q^{n-k} p^k x^{k-1}.$$

Hence, for  $x = 1$ , taking into account that  $p + q = 1$ , we find

$$np = \sum_{k=0}^n k C_n^k q^{n-k} p^k$$

Consequently,  $E(X) = np$

For the expectation of the random variable  $X^2$  we shall have

$$E(X^2) = \sum_{k=0}^n k^2 C_n^k p^k q^{n-k}.$$

$$(q + px)^n = \sum_{k=0}^n C_n^k p^{n-k} q^k x^k.$$

Differentiating twice both sides of the identity with respect to the variable  $x$ , we get

$$\begin{aligned}n(n-1)(q + px)^{n-2} p^2 \\ = \sum_{k=1}^n C_n^k p^{n-k} p^k k(k-1) x^{k-2}\end{aligned}$$

Setting  $x = 1$ , we arrive to the equality

$$n(n-1)p^2 = \sum_{k=1}^n k(k-1) C_n^k q^{n-k} p^k$$

or

$$n(n-1)p^2 = \sum_{k=1}^n k^2 C_n^k q^{n-k} p^k - \sum_{k=1}^n k C_n^k q^{n-k} p^k,$$

whence, taking into account that

$$\sum_{k=1}^n k C_n^k q^{n-k} p^k = E(X) = np, \text{ we find}$$

$$E(X^2) = \sum_{k=1}^n k^2 C_n^k p^{n-k} p^k$$

$$= n(n-1)p^2 + np = n^2 p^2 - np^2 + np$$

We now finally obtain

$$\begin{aligned}\text{Var}(X) &= E(X^2) - (E(X))^2 = n^2 p^2 - np^2 + np - n^2 p^2 \\ &= np(1-p),\end{aligned}$$

i.e.,  $\text{Var}(X) = npq$ .

► **Example 7.** Find the mean of the binomial distribution with parameters 4 and  $1/3$ .

► **Solution** Let  $X$  be the random variable for the probability distribution.

Here  $n = 4$ ,  $p = 1/3$  and  $q = 1 - 1/3 = 2/3$

We know that

$$P(X=x) = {}^4C_x \left(\frac{2}{3}\right)^{4-x} \left(\frac{1}{3}\right)^x, \quad x = 0, 1, 2, 3, 4.$$

$$\text{Now mean } E(X) = \sum_{i=1}^4 x_i p(x_i)$$

$$= 0 + {}^4C_1 \left(\frac{2}{3}\right)^3 \left(\frac{1}{3}\right) + 2 \cdot {}^4C_2 \left(\frac{2}{3}\right)^2 \left(\frac{1}{3}\right)^2$$

$$+ 3 \cdot {}^4C_3 \left(\frac{2}{3}\right) \left(\frac{1}{3}\right)^3 + 4 \cdot {}^4C_4 \left(\frac{1}{3}\right)^4$$

$$= 4 \times \frac{2^3}{3^4} + 2 \times 6 \times \frac{2^2}{3^4} + 3 \times 4 \times \frac{2}{3^4} + 4 \times 1 \times \frac{1}{3^4}$$

$$= \frac{32 + 48 + 24 + 4}{3^4} = \frac{108}{81} = \frac{4}{3}.$$

The formula  $E(X) = np$  gives the same result.

► **Example 8.** The mean and variance of a binomial variable  $X$  are 2 and 1 respectively. Find the probability that  $X$  takes values greater than 1.

► **Solution** Given mean  $np = 2$  ... (1)

and variance  $npq = 1$  ... (2)

Dividing (2) by (1), then

$$q = 1/2 \quad \therefore p = 1 - q = 1/2$$

$$\text{from (1), } n \times 1/2 = 2 \quad \therefore n = 4$$

The binomial distribution is  $\left(\frac{1}{2} + \frac{1}{2}\right)^4$

$$\text{Now } P(X > 1) = P(X = 2) + P(X = 3) + P(X = 4)$$

$$= {}^4C_2 \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^2 + {}^4C_3 \left(\frac{1}{2}\right)^1 \left(\frac{1}{2}\right)^3 + {}^4C_4 \left(\frac{1}{2}\right)^4$$

### Mode of Binomial Distribution

Mode of binomial distribution is that value of  $r$  for which  $P(X = r)$  is maximum.

$$\text{Now, } \frac{P(X = r)}{P(X = r - 1)} = \frac{{}^nC_r p^r q^{n-r}}{{}^nC_{r-1} p^{r-1} q^{n-r+1}}$$

$$= \frac{n!}{r!(n-r)!} \frac{(r-1)!(n-r+1)!}{n!} \frac{p}{q}$$

$$= \frac{n-r+1}{r} \frac{p}{q} = \frac{(n-r+1)p}{rq}$$

Putting  $r + 1$  in place of  $r$  in (1), we get

$$\frac{P(X = r + 1)}{P(X = r)} = \frac{(n-r)p}{(r+1)q}$$

For  $P(X = r)$  to be maximum,

$$\frac{P(X = r)}{P(X = r - 1)} \geq 1 \Rightarrow \frac{(n-r+1)p}{rq} \geq 1$$

$$\Rightarrow (n+1)p \geq r(q+p)$$

$$\Rightarrow r \leq (n+1)p \quad [\because p+q=1]$$

Also for  $P(X = r)$  to be maximum

$$\frac{P(X = r + 1)}{P(X = r)} \leq 1 \Rightarrow (n-r)p \leq (r+1)q$$

$$\Rightarrow np \leq q + r(p+q)$$

$$\Rightarrow np \leq q + r$$

$$\Rightarrow r \geq np - q = np - (1-p) = (n+1)p - 1$$

$$\Rightarrow r \geq (n+1)p - 1$$

From (A) and (B)

$$(n+1)p - 1 \leq r \leq (n+1)p$$

and  $r$  is a positive integer.

If  $X$  is a binomial random variable with parameters  $(n, p)$ , where  $0 < p < 1$ , then as  $r$  goes from 0 to  $n$ ,  $P\{X = r\}$  first increases monotonically and then decreases monotonically, reaching its largest value when  $r$  is the largest integer less than or equal to  $(n+1)p$ .

Two cases arise :

(i) in the case that  $(n+1)p$  is not an integer, when  $r$  is equal to the integer  $m$  satisfying the inequalities

$$(n+1)p - 1 < m < (n+1)p.$$

(ii) in the case  $(n+1)p$  is an integer, when  $r$  is equal to either  $(n+1)p - 1$  or  $(n+1)p$ .

In figure (1) we show the probabilities graphically for a binomial distribution with parameters  $n = 8$  and  $p = 3/4$ . Mode of binomial distribution i.e. the value of  $r$  for which  $P(X = r)$  is maximum is found to be  $r = 6$ .

$$\text{Here } (n+1)p - 1 = 9(3/4) - 1 = 5.75$$

$$(n+1)p = 9(3/4) = 6.75.$$

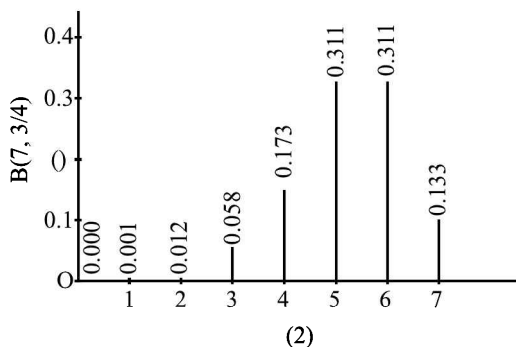
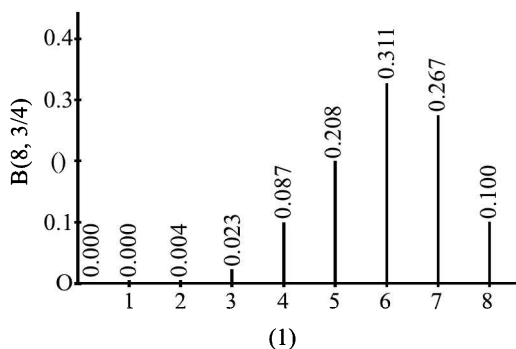
According to case (i) we have  $5.75 < 6 < 6.75$  which suggests that mode is 6.

In figure (2) we have done similarly for the case of  $n = 7$  and  $p = 3/4$ .

$$\text{Here } (n+1)p - 1 = 8(3/4) - 1 = 5$$

$$(n+1)p = 8(3/4) = 6.$$

According to case (ii) we have  $(n+1)p = 6$  as an integer. Hence the mode is 5 and 6.



### Independent trials with more than two outcomes

The extension of binomial distribution leads, for the case of independent trials with outcomes  $a_1, a_2, \dots, a_k$  occurring with probabilities  $p_1, p_2, \dots, p_k$ .

The probability for  $r_1$  occurrences of  $a_1$ ,  $r_2$  occurrences of  $a_2$ , etc., is given by

$$f(r_1, r_2, \dots, r_k; p_1, p_2, \dots, p_k) = \frac{n!}{r_1! r_2! \dots r_k!} p_1^{r_1} \cdot p_2^{r_2} \dots p_k^{r_k}.$$

For example, if a die is thrown ten times the probability that a one will occur exactly twice and a three exactly three times is given by

$$\frac{10!}{2!3!5!} \left(\frac{1}{6}\right)^2 \left(\frac{1}{6}\right)^3 \left(\frac{4}{6}\right)^5 = 0.043 \dots$$

► **Example 9.** A die is thrown 12 times. What is the probability that each number will come up twice ?

► **Solution** Here there are six outcomes, 1, 2, 3, 4, 5, 6 corresponding to the six sides of the die. We assign each outcome probability  $1/6$ . We are then asked for

$$f(2, 2, 2, 2, 2, 2; \frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}),$$

which is

$$\frac{12!}{2!2!2!2!2!2!} \left(\frac{1}{6}\right)^2 \left(\frac{1}{6}\right)^2 \left(\frac{1}{6}\right)^2 \left(\frac{1}{6}\right)^2 \left(\frac{1}{6}\right)^2 \left(\frac{1}{6}\right)^2 = 0.0034$$

### Practice Problems

[M]

1. A pair of dice is thrown 4 times. If getting a doublet is considered a success, find the probability of two successes.
2. There are 5% defective items in a large bulk of items. What is the probability that a sample of 10 times will include not more than one defective item?
3. A die is thrown 6 times. If 'getting an odd number' is a success, what is the probability of
  - (i) 5 successes
  - (ii) atleast 5 successes
  - (iii) at most 5 successes?
4. In an examination, 20 questions of true-false type are asked. Suppose a student tosses a fair coin to determine his answer to each question. If the coin falls heads, he answers 'true'; if it falls tails, he answer 'false'. Find the probability that he answers 'false'. Find the probability that he answer atleast 12 questions correctly.
5. An urn contains 25 balls of which 10 balls bear a mark 'X' and the remaining 15 bear a mark 'Y'. A ball is drawn at random from the urn, its mark is noted down and it is replaced. If 6 balls are drawn in this way, find the probability that
  - (i) all will bear 'X' mark.
  - (ii) not more than 2 will bear 'Y' mark.

- (iii) the number of balls with 'X' mark and 'Y' mark will be equal.
6. In a hurdle race, a player has to cross 10 hurdles. The probability that he will clear each hurdle is  $5/6$ . What is the probability that he will knock down fewer than 2 hurdles?
  7. How many times must a man toss a fair coin so that the probability of having atleast one head is more than 90%?
  8. If  $X$  is a binomial random variable with expected value 6 and variance 2.4, find  $P\{X = 5\}$ .
  9. Suppose  $X$  has a binomial distribution  $B(6, 1/2)$  Show that  $X = 3$  is the most likely outcome.
  10. A small factory has ten workers. The workers eat their lunch at one of two diners, and they are just as likely to eat in one as in the other. If the proprietors want to be more than 0.95 sure of having enough seats, how many seats must each of the diners have?
  11. If an ordinary die is thrown 20 times, what is the expected number of times that a six will turn up? What is the standard deviation for the number of sixes that turn up?
  12. In 16 tosses of an ordinary coin, what is the expected number of heads that turn up? What is the standard deviation for the number of heads that occur?
  13. In a large number of independent trials with probability  $p$  for success, what is the approximate probability that the number of successes will deviate from the expected number by more than one standard deviation but less than two standard deviations?
  14. Suppose that it is desired that the probability be approximately 0.95 that the fraction of sixes that turn up when a die is thrown  $n$  times does not deviate by more than 0.01 from the value  $1/6$ . How large should  $n$  be?
  15. Assume that 10% of the people in a certain city have cancer. If 900 people are selected at

random from the city, what is the expected number which will have cancer? What is the standard deviation? What is the approximate probability that more than 108 of the 900 chosen have cancer?

16. A thick coin, when tossed, will land "heads" with a probability of  $5/12$ , "tails" with a probability of  $5/12$ , and will land on edge with a probability of  $1/6$ . If it is tossed six times, what is the probability that it lands on edge exactly two times?

#### 4.14 BAYES' THEOREM

Suppose that an event has occurred which must have been due to one of the causes (hypotheses),  $A_1, A_2, \dots, A_n$ .

Let  $P(B | A_i)$  be the probability of the event  $B$  on the assumption that the cause  $A_i$  exists. Then the target is to find the probability  $P(A_i | B)$  of the existence of the cause  $A_i$ , estimated after the event  $B$  has occurred.

The Bayes' theorem updates the probability of each of the cause (hypotheses) after the arrival of new 'information' about the occurrence of an event.

If  $A_1, A_2, \dots, A_n$  are  $n$  non-empty events with given probabilities  $P(A_1), P(A_2), \dots, P(A_n)$  which constitute a partition of sample space  $S$ , (i.e.  $A_1, A_2, \dots, A_n$  are  $n$  mutually exclusive and exhaustive events so that  $A_1 \cup A_2 \cup \dots \cup A_n = S$ ) and  $B$  is any event with non-zero probability and the probabilities  $P(B|A_1), P(B|A_2), \dots, P(B|A_n)$  are known, then the Bayes' theorem states that

$$P(A_i | B) = \frac{P(A_i) P(B | A_i)}{\sum_{j=1}^n P(A_j) P(B | A_j)}$$

for any  $i = 1, 2, 3, \dots, n$

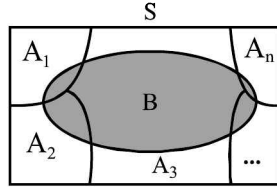
*Proof :*

The events  $B$  occurs with one of the  $n$  mutually exclusive and exhaustive events  $A_1, A_2, \dots, A_n$ .

$$B = B \cap A_1 + B \cap A_2 + B \cap A_3 + \dots + B \cap A_n$$

$$P(B) = P(B \cap A_1) + P(B \cap A_2) + \dots + P(B \cap A_n)$$

$$= \sum_{i=1}^n P(B \cap A_i)$$



Now,  $P(B \cap A_i) = P(A_i) \cdot P(B | A_i)$

$$P(B) = \sum_{i=1}^n P(A_i) \cdot P(B | A_i)$$

(Total probability theorem)

$$P(A_i | B) = \frac{P(A_i) \cdot P(B | A_i)}{P(B)}$$

$$= \frac{P(A_i) P(B | A_i)}{\sum_{j=1}^n P(A_j) P(B | A_j)}$$



The following terminology is generally used when Bayes' theorem is applied.

The events  $A_1, A_2, \dots, A_n$  are called causes (hypotheses).

The probability  $P(A_i)$  is called the priori probability of the cause  $A_i$ . The prior probability is an unconditional probability that is known to us before any observations are taken.

The conditional probability  $P(A_i | B)$  is called a posteriori probability of the cause  $E_i$ . The posterior probability is a conditional probability that is of interest to us only if it is known that the conditioning event has occurred. The joint probability  $P(B \cap A_i) = P(A_i) \cdot P(B | A_i)$  is called the antecedent probability, that the event  $B$  will occur, and that from the  $i$ th cause.

Bayes' theorem shows how to compute the conditional probabilities  $P(A_i | B)$  of the cause  $A_i$

when additional evidence  $B$  becomes available. This theorem is also called the formula for the probability of "causes". This principle is applied to determine the probability of future events.

The Bayes' theorem has its applications in variety of situations, few of which are illustrated in following examples.

► **Example 1.** Bag I contains 3 red and 4 black balls while another Bag II contains 5 red and 6 black balls. One ball is drawn at random from one of the bags and it is found to be red. Find the probability that it was drawn from Bag II.

► **Solution** Let  $A_1$  be the event of choosing the bag I,  $A_2$  the event of choosing the bag II and  $B$  be the event of drawing a red ball.

Then  $P(A_1) = P(A_2) = 1/2$

Also,  $P(B | A_1) = P(\text{drawing a red ball from Bag I}) = 3/7$

and  $P(B | A_2) = P(\text{drawing a red ball from Bag II}) = 5/11$

Now, the probability of drawing a ball from Bag II, being given that it is red, is  $P(A_2 | B)$

Using Bayes' theorem, we have

$$P(A_2 | B) = \frac{P(A_2) P(B | A_2)}{P(A_1) P(B | A_1) + P(A_2) P(B | A_2)}$$

$$= \frac{\frac{1}{2} \times \frac{5}{11}}{\frac{1}{2} \times \frac{3}{7} + \frac{1}{2} \times \frac{5}{11}} = \frac{35}{68}$$

► **Example 2.** A writes a letter to B and does not receive a reply. Assuming that one letter in  $n$  is lost in the mails, find the chance that B received the letter. It is certain that B would have replied the letter if he had received it.

► **Solution** Define events :  $N = \{A \text{ did not receive reply from B}\}$ ,  $R = \{B \text{ received the letter}\}$ .

We need to find  $P(\bar{R} | N)$ , where

$$P(N | \bar{R}) = 1, P(N | R) = 1/n,$$

$$P(\bar{R}) = 1/n \text{ (loss in mails), } P(R) = 1 - P(\bar{R}).$$

$$\begin{aligned}\therefore P(R|N) &= \frac{P(N|R) \cdot P(R)}{P(R) \cdot P(N|R) + P(\bar{R}) \cdot P(N|\bar{R})} \\ &= \frac{(1/n) \cdot (1 - n^{-1})}{(1/n) \cdot (1 - n^{-1}) + (1/n) \cdot 1} = \frac{n-1}{2n-1}\end{aligned}$$

► **Example 3.** A man is known to speak truth 3 out of 4 times. He throws a die and reports that it is a six. Find the probability that it is actually a six.

► **Solution** Let E be the event that the man reports that six occurs in the throwing of the die and let  $S_1$  be the event that six occurs and  $S_2$  be the event that six does not occur.

Then  $P(S_1)$  = Probability that six occurs =  $1/6$

$P(S_2)$  = Probability that six does not occur =  $5/6$

$P(E|S_1)$  = Probability that the man reports that six occurs when six has actually occurred on the die.

= Probability that the man speaks the truth =  $3/4$

$P(E|S_2)$  = Probability that the man does not speak the

$$\text{truth} = 1 - \frac{3}{4} = \frac{1}{4}$$

Thus, by Bayes' theorem, we get

$P(S_1|E)$  = Probability that the report of the man that six has occurred is actually a six

$$\begin{aligned}&= \frac{P(S_1) \cdot P(E|S_1)}{P(S_1) \cdot P(E|S_1) + P(S_2) \cdot P(E|S_2)} \\ &= \frac{\frac{1}{6} \times \frac{3}{4}}{\frac{1}{6} \times \frac{3}{4} + \frac{5}{6} \times \frac{1}{4}} = \frac{1}{8} \times \frac{24}{8} = \frac{3}{8}\end{aligned}$$

Hence, the required probability is  $3/8$ .

► **Example 4.** Suppose that the reliability of an HIV test is specified as follows :

Of people having HIV, 90% of the test detect the disease but 10% go undetected. Of people having HIV, 99% of the test are judged HIV+ive but 1% are diagnosed as showing HIV+ive. From a large population of which only 0.1% have HIV, one person is selected at random, given the HIV test, and the pathologist reports him/her as HIV+ive. What is the probability that the person actually has HIV?

► **Solution** Let E denote the event that the person selected is actually having HIV and A the event that the person's HIV test is diagnosed as +ive. We need to find  $P(E|A)$ .

Also  $E'$  denotes the event that the person selected is actually not having HIV.

Clearly,  $\{E, E'\}$  is a partition of the sample space of all people in the population.

We are given that

$$P(E) = 0.1\% = \frac{0.1}{100} = 0.001$$

$$P(E') = 1 - P(E) = 0.999$$

$P(A|E)$  = P(Person tested as HIV +ive given that he/she is actually having HIV)

$$= 90\% = \frac{90}{100} = 0.9$$

and  $P(A|E')$  = P(Person tested as HIV +ive given that he/she is actually not having HIV)

$$= 1\% = \frac{1}{100} = 0.01$$

Now, by Bayes' theorem

$$\begin{aligned}P(E|A) &= \frac{P(E) \cdot P(A|E)}{P(E) \cdot P(A|E) + P(E') \cdot P(A|E')} \\ &= \frac{0.001 \times 0.9}{0.001 \times 0.9 + 0.999 \times 0.01} = \frac{90}{1089} \\ &= 0.083 \text{ approx.}\end{aligned}$$

Thus, the probability that a person selected at random is actually having HIV given that he/she is tested HIV +ive is 0.083.

► **Example 5.** Given three identical boxes I, II and III, each containing two coins. In box I, both coins are gold coins, in box II, both are silver coins and in the box III, there is one gold and one silver coin. A person chooses a box at random and takes out a coin. If the coin is of gold, what is the probability that the other coin in the box is also of gold?

► **Solution** Let  $E_1$ ,  $E_2$  and  $E_3$  be the events that boxes, I, II and III are chosen, respectively.

Then,  $P(E_1) = P(E_2) = P(E_3) = 1/3$

Also, let A be the event that 'the coin drawn is of gold'

Then  $P(A|E_1) = P(\text{a gold coin from bag I}) = 2/2 = 1$

$P(A|E_2) = P(\text{a gold coin from bag II}) = 0$

$P(A|E_3) = P(\text{a gold coin from bag III}) = 1/2$

Now, the probability that the other coin in the box is of gold = the probability that gold coin is drawn from the box I =  $P(E_1|A)$

By Bayes' theorem, we know that

$$P(E_1|A) = \frac{P(E_1) P(A|E_1)}{P(E_1) P(A|E_1) + P(E_2) P(A|E_2) + P(E_3) P(A|E_3)}$$

$$= \frac{\frac{1}{3} \times 1}{\frac{1}{3} \times 1 + \frac{1}{3} \times 0 + \frac{1}{3} \times \frac{1}{2}} = \frac{2}{3}$$

► **Example 6.** In a factory which manufactures bolts, machines A, B and C manufacture respectively 25%, 35% and 40% of the bolts. Of their outputs, 5, 4 and 2 percent are respectively defective bolts. A bolt is drawn at random from the product and is found to be defective. What is the probability that it is manufactured by the machine B?

► **Solution** Let events  $B_1, B_2, B_3$  be the following:

$B_1$  : the bolt is manufactured by machine A

$B_2$  : the bolt is manufactured by machine B

$B_3$  : the bolt is manufactured by machine C

Clearly,  $B_1, B_2, B_3$  are mutually exclusive and exhaustive events and hence, they represent a partition of the sample space.

Let the event E be 'the bolt is defective'.

The event E occurs with  $B_1$  or with  $B_2$  or with  $B_3$ . Given that,  $P(B_1) = 25\% = 0.25$ ,  $P(B_2) = 0.35$  and  $P(B_3) = 0.40$

Again  $P(E|B_1)$  = Probability that the bolt drawn is defective given that it is manufactured by machine A =  $5\% = 0.05$

Similarly,  $P(E|B_2) = 0.04$ ,  $P(E|B_3) = 0.02$ .

Hence, by Bayes' Theorem, we have

$$P(B_2|E) = \frac{P(B_2) P(E|B_2)}{P(B_1) P(E|B_1) + P(B_2) P(E|B_2) + P(B_3) P(E|B_3)}$$

$$= \frac{0.35 \times 0.04}{0.25 \times 0.05 + 0.35 \times 0.04 + 0.40 \times 0.02}$$

$$= \frac{0.0140}{0.0345} = \frac{28}{69}$$

► **Example 7.** A parent particle can be divided into 0, 1 or 2 particles with probabilities  $\frac{1}{4}, \frac{1}{2}, \frac{1}{4}$  respectively. It appears after splitting. Beginning with one particle, the progenitor, let us denote by  $X_i$ , the number of particles in the  $i$  generation. Find

(i)  $P(X_2 > 0)$

(ii) the probability that  $X_1 = 2$  given that  $X_2 = 1$

► **Solution** (i)  $P(X_2 > 0) = 1 - P(X_2 = 0)$

But  $P(X_2 = 0)$

$$= P(X_1 = 0) P(X_2 = 0|X_1 = 0) + P(X_1 = 1) P(X_2 = 0|X_1 = 1) + P(X_1 = 2) P(X_2 = 0|X_1 = 2)$$

$$= \frac{1}{4} \cdot 1 + \frac{1}{2} \cdot \frac{1}{4} + \frac{1}{4} \cdot \frac{1}{16} = \frac{25}{64}$$

(ii)  $P(X_1 = 2|X_2 = 1)$

$$= \frac{P(X_1 = 2) P(X_2 = 1|X_1 = 2)}{P(X_1 = 1) P(X_2 = 1|X_1 = 1) + P(X_1 = 2) P(X_2 = 1|X_1 = 2)}$$

$$= \frac{\frac{1}{4} \cdot \frac{1}{4}}{\frac{1}{2} \cdot \frac{1}{2} + \frac{1}{4} \cdot \frac{1}{4}} = \frac{1}{5}$$

### Practice Problems

[N]

1. An urn contains 5 red and 5 black balls. A ball is drawn at random, its colour is noted and is returned to the urn. Moreover, 2 additional balls of the colour drawn are put in the urn and then a ball is drawn at random. What is the probability that the second ball is red?
2. Of the students in a college, it is known that 60% reside in hostel and 40% are day scholars (not residing in hostel). Previous year results report that 30% of all students who reside in hostel attain A grade and 20% of day scholars attain A grade in their annual examination. At the end of the year, one student is chosen at random from the college and he has an A grade. What is the probability that the student is a hostlier?



3. There are three coins. One is a two headed coin (having head on both faces), another is a biased coin that comes up heads 75% of the time and third is an unbiased coin. One of the three coins is chosen at random and tossed, it shows heads, what is the probability that it was the two headed coin?
4. A factory has two machines A and B. Past record shows that machine A produced 60% of the items of output and machine B produced 40% of the items. Further, 2% of the items produced by machine A and 1% produced by machine B were defective. All the items are put into one stockpile and then one item is chosen at random from this and is found to be defective. What is the probability that it was produced by machine B?
5. A manufacturer has three machine operators A, B and C. The first operator A produces 1% defective items, where as the other two operators B and C produces 5% and 7% defective items respectively. A is on the job for 50% of the time, B is on the job for 30% of the time and C is on the job for 20% of the time. A defective item is produced, what is the probability that it was produced by A?
6. A card from a pack of 52 cards is lost. From the remaining cards of the pack, two cards are drawn and are found to be both diamonds. Find the probability of the lost card being a diamond.
7. A letter is known to have come from either MAHARASTRA or MADRAS. On the postmark only consecutive letters RA can be read clearly. What is the chance that the letter came from MAHARASTRA?
8. A man has 10 coins and one of them is known to have two heads. He takes one at random and tosses it 5 times and it always falls head. What is the chance that it is the coin with two heads?
9. Suppose a girl throws a die. If she gets a 5 or 6, she tosses a coin three times and notes the number of heads. If she gets 1, 2, 3 or 4, she tosses a coin once and notes whether a head or tail is obtained. If she obtained exactly one head, what is the probability that she threw 1, 2, 3 or 4 with the die?
10. There are three urns. Each urn contains one white ball. In addition, urn I contains one black ball, urn II contains two, and urn III contains 3. An urn is selected and one ball is drawn. The probability for selecting the three urns is  $\frac{1}{6}$ ,  $\frac{1}{2}$  and  $\frac{1}{3}$ , respectively. If we know that a white ball is drawn, how does this alter the probability that a given urn was selected?
11. A gambler is told that one of three slot machines pays off with probability  $\frac{1}{2}$ , while each of the other two pays off with probability  $\frac{1}{3}$ .
  - (i) If the gambler selects one at random and plays it twice, what is the probability that he will lose the first time and win the second?
  - (ii) If he loses the first time and wins the second, what is the probability that he chose the favorable machine?
12. Assume that the chances of a patient having a heart attack is 40%. It is also assumed that a mediation and yoga course reduce the risk of heart attack by 30% and prescription of certain drug reduces its chances by 25%. At a time a patient can choose any one of the two options with equal probabilities. It is given that after going through one of the two options the patient selected at random suffers a heart attack. Find the probability that the patient followed a course of meditation and yoga?

#### 4.15 VALUE OF TESTIMONY

The theory of probability has been used to estimate the value of the testimony of witnesses. Such an application is open to adverse criticism. It rests on two assumptions which can hardly be justified, namely: (i) that to each witness there pertains a constant  $p$  (his credibility), which measures the average frequency with which he speaks the truth; (ii) that the statements of witness are independent of

one another in the sense required in the theory of probability.

If we are prepared to make these assumptions, the procedure is as follows:

Two independent witnesses, A and B, whose probabilities of speaking the truth are  $p_1$  and  $p_2$  respectively, agree in making a certain statement. What is the probability that the statement is true? Here the observed event is the fact that A and B make the same statement.

Before the event there are four hypotheses;

- for A and B may both speak truly;
- or A may speak truly, B falsely;
- or A may speak falsely, B truly;
- or A and B may both speak falsely.

The probabilities of these four hypotheses are

$p_1 p_2$ ,  $p_1(1 - p_2)$ ,  $p_2(1 - p_1)$ ,  $(1 - p_1)(1 - p_2)$  respectively.

The 'event' is the agreement of A and B in making a certain statement. The possibilities are :

- (i) the statement is true; (ii) is false

If it is true, the chance that they both say it is true is  $p_1 p_2$ .

If it is false, the chance that they both say it is true is  $(1 - p_1)(1 - p_2)$ .

Thus the antecedent probabilities of the event on the hypotheses (i), (ii) are

$p_1 p_2$  and  $(1 - p_1)(1 - p_2)$ ,

Hence after the observed event, in which A and B make the same statements, the probability that the statement is true is to the probability that it is false as  $p_1 p_2$  to  $(1 - p_1)(1 - p_2)$ ; that is, the probability that the joint statement is true is

$$\frac{p_1 p_2}{p_1 p_2 + (1 - p_1)(1 - p_2)}.$$

Similarly if a third person, whose probability of speaking the truth is  $p_3$ , makes the same statement, the probability that the statement is true is

$$\frac{p_1 p_2 p_3}{p_1 p_2 p_3 + (1 - p_1)(1 - p_2)(1 - p_3)};$$

and so on for any number of persons.

In the previous discussion it has been supposed that we have no knowledge of the event except the

statement made by A and B; if we have information from other sources as to the probability of the truth or falsity of the statement, this must be taken into account in estimating the probability of the various hypotheses.

For instance, if A and B agree in stating a fact, of which the priori probability is  $P$ , then we should estimate the probability of the truth and falsity of the statement by

$P p_1 p_2$  and  $(1 - P)(1 - p_1)(1 - p_2)$  respectively.  
P (their combined statement is true)

$$= \frac{P p_1 p_2}{P p_1 p_2 + (1 - P)(1 - p_1)(1 - p_2)}.$$

Here it has been assumed that the statement given by all the independent witnesses can be given in two ways only, so that if all the witnesses tell falsehoods they agree in telling the same falsehood.

If this is not the case and  $c$  is the chance of their coincidence testimony then the

Probability that the statement is true =  $P p_1 p_2$

Probability that the statement is false

$$= c(1 - P)(1 - p_1)(1 - p_2)$$

P (their combined statement is true)

$$= \frac{P p_1 p_2}{P p_1 p_2 + c(1 - P)(1 - p_1)(1 - p_2)}$$

However chance of coincidence testimony is taken only if the joint statement is not contradicted by any witness.

► **Example 1.** A tells the truth with probability  $p$  and B with probability  $p'$ .

- (i) If they make the same statement, what is the chance that the statement is true?
- (ii) If A makes a statement and B says that A lies, what is the chance that A told the truth?

► **Solution** Let  $S$  = Statement made ;  $T$  = Statement  $S$  is true,  $F$  = Statement  $S$  is false.

$C$  = A and B make same statement  $S$  ;  $D$  = A asserts  $S$  and B denies it.

Then  $P(T) = P(F) = 1/2$ ,  $P(C|T) = pp'$ ,

$P(C|F) = qq'$ ,  $P(D|T) = qp'$ ,  $P(D|F) = p'q$ .

where  $p + q = 1$ ,  $p' + q' = 1$ . Now,

$$P(C) = P(T) P(C|T) + P(F) P(C|F) = \frac{1}{2} (pp' + qq').$$

$$\therefore P(T|C) = P(T) P(C|T) / P(C) = \frac{1}{2} pp' / \frac{1}{2} (pp' + qq')$$

$$= pp' / (pp' + qq'). \text{ [Bayes' Theorem]}$$

$$\text{Further, } P(D) = P(T) P(D|T) + P(F) P(D|F)$$

$$= \frac{1}{2} (pq' + qp)$$

If A, denotes the event that A tells the truth, then by Bayes' Theorem

$$P(A|D) = P(A) P(D|A) / P(D) = qp' / (pq' + qp).$$

► **Example 2.** A speaks truth 3 times out of 4, and B 7 times out of 10; they both assert that a white ball has been drawn from a bag containing 6 balls all of different colours : find the probability of the truth of the assertion.

► **Solution** There are two hypotheses;

- (i) their coincident testimony is true,
- (ii) it is false.

$$\text{Here } P_1 = \frac{1}{6}, P_2 = \frac{5}{6};$$

$$P_1 = \frac{3}{4} \times \frac{7}{10}, P_2 = \frac{1}{25} \times \frac{1}{4} \times \frac{3}{10};$$

for in estimating  $p_2$  we must take into account the chance that A and B will both select the white ball when it has not been drawn; this chance is

$$\frac{1}{5} \times \frac{1}{5} \text{ or } \frac{1}{25}$$

Now the probabilities of the two hypotheses are as  $P_1 p_1$  to  $P_2 p_2$ , and therefore as 35 to 1; thus the

probability that the statement is true is  $\frac{35}{36}$ .

► **Example 3.** A, B are two inaccurate arithmeticians whose chance of solving a given

question correctly are  $\frac{1}{8}$  and  $\frac{1}{12}$  respectively, if they

obtain the same result and if it is 1000 to 1 against their making the same mistake, find the chance that the result is correct.

► **Solution** A's chance of solving a question correctly =  $\frac{1}{8}$

B's chance of solving a question correctly =  $\frac{1}{12}$

Hence the chance of the compound event that they

both get the correct result is  $\frac{1}{8} \times \frac{1}{12} = \frac{1}{96}$ ; the chance

that they both fail to get correct result is

$\frac{7}{8} \times \frac{11}{12} = \frac{77}{96}$ . As the chance that they both commit

the same mistake is  $\frac{1}{1001}$ , hence the chance that

they get the same incorrect result is

$$\frac{1}{1001} \times \frac{7}{8} \times \frac{11}{12} = \frac{1}{13 \cdot 8 \cdot 12}$$

Thus, the chance that their solution is correct is to the chance that it is incorrect

$$= \frac{1}{8 \cdot 12} = \frac{1}{13 \cdot 8 \cdot 12} = 13 : 1$$

Hence the required chance =  $\frac{13}{14}$ .

► **Example 4.** A bag contains a balls, one of the which is white. The probabilities that A and B speak the truth are p, p' respectively. A ball is drawn from the bag, and A and B both assert that it is white. What are the odds in favour of its being white?

► **Solution** The 'event' is the agreement of A and B in making a statement. The possibilities are (i) it is true, (ii) it is false.

The priori probabilities  $P_1, P_2$ , of (i), (ii) are  $P_1 = 1/n, P_2 = (n-1)/n$ .

If  $p_1, p_2$  are the chances of the event under (i) and (ii), we have  $p_1 = pp'$ .

In the case of (ii),  $(n-1)$  balls remain in the bag, and one of these is white. The chance that A should choose this ball and wrongly assert that it was drawn from the bag is  $(1-p)(n-1)$ . The chance that B should do the same thing is  $(1-p')(n-1)$ ; hence

$$p_2 = (1-p)(1-p')(n-1)^2,$$

and the odds in favour of a white ball having been drawn are as

$$P_1 P_1 : P_2 P_2 = \frac{1}{n} p p' : \frac{n-1}{n} \cdot (1-p) \left(1-p \frac{n-1}{n}\right) \cdot \frac{1}{(n-1)^2} \cdot \\ = (n-1) p p' : (1-p) (1-p').$$

► **Example 5.** There is a raffle with 12 tickets and two prizes of Rs. 9 and Rs. 6. A, B, C whose probabilities of speaking the truth are  $\frac{1}{2}, \frac{2}{3}, \frac{3}{5}$  respectively, report the result to D, who holds one ticket. A and B assert that they have won the Rs. 9 prize, and C asserts that he has won the Rs. 6 prize; what is D's expectation?

► **Solution** Three cases are possible; D may have won Rs. 9, Rs. 6, or nothing, for A, B, C may all have spoken falsely.

Now we have the a priori probabilities

$$P_1 = \frac{1}{12}, P_2 = \frac{1}{12}, P_3 = \frac{10}{12};$$

$$\text{also } p_1 = \frac{1}{2} \times \frac{2}{3} \times \frac{2}{5} = \frac{4}{30}, p_2 = \frac{1}{2} \times \frac{1}{3} \times \frac{3}{5} = \frac{3}{30},$$

$$p_3 = \frac{1}{2} \times \frac{1}{3} \times \frac{2}{5} = \frac{2}{30};$$

$$\frac{P_1 p_1}{4} = \frac{P_2 p_2}{3} = \frac{P_3 p_3}{20} = \frac{1}{27};$$

hence D's expectation

$$= \frac{4}{27} \text{ of Rs. 9} + \frac{3}{27} \text{ of Rs. 6} = \text{Rs. 2}$$

#### 4.16 GEOMETRICAL PROBABILITY

The classical probability formula is extended as follows to include the case of continuous sets  $S$  of elementary outcomes. The application of geometry to problems of probability introduces in a natural way a broader concept of probability which goes beyond the elementary notion of the number of happenings of discrete events. Here, all points of interest lie within some prescribed region  $R$ . The probability measure is so defined that the total probability of the admissible region is unity, whereas that of all non-admissible region is zero. Since an uncountably

infinite number of points lie in a finite region  $R$  of space  $S$ , The probability measure has to be defined in terms of the geometrical measure appropriate to the dimensionality (e.g. length, area, volume, etc.).

Thus,

$$P\{x \in R\} = \frac{\text{Measure of region } R}{\text{Measure of region } S}, R \subseteq S.$$

It is generally assumed that the probability measure of  $R$  is directly proportional to geometrical measure of  $R$ . The following statements are axiomatic :

- (i) If a point is taken at random on a given straight line  $AB$ , the chance that it falls on a particular segment  $PQ$  of the line is  $PQ/AB$ .
- (ii) If a point is taken at random on the area  $S$  which includes an area  $\sigma$ , the chance that the point falls on  $\sigma$  is  $\sigma/S$ .
- (iii) If a point is taken at random within the volume  $V$  which includes a volume  $\rho$ , the chance that the point falls in  $\rho$  is  $\rho/V$ .

► **Example 1.** Assume that a power line connecting points  $A$  and  $B$  has been damaged by a storm. What is the probability that the damage occurred in the part of the line between a points  $C$  and  $D$ , which themselves lie between  $A$  and  $B$ ?

► **Solution** In this case the set of elementary events is infinite since the break may occur at any point along the line between  $A$  and  $B$ . It is natural to assume that the probability of the break in any section of the line is proportional to the length of that section. Since the probability of the damage along the whole segment is unity (the break has occurred), the probability of the break in  $CD$  is

$$P(A) = |CD| / |AB|.$$

► **Example 2.** A telephone subscriber has ordered a call. He may get it at any time in the next hour. What is the probability of the call taking place in the last 20 minutes of the hour?

► **Solution** Assume that event  $E$  is the call taking place in the last 20 minutes of the hour. Let us represent the sample space as a line segment of length 60. The elementary events favourable to the event  $E$  lie in the last third of the segment. consequently,

$$P(E) = 1/3.$$

► **Example 3.** If  $p$  is chosen at random in the closed interval  $[0, 5]$ , what is the probability of the equation

$$x^2 + px + \frac{1}{4}(p+2) = 0 \text{ to have real roots?}$$

► **Solution** Here  $n(S) = \text{length of the interval } [0, 5] = 5$

$n(E) = \text{length of the interval } \subseteq [0, 5] \text{ in which } p \text{ belongs such that the given equation has real roots.}$

Now,  $x^2 + px + \frac{1}{4}(p+2) = 0$  will have real roots if  $D \geq 0$ .

$$D \geq 0 \Rightarrow p^2 - 4 \cdot 1 \cdot \frac{1}{4}(p+2) \geq 0$$

$$\text{or } p^2 - p - 2 \geq 0$$

$$\text{or } (p-2)(p+1) \geq 0.$$

So, by sign-scheme,  $p \leq -1$  or  $p \geq 2$ .

But  $p \in [0, 5]$ . Hence  $E = [2, 5]$ .

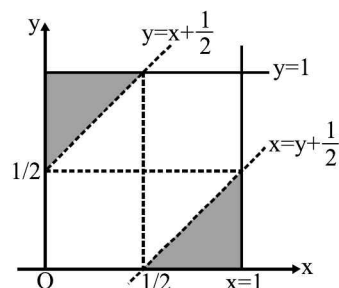
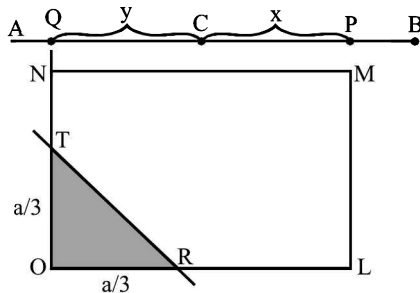
$$\therefore n(E) = \text{length of the interval } [2, 5] = 3.$$

$$\therefore \text{The required probability} = \frac{n(E)}{n(S)} = \frac{3}{5}$$

► **Example 4.** Two points are selected in a line  $AB$  of length  $a$  so as to lie on opposite sides of its middle point  $C$ . Find the probability that the distance between them is less than  $a/3$ .

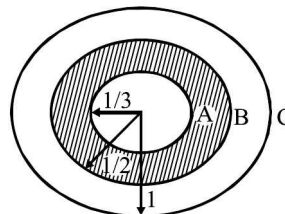
► **Solution** Choose points  $P$  and  $Q$  on the segments  $CB$  and  $CA$  and let  $CP = x$ ,  $CQ = y$ , so that the random choice provides  $0 \leq x \leq a/2$ ,  $0 \leq y \leq a/2$ . We are interested in the event  $x + y \leq a/3$ . Draw the square  $OLMN$  of side length  $a/2$  and its sub-region triangle  $ORT$ , with  $OR = OT = 1/3$ . The required probability is

$$p = \frac{1}{2} \cdot \left(\frac{1}{3}a\right) \left(\frac{1}{3}a\right) / \left(\left(\frac{1}{2}a\right) \left(\frac{1}{2}a\right)\right) = \frac{2}{9}$$



► **Example 5.** Two points are chosen independently and at random from the interval  $[0, 1]$ . What is the chance that the two numbers differ by more than  $1/2$ ?

► **Solution** Let the numbers chosen be denoted by  $X$  and  $Y$ . Then their ranges are  $0 \leq x, y \leq 1$ . We need to find  $p = P\{|X - Y| \leq \frac{1}{2}\}$ . This corresponds to the sum of two shaded areas of the unit square, obviously,  $p = 1/4$ .



► **Example 6.** A circular target is divided into three zones bounded by concentric circles of radii  $1/3$ ,  $1/2$ ,  $1$  respectively. If three shots are fired at random at the target, what is the probability that exactly one shot lands in each zone?

► **Solution** The measure of sample space [area of circle of unit radius] is  $\pi(1)^2$  i.e.  $\pi$ . Hence, the probability of a shot falling in the region of radius  $a$  ( $a < 1$ ) is  $p = \pi a^2 / \pi = a^2$ . Thus, the probability of a shot falling in region A ( $a = 1/3$ ) is  $p_1 = 1/9$ , probability of a shot falling in region B is  $p_2 = (1/2)^2 - (1/3)^2 = 5/36$  and the probability of a shot falling in region C is  $p_3 = 1 - (1/2)^2 = 3/4$ . The shots fall in the three zones independently of each other. Hence by multiplication rule, the probability of one arrangement (say first shot in A, second shot in B and

third shot in C) is  $p_1 p_2 p_3 = 5/432$ . Since there are  $3! = 6$  possible disjoint arrangements, the required probability is  $6 \times 5/432 = 5/72$ .

► **Example 7.** The sum of two positive quantities is equal to  $2n$ . Find the chance that the product of the two quantities is not less than  $3/4$  times their greatest product.

► **Solution** Let  $x > 0, y > 0$  be the given quantities so that  $x + y = 2n$ . To find the greatest value of  $xy$ , we notice that  $z = xy = x(2n - x) = 2nx - x^2$ . Now for greatest value of  $xy$ ,  $x = n$  and hence  $y = n$  as  $x + y = 2n$ .

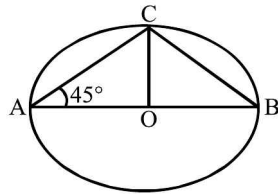
Thus, the greatest value of  $xy$  is  $n^2$ . Now, we want,  
 $p = P\{xy \geq (3n^2/4)\} = P\{xy > 3n^2/4\}$  (Put  $y = 2n - x$ )  
 $= P\{4x^2 - 8nx + 3n^2 \leq 0\} = P\{(2x - 3n)(2x - n) \leq 0\}$   
 $= P\{n/2 \leq x \leq 3n/2\}$ .

Since  $0 \leq x \leq 2n$ , the total range of  $x$  is  $2n$ .  
 The favourable range is  $(3n/2) - (n/2)$ . It follows that,  
 $p = (n/2n) = 1/2$ .

► **Example 8.** AOB is a fixed diameter of a circle of radius  $a$ . A point is chosen at random within the circle. Find the chance that the point is within an inscribed isosceles triangle with AOB as one of the three sides.

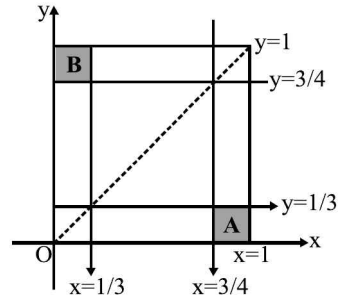
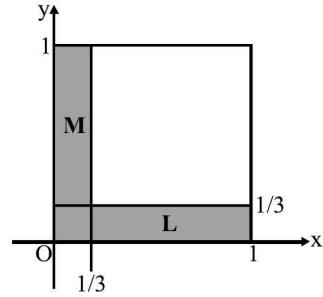
► **Solution** If ABC is isosceles triangle  
 ( $AB = 2a, AC = BC = a\sqrt{2}$ )

its area is  $\Delta = a^2$ . There are two such triangles (two opposite sides of AOB), hence  $p = 2a^2/\pi a^2 = 2/\pi$ .



► **Example 9.** Two numbers are selected independently at random in the interval  $[0, 1]$ . If the smaller one is less than  $1/3$ , prove that the chance that the larger one is greater than  $3/4$  is  $p = 3/10$ .

► **Solution** Here  $S = \{(x, y) : 0 \leq x, y \leq 1\}$ ; so that geometric configuration is a unit square.  
 Let  $M = \max(x, y)$  and  $m = \min(x, y)$ .



We need find

$$p = \{M \geq 3/4 \mid m \leq 1/3\} = P\{M \geq 3/4, m \leq 1/3\} / P\{m \leq 1/3\}.$$

Now  $P\{m \leq 1/3\} = \text{Shaded Area L}$

$$\text{and } M = \frac{1}{3} + \frac{1}{3} - \frac{1}{9} = \frac{5}{9}.$$

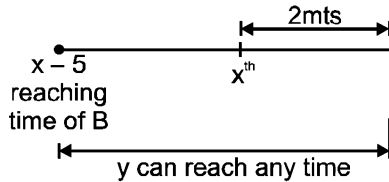
$$P\{M \geq 3/4, m \leq 1/3\}$$

$$= \text{Shaded Area A and B} = 2 \times \frac{1}{3} \cdot \frac{1}{4} = \frac{1}{6}.$$

$$\therefore p = (1/6)/(5/9) = 3/10$$

► **Example 10.** At a random moment within a period of 20 minutes, student 'A' telephones student 'B', waits for 2 minutes and then puts down the receiver. During the same 20 minutes student 'B' arrives home at a random moment, stays for 5 minutes and then leaves. Find the chance the two will have a talk. Find your answer correct upto two places of decimals.

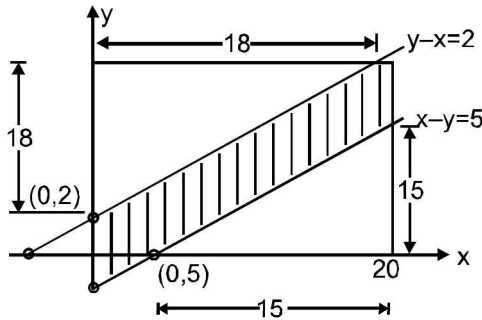
► **Solution** Let  $\left. \begin{array}{l} \text{A gives a ring at } x^{\text{th}} \text{ minute} \\ \text{B arrives at } y^{\text{th}} \text{ minute} \end{array} \right\} \text{ in}$   
the same time interval



where  $0 \leq x \leq 20$  &  $0 \leq y \leq 20$ .

Hence  $y \leq x + 2$  &  $y \geq x - 5$

or  $y - x \leq 2$  &  $x - y \leq 5$



$$\therefore n(S) = 20 \times 20 = 400$$

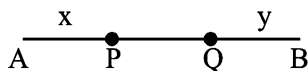
$$\begin{aligned} n(A) &= 400 - \left[ \frac{324 + 225}{2} \right] \\ &= 400 - \frac{549}{2} = \frac{251}{2} \end{aligned}$$

$$\begin{aligned} \therefore p &= \frac{\frac{251}{2}}{400} = \frac{251}{800} \\ &= \text{approximately } 0.31 \end{aligned}$$

► **Example 11.** A line is divided at random into three parts, what is the chance that they form the sides of a possible triangle?

► **Solution** Let AB be the straight line divided at P and Q, and suppose AB = a,

AP = x and BQ = y.



Now in favourable cases we must have,

$$x < \frac{a}{2}, y < \frac{a}{2}, PQ < \frac{a}{2}, \text{ or } (a - x + y) < \frac{a}{2}$$

$$\text{or } x + y > \frac{a}{2}. \quad \dots(1)$$

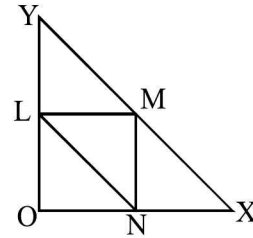
And for the possible cases, we must have,

$$x > 0 \text{ and } < a, y > 2 \text{ and } < a, x + y < a. \quad \dots(2)$$

Again we take a pair of rectangular axes OX, OY, let each be equal to a. X, Y, and bisect XU, OX, OY in M, N, L respectively.

Then LN is the line  $x + y = \frac{a}{2}$ , MN, LM are the lines

$x = \frac{a}{2}, y = \frac{a}{2}$  respectively.



The conditions (1) are only satisfied by points in the triangle LMN, while the conditions (2) are satisfied by all points within the triangle OXY.

Since the area of the triangle OXY is equal to four times the area of the triangle LMN, hence the required chance = 1/4.

► **Example 12.** A coin of radius  $r < a/2$  is thrown at random onto an infinite chessboard with a side of a square a. Find the probabilities of the following events:

► **Solution**

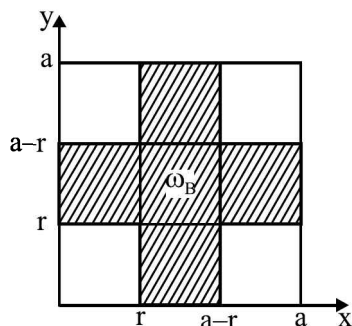
A = {the coin finds itself entirely inside one square},  
B = {the coin intersects atmost one side of a square}.

Let (x, y) be the coordinate of the centre of the coin. By virtue of infinity of the chessboard it may be assumed that the elementary outcomes of the experiment are completely determined by the position of the centre of the coin relative to the vertices of the square containing that centre. Placing the origin of coordinates into one of the vertices of the square we can write the set of elementary outcomes as

$S = \{(x, y) \mid 0 \leq x, y \leq a - r\}$ , i.e., it is a square with side  $a - 2r$ . From the geometrical probability formula

(1) we find

$$P(A) = \frac{\text{area}(\omega_A)}{\text{area}(S)} = \frac{(a-2r)^2}{a^2}.$$



The set  $\omega_B$ , which has a more complex structure, is represented in Figure.

Since  $S(\omega_B) = a^2 - 4r^2$ , we find,

$$P(B) = \frac{\text{area}(\omega_B)}{\text{area}(S)} = 1 - 4 \frac{r^2}{a^2}.$$

If a random event whose probability must be found reduces to deciding whether a point lies within a certain section of a plane figure, the boundaries of the figure and the section being the curves of known functions, then the calculation of the areas in expression (1) reduces to the calculation of definite integrals.

► **Example 13.** Two real numbers,  $x$  and  $y$ , are selected at random, given that  $0 \leq x \leq 1$ ,  $0 \leq y \leq 1$ . Find the probability that  $y^2 \leq x$ .

► **Solution** Assume that a point with coordinates  $(x, y)$  corresponds to the pair of numbers  $x$  and  $y$ . The sample space is a square whose sides are unit segments of the coordinate axes. A figure whose set of points corresponds to the outcome favourable to the event  $y^2 \leq x$  is bounded by the graphs of the functions  $y = 0$ ,  $x = 1$  and  $y^2 = x$ . This area can be calculated by the formula

$$S = \int_0^1 \sqrt{x} dx = \frac{2}{3} x^{3/2} \Big|_0^1 = \frac{2}{3}.$$

Since the area of a unit square is unity, it follows that  $P(A) = 2/3$ .

### Practice Problems



1. A speaks the truth 3 out of 4 times, and B 5 out of 6 times. What is the probability that they will contradict each other in stating the same fact?
2. A speaks the truth 2 out of 3 times, and B 4 times out of 5. They agree in the assertion that from a bag containing 6 balls of different colours a red ball has been drawn. Find the probability that the statement is true.
3. Two persons, whose probabilities of speaking the truth are  $\frac{2}{3}$  and  $\frac{5}{6}$  respectively, assert that a specified ticket has been drawn out of a bag containing 15 tickets. What is the probability of the truth of the assertion?
4. The sides of a rectangle are taken at random each less than  $a$  and all lengths are equally likely. Show that the chance that the diagonal is less than  $a$  is  $\pi/4$ .
5. A point is taken at random within an ellipse  $(x^2/a^2) + (y^2/b^2) = 1$ . Show that the chance that its distance from the centre exceeds  $b$  is  $1 - (b/a)$ .
6. A point is selected at random inside a circle. Prove that the chance that the point is closer to the centre of the circle than to its circumference is  $1/4$ .
7. A bus of line A arrives at a station every 4 minutes and a bus of line B every 6 minutes. The length of an interval between the arrival of a bus of line A and a bus of line B may be any number of minutes from 0 to 4, all equally likely. Show that (i)  $P\{\text{The first bus that arrives belongs to line A}\} = 2/3$  (ii)  $P\{\text{A bus of any line arrives within 2 minutes}\} = 2/3$ .
8. Two real numbers  $x$  and  $y$  are chosen at random and are such that  $|x| \leq 3$ ,  $|y| \leq 5$ . What is the probability of the fraction  $x/y$  being positive?
9. A real number  $x$  is selected at random from the solution set of the inequality  $y^2 - y - 6 \leq 0$ . What is the probability that  $\frac{(x+1)(x-2)}{x-4} \geq 0$ ?



10. Two positive numbers  $x$  and  $y$  are taken at random neither of them exceeding unity. What is the probability of their sum not exceeding unity if the sum of their squares exceeds  $1/4$ ?
11. A parabola touches the base of a square and passes through its upper vertices. What is the probability that a point thrown in the square at random will fall in the domain between the upper side of the square and the parabola?
12. At a random moment within a period of 20 minutes student A telephones student B, waits for 2 minutes and then puts down the receiver. During the same 20 minutes student B arrives home at a random moment, stays for 5 minutes and then leaves. What is the probability that the two will have a talk?
13. Two steamers are to come to the same pier. The times of their arrival are independent and equally possible during a given twenty-four hours. Calculate the probability that one of the steamers will have to wait for the pier to be free if the berthing time of the first steamer is one hour and that of the second, two hours.
14. Given two concentric circles of radii  $r_2 > r_1$ . Two points A and B are placed at random on the larger circle. What is the probability that the segment AB will not intersect the smaller circle?
15. Two numbers are selected at random from the interval  $[-1, 2]$ . What is the probability that their sum will be greater than unity and their product less than unity?



### Target Problems for JEE Advanced

► **Problem 1.** An urn contains  $N$  balls numbered 1 through  $N$ , where the first  $D$  balls are defective and the remaining  $N - D$  are non-defective. A sample of size  $n$  balls is drawn from the urn. Let  $A_k$  be the event that the sample of  $n$  balls contains exactly  $k$  defectives. Find  $P(A_k)$  when the sample is drawn (i) without replacement and (ii) with replacement.

► **Solution** (i) The number of subsets of size  $n$  of

the  $N$  balls is  $n(S) = {}^N C_n$  and all such selections may be treated equiprobable. Now  $A_k$  consists of those samples of size  $n$  which contain exactly  $k$  defective balls from the balls that are numbered 1 to  $D$  inclusive. Such a selection can be made in  ${}^D C_k$  ways. The remaining  $(n - k)$  balls are to be drawn from balls numbered  $D + 1$  to  $N$  and this selection can be made in  ${}^{N-D} C_{n-k}$  ways.

The favourable cases for the required events is  $n(A)$  ways. Hence

$$P(A_k) = \frac{n(A)}{n(S)} = \frac{{}^D C_k \cdot {}^{N-D} C_{n-k}}{{}^N C_n}.$$

- (ii) Under replacement scheme, the total number of drawing  $n$  balls out of  $N$  balls in  $(N)^n$ , since at each draw, the ball can be one from 1 to  $N$  inclusive.

Now  $A_k$  consists of exactly  $k$  of the balls numbered from 1 to  $D$  inclusive; these  $k$  balls can be selected in  ${}^D C_k$  ways to fill  $k$  positions out of  $n$  available positions.

For each different  ${}^D C_k$  selection, there are  $D^k (N - D)^{n-k}$

different  $n$ -tuples. Hence

$$P(A_k) = {}^D C_k D^k (N - D)^{n-k} / (N)^n = {}^D C_k q^k p^{n-k},$$

where  $p = D/N$ ,  $q = 1 - p$ .

► **Problem 2.** Two integers  $n_1$  and  $n_2$  are drawn at random out of the set  $\{1, 2, 3, \dots, 9\}$  with replacement and the equation,  $x^2 + \sqrt{2} |n_1 - n_2| x + n_1 = 0$  is formed. Find the probability that the equation has real roots.

► **Solution** For real roots  $(n_1 - n_2)^2 \geq 2 n_1$   
The possible pairs are

| $n_1$ | $n_2$               | Total |
|-------|---------------------|-------|
| 1     | 3, 4, 5, 6, 7, 8, 9 | 7     |
| 2     | 4, 5, 6, 7, 8, 9    | 6     |
| 3     | 6, 7, 8, 9          | 4     |
| 4     | 1, 7, 8, 9          | 4     |
| 5     | 1, 9                | 2     |
| 6     | 1, 2                | 2     |
| 7     | 1, 2, 3             | 3     |
| 8     | 1, 2, 3, 4          | 4     |

$$\begin{array}{ccc} 9 & 1, 2, 3, 4 & 4 \\ & & \text{-----} \\ & & 36 \end{array}$$

$$n(S) = 9 \cdot 9 = 81;$$

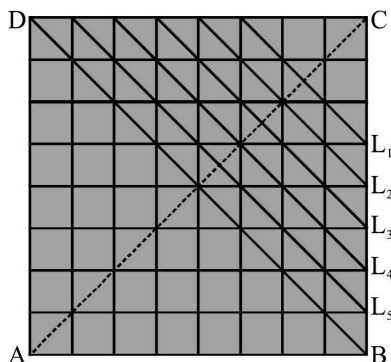
$$n(A) = 36; P(A) = \frac{36}{81} = \frac{4}{9}.$$

► **Problem 3.** If three small squares are chosen at random on a chess-board, find the chance that they are in a diagonal line.

► **Solution** In a chess-board, there are  $8 \times 8 = 64$  small squares. And any three small squares can be chosen at random in

$$n(S) = {}^{64}C_3 = 64 \times 31 \times 21 = 41664 \text{ ways.}$$

To choose 3 small squares so that they lie on a diagonal line



we observe that they lie on lines such as  $L_1, L_2, \dots, L_3, BD$  and lines on the other sides of  $BD$ , and an equal number of opportunities on the either side of the other main diagonal  $AC$ . Hence, the total number of favourable cases is

$$\begin{aligned} n(A) &= 2 {}^8C_3 + 4[{}^7C_3 + {}^6C_3 + {}^5C_3 + {}^4C_3 + {}^3C_3] \\ &= 2 \times 56 + 4(35 + 20 + 10 + 4 + 1) \\ &= 112 + 280 = 392. \end{aligned}$$

$$\text{Hence } p = n(A)/n(S) = 392/41664 = 7/744.$$

► **Problem 4.** If  $F$  is the set of all onto functions from  $A = \{a_1, a_2, \dots, a_n\}$  to  $B = \{x, y, z\}$  and  $f \in F$  is chosen randomly what is the probability that (a)  $f^{-1}(x)$  has 2 elements in it? (b)  $f^{-1}(x)$  is a singleton?

► **Solution** First let us count the functions from  $A$  to  $B$  which are onto. consider the three properties :  
Range of the function omits  $x$ ;  
Range of the function omits  $y$  ; and  
Range of the function omits  $z$ .

Number of onto functions

= Number of those functions which have none of the three properties above =  $n(0)$

= Total number of all functions –  $n(1) + n(2) - n(3)$ .

Now, total number of all functions is  $3^n$ .

Number of functions whose range omits  $x$  is  $2^n$ .

So  $n(1) = 3 \cdot 2^n$ .

Number of functions whose range omits  $x$  and  $y$  is just 1. So  $n(2) = 3$ .

Hence, the number of onto functions =  $3^n - 3 \cdot 2^n + 3$ .

(a) Among these we have now to count how many has 2 elements in  $f^{-1}(x)$ . Pick those 2 elements in the preimage of  $x$ . This can be done in  $n(n-1)/2$  ways. Map the remaining  $(n-2)$  elements of  $A$  to the two elements  $y, z$  of  $B$ . This has  $2^{n-2}$  possibilities. Of these we have to omit the 2 functions whose range is just  $\{y\}$  or  $\{z\}$  in order to stay within the population of onto functions. Hence the desired probability is

$$= \frac{\frac{n(n-1)}{2}(2^{n-2} - 2)}{3^n - 3 \cdot 2^n + 3} = \frac{n(n-1)(2^{n-2} - 1)}{3^n - 3 \cdot 2^n + 3}.$$

(b) We shall now count the onto functions which satisfy  $f^{-1}(x)$  is singleton. We can choose this singleton in  $n$  ways. The remaining  $(n-1)$  elements of  $A$  can be mapped onto  $\{y, z\}$  in  $2^{n-1} - 2$  ways. Thus the desired probability is

$$= \frac{n(2^{n-1} - 2)}{3^n - 3 \cdot 2^n + 3}.$$

► **Problem 5.** If  $n$  people are present in a room, what is the probability that no two of them celebrate their birthday on the same day of the year? How large need  $n$  be so that this probability is less than  $1/2$ ?

► **Solution** As each person can celebrate his or her birthday on any one of 365 days, there is a total of  $(365)^n$  possible outcomes. (We are ignoring the possibility of someone's having been born on February 29.) Assuming that each outcome is equally likely, we see that the desired probability is

$$\frac{(365)(364)(363) \dots (365 - n + 1)}{(365)^n}.$$

It is a rather surprising fact that when  $n \geq 23$ , this probability is less than  $1/2$ . That is, if there are 23

or more people in a room, then the probability that atleast two of them have the same birthday exceeds  $1/2$ . Many people are initially surprised by this result, since 23 seems so small in relation to 365, the number of days of the year. However, every pair of individuals

has probability  $\frac{365}{(365)^2} = \frac{1}{365}$  of having the same

birthday, and in a group of 23 people there are  ${}^{23}C_2 = 253$  different pairs of individuals. Looked at this way, the result no longer seems so surprising.

When there are 50 people in the room, the probability that atleast two share the same birthday is approximately .970. And with 100 persons in the room, the odds are better than 3,000,000 : 1 (that is,

the probability is greater than  $\frac{3 \times 10^6}{3 \times 10^6 + 1}$ ) that atleast two people have the same birthday.

► **Problem 6.** Six persons stand at random in a queue for buying cinema tickets individually. Three of them have only a five rupee note each while each of the other three has a ten rupee note only. The booking clerk has an empty cash box. Find the probability that the six persons will get tickets, each paying rupees five.

► **Solution** Here random experiment is : arranging 6 persons in a line.

Let S be the sample space and E be the required event. Then  $n(S) =$  total number of arrangements of six

persons =  $6! = 720$

Let a denote the person having 5 rupee note and b denote the person having 10 rupee note. Since all the six persons should get tickets, therefore, first place should be occupied by a person having 5 rupee note and sixth place should be occupied by a person having 10 rupee note. Therefore possible cases are the following :

|   |     |     |   |
|---|-----|-----|---|
| a | a a | a a | b |
| a | a b | a b | b |
| a | a b | a b | b |
| a | b a | b a | b |
| a | b a | b a | b |

Here a's can be arranged among themselves and b's can be arranged among themselves

$$\therefore n(E) = \underline{3} \cdot \underline{3} + \underline{3} \underline{3} + \underline{3} \underline{3} + \underline{3} \underline{3} + \underline{3} \underline{3} + \underline{3} \underline{3} = 180$$

∴ The required probability,

$$P(E) = \frac{n(E)}{n(S)} = \frac{180}{720} = \frac{1}{4}$$

► **Problem 7.** If n integers taken at random are multiplied together, show that the chance that the last

digit of the product is 1, 3, 9 is  $\frac{2^n}{5^n}$ , the chance of

its being 2, 4, 6 or 8 is  $\frac{4^n - 2^n}{5^n}$ , of its being 5 is

$\frac{5^n - 4^n}{10^n}$  and that of its being 0 is  $\frac{10^n - 8^n - 5^n + 4^n}{10^n}$ .

► **Solution** (i) If the last digit in the product is not 1, 3, 7 or 9, it must be 0, or 5 or even. Therefore none of the n numbers must end in 0, 2, 4, 5, 6 or 8. Hence out of 10 numbers in which any of the n numbers may end, four are favourable to the event that none of the n numbers end in 0, 1, 4, 5, 6 or 8.

Therefore, the chances that each of the n numbers should not end in any of these is  $\frac{4}{10}$ , and thus the

required chance =  $\left(\frac{2}{5}\right)^n = \frac{2^n}{5^n}$ .

(ii) If the last digit in the product be 2, 4, 6 or 8 none of the numbers must end in 0 or 5 and one of the last digits must be even. Now out of 10 numbers in which any one of the n numbers may end, 8 are favourable to the event that none of the numbers end in 0 and 5. Hence  $8^n$  is the number of ways in which we can exclude 0 and 5; and of these  $4^n$  are the cases in which the last digit can be selected solely from 1, 3, 7 or 9. Thus the favourable numbers of cases to the event that the last digit in the product is 2, 4, 6 or 8 is  $8^n - 4^n$ .

Again the total number of ways are  $10^n$ , as there are 10 different ways in which any of the numbers may be selected from 10 digit.

Thus the required chances

$$= \frac{8^n - 4^n}{10^n} = \frac{4^n - 2^n}{5^n}.$$

- (iii) If the last digit in the product is 5, one of numbers must end in 5 and all the rest must be odd. Now there are 5 odd digits in all and hence  $5^n$  is the number of ways in which an odd digit can be chosen to end the number in the odd digit; and of these  $4^n$  are the cases in which the odd digit can be selected solely from 1, 3, 7, or 9. Thus the favourable number of cases are

$$5^n - 4^n. \text{ Hence the required chances} = \frac{5^n - 4^n}{10^n}.$$

- (iv) To find the chance that the last digit in the product of  $n$  numbers is 0, we first find the chance that the last digit is not zero, that is 1, 2, 3, 4, 5, 6, 7, 8 or 9.

Now the chances that the last digit in the

product is 1, 3, 7 or 9 is  $\frac{2^n}{5^n}$ ; the chance that

it is 2, 4, 6 or 8 is  $\frac{4^n - 2^n}{5^n}$ ; and the chance that

it is 5 is  $\frac{5^n - 4^n}{10^n}$ .

Hence the chance that the last digit is not zero is the last digit is not zero is the sum of these three events, i.e., is equal to

$$\frac{2^n}{5^n} + \frac{4^n - 2^n}{5^n} + \frac{5^n - 4^n}{10^n} = \frac{8^n + 5^n - 4^n}{10^n}$$

Thus the required chance

$$= 1 - \frac{8^n + 5^n - 4^n}{10^n} = \frac{10^n - 8^n - 5^n + 4^n}{10^n}.$$

► **Problem 8.** Two number  $x$  and  $y$  are chosen at random from the set  $\{1, 2, 3, \dots, 3n\}$ . Find the probability that  $x^2 - y^2$  is divisible by 3.

► **Solution**  $x^2 - y^2 = (x + y)(x - y)$  and 3 is a prime number.

∴  $x^2 - y^2$  is divisible by 3 if  $x + y$  or  $x - y$  is divisible by 3.

Now,  $\{1, 2, 3, \dots, 3n\}$

$$= \{3, 6, 9, \dots, 3n\} \cup \{1, 4, 7, \dots, 3n - 2\}$$

$$\cup \{2, 5, 8, \dots, 3n - 1\} = A \cup B \cup C \text{ (say).}$$

Clearly, if  $x, y$  are selected from  $A$  or  $B$  or  $C$  then  $x + y$  or  $x - y$  are divisible by 3;

and, if  $x, y$  are selected one from  $B$  and the other from  $C$  then  $x + y$  is divisible by 3.

∴ The probability of  $x^2 - y^2$  is divisible by 3 = probability of selecting both  $x, y$  from  $A$  or  $B$  or  $C$  + probability of selecting  $x, y$  one from  $B$  and the other from  $C$

$$= \frac{{}^nC_2}{{}^{3n}C_2} \times 3 + \frac{{}^nC_1 \times {}^nC_1}{{}^{3n}C_2}$$

$$= \frac{3n(n-1)}{3n(3n-1)} + \frac{2n^2}{3n(3n-1)}$$

$$= \frac{3n-3}{3(3n-1)} + \frac{2n}{3(3n-1)} = \frac{5n-3}{3(3n-1)}$$

► **Problem 9.** Two balls are drawn at random in succession and without replacement from an urn containing three white balls and seven red balls. The events are :

$A = \{\text{the first ball is white}\}$ ,  $B = \{\text{the second ball is white}\}$ ,  $C = \{\text{atleast one of the balls drawn out is white}\}$ , Calculate the conditional probabilities  $P(B/P)$ ,  $P(A/B)$  and  $P(A/C)$ ,

► **Solution** To calculate the desired conditional probabilities, we number consecutively all the balls in the order shown in the scheme :

| white |   |   | red |   |   |   |   |   |    |
|-------|---|---|-----|---|---|---|---|---|----|
| ○     | ○ | ○ | ⊗   | ⊗ | ⊗ | ⊗ | ⊗ | ⊗ | ⊗  |
| 1     | 2 | 3 | 4   | 5 | 6 | 7 | 8 | 9 | 10 |

According to the description of the experiment we have the following scheme : drawing at random without replacement a pair of numbers from the set  $\{1, 2, \dots, 10\}$  with ordering, therefore the set of elementary outcomes can be written as

$$S = \{\omega_{ij} = (i, j) \mid i = 1, \dots, 10; j = 1, \dots, 10; i \neq j\}$$

It follows that  $n(S) = {}^{10}P_2 = 90$ ,  $P(\omega_{ij}) = 1/90$  for all allowed values of  $i$  and  $j$ .

The subsets corresponding to the events  $A$  and  $B$  have the following composition :

$$A = \{(i, j) \mid i = 1, 2, 3; j = 1, \dots, 10; i \neq j\},$$

$$B = \{(i, j) \mid i = 1, \dots, 10; j = 1, 2, 3; i \neq j\}.$$

It is not hard to compute that  $n(A) = 3 \times 9$ ,  
 $n(B) = 9 \times 3$ ,

$$P(A) = \frac{3}{10}, \quad P(B) = \frac{3}{10}.$$

Corresponding to the event  $AB$  is the subset  
 $AB = \{(i, j) \mid i = 1, 2, 3; j = 1, 2, 3; i \neq j\}$ ;  
hence  $n(AB) = 3 \times 2$ ,

$$P(AB) = \frac{1}{15}.$$

Since  $C = A + B$ , applying the addition formula yields

$$P(C) = P(A + B) = P(A) + P(B) - P(AB) = 8/15.$$

To calculate the probabilities of the product  $AC$  we notice that  $AB \subset A$ ,

$$AC = A(A + B) = A + AB = A.$$

$$\text{Hence } P(AC) = P(A) = \frac{3}{10}$$

Further, by formula we get

$$P(B/A) = \frac{P(AB)}{P(A)} = \frac{2}{9}, \quad P(A/B) = \frac{P(AB)}{P(B)} = \frac{2}{9},$$

$$P(A/C) = \frac{P(A)}{P(C)} = \frac{9}{16}.$$

► **Problem 10.** Cards are dealt one by one from a well shuffled deck until an ace appears.

- (i) Show that the chance that exactly  $n$  cards are dealt in all before the first ace, is  

$$\frac{(51 - n)(49 - n)}{51 \cdot 50 \cdot 49 \cdot 13};$$

- (ii) If the cards are continued to be dealt until a second ace, is  

$$\frac{r(51 - n)(50 - n)}{50 \cdot 49 \cdot 17 \cdot 13}.$$

► **Solution** (i) Exactly  $n$  cards are dealt before the first ace means that  $(n + 1)$ th deal is the first ace. Thus  $n$  cards are extracted from the non-ace cards  $48 (= 52 - 4)$  and  $(n + 1)$ th card is extracted from  $52 - n$  cards. Let the events be

$$A = \{n \text{ cards out of } 48 \text{ non-ace cards}\};$$

$$B = \{(n + 1) \text{th card from } 52 - n \text{ cards}\}.$$

$$P(AB) = P(A)P(B|A) = \frac{{}^{48}C_n}{52} \cdot \frac{4}{52 - n}$$

$$= \frac{48!}{52!} \frac{(52 - n)!}{(48 - n)!} \frac{4}{52 - n} = \frac{(51 - n)(49 - n)}{51 \cdot 50 \cdot 49 \cdot 13}$$

- (ii) Here  $(r + 1)$ th card is to be the second ace; thus there is one ace in the first  $r$  cards. Define  
 $C = \{(r - 1) \text{ cards out of } 48 \text{ cards and one ace}\},$   
 $D = \{(r + 1) \text{th card second ace}\}$

$$\begin{aligned} \therefore P(CD) &= P(C)P(D|C) = \frac{{}^4C_1 {}^{52}C_{r-1}}{{}^{52}C_r} \frac{3}{52 - r} \\ &= 12 \cdot \frac{48!}{52!} \frac{(52 - r)!}{(49 - r)!} \frac{r!}{(r - 1)!} \\ &= \frac{r(51 - r)(50 - r)}{50 \cdot 49 \cdot 17 \cdot 13} \end{aligned}$$

### Multiplication rule of probability for more than two events

If  $A$ ,  $B$  and  $C$  are three events of sample space, we have

$$P(A \cap B \cap C) = P(A)P(B|A)P(C|AB)$$

The following example illustrates the extension of multiplication rule of probability for three events.

► **Problem 11.** Two drawings, each of 3 balls, are made from a bag containing 5 white and 8 black balls, the balls not being replaced before the second trial: find the chance that the first drawing will give 3 white and the second 3 black balls.

► **Solution** At the first trial, 3 balls may be drawn in  ${}^{13}C_3$  ways; and 3 white balls may be drawn in  ${}^5C_3$  ways; therefore the chance of 3 white at first trial

$$= \frac{5.4}{1.2.3} \frac{13.12.11}{1.2.3} = \frac{5}{143}$$

When 3 white balls have been drawn and removed, the bag contains 2 white and 8 black balls; therefore at the second trial 3 balls may be drawn in  ${}^{10}C_3$  ways; and 3 black balls may be drawn in  ${}^8C_3$  ways; therefore the chance of 3 black at the second trial

$$= \frac{8.7.6}{1.2.3} \frac{10.9.8}{1.2.3} = \frac{7}{15};$$

therefore the chance of the compound event

$$= \frac{15}{143} \times \frac{7}{15} = \frac{7}{429}.$$

► **Problem 12.** There are 100 students in a lecture-hall, 50 know Russian, 40 know French and 35 know

German. Twenty students know Russian and French, eight Russian and German, ten French and German. Five students know Russian, French and German. One of the students leaves the hall. Consider the following events :  $R = \{\text{the student knows Russian}\}$ ,  $F = \{\text{the student knows French}\}$ ,  $D = \{\text{the student knows German}\}$ .

Calculate the probabilities of the following events:  
 $A = \{\text{the student knows either Russian or French}\}$ ,  
 $B = \{\text{the student knows only Russian}\}$ ,  
 $C = \{\text{the student knows none of the languages}\}$ .

► **Solution** Since  $A = R + F$ , we get, using the addition formula

$$P(A) = P(R) + P(F) - P(RF) = 0.5 + 0.4 - 0.2 = 0.7.$$

The event  $B$  can be represented as  $B = R\overline{D}\overline{F}$ .

$$\begin{aligned} P(R\overline{D}\overline{F}) &= P(R) P(\overline{D}\overline{F} / R) = P(R) [1 - P(D + F/R)] \\ &= P(R) - P(R) P(D + F/R) = P(R) - P(DR + FR) \\ &= P(R) - [P(DR) + P(FR) - P(DRF)] \\ &= 0.5 - 0.08 - 0.2 + 0.05 = 0.27. \end{aligned}$$

Check the result by means of Venn diagram :

$$P(R\overline{D}\overline{F}) = \frac{50 - 20 - 3}{100} = 0.27.$$

To calculate the probability of the event  $C = \overline{R}\overline{D}\overline{F}$  we use De Morgan's rules and the addition formula for three events :

$$\begin{aligned} P(C) &= P(\overline{R}\overline{D}\overline{F}) = P(\overline{R + D + F}) = 1 - P(R + D + F) \\ &= 1 - [P(R) + P(D) + P(F) - P(RD) - P(RF) \\ &\quad - P(DF) + P(RDF)] = 0.08. \end{aligned}$$

► **Problem 13.** There are two local factories that produce radios. Each radio produced at factory A is defective with probability .05, whereas each one produced at factory B is defective with probability .01. Suppose you purchase two radios that were produced at the same factory. which is equally likely to have been either factory A or factory B. If the first radio that you check is defective, what is the conditional probability that the other one is also defective?

► **Solution** Let  $D_i$ ,  $i = 1, 2$ , denote the event that radio  $i$  is defective. Also, let  $A$  and  $B$  be the events that the radios were produced at factory A and at factory B, respectively. Then

$$\begin{aligned} P(D_2 | D_1) &= \frac{P(D_1 D_2)}{P(D_1)} \\ &= \frac{P(D_1 D_2 | A)P(A) + P(D_1 D_2 | B)P(B)}{P(D_1 | A)P(A) + P(D_1 | B)P(B)} \\ &= \frac{(.05)^2(1/2) + (.01)^2(1/2)}{(.05)(1/2) + (.01)(1/2)} = 13/300 \end{aligned}$$

► **Problem 14.** There are  $n$  students in a class and probability that exactly  $\lambda$  out of  $n$  pass the examination is directly proportional to  $\lambda^2$  ( $0 \leq \lambda \leq n$ ).

- Find out the probability that a student selected at random has passed the examination.
- If a selected student has been found to pass the examination then find out the probability that he is the only student to have passed the examination.

► **Solution** Let  $E_\lambda$  be the event that exactly " $\lambda$ " out of  $n$  pass the examinations and  $A$  is the event that a student pass the examination.

$$P(E_\lambda) \propto \lambda^2$$

$$\Rightarrow P(E_\lambda) = k\lambda^2 \text{ (k is proportionality constant)}$$

$$\therefore E_0, E_1, E_2, \dots, E_n$$

are mutually exclusive and exhaustive events

$$\Rightarrow P(E_0) + P(E_1) + P(E_2) + \dots + P(E_n) = 1$$

$$\Rightarrow 0 + k(1)^2 + k(2)^2 + \dots + k(n)^2 = 1$$

$$\Rightarrow k(1^2 + 2^2 + 3^2 + \dots + n^2) = 1$$

$$\Rightarrow \frac{kn(n+1)(2n+1)}{6} = 1$$

$$\therefore k = \frac{6}{n(n+1)(2n+1)}$$

$$(i) P(A) = \sum_{\lambda=0}^n P(E_\lambda) P(A/E_\lambda)$$

$$= \sum_{\lambda=1}^n k\lambda^2 \times \frac{\lambda}{n} = \frac{k}{n} \sum_{\lambda=1}^n \lambda^3 = \frac{k}{n} \left[ \frac{n(n+1)}{2} \right]^2$$

$$= \frac{6}{n^2(n+1)(2n+1)} \cdot \frac{n^2(n+1)^2}{4} = \frac{3(n+1)}{2(2n+1)}$$

$$(ii) P(E_1 / A) = \frac{P(E_1) P(A/E_1)}{P(A)}$$

$$= \frac{\frac{6}{n(n+1)(2n+1)} \times \frac{1}{n}}{\frac{3(n+1)}{2(2n+1)}} = \frac{4}{[n(n+1)]^2}$$

► **Problem 15.** Sixteen players  $S_1, S_2, \dots, S_{16}$  play in a tournament. They are divided into eight pairs at random. From each pair a winner is decided on the basis of a game played between the two players of the pair. Assume that all the players are of equal strength.

- (i) Find the probability that the player  $S_1$  is among the eight winners.
- (ii) Find the probability that exactly one of the two players  $S_1$  and  $S_2$  is among the eight winners.

► **Solution** (i) Since all players are of equal strength, With whom  $S_1$  is paired, will have no bearing on the probability of his winning.

The number of ways of choosing 8 winners out of 16 is  ${}^{16}C_8$ . The number of ways of choosing  $S_1$  and 7 other winners out of 15 is  ${}^{15}C_7$ .

∴ Probability that  $S_1$  will win

$$= \frac{{}^{15}C_7}{{}^{16}C_8} = \frac{{}^{15}C_7}{\frac{16}{8} \cdot {}^{15}C_7} = \frac{8}{16} = \frac{1}{2}$$

[Similarly, we can show that the probability that  $S_2$  will win is  $1/2$ ]

- (ii) Let  $E_1(E_2)$  denote the event that  $S_1$  and  $S_2$  are paired (not paired) together and let  $A$  denote the event that one of two players  $S_1$  and  $S_2$  is among the winners.

Since  $S_1$  can be paired with any of the remaining 15 players,

$$\text{we have } P(E_1) = \frac{1}{15}$$

$$\text{and } P(E_2) = 1 - P(E_1) = 1 - \frac{1}{15} = \frac{14}{15}$$

In case  $E_1$  occurs, it is certain that one of  $S_1$  and  $S_2$  will be among the winners. In case  $E_2$  occurs, the probability that exactly one of  $S_1$  and  $S_2$  is among the winners is

$$\begin{aligned} & P[(S_1 \cup S_2') \cup (S_1' \cap S_2)] \\ &= P(S_1 \cap S_2') + P(S_1' \cap S_2) \\ &= P(S_1)(S_2') + P(S_1')P(S_2) \\ &= \left(\frac{1}{2}\right)\left(1 - \frac{1}{2}\right) + \left(1 - \frac{1}{2}\right)\left(\frac{1}{2}\right) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2} \end{aligned}$$

$$\text{i.e. } P(A/E_1) = 1 \text{ and } P(A/E_2) = 1/2$$

By the total probability rule,

$$P(A) = P(E_1)P(A/E_1) + P(E_2)P(A/E_2)$$

$$= \frac{1}{15}(1) + \frac{14}{15}\left(\frac{1}{2}\right) = \frac{8}{15}$$

► **Problem 16.** A deck of cards is dealt out, (a) what is the probability that the tenth card dealt is (i) a king (ii) a spade (iii) the king of spades ? (b) What is the probability that (i) the first king (ii) the first spade (iii) the king of spades occurs on the tenth card?

► **Solution** (a) (i) There are four kings and there are 52 cards. So the probability of a king occurring is  $4/52 = 1/13$ . This is the probability even for the tenth card, in fact, even for the last card because, we are not given any information about what has happened in the previous dealings. The problem will be better understood if you keep tossing a coin and ask what is the probability of 'heads' at the 10th toss. It is again  $1/2$  as it always was.

In the dealing of cards, if on the other hand we are given such and such a thing happened in the first 9 deals, then the probability at the 10th deal might be different. Otherwise, it is the same  $4/52 = (1/13)$  for all the deals.

- (ii) Similarly the answer for the probability of a spade occurring at the 10th card is  $13/52 = 1/4$ .
- (iii) There is only one king of spades. So the required probability is  $1/52$ .
- (b) (i) Now we are told that the first 9 cards should not be a king and then the tenth should be a king. The probability of this happening is

$$\left(1 - \frac{1}{13}\right)^9 \times \frac{4}{52}.$$

- (ii) Similar to the above. The answer is

$$\left(1 - \frac{1}{4}\right)^9 \times \frac{13}{52}.$$

► **Problem 17.** A sportsman's chance of shooting an animal at a distance  $r$  ( $> a$ ) is  $a^2/r^2$ . He fires when  $r = 2a$  and if he misses, he reloads and fires when  $r = 3a, 4a, \dots$ . If he misses at distance  $na$ , the animal escapes. Find the odds against the sportsman.

► **Solution** Chances of the animal being shot dead at the successive ranges are  $(1/2)^2, (1/3)^2, (1/4)^2, \dots, (1/n)^2$ . The corresponding chances of the animal's escape are

$1 - (1/2)^2, 1 - (1/3)^2, 1 - (1/4)^2, \dots, 1 - (1/n)^2$ . Thus, the chance of escape of the animal at the end ( $r = na$ ) is,

$$p = \left(1 - \frac{1}{2^2}\right) \left(1 - \frac{1}{3^2}\right) \left(1 - \frac{1}{4^2}\right) \dots$$

$$\left(1 - \frac{1}{n^2}\right) = \prod_{k=2}^n \left(1 - \frac{1}{k^2}\right)$$

$$= \prod_{k=2}^n \left(\frac{k-1}{k}\right) \cdot \left(\frac{k+1}{k}\right)$$

$$= \left(\frac{1}{2} \cdot \frac{3}{2}\right) \left(\frac{2}{3} \cdot \frac{4}{3}\right) \left(\frac{3}{4} \cdot \frac{5}{4}\right) \dots \left(\frac{n-1}{n} \cdot \frac{n+1}{n}\right)$$

$$= \frac{n+1}{2n}$$

$$\text{So } q = 1 - p = 1 - \left(\frac{n+1}{2n}\right) = \frac{n-1}{2n}$$

(chance of animal being killed)

Thus the odds against the sportsman are

$$p : q = n+1 : n-1.$$

► **Problem 18.** A and B participate in a tournament of "best of 7 games". It is equally likely that either A wins or B wins or the game ends in a draw. What is the probability that A wins the tournament?

► **Solution** Let the probability that A wins the tournament =  $x$

The probability that B wins the tournament =  $x$

The probability that the game ends in a draw =  $y$  so

$$\text{that } 2x + y = 1 \Rightarrow x = \frac{1-y}{2}$$

Probability that the game ends in a draw

$$\text{Case I : one draw, 3 wins and 3 losses} = \frac{7!}{3!3!} \left(\frac{1}{3}\right)^7$$

$$\text{Case II : three draws, 2 win and 2 loss} = \frac{7!}{3!2!2!} \left(\frac{1}{3}\right)^7$$

$$\text{Case III : all draws} = \frac{7!}{5!} \left(\frac{1}{3}\right)^7$$

$$y = \frac{1}{3^7} \left( \frac{7!}{3!3!} + \frac{7!}{3!2!2!} + \frac{7!}{5!} + 1 \right) = \frac{131}{729}$$

$$\Rightarrow x = \frac{1}{2} \left( 1 - \frac{131}{729} \right) = \frac{299}{729}$$

► **Problem 19.** In a lottery all the tickets are blanks, except one. Each person draws a ticket and retains it. Show that each person has an equal chance of drawing the prize.

► **Solution** Let there be  $n$  tickets in all in the lottery, one of which is a prize and the others are blank.

Thus the chance of drawing a ticket containing prize

$$= \frac{1}{n}, \text{ and hence the chance of drawing a blank ticket}$$

is  $\frac{n-1}{n}$ . If the first person fails to draw the prize, there

are now  $n-1$  tickets one of which is prize, hence the chance of drawing the prize by second person is

$$\frac{n-1}{n} \cdot \frac{1}{n-1} = \frac{1}{n}. \text{ If the first two persons fail to draw}$$

the prize, the third person will draw from  $n-2$  tickets, and hence his chance of drawing the prize

$$= \frac{n-1}{n} \cdot \frac{n-2}{n-1} \cdot \frac{1}{n-2} = \frac{1}{n};$$

and so on. Thus each chance is  $\frac{1}{n}$  and hence each person has an equal chance of drawing the prize.

► **Problem 20.** Two players A, B of equal skills are playing a set of games; A wants 2 games to complete the set and B wants 3 games, compare their chances of winning.

► **Solution** As the two players, A and B, are of equal skill, the chance of A's success is  $1/2$  in a single game. Now the player A will win the set, if he wins 2 points before B wins 3. Again the set will necessarily be



decided in 4 games, and A may win his 2 points in exactly 2 games, or 3 games, or 4 games.

The chance that A will win in exactly 2 games in  $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$ , the chance that A wins in exactly 3 games is  $2 \times \left(\frac{1}{2}\right)^3$ , because A will win in 3 games when he wins the last game and one out of the preceding two games.

One out of the preceding two games can be won in 2 ways either by winning the first game or the second

game, the chance that A wins in 4 games is  ${}^3C_1 \left(\frac{1}{2}\right)^4$ , because A will win in exactly 4 games when he wins the fourth game and one out of the preceding 3 games, one out of the preceding 3 games can be won in  ${}^3C_1$  ways.

Since these three events are mutually exclusive hence the chance of A's success

$$= \frac{1}{4} + 2\left(\frac{1}{2}\right)^3 + {}^3C_1 \left(\frac{1}{2}\right)^4$$

$$= \frac{1}{4} + \frac{1}{4} + \frac{3 \cdot 2 \cdot 1}{2 \cdot 1 \cdot 16} = \frac{11}{16}$$

and hence B's chance of success =  $\frac{5}{18}$ .

Therefore A's chance is to B's as 11: 5.

► **Problem 21.** Independent trials, consisting of rolling a pair of fair dice, are performed. What is the probability that an outcome of 5 appears before an outcome of 7 when the outcome of a roll is the sum of the dices?

► **Solution** If we let  $E_n$  denote the event that no. 5 or 7 appears on the first  $n - 1$  trials and a 5 appears on the  $n$ th trial, then the desired probability is

$$P\left(\bigcup_{n=1}^{\infty} E_n\right) = \sum_{n=1}^{\infty} P(E_n)$$

Now, since  $P\{5 \text{ on any trial}\} = \frac{4}{36}$  and

$P\{7 \text{ on any trial}\} = \frac{6}{36}$ , we obtain,

by the independence of trials

$$P(E_n) = \left(1 - \frac{10}{36}\right)^{n-1} \frac{4}{36}$$

and thus

$$\left(\bigcup_{n=1}^{\infty} E_n\right) = \frac{1}{9} \sum_{n=1}^{\infty} \left(\frac{13}{18}\right)^{n-1} = \frac{1}{9} \frac{1}{1 - \frac{13}{18}} = \frac{2}{5}.$$

► **Problem 22.** A random variable  $X$  takes values 0, 1, 2, 3, .... with probability proportional to  $(x + 1)(1/5)^x$ . Find  $P\{X \leq 5\}$ .

► **Solution** Here  $X$  is discrete and

$$P\{X = x\} = k(x + 1)(1/5)^x.$$

Since  $\sum p(x) = 1$ ,  $x = 0, 1, 2, \dots$

we get, putting  $1/5 = r$ .

$$1 = k \sum (x + 1)r^x = k(1 + 2r + 3r^2 + 4r^3 + \dots)$$

$$= k(1 - r)^{-2} = k(25/16) \Rightarrow k = 16/25.$$

∴  $p(x) = (16/25)(x + 1)(1/5)^x$ ,  $x = 0, 1, 2, \dots$

$$P(X \leq 5) = P(X = 0) + P(X = 1) + \dots + P(X = 5)$$

$$= \frac{16}{25} \left[ 1 + \frac{2}{5} + \frac{3}{5^2} + \dots + \frac{6}{5^5} \right]$$

$$= 0.64(1 + 0.4 + 0.12 + 0.032 + 0.008 + 0.00192)$$

$$= 0.64 \times 1.562 = 0.9997.$$



Since  $P(5) = 0.9997$ , it follows that, for  $X > 5$ ,  $P(x)$  will change very slowly.

► **Problem 23.** Independent trials, consisting of the flipping of a coin having probability  $p$  of coming up heads, are continually performed until either a head occurs or a total of  $n$  flips is made. If we let  $X$  denote the number of times the coin is flipped, then  $X$  is a random variable taking on one of the values 1, 2, 3, ...,  $n$  with respective probabilities. Find the probability distribution and verify that the sum of the probabilities is one.

► **Solution**  $P\{X = 1\} = P\{H\} = p$

$$P\{X = 2\} = P\{(T, H)\} = (1 - p)p$$

$$P\{X = 3\} = P\{(T, T, H)\} = (1 - p)^2 p$$

⋮

$$P\{X = n - 1\} = P\{(\underbrace{T, T, \dots, T}_{n-2}, H)\} = (1 - p)^{n-2}p$$

$$P\{X = n\} = P\{(\underbrace{T, T, \dots, T}_{n-1}, T), (\underbrace{T, T, \dots, T}_{n-1}, H)\}$$

$$= (1 - p)^{n-1}$$

As a check, note that

$$P\left(\bigcup_{i=1}^n \{X = i\}\right) = \sum_{i=1}^n P\{X = i\}$$

$$= \sum_{i=1}^{n-1} p(1-p)^{i-1} + (1-p)^{n-1}$$

$$= p \left[ \frac{1 - (1-p)^{n-1}}{1 - (1-p)} \right] + (1-p)^{n-1}$$

$$= 1 - (1-p)^{n-1} + (1-p)^{n-1} = 1$$

► **Problem 24.** A contestant on a quiz show is presented with two questions, questions 1 and 2, which he is to attempt to answer in some order chosen by him. If he decides to try question  $i$  first, then he will be allowed to go on to question  $j$ ,  $j \neq i$ , only if his answer to  $i$  is correct. If his initial answer is incorrect, he is not allowed to answer question  $i$  correctly,  $i = 1, 2$ . Thus, for instance, he will receive  $V_1 + V_2$  dollars if both questions are correctly answered. If the probability that he knows the answer to question  $i$  is  $P_i$ ,  $i = 1, 2$  which question should he attempt first so as to maximize his expected winnings? Assume that the events  $E_i$ ,  $i = 1, 2$ , that he knows the answer to question  $i$ , are independent events.

► **Solution** If he attempts question 1 first, then he will win

$$\begin{array}{ll} 0 & \text{with probability } 1 - P_1 \\ V_1 & \text{with probability } P_1(1 - P_2) \\ V_1 + V_2 & \text{with probability } P_1P_2 \end{array}$$

Hence his expected winnings in this case will be

$$V_1P_1(1 - P_2) + (V_1 + V_2)P_1P_2$$

On the other hand, if he attempts question 2 first, his expected winnings will be

$$V_2P_2(1 - P_1) + (V_1 + V_2)P_1P_2$$

Therefore, it is better to try question 1 first if

$$V_1P_1(1 - P_2) \geq V_2P_2(1 - P_1)$$

or, equivalently, if

$$\frac{V_1P_1}{1 - P_1} \geq \frac{V_2P_2}{1 - P_2}$$

Thus, for instance, if he is 60 percent certain of answering question 1, worth \$200, correctly and he is 80 percent certain of answering question 2, worth \$100, correctly, then he should attempt question 2 first because

$$400 = \frac{(100)(.8)}{.2} > \frac{(200)(.6)}{.4} = 300$$

► **Problem 25.** Two cards are drawn simultaneously (or successively without replacement) from a well shuffled pack of 52 cards. Find the mean, variance and standard deviation of the number of kings.

► **Solution** Let  $X$  denote the number of kings in a draw of two cards.  $X$  is a random variable which can assume the values 0, 1 or 2.

Now  $P(X = 0) = P(\text{no king})$

$$= \frac{{}^{48}C_2}{{}^{52}C_2} = \frac{\frac{48!}{2!(48-2)!}}{\frac{52!}{2!(52-2)!}} = \frac{48 \times 47}{52 \times 51} = \frac{188}{221}$$

$$P(X = 1) = P(\text{one king and one non-king})$$

$$= \frac{{}^4C_1 {}^{48}C_1}{{}^{52}C_2} = \frac{4 \times 48 \times 2}{52 \times 51} = \frac{32}{221}$$

and  $P(X = 2) = P(\text{two kings})$

$$= \frac{{}^4C_2}{{}^{52}C_2} = \frac{4 \times 3}{52 \times 51} = \frac{1}{221}$$

Thus, the probability distribution of  $X$  is

| $X$    | 0                          | 1                           | 2                                           |
|--------|----------------------------|-----------------------------|---------------------------------------------|
| $P(X)$ | $\frac{188}{221}$          | $\frac{32}{221}$            | $\frac{1}{221}$                             |
|        | $0 \times \frac{188}{221}$ | $+ 1 \times \frac{32}{221}$ | $+ 2 \times \frac{1}{221} = \frac{34}{221}$ |

$$\text{Also } E(X^2) = \sum_{i=1}^n x_i^2 p(x_i)$$

$$= 0^2 \times \frac{188}{221} + 1^2 \times \frac{32}{221} + 2^2 \times \frac{1}{221} = \frac{36}{221}$$

$$\text{Now } \text{Var}(X) = E(X^2) - [E(X)]^2$$

$$= \frac{36}{221} - \left(\frac{34}{221}\right)^2 = \frac{6800}{(221)^2}$$

Therefore

$$\sigma_x = \sqrt{\text{Var}(X)} = \frac{\sqrt{6800}}{221} = 0.37$$

► **Problem 26.** A bag contains  $\frac{n(n+1)}{2}$  counters

of which one is marked '1', two are marked '4', three are marked 9, and so on; a man puts in his hand and draws out a counter at random and is to receive as many rupees as the number marked upon it, find the value of his expectation.

► **Solution** The total number of different possible

ways of drawing a counter from the bag is  $\frac{n(n+1)}{2}$

out of which one is favourable to the event that the drawn counter is marked 1, two to the event that the counter marked is 4, and so on.

Therefore the chance of the event that the drawn counter is marked one is  $\frac{2}{n(n+1)}$ .

The chance of the event that the counter is marked 4 is

$$\frac{2}{n(n+1)/2} = \frac{4}{n(n+1)},$$

The chance of drawing one of the three counters marked 9 is

$$\frac{3}{n(n+1)/2} = \frac{6}{n(n+1)}, \text{ and so on.}$$

∴ The required expectation in rupees

$$\begin{aligned} &= \frac{2}{n(n+1)} \times 1 + \frac{4}{n(n+1)} \times 4 \\ &+ \frac{6}{n(n+1)} \times 9 + \dots + \frac{2n}{n(n+1)} \times n^2 \\ &= \frac{2}{n(n+1)} \{1^3 + 2^3 + 3^3 + \dots + n^3\} \\ &= \frac{2}{n(n+1)} \times \left\{ \frac{n(n+1)}{2} \right\}^2 = \frac{n(n+1)}{2} \text{ rupees.} \end{aligned}$$

► **Problem 27.** The probability of a shooter hitting a target is  $3/4$ . How many minimum number of times must he fire so that the probability of hitting the target at least once is more than 0.99?

► **Solution** Let the shooter fire  $n$  times. Obviously,  $n$  fires are  $n$  Bernoulli trials. In each trial,  $p$  = probability of hitting the target =  $3/4$  and  $q$  = probability of not hitting the target =  $1/4$ . Then

$$\begin{aligned} P(X = x) &= {}^nC_x q^{n-x} p^x \\ &= {}^nC_x \left(\frac{1}{4}\right)^{n-x} \left(\frac{3}{4}\right)^x = {}^nC_x \frac{3^x}{4^n} \end{aligned}$$

Now, given that,

$$P(\text{hitting the target at least once}) > 0.99$$

$$\text{i.e. } P(x \leq 1) > 0.99$$

Therefore,

$$1 - P(x = 0) > 0.99$$

$$\text{or } 1 - {}^nC_0 \frac{1}{4^n} > 0.99$$

$$\text{or } {}^nC_0 \frac{1}{4^n} < 0.01 \text{ i.e. } \frac{1}{4^n} < 0.01$$

$$\text{or } 4n > \frac{1}{0.01} = 100 \quad \dots(1)$$

The minimum value of  $n$  to satisfy the inequality (1) is 4.

Thus, the shooter must fire 4 times.

► **Problem 28.** Consider a game played by five people in which each flips a fair coin at the same time. If all but one of the coins comes up the same, then the odd person wins (for example, if there are four heads and one tail then the 'tail' wins). If such a situation does not occur, then the players flip again. Find the probability that game is settled on or after  $n^{\text{th}}$  toss.

► **Solution** Let ' $E_1$ ' be the event that game is settled on the first toss.

⇒  $P(E_1)$  = (Probability of one person getting tail and all other getting head) + probability of one getting head and all others getting tail.

$$\begin{aligned} &= {}^5C_4 \cdot \frac{1}{2} \cdot \left(\frac{1}{2}\right)^4 + {}^5C_4 \cdot \frac{1}{2} \cdot \left(\frac{1}{2}\right)^4 \\ &= {}^5C_4 \cdot \left(\frac{1}{2}\right)^4 = \frac{5}{16}. \end{aligned}$$

$$\text{Thus } P(\bar{E}_1) = 1 - \frac{5}{16} = \frac{11}{16}$$

If ' $E_i$ ' be the event that game is settled on  $i^{\text{th}}$  toss,

$$\text{where } i \geq 1, \text{ then } P(E_i) = \left(\frac{11}{16}\right)^{i-1} \cdot \frac{5}{16}$$

⇒ Required probability

$$\begin{aligned}
 &= \sum_{i=n}^{\infty} P(E_i) = \frac{5}{16} \sum_{i=n}^{\infty} \left(\frac{11}{16}\right)^{n-1} = \frac{5}{16} \cdot \frac{\left(\frac{11}{16}\right)^{n-1}}{1 - \frac{11}{16}} \\
 &= \frac{5}{16} \cdot \frac{\left(\frac{11}{16}\right)^{n-1}}{\frac{5}{16}} = \left(\frac{11}{16}\right)^{n-1}.
 \end{aligned}$$

► **Problem 29.** If a fair coin is flipped  $2n$  times, show that the probability of obtaining more heads than tails

$$\text{is } \frac{1}{2} \left\{ 1 - {}^{2n}C_n \left(\frac{1}{2}\right)^{2n} \right\}.$$

► **Solution** We consider the sample space  $S$  where  $S = \{H > T, H = T, H < T\}$ , where  $H > T$  means more heads than tails and so on. By symmetry,  $P(H > T) = P(H < T)$ .

Now  $P(S) = P(H > T) + P(H = T) + P(H < T)$

$$\Rightarrow 1 = 2P(H > T) + P(H = T).$$

$$\text{Thus } P(H > T) = (1/2) [1 - P(H = T)] \quad \dots(1)$$

Now  $P(H) = 1/2$ , and to obtain equal number of heads and tails out of  $2n$  flips, we require  $n$  heads,

$$P(X=n) = {}^{2n}C_n \left(\frac{1}{2}\right)^{2n-n} \left(\frac{1}{2}\right)^n = {}^{2n}C_n \left(\frac{1}{2}\right)^{2n} \quad \dots(2)$$

$$\begin{aligned}
 P(H > T) &= P(H < T) = \frac{1}{2} \left[ 1 - {}^{2n}C_n \left(\frac{1}{2}\right)^{2n} \right] \\
 &\quad [\text{by (1) \& (2)}]
 \end{aligned}$$

► **Problem 30.** If  $X$  and  $Y$  are independent binomial variates  $B(5, 1/2)$  and  $B(7, 1/2)$  then find the value of  $P(X + Y = 3)$ .

► **Solution** We have

$$\begin{aligned}
 P(X + Y = 3) &= P(X = 0, Y = 3) + P(X = 1, Y = 2) \\
 &\quad + P(X = 2, Y = 1) + P(X = 3, Y = 0) \\
 &= P(X = 0) P(Y = 3) + P(X = 1) P(Y = 2) \\
 &\quad + P(X = 2) P(Y = 1) + P(X = 3) P(Y = 0) \\
 &\quad \{ \because X \text{ and } Y \text{ are independent} \}
 \end{aligned}$$

$$\therefore {}^5C_0 \left(\frac{1}{2}\right)^5 \cdot {}^7C_3 \left(\frac{1}{2}\right)^7 + {}^5C_1 \left(\frac{1}{2}\right)^5 \cdot {}^7C_2 \left(\frac{1}{2}\right)^7$$

$$\begin{aligned}
 &+ {}^5C_2 \left(\frac{1}{2}\right)^5 \cdot {}^7C_1 \left(\frac{1}{2}\right)^7 + {}^5C_3 \left(\frac{1}{2}\right)^5 \cdot {}^7C_0 \left(\frac{1}{2}\right)^7 \\
 &\left(\frac{1}{2}\right)^{12} [(1)(35) + (5)(21) + (10)(7) + (10)(1)] \\
 &= \frac{220}{2^{12}} = \frac{55}{1024}
 \end{aligned}$$

► **Problem 31.** A man takes a step forward with probability 0.4 and backward with probability 0.6. Find the probability that at the end of eleven steps he is one step away from the starting point.

► **Solution** As  $0.4 + 0.6 = 1$ , the man either takes a step forward or a step backward. Let a step forward be a success and a step backward be a failure.

Then, the probability of success in one step

$$= p = 0.4 = 2/5$$

and the probability of failure in one step

$$= q = 0.6 = 3/5.$$

In 11 steps he will be one step away from the starting point if the numbers of successes and failures differ by 1.

So, the number of successes = 6,

the number of failures = 5

or the number of successes = 5,

the number of failures = 6

∴ The required probability

$$\begin{aligned}
 &= {}^{11}C_6 p^6 q^5 + {}^{11}C_5 p^5 q^6 \\
 &= {}^{11}C_6 \left(\frac{2}{5}\right)^6 \cdot \left(\frac{3}{5}\right)^5 + {}^{11}C_5 \left(\frac{2}{5}\right)^5 \cdot \left(\frac{3}{5}\right)^6 \\
 &= \frac{11!}{6!5!} \cdot \left(\frac{2}{5}\right)^5 \cdot \left(\frac{3}{5}\right)^5 \cdot \left\{ \frac{2}{5} + \frac{3}{5} \right\} \\
 &= \frac{11 \cdot 10 \cdot 9 \cdot 8 \cdot 7}{120} \cdot \frac{2^5 \cdot 3^5}{5^{10}} = 462 \times \left(\frac{6}{25}\right)^5
 \end{aligned}$$

► **Problem 32.** In a sequence of independent trials, the probability of success in one trial is  $1/4$ . Find the probability that the second success takes place on or after the fourth trial.

► **Solution** The probability of the second success taking place on the fourth trial

= probability of one success in the first three trials and success on the fourth trial

$$= \left\{ {}^3C_1 \times \frac{1}{4} \times \left(\frac{3}{4}\right)^2 \right\} \times \frac{1}{4}.$$

The probability of the second success taking place on the fifth trial

= probability of one success in the first four trials and success on the fifth trial

$$= \left\{ {}^4C_1 \times \frac{1}{4} \times \left(\frac{3}{4}\right)^3 \right\} \times \frac{1}{4} \text{ and so on.}$$

∴ The required probability

$$\begin{aligned} &= {}^3C_1 \cdot \frac{1}{4} \cdot \left(\frac{3}{4}\right)^2 \cdot \frac{1}{4} + {}^4C_1 \cdot \frac{1}{4} \cdot \left(\frac{3}{4}\right)^3 \cdot \frac{1}{4} \\ &\quad + {}^5C_1 \cdot \frac{1}{4} \cdot \left(\frac{3}{4}\right)^4 \cdot \frac{1}{4} + \dots \\ &= \frac{9}{256} \left\{ {}^3C_1 + {}^4C_1 \cdot \frac{3}{4} + {}^5C_1 \cdot \left(\frac{3}{4}\right)^2 + \dots \text{to } \infty \right\} \\ &= \frac{9}{256} \left\{ 3 + 4 \cdot \frac{3}{4} + 5 \cdot \left(\frac{3}{4}\right)^2 + \dots \text{to } \infty \right\} \dots (1) \end{aligned}$$

$$\text{Let } S = 3 + 4 \cdot \frac{3}{4} + 5 \cdot \left(\frac{3}{4}\right)^2 + \dots \text{to } \infty$$

$$\therefore \frac{3}{4}S = 3 \cdot \frac{3}{4} + 4 \cdot \left(\frac{3}{4}\right)^2 + \dots \text{to } \infty$$

Subtracting

$$\frac{1}{4}S = 3 + \frac{3}{4} + \left(\frac{3}{4}\right)^2 + \dots \text{to } \infty = 3 + \frac{\frac{3}{4}}{1 - \frac{3}{4}} = 6$$

$$\therefore S = 24$$

∴ From (i), the required probability

$$= \frac{9}{256} \times 24 = \frac{27}{32}.$$

► **Problem 33.** Of two equally strong ping-pong players A and B, is it more probable that (i) A will beat B in 3 games out of 4, or in 5 games out of 8, (ii) atleast 3 games out of 4 or atleast 5 out of 8?

► **Solution** (i) A will beat B if A wins three games out of four games played, and the corresponding probability  $p_1$  is

$$p_1 = {}^4C_3 (1/2)^{4-3} (1/2)^3 = 1/4 \quad (p = q = 1/2)$$

Again, A will beat B if A wins five games out of eight games played, and the corresponding probability  $p_2$  is

$$p_2 = {}^8C_5 (1/2)^3 (1/2)^5 = 56/(2)^8 = 7/32.$$

Since  $p_1 > p_2$ , A is more likely to win 3 games out of 4 than to win 5 games out of 8.

(ii) The probability of winning atleast 3 games out of 4 is

$$P_3 = {}^4C_3 \left(\frac{1}{2}\right)^4 + {}^4C_4 \left(\frac{1}{2}\right)^4 = \frac{1}{4} + \frac{1}{16} = \frac{5}{16}.$$

The probability of winning atleast 5 games out of 8 is

$$P_4 = \left[ {}^8C_5 + {}^8C_6 + {}^8C_7 + {}^8C_8 \right] \left(\frac{1}{2}\right)^8 = \frac{93}{256}$$

Since  $p_4 > p_3$ , it is more probable to win atleast 5 games out of 8.

► **Problem 34.** A doctor is to visit a patient. From the past experience, it is known that the probabilities that he will come by train, bus, scooter or by other means of transport are respectively

$\frac{3}{10}$ ,  $\frac{1}{5}$ ,  $\frac{1}{10}$  and  $\frac{2}{5}$ . The probabilities that he will be late are  $\frac{1}{4}$ ,  $\frac{1}{3}$  and  $\frac{1}{12}$ , if he comes by train, bus and scooter respectively, but if he comes by other means of transport, then he will not be late. When he arrives, he is late. What is the probability that he comes by train?

► **Solution** Let E be the event the doctor visits the patient late and let  $T_1, T_2, T_3, T_4$  be the events that the doctor comes by train, bus, scooter, and other means of transport respectively.

$$\text{Then } P(T_1) = \frac{3}{10}, P(T_2) = \frac{1}{5},$$

$$P(T_3) = \frac{1}{10} \text{ and } P(T_4) = \frac{2}{5}$$

$P(E|T_1)$  = Probability that the doctor arriving late comes by train =  $1/4$

Similarly,  $P(E|T_1) = 1/3$ ,  $P(E|T_3) = 1/12$  and  $P(E|T_4) = 0$ , since he is not late if he comes by other means of transport.

Therefore, by Bayes' Theorem, we have  
 $P(T_1|E)$  = Probability that the doctor arriving late comes by train

$$\begin{aligned} &= \frac{P(T_1) P(E|T_1)}{P(T_1) P(E|T_1) + P(T_2) P(E|T_2) + P(T_3) P(E|T_3) + P(T_4) P(E|T_4)} \\ &= \frac{\frac{3}{10} \times \frac{1}{4}}{\frac{3}{10} \times \frac{1}{4} + \frac{1}{5} \times \frac{1}{3} + \frac{1}{10} \times \frac{1}{12} + \frac{2}{5} \times 0} \\ &= \frac{\frac{3}{40} \times \frac{120}{18} = \frac{1}{2}}{\frac{3}{10} \times \frac{1}{4} + \frac{1}{5} \times \frac{1}{3} + \frac{1}{10} \times \frac{1}{12} + \frac{2}{5} \times 0} \end{aligned}$$

Hence, the required probability is  $1/2$ .

► **Problem 35.** A purse contains 5 coins, each either a fifty-paisa coin or one-rupee coin. Two coins are drawn at random and are found to be one-rupee coins. Find the expected total value of the remaining coins.

► **Solution** Clearly, atleast two of the coins are one-rupee coins. Let  $r$  stand for one-rupee coin and  $f$  stand for fifty-paisa coin.

The coins in the purse can be any one of the following:

$2r, 3f$ ;  $3r, 2f$ ;  $4r, 1f$ ;  $5r$ .

Let  $E_2$  = the event of the purse containing  $2r$  and  $3f$ .

$E_3$  = the event of the purse containing  $3r$  and  $2f$ ; etc.

Clearly, the events are  $E_2, E_3, E_4$  and  $E_5$ .

They are equally probable, exhaustive and mutually exclusive.

$\therefore P(E_2) = P(E_3) = P(E_4) = P(E_5) = 1/4$ .

Now,  $P(E/E_2)$  = probability of drawing two one-rupee coins when  $E_2$  happens

$$= \frac{{}^2C_2}{{}^5C_2} = \frac{1}{10}$$

$$\text{Similarly, } P(E/E_3) = \frac{{}^3C_2}{{}^5C_2} = \frac{3}{10}$$

$$P(E/E_4) = \frac{{}^4C_2}{{}^5C_2} = \frac{6}{10} = \frac{3}{5}$$

$$P(E/E_5) = \frac{{}^5C_2}{{}^5C_2} = 1.$$

Now,  $P(E_2/E)$  = probability of the purse containing  $2r$  and  $3f$  when  $2r$  are drawn etc.

By Bayes' theorem,

$$\begin{aligned} P(E_2/E) &= \frac{P(E/E_2)}{P(E/E_2) + P(E/E_3) + P(E/E_4) + P(E/E_5)} \\ &= \frac{\frac{1}{10}}{\frac{1}{10} + \frac{3}{10} + \frac{3}{5} + 1} = \frac{1}{10} \cdot \frac{10}{20} = \frac{1}{20} \end{aligned}$$

$$\begin{aligned} \text{Similarly, } P(E_3/E) &= \frac{\frac{3}{10}}{\frac{1}{10} + \frac{3}{10} + \frac{3}{5} + 1} \\ &= \frac{3}{10} \cdot \frac{10}{20} = \frac{3}{20} \end{aligned}$$

$$P(E_4/E) = \frac{\frac{3}{5}}{\frac{1}{10} + \frac{3}{10} + \frac{3}{5} + 1} = \frac{3}{5} \cdot \frac{10}{20} = \frac{3}{10}$$

$$P(E_5/E) = \frac{1}{\frac{1}{10} + \frac{3}{10} + \frac{3}{5} + 1} = \frac{10}{20} = \frac{1}{2}.$$

$\therefore$  The expected total value of the remaining coins

$$\begin{aligned} &= P(E_2/E) \cdot (\text{value of } 3f) \\ &+ P(E_3/E) \cdot (\text{value of } 1r \text{ and } 2f) \\ &+ P(E_4/E) \cdot (\text{value of } 2r \text{ and } 1f) \\ &+ P(E_5/E) \cdot (\text{value of } 3r) \\ &= \frac{1}{20} \cdot (\text{Rs. } 1.50) + \frac{3}{20} \cdot (\text{Rs. } 2) + \frac{3}{10} \cdot (\text{Rs. } 2.50) \\ &+ \frac{1}{2} \cdot (\text{Rs. } 3) \end{aligned}$$

$$= \text{Rs.} \left( \frac{1}{20} \times \frac{3}{2} + \frac{3}{20} \times 2 + \frac{3}{10} \times \frac{5}{2} + \frac{3}{2} \right)$$

$$= \text{Rs.} \left( \frac{3}{40} \times \frac{3}{10} + \frac{3}{4} + \frac{3}{2} \right) = \text{Rs.} 2.625$$

► **Problem 36.** Three biased coins  $C_1$ ,  $C_2$  and  $C_3$  lie on a table. Their respective probabilities of falling heads when tossed once are  $1/3$ ,  $2/3$  and  $1$ . A coin is picked up randomly, tossed twice and observed to fall heads on both the occasions.

Now given that a coin has been tossed twice and observed to fall heads both times. If the probability that a 3<sup>rd</sup> throw of the same coin will also produce

heads is  $\frac{a}{b}$  where  $a$  and  $b$  are relatively prime, find the value of  $(a^2 + b^2)$ .

► **Solution** A: coin tossed twice fell head wise

$$\left. \begin{array}{l} B_1: \text{It is the coin } C_1; \quad P(B_1) = \frac{1}{3} \\ B_2: \text{It is the coin } C_2; \quad P(B_2) = \frac{1}{3} \end{array} \right\}$$

(random selection)

$$B_3: \text{It is the coin } C_3; \quad P(B_3) = \frac{1}{3}$$

$$P(A/B_1) = \frac{1}{9}; \quad P(A/B_2) = \frac{4}{9}; \quad P(A/B_3) = \frac{9}{9}$$

(given)

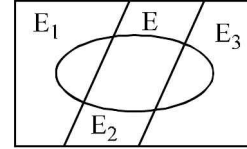
$$\therefore P(B_1/A) = \frac{\frac{1}{9}}{\frac{1}{9} + \frac{4}{9} + \frac{9}{9}} = \frac{1}{14}$$

$$\text{Similarly } P(B_2/A) = \frac{4}{14} \text{ and } P(B_3/A) = \frac{9}{14}$$

now Let  $E$ : event that the same coin tossed again will fall head wise suppose

$$\begin{aligned} B_1/A &= E_1; B_2/A = E_2 \text{ and } B_3/A = E_3 \\ \therefore P(E) &= P(E \cap E_1) + P(E \cap E_2) + P(E \cap E_3) \\ &= P(E_1) \cdot P(E/E_1) + P(E_2) \cdot P(E/E_2) \\ &\quad + P(E_3) \cdot P(E/E_3) \\ &= \frac{1}{14} \cdot \frac{1}{3} + \frac{4}{14} \cdot \frac{2}{3} + \frac{9}{14} \cdot \frac{3}{3} \end{aligned}$$

$$= \frac{1+8+27}{42} = \frac{36}{42} = \frac{6}{7}$$



$$\therefore \frac{a}{b} = \frac{6}{7} \Rightarrow a = 6 \text{ and } b = 7$$

$$\text{hence } a^2 + b^2 = 36 + 49 = 85.$$

► **Problem 37.** Three shots are fired at a target in succession. The probabilities of a hit in the first

shot is  $\frac{1}{2}$ , in the second  $\frac{2}{3}$  and in the third shot is

$\frac{3}{4}$ . In case of one hit the probability of destroying

the target is  $\frac{2}{3}$  and in the case of two hits  $\frac{7}{11}$  and

in the case of three hits is  $1.0$ . Find the probability of destroying the target in three shots.

► **Solution**  $P(A) = \frac{1}{2}$ ,  $P(B) = \frac{2}{3}$ ,  $P(C) = \frac{3}{4}$

$E_1$ : Exactly one shot hit

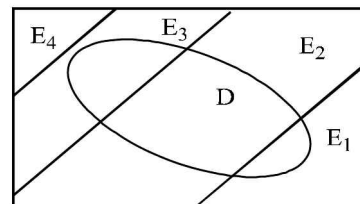
$E_2$ : Exactly two shots hit

$E_3$ : Exactly three shots hit

$$P(E_1) = \frac{1}{2} \cdot \frac{1}{3} \cdot \frac{1}{4} + \frac{2}{3} \cdot \frac{1}{2} \cdot \frac{1}{4} + \frac{3}{4} \cdot \frac{1}{2} \cdot \frac{1}{3}$$

$$= \frac{1}{24} + \frac{2}{24} + \frac{3}{24} = \frac{6}{24} = \frac{1}{4}$$

$$P(E_2) = \frac{1}{2} \cdot \frac{2}{3} \cdot \frac{1}{4} + \frac{2}{3} \cdot \frac{3}{4} \cdot \frac{1}{2} + \frac{3}{4} \cdot \frac{1}{2} \cdot \frac{1}{3}$$



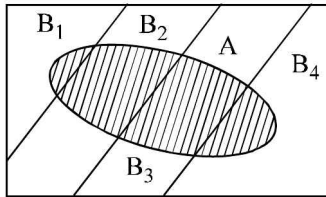
$$\begin{aligned}
 &= \frac{2}{24} + \frac{6}{24} + \frac{3}{24} = \frac{11}{24} \\
 P(E_3) &= \frac{1}{2} \cdot \frac{2}{3} \cdot \frac{3}{4} = \frac{6}{24} = \frac{1}{4} \\
 P(E_4) &= \frac{1}{2} \cdot \frac{1}{3} \cdot \frac{1}{4} = \frac{1}{24} \\
 P(D) &= P(D \cap E_1) + P(D \cap E_2) + P(D \cap E_3) \\
 &= P(E_1) \cdot P(D|E_1) + P(E_2) \cdot P(D|E_2) \\
 &\quad + P(E_3) \cdot P(D|E_3) \\
 &= \frac{6}{24} \cdot \frac{1}{3} + \frac{11}{24} \cdot \frac{7}{11} + \frac{6}{24} \cdot 1 \\
 &= \frac{2+7+6}{24} = \frac{15}{24} = \frac{5}{8}
 \end{aligned}$$

► **Problem 38.** With respect to a particular question on a multiple choice test (having 4 alternatives with only 1 correct) a student knows the answer and therefore can eliminate 3 of the 4 choices from consideration with probability  $2/3$ , can eliminate 2 of the 4 choices from consideration with probability  $1/6$ , can eliminate 1 choice from consideration with probability  $1/9$ , and can eliminate none with probability  $1/18$ . If the student knows the answer, he answers correctly, otherwise he guesses from among the choices not eliminated.

If the answer given by the student was found correct, then the probability that he knew the answer is  $a/b$  where  $a$  and  $b$  are relatively prime. Find the value of  $(a + b)$ .

► **Solution** A : his answer was correct  
 $B_1$  : he can eliminate 3 out of 4 alternatives  
 $B_2$  : he can eliminate 2 out of 4 alternatives  
 $B_3$  : he can eliminate 1 out of 4 alternatives  
 $B_4$  : he can eliminate none out of 4 alternatives

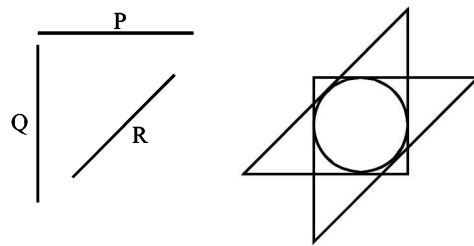
$$P(B_1) = \frac{2}{3}; P(B_2) = \frac{1}{6}; P(B_3) = \frac{1}{9}; P(B_4) = \frac{1}{18}$$



$$\begin{aligned}
 P(A/B_1) &= 1; P(A/B_2) = \frac{1}{2}; P(A/B_3) = \frac{1}{3}; \\
 P(A/B_4) &= \frac{1}{4} \\
 P(B_1/A) &= \frac{P(B_1 \cap A)}{P(A)} \\
 &= \frac{P(B_1 \cap A)}{P(A \cap B_1) + P(A \cap B_2) + P(A \cap B_3) + P(A \cap B_4)} \\
 &= \frac{P(B_1) \cdot P(A/B_1)}{\sum_{i=1}^4 P(B_i) \cdot P(A/B_i)} \\
 &= \frac{\frac{2}{3} \cdot 1}{\frac{2}{3} \cdot 1 + \frac{1}{6} \cdot \frac{1}{2} + \frac{1}{9} \cdot \frac{1}{3} + \frac{1}{18} \cdot \frac{1}{4}} \\
 &= \frac{\frac{2}{3} \cdot \frac{216}{173}}{\frac{144}{173}} = \frac{a}{b} \\
 \therefore a + b &= 144 + 173 = 317
 \end{aligned}$$

► **Problem 39.** Three tangents are drawn at random to a given circle. Show that the odds are 3 to 1 against the circle being inscribed in the triangle formed by them.

► **Solution**



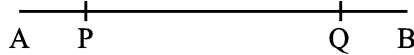
Draw three random lines  $P, Q, R$  in the same plane as the circle, and draw to the circle the six tangents parallel to these lines.

Then of the 8 triangles so formed it is evident that the circle will be escribed to 6 and inscribed in 2; and as this it is true that the odds are 3 to 1 whatever be the original directions of  $P, Q, R$ .



► **Problem 40.** On a straight line of length  $a$  two points are taken at random. Find the chance that the distance between them is greater than  $b$ .

► **Solution** Let  $AB$  be the straight line of length  $a$ , and let  $P, Q$  be the random points, being at a distance of  $x, y$  from the end  $A$  of the line.



Now in favourable cases, we must have,  
 $x > b + y$ , or  $y > b + x$ . ... (1)

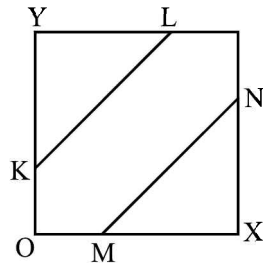
Again for all the cases possible, we must have,

$$\left. \begin{array}{l} x > 0 \text{ and } x < a \\ y > 0 \text{ and } y < a \end{array} \right\} \quad \dots (2)$$

Let us now take a pair of rectangular axes and make  $OX, OY$  each equal to  $a$ .

Draw the line  $y = b + x$ , represented by  $KL$  in the figure, and the line  $x = b + y$ , represented by  $MN$ .

Then  $OM, OK$  are each equal to  $b$ ,  $MX, KY$  are each equal to  $a - b$ .



The conditions (1) are only satisfied by points in the triangle  $MNX$  and  $KLY$ , while the conditions (2) are satisfied by any points within the rectangle  $OX, OY$ .

$$\therefore \text{ the required chance } = \left( \frac{a-b}{a} \right)^2.$$

### Things to Remember

1. A list of rules which follow from the definitions of sum and product of events:

$$\begin{array}{ll} A \cup S = S, & A \cap S = A, \\ A \cup \Phi = A, & A \cap \Phi = \Phi. \end{array}$$

If  $A \subseteq B$ , then  $A \cup B = B$ ,  $A \cap B = A$ .

$$(\overline{\overline{A}}) = A, \quad \overline{\overline{S}} = \Phi, \quad \overline{\overline{\Phi}} = S.$$

**Commutative laws**  $A \cup B = B \cup A$   
 $AB = BA$

**Associative laws**  $(A \cup B) \cup C = A \cup (B \cup C)$   
 $(AB)C = A(BC)$

**Distributive laws**  $(A \cup B)C = AC \cup BC$   
 $AB \cup C = (A \cap C) \cup (B \cap C)$

**Idempotency law**  $A \cup A = A$   $AA = A$

**De Morgan's Law** If  $A$  &  $B$  are two subsets of the sample space  $S$ , then

- (a)  $(A \cup B)^c = A^c \cap B^c$  and
- (b)  $(A \cap B)^c = A^c \cup B^c$  and for  $n$  events,  $A_1, A_2, \dots, A_n$ ,
- (c)  $(A_1 \cup A_2 \cup \dots \cup A_n)^c = A_1^c \cap A_2^c \cap \dots \cap A_n^c$ ,
- (d)  $(A_1 \cap A_2 \cap \dots \cap A_n)^c = A_1^c \cup A_2^c \cup \dots \cup A_n^c$ .

$$2. P(A) = \frac{\text{number of points in } A}{\text{number of points in } S} = \frac{n(A)}{n(S)}.$$

$$3. \text{ (i) Odds in favour of event } A = \frac{\text{number of cases favourable to event } A}{\text{number of cases against } A}$$

$$= \frac{n(A)}{n(A')} = \frac{n(A)/n(S)}{n(A')/n(S)} = \frac{P(A)}{P(A')}$$

(ii) Odds against an event  $A$

$$= \frac{\text{number of cases against the event } A}{\text{number of cases favourable to event } A} = \frac{n(A')}{n(A)} = \frac{P(A')}{P(A)}$$

4. (i) If  $A^c$  is the event complementary to  $A$ , then  $P(A^c) = 1 - P(A)$ .

(ii) If  $A \subset B$ , then  $P(A) \leq P(B)$ .

(iii) If  $A$  and  $B$  are any two events in a sample space  $S$ , then  $P(A \cup B) = P(A) + P(B) - P(AB)$

(iv) For three events  $A, B, C$   
 $P(A \cup B \cup C) = P(A) + P(B) + P(C) - [P(AB) + P(BC) + P(AC)] + P(ABC)$ .

(v)  $P$  (atleast two of  $A, B, C$  occur)  
 $= P(B \cap C) + P(C \cap A) + P(A \cap B) - 2P(A \cap B \cap C)$

(vi)  $P$  (exactly two of  $A, B, C$  occur)  
 $= P(B \cap C) + P(C \cap A) + P(A \cap B) - 3P(A \cap B \cap C)$

- (vii)  $P(\text{exactly one of } A, B, C \text{ occurs})$   
 $= P(A) + P(B) + P(C) - 2P(B \cap C)$   
 $- 2P(C \cap A) - 2P(A \cap B) + 3P(A \cap B \cap C)$
5. (i) If  $A \subset B$ , then  $P(A) \leq P(B)$ .  
 (ii)  $P[AB] \leq P[A] \leq P[A \cup B] \leq P[A] + P[B]$ .  
 (iii)  $\max\{P(A), P(B)\} \leq P(A \cup B)$   
 $\leq \min\{1, P(A) + P(B)\}$ .  
 (iv) Bon-ferroni's inequality :  
 $P(A) + P(B) - 1 \leq P(AB) \leq \min\{P(A), P(B)\}$   
 (v)  $\max\{0, 1 - P(\bar{A}) - P(\bar{B})\} \leq P(AB)$   
 (vi) Boole's inequality : For a finite set of events,  
 $A_1, A_2, \dots, A_n$ ,  
 $P[A_1 \cup A_2 \cup \dots \cup A_n] \leq P[A_1] + P[A_2] + \dots + P[A_n]$ .
6. If  $P(A) > 0$ , then  $P(B|A) = \frac{n(A \cap B)}{n(A)}$   
 $= \frac{P(A \cap B)}{P(A)}$ , when  $P(A) \neq 0$ .
7. Multiplication theorem  $P(A \cap B) = P(A) \cdot P(B|A)$   
 $= P(B) \cdot P(A|B)$  provided  $P(A) \neq 0$  and  $P(B) \neq 0$ .
8. Generalized multiplication theorem  
 $P(A_1 A_2 A_3 \dots A_n) = P(A_1) P(A_2 | A_1) P(A_3 | A_1 A_2)$   
 $\dots P(A_n | A_1 \dots A_{n-1})$
9. Total probability theorem  
 $P(B) = P(B|A) P(A) + P(B|\bar{A}) P(\bar{A})$   
 Let  $\{A_1, A_2, \dots, A_n\}$  be a partition of the sample space  $S$ , and suppose that each of the events  $A_1, A_2, \dots, A_n$  has non-zero probability of occurrence. Let  $B$  be any event associated with  $S$ , then  
 $P(B) = P(A_1) P(B|A_1) + P(A_2) P(B|A_2) + \dots + P(A_n) P(B|A_n)$
10. (i)  $A$  and  $B$  are independent events if and only if  $P(A \cap B) = P(A) \times P(B)$   
 (ii) If any one of the pairs  $\{A, B\}$ ,  $\{A, B^c\}$ ,  $\{A^c, B\}$ , or  $\{A^c, B^c\}$  is an independent pair, then all the pairs are independent pairs.  
 (iii) Mutually exclusive events (none of which is an impossible event) are not independent and non-impossible independent events are not mutually exclusive.  
 (iv) Three events  $A, B$  and  $C$  are said to be mutually independent, if
- $P(A \cap B) = P(A) P(B)$  ... (1)  
 $P(A \cap C) = P(A) P(C)$  ... (2)  
 $P(B \cap C) = P(B) P(C)$  ... (3)  
 and  $P(A \cap B \cap C) = P(A) P(B) P(C)$  ... (4)
11. If  $A$  and  $B$  are two independent events then  
 (i)  $P(A \cup B) = P(A) + P(B) - P(A \cap B)$   
 $= P(A) + P(B) - P(A) \cdot P(B)$   
 (ii)  $P(A \cup B) = 1 - P(\overline{A \cup B})$   
 $= 1 - P(\bar{A} \cap \bar{B})$   
 $= 1 - P(\bar{A}) P(\bar{B})$
12. If  $A, B$  and  $C$  are three independent events then  
 (i) the probability of intersection of these is equal to the product of the probabilities of these events, i.e.  
 $P(A \cap B \cap C) = P(A) P(B) P(C)$ .  
 (ii) the probability that atleast one of the three events  $A, B$  or  $C$  occur is  
 $P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A) \cdot P(B) - P(B) \cdot P(C) - P(A) \cdot P(C) + P(A) P(B) P(C)$ .
13. If  $A_1, A_2, \dots, A_n$  are  $n$  independent events then  
 $P[A_1 \cup A_2 \cup \dots \cup A_n]$   
 $= 1 - P(\overline{A_1 \cup A_2 \cup \dots \cup A_n})$   
 $= 1 - P(\bar{A}_1) P(\bar{A}_2) \dots P(\bar{A}_n)$
14. Let the events  $A_1, A_2, \dots, A_n$  be independent and  $P(A_i) = p_i$  for  $i = 1, \dots, n$ .  
 (i) the probability that  $A_1$  happens and  $A_2$  fails is  $p_1 (1 - p_2)$ .  
 (ii) the probability that  $A_2$  happens and  $A_1$  fails is  $p_2 (1 - p_1)$ .  
 (iii) the probability that both  $A_1$  and  $A_2$  fail is  $(1 - p_1) (1 - p_2)$ .  
 (iv) the probability that one and only one of the events  $A_1$  and  $A_2$  happen is  
 $p_1 (1 - p_2) + p_2 (1 - p_1) = p_1 + p_2 - 2p_1 p_2$ .  
 (v) the probability that all the events  $A_1, A_2, \dots, A_n$  occur is  $p_1 p_2 \dots p_n$ .  
 (vi) the probability that none of the events  $A_1, A_2, \dots, A_n$  occur is  $(1 - p_1) (1 - p_2) \dots (1 - p_n)$ .

- (vii) the probability that the first two events happen and rest fail is  $p_1 p_2 (1 - p_3) (1 - p_4) \dots (1 - p_n)$ .

15. (i) Expectation  $E(X) = \mu = \sum_{i=1}^n x_i p_i$   
 $= x_1 p_1 + x_2 p_2 + \dots + x_n p_n$ .

(ii)  $\sigma_x^2 = \text{Var}(X) = \sum_{i=1}^n (x_i - \mu)^2 p(x_i)$   
 $= E(X^2) - [E(X)]^2$

16. (i) The probability of  $x$  successes in  $n$ -Bernoulli trials is  ${}^n C_x p^x q^{n-x}$

- (ii) The probability of atleast  $r$  successes in  $n$  trials  
 $= {}^n C_r p^r q^{n-r} + {}^n C_{r+1} p^{r+1} q^{n-(r+1)} + \dots + {}^n C_n p^n$ .

- (iii) The probability of atmost  $r$  successes in  $n$  trials  
 $= {}^n C_0 q^n + {}^n C_1 p q^{n-1} + \dots + {}^n C_r p^r q^{n-r}$ .

17. If  $X$  is a binomial random variable with parameters  $n$  and  $p$ , then  
 $E[X] = np$  and  $\text{Var}(X) = np(1 - p)$ .

18. Mode of binomial distribution with parameters  $n$  and  $p$  :

- (i) in the case that  $(n + 1)p$  is not an integer, when  $r$  is equal to the integer  $m$  satisfying the inequalities  
 $(n + 1)p - 1 < m < (n + 1)p$  and  
(ii) in the case  $(n + 1)p$  is an integer, when  $r$  is equal to either  $(n + 1)p - 1$  or  $(n + 1)p$ .

19. If  $A_1, A_2, \dots, A_n$  are  $n$  non-empty events with given probabilities  $P(A_1), P(A_2), \dots, P(A_n)$  which constitute a partition of sample space  $S$ , (i.e.  $A_1, A_2, \dots, A_n$  are  $n$  mutually exclusive and exhaustive events so that  $A_1 \cup A_2 \cup \dots \cup A_n = S$ ) and  $B$  is any event with non-zero probability and the probabilities  $P(B|A_1), P(B|A_2), \dots, P(B|A_n)$  are known, then the Bayes' theorem states that

$$P(A_i | B) = \frac{P(A_i) P(B | A_i)}{\sum_{j=1}^n P(A_j) P(B | A_j)}$$

for any  $i = 1, 2, 3, \dots, n$

20. Value of testimony:  $P$  (the combined of  $A$  &  $B$  statement is true)

$$= \frac{P p_1 p_2}{P p_1 p_2 + c(1 - P)(1 - p_1)(1 - p_2)}$$

21. Geometrical Probability

- (i) If a point is taken at random on a given straight line  $AB$ , the chance that it falls on a particular segment  $PQ$  of the line is  $PQ/AB$ .  
(ii) If a point is taken at random on the area  $S$  which includes an area  $\sigma$ , the chance that the point falls on  $\sigma$  is  $\sigma/S$ .  
(iii) If a point is taken at random within the volume  $V$  which includes a volume  $\rho$ , the chance that the point falls in  $\rho$  is  $\rho/V$ .

### [Objective Exercises]

#### Single Correct Answer Type

- Three numbers are selected at random without replacement from the set of numbers  $\{1, 2, \dots, N\}$ . The conditional probability that the third number lies between the first two, if the first number is known to be smaller than the second, is  
 (A)  $1/6$  (B)  $1/3$   
 (C)  $1/2$  (D) None of these
- The probability that a man can hit a target is  $3/4$ . He tries 5 times. The probability that he will hit the target atleast 3 times is  
 (A)  $\frac{291}{364}$  (B)  $\frac{371}{464}$   
 (C)  $\frac{471}{562}$  (D)  $\frac{459}{512}$

3. Let A and B be two events such that  $P(A) = 7/20$ ,  $P(B) = 9/20$ ,  $P(A \cup B) = 11/20$ , then the value of  $P(\bar{A} \cup B)$  is equal to  
 (A)  $1/4$  (B)  $1/5$   
 (C)  $1/10$  (D) None of these
4. Ram and Rahim throw two dice till one wins. Ram wins if he throws a sum of 6 before Rahim throws a sum of 7 and Rahim wins if he throws a sum of 7 before Ram throws a sum of 6. Ram starts the game. Then :  
 (A) Ram is more likely to win  
 (B) Rahim is more likely to win  
 (C) Both have equal chance to win  
 (D) None of these
5. A random variable X is distributed normally with mean 10 and variance 1 then the value of  $P(X > 11.96)$  is :  
 (A) 0.25 (B) 0.010  
 (C) 0.050 (D) 0.100
6. Two friends Ashok and Baldev have equal number of sons. There are 3 tickets for a cricket match which are to be distributed among the sons. The probability that 2 tickets go to sons of the one and one ticket go to the sons of the other is  $6/7$ . The number of sons each of the two friends have :  
 (A) 3 (B) 4  
 (C) 5 (D) None of these
7. The probability of loss in a single game is equal to 0.5. If the events A is "winning not less than three games of four" and the events B is "winning not less five games of either". then:  
 (A) the events A and B are equally probable  
 (B) the event A is more probable than B  
 (C) the event B is more probable than A  
 (D) None of above
8. Three numbers are selected one by one from whole numbers 1 to 20. The probability that they are consecutive integers is :  
 (A)  $1/3$  (B)  $5/190$   
 (C)  $3/190$  (D)  $7/190$
9. A binary number is made up of 16 bits. The probability of an incorrect bit appearing is p and the errors in different bits are independent of one another. The probability of forming an incorrect number is :  
 (A)  $\frac{p}{16}$  (B)  $p^{16}$   
 (C)  ${}^{16}C_1 p^{16}$  (D)  $1 - (1 - p)^{16}$
10. A subset A of the set  $X = \{1, 2, \dots, 100\}$  is chosen at random. The set X is reconstructed by replacing the elements of A, and another subset B of X is chosen at random. The probability that  $A \cap B$  contains exactly 10 elements is  
 (A)  ${}^{100}C_{10} \left(\frac{3}{4}\right)^{90}$  (B)  ${}^{100}C_{10} \left(\frac{1}{2}\right)^{100}$   
 (C)  ${}^{100}C_{10} \frac{3^{90}}{r^{100}}$  (D) None of these
11. Three six-faced fair dice are thrown together. The probability that the sum of the numbers appearing on the dice is k ( $3 \leq k \leq 8$ ) is  
 (A)  $\frac{(k-1)(k-2)}{432}$  (B)  $\frac{k(k-1)}{432}$   
 (C)  $\frac{k^2}{432}$  (D) None of these
12. The minimum number of times a fair coin must be tossed so that the probability of getting atleast one head is atleast 0.8, is  
 (A) 7 (B) 6  
 (C) 5 (D) 3
13. One ticket is selected at random from 100 tickets numbered 00, 01, 02, ..., 99. Suppose X and Y are the sum and product of the digit found on the ticket  $P(X = 7/Y = 0)$  is given by  
 (A)  $2/3$  (B)  $2/19$   
 (C)  $1/50$  (D)  $3/19$

14. A bag contains four tickets numbered 00, 01, 10, 11. Four tickets are chosen at random with replacement, the probability that sum of the numbers on the tickets is 23, is  
 (A)  $\frac{3}{32}$  (B)  $\frac{1}{64}$   
 (C)  $\frac{5}{256}$  (D)  $\frac{7}{256}$
15. Suppose  $X$  is a binomial variate  $B(5, p)$  and  $P(X = 2) = P(X = 3)$ , then  $p$  is equal to  
 (A)  $\frac{1}{2}$  (B)  $\frac{1}{3}$   
 (C)  $\frac{1}{4}$  (D)  $\frac{1}{5}$
16. Mr. A gave his telephone number to Mr. B. Mr. B remembers that the first two digits were 40 and the remaining four digits were two 3s, one 6 and one 8. He is not certain about the order of the digits. Mr. B dials 403638. The probability that he will get Mr. A's number is  
 (A)  $\frac{1}{2}$  (B)  $\frac{1}{4}$   
 (C)  $\frac{1}{8}$  (D)  $\frac{1}{12}$
17. A soldier is firing at a moving target. He fires four shots. The probability of hitting the target at the first, second, third and fourth shots are 0.6, 0.4, 0.2 and 0.1 respectively. The probability that he hits the target is  
 (A)  $\frac{517}{625}$  (B)  $\frac{3}{625}$   
 (C)  $\frac{105}{625}$  (D) None
18. Events  $A$  and  $B$  are such that  $P(A) = \frac{1}{4}$ ,  $P(B|A) = \frac{1}{2}$ ,  $P(A|B) = \frac{1}{4}$ . Then  
 (A)  $A$  and  $B$  are mutually exclusive.  
 (B)  $A$  is a subevent of  $B$ .  
 (C)  $P(\bar{A} | \bar{B}) = \frac{3}{4}$   
 (D)  $P(A|B) + P(A|\bar{B}) = 1$ .
19. The probability that a rectangle picked up from a chessboard has the area  $6 \text{ cm}^2$  where the distance between consecutive parallel lines on the board is 1 cm, is  
 (A)  $\frac{3}{56}$  (B)  $\frac{3}{28}$   
 (C)  $\frac{9}{56}$  (D) None
20. A natural number is selected at random from the set  $X = \{x \mid 1 \leq x \leq 100\}$ . The probability that the number satisfies the inequation  $x^2 - 13x \leq 30$ , is  
 (A)  $\frac{9}{50}$  (B)  $\frac{3}{20}$   
 (C)  $\frac{2}{11}$  (D) None
21. An unbiased ordinary dice is rolled four times. Out of the four face-values obtained, the probability that the minimum face-value is not less than 2 and the maximum face-value is not greater than 5, is  
 (A)  $\frac{16}{81}$  (B)  $\frac{1}{81}$   
 (C)  $\frac{80}{81}$  (D)  $\frac{65}{81}$
22. The mean of the numbers obtained on throwing a die having written 1 on three faces, 2 on two faces and 5 on one face is  
 (A) 1 (B) 2  
 (C) 5 (D)  $\frac{8}{3}$
23. Suppose that two cards are drawn at random from a deck of cards. Let  $X$  be the number of aces obtained. Then the value of  $E(X)$  is  
 (A)  $\frac{37}{221}$  (B)  $\frac{5}{13}$   
 (C)  $\frac{1}{13}$  (D)  $\frac{2}{13}$
24. An ellipse is inscribed in a circle and a point within the circle is chosen at random. If the probability that this point lies outside the ellipse is  $\frac{2}{3}$  then the eccentricity of the ellipse is:  
 (A)  $\frac{2\sqrt{2}}{3}$  (B)  $\frac{\sqrt{5}}{3}$   
 (C)  $\frac{8}{9}$  (D)  $\frac{2}{3}$
25. A fair coin is tossed 12 times. The probability that two heads do not occur consecutively is  
 (A)  $\frac{375}{4096}$  (B)  $\frac{376}{4096}$   
 (C)  $\frac{377}{4096}$  (D) none of these
26. The probability  $P_r$ , that a family has  $r$  children is given by  $P_0 = P_1 = a$ ,  $P_r = (1 - 2a) 2^{-(r-1)}$  ( $r > 2$ ). It is known that a family has two boys. Then the probability that the family has two girls as well is  
 (A)  $\frac{81}{512}$  (B)  $\frac{79}{512}$

- (C)  $\frac{77}{512}$  (D) None of these
27. The probability that a certain beginner at golf gets a good shot if he uses the correct club is  $\frac{1}{3}$  and the probability of a good shot with an incorrect club is  $\frac{1}{4}$ . In his bag there are 5 different clubs, only one of which is correct for the shot in question. If he chooses a club at random and takes a stroke, the probability that he gets a good shot is  
 (A)  $\frac{2}{3}$  (B)  $\frac{1}{15}$   
 (C)  $\frac{4}{15}$  (D)  $\frac{7}{15}$
28. A bag has 13 red, 14 green and 15 black balls. The probability of getting exactly 2 blacks on pulling out 4 balls is  $p_1$ . Now the number of each colour balls is doubled and 8 balls are pulled out. The probability of getting exactly 4 blacks is  $p_2$ . Then  
 (A)  $p_1 < p_2$  (B)  $p_1 = p_2$   
 (C)  $p_1 > p_2$  (D) none of these
29. Two persons A and B have respectively  $n+1$  and  $n$  coins, which they toss simultaneously. Then the probability that A will have more heads than B is  
 (A)  $\frac{1}{2}$  (B)  $> \frac{1}{2}$   
 (C)  $< \frac{n-1}{2n}$  (D) none of these
30. A player randomly chooses one of the coins A, B. Coin A has probability of heads  $\frac{3}{4}$  and coin B has probability of heads  $\frac{1}{4}$ . He tosses the coin twice. The probability that he obtains heads only once is  
 (A)  $\frac{3}{8}$  (B)  $\frac{5}{8}$   
 (C)  $\frac{7}{8}$  (D) None
31. In a game "odd man out" each of  $m \geq 2$  person tosses a coin to determine who will buy refreshments for the entire group. The odd man out is the one with a different outcome from the rest. The probability that there is a loser in any game is  
 (A)  $\frac{1}{2^{m-1}}$  (B)  $\frac{m-1}{2^{m-1}}$   
 (C)  $\frac{m}{2^{m-1}}$  (D) none of these
32. A bag contains  $n$  white and  $n$  red balls. Pairs of balls are drawn without replacement until the bag is empty. The probability that each pair consists of one white and one red ball is  
 (A)  $\frac{2n}{2^n C_n}$  (B)  $\frac{2^n}{2^n C_n}$   
 (C)  $\frac{1}{2^n C_n}$  (D) none of these
33. A contest consists of predicting the results (win, draw or defeat) of 10 football matches. The probability that one entry contains at least 5 correct answers is  
 (A)  $\frac{12585}{3^{10}}$  (B)  $\frac{12385}{3^{10}}$   
 (C)  $\frac{9385}{3^{10}}$  (D) none of these
34. A pack of cards is counted with face downwards. It is found that one card is missing. One card is drawn and is found to be red. The probability that the missing card is red, is  
 (A)  $\frac{12}{51}$  (B)  $\frac{18}{51}$   
 (C)  $\frac{25}{51}$  (D) none of these
35. A man firing at a distant target has 10% chance of hitting the target in one shot. The number of times he must fire at the target to have about 50% chance of hitting the target is  
 (A) 10 (B) 9  
 (C) 7 (D) 8
36. All the spades are taken out from a pack of cards. From these cards, cards are drawn one by one without replacement till the ace of spade comes. The probability that the ace comes in the 4th draw is  
 (A)  $\frac{1}{13}$  (B)  $\frac{10}{13}$   
 (C)  $\frac{3}{13}$  (D) none of these

37. Ram's gardener is not dependable. The probability that he will forget to water the rose bush is  $\frac{2}{3}$ . The rose bush is in questionable condition any how, if watered the probability of its withering is  $\frac{1}{2}$ , if not watered, the probability of its withering is  $\frac{3}{4}$ . Ram went out of station and upon returning, he finds that the rose bush has withered. The probability that the gardener did not water the bush is  
 (A)  $\frac{1}{2}$  (B)  $\frac{1}{3}$   
 (C)  $\frac{3}{4}$  (D) None
38. Six girls are to enter a dance with 10 boys to form a ring so that every girl is between two boys. The probability that some specified boy remains between 2 boys is  
 (A)  $\frac{1}{15}$  (B)  $\frac{7}{15}$   
 (C)  $\frac{2}{15}$  (D) None
39. Suppose  $X$  follows a binomial distribution with parameters  $n$  and  $p$ , where  $0 < p < 1$ . If  $P(X = r)/P(X = n - r)$  is independent of  $n$  and  $r$ , then  $p$  is equal to  
 (A)  $\frac{1}{4}$  (B)  $\frac{1}{3}$   
 (C)  $\frac{1}{2}$  (D) None
40. 10% of a certain population suffer from a serious disease. A person suspected of the disease is given two independent tests. Each test makes a correct diagnosis 90% of the time. The probability that the person really has the illness given that both tests are positive is  
 (A) 0.5 (B) 0.9  
 (C) 0.6 (D) None
41. A drawn two cards at random from a pack of 52 cards. After returning them to the pack and shuffling it, B drawn two cards at random. The probability that their draws contain exactly one common cards is  
 (A)  $\frac{25}{546}$  (B)  $\frac{50}{663}$   
 (C)  $\frac{25}{663}$  (D) None
42. A die is thrown as long as necessary for ace or a 6 to turn up. Given that no ace turned up at the first two throws. The probability that atleast three throws will be necessary is  
 (A)  $\frac{11}{25}$  (B)  $\frac{13}{25}$   
 (C)  $\frac{16}{25}$  (D) None
43. There are two balls in an urn whose colours are not known (each ball can be either white or black). A white ball is put into the urn. A ball is drawn from the urn. The probability that it is white is  
 (A)  $\frac{1}{4}$  (B)  $\frac{1}{3}$   
 (C)  $\frac{2}{3}$  (D)  $\frac{1}{6}$
44. The altitude through A of  $\triangle ABC$  meets BC at D and the circumscribed circle at E. If  $D \equiv (2, 3)$ ,  $E \equiv (5, 5)$  and the ordinate of the orthocentre being a natural number. The probability that the orthocenter lies on the lines  $y = 1, y = 2, y = 3, \dots, y = 10$  is  
 (A)  $\frac{2}{5}$  (B)  $\frac{1}{5}$   
 (C)  $\frac{3}{5}$  (D) None
45. Let a die be loaded in such a way that prime number faces are twice as likely to occur as non prime number faces. The probability that an odd number will show up when the die is tossed is  
 (A)  $\frac{1}{3}$  (B)  $\frac{4}{9}$   
 (C)  $\frac{2}{3}$  (D)  $\frac{5}{9}$
46. A person draws a card from a pack of playing cards, replaces it and shuffles the pack. He continues doing this until he shown a spade. The chance that he will fail in the first two trial is  
 (A)  $\frac{24}{41}$  (B)  $\frac{25}{41}$   
 (C)  $\frac{16}{41}$  (D)  $\frac{17}{41}$
47. A bag contains 15 red, 10 white and 5 black balls. Two balls are chosen at random and one is found to be red. The probability that the second one is also red is  
 (A)  $\frac{14}{15}$  (B)  $\frac{29}{30}$   
 (C)  $\frac{7}{29}$  (D)  $\frac{14}{29}$

48. A fair coin is tossed repeatedly. If tail appears on first four tosses, then the probability of head appearing on the fifth toss is  
 (A)  $1/2$  (B)  $1/32$   
 (C)  $31/32$  (D)  $1/5$
49. There are four machines and it is known that exactly two of them are faulty. They are tested, one by one in random order till both the faulty machines are identified. Then the probability that only two tests are needed is  
 (A)  $1/3$  (B)  $1/6$   
 (C)  $1/2$  (D)  $1/4$
50. For three events A, B and C,  $P(\text{exactly one of the events A or B occurs}) = P(\text{exactly one of the events B or C occurs}) = P(\text{exactly one of the events C or A occurs}) = p$  and  $P(\text{all the three events occur simultaneously}) = p^2$ , where  $0 < p < 1/2$ . Then the probability of atleast one of the three events A, B and C occurring is  
 (A)  $\frac{3p + 2p^2}{2}$  (B)  $\frac{p + 3p^2}{2}$   
 (C)  $\frac{3p + p^2}{2}$  (D)  $\frac{3p + 2p^2}{4}$
- (A)  $p_1 = p_2$  (B)  $p_1 > p_2$   
 (C)  $p_1 < p_2$  (D)  $p_1^2 - p_2^2 < \frac{1}{3}$
54. There are two planes one having 3 engines and other having 5 engines, each engine having a probability P of failing on any given flight. A plane crashes if more than half of its engines fail. The value of P so that the two planes have the same probability of crashing is  
 (A) 0 (B)  $1/3$   
 (C)  $1/9$  (D)  $1/2$
55. There are 2 red, 2 black, 2 white and x blue socks in a drawer. If two socks are taken from the drawer without replacement, the probability that they are of  
 (A) red colour is  $\frac{2}{(6+x)(5+x)}$   
 (B) blue colour is  $\frac{x(x-1)}{(6+x)(5+x)}$   
 (C) same colour is  $\frac{x^2 - x + 6}{(6+x)(5+x)}$   
 (D) same colour is  $1/5$  when  $x = 4$ .

### Multiple Correct Answer Type for JEE Advanced

51. For two events A and B,  $P(A \cap B)$  is  
 (A) equal to  $P(A) + P(B) + P(A \cup B)$   
 (B) not less than  $P(A) + P(B) - 1$   
 (C) not greater than  $P(A) + P(B) - 1$   
 (D) equal to  $P(A) + P(B) - P(A \cup B)$
52. A fair coin is tossed n times. Let X = the number of times head occurs  $P(X = 4)$ ,  $P(X = 5)$  and  $P(X = 6)$  are in A.P., then the value of n can be  
 (A) 7 (B) 10  
 (C) 12 (D) 14
53. A square is inscribed in a circle. If  $p_1$  is the probability that a randomly chosen point of the circle lies within the square and  $p_2$  is the probability that the point lies outside the square, then  
 (A)  $P(A/B^C) = \frac{P(A) - P(A \cap B)}{1 - P(B)}$   
 (B)  $P(A \cap B) \geq P(A) + P(B) - 1$
56. A pair of fair dice having six faces numbered from 1 to 6 are thrown once, suppose two events E and F are defined as  
 E : Product of the two numbers appearing is divisible by 5.  
 F : At least one of the dice shows up the face one.  
 Then the events E and F are  
 (A) mutually exclusive  
 (B) independent  
 (C) neither exclusive  
 (D) are equiprobable
57. If A & B are two events such that  $P(B) \neq 1$ ,  $B^C$  denotes the event complementary to B, then  
 (A)  $P(A/B^C) = \frac{P(A) - P(A \cap B)}{1 - P(B)}$   
 (B)  $P(A \cap B) \geq P(A) + P(B) - 1$



- (C)  $P(A) > P(A/B)$  according as  $P(A/B^C) > P(A)$   
 (D)  $P(A/B^C) + P(A^C/B^C) = 1$
- 58.** Two real numbers,  $x$  &  $y$  are selected at random. Given that  $0 \leq x \leq 1$ ;  $0 \leq y \leq 1$ . Let  $A$  be the event that  $y^2 \leq x$ ;  $B$  be the event that  $x^2 \leq y$ , then :  
 (A)  $P(A \cap B) = 1/3$   
 (B)  $A$  &  $B$  are exhaustive events  
 (C)  $A$  &  $B$  are mutually exclusive  
 (D)  $A$  &  $B$  are independent events.
- 59.** In which of the following cases the events  $E$  and  $F$  are independent?  
 (A) Fair die is tossed twice.  
 $E$  : Denotes the event that the sum of 2 rolls equals 5.  
 $F$  : denotes the event that an even number comes up in the first throw.  
 (B) 3 fair coins are tossed.  
 $E$  : denotes the event all heads or all tails.  
 $F$  : at least 2 tails  
 (C) A card is drawn from a well shuffled pack of 52 playing cards.  
 $E$  : denotes the event that the card drawn in either red or black.  
 $F$  : denotes the event that the card drawn is the face card.  
 (D)  $P(E \cup F) = 3/4$ ;  $P(E^C) = 2/3$  and  $P(E \cap F) = 1/4$
- 60.** If  $E_1$  and  $E_2$  are two events such that  $P(E_1) = 1/4$ ,  $P(E_2/E_1) = 1/2$  and  $P(E_1/E_2) = 1/4$   
 (A) then  $E_1$  and  $E_2$  are independent  
 (B)  $E_1$  and  $E_2$  are exhaustive  
 (C)  $E_2$  is twice as likely to occur as  $E_1$   
 (D) Probabilities of the events  $E_1 \cap E_2$  are in G.P.
- 61.** Let  $X$  be the number of times heads occur in a tosses of a fair coin. If  $P(X = 4)$ ,  $P(X = 5)$  and  $P(X = 6)$  are in A.P., the value of  $n$  is  
 (A) 7 (B) 10  
 (C) 12 (D) 14
- 62.** If  $X$  and  $Y$  are independent events such that  $0 < P(X) < 1$  and  $0 < P(Y) < 1$ , then  
 (A)  $X$  and  $Y$  are mutually exclusive  
 (B)  $X$  and  $Y^C$  are independent  
 (C)  $X$  and  $Y^C$  are independent  
 (D)  $P(X/Y) + P(X^C/Y) = 1$
- 63.** If  $A$  and  $B$  are two events such that  $P(A) = 5/8$  and  $P(B) = 3/4$ , then ;  
 (A)  $P(A' \cup B) \geq \frac{1}{2}$   
 (B)  $\frac{1}{8} \leq P(A' \cup B) \leq \frac{3}{8}$   
 (C)  $\frac{3}{8} \leq P(A \cap B) \leq \frac{5}{8}$   
 (D)  $P(A' \cup B) \geq \frac{3}{4}$
- 64.** Two events  $A$  and  $B$  are such that the probability that at least one of them occurs is  $5/6$  and both of them occurring simultaneously is  $1/3$ . If the probability of not occurrence of  $B$  is  $1/12$  then  
 (A)  $A$  and  $B$  are equally likely  
 (B)  $A$  and  $B$  are independent  
 (C)  $P(A/B) = 2/3$   
 (D)  $3 P(A) = 4 P(B)$
- 65.** In a single throw with 3 ordinary unbiased dice, the chance of :  
 (A) throwing a "4-5-6" is equal to  $1/216$   
 (B) throwing a total of 11 is  $1/8$   
 (C) all the 3 dice showing up with different faces is  $4/9$   
 (D) all the 3 dice showing up alike faces is  $1/36$
- 66.** The probabilities of events,  $A \cap B$ ,  $A$ ,  $B$  &  $A \cup B$  are respectively A.P. with probability of second term equal to the common difference. Therefore the events  $A$  and  $B$  are  
 (A) mutually exclusive  
 (B) independent  
 (C) such that one of them must occur  
 (D) such that one is twice as likely as the other

67. A box contains 11 tickets numbered from 1 to 11. Six tickets are drawn simultaneously at random.

Let  $E_1$  denotes the event that the sum of the numbers on the tickets drawn is even and  $E_2$  denotes the event that the sum of the numbers on the tickets drawn is odd. Which of the following hold good?

- (A)  $E_1$  and  $E_2$  are equally likely  
 (B)  $E_1$  and  $E_2$  are exhaustive  
 (C)  $P(E_2) > P(E_1)$   
 (D)  $P(E_1/E_2) = P(E_2/E_1)$
68. If  $\bar{E}$  &  $\bar{F}$  are the complementary events of events  $E$  &  $F$  respectively & if  $0 < P(F) < 1$ , then  
 (A)  $P(E|F) + P(\bar{E}|\bar{F}) = 1$   
 (B)  $P(E|F) + P(E|\bar{F}) = 1$   
 (C)  $P(\bar{E}|F) + P(E|\bar{F}) = 1$   
 (D)  $P(E|\bar{F}) + P(\bar{E}|\bar{F}) = 1$
69. If  $E_1$  and  $E_2$  are two events such that  $P(E_1) = 1/4$ ,  $P(E_2/E_1) = 1/2$  and  $P(E_1/E_2) = 1/4$   
 (A) then  $E_1$  and  $E_2$  are independent  
 (B)  $E_1$  and  $E_2$  are exhaustive  
 (C)  $E_2$  is twice as likely to occur as  $E_1$   
 (D) Probabilities of the events  $E_1 \cap E_2$ ,  $E_1$  and  $E_2$  are in G.P.
70. A bag initially contains one red & two blue balls. An experiment consisting of selecting a ball at random, noting its colour & replacing it together with an additional ball of the same colour. If three such trials are made, then :  
 (A) probability that atleast one blue ball is drawn is 0.9  
 (B) probability that exactly one blue ball is drawn is 0.2  
 (C) probability that all the drawn balls are red given that all the drawn balls are of same colour is 0.2  
 (D) probability that atleast one red ball is drawn is 0.6 .

### Comprehension - 1

There are four boxes  $A_1, A_2, A_3$  and  $A_4$ . Box  $A_i$  has  $i$  cards and on each card a number is printed, the numbers are from 0 to  $i$ , A box is selected randomly,

the probability of selection of box  $A_i$  is  $\frac{i}{10}$  and then a card is drawn. Let  $E_i$  represents the event that a card with number 'i' is drawn.

71.  $P(E_1)$  is equal to  
 (A)  $1/5$  (B)  $1/10$   
 (C)  $2/5$  (D)  $1/4$
72.  $P(A_3/E_2)$  is equal to  
 (A)  $1/4$  (B)  $1/3$   
 (C)  $1/2$  (D)  $2/3$
73. Expectation of the number on the card is  
 (A) 2 (B) 2.5  
 (C) 3 (D) 3.5

### Comprehension - 2

A bag contains 6 balls out of which 3 are blue and 3 red. A sample of 4 balls is taken from this bag. Let  $x$  denote the number of red balls in the sample. Then  $x$  is a random variable. Let  $x$  takes the values  $x_1, x_2, x_3, \dots, x_n$  and  $p_1$  be the probability that  $x$  takes the value  $x_1$ . Then mean of the variable  $x$  is denoted by

$\bar{x}$  and  $\bar{x} = \sum_{i=1}^n p_i x_i$ . Further variance of the variable  $x$  is denoted by  $\sigma^2 = \sum_{i=1}^n p_i x_i^2$ .

74. Which of the following represents the correct probability distribution for  $x$

|     |     |        |     |     |        |
|-----|-----|--------|-----|-----|--------|
| (A) | $x$ | $P(x)$ | (B) | $x$ | $P(x)$ |
|     | 0   | 0      |     | 0   | 0      |
|     | 1   | 5/15   |     | 1   | 3/15   |
|     | 2   | 5/15   |     | 2   | 9/15   |
|     | 3   | 5/15   |     | 3   | 3/15   |

|     |     |        |     |     |        |
|-----|-----|--------|-----|-----|--------|
| (C) | $x$ | $P(x)$ | (D) | $x$ | $P(x)$ |
|     | 0   | 0      |     | 0   | 0      |
|     | 1   | 3/15   |     | 1   | 9/15   |
|     | 2   | 6/15   |     | 2   | 3/15   |
|     | 3   | 6/15   |     | 3   | 3/15   |

75. Mean of the given probability distribution of  $x$  is

- (A) 1 (B)  $\frac{3}{2}$   
(C) 2 (D)  $\frac{5}{2}$

76. Variance of the given probability distribution of  $x$  is

- (A)  $\frac{2}{5}$  (B)  $\frac{3}{5}$   
(C)  $\frac{4}{5}$  (D)  $\frac{1}{5}$

### Comprehension - 3

A box contains  $ab$  red balls, ' $2b$ ' white balls and ' $3b$ ' blue balls where  $b$  is a positive integer. 3 balls are selected at random from the box.

77. If balls are drawn without replacement and 'A' denotes the event that "No two of the selected balls have the same colour" then

- (A) There is no value of  $b$  for which  $P(A) = 0.3$   
(B) There is exactly one value of  $b$  for which  $P(A) = 0.3$  and this value is less than 10.  
(C) There is exactly one value of  $b$  for which  $P(A) = .3$  and this value is greater than 10.  
(D) There is more than one value of  $b$  for which  $P(A) = 0.3$

78. If balls are drawn without replacement and 'B' denotes the event that "No two of the 3 drawn balls are blue" then

- (A)  $P(B) = \frac{1}{3}$  if  $b = 1$   
(B)  $P(B) = \frac{2}{3}$  if  $b = 2$   
(C)  $P(B) = \frac{1}{4}$  if  $b = 4$   
(D)  $P(B) = \frac{1}{2}$  for all value of  $b$ .

79. The value of  $P(A/B)$  equals

- (A)  $\frac{3}{5}$  (B)  $\frac{3}{10}$   
(C)  $\frac{1}{2}$  (D)  $\frac{2}{3}$

### Comprehension - 4

A biased coin shows head with a probability of  $\frac{3}{4}$  and

tails with a probability of  $\frac{1}{4}$ . Let  $p_n$  denotes the probability that no three or more heads appear consecutively in ' $n$ ' throws of the coins. Answer the question that follow

80. If  $p_n = \alpha p_{n-1} + \beta p_{n-2} + \gamma p_{n-3}$  then  $64(\alpha + \beta + \gamma)$  is equal to

- (A) 37 (B) 23  
(C) 121 (D) 119

81. The value of  $p_4$  is

- (A)  $\frac{121}{256}$  (B)  $\frac{23}{64}$   
(C)  $\frac{37}{64}$  (D)  $\frac{119}{256}$

82. If the coin is tossed four times and let 'A' be the event that three or more heads occurs in four tosses and 'B' is the event that three heads do not occur in first three tosses, then  $P(A/B)$

- (A)  $\frac{27}{148}$  (B)  $\frac{4}{37}$   
(C)  $\frac{81}{148}$  (D) None of these

### Comprehension - 5

A JEE aspirant estimates that she will be successful with an 80 percent chance if she studies 10 hours per day, with 60 percent chance if she studies 7 hours per day and with a 40 percent chance if she studies 4 hours per day. She further believes that she will study 10 hours, 7 hours and 4 hours per day with probabilities 0.1, 0.2 and 0.7, respectively.

83. The chance she will be successful, is

- (A) 0.28 (B) 0.38  
(C) 0.48 (D) 0.58

84. Given that she is successful, the chance she studied for 4 hours, is

- (A)  $\frac{1}{2}$  (B)  $\frac{7}{12}$   
(C)  $\frac{2}{3}$  (D)  $\frac{1}{4}$

85. Given that she does not achieve success, the chance she studied for 4 hour, is

- (A)  $\frac{18}{26}$  (B)  $\frac{19}{26}$   
(C)  $\frac{20}{26}$  (D)  $\frac{21}{26}$

**Assertion (A) and Reason (R)**

- (A) Both A and R are true and R is the correct explanation of A.  
 (B) Both A and R are true but R is not the correct explanation of A.  
 (C) A is true, R is false.  
 (D) A is false, R is true.

**86. Assertion (A) :** If a is chosen randomly from the set {1, 2, 3, 4, 5} and b is chosen randomly from the set {6, 7, 8}, then the probability that  $\frac{a}{b}$  is as even number is  $\frac{21}{32}$ .

**Reason (R) :** The probability does not depend as the choice of b.

**87. Assertion (A) :** Alisha has four red socks and four blue socks. If she divides these socks into four pairs, then the probability that each pair will consist either of two red socks or two blue socks is  $\frac{3}{35}$ .

**Reason (R) :** The number of ways Alisha can divide the four red socks into two pairs is 3 and the total number of way she can divide the 8

$$\text{socks into 4 pair is } \frac{1 \cdot 8}{(1 \cdot 2)^4 \cdot 4}$$

**88. Assertion (A) :** At a nursery, 2008 babies sit in a circle. Suddenly each baby pokes the baby immediately to either its left or its right, with equal probability. The expected number of unpoked babies is 502.

**Reason (R) :** The probability that any given baby goes unpoked is  $\frac{1}{4}$ . So the answer is

$$\frac{2008}{4},$$

**89. Assertion (A) :** Two coins are simultaneously flipped until atleast one coin comes up head. The probability that the other coin also came up head on the last flip is  $\frac{1}{3}$ .

**Reason (R) :** The desired probability is

$$\frac{1}{4} + \frac{1}{4^2} + \frac{1}{4^3} + \dots = \frac{1}{3}.$$

**90. Assertion (A) :** If  $\frac{1+4P}{4}, \frac{1-P}{4}, \frac{1-2P}{4}$  are probabilities of three pair-wise mutually exclusive events, then the possible values of P belong to the set  $\left[-\frac{1}{4}, \frac{1}{2}\right]$ .

**Reason (R) :** If three events are pair wise mutually exclusive and exhaustive then sum of their probabilities is equal to 1.

**91. Assertion (A) :** One fair dice is rolled; let 'a' denote the number that comes up. We then roll 'a' dices; let the sum of the resulting 'a' numbers be 'b'. Finally, we roll 'b' dices, and let 'c' be the sum of the resulting 'b' numbers.

The expected value of c is  $\frac{343}{8}$ .

**Reason (R) :** The expected value of one dice roll is  $(1+2+3+\dots+6)/6 = 7/2$ . For any m rolls the expected value is  $7/2 \cdot m$ .

**92. Assertion (A) :** A coin has the property that when flipped four times, it has the same probability of turning up 2 heads and 2 tails as 3 heads and 1 tail. It means that the coin is biased.

**Reason (R) :** Let p be the probability of getting a head in one flip. Now  $6p^2(1-p)^2 = 4p^3(1-p)$  has solution p = 0, 1,  $\frac{3}{5}$ . Clearly p is not 0 or 1.

**93. Assertion (A) :** A point is chosen randomly in the interior of a circle of radius 1. Its expected distance from the centre of the circle is  $\frac{2}{3}$ .

**Reason (R) :** The probability of the point falling between a distance r and r + dr from the

centre is  $\frac{\pi[(r+dr)^2 - r^2]}{\pi} = 2rdr$ . The

expected distance is  $\int_0^1 r \cdot 2rdr = \frac{2}{3}$ .

94. A fair coin is tossed 3 times consider the events

A : first toss is head

B : second toss is head

C : exactly two consecutive heads or exactly two consecutive tails.

**Assertion (A)** : A, B, C are independent events.

**Reason (R)** : A, B, C are pairwise independent.

95. From a well shuffled pack of 52 playing cards a card is drawn at random. Two events A and B are defined as

A : Red card is drawn.

B : Card drawn is either a Diamond or Heart

**Assertion (A)** :  $P(A + B) = P(AB)$

**Reason (R)** :  $A \subseteq B$  and  $B \subseteq A$

### Match the Columns for JEE Advanced

96. Five identical pieces of paper have the digits 1 to 5 written on them. The experiment consists in choosing at random three pieces and arranging them in a row from left to right in the order of arrival.

#### Column-I

(A) If A = {the number 123 will appear} then  $P(A) =$

(B) If B = {a number will appear that contains no 3} then  $P(B) =$

(C) If C = {a number will appear that consists of consecutive digits} then  $P(C) =$

(D) If D = {an even number will appear} then  $P(D) =$

#### Column-II

(P) 1/20

(Q) 9/10

(R) 1/60

(S) 1/5

(T) 2/5

97. From an urn containing  $m_1 + m_2$  balls, where  $m_1$  is the number of white balls and  $m_2$  the number of black balls,  $m$  balls are chosen at random and without replacement ( $m \leq \min(m_1, m_2)$ ).

#### Column-I

(A) If A = {all the chosen balls are white} then  $P(A) =$

(B) If B = {there are precisely k white balls

among the chosen ones;  $k \leq m$ } then  $P(B) =$

(C) If C = {atleast one white ball is drawn out} then  $P(C) =$

(D) If D = {atleast k white ball are

drawn out;  $k \leq m$ } then  $P(D) =$

#### Column-II

(P)  $\frac{{}^{m_1}C_k {}^{m_2}C_{m-k}}{{}^{m_1+m_2}C_m}$

(Q)  $1 - \frac{{}^{m_2}C_m}{{}^{m_1+m_2}C_m}$

(R)  $\frac{{}^{m_1}C_m}{{}^{m_1+m_2}C_m}$

(S)  $1 - \frac{{}^{m_1}C_m}{{}^{m_1+m_2}C_m}$

(T)  $\frac{1}{{}^{m_1+m_2}C_m} \sum_{n=k}^m {}^{m_1}C_n {}^{m_2}C_{m-n}$

98. A device S consists of  $n$  units, each of which may end up in one of the following states as a result of its operation for the time  $\tau$  :

$s_1 = \{\text{the unit is in good order}\}$ ;  $s_2 = \{\text{the unit must be adjusted}\}$ ;

$s_3 = \{\text{the unit must be repaired}\}$ ;  $s_4 = \{\text{the unit is out of order}\}$ .

$P(s_1) = p_1, P(s_2) = p_2, P(s_3) = p_3, P(s_4) = p_4, p_1 + p_2 + p_3 + p_4 = 1$ .

The states of separate units are independent.

**Column-I**

- (A) If  $A = \{\text{all the units are in good order}\}$  then  $P(A) =$   
 (B) If  $B = \{\text{all the units must be adjusted}\}$  then  $P(B) =$   
 (C) If  $C = \{\text{one unit must be repaired and the other units must be adjusted}\}$  then  $P(C) =$   
 (D) If  $D = \{\text{atleast one unit is completely out of order}\}$  then  $P(D) =$

**Column-II**

- (P)  $np_3p_1^{n-1}$   
 (Q)  $1 - (1 - p_4)^n$   
 (R)  $(n - 1) p_2^2 p_1^{n-3}$   
 (S)  $p_2^n$   
 (T)  $p_1^n$

99. An elevator starts with 7 passengers and stops at 10 floors.

**Column-I**

- (A) If  $A = \{\text{No two passengers leave at the same floor}\}$  then  $P(A) =$   
 (B) If  $B = \{\text{All the passengers are discharged on the 5th floor}\}$  then  $P(B) =$   
 (C) If  $C = \{\text{All of the passengers are discharged on the same unspecified floor}\}$  then  $P(C) =$   
 (D) If  $D = \{\text{Exactly three of them are discharged on the 2nd floor}\}$  then  $P(D) =$

**Column-II**

- (P)  $(10)^{-7}$   
 (Q)  $(10)^{-6}$   
 (R)  $10.9.....5.4/(10)^7$   
 (S)  ${}^7C_3(7)^3/(10)^7$   
 (T)  ${}^7C_3(7)^4/(10)^7$

100. Over some time interval an amoeba may perish with probability  $1/4$ , survive with probability  $1/4$  or divide into two with probability  $1/2$ . Over the next time interval of the same duration, whatever its "origin" may be, every amoeba has the same things happen to it. At the end of the second interval match the number of amoebas with the probabilities :

**Column-I**

- (A) There may exist 1 amoeba with probability  
 (B) There may exist 2 amoebas with probability  
 (C) There may exist 3 amoebas with probability  
 (D) There may exist 4 amoebas with probability

**Column-II**

- (P)  $4/32$   
 (Q)  $5/32$   
 (R)  $7/32$   
 (S)  $9/32$   
 (T)  $11/32$

### Review Exercises for JEE Advanced

1. A group of four homogeneous objects is being tracked. Each of the objects can be either detected or not detected. We consider the following events;  
 $A = \{\text{exactly one of the four objects is detected}\}$ ;  
 $B = \{\text{atleast one object is detected}\}$ ;  
 $C = \{\text{not less than two objects are detected}\}$ ;

$D = \{\text{exactly two objects are detected}\}$ ;

$E = \{\text{exactly three objects are detected}\}$ ;

$F = \{\text{all four objects are detected}\}$ .

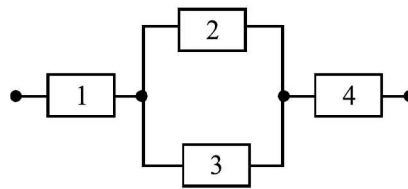
Find what the following events consist of:

- (i)  $A + B$  (ii)  $AB$   
 (iii)  $B + C$  (iv)  $BC$   
 (v)  $D + E + F$  (vi)  $BF$ .

Are the events  $BF$  and  $CF$  the same? Are the events  $BC$  and  $D$  the same?

2. Given below are experiments and the events that can occur in them. Indicate their complementary events.
  - (i) Two messages are sent over a communication channel : an event  $A = \{\text{both messages are sent correctly}\}$ .
  - (ii) A ball is drawn from an urn which contains two white, three black and four red balls; and event  $B = \{\text{B white ball is drawn}\}$ .
  - (iii) Five messages are sent; and event  $C = \{\text{not less than three messages are sent correctly}\}$ .
  - (iv)  $n$  shots are fired at a target ; an event  $D = \{\text{at least one hit is registered}\}$ .
  - (v) A preventive inspection of an instrument is made, the instrument consisting of  $k$  units, each of which can be either put in order at once or sent for repair. An event  $E = \{\text{none of the units is sent for repair}\}$ .
  - (vi) Two people play chess; and event  $F = \{\text{the whites win}\}$ .
3. To reduce the number of games,  $2n$  football teams, among which there are two prize-winners of the previous championship tournament, are split by means of a toss-up into two groups (the first and the second),  $n$  teams each. What is the probability,  $q_n$ , that both prize-winner teams will find themselves in different groups?
4. Ten variants of a test paper, each written on a separate card, are mixed and distributed randomly among seven students sitting in the same row, each receiving on variant. Find the probabilities of the following events :  $A = \{\text{variants 1 and 2 remain unused}\}$ ,  $B = \{\text{variants 1 and 2 are received by students sitting side by side}\}$ ,  $C = \{\text{consecutive number of the variants will be distributed}\}$ .
5. A draws a card from a pack of  $n$  cards marked  $1, 2, 3, \dots, n$ ; the card is replaced in the pack and then B draws a card. Find the chance that A draws (i) the same card as B ; (ii) a higher card than B ; (iii) a lower card than B. Verify that the sum of these chances is unity.
6. If 10 married couples are seated at random at a round table compute the probability that no wife sits next to her husband.
7. A 5-card hand is dealt from a well-shuffled deck of 52 playing cards. What is the probability that the hand contains atleast one card from each of the four suits?
8. If two squares are chosen at random on a chess board, show that
  - (i) the chance that they have a side common is  $\frac{1}{16}$  ;
  - (ii) the chance that they have contact at a corner is  $\frac{7}{144}$  .
9. Two dice are tossed at random. Find the probabilities of the following events :  $A = \{\text{the sum of the points on the two dice is even}\}$ ,  $B = \{\text{the product of the points is even}\}$ ,  $C = \{\text{the number of points is even on one die and odd on the other}\}$ ,  $D = \{\text{a six has turned up on neither of the dice}\}$ .
10. Seven apples, three oranges and five lemons are put at random in three bags, but it is stipulated that there should be an equal quantity of fruit in each bag. Find the probabilities of the following events :  $A = \{\text{there is one orange in each bag}\}$ ,  $B = \{\text{there are no oranges in a certain bag}\}$ .
11. The probability that a person stopping at a petrol pump, will ask to have its tyres checked is 0.12, the probability that he will ask to have its oil checked is 0.29 and the probability that he will ask to have them both checked is 0.07.
  - (a) What is the probability that a person stopping at the petrol pump will have :

- (i) Either his tyres or his oil checked?  
 (ii) Either his tyres or his oil checked but not both?
- (b) What is the probability that a person who has his tyres (oil) checked will also have his oil (tyres) checked?
12. Suppose that a person chooses a letter at random from R E S E R V E and then chooses one at random from V E R T I C A L. What is the probability the same letter is chosen?
13. The probability that a candidate attempts one particular question in an examination is  $\frac{9}{10}$  and, having done so, the probability of his or her success is  $\frac{2}{3}$ . What is the probability that the examiner will find atleast one correct solution in the first three scripts which he marks.
14. There are 20 tennis balls in a box, 15 new and five used ones. Two balls are chosen at random for a play and replaced after it. Two more balls are then chosen also at random for another play. What is the probability that new balls will be used in the second play?
15. Three shooters whose probabilities of hitting a target are constant under invariable conditions and equal to  $p_1 = 0.8$ ,  $p_2 = 0.7$ ,  $p_3 = 0.6$  respectively, each fire a single shot at the same target. Calculate the probability of event  $A = \{\text{there are exactly two holes found in the target}\}$ .
16. Each of three shooters fires a shot at three emerging targets choosing a target at random and independently of the other shooters. The probability of hitting on one shot is the same for all the shooters and equal to  $p$ . Evaluate the probability of the following events :  
 $A = \{\text{exactly one of the targets will have no holes}\}$ ,  
 $B = \{\text{each of the targets will have atleast one hole}\}$ ,
- $C = \{\text{all the three holes will happen to be in the same target}\}$ .
17. There are numbers are drawn from the set of numbers  $\{1, 2, \dots, N\}$  according to the scheme of random sampling without replacement. Find the probability that the third number will fall in the interval between the first two if it is known that the first number is less than the second number.
18. One card is drawn at random from a full pack of 52 playing cards. The events are as follows:  $A = \{\text{an ace}\}$ ,  $B = \{\text{a card of a black suit}\}$ ,  $F = \{\text{a picture-card (i.e., a jack, a queen, a king or an ace)}\}$ .  
 (i) Determine whether the following pairs of events are dependent or independent :  $A$  and  $B$ ,  $A$  and  $F$ ,  $F$  and  $B$ ;  
 (ii) calculate the probabilities of the events  $BF$ ,  $AF$  and  $ABF$  using the multiplication formula;  
 (iii) calculate the probabilities of the events  $B + F$ ,  $F - A$ ,  $F - AB$ .
19. A fighter attacks the enemy's fighting formation in which there are eight carriers and two noise suppliers that are similar in appearance. The fighter attacked four aircraft in succession, choosing each target at random. Find the probabilities of the following events :  $A = \{\text{most of the aircraft attacked are carriers}\}$ ,  $B = \{\text{all the noise suppliers were attacked}\}$ ,  $C = \{\text{the aircrafts of the two groups were attacked alternatively}\}$ ,  $D = \{\text{a noise suppliers was attacked atleast on the third turn}\}$ .
20. The reliability  $p_k$  of the  $k$ th elements is assumed to be known (correspondingly  $q_k = 1 - p_k$  is the probability of the failure of the  $k$ th element). A failure of any one of the elements results in a signal being interrupted in the branch of the circuit containing the element. Calculate the reliability  $p$  of the circuit.





21. A, B, C alternate firing at a target in this order. The probabilities that A, B, C strike the target are  $p_1, p_2, p_3$  respectively. Assuming Bernoulli trials, show that the chance that A will hit the target first is  $p_1/(1 - q_1q_2q_3)$  and the chance that B will hit the target first is  $q_1p_2/(1 - q_1q_2q_3)$ . [ $q = 1 - p$ ]
22. The destruction of warplane may occur either as a result of its both engines being damaged (events  $D_1$  and  $D_2$ ) or as a result of the pilot's cabin being hit (event  $K$ ). The plane is fired at many times by an anti-aircraft gun. Any hit in the corresponding unit causes it to be damaged. Find the probability of destroying the plane if  $P(D_1) = P(D_2) = p_1, P(K) = p_2$ .
23. A test is given to the students in the classroom of controlling machines. It consists of three problems. It is enough to solve two problems to get a positive mark. Each problem has five different answers encoded, of which only one is correct. John knows the material poorly, so he chooses his answers at random. What is the probability of him getting a positive mark ?
24. Let the probability that the weather on one day is of the same kind (rain or no rain) as the previous day be  $p$ . Let  $P_1$  be the probability of rain on the first day of the year. What is the probability  $P_n$  of rain on the  $n$ th day ?
25. A variable  $X$  has the following probability mass function :
- |          |   |     |      |      |      |       |        |            |
|----------|---|-----|------|------|------|-------|--------|------------|
| $x :$    | 0 | 1   | 2    | 3    | 4    | 5     | 6      | 7          |
| $f(x) :$ | 0 | $k$ | $2k$ | $2k$ | $3k$ | $k^2$ | $2k^2$ | $7k^2 + k$ |
- (a) Find  $k$  and evaluate  $P(X < 6), P(X^2 < 6)$  and  $P(1 < X^2 < 17)$ ,
- (b) Find the smallest value of  $x$  for which  $P(X \leq x) > 1/2$ .
26.  $n$  shells are fired independently at a fuel tank. Each shell hits the tank with probability  $p$ . If the tank is hit by one shell, then the fuel catches fire with probability  $p_1$ . Find the probability that the fuel will catch fire if  $n$  shells are fired.
27. A radar set receives a mixture of a desired signal and a noise disturbance with a 0.8 probability and a noise disturbance alone with a 0.2 probability. If it receives a desired signal with a noise disturbance it records the presence of some signal with a 0.7 probability, if a noise disturbance alone is received, it records some signal with a 0.3 probability. It is known that the set has recorded the presence of some signal. Find the probability of this set containing a desired signal.
28. In a room there are three chests, each chest contains two drawers, and each drawer contains one coin. In one chest each drawer contains a gold coin; in the second chest each drawer contains a silver coin; and in the last chest one drawer contains a gold coin and the other contains a silver coin. A chest is picked at random and then a drawer is picked at random from that chest. When the drawer is opened, it is found to contain a gold coin. What is the probability that the other drawer of that same chest will also contain a gold coin?
29. There are five urns and they are numbered 1 to 5. Each urn contains 10 balls. Urn  $i$  has  $i$  defective balls and  $10 - i$  non-defective balls,  $i = 1, 2, \dots, 5$ . An urn is chosen at random and a ball is selected from it. What is the probability that a defective ball is selected ? If the selected ball is defective, show that the probability that it came from urn 5 is  $1/3$ .
30. The probability  $P_k$  that a family has  $k$  children is given by  $P_k = (1 - 2p)/2^{k+1}, k \geq 2, P_0 = P_1 = p$ . It is known that the family has 2 boys. Show that
- (i)  $P\{\text{Family has only 2 children}\} = 27/64$ ,
  - (ii)  $P\{\text{Family has 2 girls as well}\} = 81/512$ .
31. Three prisoners A, B, C are told that any two of them are to be released on parole. A warder friend of prisoner A knows who are to be released. Prisoner A thinks of asking the warder for the name of one prisoner other than himself, who are to be released. He argues to himself that before he asks, his chances of release are  $2/3$ .

After knowing that B (say) shall be released, his own chances will fall to  $1/2$ , because either A & B or B & C are to be released. And so A decides not to reduce his chances by asking. Show that A is mistaken in his calculations.

32. On a straight line of length  $a$ , two points are taken at random. Find the chance that the distance between them is greater than  $b$ .
33. A domain is bounded by the curves  $x = 0$ ,  $x = \pi/2$ ,  $y = \sin x$ ,  $y = \sin a$ . A point is thrown into a rectangle whose sides are the interval  $[0, \pi/2]$  of the  $x$ -axis and the interval  $[0, 1]$  of the  $y$ -axis. For what value of  $a$  is the probability of the point falling into the domain is the least?
34. Two positive numbers  $x$  and  $y$  each of which does not exceed two are taken at random. Find the probability that  $xy \leq 1$ ,  $y/x \leq 2$ .
35. A parabola touches a semicircle and passes through the end point thrown into the semicircle at random will fall in the domain bounded by the arc of the semicircle at random will fall in the domain bounded by the arc of the semicircle and the parabola?



### Target Exercises for JEE Advanced

1. There are  $n$  tickets in a lottery, with  $m$  winning ones.  $k$  tickets are bought. What is the probability of the following events :  $A = \{\text{of } k \text{ tickets atleast one is a winning one}\}$ ,  $B = \{\text{of } k \text{ tickets exactly one is winning ticket}\}$ ,  $C = \{\text{of } k \text{ tickets exactly } k_1 \text{ tickets are winning ones}\}$ ?
2. The register of a calculator contains eight digits. Assuming that any number has equal chances to appear on the register, compute the probability of the following events :  $A = \{\text{there are zeros in all digit positions}\}$ ,  $B = \{\text{there are the same digits in all digits positions}\}$ ,  $C = \{\text{the register contains exactly two identical digits}\}$ ,  $D = \{\text{the register contains exactly two pairs of identical digits}\}$ ,  $E = \{\text{the register contains exactly three identical digits}\}$ ,  
 $F = \{\text{the register contains atmost three different digits}\}$ .
3. Four cars arrive at a cross-roads, one from each of the four directions. Each car is equally likely to perform one of the four maneuvers there : to turn around and go back, to go forward, to the left, or to the right. After some time all the cars have left the crossing. Find the probabilities of the following events :  $A = \{\text{all the cars will go along the same street}\}$ ,  $B = \{\text{three cars will go along the same street}\}$ ,  $C = \{\text{the cars will go, one along each of the four streets}\}$   $D = \{\text{no car will go along atleast one of the streets}\}$ .
4. When preparing for an examination in mathematics a student must be ready with the answers to 20 questions in the fundamentals of mathematical analysis and 25 questions in geometry. However, he had only time enough to prepare answers to 15 questions in the analysis and 20 question in geometry. An examination paper contains three questions, two of which are in analysis and one in geometry.  
 What is the probability:
  - (a) that the student will pass the exam with an excellent mark (will answer all three questions);
  - (b) with a good mark (will answer any two of the three questions)?
5. Each of a  $n$  sticks is broken at random into two pieces, a short and a long one. Then the resulting  $2n$  pieces are united at a random into  $n$  pairs, each forming a new stick. Find the probabilities of the following events:  $A = \{\text{all the pieces will be united in the original order}\}$ ,  $B = \{\text{all long pieces will become united with short ones}\}$ .
6. The experiment consists in distributing  $n$  different objects randomly among  $m$  people ( $m < n$ ), this being done in such a way that every one is likely to get any number of objects from those available. What is the probability of the

following events :  $A = \{\text{all objects will go to one of the participants}\}$ ,  $B = \{\text{a certain person will get no object}\}$ ,  $C = \{\text{certain } m_1 \text{ persons will get an object each}\}$ ,  $D = \{\text{certain } n_1 \text{ objects will go to one of the participants}\}$ ?

7. (i) Show that the probability that, in a group of  $r$  people, exactly one pair has the same birthday is

$$p_r = {}^r C_2 \frac{365 \cdot 364 \cdots (365 - r + 2)}{365^r}$$

- (ii) Show that  $p_r = {}^r C_2 \frac{q_r}{366 - r}$ , where  $p_r$  is defined in (i) and  $q_r$  is the probability that no pair has the same birthday.

8. Suppose that each of  $n$  sticks is broken into one long and one short part. The  $2n$  parts are then shuffled and arranged into  $n$  pairs from which new sticks are formed. Show that
- $P\{\text{all parts are joined in original order}\} = n! 2^n / (2n)!$ .
  - $P\{\text{all long parts are paired with short parts}\} = (n!)^2 2^n / (2n)!$ .
  - $P\{\text{atleast one of the original sticks is formed}\} = \sum (-1)^{k-1} {}^n C_k 2^k \cdot (2n - 2k)! / (2n)!$ .
9. Three fair cubical dice are thrown. Find the probability that :
- the sum of the scores is 18;
  - the sum of the scores is 5;
  - none of the three dice shows a 6;
  - the product of the scores is 90.
10. A box contains 4 balls numbered 0 through 3. One ball is chosen at random and kept aside. All balls with non-zero numbers less than that of the kept-aside ball are also removed from the box. Now a second ball is chosen at random from those remaining in the box. Find the probability that the 2nd ball chosen is numbered 3.
11. There are two good transistors and two defective ones in a box. These are tested one by one. What

is the probability that the second defective unit to be tested is the second unit tested? The third tested? The fourth tested?

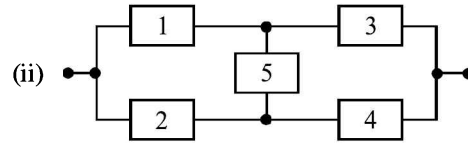
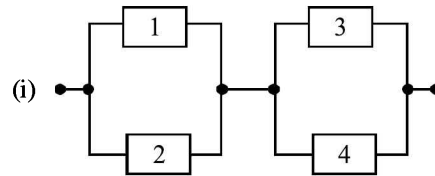
12. Given the following probabilities :

$$\begin{aligned} P(C) &= 0.40 & P(AB^c) &= 0.25 \\ P(A^c B) &= 0.30 & P(A^c C) &= 0.40 \\ P(A^c C) &= 0.15 & P(ABC) &= 0.10 \\ P(A^c B C^c) &= 0.20 \end{aligned}$$

Determine which, if any, of the events  $A$ ,  $B$ , and  $C$  are pairwise independent.

13. A man claims he can pick the winning horse in a race 20% of the time. In order to test his claim, he picks horses to win in 10 races. We may assume the results to be Bernoulli trials.
- What is the probability of his picking atleast 9 of the 10 races if his probability  $p$  of a correct choice on any race is 0.90?
  - Suppose the man is in effect guessing and that there are five horses in each race from which he chooses at random. What is the probability that he will pick as many as five races correctly?
14. A student knows 20 out of 25 questions of the syllabus. A student is considered to have passed the exam if he answers atleast three out of four questions in a question card. What is the probability of the student passing the exam?
15. A family who got a new flat installed  $2k$  new electric lamps. In the course of a year each lamp may burn out with probability  $r$ . Find the probability of an event  $A = \{\text{in the course of a year no less than half of the lamps will have to be replaced}\}$ .
16. An aircraft consists of three differently vulnerable parts : (i) the pilot's cabin and the engines, (ii) the fuel tanks and (iii) the airframe. It suffices to make one hit at the first part, two hits at the second part or three hits at the third part to destroy the aircraft. When one shell hits the aircraft it hits, with probability  $p_k$ , and independently of the others, the  $k$ th part

- ( $k = 1, 2, 3$ ). The aircraft was fired at. The events are:  $A_m = \{\text{the aircraft has been hit by } m \text{ shells}\}$ ,  $B = \{\text{the aircraft is destroyed}\}$ . Find the conditional probability  $P(B/A_m)$  for  $m = 2, 3, 4$ .
17.  $2n$  people are queued up at a theatre box office;  $n$  of them have only 5-rupee bill and the remaining  $n$  only one 10-rupee bill. There is no cash in the box office when it opens and each customer in turn is going to purchase a single 5-rupee ticket. Show that the probability that no customer will have to wait for change is  $1/(1 + n)$ .
18. The probability of hitting a target with a single shot equals  $p_1$ . How many independent shots must be made under invariable conditions in order to hit the target atleast once with probability not less than  $p_2$ ? Write the general expression for the least number of shots  $n(p_1, p_2)$  and find the following numerical values:  $n(0.3, 0.9)$ ,  $n(0.3, 0.95)$ .
19. How many times must a pair of dice be thrown for a sum of points equal to 12 to appear at least once with probability not less than 0.5?
20. An air target is being fired at with an anti-aircraft gun. Hits on individual shots are independent and have probabilities  $p$ . If a shell hits the target, it is destroyed with probability  $p_1$ . The ammunition of the gun is  $n$  shells. The firing is continued until the target is destroyed or the ammunition is spent to the last shell. Find the probabilities of the following events :  
 $A = \{\text{the ammunition will not all be spent}\}$ ,  
 $B = \{\text{atleast } k \text{ shells will remain unused}\}$ .
21. Piya and Mona each toss two dice in turn  $m$  times. She who is the first to get a sum of points equalling eight on both dice wins. Piya is the first to toss. Find the probabilities  $p_1$  and  $p_2$  of winning for each of the players and determine by what factor Piya's chances of winning are higher than Mona's if :  
 (i) the number of tosses is not limited and  $m = 1$ ;  
 (ii) the number of tosses is not limited but  $m > 1$ .
22. The reliability  $p_k$  of the  $k$ th elements is assumed to be known (correspondingly  $q_k = 1 - p_k$  is the probability of the failure of the  $k$ th element). A failure of any one of the elements results in a signal being interrupted in the branch of the circuit containing the element. Calculate the reliability  $p$  of each of the circuit.



23. The probability of an instrument failing after having been used  $k$  times is equal to  $p(k)$ . It is known that it did not fail during the first  $m$  times. What is the probability of its failing during the next  $n$  applications?
24. Show that the probability [denoted by  $P(r + 1)$ , say] of atleast  $(r + 1)$  successes in  $(n + 1)$  independent repeated Bernoulli trials, with probability  $p$  of success at each trial, is equal to
- $$(r + 1) \binom{n + 1}{r + 1} \int_0^p x^r (1 - x)^{n-r} dx$$
25. A pair of dice is rolled. Each die has the number 1 on two opposite faces, the number 2 on two opposite faces, and the number 3 on two opposite faces. The "roller" wins a dollar if  
 (i) the sum of four occurs on the first roll ;or  
 (ii) the sum of three or five occurs on the first roll and the same sum occurs on a subsequent roll before the sum of four occurs.

Otherwise he loses a dollar.

- (a) What is the probability that the person rolling the dice wins?
  - (b) What is the expected value of the game?
26. A bag contains 3 balls, and it is equally likely that 1, 2 or all of them are white. A ball is drawn and is found to be white.
- (i) What is the chance that this is the only white ball?
  - (ii) What is the chance that another drawing will give a white ball?
27. A bag contains 5 balls, and it is not known how many of these are white. Two balls are drawn, and these are white. What is the chance that all are white?
28. A box contains two dice similar in appearance and weight, one fair with equal probabilities of turning up for all the six digits on a random throw and one unfair with a nonuniform distribution of weight over the bulk. On a random throw of the unfair die the six turns up with probability  $1/3$ , the one does with probability  $1/9$ , the remaining digits turning up with equal probability. A die is drawn out of the box at random and tossed,  $k$  points turning up as a result. Find the probability  $p_k$  of the fair die having been tossed if :
- (a)  $k = 1$ , (b)  $k = 6$ , (c)  $k = 5$ .
29. A device consists of two series connected units. The reliability (the probability of no-failure operation during time  $T$ ) of the first unit is 0.9 and that of the second is 0.8. A failure of the device is recorded during time  $T$ . Find the probability of the following events :  $A_1 = \{\text{only the first unit has failed}\}$ ,  $A_2 = \{\text{both units have failed}\}$ .
30. The target being fired at consists of two parts differing in vulnerability. One hit at the first part or two hits at the second are sufficient to destroy the target. If a shell hits the target, then the probability of its having hit the first part is 0.4 and that of having hit the second part is 0.6. Three independent shots have been fired at the target, with the probability of hitting the target on each shot equal to 0.7 and with the destruction of the target as a result. What is the probability of the target having been destroyed only by hits at the second part?
31. Out of a group of 25 students to take an exam in probability theory ten have an excellent knowledge of the subject, seven a good knowledge, five a satisfactory one and three a poor one. The excellent students know the answers to all the 25 questions, the satisfactorily prepared students to 15 and the poorly prepared to only 10. A student called upon at random answered two of the questions he was asked. Find the probabilities of the following events :  $S_1 = \{\text{the student has an excellent or good knowledge of the subject}\}$ ,  $S_2 = \{\text{the student has a satisfactory knowledge of the subject}\}$ ,  $S_2 = \{\text{the student has a satisfactory knowledge}\}$ ,  $S_3 = \{\text{the student has a poor knowledge}\}$ .
32. A numerical text is transmitted through a communication channel which contains only three digits 1, 2, 3 that may appear in the text with equal probability. Due to the presence of noise every transmitted digit is received correctly with probability  $p$  and mistaken for some other digit with probability  $\frac{1}{2}(1-p)$ . It is assumed that the digits are distorted independently. Find the probability that 111 was transmitted if 123 is received.
33. Suppose a function  $f(x)$  given on the segment  $[0, 1]$  is such that,  $f'(x) > 0$  for  $x \in [0, 1]$ , with  $f(0) = 0$ ,  $f(1) = 1$ . Prove that when a point is thrown into a square whose sides are the

interval  $[0, 1]$  of the  $x$ -axis and the interval  $[0, 1]$  of the  $y$ -axis the greatest probability that it will fall in the domain bounded by the curves  $y = f(x)$ ,  $y = f(a)$ ,  $y = 0$ ,  $y = 1$  is attained for  $a = 1/2$ .

34. A point  $M(x, y)$  is chosen at random inside a square with vertices  $(0, 0)$ ,  $(1, 0)$ ,  $(1, 1)$  and  $(0, 1)$ . Find the probabilities of the following events:

$$A = \{(x, y) \mid x \times \max(x, y) < a, a > 0\},$$

$$B = \{(x, y) \mid \min(x, y) < a \leq 1\},$$

$$C = \{(x, y) \mid x^2 + y^2 \leq a^2, a > 0\},$$

$$D = \{(x, y) \mid xy < a, a > 0\}.$$

35. The values of  $a$  and  $b$  are equally possible in the square  $|a| \leq 1, |b| \leq 1$ . Find the probabilities of the following events :

$$A = \{\text{the roots of the quadratic trinomial } x^2 + 2ax + b \text{ are real}\},$$

$$B = \{\text{the roots of } x^2 + 2ax + b \text{ are positive}\}.$$

### Previous Years' Questions (JEE Advanced)

#### A. Fill in the blanks

1. For a biased die the probabilities for the different faces to turn up are given below :

| Face  | 1   | 2    | 3    | 4    | 5    | 6    |
|-------|-----|------|------|------|------|------|
| Prob. | 0.1 | 0.32 | 0.21 | 0.15 | 0.05 | 0.17 |

This die is tossed and you are told that either face 1 or face 2 has turned up. Then the probability that it is face 1 is.....[IIT - 1981]

2.  $P(A \cup B) = P(A \cap B)$  if and only if the relation between  $P(A)$  and  $P(B)$  is..... [IIT - 1985]
3. A box contains 100 tickets numbered 1, 2,.....,100. Two tickets are chosen at random. It is given that the maximum number on the two

chosen tickets is not more than 10. The minimum number on them is 5 with probability..... [IIT - 1985]

4. If  $\frac{1+3p}{3}$ ,  $\frac{1-p}{4}$  and  $\frac{1-2p}{2}$  are the probabilities of three mutually exclusive events, then the set of all values of  $p$  is..... [IIT - 1986]

5. Urn A contains 6 red and 4 black balls and urn B contains 4 red and 6 black balls. One ball is drawn at random from urn A and placed in urn B. Then one ball is drawn at random from urn B and placed in urn A. If one ball is now drawn at random from urn A. The probability that it is found to be red is..... [IIT - 1988]

6. A pair of fair dice is rolled together till a sum of either 5 or 7 is obtained. Then the probability that 5 comes before 7 is..... [IIT - 1989]

7. Let  $A$  &  $B$  be two events such that  $P(A) = 0.3$  &  $P(A \cup \bar{B}) = 0.8$ . If  $A$  &  $B$  are independent events then  $P(B) =$  ..... [IIT - 1990]

8. If the mean and the variance of a binomial variate  $X$  are 2 and 1 respectively, then the probability that  $X$  takes a value greater than one is equal to ..... [IIT - 1991]

9. Three faces of a fair die are yellow, two faces red & one blue. The die is tossed 3 times. The probability that the colours yellow, red & blue appear in the first, second & the third tosses respectively is ..... [IIT - 1992]

10. If two events  $A$  &  $B$  are such that  $P(A^c) = 0.3$ ,  $P(B) = 0.4$  &  $P(AB^c) = 0.5$  then  $P[B / (A \cup B^c)] =$  ..... [IIT - 1994]

11. Three numbers are chosen at random without replacement from  $\{1, 2, \dots, 10\}$ . The probability that the minimum of the chosen numbers is 3, or their maximum is 7, is..... [IIT - 1997]

**B. True/False**

12. If the letters of the word "Assassin" are written down at random in a row, the probability that no two S's occur together is  $1/35$ . [IIT - 1983]

13. If the probability for A to fail in an examination is 0.2 and that for B is 0.3, then the probability that either A or B fails is 0.5. [IIT - 1989]

**C. Multiple Choice Questions with ONE correct answer**

14. Two fair dice are tossed. Let  $x$  be the event that the first die shows an even number and  $y$  be the event that the second die shows an odd number. The two events  $x$  and  $y$  are : [IIT - 1979]

- (A) Mutually exclusive  
(B) Independent and mutually exclusive  
(C) Dependent  
(D) None of these

15. Two events A and B have probabilities 0.25 and 0.50 respectively. The probability that both A and B occur simultaneously is 0.14. Then the probability that neither A nor B occurs is

[IIT - 1980]

- (A) 0.39 (B) 0.25  
(C) 0.11 (D) None of these

16. The probability that an event A happens in one trial of an experiment is 0.4. Three independent trials of the experiment are performed. The probability that the event A happens at least once is

[IIT - 1980]

- (A) 0.936 (B) 0.784  
(C) 0.904 (D) None of these

17. If A and B are two events such that  $P(A) > 0$ , and

$P(B) \neq 1$ , then  $P\left(\frac{\bar{A}}{B}\right)$  is equal to [IIT - 1982]

- (A)  $1 - P\left(\frac{A}{B}\right)$  (B)  $1 - P\left(\frac{\bar{A}}{B}\right)$

(C)  $\frac{1 - P(A \cup B)}{P(\bar{B})}$  (D)  $\frac{P(\bar{A})}{P(\bar{B})}$

18. Fifteen coupons are numbered 1, 2, ..., 15, respectively. Seven coupons are selected at random one at a time with replacement. The probability that the largest number appearing on a selected coupon is 9, is [IIT - 1983]

(A)  $\left(\frac{9}{16}\right)^6$  (B)  $\left(\frac{8}{15}\right)^7$

(C)  $\left(\frac{3}{5}\right)^7$  (D) none of these

19. Three identical dice are rolled. The probability that the same number will appear on each of them is [IIT - 1984]

- (A)  $1/6$  (B)  $1/36$   
(C)  $1/18$  (D)  $3/28$

20. A box contains 24 identical balls of which 12 are white and 12 are black. The balls are drawn at random from the box one at a time with replacement. The probability that a white ball is drawn for the 4th time on the 7th draw is

[IIT - 1984]

- (A)  $5/64$  (B)  $27/32$   
(C)  $5/32$  (D)  $1/2$

21. A student appears for tests I, II and III. The student is successful if he passes either in test I and II or test I and III. The probabilities of the student passing in tests I, II and III are  $p$ ,  $q$  and  $1/2$  respectively. If the probability that student is successful is  $1/2$ , then [IIT - 1986]

- (A)  $p = q = 1$  (B)  $p = q = 1/2$   
(C)  $p = 1, q = 0$  (D) none of these

22. The probability that at least one of the events A and B occurs is 0.6. If A and B occur simultaneously with probability 0.2, then

$P(\bar{A}) + P(\bar{B})$  is [IIT - 1987]

- (A) 0.4 (B) 0.8  
(C) 1.2 (D) 1.4

23. One hundred identical coins, each with probability,  $p$ , of showing up heads are tossed once. If  $0 < p < 1$  and the probability of heads showing on 50 coins is equal to that of heads showing on 51 coins, then the value of  $p$  is [IIT - 1988]  
 (A)  $1/2$  (B)  $49/101$   
 (C)  $50/101$  (D)  $51/101$
24. India plays two match each with West-Indies & Australia. In any match the probability of India getting points 0, 1 & 2 are 0.45, 0.05 & 0.50 respectively. Assuming that the outcomes are independent, the probability of India getting at least 7 points is [IIT - 1992]  
 (A) 0.8750 (B) 0.0875  
 (C) 0.0625 (D) 0.0250
25. An unbiased dice with faces marked 1, 2, 3, 4, 5 and 6 is rolled four times. Out of four face values obtained, the probability that the minimum face value is not less than 2 & the maximum face value is not greater than 5 is [IIT - 1993]  
 (A)  $16/81$  (B)  $1/81$   
 (C)  $80/81$  (D)  $65/81$
26. Let A, B, C be three mutually independent events. Consider the two statements  $S_1$  and  $S_2$   
 $S_1$  : A and  $B \cup C$  are independent  
 $S_2$  : A and  $B \cap C$  are independent  
 Then [IIT - 1994]  
 (A) Both  $S_1$  and  $S_2$  are two  
 (B) Only  $S_1$  is true  
 (C) Only  $S_2$  is true  
 (D) Neither  $S_1$  nor  $S_2$  is true
27. The probability of India wining a test match against West Indies is  $1/2$ . Assuming independence from the match to match the probability that in a 5 match series India's second win occurs at the third test is - [IIT - 1995]  
 (A)  $1/8$  (B)  $1/4$   
 (C)  $1/2$  (D)  $2/3$
28. Three of the six vertices of a regular hexagon are chosen at random. The probability that the triangle with these three vertices is equilateral, equals [IIT - 1995]  
 (A)  $1/2$  (B)  $1/5$   
 (C)  $1/10$  (D)  $1/20$
29. For the three events A, B & C,  $P$  (exactly one of the events A or B occurs) =  $P$  (exactly one of the events B or C occurs) =  $P$  (exactly one of the events C or A occurs) =  $p$  &  $P$  (all the three events occur simultaneously) =  $p^2$ , where  $0 < p < 1/2$ . Then the probability of at least one of the three events A, B & C occurring is [IIT - 1996]  
 (A)  $\frac{3p + 2p^2}{2}$  (B)  $\frac{p + 3p^2}{4}$   
 (C)  $\frac{p + 3p^2}{2}$  (D)  $\frac{3p + 2p^2}{4}$
30. If from each of the 3 boxes containing 3 white & 1 black 2 white & 2 black, 1 white & 3 black balls, one ball is drawn at random, then the probability that 2 white & 1 black ball will be drawn is [IIT - 1998]  
 (A)  $13/32$  (B)  $1/14$   
 (C)  $1/32$  (D)  $3/16$
31. There are 4 machines & it is known that exactly 2 of them are faulty. They are tested, one by one, in a random order till both the faulty machines are identified. Then the probability that only 2 tests are needed is [IIT - 1998]  
 (A)  $1/3$  (B)  $1/6$   
 (C)  $1/2$  (D)  $1/4$
32. If E & F are events with  $P(E) \leq P(F)$  &  $P(E \cap F) > 0$ , then - [IIT - 1998]  
 (A) occurrence of E  $\Rightarrow$  occurrence of F  
 (B) occurrence of F  $\Rightarrow$  occurrence of E  
 (C) non-occurrence of E  $\Rightarrow$  non occurrence of F  
 (D) none of the above implications holds



33. A fair coin is tossed repeatedly. If tail appears on first four tosses, then the probability of head appearing on fifth toss equals - [IIT - 1998]  
 (A)  $1/2$  (B)  $1/32$   
 (C)  $31/32$  (D)  $1/5$
34. 7 white balls & 3 black balls are randomly placed in a row. The probability that no two black balls are placed adjacently equals [IIT - 1998]  
 (A)  $1/2$  (B)  $7/15$   
 (C)  $2/15$  (D)  $1/3$
35. If the integers  $m$  and  $n$  are chosen at random between 1 and 100, then the probability that a number of the form  $7^m + 7^n$  is divisible by 5 equals [IIT - 1999]  
 (A)  $1/4$  (B)  $1/7$   
 (C)  $1/8$  (D)  $1/49$
36. Two numbers are selected randomly from the set  $S = \{1, 2, 3, 4, 5, 6\}$  without replacement one by one. The probability that minimum of the two numbers is less than 4 is [IIT - 2003]  
 (A)  $1/15$  (B)  $14/15$   
 (C)  $1/5$  (D)  $4/5$
37. If  $P(B) = 3/4$ ,  $P(A \cap B \cap \bar{C}) = 1/3$  and  $P(\bar{A} \cap B \cap \bar{C}) = 1/3$ , then  $P(B \cap C)$  is [IIT - 2003]  
 (A)  $1/12$  (B)  $1/6$   
 (C)  $1/15$  (D)  $1/9$
38. If three distinct numbers are chosen randomly from the first 100 natural numbers, then the probability that all three of them are divisible by both 2 and 3 is [IIT - 2004]  
 (A)  $4/25$  (B)  $4/35$   
 (C)  $4/33$  (D)  $4/1155$
39. A six faced fair dice is thrown until 1 comes, then the probability that 1 comes in even no. of trials is [IIT - 2005]  
 (A)  $5/11$  (B)  $5/6$   
 (C)  $6/11$  (D)  $1/6$
40. One Indian and four American men and the their wives are to be seated randomly around a circular table. Then the conditional probability that the Indian man is seated adjacent to his wife given that each American man is seated adjacent to his wife is [IIT - 2007]  
 (A)  $1/2$  (B)  $1/3$   
 (C)  $2/5$  (D)  $1/5$
41. An experiment has 10 equally likely outcomes. Let A and B be two non-empty events of the experiment. If A consists of 4 outcomes, the number of outcomes that B must have so that A and B are independent, is [IIT - 2008]  
 (A) 2, 4 or 8 (B) 3, 6 or 9  
 (C) 4 or 8 (D) 5, or 10
42. A signal which can be green or red with probability  $\frac{4}{5}$  and  $\frac{1}{5}$  respectively, is received by station A and then transmitted to station B. The probability of each station receiving the signal correctly is  $\frac{3}{4}$ . If the signal received at station B is green, then the probability that the original signal was green is [IIT - 2010]  
 (A)  $\frac{3}{5}$  (B)  $\frac{6}{7}$   
 (C)  $\frac{20}{23}$  (D)  $\frac{9}{20}$
43. Let  $\omega$  be a complex cube root of unity with  $\omega \neq 1$ . A fair die is thrown three times. If  $r_1, r_2$  and  $r_3$  are the numbers obtained on the die, then the probability that  $\omega^{r_1} + \omega^{r_2} + \omega^{r_3} = 0$  is [IIT - 2010]  
 (A)  $1/18$  (B)  $1/9$   
 (C)  $2/9$  (D)  $1/36$
44. Four fair dice  $D_1, D_2, D_3$  and  $D_4$ , each having six faces numbered 1, 2, 3, 4, 5 and 6 are rolled simultaneously. The probability that  $D_4$  shows a number appearing on one of  $D_1, D_2$  and  $D_3$  is [IIT 2012]

- (A)  $\frac{91}{216}$  (B)  $\frac{125}{216}$   
 (C)  $\frac{108}{216}$  (D)  $\frac{127}{216}$

**D. Multiple Choice Questions with ONE or MORE THAN ONE correct answers**

45. If M and N are any two events, the probability that exactly one of them occurs is

[IIT - 1984]

- (A)  $P(M) + P(N) - 2P(M \cap N)$   
 (B)  $P(M) + P(N) - P(M \cap N)$   
 (C)  $P(M^c) + P(N^c) - 2P(M^c \cap N^c)$   
 (D)  $P(M \cap N^c) + P(M^c \cap N)$

46. For two given events A and B,  $P(A \cap B)$

[IIT - 1988]

- (A) not less than  $P(A) + P(B) - 1$   
 (B) not greater than  $P(A) + P(B)$   
 (C) equal to  $P(A) + P(B) - P(A \cup B)$   
 (D) equal to  $P(A) + P(B) + P(A \cup B)$

47. If E and F is independent events such that  $0 < 1$  and  $0 < P(F) < 1$ , then [IIT - 1989]

- (A) E and F are mutually exclusive  
 (B) E and  $F^c$  (the complement of the event F) are independent  
 (C)  $E^c$  and  $F^c$  are independent  
 (D)  $P(E | F) + P(E^c | F) = 1$

48. For any 2 events A & B in a sample space - [IIT - 1991]

- (A)  $P(A/B) \geq \frac{P(A) + P(B) - 1}{P(B)}$ , is always true  
 (B)  $P(A \cap \bar{B}) = P(A) - P(A \cap B)$ , does not hold  
 (C)  $P(A \cup B) = 1 - P(\bar{A}) \cdot P(\bar{B})$ , if A & B are independent  
 (D)  $P(A \cup B) = 1 - P(\bar{A}) \cdot P(\bar{B})$ , if A & B are disjoint

49. E and F are two independent events. The probability that both E & F happen is  $1/12$  & the probability that neither E nor F happens is  $1/2$ . Then [IIT - 1993]

- (A)  $P(E) = 1/3, P(F) = 1/4$   
 (B)  $P(E) = 1/2, P(F) = 1/6$   
 (C)  $P(E) = 1/6, P(F) = 1/2$   
 (D)  $P(E) = 1/4, P(F) = 1/3$

50. Let  $0 < P(A) < 1, 0 < P(B) < 1$  &  $P(A \cup B) = P(A) + P(B) - P(A) \cdot P(B)$ , then [IIT - 1995]

- (A)  $P(B/A) = P(B) - P(A)$   
 (B)  $P(A^c \cup B^c) = P(A^c) + P(B^c)$   
 (C)  $P((A \cup B)^c) = P(A^c) \cdot P(B^c)$   
 (D)  $P(A/B) = P(A)$

51. If  $\bar{E}$  &  $\bar{F}$  are the complementary events E & F respectively & if  $0 < P(F) < 1$ , then [IIT - 1998]

- (A)  $P(E | F) + P(\bar{E} | F) = 1$   
 (B)  $P(E | F) + P(E | \bar{F}) = 1$   
 (C)  $P(\bar{E} | F) + P(E | \bar{F}) = 1$   
 (D)  $P(E | \bar{F}) + P(\bar{E} | \bar{F}) = 1$

52. The probability that a student passes in Mathematics, Physics and Chemistry are m, p and c respectively. Of these subjects, the student has a 75% chance of passing in at least one, a 50% chance of passing in at least two and a 40% chance of passing in exactly two, which of the following are true - [IIT - 1999]

- (A)  $p + m + c = 19/20$   
 (B)  $p + m + c = 27/20$   
 (C)  $pmc = 1/10$   
 (D)  $pmc = 1/4$

53. Let E and F be two independent events. The probability that exactly one of the occurs is  $\frac{11}{25}$  and the probability of none of them occurring is  $\frac{2}{25}$ . If P(T) denotes the probability of occurrence of the event T, then

- (A)  $P(E) = \frac{4}{5}, P(F) = \frac{3}{5}$   
 (B)  $P(E) = \frac{2}{5}, P(F) = \frac{2}{5}$

(C)  $P(E) = \frac{2}{5}, P(F) = \frac{1}{5}$

(D)  $P(E) = \frac{3}{5}, P(F) = \frac{4}{5}$  [IIT - 2011]

54. A ship is fitted with three engines  $E_1, E_2$  and  $E_3$ . The engines function independently of each

other with respective probabilities  $\frac{1}{2}, \frac{1}{4}$  and  $\frac{1}{4}$ .

For the ship to be operational at least two of its engines must function. Let  $X$  denote the event that the ship is operational and let  $X_1, X_2$  and  $X_3$  denote respectively the events that the engines  $E_1, E_2$  and  $E_3$  are functioning. Which of the following is (are) true ?

(A)  $P[X_1^0 / X] = \frac{3}{16}$

(B)  $P[\text{Exactly two engines of the ship are functioning} / X] = \frac{7}{8}$

(C)  $P[X / X_2] = \frac{5}{16}$

(D)  $P[X / X_1] = \frac{7}{16}$  [IIT - 2012]

55. Four fair dice  $D_1, D_2, D_3$  and  $D_4$  each having six faces numbered 1, 2, 3, 4, 5 and 6 are rolled simultaneously. The probability that  $D_4$  shows a number appearing on one of  $D_1, D_2$  and  $D_3$  is [IIT - 2012]

(A)  $\frac{91}{216}$  (B)  $\frac{108}{216}$

(C)  $\frac{125}{216}$  (D)  $\frac{127}{216}$

56. Let  $X$  and  $Y$  be two events such that  $P(X/Y) = \frac{1}{2}$ ,

$P(Y/X) = \frac{1}{3}$  and  $P(X \cap Y) = \frac{1}{6}$ . Which of the following is (are) correct ? [IIT - 2012]

(A)  $P(X \cap Y) = \frac{2}{3}$

(B)  $X$  and  $Y$  are independent

(C)  $X$  and  $Y$  are not independent

(D)  $P(X^c \cap Y) = \frac{1}{3}$

### E. Subjective Problems

57. Balls are drawn one-by one without replacement from a box containing 2 black, 4 white and 3 red balls till all the balls are drawn. Find the probability that the balls drawn are in the order 2 black, 4 white and 3 red. [IIT - 1978]

58. Six boys and six girls sit in a row randomly. Find the probability that [IIT - 1979]  
(i) the six girls sit together  
(ii) the boys and girls sit alternately.

59. An anti-aircraft gun can take a maximum of four shots at an enemy plane moving away from it. The probabilities of hitting the plane at the first, second, third and fourth shot are 0.4, 0.3, 0.2 and 0.1 respectively. What is the probability that the gun hits the plane? [IIT - 1981]

60. A and B are two candidates seeking admission in IIT. The probability that A is selected is 0.5 and the probability that both A and B are selected is at most 0.3. Is it possible that the probability of B getting selected is 0.9? [IIT - 1982]

61. Cards are drawn one by one at random from a well-shuffled full pack of 52 playing cards until 2 aces are obtained for the first time. If  $N$  is the number of cards required to be drawn, then show that

$$P\{N=n\} = \frac{(n-1)(52-n)(51-n)}{50 \times 49 \times 17 \times 13}$$

where  $2 \leq n \leq 50$  [IIT - 1983]

62. A, B, C are events such that  
 $P_r(A) = 0.3, P_r(B) = 0.4, P_r(C) = 0.8$   
 $P_r(AB) = 0.08, P_r(AC) = 0.28; P_r(ABC) = 0.09$   
If  $P_r(A \cup B \cup C) \geq 0.75$ ,

then show that  $P_r(BC)$  lies in the interval  $0.23 \leq x \leq 0.48$  [IIT - 1983]

63. A and B are two independent events. This probability that both A and B occur is  $1/6$  and the probability that neither of them occurs is  $1/3$ . Find the probability of the occurrence of A. [IIT - 1984]
64. In a certain city only two newspapers A and B are published, it is known that 25% of the city population reads A and 20% reads B while 8% reads both A and B. It is also known that 30% of those who read A but not B look into advertisements and 40% of those who read B but not A look into advertisement while 50% of those whose read both A and B look into advertisements. What is the percentage of the population that reads an advertisement? [IIT - 1984]
65. In a multiple-choice question there are four alternative answers, of which one or more are correct. A candidate will get marks in the question only if he ticks the correct answers. The candidate decides to tick the answers at random, if he is allowed upto three chances to answer the questions, find the probability that he will get marks in the questions. [IIT - 1985]
66. A lot contains 20 articles. the probability that the lot contains exactly 2 defective articles is 0.6. Articles are drawn from the lot at random one by one without replacement and are tested till all defective articles are found. What is the probability that the testing procedure ends at the twelfth testing. [IIT - 1986]
67. A man takes a step forward with probability 0.4 and backwards with probability 0.6. Find the probability that at the end of eleven steps he is one step away from the starting point. [IIT - 1987]
68. An urn contains 2 white and 2 black balls. A ball is drawn at random. If it is white it is not replaced into the urn. Otherwise It is replaced along with another ball of the same colour. The process is repeated. Find the probability that the third ball drawn is black. [IIT - 1987]
69. A box contains 2 fifty paise coins, 5 twenty five paise coins and a certain fixed number  $N$  ( $\geq 2$ ) of ten and five paise coins. Five coins are taken out of the box at random. Find the probability that the total value of these 5 coins is less than one rupee and fifty paise. [IIT - 1988]
70. Suppose the probability for A to win a game against B is 0.4. If A has an option of playing either a "best of 3 games" or a "best of 5 games" much against B, which option should be choose so that the probability of his winning the match is higher? (No game ends in a draw). [IIT - 1989]
71. A is a set containing  $n$  elements. A subset  $P$  of  $A$  is chosen at random. The set  $A$  is reconstructed by replacing the elements of  $P$ . A subset  $Q$  of  $A$  is again chosen at random. Find the probability that  $P$  and  $Q$  have no common elements. [IIT - 1990]
72. In a test an examines either guesses or copies or knows the answer to a multiple choice question with 4 choices. The probability that he makes a guess is  $1/3$  & the probability that he copies the answer is  $1/6$ . The probability that his answer is correct given that he copied it, is  $1/8$ . Find the probability that he knew the answer to the question given that he correctly answered it. [IIT- 1991]
73. A lot contains 50 defective and 50 non defective bulbs. Two bulbs are drawn at random, one at a time, with replacement. Then events A, B, C are defined as

A = (the first bulb is defective)  
 B = (the second bulb is nondefective)  
 C = (the two bulbs are both defective or both non defective)

Determine whether

- (i) A, B, C are pairwise independent  
 (ii) A, B, C are independent. [IIT - 1992]

74. Numbers are selected at random, one at a time, from the two digit numbers 00, 01, 02,..... 99 with replacement. An event E occurs if and only if the product of the two digits of a selected number is 18. If four number are selected, find the probability that the event E occurs at least 3 times. [IIT - 1993]
75. An unbiased coin is tossed. If the result is a head, a pair of unbiased dice is rolled & the number of obtained by adding the numbers on the two faces is noted. If the result is a tail a card from a well shuffled pack of eleven cards numbered 2, 3, 4,....., 12 is picked & the number on the card is noted. What is the probability that the noted number is either 7 or 8? [IIT - 1994]
76. Three groups A, B & C are competing for positions are the Board of Directors of a company. The probabilities of their winning 0.5, 0.3, 0.2 respectively. If the group A wins, the probability of introducing a new product is 0.7 and the corresponding probabilities for groups B & C are 0.6 & 0.5 respectively. Find the probability that the new product will be introduced. [IIT - 1994]
77. In how many ways 3 girls & 9 boys can be seated in two vans, each having numbered seat, 3 in the front & 4 at the back? How many seating arrangements are possible if 3 girls should sit together in a back row on adjacent seats? Now, if all the seating arrangements are equally likely, what is the probability of 3 girls sitting together in a back row on adjacent seats? [IIT - 1996]
78. Sixteen players  $S_1, S_2, \dots, S_{16}$  play in a tournament. They are divided into eight pairs at random. From each pair a winner is decided on the basis of a game played between the two players of the pair. Assume that all the players are of equal strength.
- (a) Find the probability that the player  $S_1$  is among the eight winners.
- (b) Find the probability that exactly one of the two players  $S_1$  and  $S_2$  is among the eight winners. [IIT - 1997]
79. If  $p$  &  $q$  are chosen randomly from the set  $\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$  with replacement. Determine the probability that the roots of the equation  $x^2 + px + q = 0$  are real. [IIT - 1997]
80. 3 players A, B and C toss a coin cyclically in the order (that is A, B, C, A, B, C, A, B ..... ) till a head shows. Let  $p$  be the probability that the coin show a head. Let  $\alpha, \beta$  &  $\gamma$  be respectively the probability that A, B and C gets the first head. Prove that  $\beta = (1 - p) \alpha$ . Determine  $\alpha, \beta$  &  $\gamma$  (in terms of  $p$ ) [IIT - 1998]
81. Eight players  $P_1, P_2, P_3, \dots, P_8$  play a knock-out tournament. It is known that when the players  $P_i$  and  $P_j$  play, the player  $P_i$  will win if  $i < j$ . Assuming that the players are paired at random in each round, what is the probability that the players  $P_4$  reaches to final. [IIT - 1999]
82. A and B are two independent events. The probability that both occur simultaneously is  $1/6$  and the probability that neither occurs is  $1/3$ . Find the probabilities of occurrence the events A and B separately. [IIT - 2000]
83. A coin has probability ' $p$ ' of showing head when tossed. It is tossed ' $n$ ' times. Let  $p_n$  denote the probability that no two (or more) consecutive heads occur. Prove that  $p_1=1, p_2=1-p^2$  &  $p_n=(1-p)p_{n-1}+p(1-p)p_{n-2}$  for all  $n \geq 3$ . [IIT - 2000]
84. An urn contains  $m$  white and  $n$  black balls. A ball is drawn at random and is put back into the urn along with  $k$  additional balls of the same colour as that of the ball drawn. A ball is again drawn at random. What is the probability that the ball drawn now is white? [IIT - 2001]

85. An unbiased die, with faces numbered, 1, 2, 3, 4, 5, 6 is thrown  $n$  times and the list of  $n$  numbers showing up is noted. What is the probability that, among the number 1, 2, 3, 4, 5, 6, only three numbers appear in this list? [IIT - 2001]
86. A box contains  $N$  coins,  $m$  of which are fair and the rest are biased. The probability of getting a head when a fair coin is tossed is  $1/2$ , while it is  $2/3$  when a biased coin is tossed. A coin is drawn from the box at random and is tossed twice. The first time it shows head and the second time it shows tail. What is the probability that the coin drawn is fair? [IIT - 2002]
87. For a student to qualify, he must pass at least two out of three exams. The probability that he will pass the 1st exam is  $P$ . If he fails in one of the exams then the probability of his passing in the next exam is  $P/2$  otherwise it remains the same. Find the probability that he will qualify. [IIT - 2003]
88. A is targeting to B, B and C are targeting to A. Probability of hitting the target by A, B and C are  $2/3$ ,  $1/2$  and  $1/3$  respectively. If A is hit then find the probability that B hits the target and C does not. [IIT - 2003]
89. A and B are two independent events and C be an event that exactly one of A & B occurs, prove that  $P(A \cup B) \cdot P(A' \cap B') \leq P(C)$ . [IIT - 2004]
90. A box contains 18 balls, 12 red and 6 white. Six balls are drawn one by one without replacement of which at least 4 are white. Find the probability that in the next two draws exactly one white ball is drawn. (Leave the answer in terms of  ${}^nC_r$ ) [IIT - 2004]
91. A person goes to office either by car, scooter, bus or train, the probability of which being  $\frac{1}{7}$ ,  $\frac{3}{7}$ ,  $\frac{2}{7}$  and  $\frac{11}{7}$  respectively. Probability that he reaches office late, if he takes car, scooter, bus or train is  $\frac{2}{9}$ ,  $\frac{1}{9}$ ,  $\frac{4}{9}$  and  $\frac{1}{9}$  respectively. Given

that he reached office in time, then what is the probability that he reached office in time, then what is the probability that he travelled by a car. [IIT - 2005]

### F. Comprehension Based Questions

Read the passage given below and answer the questions that follows.

#### PASSAGE

There are  $n$  urns, each of these contain  $n + 1$  balls. The  $i$ th urn contains  $i$  white ball and  $(n + 1 - i)$  red balls. Let  $u_i$  be the event of selecting  $i$ th urn,  $i = 1, 2, 3, \dots, n$  and  $w$  the event of getting a white balls.

92. If  $P(u_i) \propto i$ , where  $i = 1, 2, 3, \dots, n$ , then  $\lim_{n \rightarrow \infty} P(w) =$  [IIT - 2006]  
 (A) 1 (B)  $2/3$   
 (C)  $3/4$  (D)  $1/4$

93. If  $P(u_i) = c$ , (a constant) then  $P(u_n / w) =$  [IIT - 2006]  
 (A)  $\frac{2}{n+1}$  (B)  $\frac{1}{n+1}$   
 (C)  $\frac{n}{n+1}$  (D)  $\frac{1}{2}$

94. Let  $P(u_i) = \frac{1}{n}$ , if  $n$  is even and  $E$  denotes the event of choosing even numbered urn, then the value of  $P(w/E)$  is [IIT - 2006]

- (A)  $\frac{n+2}{2n+1}$  (B)  $\frac{n+2}{2(n+1)}$   
 (C)  $\frac{n}{n+1}$  (D)  $\frac{1}{n+1}$

#### PASSAGE

A fair die is tossed repeatedly until a six is obtained. Let  $X$  denote the number of tosses required

95. The probability that  $X = 3$  equals [IIT - 2009]  
 (A)  $\frac{25}{216}$  (B)  $\frac{25}{36}$   
 (C)  $\frac{5}{36}$  (D)  $\frac{125}{216}$

96. The probability that  $X \geq 3$  equals [IIT - 2009]

- (A)  $\frac{125}{216}$  (B)  $\frac{25}{36}$   
 (C)  $\frac{5}{36}$  (D)  $\frac{25}{216}$

97. The conditional probability that  $X \geq 6$  given  $X \geq 3$  equals [IIT - 2009]

- (A) 48 (B) 32  
 (C) 40 (D) 80

**Paragraph for question Nos. 98 and 99**

Let  $U_1$  and  $U_2$  be two urns such that  $U_1$  contains 3 white and 2 red balls, and  $U_2$  contains only 1 white ball. A fair coin is tossed. If head appears then 1 ball is drawn at random from  $U_1$  and put into  $U_2$ . However, if tail appears then 2 balls are drawn at random from  $U_1$  and put into  $U_2$ . Now 1 ball is drawn at random from  $U_2$ .

98. The probability of the drawn ball from  $U_2$  being white is [IIT - 2012]

- (A)  $\frac{13}{30}$  (B)  $\frac{23}{30}$   
 (C)  $\frac{19}{30}$  (D)  $\frac{11}{30}$

99. Given that the drawn ball from  $U_2$  is white, the probability that head appeared on the coin is [IIT - 2012]

- (A)  $\frac{17}{23}$  (B)  $\frac{11}{23}$   
 (C)  $\frac{15}{23}$  (D)  $\frac{12}{23}$

**G. Assertion (A) & Reason (R)**

100. Let  $H_1, H_2, \dots, H_n$  be mutually exclusive and exhaustive events with  $P(H_i) > 0, i = 1, 2, \dots, n$ . Let  $E$  be any other event with  $0 < P(E) < 1$ .  
**Assertion (A) :**  $P(H_i | E) > P(E | H_i), P(H_i)$  for  $i = 1, 2, \dots, n$  because

**Reason (R) :**  $\sum_{i=1}^n P(H_i) = 1$ . [IIT - 2007]

101. Consider the system of equations

$ax + by = 0, cx + dy = 0$ , where  $a, b, c, d \in \{0, 1\}$ .

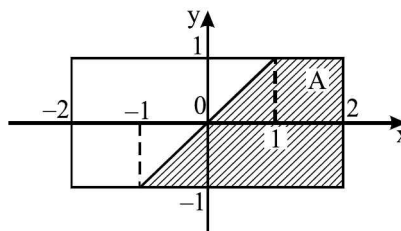
**Assertion (A) :** The probability that the system of equations has a unique solution is  $\frac{3}{8}$ .

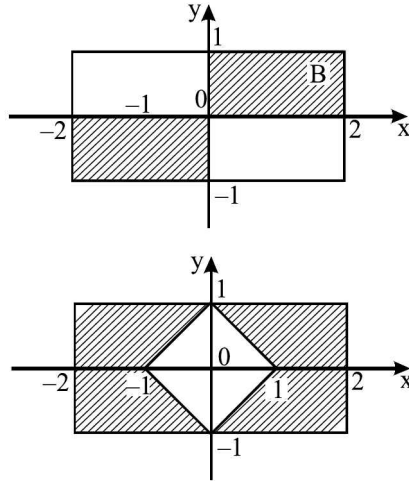
**Reason (R) :** The probability that the system of equations has a solution is 1. [IIT - 2008]

## ANSWERS

### Practice Problems—A

- $S = \{(k, m) \mid 1 \leq k, m \leq 6\}$ ,  
 $A = \{(3, 3), (6, 3), (3, 6), (6, 6)\}$ ,  
 $B = \{(k, m) \mid 1 \leq k, m \leq 5\}$ ,  
 $C = \{(k, m) \mid 4 \leq k, m \leq 6\}$ ,  
 $D = \{(k, k) \mid 1 \leq k \leq 6\}$ .
- $S = \{(hhh), (thh), (hth), (hht), (htt), (tth), (tth), (ttt)\}$ ,  
 $A = \{(htt), (tth), (tth)\}$ ,  
 $B = \{(hhh)\}$ ,  
 $C = \{(hhh), (thh), (hth), (hht)\}$ ,  
 $D = \{(hhh), (thh), (hht)\}$
- $S = \{n \mid n \in \mathbb{N}\}$ ,  
 $A = \{3\}, B = \{n \mid n = 3, 4, \dots\} = S - \{1, 2\}$
- $S = \{(x, y) \mid -2 \leq x \leq 2, -1 \leq y \leq 1\}$ .  
 The sets of corresponding to the indicated events are represented in figure.





### Practice Problems—B

- (i) Yes (ii) No
- (i) No (ii) Yes
- (i) No (ii) Yes
- (iii) No
- (i) No (ii) Yes
- (iii) No
- $E_1 = \{1, 2, 4, 5\}$  (the number is not a multiple of 3);  $E_2 = \{1, 2, 3\}$  (the number of points is at most three);  
 $E_3 = \{3\}$  (the number is odd and a multiple of three);  
 $E_4 = \{1, 2, 5, 6\}$  (the number of points is either odd or a multiple of three);  
 $E_5 = \overline{A\overline{B}} = AB = E_3$ ;  $E_6 = \Phi \cup D = D = S$ .
- $E_1 = \overline{A\overline{B}\overline{C}} \cup \overline{A\overline{B}C} \cup \overline{A\overline{B}C}$ ,  $E_2 = A \cup B \cup C$ ,  
 $E_3 = \overline{A\overline{B}\overline{C}} \cup \overline{A\overline{B}C} \cup \overline{A\overline{B}C}$ ,  $E_4 = E_3 \cup ABC$ ,  
 $E_5 = \overline{A\overline{B}\overline{C}}$ ,  $E_6 = E_4$ ,  $E_7 = \overline{A} \cup \overline{B} \cup \overline{C} = \overline{ABC}$ .
- (i)  $A \cup C = E$ , (ii)  $AC = K$ ,  
 (iii)  $\overline{E} = G$ , (iv)  $G \cup E = E$ ,  
 (v)  $GE = G$ , (vi)  $BD = H$ ,  
 (vii)  $E \cup K = E$ .
- $X = \overline{B}$
- (i)  $S = \{\omega_1, \overline{\omega_1}\omega_2, \overline{\omega_1}\omega_2\omega_3, \dots, \overline{\omega_1} \dots \overline{\omega_7}\omega_8\}$ .

$$(ii) A = \overline{\omega_1}\overline{\omega_2}\omega_3 \cup \overline{\omega_1}\overline{\omega_2}\overline{\omega_3}\omega_4 \cup \dots \cup \overline{\omega_1}$$

$$\dots \overline{\omega_7}\omega_8 = \overline{\omega_1}\overline{\omega_2}$$

$$14. S = \{A_1, \overline{A_1}B_1, \overline{A_1}\overline{B_1}A_2, \dots\},$$

$$A = A_1 \cup \overline{A_1}\overline{B_1}A_2 + \dots,$$

$$B = \overline{A_1}B_1 \cup \overline{A_1}\overline{B_1}\overline{A_2}B_2 + \dots$$

### Practice Problems—C

- 12/365
- 1/3
- $\frac{1}{3}$
- $P(A) = 1/3$ ,  $P(B) = 3/8$ , B is more probable.
- 0.665
- 5/9
- $P(A) = 0.01$ ,  $P(B) = 0.2$ ,  $P(C) = 5/144$
- 5/126
- 1/9, 1/3.
- $\frac{4}{7}$
- (a) 1/2; (b)  $\frac{n-1}{2n}$
- $\left(\frac{3}{4}\right)^7$

### Practice Problems—D

- $1 - \frac{2^n}{3^n}$
- 7 : 5
- 0.3
- 67/91
- (i) 6/55 (ii) 4/11
- (i)  $\frac{7}{29}$  (ii)  $\frac{5}{13}$
- (i)  $\frac{1}{6}$  (ii) No
- $\frac{1}{6}, \frac{1}{2}, \frac{1}{3}$
- 432
- 3/5
- $P(A) = 17/24$ ,  $P(B) = 1/120$ ,  $P(C) = 7/40$
- 21/100



**Practice Problems—E**

4.  $0.1 \leq P(AB) \leq 0.4$   
 5. Inconsistent

**Practice Problems—F**

1. (i) 0.32 (ii) 0.64  
 (iii) 0.98
2. (i)  $\frac{4}{11}$  (ii)  $\frac{4}{5}$   
 (iii)  $\frac{2}{3}$
3. (i)  $\frac{1}{2}$  (ii)  $\frac{3}{7}$   
 (iii)  $\frac{6}{7}$
4. (i)  $\frac{1}{3}$  (ii)  $\frac{1}{9}$
5. (i)  $\frac{1}{2}, \frac{1}{3}$  (ii)  $\frac{1}{2}, \frac{2}{3}$   
 (iii)  $\frac{3}{4}, \frac{1}{4}$
6.  $\frac{5}{9}$  7.  $\frac{1}{15}$
10. 0 11.  $\frac{3}{11}; \frac{1}{3}$
12.  $\frac{1}{8}$
13.  $P(A) = 0.24, P(\bar{A} \bar{B}) = 0.48, P(\bar{A} + \bar{B}) = 0.88$ .
15.  $4/5, 1/4, 3/10$

**Practice Problems—G**

1.  $\frac{25}{102}$  2.  $\frac{44}{91}$
3. (i)  $\frac{1}{8}$  (ii)  $\frac{1}{8}$   
 (iii)  $\frac{3}{8}$

4.  $\frac{11}{850}$
5. (i)  $1/343$  (ii)  $120 / 343$
6.  $\frac{6426}{13800}$  7.  $\frac{7}{8}$
8.  $\frac{2}{5}$  10. 0.596

**Practice Problems—H**

1.  $2/3$  2. 0.85  
 3. 0.84 4. 0.895
6.  $\frac{10}{19}$  7.  $13 : 5$
8.  $\frac{3}{4}$

**Practice Problems—I**

1. A and B are independent.  
 2. A and B are independent.
3. (i)  $P = \frac{1}{10}$  (ii)  $P = \frac{1}{5}$
4. A and B are not independent.  
 5. No 6. 0.3, 0.5, 0.7, 0.5  
 7.  $1/3, 1/2, 2/3$   
 8.  $P(A/B) = 1/3, P(B/A) = 1$ . They are dependent if  $B \neq S$ .
10. C and D are dependent events,  $P(D/C) = 1/4$

**Practice Problems—J**

2.  $1/12, 1/2, 1/2, 1/4, 1/6$   
 3.  $1/4$ .
4. (i)  $\frac{16}{81}$  (ii)  $\frac{20}{81}$   
 (iii)  $\frac{40}{81}$
5. (i)  $\frac{2}{3}$  (ii)  $\frac{1}{2}$
6. (i)  $\frac{1}{5}$  (ii)  $\frac{1}{3}$   
 (iii)  $\frac{1}{2}$

9.  $1/2, 2/3, 1/3, 1/2, 5/6$  13.  $\frac{169}{324}, \frac{155}{324}$   
 14. (i)  $(1-p)p$ ;  
 (ii)  $1 - (1-p)^2 = (2-p)p$ .  
 15. (i) 0.216 (ii)  $1/6$   
 16. 0.22  
 17. (i)  $1 - q_1q_2q_3$   
 (ii)  $1 - (1-p_1p_2p_3)(1-p_4p_5p_6)$

### Practice Problems—K

1. (ii), (iii) and (iv) 2.  $X = 0, 1, 2$ ; yes  
 3.  $X = 6, 4, 2, 0$   
 4. (i) 

|      |               |               |               |
|------|---------------|---------------|---------------|
| X    | 0             | 1             | 2             |
| P(X) | $\frac{4}{9}$ | $\frac{4}{9}$ | $\frac{1}{9}$ |

  
 (ii) 

|      |                 |                 |
|------|-----------------|-----------------|
| X    | 0               | 1               |
| P(X) | $\frac{25}{36}$ | $\frac{11}{36}$ |

  
 5. 

|      |                   |                   |                  |                  |                 |
|------|-------------------|-------------------|------------------|------------------|-----------------|
| X    | 0                 | 1                 | 2                | 3                | 4               |
| P(X) | $\frac{256}{625}$ | $\frac{256}{625}$ | $\frac{96}{625}$ | $\frac{16}{625}$ | $\frac{1}{625}$ |

  
 6. 

|      |                |                |                |
|------|----------------|----------------|----------------|
| X    | 0              | 1              | 2              |
| P(X) | $\frac{9}{16}$ | $\frac{6}{16}$ | $\frac{1}{16}$ |

  
 7. (i)  $k = \frac{1}{10}$  (ii)  $P(X < 3) = \frac{3}{10}$   
 (iii)  $P(X > 6) = \frac{17}{100}$  (iv)  $P(0 < X < 3) = \frac{3}{10}$   
 8. (i)  $k = \frac{1}{6}$   
 (ii)  $P(X < 2) = \frac{1}{2}, P(X \leq 2) = 1, P(X \geq 2) = \frac{1}{2}$

### Practice Problems—L

1.  $\frac{1}{3}$  2.  $\frac{14}{3}$   
 3.  $\text{Var}(X) = 5.833, \text{S.D.} = 2.415$   
 4. 10 nP. coins

5.  $\frac{1}{3}$   
 6. mean = 6; variance = 9  
 7. mean =  $4/5$ ; variance =  $9/25$   
 8. 

|               |               |                |                |
|---------------|---------------|----------------|----------------|
| 1             | 2             | 3              | 4              |
| $\frac{2}{3}$ | $\frac{2}{9}$ | $\frac{2}{27}$ | $\frac{1}{27}$ |

  
 mean =  $\frac{40}{27}$ ; variance =  $\frac{452}{729}$

9. Rs. 1 10.  $\frac{11}{216}$

### Practice Problems—M

1.  $\frac{25}{216}$  2.  $\left(\frac{29}{20}\right)\left(\frac{19}{20}\right)^9$   
 3. (i)  $\frac{3}{32}$  (ii)  $\frac{7}{64}$   
 (iii)  $\frac{63}{64}$   
 4.  $\left(\frac{1}{20}\right)^{20} [{}^{20}C_{12} + {}^{20}C_{13} + \dots + {}^{20}C_{20}]$   
 5. (i)  $\left(\frac{2}{5}\right)^6$  (ii)  $7\left(\frac{2}{5}\right)^4$   
 (iii)  $\frac{864}{3125}$   
 6.  $\frac{5^{10}}{2 \times 6^9}$  7.  $n \geq 4$   
 8.  ${}^{10}C_5(0.24)^5$  10. Eight seats  
 11.  $\frac{10}{3}, \frac{5}{3}$  12. 8; 2  
 13. 0.272  
 14. Approximately 5555 15. 90; 9; 0.023  
 16. 0.2009

### Practice Problems—N

1.  $\frac{1}{2}$  2.  $\frac{9}{13}$   
 3.  $\frac{4}{9}$  4.  $\frac{1}{4}$   
 5.  $\frac{5}{34}$  6.  $\frac{11}{50}$

- |                          |                                             |          |         |          |
|--------------------------|---------------------------------------------|----------|---------|----------|
| 7. $\frac{10}{19}$       | 8. $\frac{32}{41}$                          | 43. C    | 44. C   | 45. D    |
| 9. $\frac{8}{11}$        | 10. $\frac{1}{4}, \frac{1}{2}, \frac{1}{4}$ | 46. A    | 47. D   | 48. A    |
| 11. (i) $\frac{25}{108}$ | (ii) $\frac{9}{25}$                         | 49. A    | 50. A   | 51. BCD  |
| 12. $\frac{14}{29}$      |                                             | 52. AD   | 53. BD  | 54. AD   |
|                          |                                             | 55. ABCD | 56. CD  | 57. ABCD |
|                          |                                             | 58. AB   | 59. ABC | 60. ACD  |

**Practice Problems—O**

- |                                                      |                                             |
|------------------------------------------------------|---------------------------------------------|
| 1. $\frac{1}{3}$                                     | 2. $\frac{40}{41}$                          |
| 3. $\frac{140}{141}$                                 | 8. $1/2$                                    |
| 9. $\frac{3}{5}$                                     | 10. $\frac{1}{2} - \frac{\pi}{16}$          |
| 11. $2/3$                                            | 12. 0.314 approx.                           |
| 13. 139/1152                                         | 14. $\frac{2}{\pi} \arccos \frac{r_1}{r_2}$ |
| 15. $\frac{2}{9} \ln 2 + \frac{1}{12} \approx 0.237$ |                                             |

**Objective Exercises**

- |       |       |       |
|-------|-------|-------|
| 1. B  | 2. D  | 3. B  |
| 4. B  | 5. A  | 6. B  |
| 7. C  | 8. C  | 9. D  |
| 10. C | 11. A | 12. D |
| 13. B | 14. A | 15. A |
| 16. D | 17. A | 18. C |
| 19. D | 20. B | 21. A |
| 22. B | 23. D | 24. A |
| 25. C | 26. A | 27. C |
| 28. C | 29. A | 30. A |
| 31. C | 32. B | 33. A |
| 34. C | 35. C | 36. A |
| 37. C | 38. C | 39. B |
| 40. B | 41. B | 42. C |

- |          |         |          |
|----------|---------|----------|
| 43. C    | 44. C   | 45. D    |
| 46. A    | 47. D   | 48. A    |
| 49. A    | 50. A   | 51. BCD  |
| 52. AD   | 53. BD  | 54. AD   |
| 55. ABCD | 56. CD  | 57. ABCD |
| 58. AB   | 59. ABC | 60. ACD  |
| 61. AD   | 62. BCD | 63. ABCD |
| 64. BCD  | 65. BD  | 66. AD   |
| 67. BCD  | 68. AD  | 69. ACD  |
| 70. ABCD | 71. C   | 72. B    |
| 73. A    | 74. B   | 75. C    |
| 76. A    | 77. B   | 78. D    |
| 79. A    | 80. A   | 81. A    |
| 82. C    | 83. C   | 84. B    |
| 85. D    | 86. D   | 87. A    |
| 88. A    | 89. A   | 90. B    |
| 91. A    | 92. A   | 93. A    |
| 94. B    | 95. A   |          |

96. (A)–(R), (B)–(T), (C)–(P), (D)–(T)

97. (A)–(R), (B)–(P), (C)–(Q), (D)–(T)

98. (A)–(T), (B)–(S), (C)–(P), (D)–(Q)

99. (A)–(R), (B)–(P), (C)–(Q), (D)–(T)

100. (A)–(P), (B)–(S), (C)–(P), (D)–(P)

**Review Exercises for JEE Advanced**

1. (i)  $A + B = B$ ; (ii)  $AB = A$ ;  
 (iii)  $B + C = B$ ; (iv)  $BC = C$ ;  
 (v)  $D + E + F = C$ ; (vi)  $BF = F$ .

The events BF and CF are the same; BC and D are not the same.

2. (i)  $\bar{A} = \{\text{at least one message is distorted}\}$ ;  
 (ii)  $\bar{B} = \{\text{a black or a red ball is drawn}\}$ ;  
 (iii)  $\bar{C} = \{\text{not more than two messages are sent correctly}\}$ ,

- (iv)  $\bar{D} = \{\text{no hits are registered}\};$   
 (v)  $\bar{E} = \{\text{at least one unit must be repaired}\};$   
 (vi)  $\bar{F} = \{\text{the blacks win or the game is drawn}\}.$
3.  $q_n = \frac{n}{2n-1}.$
4.  $P(A) = 1/45, P(B) = 7/45, P(C) = 1/15$
5. (i)  $1/n$  (ii)  $(n-1)/2n$   
 (iii)  $(n-1)/2n$
6.  $\frac{2^n(19-n)!}{19!}$  7. 0.2637
9.  $P(A) = 1/2, P(B) = 3/4, P(C) = 1/2, P(D) = 25/36.$
10.  $P(A) = 25/546, P(B) = 72/273$
11. (a) 0.34, 0.41, (b)  $7/29, 7/12$
12.  $3/28$  13.  $\frac{117}{125}$
14. 0.445 15. 0.452
16.  $P(A) = \frac{2}{3}p^2(3-2p), P(B) = \frac{2}{9}p^3, P(C) = \frac{1}{9}p^3.$
17.  $\frac{1}{3}$
18. (i) A and B, F and B are independent, A and F are dependent.  
 (ii)  $P(BF) = 2/13, P(A/F) = 1/13, P(ABF) = 1/26$   
 (iii)  $P(B+F) = 17/26, P(F-A) = 3/13, P(F-AB) = 7/26.$
19.  $P(A) = 13/15, P(B) = 2/15, P(C) = 2/45, P(D) = 28/45.$
20.  $p_1p_4(1-q_2q_3)$
22.  $1 - (1-p_1^2)(1-p_2)$  23. 0.104
24.  $P_n = \frac{1}{2}[1 + (2P_1 - 1)(2p - 1)^{n-1}]$
25. (a)  $k = 1/10, 0.81, 0.30, 0.70$   
 (b)  $\min x = 4$
26.  $1 - npq_1q^{n-1} - q^n$ , where  $q_1 = 1 - p_1, q = 1 - p.$
27.  $\approx 0.903$  28.  $\frac{2}{3}$
32.  $\left(\frac{a-b}{a}\right)^2$  33.  $1/2$
34.  $\frac{1+3\ell n2}{8}$  35.  $\frac{3\pi-8}{\pi}$

### Target Exercises for JEE Advanced

1.  $P(A) = 1 - \frac{{}^{n-m}C_k}{{}^nC_k}$  if  $k \leq n-m$ ; 1 if  $k > n-m$ ;  
 $P(B) = \frac{{}^3C_1 {}^{n-m}C_{k-1}}{{}^nC_k}$  if  $k \leq n-m+1$ ; 0 if  $k > n-m+1$ ;  
 $P(C) = \frac{{}^nC_{k_1} {}^{n-m}C_{k-k_1}}{{}^nC_k}$  if  $k \leq n-m+k_1$ ; 0 if  $k > n-m+k_1$
2.  $P(A) = 10^{-8}, P(B) = 10^{-7},$   
 $P(C) = 28 {}^9P_6 \times 10^{-7} \approx 0.17,$   
 $P(D) = 378 {}^8P_4 \times 10^{-6} \approx 0.64,$   
 $P(E) = 56 {}^9P_5 \times 10^{-7} \approx 0.08,$   
 $P(F) = 10^{-8} \times {}^{10}C_3 \times 3^8 \approx 0.008$
3.  $P(A) = 1/64, P(B) = 3/16, P(C) = 3/32, P(D) = 29/32$
4. (a)  $42/95$  (b)  $81/190$
5.  $P(A) = \frac{2n}{(2n)!}, P(B) = \frac{1}{1 \times 3 \times \dots \times (2n-1)}$
6.  $P(A) = \frac{1}{m^{n-1}}, P(B) = \left(1 - \frac{1}{m}\right)^n,$   
 $P(C) = \frac{{}^nP_{m_1} (m-m_1)^{n-m_1}}{m^n}, P(D) = \frac{(m-1)^{n-n_1}}{m^{n-1}}$
9. (i)  $\frac{1}{216}$  (ii)  $\frac{1}{36}$   
 (iii)  $\frac{125}{216}$  (iv)  $\frac{1}{36}$
10.  $7/23$  11.  $\frac{1}{6}, \frac{1}{3}, \frac{1}{2}$
12. B and C are independent.
13. (i) 0.736, (ii) 0.033
14.  $\frac{2109}{2530} \approx 0.8336$
15.  $P(A) = 1 - \sum_{m=0}^{k-1} {}^{2k}C_m r^m (1-r)^{2k-m^3}$

16.  $P(B/A_2) = 1 - (p_3^2 + 2p_2p_3)$ ,  $P(B/A_3) = 1 - 3p_2p_3^2$ ,  $P(B/A_4) = 1$

18.  $n = \left\lceil \frac{\ln(1 - p_2)}{\ln(1 - p_1)} + 1 \right\rceil$ , where  $[x]$  is the integral part of number  $x$ ,  $n(0.3; 0.9) = 7$ ,  $n(0.3; 0.95) = 9$

19.  $n \geq 25$

20.  $P(A) = 1 - (1 - pp_1)^{n-1}$ ,  $P(B) = 1 - (1 - pp_1)^{n-k}$

21. (i)  $p_1 = 36/67$ ,  $p_2 = 31/67$ ,  $p_1 : p_2 = 1.16$

(ii)  $P_1 = \frac{36^m}{36^m + 31^m}$ ,  $P_2 = \frac{31^m}{36^m + 31^m}$ ,

$$\frac{P_1}{P_2} = \left( \frac{36}{31} \right)^m$$

22. (i)  $(1 - q_1q_2)(1 - q_3q_4)$

(ii)  $p_5(1 - q_1q_2)(1 - q_3q_4) + q_5(p_1p_3 + p_2p_4 - p_1p_3p_2p_4)$

23.  $(p(m+n) - p(m))/(1 - p(m))$ .

25. (a)  $\frac{23}{45}$  (b)  $\frac{1}{45}$

26. (i)  $1/6$  (ii)  $7/9$

27.  $1/2$

28. (a)  $p_1 = 3/5$  (b)  $p_6 = 1/3$

(c)  $p_5 = 6/11$

29.  $P(A_1) \approx 0.285$ ,  $P(A_2) \approx 0.0714$

30.  $\approx 0.6084$ .

31.  $P(S_1) \approx 0.8677$ ,  $P(S_2) \approx 0.1052$ ,  $P(S_2) \approx 0.0271$

32.  $\frac{1}{4}p(1-p)^2$ .

34.  $P(A) = \begin{cases} a^2, & 0 < a \leq 1 \\ 1, & 1 < a \end{cases}$ ;  $P(B) = a(2-a)$ ;

$$P(C) = \begin{cases} \frac{\pi a^2}{4} & \text{if } 0 < a \leq 1, \\ \sqrt{a^2 - 1} + a^2 \left( \frac{\pi}{4} - \arccos \frac{1}{a} \right) & \text{if } 1 < a \leq \sqrt{2} \\ 1 & \text{if } \sqrt{2} < a \end{cases}$$

$$P(D) = \begin{cases} a(1 - \ln a) & \text{if } 0 < a \leq 1, \\ 1 & \text{if } 1 < a \end{cases}$$

35.  $P(A) = 2/3$ ,  $P(B) = 1/12$ .

### Previous Years' Questions (JEE Advanced)

1.  $\frac{5}{21}$

2.  $P(A) = P(B)$

3.  $1/9$

4.  $\frac{1}{3} \leq p \leq \frac{1}{2}$

5.  $32/55$

6.  $2/5$

7.  $2/7$

8.  $11/16$

9.  $1/36$

10.  $0.25$

11.  $11/40$

12. F

13. F

14. D

15. A

16. B

17. C

18. D

19. B

20. C

21. C

22. C

23. D

24. B

25. A

26. A

27. B

28. C

29. A

30. A

31. A

32. D

33. A

34. B

35. A

36. D

37. A

38. D

39. A

40. C

41. D

42. C

43. C

44. A

45. ACD

46. ABC

47. BCD

48. AC

49. AD

50. CD

51. AD

52. BC

53. A, B

54. B, D

55. A

56. A, B

57.  $\frac{1}{1260}$

58. (i)  $\frac{1}{132}$

(ii)  $\frac{1}{462}$

59.  $0.6976$

60. No

63.  $\frac{1}{2}$  and  $\frac{1}{3}$  or  $\frac{1}{3}$  and  $\frac{1}{2}$

64.  $13.9\%$

65.  $1/5$

66.  $99/1900$

67.  $0.37$

68.  $23/30$

69.  $1 - \frac{10(N+2)}{N+7} C_5$

70. best of three games

71.  $(3^n - 2^{n+1} + 1) / (4^n - 2^{n+1} + 1)$

72.  $24/29$

73. A, B, C are pairwise independent but A, B, C are dependent.

74.  $97/(25)^4$

75.  $193/792$

76.  $0.63$

77.  ${}^{14}P_{12}, 4 \times 3! \times {}^{11}P_9, 1/91$

78. (a)  $1/2$ , (b)  $8/15$

79.  $31/50$

**4.140** *Comprehensive Algebra Vol-2*

$$80. \alpha = \frac{p}{1-(1-p)^3}, \beta = \frac{(1-p)p}{1-(1-p)^3}, \gamma = \frac{(1-p)^2 p}{1-(1-p)^3}$$

$$81. 4/35$$

$$82. 1/2 \text{ \& } 1/3 \text{ or } 1/3 \text{ \& } 1/2$$

$$84. \frac{m}{m+n}$$

$$85. ({}^6C_3) \left( \frac{1}{2} \right)^n$$

$$86. \frac{9m}{m+8N}$$

$$87. 2P^2 - P^3$$

$$88. 1/2$$

$$90. \frac{{}^{10}C_1 \times {}^2C_2}{{}^{12}C_2} \times \frac{{}^{12}C_2 \times {}^6C_4}{{}^{18}C_6} \times \frac{{}^{11}C_1 \times {}^1C_1}{{}^{12}C_2} \times \frac{{}^{12}C_1 \times {}^6C_5}{{}^{18}C_6}$$

$$91. \frac{1}{7}$$

$$92. B$$

$$93. A$$

$$94. B$$

$$95. A$$

$$96. B$$

$$97. A$$

$$98. B$$

$$99. D$$

$$100. D$$

$$101. B$$